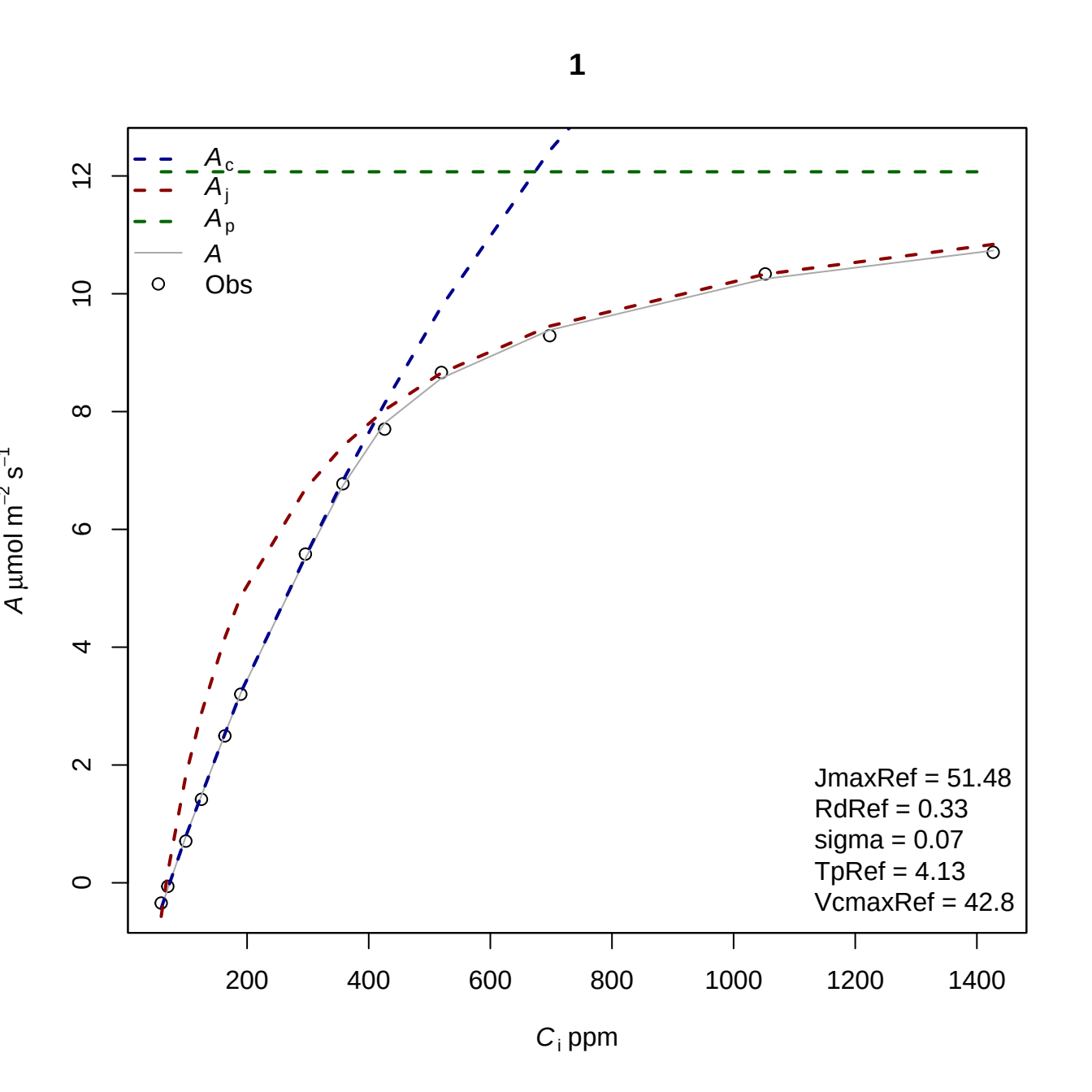
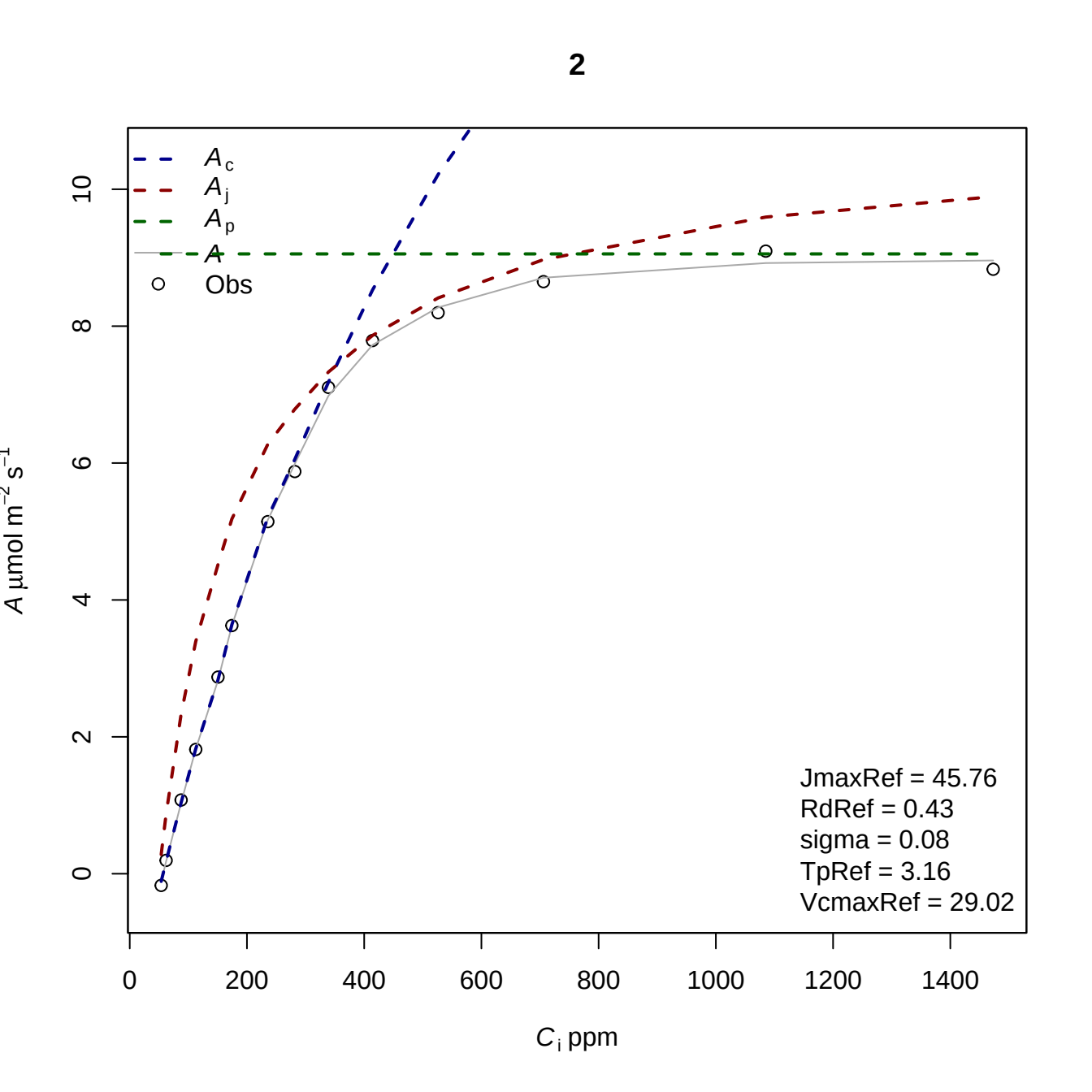
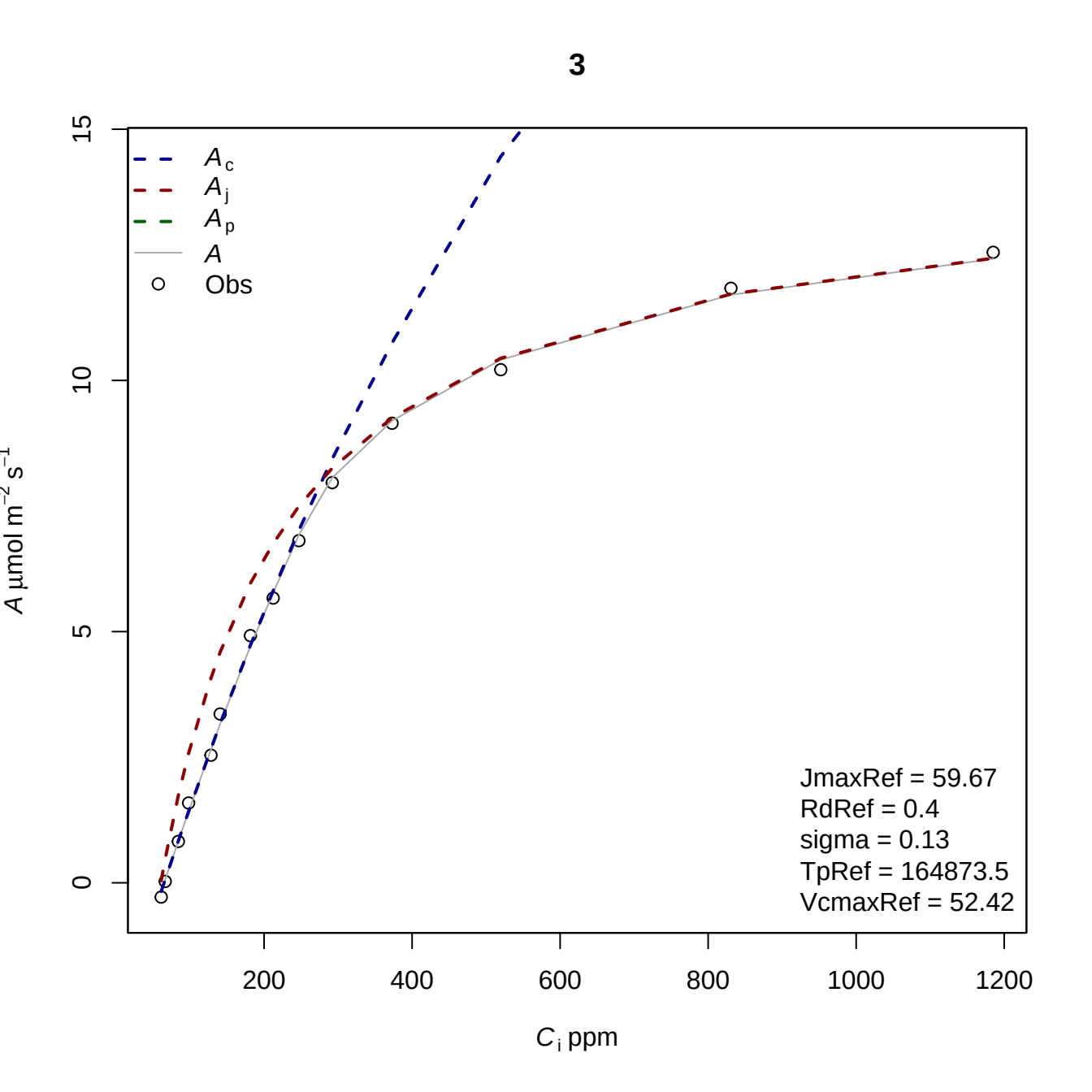
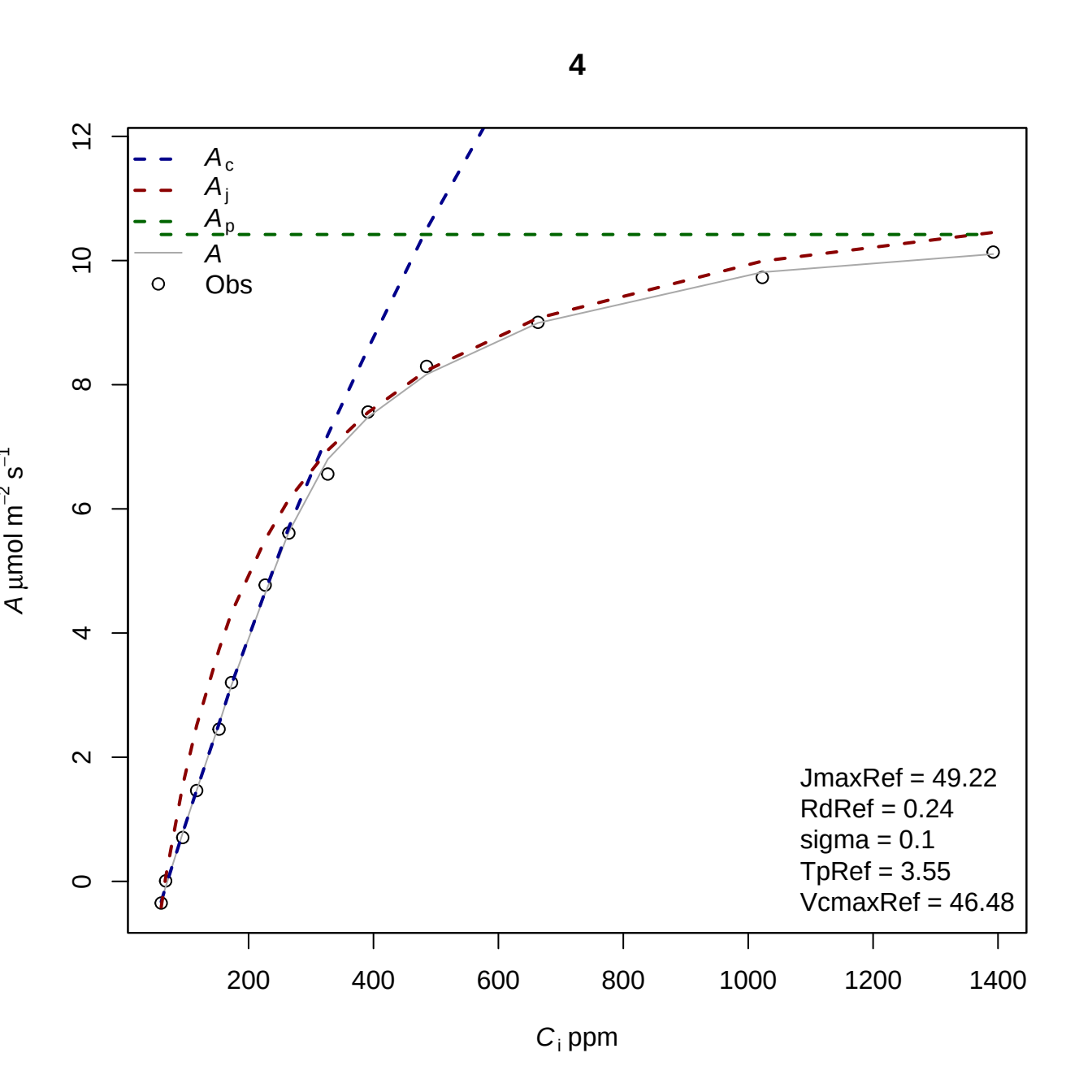
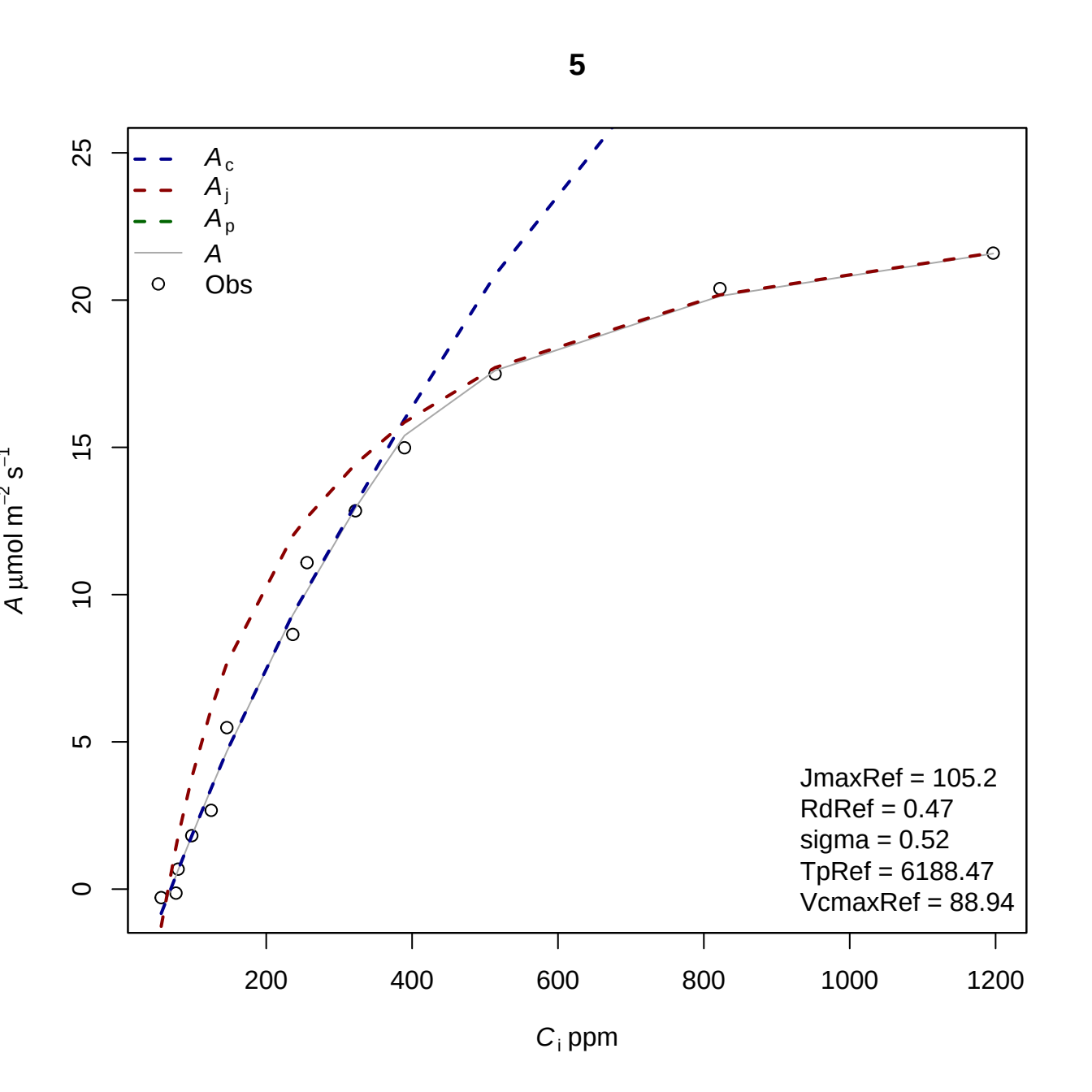
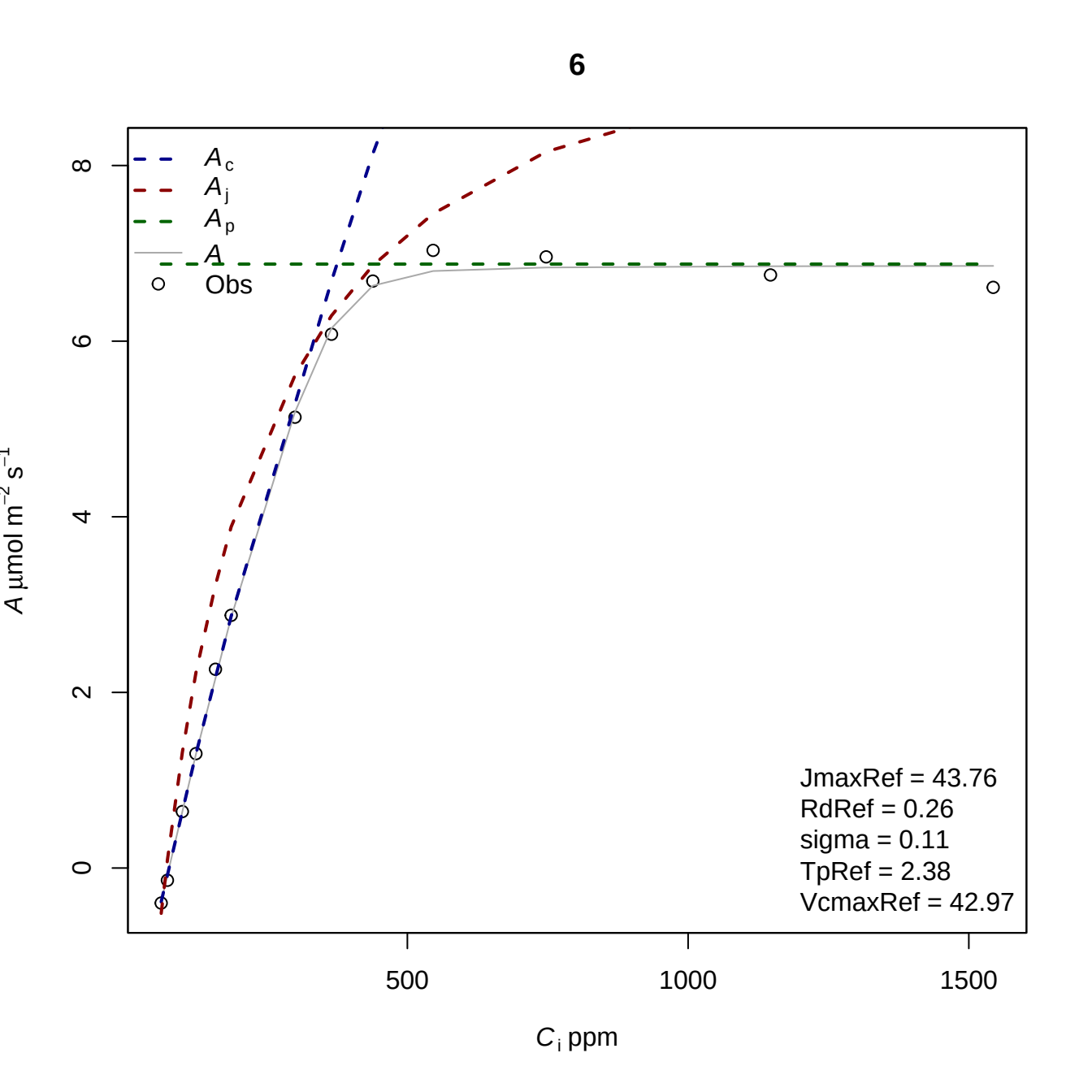

1
A_c
A_j
A_p
A
Obs
JmaxRef = 51.48
RdRef = 0.33
sigma = 0.07
TpRef = 4.13
VcmaxRef = 42.8
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
200
400
600
800
1000
1200
1400

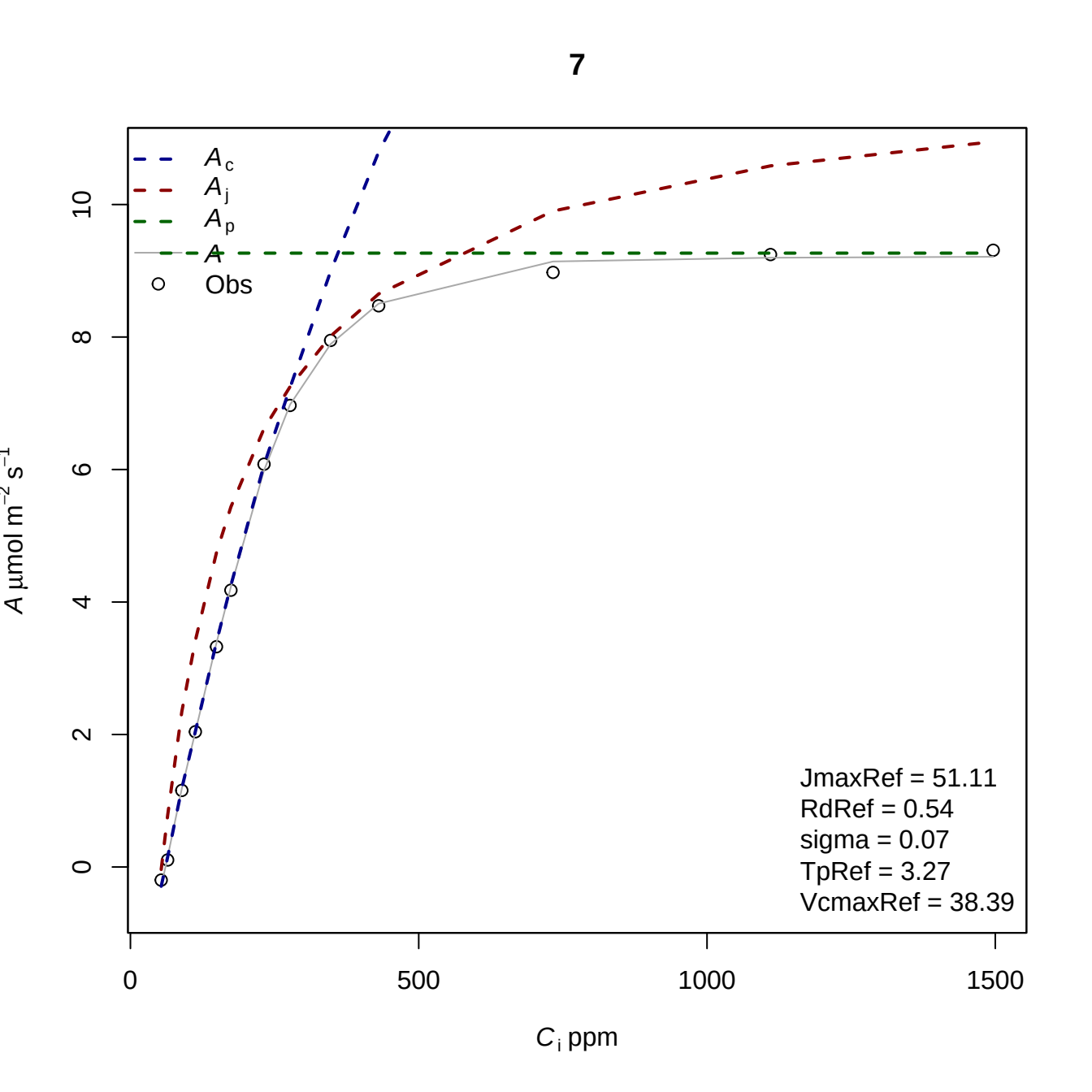

2
A_c
A_j
A_p
A
Obs
JmaxRef = 45.76
RdRef = 0.43
sigma = 0.08
TpRef = 3.16
VcmaxRef = 29.02
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

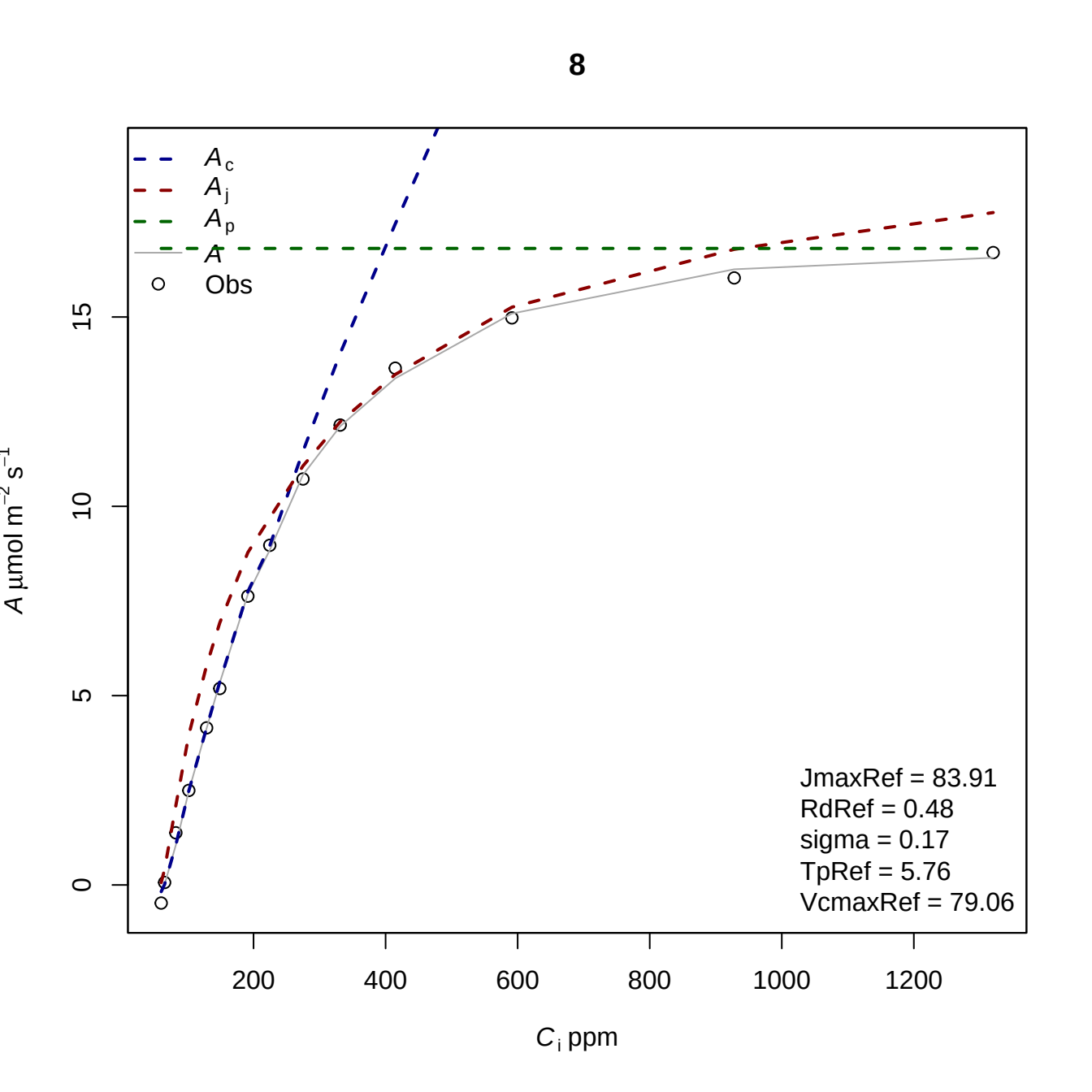

3
A_c
A_j
A_p
A
Obs
JmaxRef = 59.67
RdRef = 0.4
sigma = 0.13
TpRef = 164873.5
VcmaxRef = 52.42
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
200
400
600
800
1000
1200

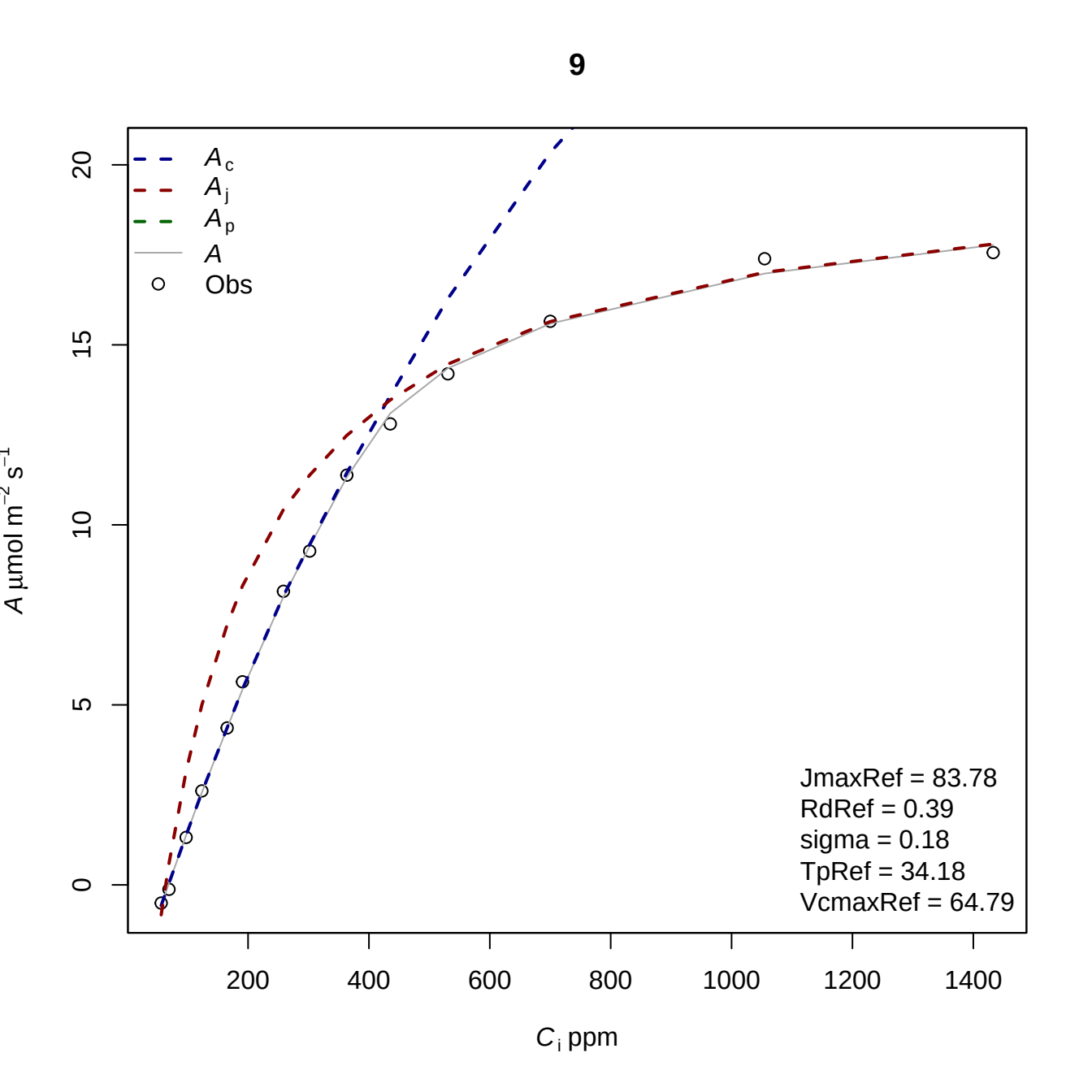

4
A_c
A_j
A_p
A
Obs
JmaxRef = 49.22
RdRef = 0.24
sigma = 0.1
TpRef = 3.55
VcmaxRef = 46.48
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
200
400
600
800
1000
1200
1400

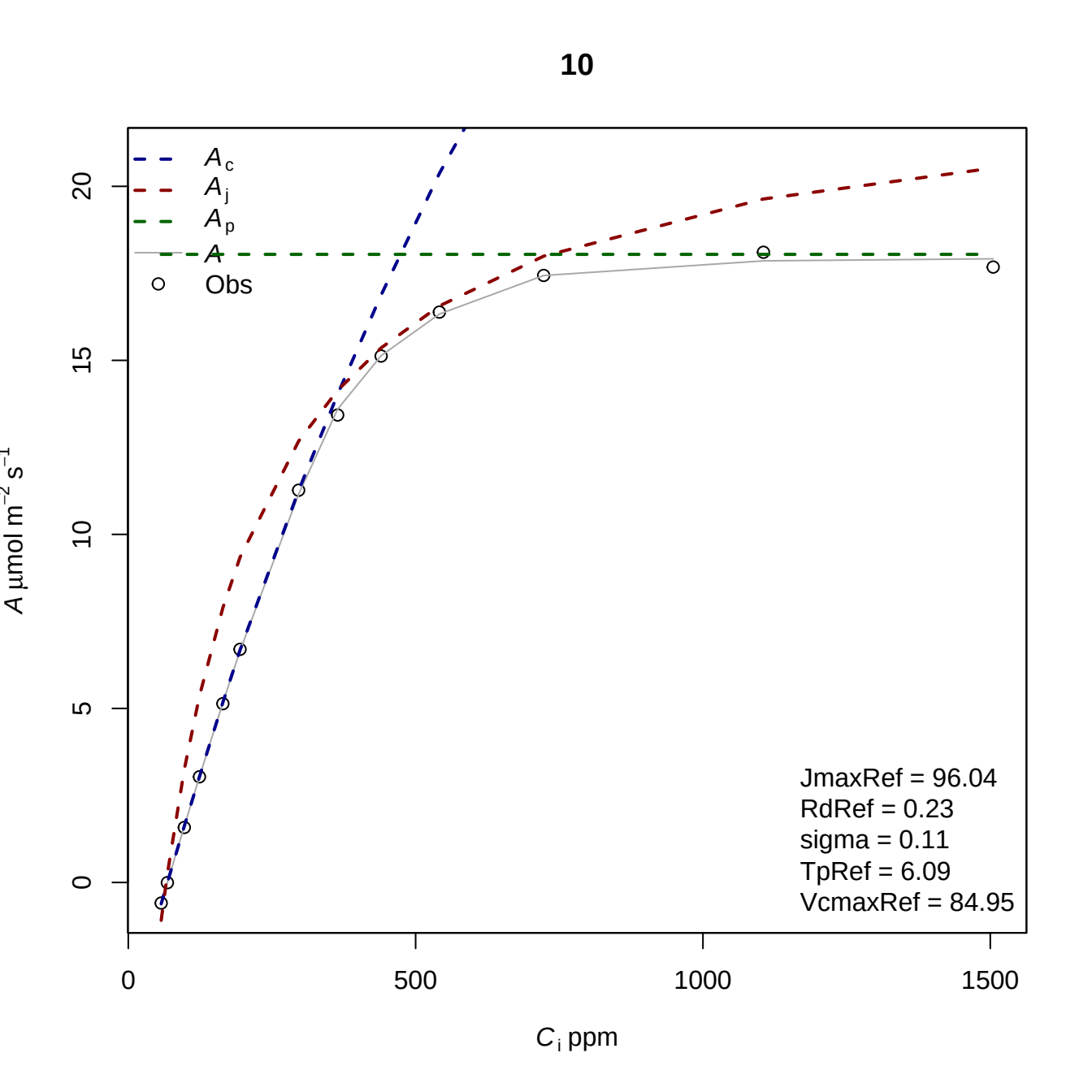

5
A_c
A_j
A_p
A
Obs
JmaxRef = 105.2
RdRef = 0.47
sigma = 0.52
TpRef = 6188.47
VcmaxRef = 88.94
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
200
400
600
800
1000
1200

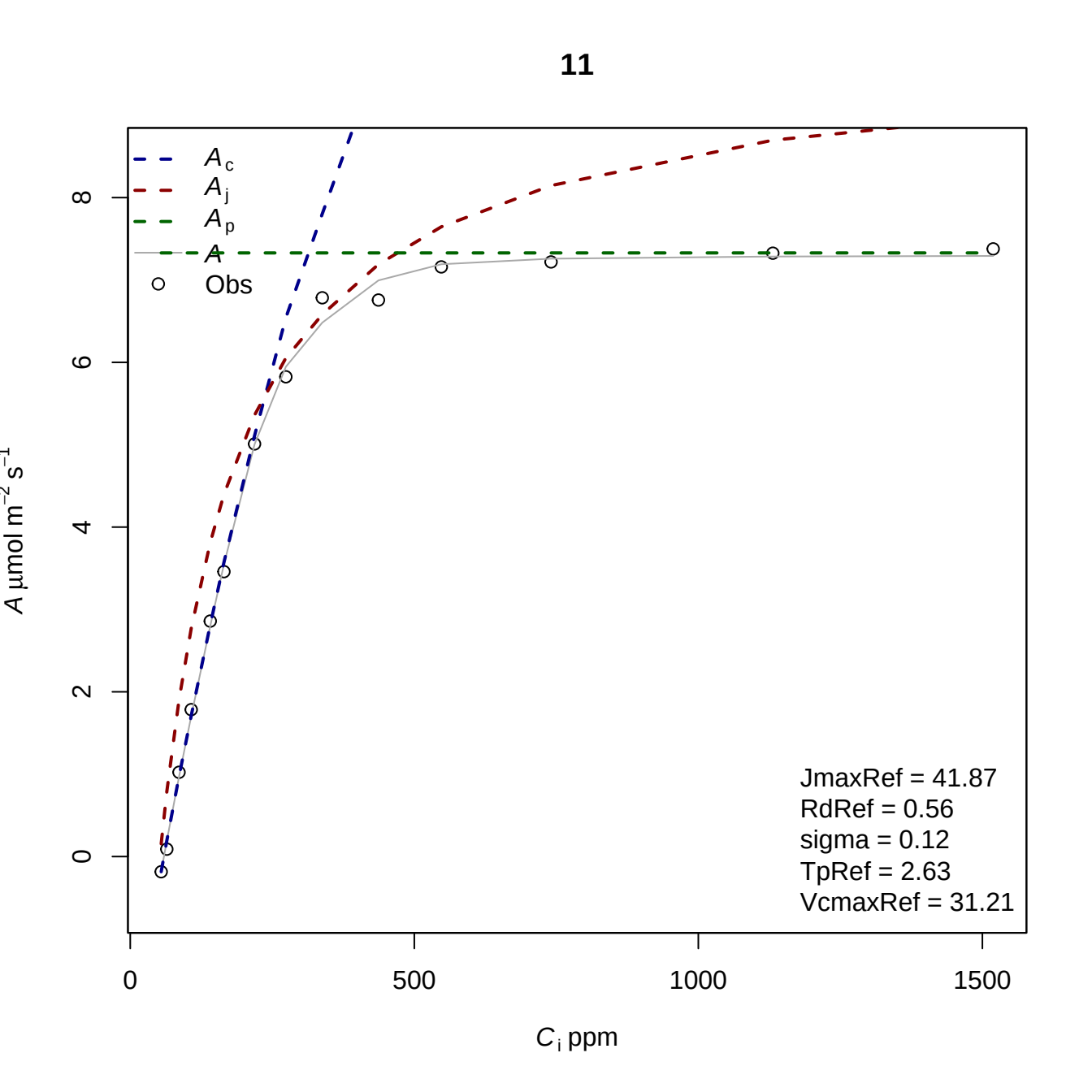

6
A_c
A_j
A_p
A
Obs
JmaxRef = 43.76
RdRef = 0.26
sigma = 0.11
TpRef = 2.38
VcmaxRef = 42.97
0
2
4
6
8
500
1000
1500
A µmol m^{-2} s^{-1}
C_i ppm

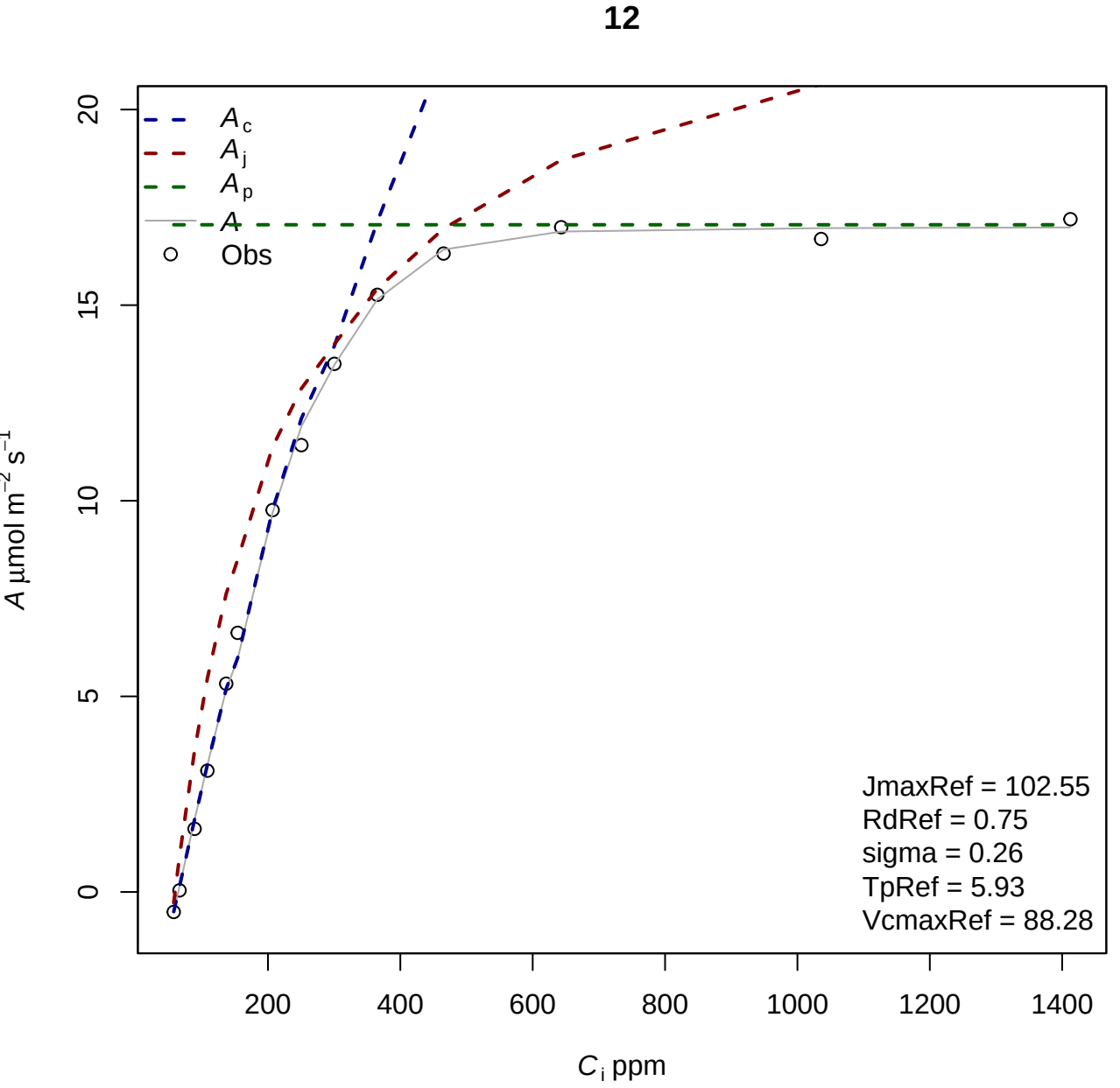

7
A_c
A_j
A_p
A
Obs
JmaxRef = 51.11
RdRef = 0.54
sigma = 0.07
TpRef = 3.27
VcmaxRef = 38.39
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
0
500
1000
1500

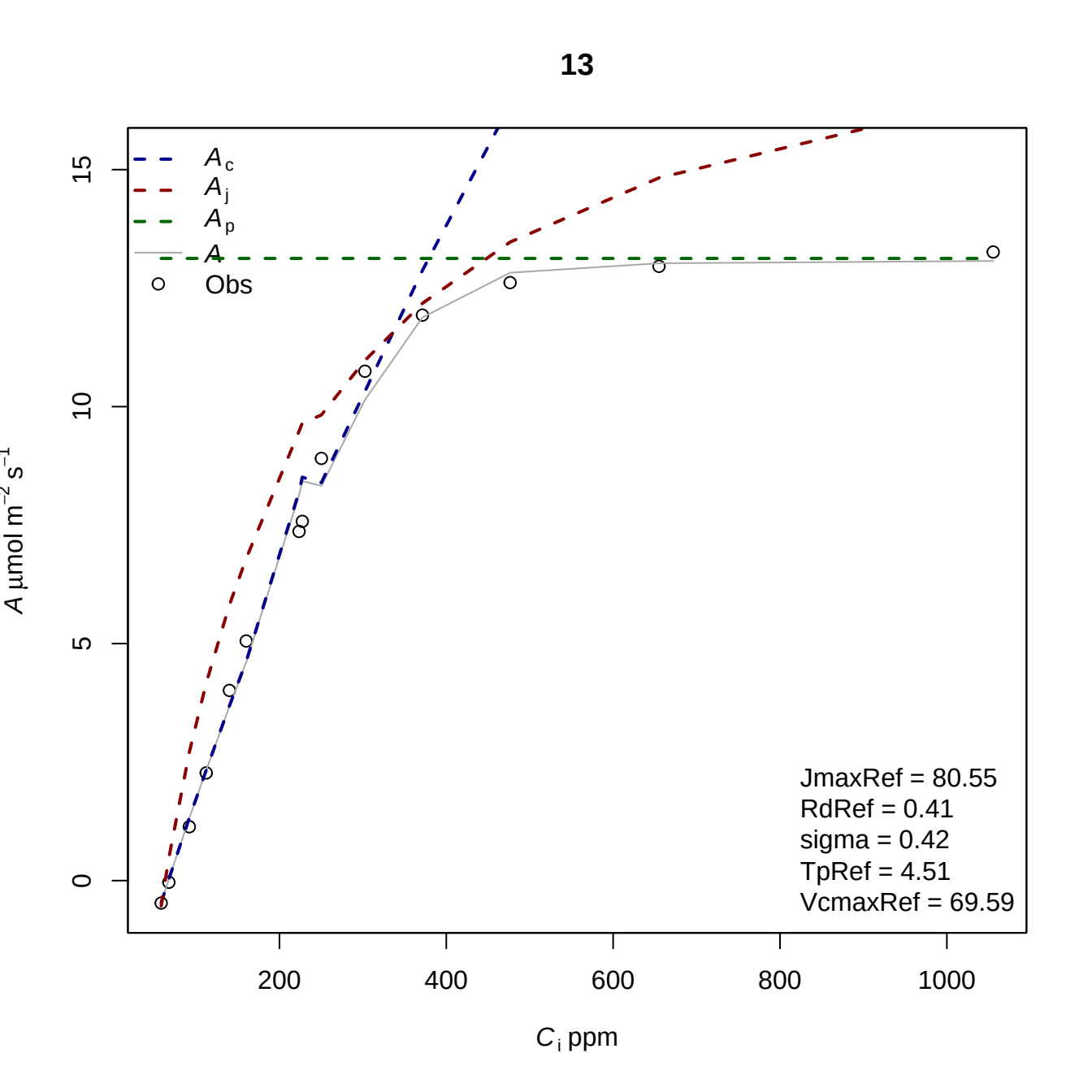

8
A_c
A_j
A_p
A
Obs
JmaxRef = 83.91
RdRef = 0.48
sigma = 0.17
TpRef = 5.76
VcmaxRef = 79.06
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
200
400
600
800
1000
1200

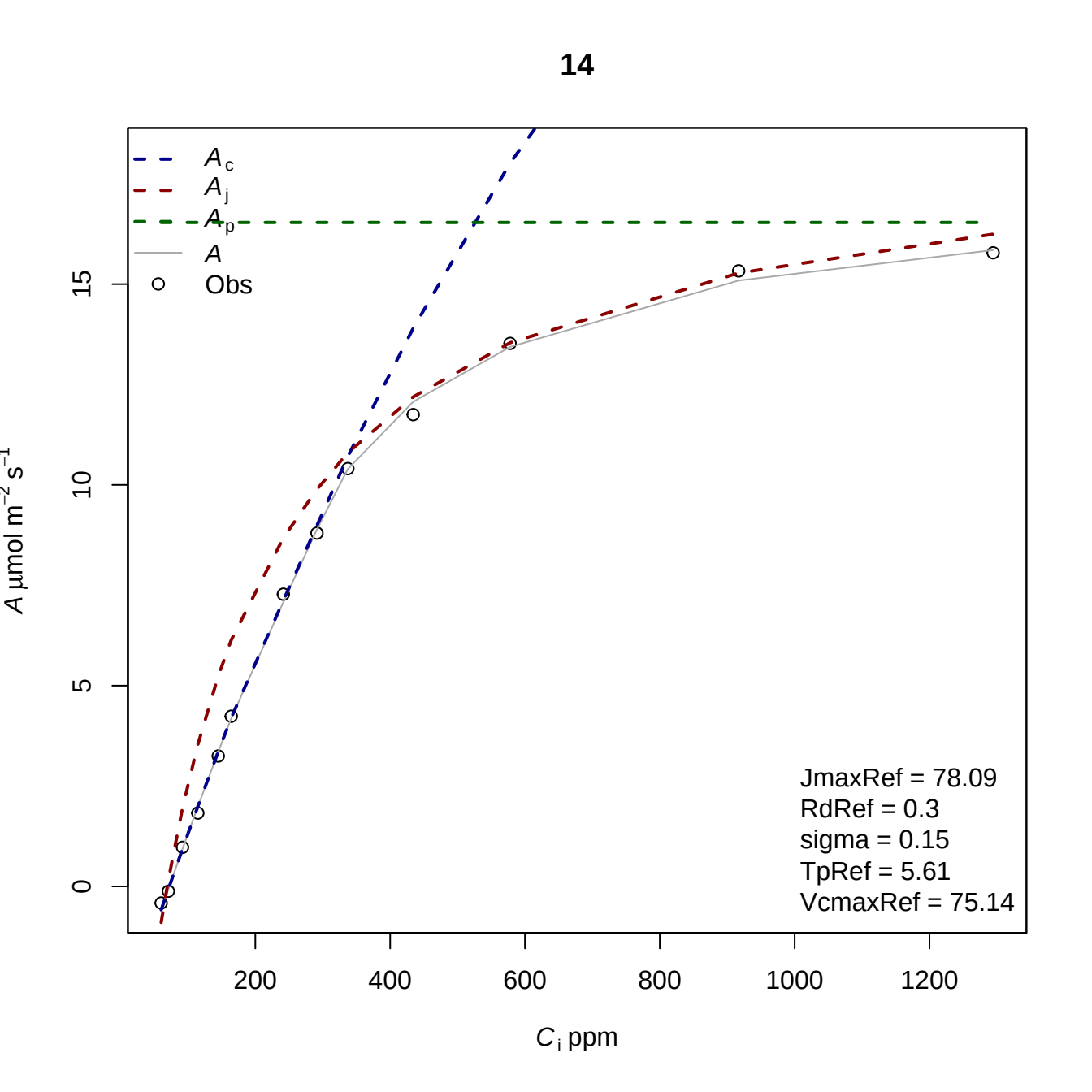

9
A_c
A_j
A_p
A
Obs
JmaxRef = 83.78
RdRef = 0.39
sigma = 0.18
TpRef = 34.18
VcmaxRef = 64.79
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
200
400
600
800
1000
1200
1400

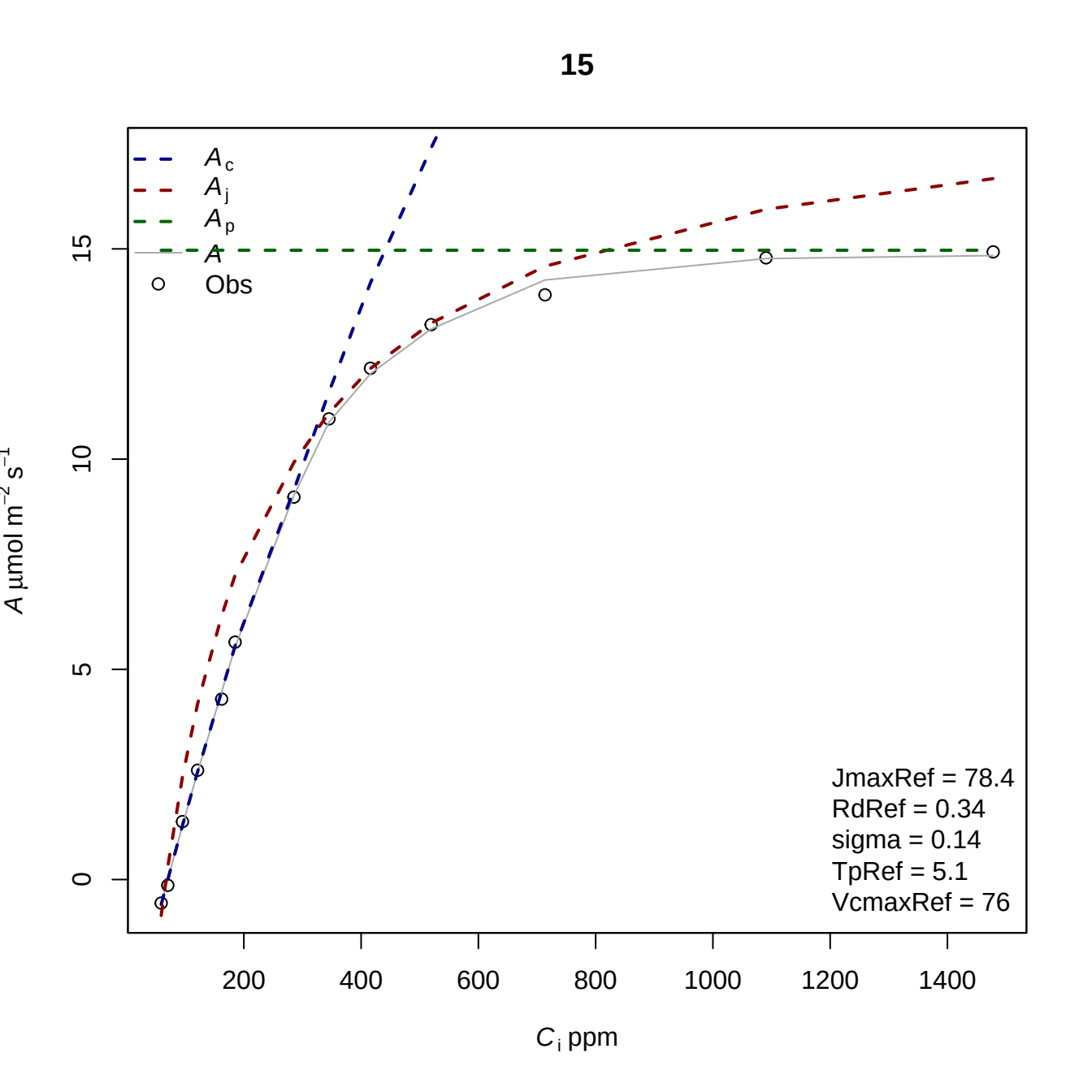

10
A_c
A_j
A_p
A
Obs
JmaxRef = 96.04
RdRef = 0.23
sigma = 0.11
TpRef = 6.09
VcmaxRef = 84.95
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
C_i ppm

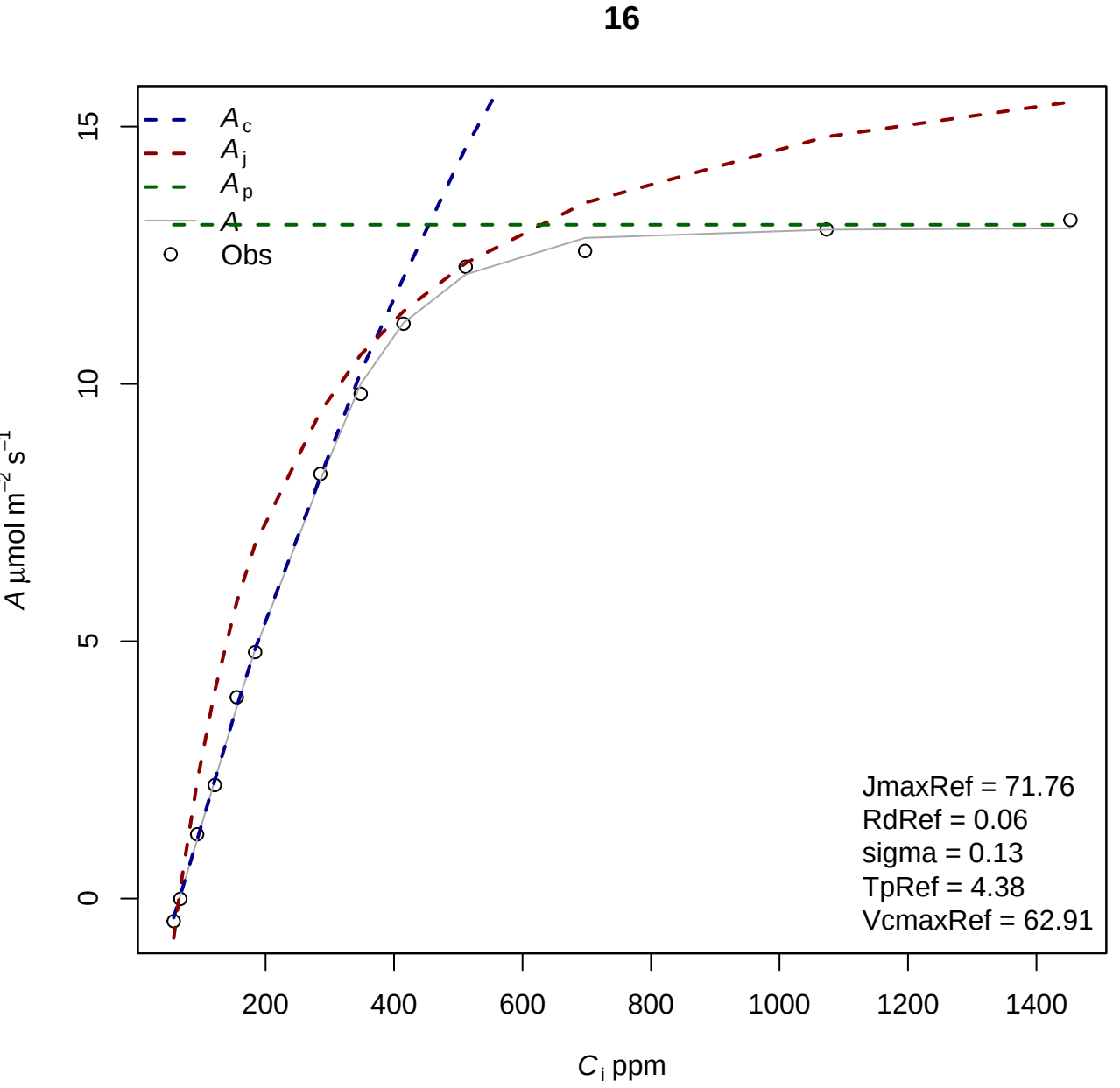

11
A_c
A_j
A_p
A
Obs
JmaxRef = 41.87
RdRef = 0.56
sigma = 0.12
TpRef = 2.63
VcmaxRef = 31.21
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
C_i ppm

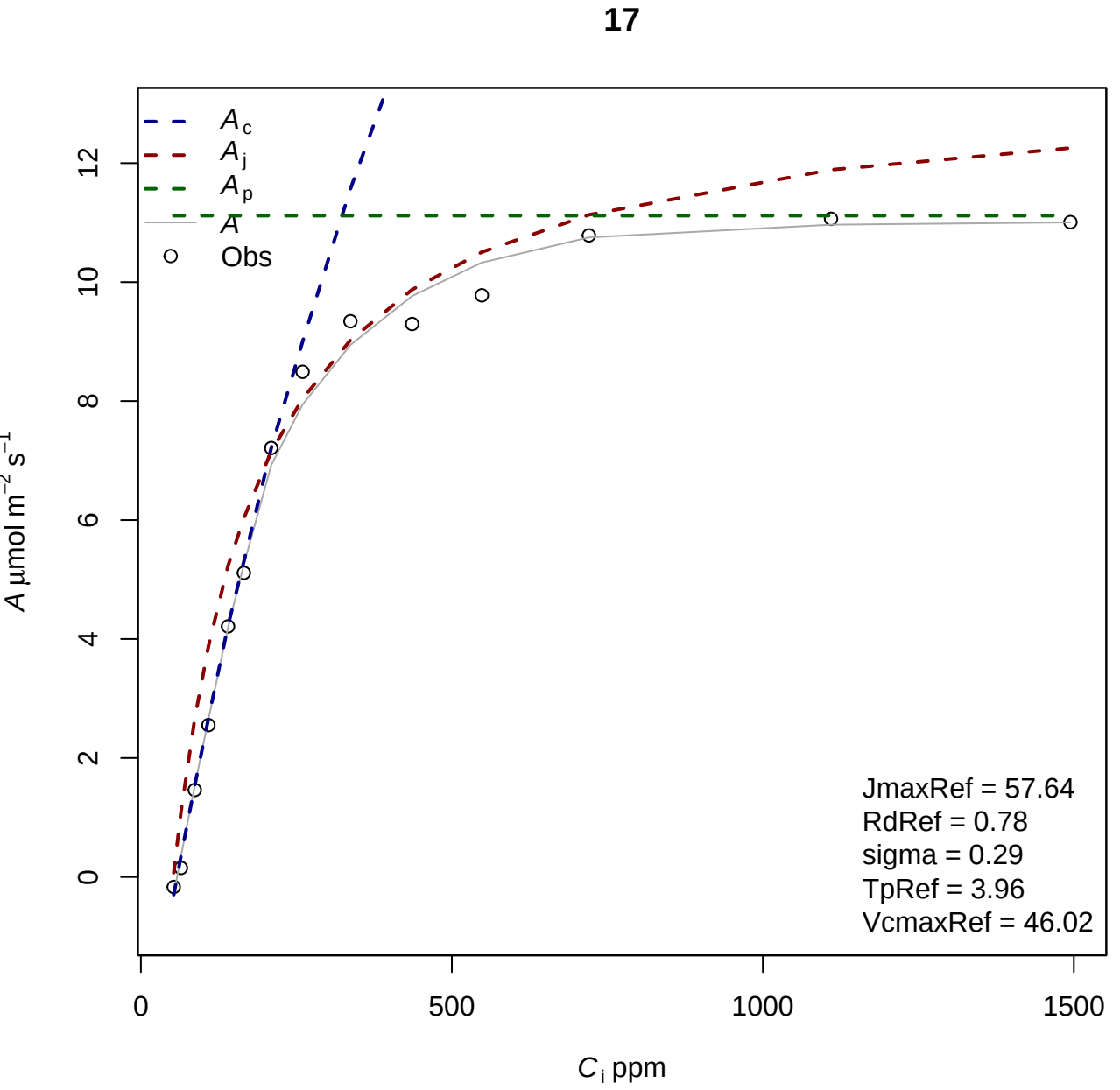

12
A_c
A_j
A_p
A
Obs
JmaxRef = 102.55
RdRef = 0.75
sigma = 0.26
TpRef = 5.93
VcmaxRef = 88.28
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
200
400
600
800
1000
1200
1400

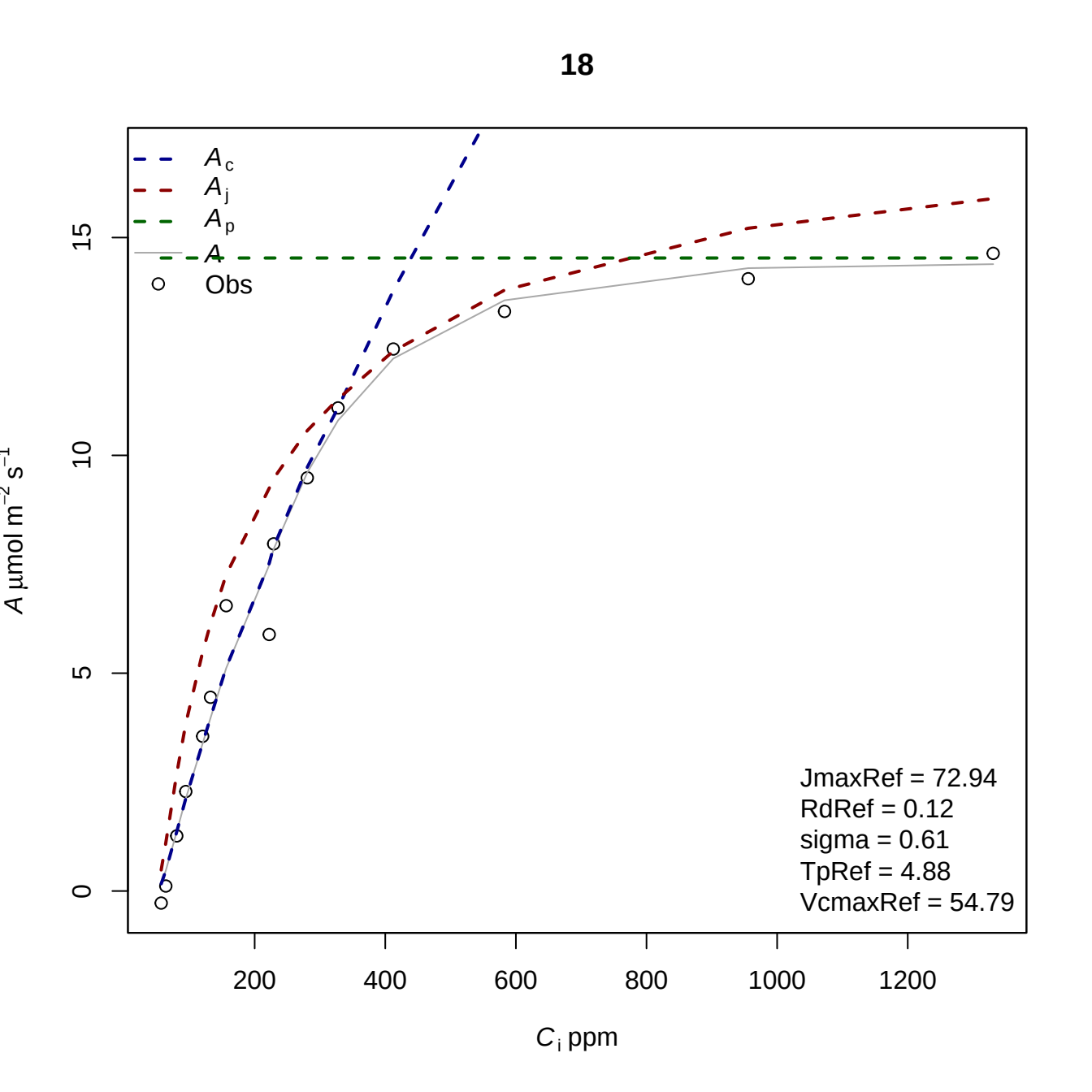

13
A_c
A_j
A_p
A
Obs
JmaxRef = 80.55
RdRef = 0.41
sigma = 0.42
TpRef = 4.51
VcmaxRef = 69.59
0
5
10
15
200
400
600
800
1000
A µmol m^{-2} s^{-1}
C_i ppm

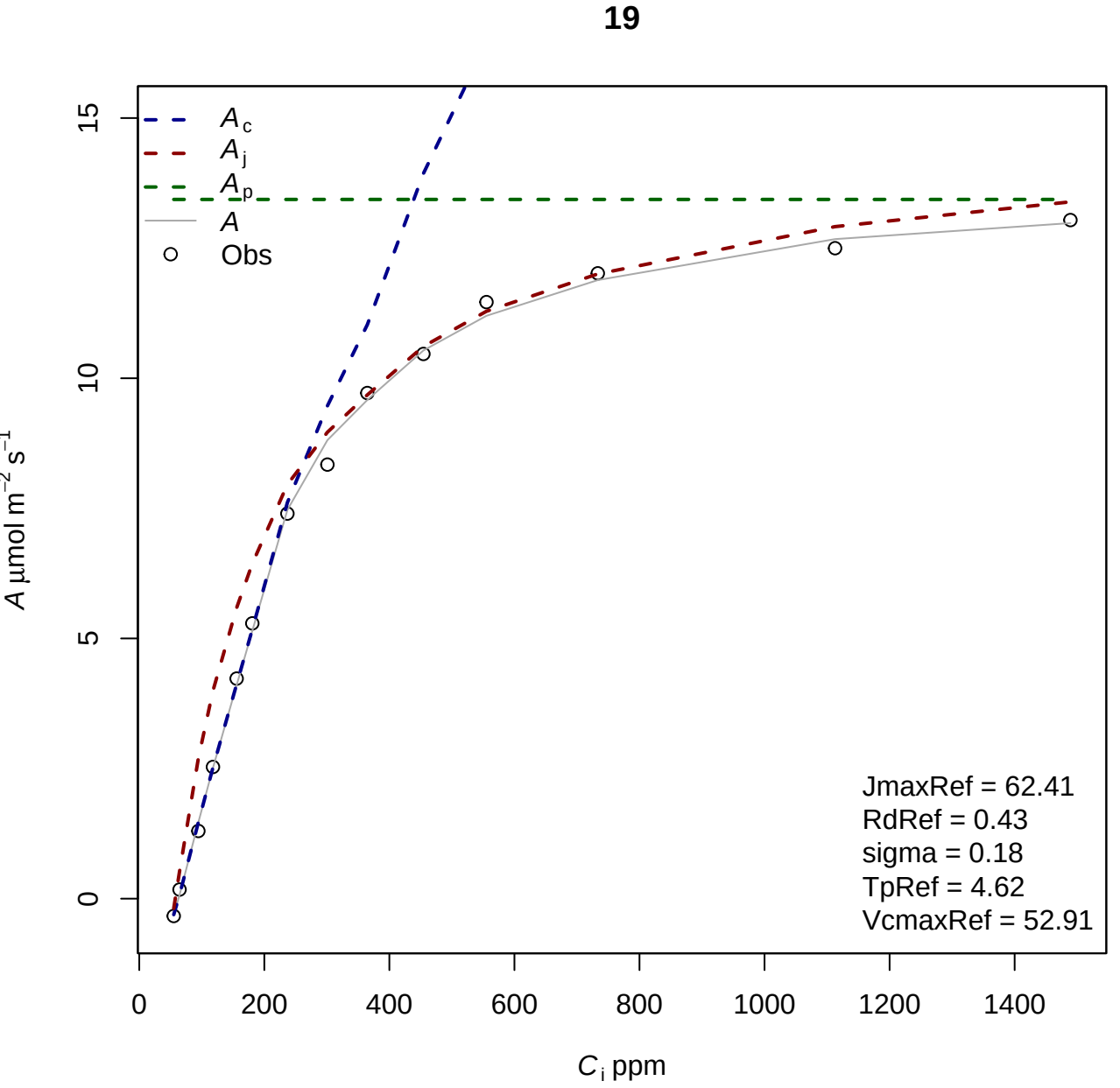

14
A_c
A_j
A_p
A
Obs
JmaxRef = 78.09
RdRef = 0.3
sigma = 0.15
TpRef = 5.61
VcmaxRef = 75.14
0
5
10
15
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

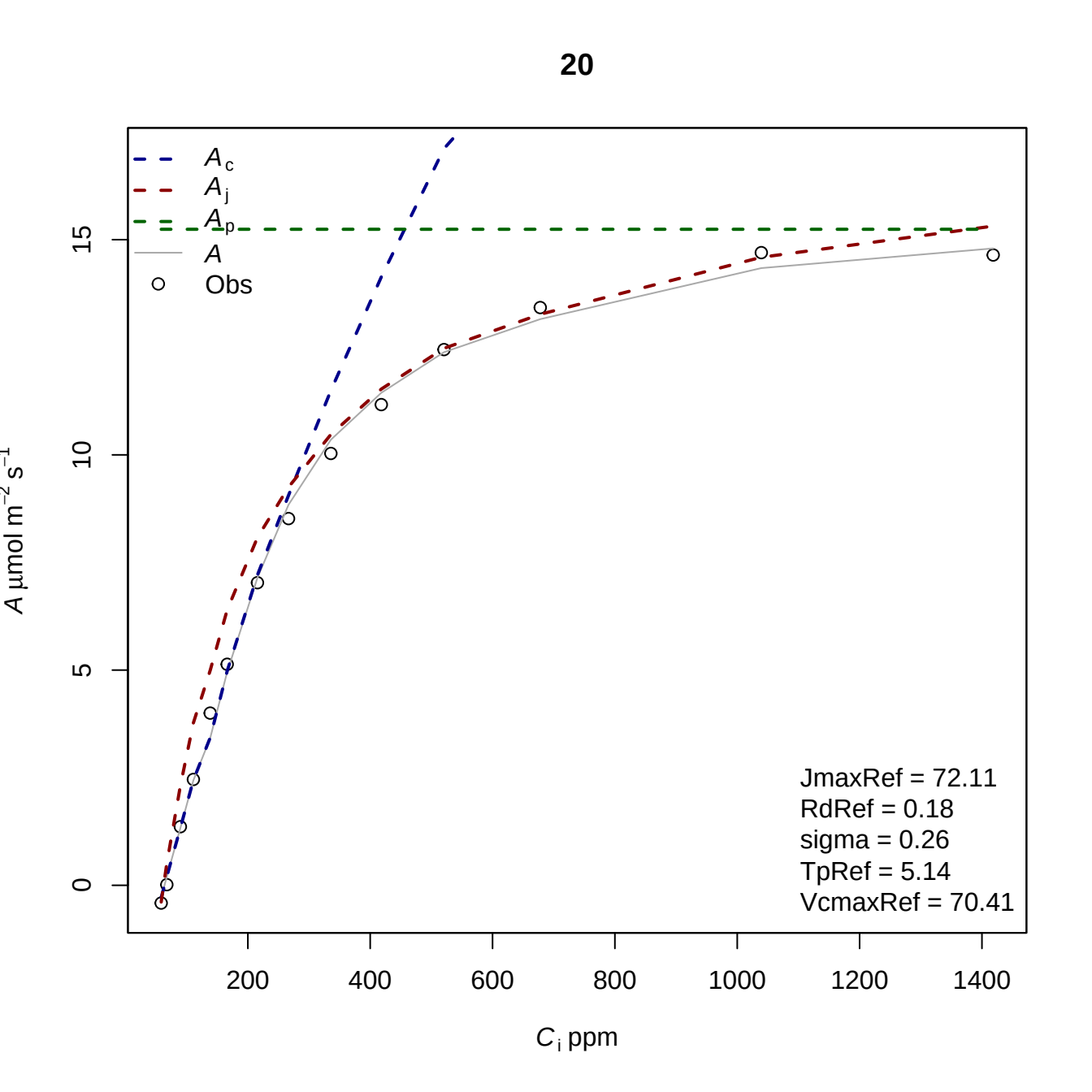

15
A_c
A_j
A_p
A
Obs
JmaxRef = 78.4
RdRef = 0.34
sigma = 0.14
TpRef = 5.1
VcmaxRef = 76
0
5
10
15
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
C_i ppm

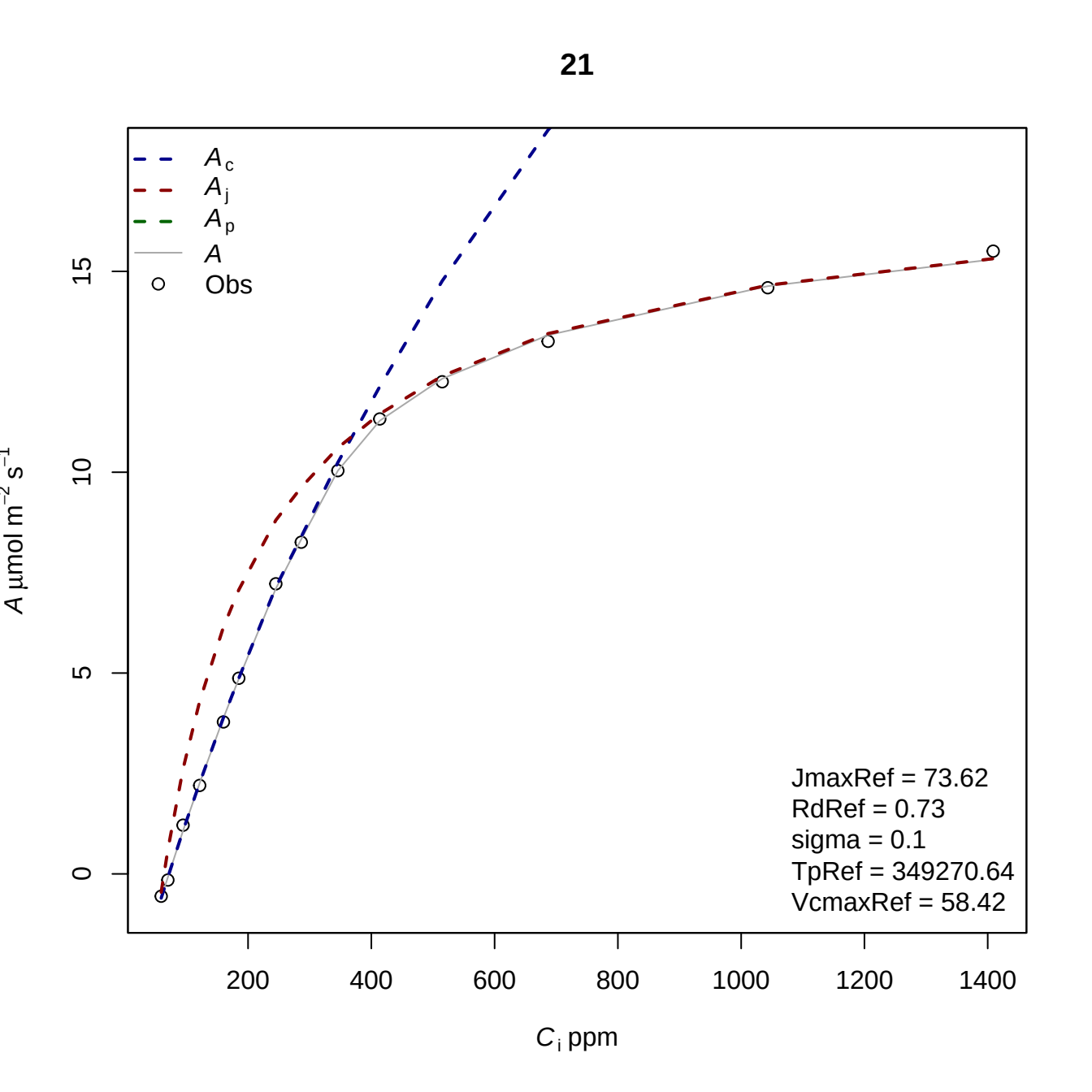

16
A_c
A_j
A_p
A
Obs
JmaxRef = 71.76
RdRef = 0.06
sigma = 0.13
TpRef = 4.38
VcmaxRef = 62.91
0
5
10
15
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
C_i ppm

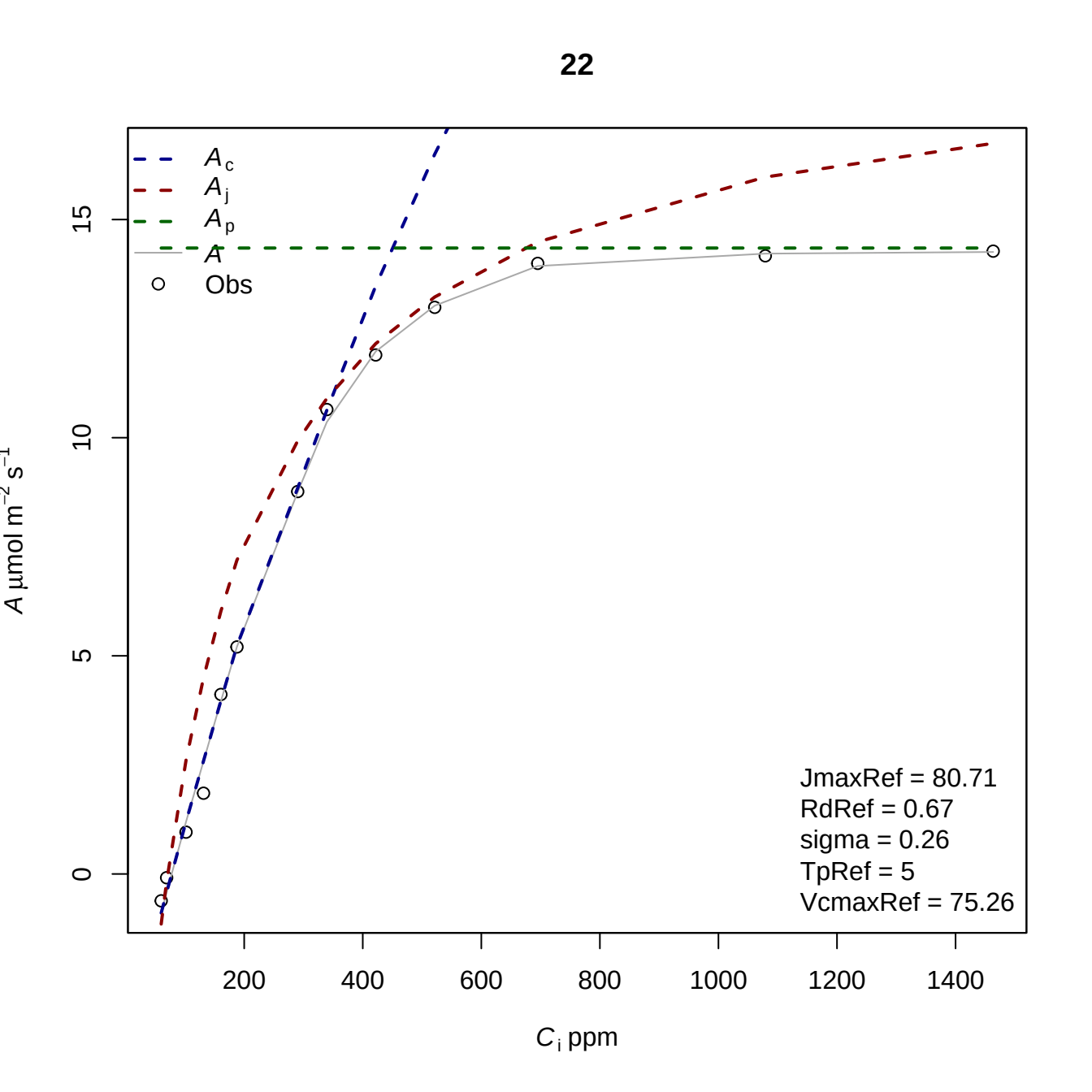

17
A_c
A_j
A_p
A
Obs
JmaxRef = 57.64
RdRef = 0.78
sigma = 0.29
TpRef = 3.96
VcmaxRef = 46.02
A µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
C_i ppm

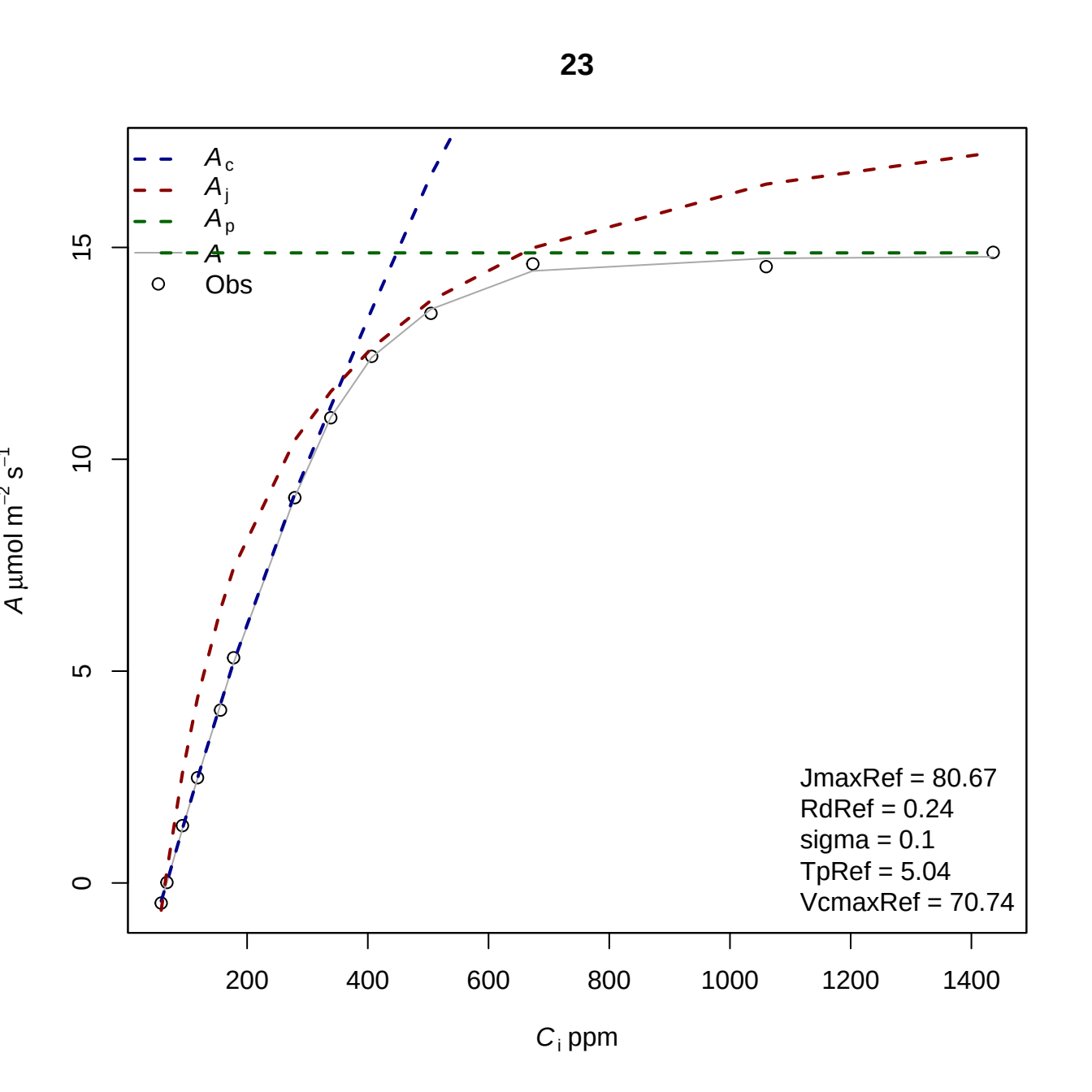

18
A_c
A_j
A_p
A
Obs
JmaxRef = 72.94
RdRef = 0.12
sigma = 0.61
TpRef = 4.88
VcmaxRef = 54.79
0
5
10
15
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

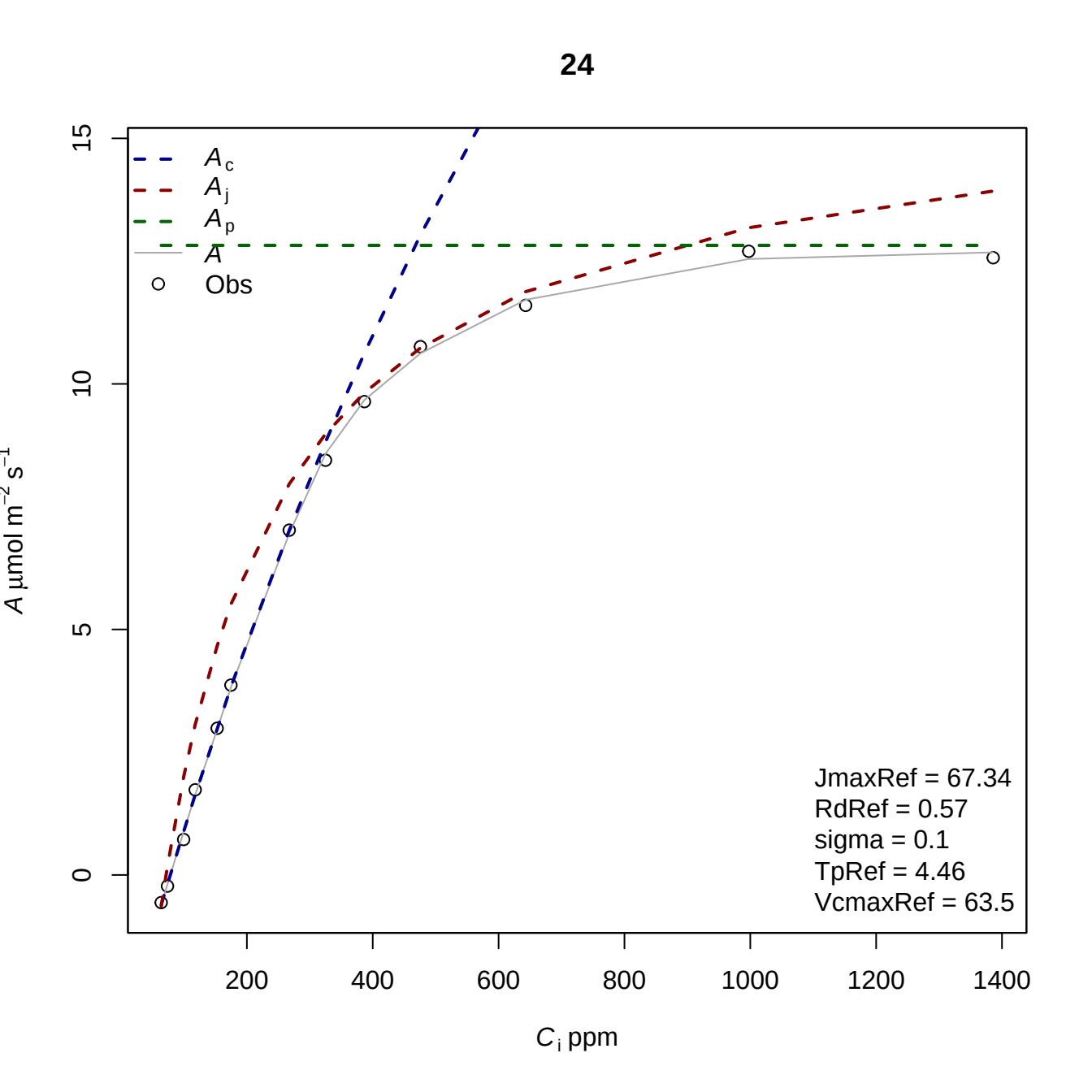

19
A_c
A_j
A_p
A
Obs
JmaxRef = 62.41
RdRef = 0.43
sigma = 0.18
TpRef = 4.62
VcmaxRef = 52.91
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
200
400
600
800
1000
1200
1400

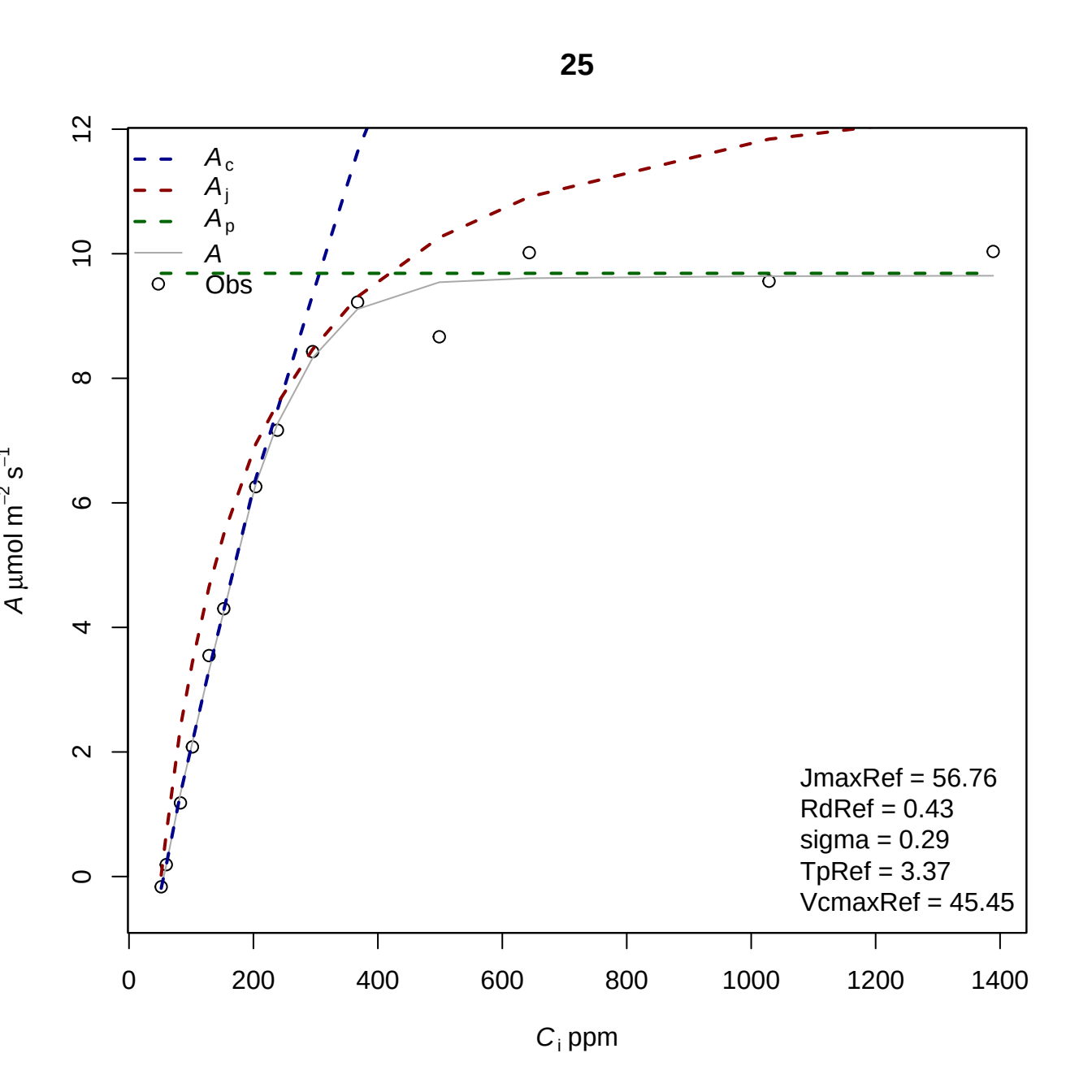

20
A_c
A_j
A_p
A
Obs
JmaxRef = 72.11
RdRef = 0.18
sigma = 0.26
TpRef = 5.14
VcmaxRef = 70.41
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
200
400
600
800
1000
1200
1400

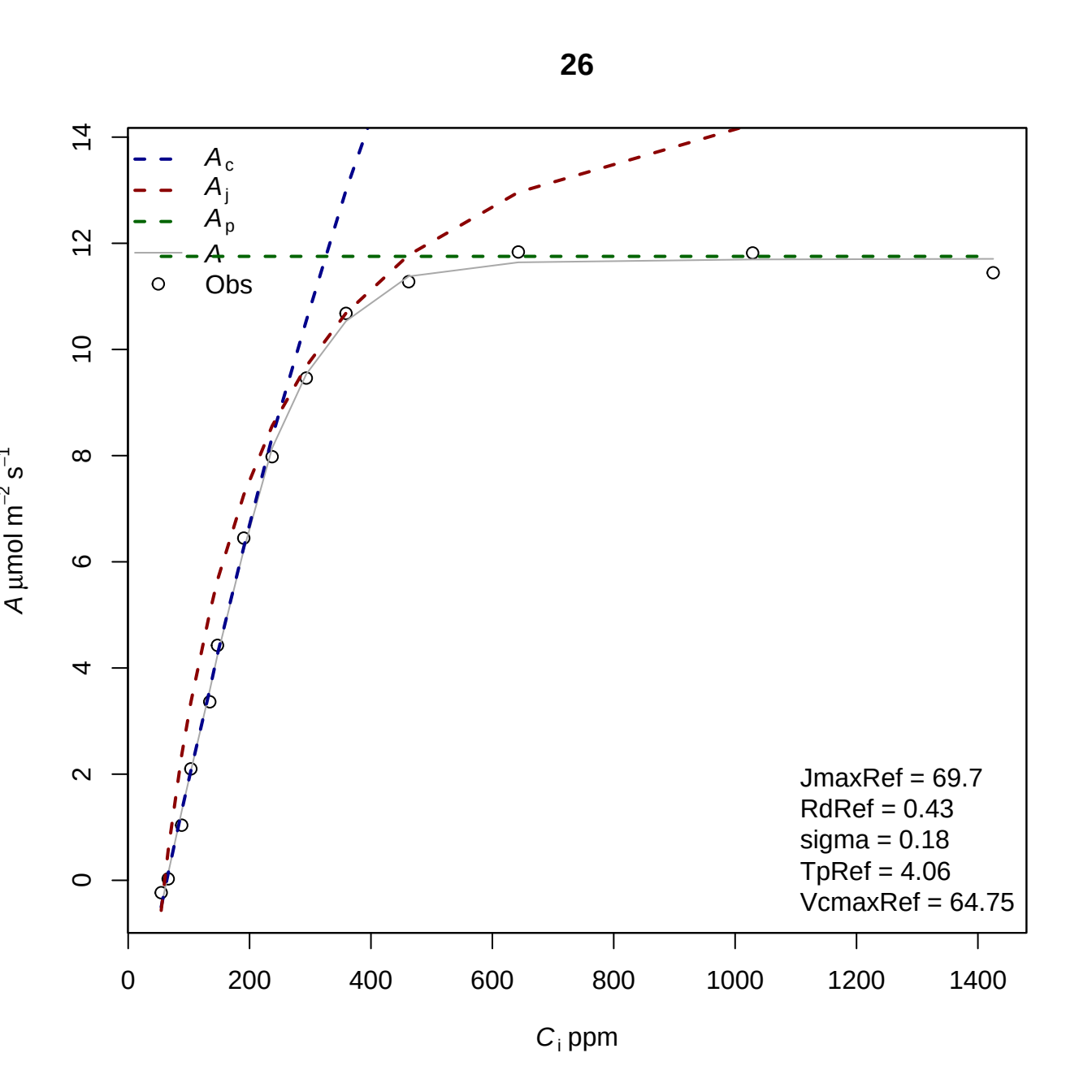

21
A_c
A_j
A_p
A
Obs
JmaxRef = 73.62
RdRef = 0.73
sigma = 0.1
TpRef = 349270.64
VcmaxRef = 58.42
0
5
10
15
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
C_i ppm

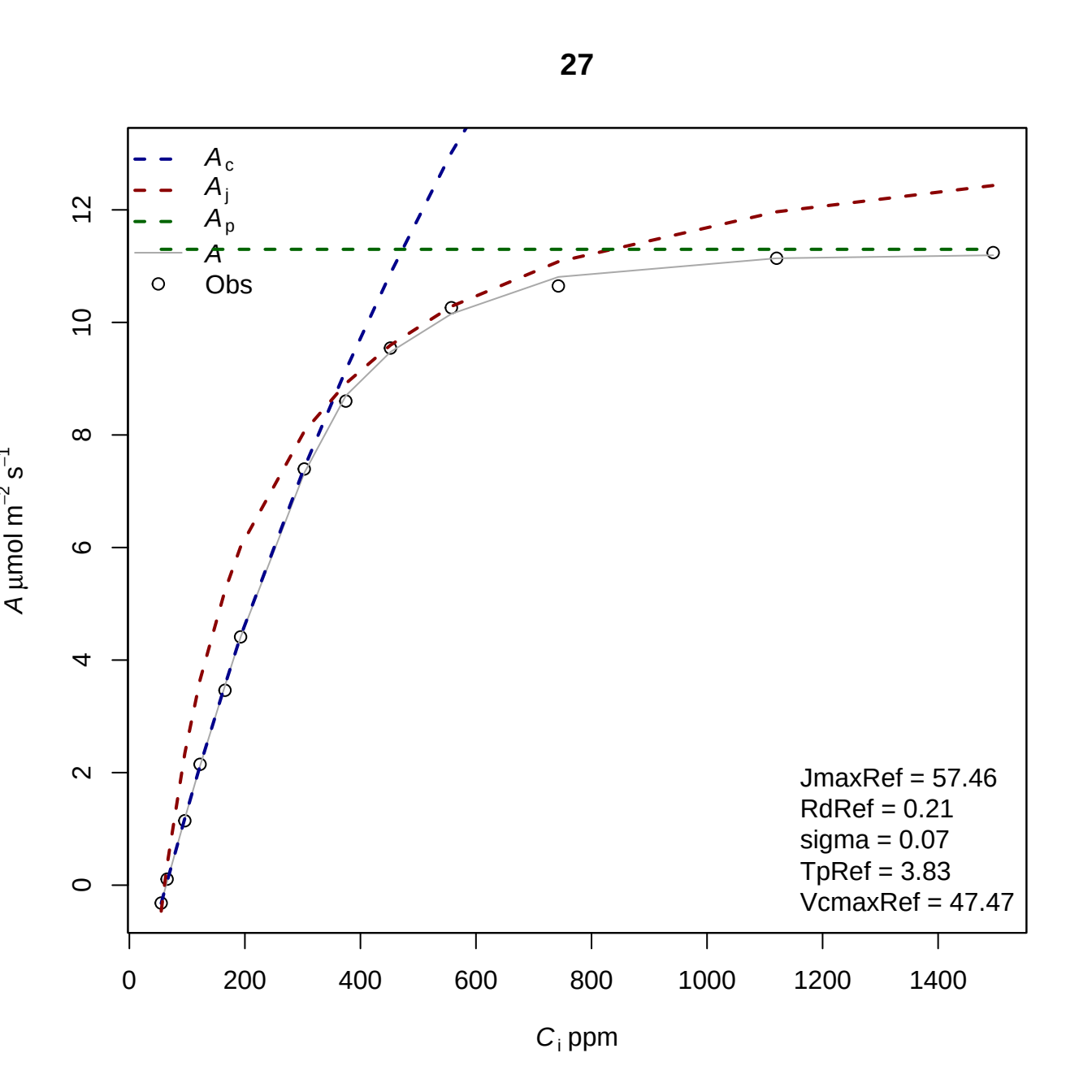

22
A_c
A_j
A_p
A
Obs
JmaxRef = 80.71
RdRef = 0.67
sigma = 0.26
TpRef = 5
VcmaxRef = 75.26
0
5
10
15
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
C_i ppm

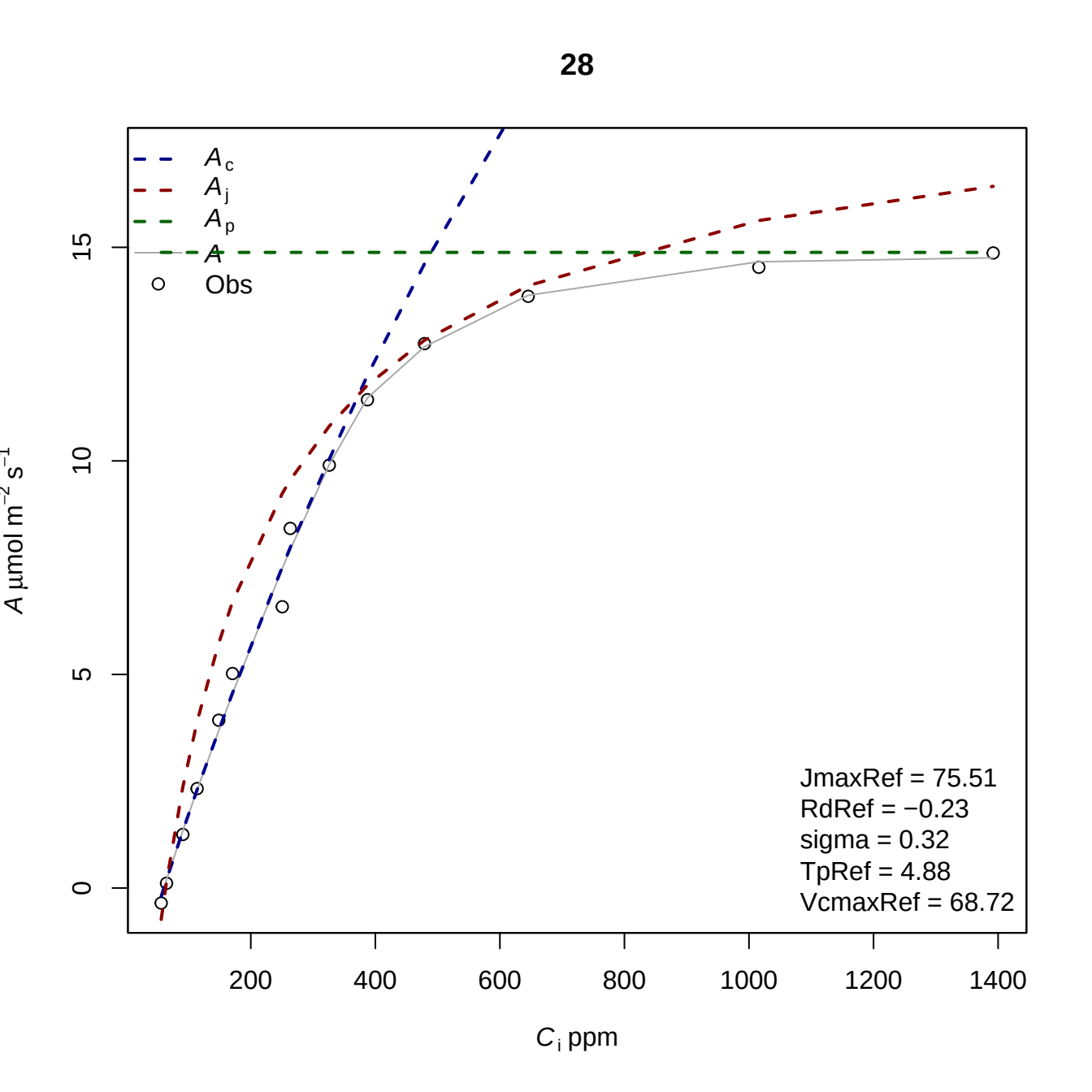

23
A_c
A_j
A_p
A
Obs
JmaxRef = 80.67
RdRef = 0.24
sigma = 0.1
TpRef = 5.04
VcmaxRef = 70.74
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
200
400
600
800
1000
1200
1400

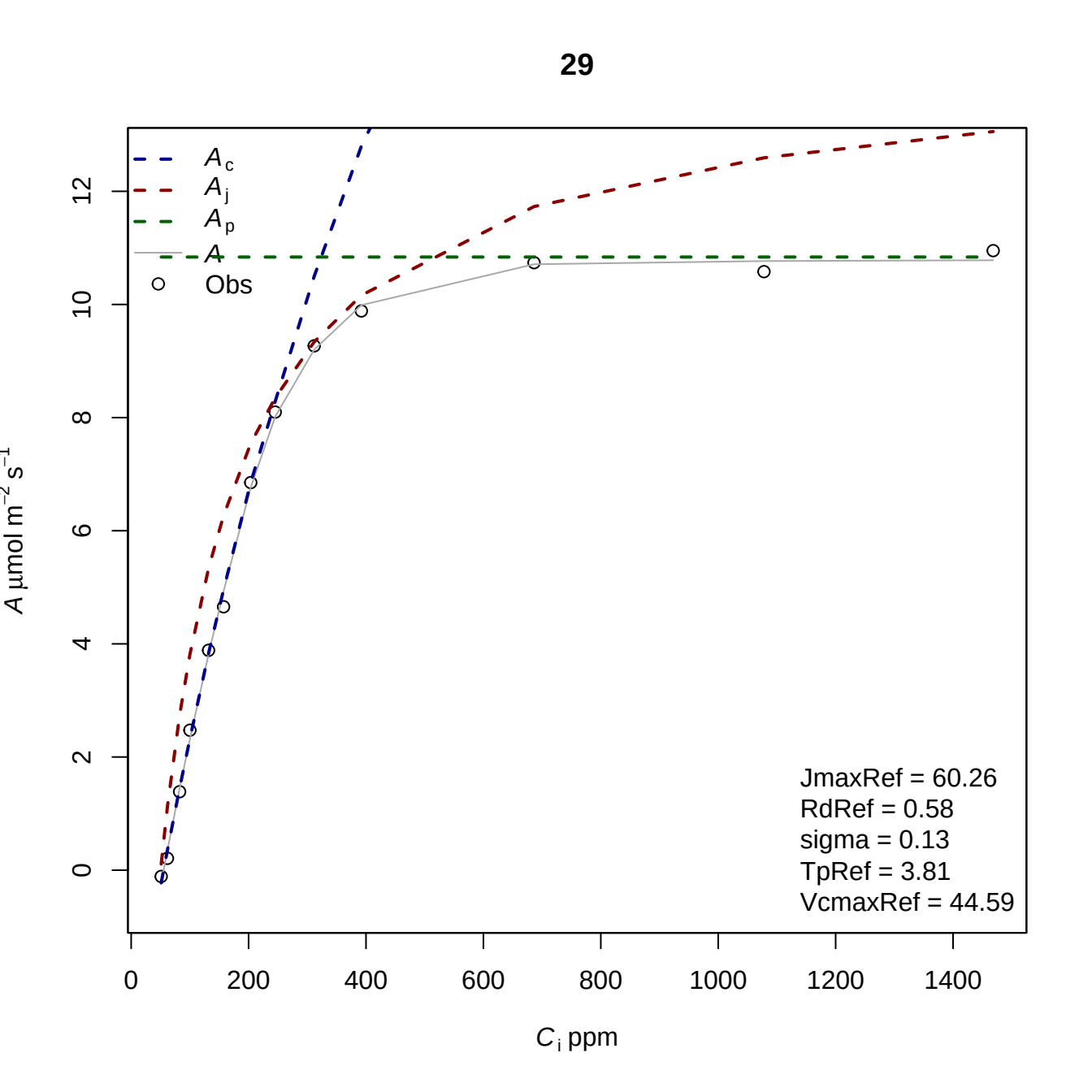

24
A_c
A_j
A_p
A
Obs
JmaxRef = 67.34
RdRef = 0.57
sigma = 0.1
TpRef = 4.46
VcmaxRef = 63.5
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
200
400
600
800
1000
1200
1400

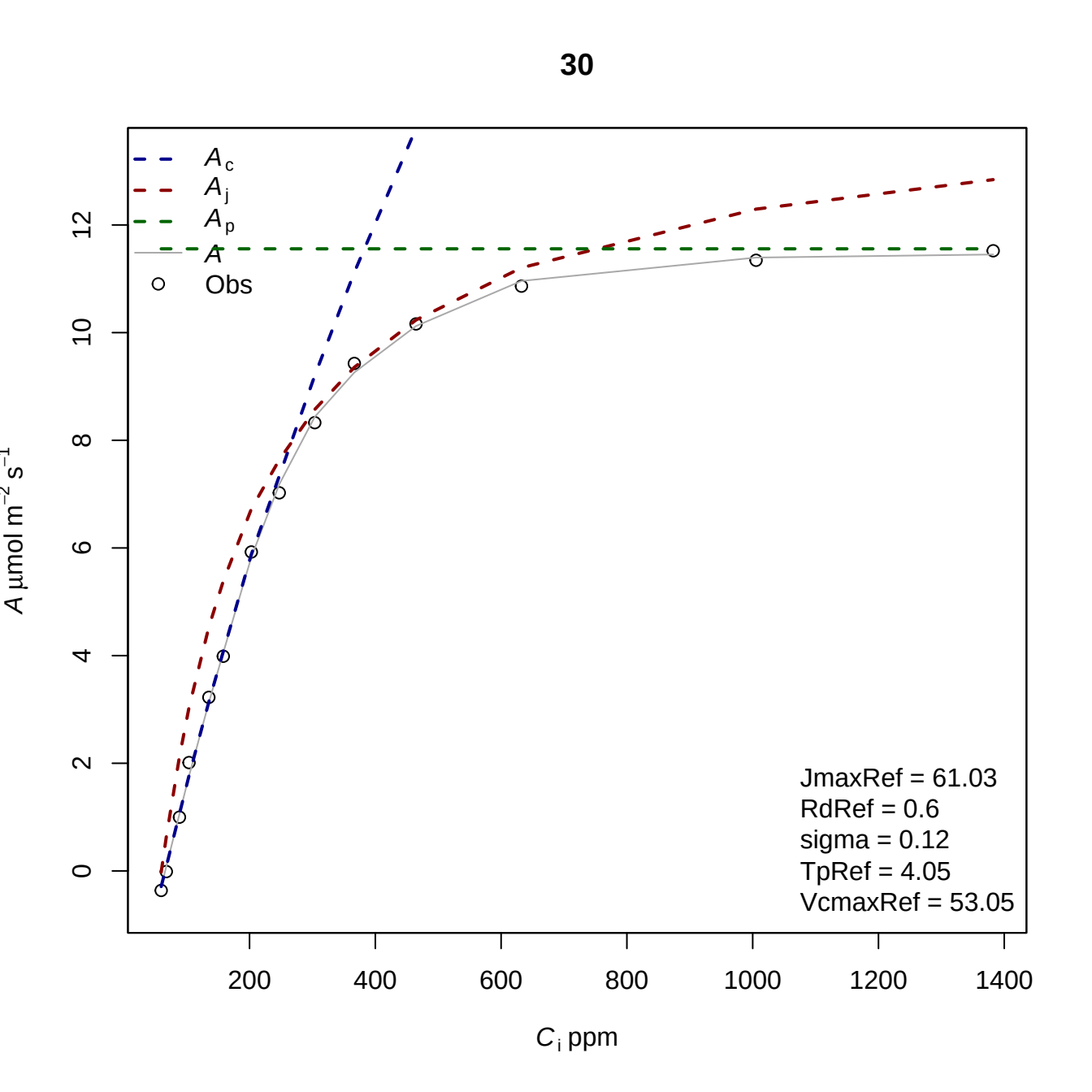

25
A_c
A_j
A_p
A
Obs
JmaxRef = 56.76
RdRef = 0.43
sigma = 0.29
TpRef = 3.37
VcmaxRef = 45.45
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

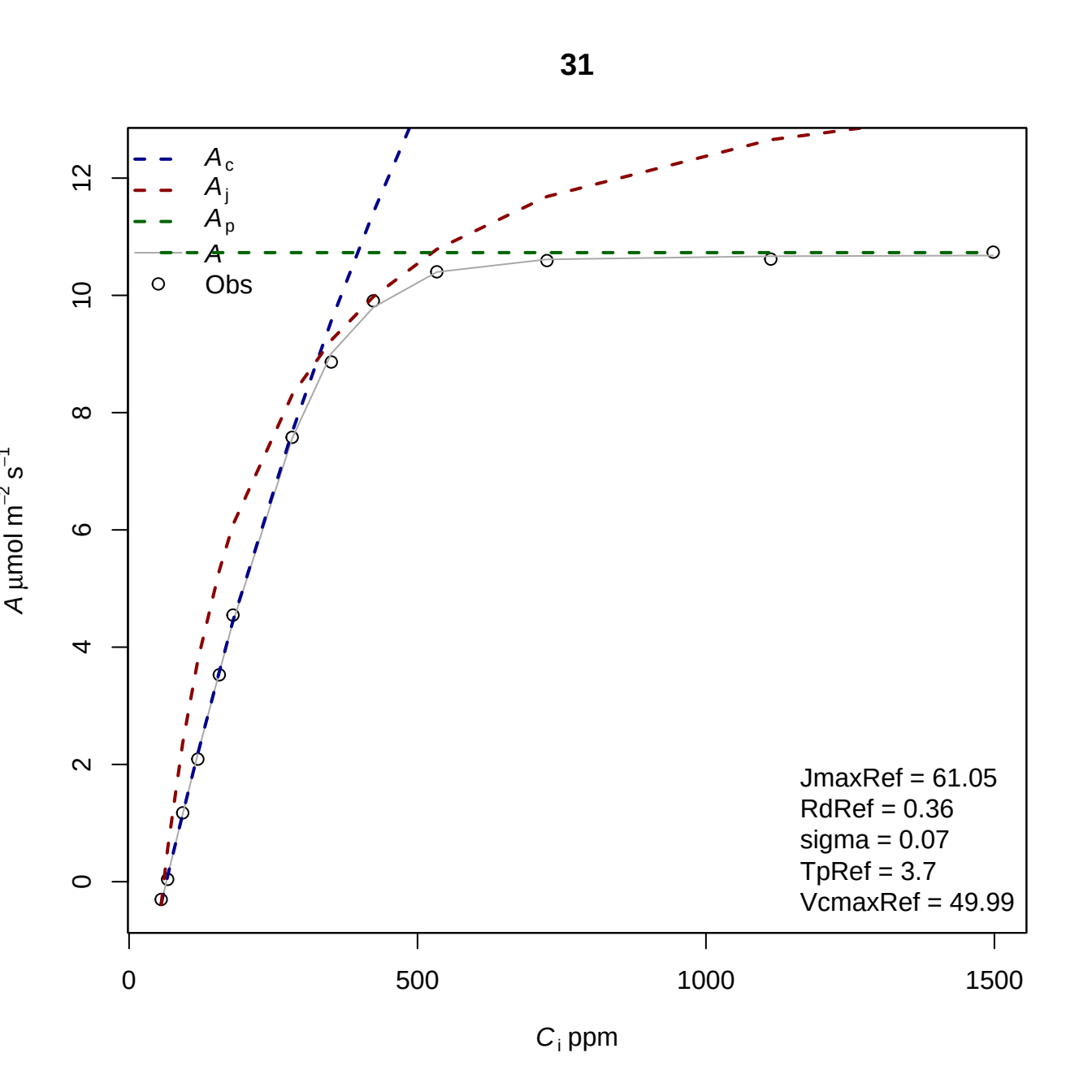

26
A_c
A_j
A_p
A
Obs
JmaxRef = 69.7
RdRef = 0.43
sigma = 0.18
TpRef = 4.06
VcmaxRef = 64.75
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
14
0
200
400
600
800
1000
1200
1400

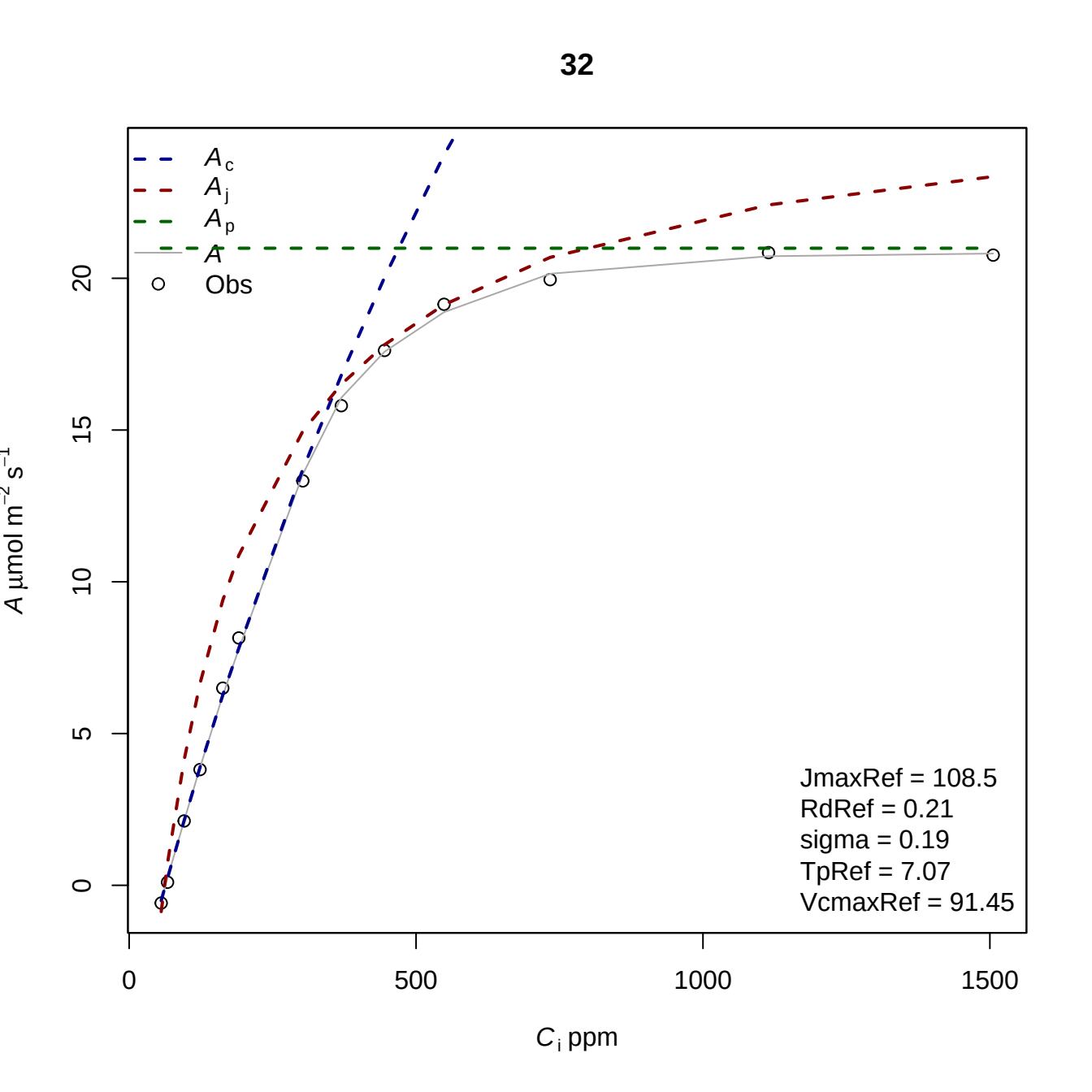

27
A_c
A_j
A_p
A
Obs
JmaxRef = 57.46
RdRef = 0.21
sigma = 0.07
TpRef = 3.83
VcmaxRef = 47.47
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

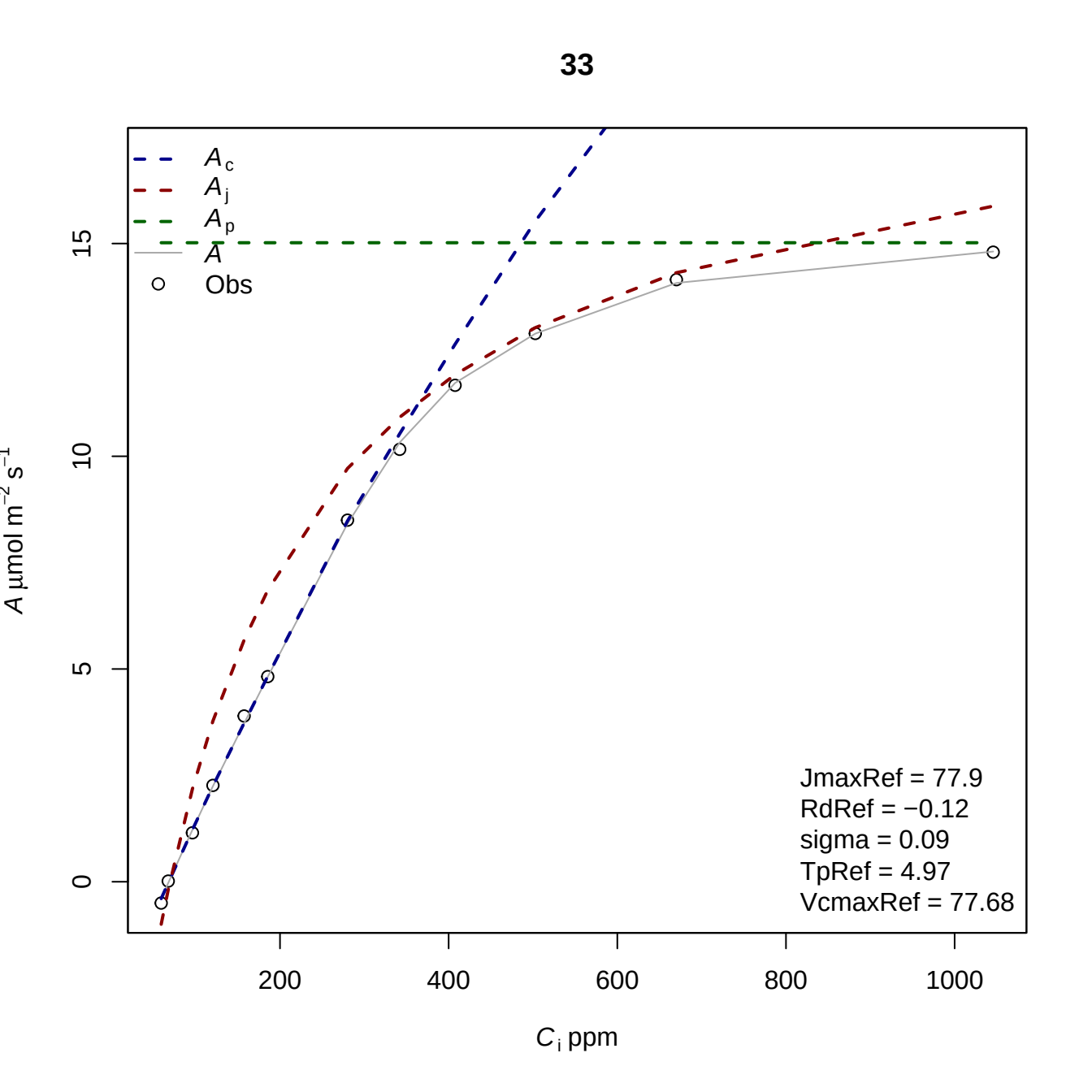

28
A_c
A_j
A_p
A
Obs
JmaxRef = 75.51
RdRef = −0.23
sigma = 0.32
TpRef = 4.88
VcmaxRef = 68.72
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
200
400
600
800
1000
1200
1400

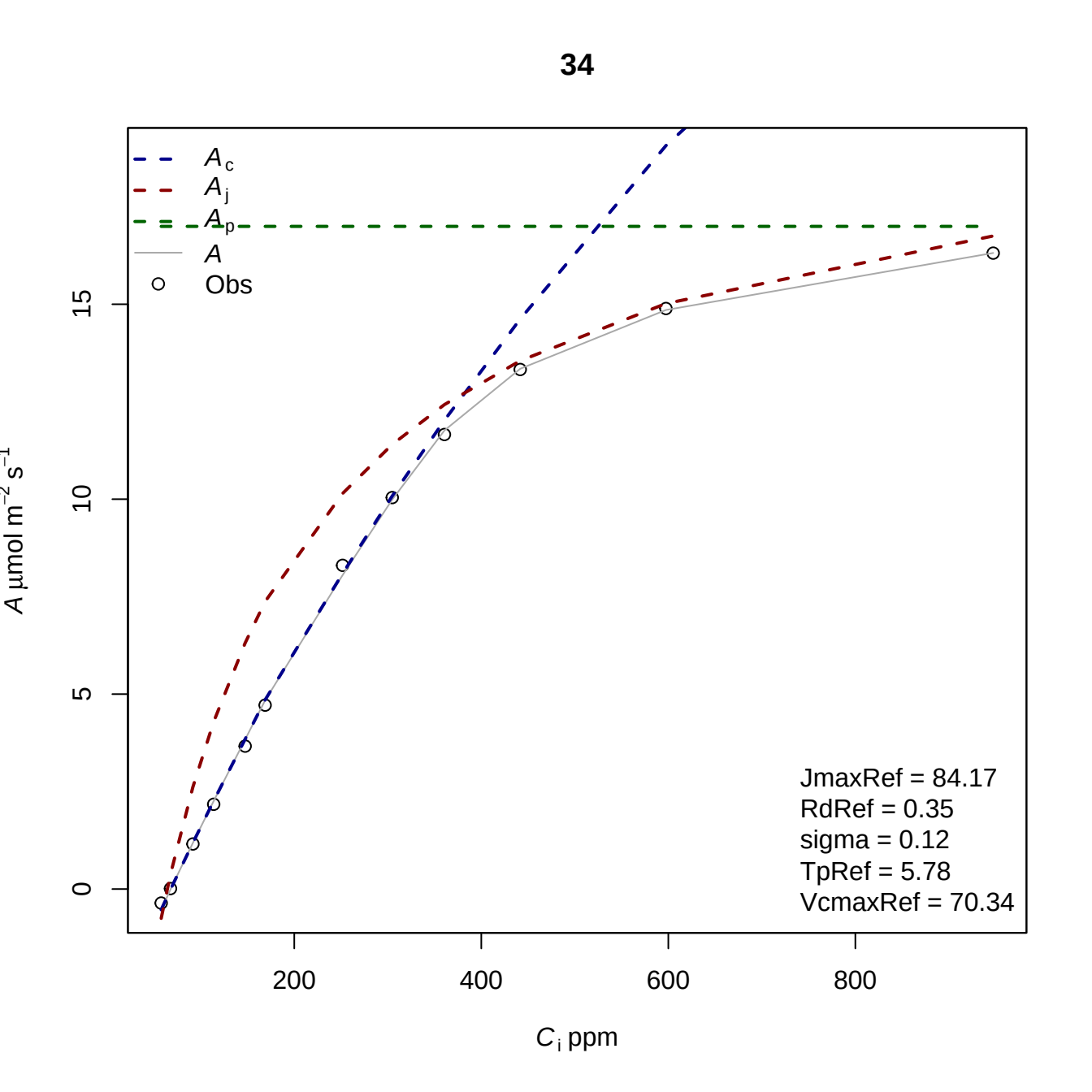

29
A_c
A_j
A_p
A
Obs
JmaxRef = 60.26
RdRef = 0.58
sigma = 0.13
TpRef = 3.81
VcmaxRef = 44.59
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

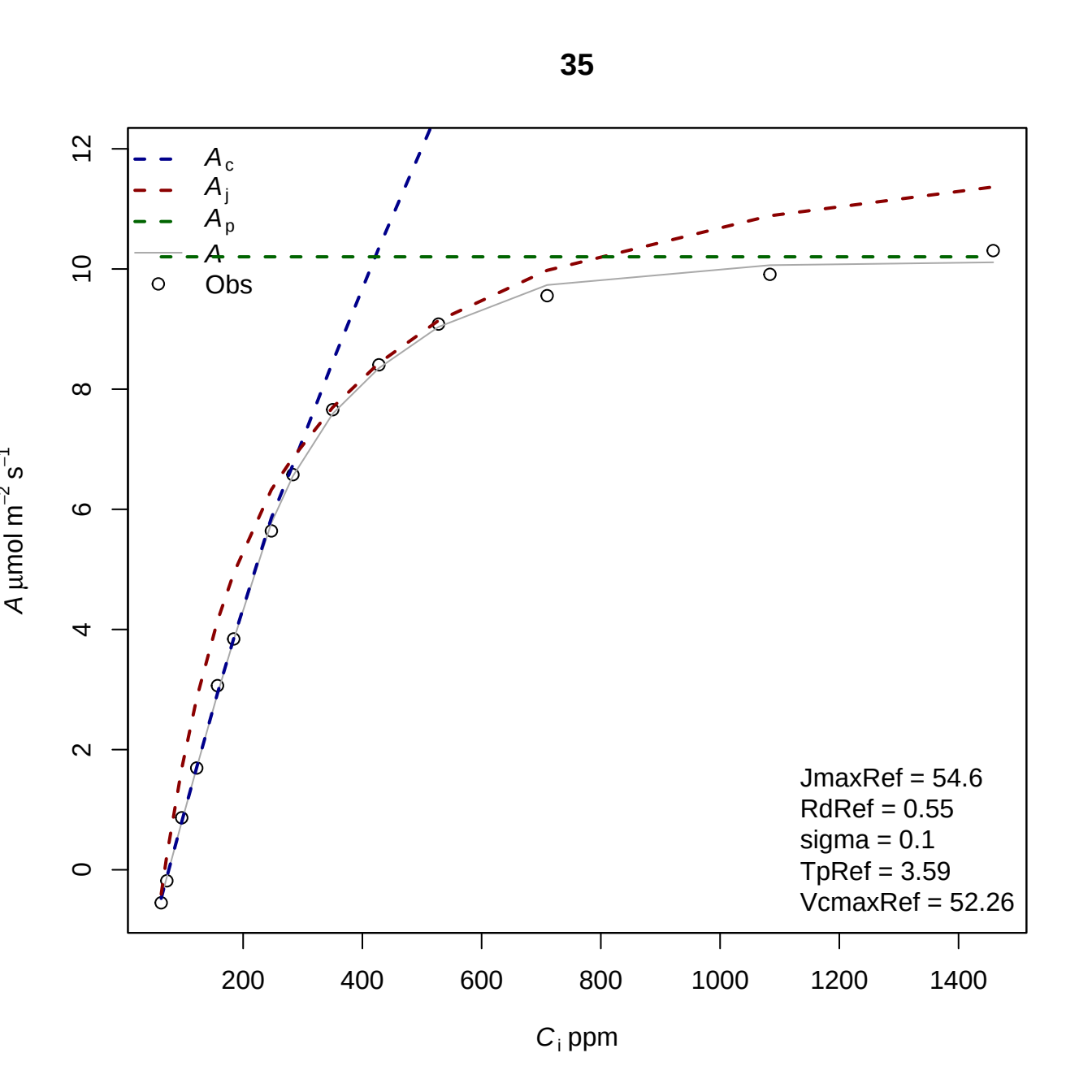

30
A_c
A_j
A_p
A
Obs
JmaxRef = 61.03
RdRef = 0.6
sigma = 0.12
TpRef = 4.05
VcmaxRef = 53.05
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
200
400
600
800
1000
1200
1400

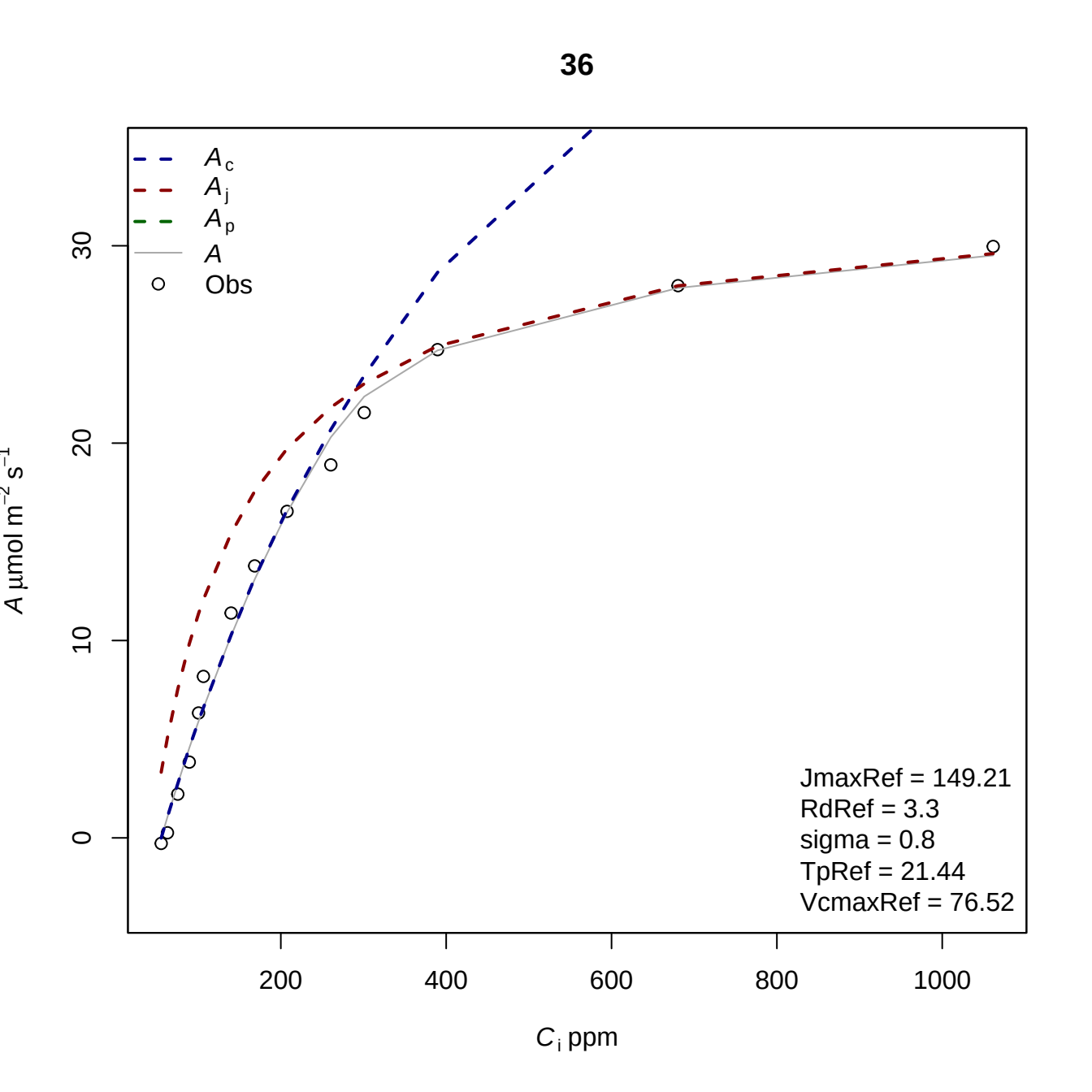

31
A_c
A_j
A_p
A
Obs
JmaxRef = 61.05
RdRef = 0.36
sigma = 0.07
TpRef = 3.7
VcmaxRef = 49.99
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
C_i ppm

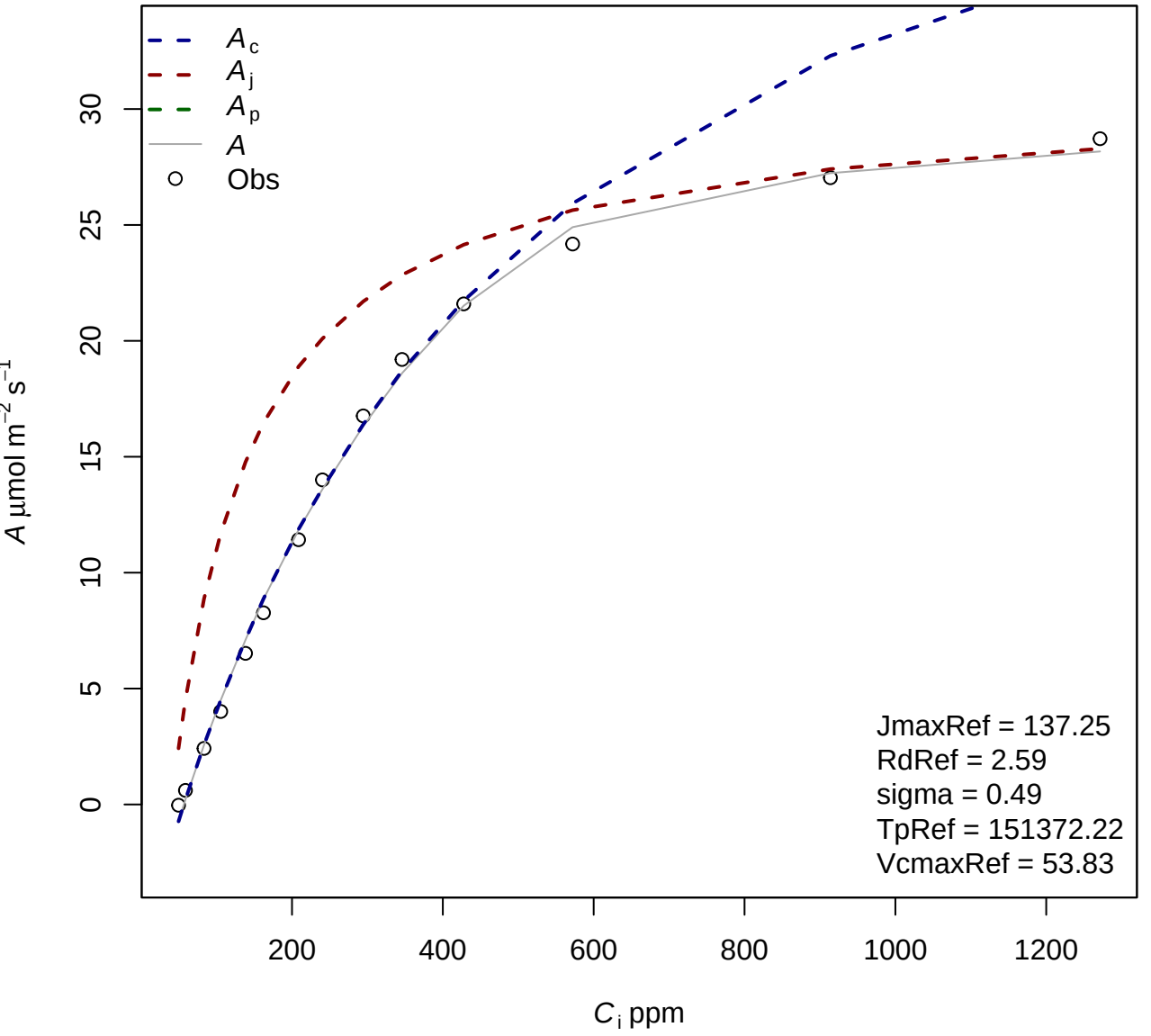

32
A_c
A_j
A_p
A
Obs
JmaxRef = 108.5
RdRef = 0.21
sigma = 0.19
TpRef = 7.07
VcmaxRef = 91.45
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
500
1000
1500

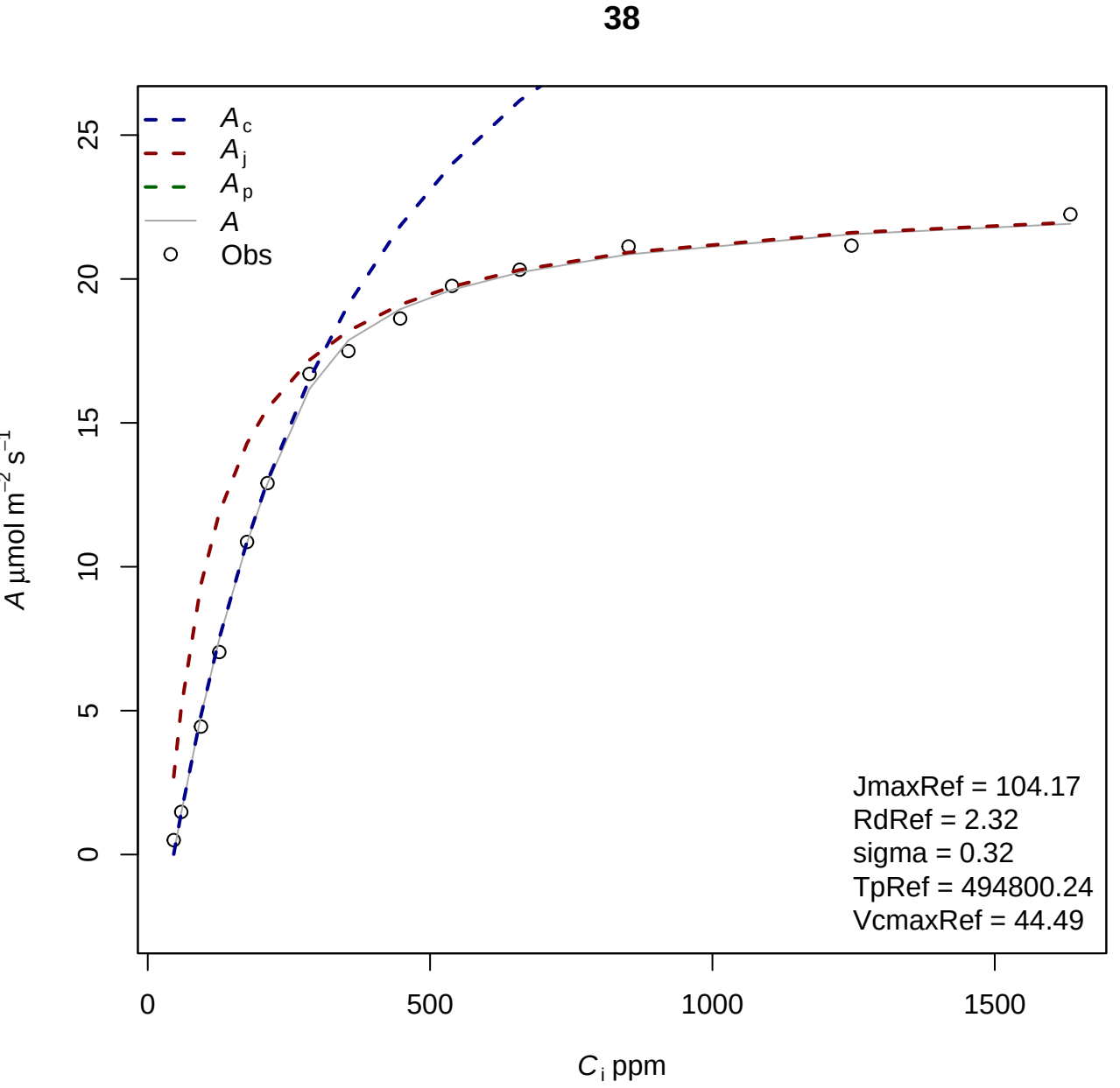

33
A_c
A_j
A_p
A
Obs
JmaxRef = 77.9
RdRef = −0.12
sigma = 0.09
TpRef = 4.97
VcmaxRef = 77.68
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
200
400
600
800
1000

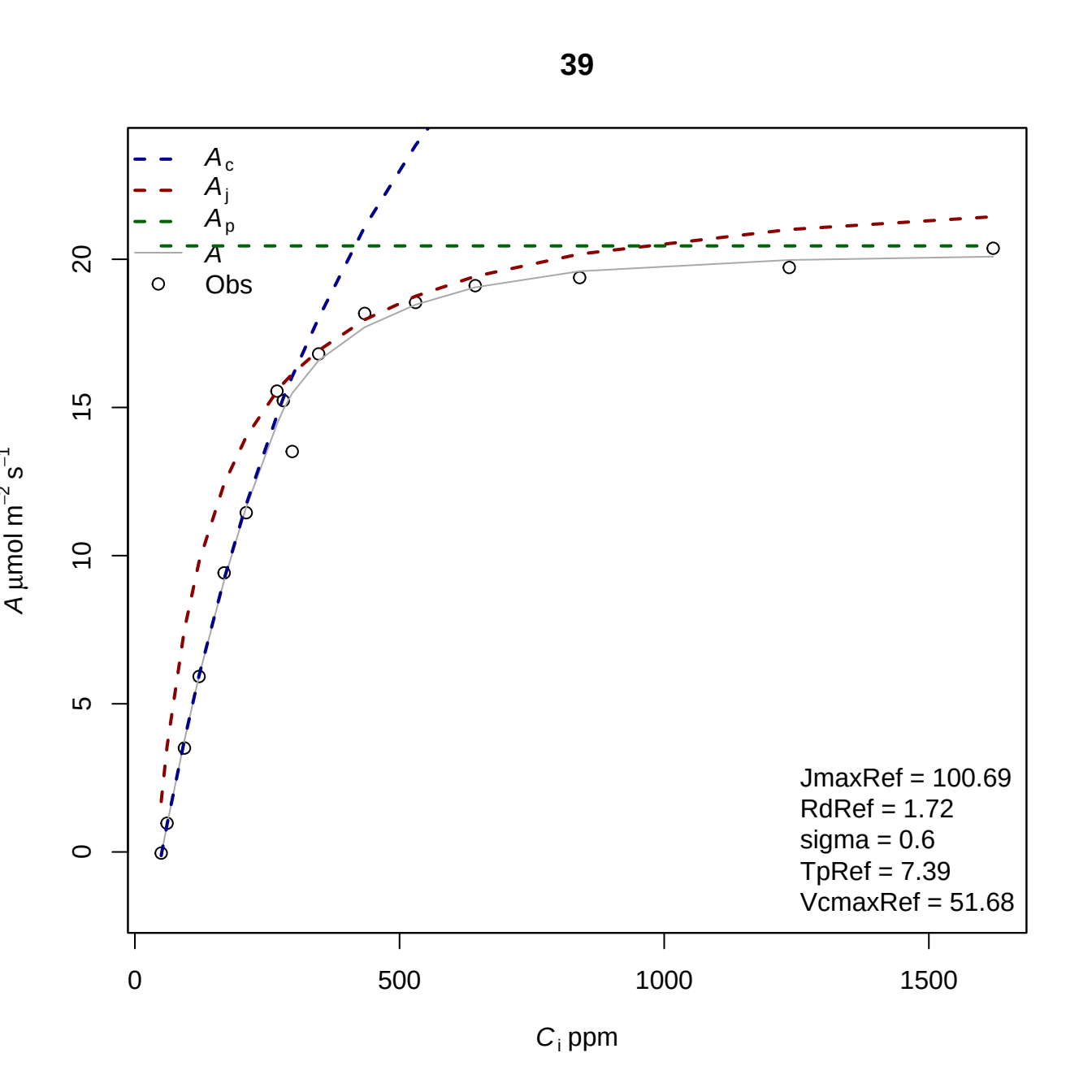

34
A_c
A_j
A_p
A
Obs
JmaxRef = 84.17
RdRef = 0.35
sigma = 0.12
TpRef = 5.78
VcmaxRef = 70.34
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
200
400
600
800

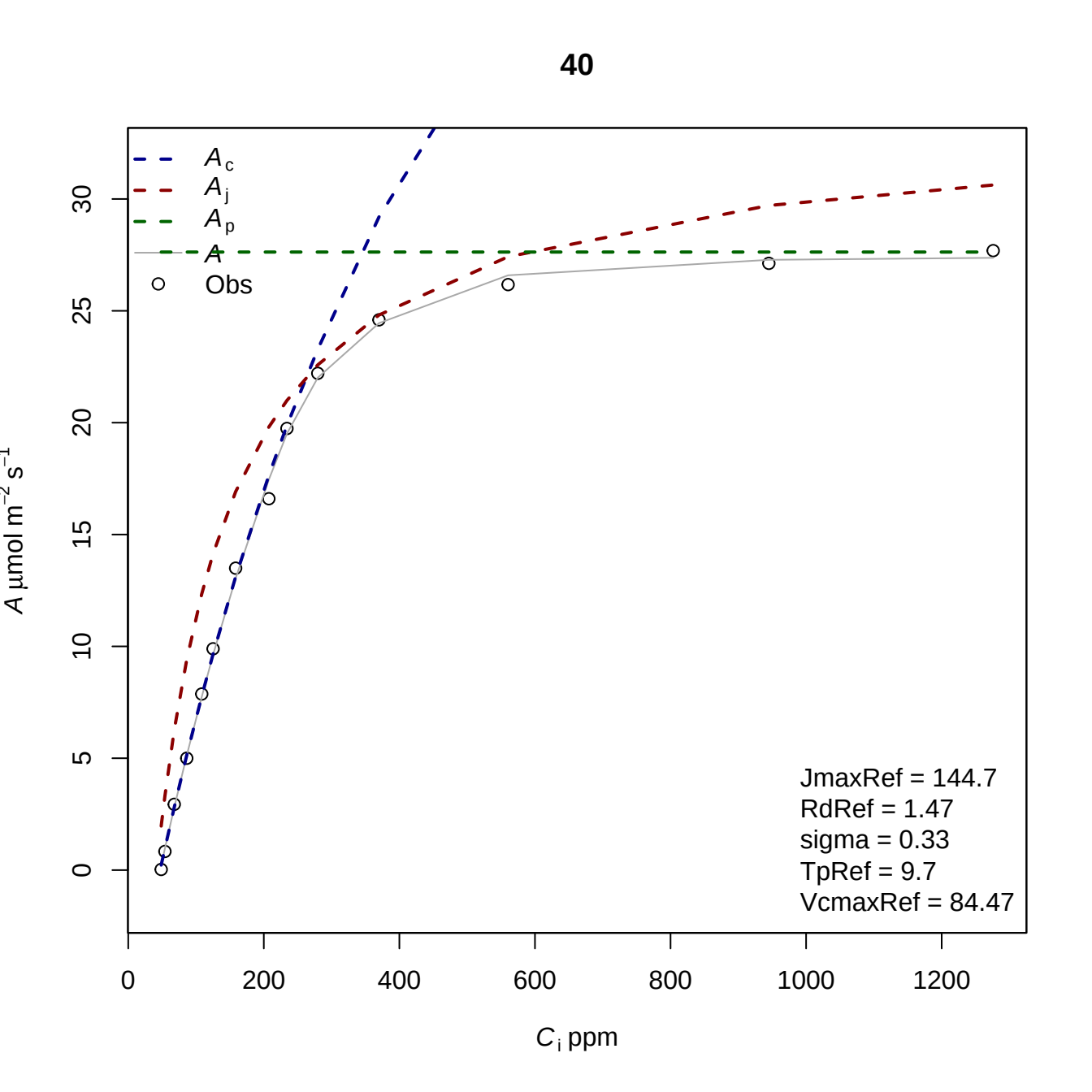

35
A_c
A_j
A_p
A
Obs
JmaxRef = 54.6
RdRef = 0.55
sigma = 0.1
TpRef = 3.59
VcmaxRef = 52.26
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
200
400
600
800
1000
1200
1400

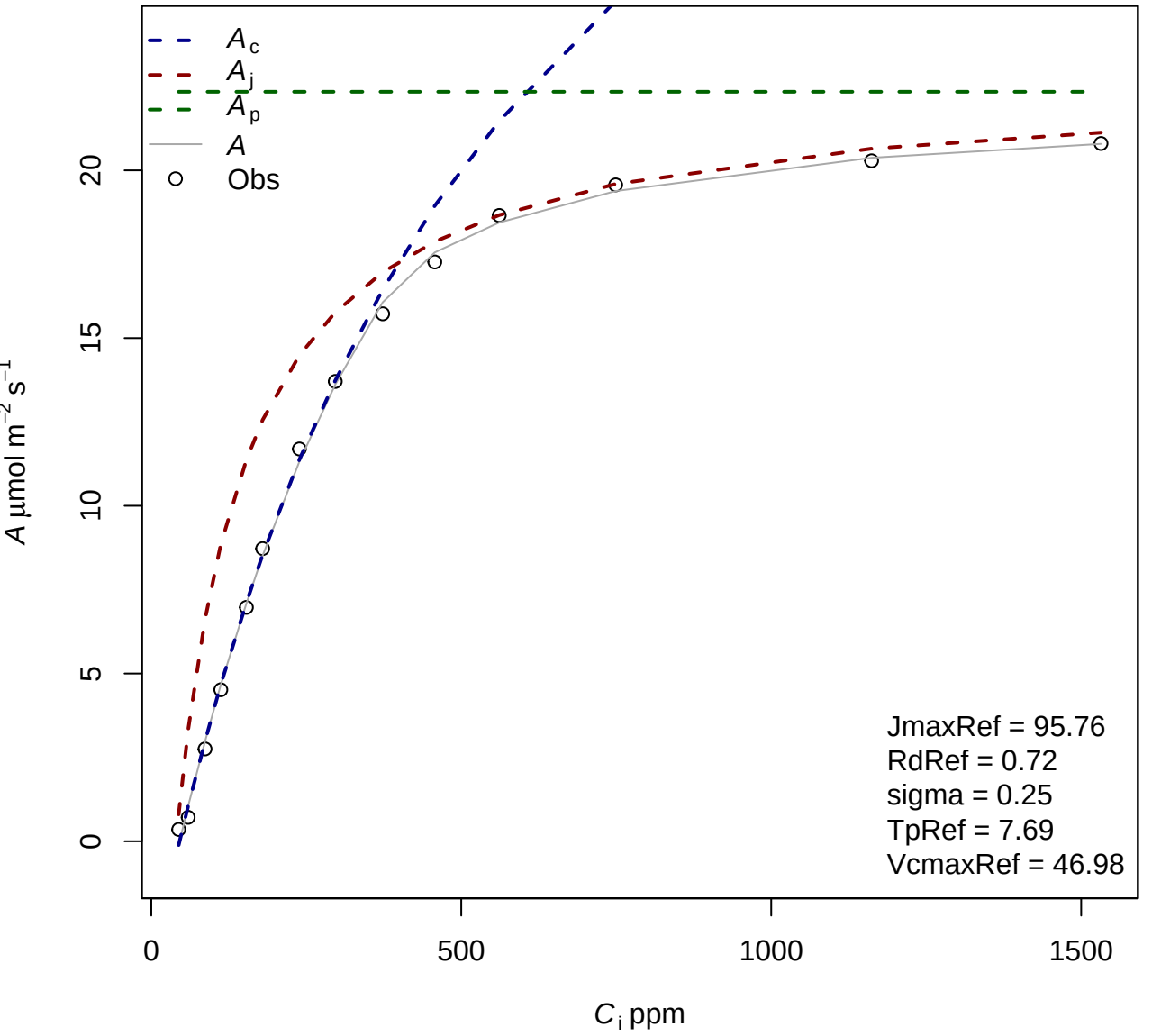

36
A_c
A_j
A_p
A
Obs
JmaxRef = 149.21
RdRef = 3.3
sigma = 0.8
TpRef = 21.44
VcmaxRef = 76.52
0
10
20
30
200
400
600
800
1000
A µmol m^{-2} s^{-1}
C_i ppm

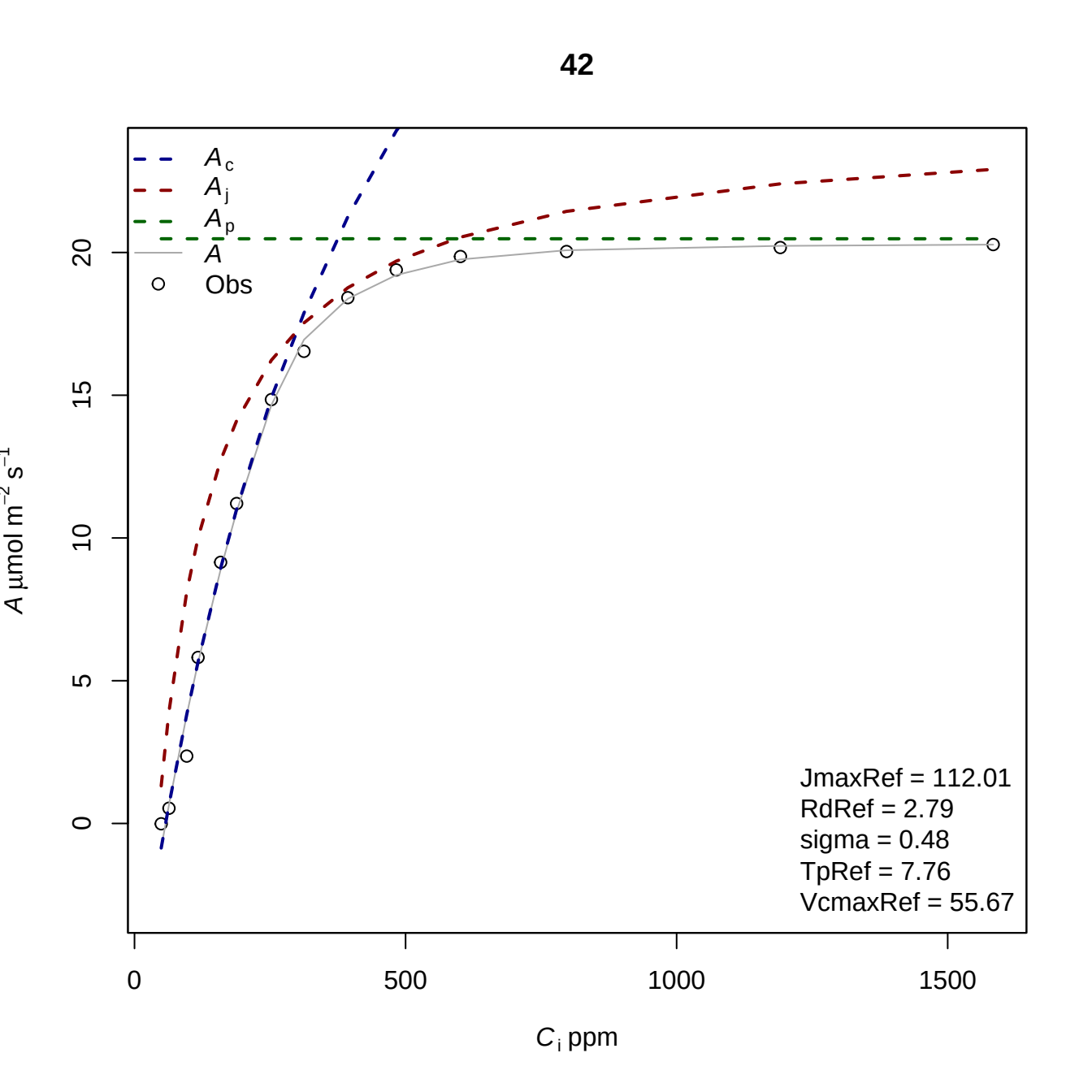

A_c
A_j
A_p
A
Obs
JmaxRef = 137.25
RdRef = 2.59
sigma = 0.49
TpRef = 151372.22
VcmaxRef = 53.83
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
200
400
600
800
1000
1200

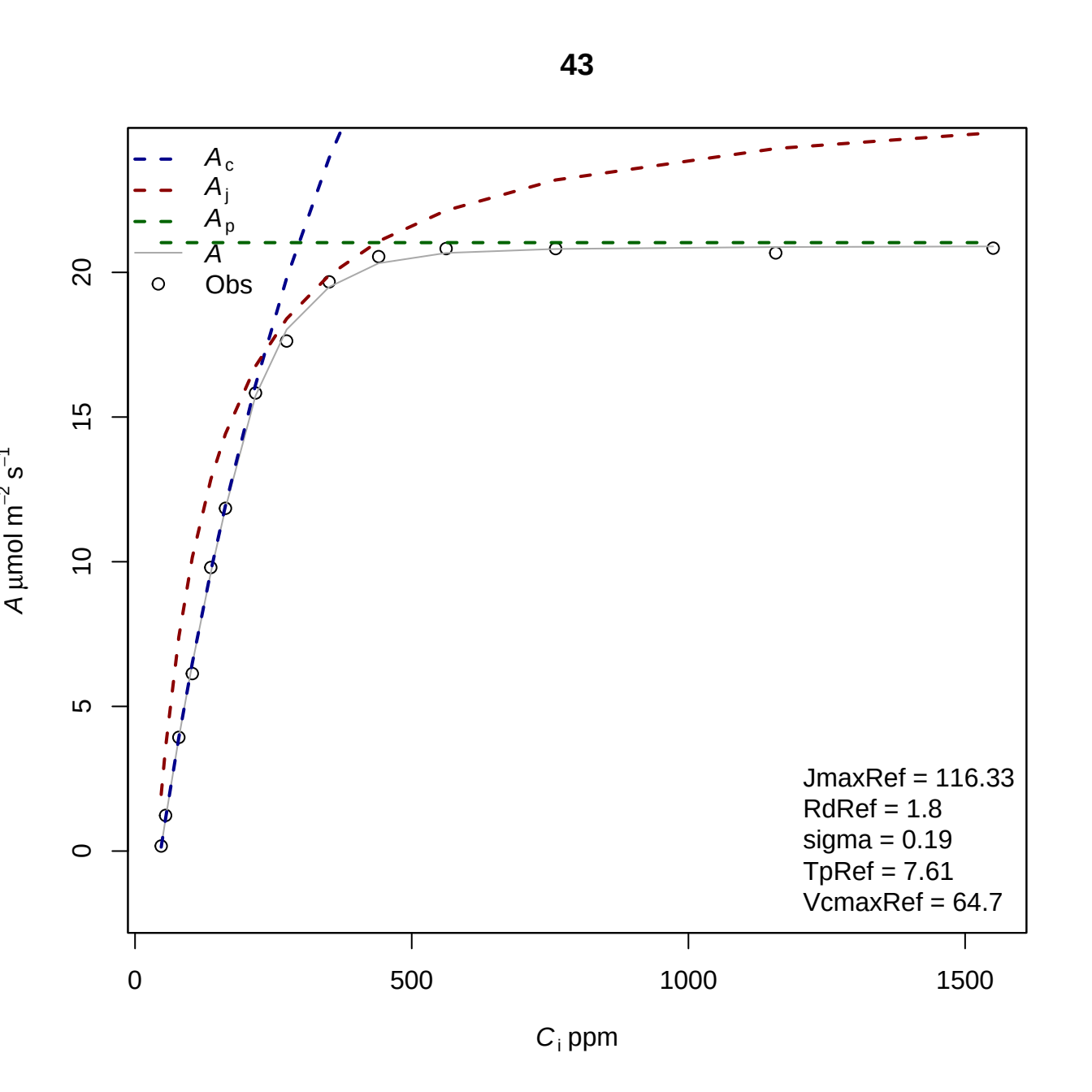

38
A_c
A_j
A_p
A
Obs
JmaxRef = 104.17
RdRef = 2.32
sigma = 0.32
TpRef = 494800.24
VcmaxRef = 44.49
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
500
1000
1500

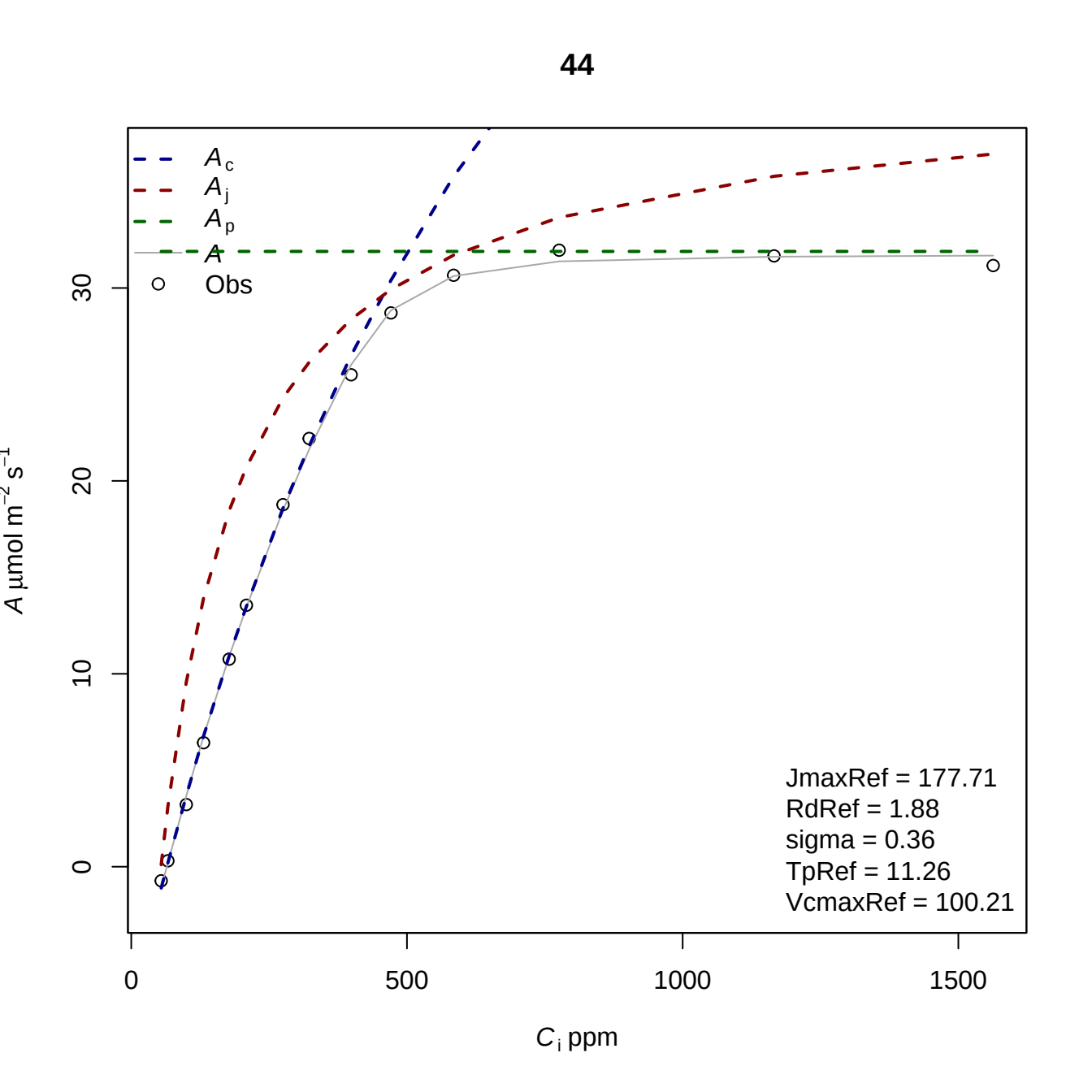

39
A_c
A_j
A_p
A
Obs
JmaxRef = 100.69
RdRef = 1.72
sigma = 0.6
TpRef = 7.39
VcmaxRef = 51.68
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
500
1000
1500

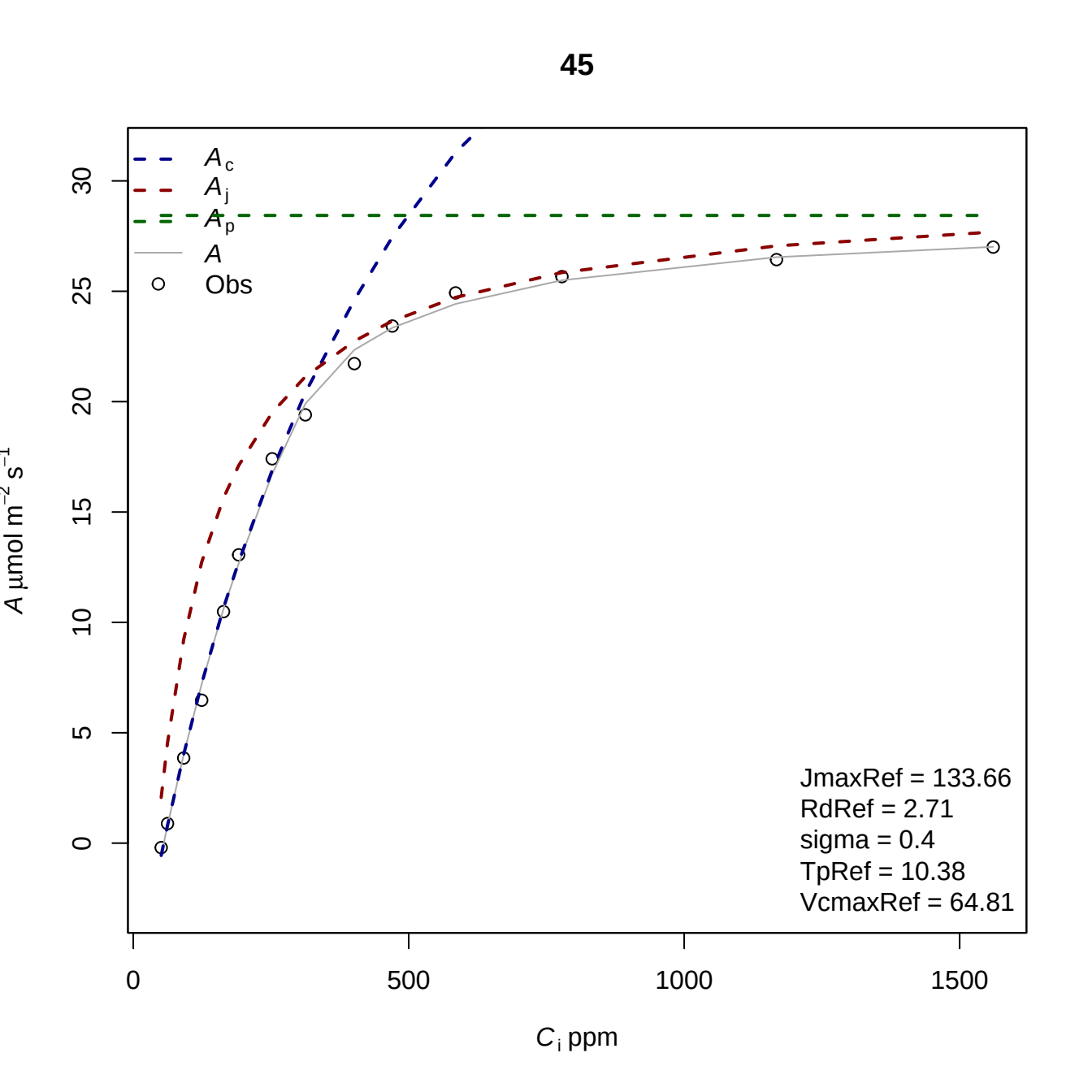

40
A_c
A_j
A_p
A
Obs
JmaxRef = 144.7
RdRef = 1.47
sigma = 0.33
TpRef = 9.7
VcmaxRef = 84.47
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

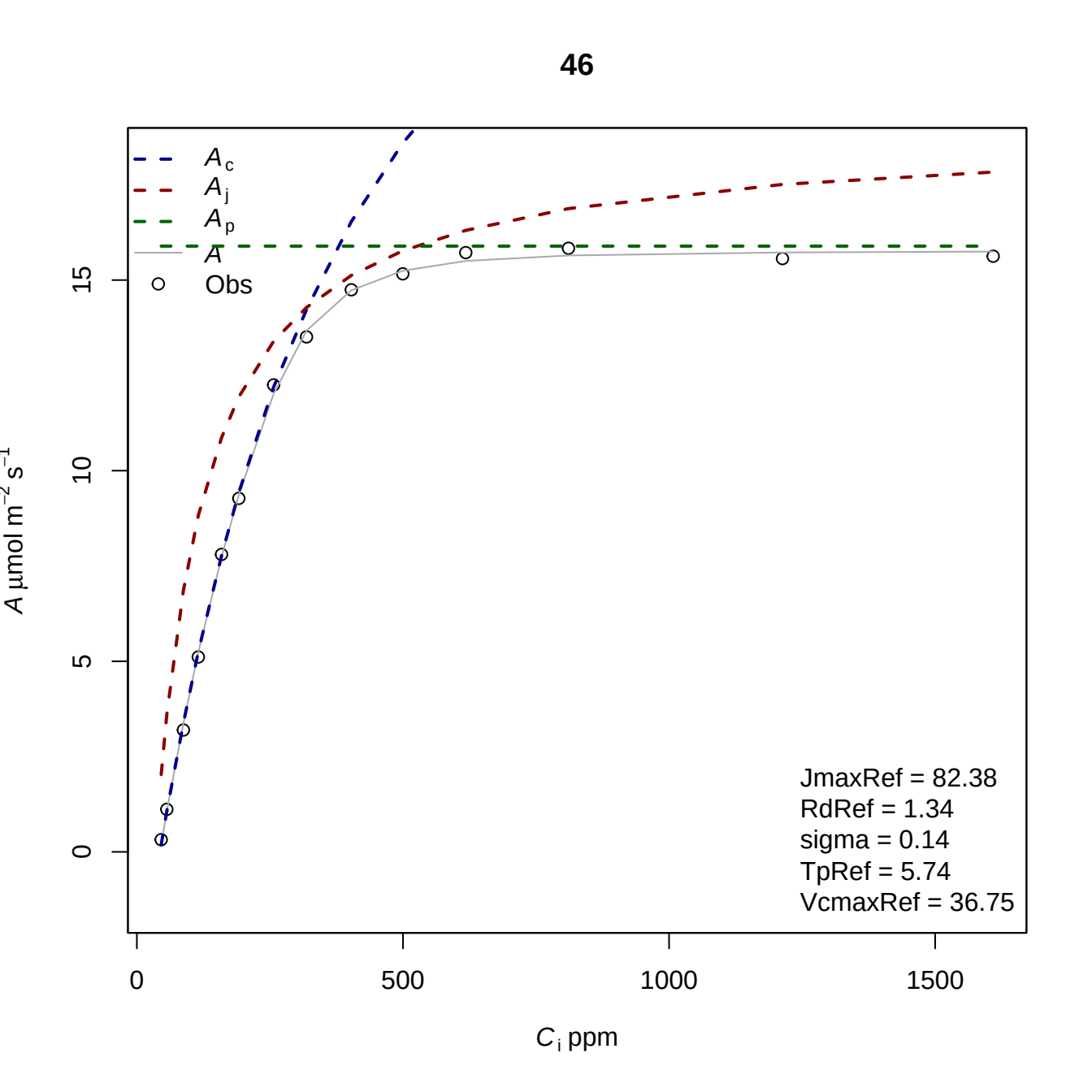

A_c
A_j
A_p
A
Obs
JmaxRef = 95.76
RdRef = 0.72
sigma = 0.25
TpRef = 7.69
VcmaxRef = 46.98
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
C_i ppm

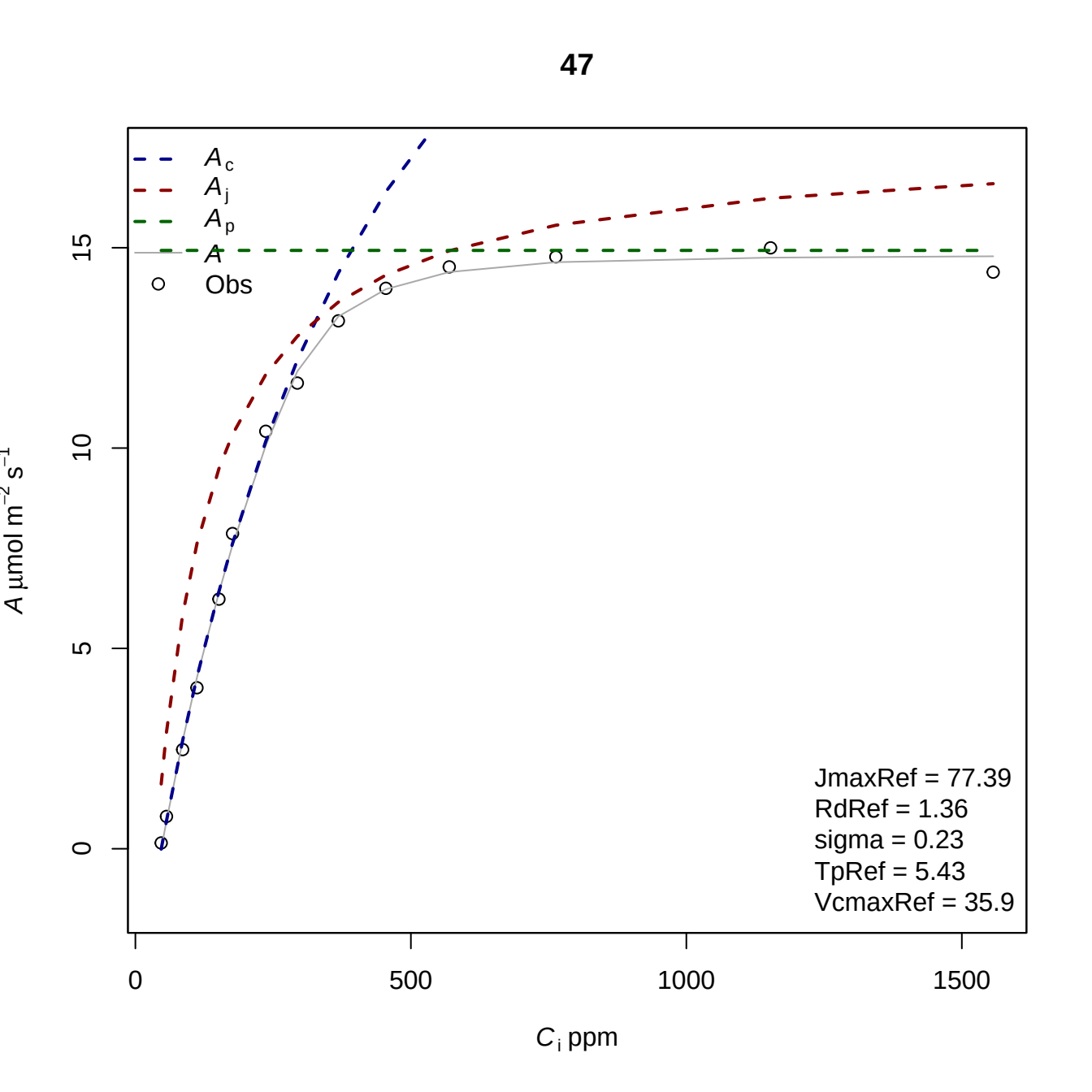

42
A_c
A_j
A_p
A
Obs
JmaxRef = 112.01
RdRef = 2.79
sigma = 0.48
TpRef = 7.76
VcmaxRef = 55.67
A µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
C_i ppm

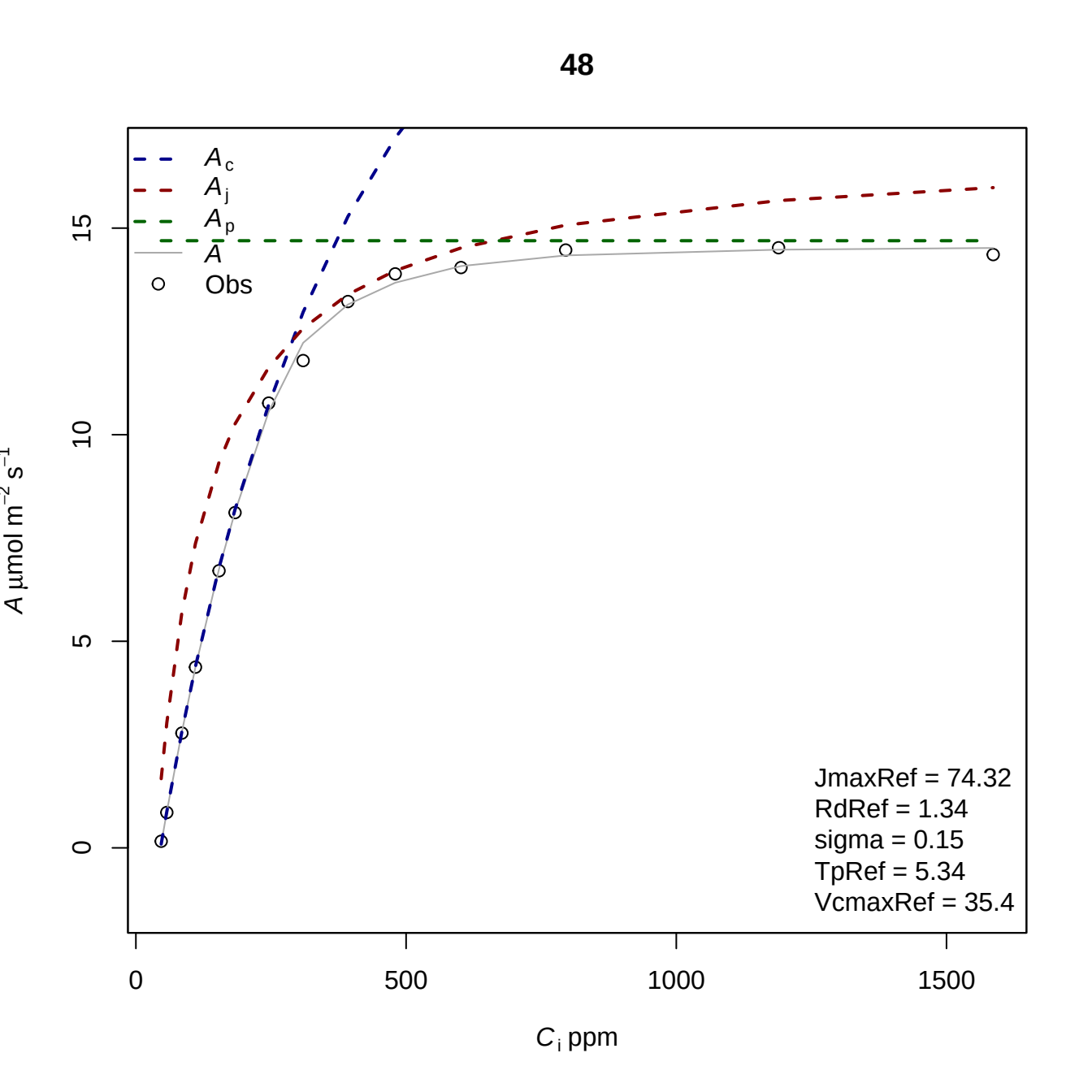

43
A_c
A_j
A_p
A
Obs
JmaxRef = 116.33
RdRef = 1.8
sigma = 0.19
TpRef = 7.61
VcmaxRef = 64.7
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
C_i ppm

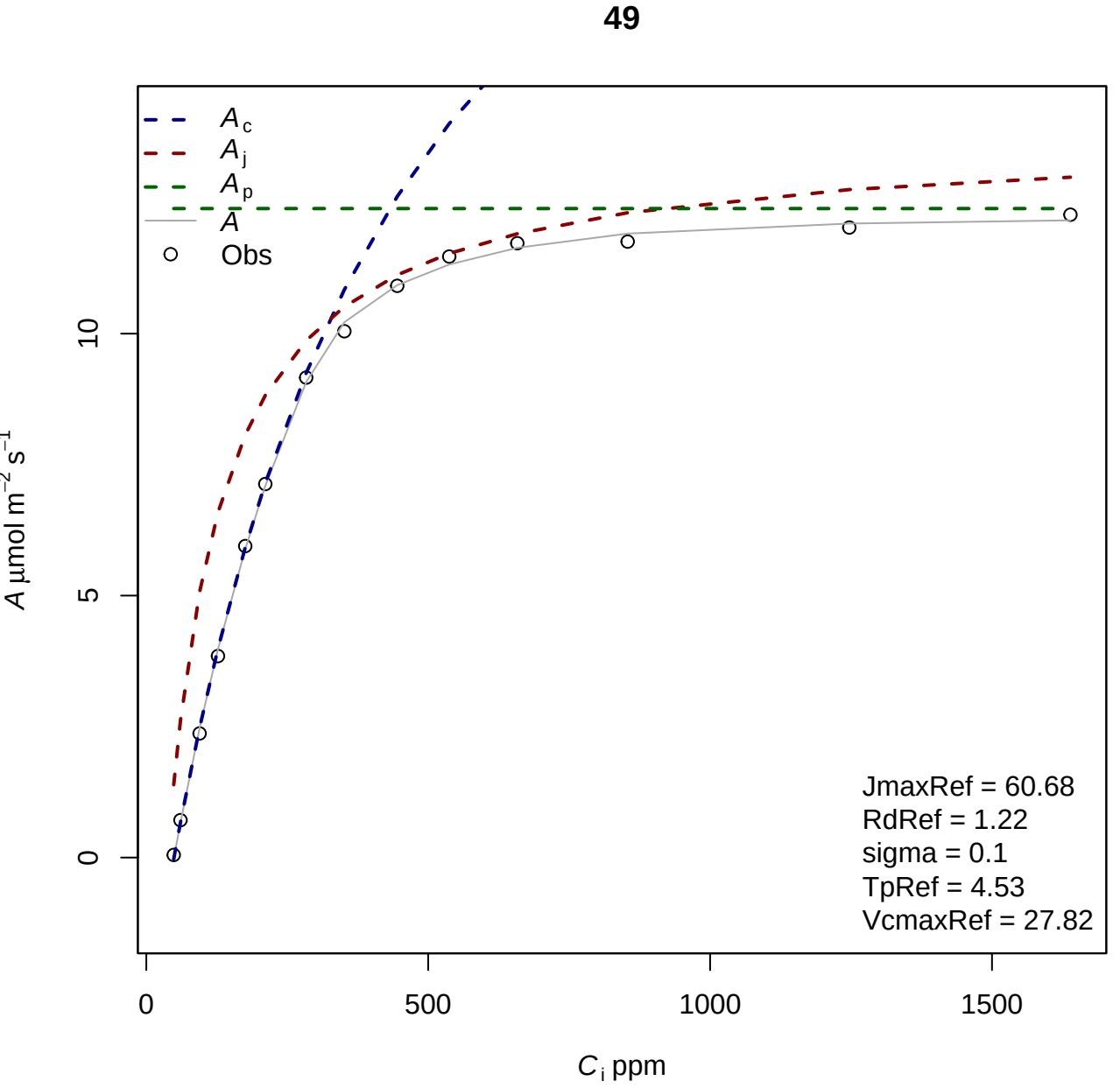

44
A_c
A_j
A_p
A
Obs
JmaxRef = 177.71
RdRef = 1.88
sigma = 0.36
TpRef = 11.26
VcmaxRef = 100.21
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
C_i ppm

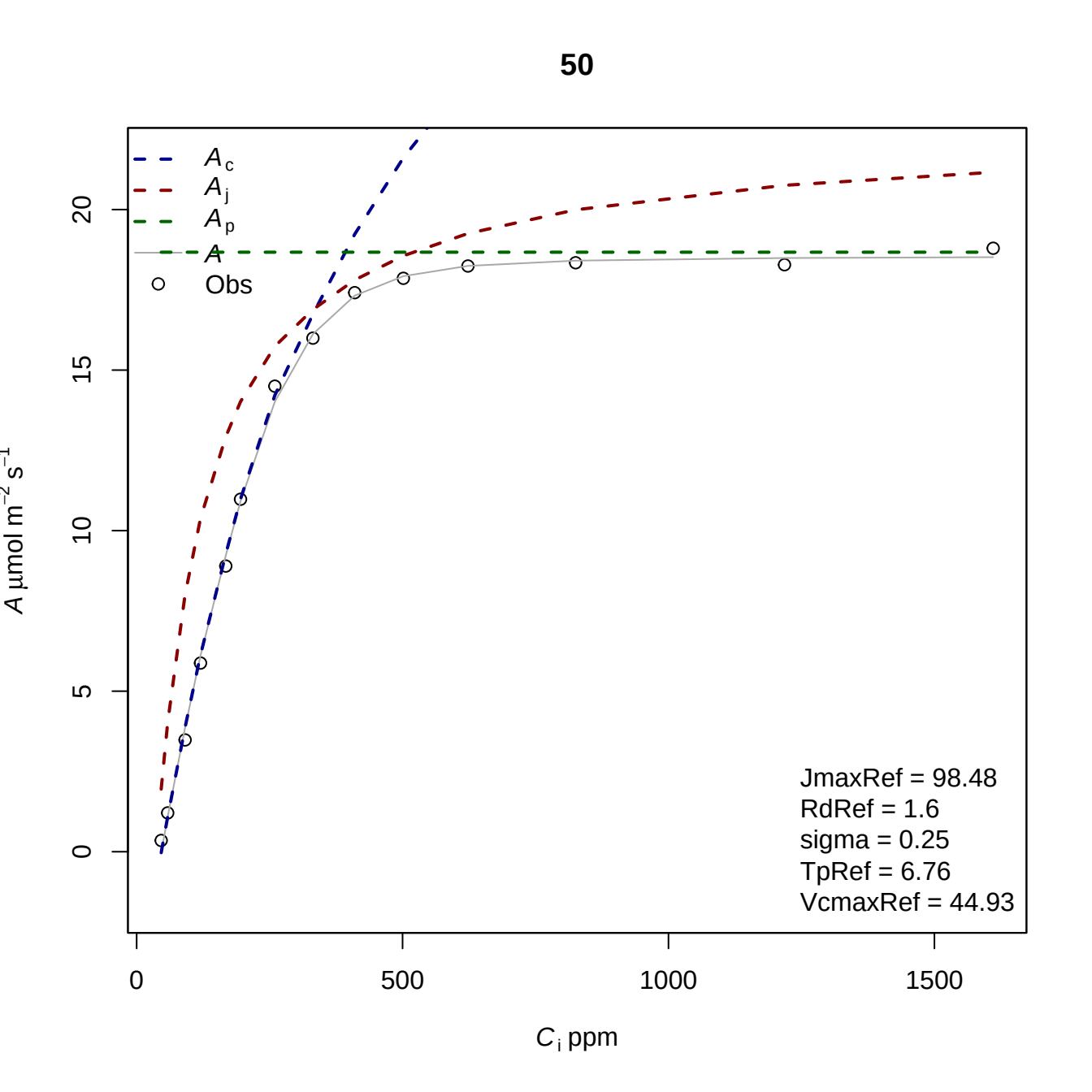

45
A_c
A_j
A_p
A
Obs
JmaxRef = 133.66
RdRef = 2.71
sigma = 0.4
TpRef = 10.38
VcmaxRef = 64.81
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
0
500
1000
1500

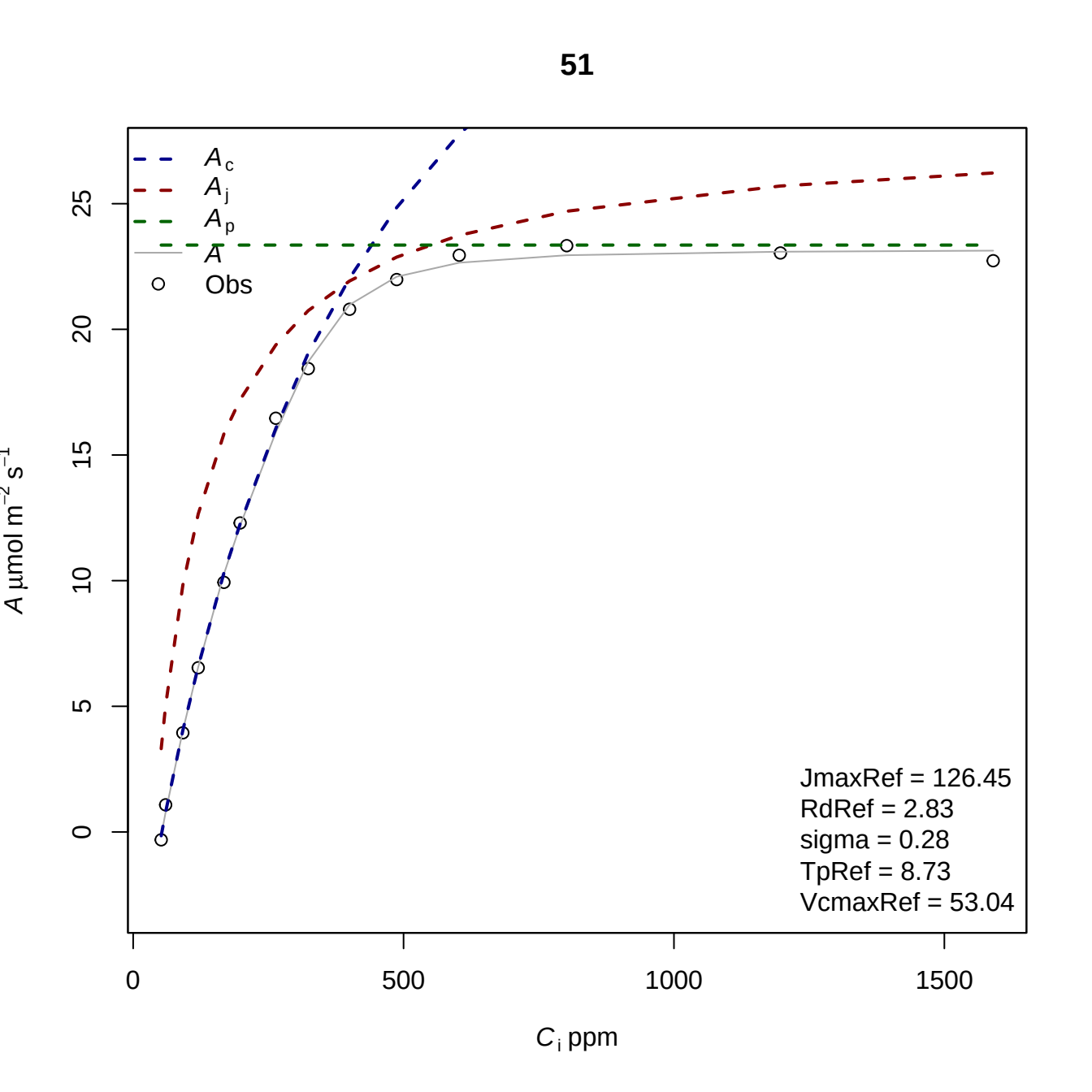

46
A_c
A_j
A_p
A
Obs
JmaxRef = 82.38
RdRef = 1.34
sigma = 0.14
TpRef = 5.74
VcmaxRef = 36.75
A µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
C_i ppm

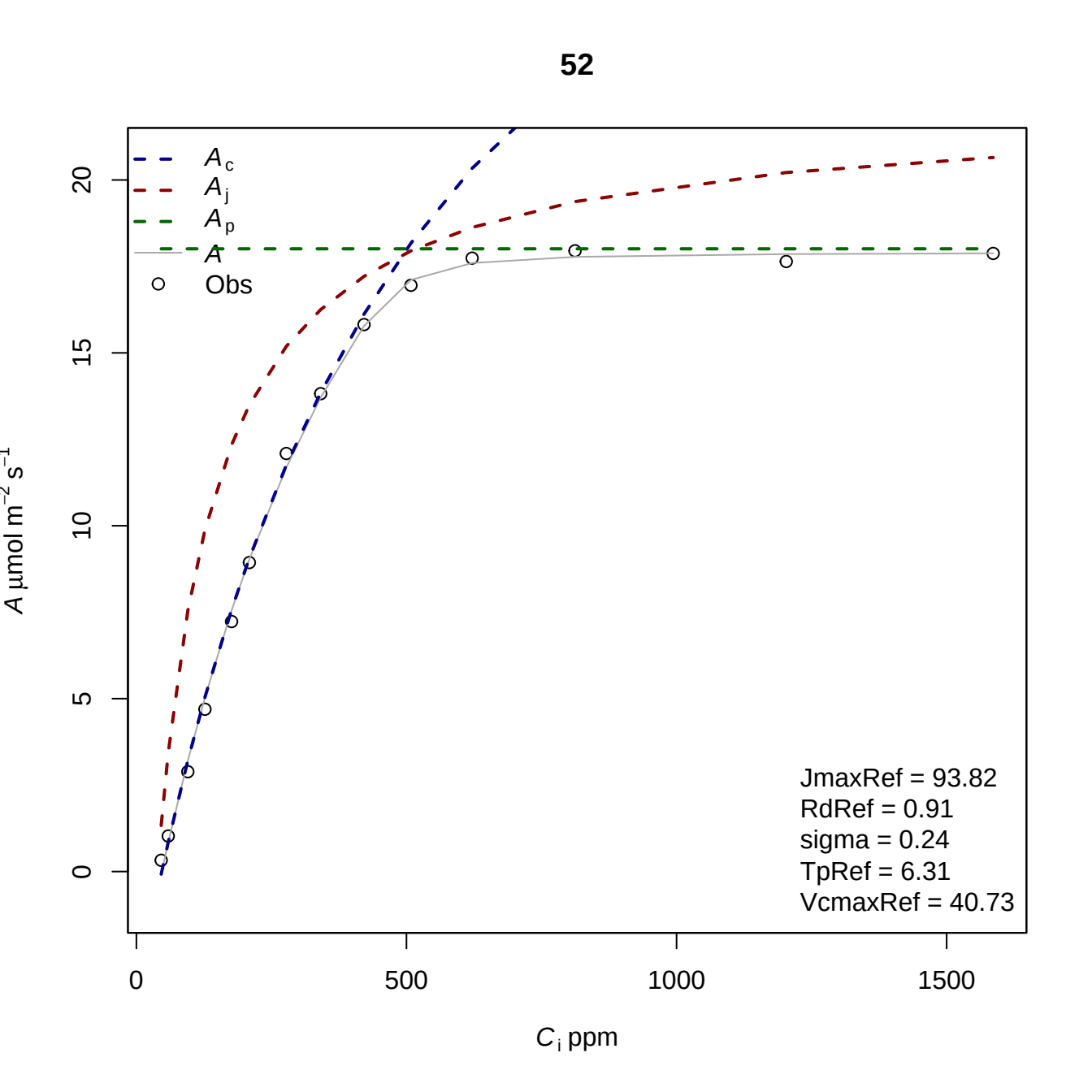

47
A_c
A_j
A_p
A
Obs
JmaxRef = 77.39
RdRef = 1.36
sigma = 0.23
TpRef = 5.43
VcmaxRef = 35.9
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
500
1000
1500

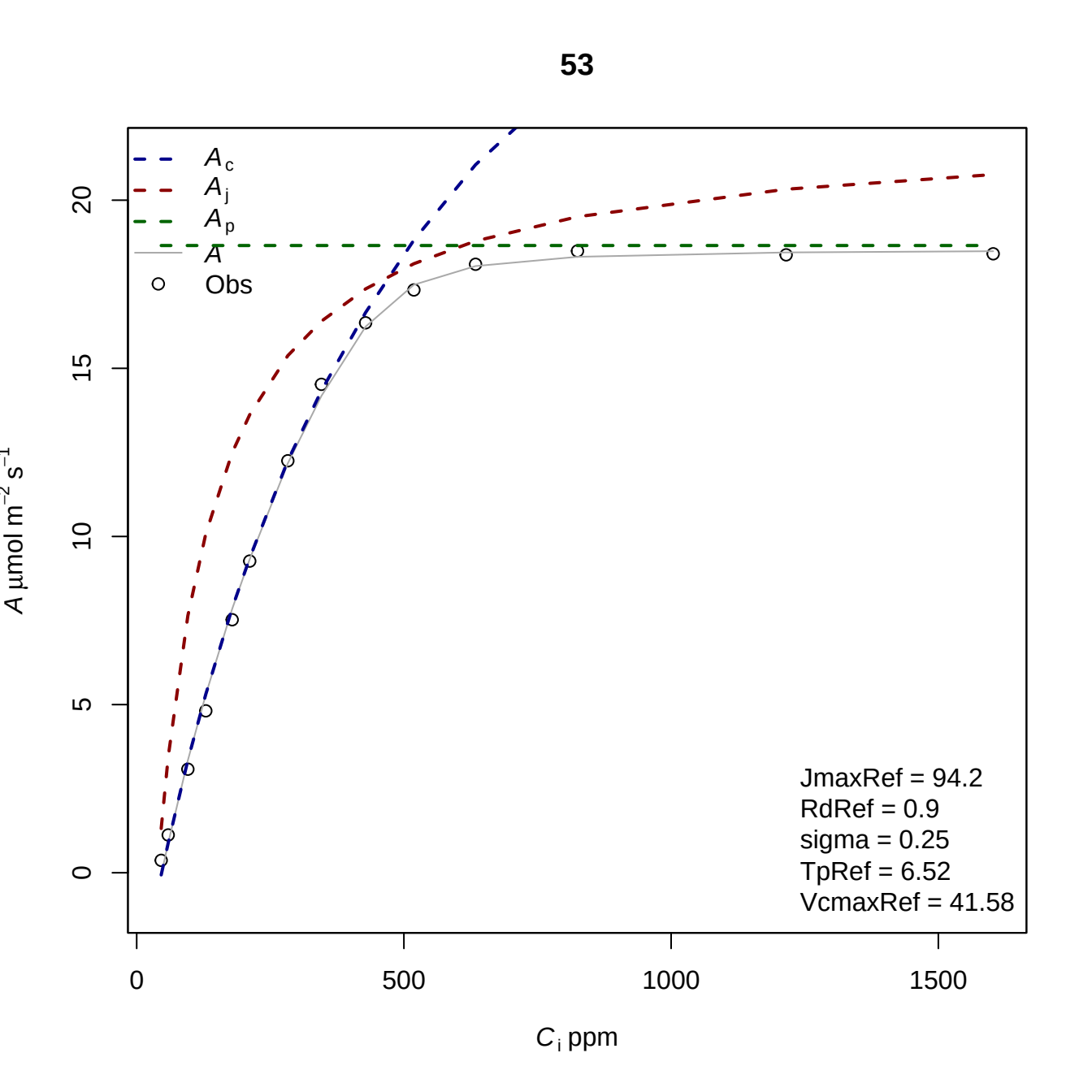

48
A_c
A_j
A_p
A
Obs
JmaxRef = 74.32
RdRef = 1.34
sigma = 0.15
TpRef = 5.34
VcmaxRef = 35.4
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
500
1000
1500

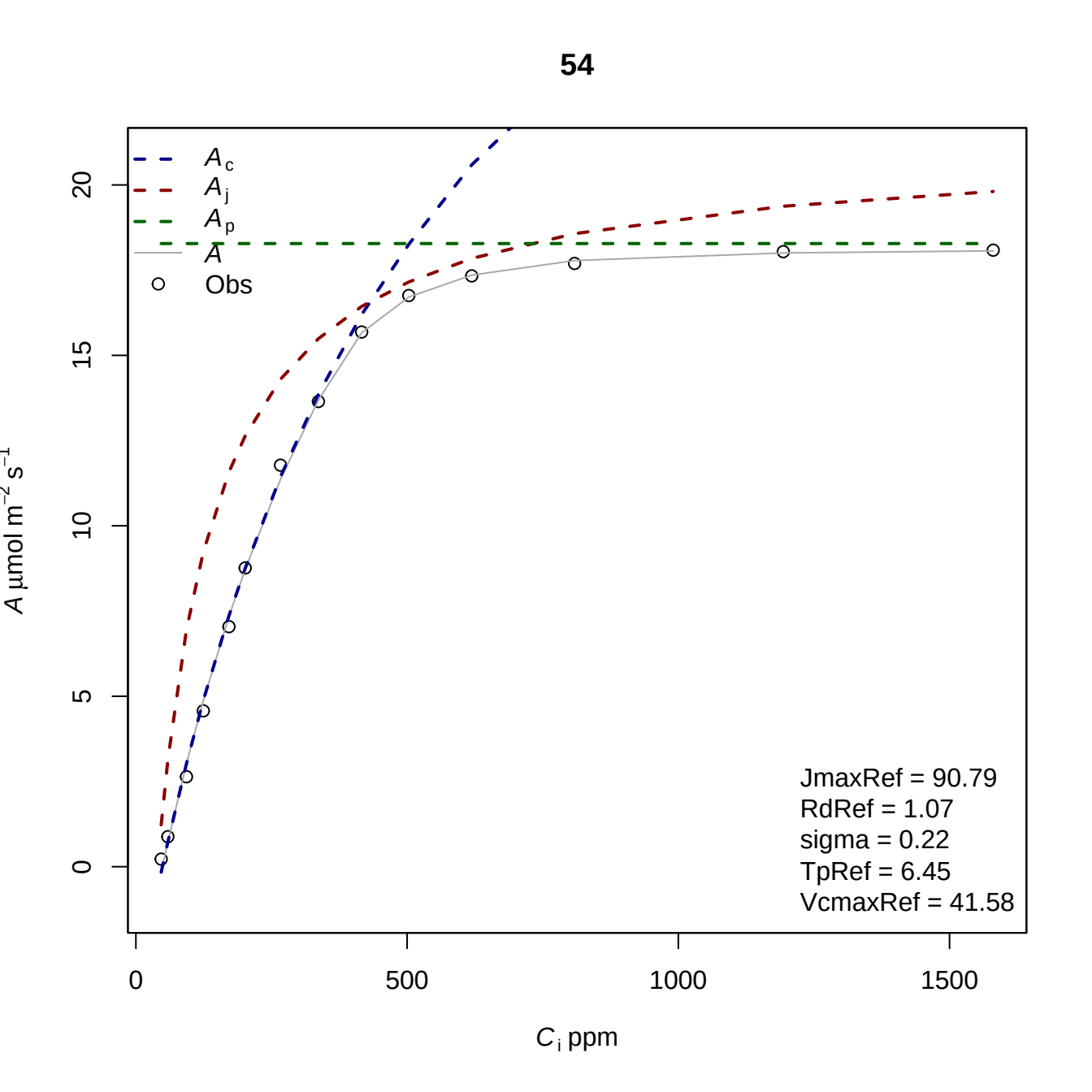

49
A_c
A_j
A_p
A
Obs
JmaxRef = 60.68
RdRef = 1.22
sigma = 0.1
TpRef = 4.53
VcmaxRef = 27.82
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
C_i ppm

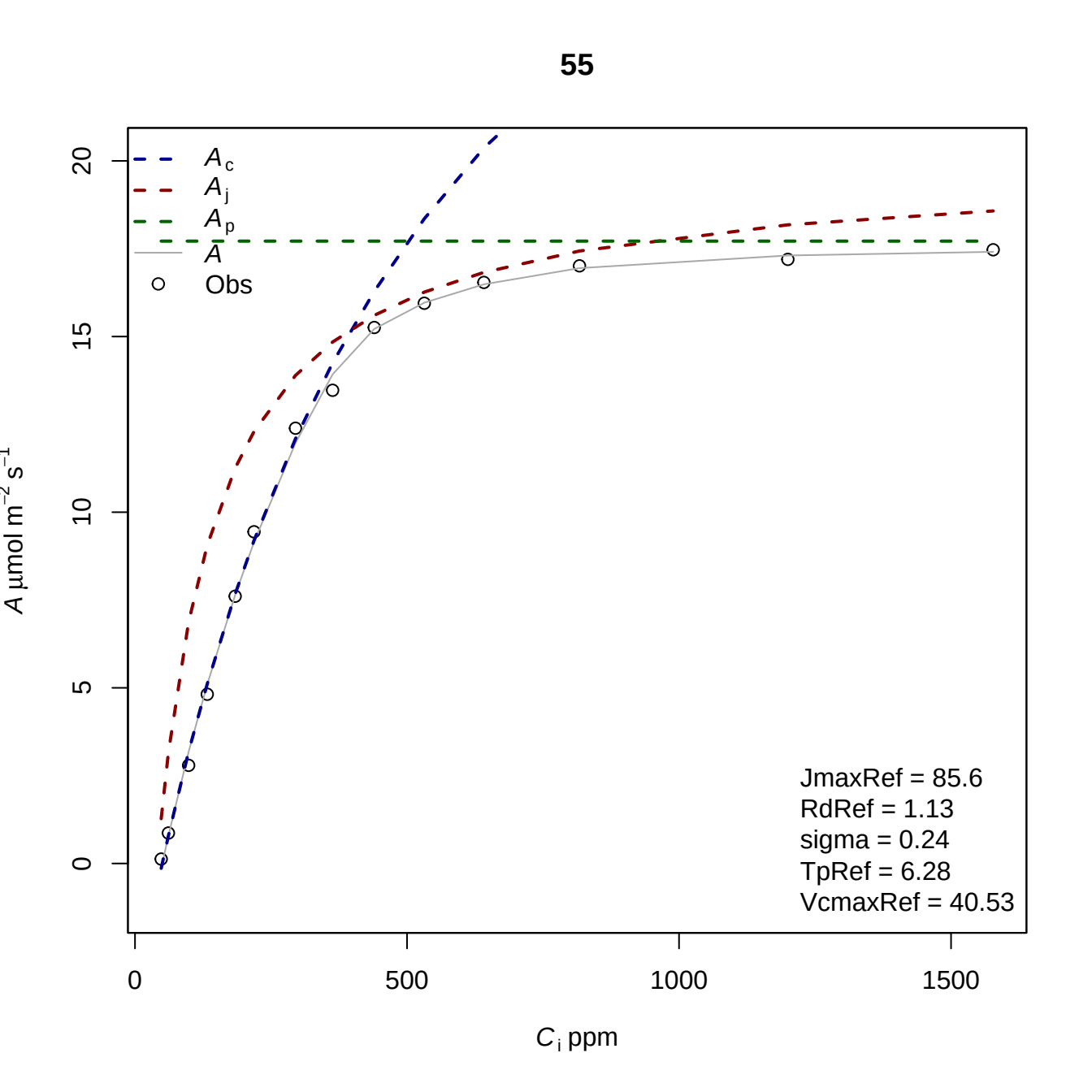

50
A_c
A_j
A_p
A
Obs
JmaxRef = 98.48
RdRef = 1.6
sigma = 0.25
TpRef = 6.76
VcmaxRef = 44.93
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
C_i ppm

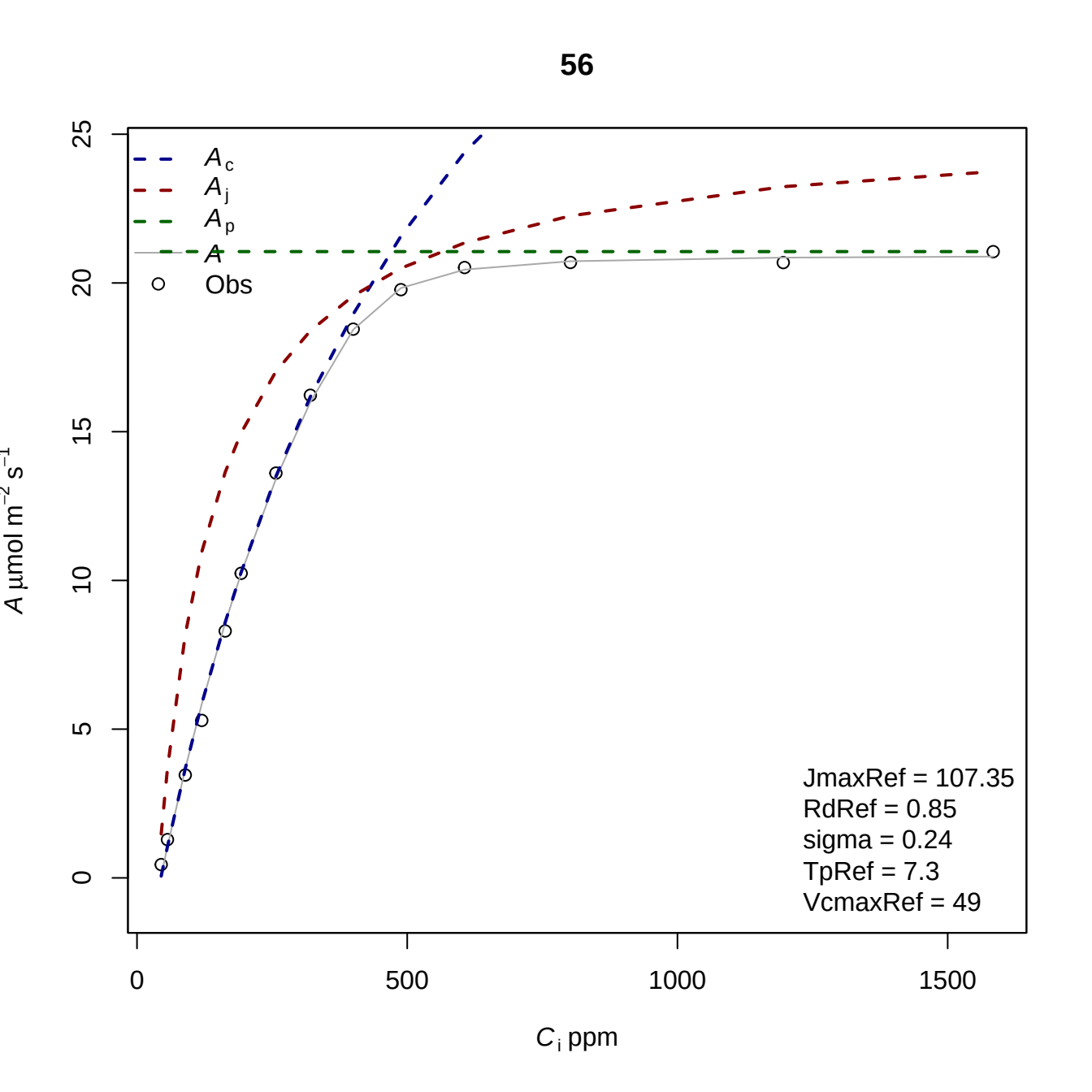

51
A_c
A_j
A_p
A
Obs
JmaxRef = 126.45
RdRef = 2.83
sigma = 0.28
TpRef = 8.73
VcmaxRef = 53.04
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
500
1000
1500

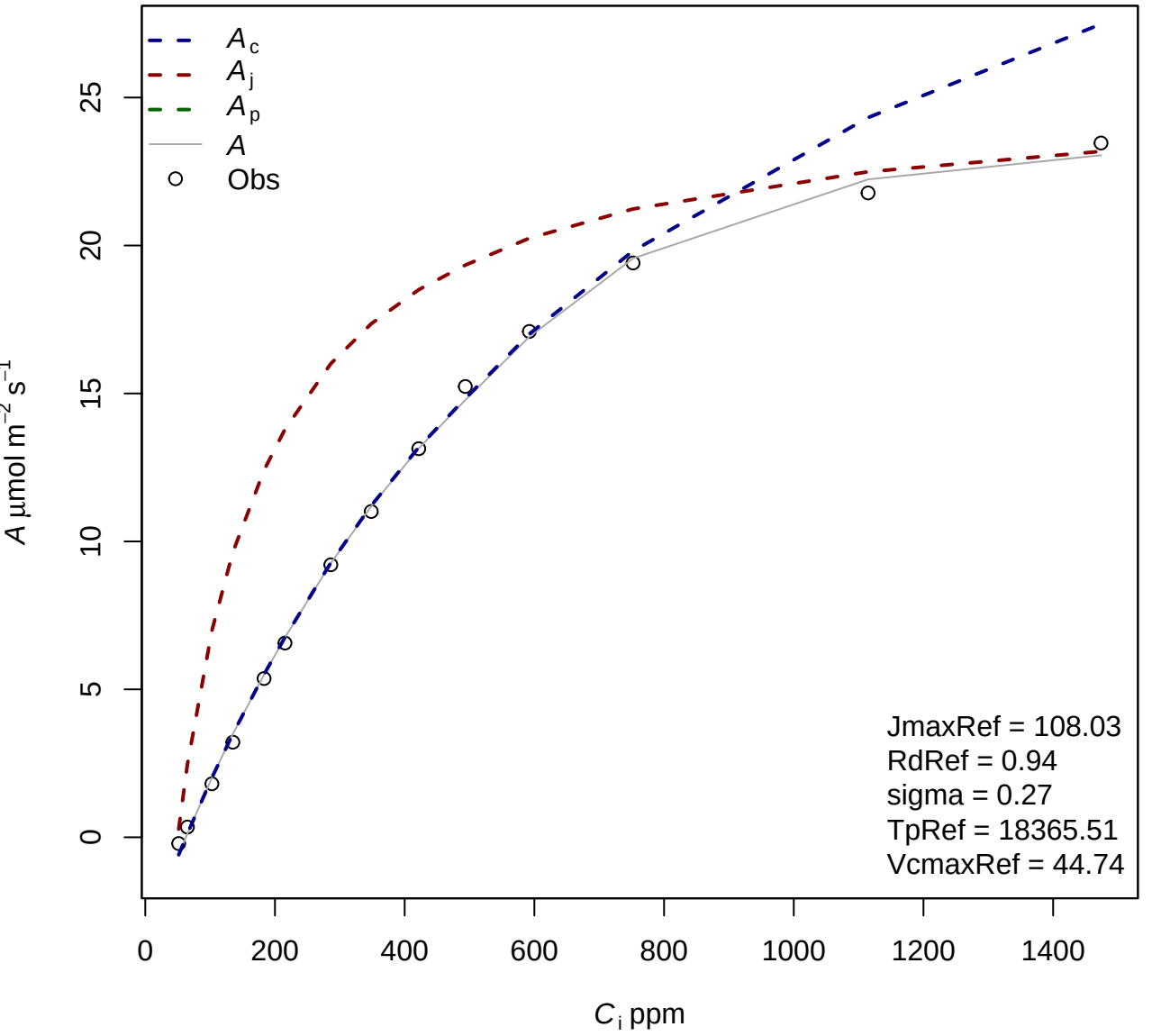

52
A_c
A_j
A_p
A
Obs
JmaxRef = 93.82
RdRef = 0.91
sigma = 0.24
TpRef = 6.31
VcmaxRef = 40.73
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
500
1000
1500

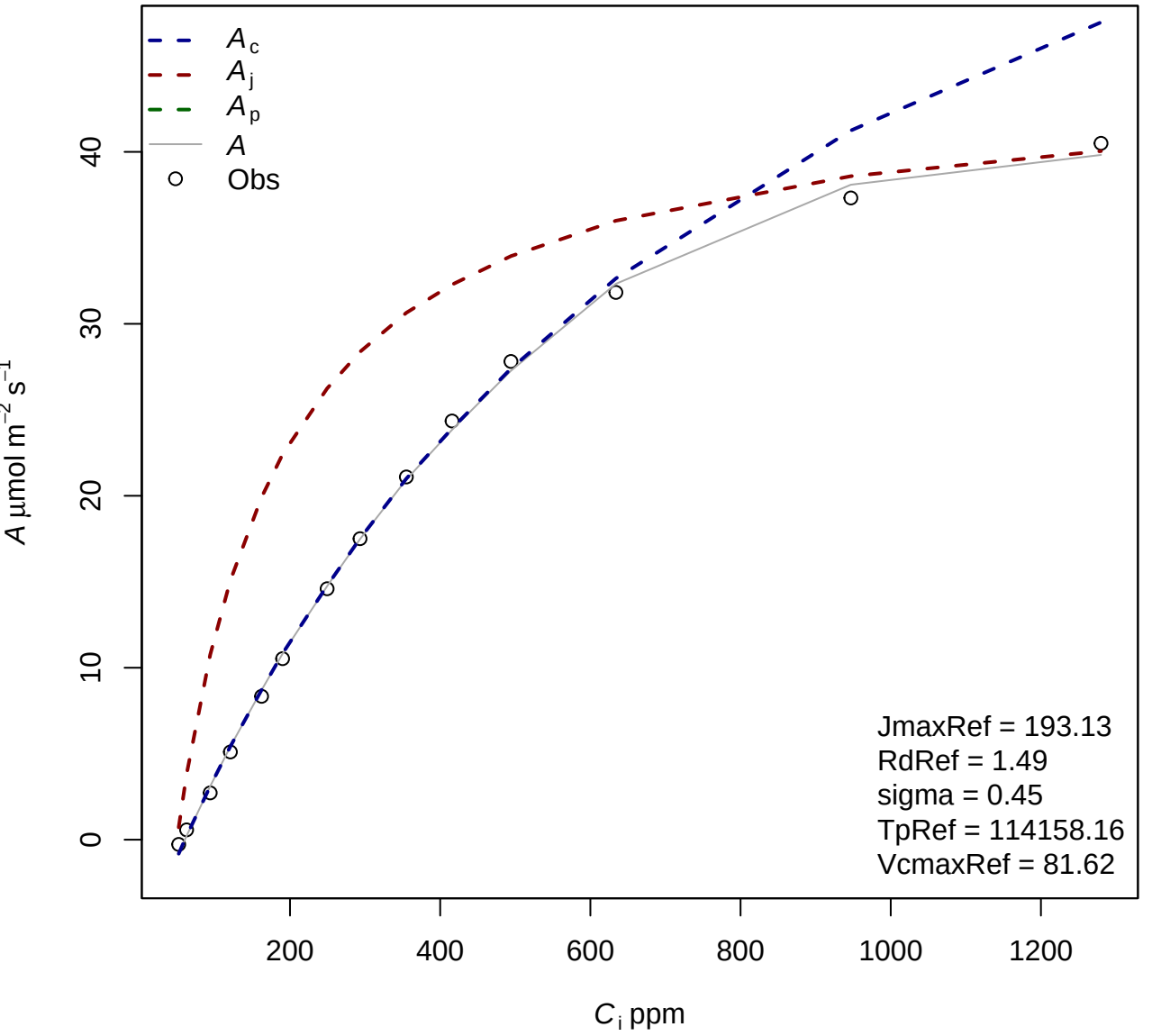

53
A_c
A_j
A_p
A
Obs
JmaxRef = 94.2
RdRef = 0.9
sigma = 0.25
TpRef = 6.52
VcmaxRef = 41.58
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
C_i ppm

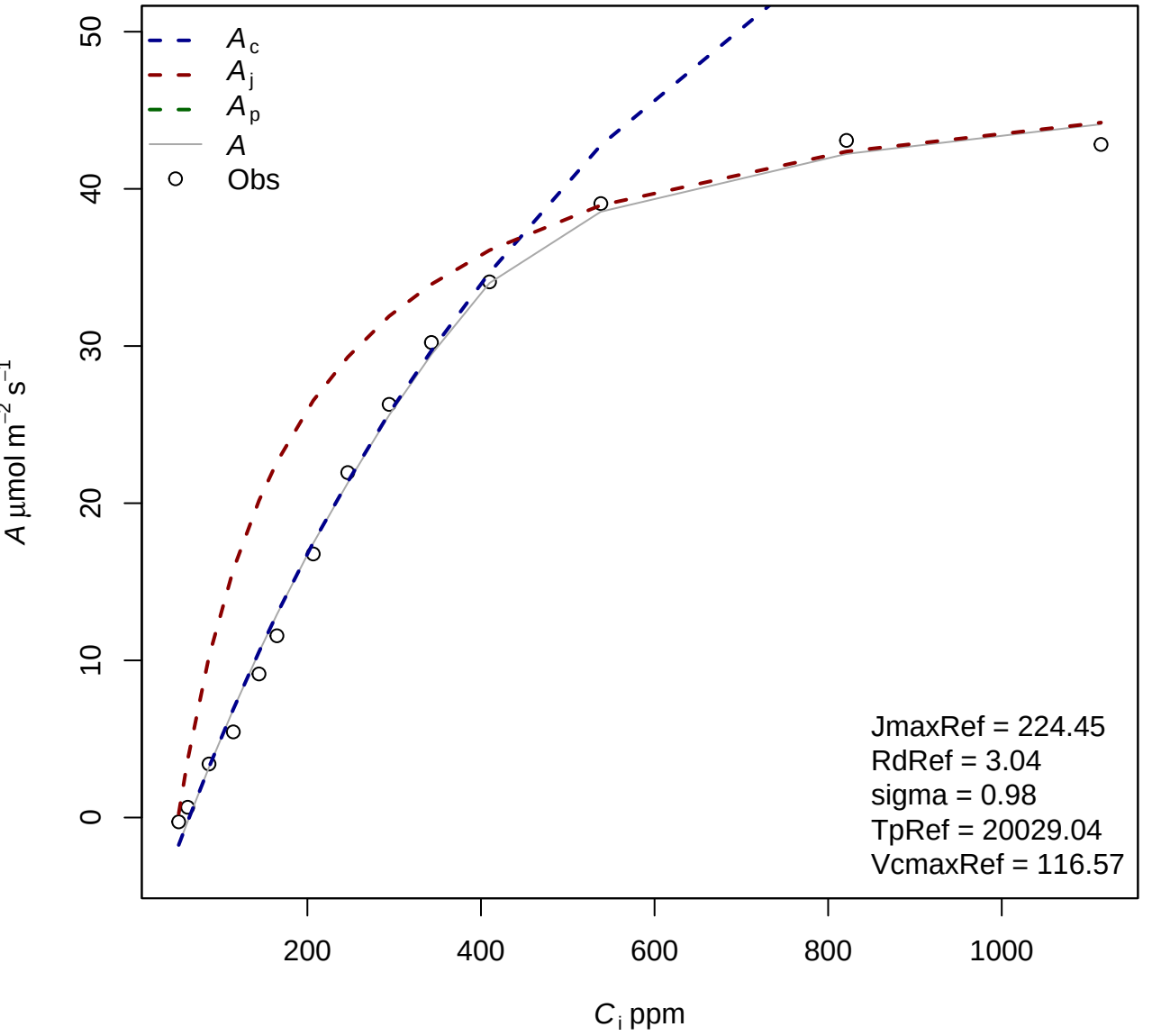

54
A_c
A_j
A_p
A
Obs
JmaxRef = 90.79
RdRef = 1.07
sigma = 0.22
TpRef = 6.45
VcmaxRef = 41.58
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
500
1000
1500

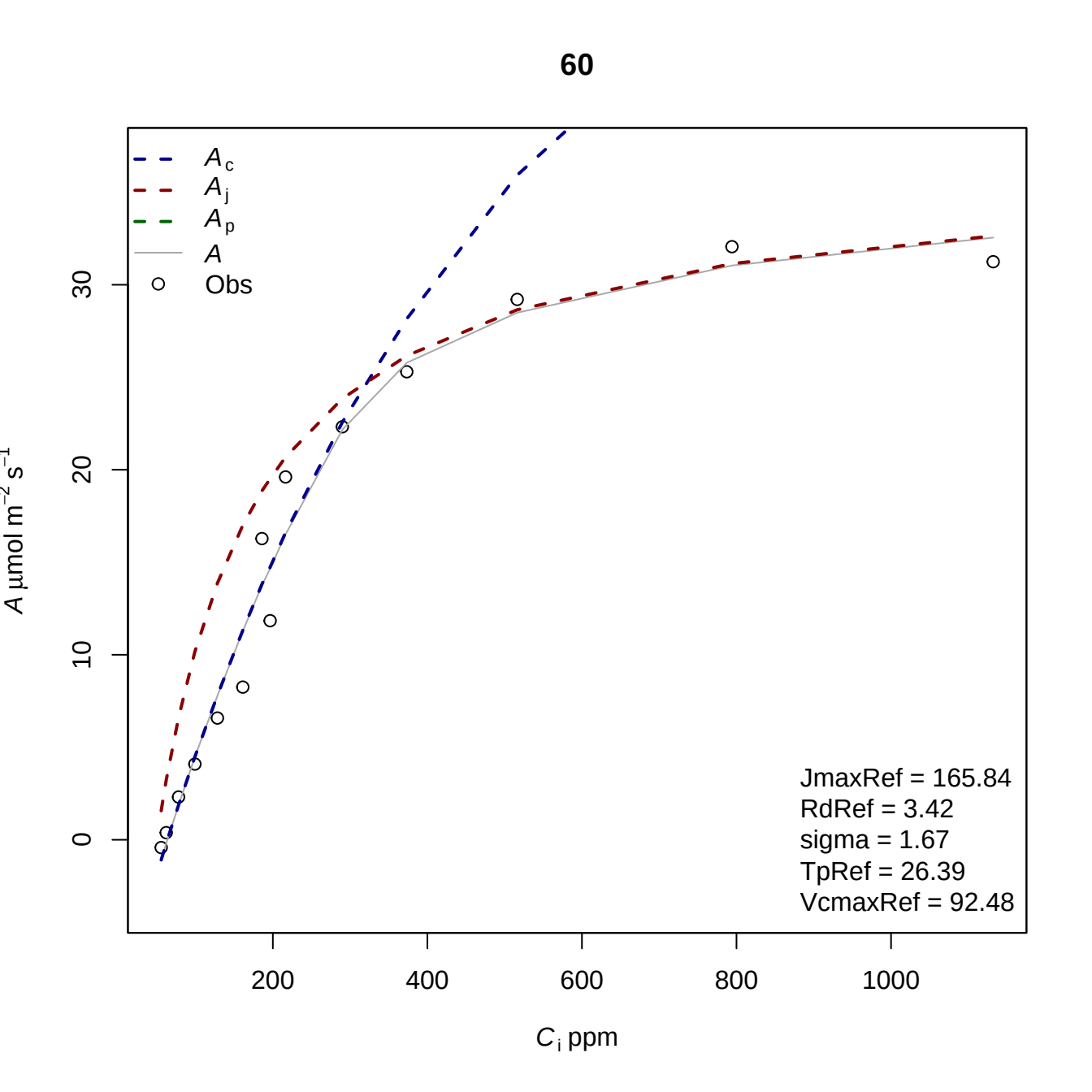

55
A_c
A_j
A_p
A
Obs
JmaxRef = 85.6
RdRef = 1.13
sigma = 0.24
TpRef = 6.28
VcmaxRef = 40.53
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
500
1000
1500

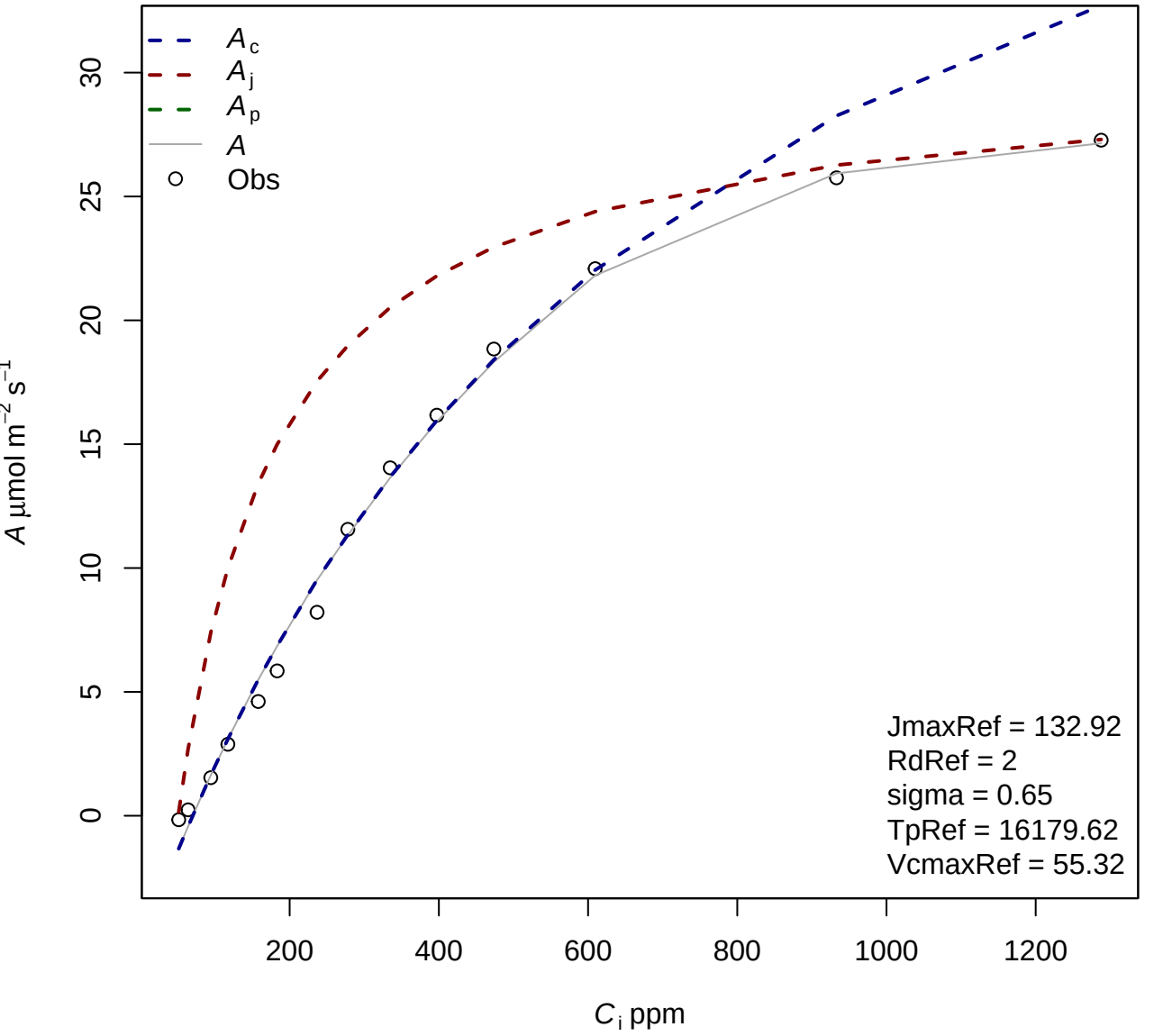

56
A_c
A_j
A_p
A
Obs
JmaxRef = 107.35
RdRef = 0.85
sigma = 0.24
TpRef = 7.3
VcmaxRef = 49
0
5
10
15
20
25
0
500
1000
1500
A µmol m^{-2} s^{-1}
C_i ppm

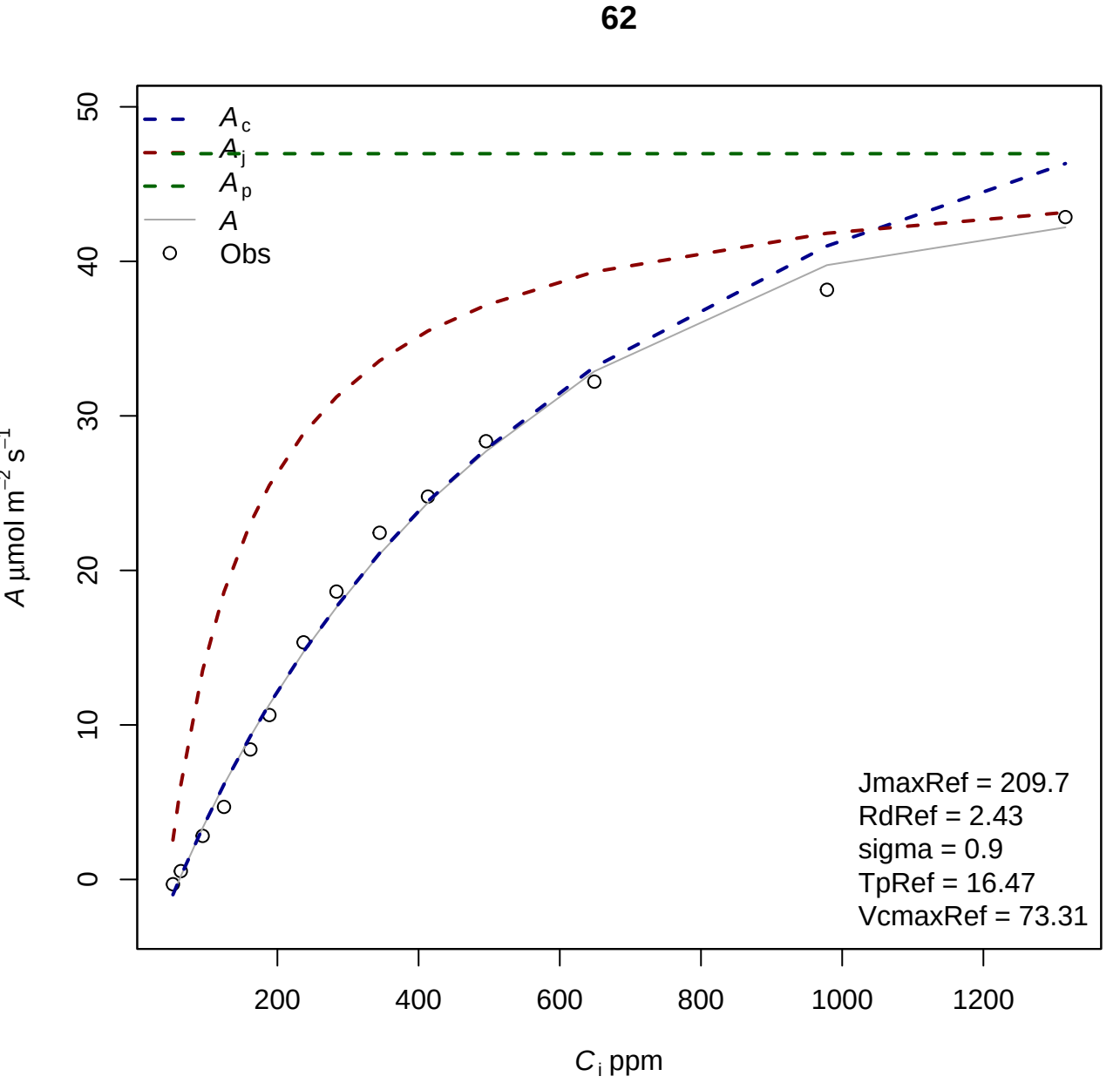

A_c
A_j
A_p
A
Obs
JmaxRef = 108.03
RdRef = 0.94
sigma = 0.27
TpRef = 18365.51
VcmaxRef = 44.74
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200
1400

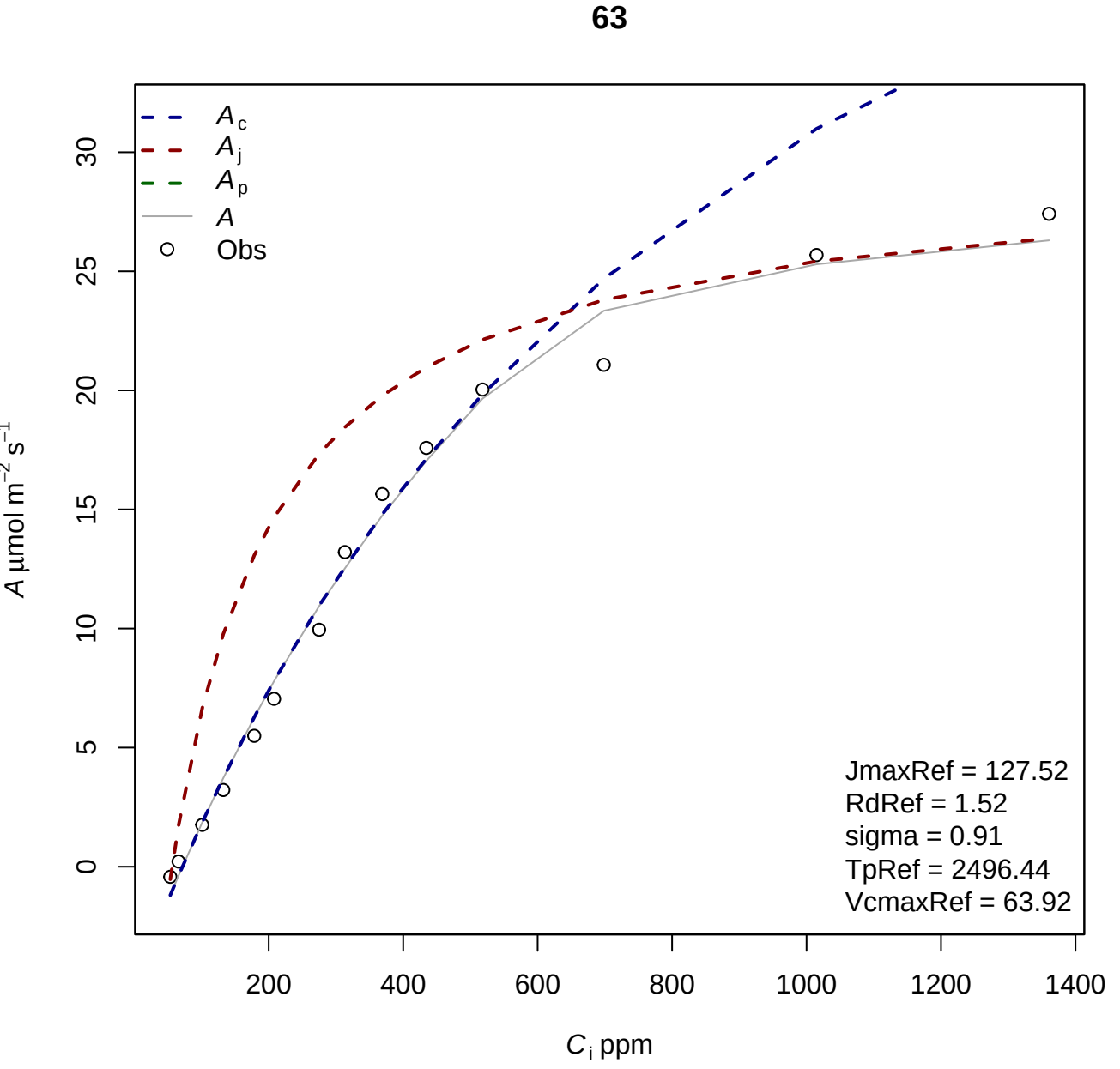

A_c
A_j
A_p
A
Obs
JmaxRef = 193.13
RdRef = 1.49
sigma = 0.45
TpRef = 114158.16
VcmaxRef = 81.62
0
10
20
30
40
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

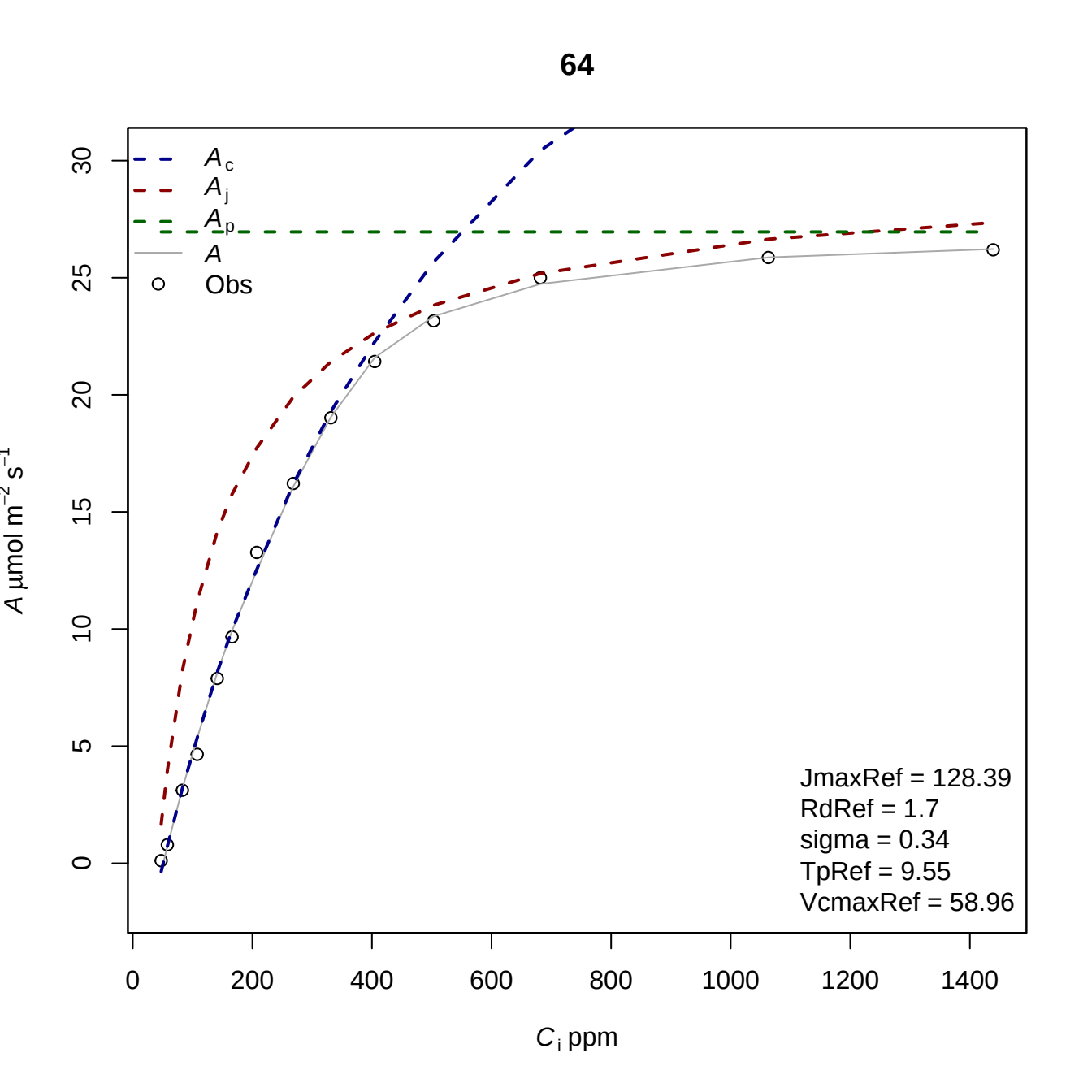

A_c
A_j
A_p
A
Obs
JmaxRef = 224.45
RdRef = 3.04
sigma = 0.98
TpRef = 20029.04
VcmaxRef = 116.57
0
10
20
30
40
50
200
400
600
800
1000
A µmol m^{-2} s^{-1}
C_i ppm

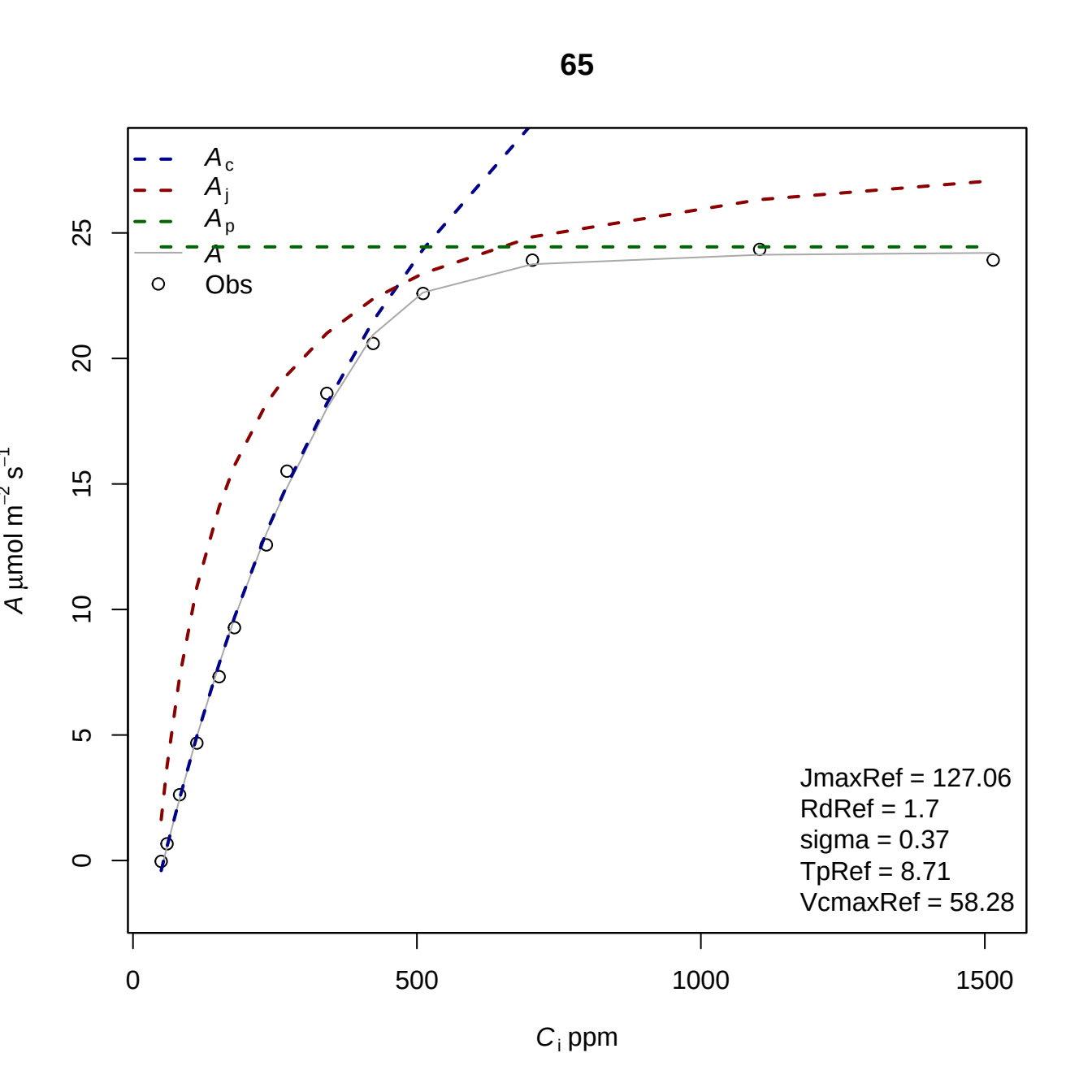

60
A_c
A_j
A_p
A
Obs
JmaxRef = 165.84
RdRef = 3.42
sigma = 1.67
TpRef = 26.39
VcmaxRef = 92.48
A µmol m^{-2} s^{-1}
0
10
20
30
200
400
600
800
1000
C_i ppm

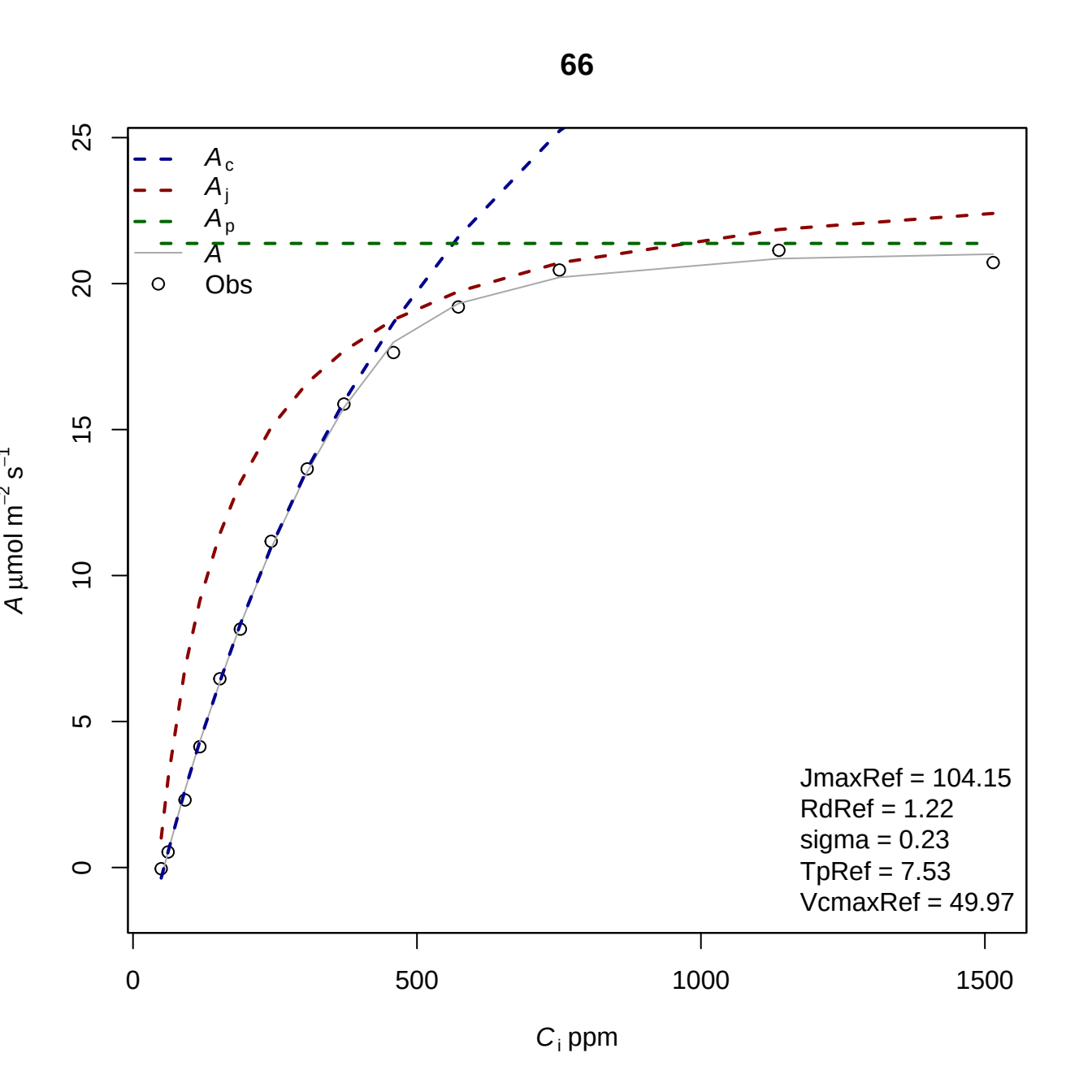

A_c
A_j
A_p
A
Obs
JmaxRef = 132.92
RdRef = 2
sigma = 0.65
TpRef = 16179.62
VcmaxRef = 55.32
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
200
400
600
800
1000
1200

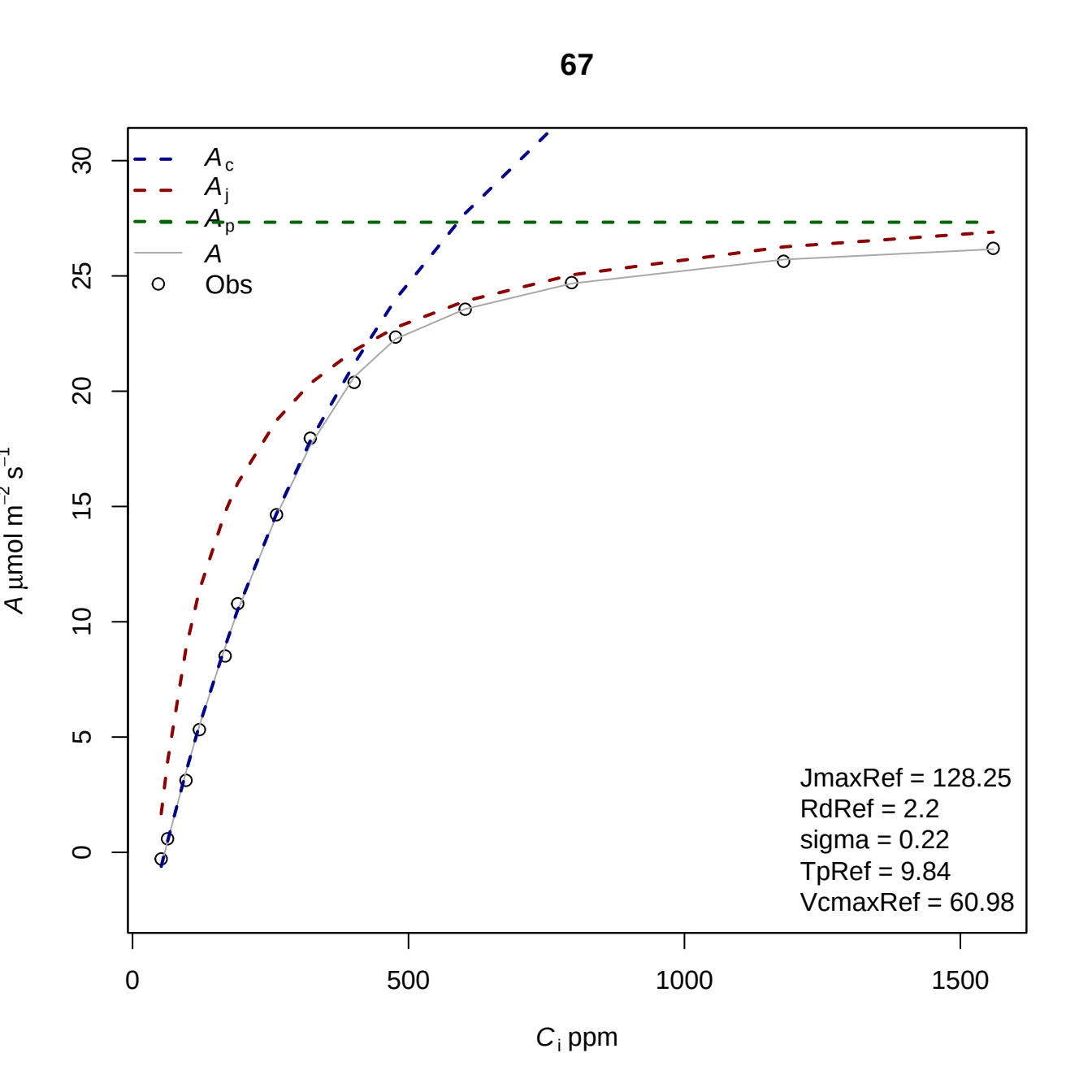

62
A_c
A_j
A_p
A
Obs
JmaxRef = 209.7
RdRef = 2.43
sigma = 0.9
TpRef = 16.47
VcmaxRef = 73.31
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
50
200
400
600
800
1000
1200

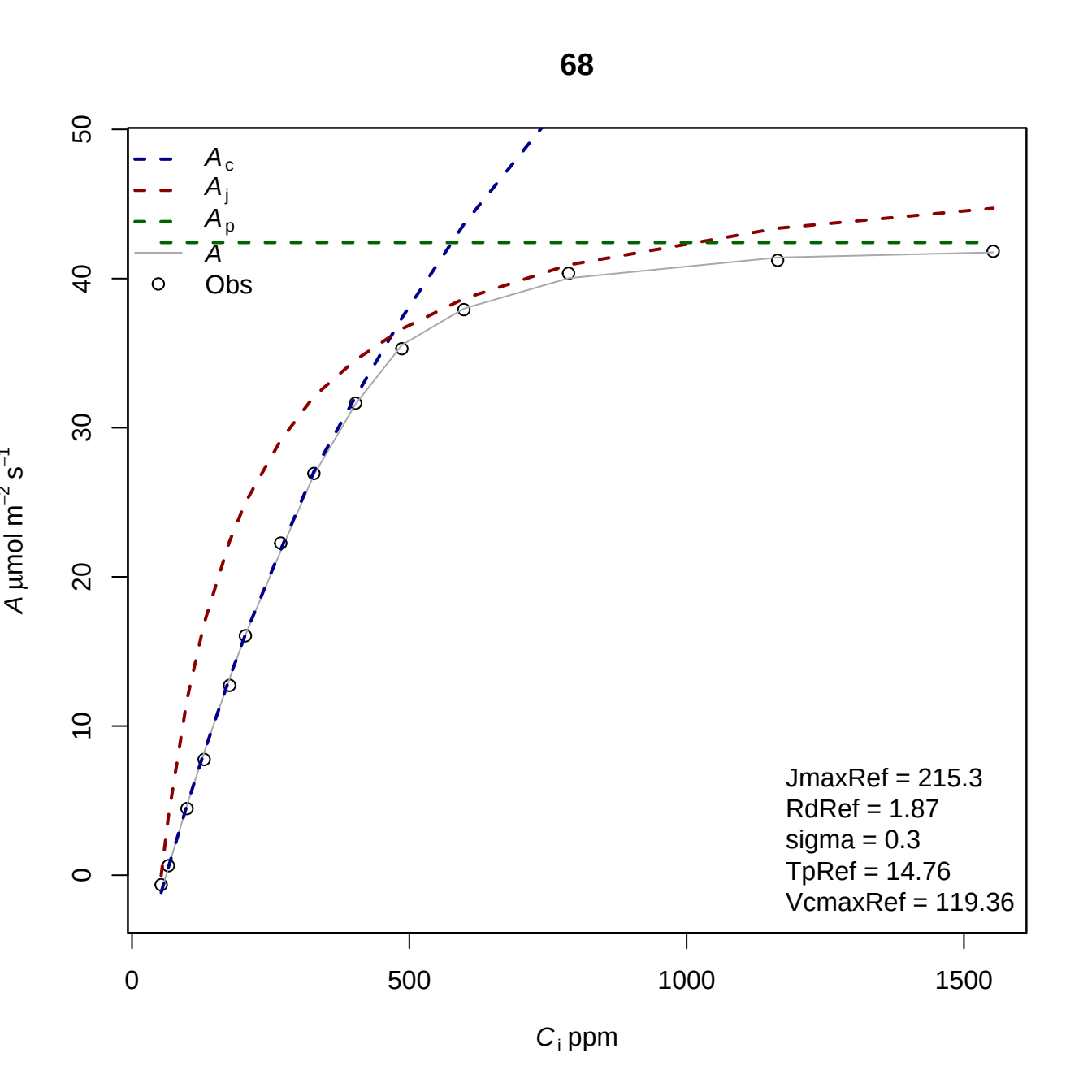

63
A_c
A_j
A_p
A
Obs
JmaxRef = 127.52
RdRef = 1.52
sigma = 0.91
TpRef = 2496.44
VcmaxRef = 63.92
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
200
400
600
800
1000
1200
1400

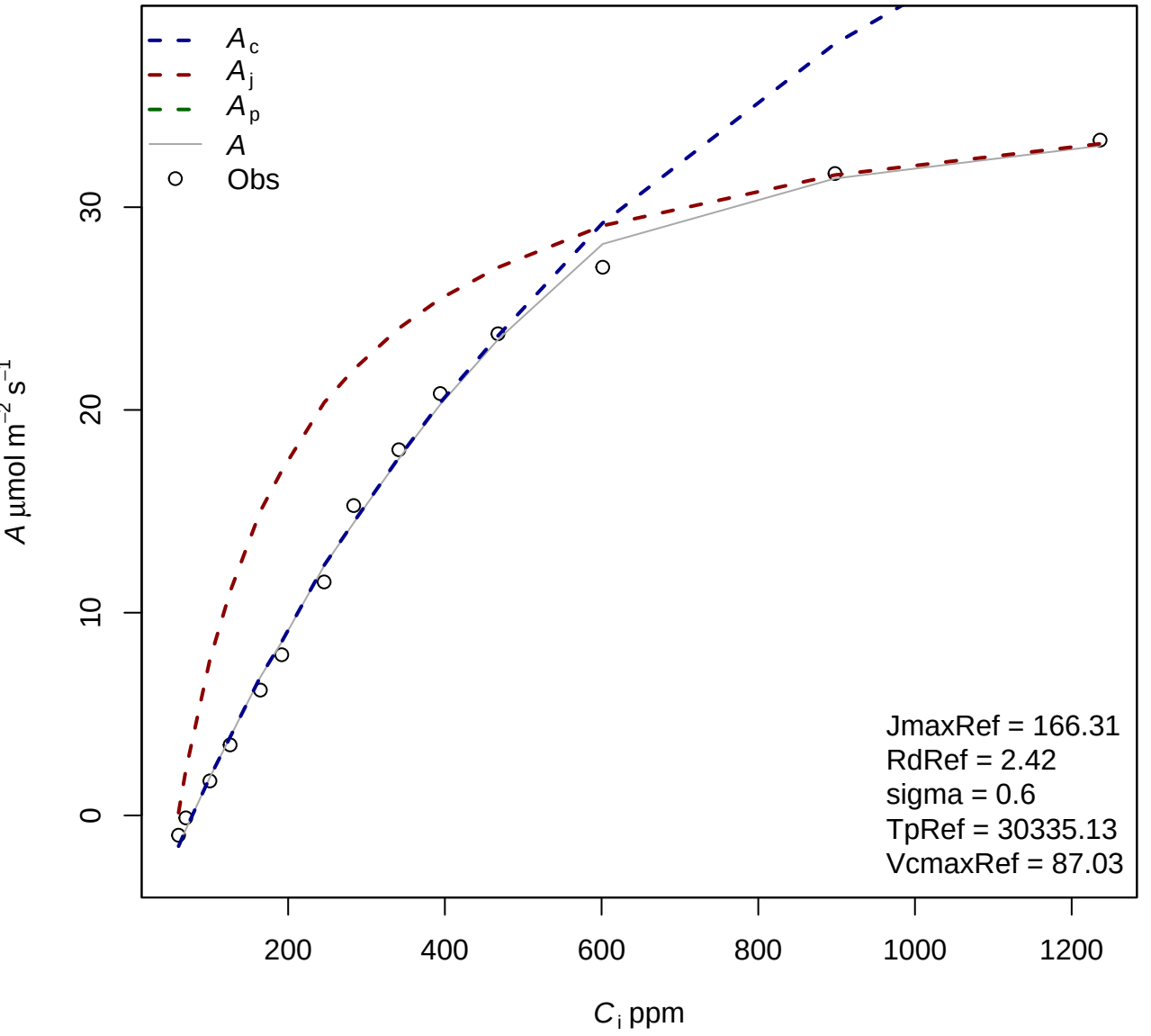

64
A_c
A_j
A_p
A
Obs
JmaxRef = 128.39
RdRef = 1.7
sigma = 0.34
TpRef = 9.55
VcmaxRef = 58.96
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
1400

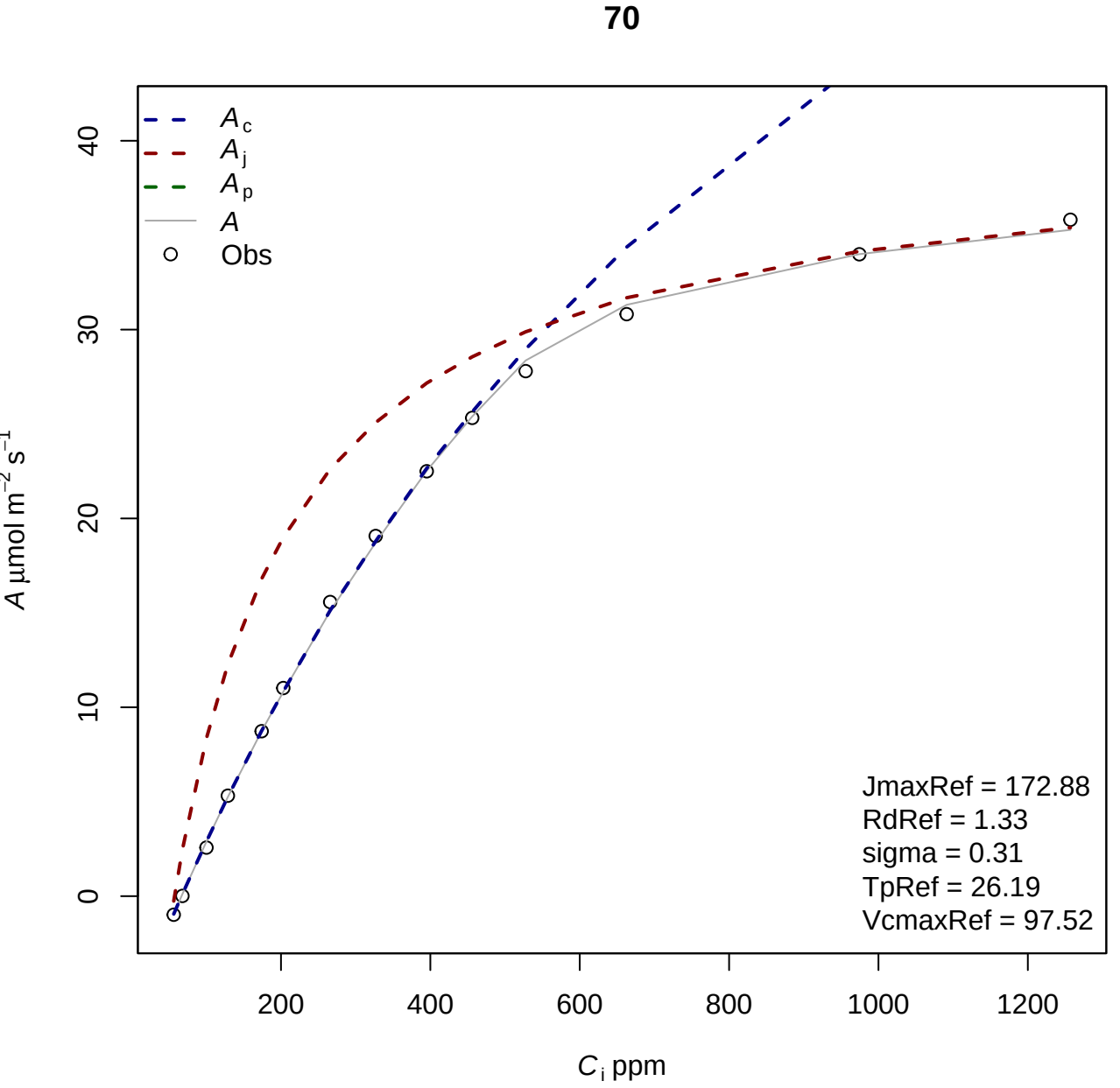

65
A_c
A_j
A_p
A
Obs
JmaxRef = 127.06
RdRef = 1.7
sigma = 0.37
TpRef = 8.71
VcmaxRef = 58.28
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
500
1000
1500

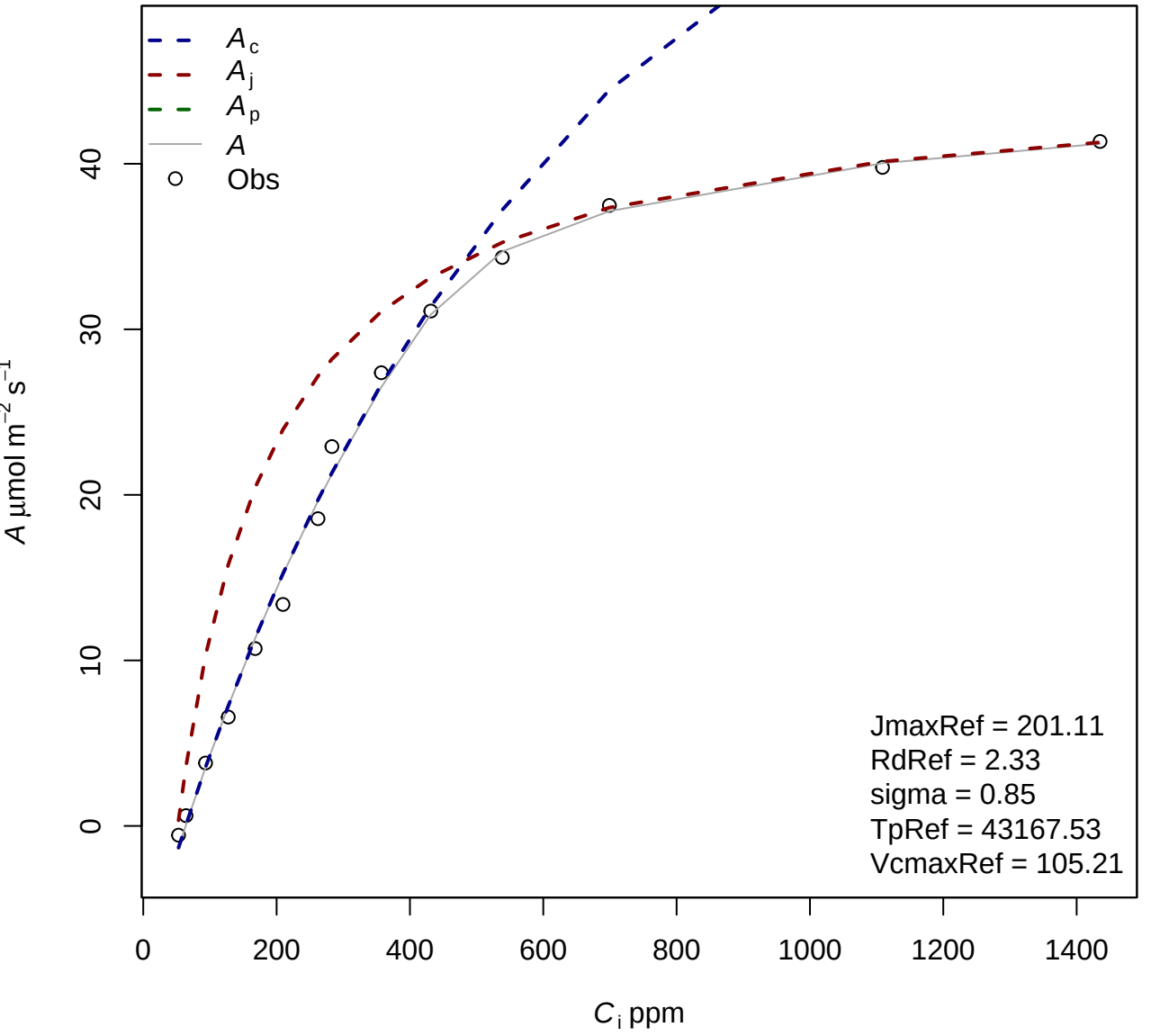

66
A_c
A_j
A_p
A
Obs
JmaxRef = 104.15
RdRef = 1.22
sigma = 0.23
TpRef = 7.53
VcmaxRef = 49.97
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
500
1000
1500

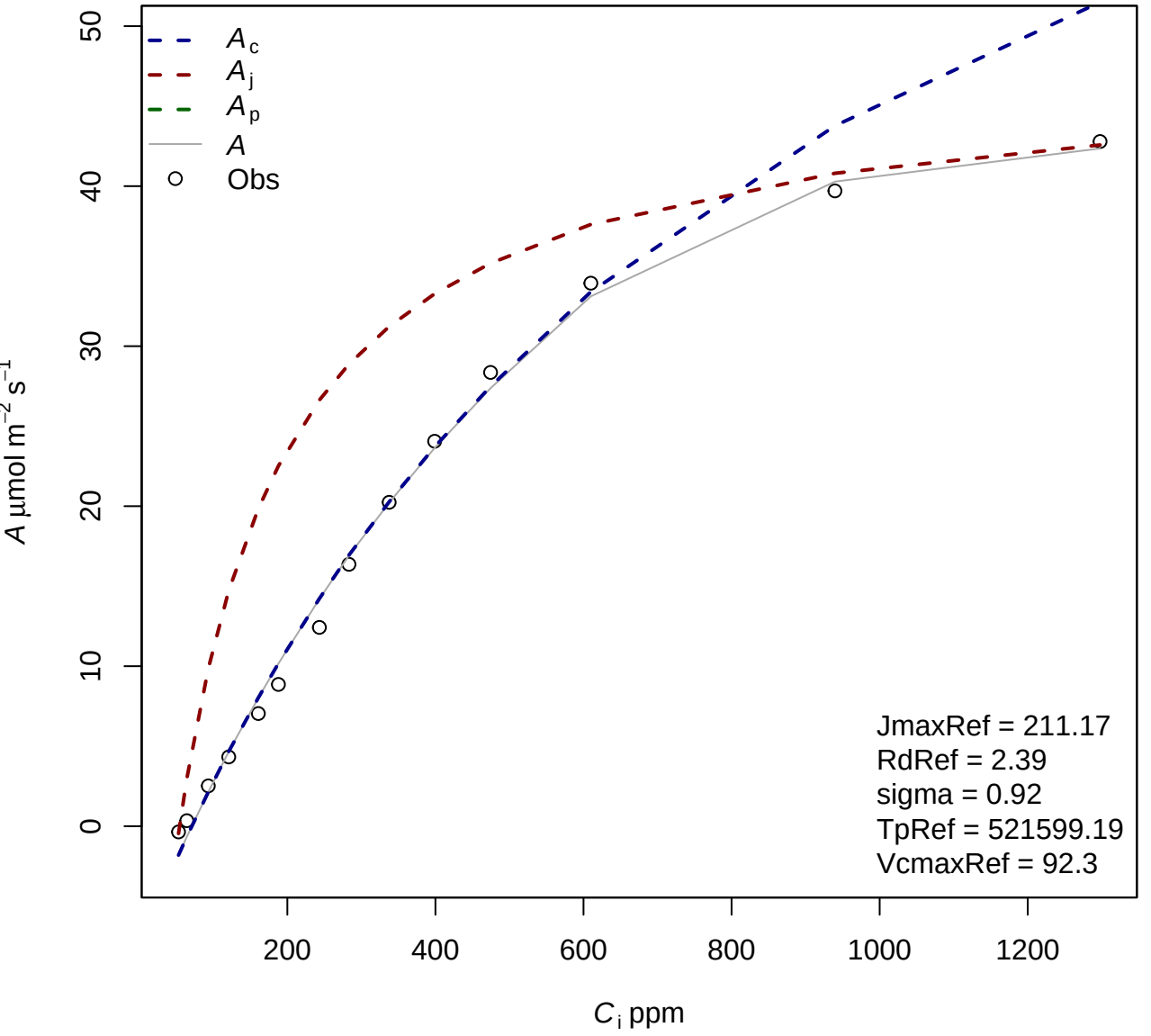

67
A_c
A_j
A_p
A
Obs
JmaxRef = 128.25
RdRef = 2.2
sigma = 0.22
TpRef = 9.84
VcmaxRef = 60.98
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
0
500
1000
1500

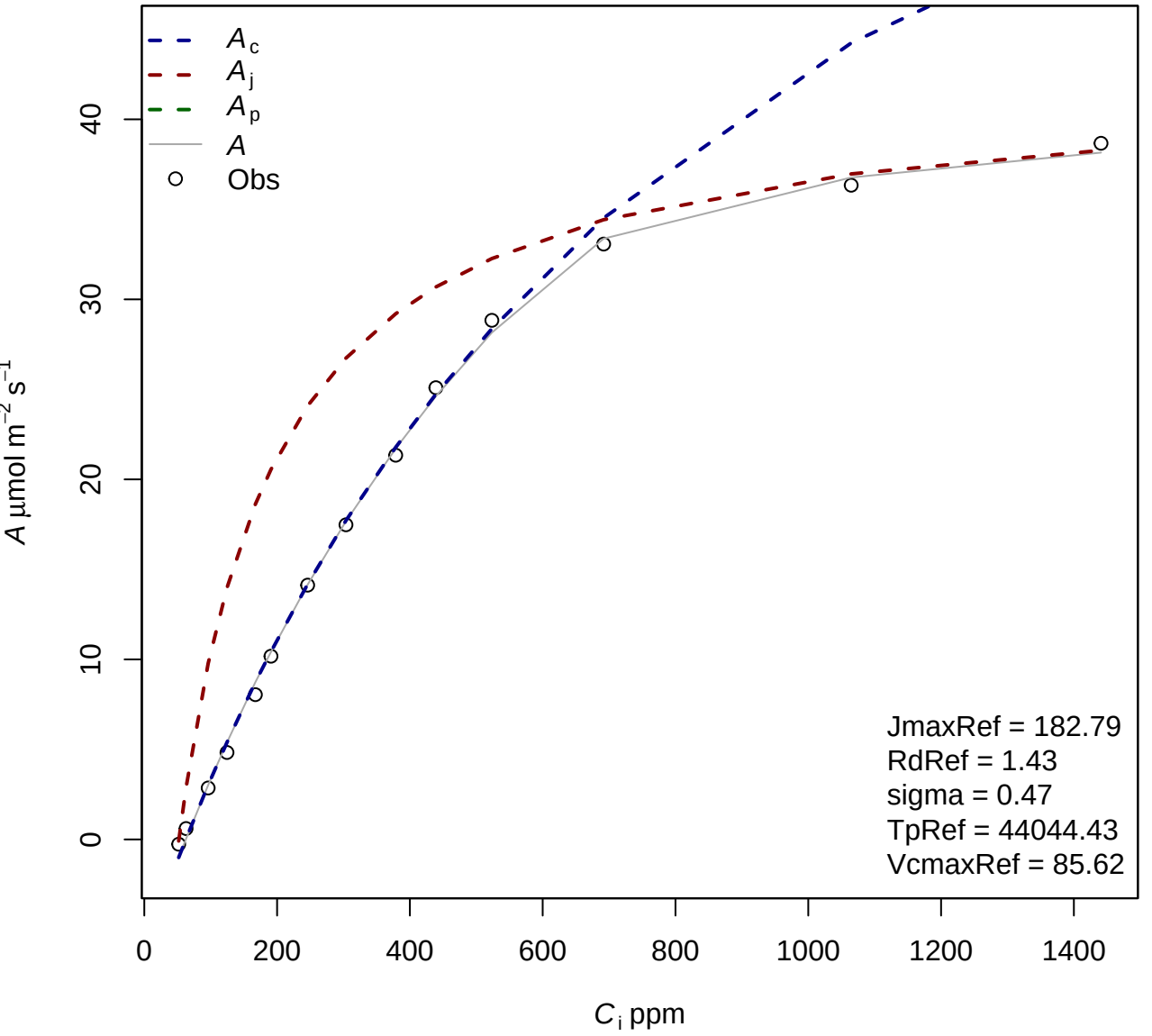

68
A_c
A_j
A_p
A
Obs
JmaxRef = 215.3
RdRef = 1.87
sigma = 0.3
TpRef = 14.76
VcmaxRef = 119.36
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
50
0
500
1000
1500

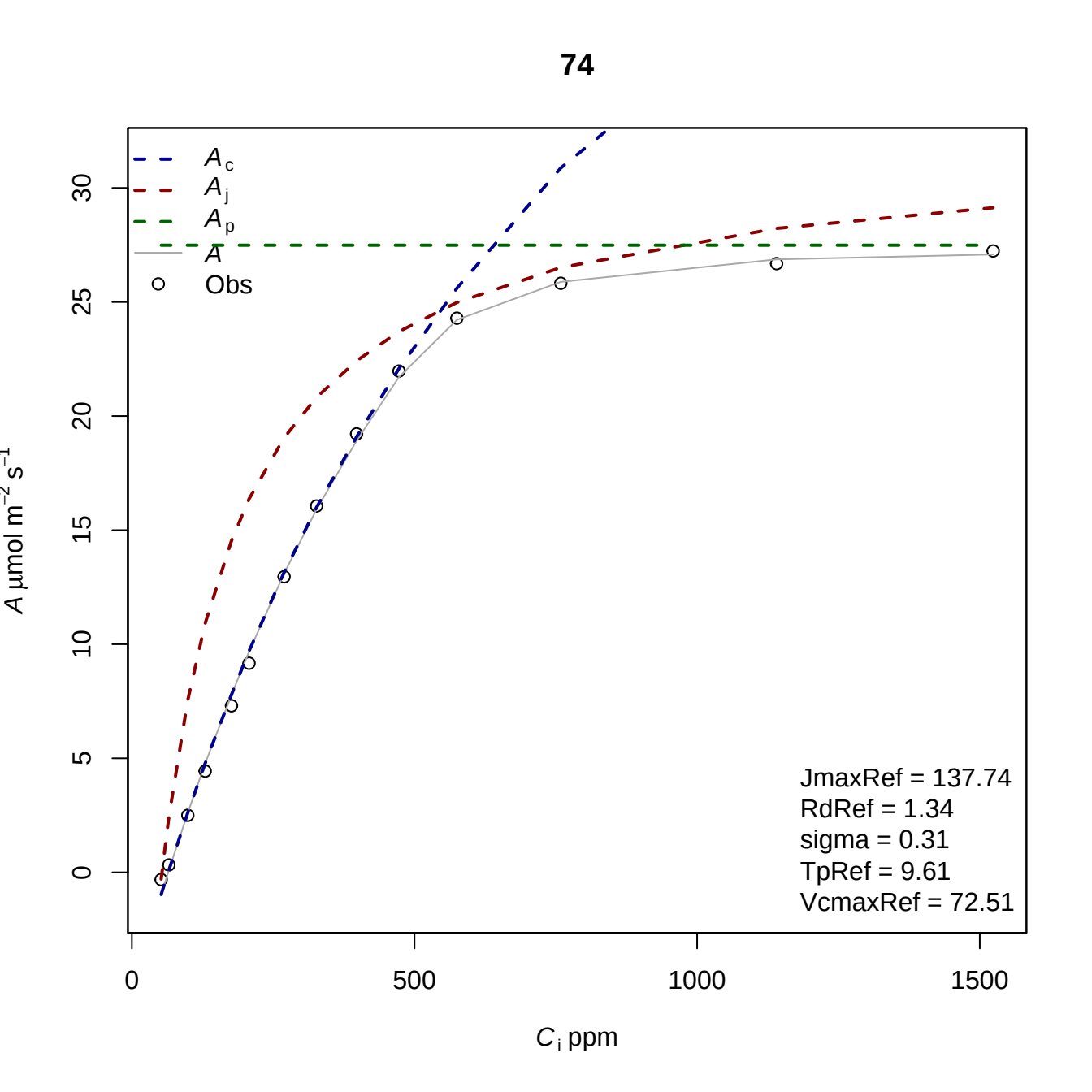

A_c
A_j
A_p
A
Obs
JmaxRef = 166.31
RdRef = 2.42
sigma = 0.6
TpRef = 30335.13
VcmaxRef = 87.03
0
10
20
30
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

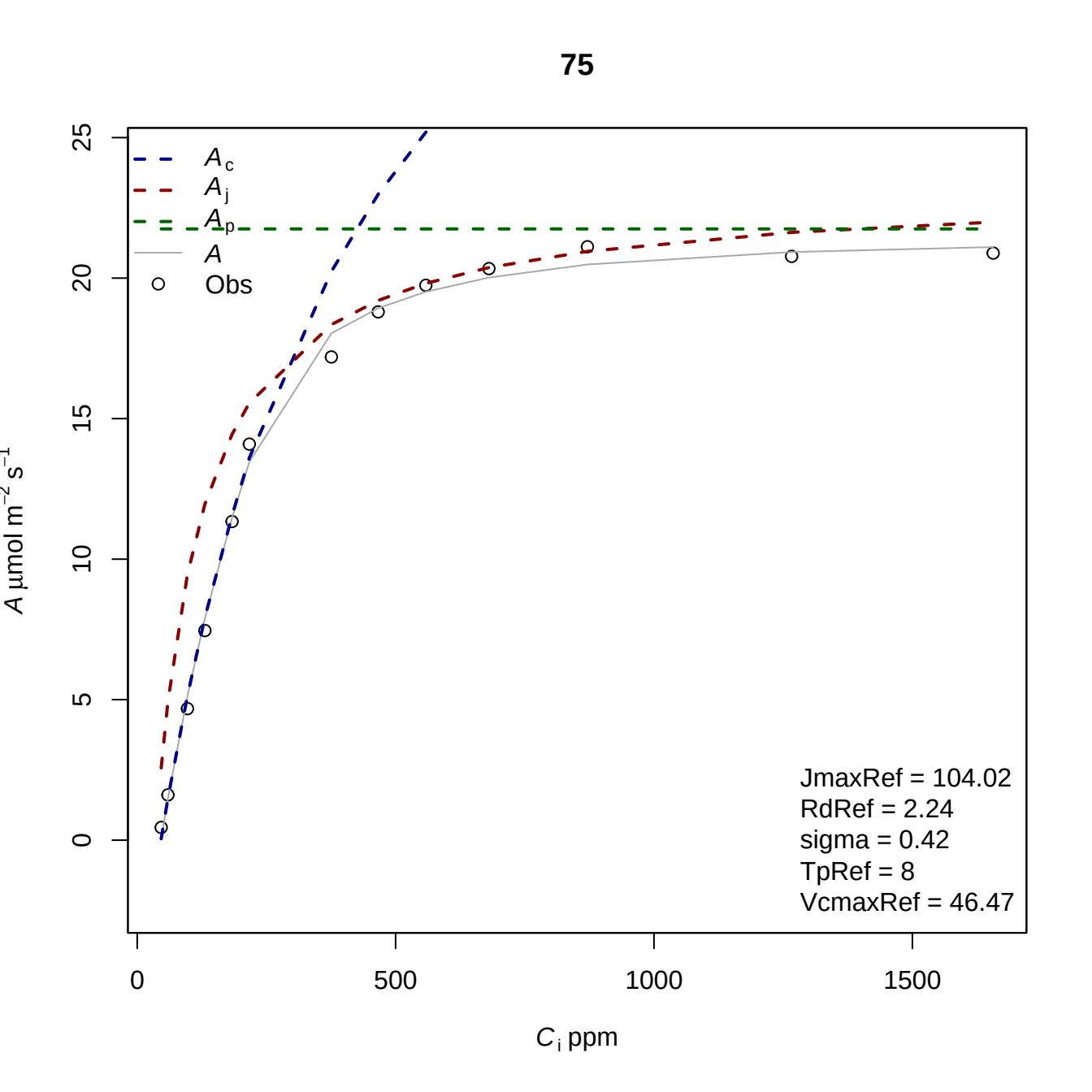

70
A_c
A_j
A_p
A
Obs
JmaxRef = 172.88
RdRef = 1.33
sigma = 0.31
TpRef = 26.19
VcmaxRef = 97.52
0
10
20
30
40
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

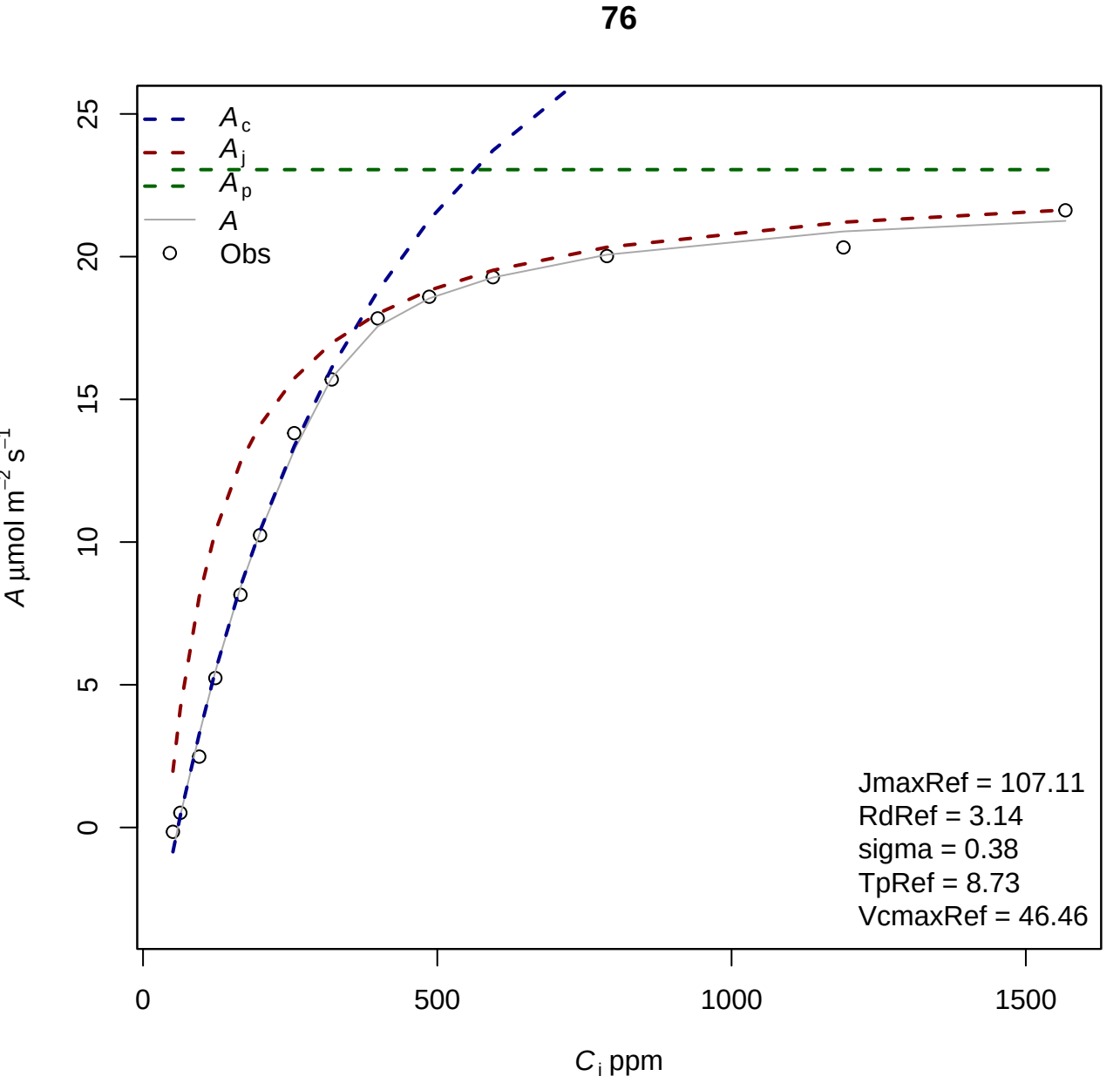

A_c
A_j
A_p
A
Obs
JmaxRef = 201.11
RdRef = 2.33
sigma = 0.85
TpRef = 43167.53
VcmaxRef = 105.21
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
0
200
400
600
800
1000
1200
1400

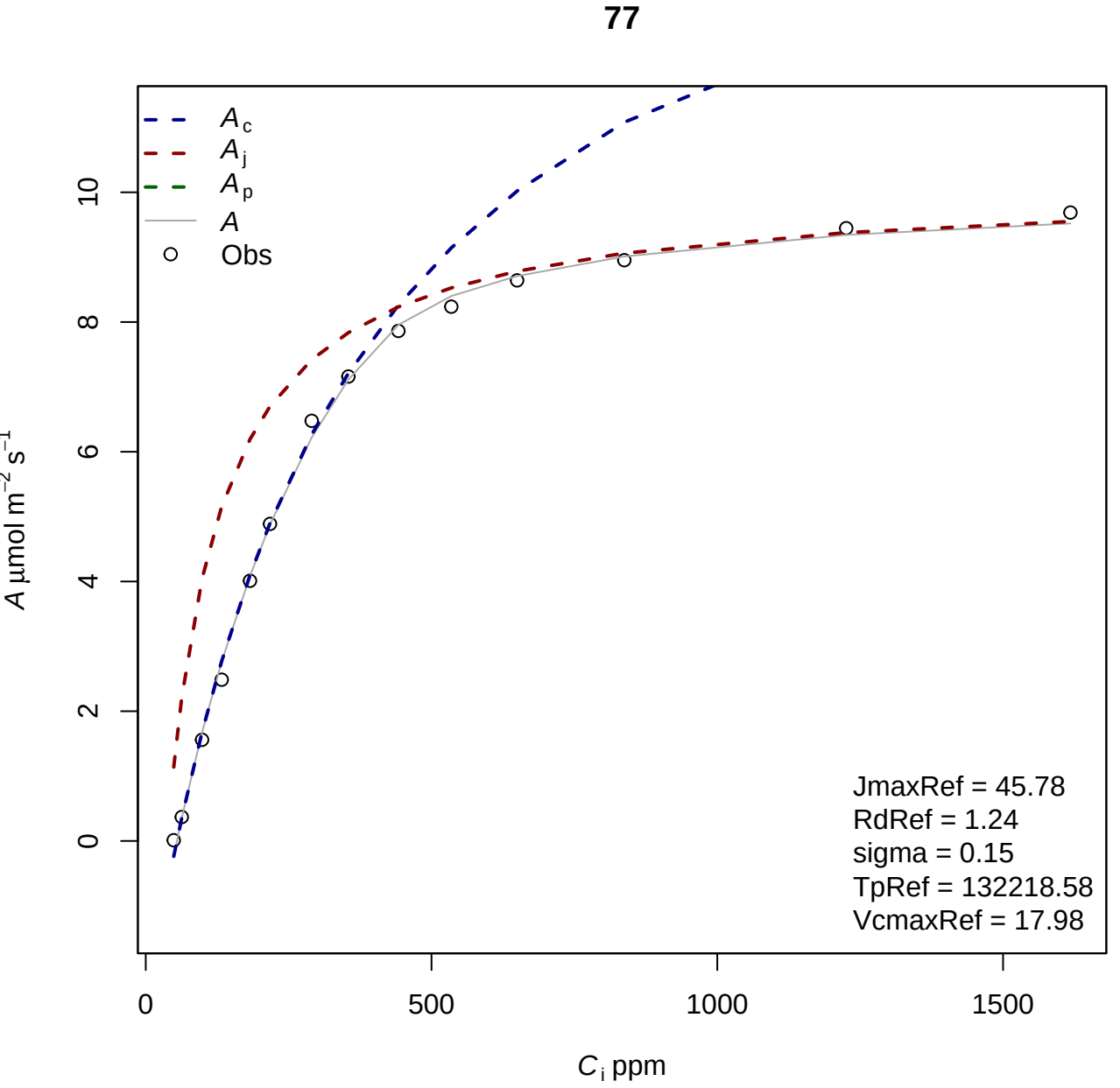

A_c
A_j
A_p
A
Obs
JmaxRef = 211.17
RdRef = 2.39
sigma = 0.92
TpRef = 521599.19
VcmaxRef = 92.3
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
50
200
400
600
800
1000
1200

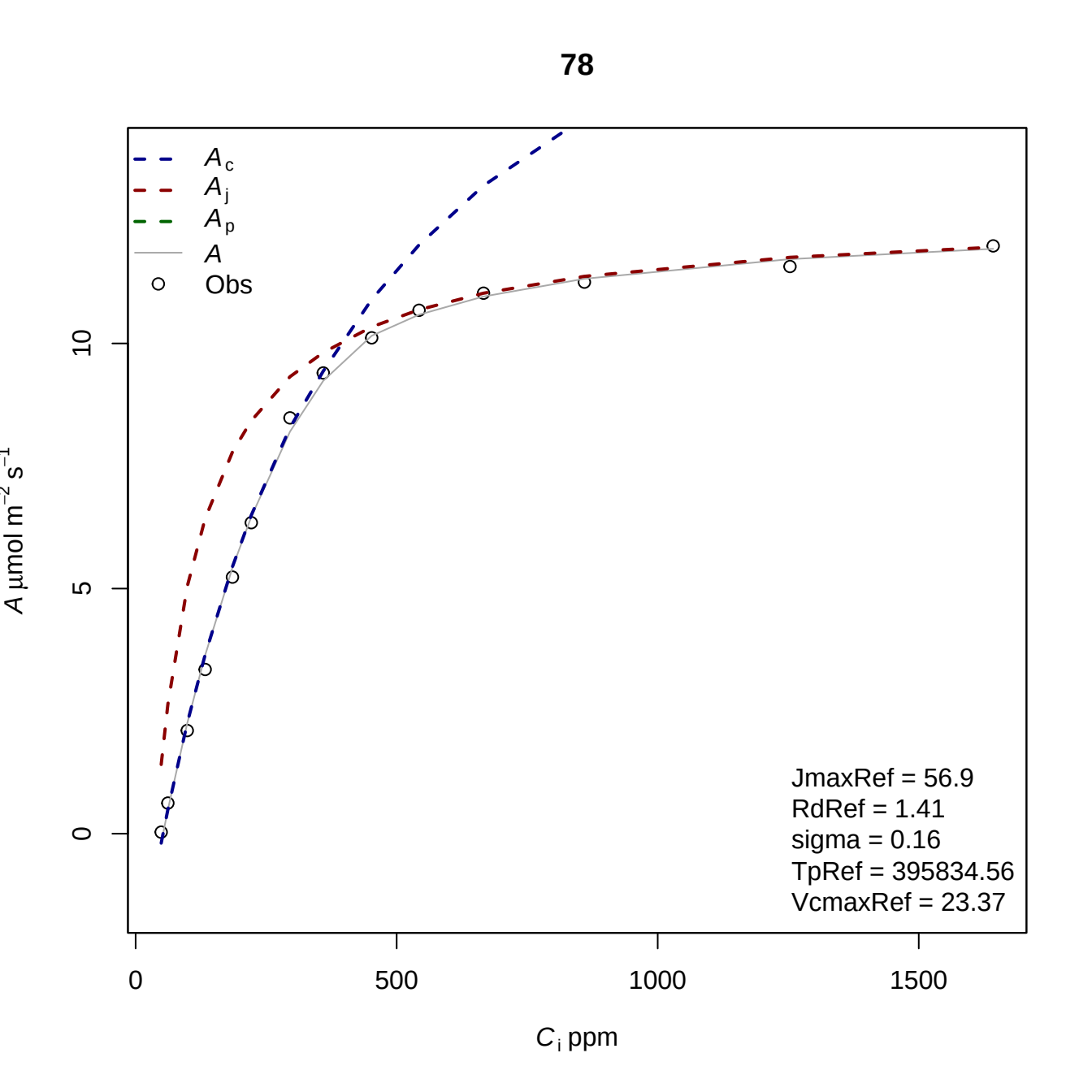

A_c
A_j
A_p
A
Obs
JmaxRef = 182.79
RdRef = 1.43
sigma = 0.47
TpRef = 44044.43
VcmaxRef = 85.62
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
0
200
400
600
800
1000
1200
1400

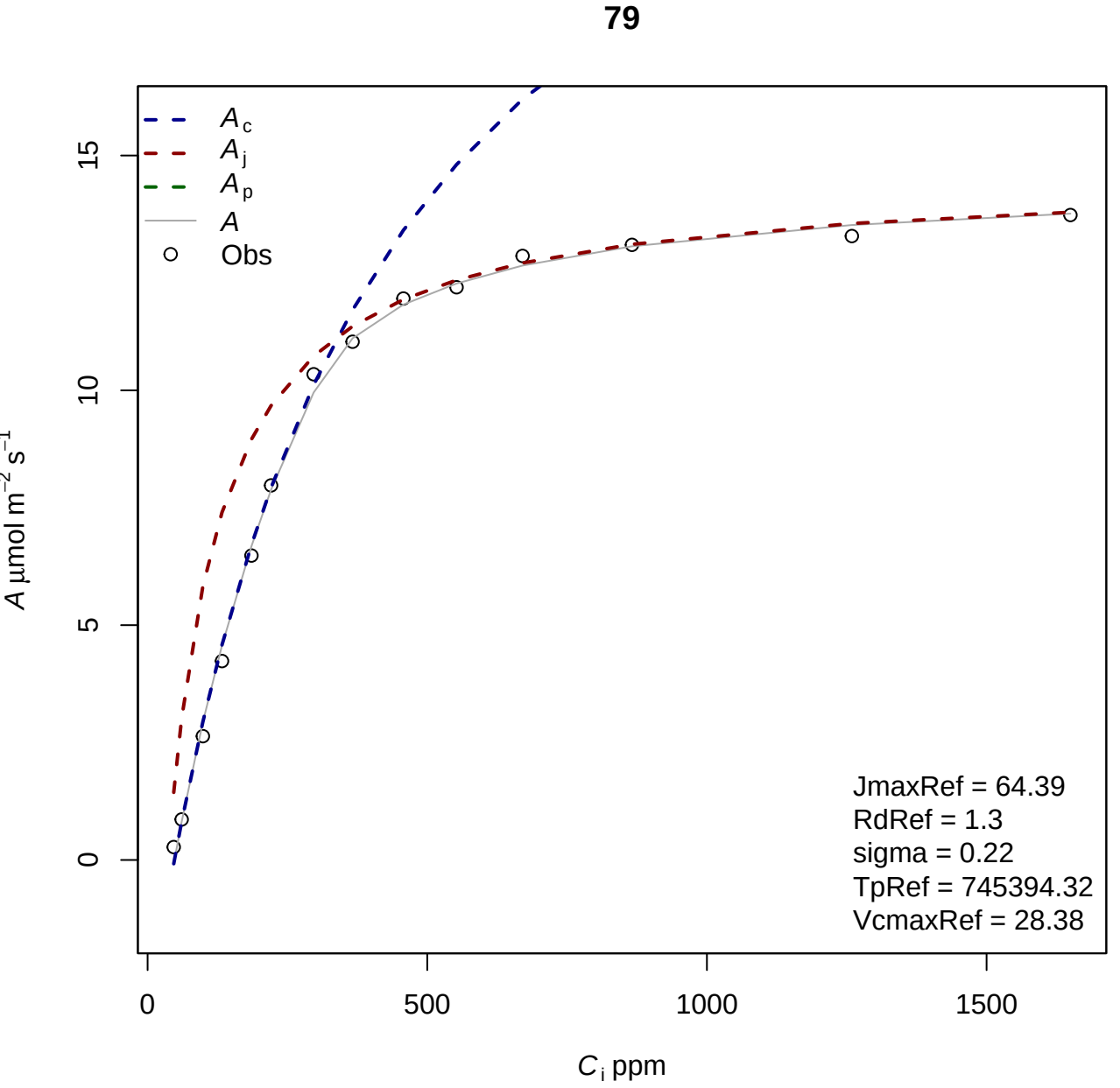

74
A_c
A_j
A_p
A
Obs
JmaxRef = 137.74
RdRef = 1.34
sigma = 0.31
TpRef = 9.61
VcmaxRef = 72.51
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
0
500
1000
1500

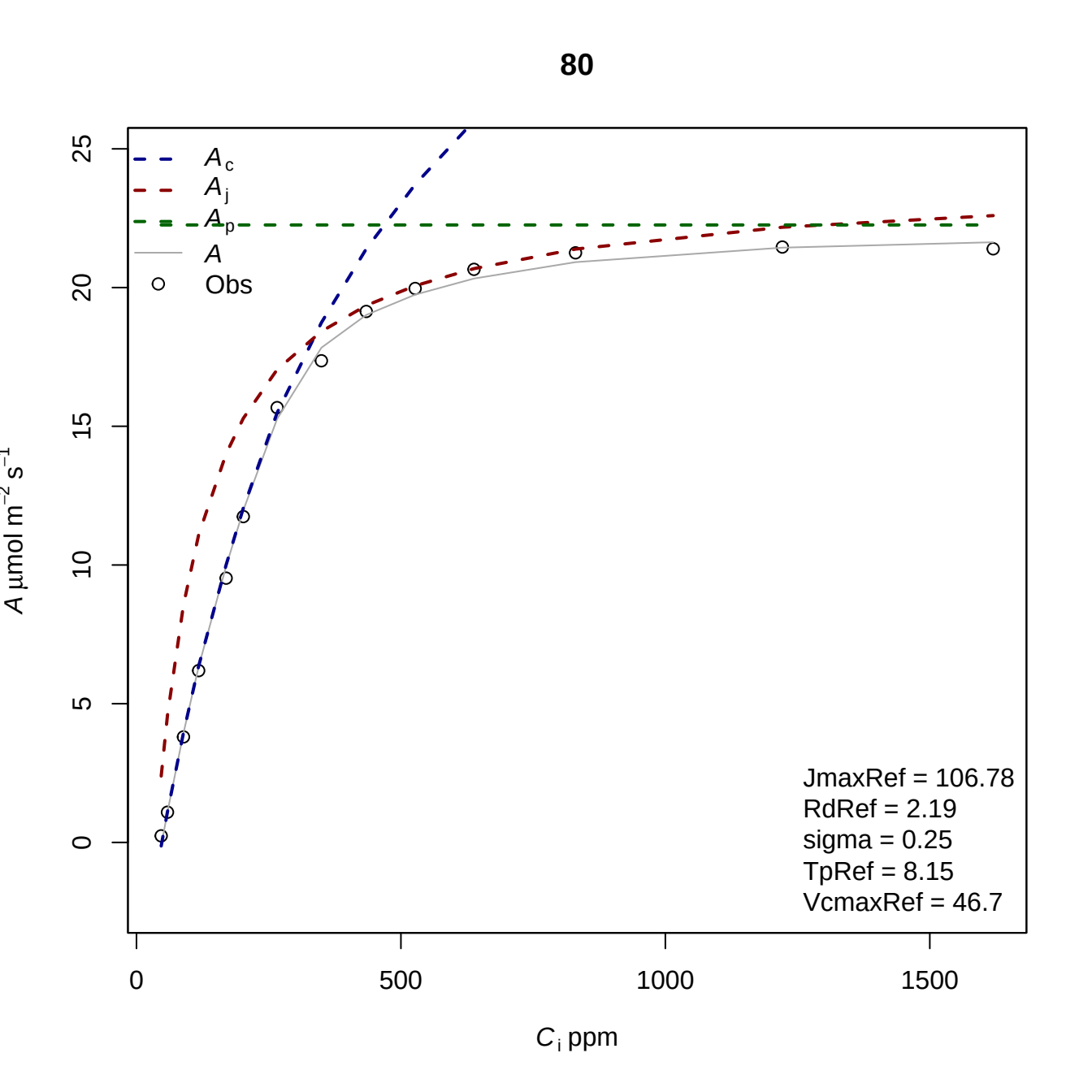

75
A_c
A_j
A_p
A
Obs
JmaxRef = 104.02
RdRef = 2.24
sigma = 0.42
TpRef = 8
VcmaxRef = 46.47
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
500
1000
1500

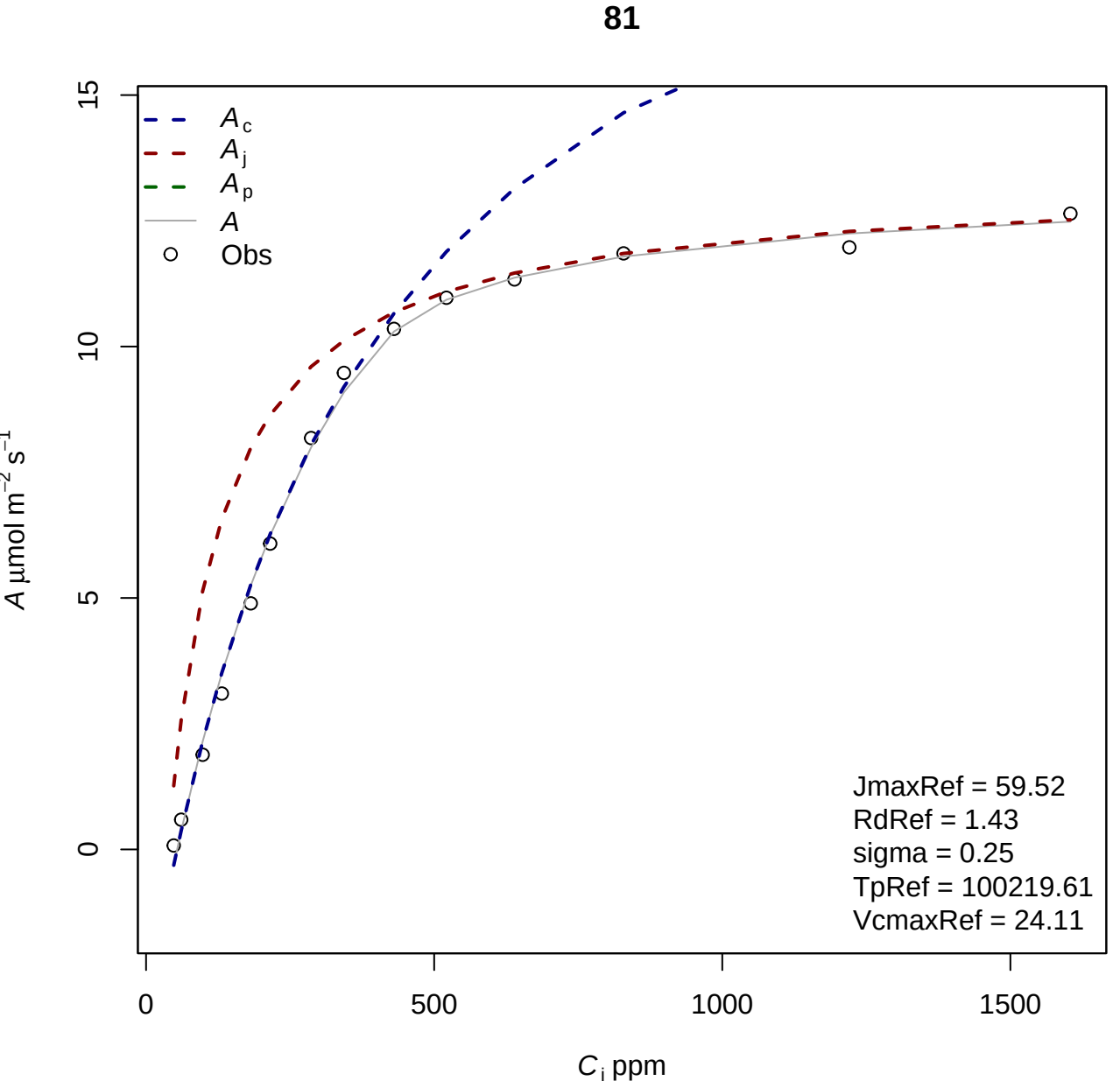

76
A_c
A_j
A_p
A
Obs
JmaxRef = 107.11
RdRef = 3.14
sigma = 0.38
TpRef = 8.73
VcmaxRef = 46.46
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
500
1000
1500

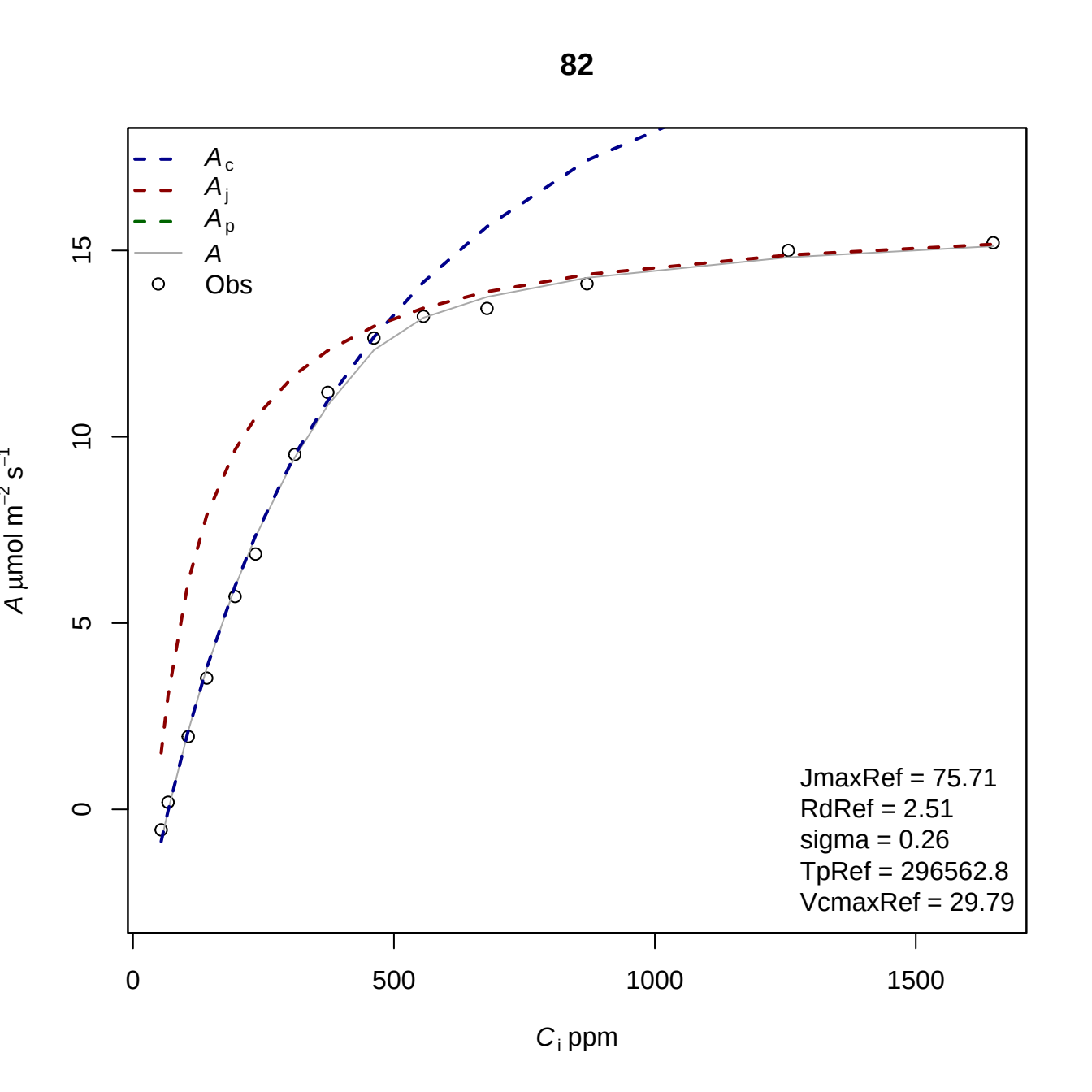

77
A_c
A_j
A_p
A
Obs
JmaxRef = 45.78
RdRef = 1.24
sigma = 0.15
TpRef = 132218.58
VcmaxRef = 17.98
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
0
500
1000
1500

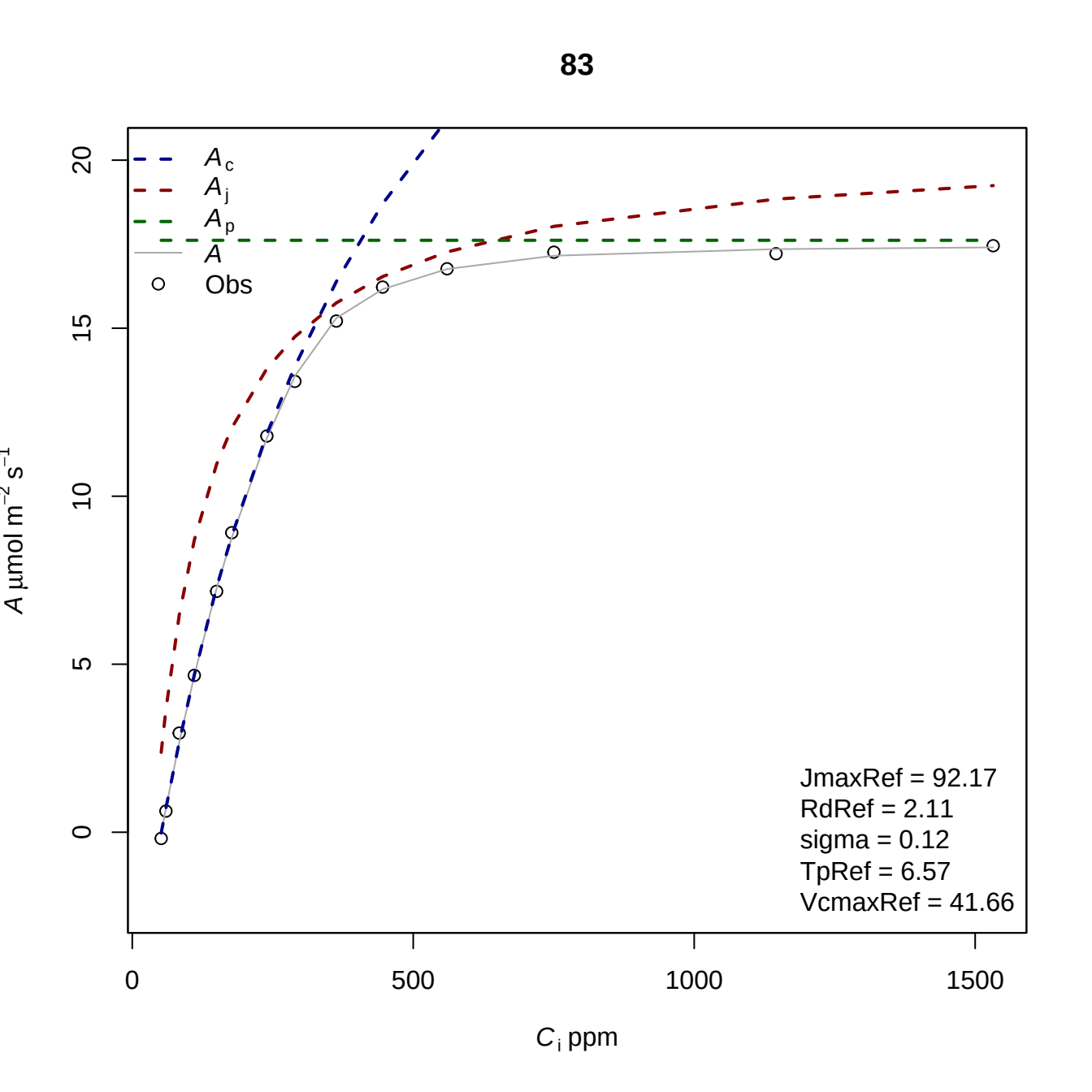

78
A_c
A_j
A_p
A
Obs
JmaxRef = 56.9
RdRef = 1.41
sigma = 0.16
TpRef = 395834.56
VcmaxRef = 23.37
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
C_i ppm

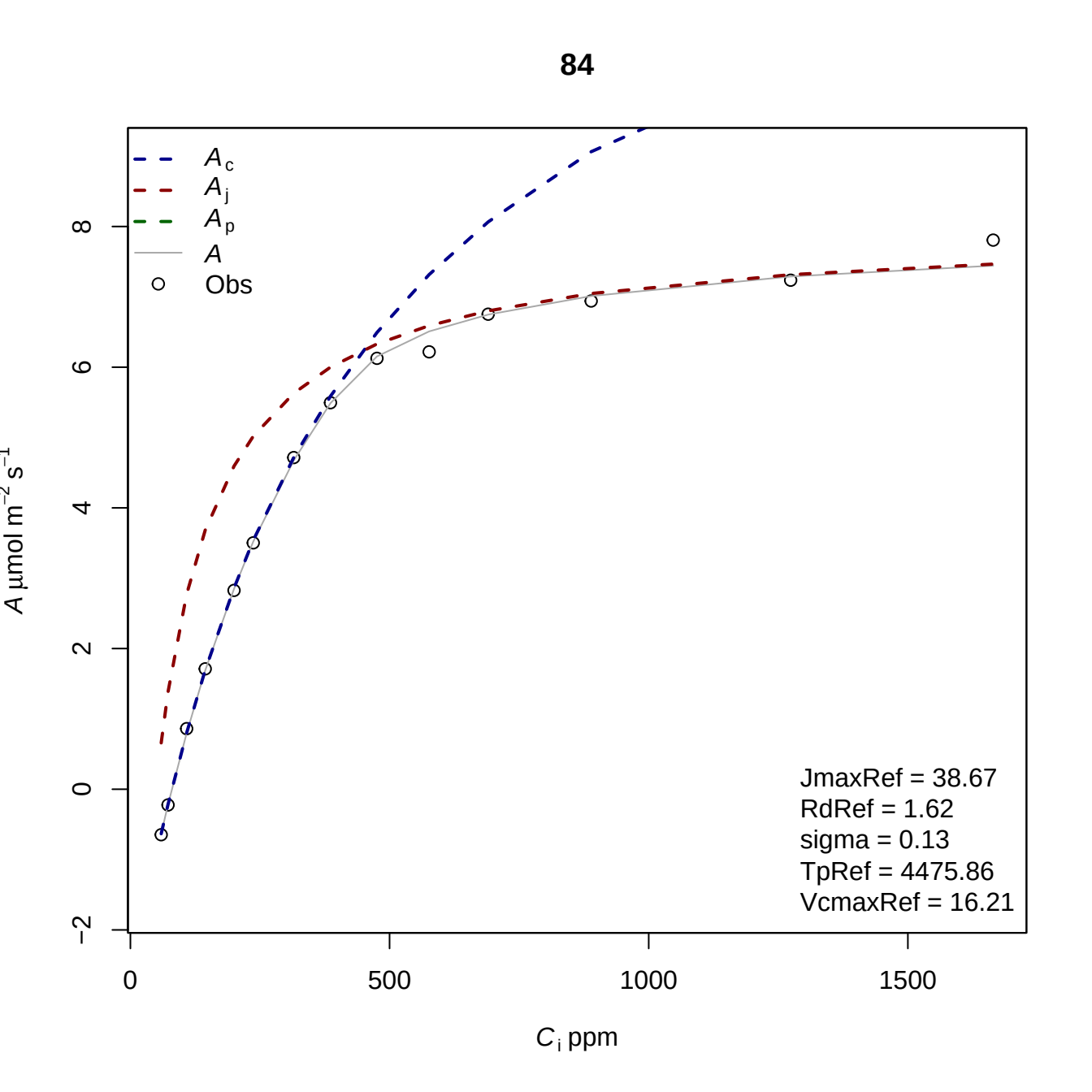

79
A_c
A_j
A_p
A
Obs
JmaxRef = 64.39
RdRef = 1.3
sigma = 0.22
TpRef = 745394.32
VcmaxRef = 28.38
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
500
1000
1500

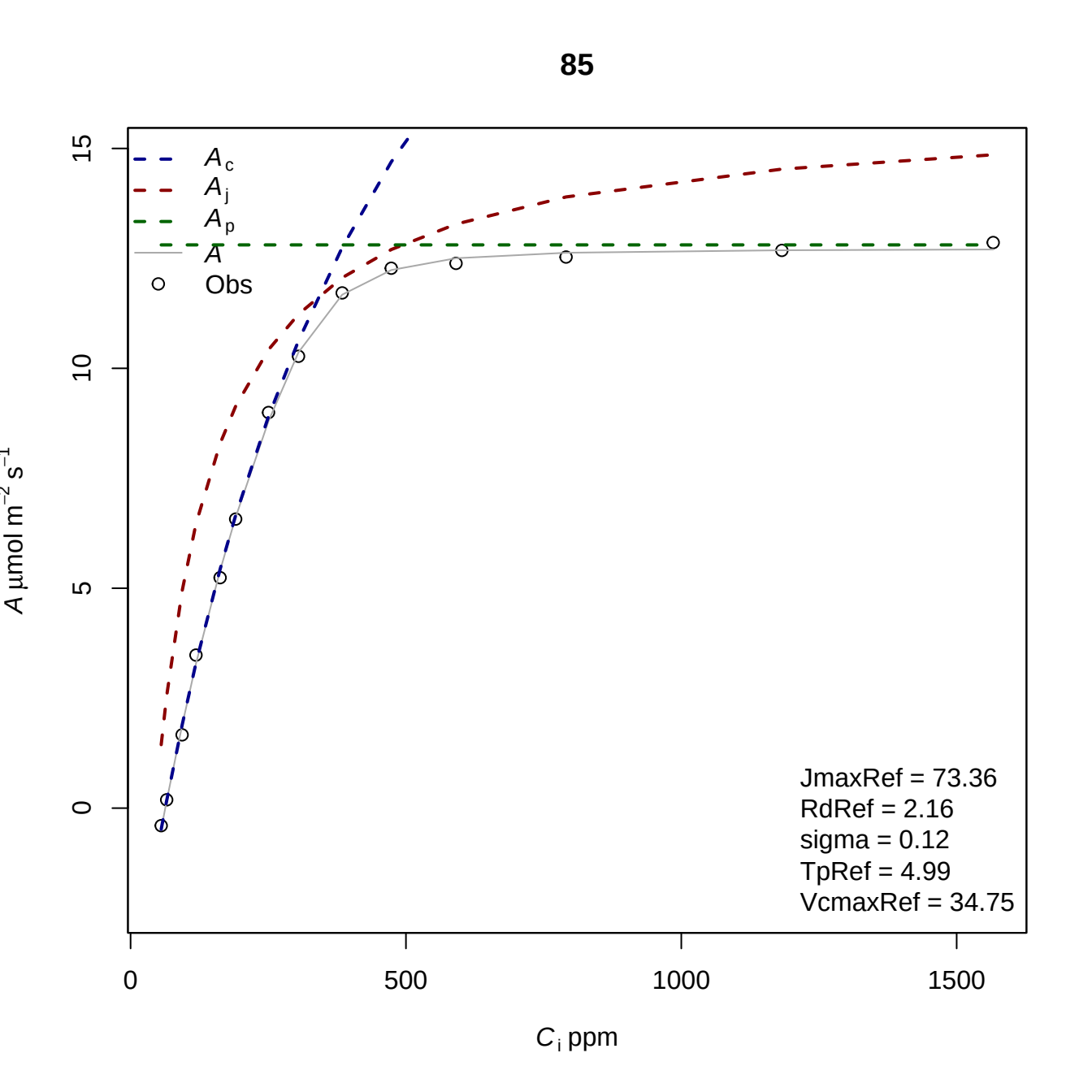

80
A_c
A_j
A_p
A
Obs
JmaxRef = 106.78
RdRef = 2.19
sigma = 0.25
TpRef = 8.15
VcmaxRef = 46.7
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
500
1000
1500

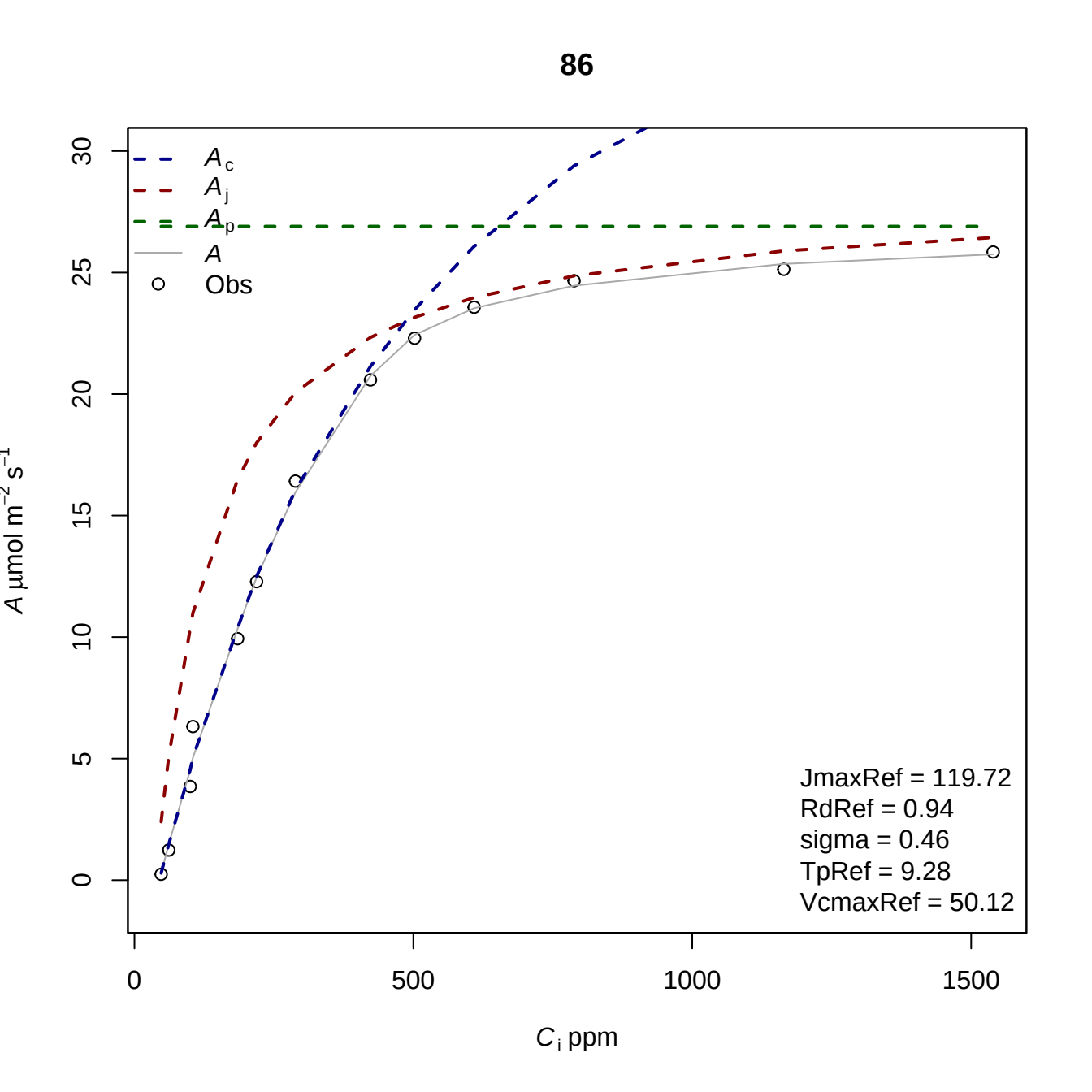

81
A_c
A_j
A_p
A
Obs
JmaxRef = 59.52
RdRef = 1.43
sigma = 0.25
TpRef = 100219.61
VcmaxRef = 24.11
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
500
1000
1500

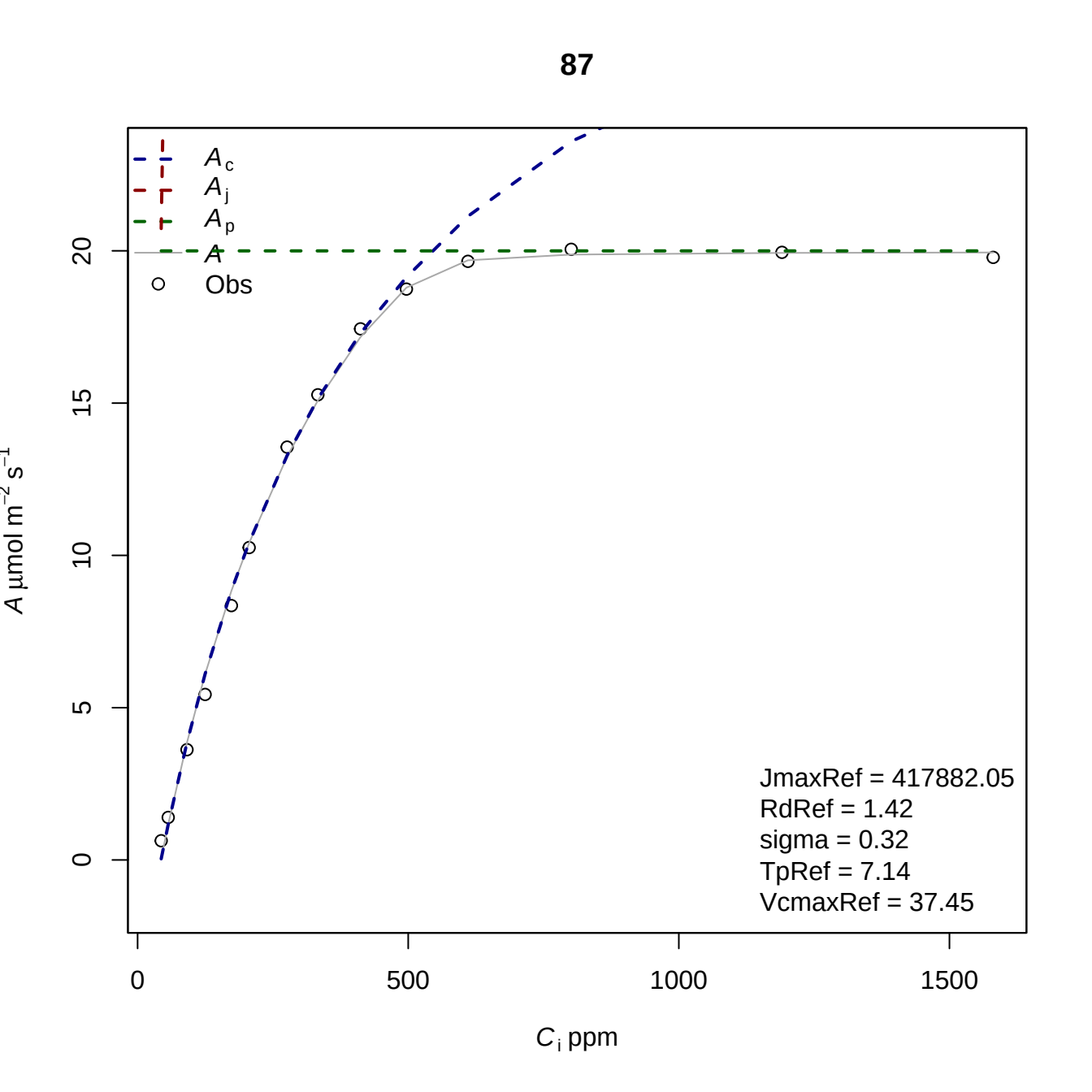

82
A_c
A_j
A_p
A
Obs
JmaxRef = 75.71
RdRef = 2.51
sigma = 0.26
TpRef = 296562.8
VcmaxRef = 29.79
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
500
1000
1500

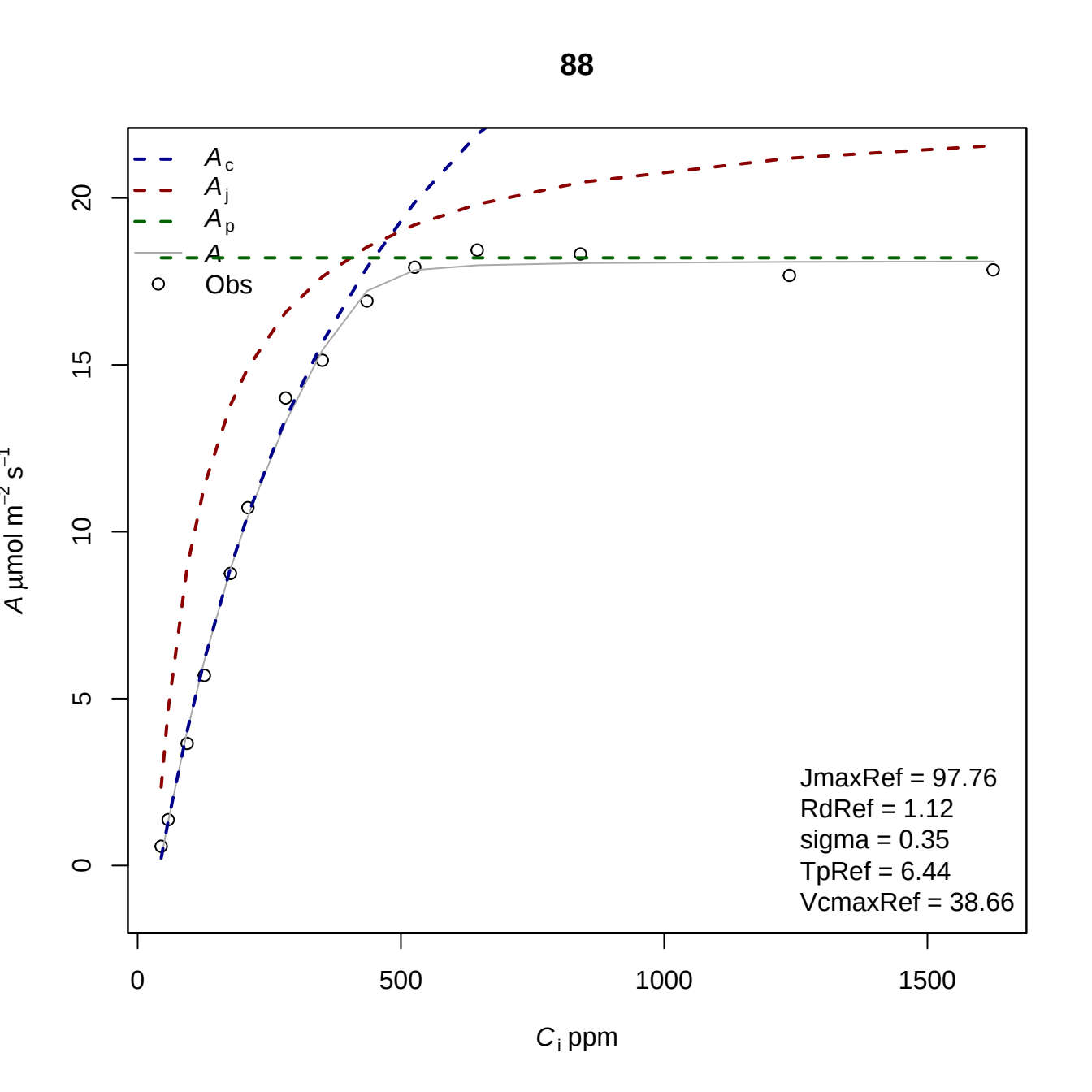

83
A_c
A_j
A_p
A
Obs
JmaxRef = 92.17
RdRef = 2.11
sigma = 0.12
TpRef = 6.57
VcmaxRef = 41.66
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
500
1000
1500

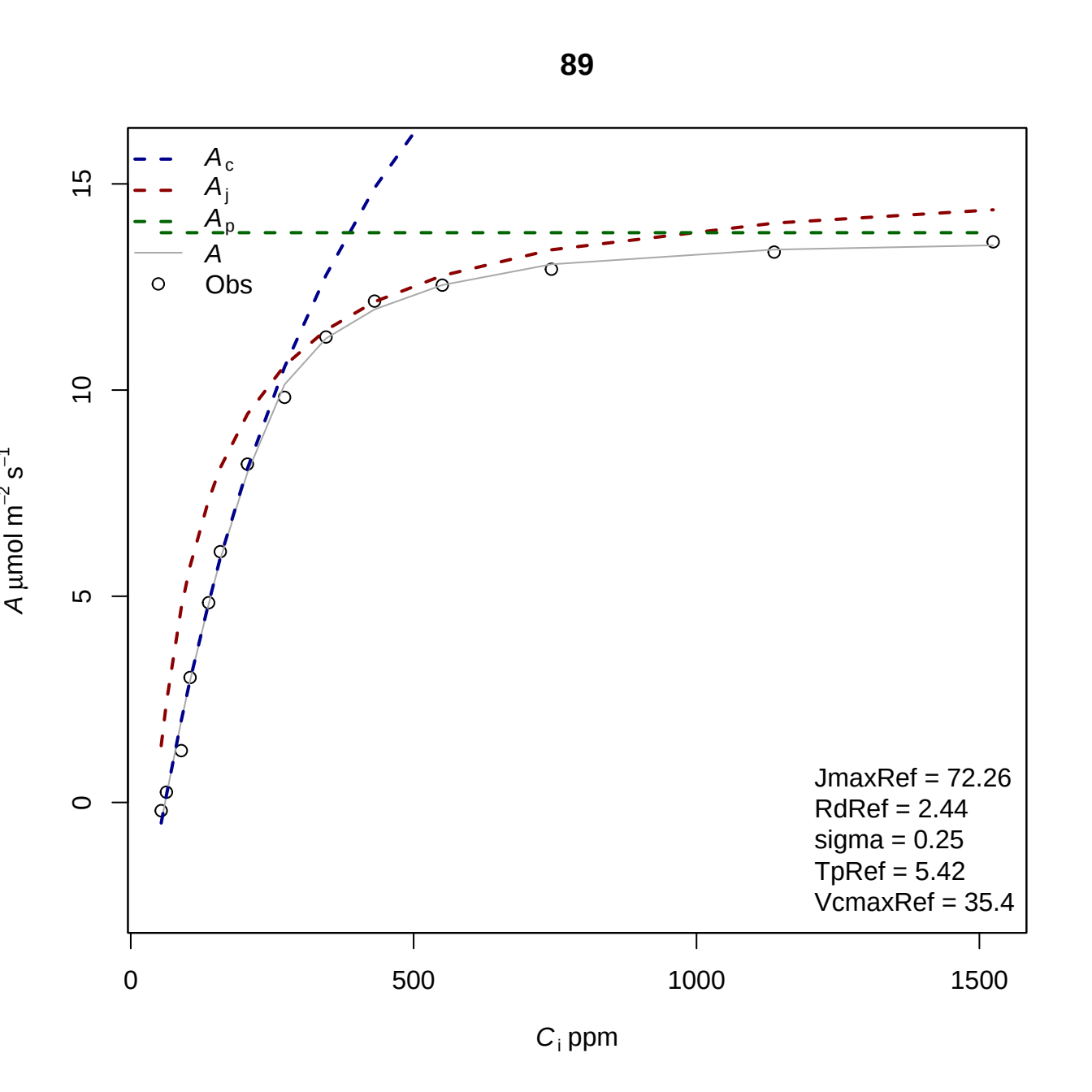

84
A_c
A_j
A_p
A
Obs
JmaxRef = 38.67
RdRef = 1.62
sigma = 0.13
TpRef = 4475.86
VcmaxRef = 16.21
A µmol m^{-2} s^{-1}
C_i ppm
−2
0
2
4
6
8
0
500
1000
1500

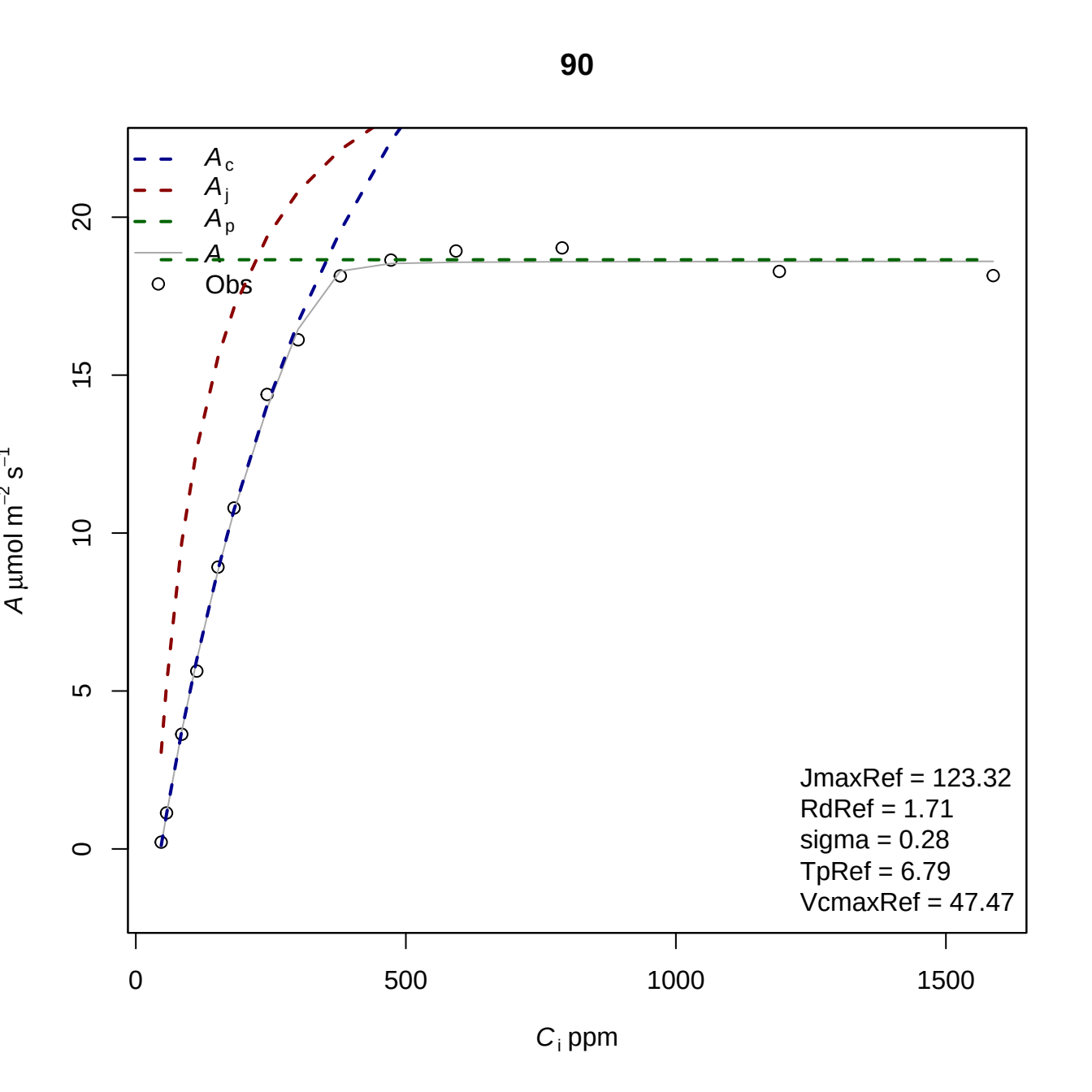

85
A_c
A_j
A_p
A
Obs
JmaxRef = 73.36
RdRef = 2.16
sigma = 0.12
TpRef = 4.99
VcmaxRef = 34.75
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
500
1000
1500

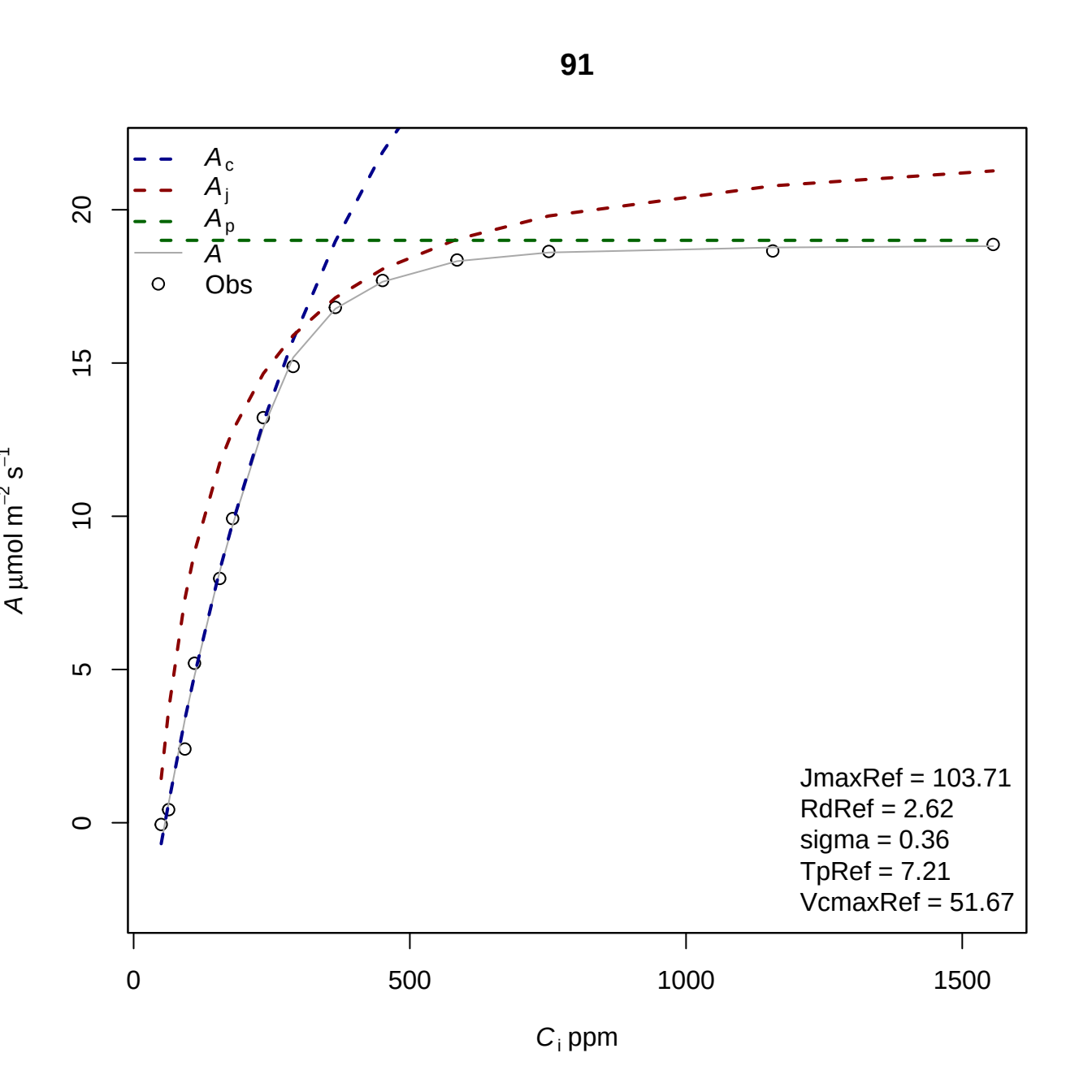

86
A_c
A_j
A_p
A
Obs
JmaxRef = 119.72
RdRef = 0.94
sigma = 0.46
TpRef = 9.28
VcmaxRef = 50.12
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
0
500
1000
1500

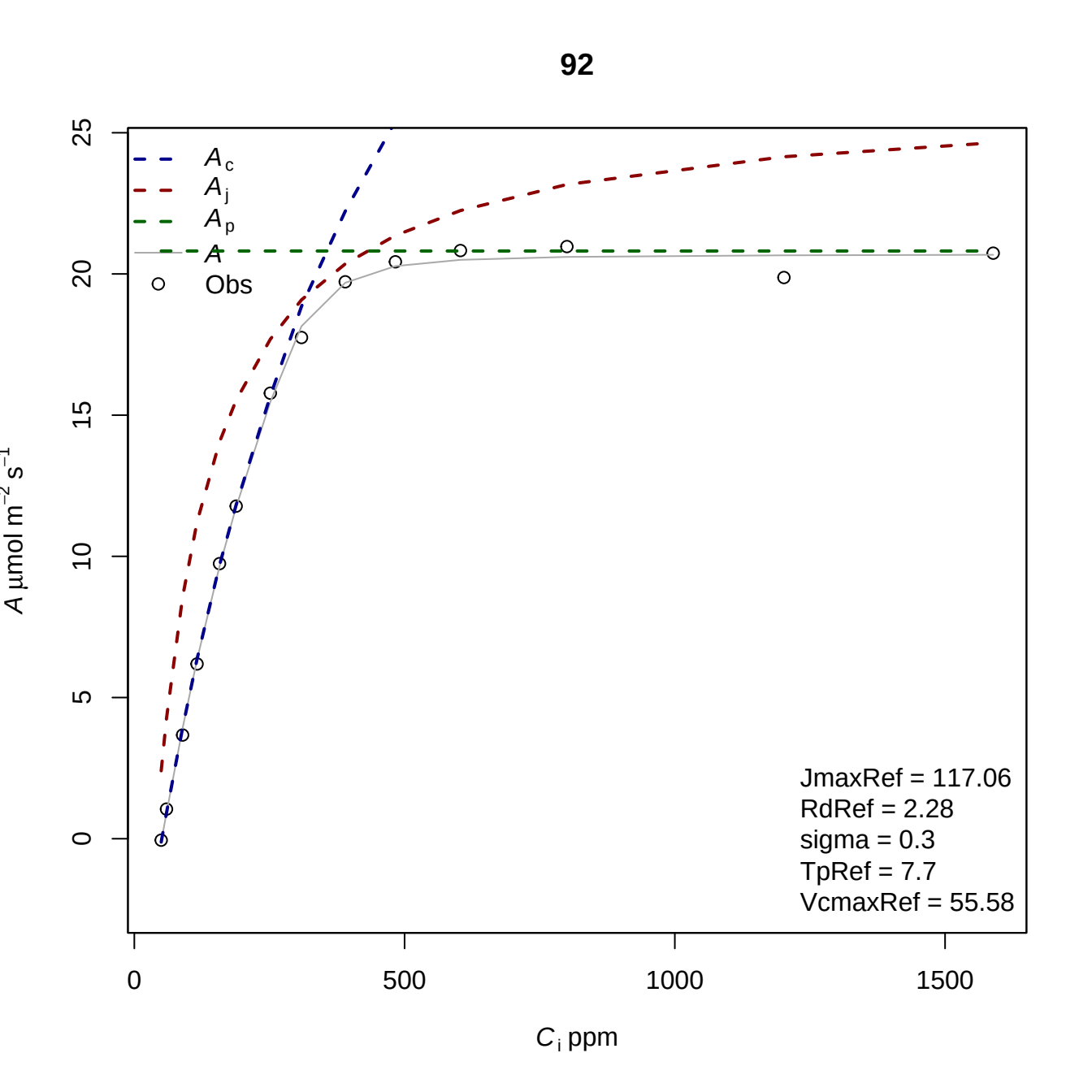

87
A_c
A_j
A_p
A
Obs
JmaxRef = 417882.05
RdRef = 1.42
sigma = 0.32
TpRef = 7.14
VcmaxRef = 37.45
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
500
1000
1500

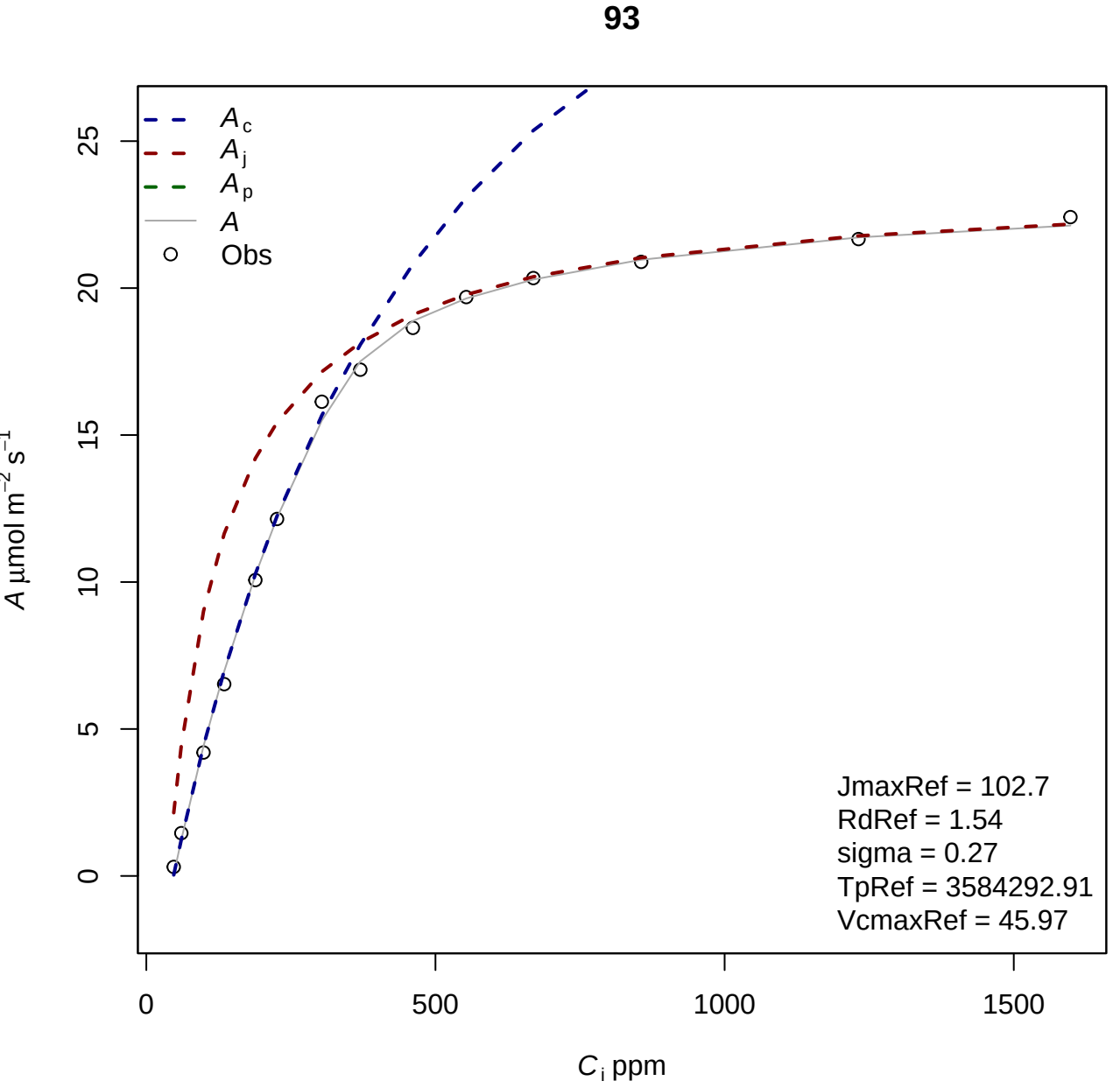

88
A_c
A_j
A_p
A
Obs
JmaxRef = 97.76
RdRef = 1.12
sigma = 0.35
TpRef = 6.44
VcmaxRef = 38.66
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
500
1000
1500

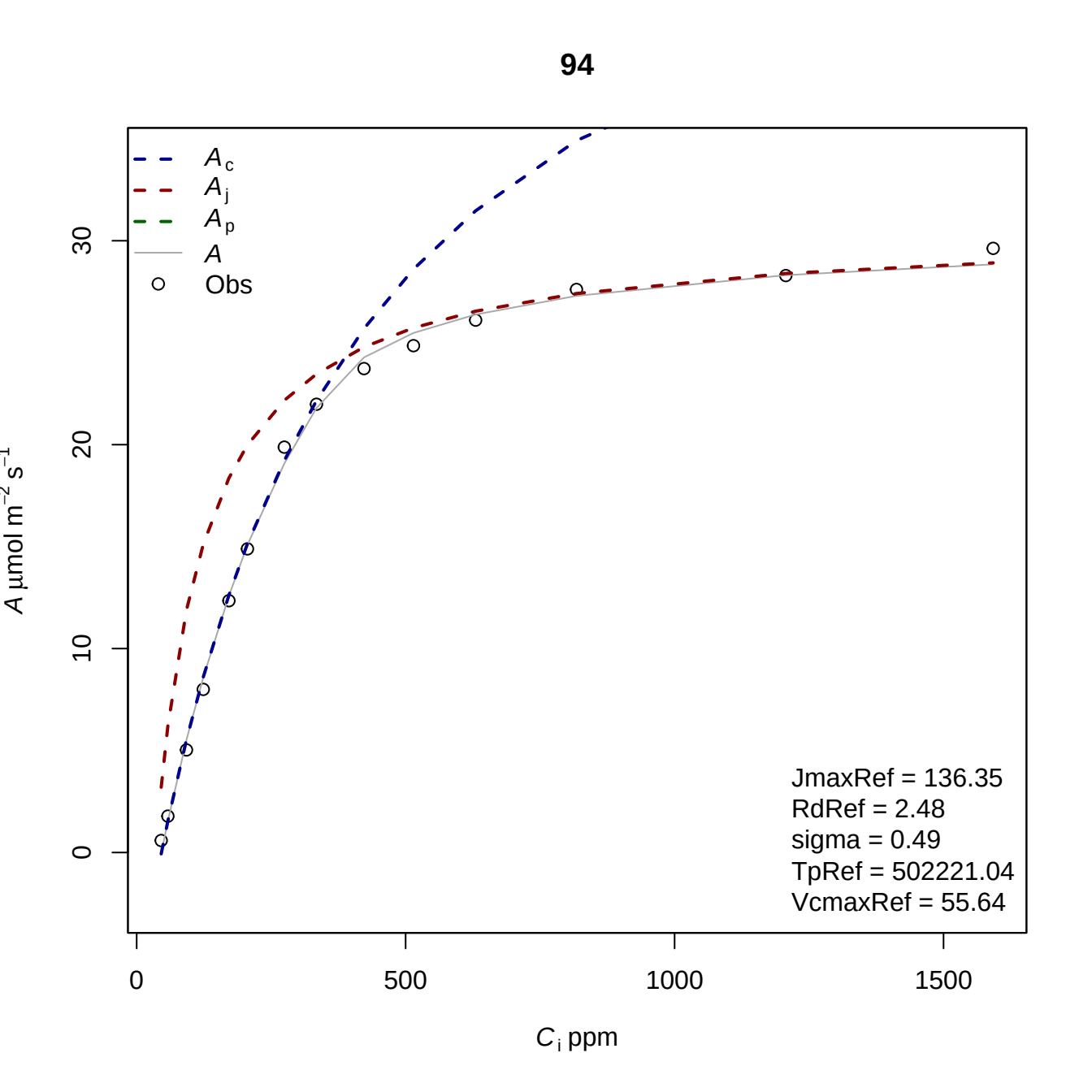

89
A_c
A_j
A_p
A
Obs
JmaxRef = 72.26
RdRef = 2.44
sigma = 0.25
TpRef = 5.42
VcmaxRef = 35.4
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
500
1000
1500

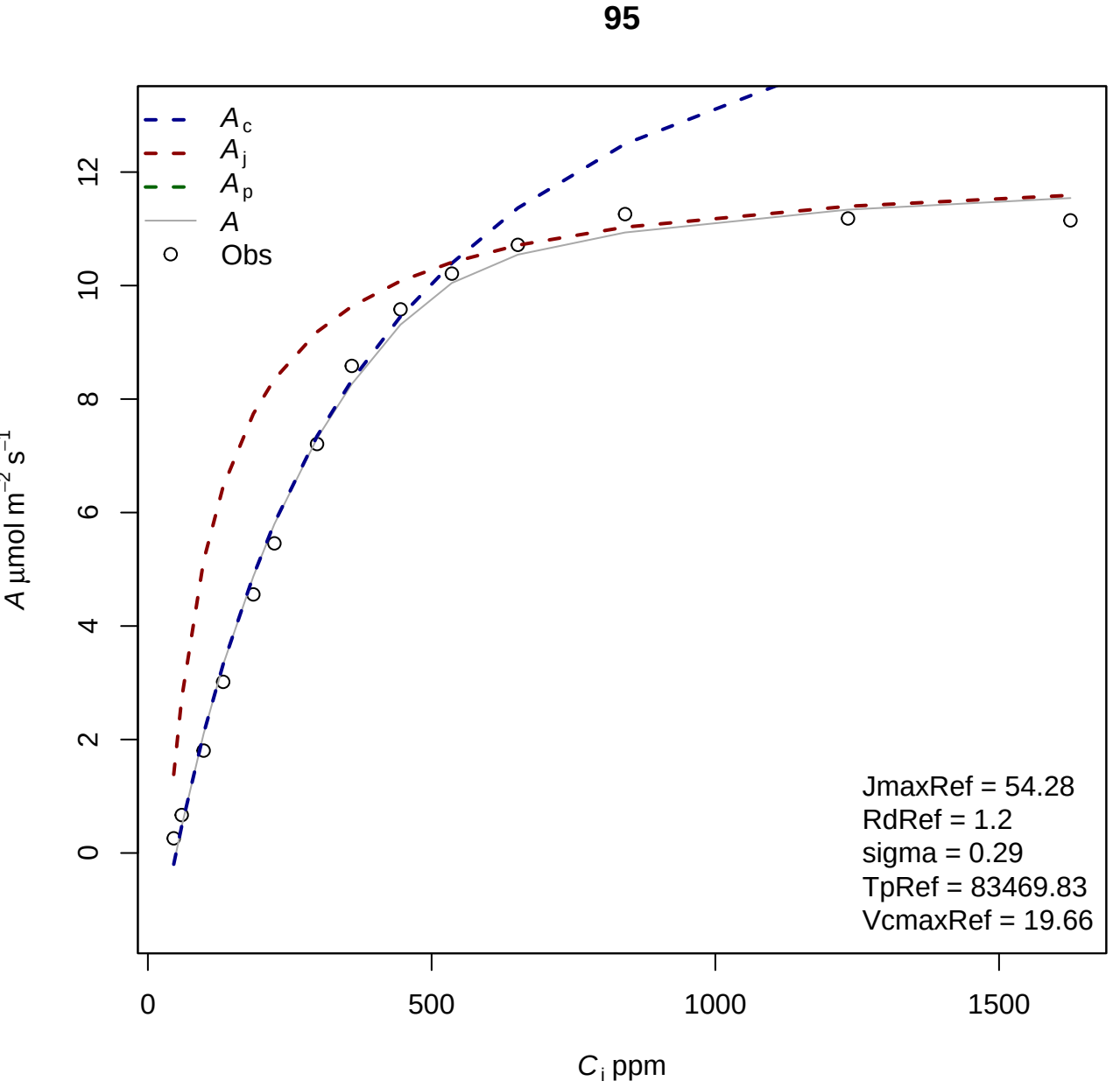

90
A_c
A_j
A_p
A
Obs
JmaxRef = 123.32
RdRef = 1.71
sigma = 0.28
TpRef = 6.79
VcmaxRef = 47.47
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
500
1000
1500

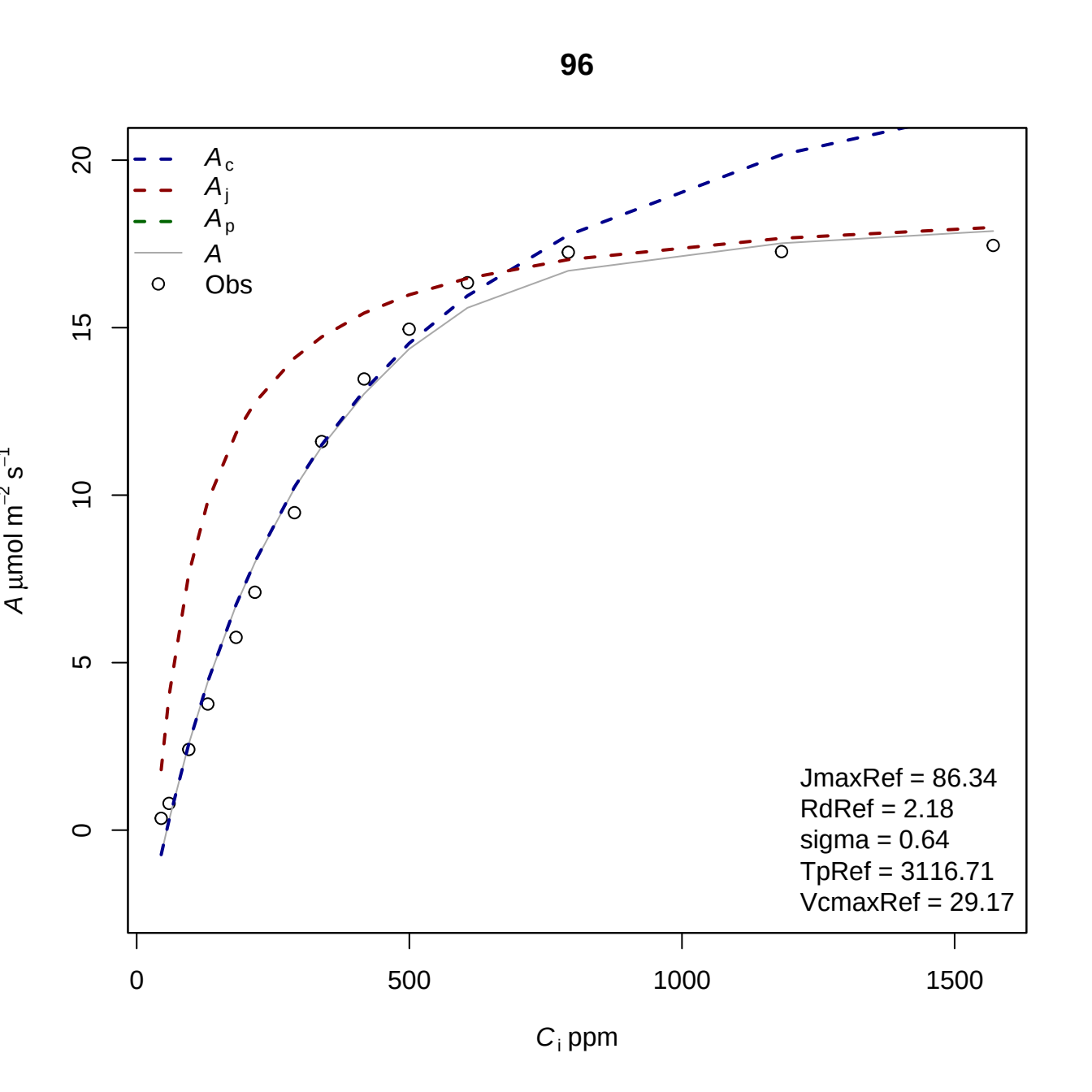

91
A_c
A_j
A_p
A
Obs
JmaxRef = 103.71
RdRef = 2.62
sigma = 0.36
TpRef = 7.21
VcmaxRef = 51.67
A µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
C_i ppm

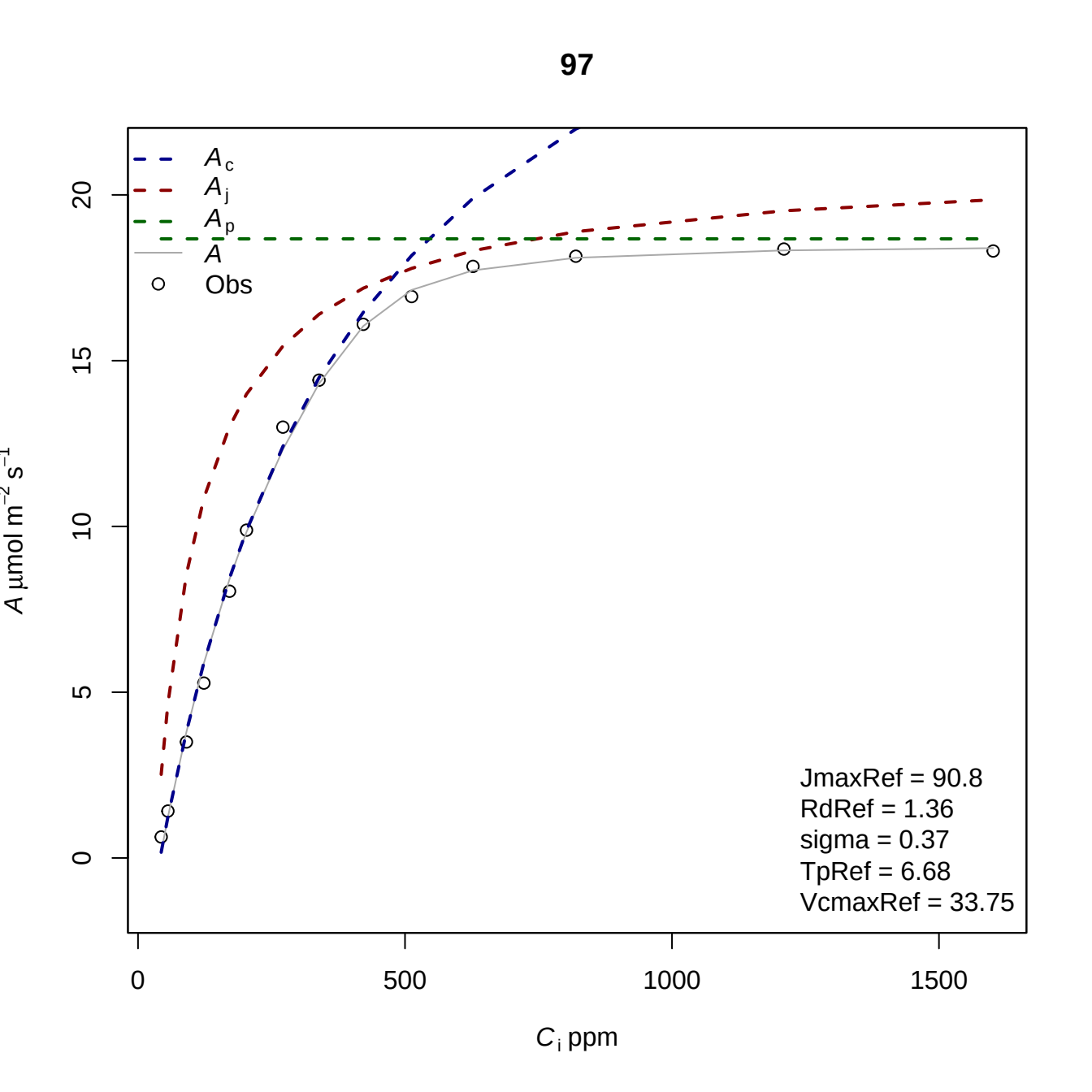

92
A_c
A_j
A_p
A
Obs
JmaxRef = 117.06
RdRef = 2.28
sigma = 0.3
TpRef = 7.7
VcmaxRef = 55.58
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
500
1000
1500

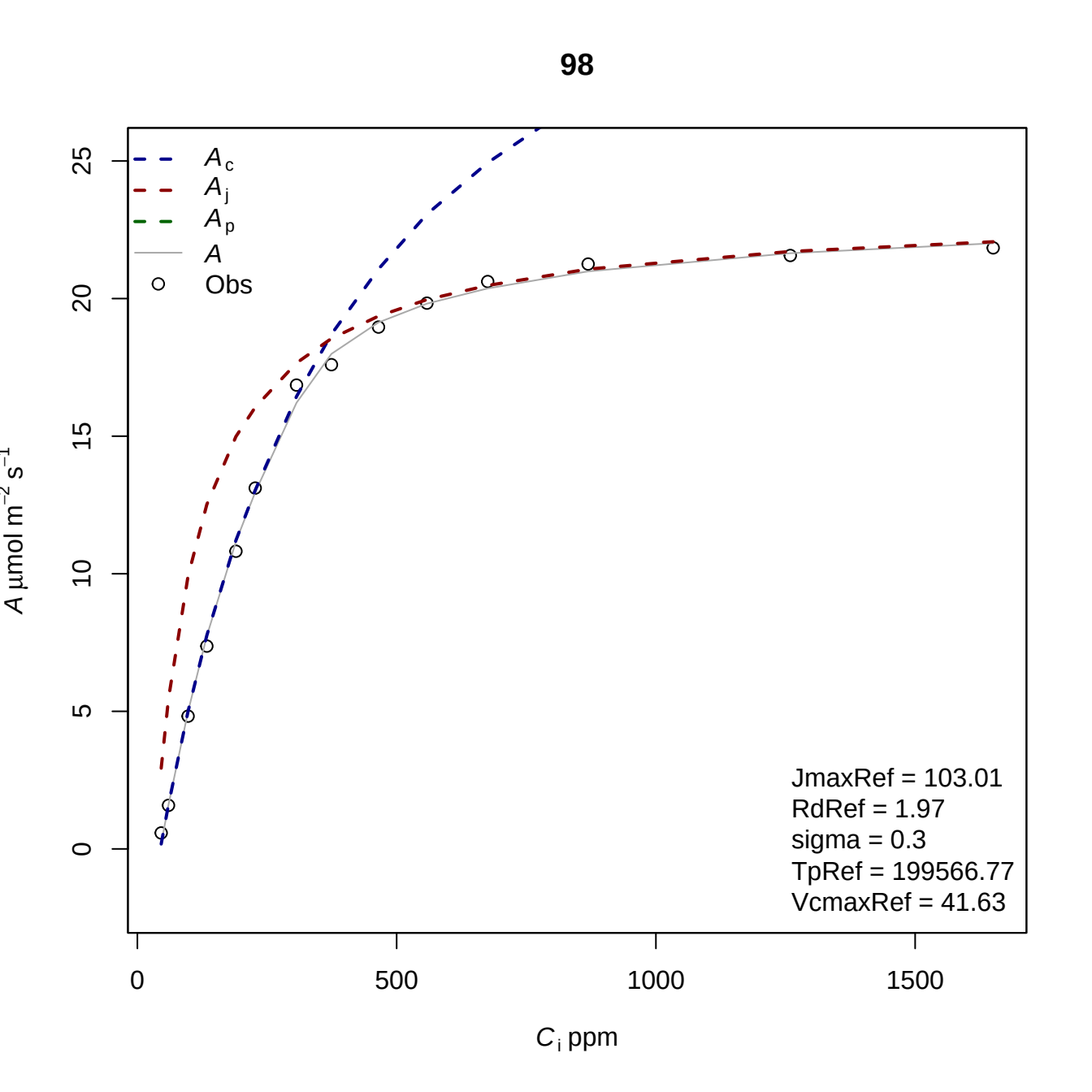

93
A_c
A_j
A_p
A
Obs
A µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
C_i ppm
JmaxRef = 102.7
RdRef = 1.54
sigma = 0.27
TpRef = 3584292.91
VcmaxRef = 45.97

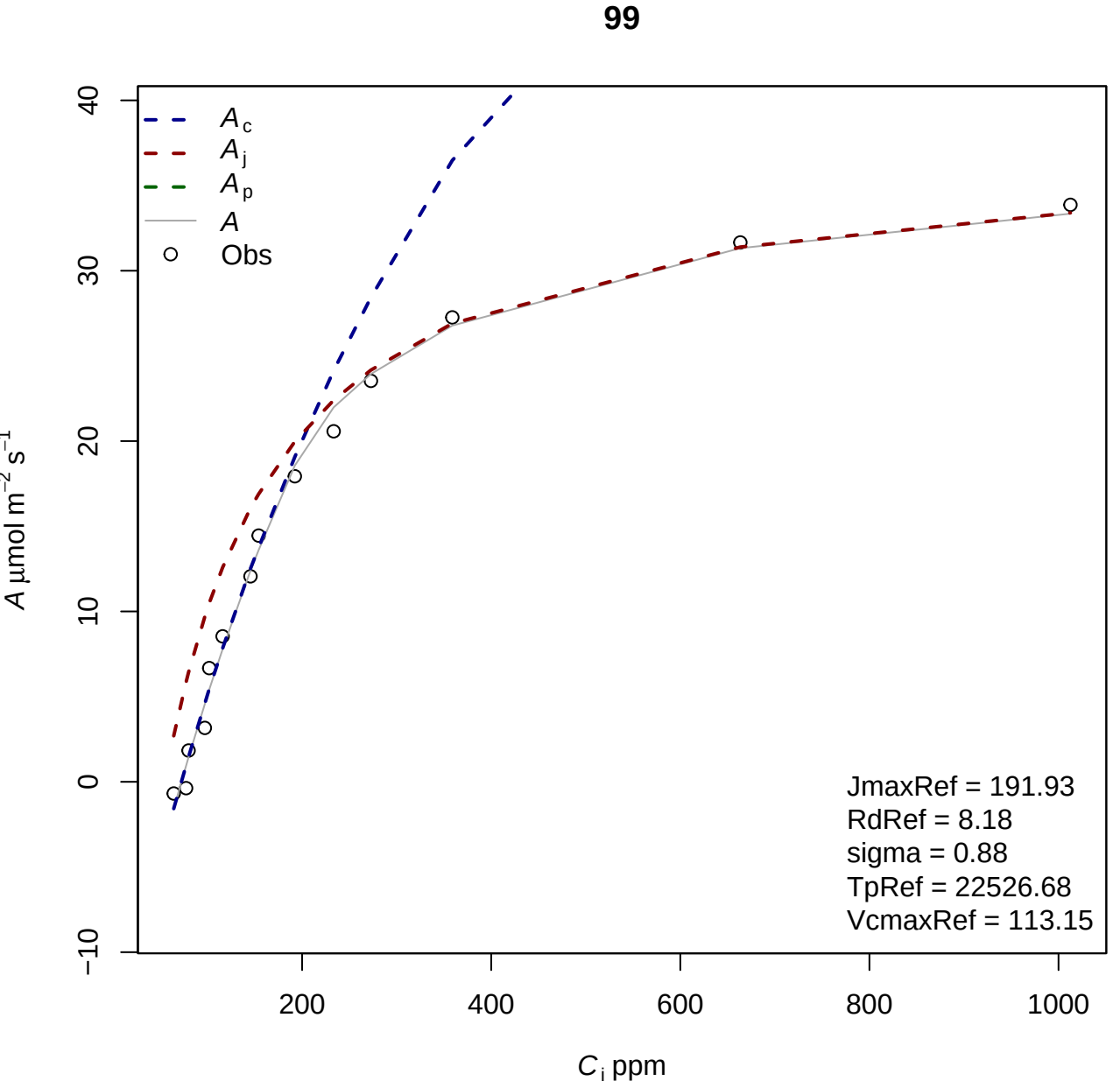

94
A_c
A_j
A_p
A
Obs
JmaxRef = 136.35
RdRef = 2.48
sigma = 0.49
TpRef = 502221.04
VcmaxRef = 55.64
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
0
500
1000
1500

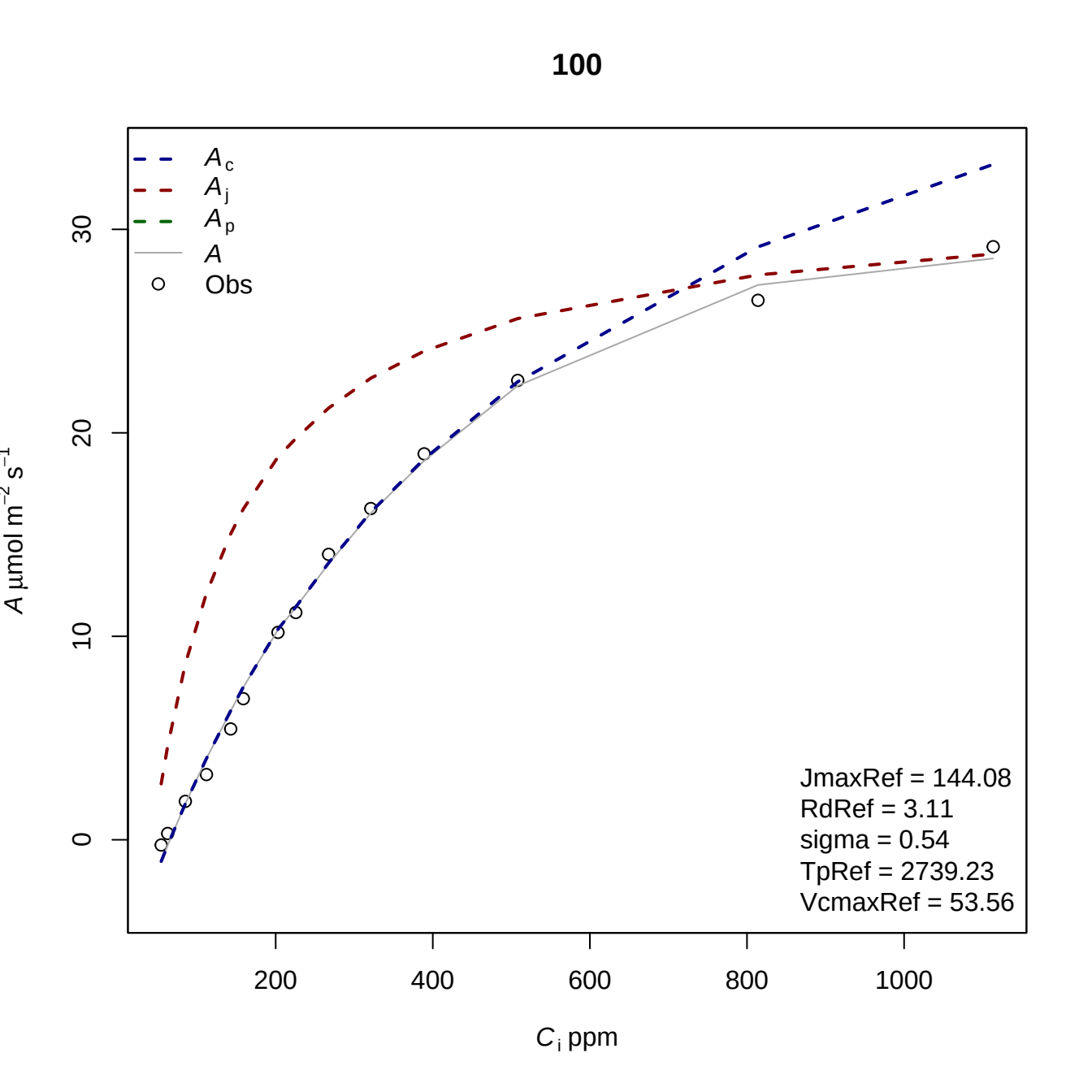

95
A_c
A_j
A_p
A
Obs
JmaxRef = 54.28
RdRef = 1.2
sigma = 0.29
TpRef = 83469.83
VcmaxRef = 19.66
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
0
500
1000
1500

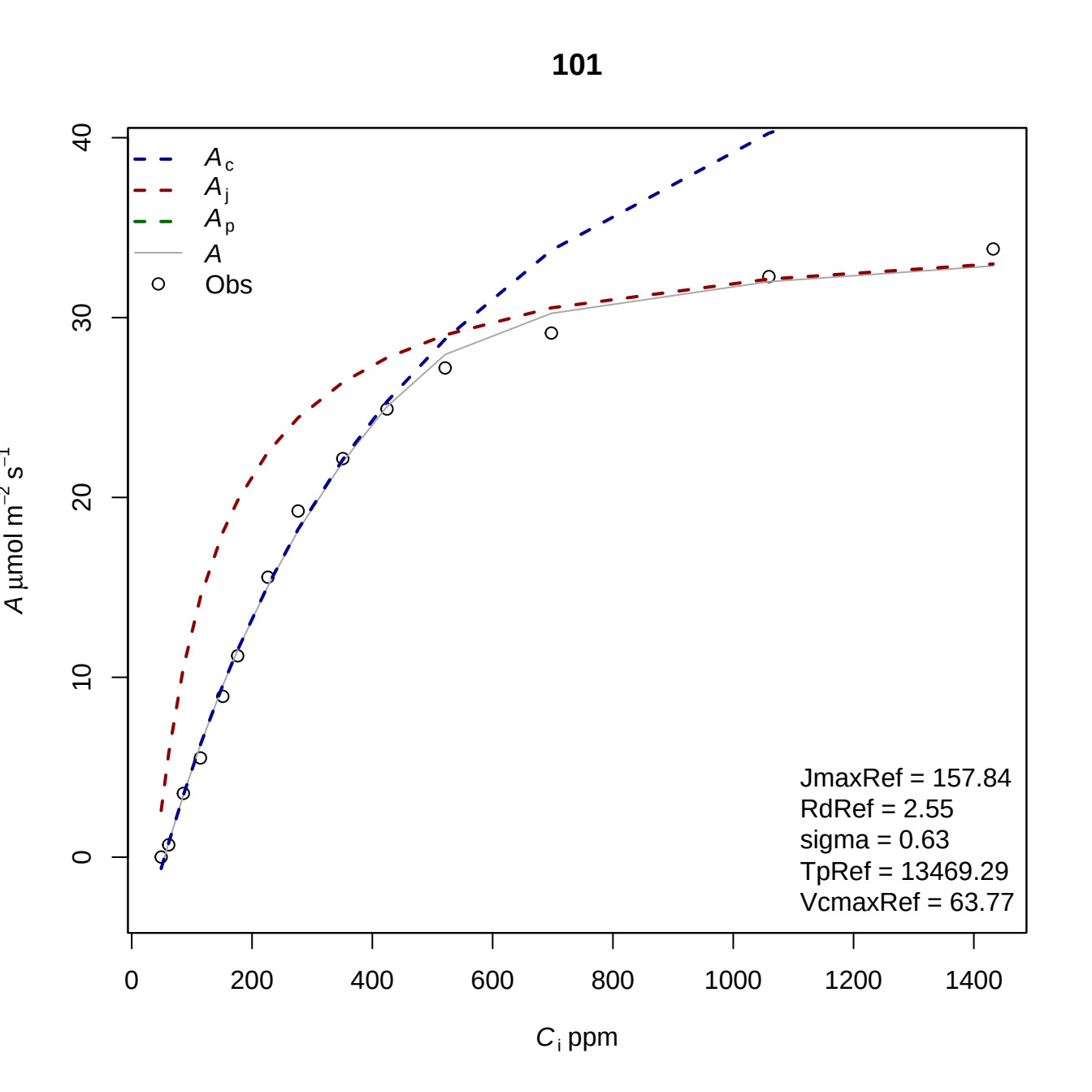

96
A_c
A_j
A_p
A
Obs
JmaxRef = 86.34
RdRef = 2.18
sigma = 0.64
TpRef = 3116.71
VcmaxRef = 29.17
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
500
1000
1500

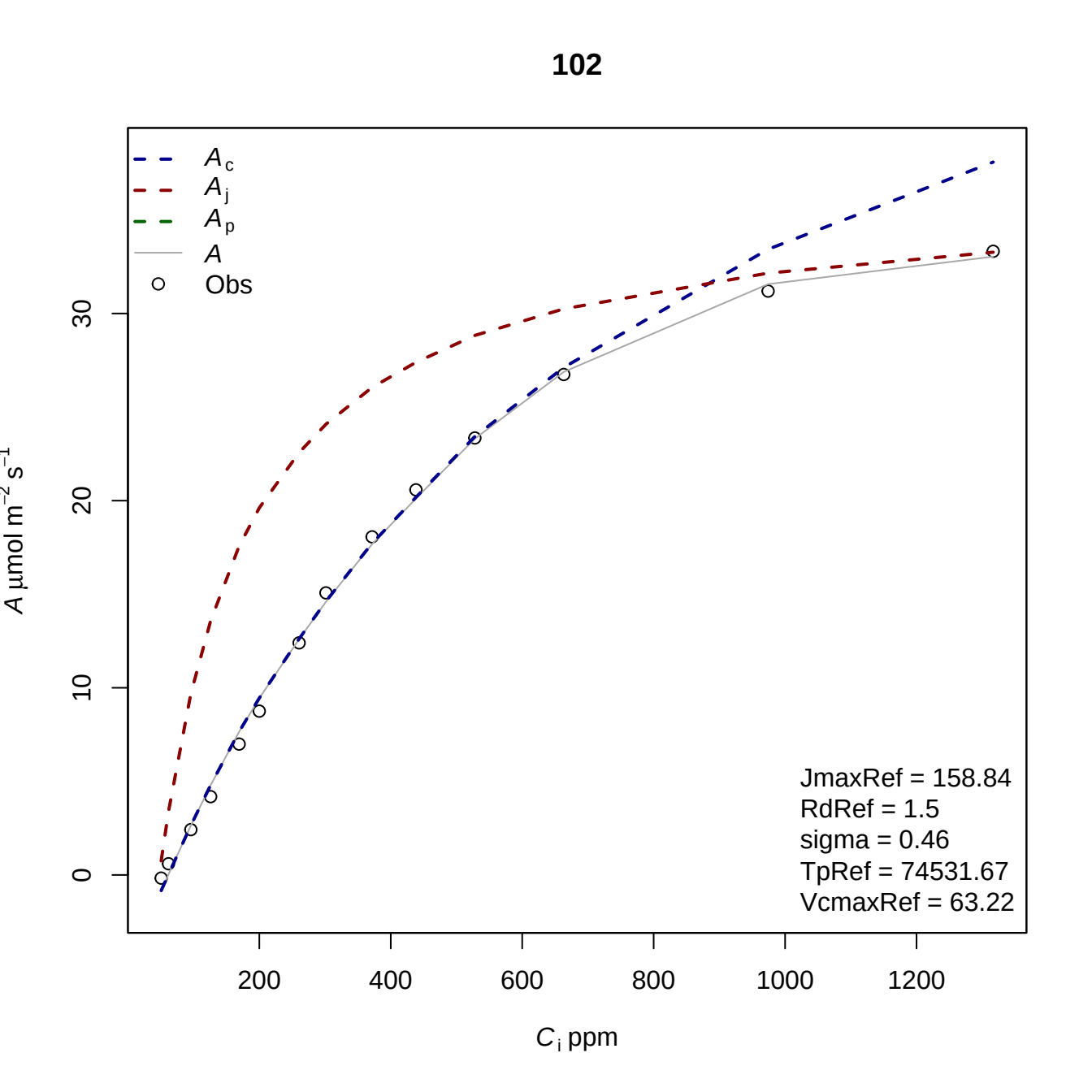

97
A_c
A_j
A_p
A
Obs
JmaxRef = 90.8
RdRef = 1.36
sigma = 0.37
TpRef = 6.68
VcmaxRef = 33.75
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
C_i ppm

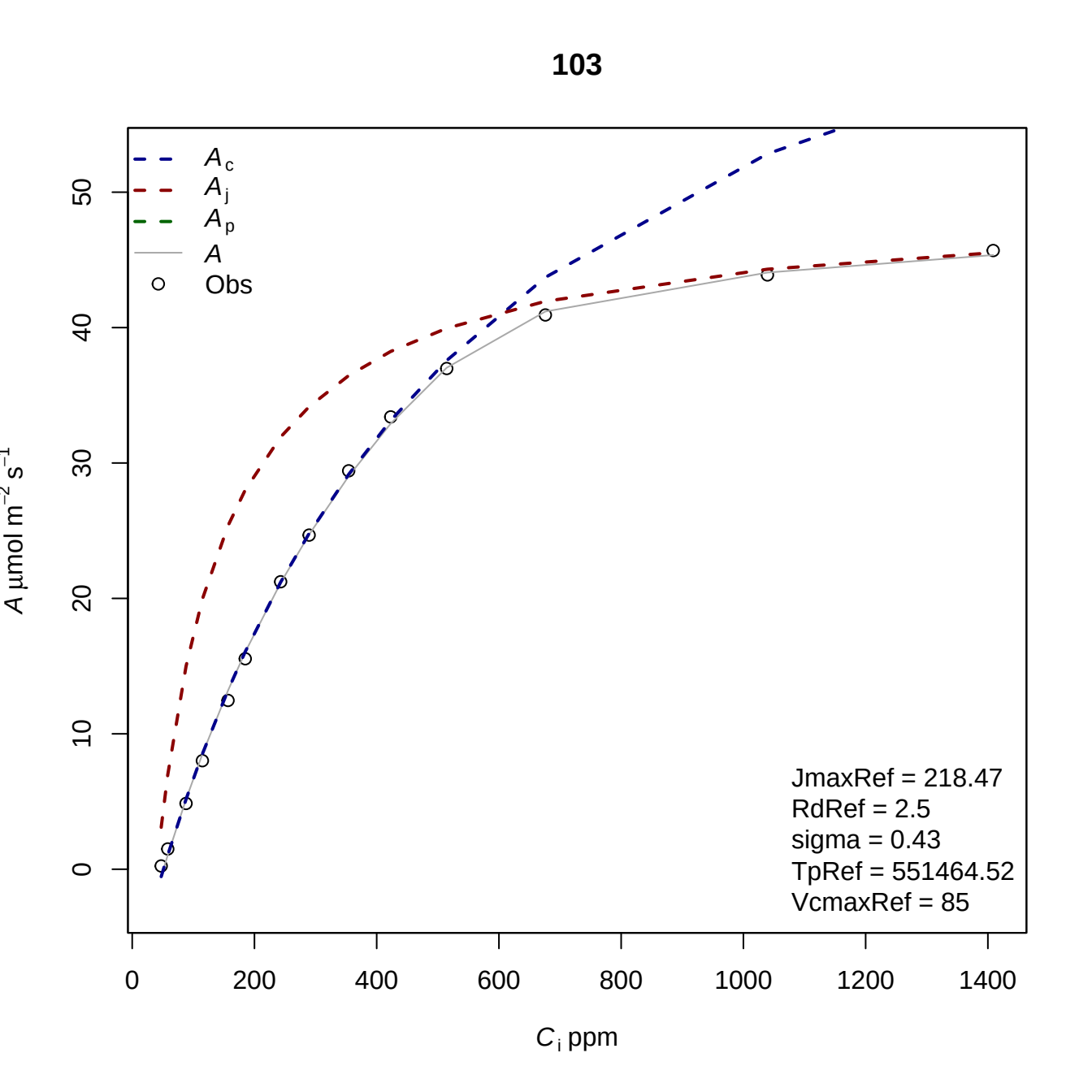

98
A_c
A_j
A_p
A
Obs
JmaxRef = 103.01
RdRef = 1.97
sigma = 0.3
TpRef = 199566.77
VcmaxRef = 41.63
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
500
1000
1500

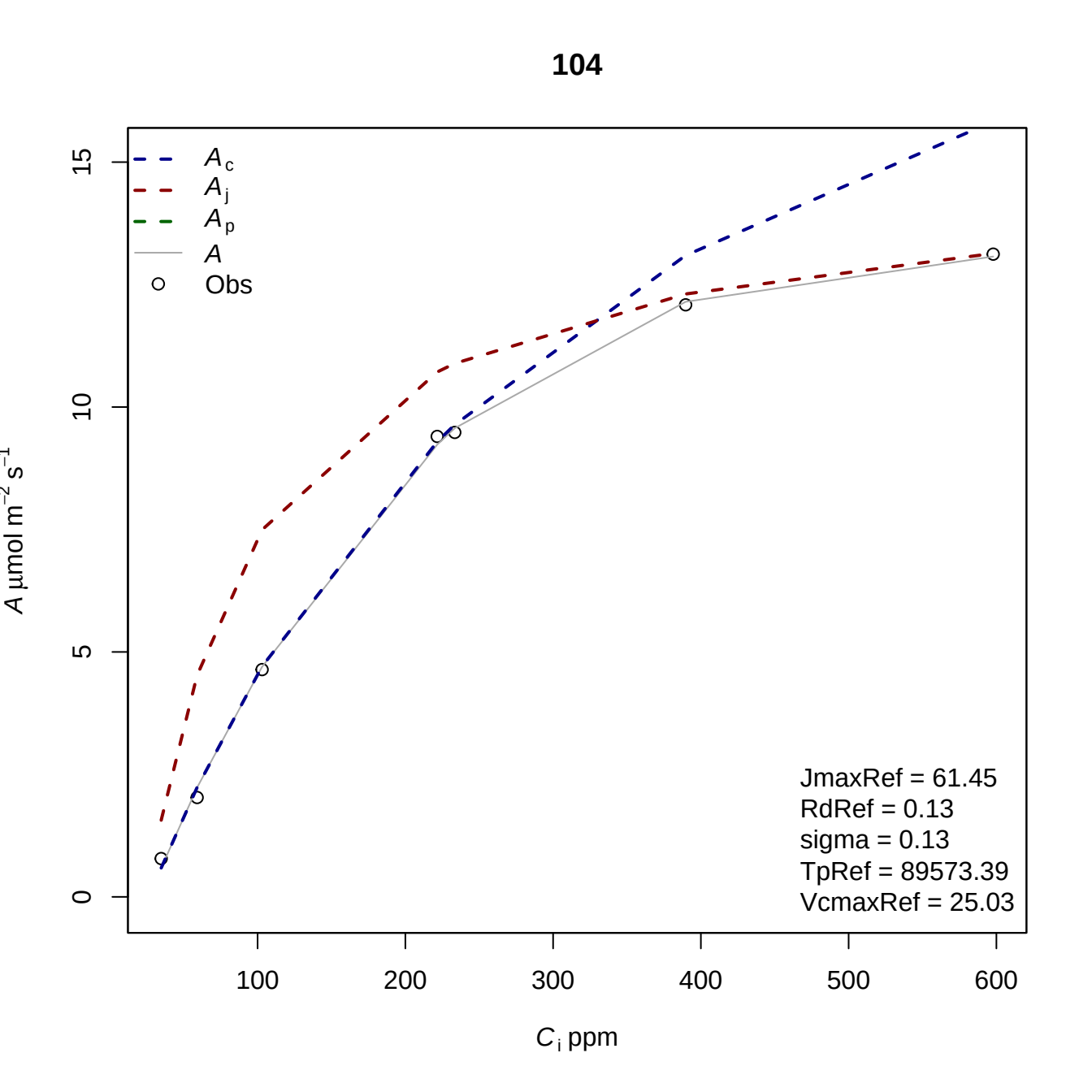

99
A_c
A_j
A_p
A
Obs
JmaxRef = 191.93
RdRef = 8.18
sigma = 0.88
TpRef = 22526.68
VcmaxRef = 113.15
A µmol m^{-2} s^{-1}
C_i ppm
40
30
20
10
0
−10
200
400
600
800
1000

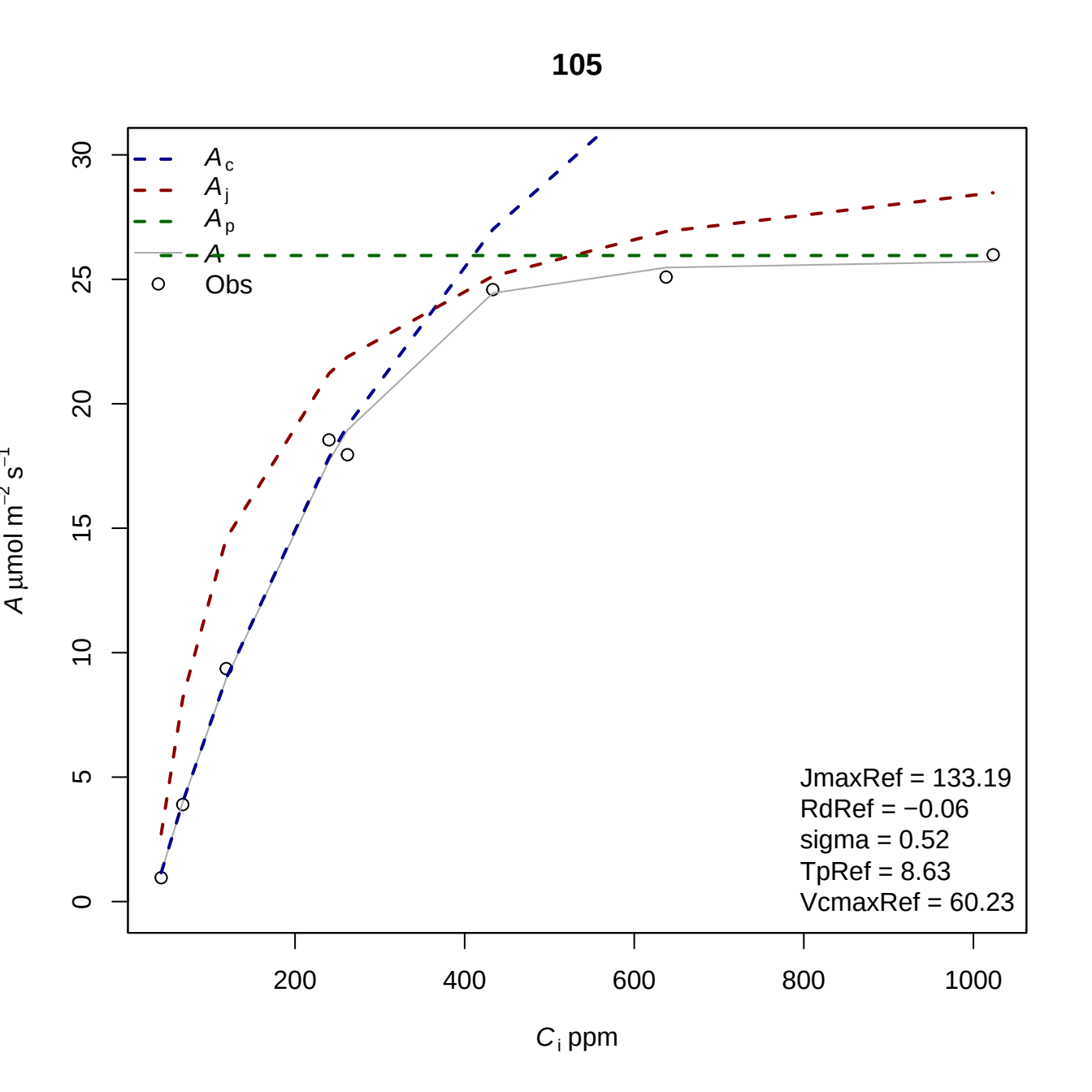

100
A_c
A_j
A_p
A
Obs
JmaxRef = 144.08
RdRef = 3.11
sigma = 0.54
TpRef = 2739.23
VcmaxRef = 53.56
0
10
20
30
200
400
600
800
1000
A µmol m^{-2} s^{-1}
C_i ppm

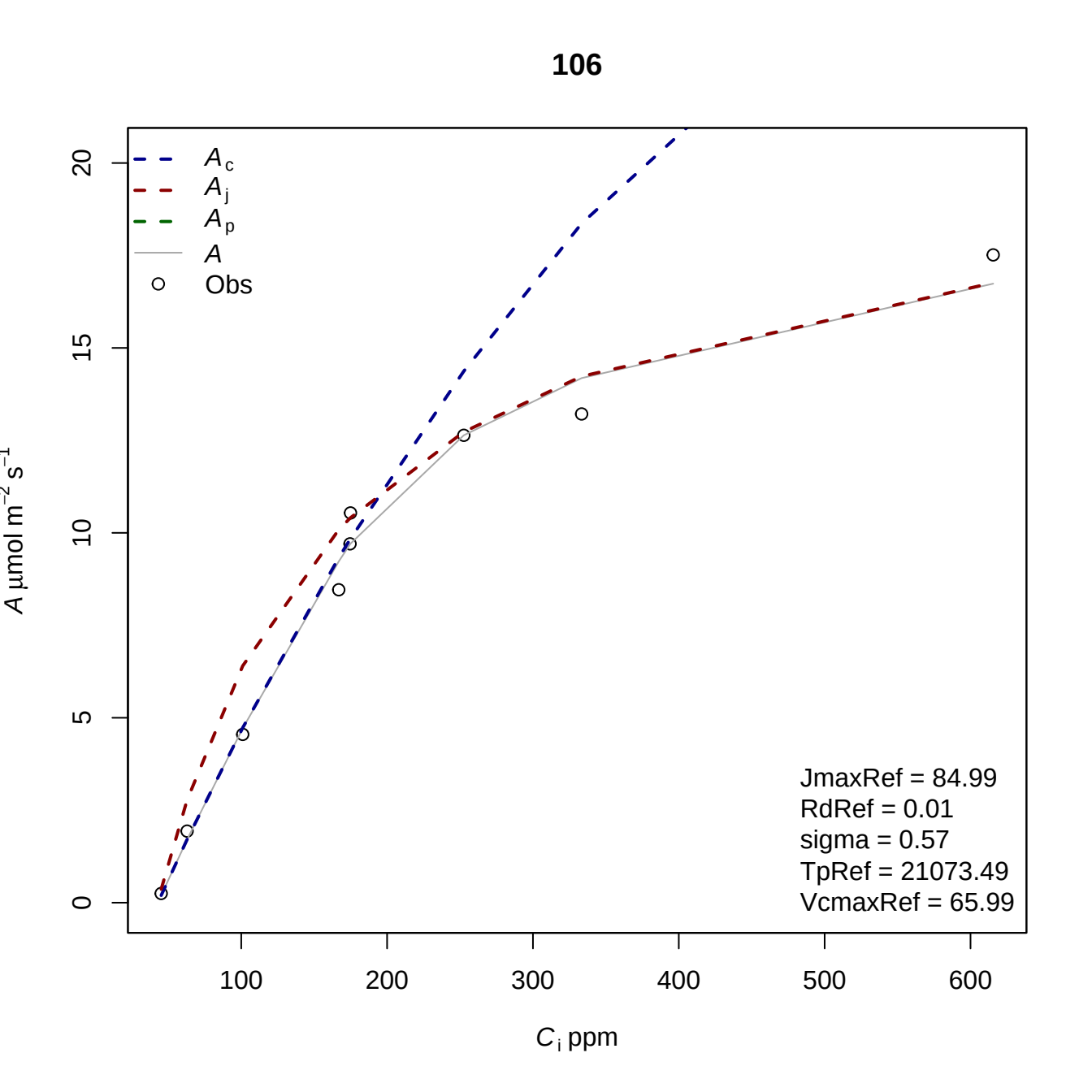

101
A_c
A_j
A_p
A
Obs
JmaxRef = 157.84
RdRef = 2.55
sigma = 0.63
TpRef = 13469.29
VcmaxRef = 63.77
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
0
200
400
600
800
1000
1200
1400

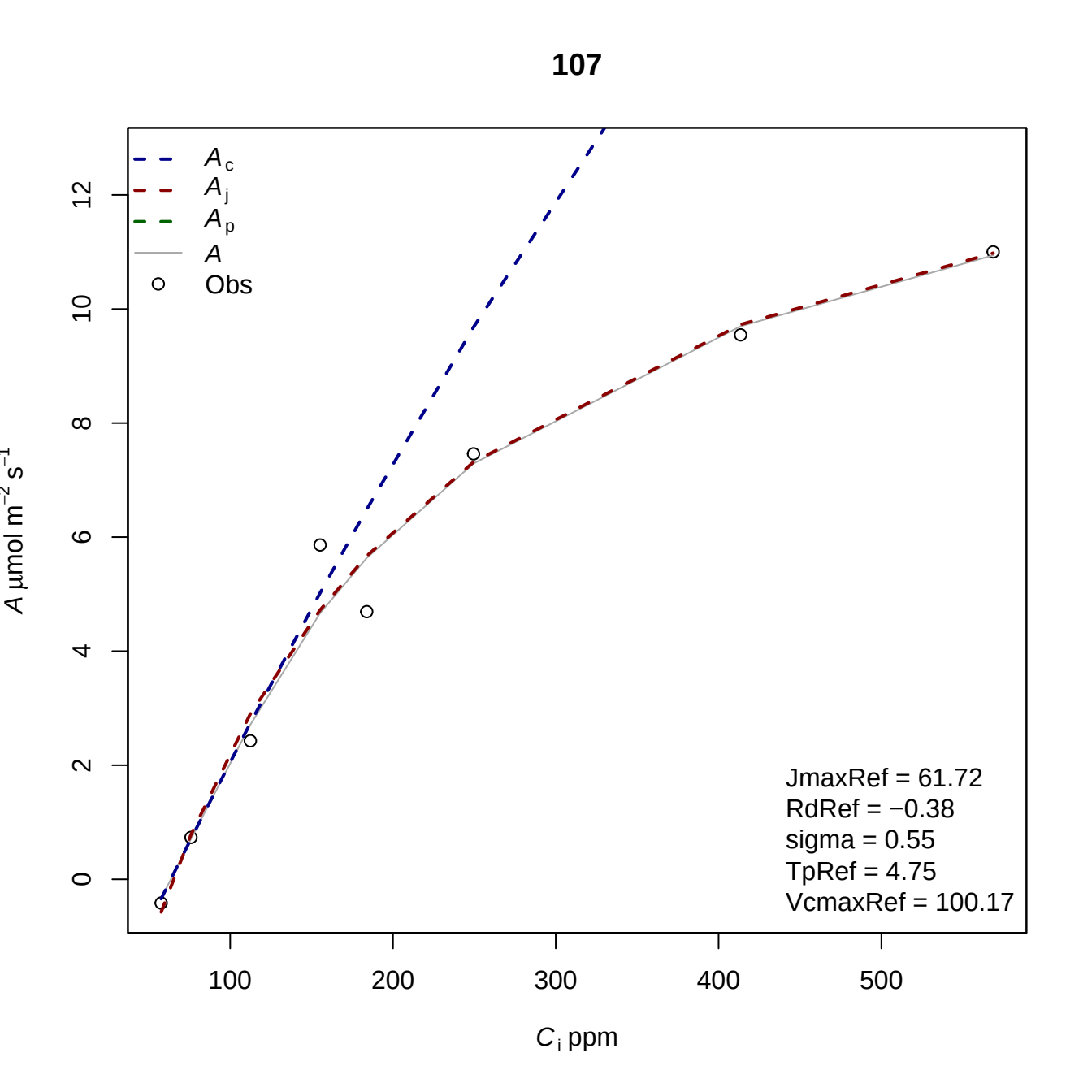

102
A_c
A_j
A_p
A
Obs
JmaxRef = 158.84
RdRef = 1.5
sigma = 0.46
TpRef = 74531.67
VcmaxRef = 63.22
0
10
20
30
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

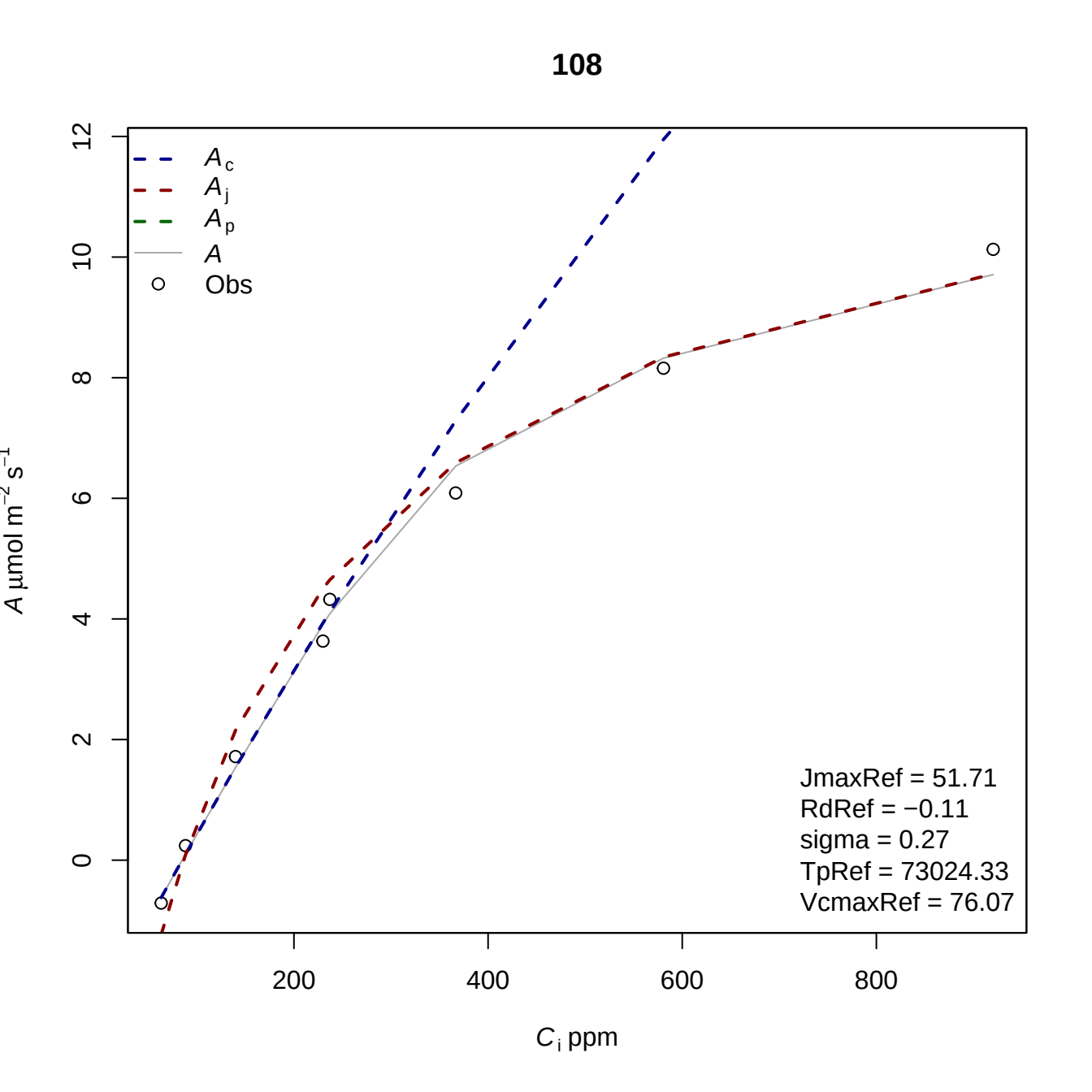

103
A_c
A_j
A_p
A
Obs
JmaxRef = 218.47
RdRef = 2.5
sigma = 0.43
TpRef = 551464.52
VcmaxRef = 85
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
50
0
200
400
600
800
1000
1200
1400

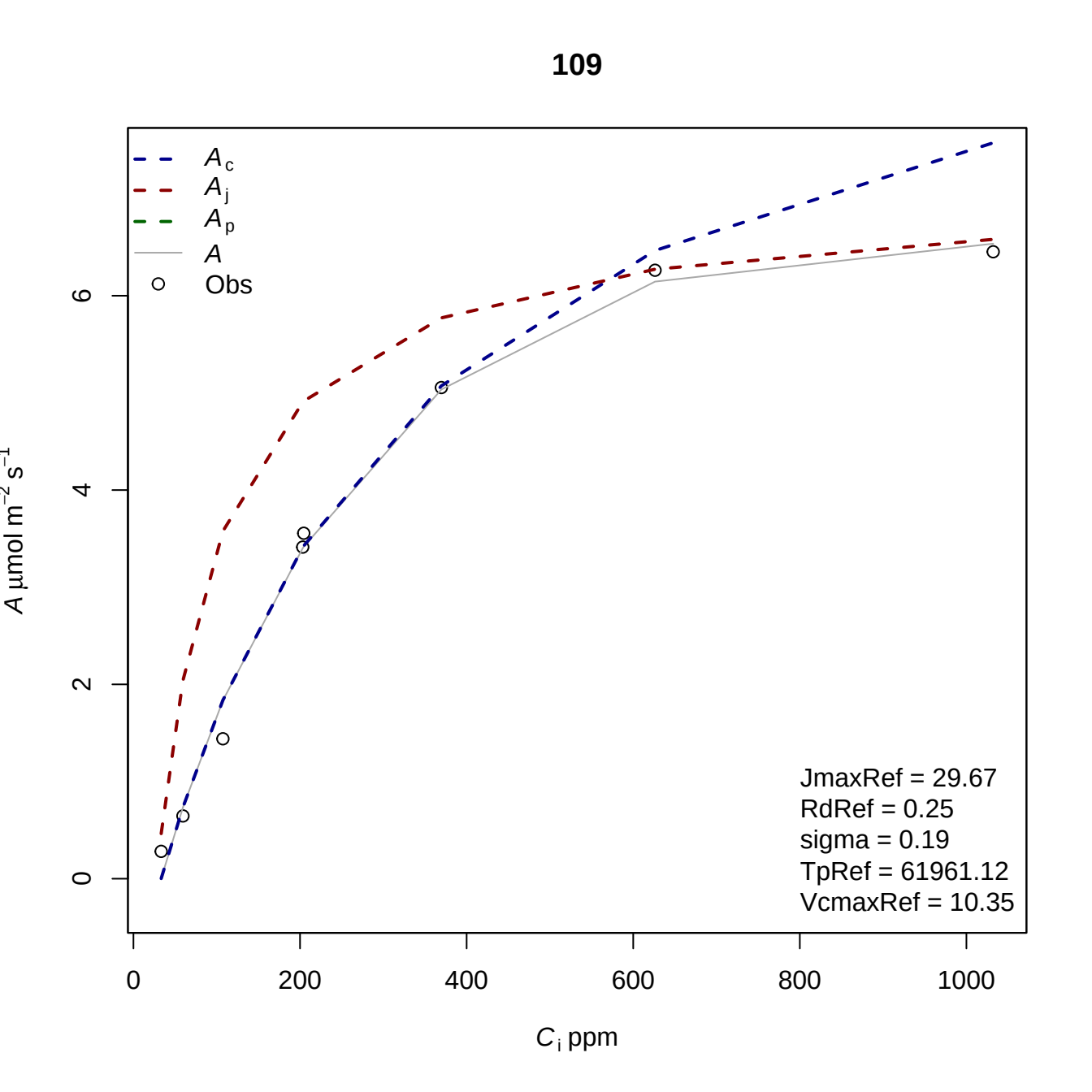

104
A_c
A_j
A_p
A
Obs
JmaxRef = 61.45
RdRef = 0.13
sigma = 0.13
TpRef = 89573.39
VcmaxRef = 25.03
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
100
200
300
400
500
600

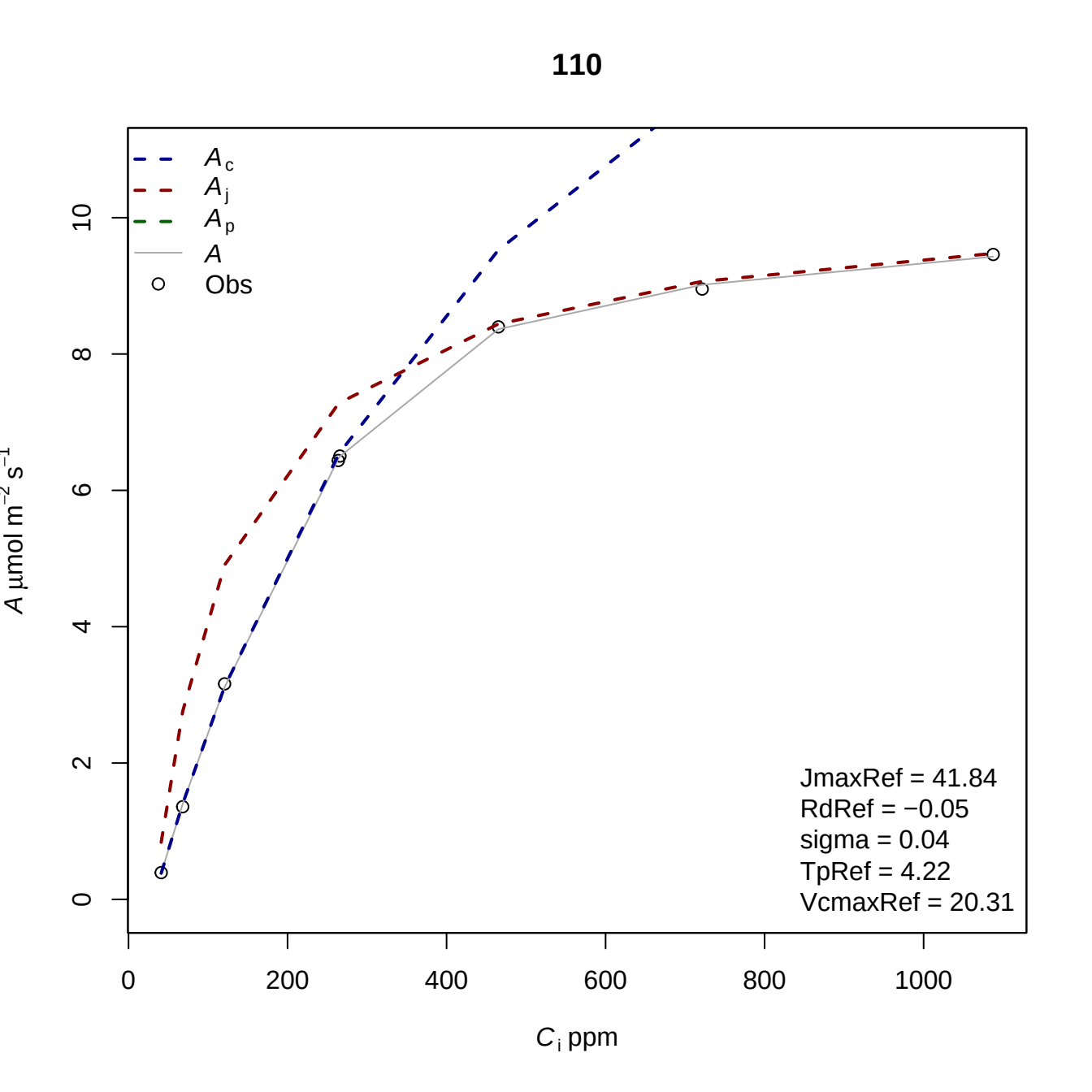

105
A_c
A_j
A_p
A
Obs
JmaxRef = 133.19
RdRef = −0.06
sigma = 0.52
TpRef = 8.63
VcmaxRef = 60.23
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
200
400
600
800
1000

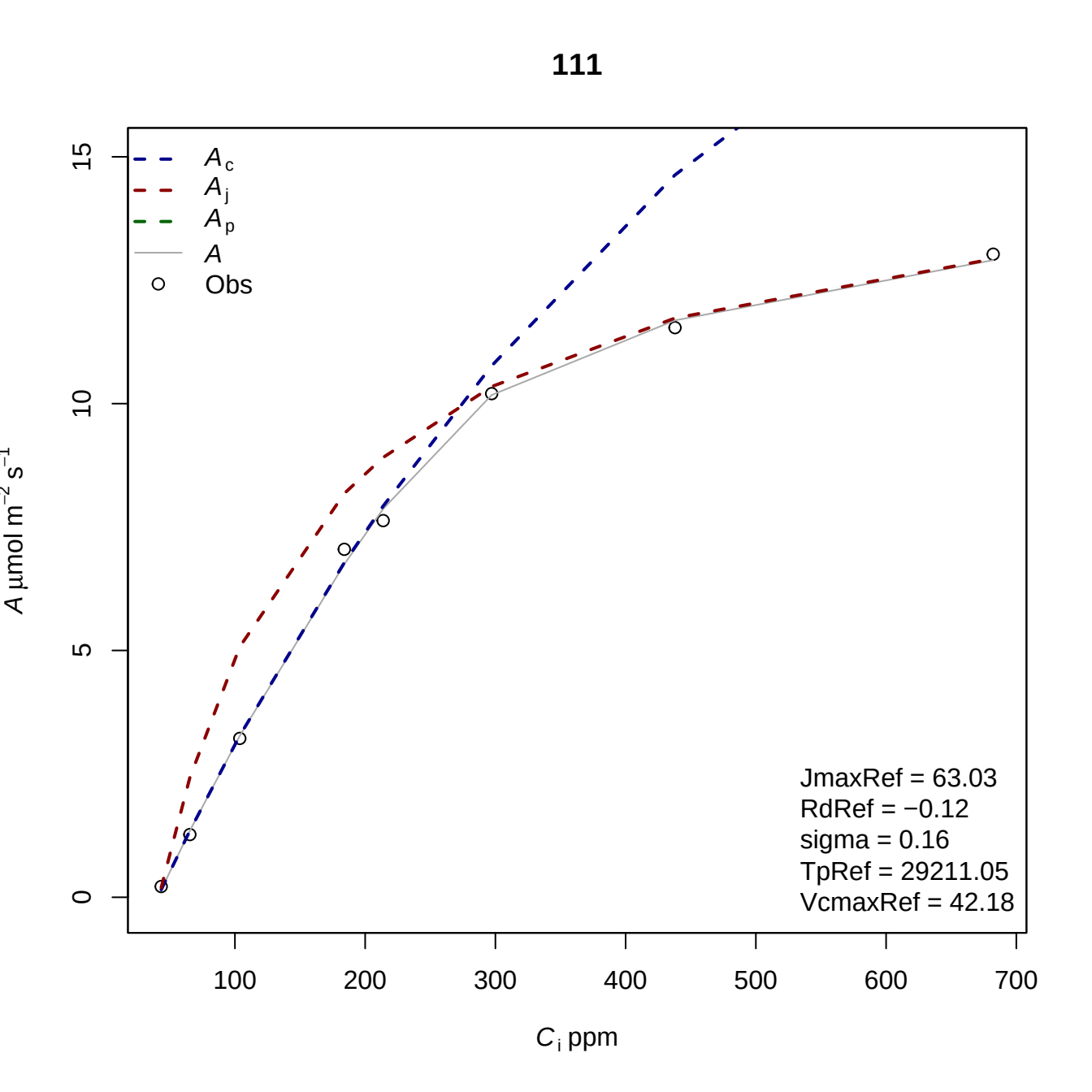

106
A_c
A_j
A_p
A
Obs
JmaxRef = 84.99
RdRef = 0.01
sigma = 0.57
TpRef = 21073.49
VcmaxRef = 65.99
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
100
200
300
400
500
600

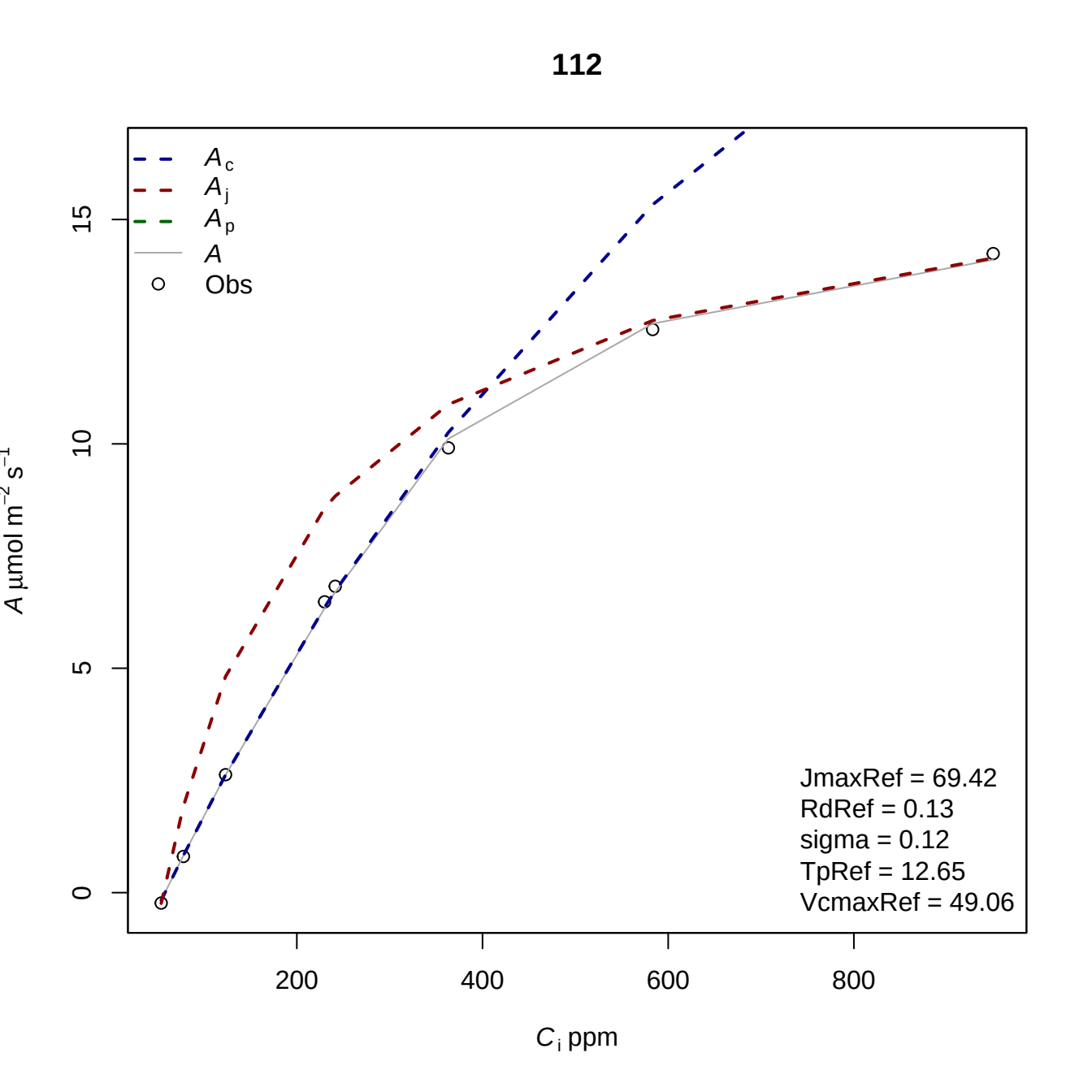

107
A_c
A_j
A_p
A
Obs
JmaxRef = 61.72
RdRef = −0.38
sigma = 0.55
TpRef = 4.75
VcmaxRef = 100.17
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
100
200
300
400
500

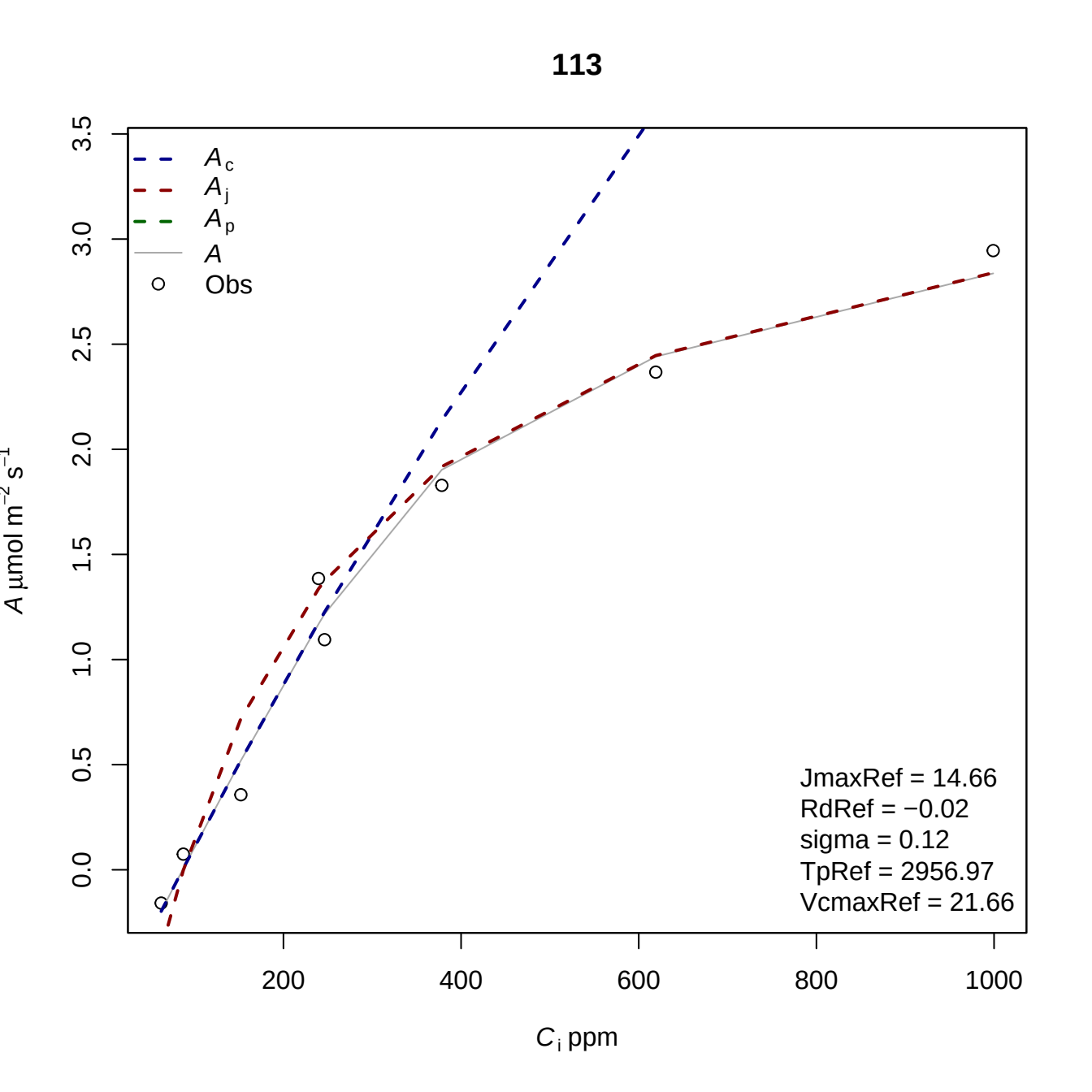

108
A_c
A_j
A_p
A
Obs
JmaxRef = 51.71
RdRef = −0.11
sigma = 0.27
TpRef = 73024.33
VcmaxRef = 76.07
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
200
400
600
800

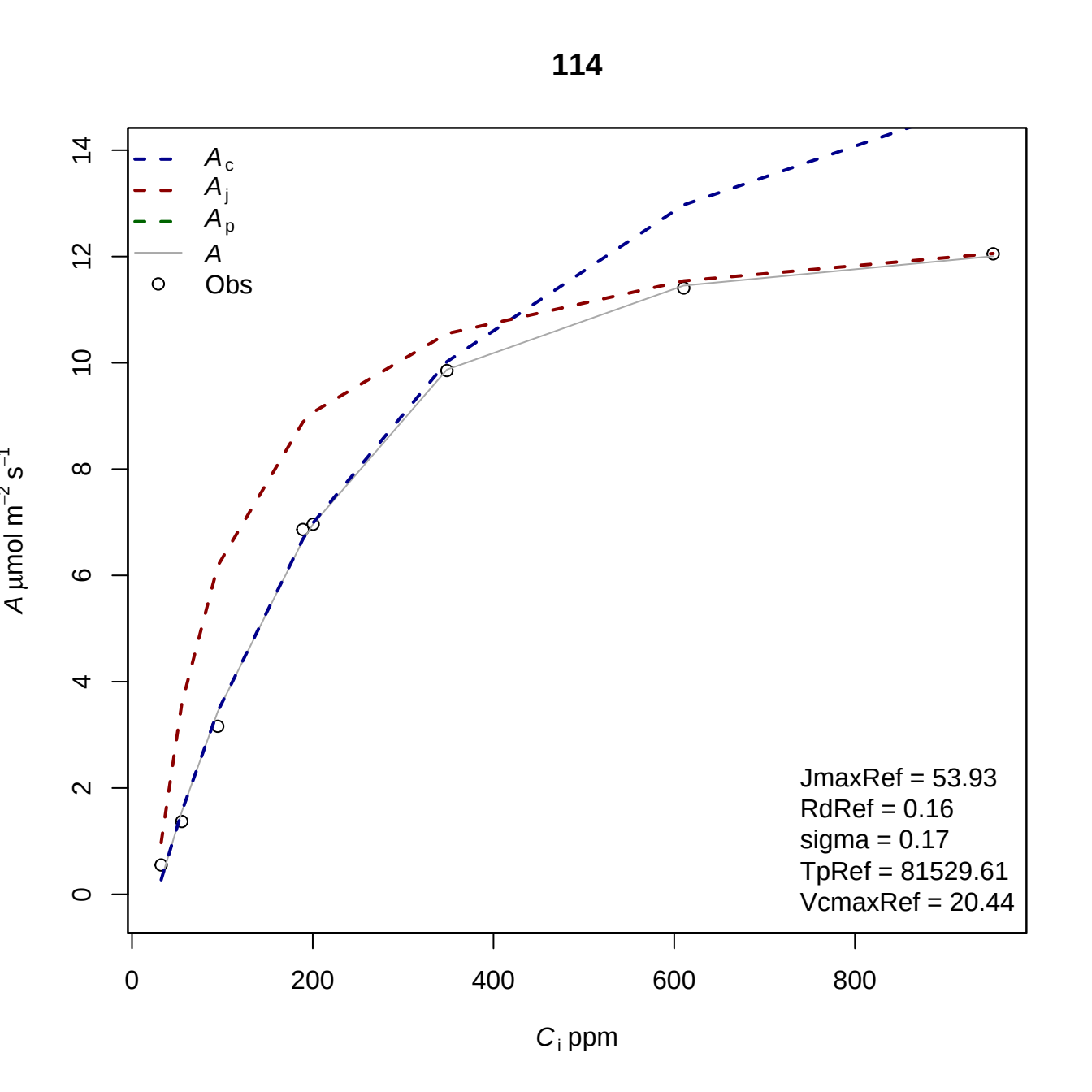

109
A_c
A_j
A_p
A
Obs
JmaxRef = 29.67
RdRef = 0.25
sigma = 0.19
TpRef = 61961.12
VcmaxRef = 10.35
0
2
4
6
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
C_i ppm

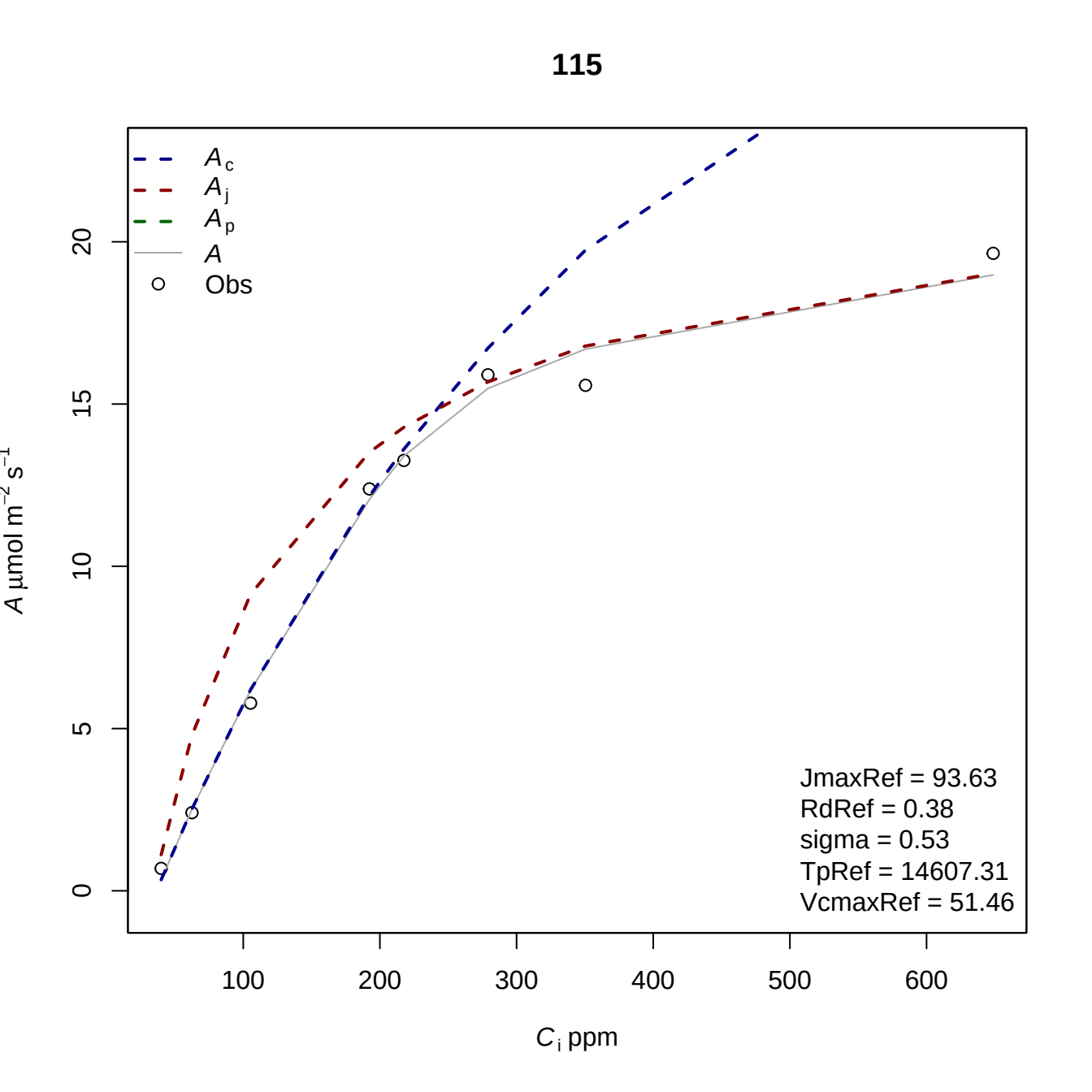

110
A_c
A_j
A_p
A
Obs
JmaxRef = 41.84
RdRef = −0.05
sigma = 0.04
TpRef = 4.22
VcmaxRef = 20.31
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
0
200
400
600
800
1000

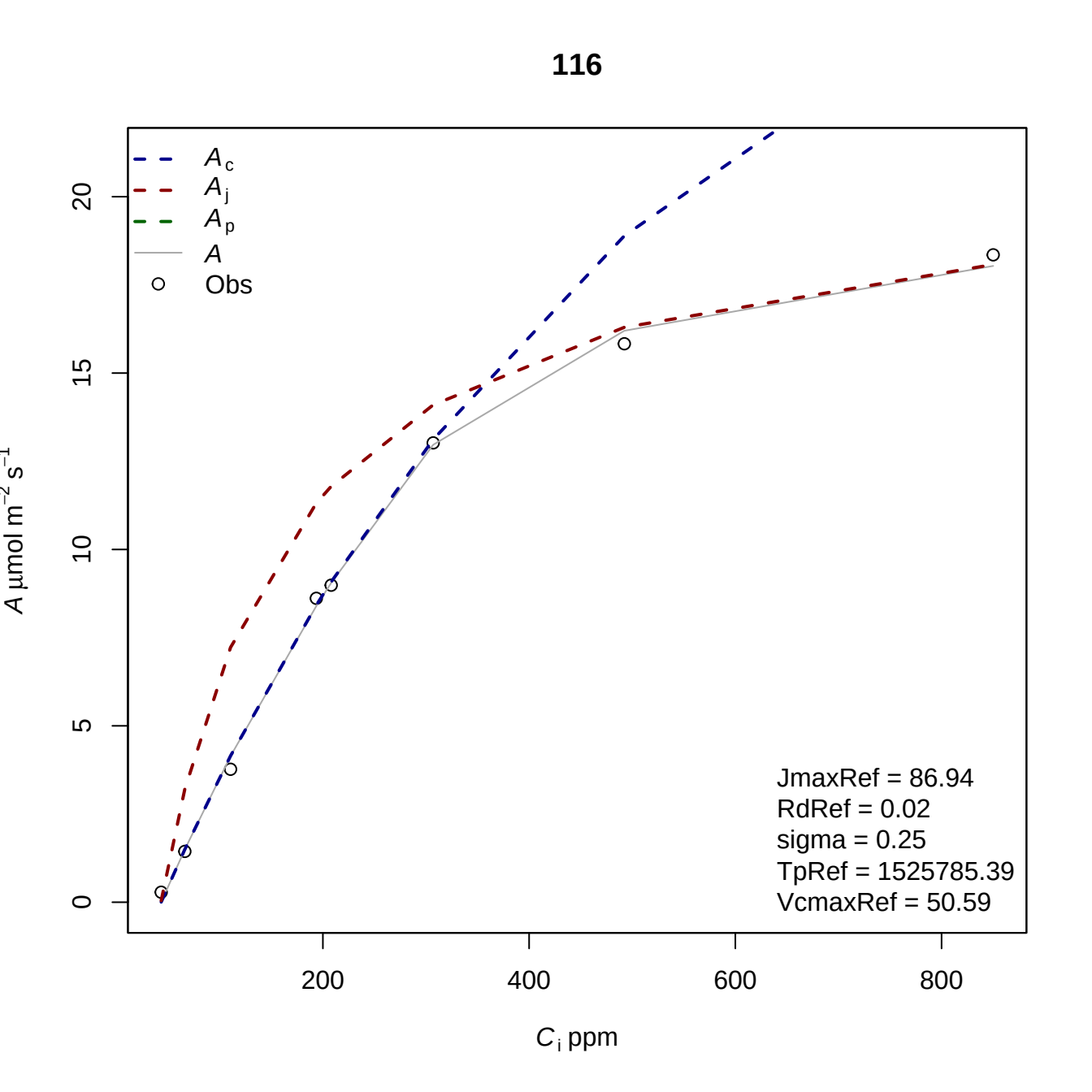

111
A_c
A_j
A_p
A
Obs
JmaxRef = 63.03
RdRef = −0.12
sigma = 0.16
TpRef = 29211.05
VcmaxRef = 42.18
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
100
200
300
400
500
600
700

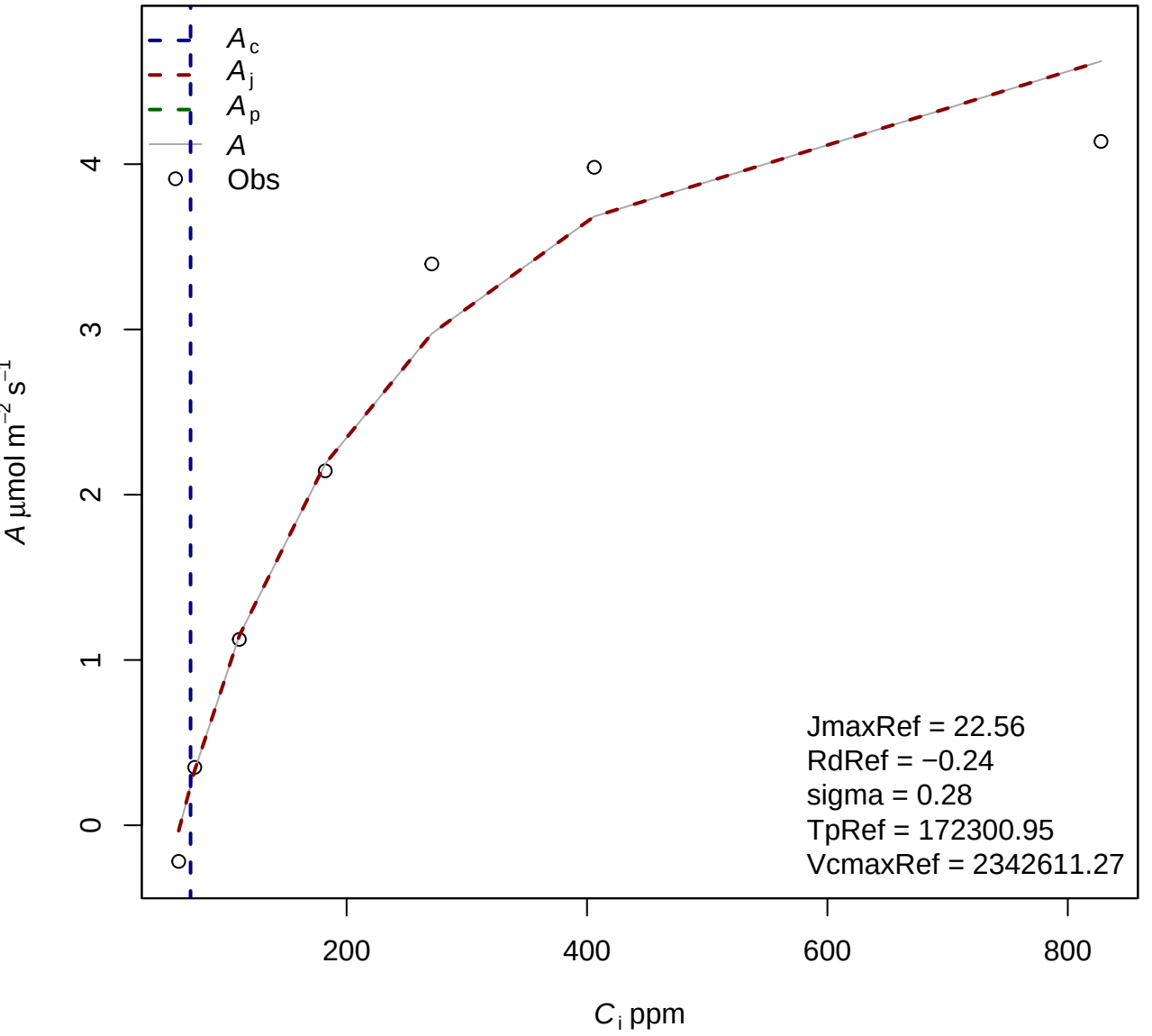

112
A_c
A_j
A_p
A
Obs
JmaxRef = 69.42
RdRef = 0.13
sigma = 0.12
TpRef = 12.65
VcmaxRef = 49.06
0
5
10
15
200
400
600
800
A µmol m^{-2} s^{-1}
C_i ppm

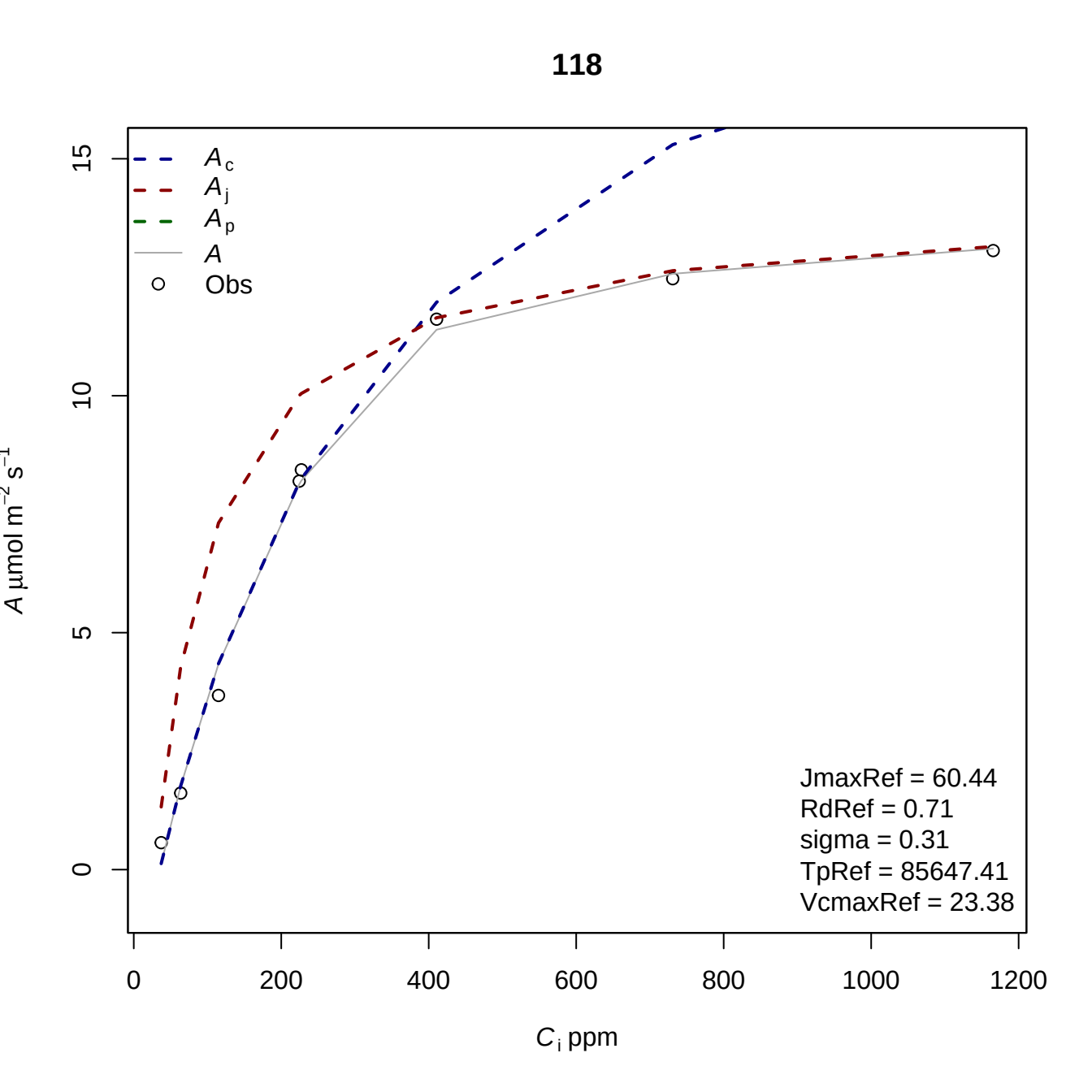

113
A_c
A_j
A_p
A
Obs
JmaxRef = 14.66
RdRef = −0.02
sigma = 0.12
TpRef = 2956.97
VcmaxRef = 21.66
A µmol m^{-2} s^{-1}
C_i ppm
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
200
400
600
800
1000

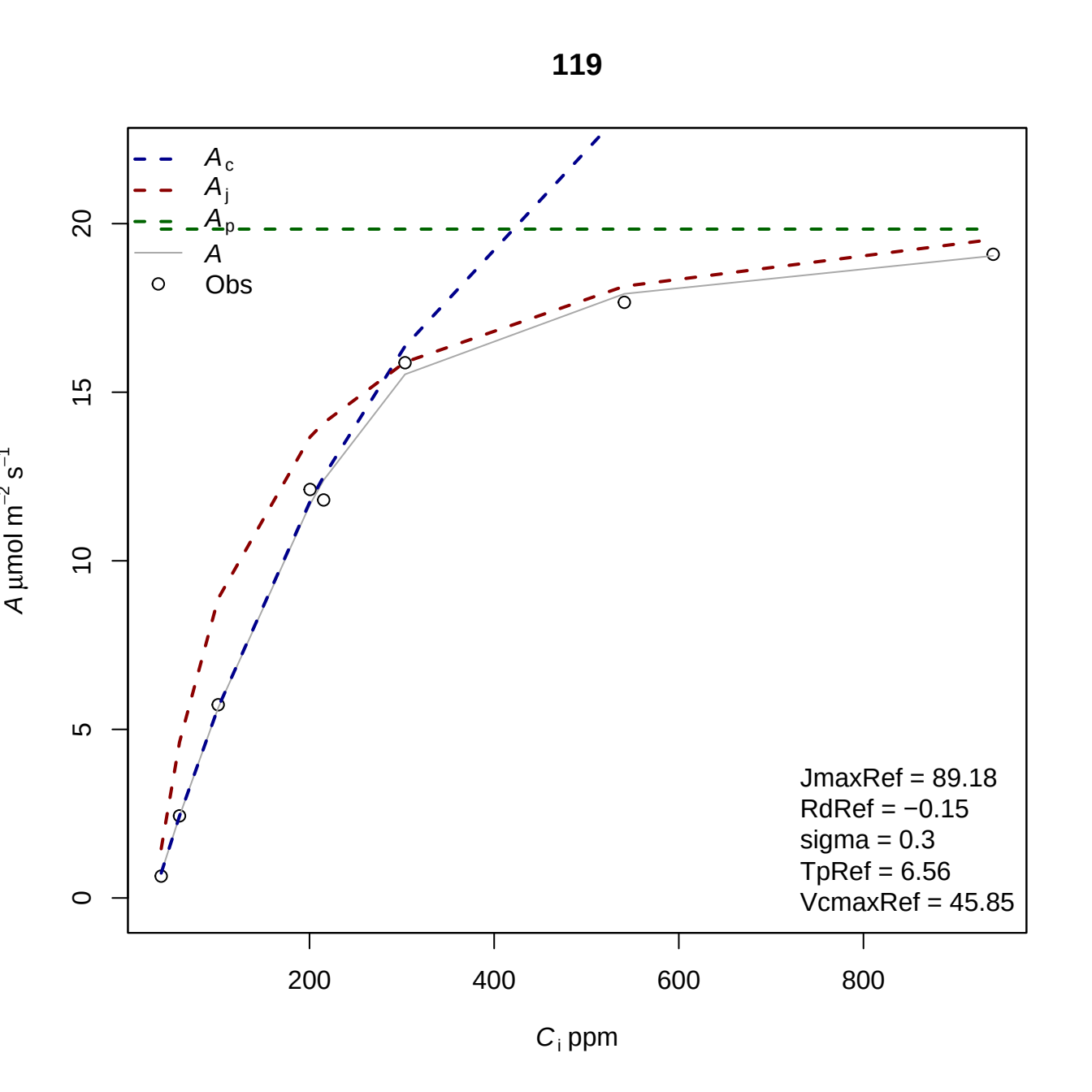

114
A_c
A_j
A_p
A
Obs
JmaxRef = 53.93
RdRef = 0.16
sigma = 0.17
TpRef = 81529.61
VcmaxRef = 20.44
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
14
0
200
400
600
800

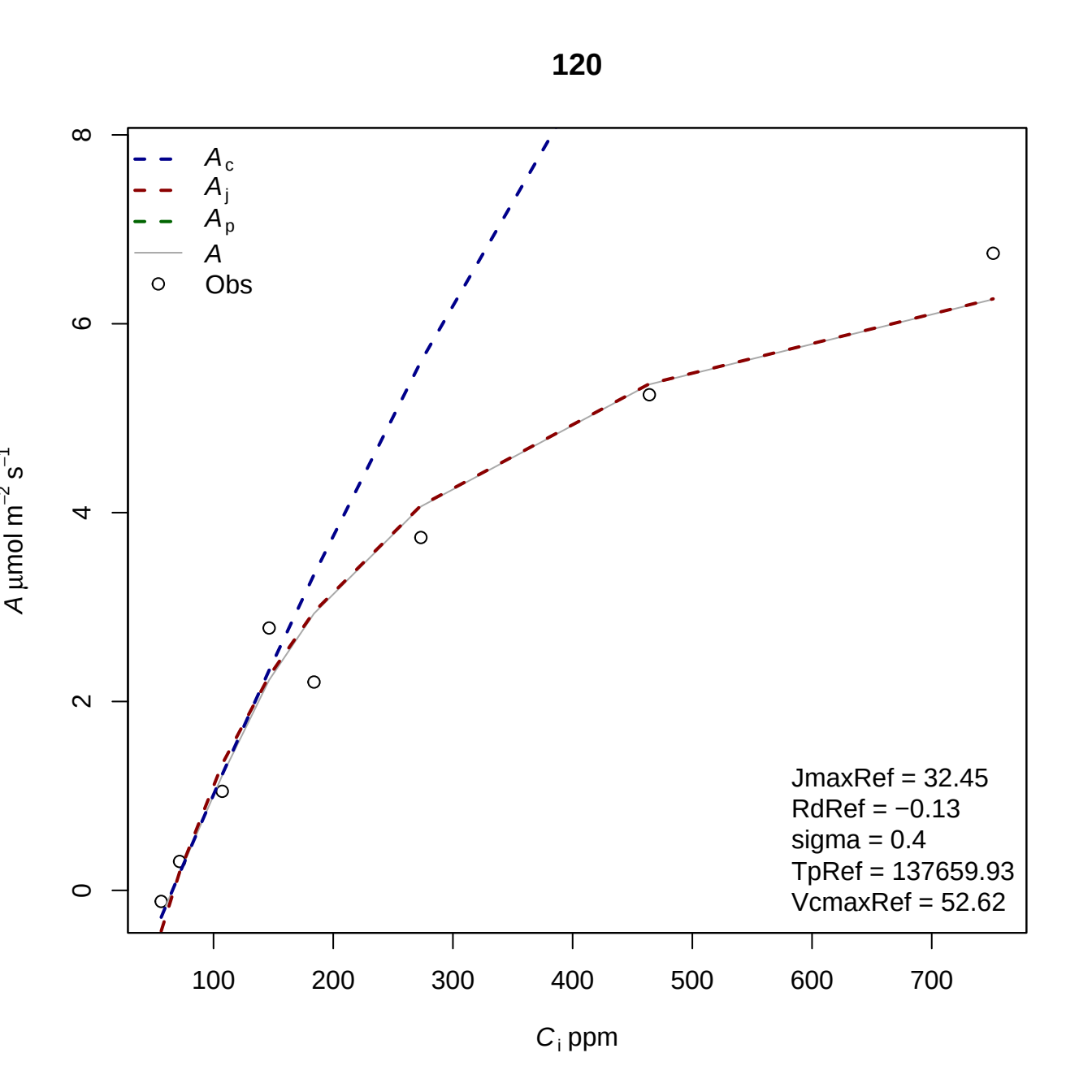

115
A_c
A_j
A_p
A
Obs
JmaxRef = 93.63
RdRef = 0.38
sigma = 0.53
TpRef = 14607.31
VcmaxRef = 51.46
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
100
200
300
400
500
600

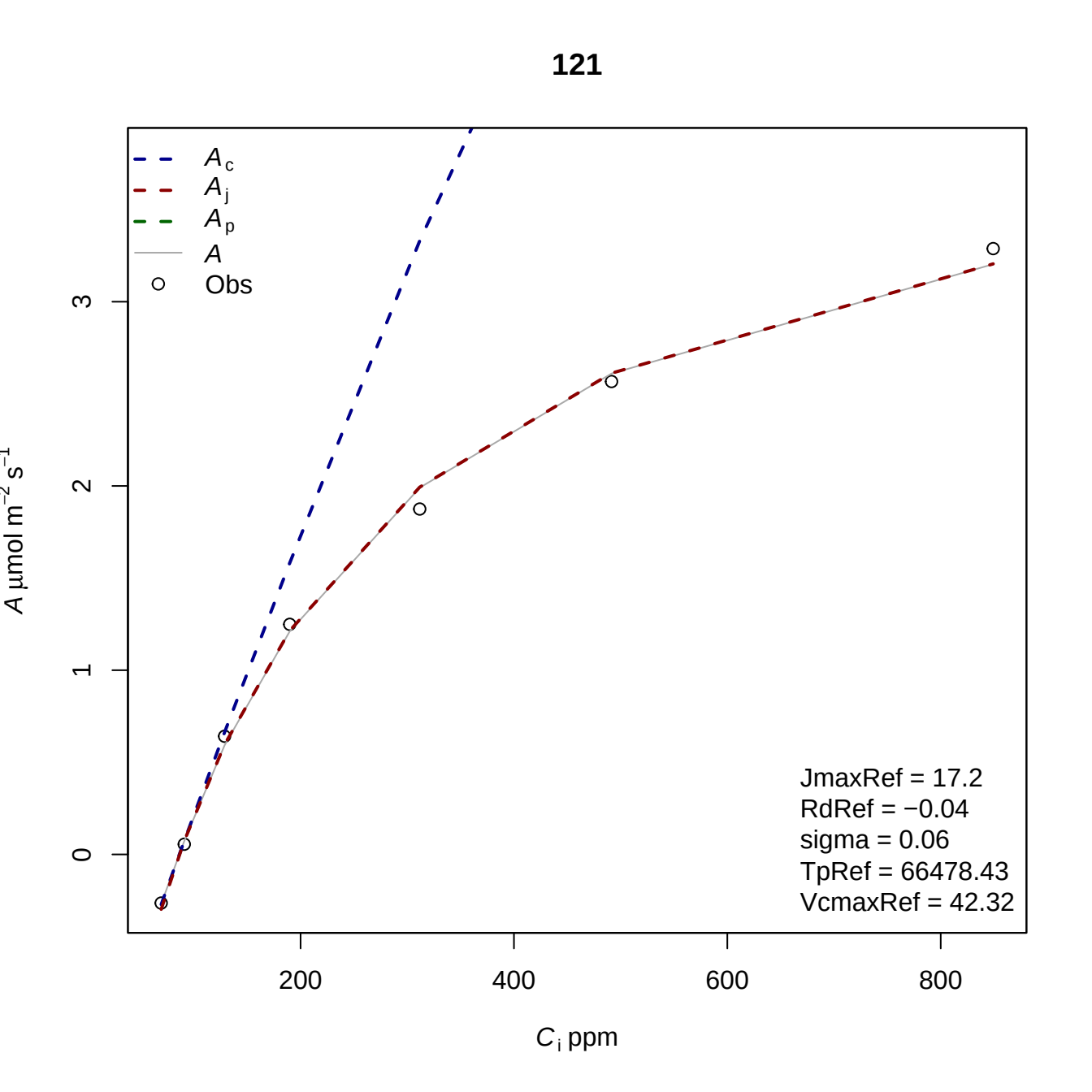

116
A_c
A_j
A_p
A
Obs
JmaxRef = 86.94
RdRef = 0.02
sigma = 0.25
TpRef = 1525785.39
VcmaxRef = 50.59
0
5
10
15
20
200
400
600
800
A µmol m^{-2} s^{-1}
C_i ppm

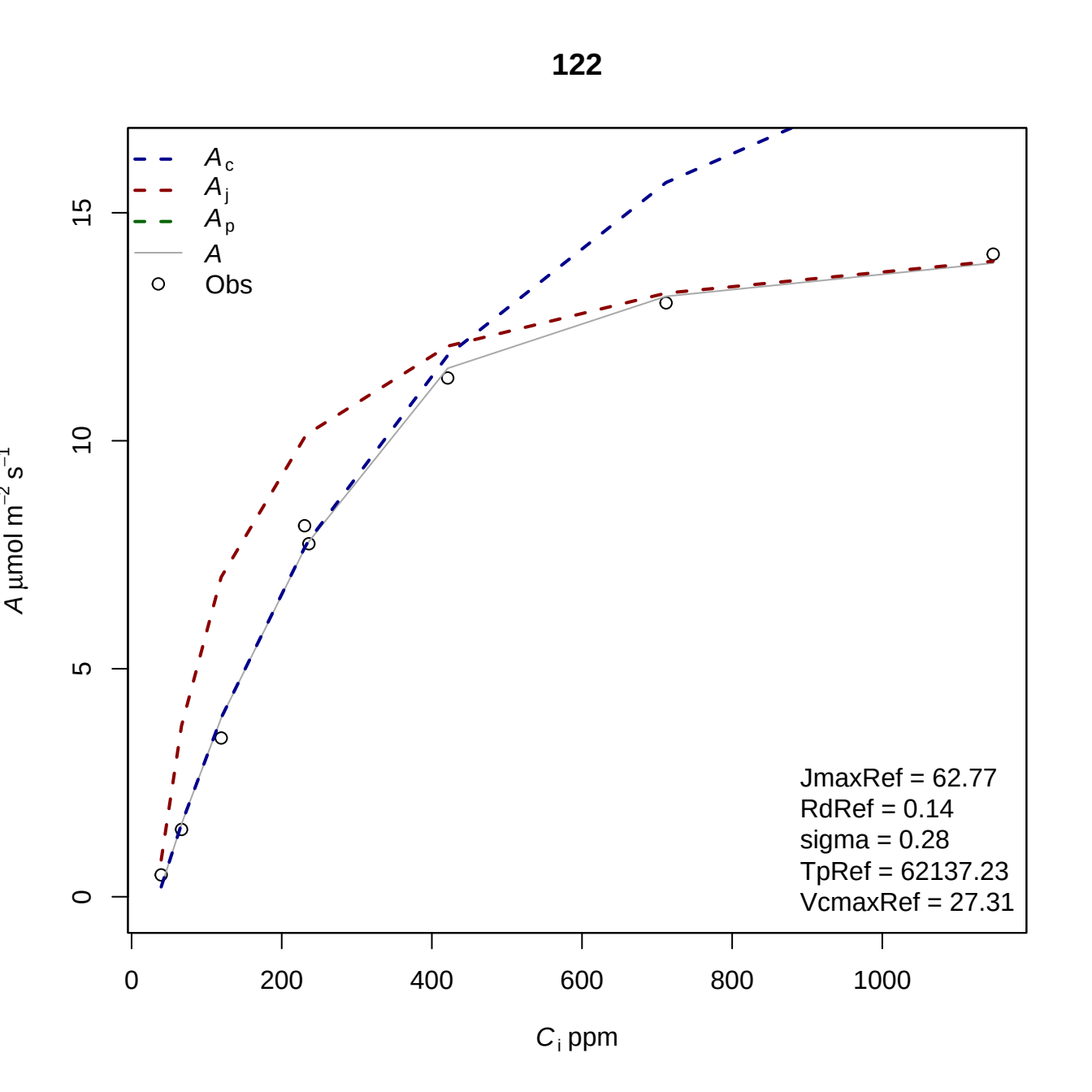

A_c
A_j
A_p
A
Obs
JmaxRef = 22.56
RdRef = −0.24
sigma = 0.28
TpRef = 172300.95
VcmaxRef = 2342611.27
0
1
2
3
4
200
400
600
800
A µmol m^{-2} s^{-1}
C_i ppm

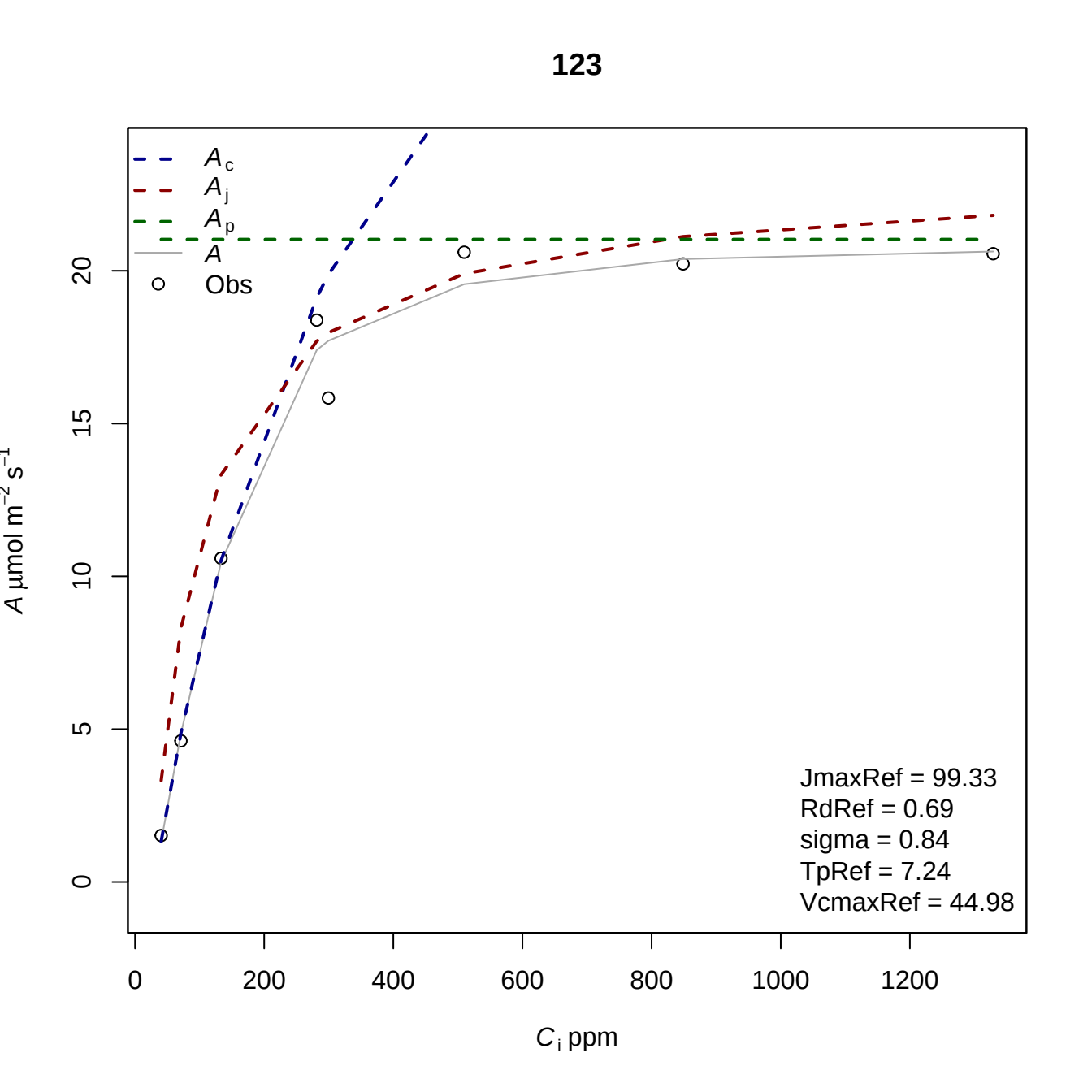

118
A_c
A_j
A_p
A
Obs
JmaxRef = 60.44
RdRef = 0.71
sigma = 0.31
TpRef = 85647.41
VcmaxRef = 23.38
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
200
400
600
800
1000
1200

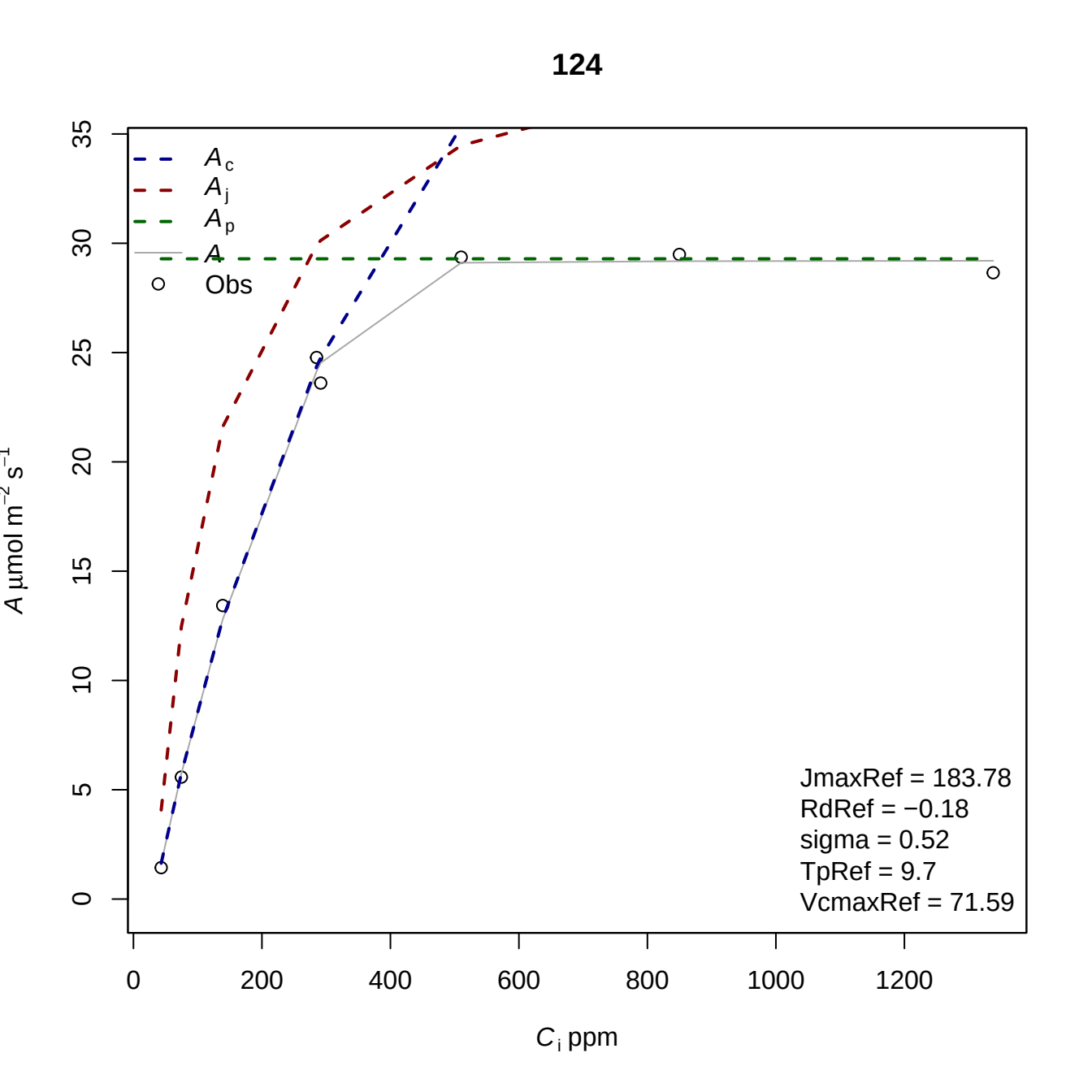

119
A_c
A_j
A_p
A
Obs
JmaxRef = 89.18
RdRef = −0.15
sigma = 0.3
TpRef = 6.56
VcmaxRef = 45.85
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
200
400
600
800

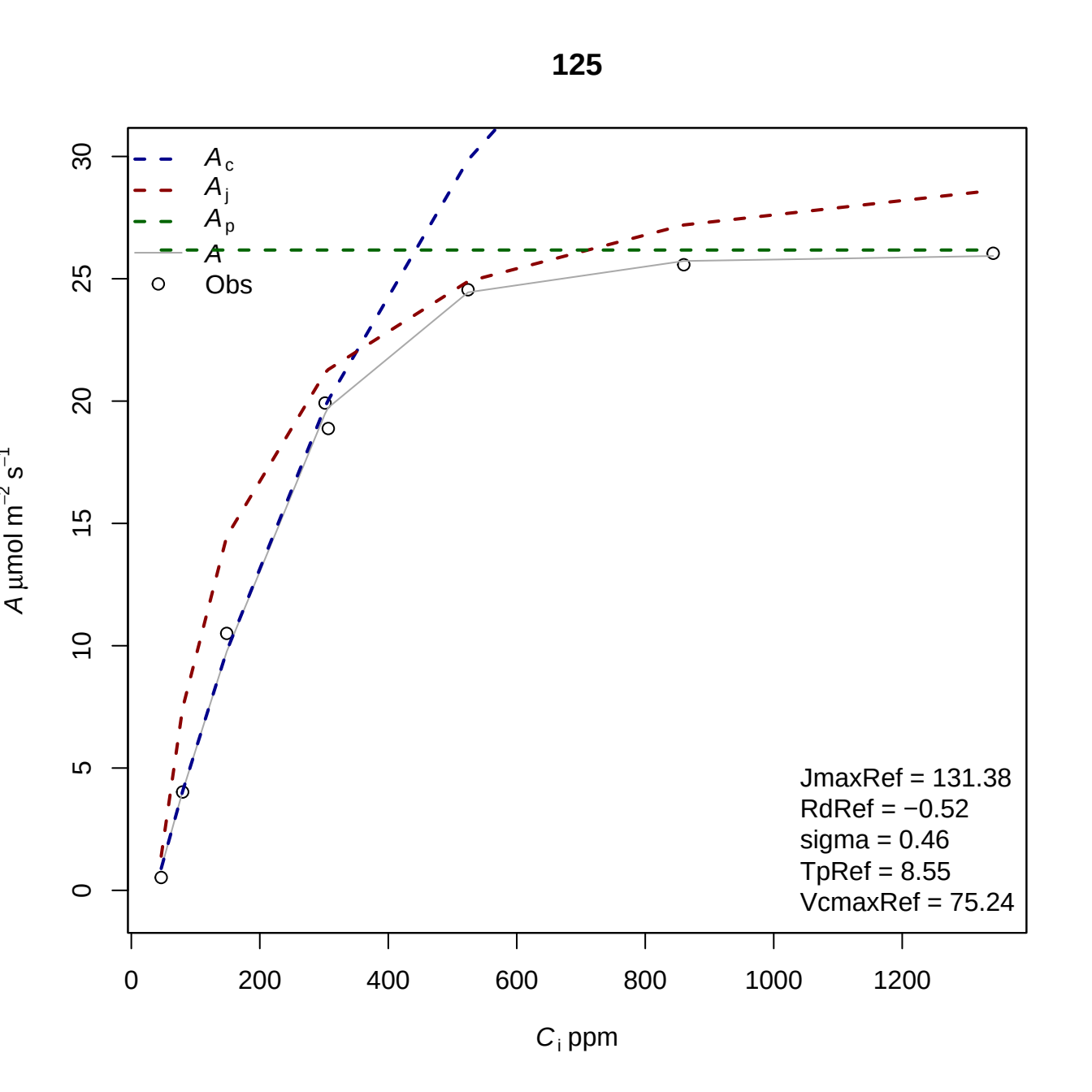

120
A_c
A_j
A_p
A
Obs
JmaxRef = 32.45
RdRef = −0.13
sigma = 0.4
TpRef = 137659.93
VcmaxRef = 52.62
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
100
200
300
400
500
600
700

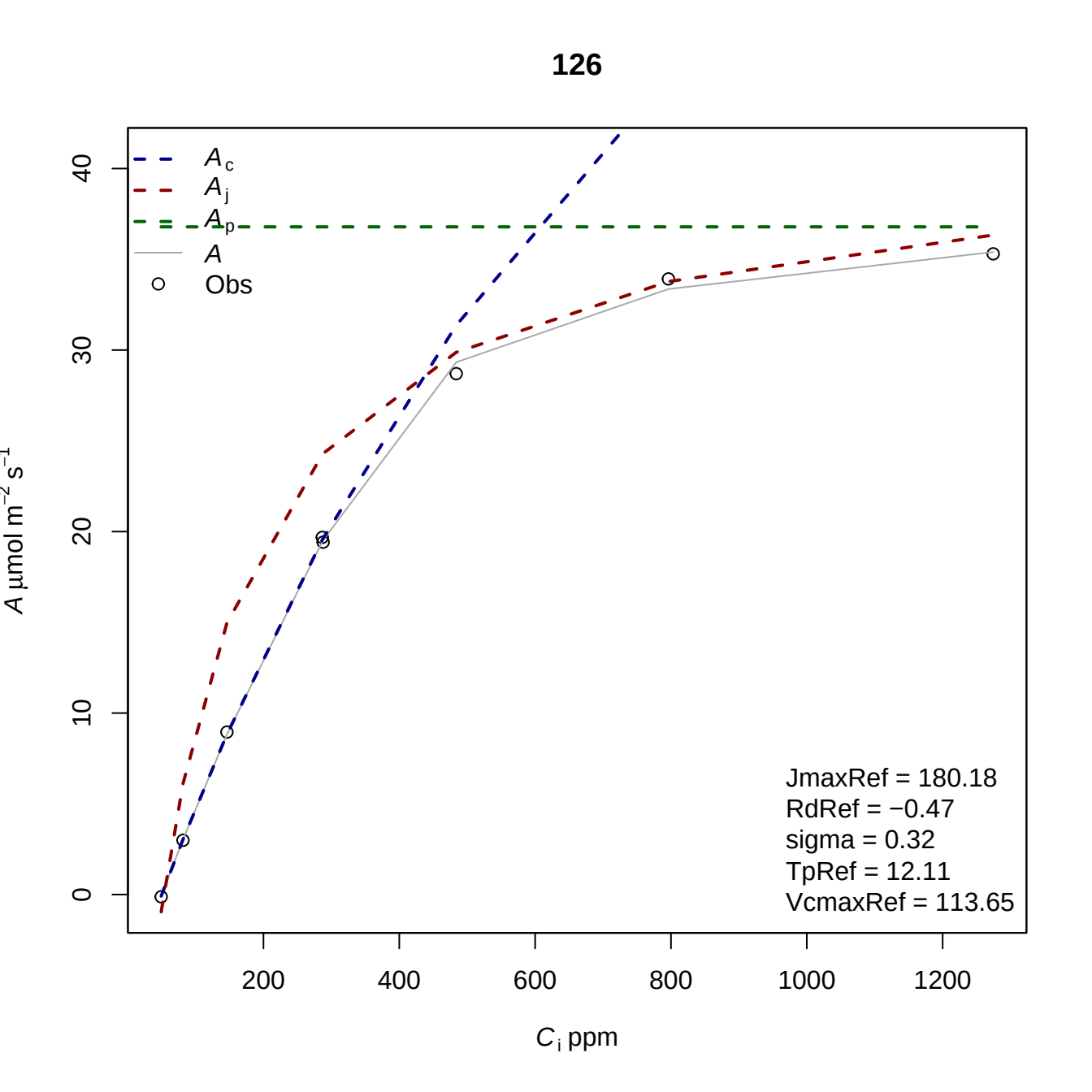

121
A_c
A_j
A_p
A
Obs
JmaxRef = 17.2
RdRef = −0.04
sigma = 0.06
TpRef = 66478.43
VcmaxRef = 42.32
0
1
2
3
200
400
600
800
A µmol m^{-2} s^{-1}
C_i ppm

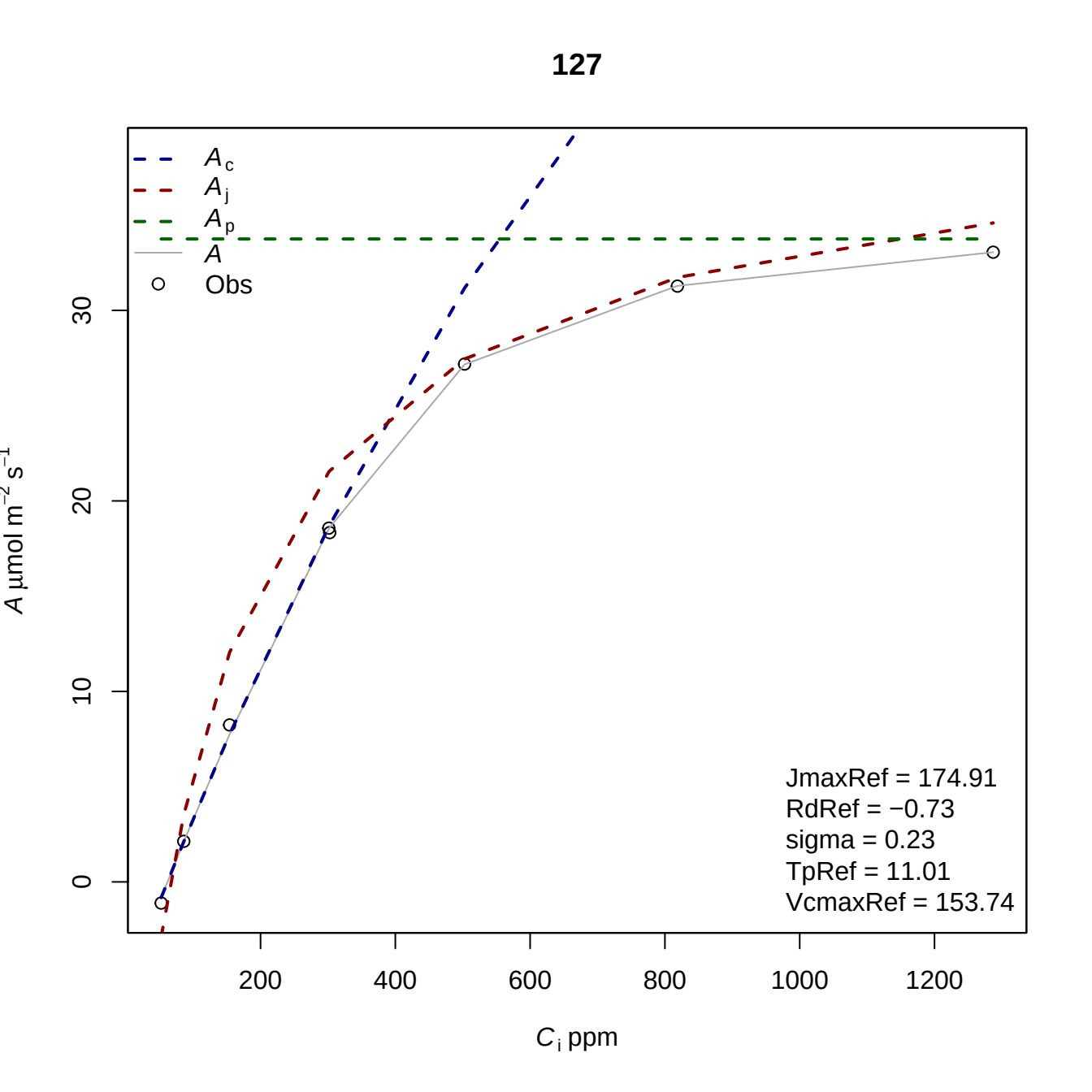

122
A_c
A_j
A_p
A
Obs
JmaxRef = 62.77
RdRef = 0.14
sigma = 0.28
TpRef = 62137.23
VcmaxRef = 27.31
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
200
400
600
800
1000

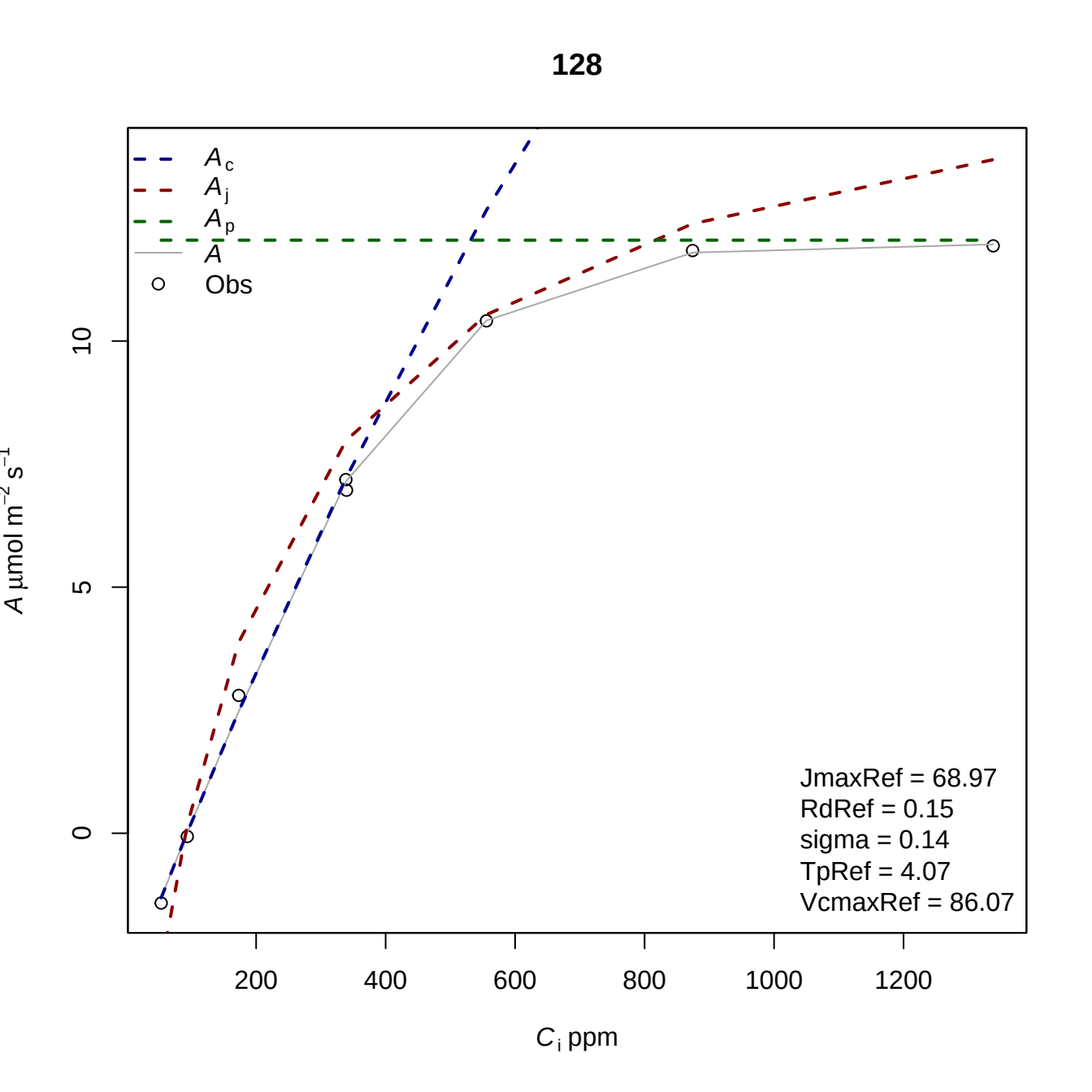

123
A_c
A_j
A_p
A
Obs
JmaxRef = 99.33
RdRef = 0.69
sigma = 0.84
TpRef = 7.24
VcmaxRef = 44.98
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

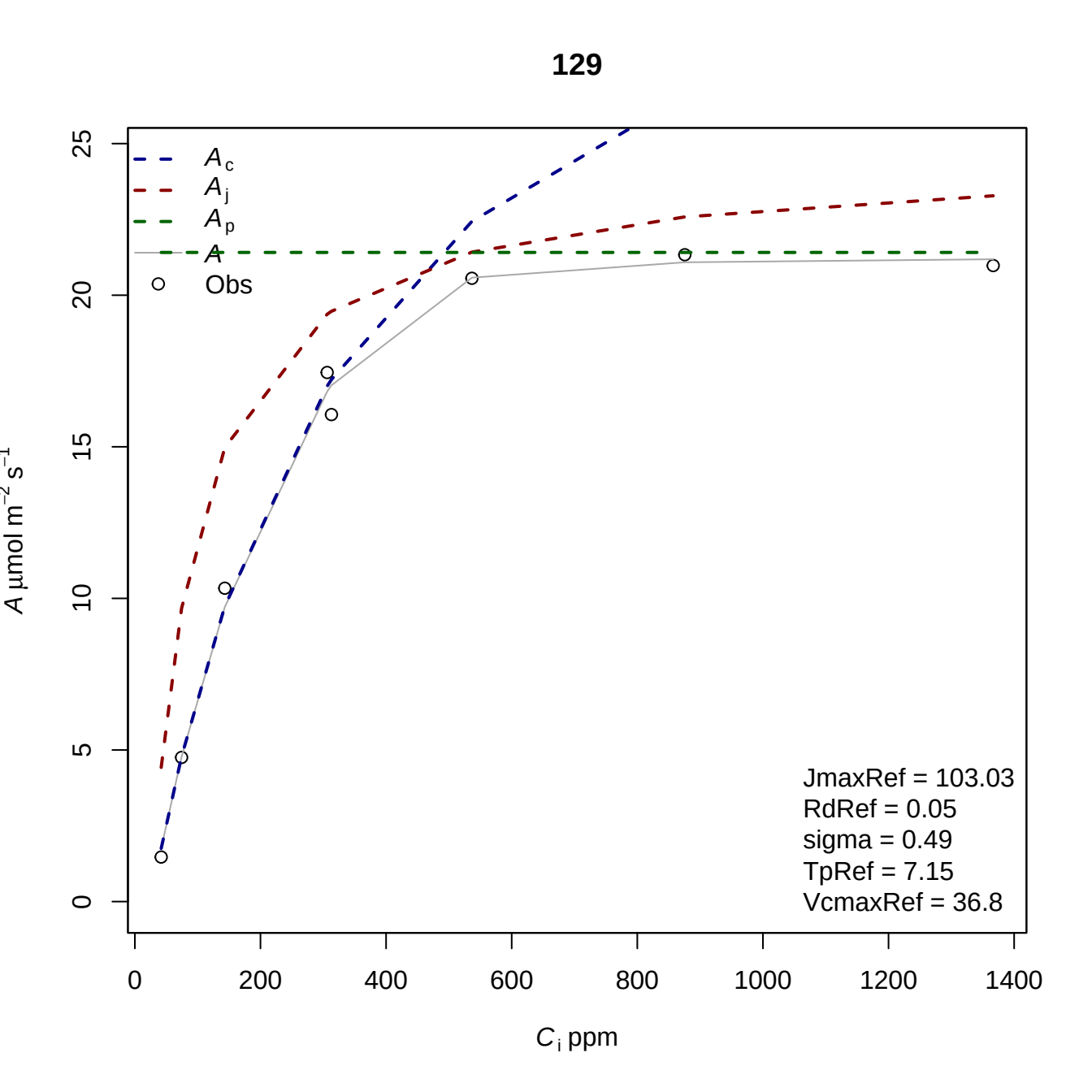

124
A_c
A_j
A_p
A
Obs
JmaxRef = 183.78
RdRef = −0.18
sigma = 0.52
TpRef = 9.7
VcmaxRef = 71.59
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200

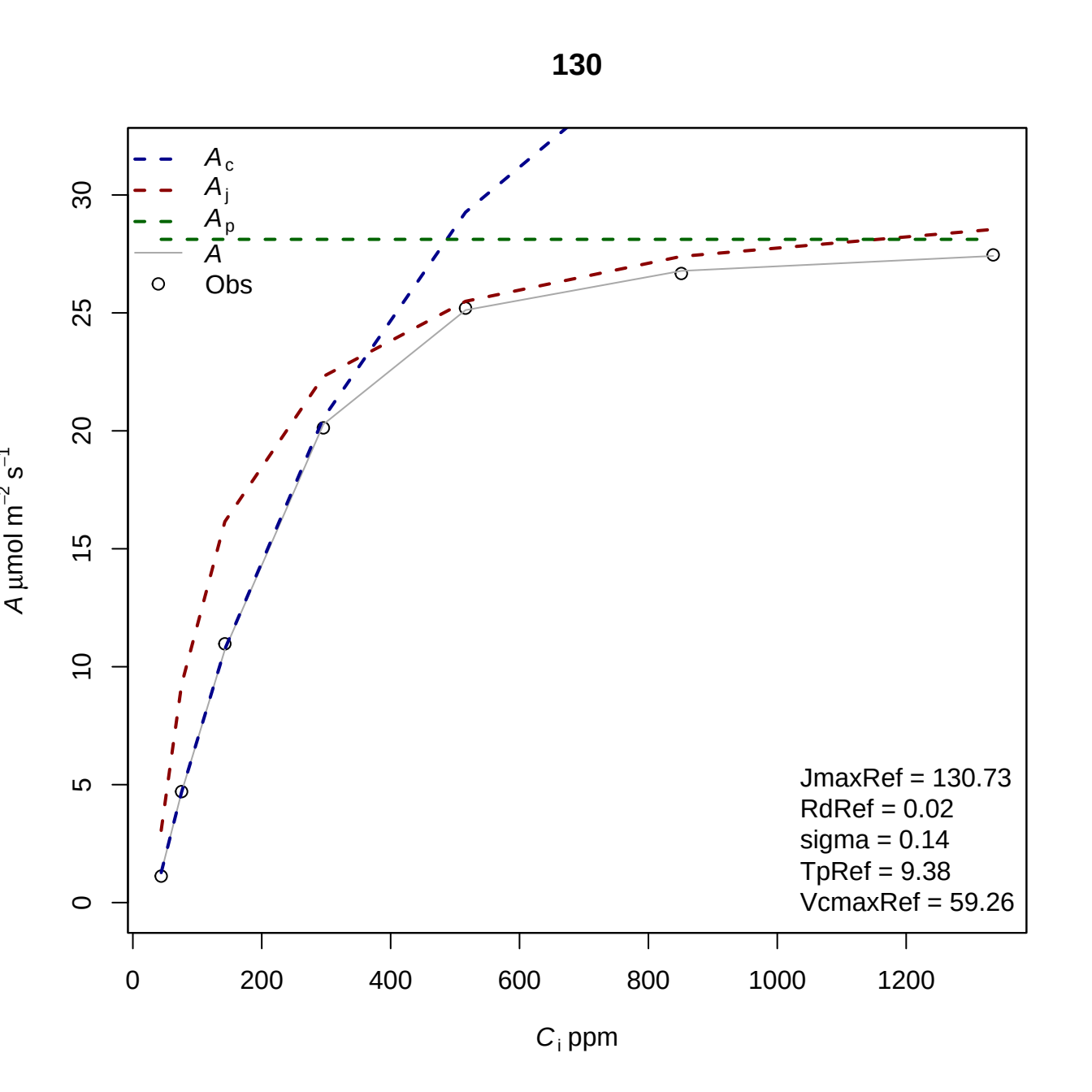

125
A_c
A_j
A_p
A
Obs
JmaxRef = 131.38
RdRef = −0.52
sigma = 0.46
TpRef = 8.55
VcmaxRef = 75.24
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

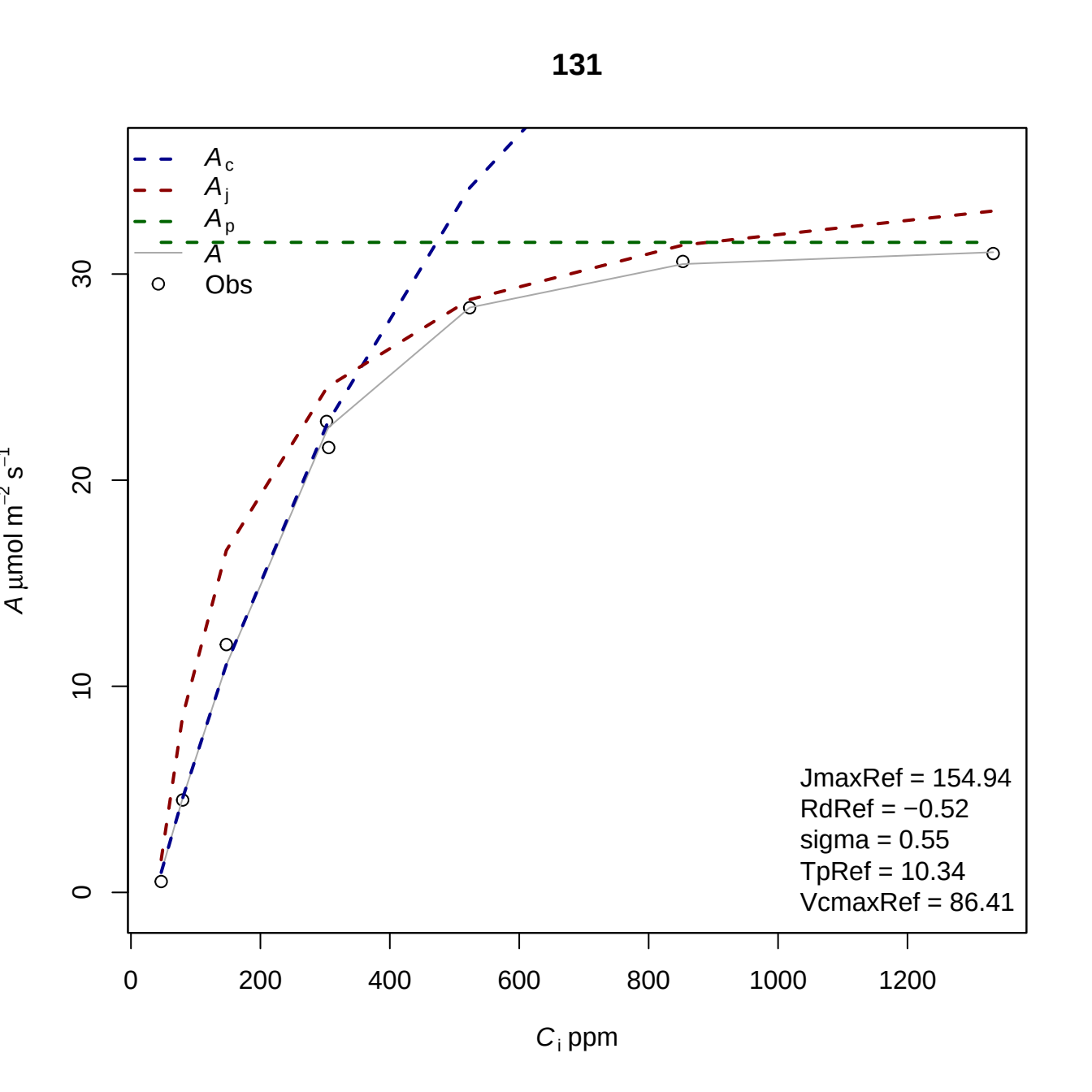

126
A_c
A_j
A_p
A
Obs
JmaxRef = 180.18
RdRef = −0.47
sigma = 0.32
TpRef = 12.11
VcmaxRef = 113.65
0
10
20
30
40
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

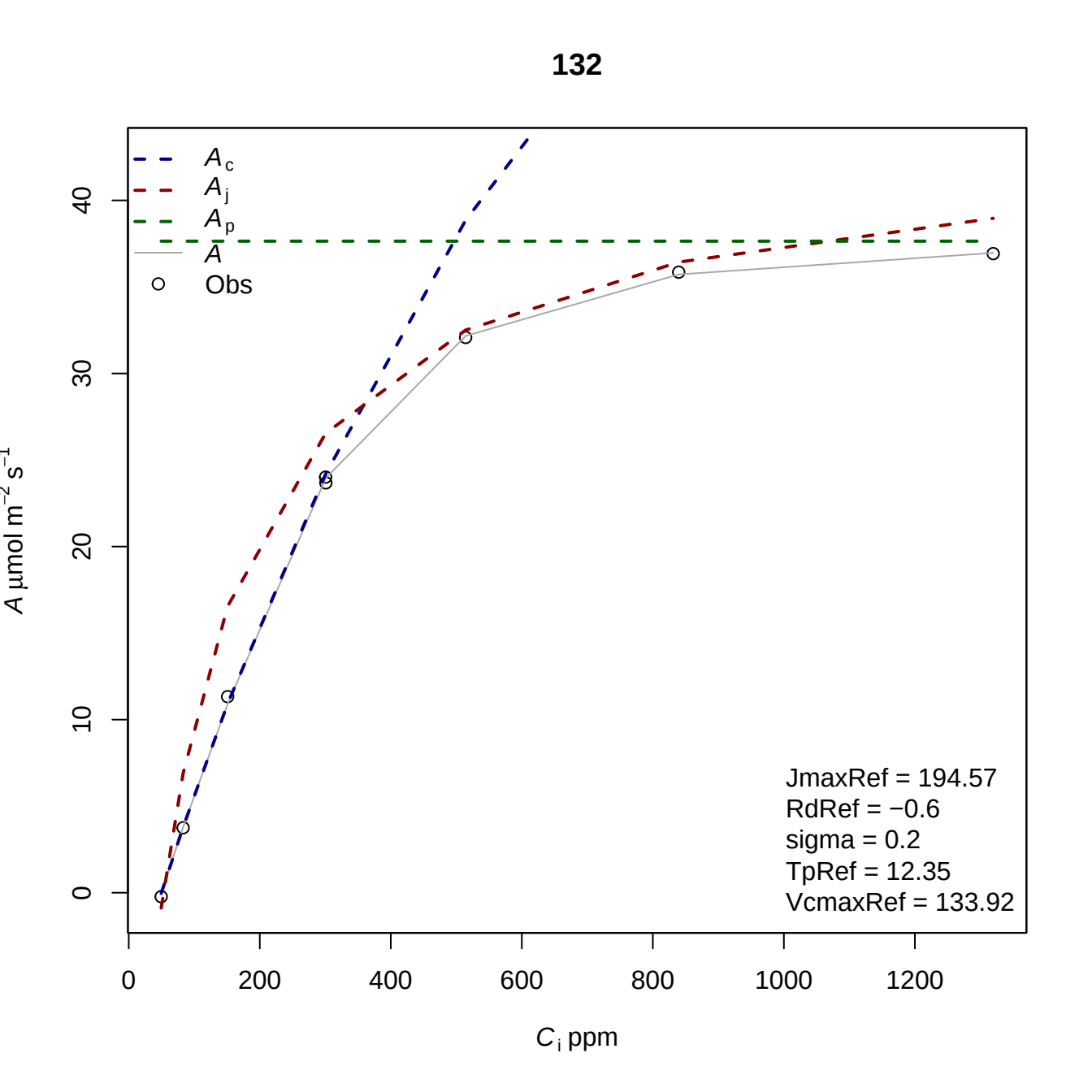

127
A_c
A_j
A_p
A
Obs
JmaxRef = 174.91
RdRef = −0.73
sigma = 0.23
TpRef = 11.01
VcmaxRef = 153.74
0
10
20
30
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

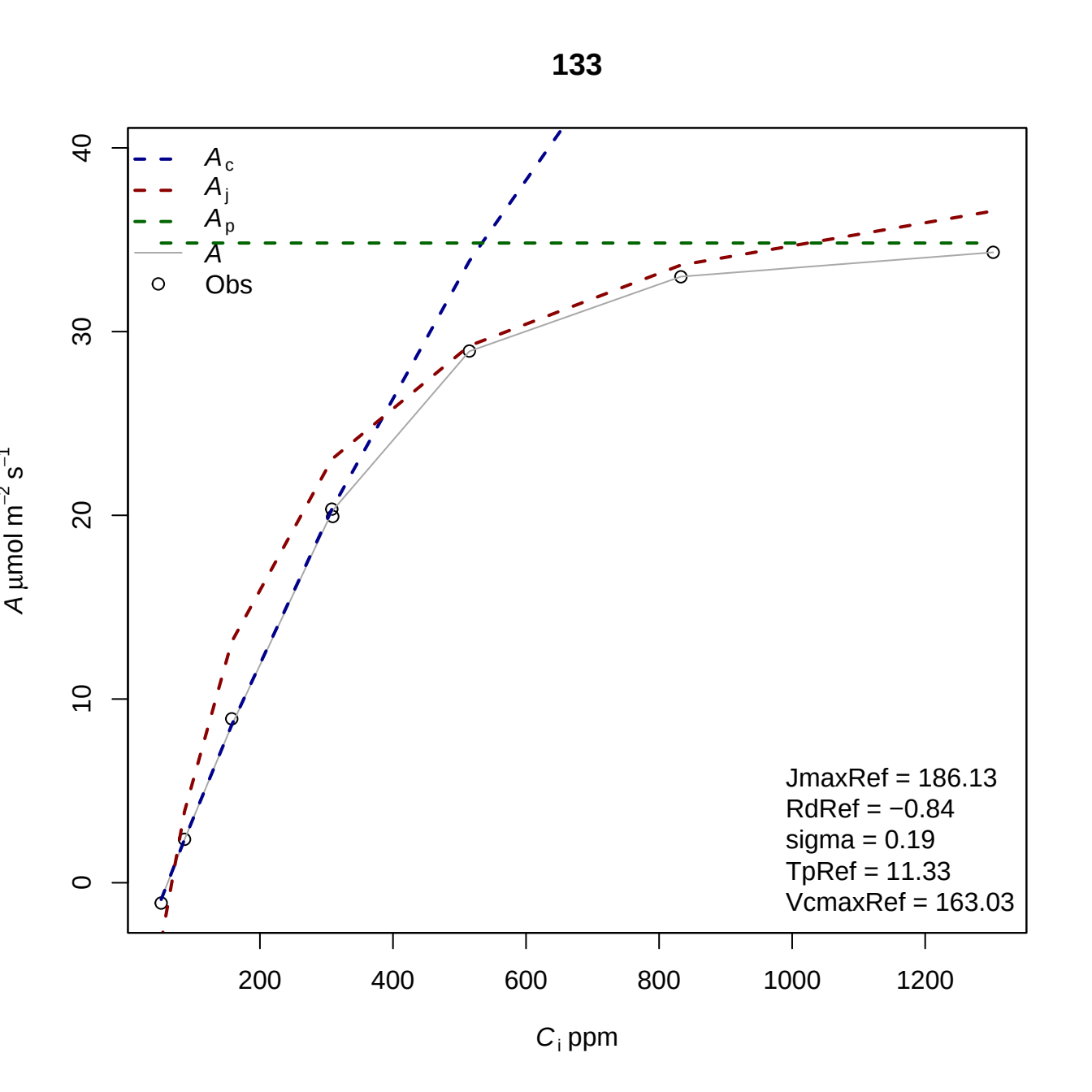

128
A_c
A_j
A_p
A
Obs
JmaxRef = 68.97
RdRef = 0.15
sigma = 0.14
TpRef = 4.07
VcmaxRef = 86.07
0
5
10
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

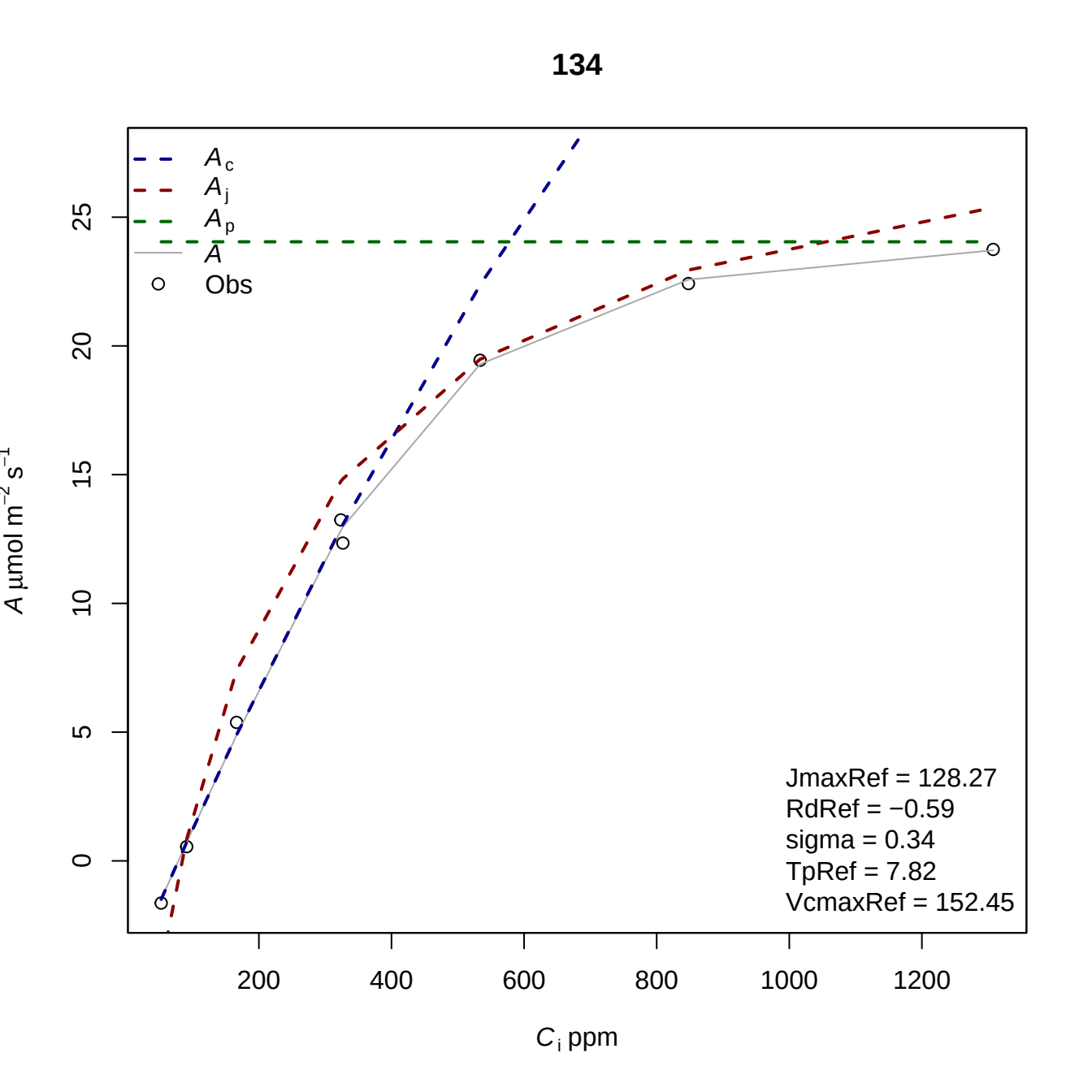

129
A_c
A_j
A_p
A
Obs
JmaxRef = 103.03
RdRef = 0.05
sigma = 0.49
TpRef = 7.15
VcmaxRef = 36.8
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200
1400

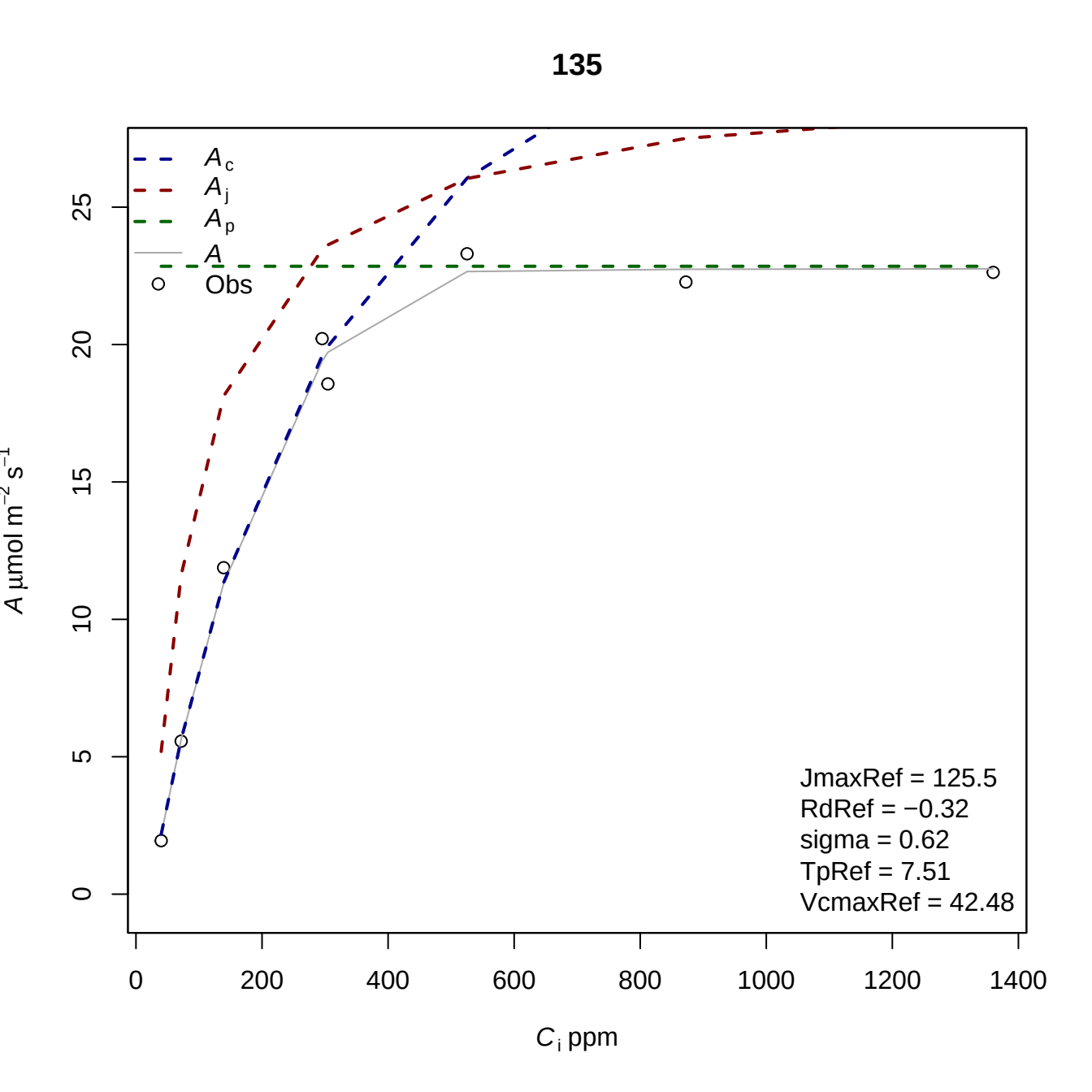

130
A_c
A_j
A_p
A
Obs
JmaxRef = 130.73
RdRef = 0.02
sigma = 0.14
TpRef = 9.38
VcmaxRef = 59.26
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

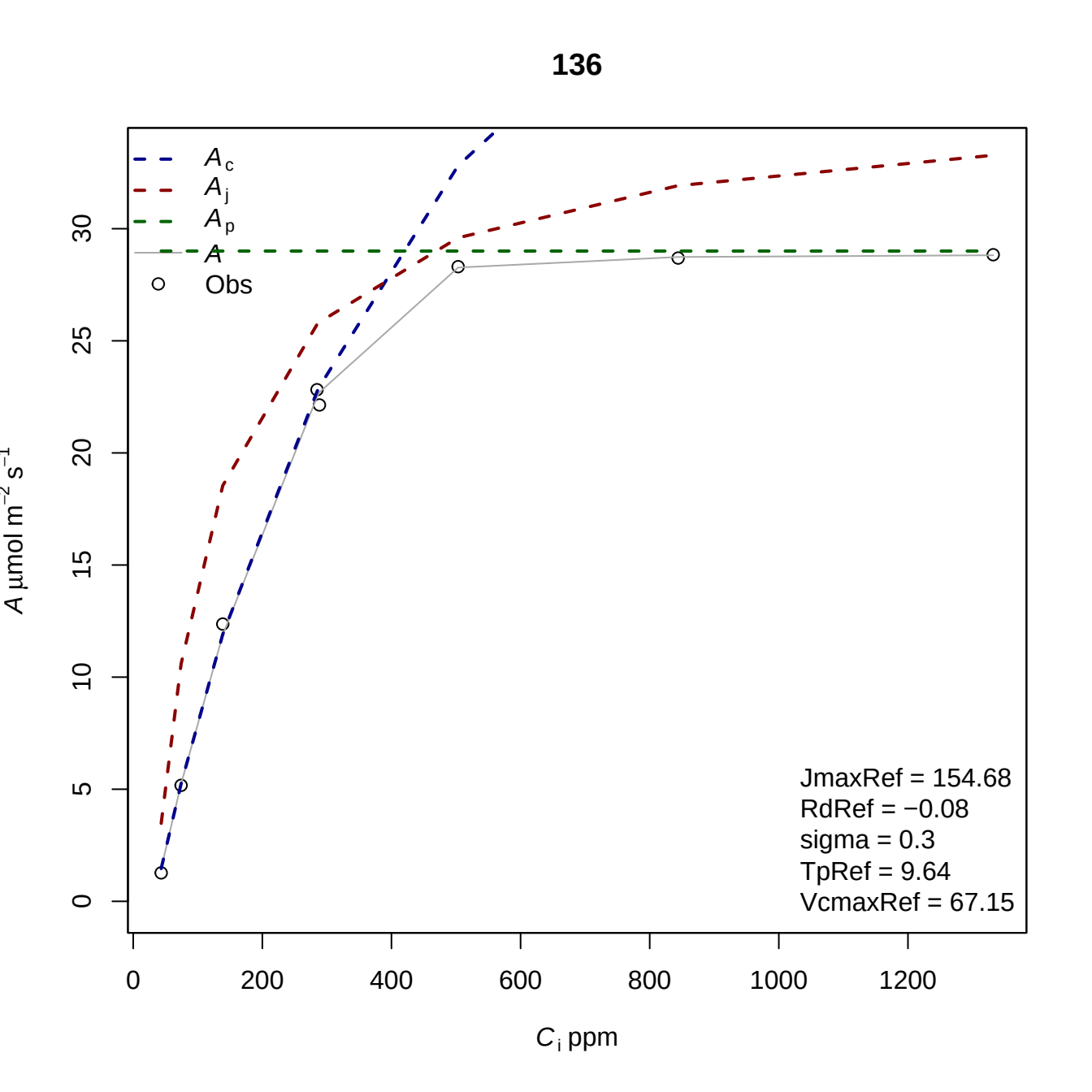

131
A_c
A_j
A_p
A
Obs
JmaxRef = 154.94
RdRef = −0.52
sigma = 0.55
TpRef = 10.34
VcmaxRef = 86.41
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
0
200
400
600
800
1000
1200

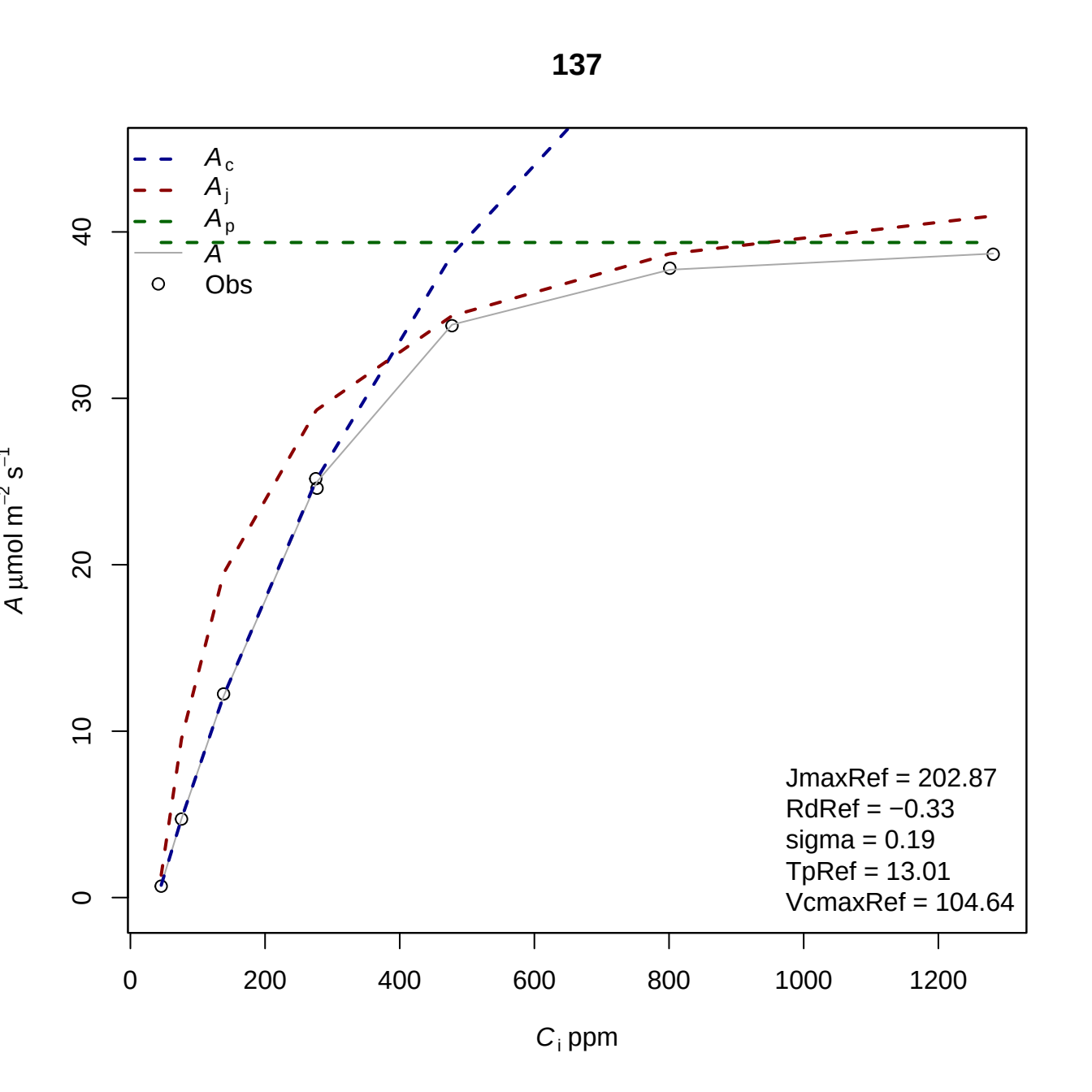

132
A_c
A_j
A_p
A
Obs
JmaxRef = 194.57
RdRef = −0.6
sigma = 0.2
TpRef = 12.35
VcmaxRef = 133.92
0
10
20
30
40
0
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

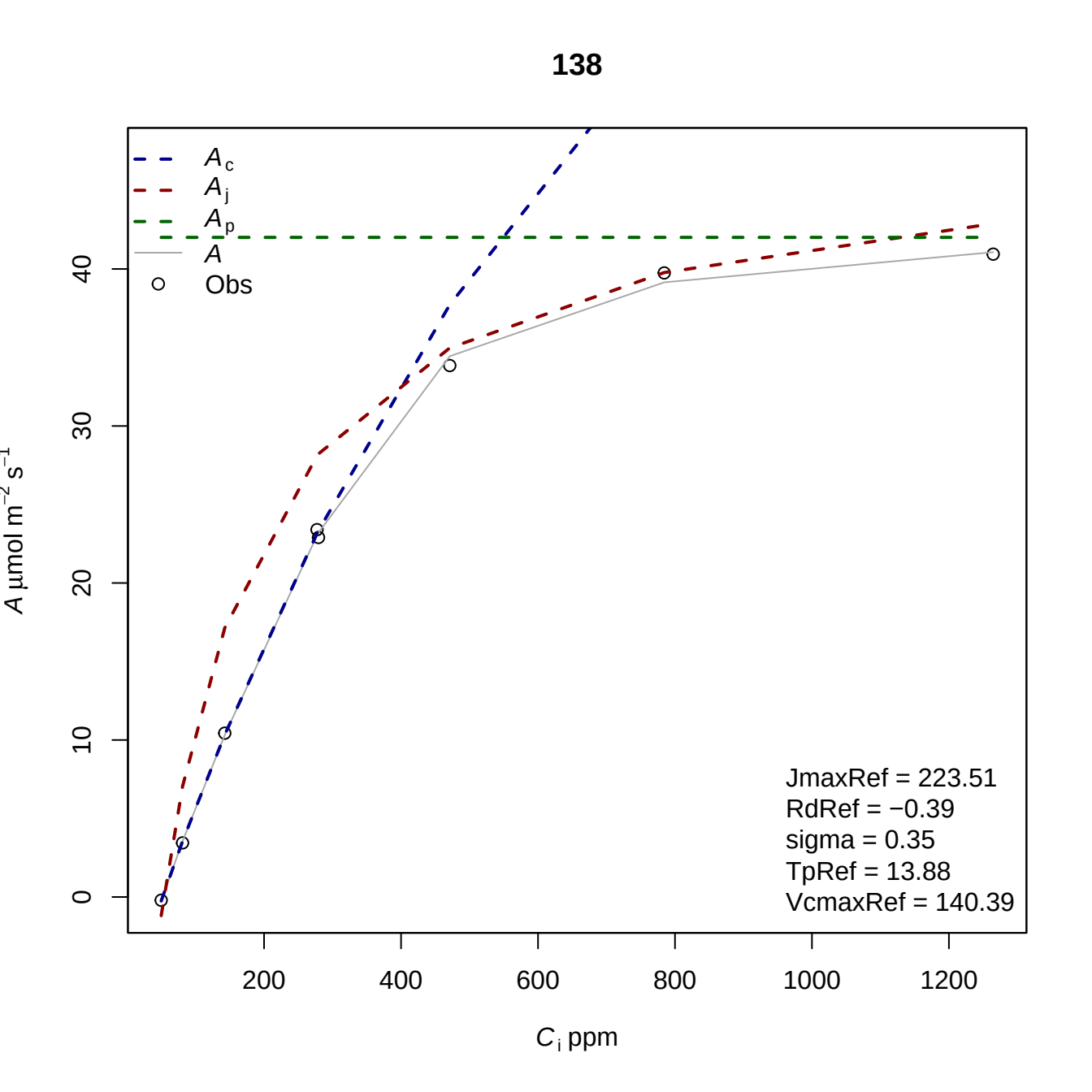

133
A_c
A_j
A_p
A
Obs
JmaxRef = 186.13
RdRef = −0.84
sigma = 0.19
TpRef = 11.33
VcmaxRef = 163.03
0
10
20
30
40
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

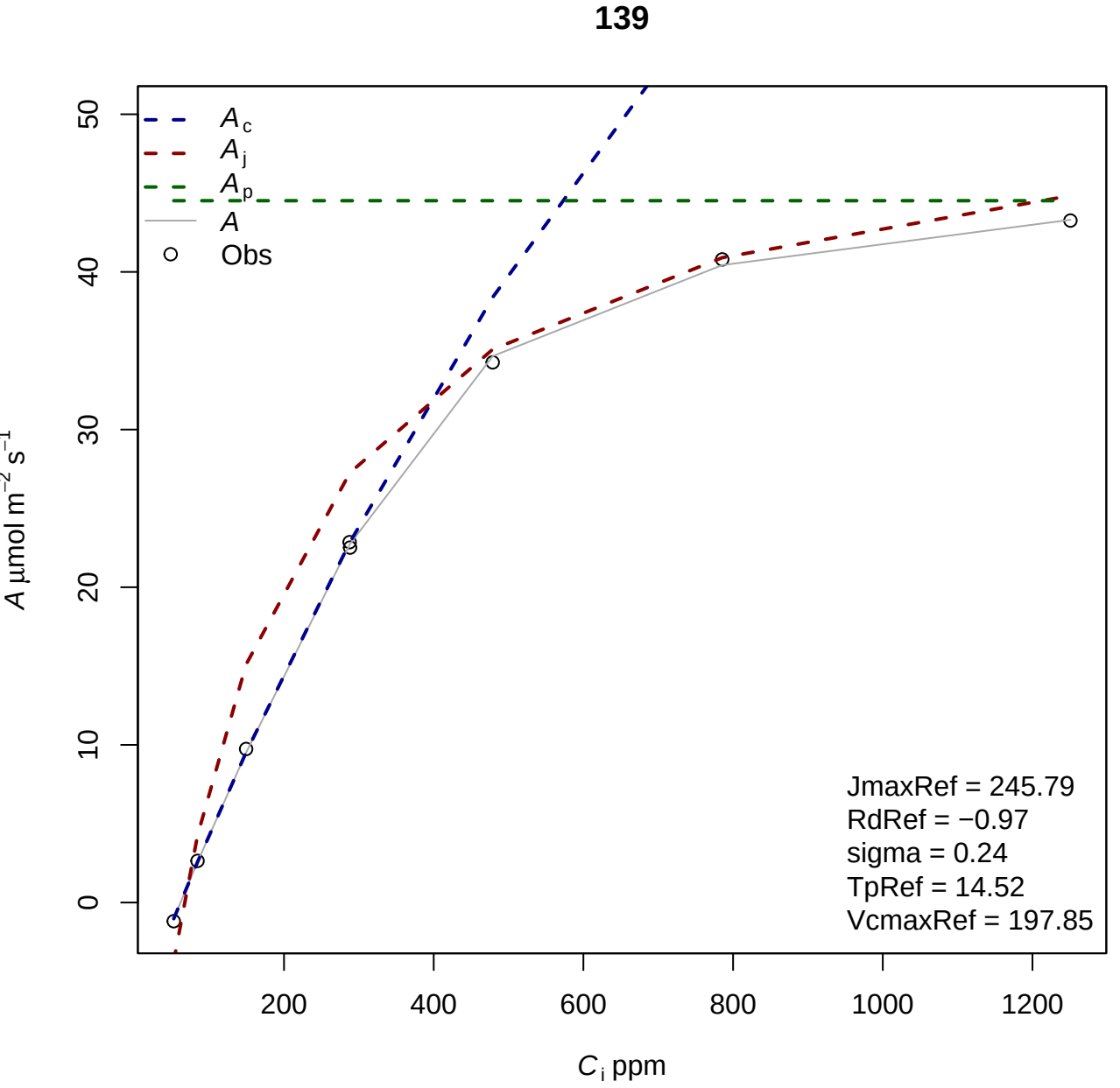

134
A_c
A_j
A_p
A
Obs
JmaxRef = 128.27
RdRef = −0.59
sigma = 0.34
TpRef = 7.82
VcmaxRef = 152.45
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
200
400
600
800
1000
1200

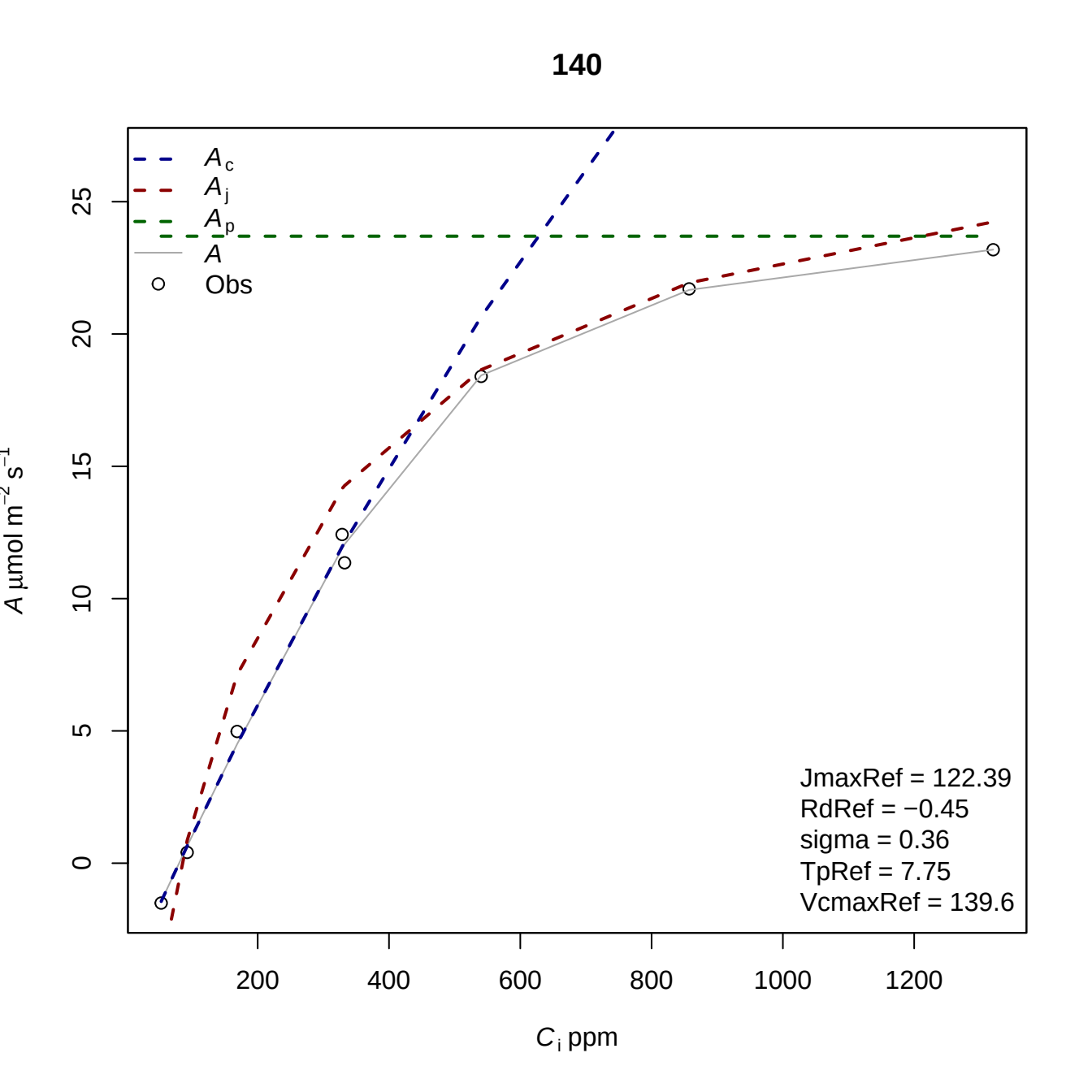

135
A_c
A_j
A_p
A
Obs
JmaxRef = 125.5
RdRef = −0.32
sigma = 0.62
TpRef = 7.51
VcmaxRef = 42.48
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200
1400

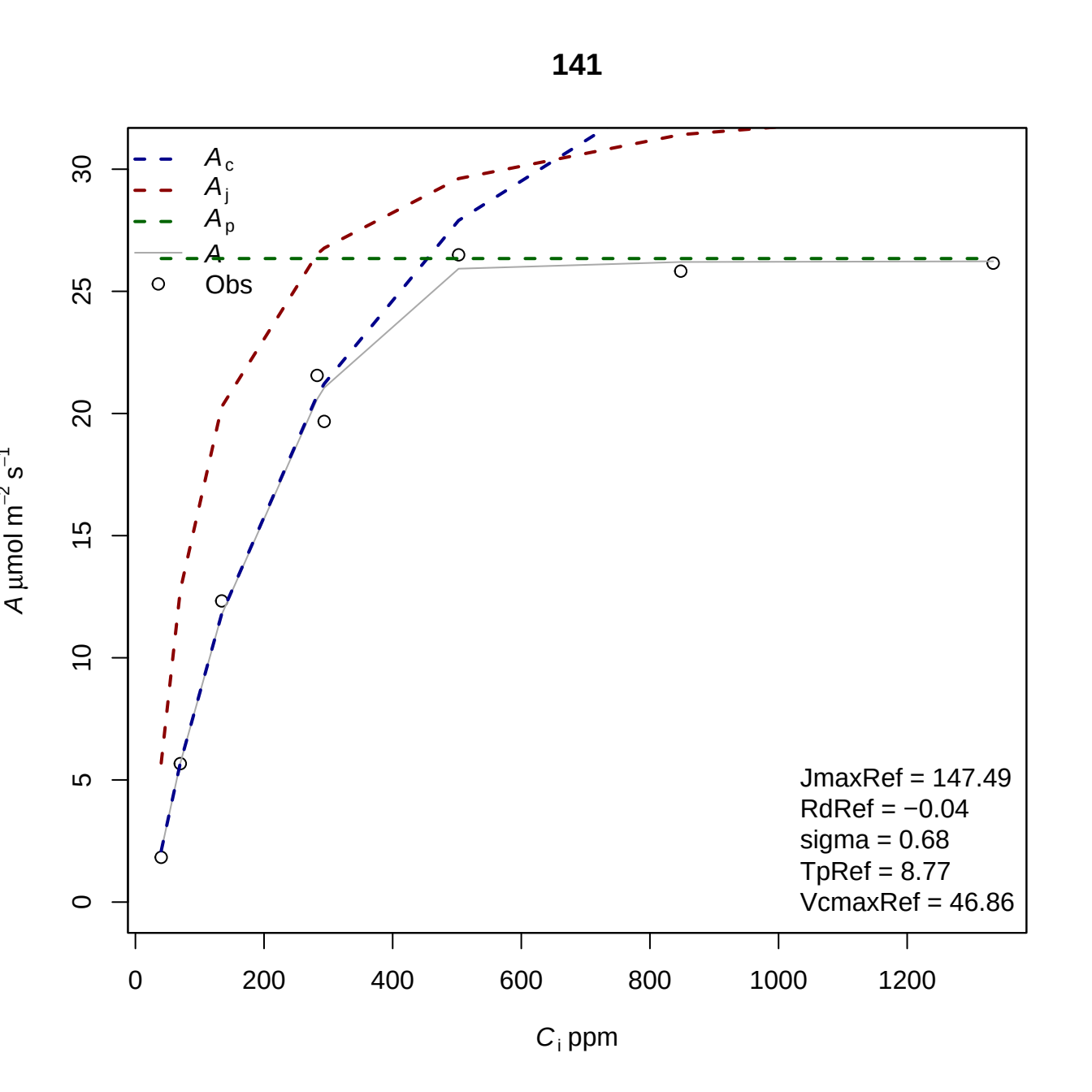

136
A_c
A_j
A_p
A
Obs
JmaxRef = 154.68
RdRef = −0.08
sigma = 0.3
TpRef = 9.64
VcmaxRef = 67.15
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

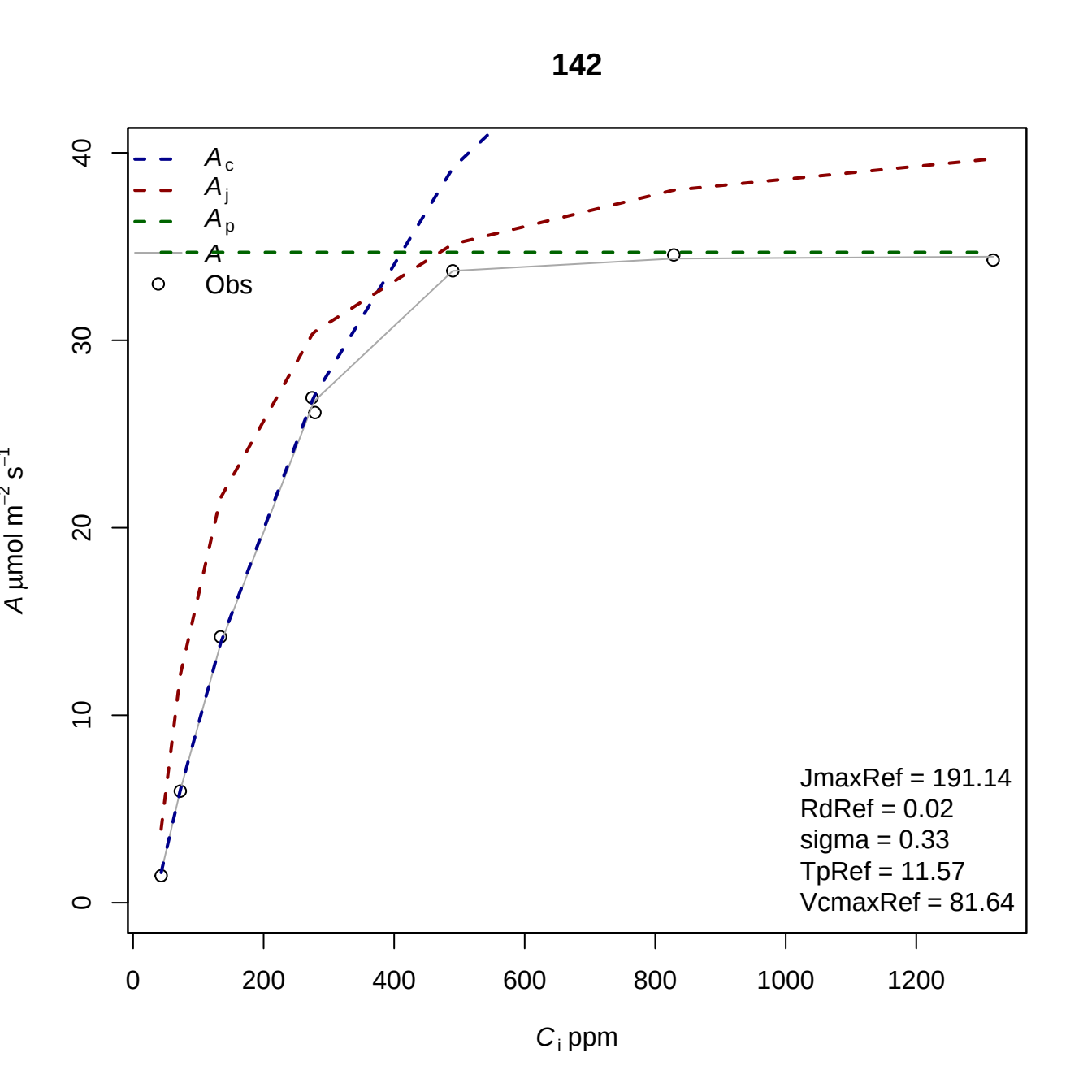

137
A_c
A_j
A_p
A
Obs
JmaxRef = 202.87
RdRef = −0.33
sigma = 0.19
TpRef = 13.01
VcmaxRef = 104.64
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

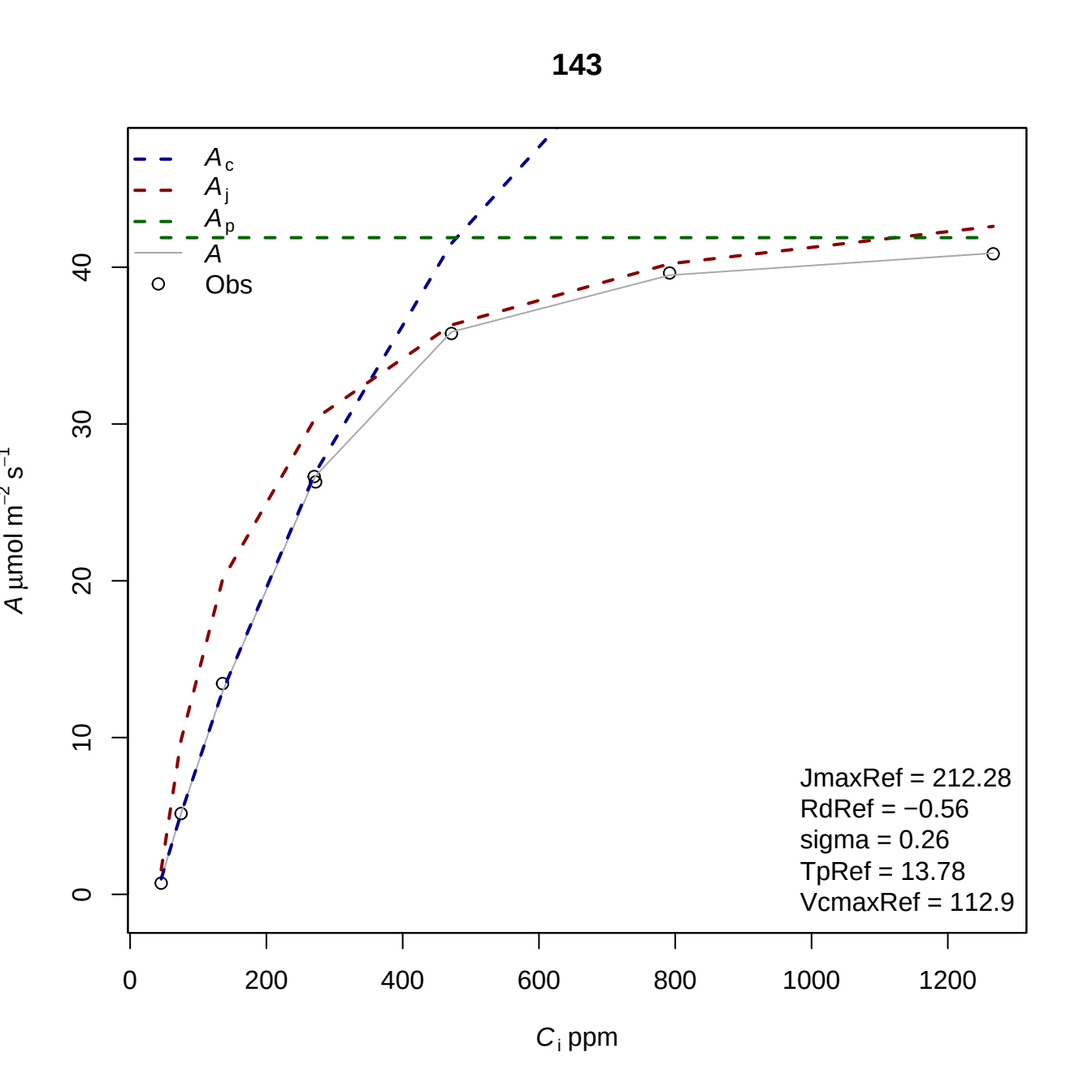

138
A_c
A_j
A_p
A
Obs
JmaxRef = 223.51
RdRef = −0.39
sigma = 0.35
TpRef = 13.88
VcmaxRef = 140.39
0
10
20
30
40
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

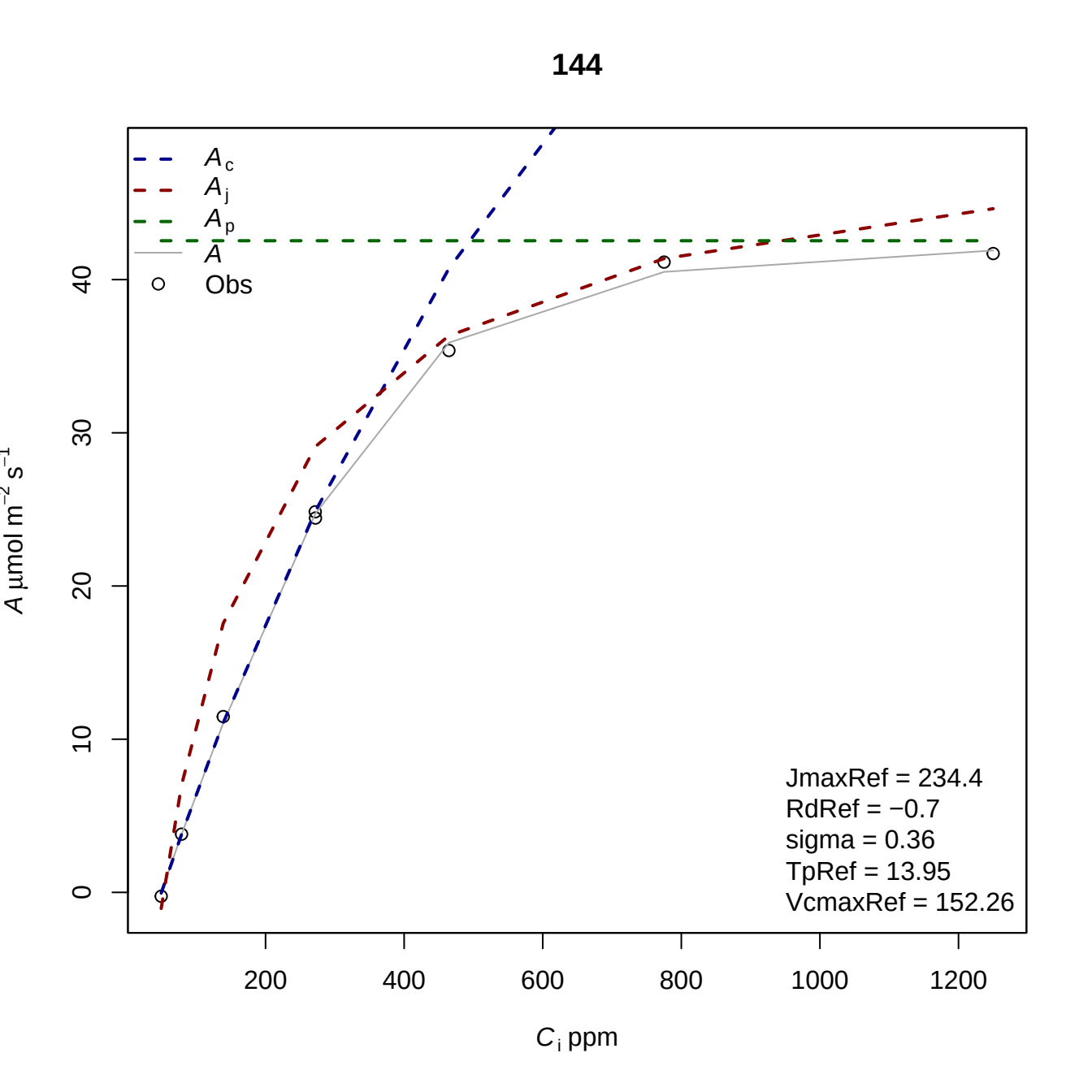

139
A_c
A_j
A_p
A
Obs
JmaxRef = 245.79
RdRef = −0.97
sigma = 0.24
TpRef = 14.52
VcmaxRef = 197.85
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
50
200
400
600
800
1000
1200

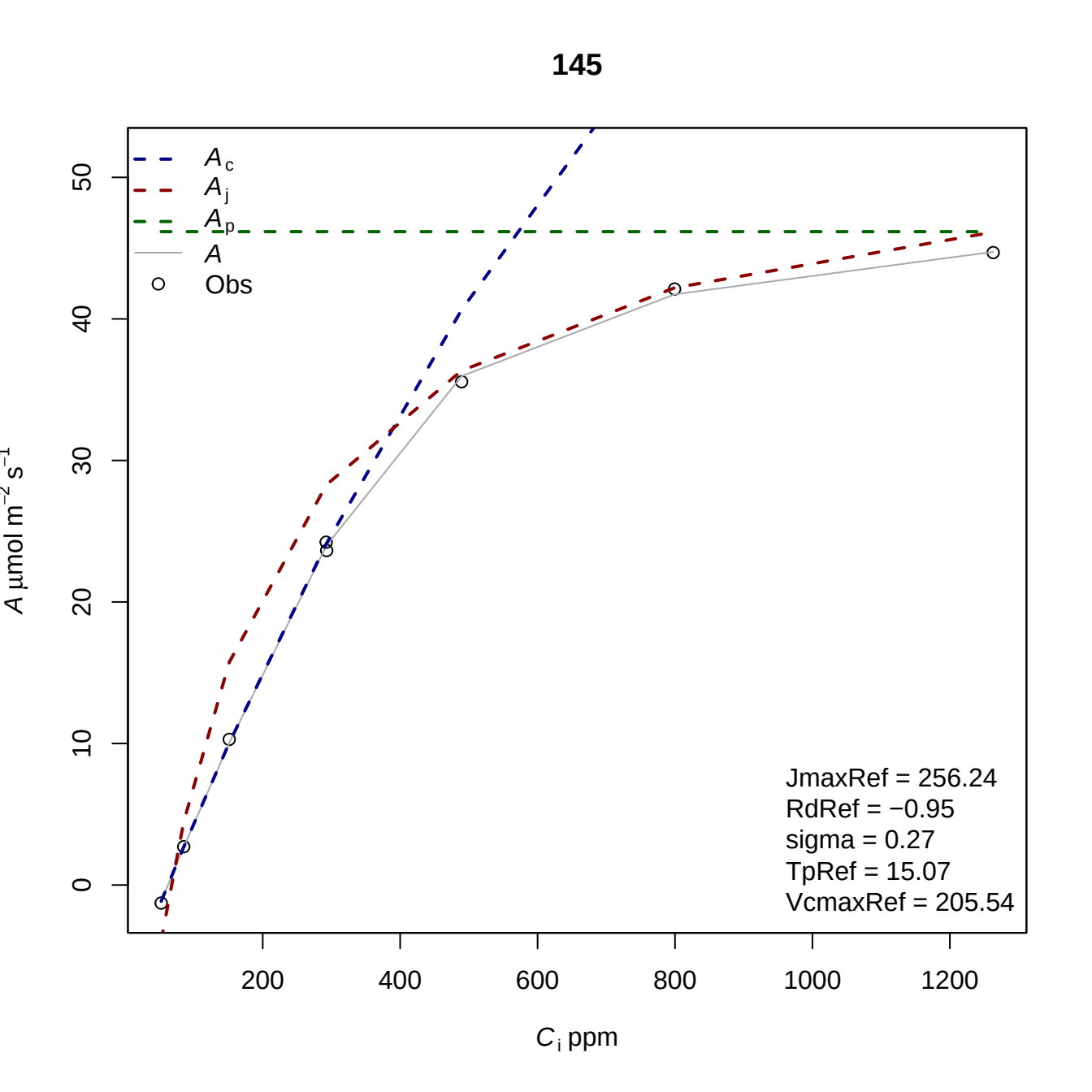

140
A_c
A_j
A_p
A
Obs
JmaxRef = 122.39
RdRef = −0.45
sigma = 0.36
TpRef = 7.75
VcmaxRef = 139.6
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
200
400
600
800
1000
1200

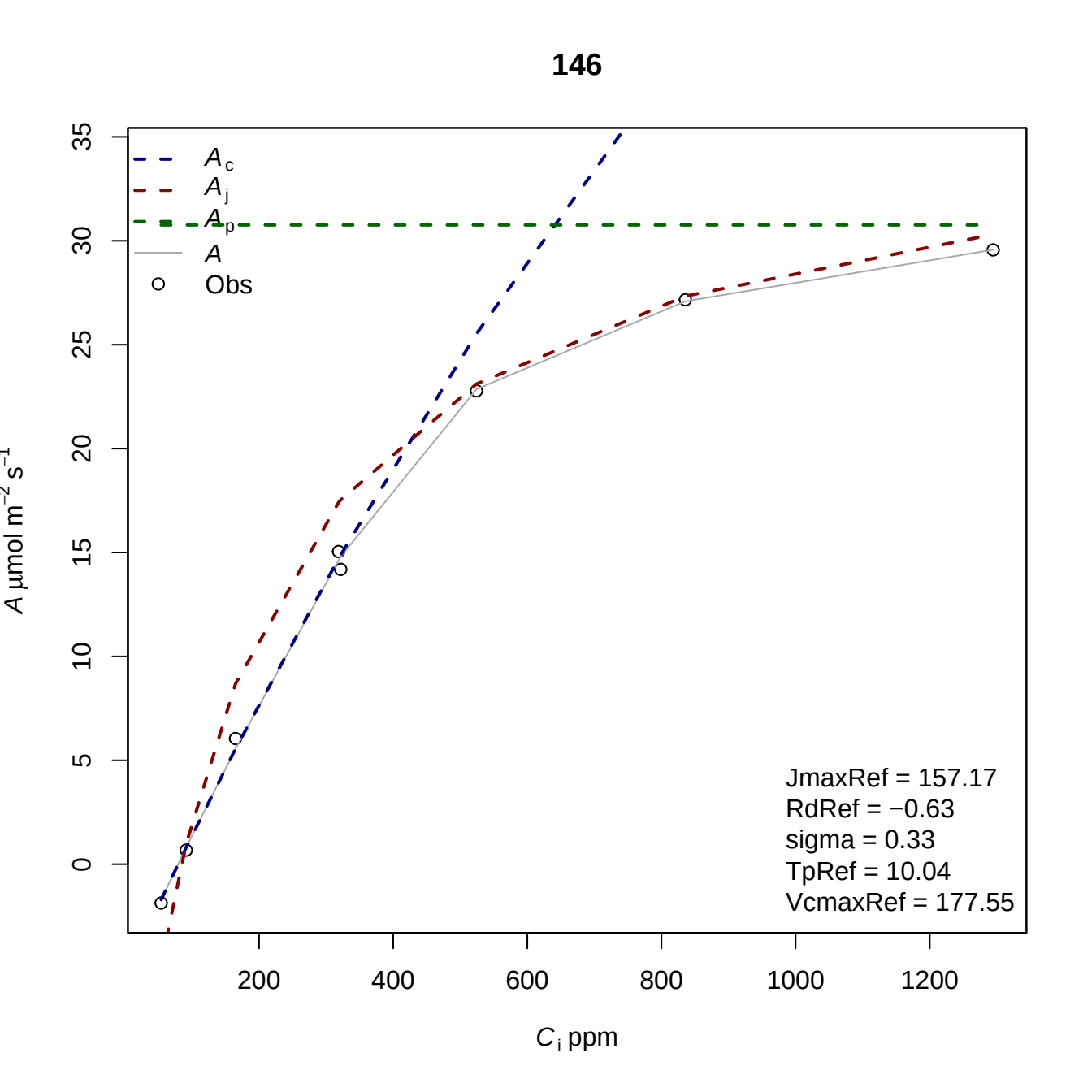

141
A_c
A_j
A_p
A
Obs
JmaxRef = 147.49
RdRef = −0.04
sigma = 0.68
TpRef = 8.77
VcmaxRef = 46.86
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

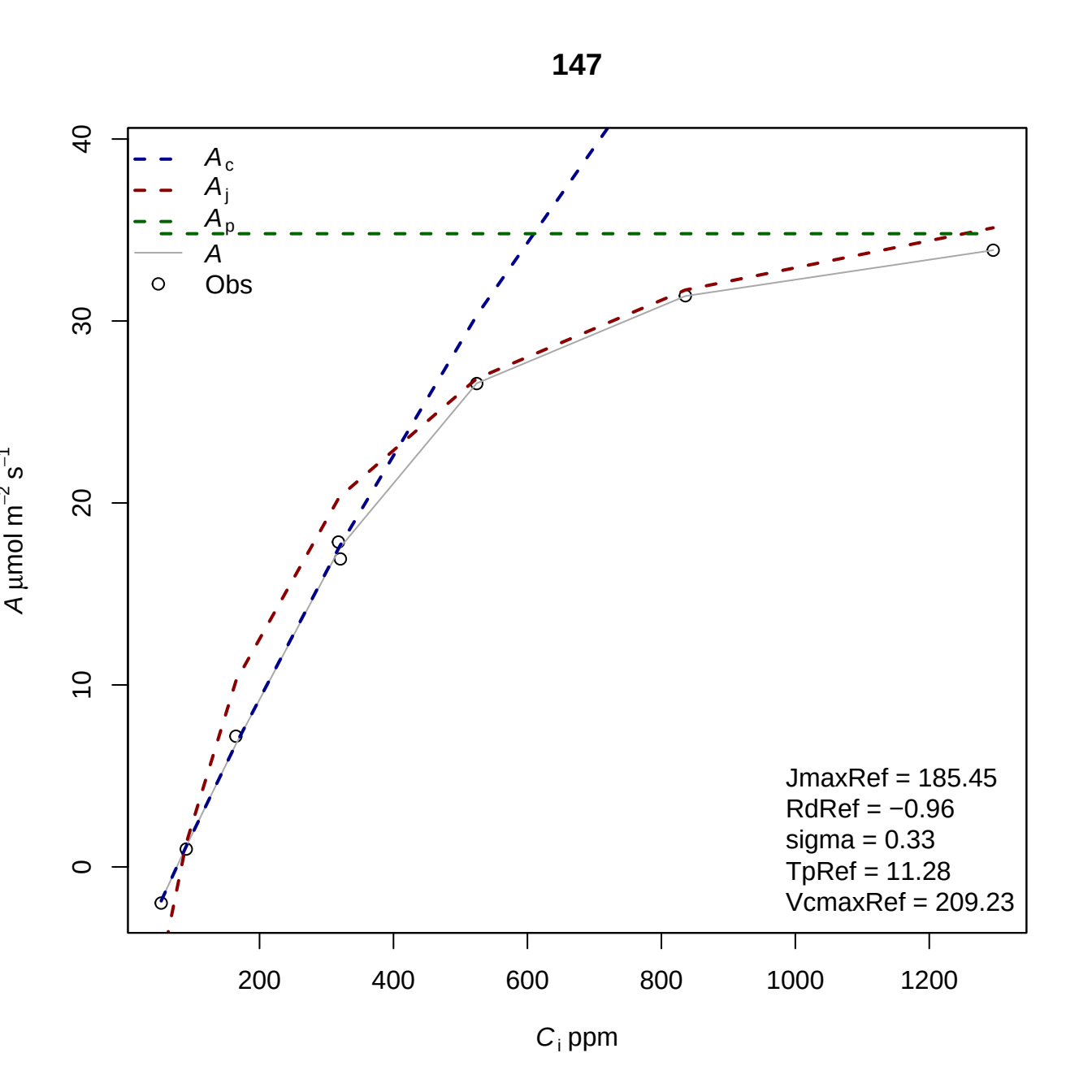

142
A_c
A_j
A_p
A
Obs
JmaxRef = 191.14
RdRef = 0.02
sigma = 0.33
TpRef = 11.57
VcmaxRef = 81.64
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
1200
C_i ppm

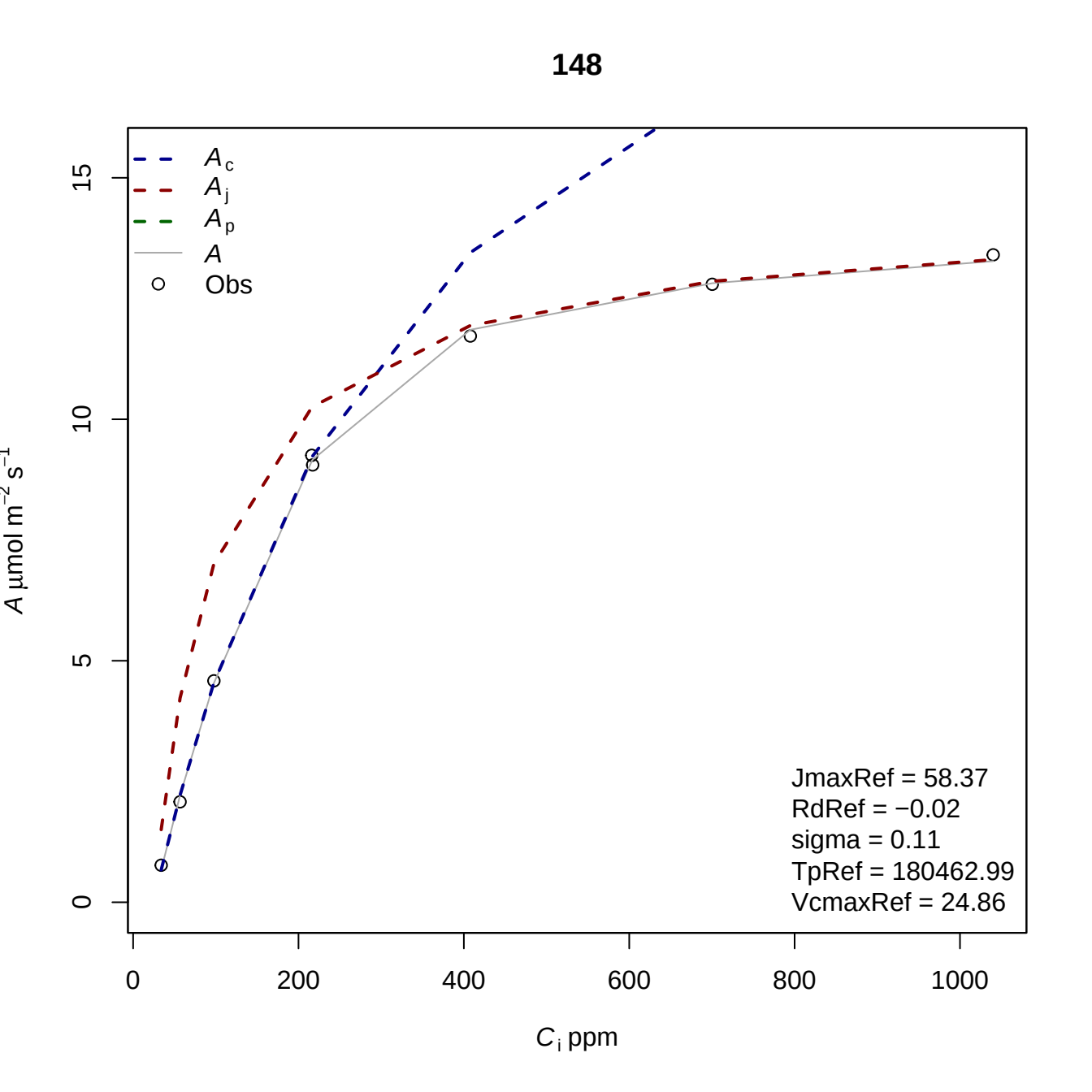

143
A_c
A_j
A_p
A
Obs
JmaxRef = 212.28
RdRef = −0.56
sigma = 0.26
TpRef = 13.78
VcmaxRef = 112.9
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

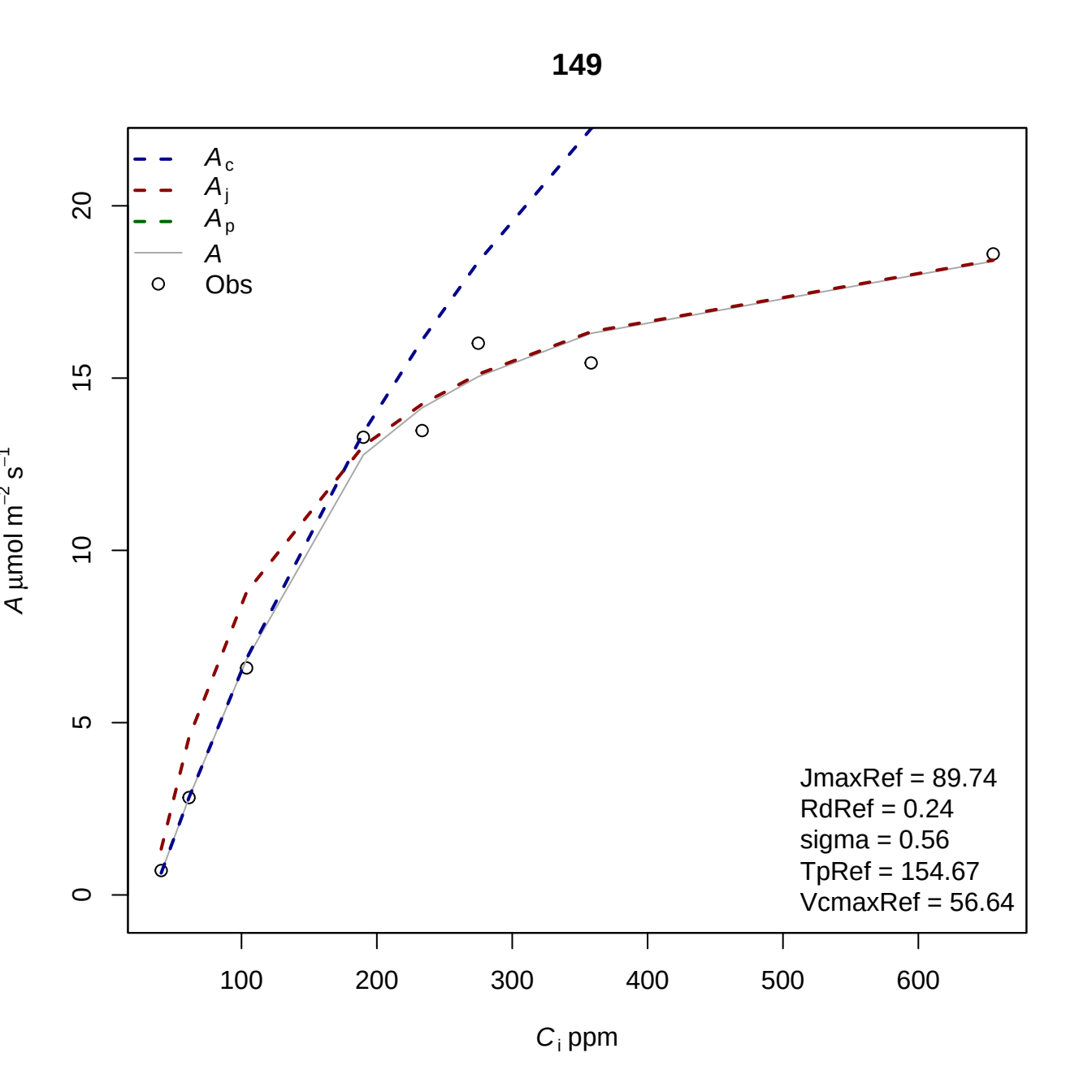

144
A_c
A_j
A_p
A
Obs
JmaxRef = 234.4
RdRef = −0.7
sigma = 0.36
TpRef = 13.95
VcmaxRef = 152.26
0
10
20
30
40
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

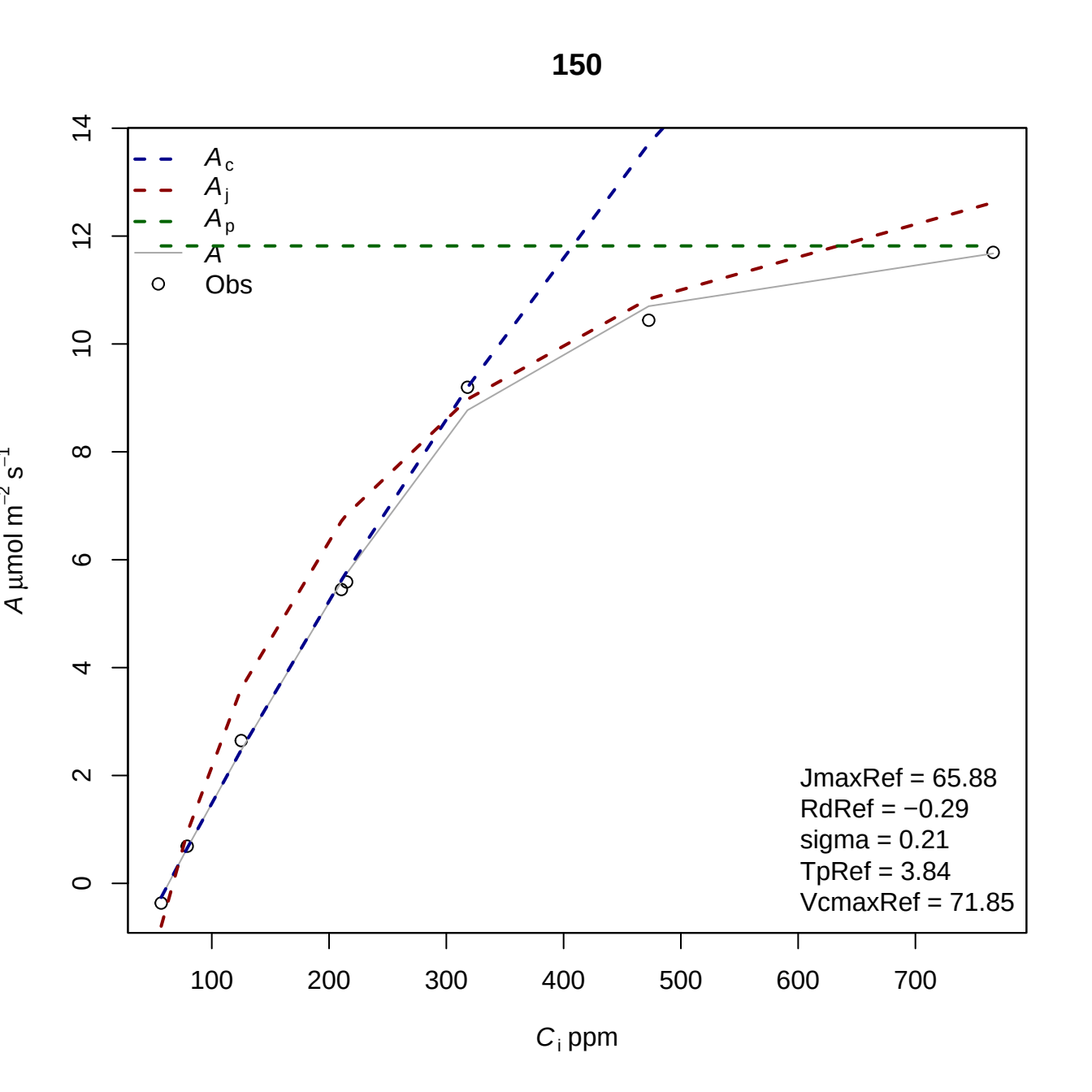

145
A_c
A_j
A_p
A
Obs
JmaxRef = 256.24
RdRef = −0.95
sigma = 0.27
TpRef = 15.07
VcmaxRef = 205.54
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
50
200
400
600
800
1000
1200

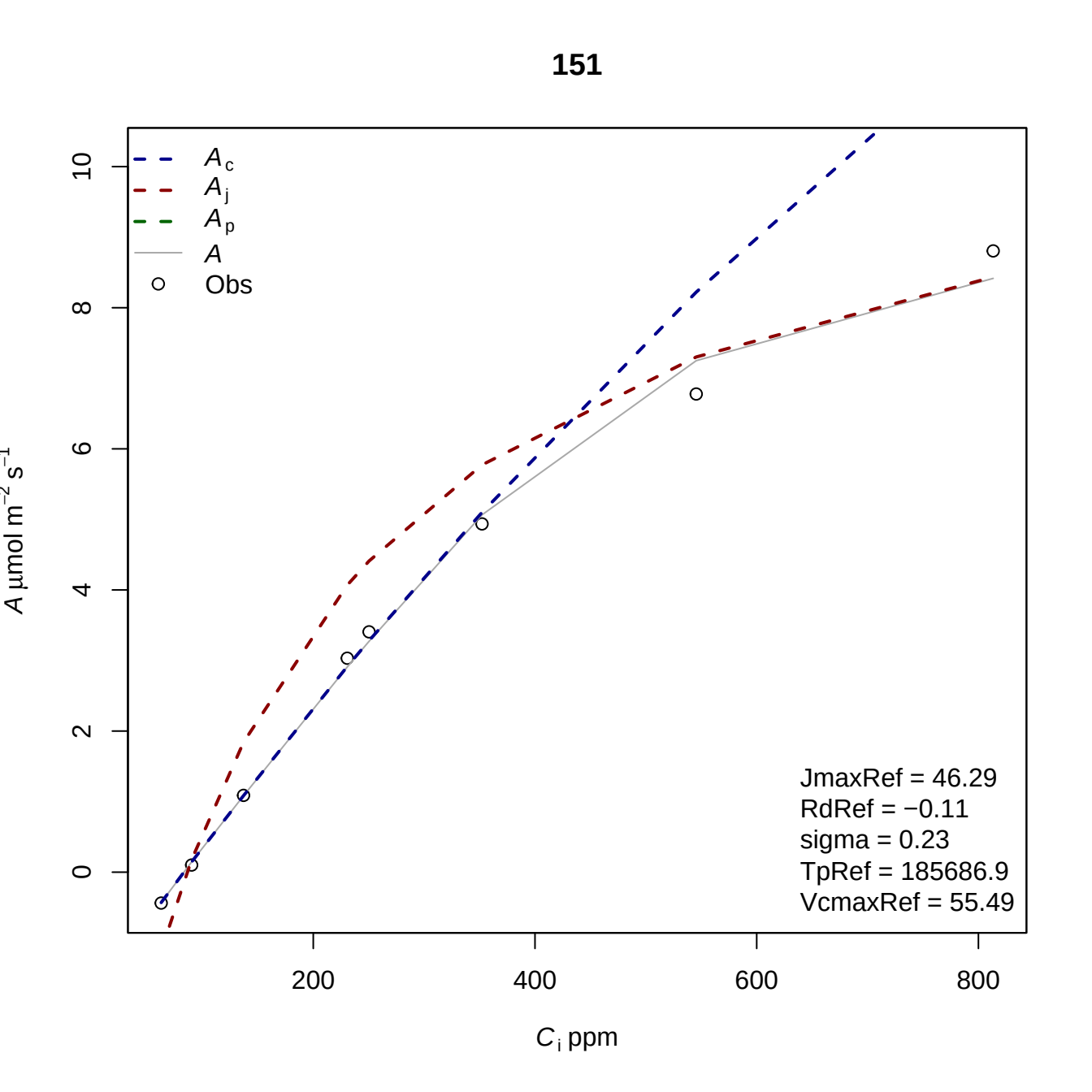

146
A_c
A_j
A_p
A
Obs
JmaxRef = 157.17
RdRef = −0.63
sigma = 0.33
TpRef = 10.04
VcmaxRef = 177.55
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
35
200
400
600
800
1000
1200

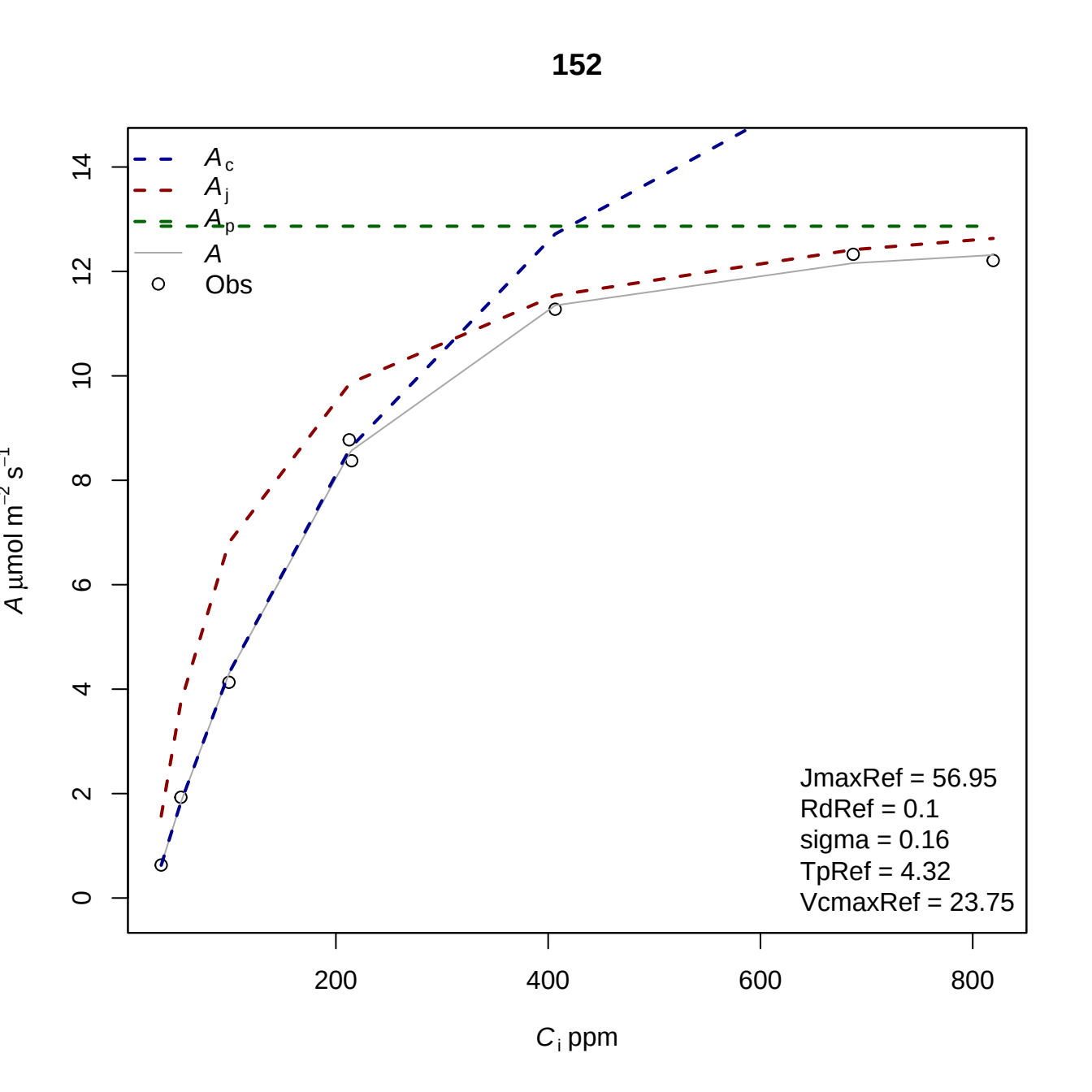

147
A_c
A_j
A_p
A
Obs
JmaxRef = 185.45
RdRef = −0.96
sigma = 0.33
TpRef = 11.28
VcmaxRef = 209.23
0
10
20
30
40
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

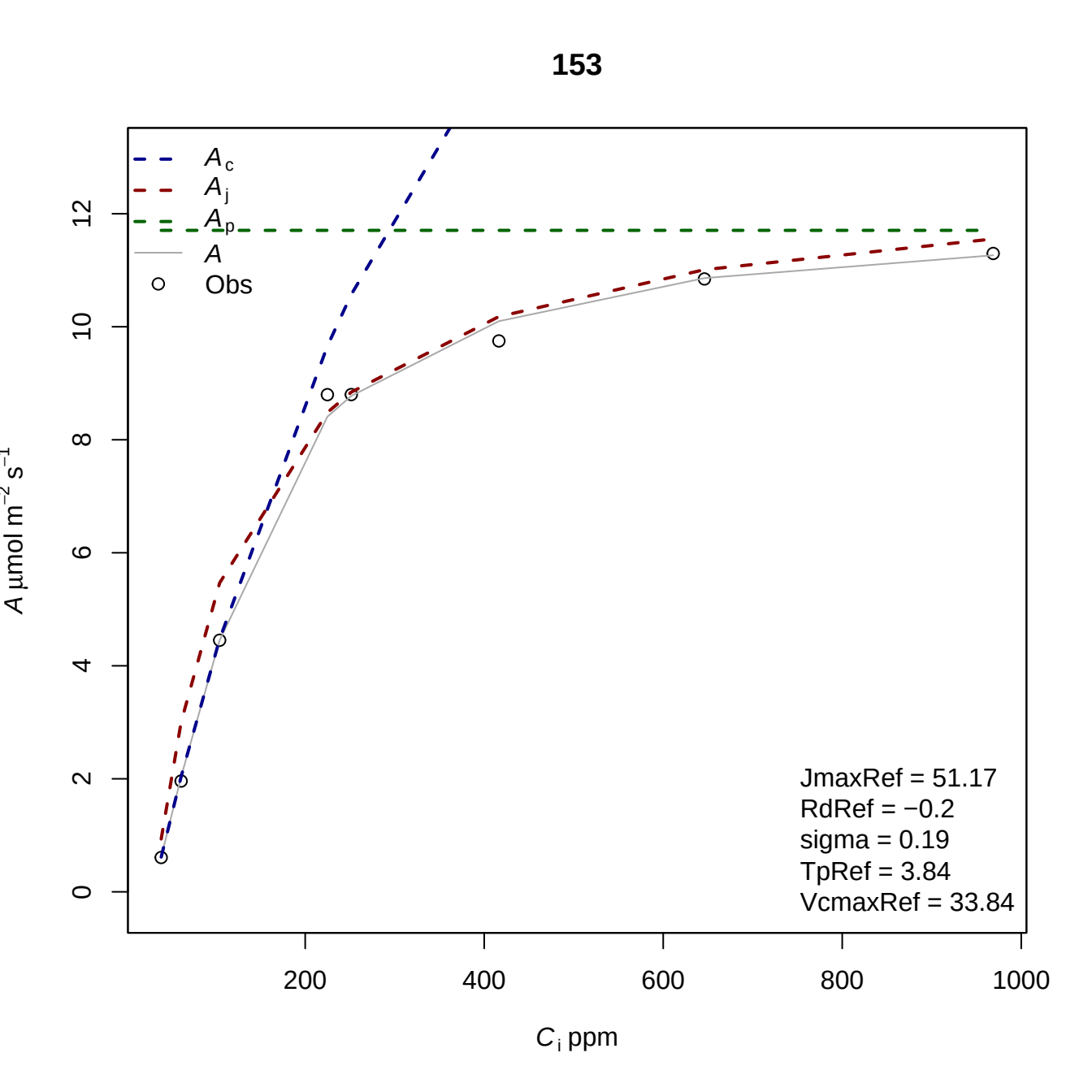

148
A_c
A_j
A_p
A
Obs
JmaxRef = 58.37
RdRef = −0.02
sigma = 0.11
TpRef = 180462.99
VcmaxRef = 24.86
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
200
400
600
800
1000

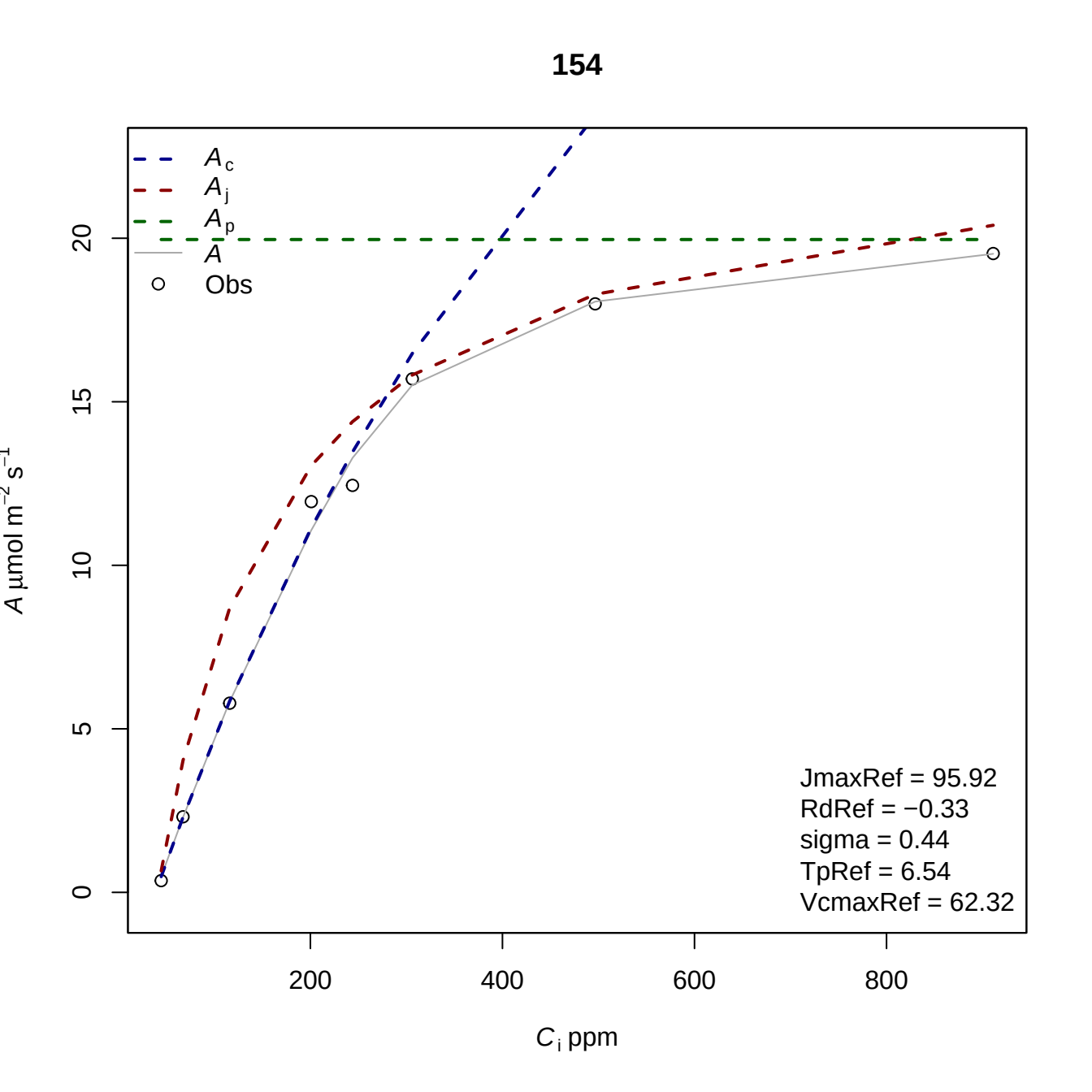

149
A_c
A_j
A_p
A
Obs
JmaxRef = 89.74
RdRef = 0.24
sigma = 0.56
TpRef = 154.67
VcmaxRef = 56.64
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
100
200
300
400
500
600

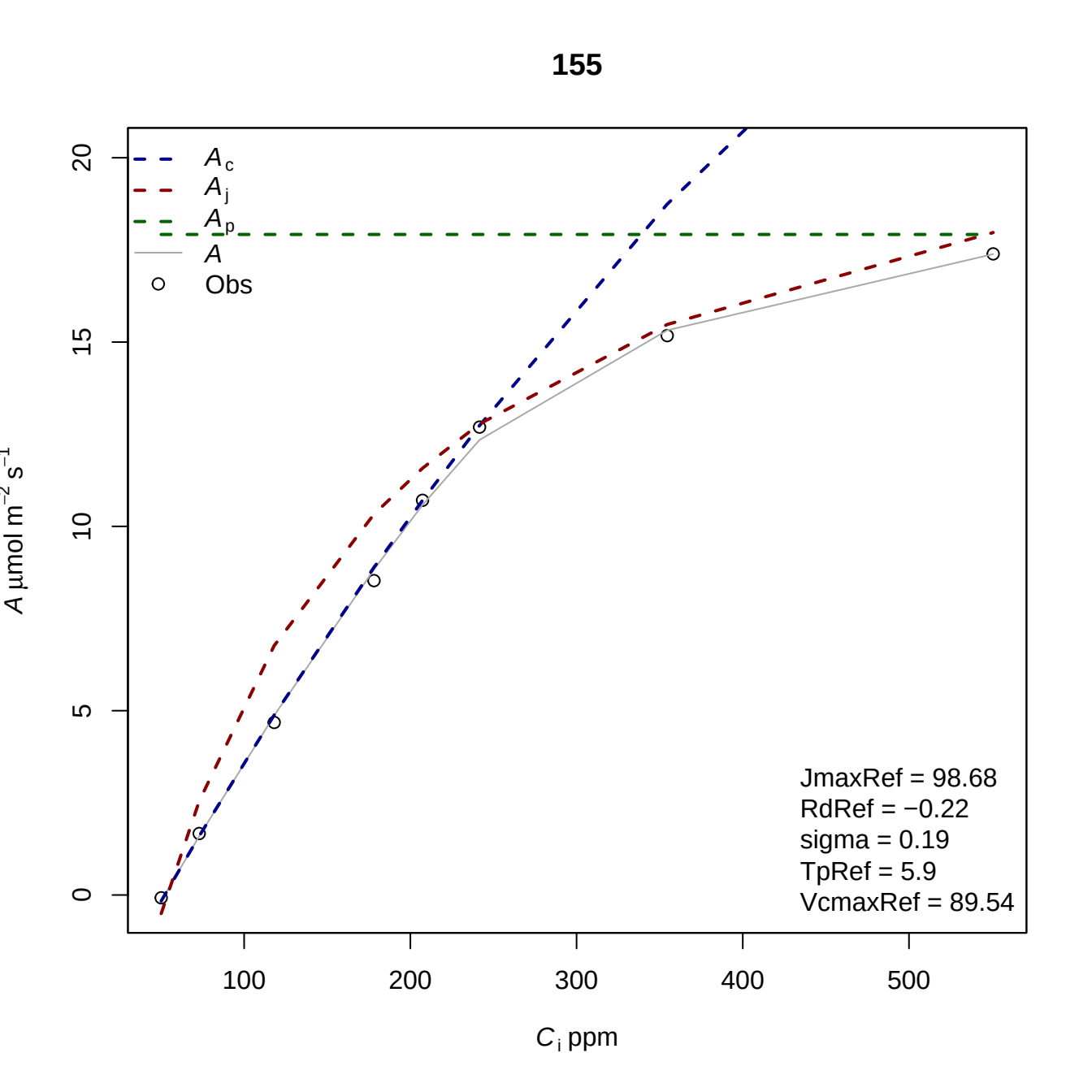

150
A_c
A_j
A_p
A
Obs
JmaxRef = 65.88
RdRef = −0.29
sigma = 0.21
TpRef = 3.84
VcmaxRef = 71.85
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
14
100
200
300
400
500
600
700

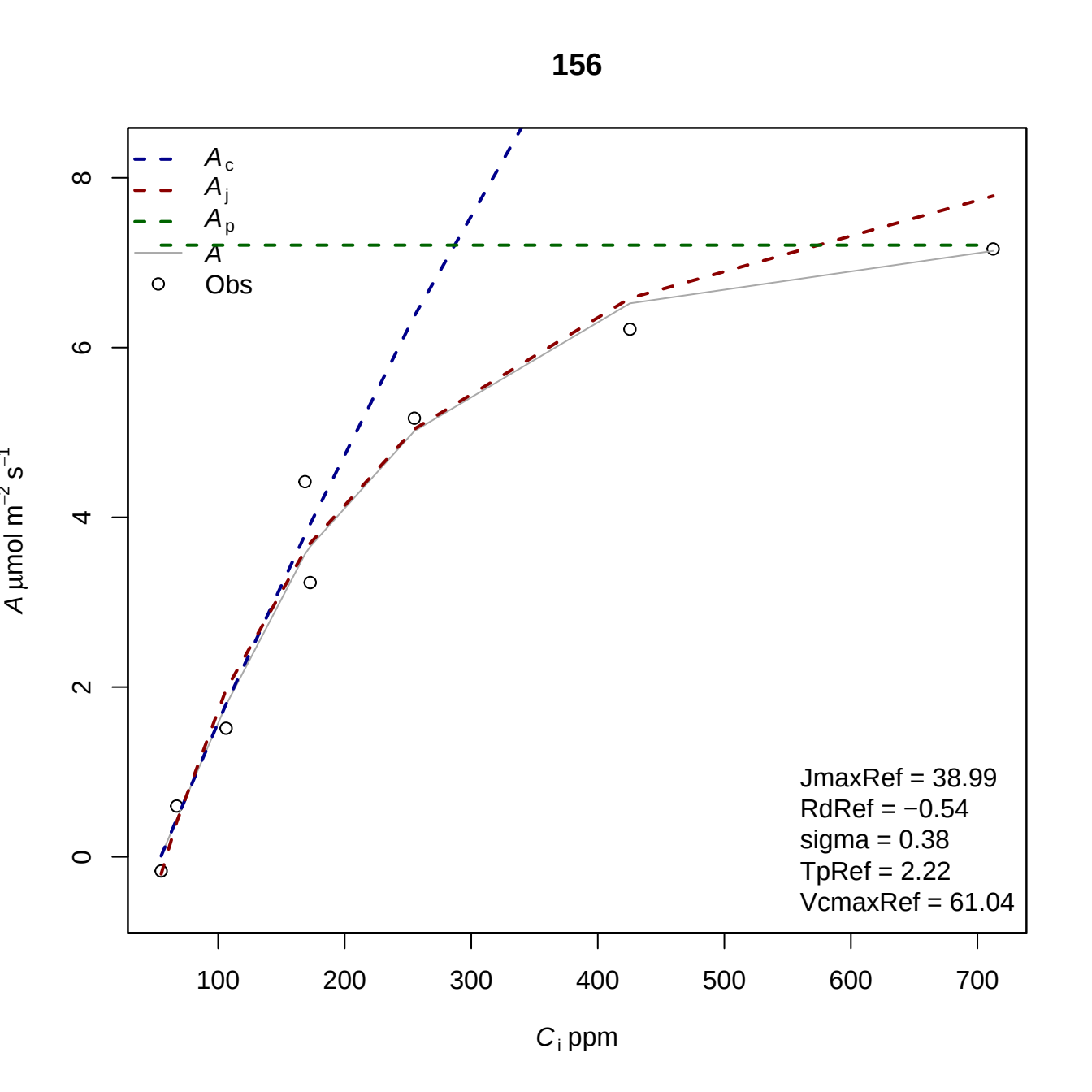

151
A_c
A_j
A_p
A
Obs
JmaxRef = 46.29
RdRef = −0.11
sigma = 0.23
TpRef = 185686.9
VcmaxRef = 55.49
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
200
400
600
800

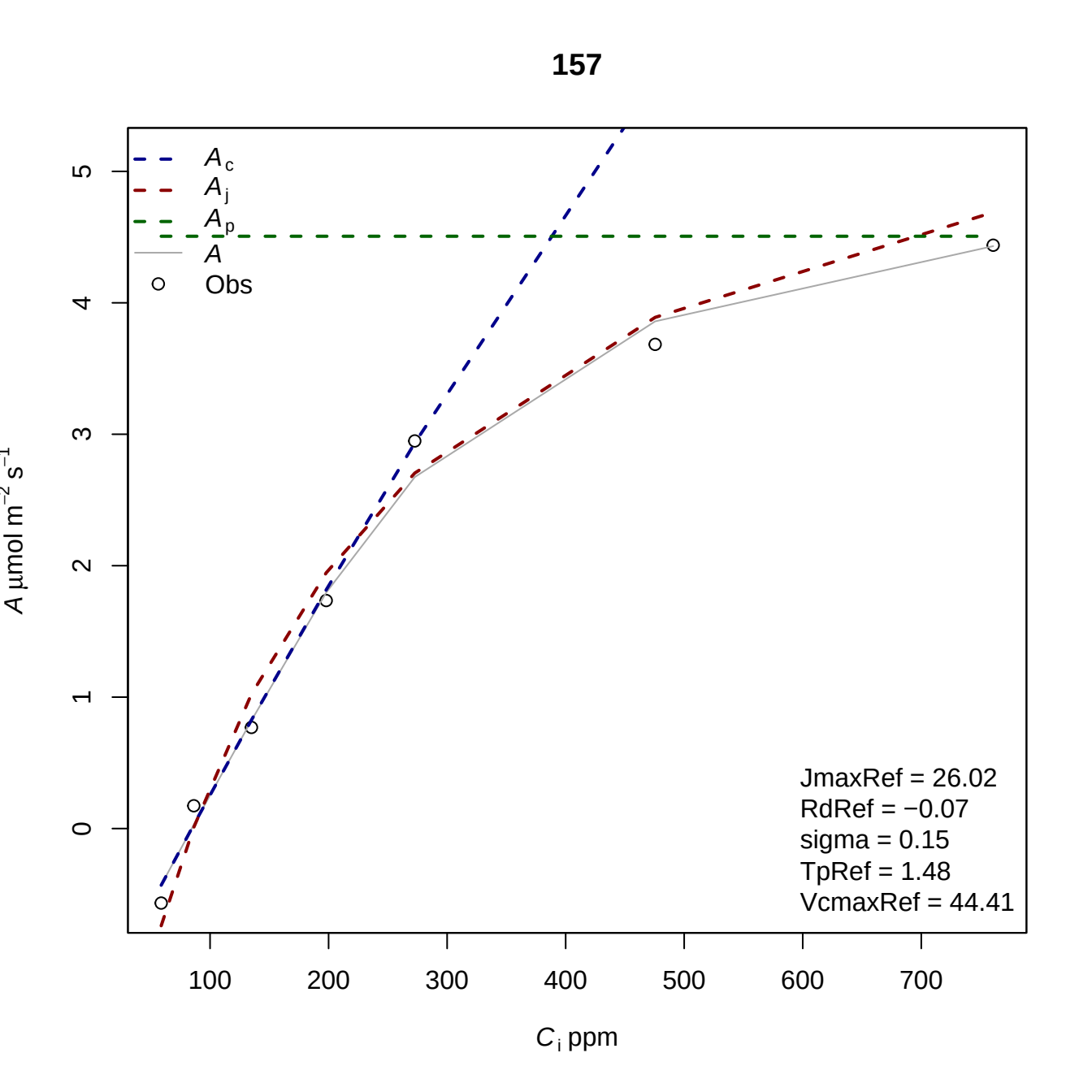

152
A_c
A_j
A_p
A
Obs
JmaxRef = 56.95
RdRef = 0.1
sigma = 0.16
TpRef = 4.32
VcmaxRef = 23.75
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
14
200
400
600
800

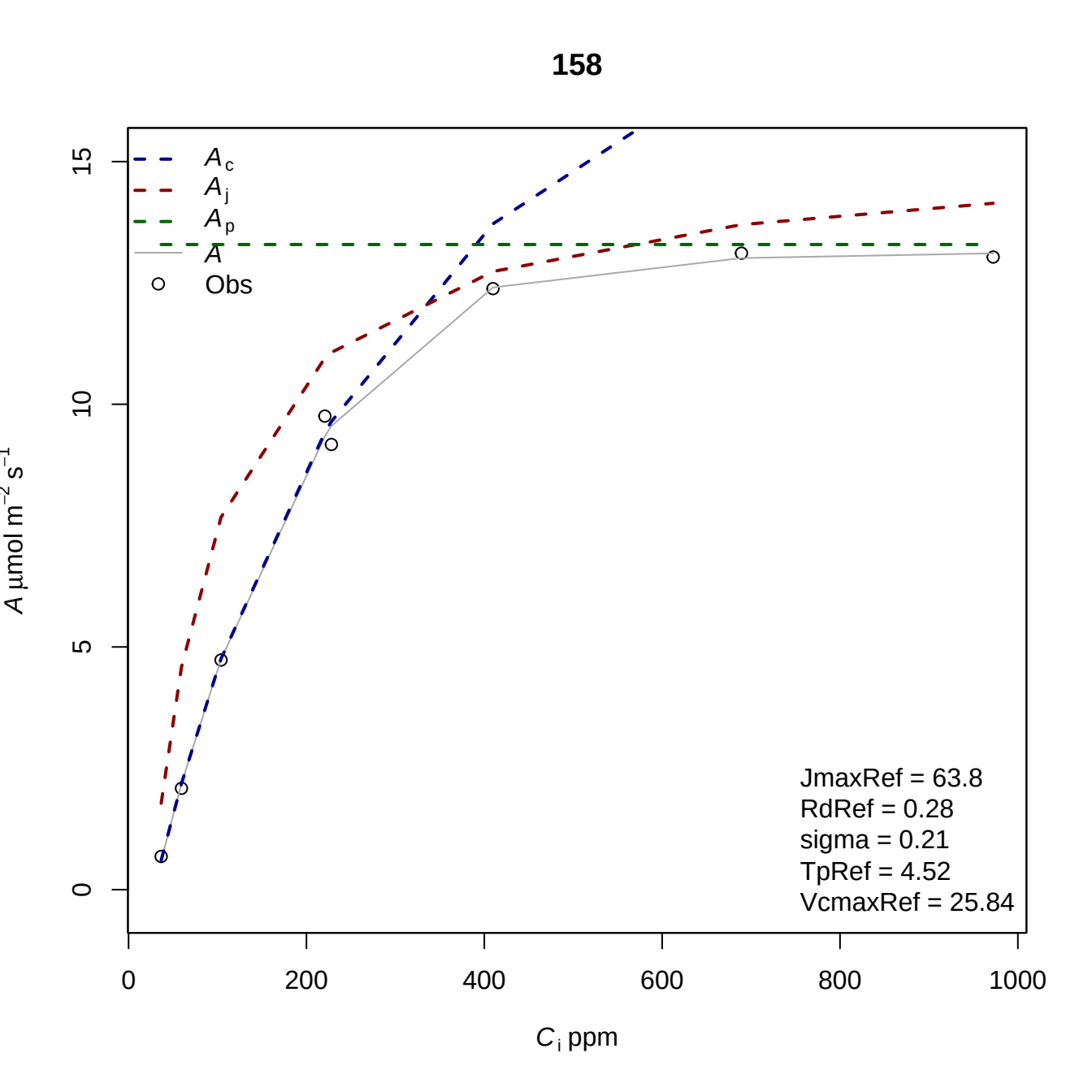

153
A_c
A_j
A_p
A
Obs
JmaxRef = 51.17
RdRef = −0.2
sigma = 0.19
TpRef = 3.84
VcmaxRef = 33.84
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
200
400
600
800
1000

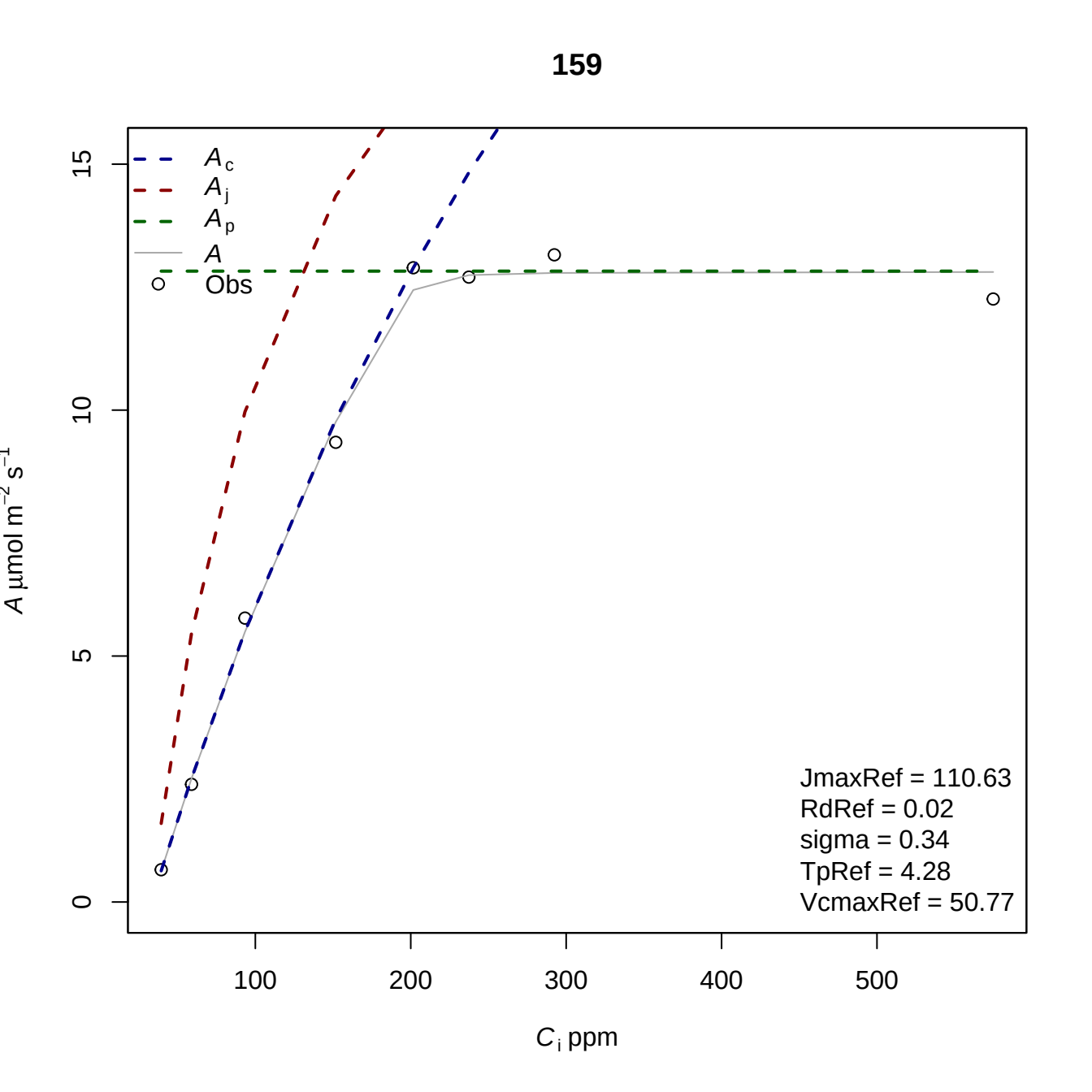

154
A_c
A_j
A_p
A
Obs
JmaxRef = 95.92
RdRef = −0.33
sigma = 0.44
TpRef = 6.54
VcmaxRef = 62.32
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
200
400
600
800

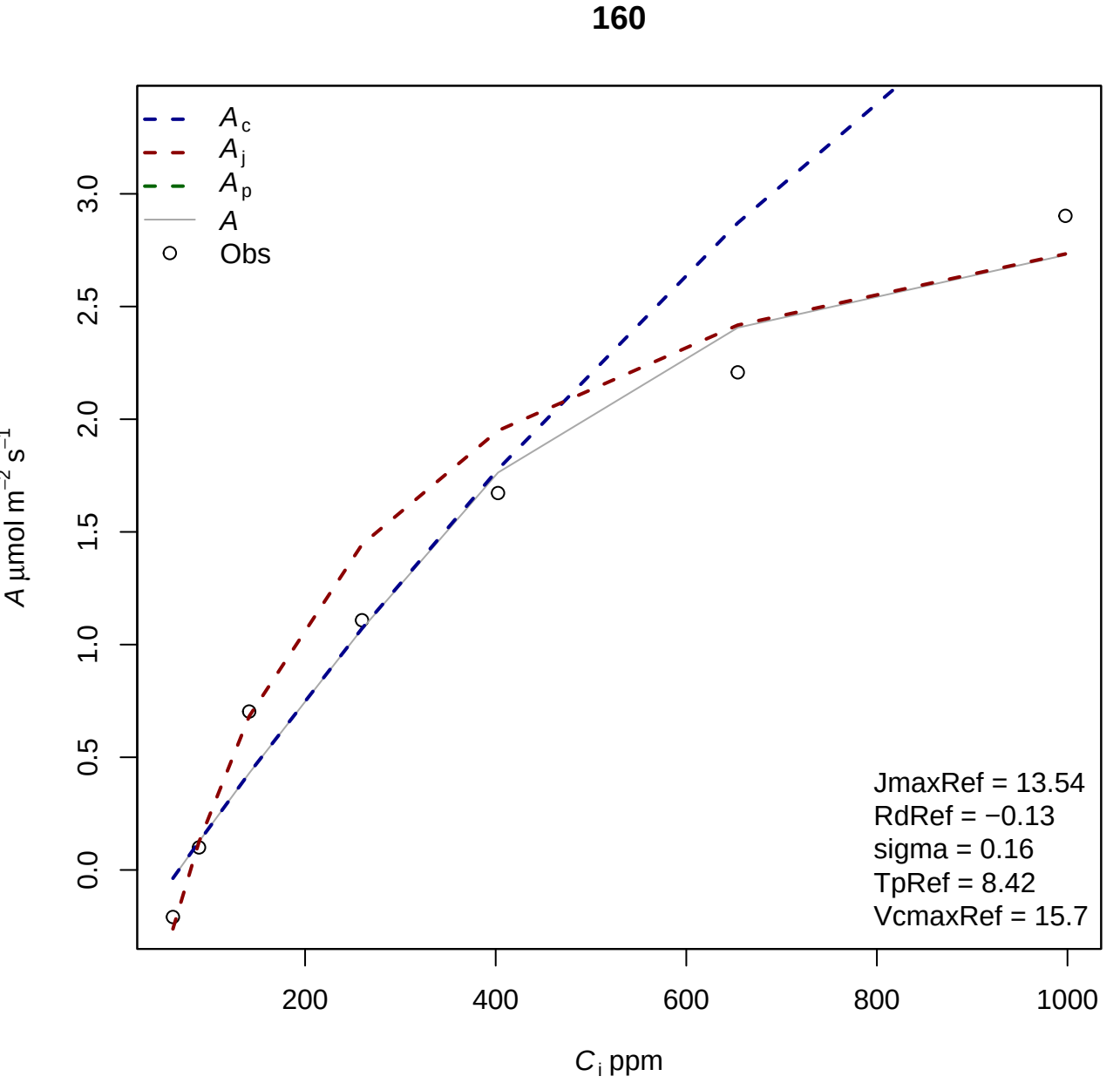

155
A_c
A_j
A_p
A
Obs
JmaxRef = 98.68
RdRef = −0.22
sigma = 0.19
TpRef = 5.9
VcmaxRef = 89.54
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
100
200
300
400
500

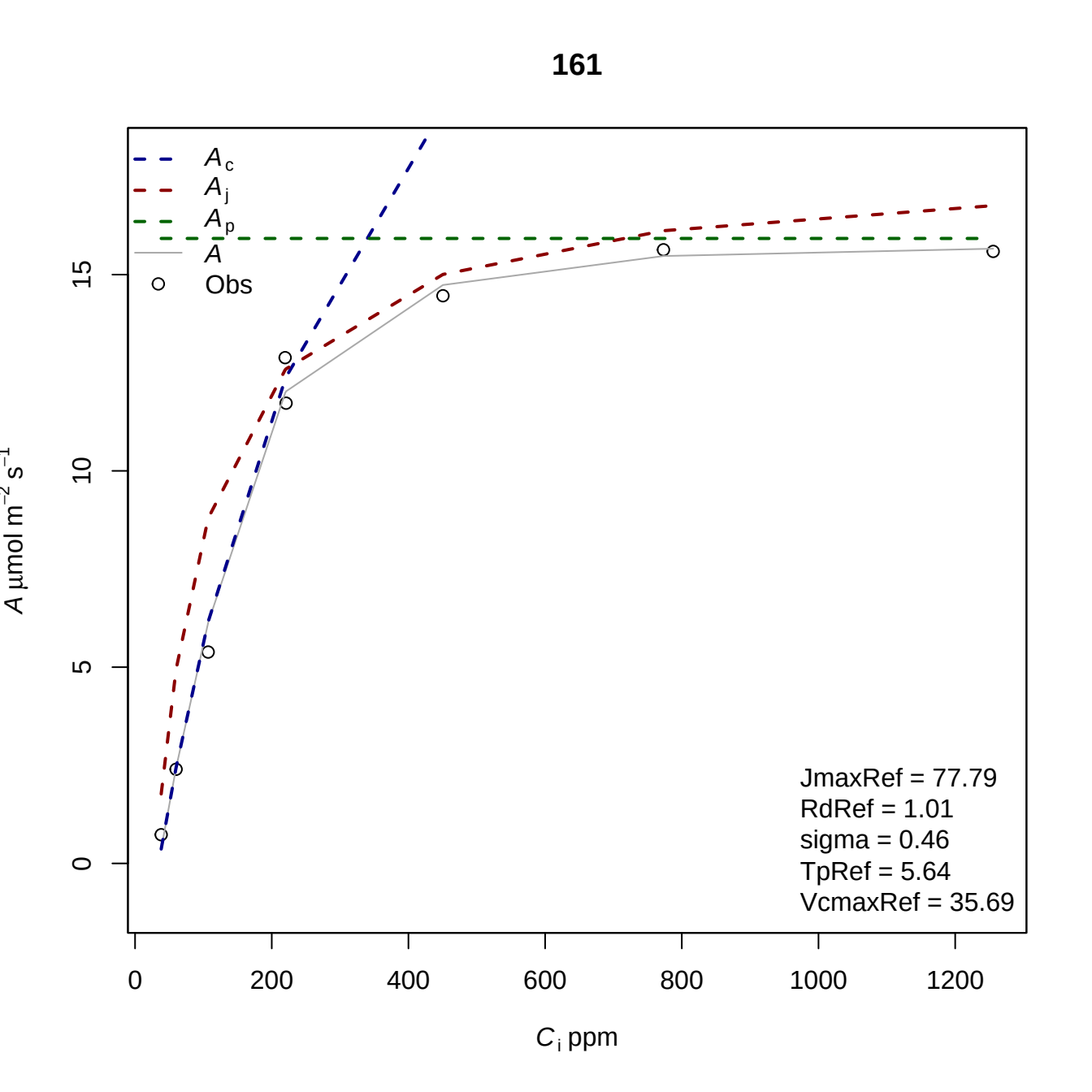

156
A_c
A_j
A_p
A
Obs
JmaxRef = 38.99
RdRef = −0.54
sigma = 0.38
TpRef = 2.22
VcmaxRef = 61.04
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
100
200
300
400
500
600
700

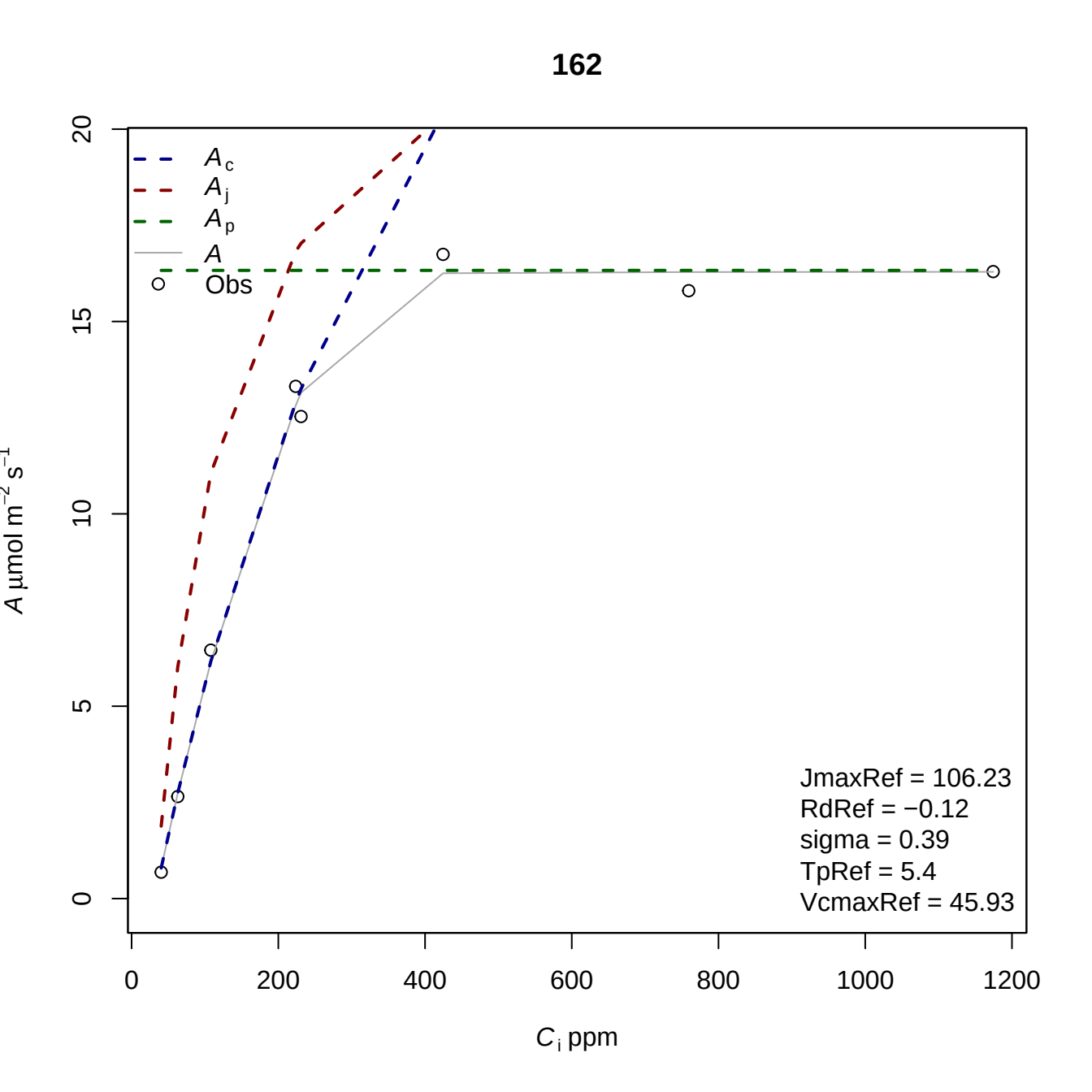

157
A_c
A_j
A_p
A
Obs
JmaxRef = 26.02
RdRef = −0.07
sigma = 0.15
TpRef = 1.48
VcmaxRef = 44.41
A µmol m^{-2} s^{-1}
C_i ppm
0
1
2
3
4
5
100
200
300
400
500
600
700

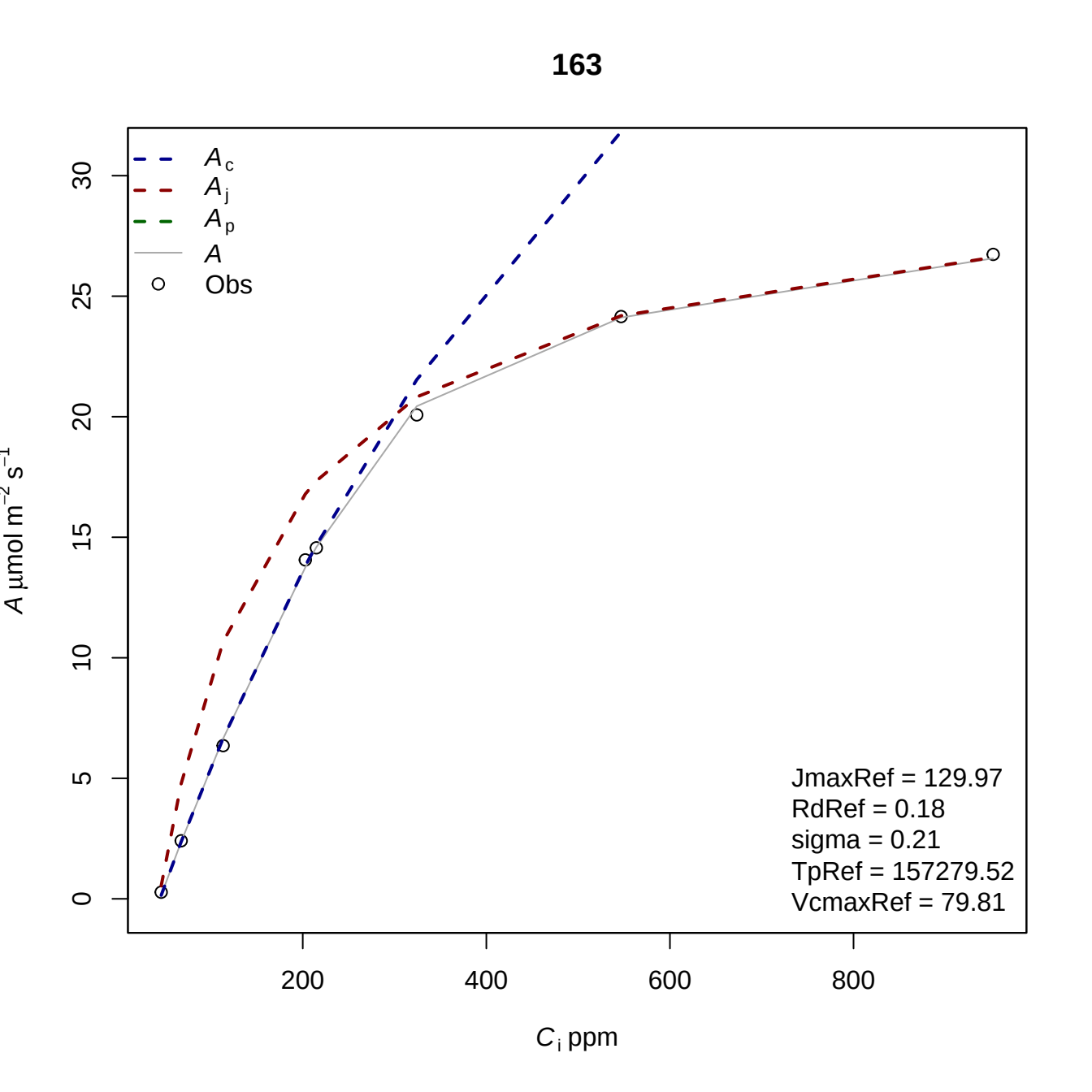

158
A_c
A_j
A_p
A
Obs
JmaxRef = 63.8
RdRef = 0.28
sigma = 0.21
TpRef = 4.52
VcmaxRef = 25.84
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
200
400
600
800
1000

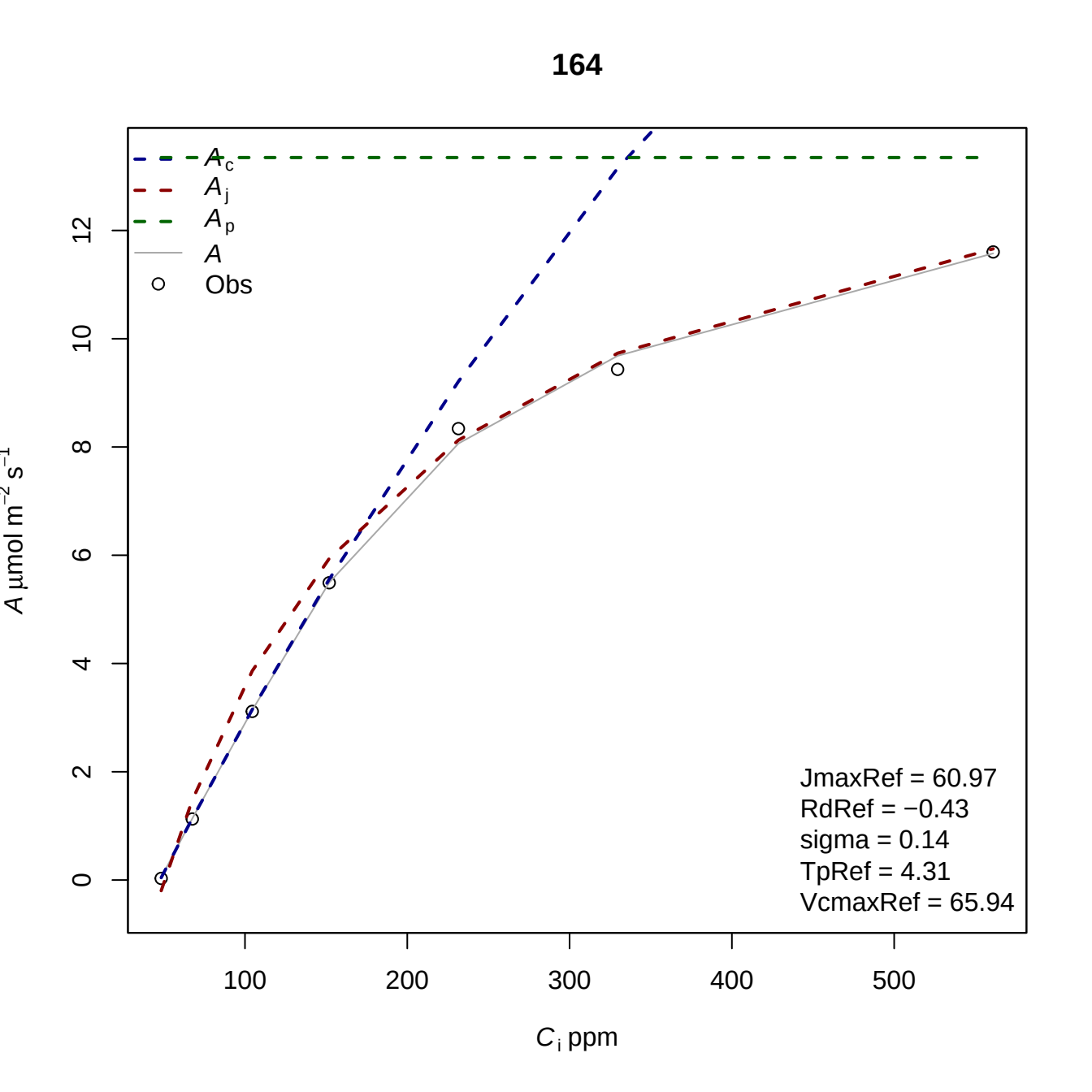

159
A_c
A_j
A_p
A
Obs
JmaxRef = 110.63
RdRef = 0.02
sigma = 0.34
TpRef = 4.28
VcmaxRef = 50.77
0
5
10
15
100
200
300
400
500
A µmol m^{-2} s^{-1}
C_i ppm

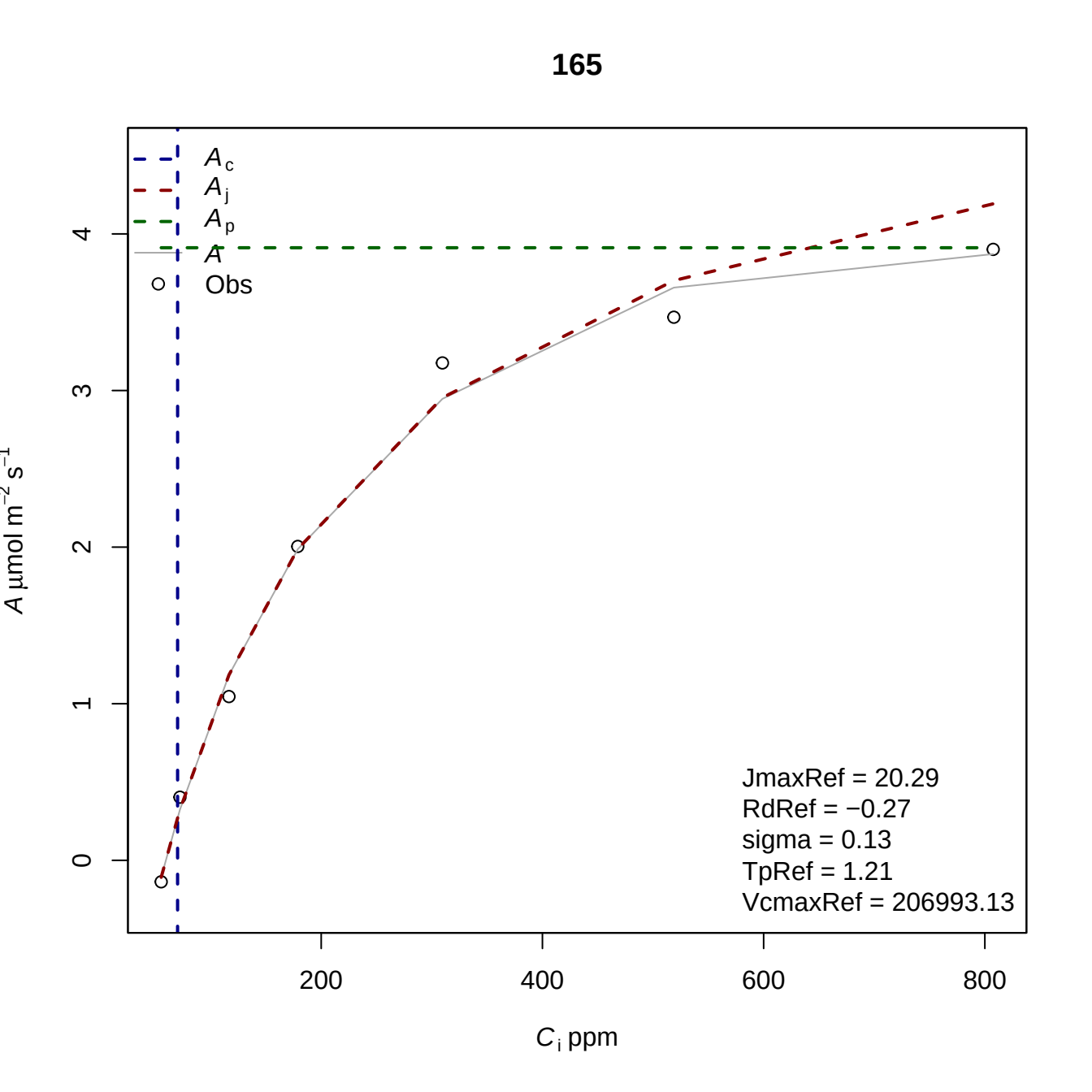

160
A_c
A_j
A_p
A
Obs
JmaxRef = 13.54
RdRef = −0.13
sigma = 0.16
TpRef = 8.42
VcmaxRef = 15.7
A µmol m^{-2} s^{-1}
C_i ppm
0.0
0.5
1.0
1.5
2.0
2.5
3.0
200
400
600
800
1000

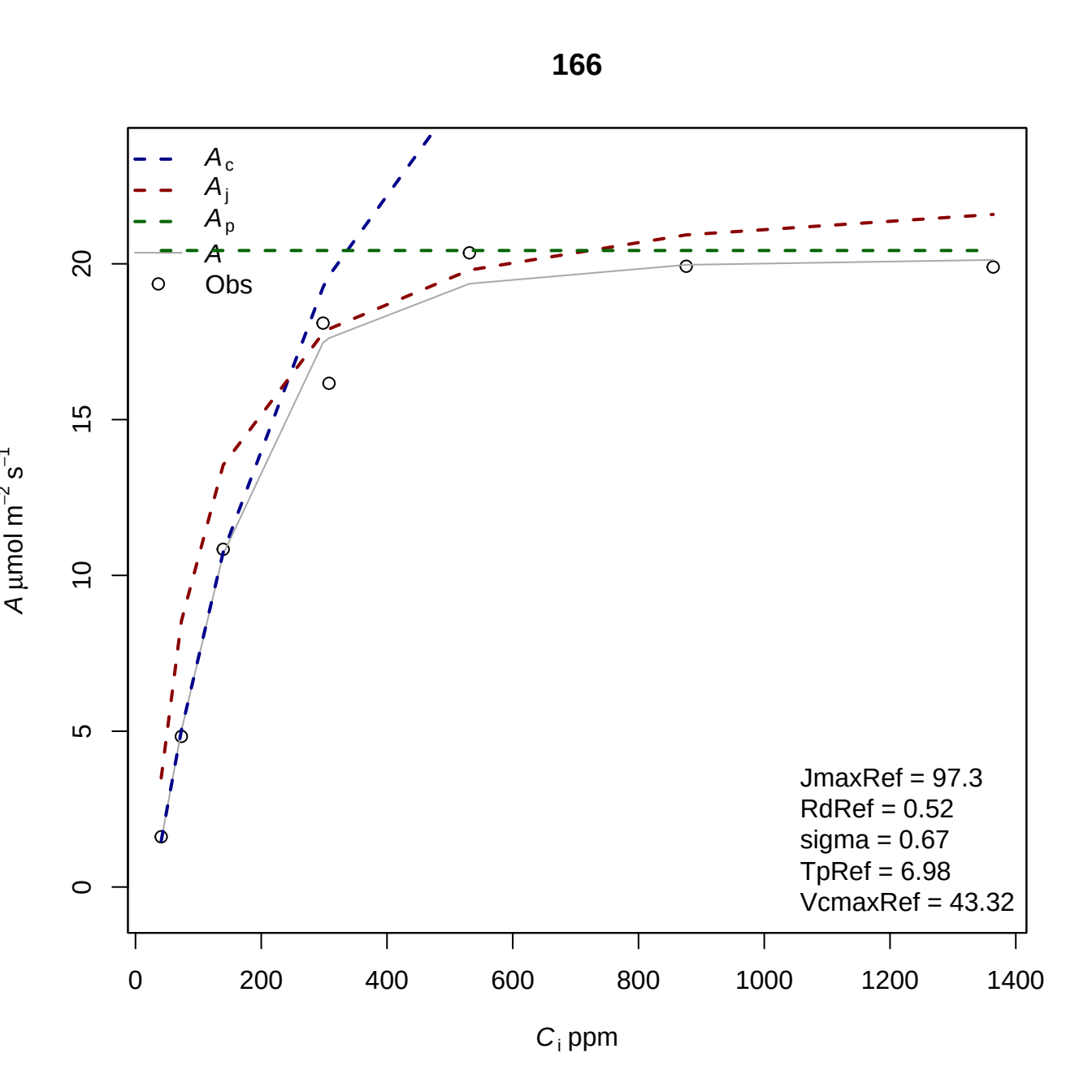

161
A_c
A_j
A_p
A
Obs
JmaxRef = 77.79
RdRef = 1.01
sigma = 0.46
TpRef = 5.64
VcmaxRef = 35.69
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
200
400
600
800
1000
1200

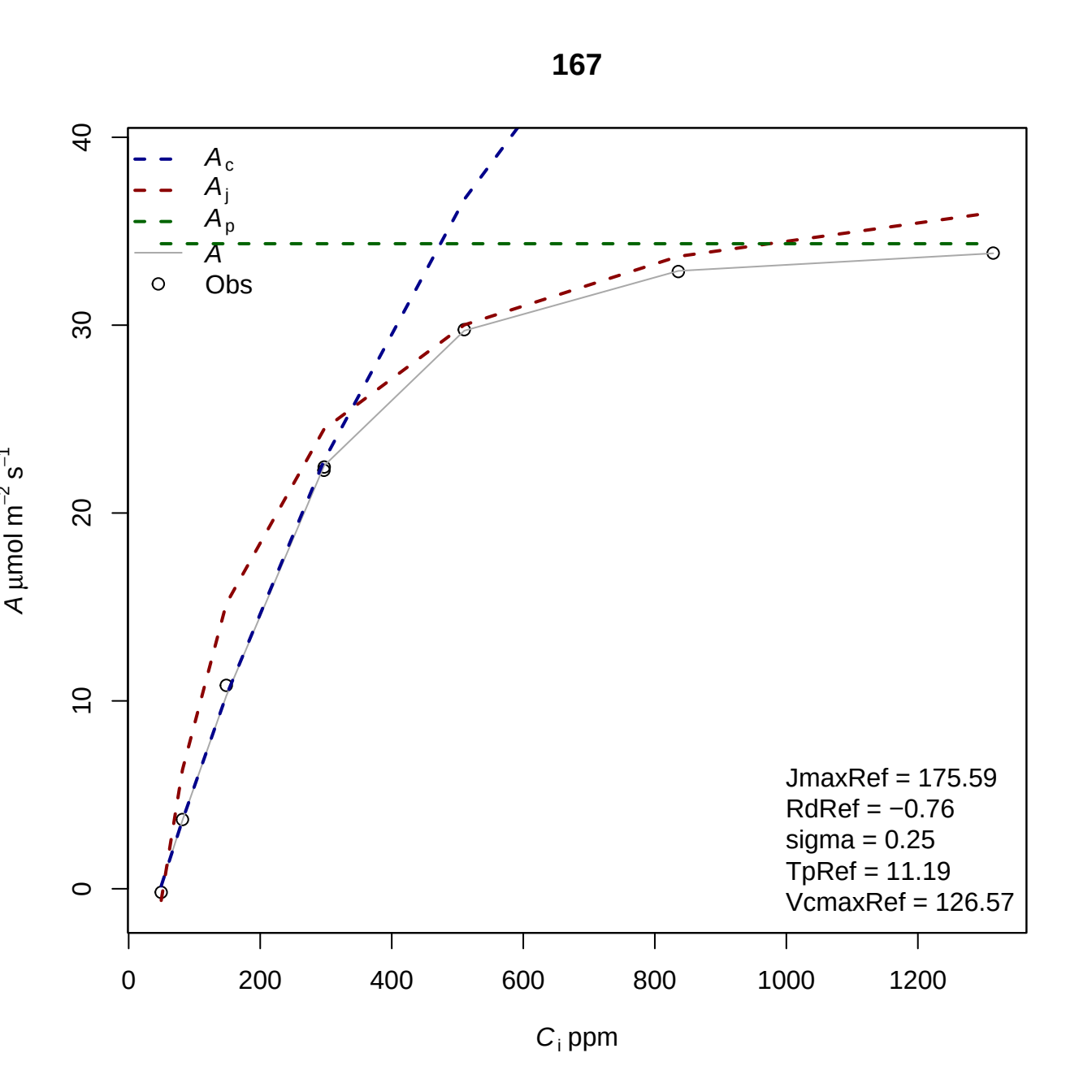

162
A_c
A_j
A_p
A
Obs
JmaxRef = 106.23
RdRef = −0.12
sigma = 0.39
TpRef = 5.4
VcmaxRef = 45.93
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

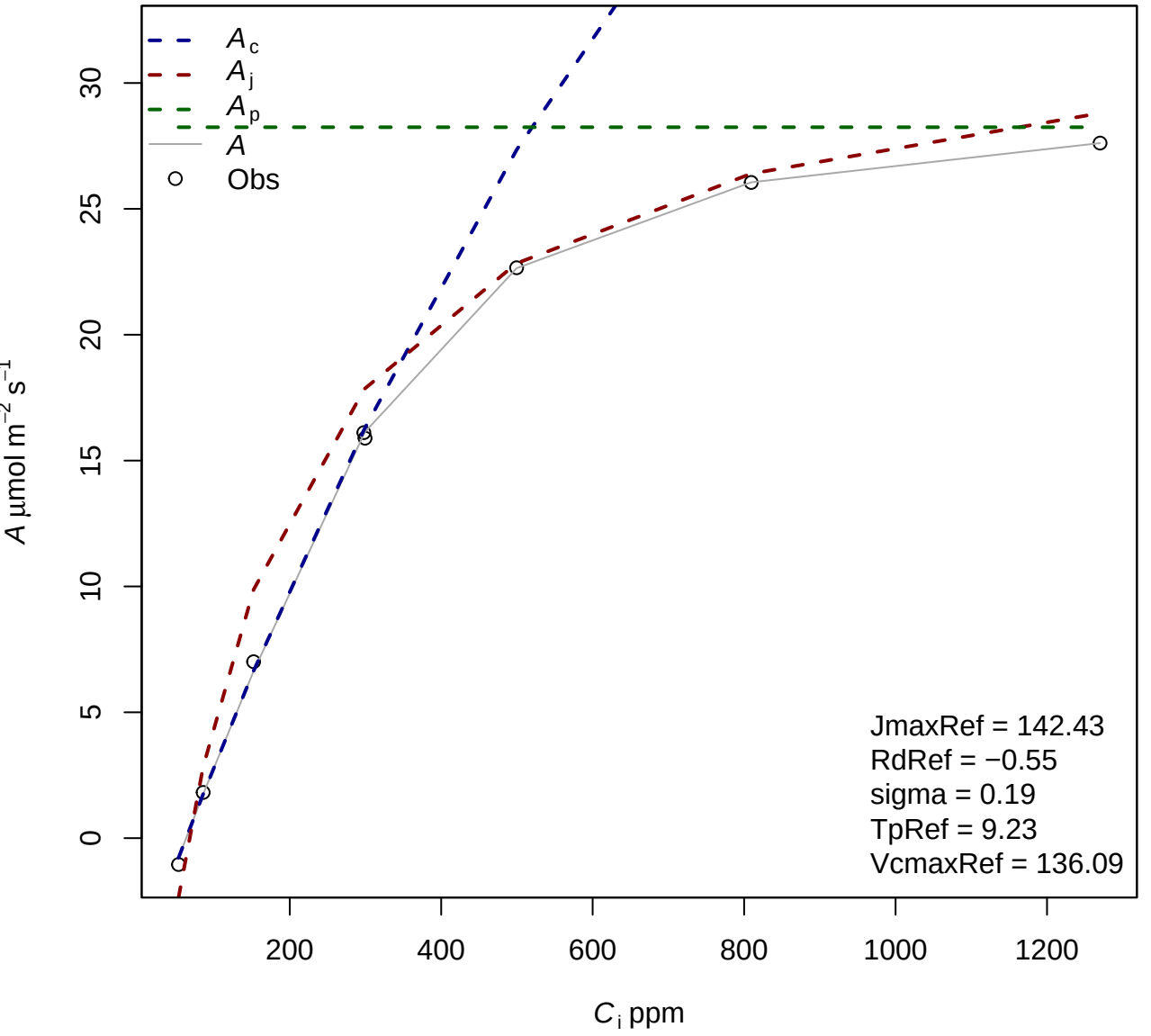

163
A_c
A_j
A_p
A
Obs
JmaxRef = 129.97
RdRef = 0.18
sigma = 0.21
TpRef = 157279.52
VcmaxRef = 79.81
0
5
10
15
20
25
30
200
400
600
800
A µmol m^{-2} s^{-1}
C_i ppm

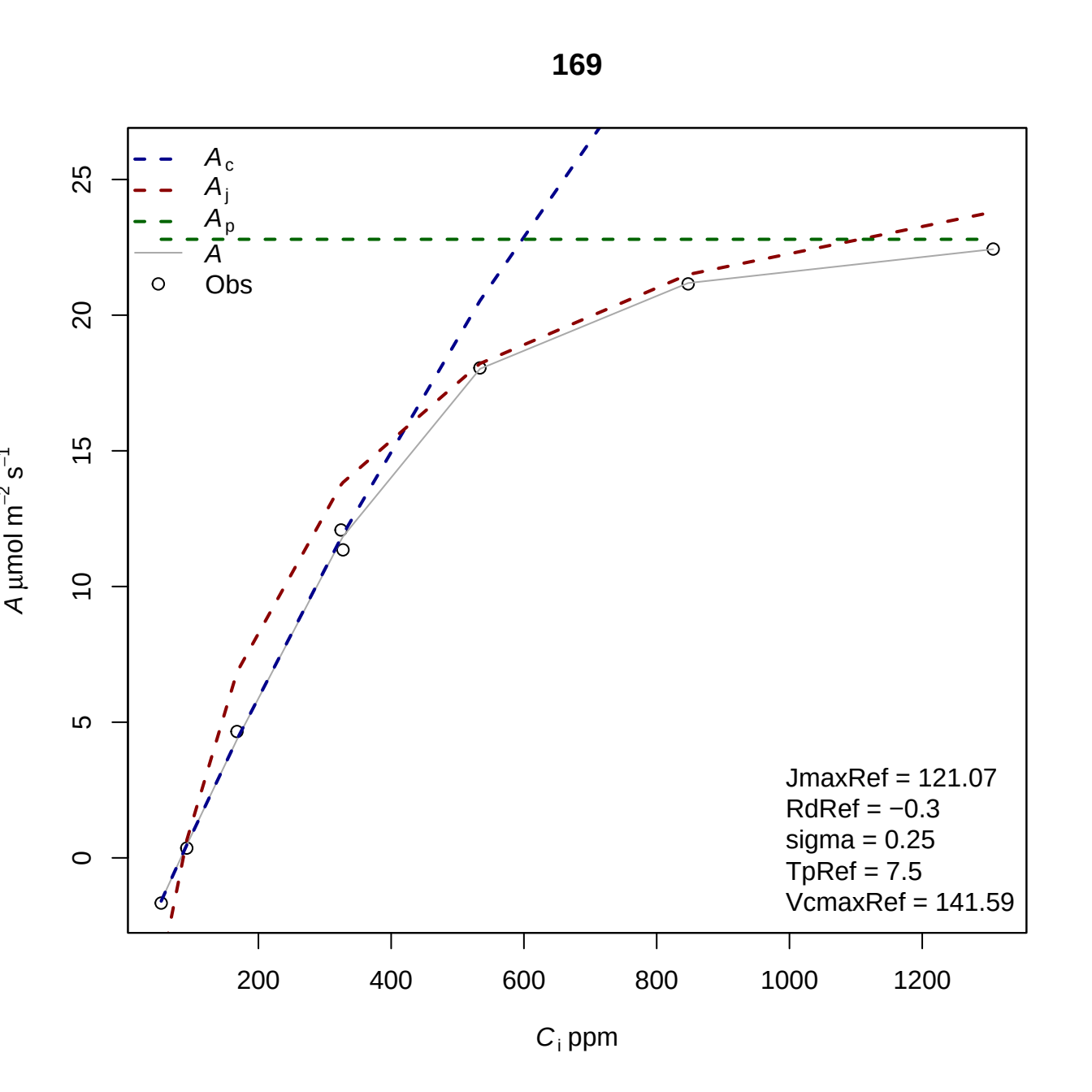

164
A_c
A_j
A_p
A
Obs
JmaxRef = 60.97
RdRef = −0.43
sigma = 0.14
TpRef = 4.31
VcmaxRef = 65.94
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
100
200
300
400
500

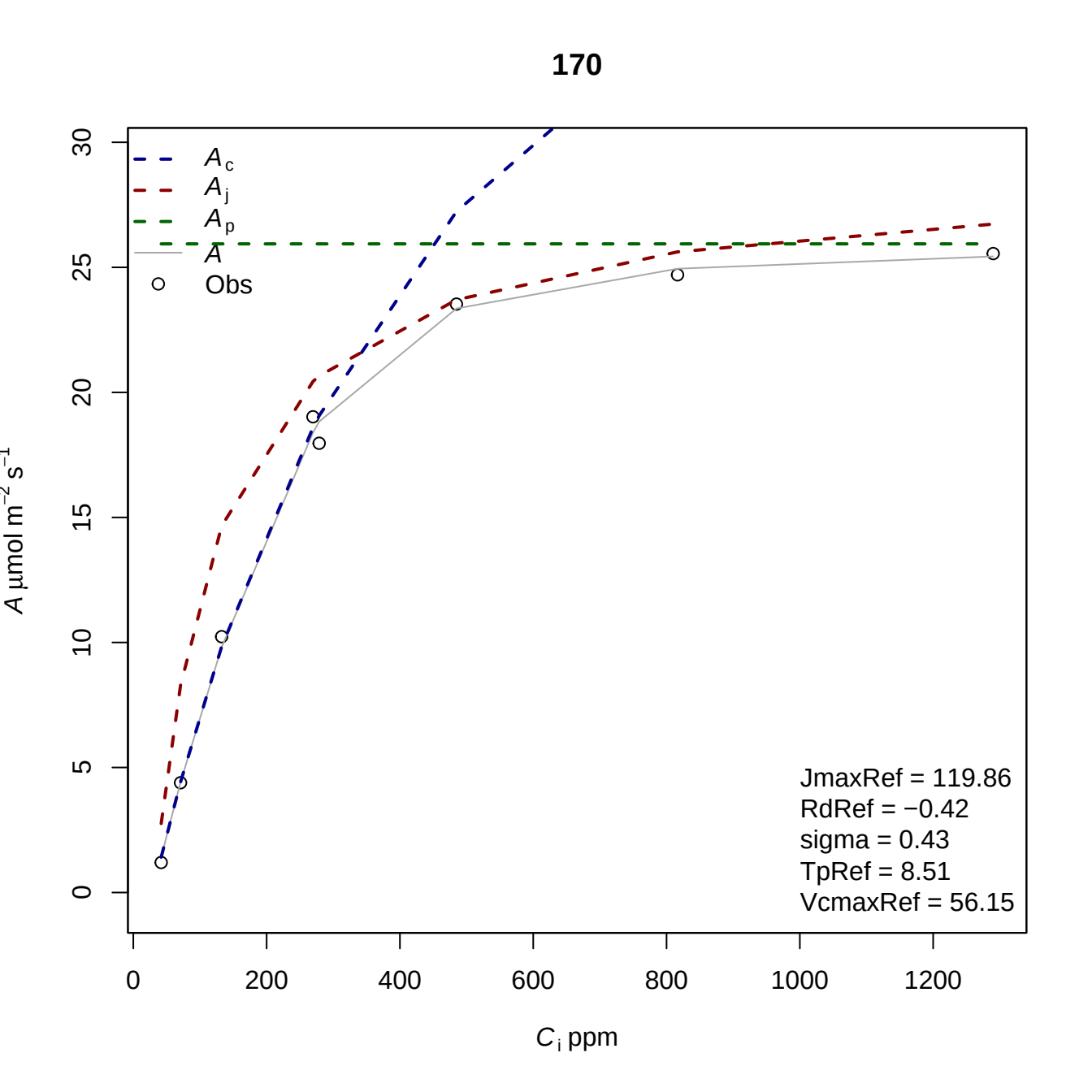

165
A_c
A_j
A_p
A
Obs
JmaxRef = 20.29
RdRef = −0.27
sigma = 0.13
TpRef = 1.21
VcmaxRef = 206993.13
A µmol m^{-2} s^{-1}
C_i ppm
0
1
2
3
4
200
400
600
800

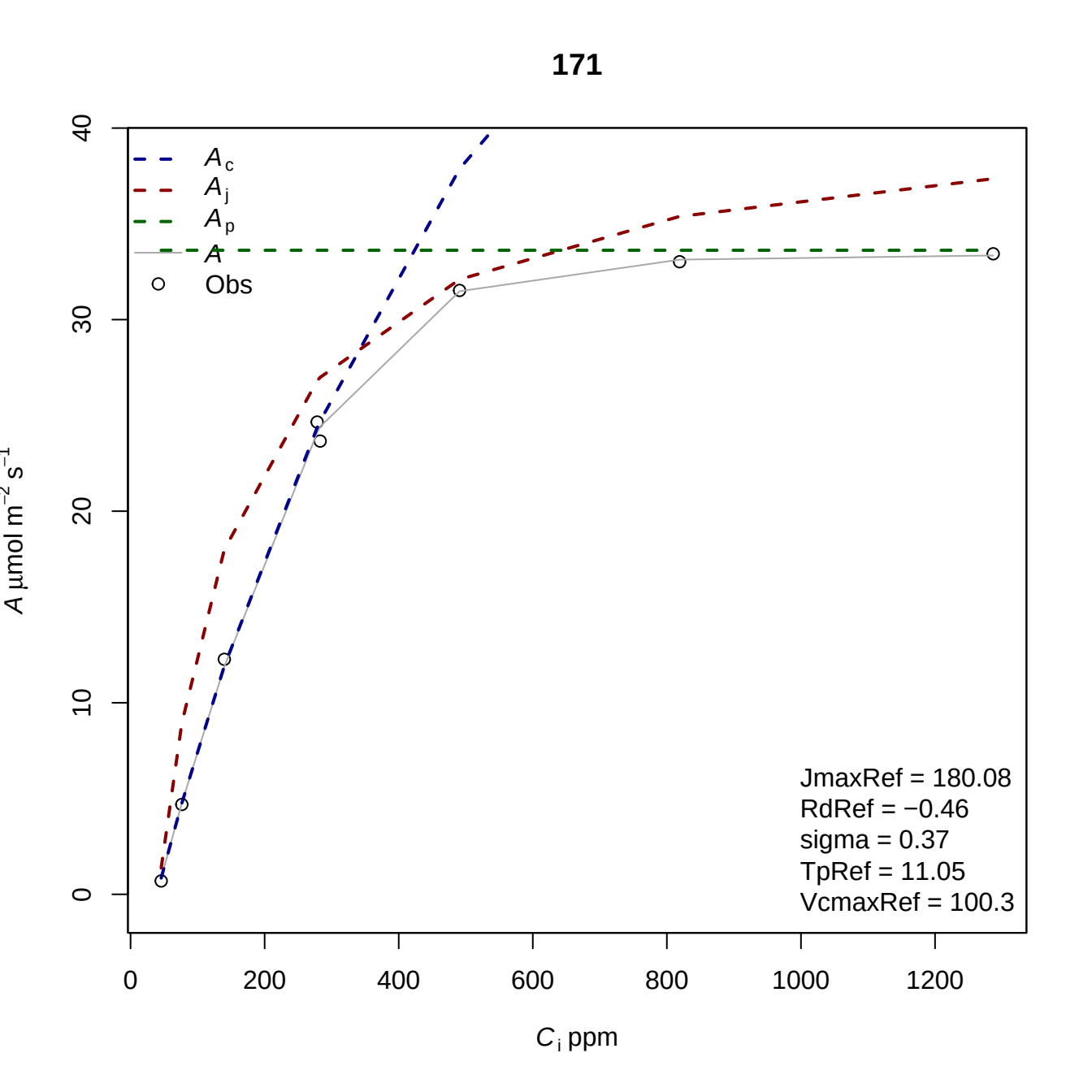

166
A_c
A_j
A_p
A
Obs
JmaxRef = 97.3
RdRef = 0.52
sigma = 0.67
TpRef = 6.98
VcmaxRef = 43.32
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000
1200
1400

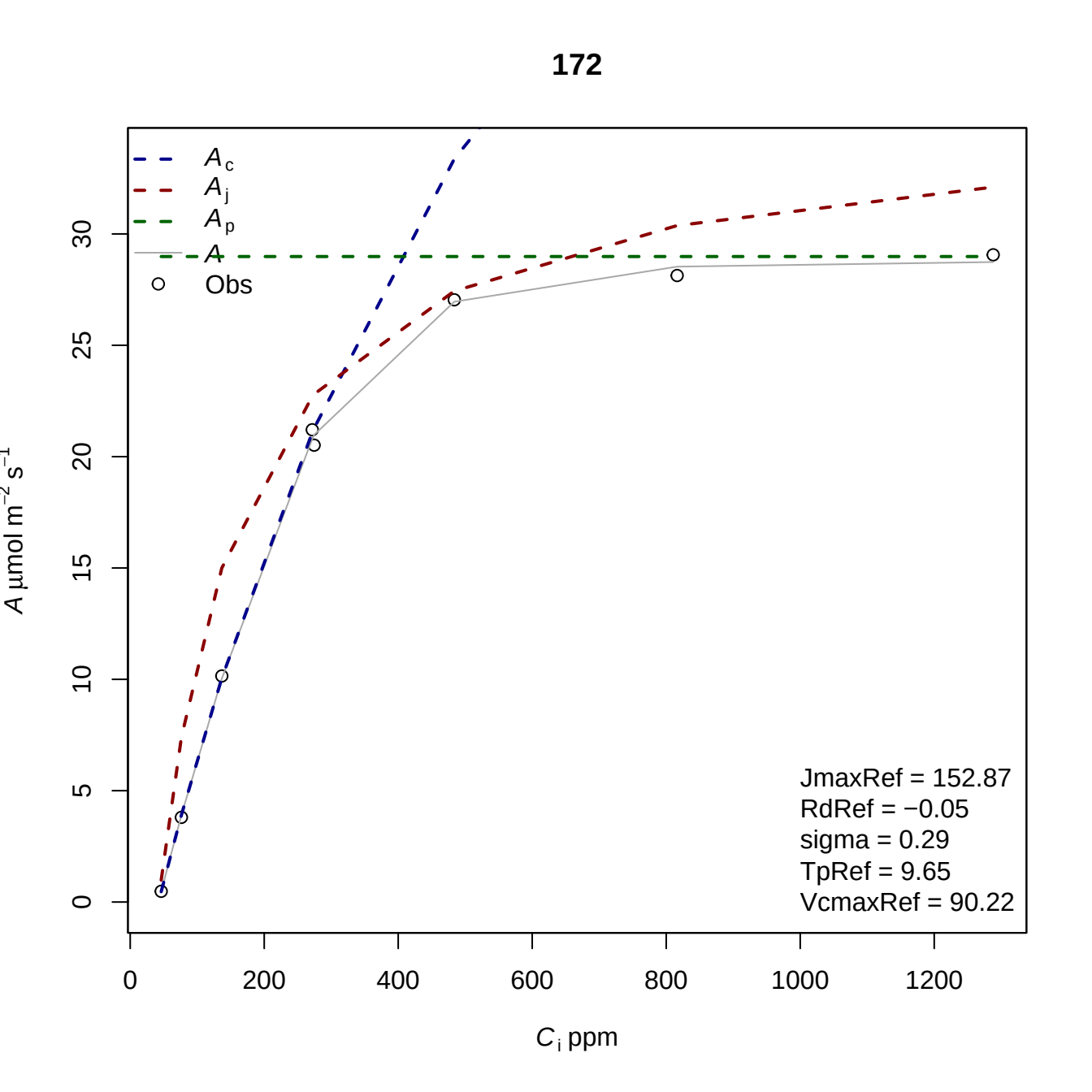

167
A_c
A_j
A_p
A
Obs
JmaxRef = 175.59
RdRef = −0.76
sigma = 0.25
TpRef = 11.19
VcmaxRef = 126.57
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

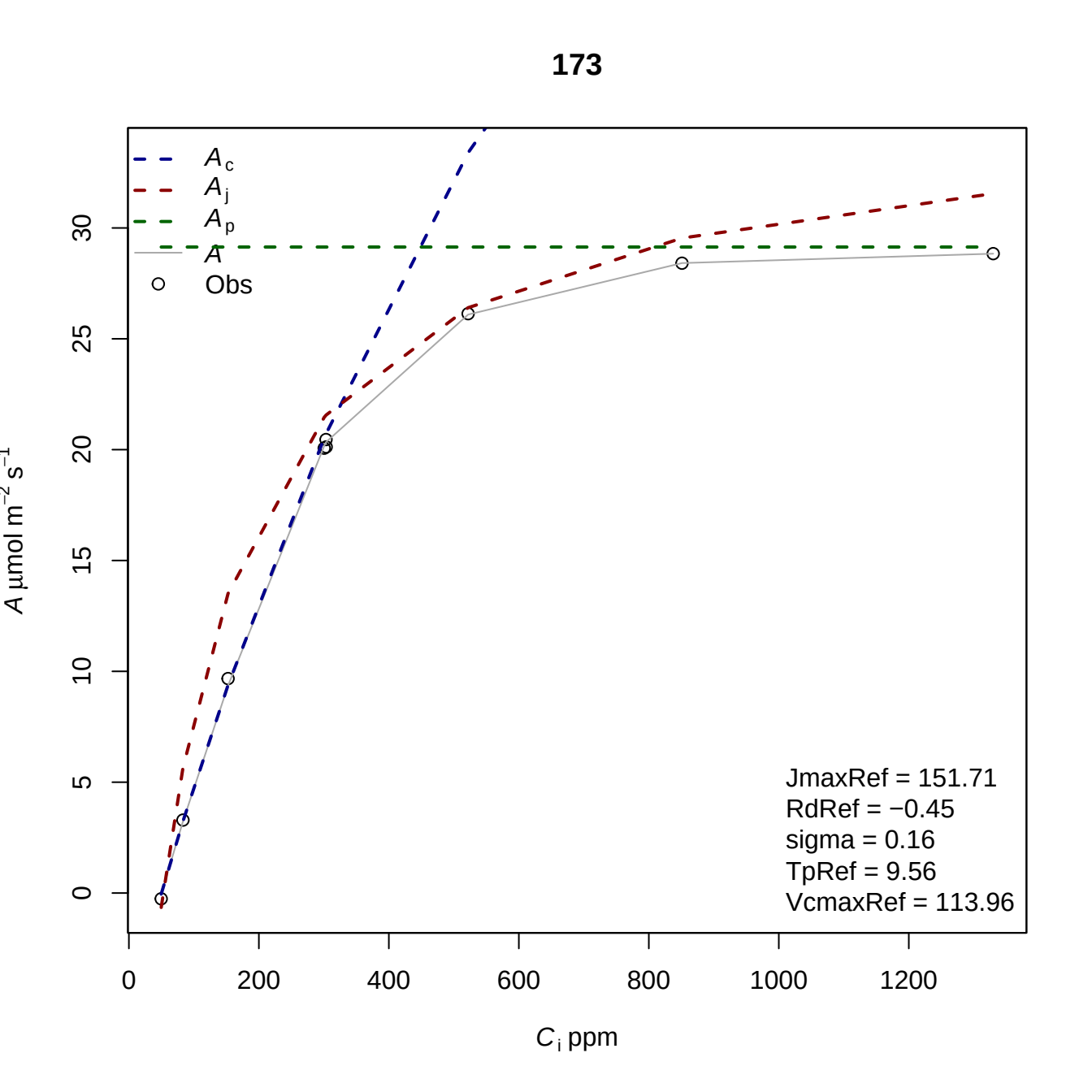

A_c
A_j
A_p
A
Obs
JmaxRef = 142.43
RdRef = −0.55
sigma = 0.19
TpRef = 9.23
VcmaxRef = 136.09
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
200
400
600
800
1000
1200

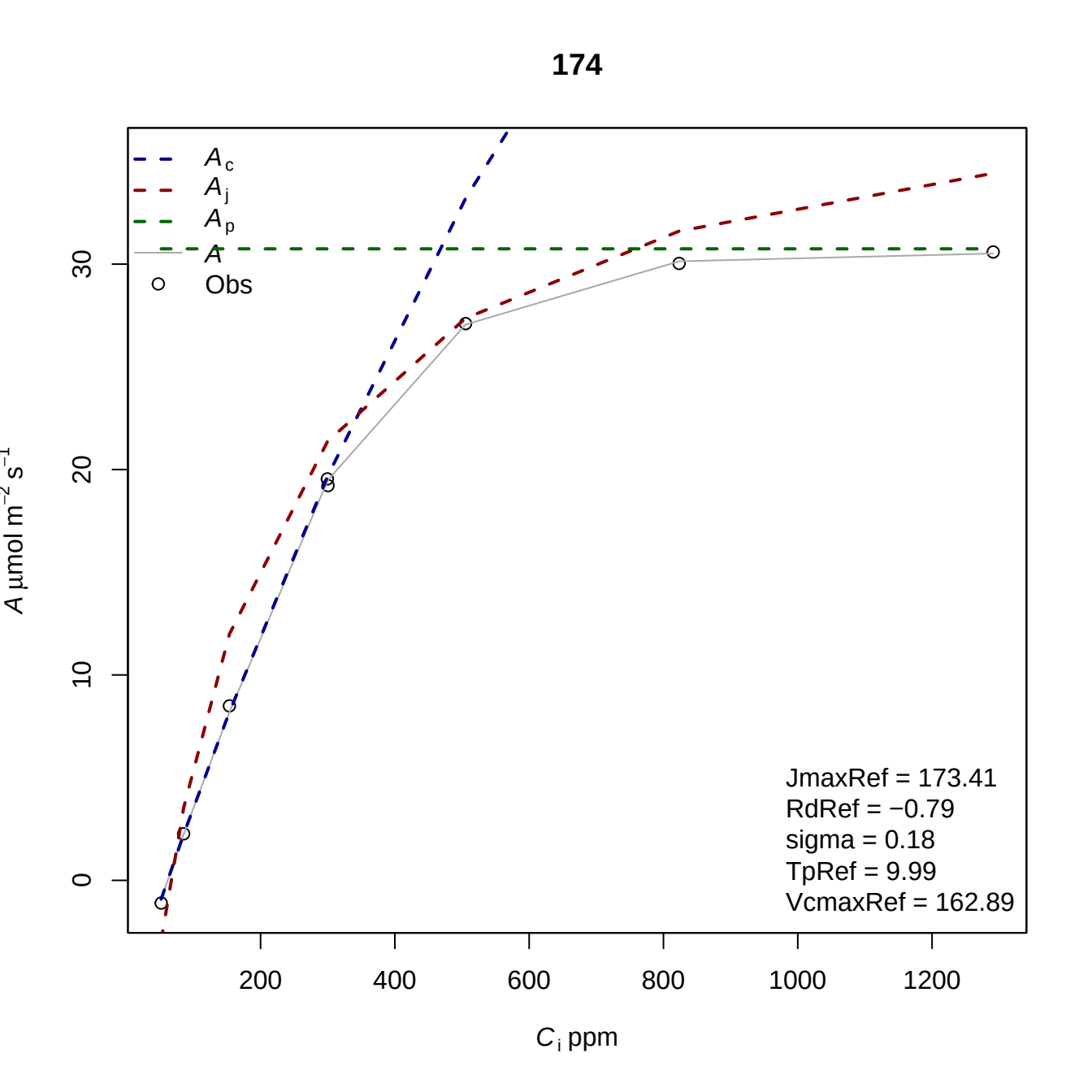

169
A_c
A_j
A_p
A
Obs
JmaxRef = 121.07
RdRef = −0.3
sigma = 0.25
TpRef = 7.5
VcmaxRef = 141.59
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
200
400
600
800
1000
1200

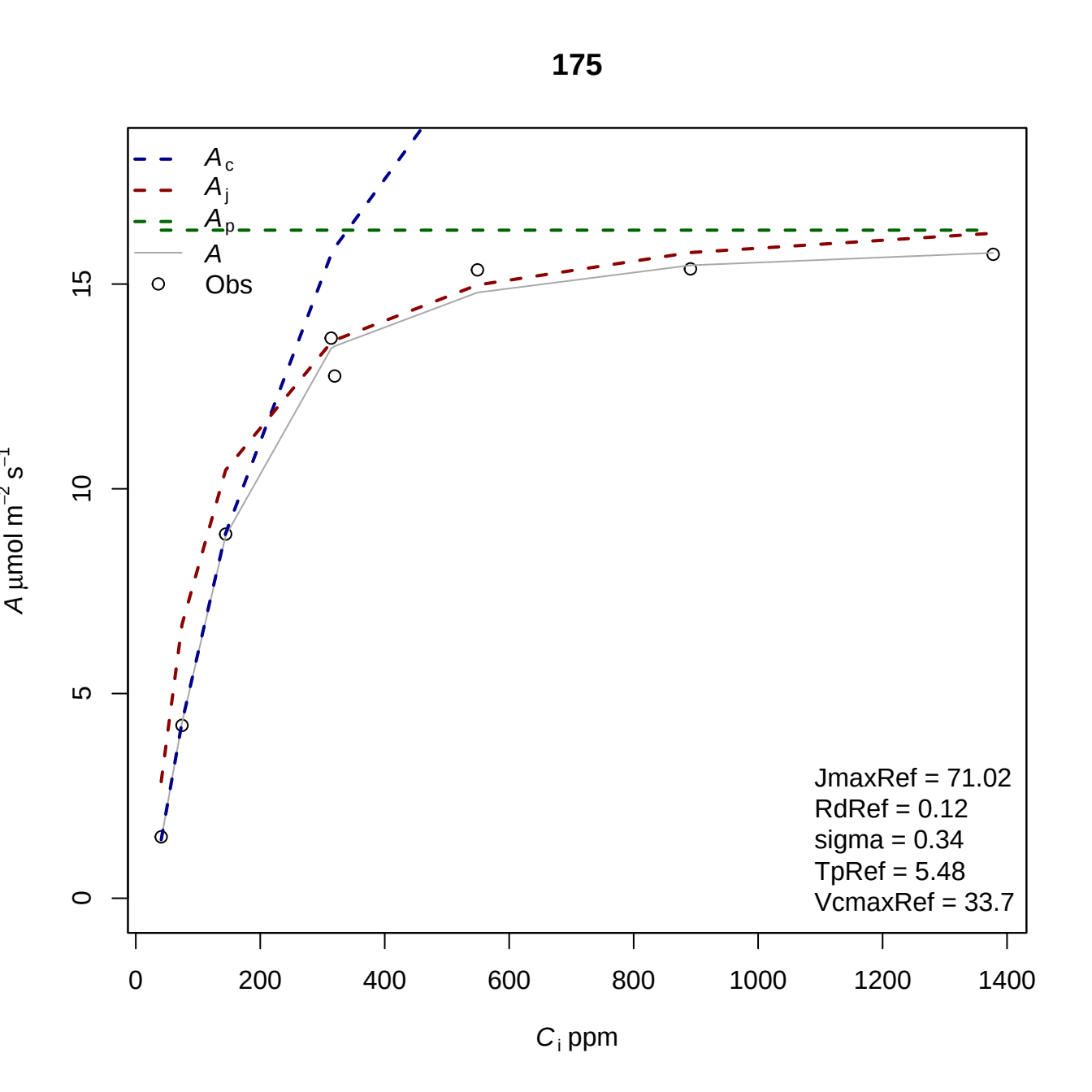

170
A_c
A_j
A_p
A
Obs
JmaxRef = 119.86
RdRef = −0.42
sigma = 0.43
TpRef = 8.51
VcmaxRef = 56.15
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

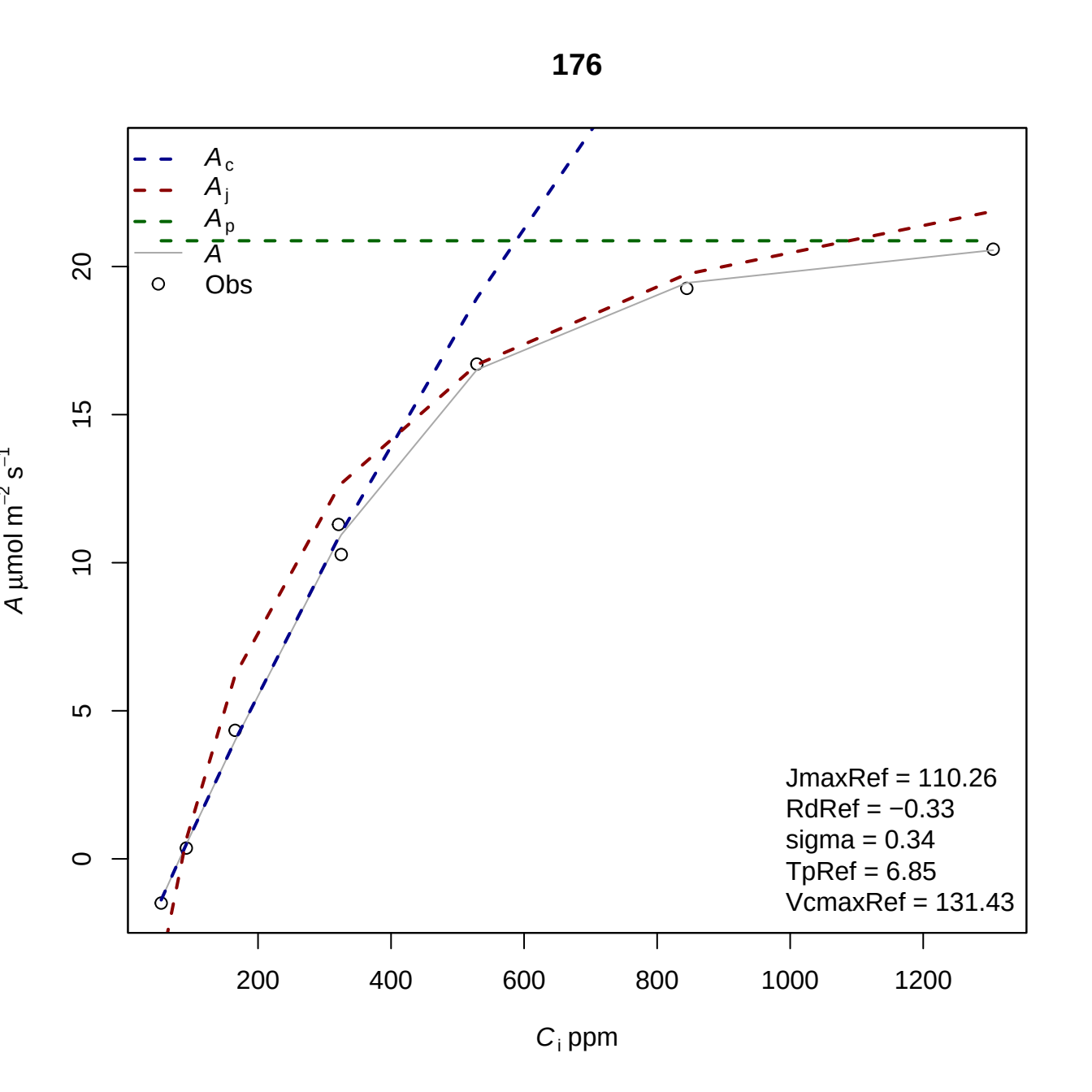

171
A_c
A_j
A_p
A
Obs
JmaxRef = 180.08
RdRef = −0.46
sigma = 0.37
TpRef = 11.05
VcmaxRef = 100.3
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

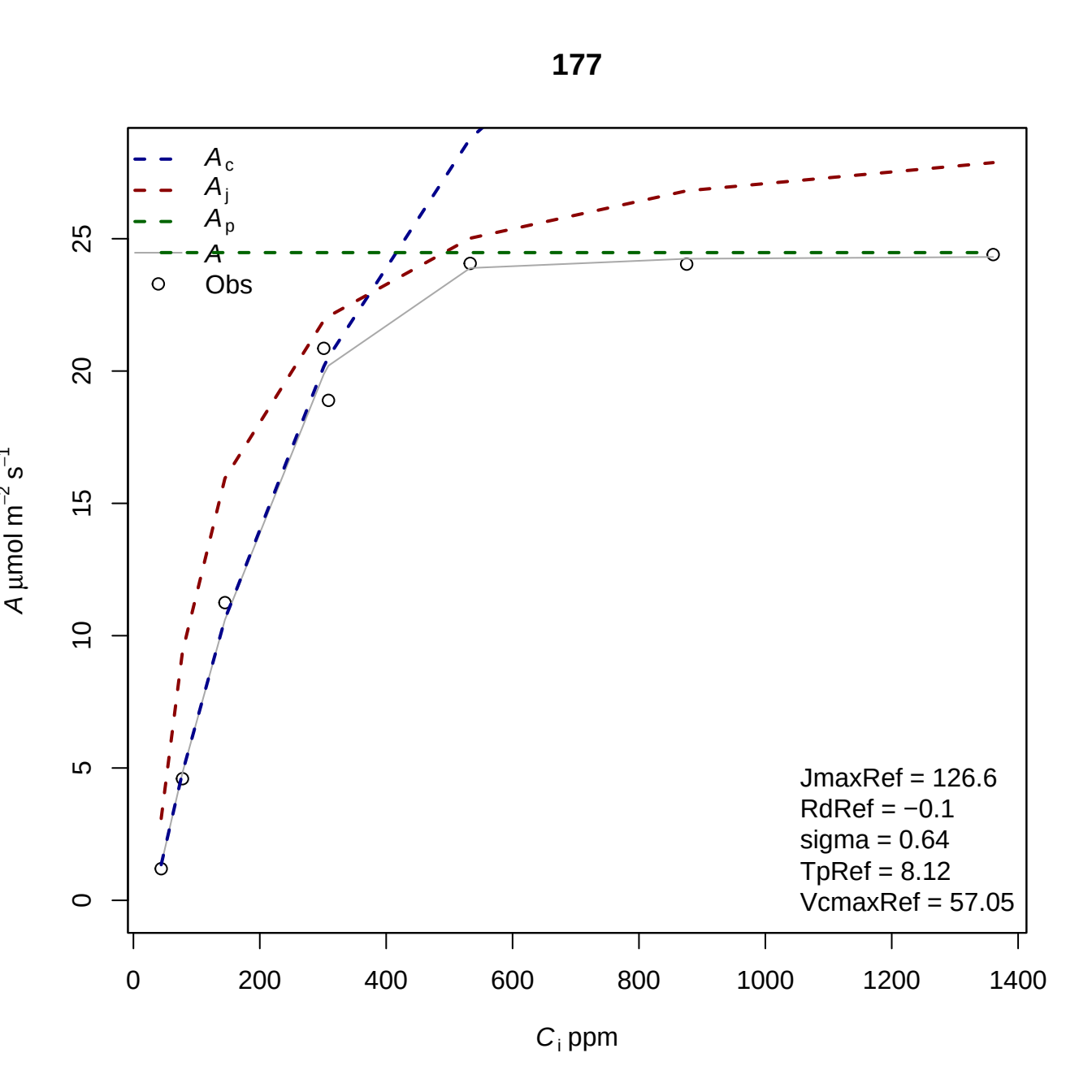

172
A_c
A_j
A_p
A
Obs
JmaxRef = 152.87
RdRef = −0.05
sigma = 0.29
TpRef = 9.65
VcmaxRef = 90.22
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

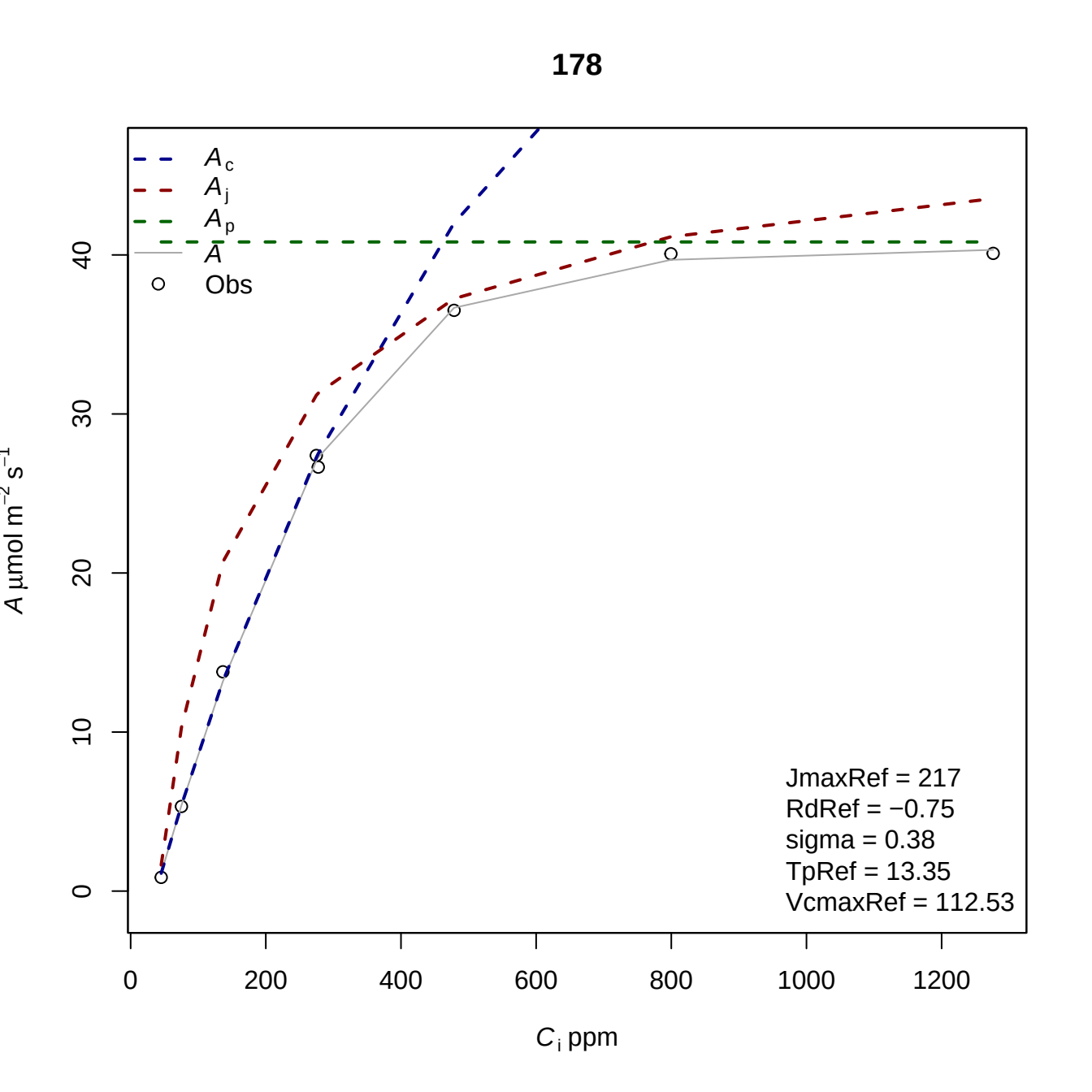

173
A_c
A_j
A_p
A
Obs
JmaxRef = 151.71
RdRef = −0.45
sigma = 0.16
TpRef = 9.56
VcmaxRef = 113.96
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

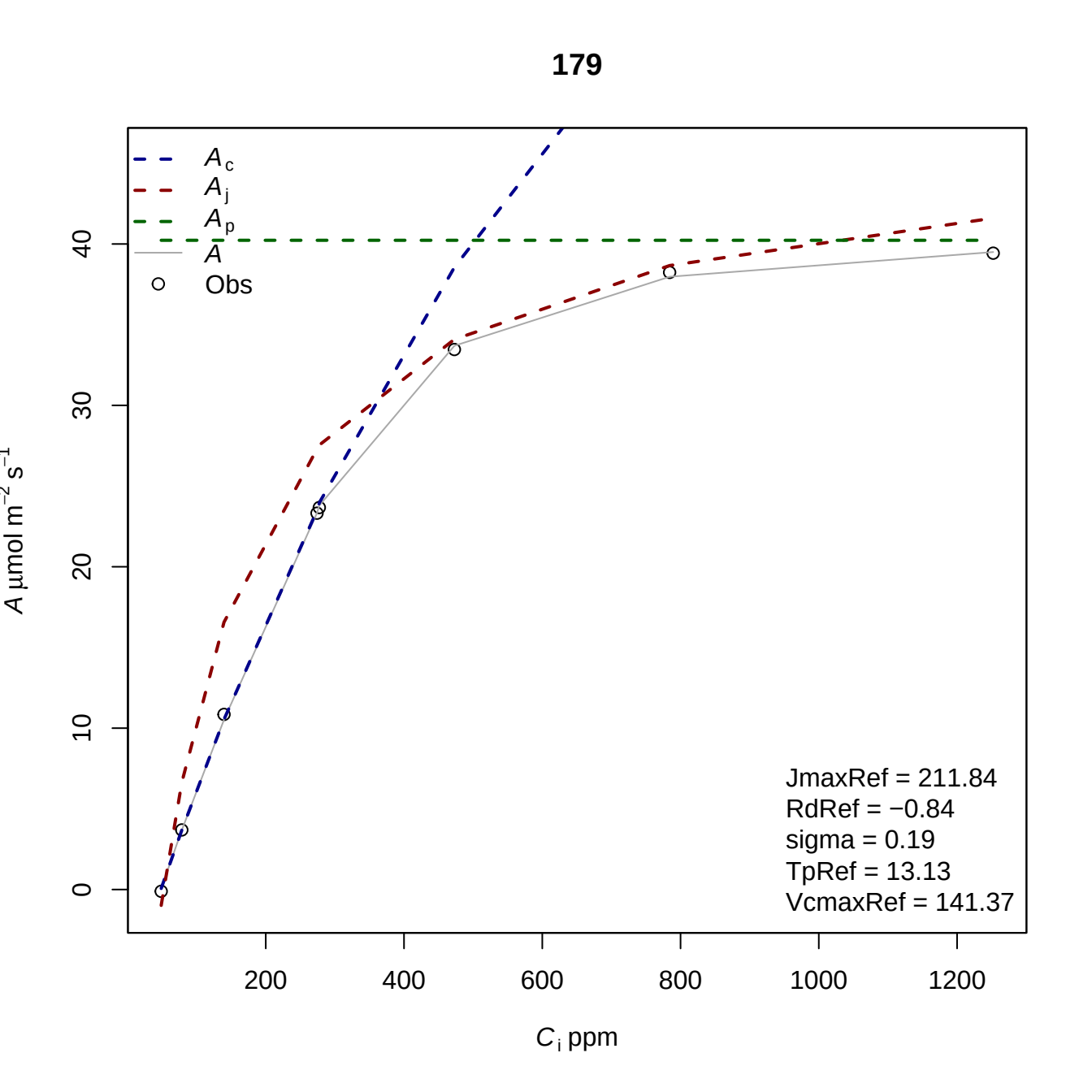

174
A_c
A_j
A_p
A
Obs
JmaxRef = 173.41
RdRef = −0.79
sigma = 0.18
TpRef = 9.99
VcmaxRef = 162.89
0
10
20
30
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

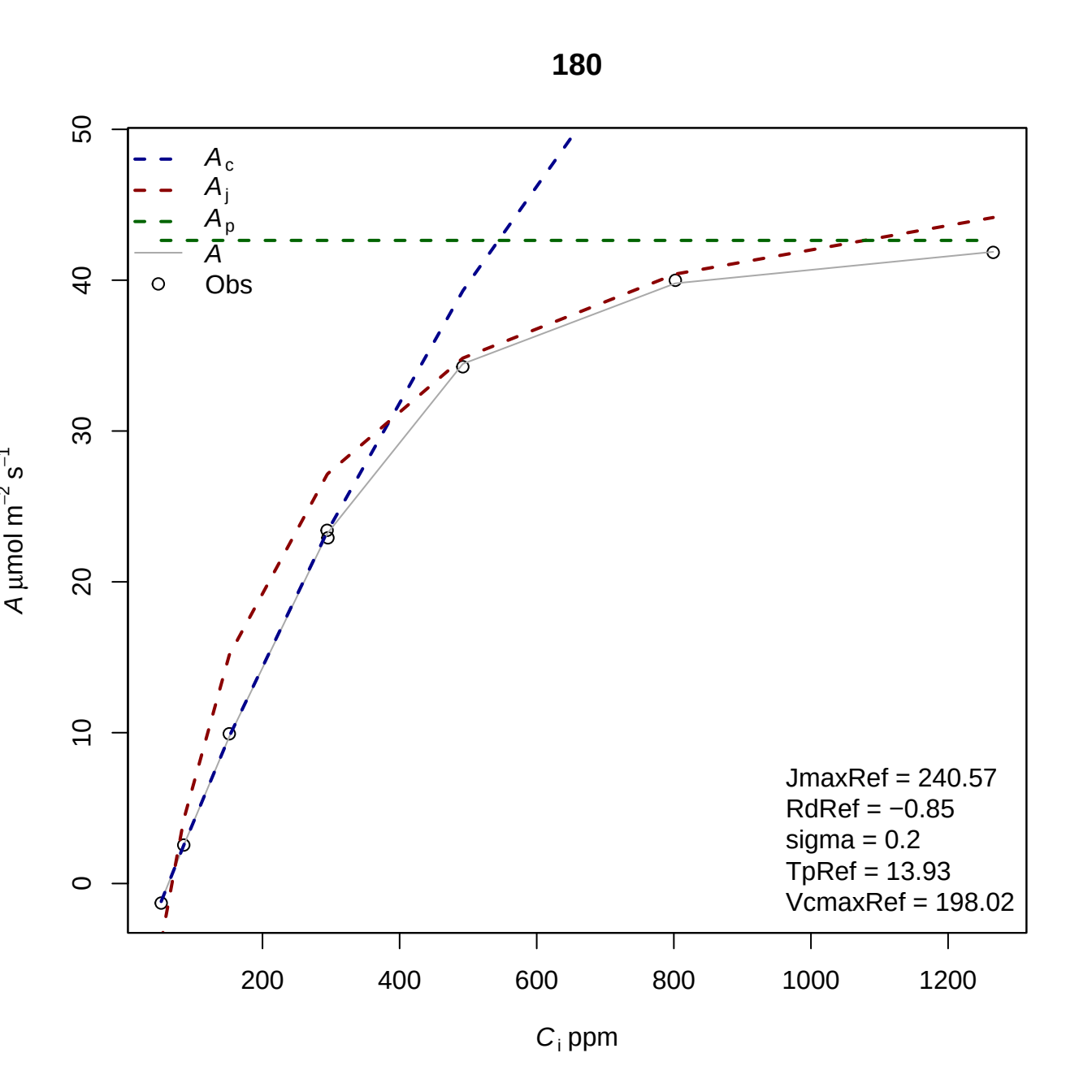

175
A_c
A_j
A_p
A
Obs
JmaxRef = 71.02
RdRef = 0.12
sigma = 0.34
TpRef = 5.48
VcmaxRef = 33.7
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
200
400
600
800
1000
1200
1400

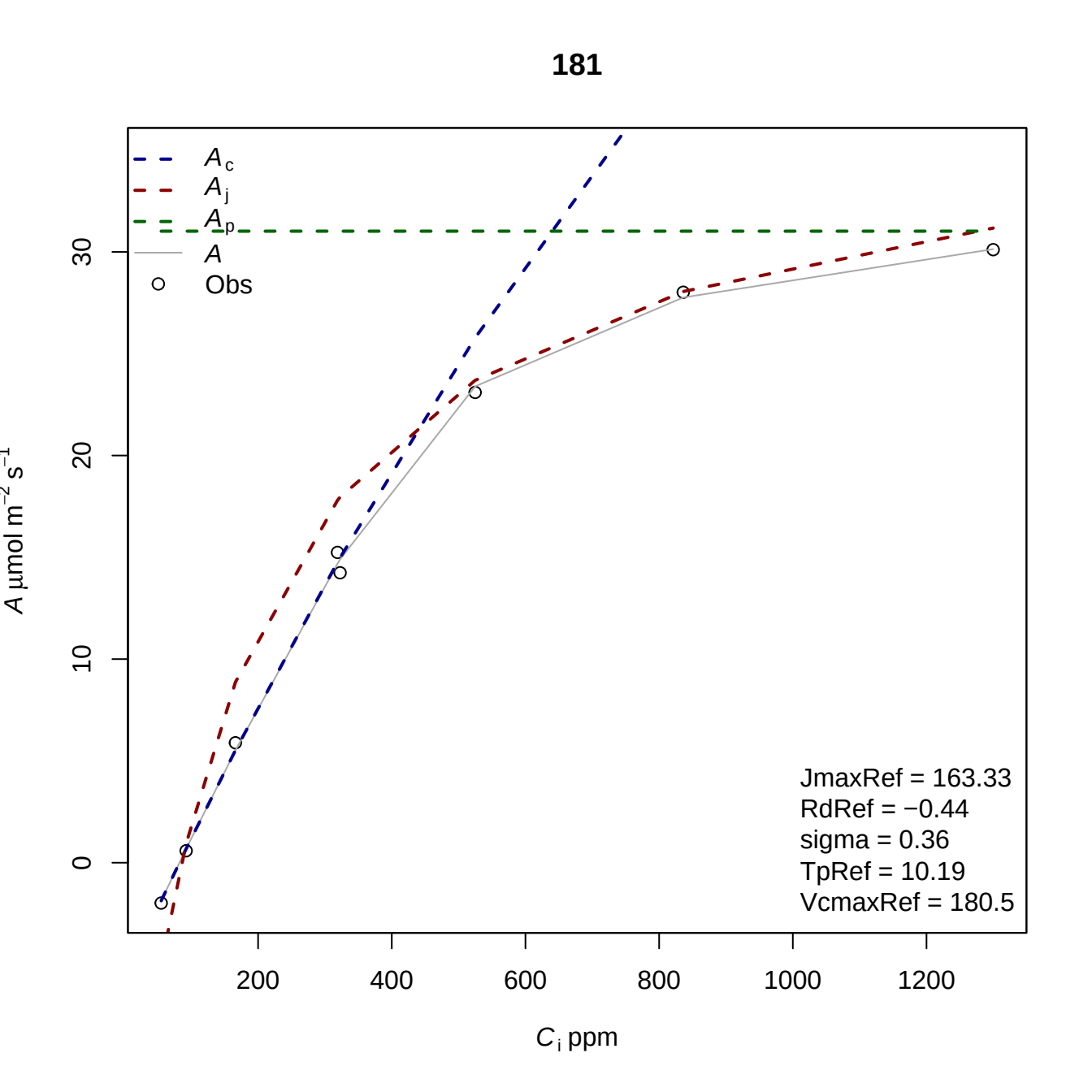

176
A_c
A_j
A_p
A
Obs
JmaxRef = 110.26
RdRef = −0.33
sigma = 0.34
TpRef = 6.85
VcmaxRef = 131.43
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
200
400
600
800
1000
1200

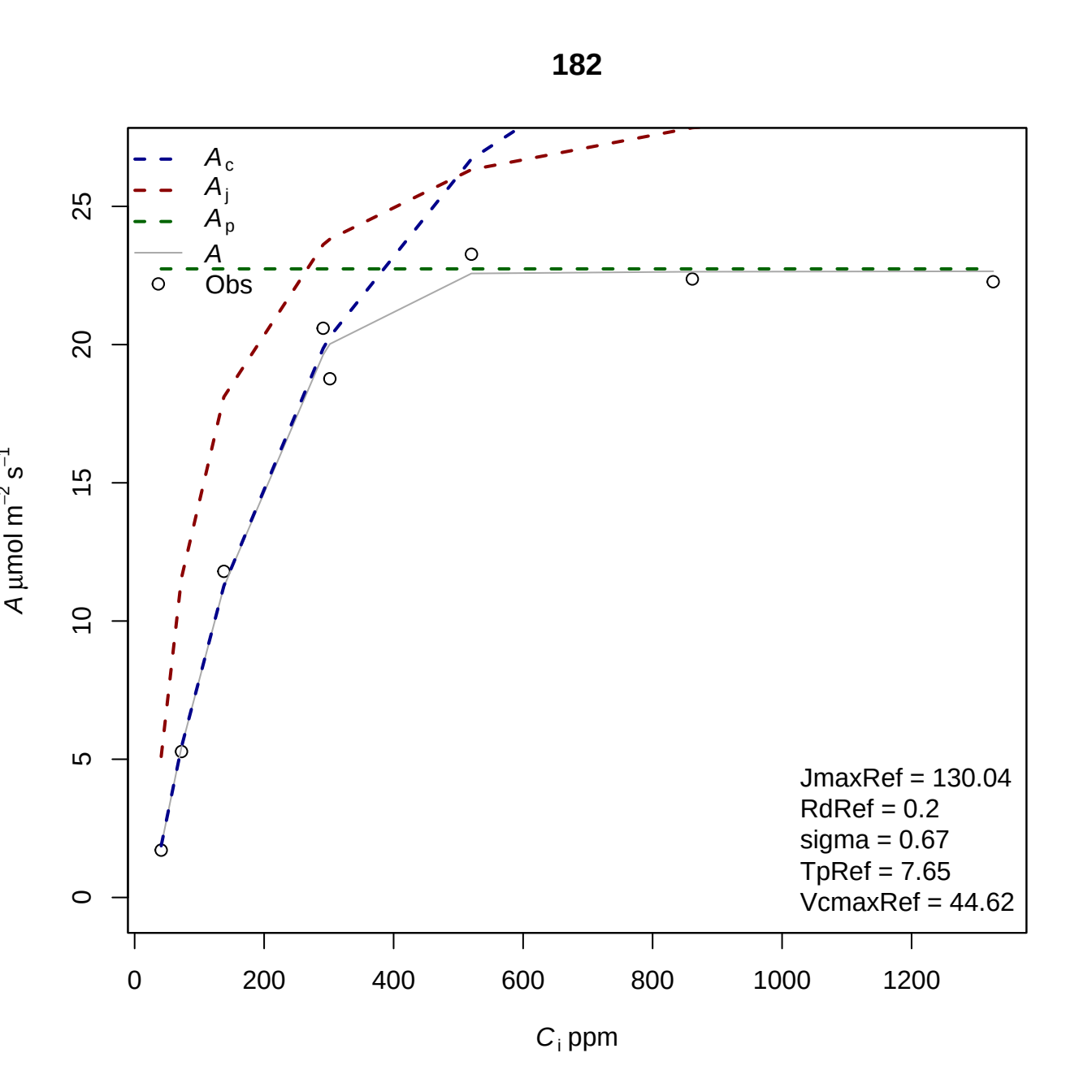

177
A_c
A_j
A_p
A
Obs
JmaxRef = 126.6
RdRef = −0.1
sigma = 0.64
TpRef = 8.12
VcmaxRef = 57.05
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200
1400

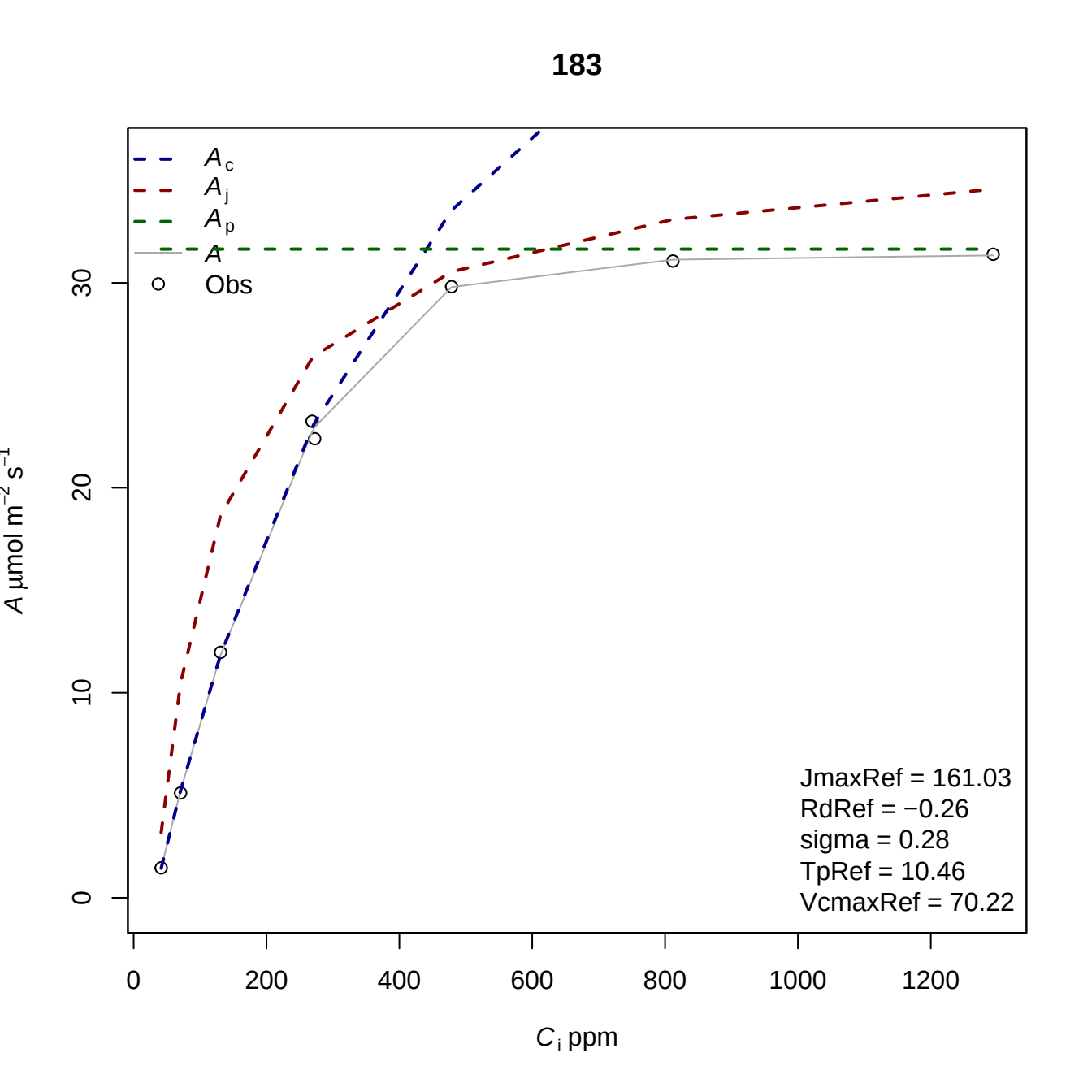

178
A_c
A_j
A_p
A
Obs
JmaxRef = 217
RdRef = −0.75
sigma = 0.38
TpRef = 13.35
VcmaxRef = 112.53
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

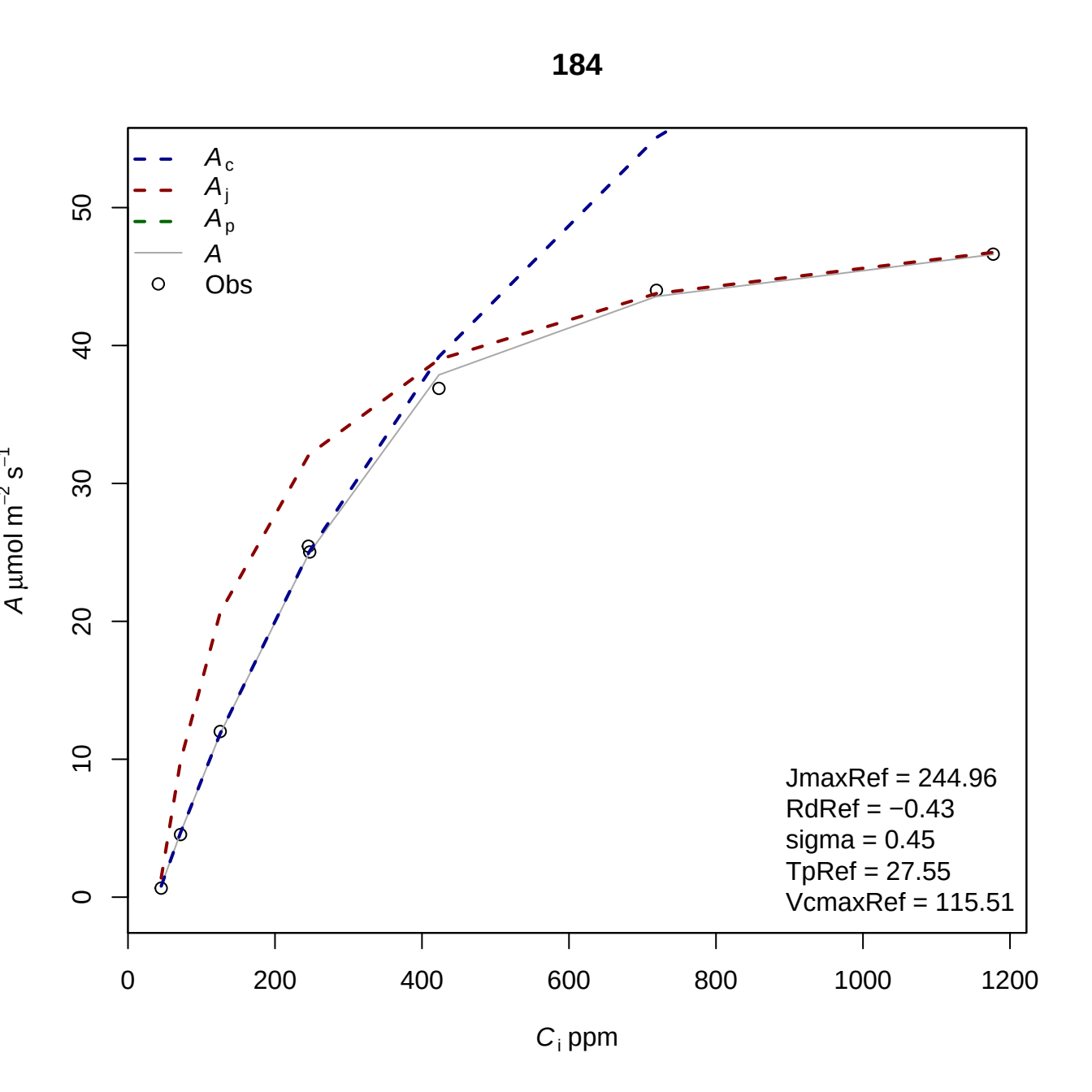

179
A_c
A_j
A_p
A
Obs
JmaxRef = 211.84
RdRef = −0.84
sigma = 0.19
TpRef = 13.13
VcmaxRef = 141.37
0
10
20
30
40
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

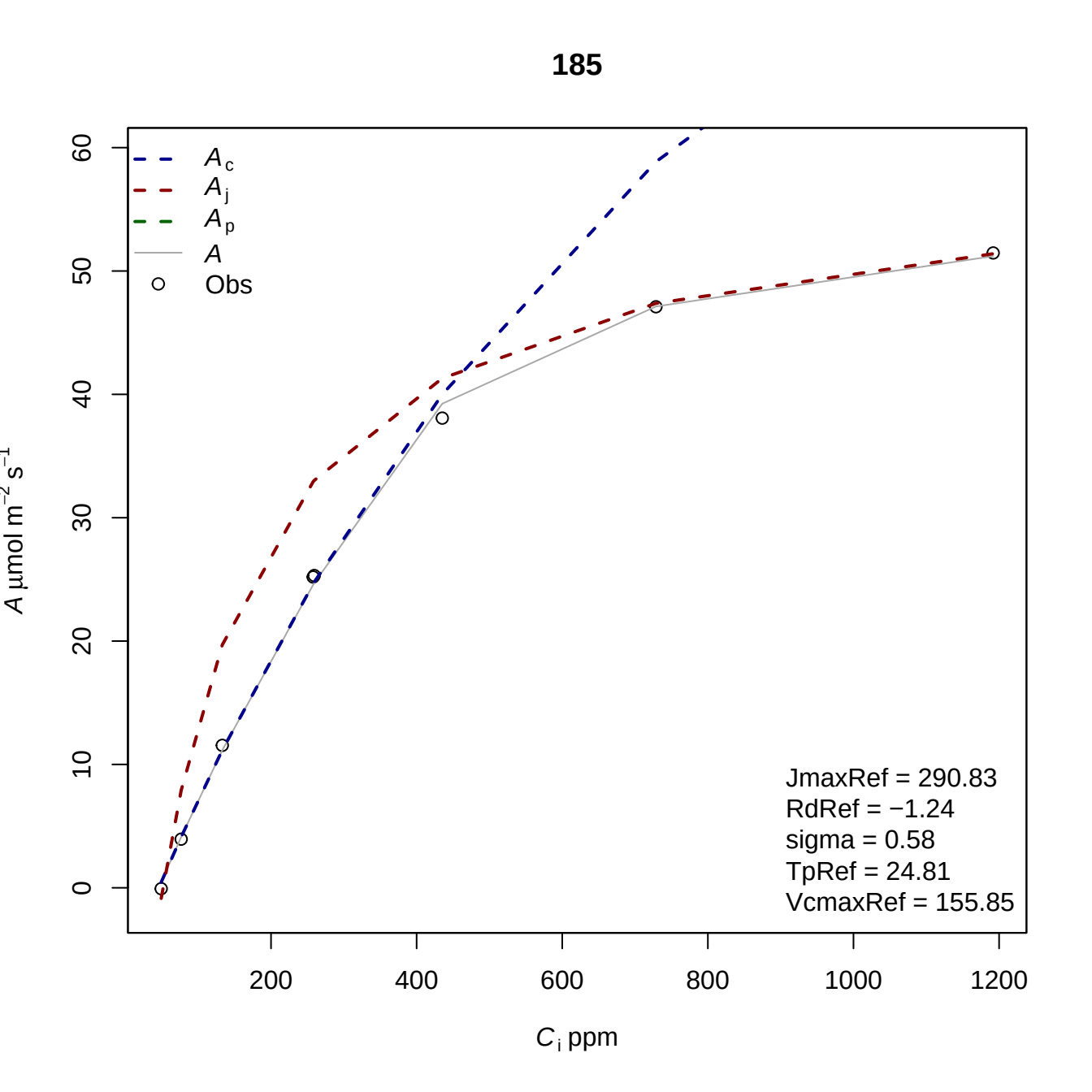

180
A_c
A_j
A_p
A
Obs
JmaxRef = 240.57
RdRef = −0.85
sigma = 0.2
TpRef = 13.93
VcmaxRef = 198.02
0
10
20
30
40
50
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

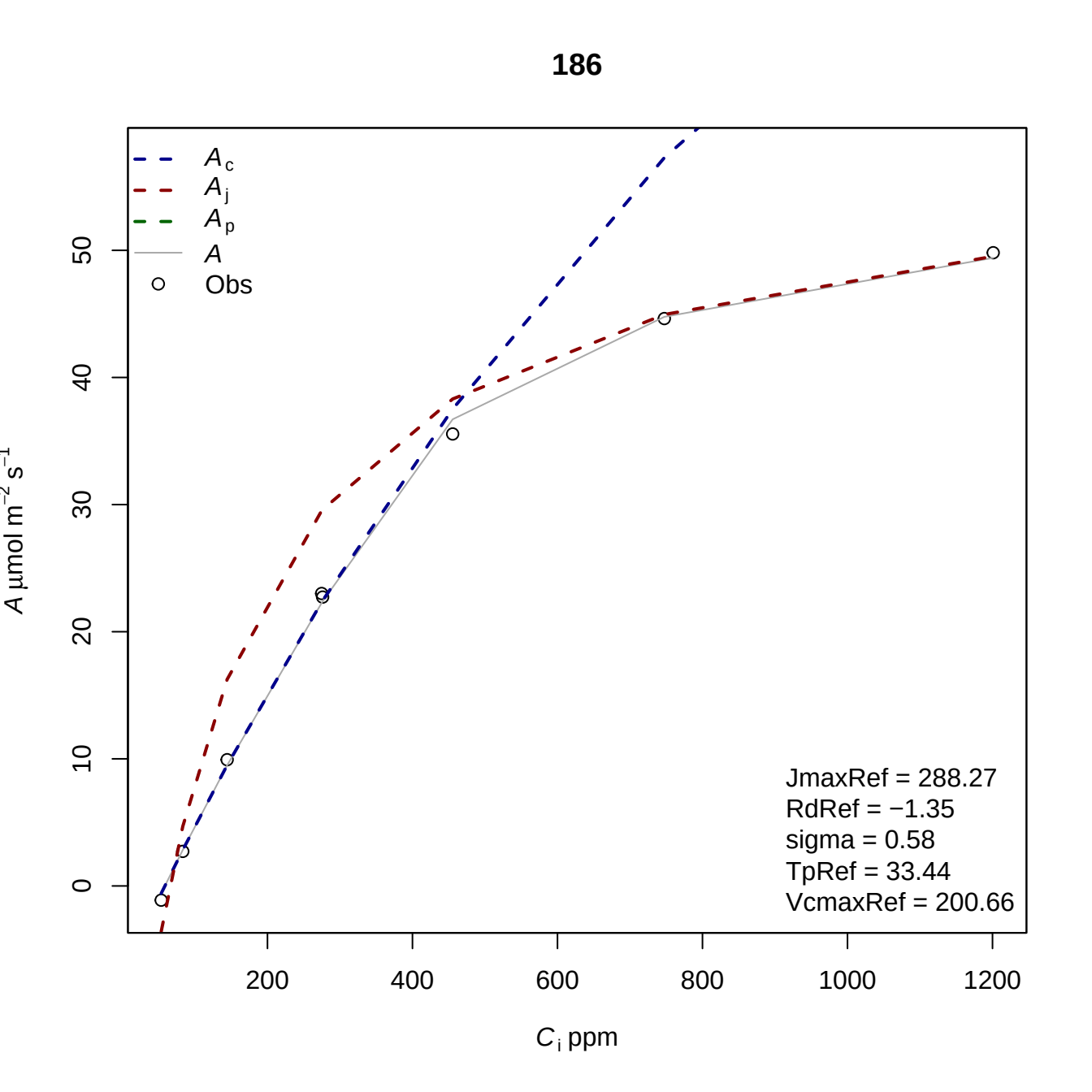

181
A_c
A_j
A_p
A
Obs
JmaxRef = 163.33
RdRef = −0.44
sigma = 0.36
TpRef = 10.19
VcmaxRef = 180.5
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
200
400
600
800
1000
1200

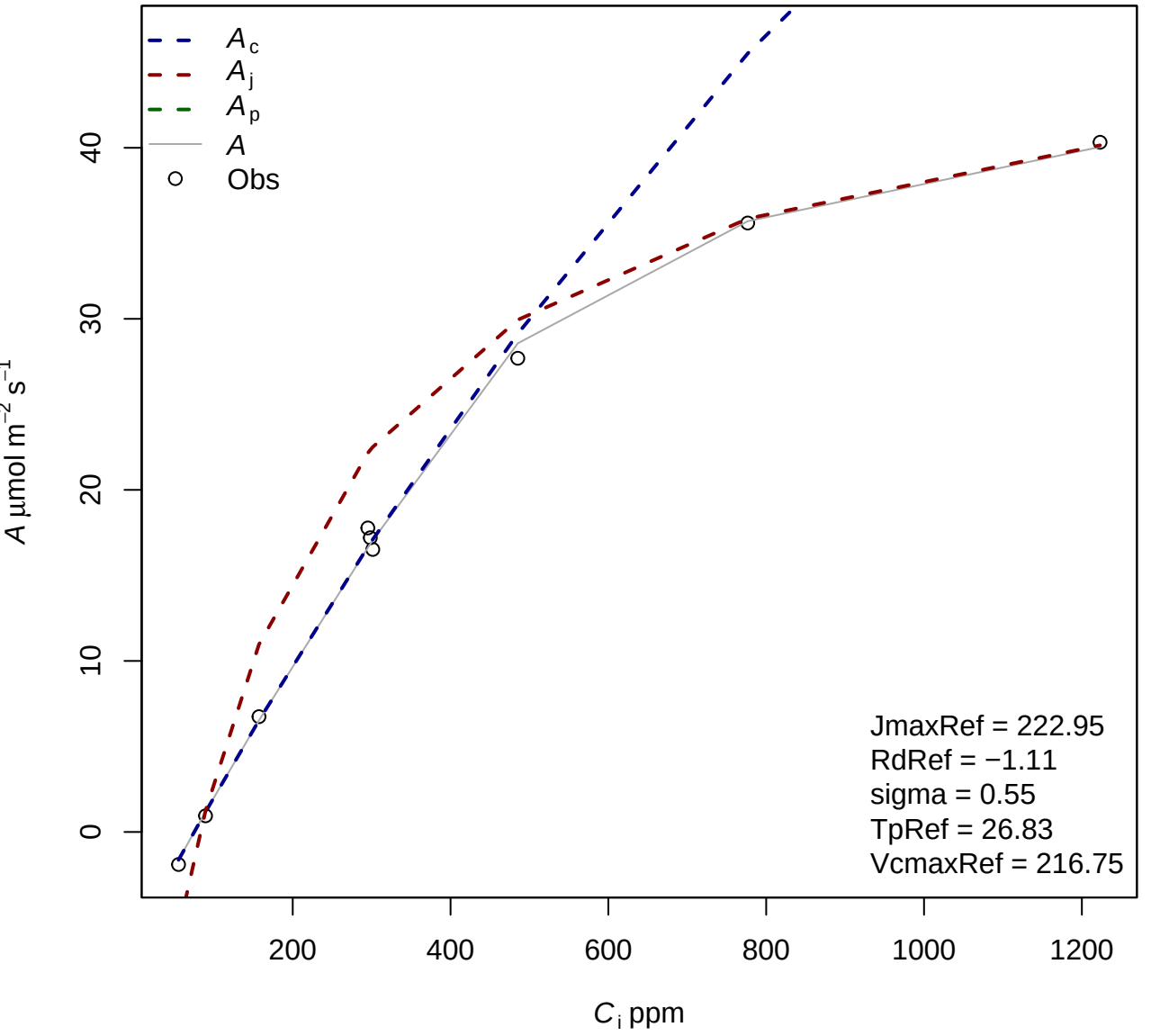

182
A_c
A_j
A_p
A
Obs
JmaxRef = 130.04
RdRef = 0.2
sigma = 0.67
TpRef = 7.65
VcmaxRef = 44.62
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200

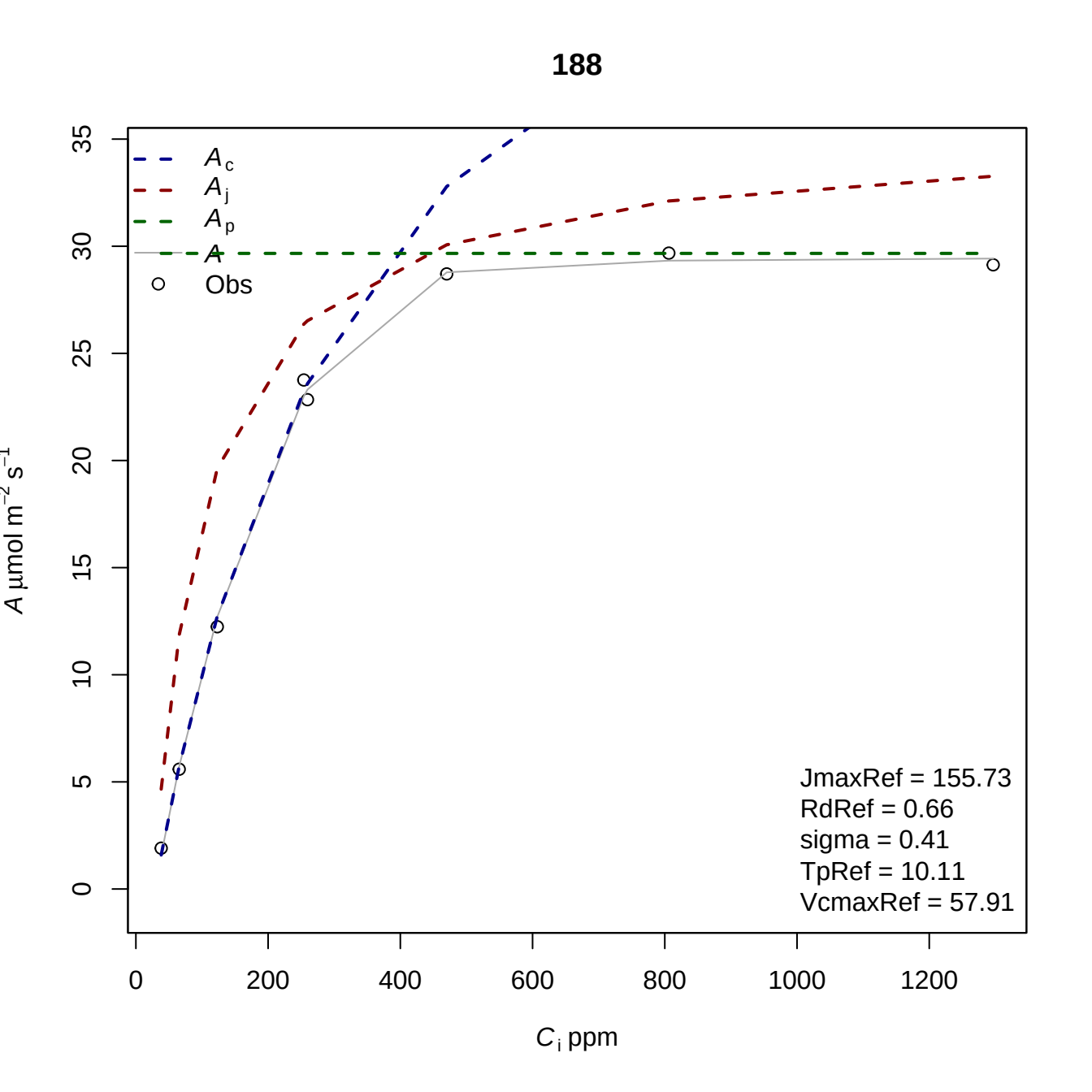

183
A_c
A_j
A_p
A
Obs
JmaxRef = 161.03
RdRef = −0.26
sigma = 0.28
TpRef = 10.46
VcmaxRef = 70.22
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
0
200
400
600
800
1000
1200

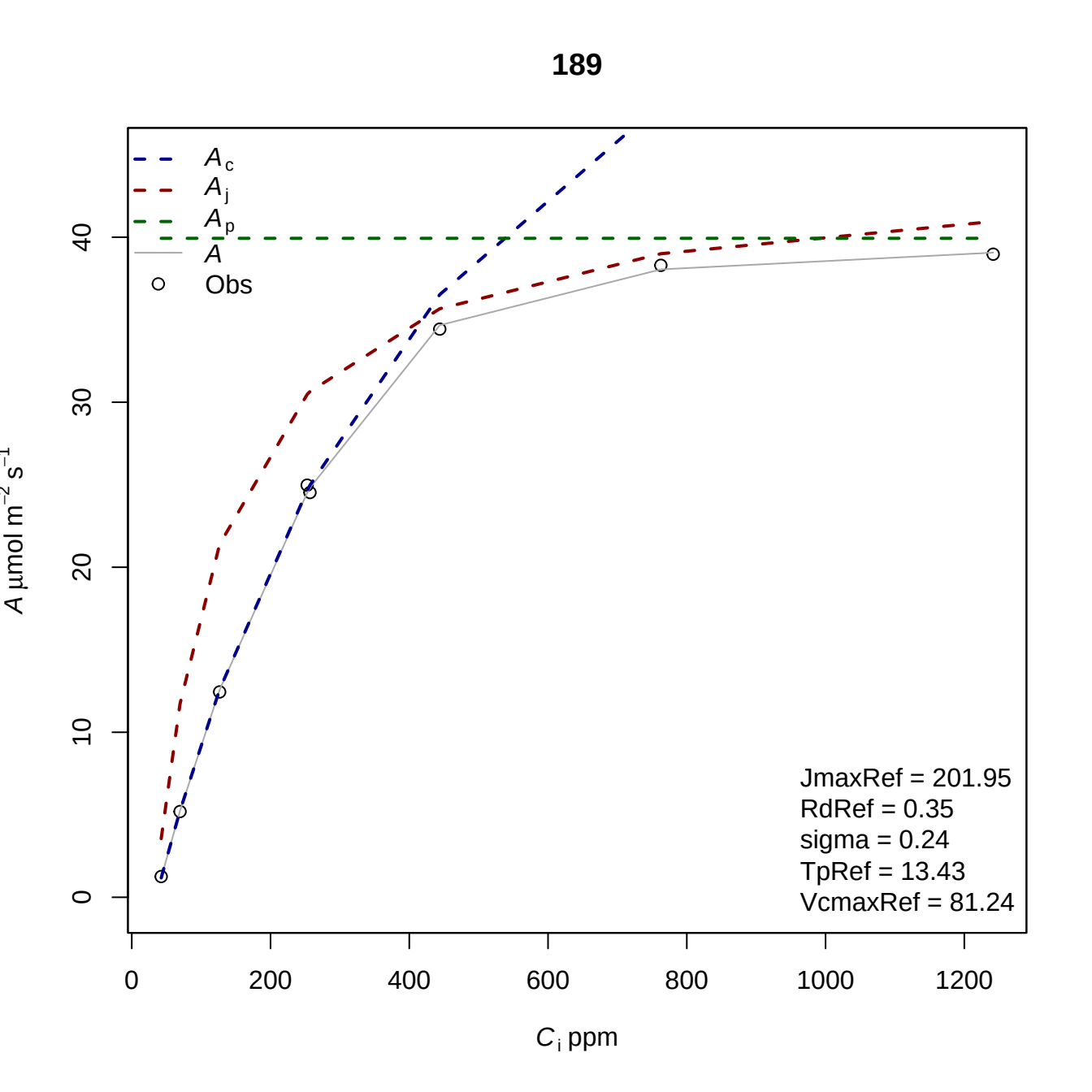

184
A_c
A_j
A_p
A
Obs
JmaxRef = 244.96
RdRef = −0.43
sigma = 0.45
TpRef = 27.55
VcmaxRef = 115.51
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
50
0
200
400
600
800
1000
1200

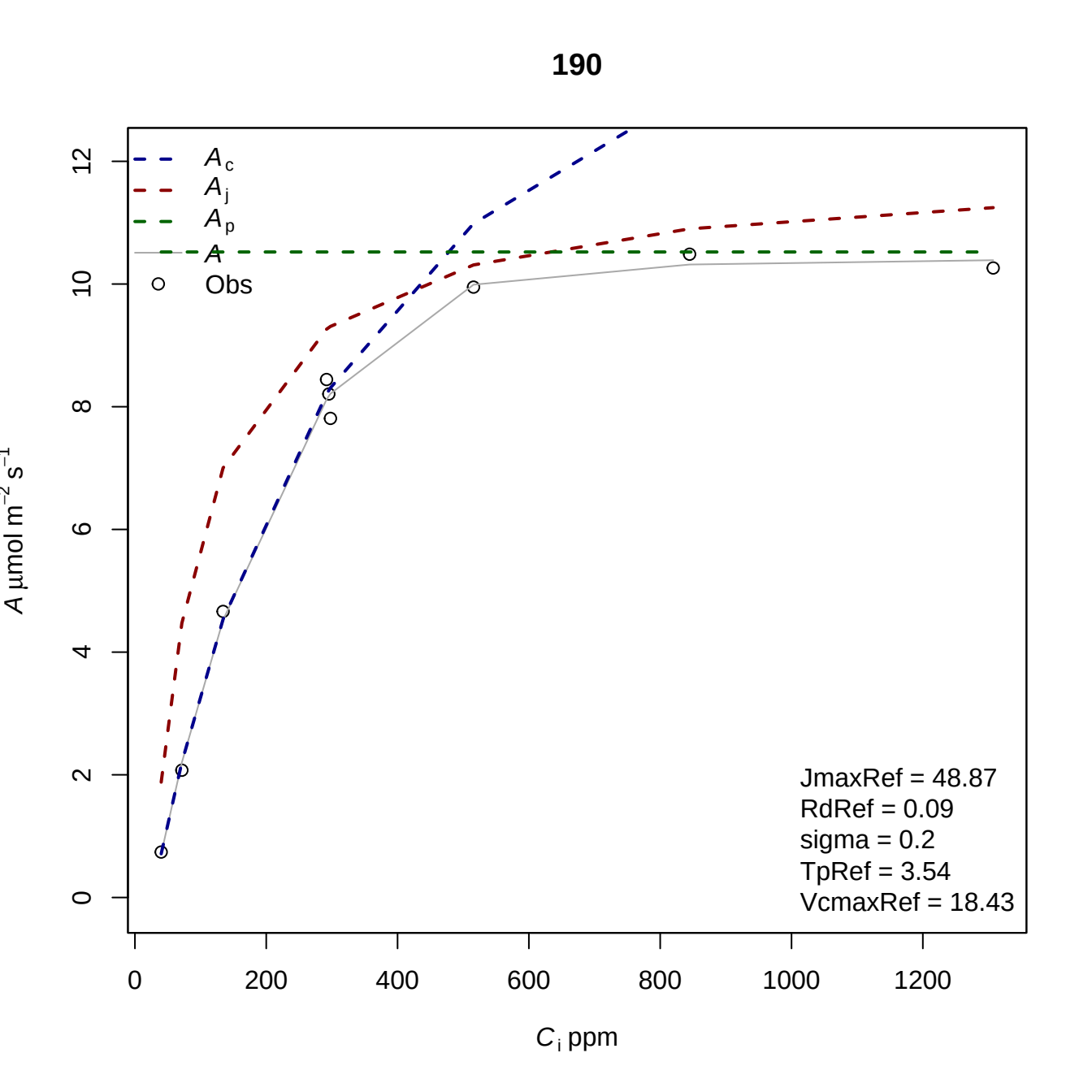

185
A_c
A_j
A_p
A
Obs
JmaxRef = 290.83
RdRef = −1.24
sigma = 0.58
TpRef = 24.81
VcmaxRef = 155.85
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
50
60
200
400
600
800
1000
1200

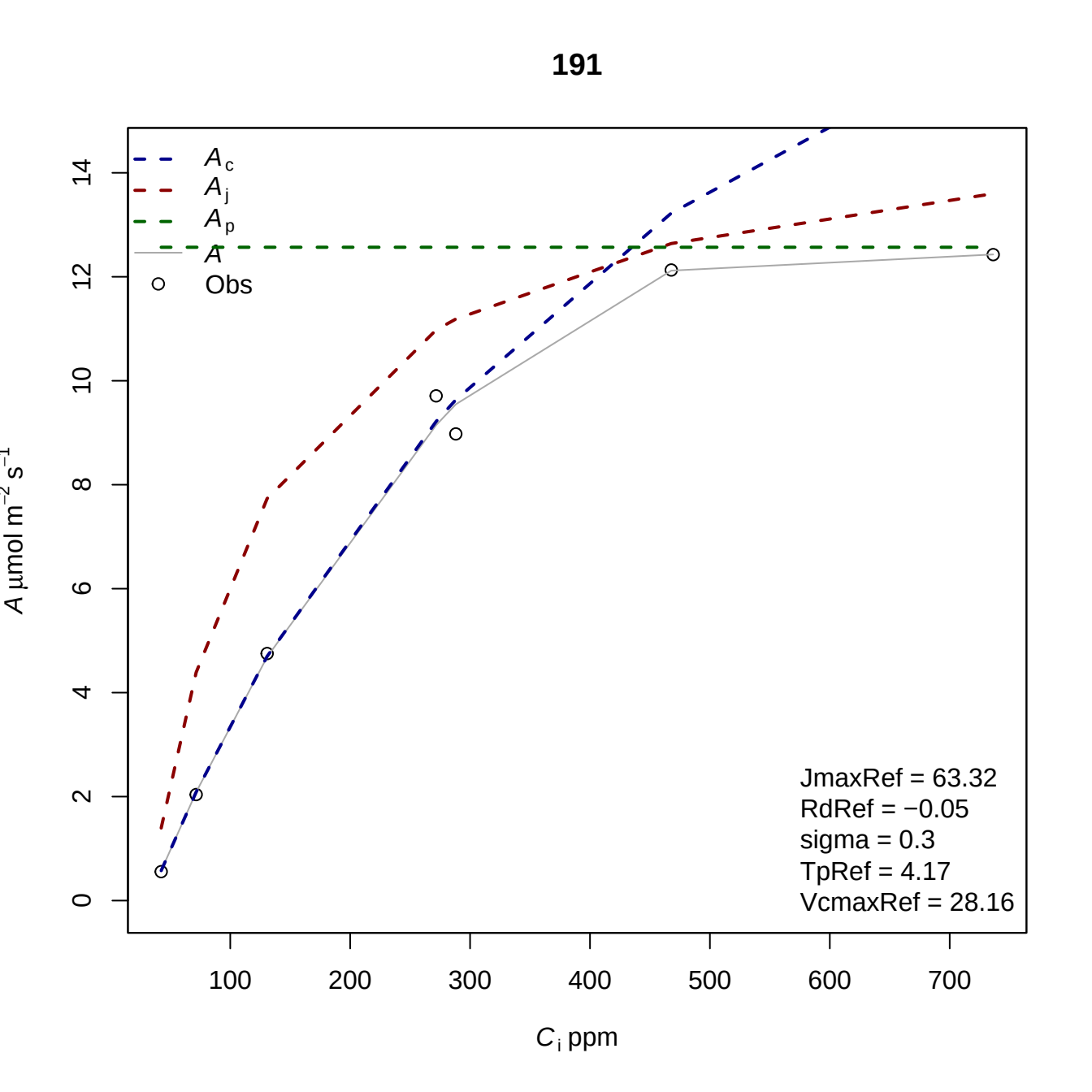

186
A_c
A_j
A_p
A
Obs
JmaxRef = 288.27
RdRef = −1.35
sigma = 0.58
TpRef = 33.44
VcmaxRef = 200.66
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
50
200
400
600
800
1000
1200

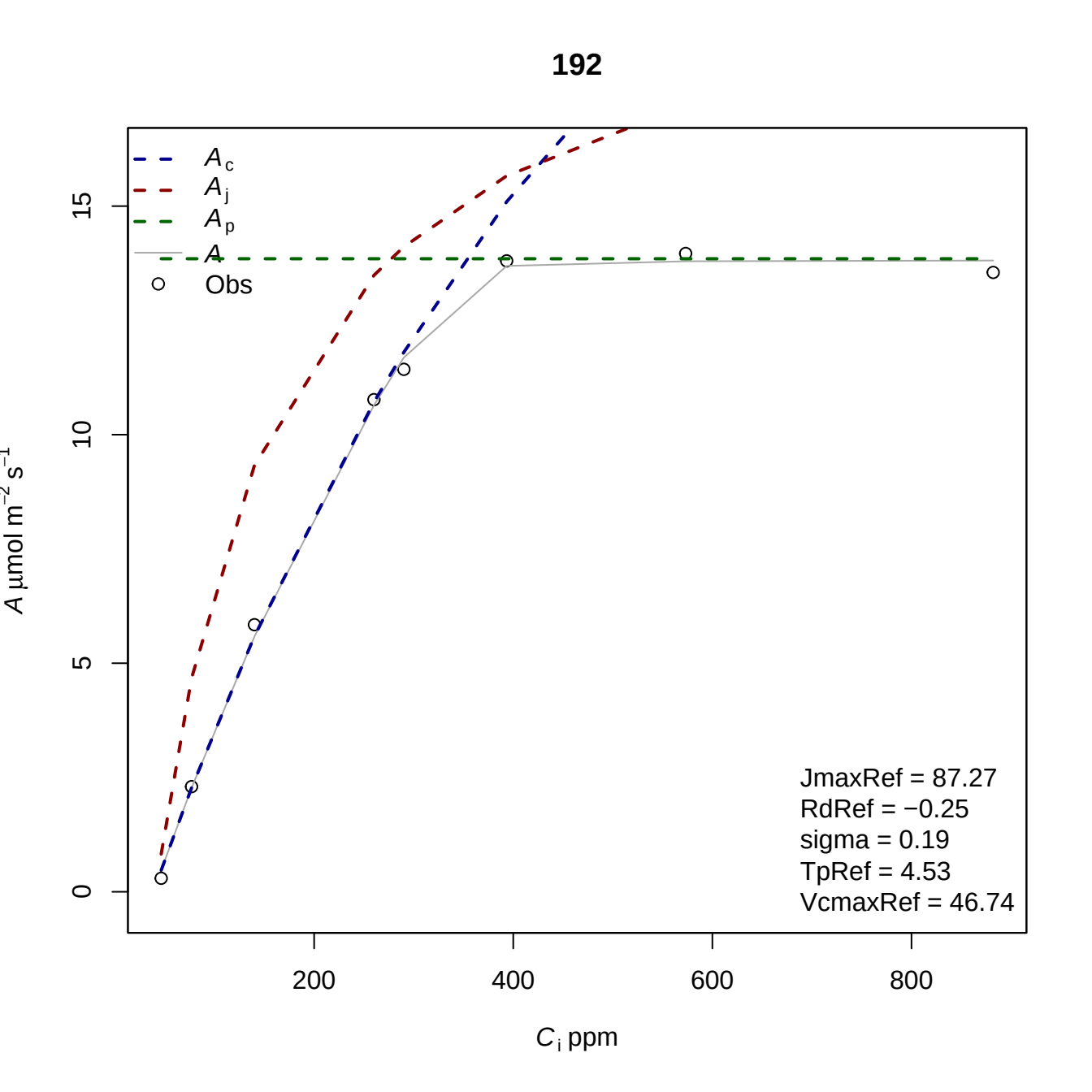

A_c
A_j
A_p
A
Obs
JmaxRef = 222.95
RdRef = −1.11
sigma = 0.55
TpRef = 26.83
VcmaxRef = 216.75
0
10
20
30
40
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

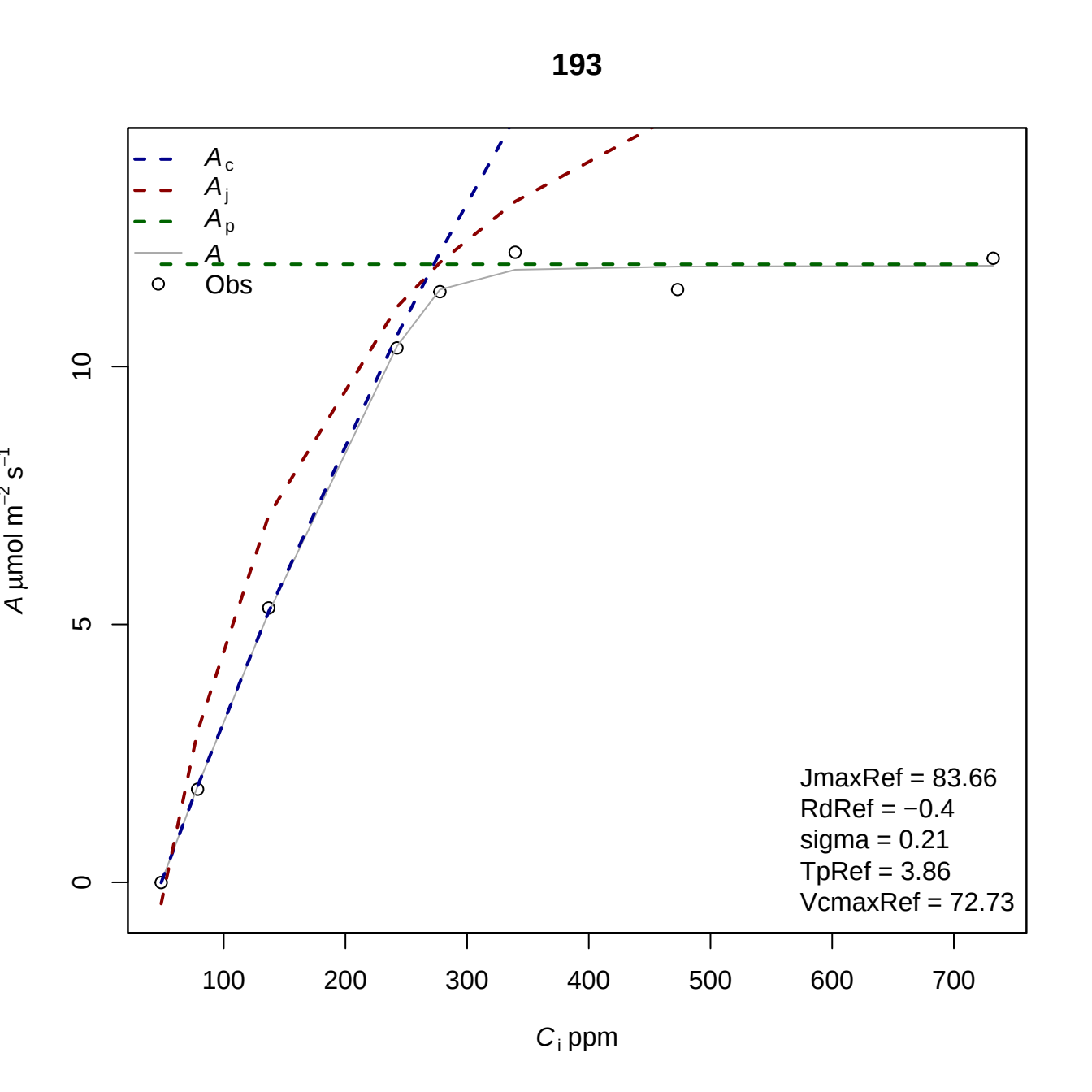

188
A_c
A_j
A_p
A
Obs
JmaxRef = 155.73
RdRef = 0.66
sigma = 0.41
TpRef = 10.11
VcmaxRef = 57.91
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200

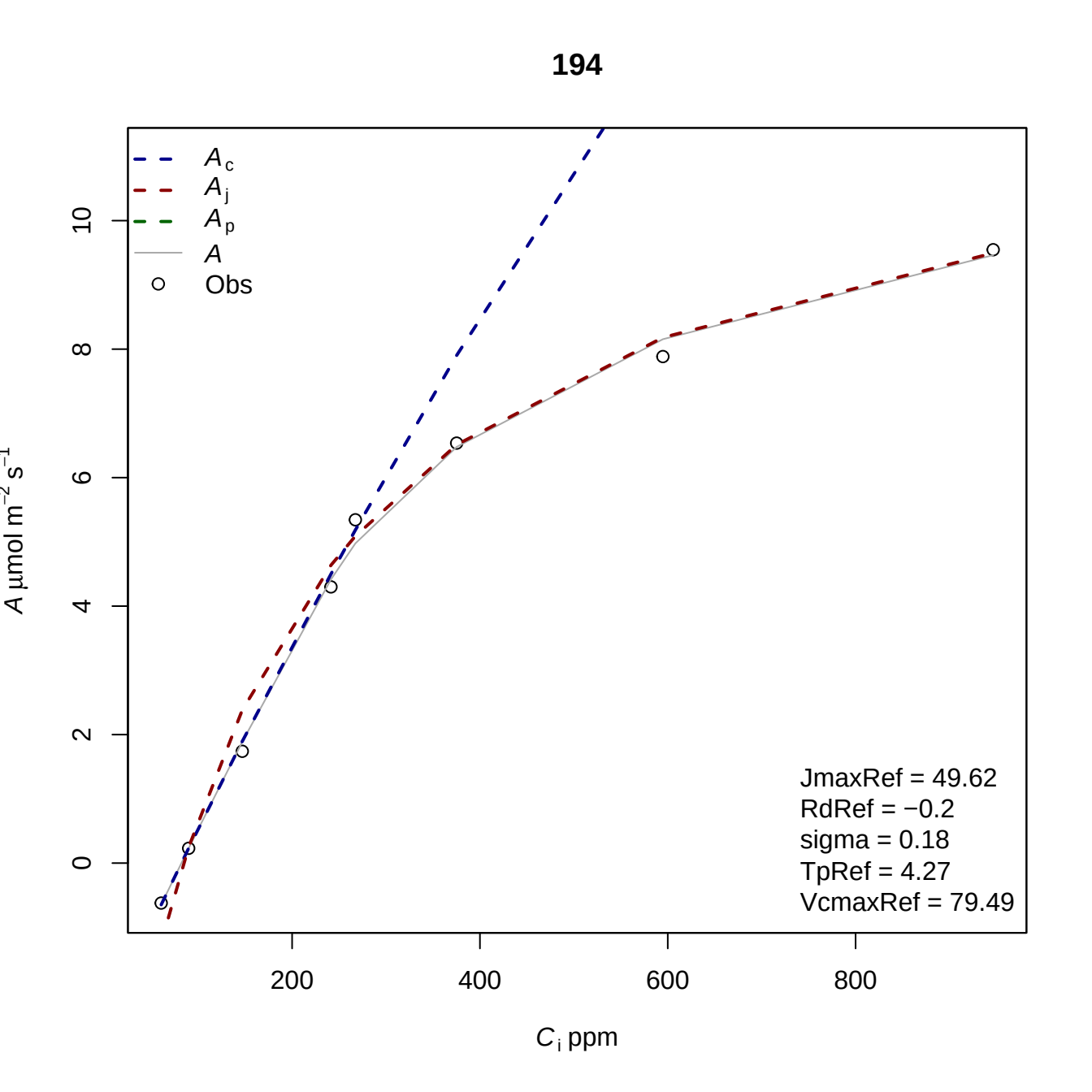

189
A_c
A_j
A_p
A
Obs
JmaxRef = 201.95
RdRef = 0.35
sigma = 0.24
TpRef = 13.43
VcmaxRef = 81.24
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
1200
C_i ppm

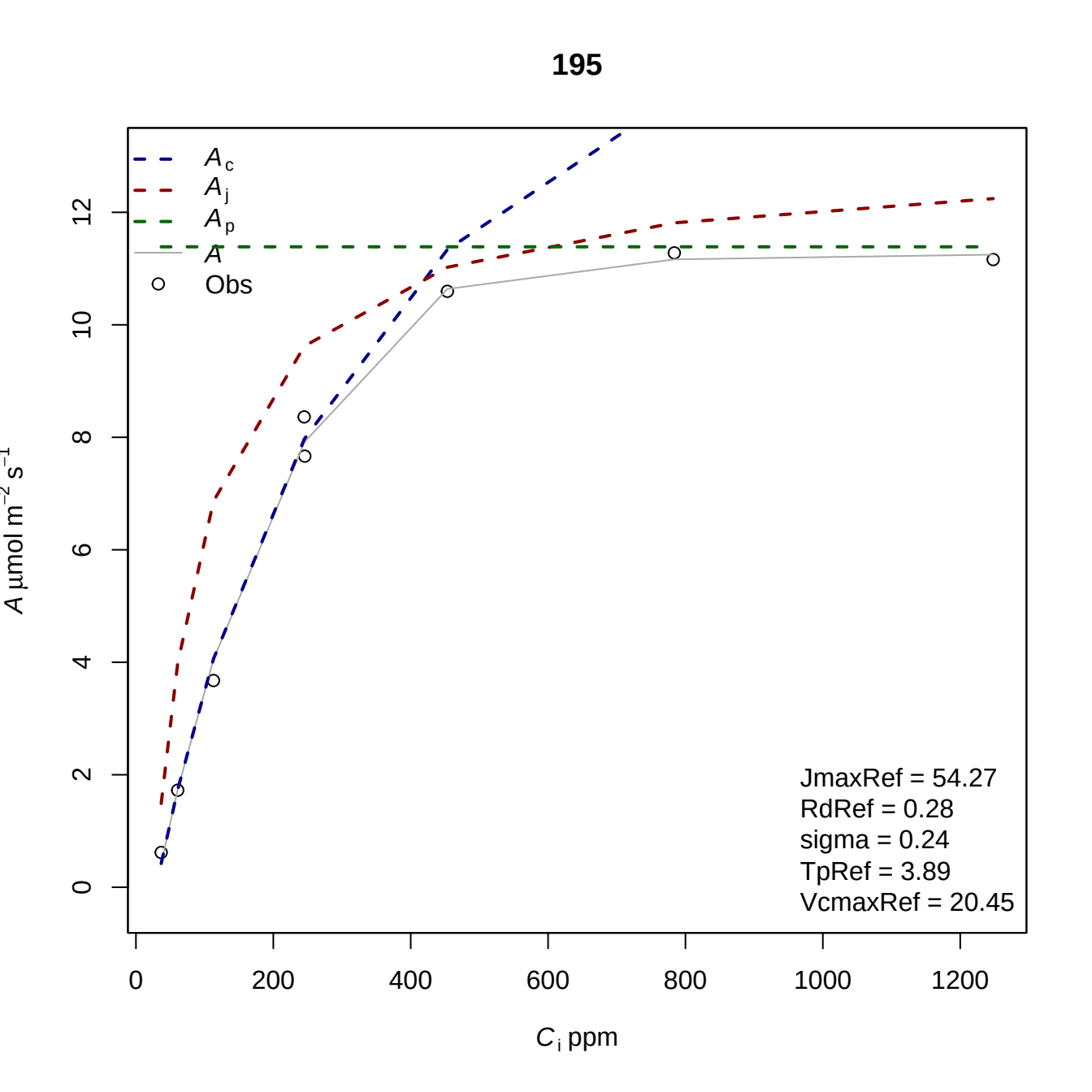

190
A_c
A_j
A_p
A
Obs
JmaxRef = 48.87
RdRef = 0.09
sigma = 0.2
TpRef = 3.54
VcmaxRef = 18.43
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

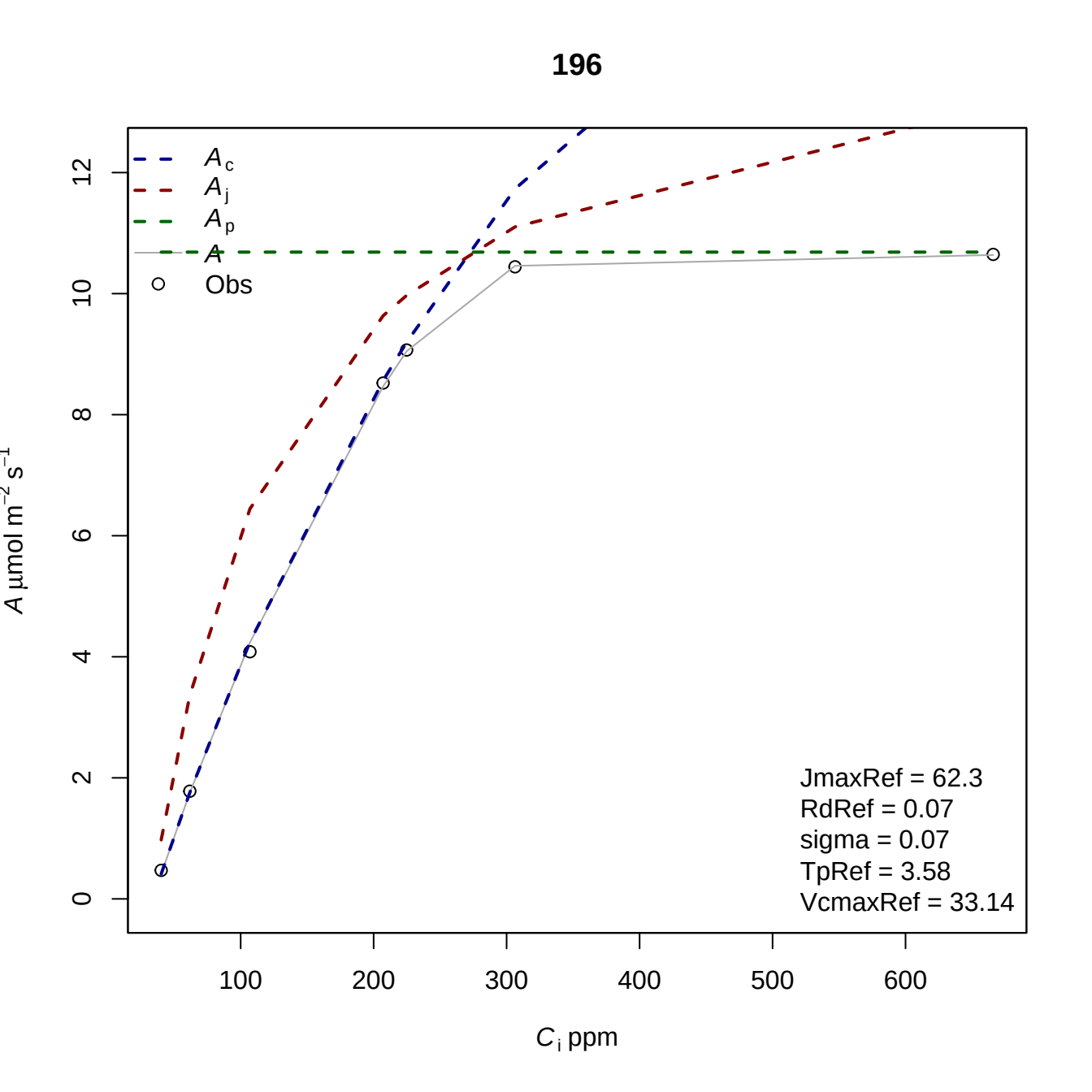

191
A_c
A_j
A_p
A
Obs
JmaxRef = 63.32
RdRef = −0.05
sigma = 0.3
TpRef = 4.17
VcmaxRef = 28.16
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
14
100
200
300
400
500
600
700

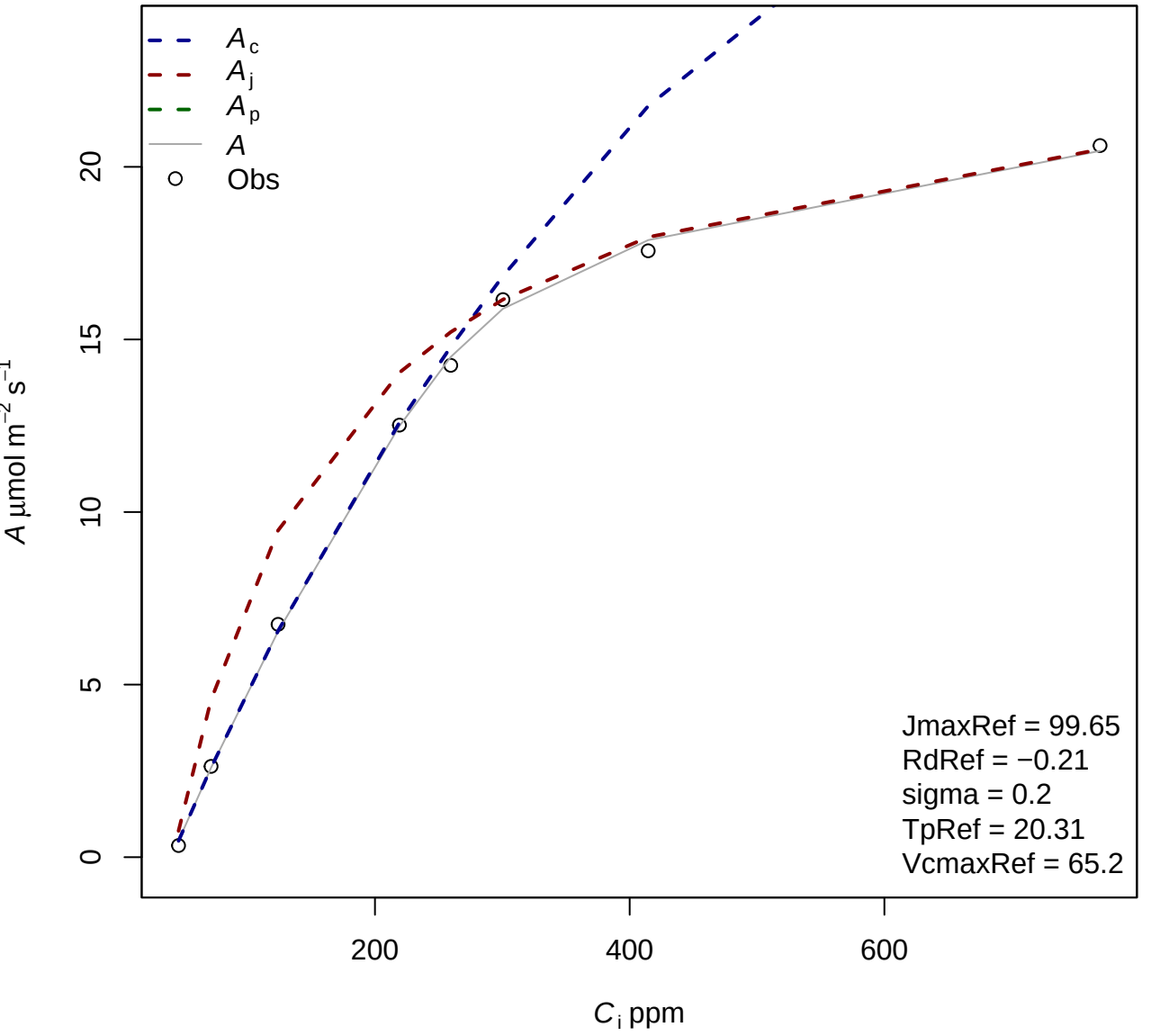

192
A_c
A_j
A_p
A
Obs
JmaxRef = 87.27
RdRef = −0.25
sigma = 0.19
TpRef = 4.53
VcmaxRef = 46.74
0
5
10
15
200
400
600
800
A µmol m^{-2} s^{-1}
C_i ppm

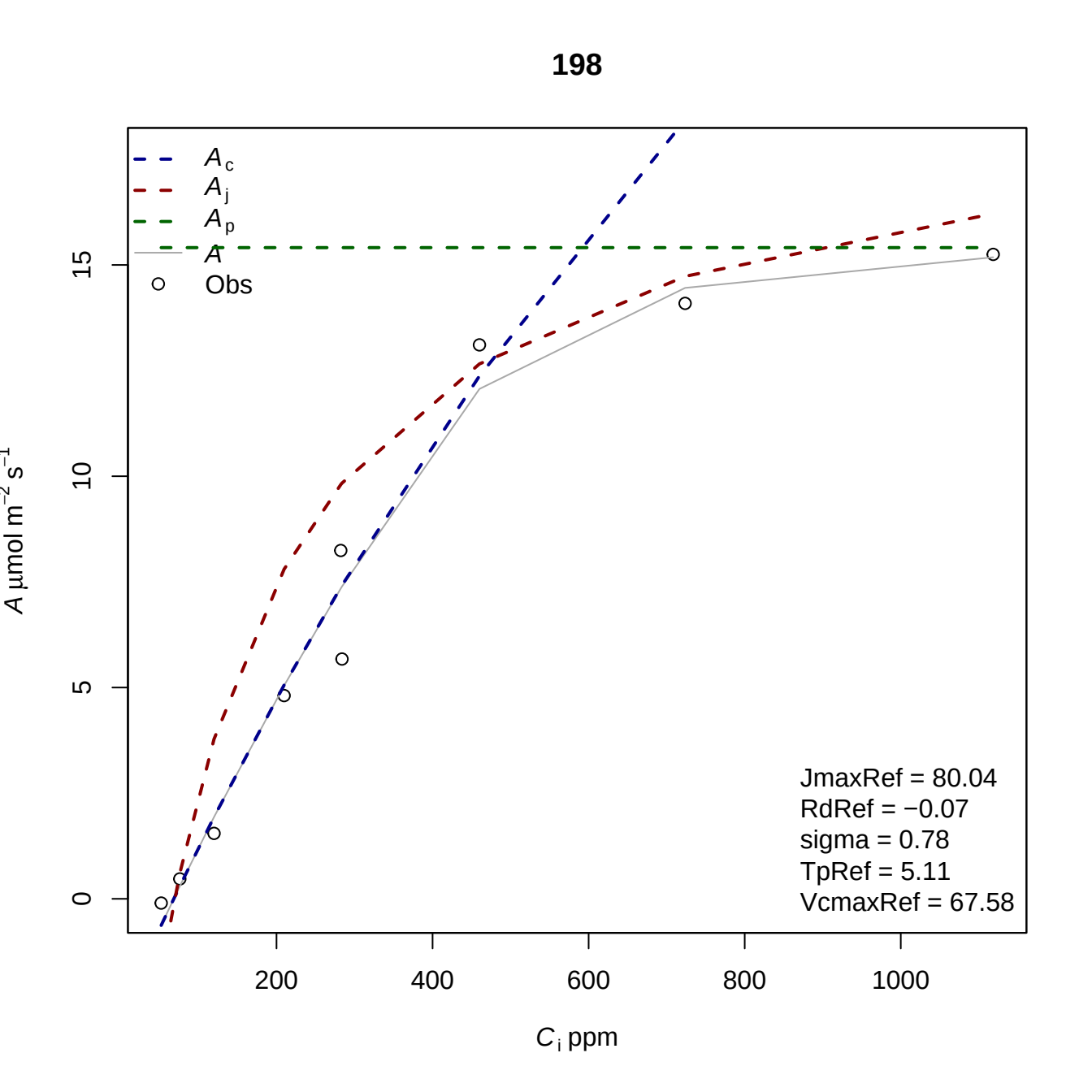

193
A_c
A_j
A_p
A
Obs
JmaxRef = 83.66
RdRef = −0.4
sigma = 0.21
TpRef = 3.86
VcmaxRef = 72.73
0
5
10
100
200
300
400
500
600
700
A µmol m^{-2} s^{-1}
C_i ppm

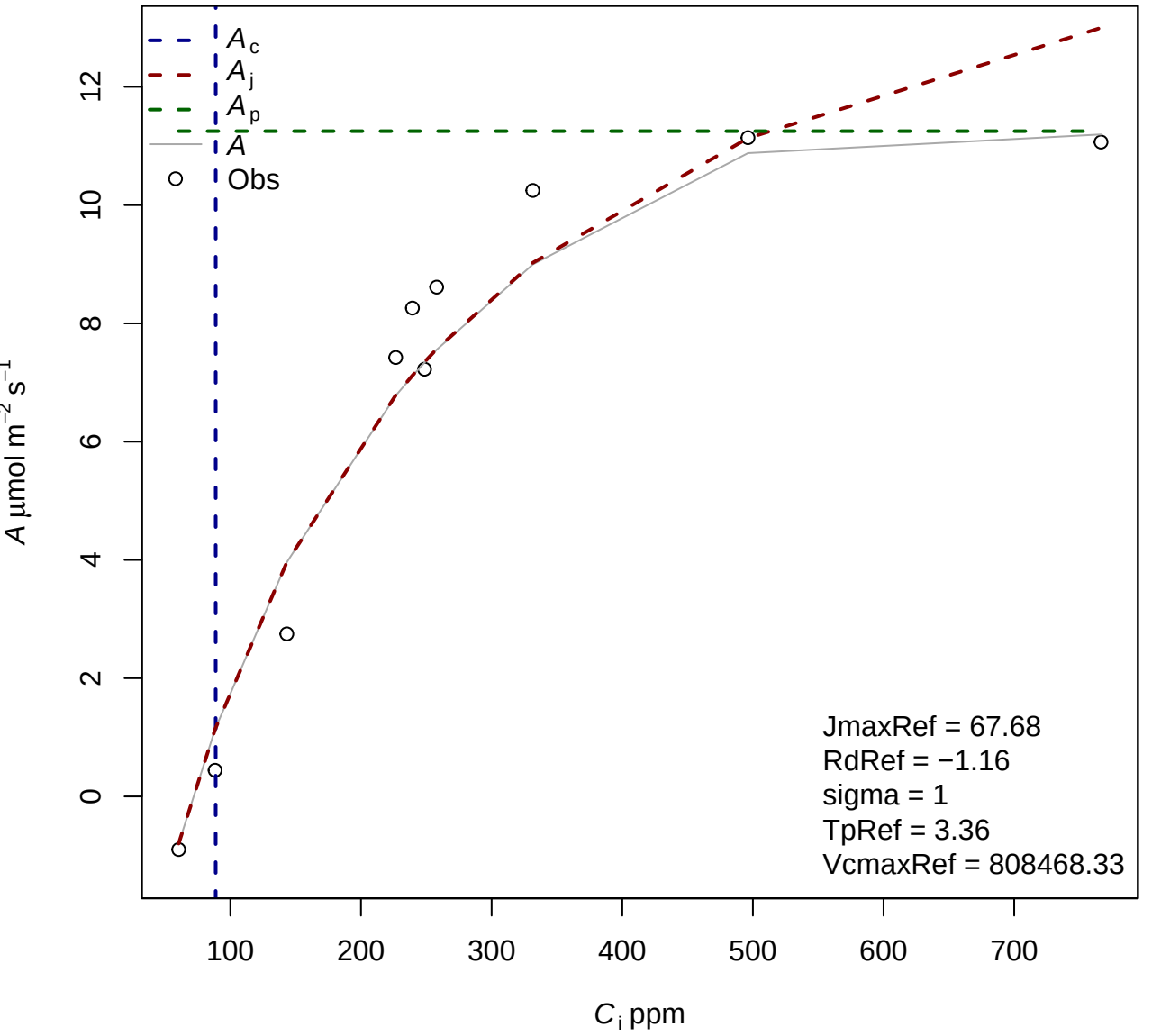

194
A_c
A_j
A_p
A
Obs
JmaxRef = 49.62
RdRef = −0.2
sigma = 0.18
TpRef = 4.27
VcmaxRef = 79.49
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
200
400
600
800

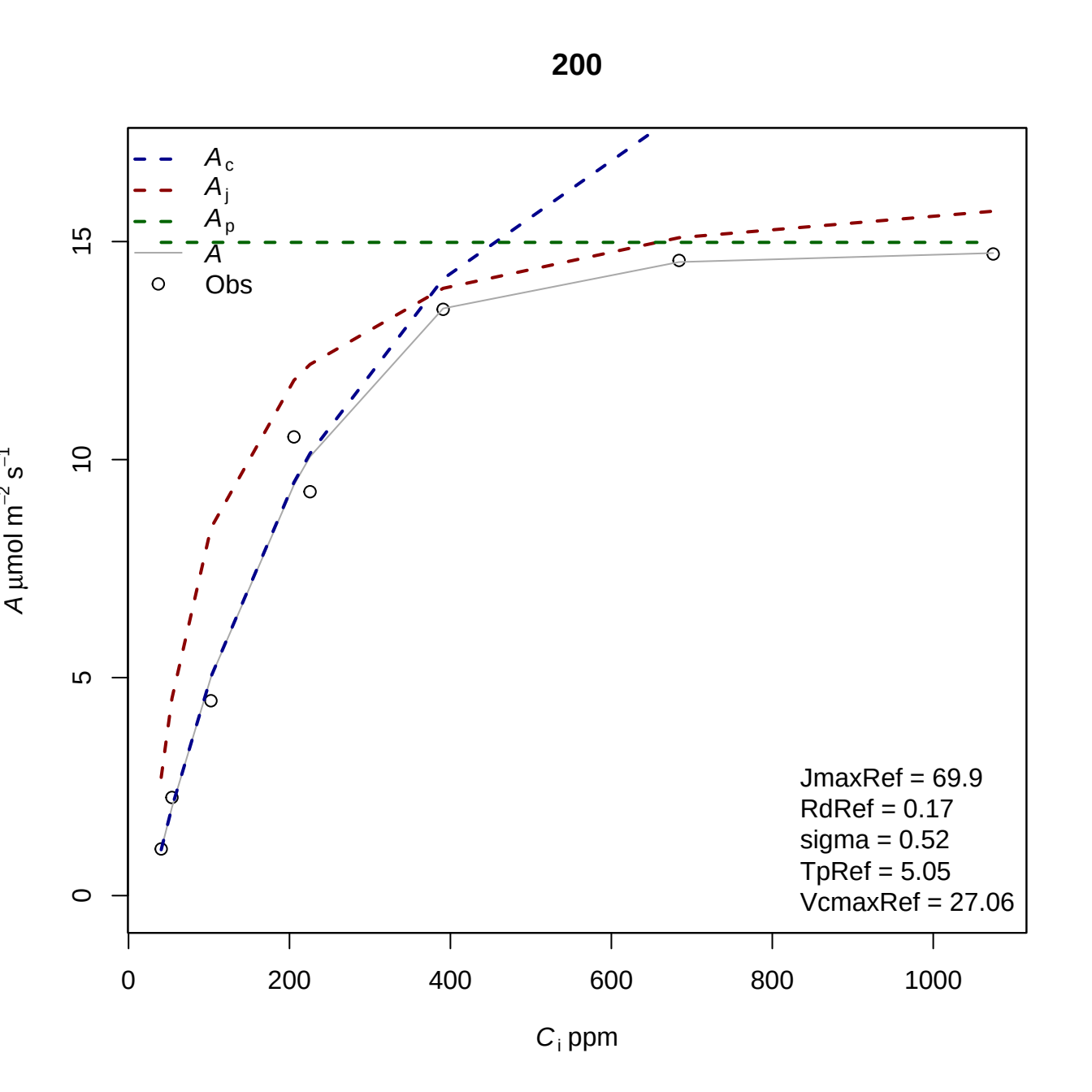

195
A_c
A_j
A_p
A
Obs
JmaxRef = 54.27
RdRef = 0.28
sigma = 0.24
TpRef = 3.89
VcmaxRef = 20.45
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

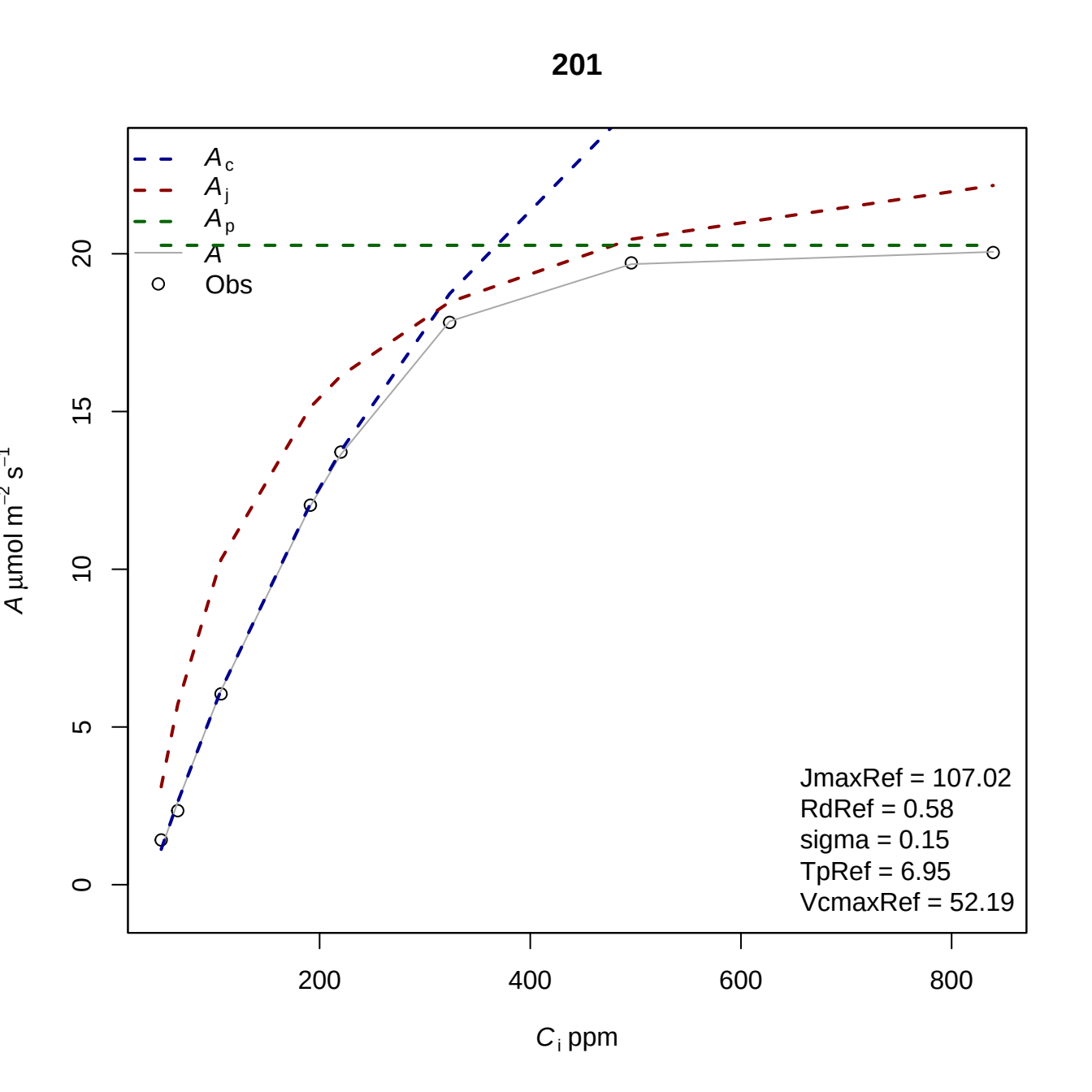

196
A_c
A_j
A_p
A
Obs
JmaxRef = 62.3
RdRef = 0.07
sigma = 0.07
TpRef = 3.58
VcmaxRef = 33.14
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
100
200
300
400
500
600

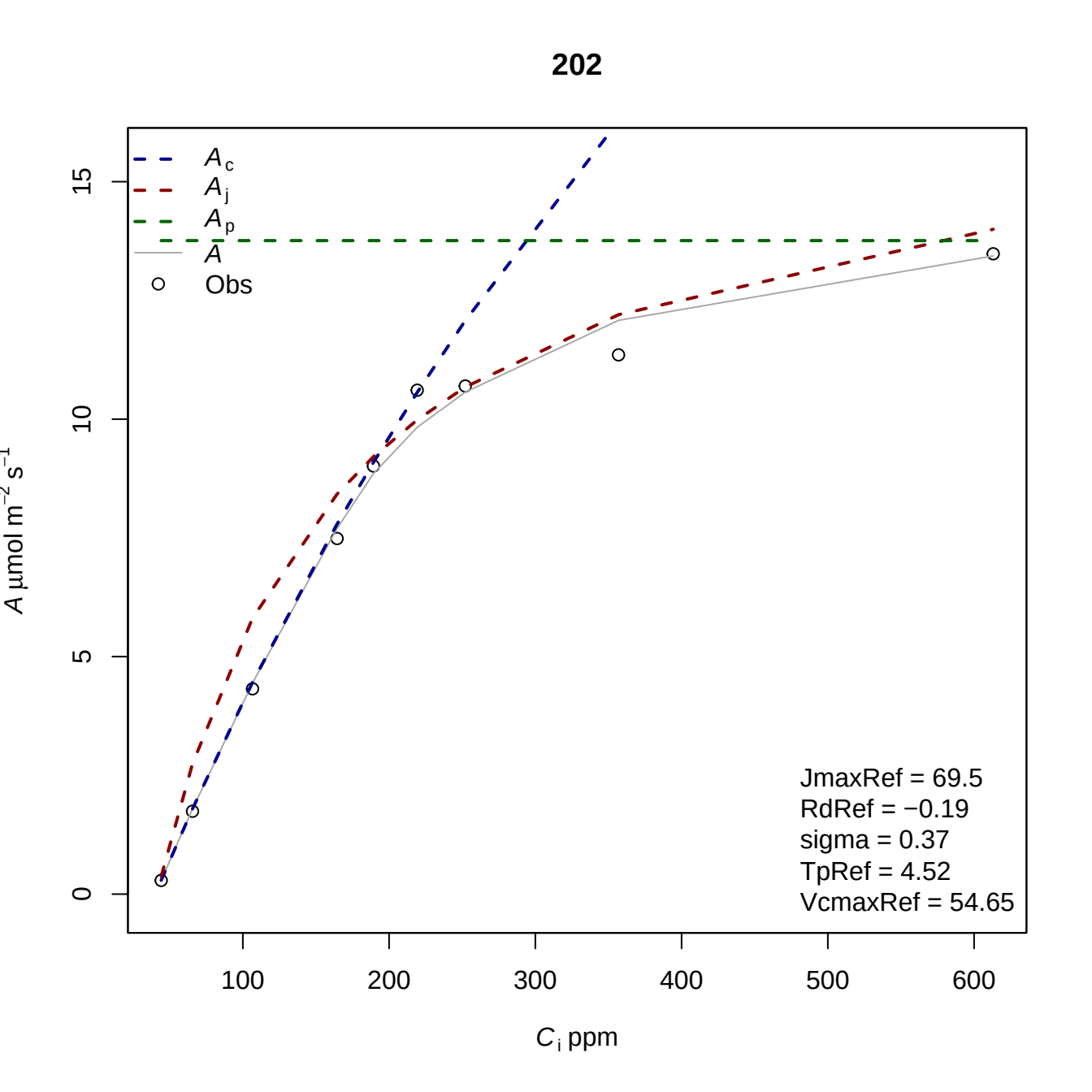

A_c
A_j
A_p
A
Obs
JmaxRef = 99.65
RdRef = −0.21
sigma = 0.2
TpRef = 20.31
VcmaxRef = 65.2
0
5
10
15
20
200
400
600
A µmol m^{-2} s^{-1}
C_i ppm

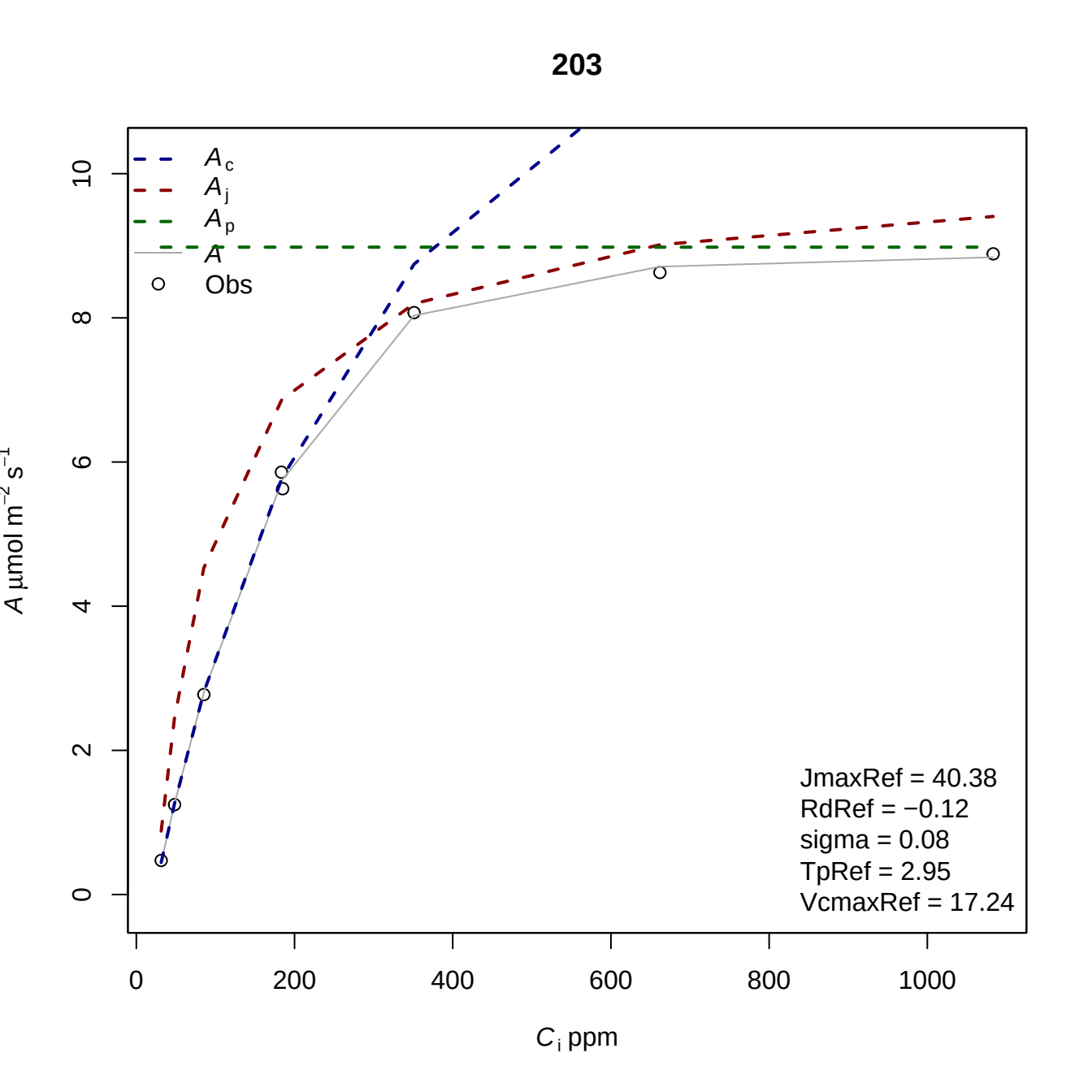

198
A_c
A_j
A_p
A
Obs
JmaxRef = 80.04
RdRef = −0.07
sigma = 0.78
TpRef = 5.11
VcmaxRef = 67.58
0
5
10
15
200
400
600
800
1000
A µmol m^{-2} s^{-1}
C_i ppm

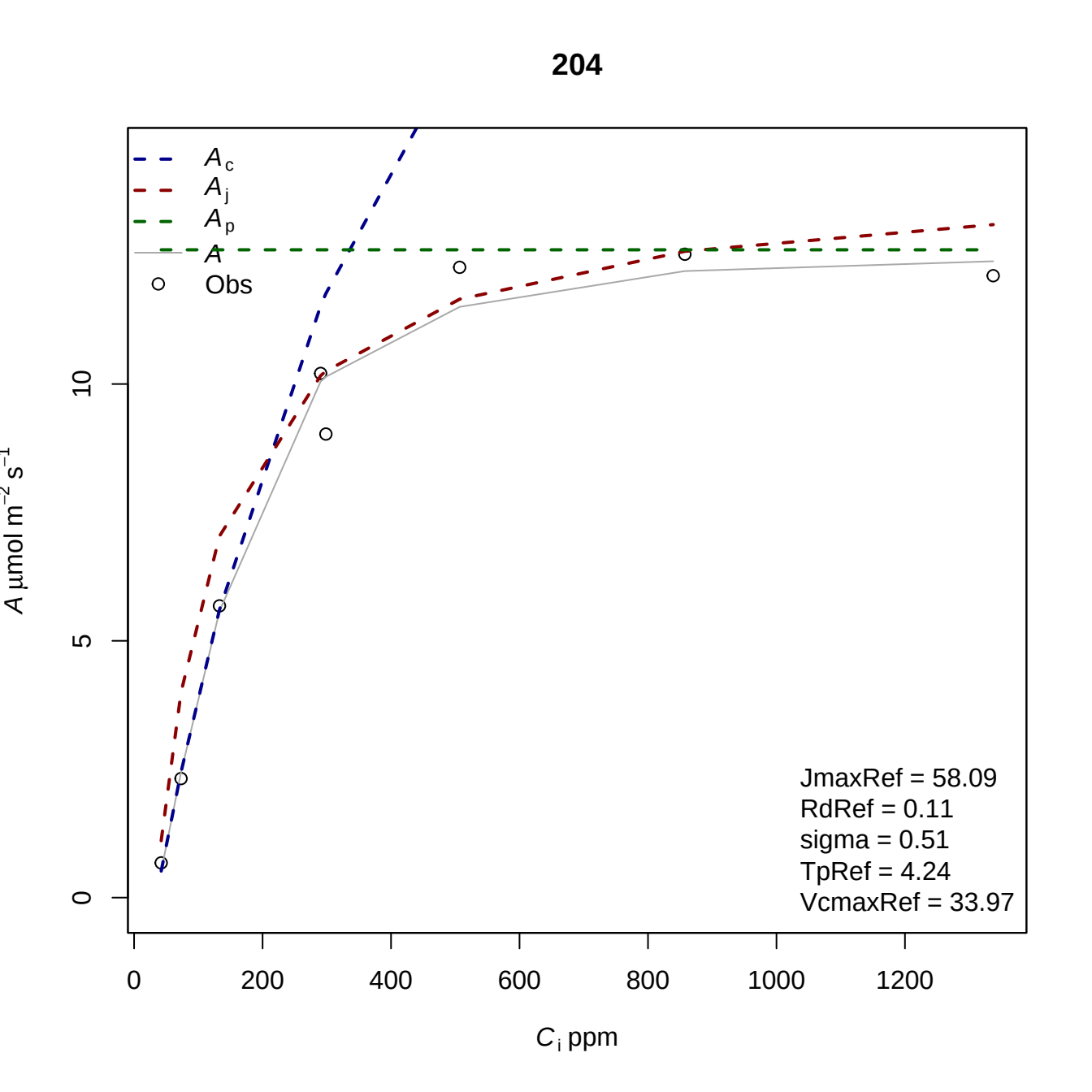

A_c
A_j
A_p
A
Obs
JmaxRef = 67.68
RdRef = −1.16
sigma = 1
TpRef = 3.36
VcmaxRef = 808468.33
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
100
200
300
400
500
600
700

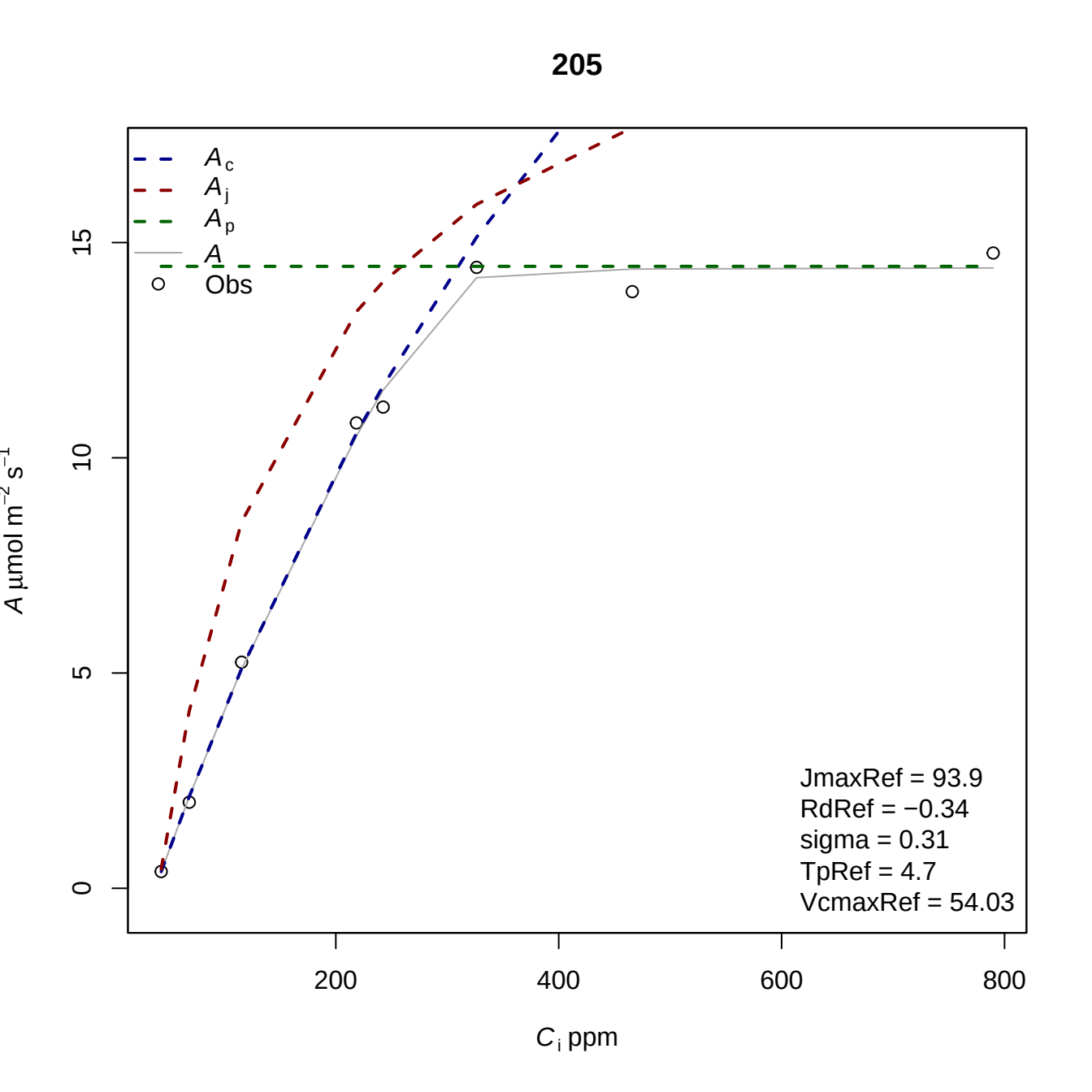

200
A_c
A_j
A_p
A
Obs
JmaxRef = 69.9
RdRef = 0.17
sigma = 0.52
TpRef = 5.05
VcmaxRef = 27.06
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
200
400
600
800
1000

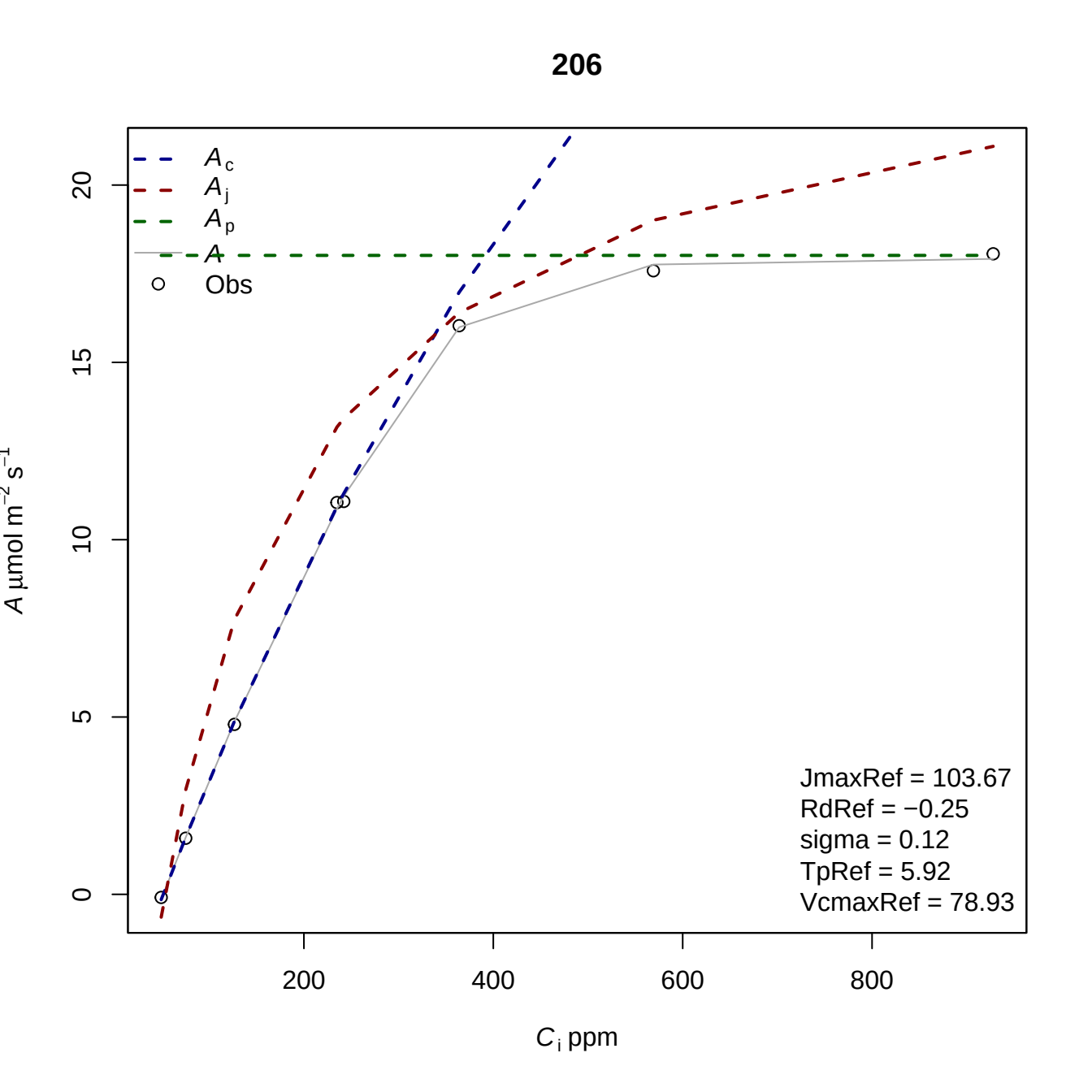

201
A_c
A_j
A_p
A
Obs
JmaxRef = 107.02
RdRef = 0.58
sigma = 0.15
TpRef = 6.95
VcmaxRef = 52.19
0
5
10
15
20
200
400
600
800
A µmol m^{-2} s^{-1}
C_i ppm

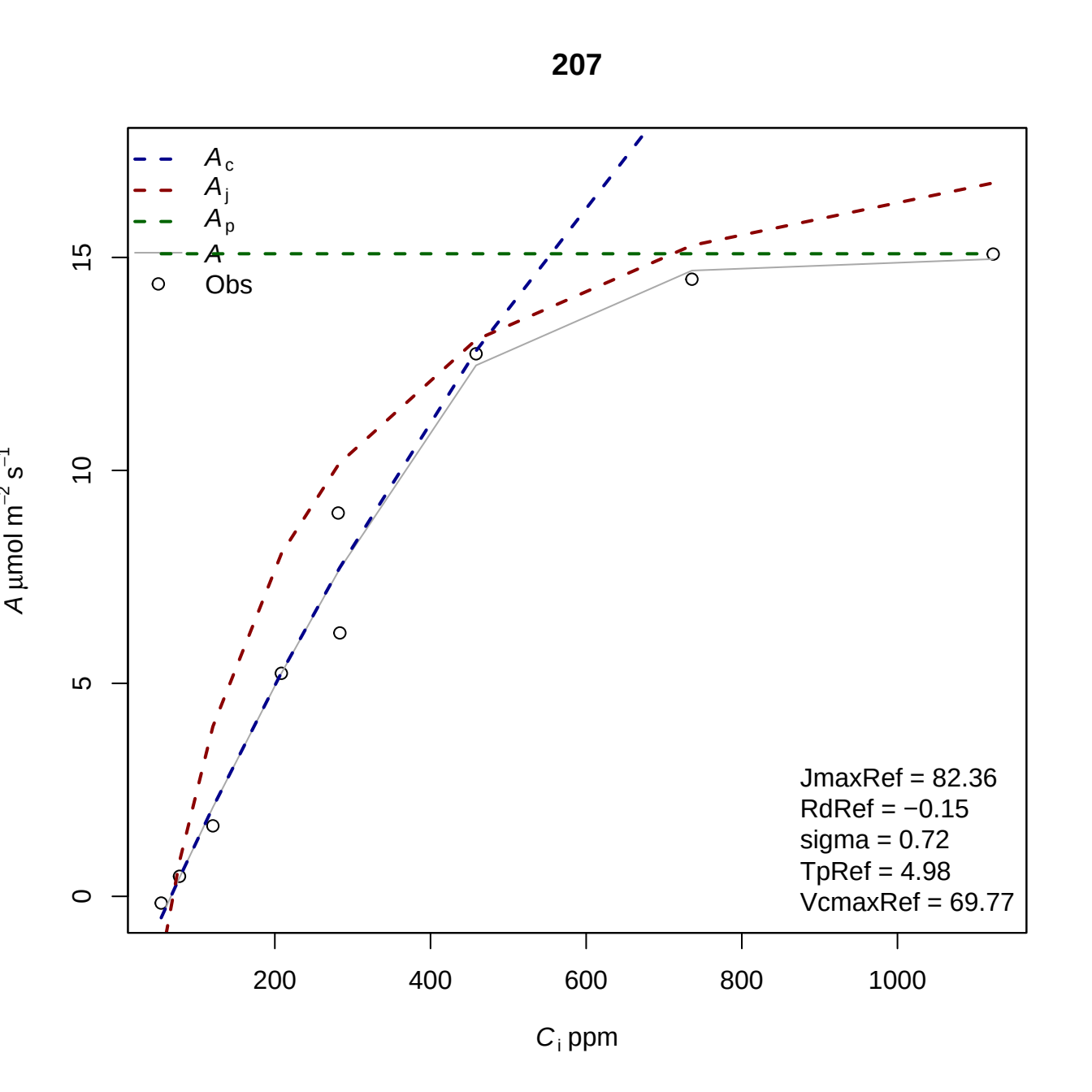

202
A_c
A_j
A_p
A
Obs
JmaxRef = 69.5
RdRef = −0.19
sigma = 0.37
TpRef = 4.52
VcmaxRef = 54.65
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
100
200
300
400
500
600

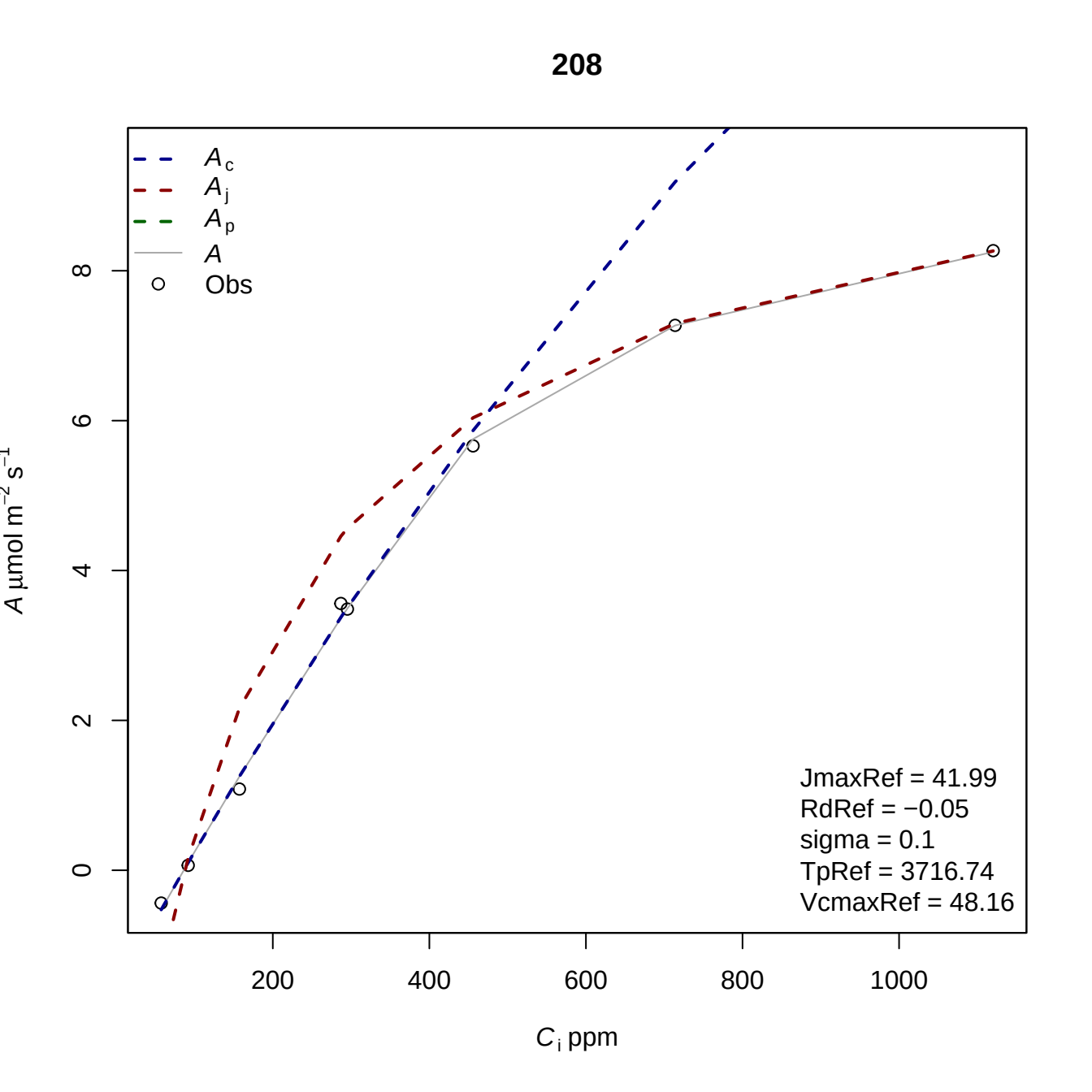

203
A_c
A_j
A_p
A
Obs
JmaxRef = 40.38
RdRef = −0.12
sigma = 0.08
TpRef = 2.95
VcmaxRef = 17.24
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
0
200
400
600
800
1000

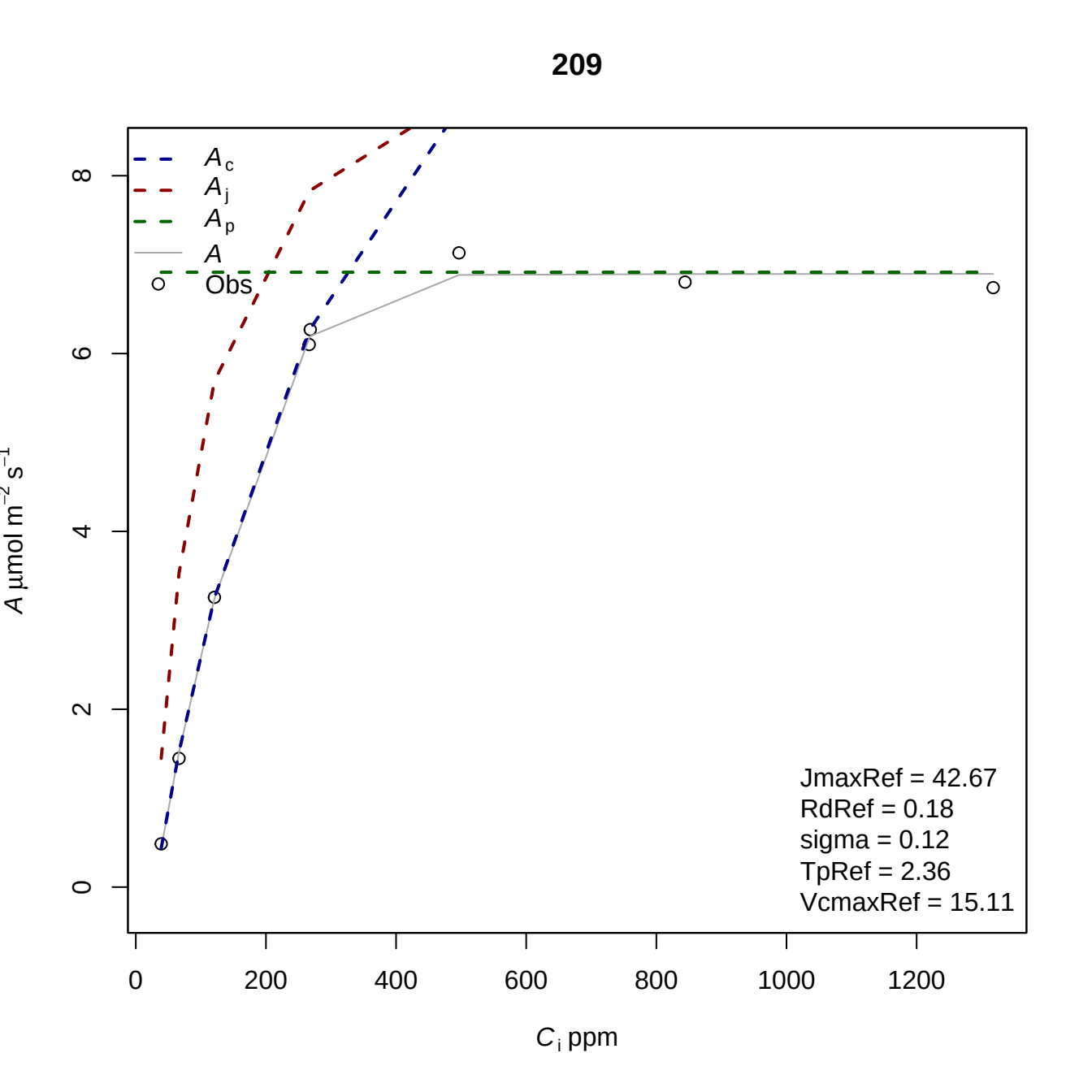

204
A_c
A_j
A_p
A
Obs
JmaxRef = 58.09
RdRef = 0.11
sigma = 0.51
TpRef = 4.24
VcmaxRef = 33.97
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
0
200
400
600
800
1000
1200

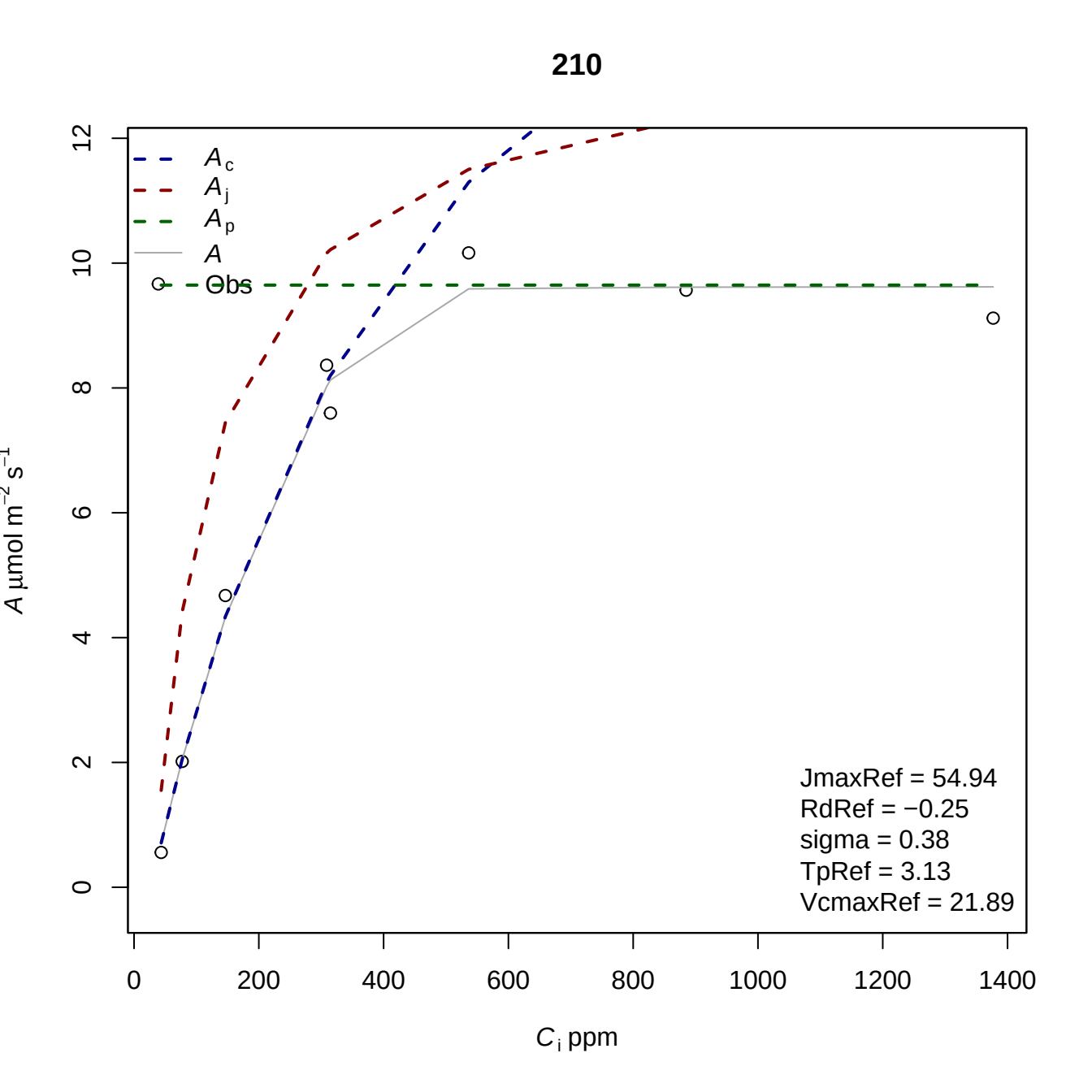

205
A_c
A_j
A_p
A
Obs
JmaxRef = 93.9
RdRef = −0.34
sigma = 0.31
TpRef = 4.7
VcmaxRef = 54.03
0
5
10
15
200
400
600
800
A µmol m^{-2} s^{-1}
C_i ppm

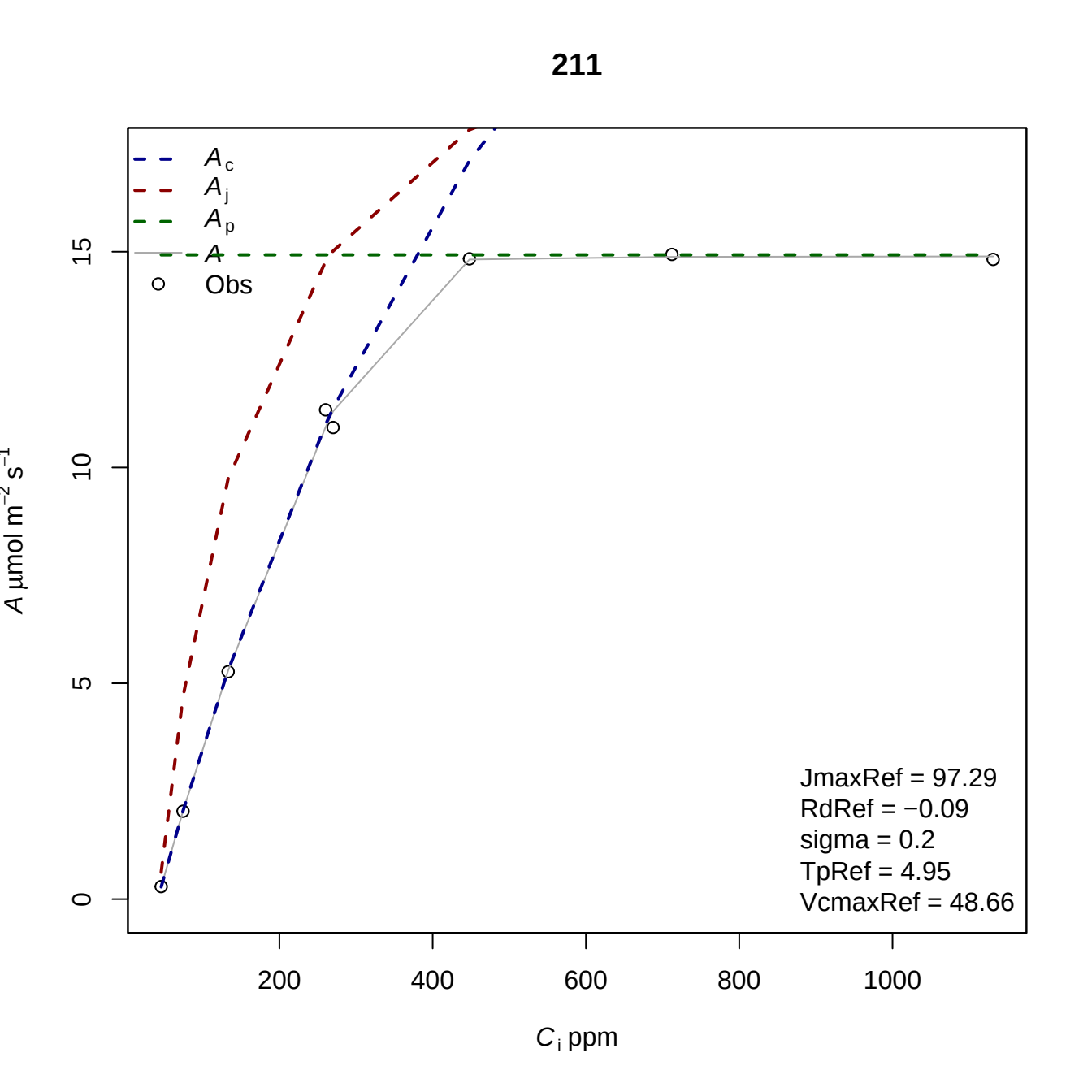

206
A_c
A_j
A_p
A
Obs
JmaxRef = 103.67
RdRef = −0.25
sigma = 0.12
TpRef = 5.92
VcmaxRef = 78.93
0
5
10
15
20
200
400
600
800
A µmol m^{-2} s^{-1}
C_i ppm

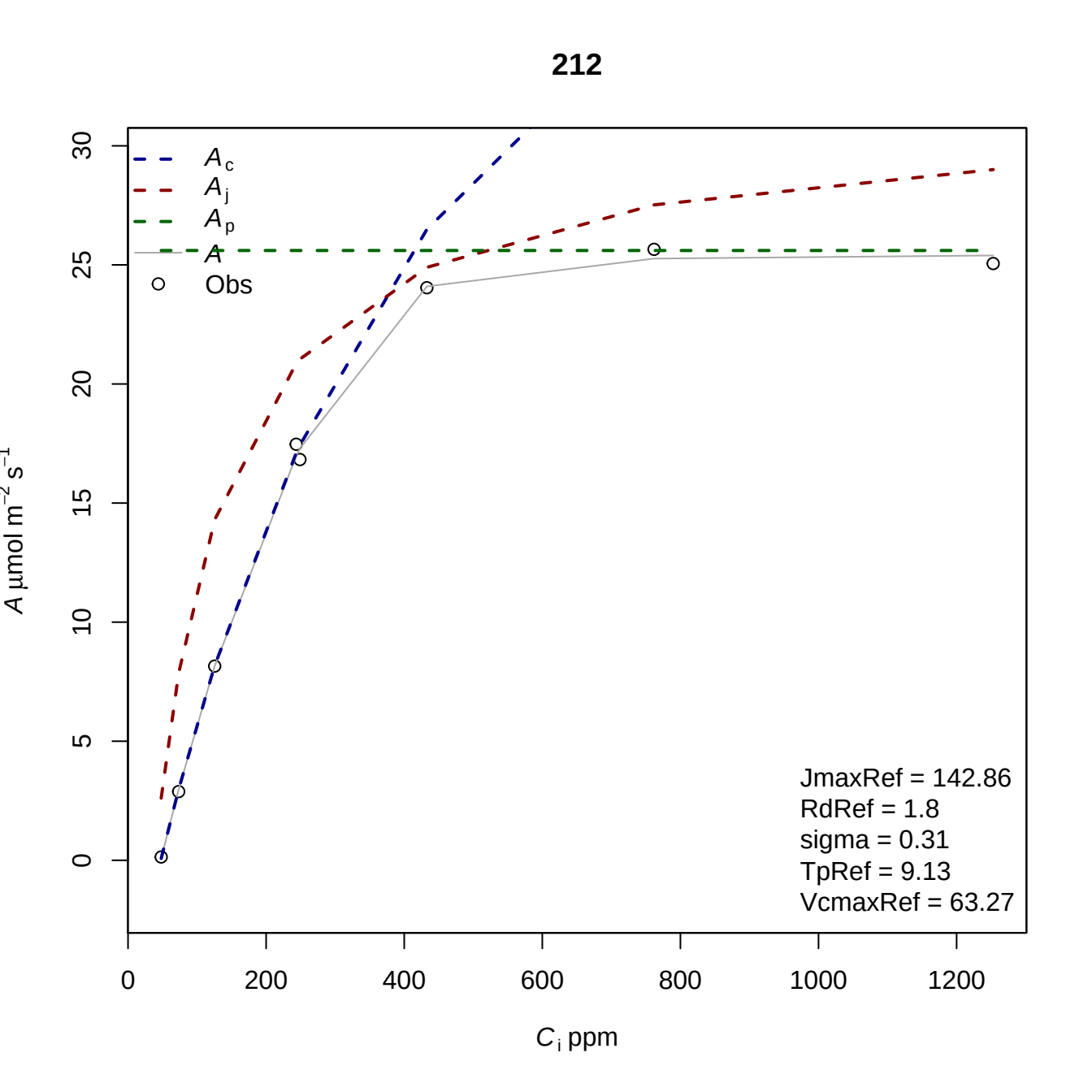

207
A_c
A_j
A_p
A
Obs
JmaxRef = 82.36
RdRef = −0.15
sigma = 0.72
TpRef = 4.98
VcmaxRef = 69.77
0
5
10
15
200
400
600
800
1000
A µmol m^{-2} s^{-1}
C_i ppm

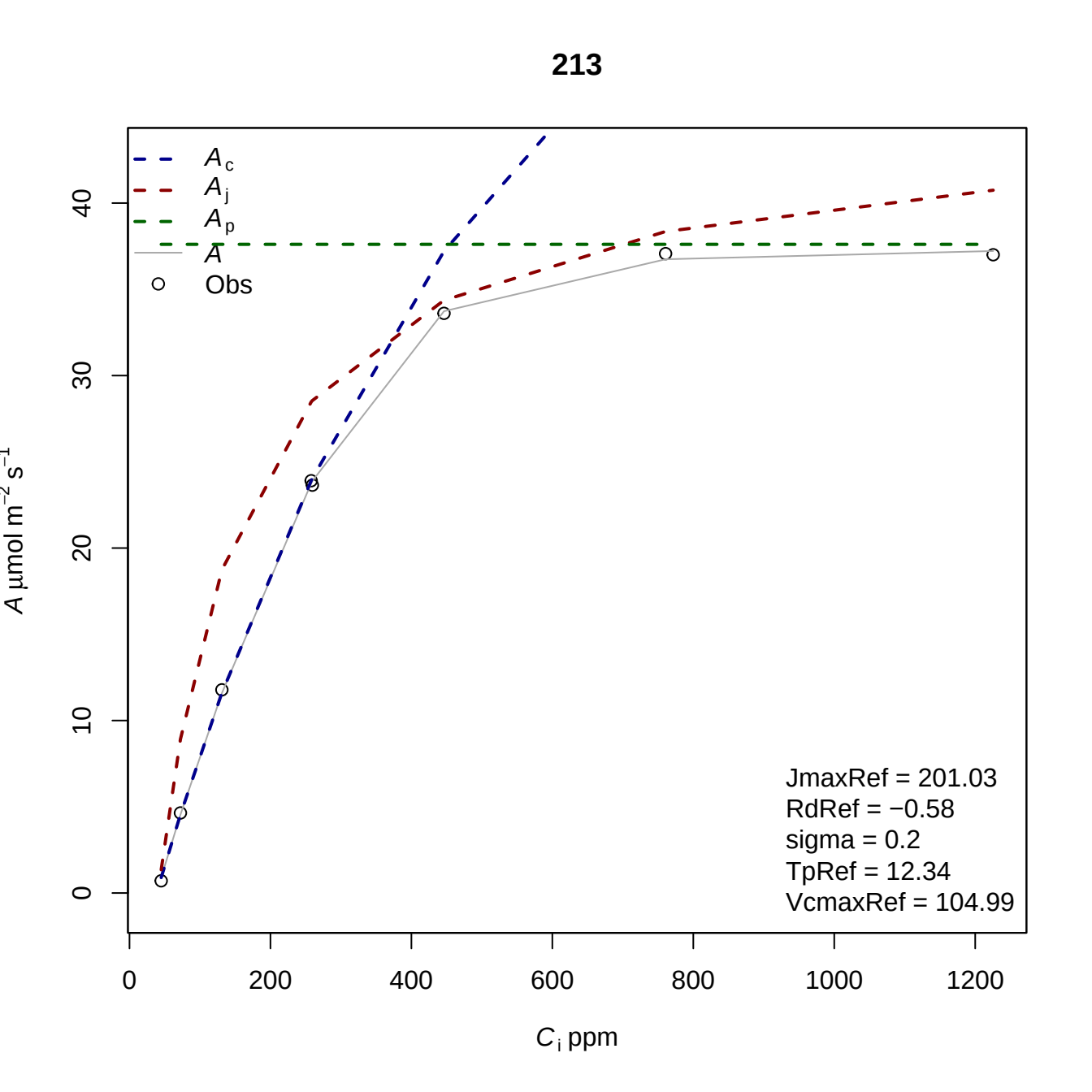

208
A_c
A_j
A_p
A
Obs
JmaxRef = 41.99
RdRef = −0.05
sigma = 0.1
TpRef = 3716.74
VcmaxRef = 48.16
0
2
4
6
8
200
400
600
800
1000
A µmol m^{-2} s^{-1}
C_i ppm

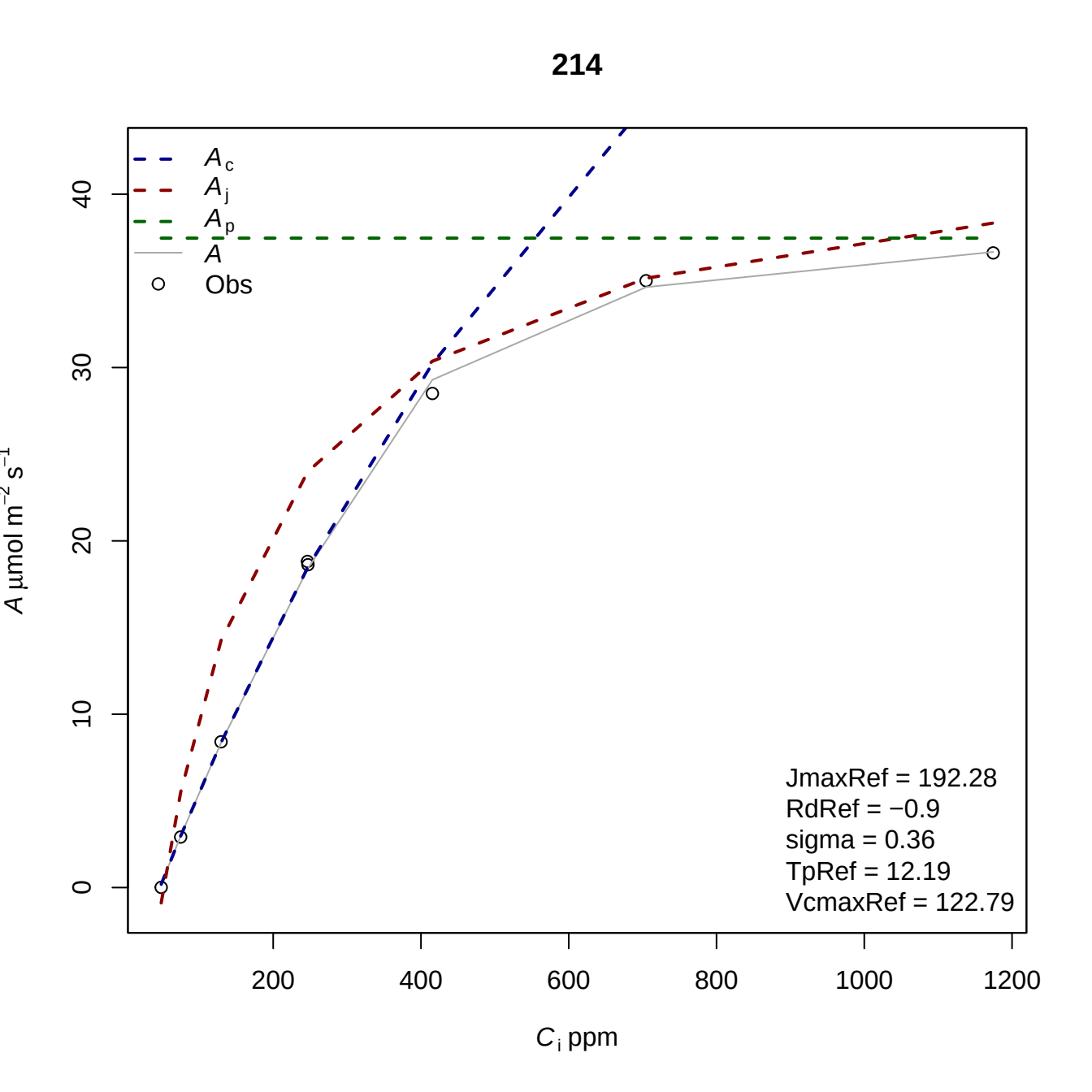

209
A_c
A_j
A_p
A
Obs
JmaxRef = 42.67
RdRef = 0.18
sigma = 0.12
TpRef = 2.36
VcmaxRef = 15.11
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
1200
C_i ppm

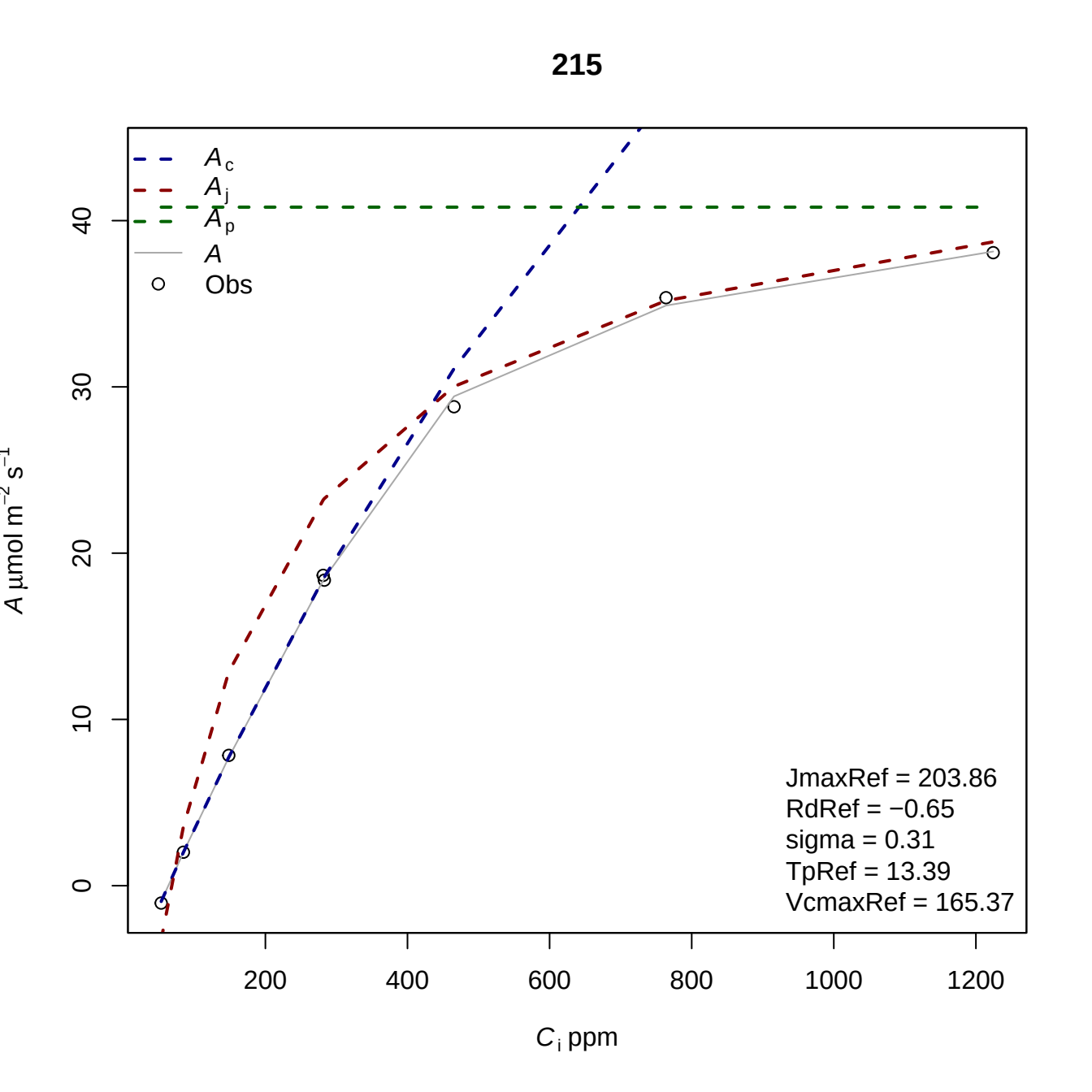

210
A_c
A_j
A_p
A
Obs
JmaxRef = 54.94
RdRef = −0.25
sigma = 0.38
TpRef = 3.13
VcmaxRef = 21.89
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

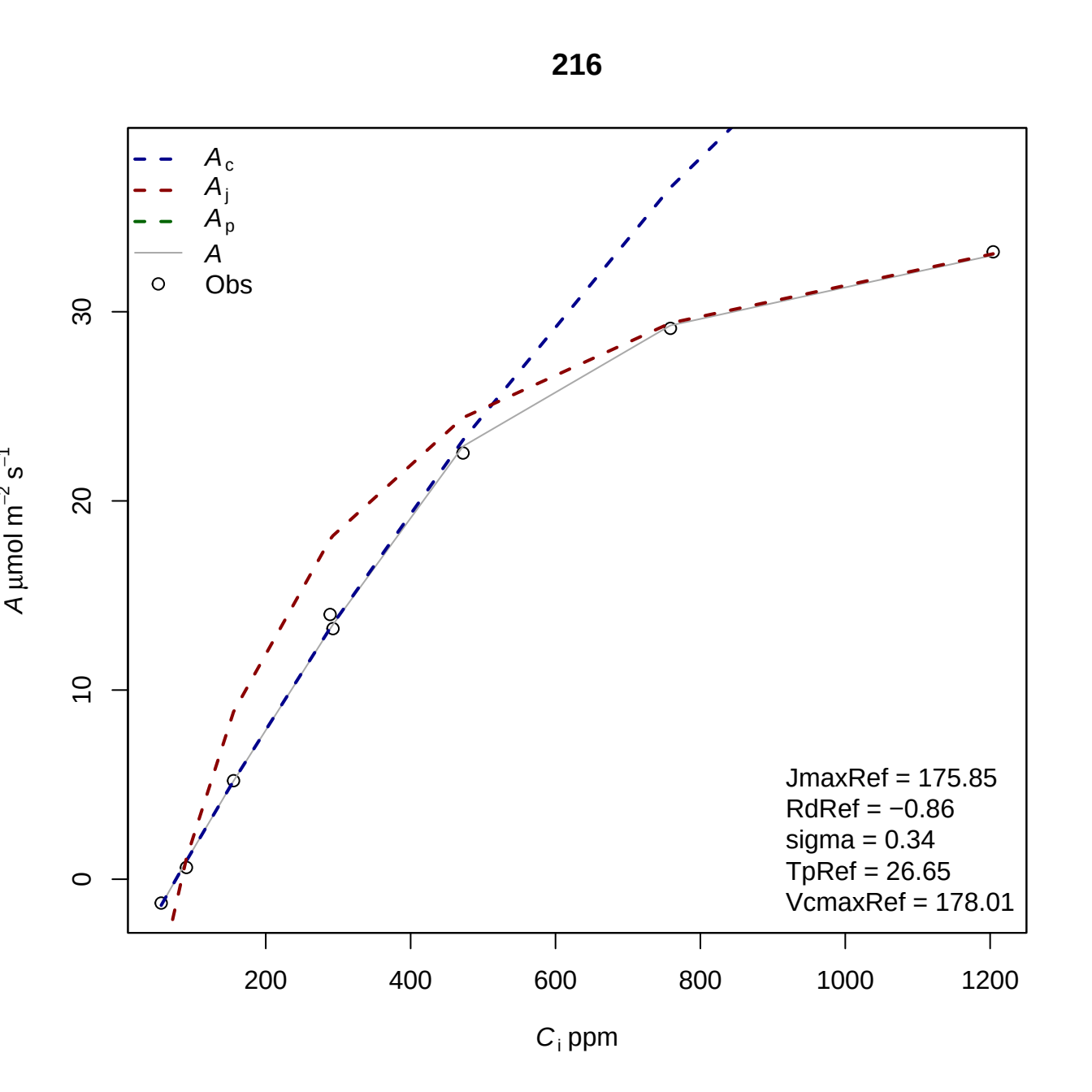

211
A_c
A_j
A_p
A
Obs
JmaxRef = 97.29
RdRef = −0.09
sigma = 0.2
TpRef = 4.95
VcmaxRef = 48.66
0
5
10
15
200
400
600
800
1000
A µmol m^{-2} s^{-1}
C_i ppm

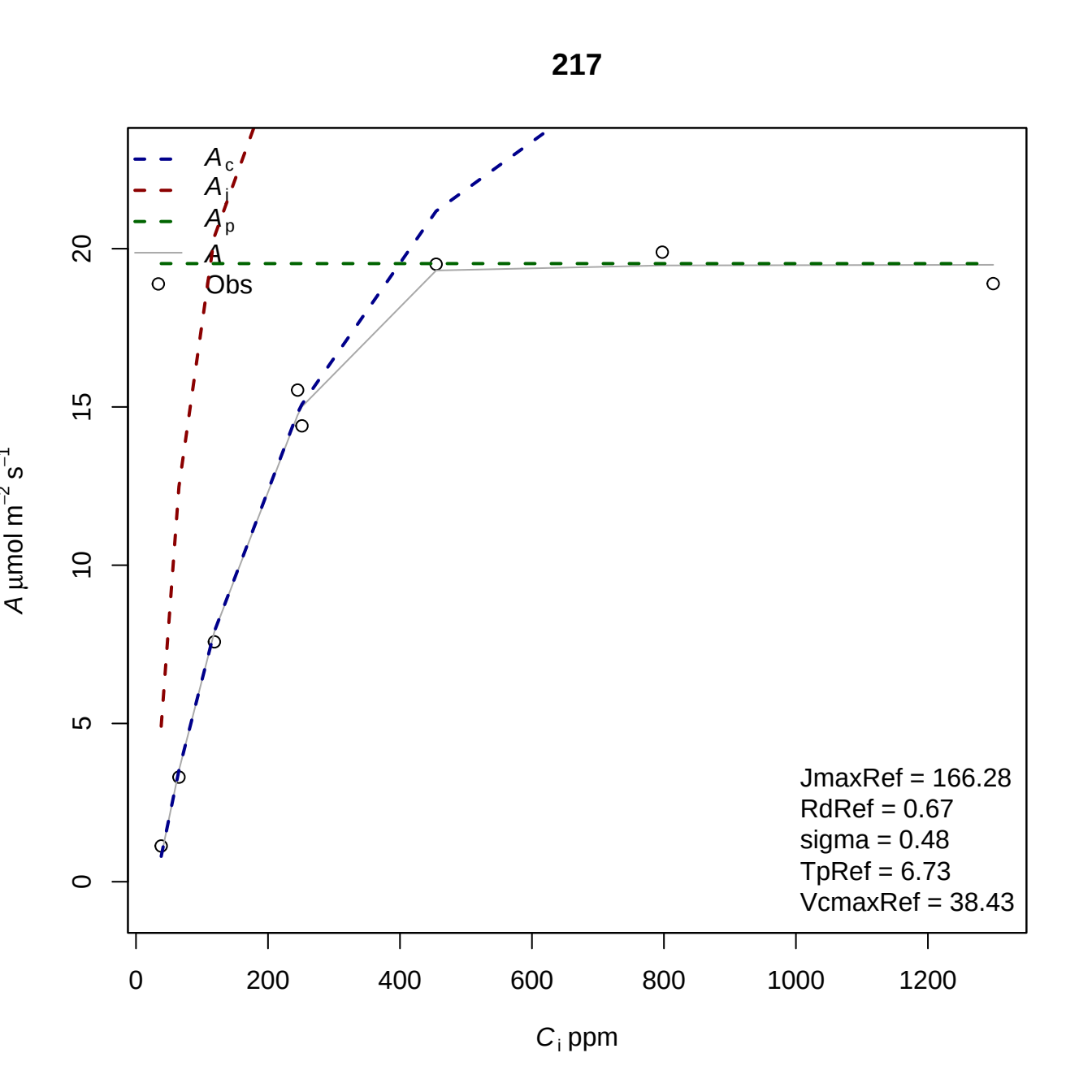

212
A_c
A_j
A_p
A
Obs
JmaxRef = 142.86
RdRef = 1.8
sigma = 0.31
TpRef = 9.13
VcmaxRef = 63.27
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

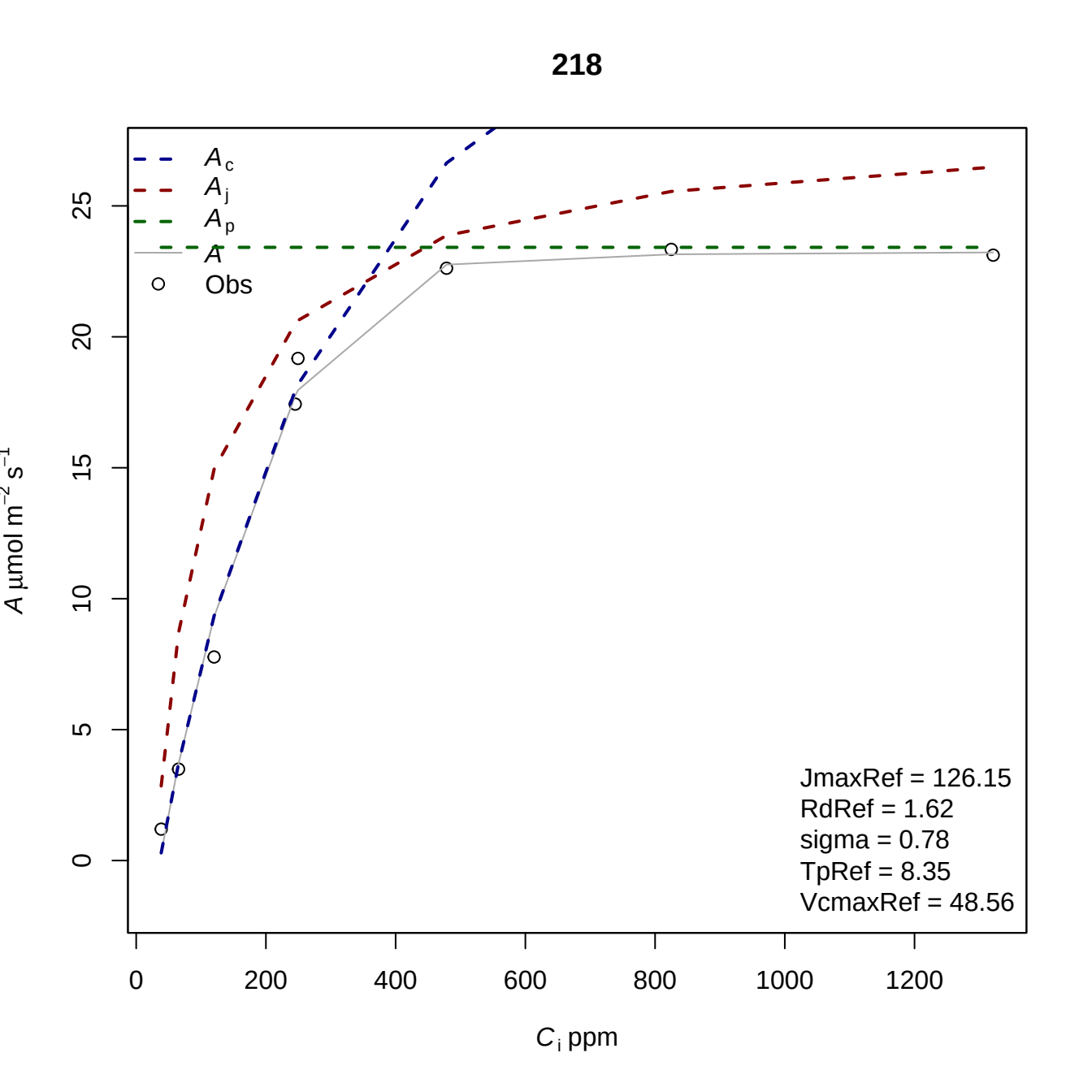

213
A_c
A_j
A_p
A
Obs
JmaxRef = 201.03
RdRef = −0.58
sigma = 0.2
TpRef = 12.34
VcmaxRef = 104.99
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

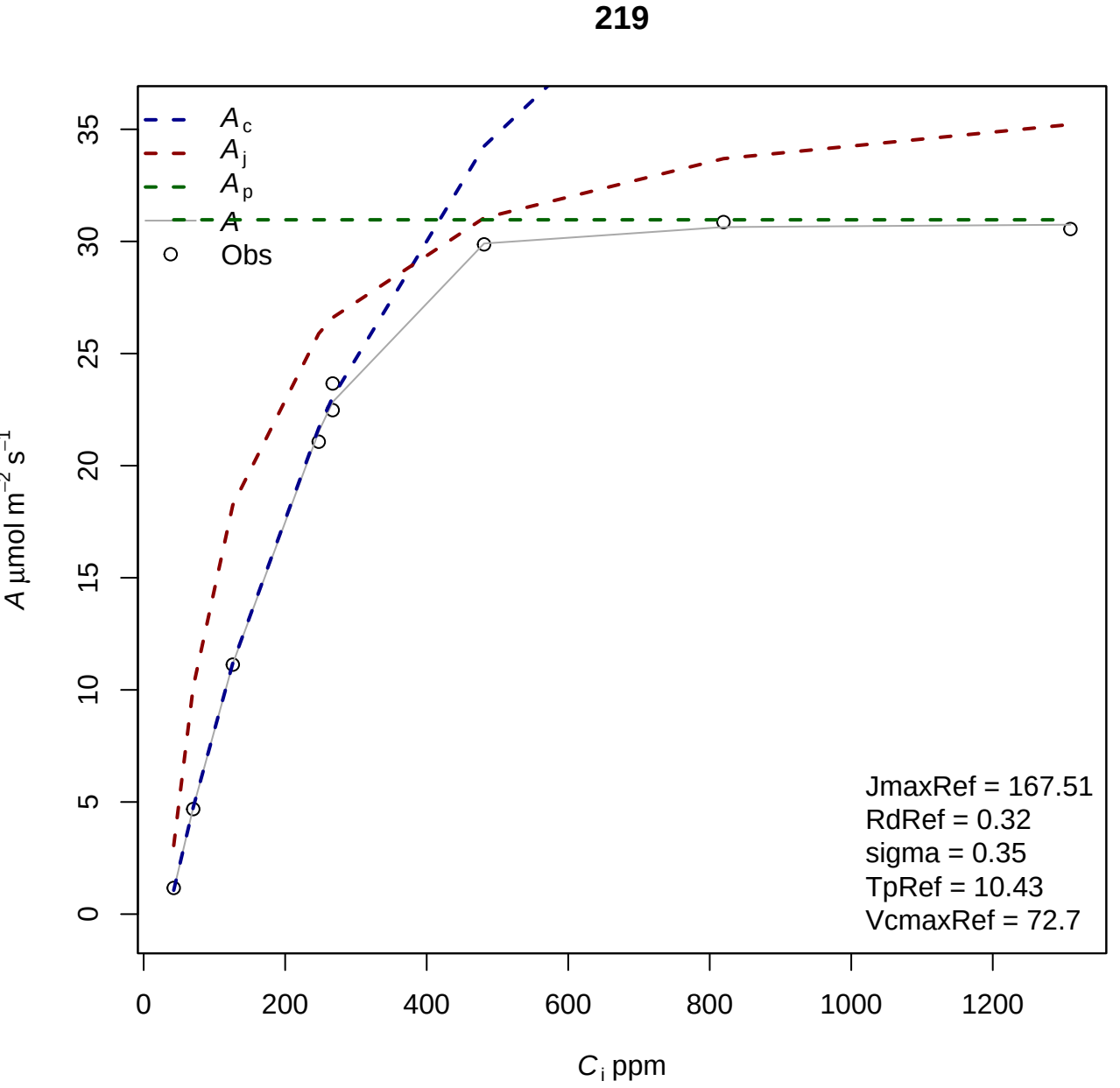

214
A_c
A_j
A_p
A
Obs
JmaxRef = 192.28
RdRef = −0.9
sigma = 0.36
TpRef = 12.19
VcmaxRef = 122.79
0
10
20
30
40
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

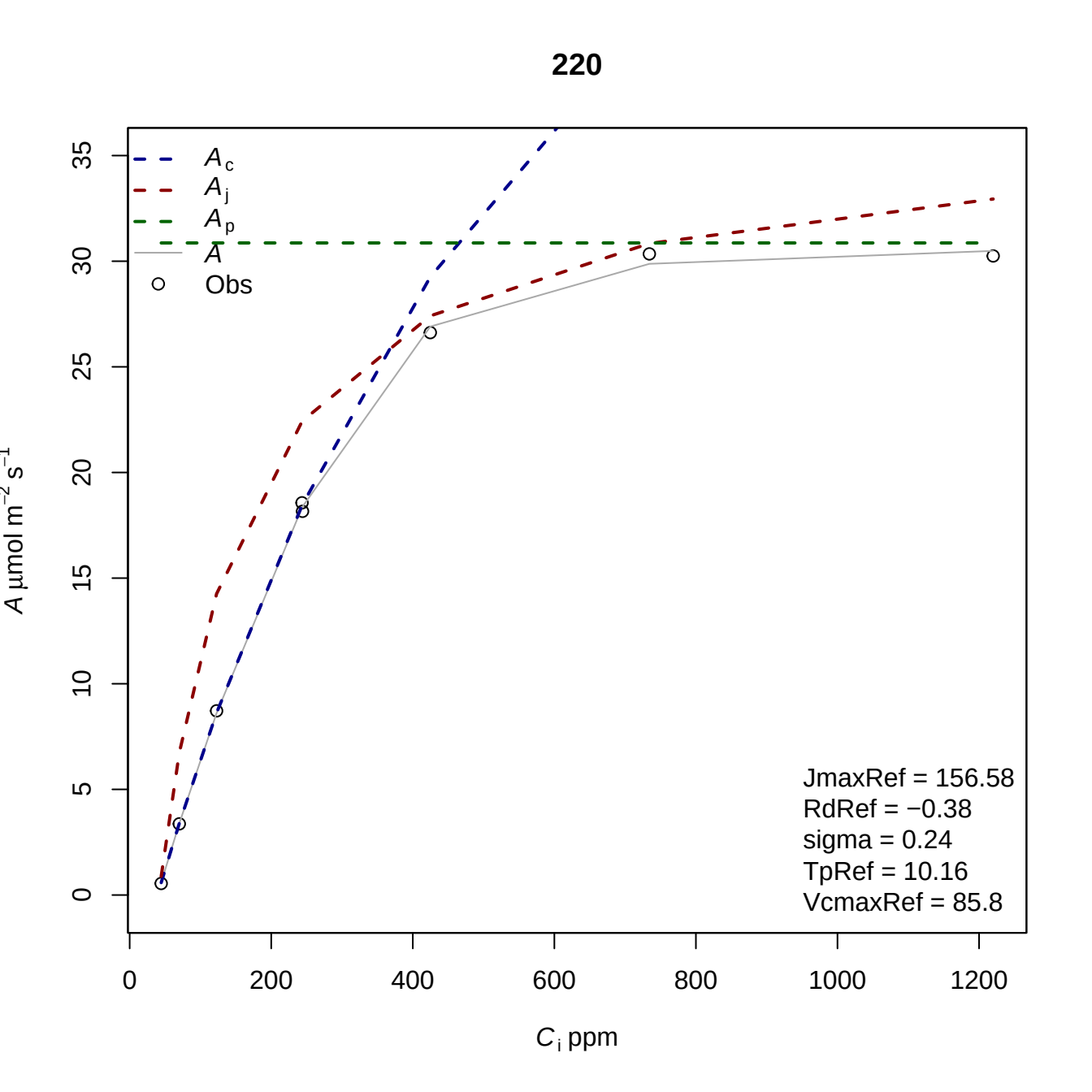

215
A_c
A_j
A_p
A
Obs
JmaxRef = 203.86
RdRef = −0.65
sigma = 0.31
TpRef = 13.39
VcmaxRef = 165.37
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
200
400
600
800
1000
1200

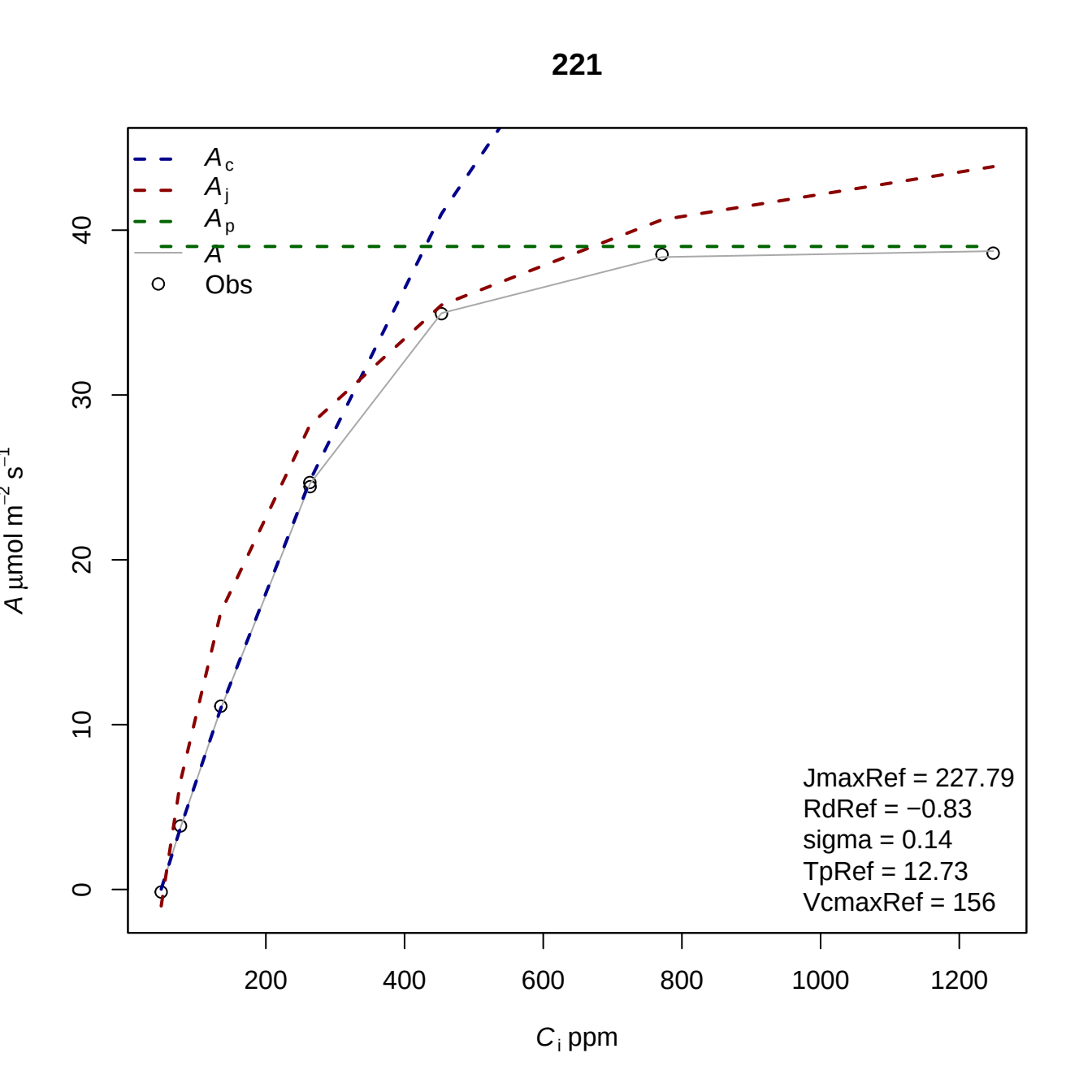

216
A_c
A_j
A_p
A
Obs
JmaxRef = 175.85
RdRef = −0.86
sigma = 0.34
TpRef = 26.65
VcmaxRef = 178.01
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
200
400
600
800
1000
1200

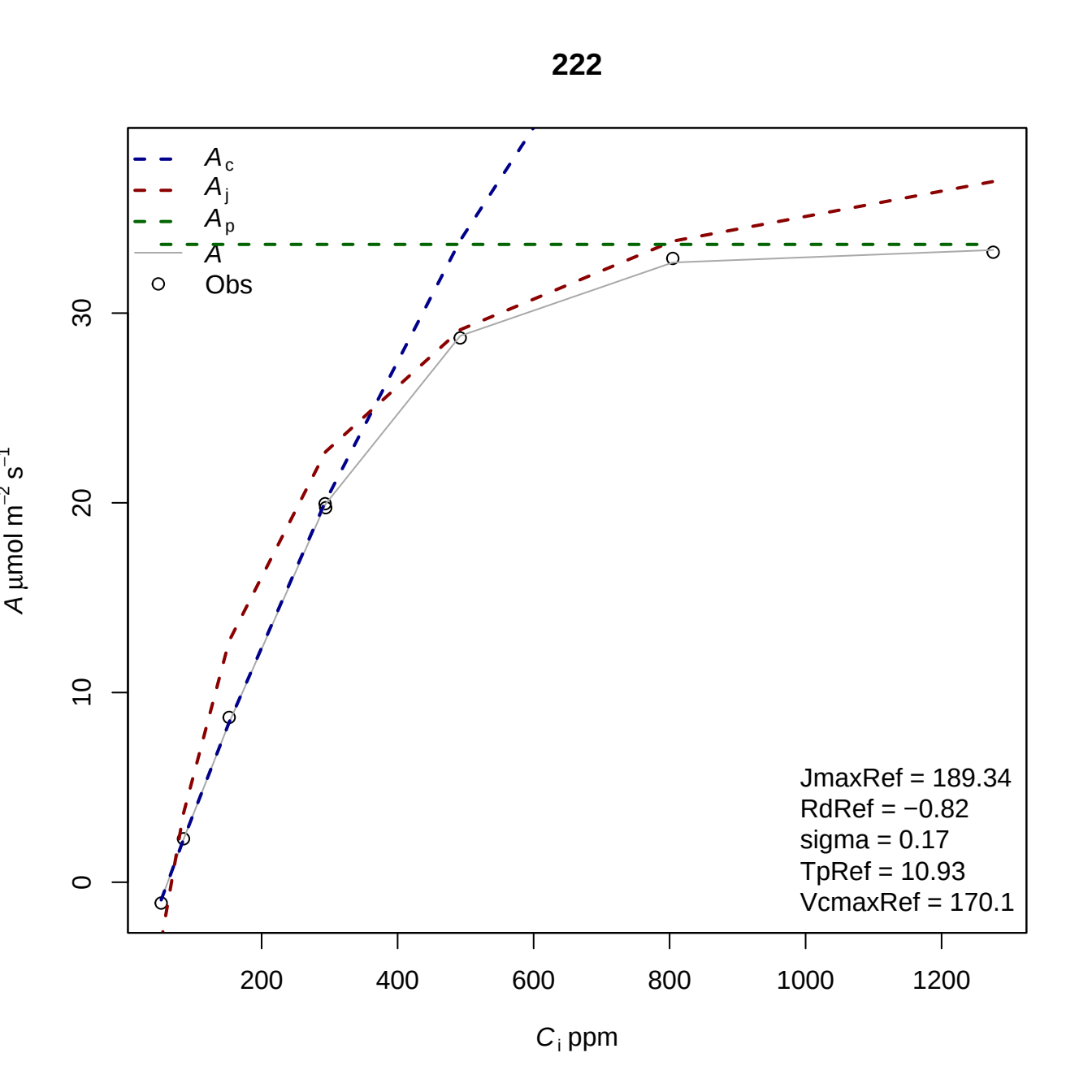

217
A_c
A_j
A_p
A
Obs
JmaxRef = 166.28
RdRef = 0.67
sigma = 0.48
TpRef = 6.73
VcmaxRef = 38.43
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

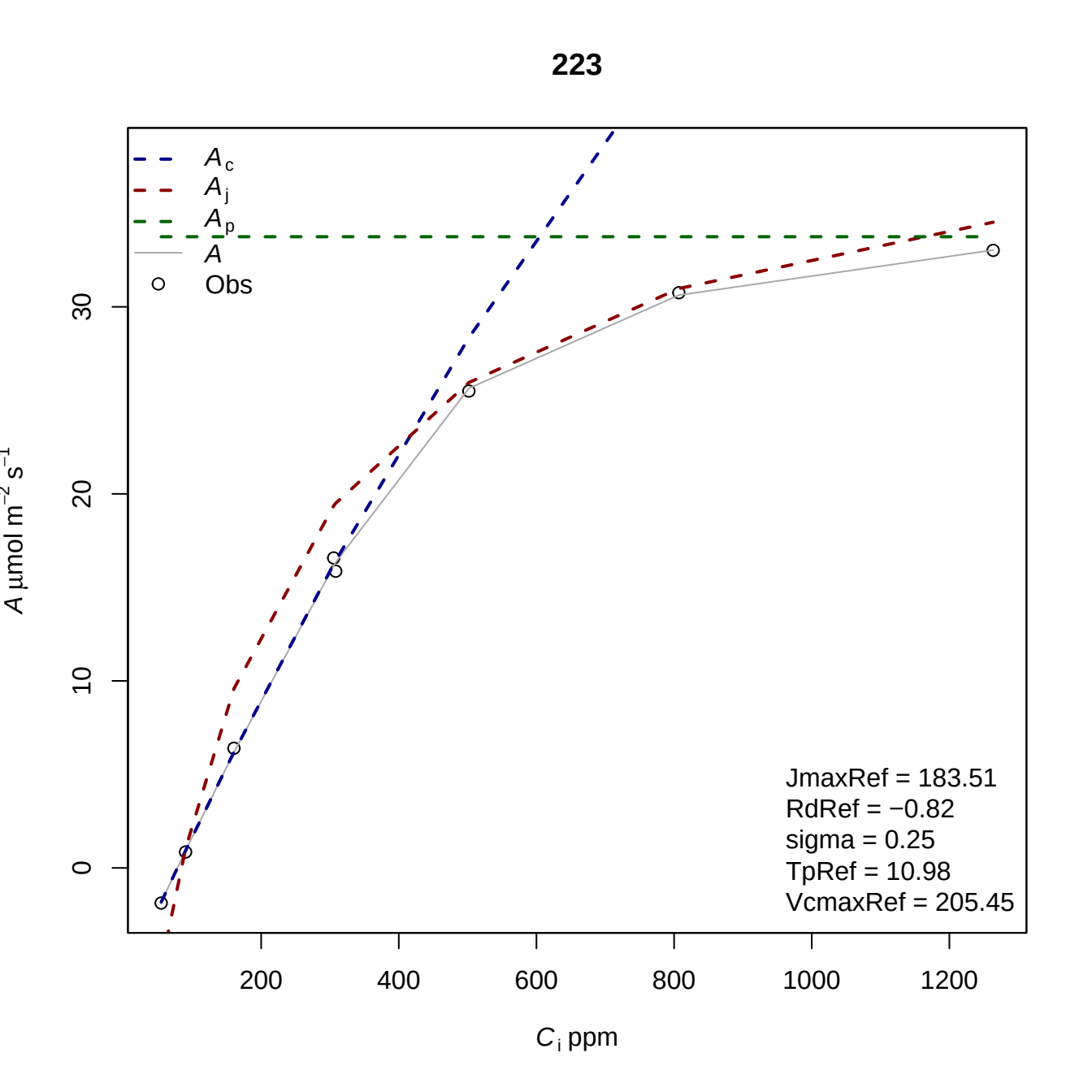

218
A_c
A_j
A_p
A
Obs
JmaxRef = 126.15
RdRef = 1.62
sigma = 0.78
TpRef = 8.35
VcmaxRef = 48.56
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200

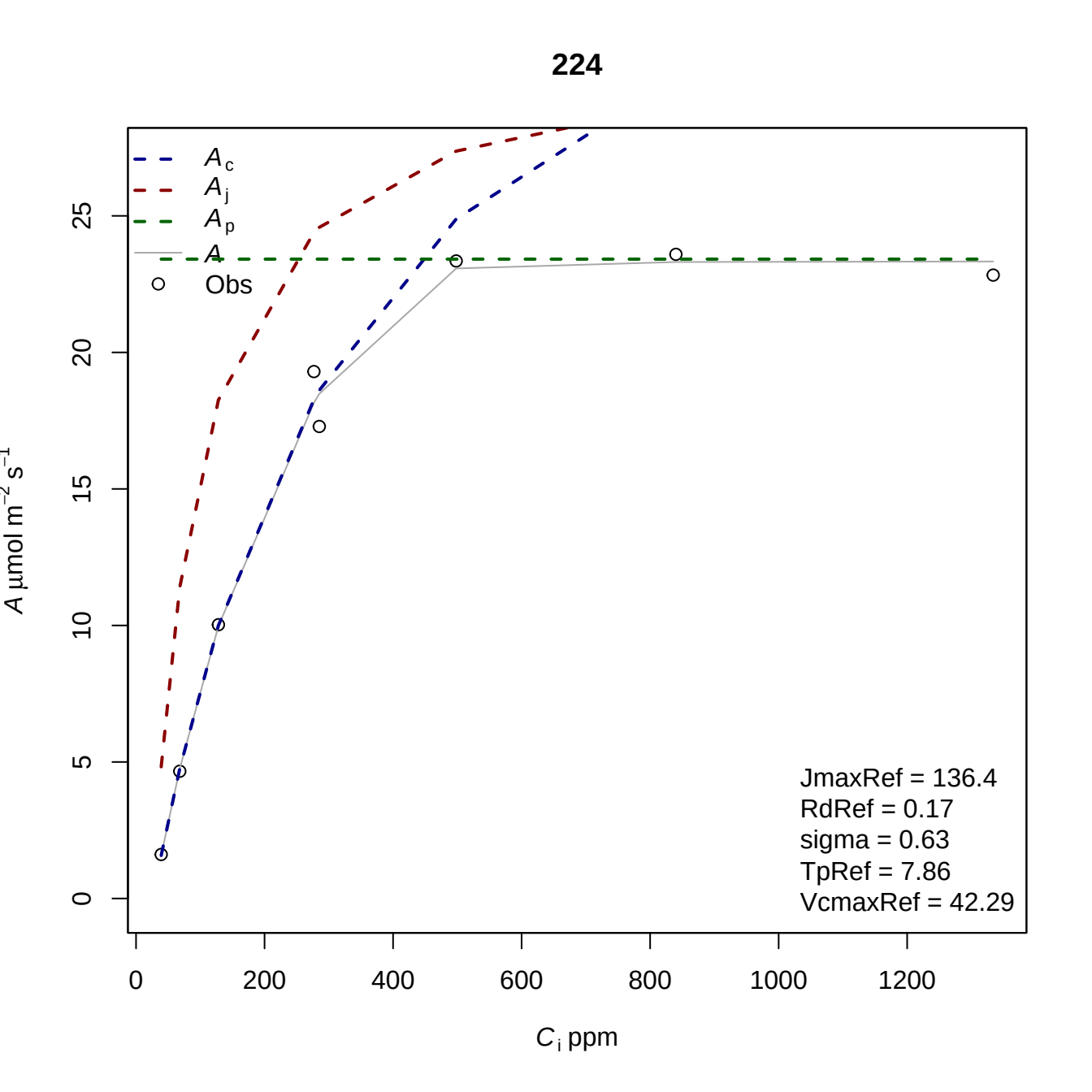

219
A_c
A_j
A_p
A
Obs
JmaxRef = 167.51
RdRef = 0.32
sigma = 0.35
TpRef = 10.43
VcmaxRef = 72.7
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200

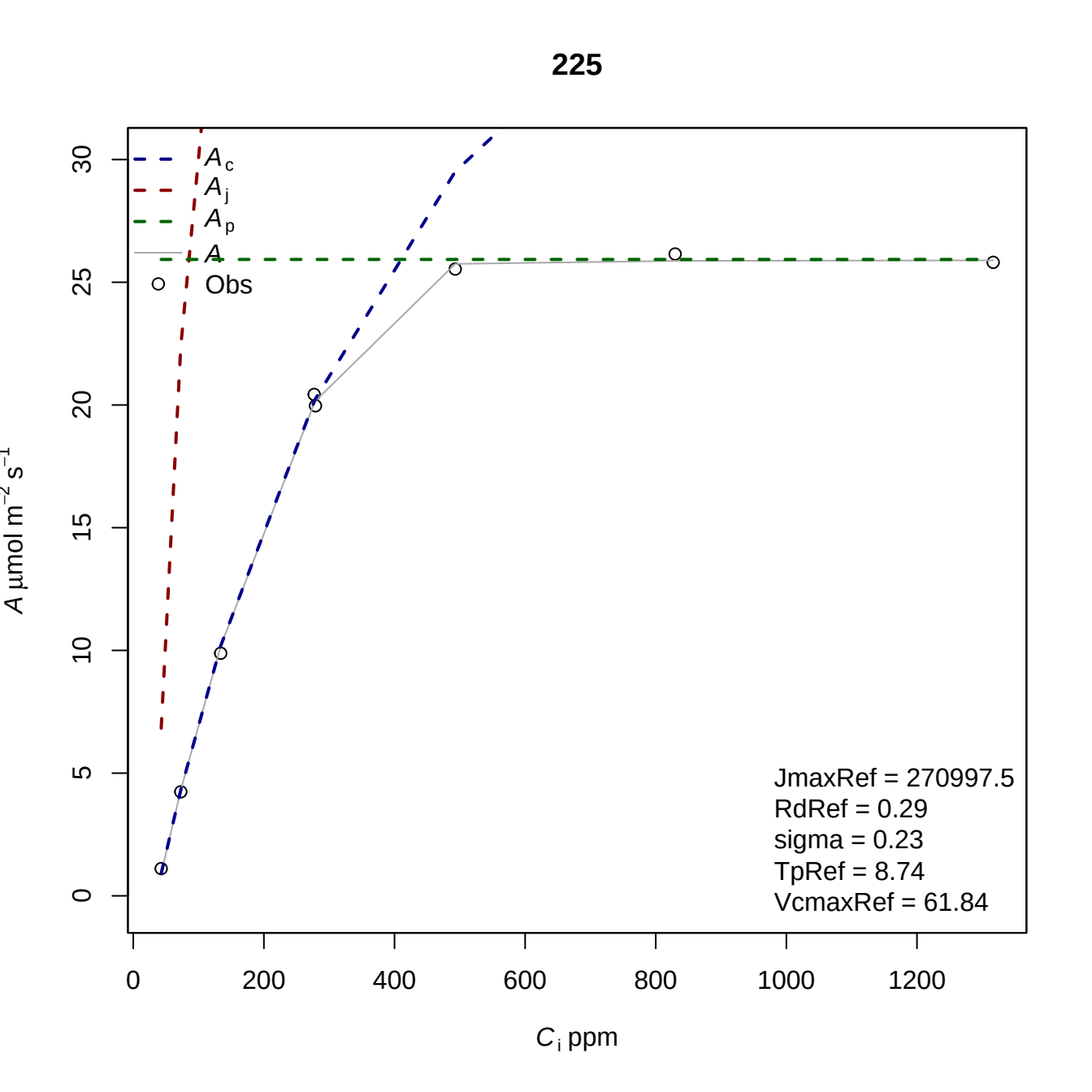

220
A_c
A_j
A_p
A
Obs
JmaxRef = 156.58
RdRef = −0.38
sigma = 0.24
TpRef = 10.16
VcmaxRef = 85.8
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200

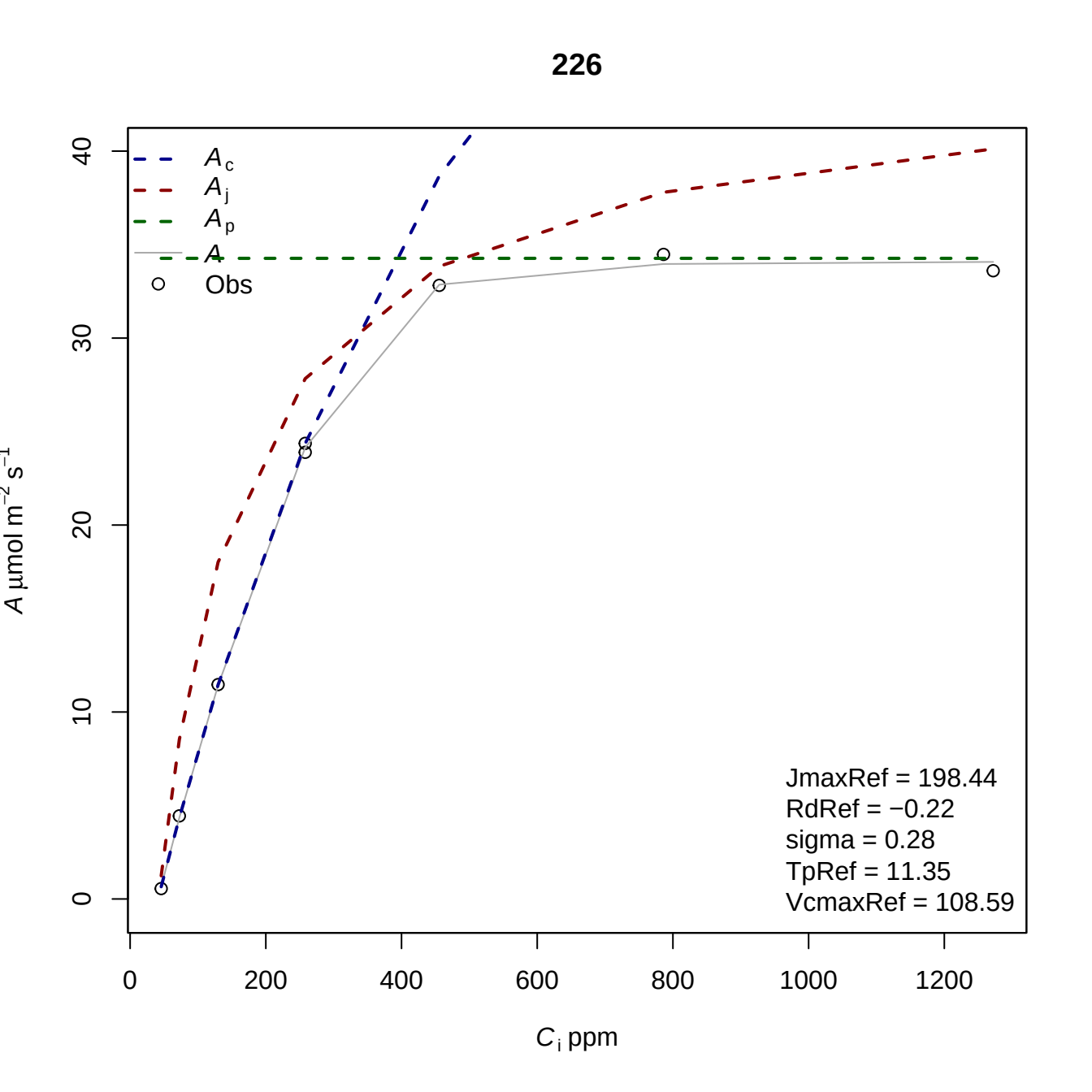

221
A_c
A_j
A_p
A
Obs
JmaxRef = 227.79
RdRef = −0.83
sigma = 0.14
TpRef = 12.73
VcmaxRef = 156
0
10
20
30
40
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

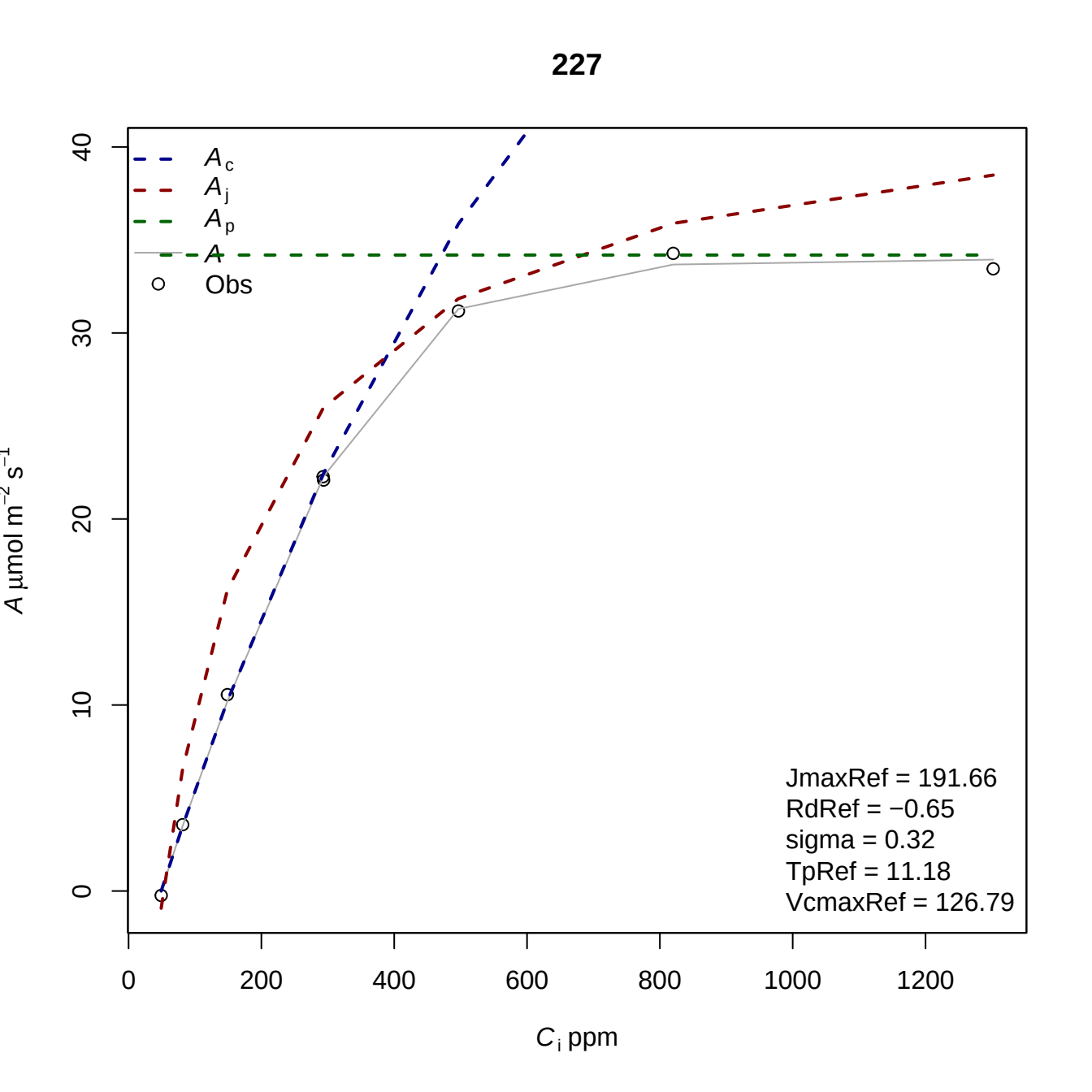

222
A_c
A_j
A_p
A
Obs
JmaxRef = 189.34
RdRef = −0.82
sigma = 0.17
TpRef = 10.93
VcmaxRef = 170.1
0
10
20
30
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

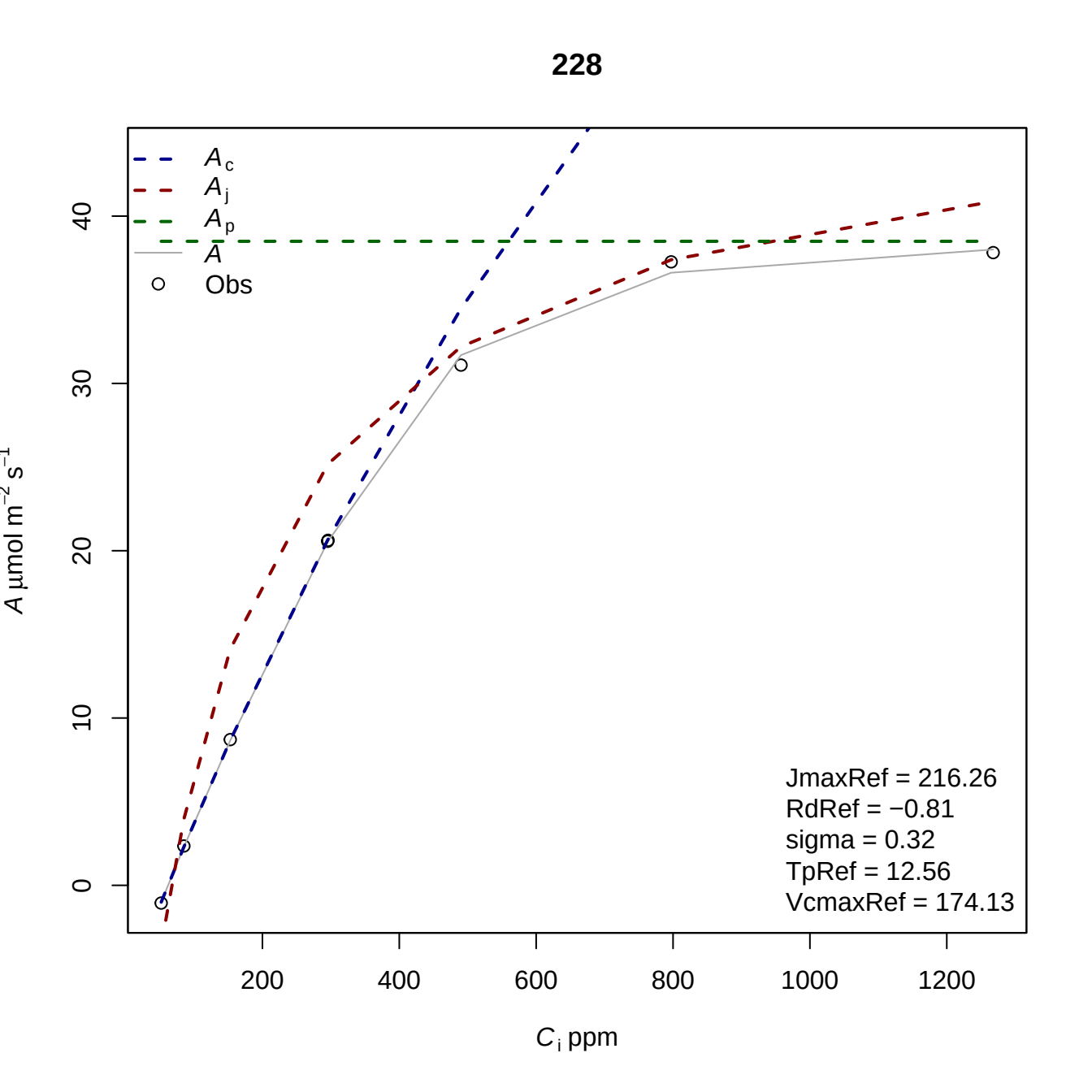

223
A_c
A_j
A_p
A
Obs
JmaxRef = 183.51
RdRef = −0.82
sigma = 0.25
TpRef = 10.98
VcmaxRef = 205.45
0
10
20
30
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

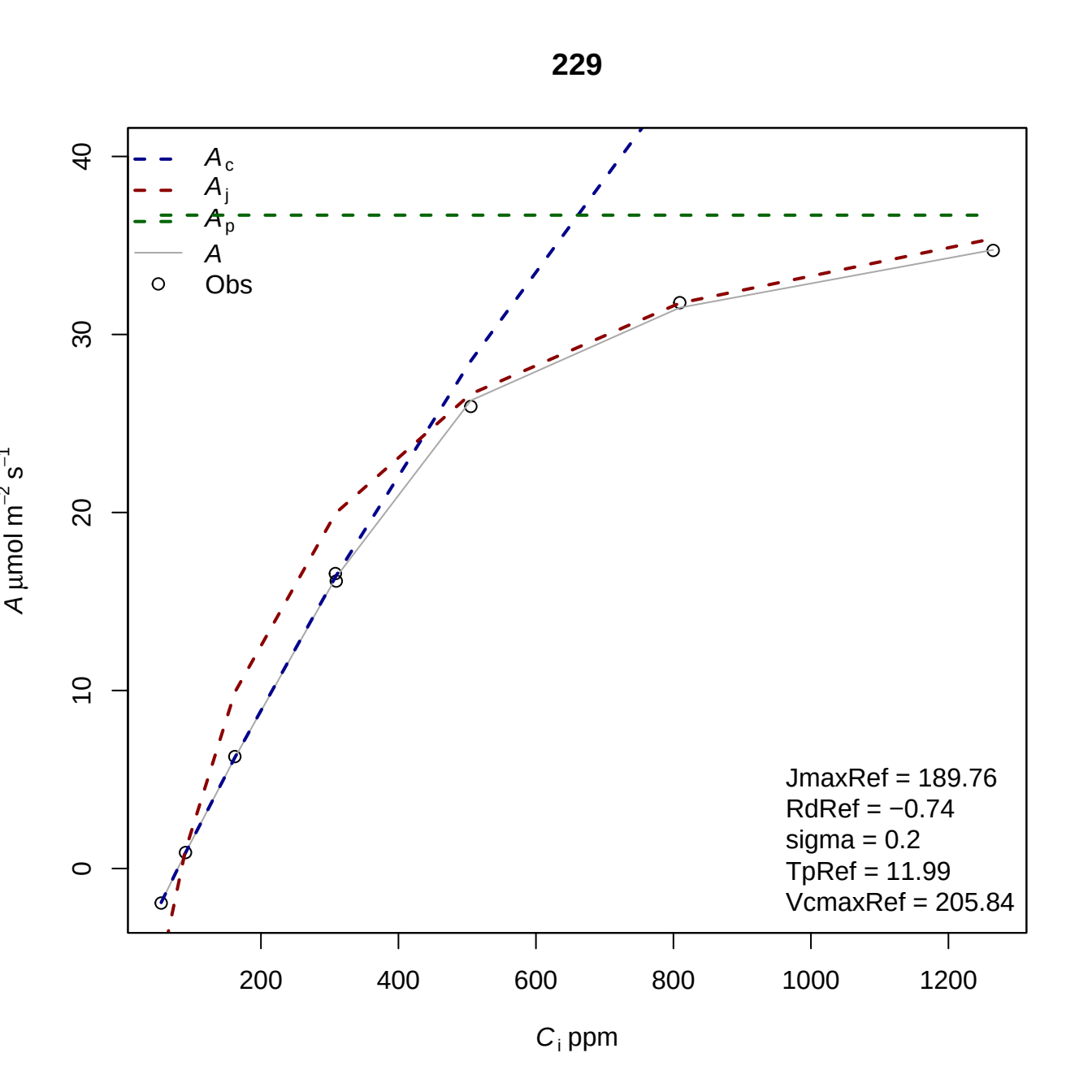

224
A_c
A_j
A_p
A
Obs
JmaxRef = 136.4
RdRef = 0.17
sigma = 0.63
TpRef = 7.86
VcmaxRef = 42.29
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
1200
C_i ppm

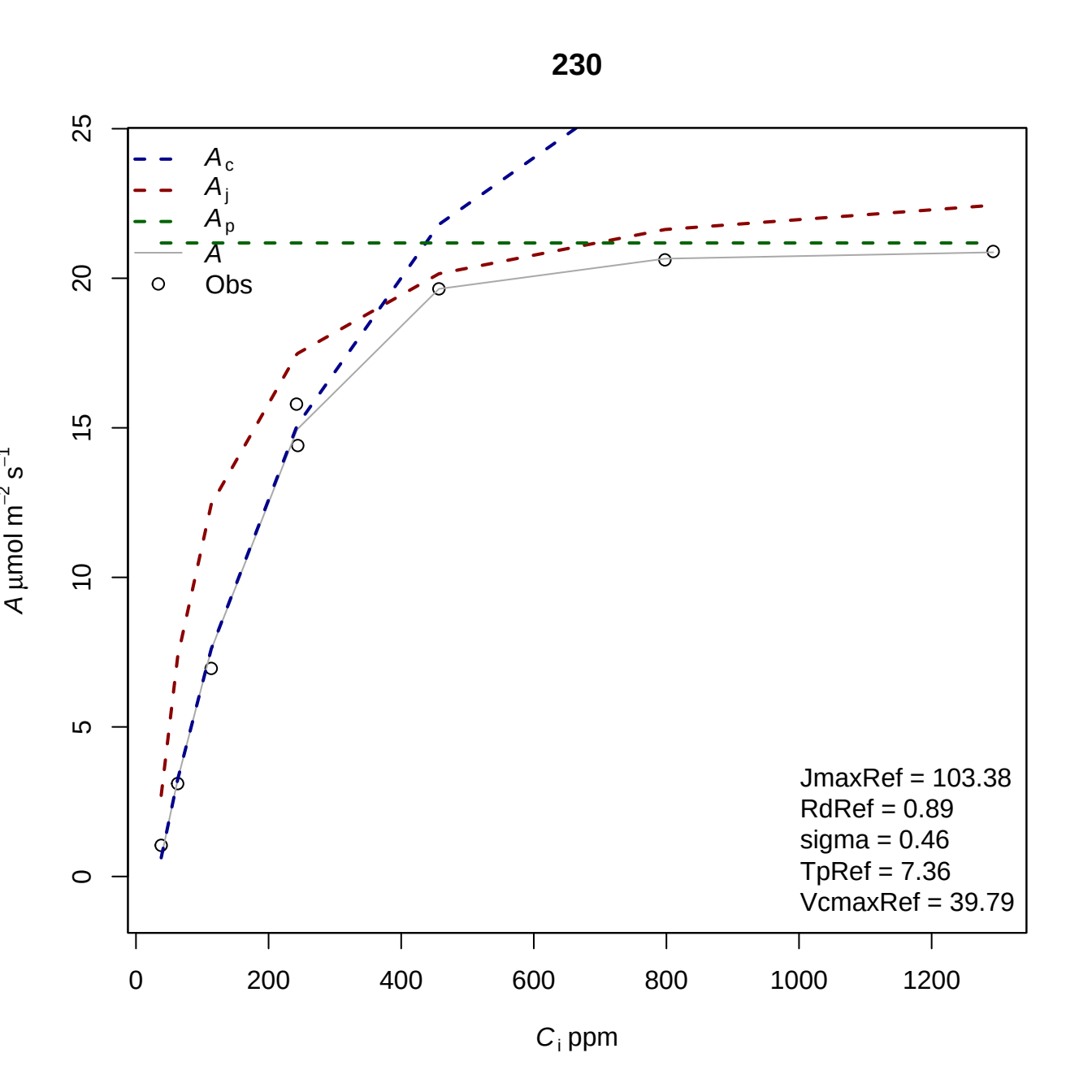

225
A_c
A_j
A_p
A
Obs
JmaxRef = 270997.5
RdRef = 0.29
sigma = 0.23
TpRef = 8.74
VcmaxRef = 61.84
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

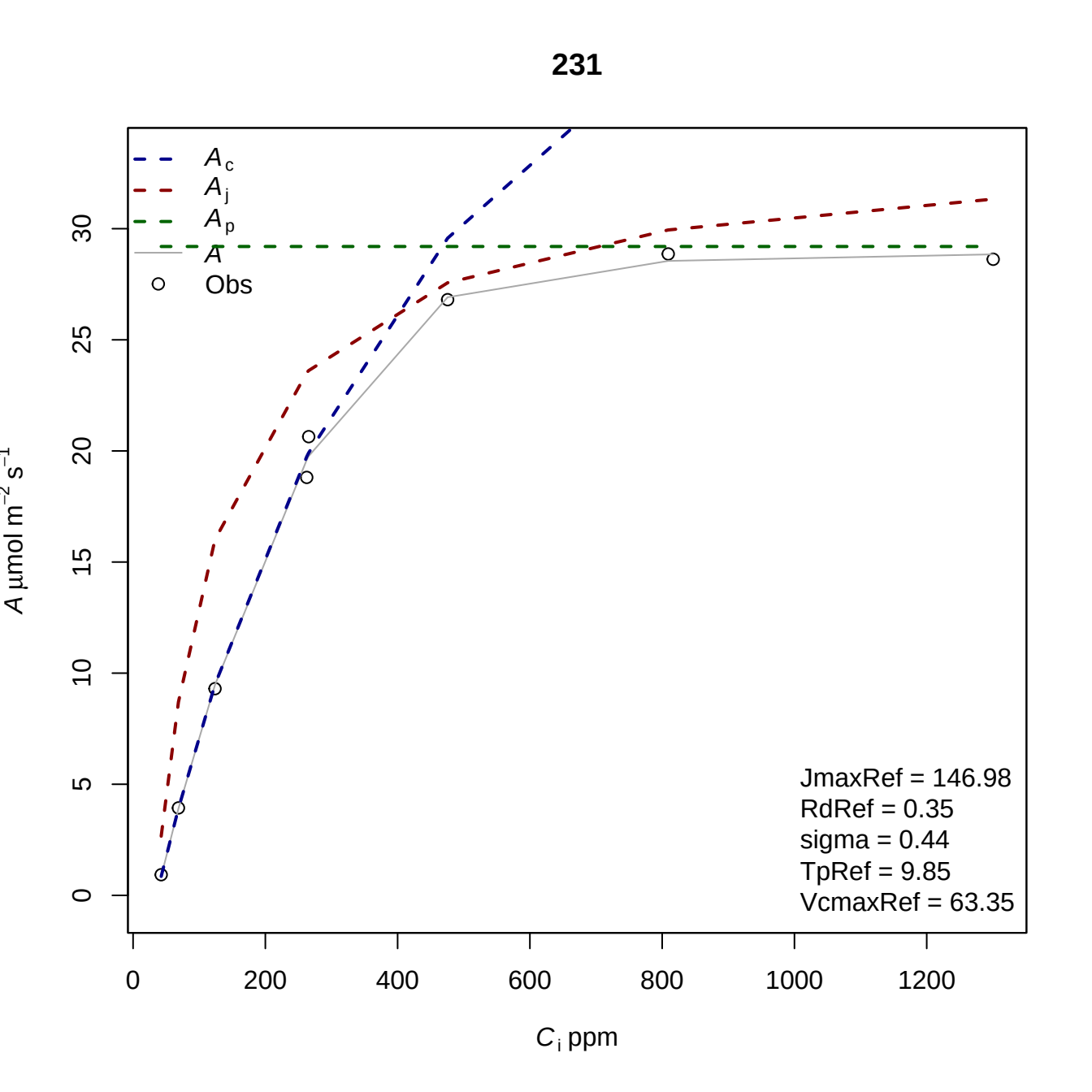

226
A_c
A_j
A_p
A
Obs
JmaxRef = 198.44
RdRef = −0.22
sigma = 0.28
TpRef = 11.35
VcmaxRef = 108.59
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

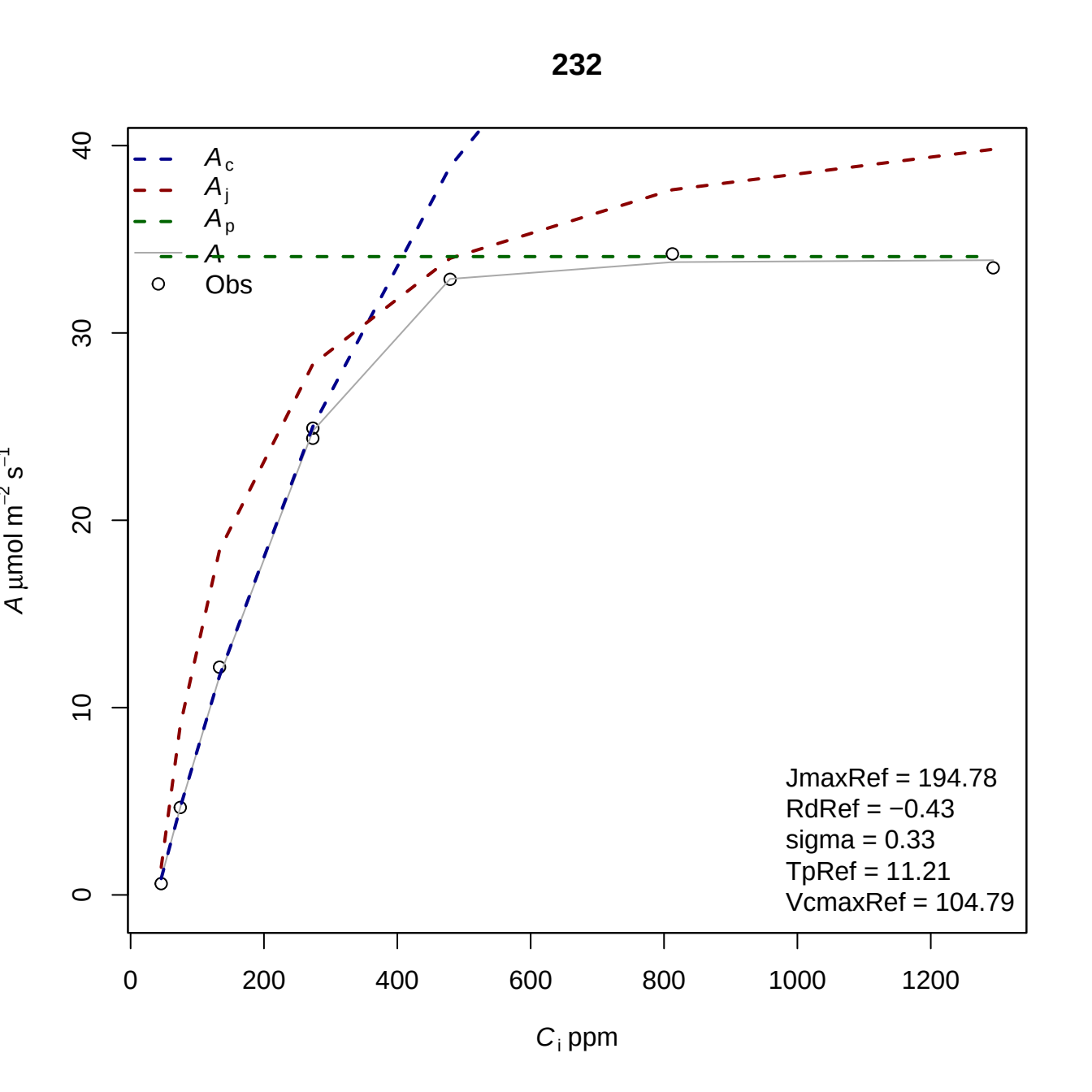

227
A_c
A_j
A_p
A
Obs
JmaxRef = 191.66
RdRef = −0.65
sigma = 0.32
TpRef = 11.18
VcmaxRef = 126.79
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

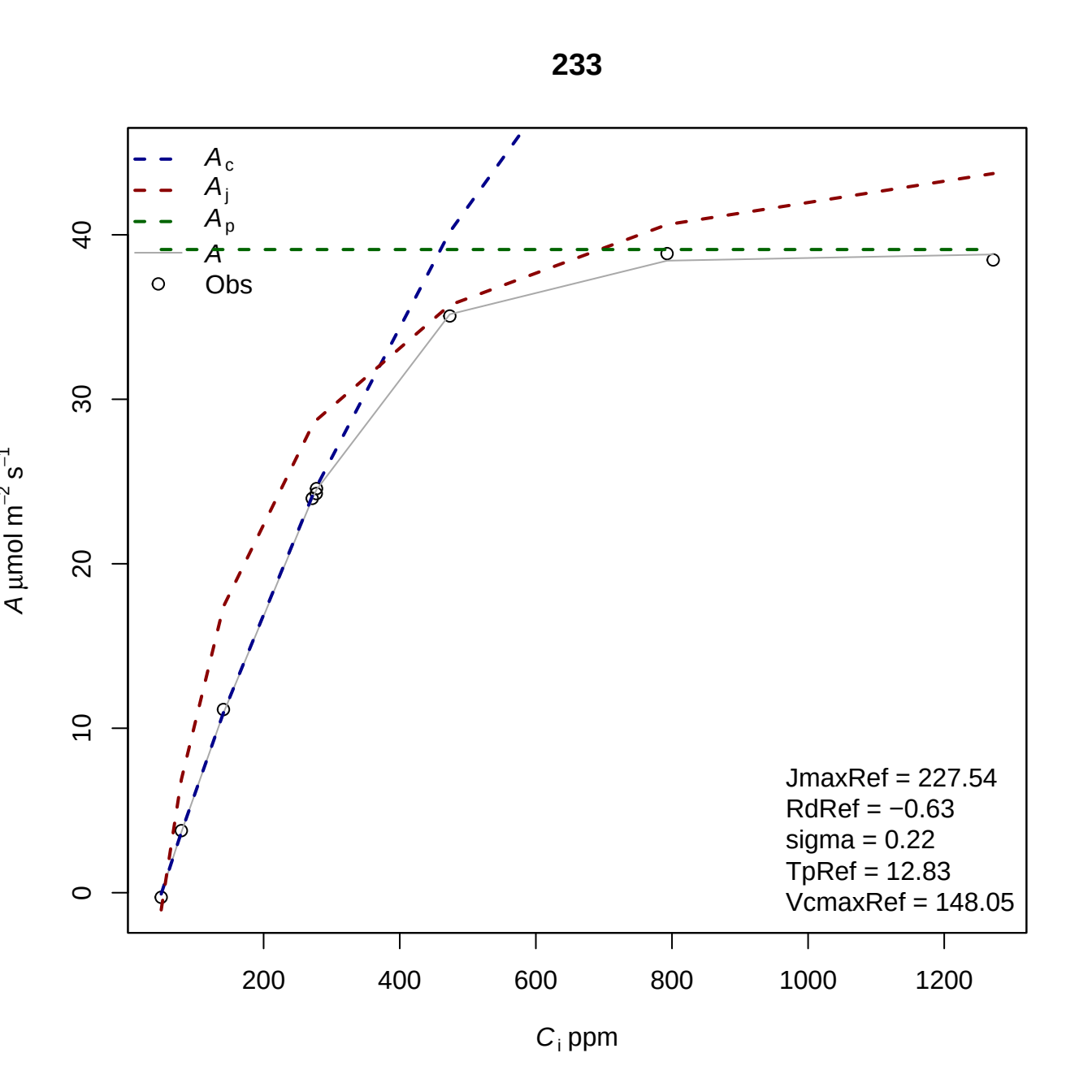

228
A_c
A_j
A_p
A
Obs
JmaxRef = 216.26
RdRef = −0.81
sigma = 0.32
TpRef = 12.56
VcmaxRef = 174.13
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
200
400
600
800
1000
1200

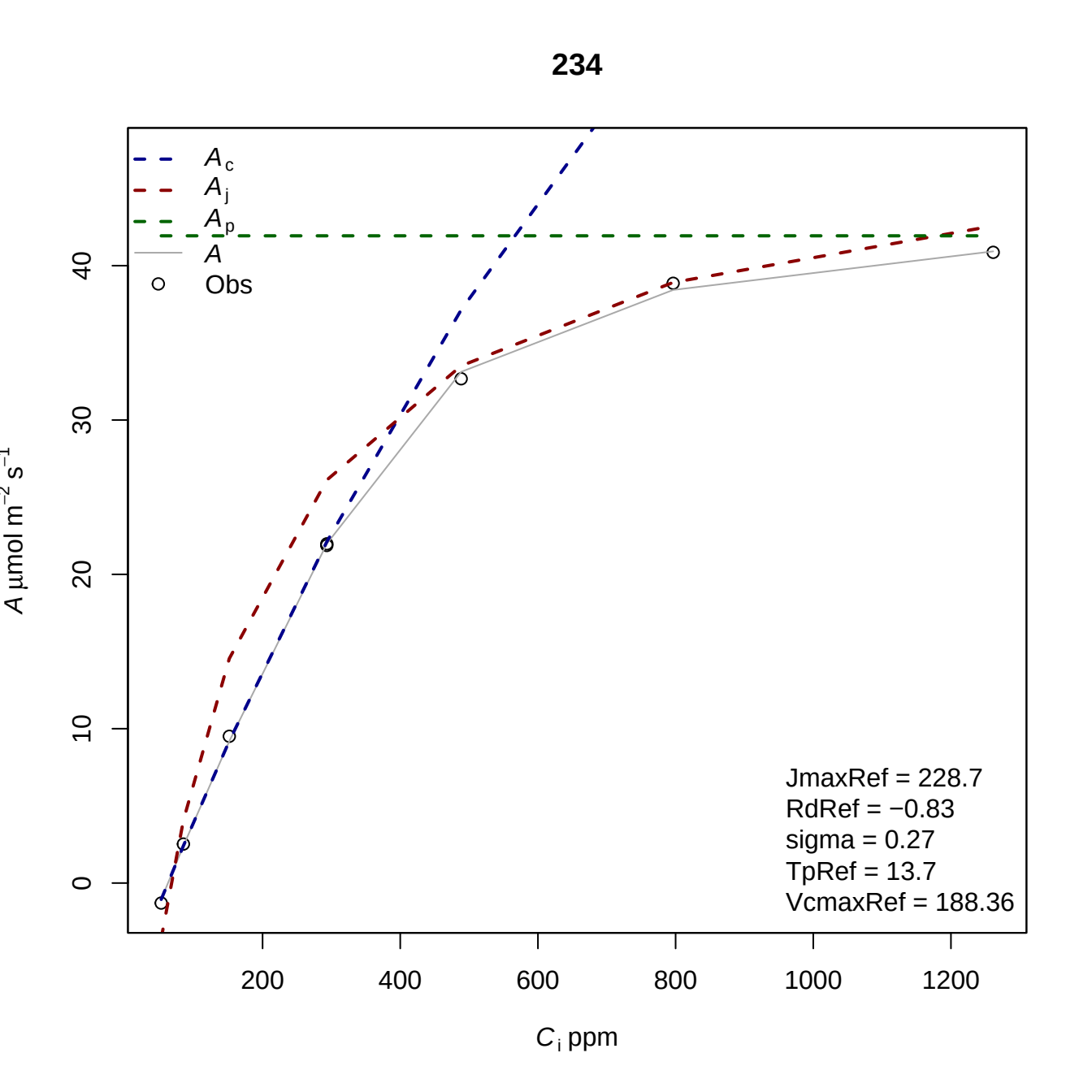

229
A_c
A_j
A_p
A
Obs
JmaxRef = 189.76
RdRef = −0.74
sigma = 0.2
TpRef = 11.99
VcmaxRef = 205.84
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
200
400
600
800
1000
1200

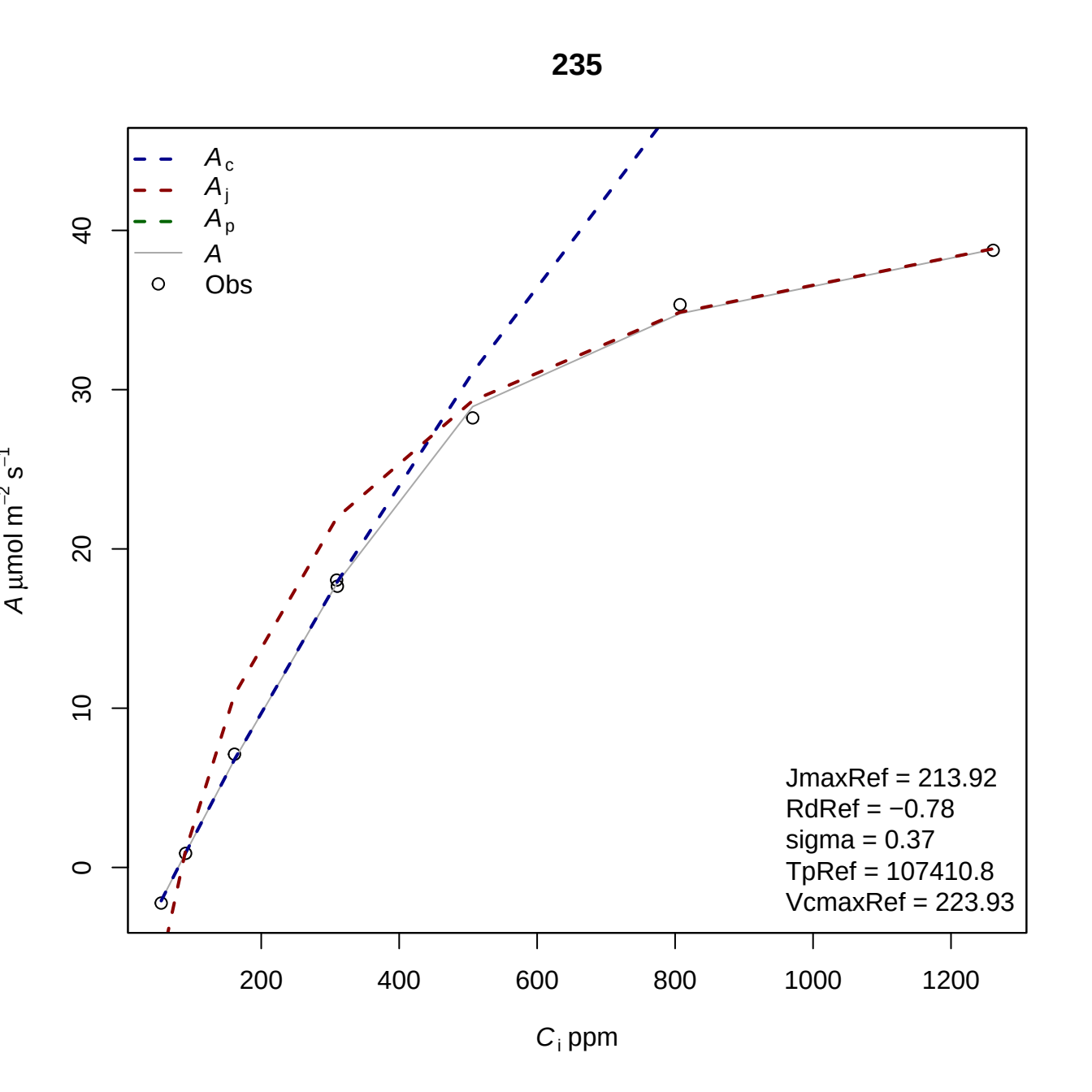

230
A_c
A_j
A_p
A
Obs
JmaxRef = 103.38
RdRef = 0.89
sigma = 0.46
TpRef = 7.36
VcmaxRef = 39.79
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200

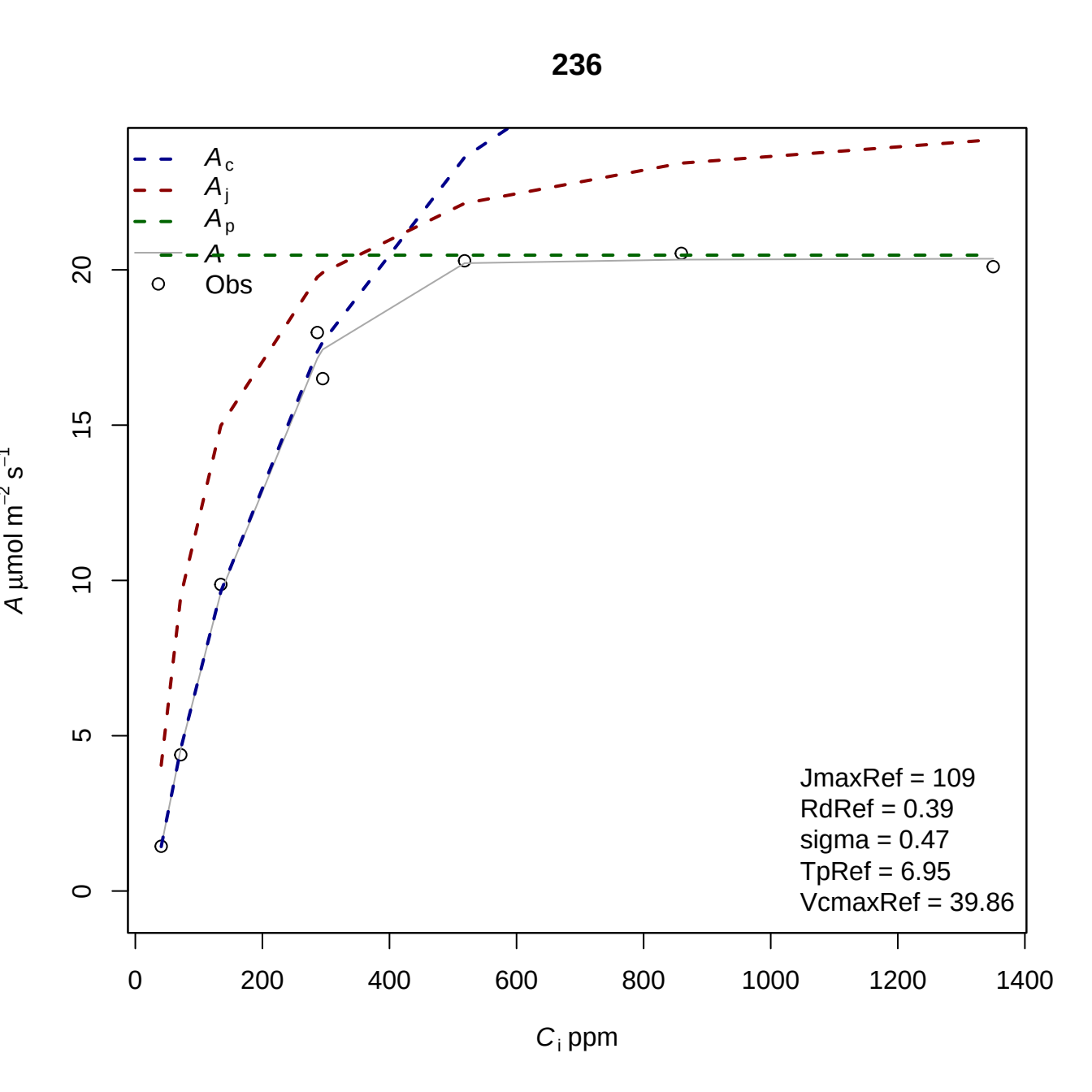

231
A_c
A_j
A_p
A
Obs
JmaxRef = 146.98
RdRef = 0.35
sigma = 0.44
TpRef = 9.85
VcmaxRef = 63.35
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

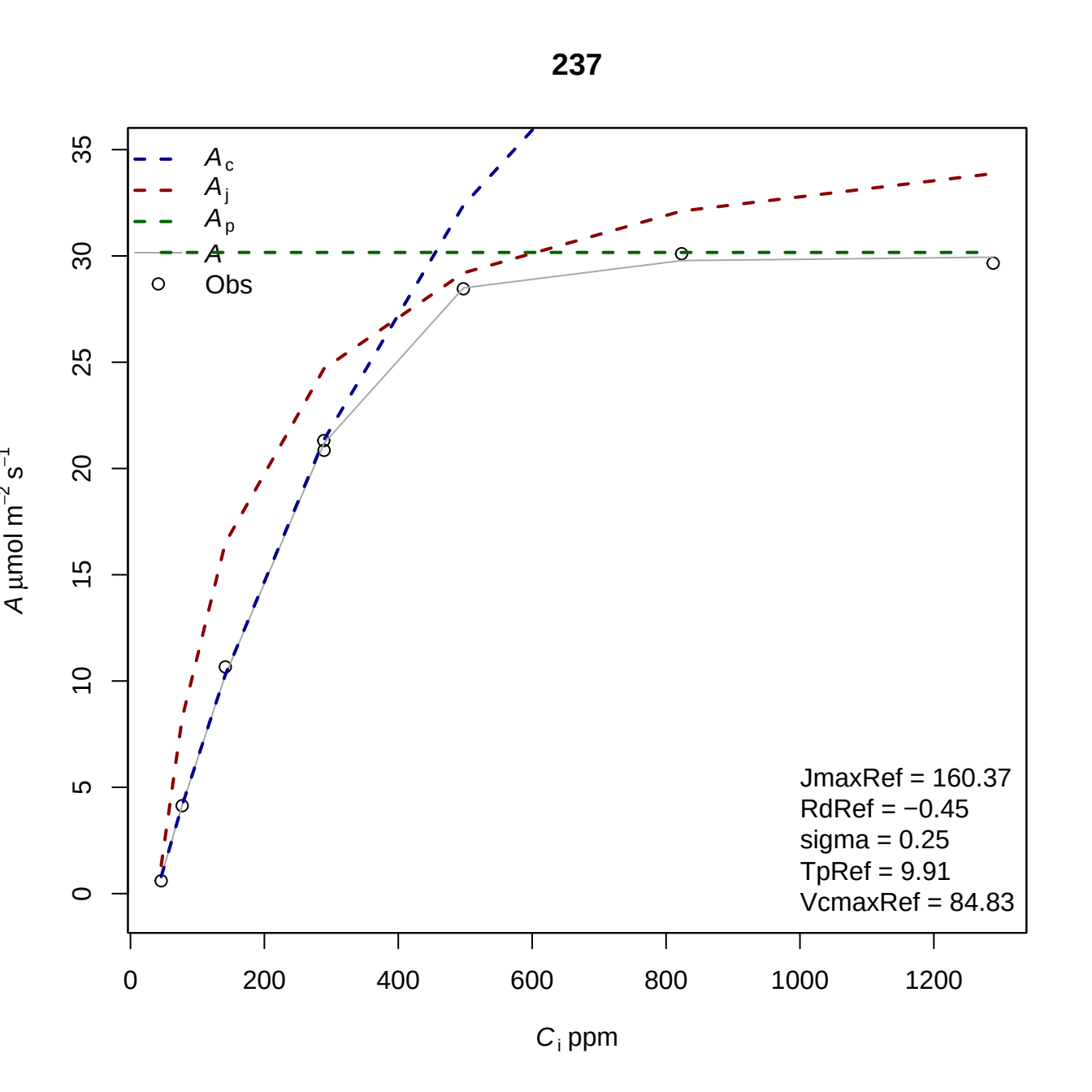

232
A_c
A_j
A_p
A
Obs
JmaxRef = 194.78
RdRef = −0.43
sigma = 0.33
TpRef = 11.21
VcmaxRef = 104.79
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

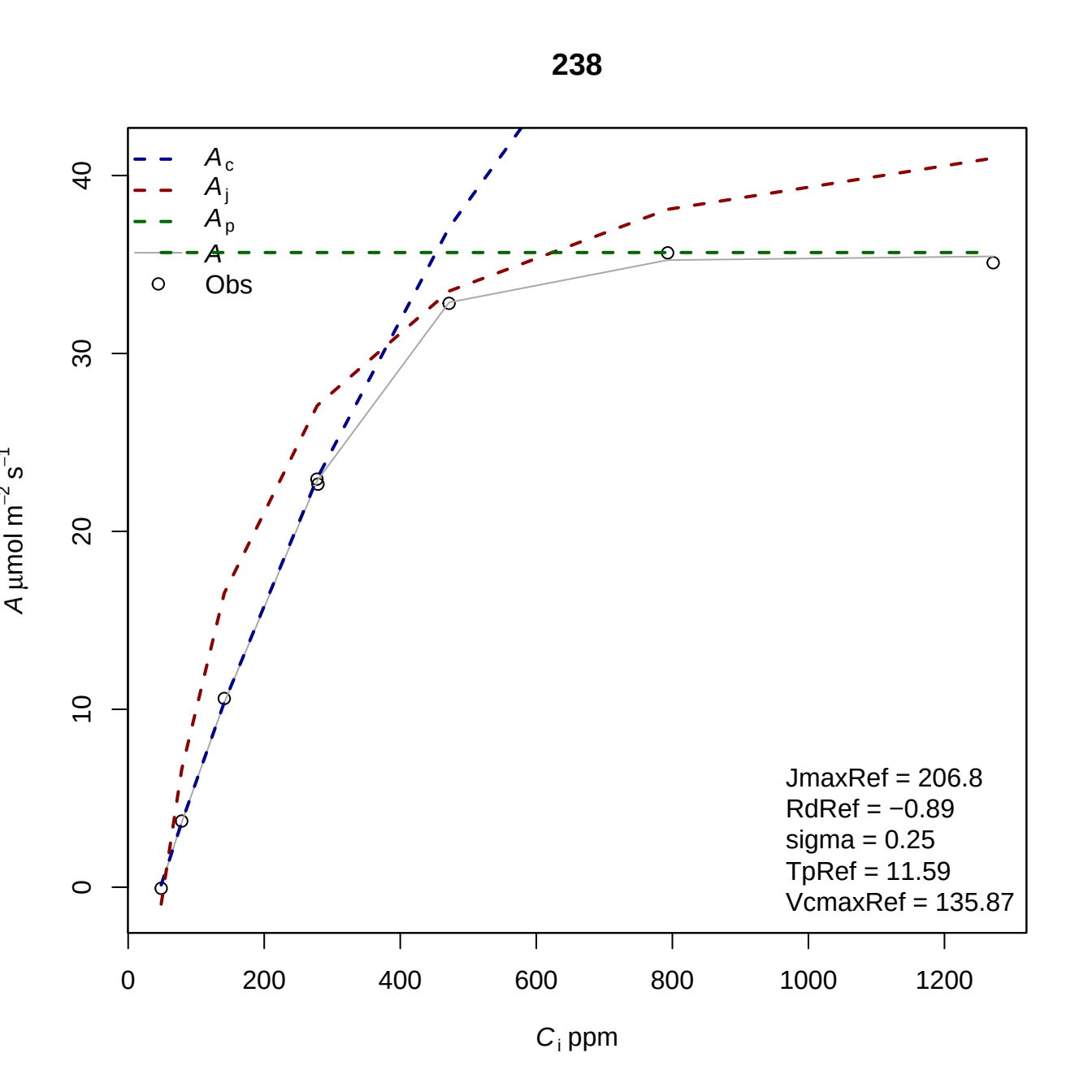

233
A_c
A_j
A_p
A
Obs
JmaxRef = 227.54
RdRef = −0.63
sigma = 0.22
TpRef = 12.83
VcmaxRef = 148.05
0
10
20
30
40
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

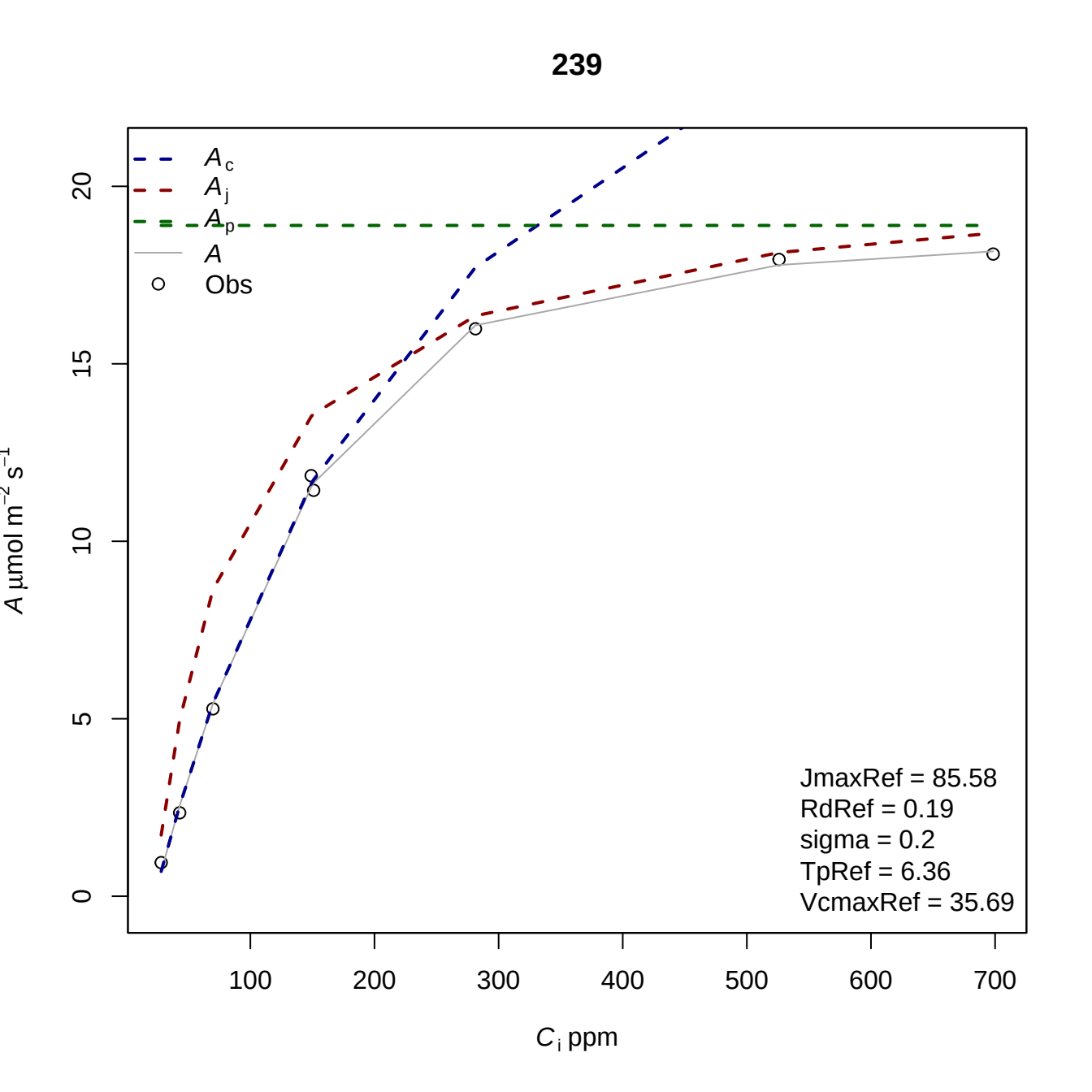

234
A_c
A_j
A_p
A
Obs
JmaxRef = 228.7
RdRef = −0.83
sigma = 0.27
TpRef = 13.7
VcmaxRef = 188.36
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
200
400
600
800
1000
1200

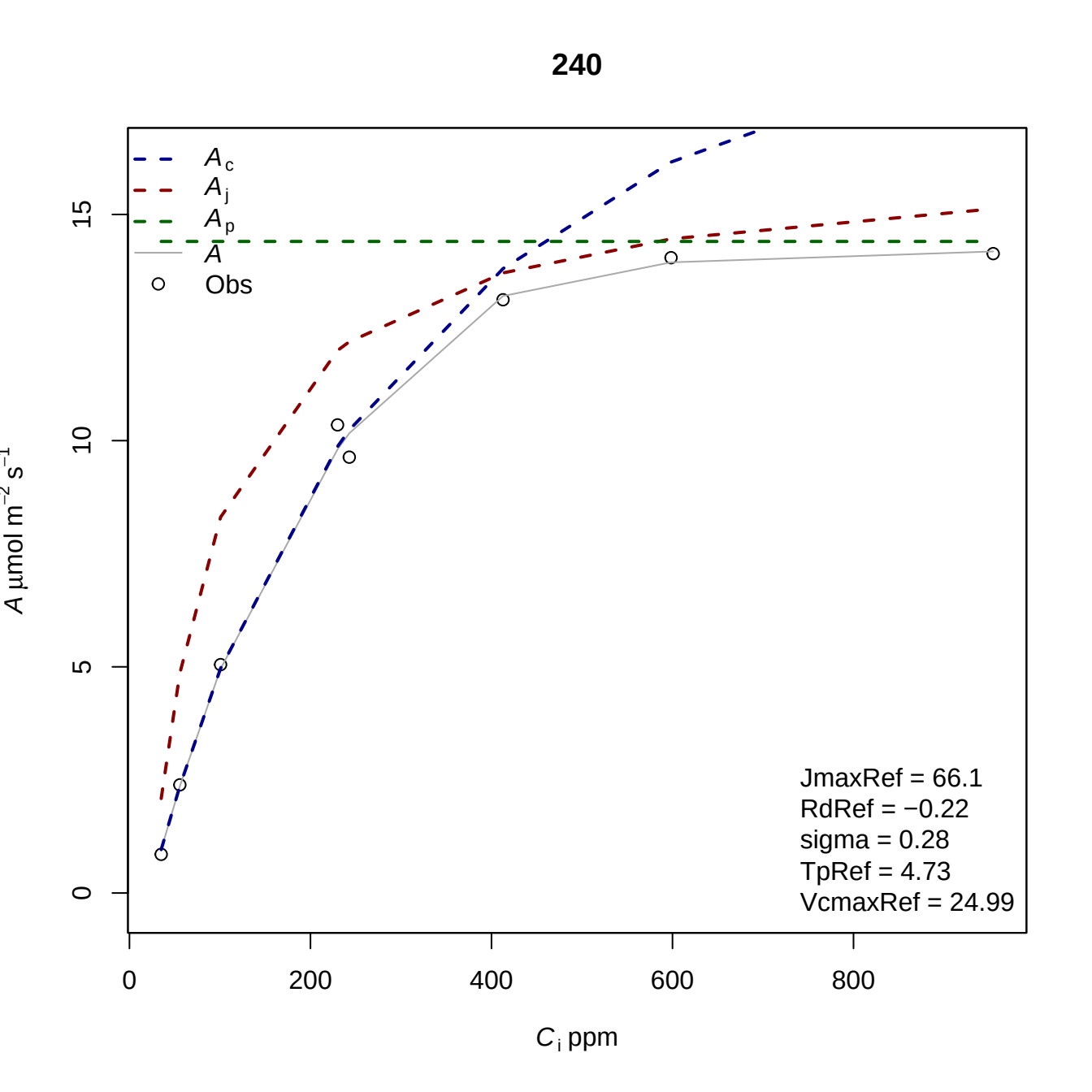

235
A_c
A_j
A_p
A
Obs
JmaxRef = 213.92
RdRef = −0.78
sigma = 0.37
TpRef = 107410.8
VcmaxRef = 223.93
0
10
20
30
40
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

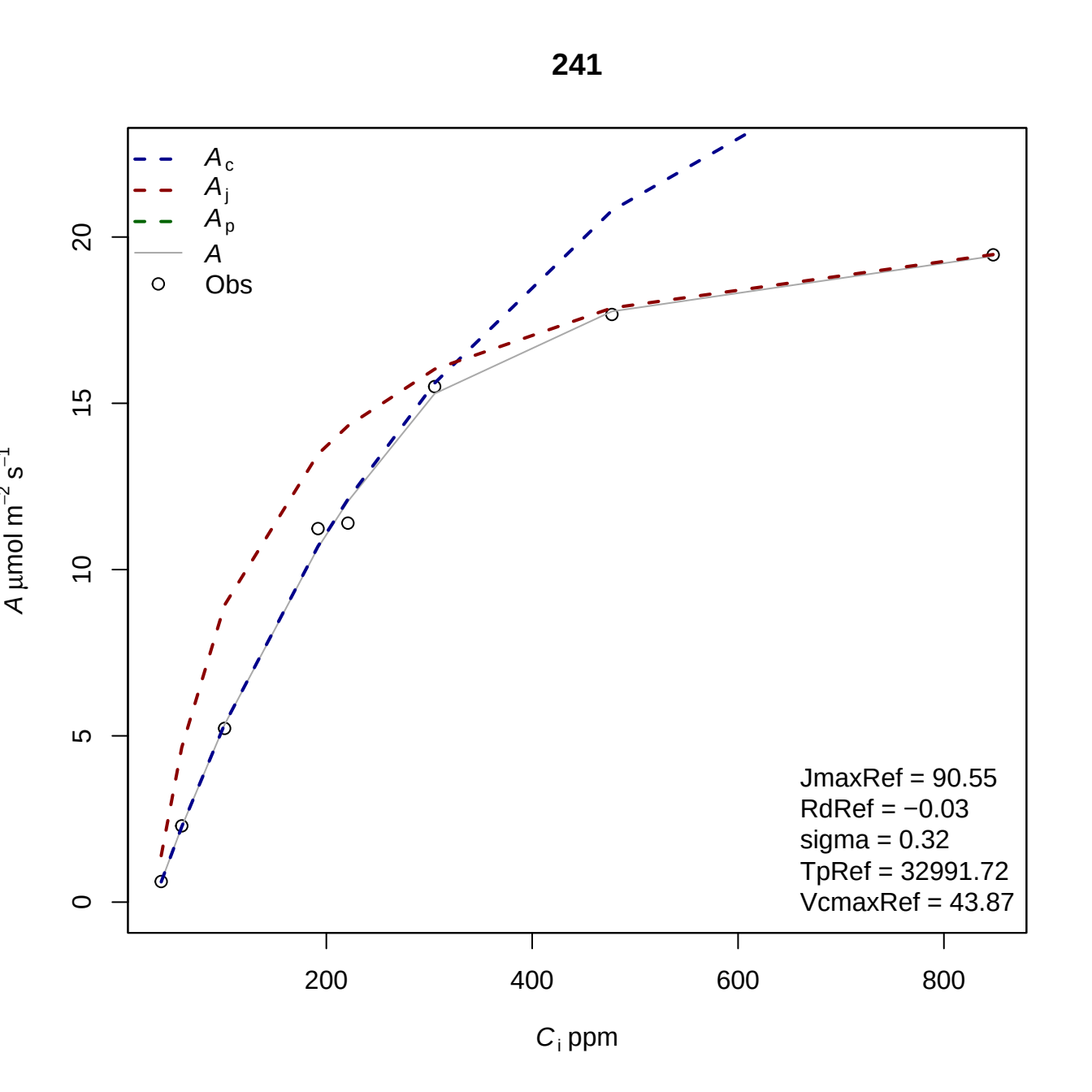

236
A_c
A_j
A_p
A
Obs
JmaxRef = 109
RdRef = 0.39
sigma = 0.47
TpRef = 6.95
VcmaxRef = 39.86
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000
1200
1400

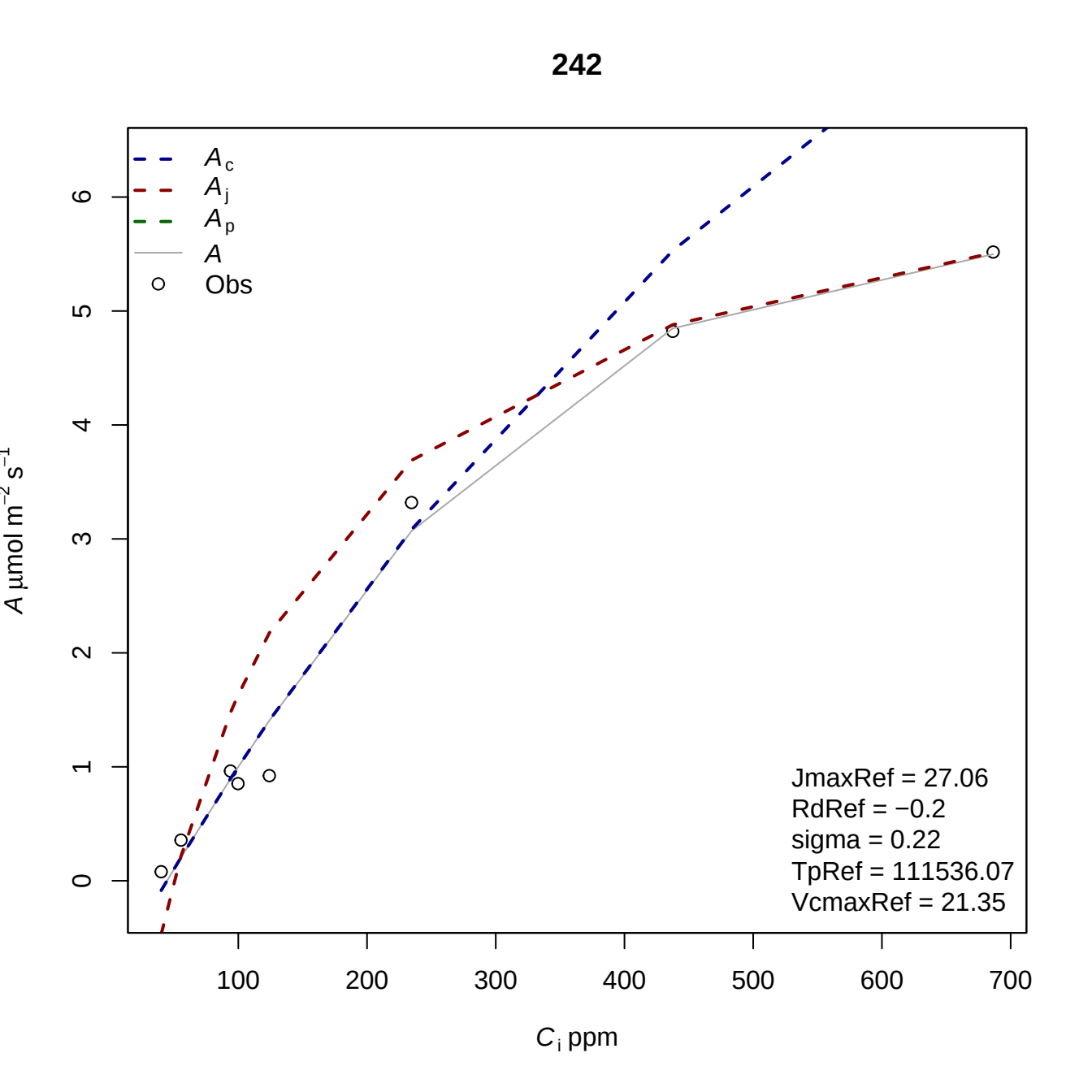

237
A_c
A_j
A_p
A
Obs
JmaxRef = 160.37
RdRef = −0.45
sigma = 0.25
TpRef = 9.91
VcmaxRef = 84.83
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200

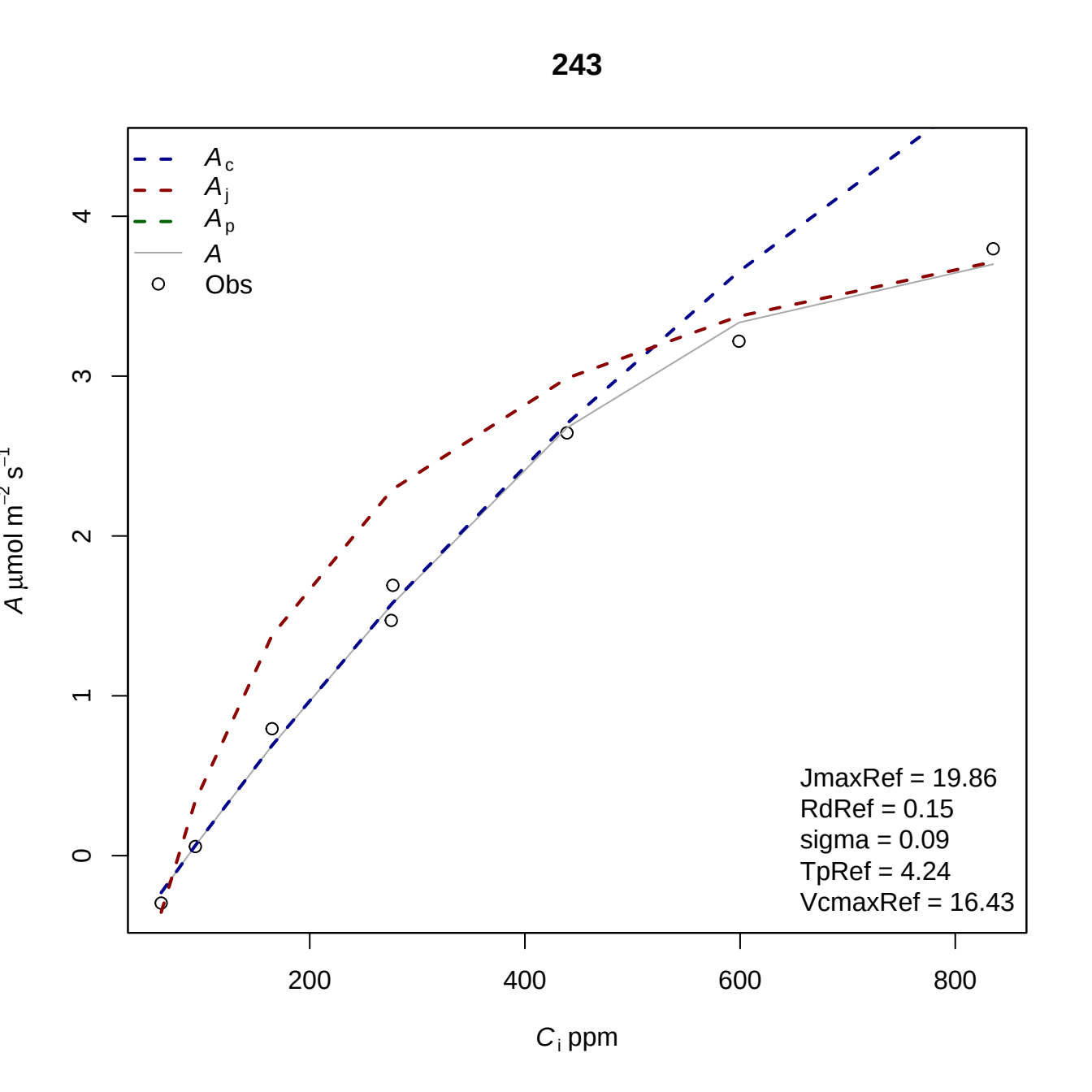

238
A_c
A_j
A_p
A
Obs
JmaxRef = 206.8
RdRef = −0.89
sigma = 0.25
TpRef = 11.59
VcmaxRef = 135.87
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

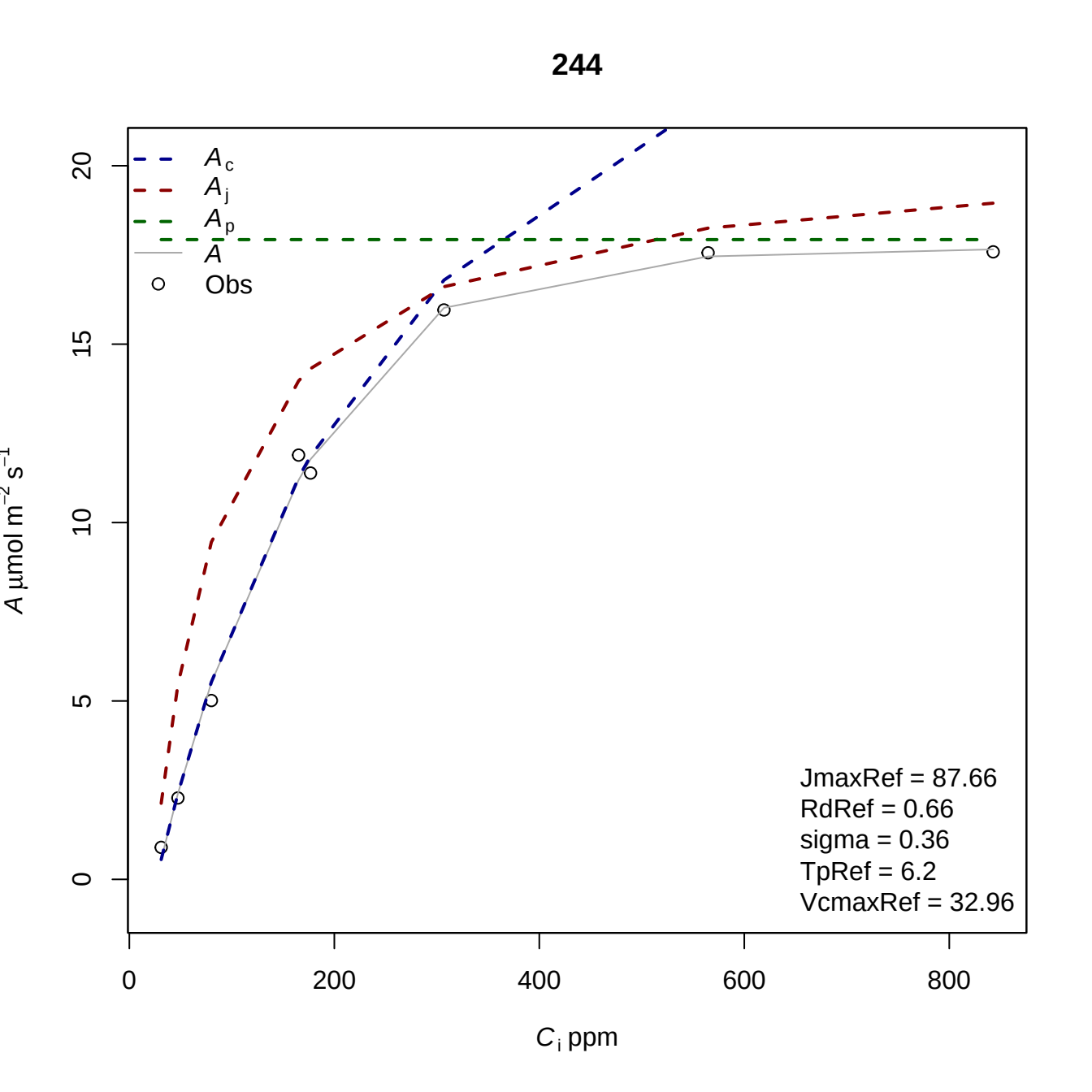

239
A_c
A_j
A_p
A
Obs
JmaxRef = 85.58
RdRef = 0.19
sigma = 0.2
TpRef = 6.36
VcmaxRef = 35.69
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
100
200
300
400
500
600
700

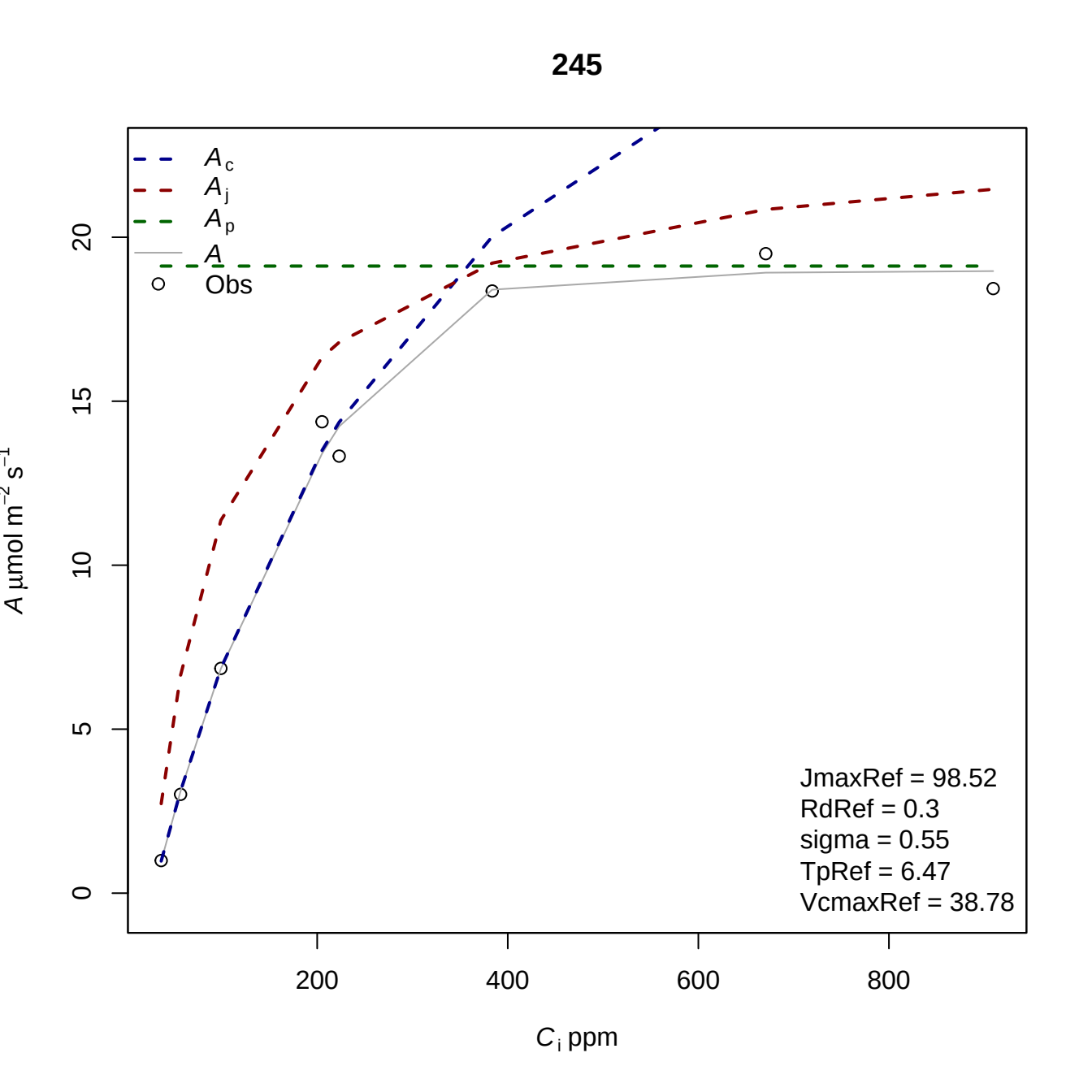

240
A_c
A_j
A_p
A
Obs
JmaxRef = 66.1
RdRef = −0.22
sigma = 0.28
TpRef = 4.73
VcmaxRef = 24.99
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
200
400
600
800

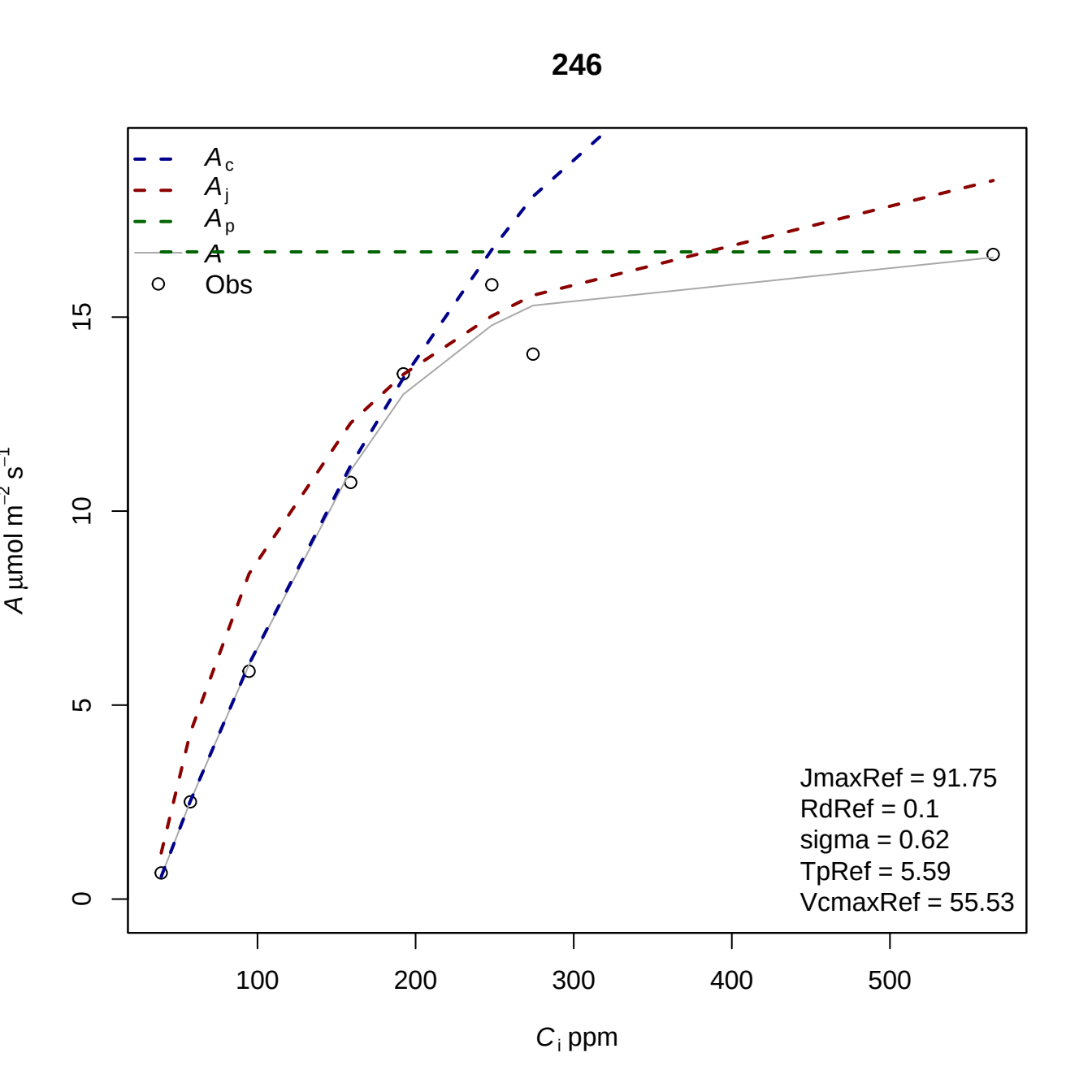

241
A_c
A_j
A_p
A
Obs
JmaxRef = 90.55
RdRef = −0.03
sigma = 0.32
TpRef = 32991.72
VcmaxRef = 43.87
0
5
10
15
20
200
400
600
800
A µmol m^{-2} s^{-1}
C_i ppm

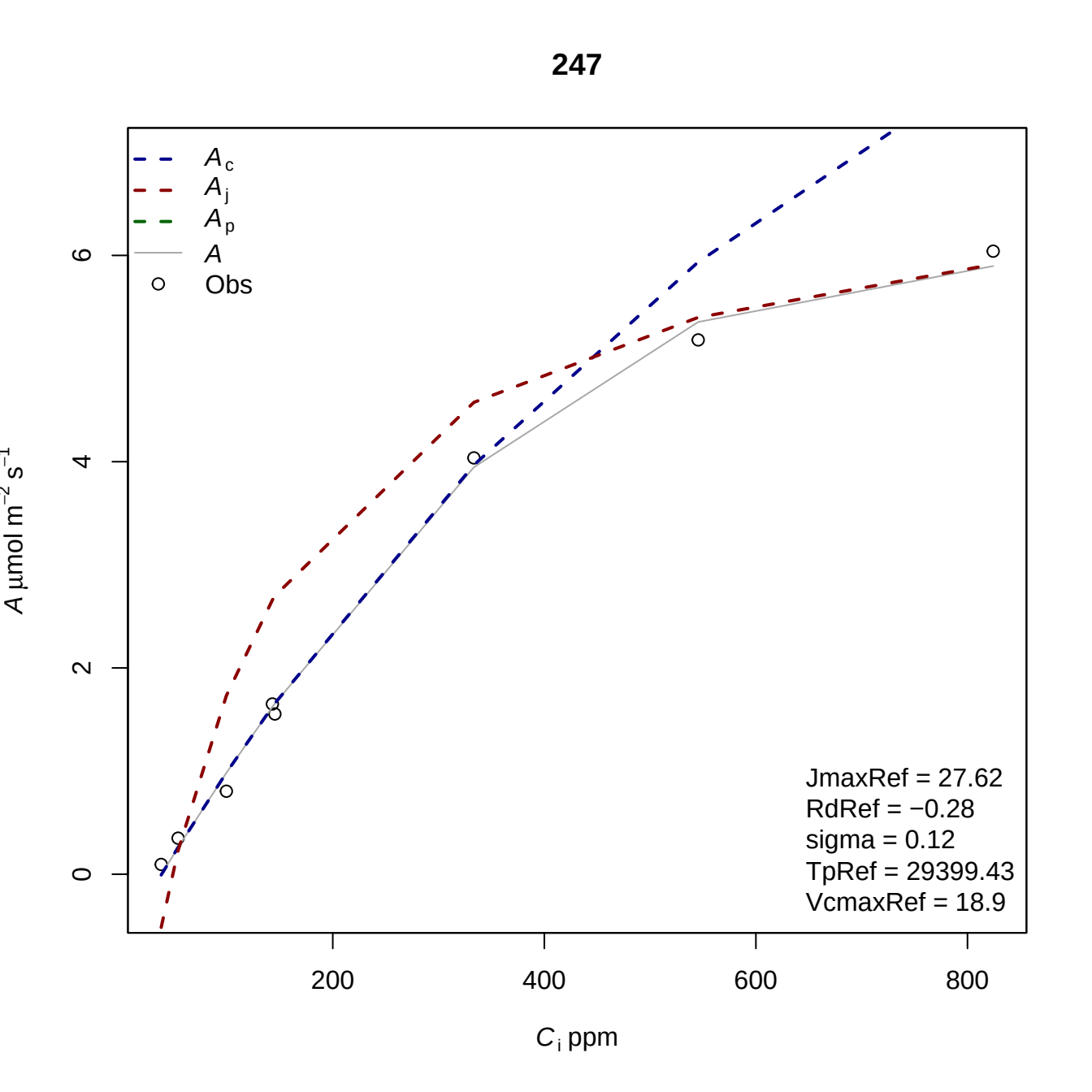

242
A_c
A_j
A_p
A
Obs
JmaxRef = 27.06
RdRef = −0.2
sigma = 0.22
TpRef = 111536.07
VcmaxRef = 21.35
A µmol m^{-2} s^{-1}
C_i ppm
0
1
2
3
4
5
6
100
200
300
400
500
600
700

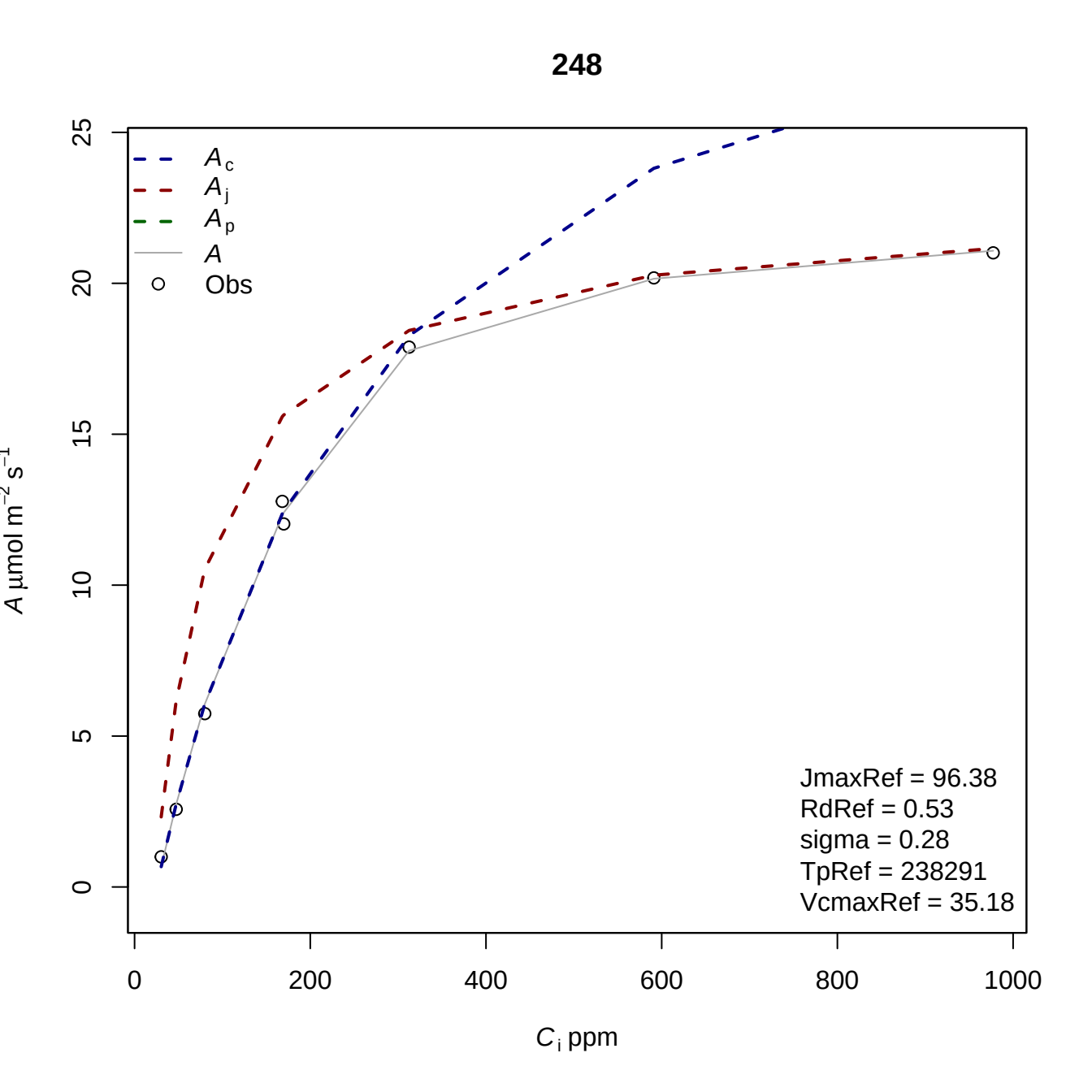

243
A_c
A_j
A_p
A
Obs
JmaxRef = 19.86
RdRef = 0.15
sigma = 0.09
TpRef = 4.24
VcmaxRef = 16.43
0
1
2
3
4
200
400
600
800
A µmol m^{-2} s^{-1}
C_i ppm

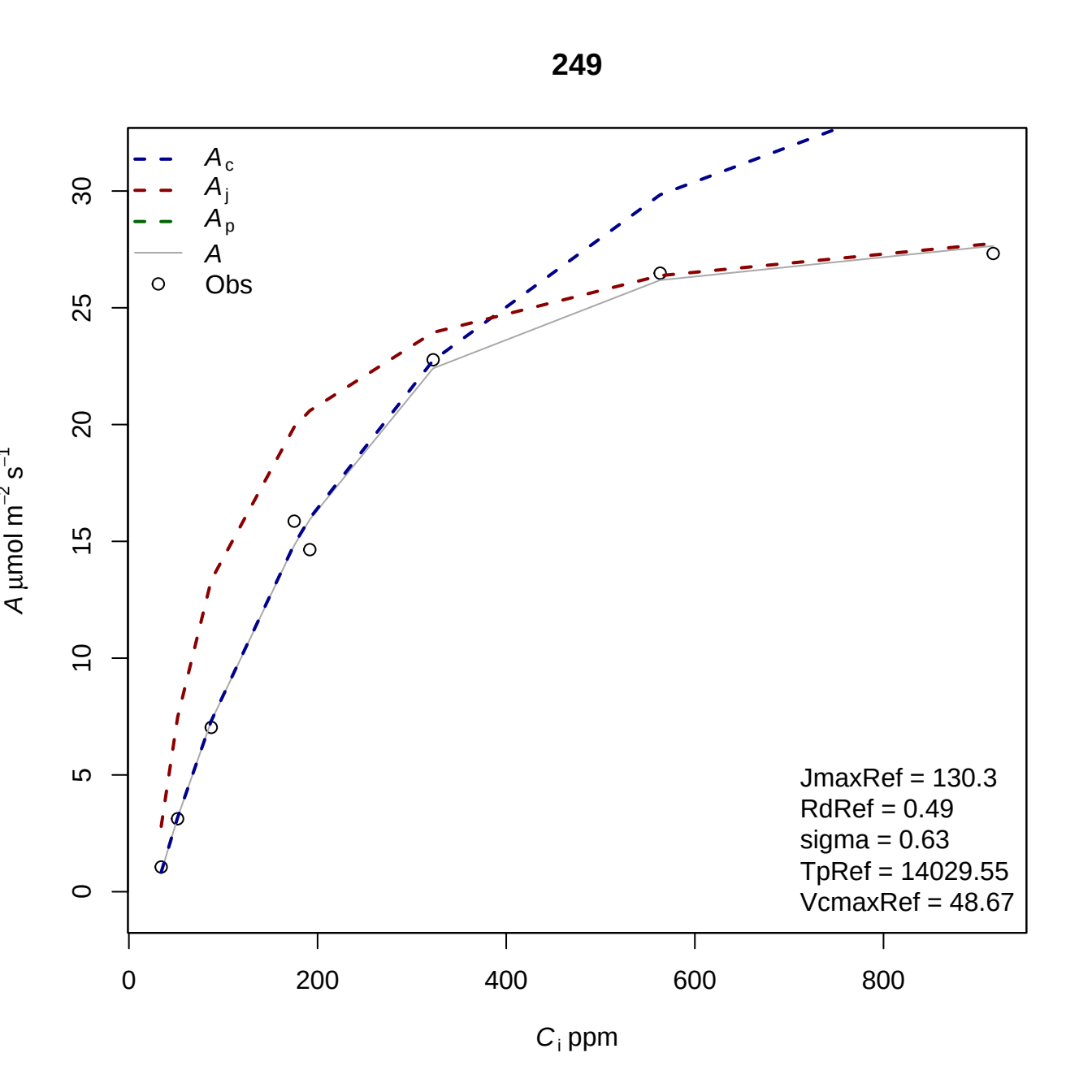

244
A_c
A_j
A_p
A
Obs
JmaxRef = 87.66
RdRef = 0.66
sigma = 0.36
TpRef = 6.2
VcmaxRef = 32.96
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800

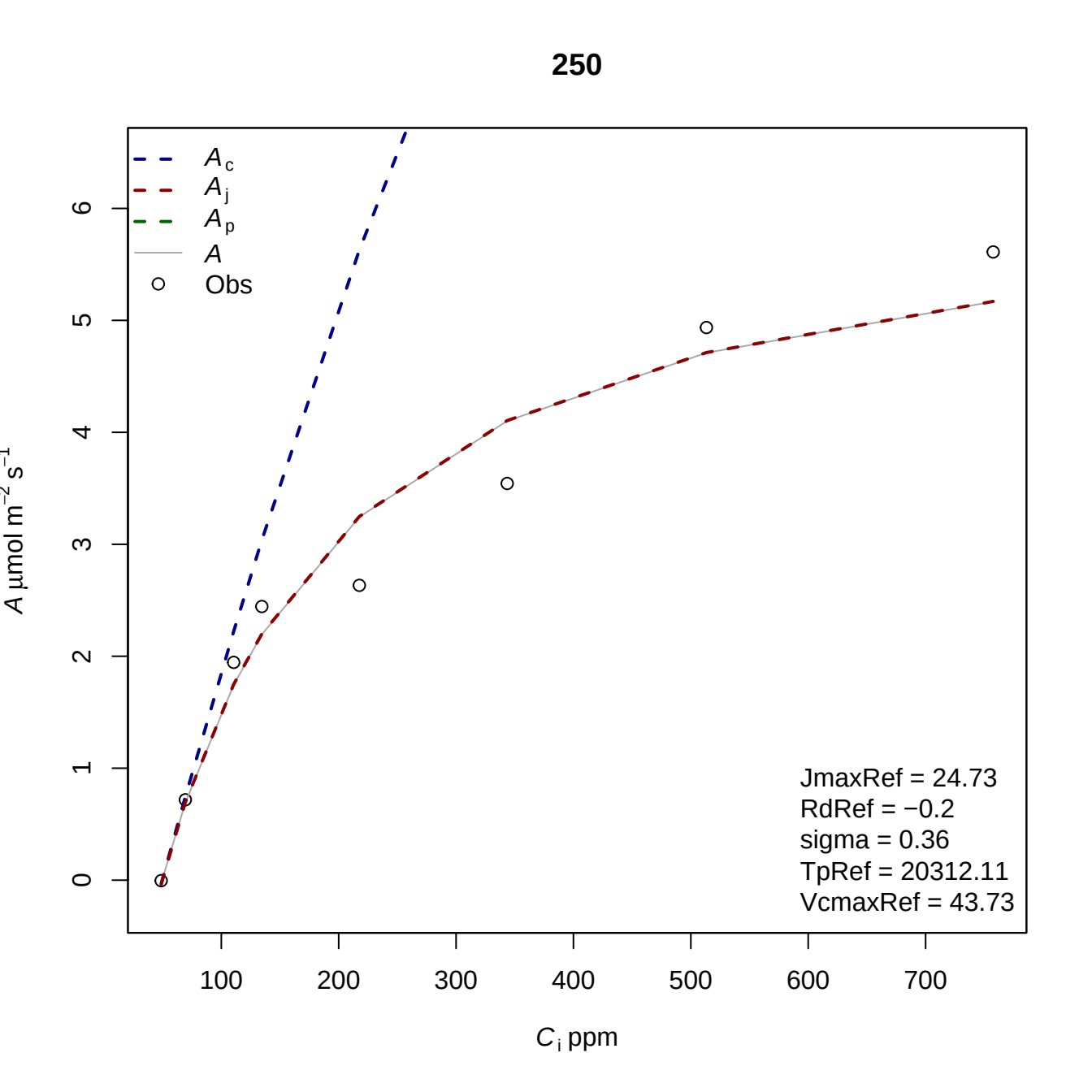

245
A_c
A_j
A_p
A
Obs
JmaxRef = 98.52
RdRef = 0.3
sigma = 0.55
TpRef = 6.47
VcmaxRef = 38.78
0
5
10
15
20
200
400
600
800
A µmol m^{-2} s^{-1}
C_i ppm

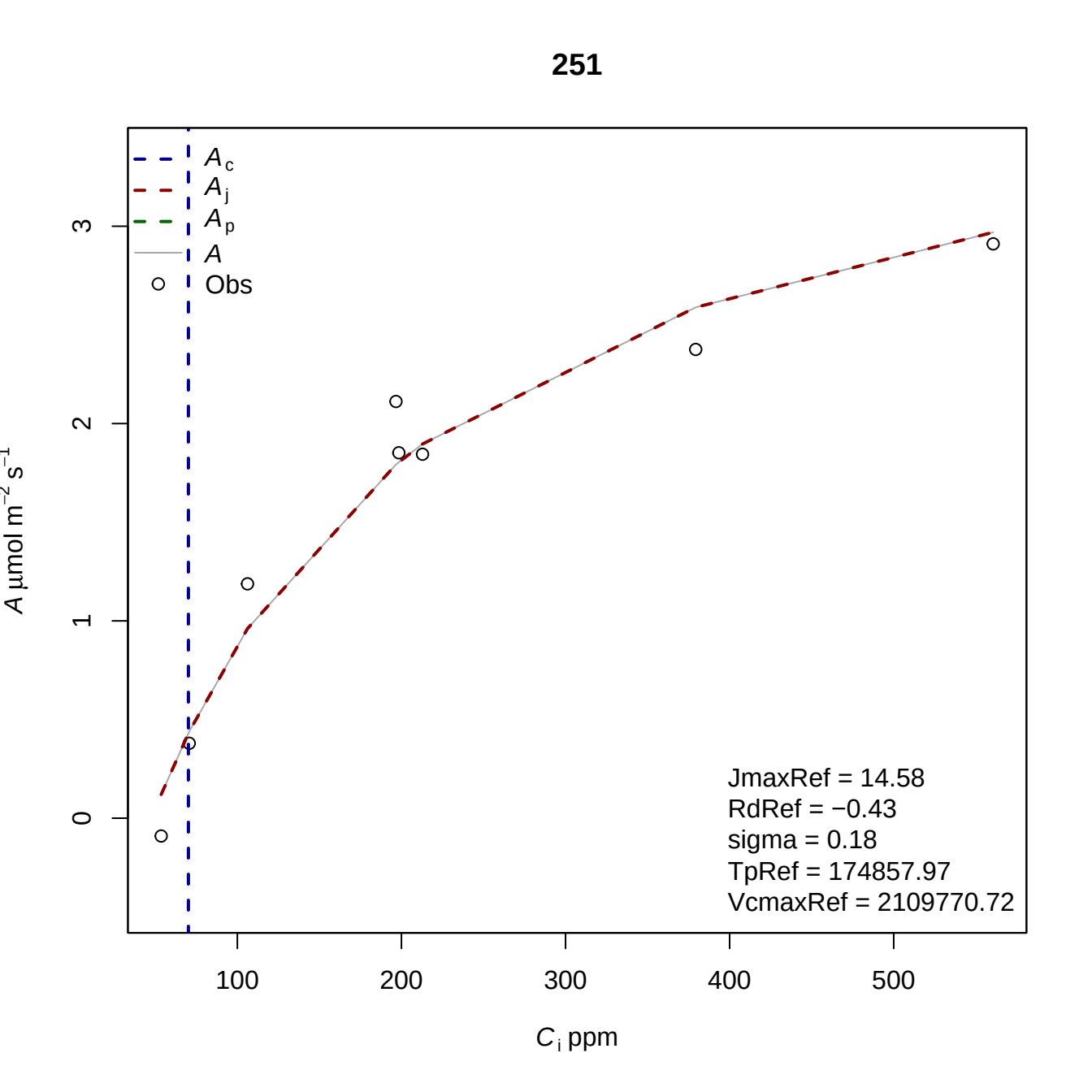

246
A_c
A_j
A_p
A
Obs
JmaxRef = 91.75
RdRef = 0.1
sigma = 0.62
TpRef = 5.59
VcmaxRef = 55.53
0
5
10
15
100
200
300
400
500
A µmol m^{-2} s^{-1}
C_i ppm

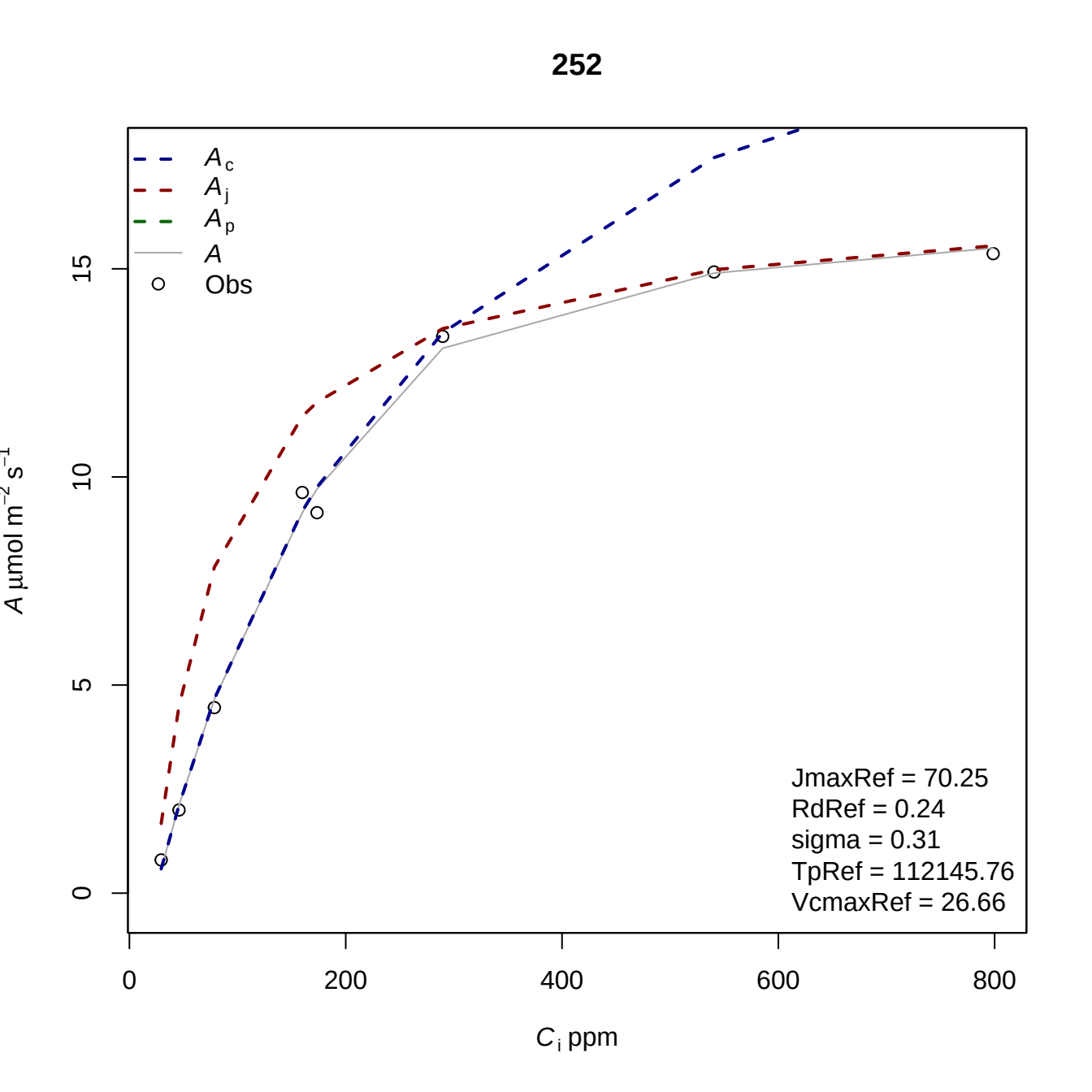

247
A_c
A_j
A_p
A
Obs
JmaxRef = 27.62
RdRef = −0.28
sigma = 0.12
TpRef = 29399.43
VcmaxRef = 18.9
0
2
4
6
200
400
600
800
A µmol m^{-2} s^{-1}
C_i ppm

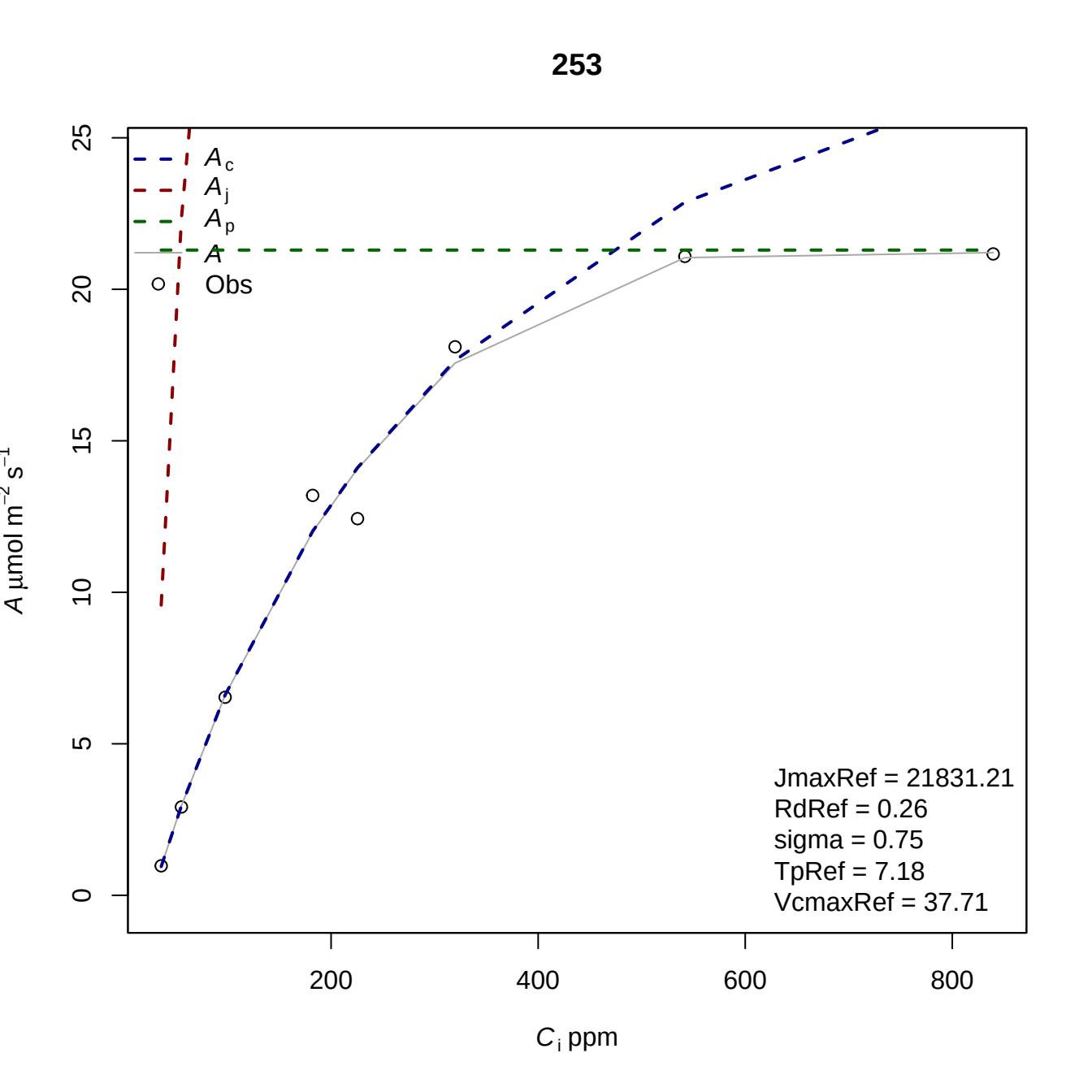

248
A_c
A_j
A_p
A
Obs
JmaxRef = 96.38
RdRef = 0.53
sigma = 0.28
TpRef = 238291
VcmaxRef = 35.18
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
200
400
600
800
1000

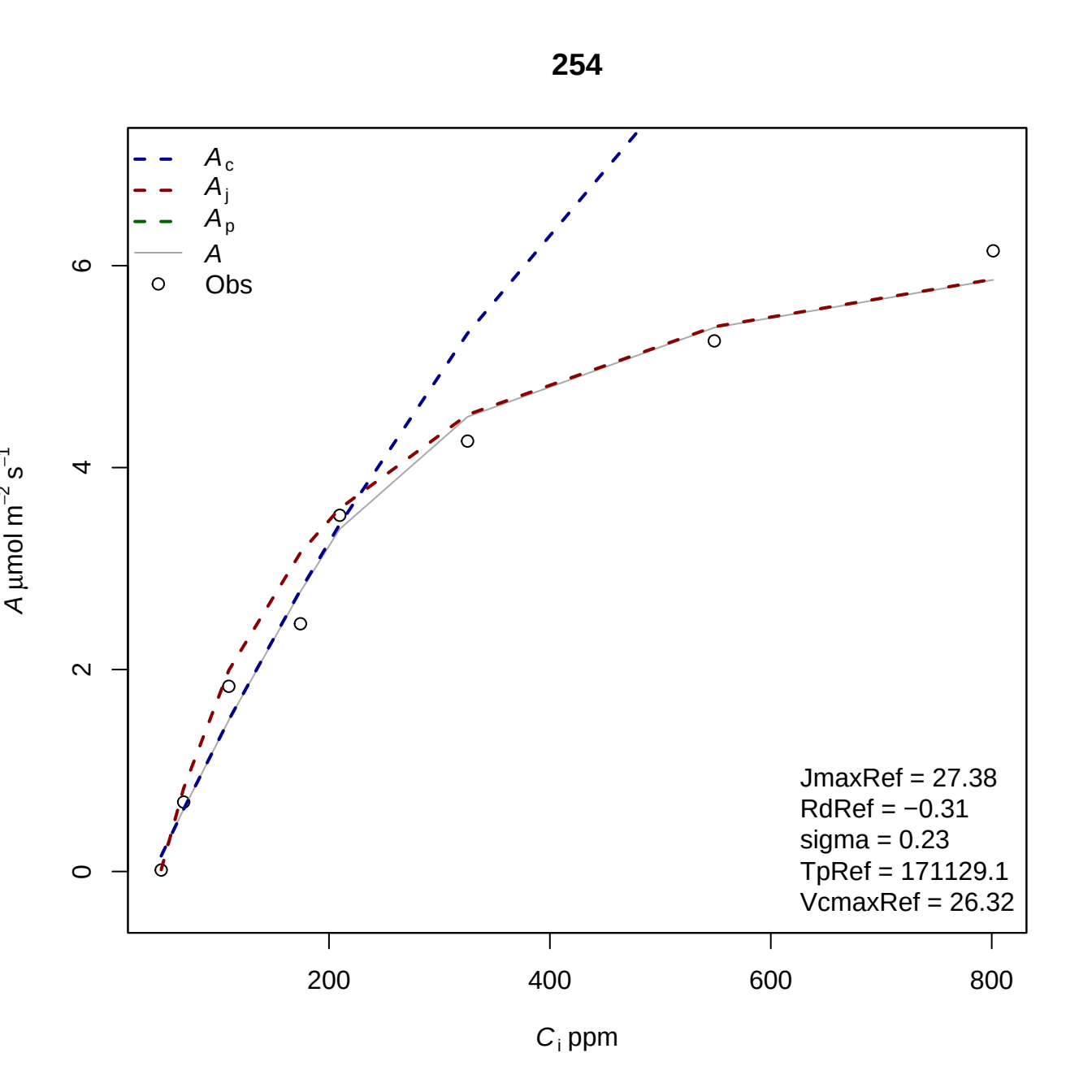

249
A_c
A_j
A_p
A
Obs
JmaxRef = 130.3
RdRef = 0.49
sigma = 0.63
TpRef = 14029.55
VcmaxRef = 48.67
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
0
200
400
600
800

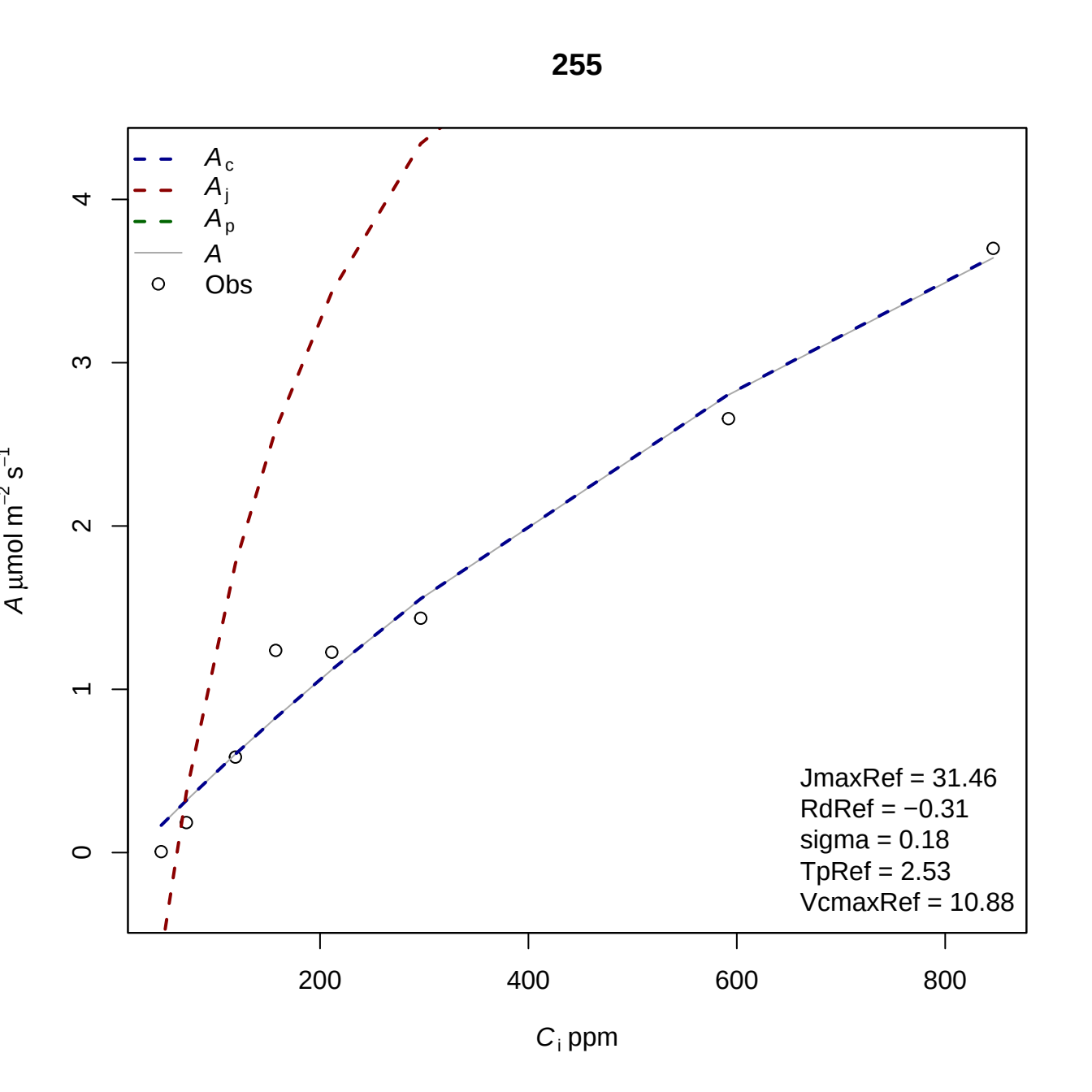

250
A_c
A_j
A_p
A
Obs
JmaxRef = 24.73
RdRef = −0.2
sigma = 0.36
TpRef = 20312.11
VcmaxRef = 43.73
A µmol m^{-2} s^{-1}
C_i ppm
0
1
2
3
4
5
6
100
200
300
400
500
600
700

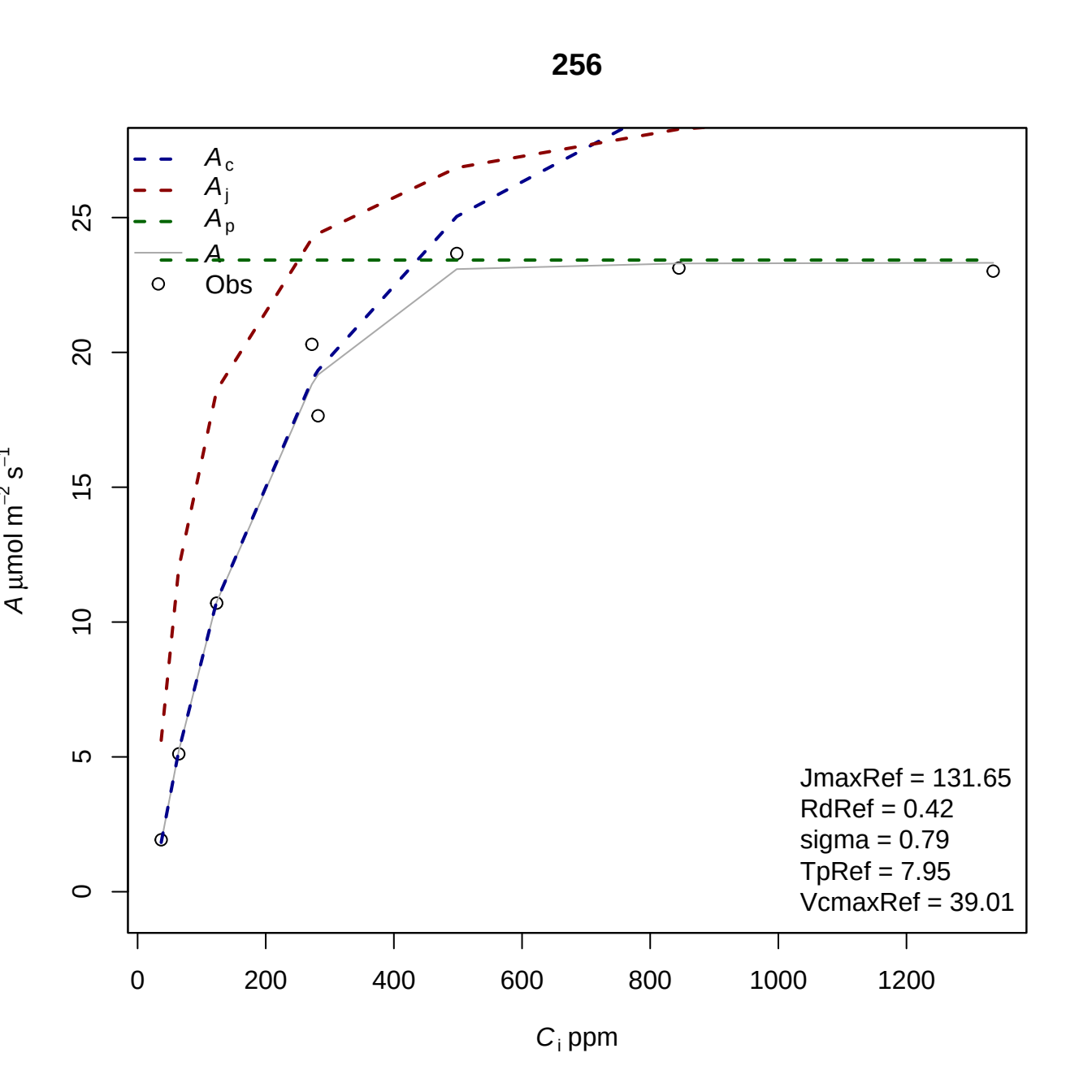

251
A_c
A_j
A_p
A
Obs
A µmol m^{-2} s^{-1}
0
1
2
3
100
200
300
400
500
C_i ppm
JmaxRef = 14.58
RdRef = −0.43
sigma = 0.18
TpRef = 174857.97
VcmaxRef = 2109770.72

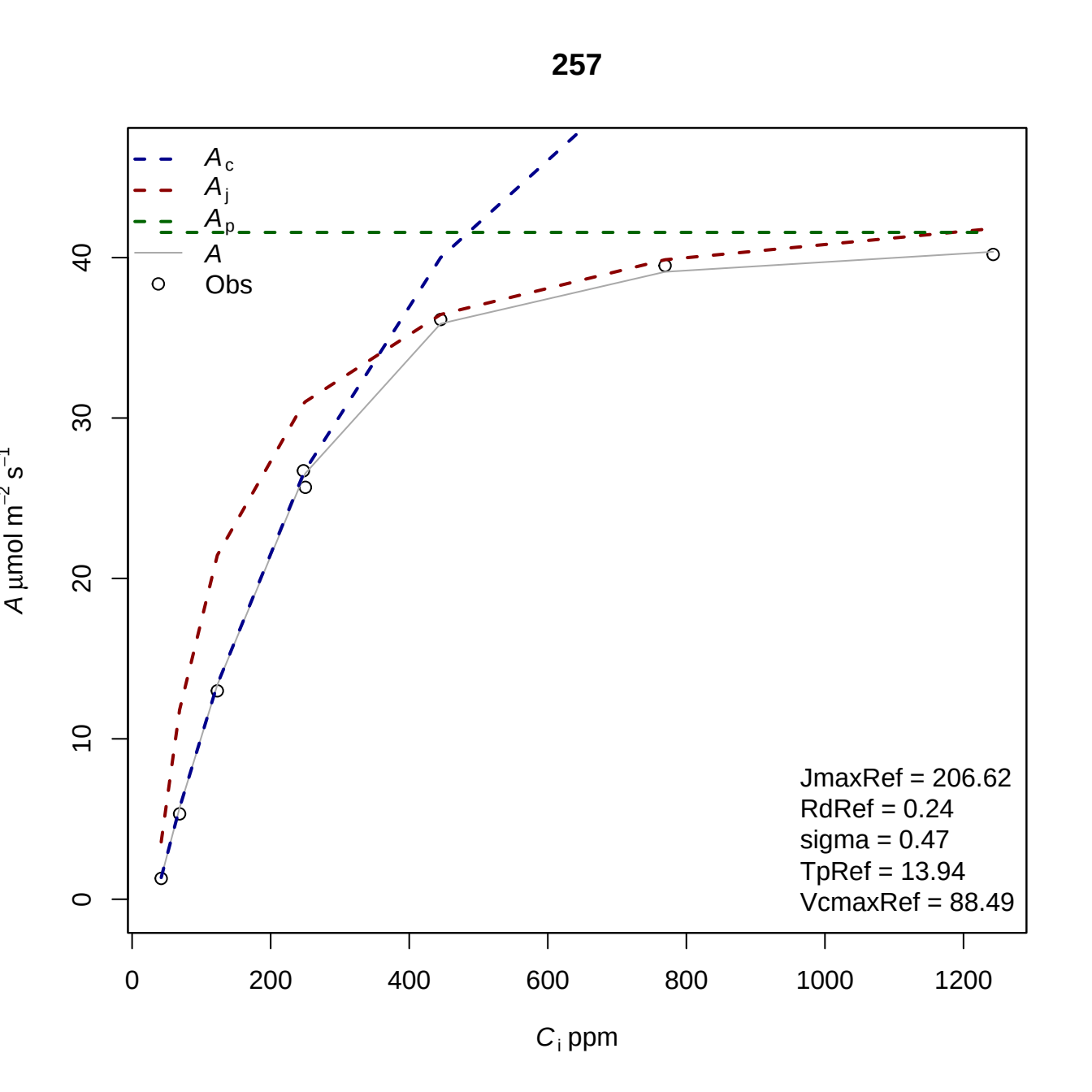

252
A_c
A_j
A_p
A
Obs
JmaxRef = 70.25
RdRef = 0.24
sigma = 0.31
TpRef = 112145.76
VcmaxRef = 26.66
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
200
400
600
800

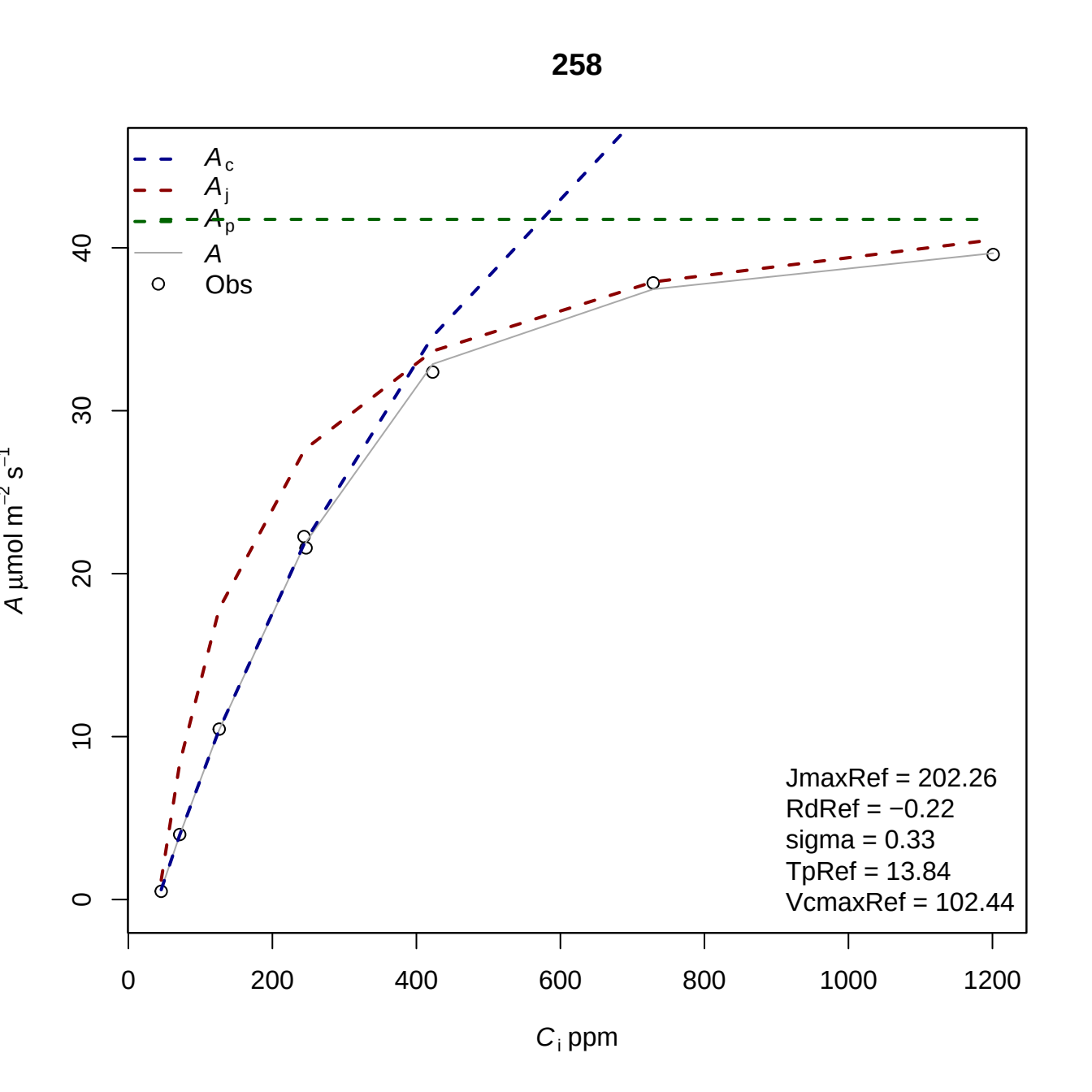

253
A_c
A_j
A_p
A
Obs
JmaxRef = 21831.21
RdRef = 0.26
sigma = 0.75
TpRef = 7.18
VcmaxRef = 37.71
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
200
400
600
800

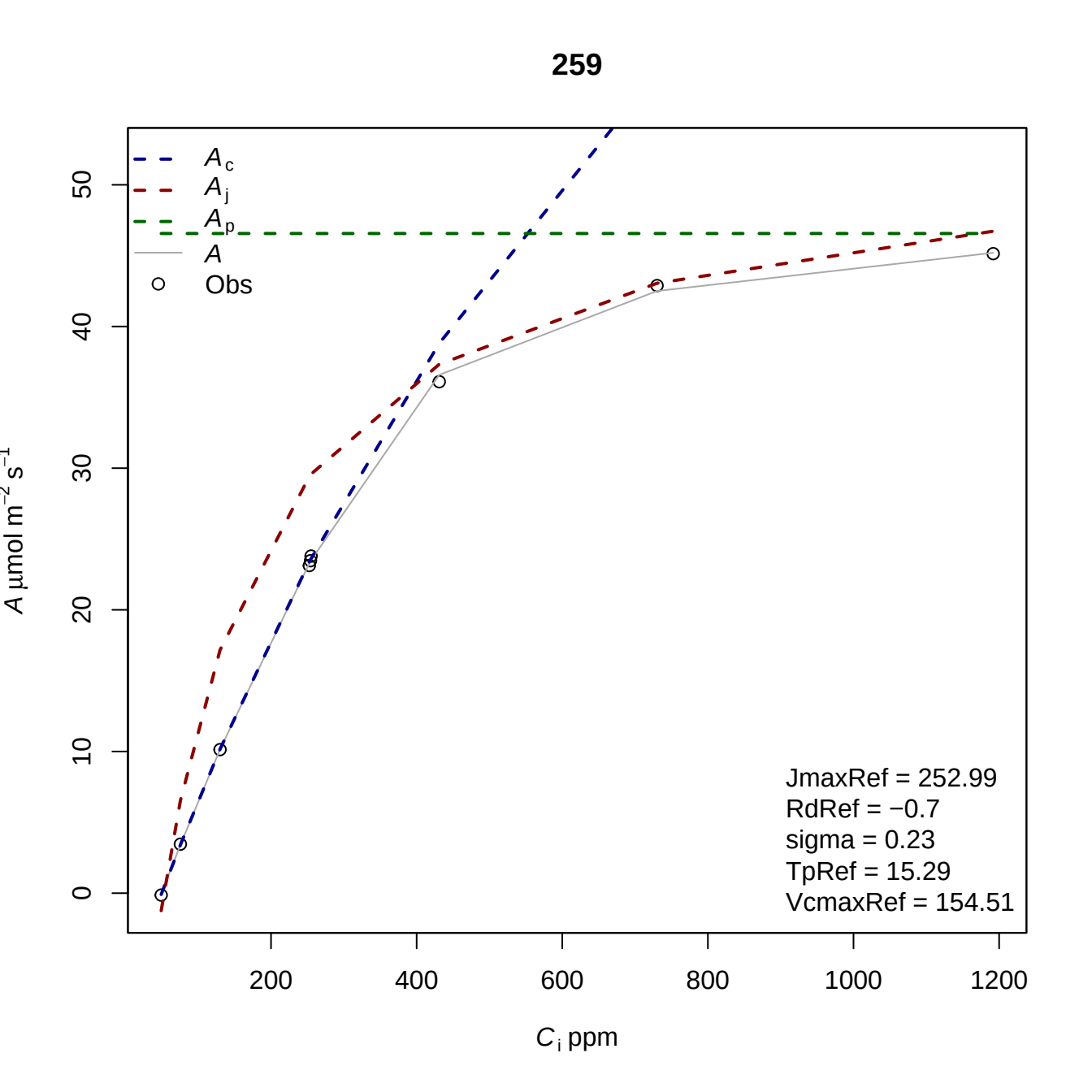

254
A_c
A_j
A_p
A
Obs
JmaxRef = 27.38
RdRef = −0.31
sigma = 0.23
TpRef = 171129.1
VcmaxRef = 26.32
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
200
400
600
800

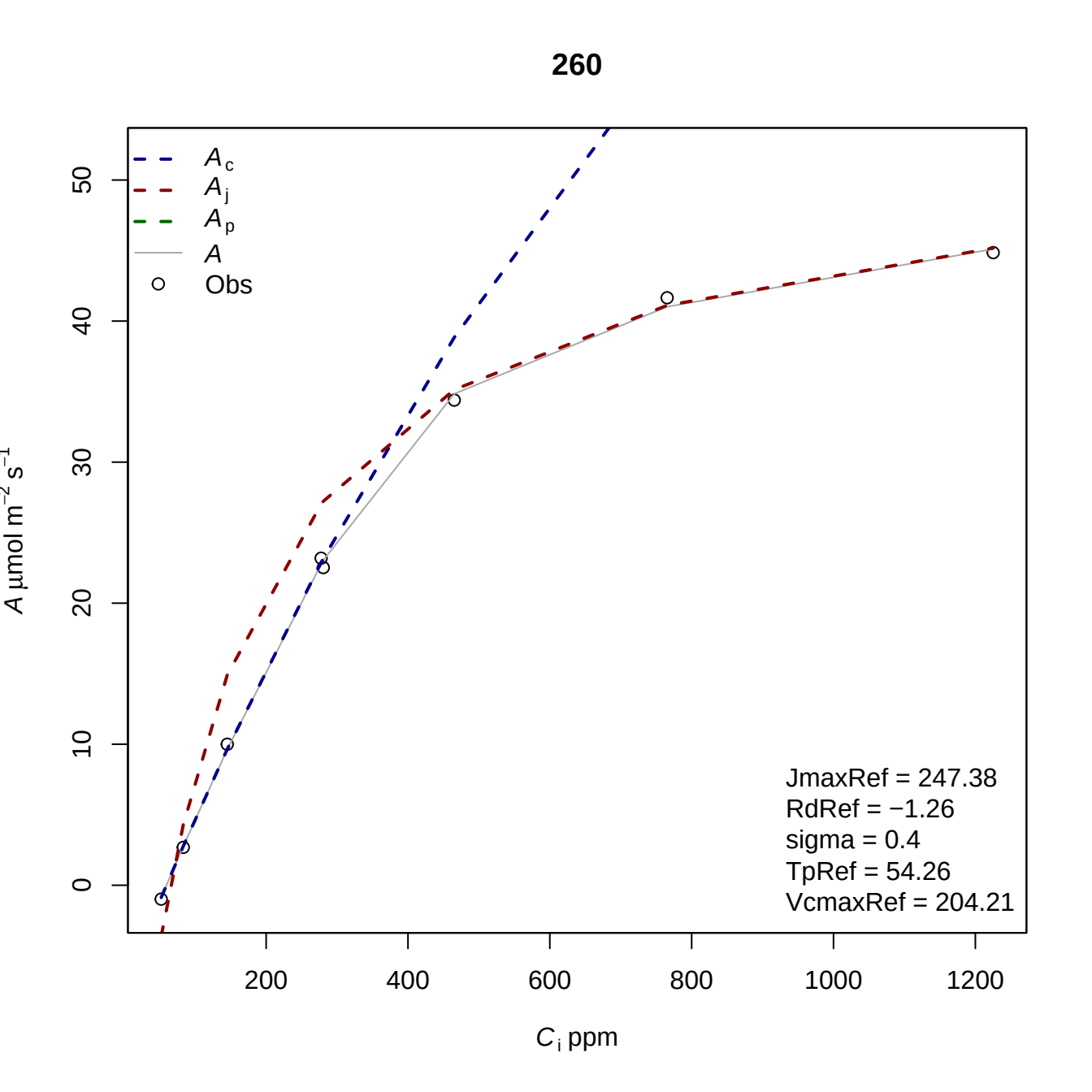

255
A_c
A_j
A_p
A
Obs
JmaxRef = 31.46
RdRef = −0.31
sigma = 0.18
TpRef = 2.53
VcmaxRef = 10.88
A µmol m^{-2} s^{-1}
C_i ppm
0
1
2
3
4
200
400
600
800

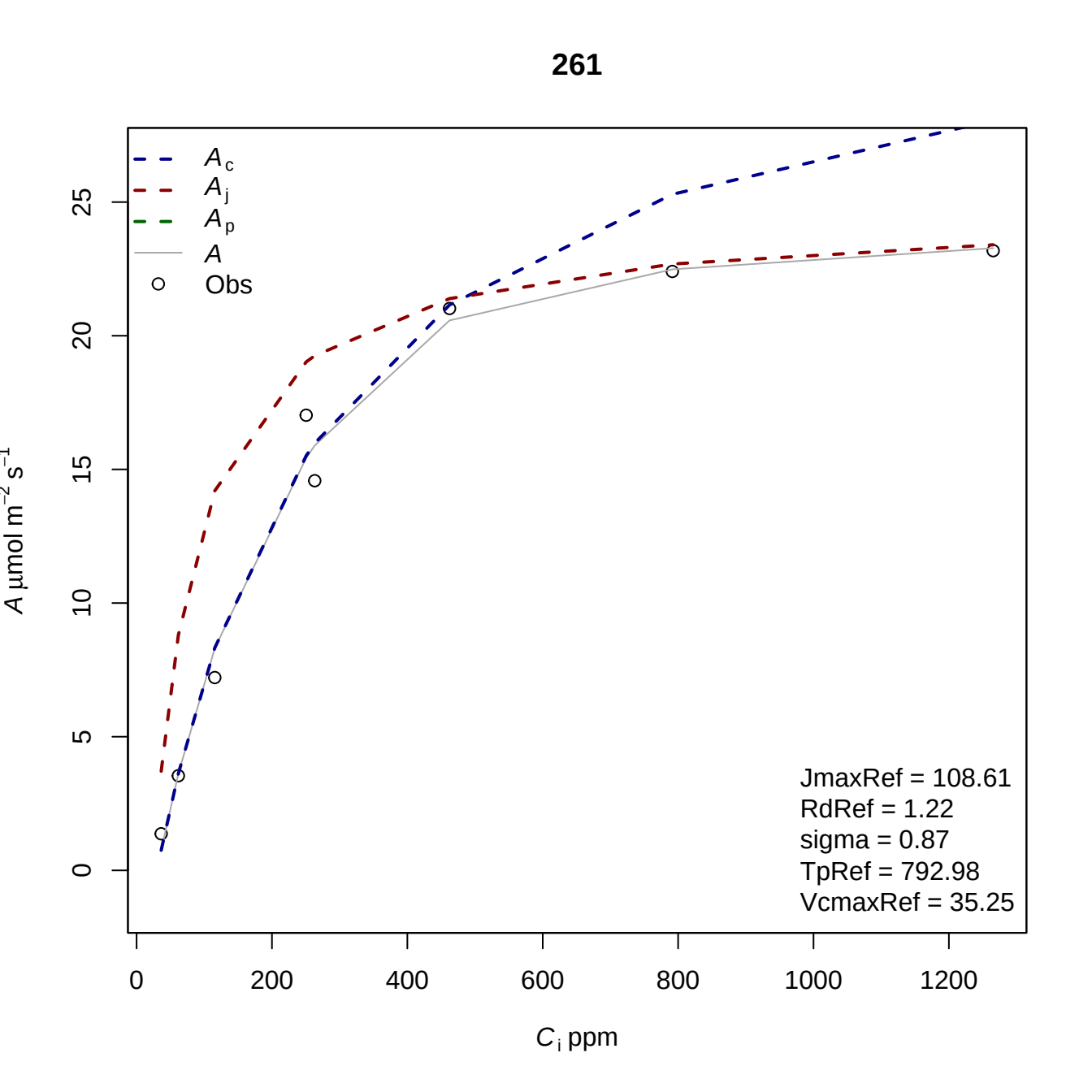

256
A_c
A_j
A_p
A
Obs
JmaxRef = 131.65
RdRef = 0.42
sigma = 0.79
TpRef = 7.95
VcmaxRef = 39.01
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200

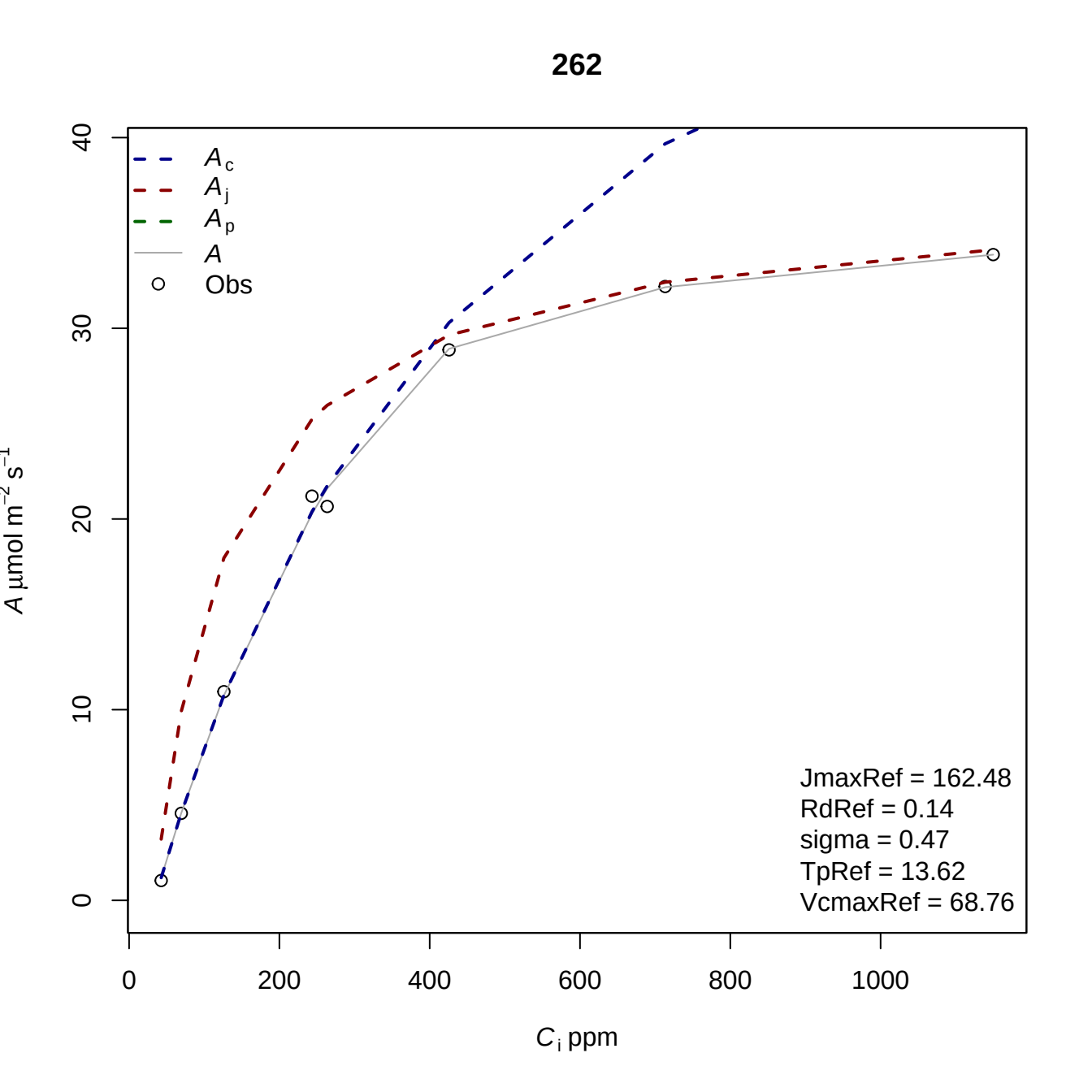

257
A_c
A_j
A_p
A
Obs
JmaxRef = 206.62
RdRef = 0.24
sigma = 0.47
TpRef = 13.94
VcmaxRef = 88.49
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

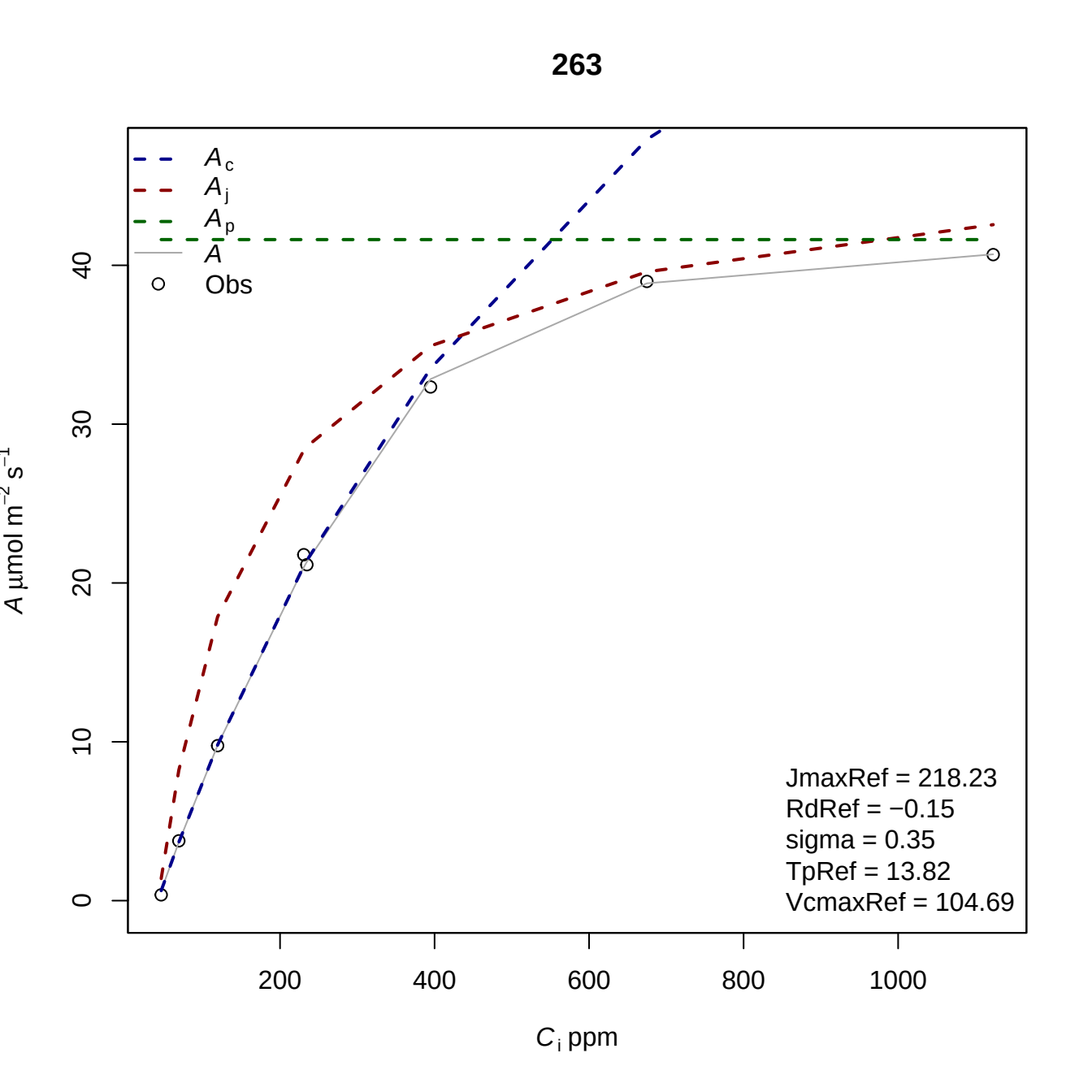

258
A_c
A_j
A_p
A
Obs
JmaxRef = 202.26
RdRef = −0.22
sigma = 0.33
TpRef = 13.84
VcmaxRef = 102.44
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

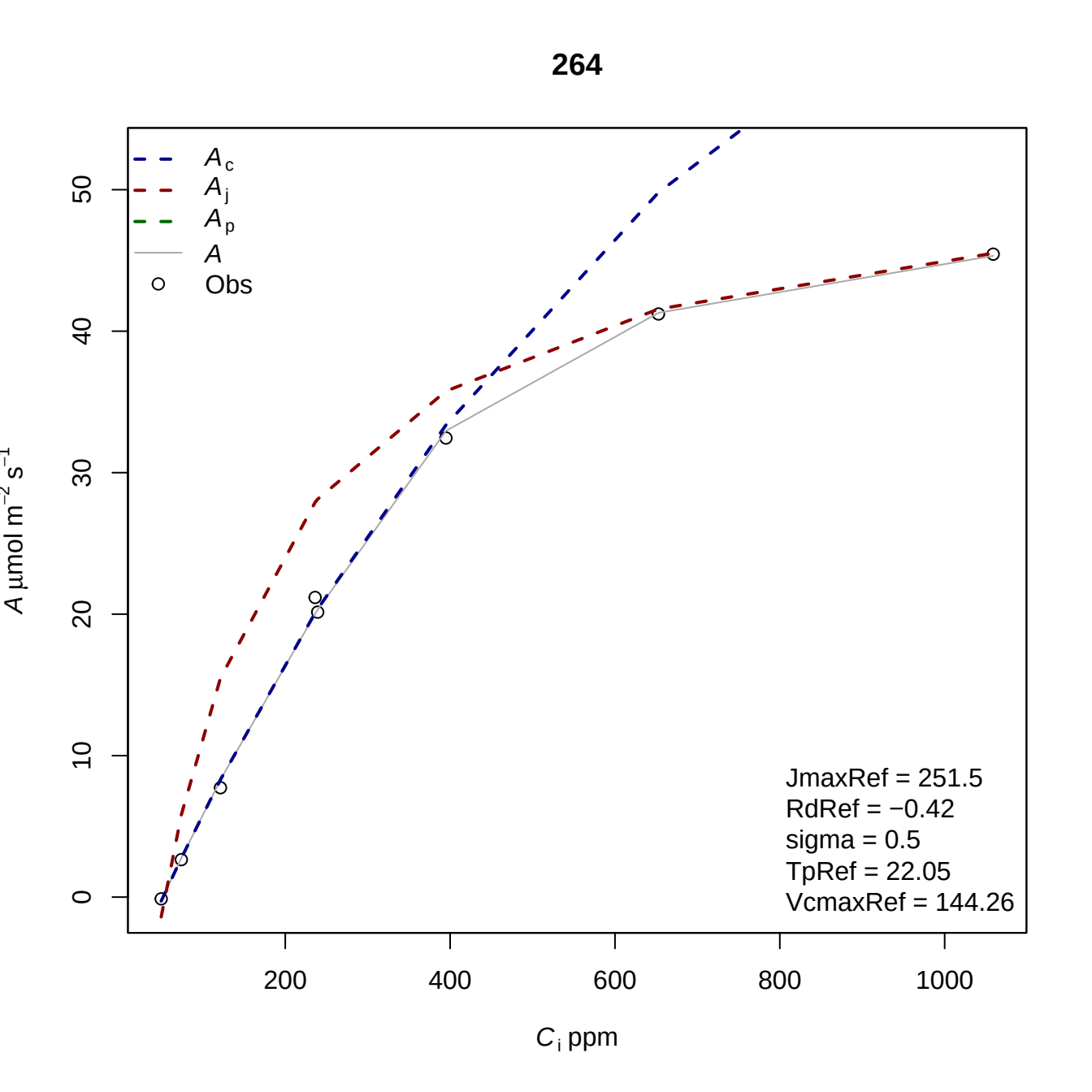

259
A_c
A_j
A_p
A
Obs
JmaxRef = 252.99
RdRef = −0.7
sigma = 0.23
TpRef = 15.29
VcmaxRef = 154.51
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
50
200
400
600
800
1000
1200

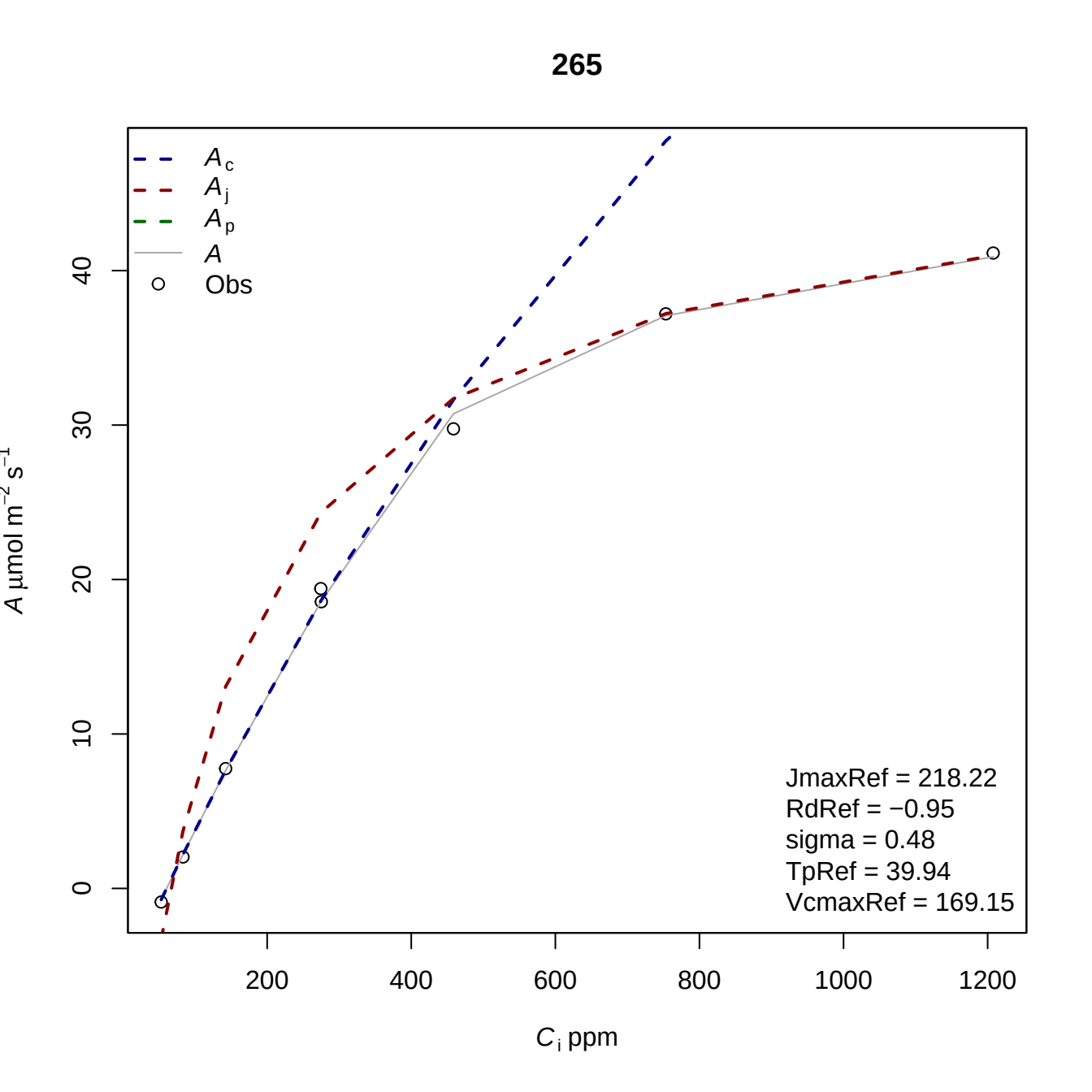

260
A_c
A_j
A_p
A
Obs
JmaxRef = 247.38
RdRef = −1.26
sigma = 0.4
TpRef = 54.26
VcmaxRef = 204.21
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
50
200
400
600
800
1000
1200

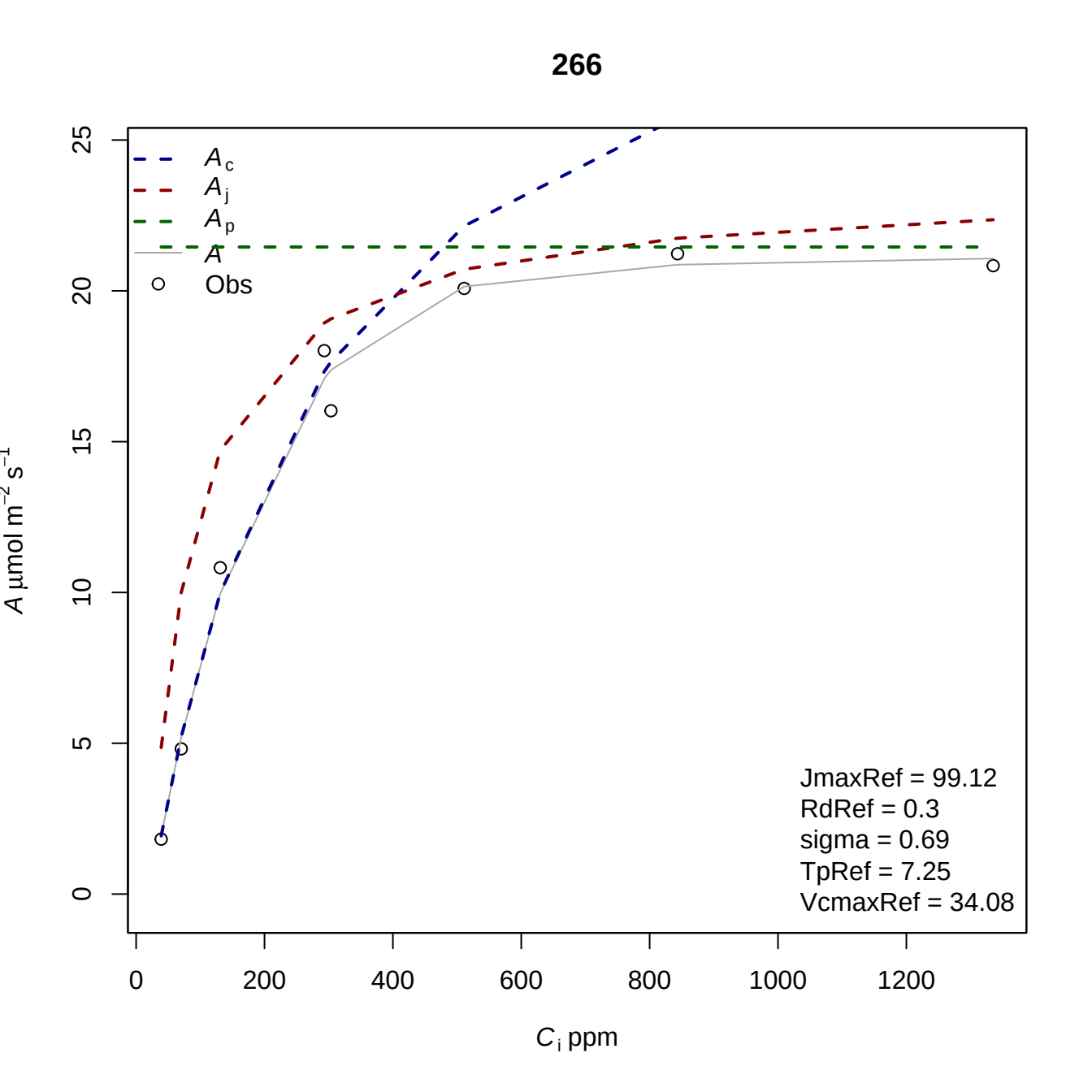

261
A_c
A_j
A_p
A
Obs
JmaxRef = 108.61
RdRef = 1.22
sigma = 0.87
TpRef = 792.98
VcmaxRef = 35.25
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200

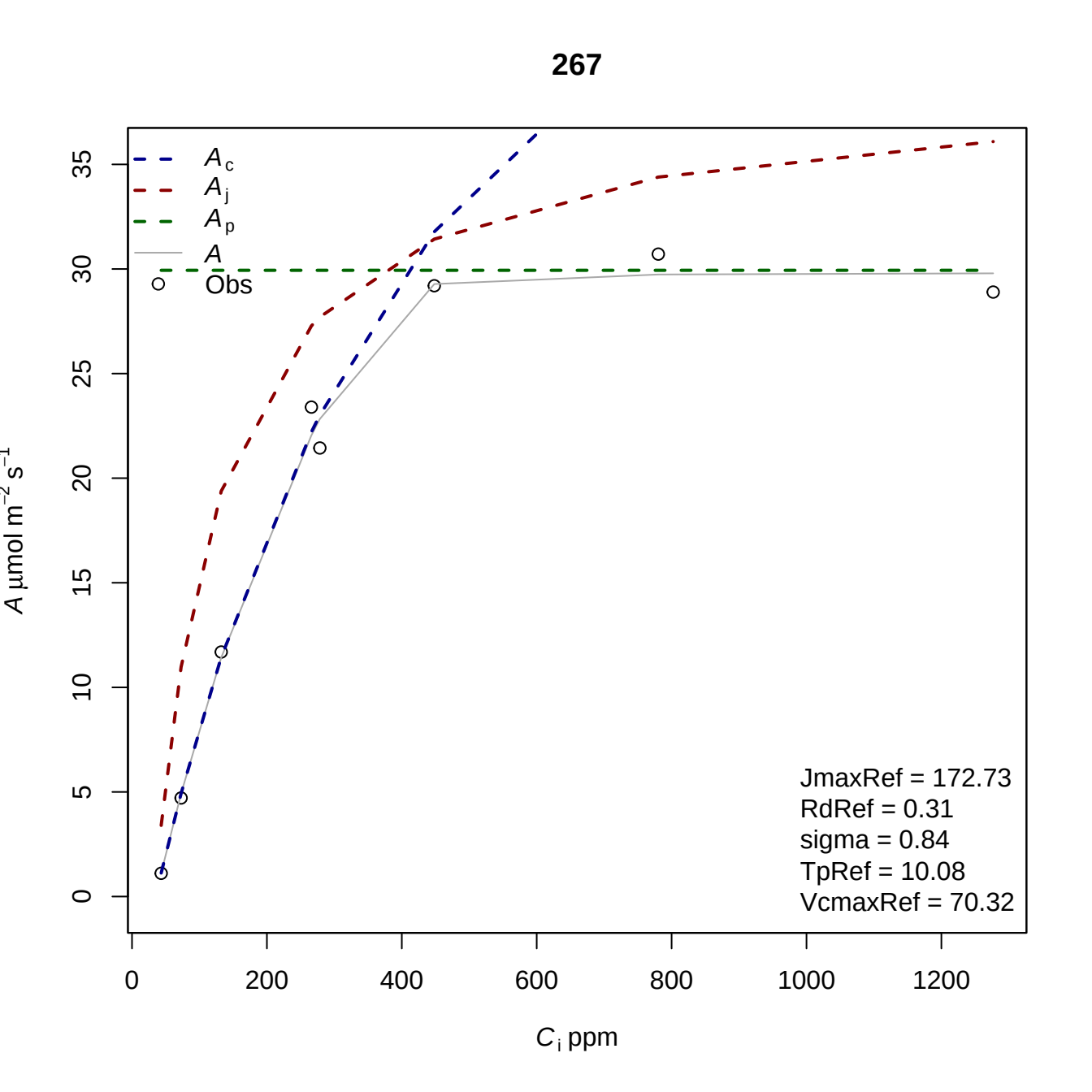

262
A_c
A_j
A_p
A
Obs
JmaxRef = 162.48
RdRef = 0.14
sigma = 0.47
TpRef = 13.62
VcmaxRef = 68.76
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
0
200
400
600
800
1000

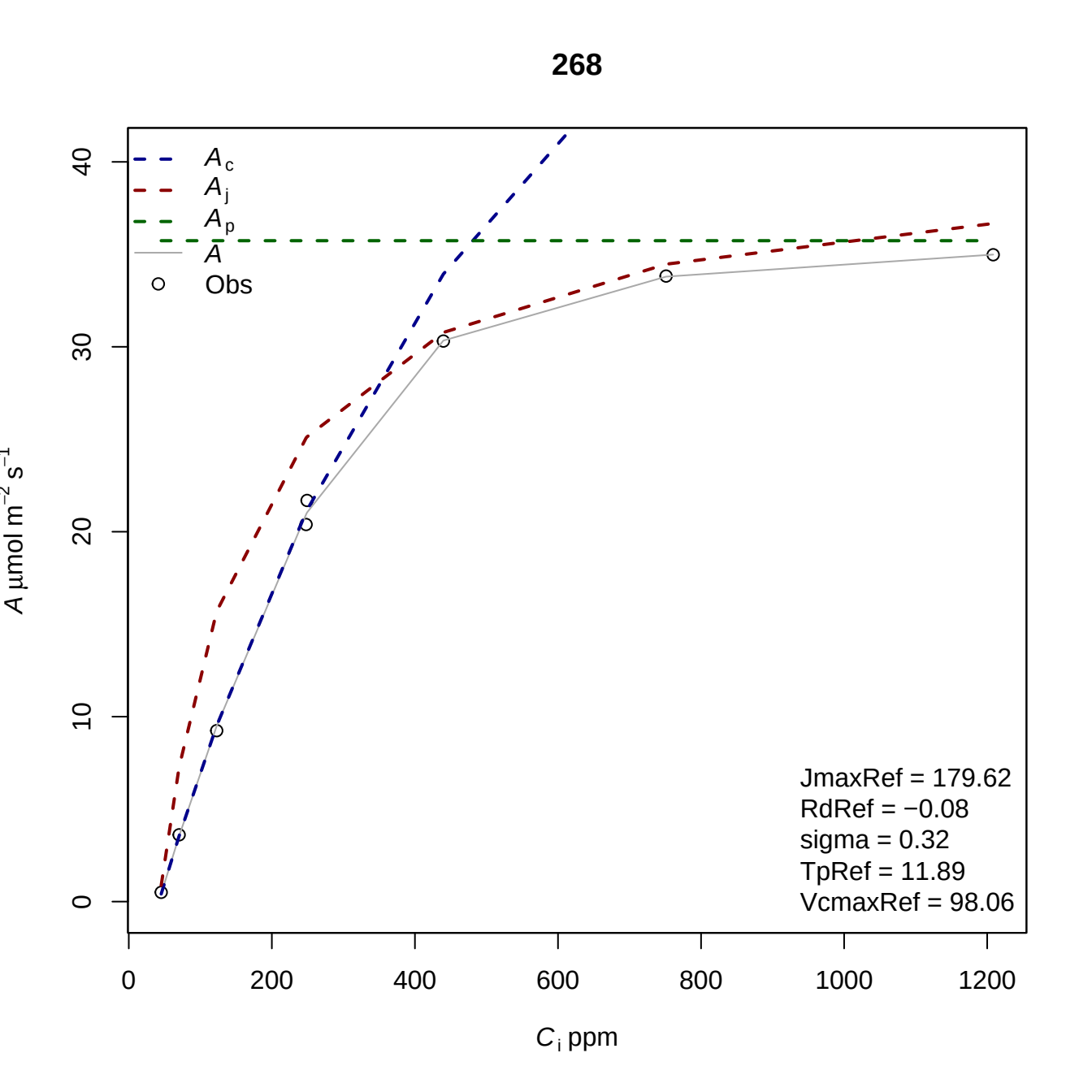

263
A_c
A_j
A_p
A
Obs
JmaxRef = 218.23
RdRef = −0.15
sigma = 0.35
TpRef = 13.82
VcmaxRef = 104.69
0
10
20
30
40
200
400
600
800
1000
A µmol m^{-2} s^{-1}
C_i ppm

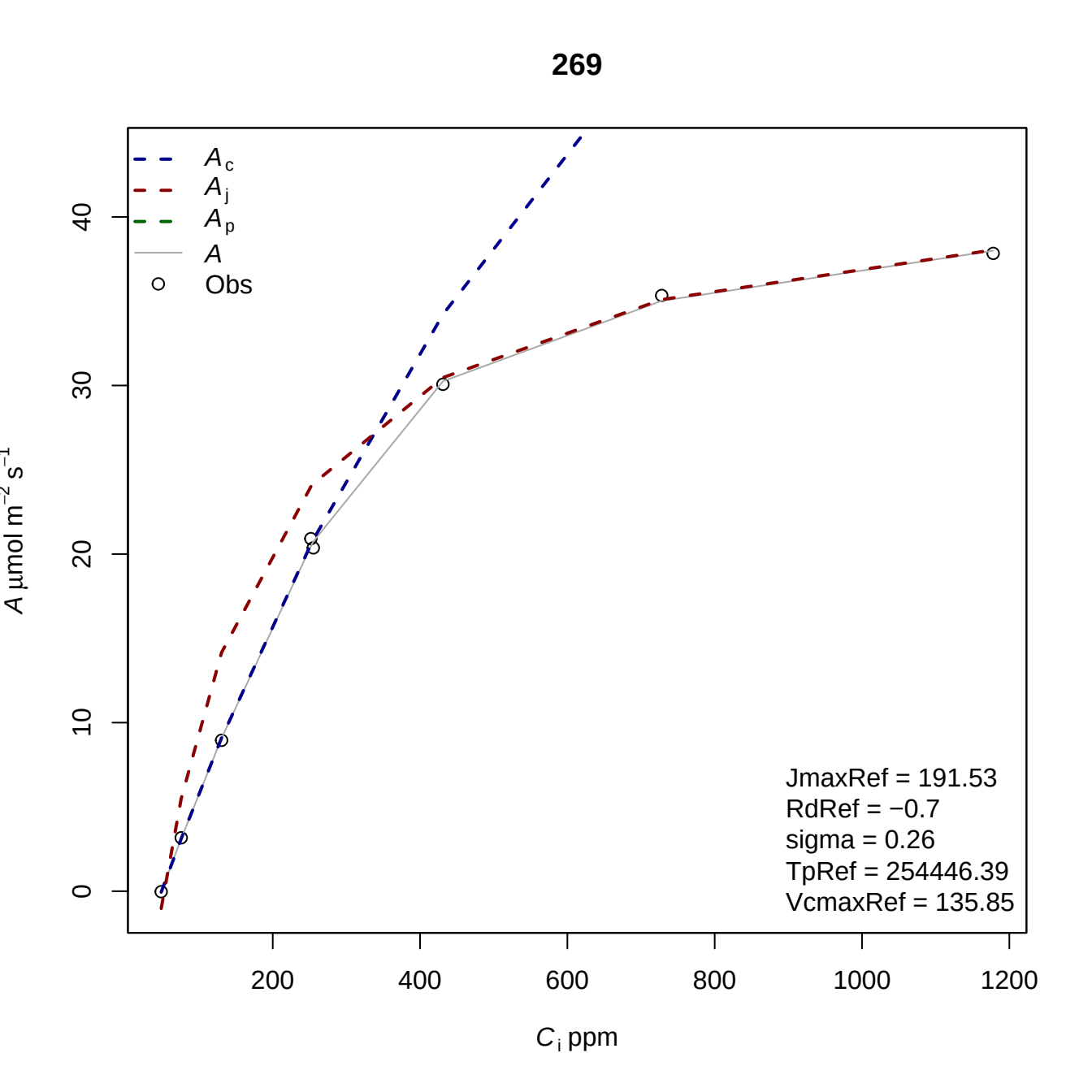

264
A_c
A_j
A_p
A
Obs
JmaxRef = 251.5
RdRef = −0.42
sigma = 0.5
TpRef = 22.05
VcmaxRef = 144.26
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
50
200
400
600
800
1000

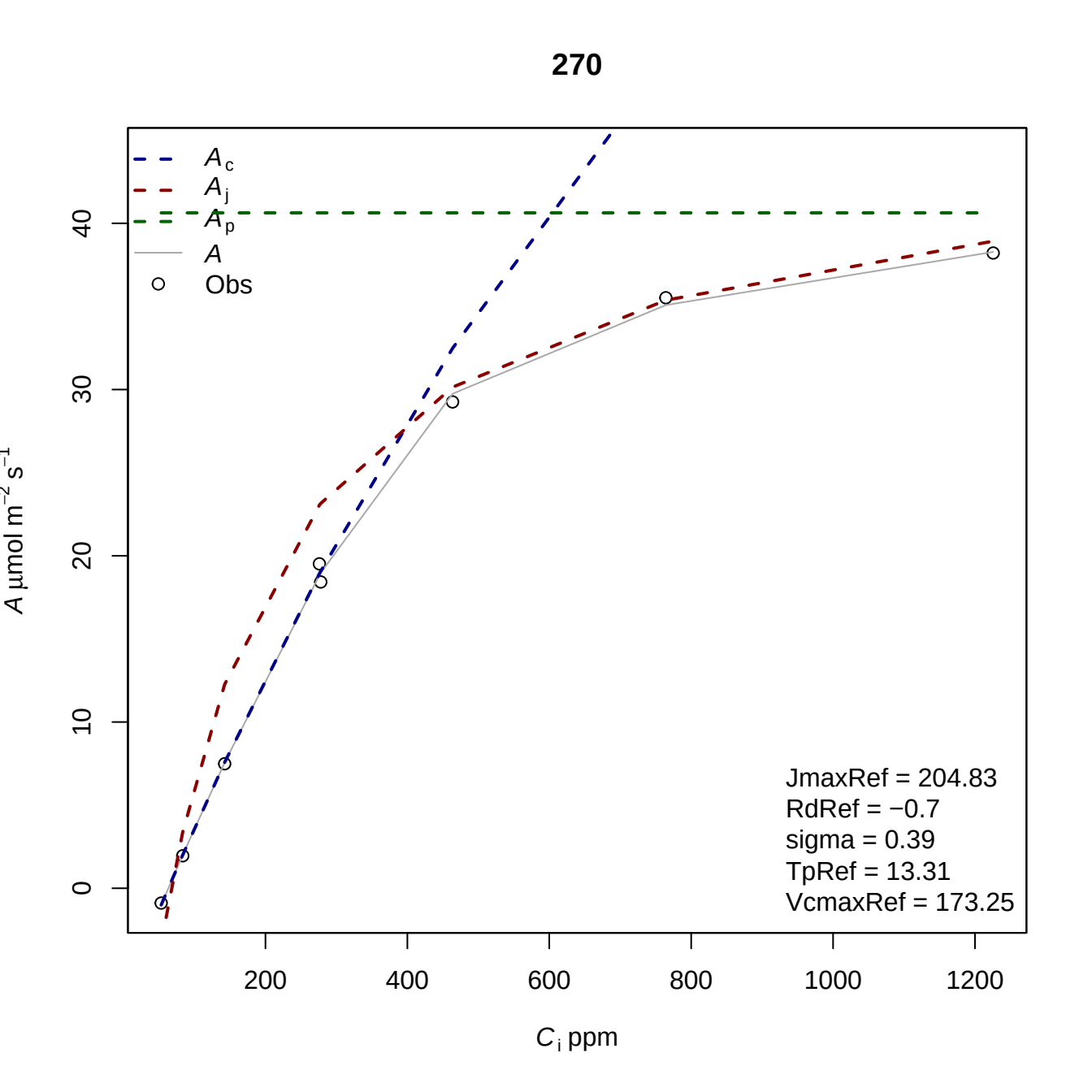

265
A_c
A_j
A_p
A
Obs
JmaxRef = 218.22
RdRef = −0.95
sigma = 0.48
TpRef = 39.94
VcmaxRef = 169.15
A µmol m^{-2} s^{-1}
0
10
20
30
40
200
400
600
800
1000
1200
C_i ppm

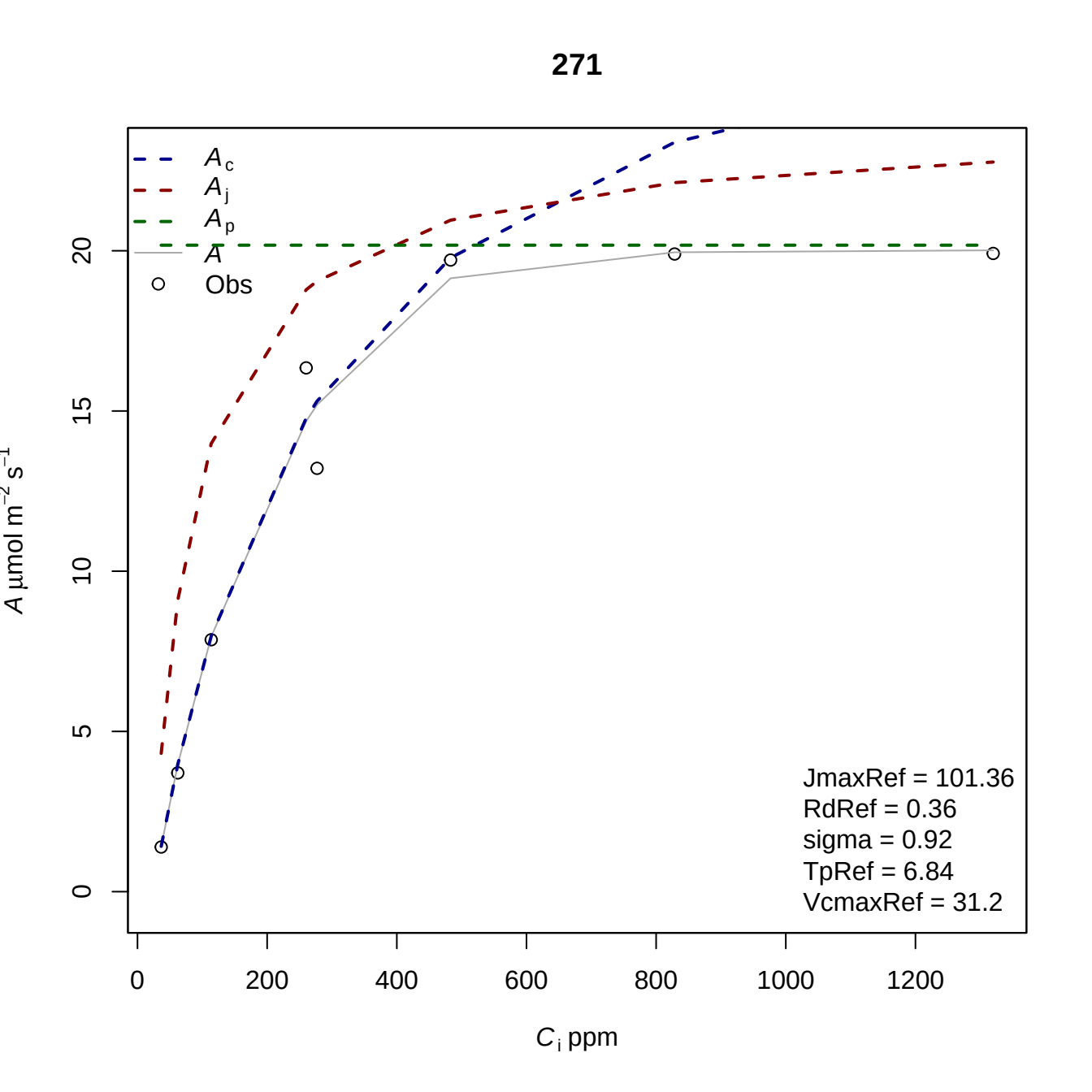

266
A_c
A_j
A_p
A
Obs
JmaxRef = 99.12
RdRef = 0.3
sigma = 0.69
TpRef = 7.25
VcmaxRef = 34.08
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200

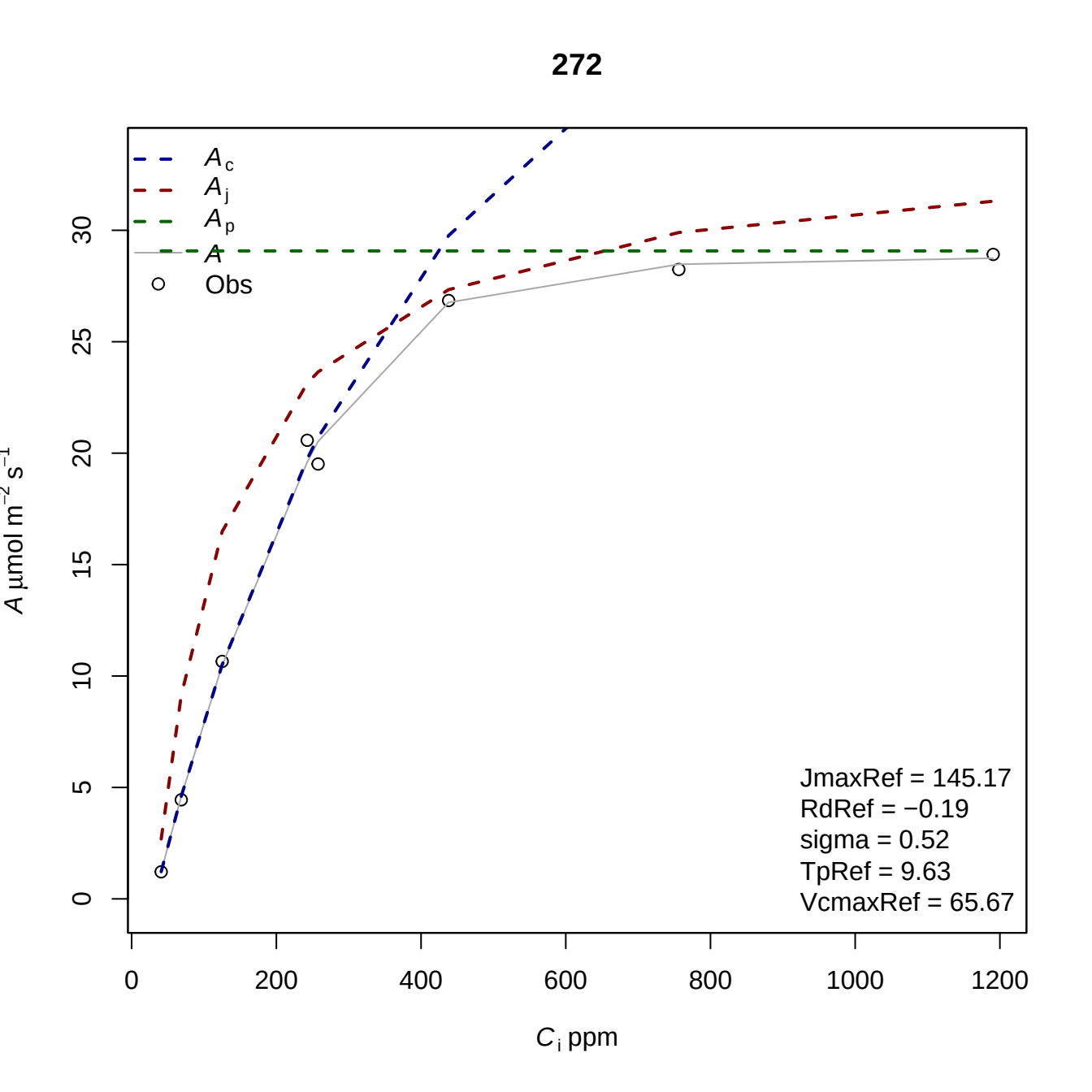

267
A_c
A_j
A_p
A
Obs
JmaxRef = 172.73
RdRef = 0.31
sigma = 0.84
TpRef = 10.08
VcmaxRef = 70.32
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200

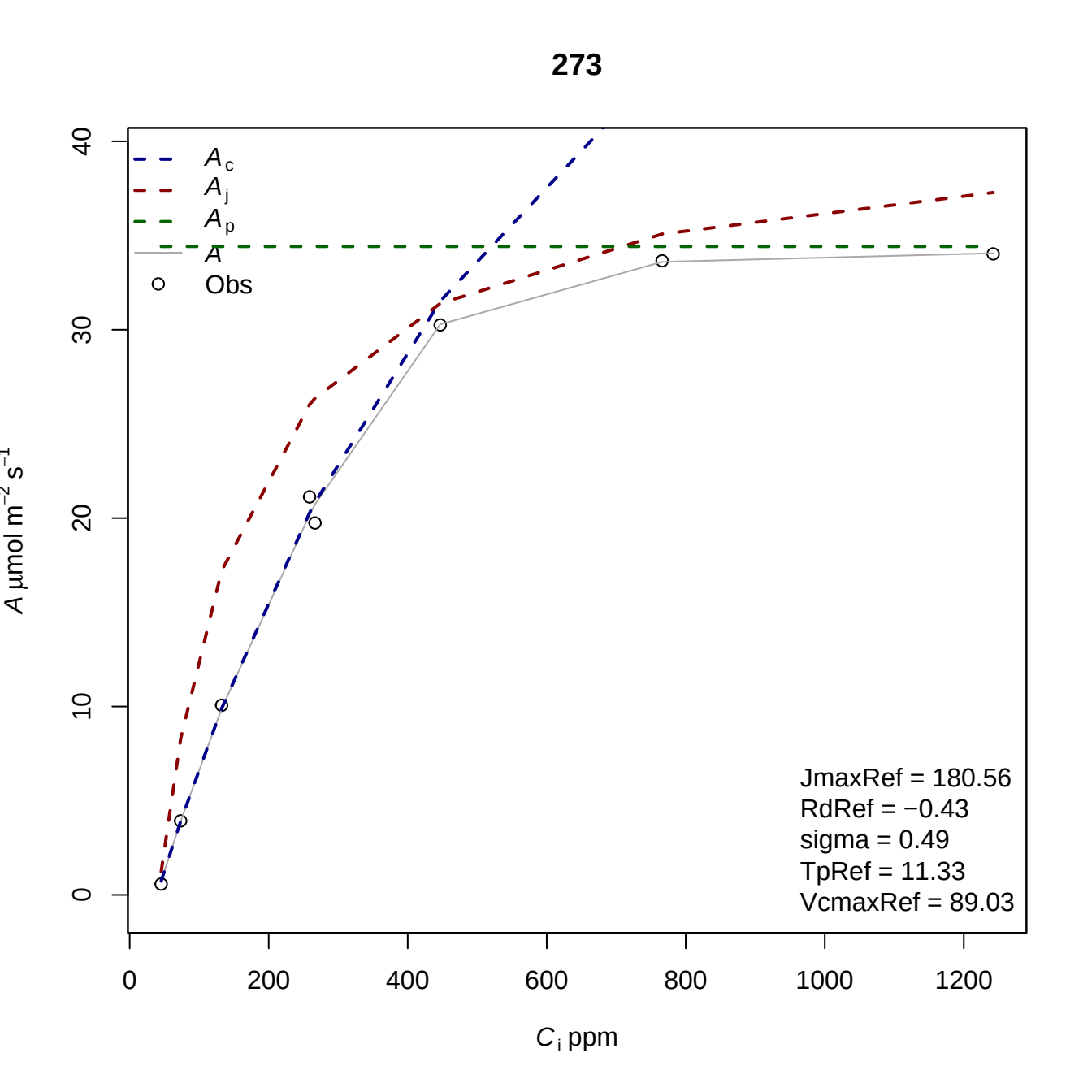

268
A_c
A_j
A_p
A
Obs
JmaxRef = 179.62
RdRef = −0.08
sigma = 0.32
TpRef = 11.89
VcmaxRef = 98.06
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

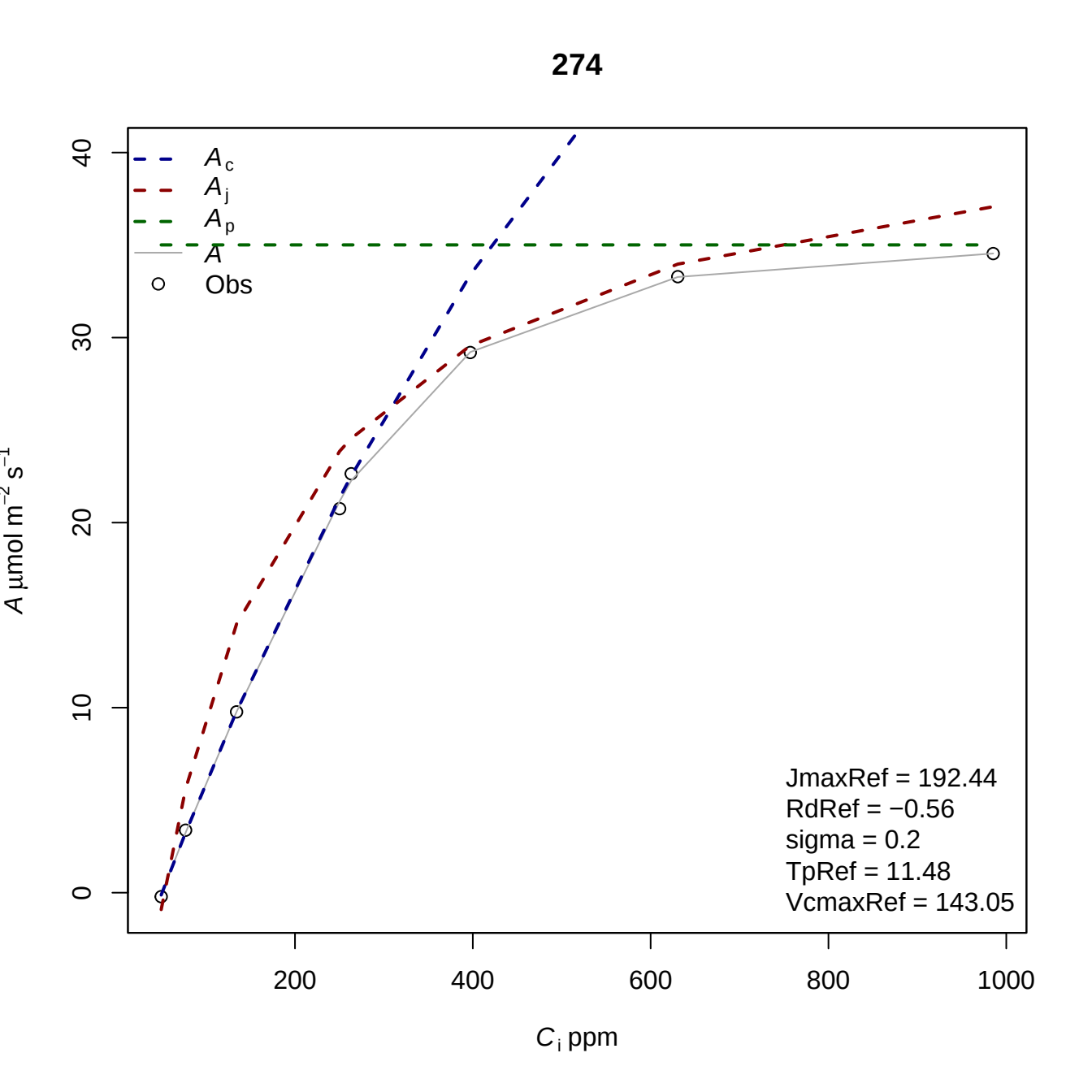

269
A_c
A_j
A_p
A
Obs
JmaxRef = 191.53
RdRef = −0.7
sigma = 0.26
TpRef = 254446.39
VcmaxRef = 135.85
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
200
400
600
800
1000
1200

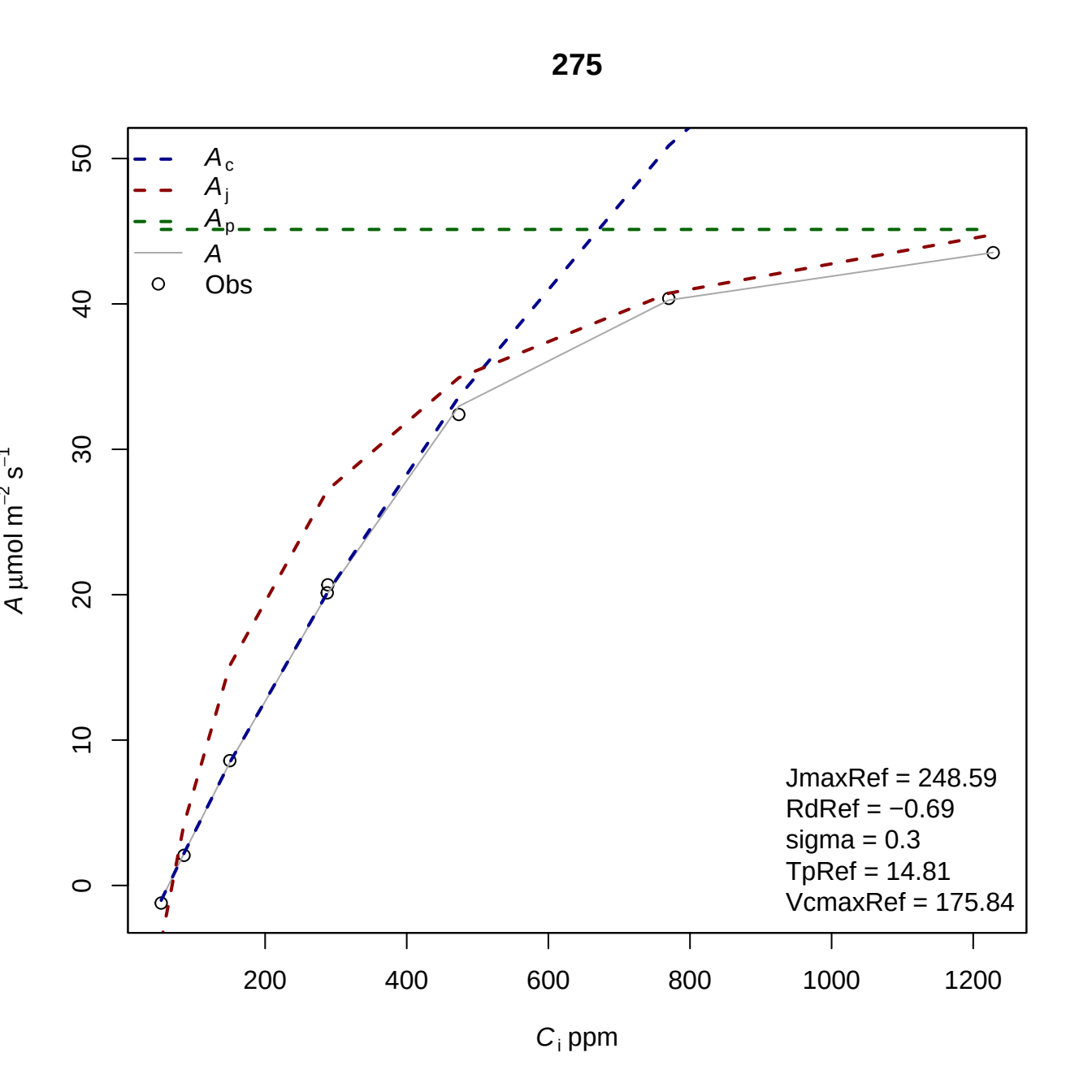

270
A_c
A_j
A_p
A
Obs
JmaxRef = 204.83
RdRef = −0.7
sigma = 0.39
TpRef = 13.31
VcmaxRef = 173.25
0
10
20
30
40
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

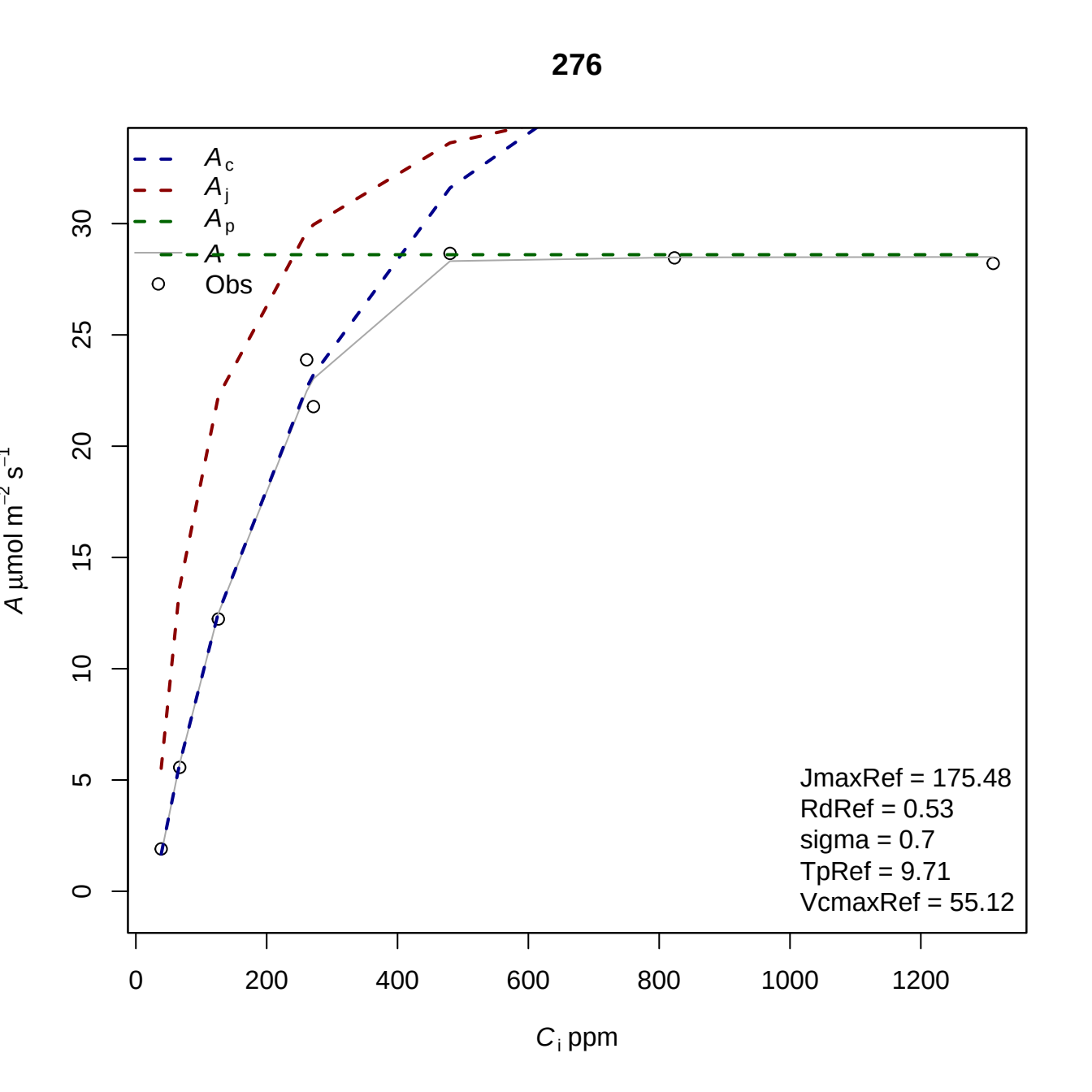

271
A_c
A_j
A_p
A
Obs
JmaxRef = 101.36
RdRef = 0.36
sigma = 0.92
TpRef = 6.84
VcmaxRef = 31.2
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

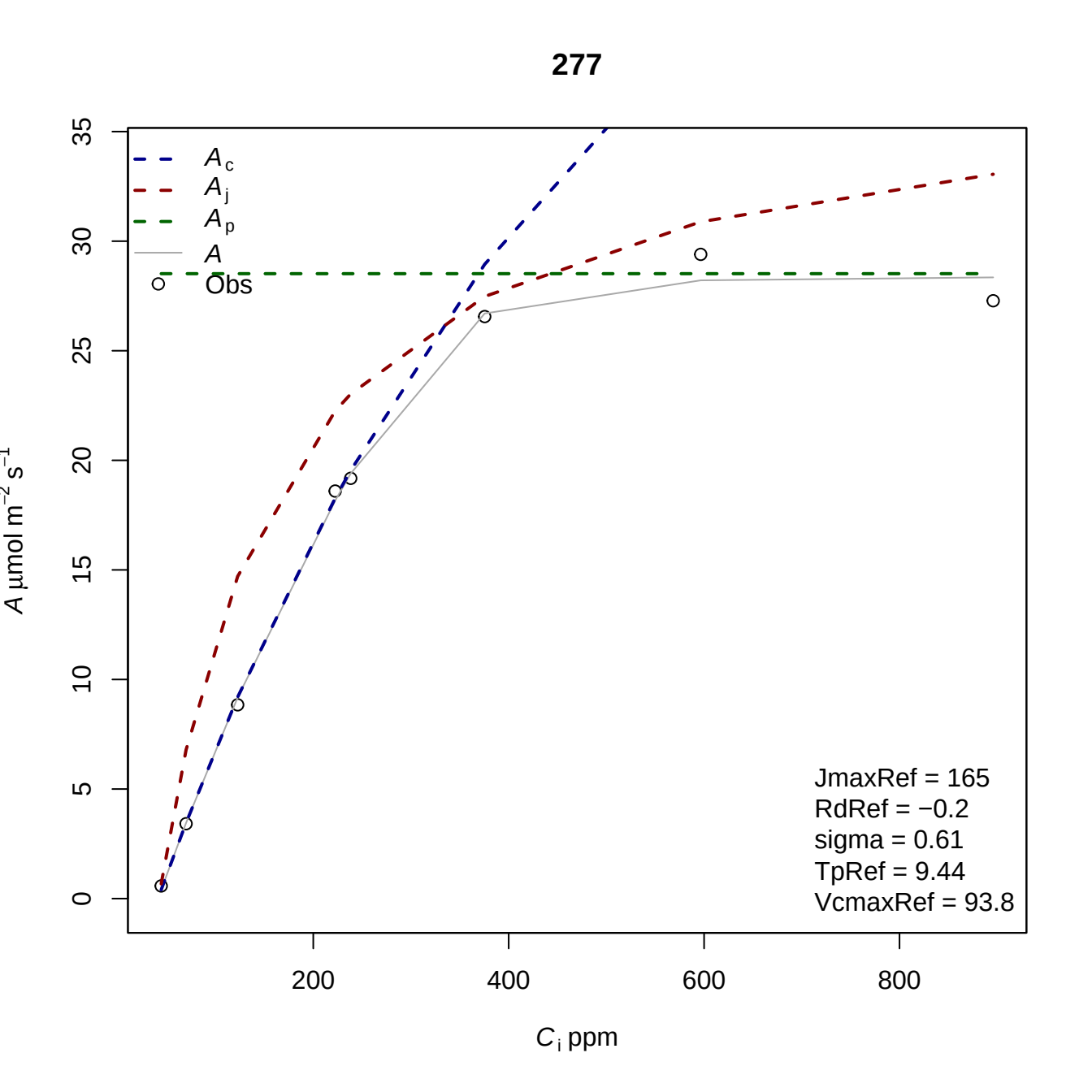

272
A_c
A_j
A_p
A
Obs
JmaxRef = 145.17
RdRef = −0.19
sigma = 0.52
TpRef = 9.63
VcmaxRef = 65.67
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

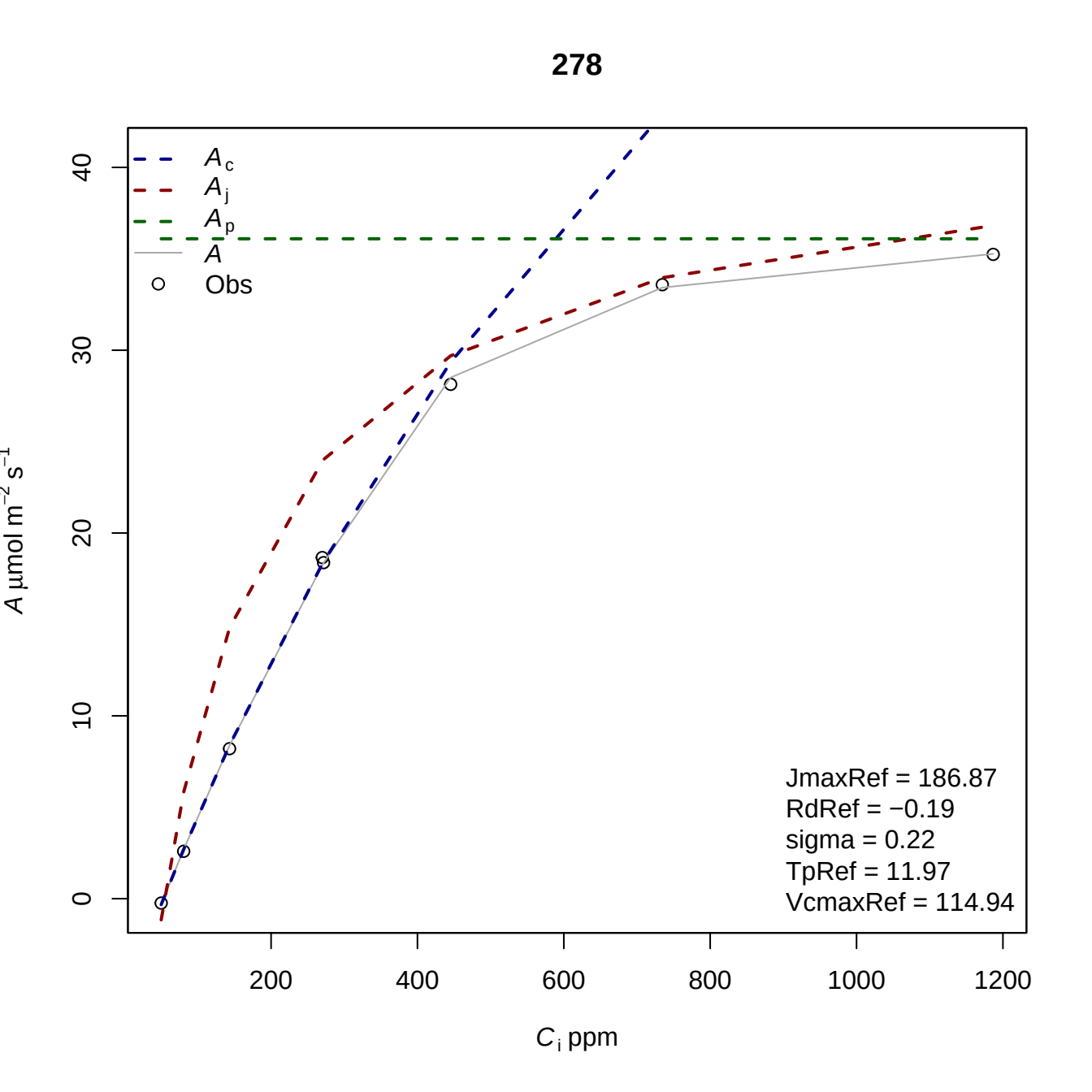

273
A_c
A_j
A_p
A
Obs
JmaxRef = 180.56
RdRef = −0.43
sigma = 0.49
TpRef = 11.33
VcmaxRef = 89.03
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

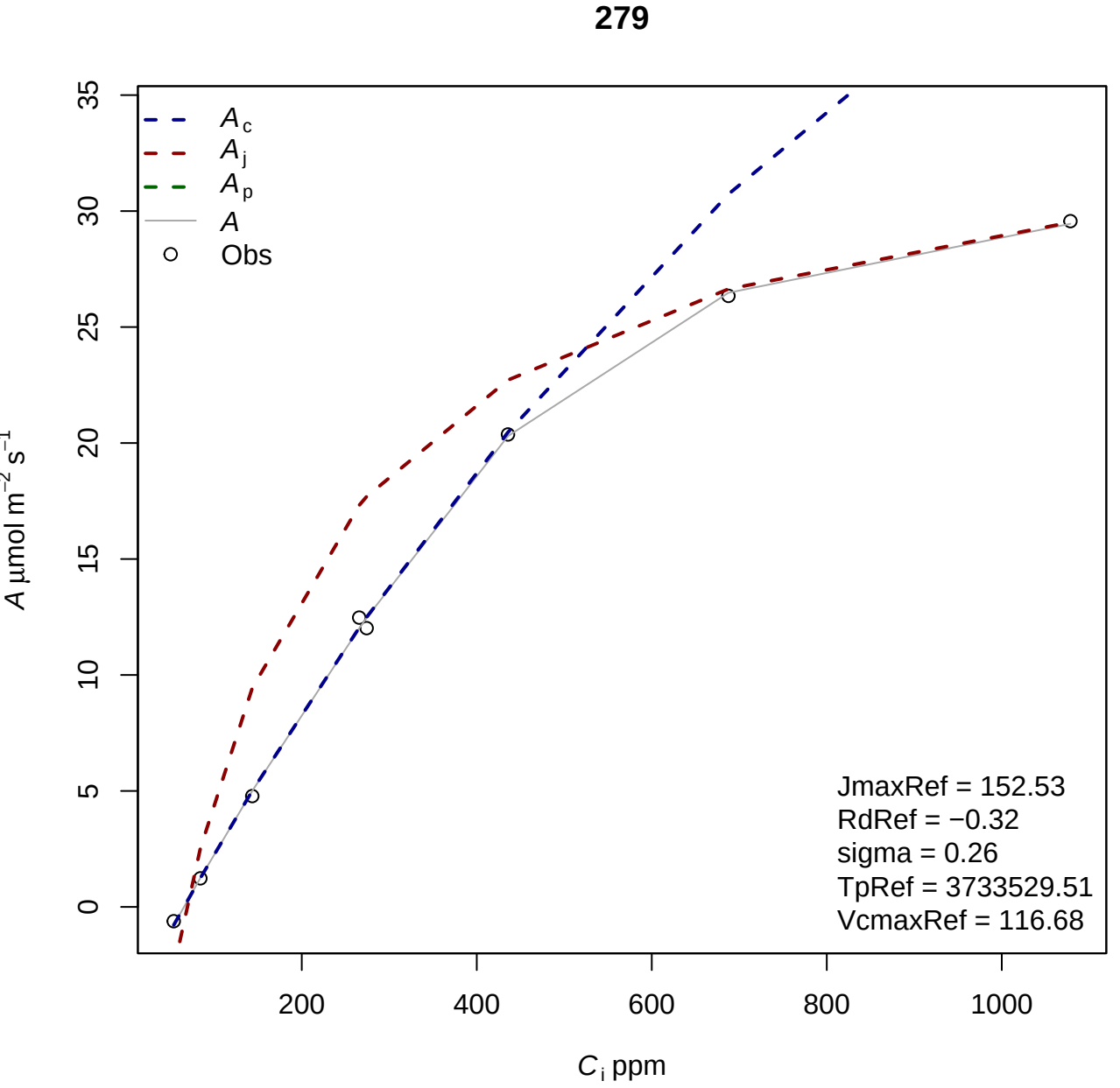

274
A_c
A_j
A_p
A
Obs
JmaxRef = 192.44
RdRef = −0.56
sigma = 0.2
TpRef = 11.48
VcmaxRef = 143.05
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
200
400
600
800
1000

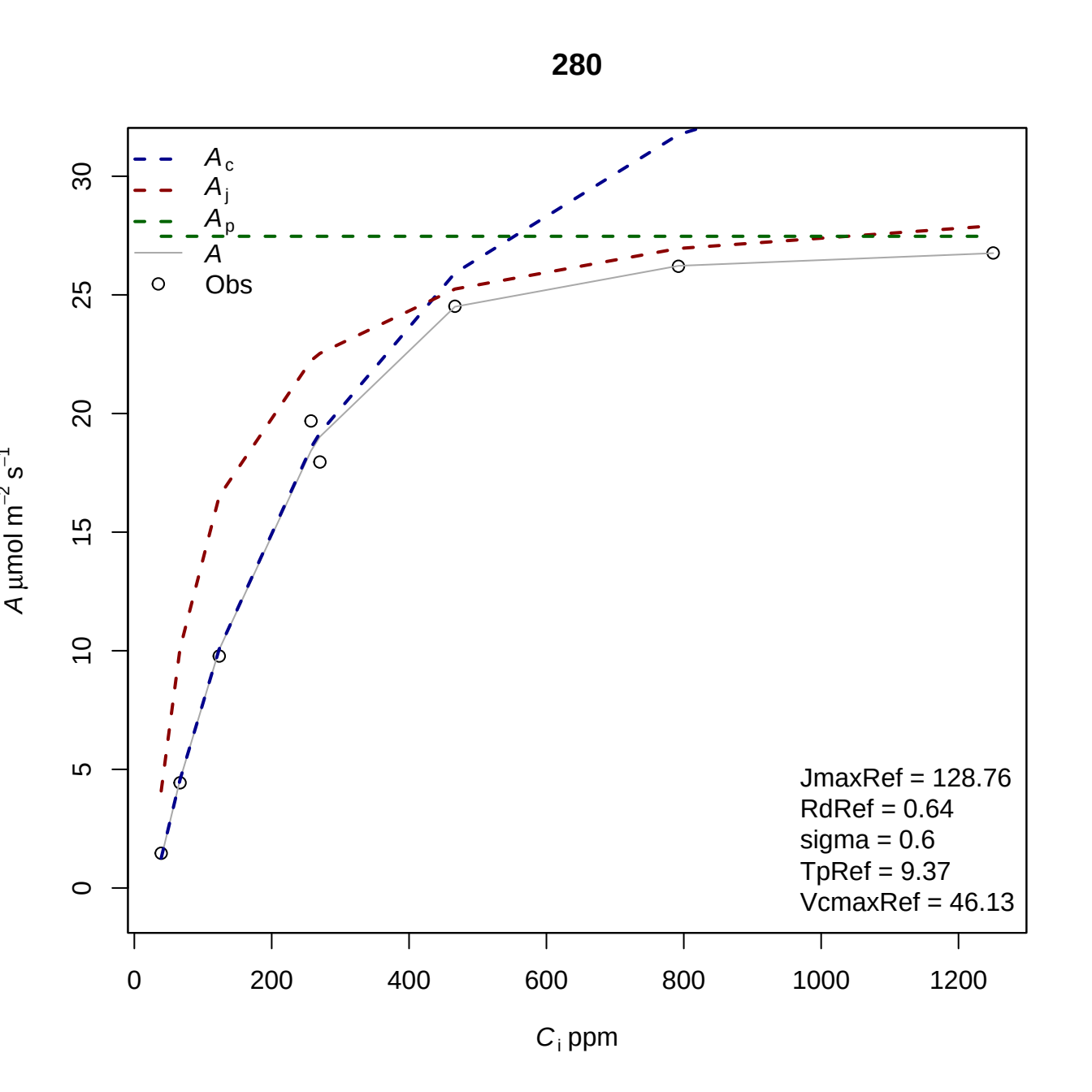

275
A_c
A_j
A_p
A
Obs
JmaxRef = 248.59
RdRef = −0.69
sigma = 0.3
TpRef = 14.81
VcmaxRef = 175.84
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
50
200
400
600
800
1000
1200

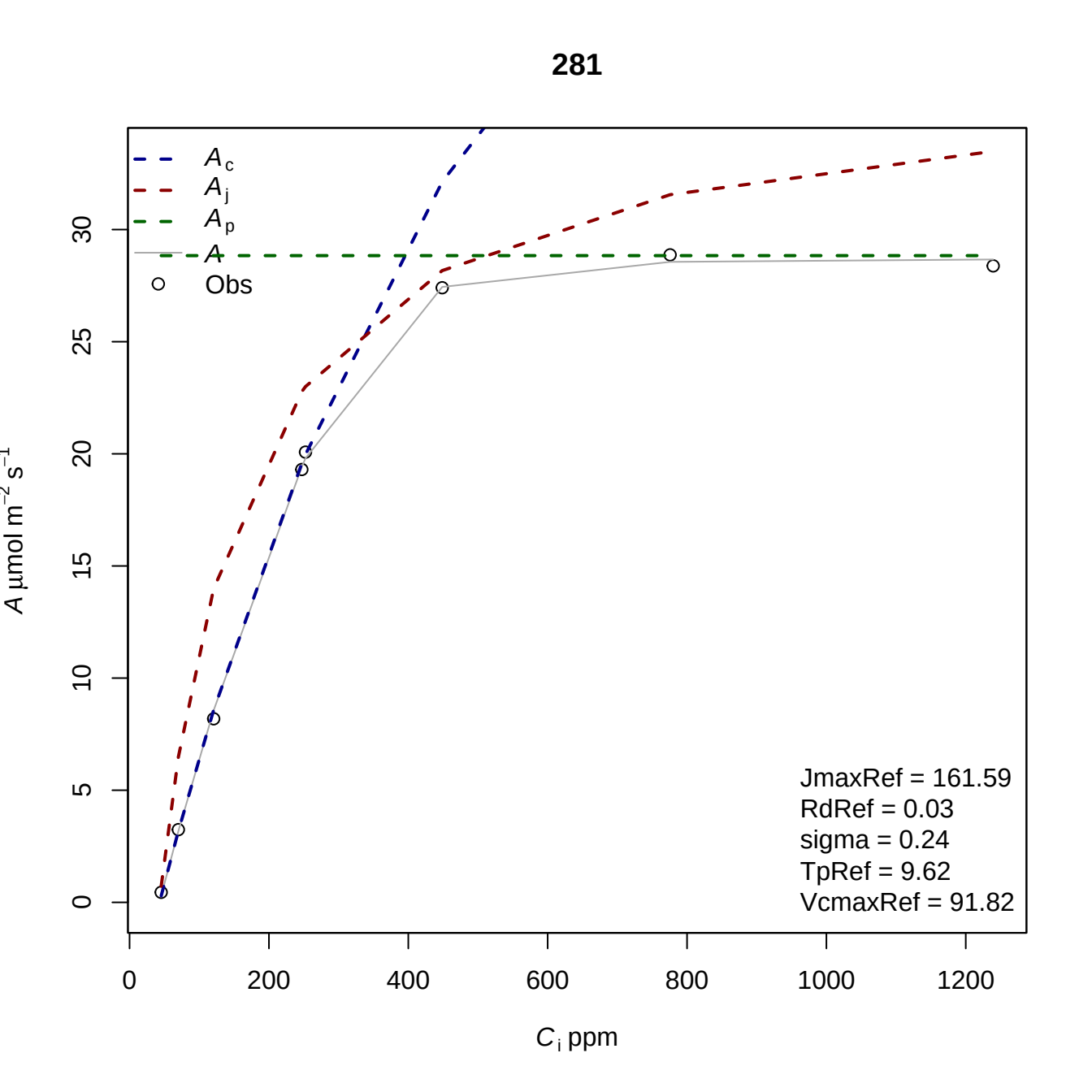

276
A_c
A_j
A_p
A
Obs
JmaxRef = 175.48
RdRef = 0.53
sigma = 0.7
TpRef = 9.71
VcmaxRef = 55.12
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
1200
C_i ppm

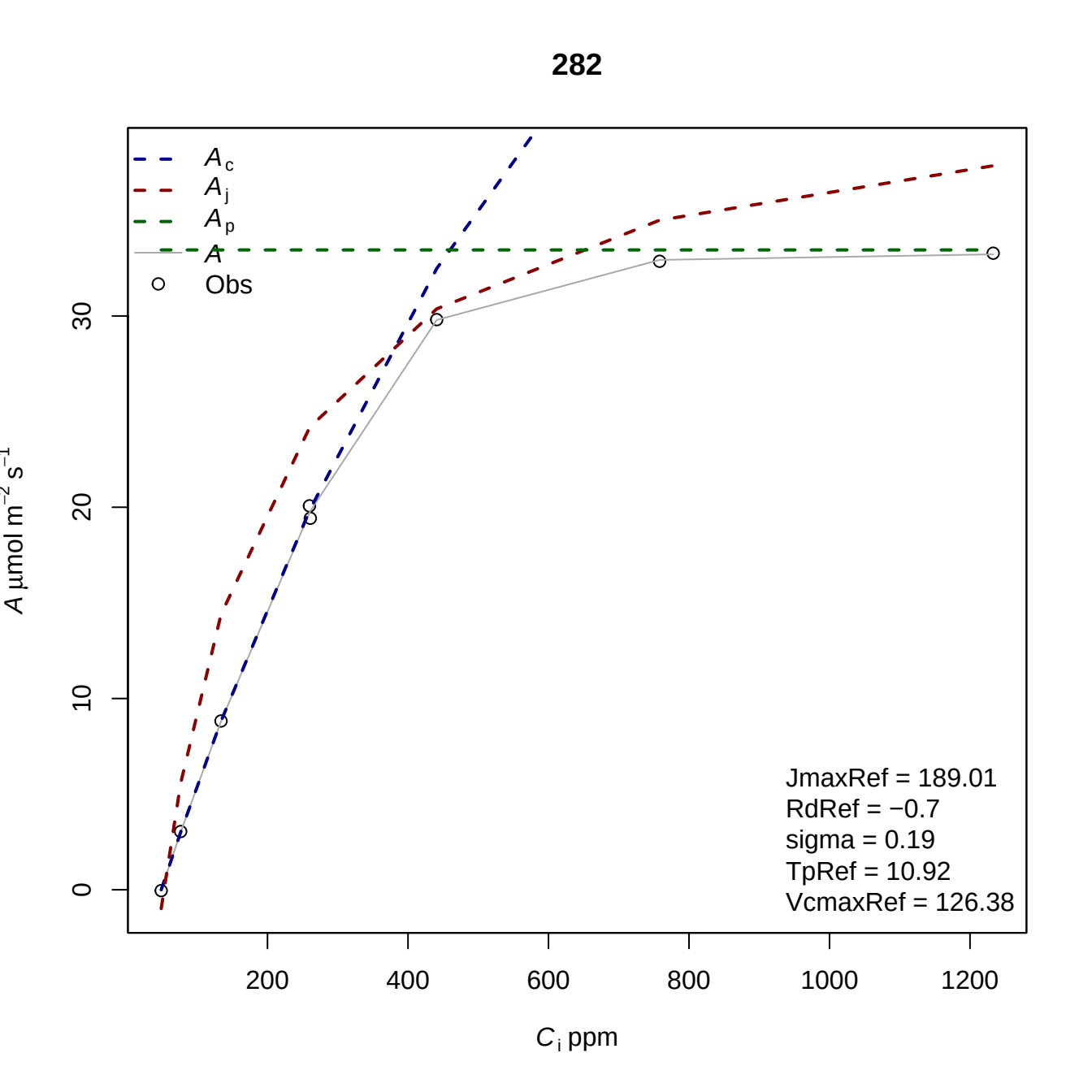

277
A_c
A_j
A_p
A
Obs
JmaxRef = 165
RdRef = −0.2
sigma = 0.61
TpRef = 9.44
VcmaxRef = 93.8
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
35
200
400
600
800

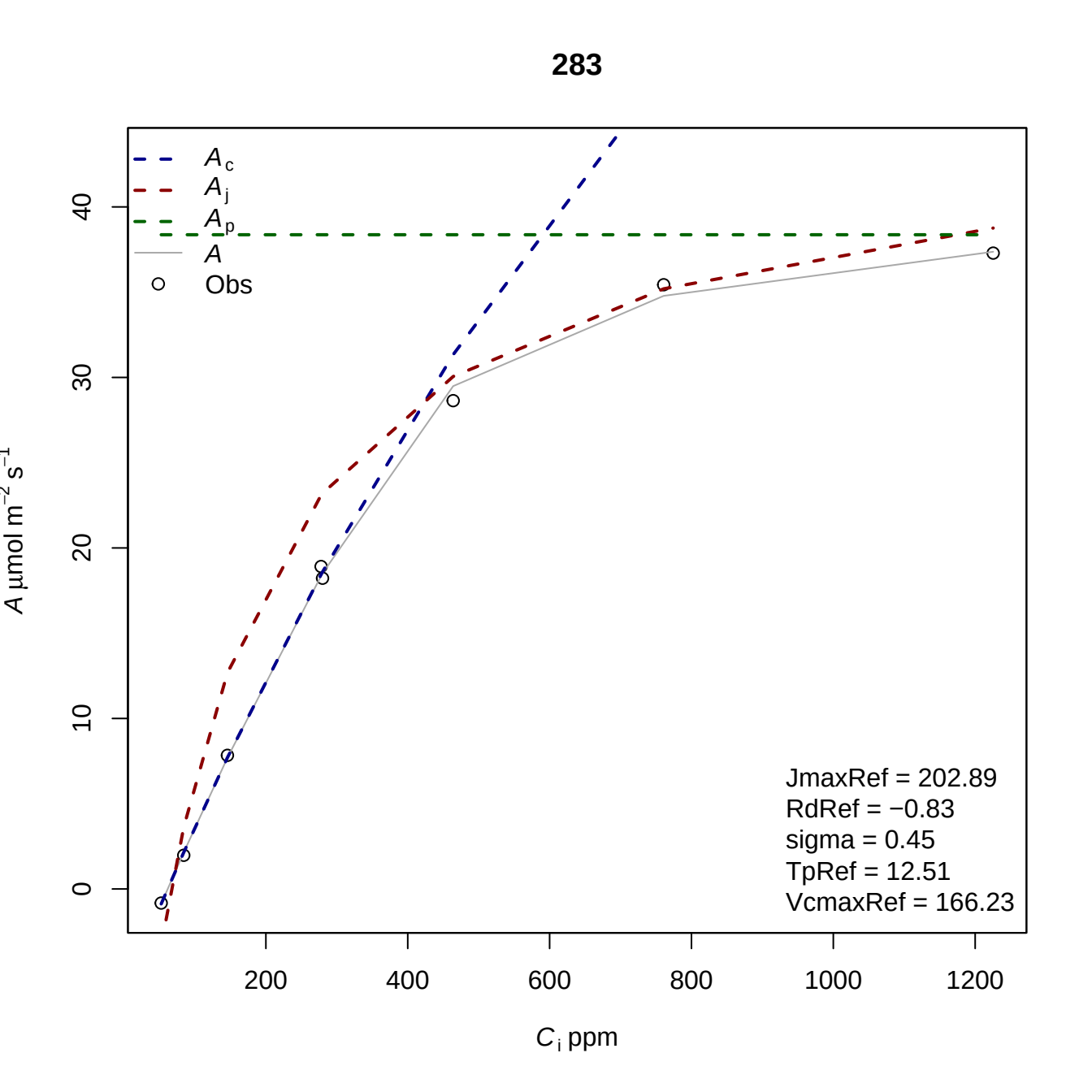

278
A_c
A_j
A_p
A
Obs
JmaxRef = 186.87
RdRef = −0.19
sigma = 0.22
TpRef = 11.97
VcmaxRef = 114.94
0
10
20
30
40
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

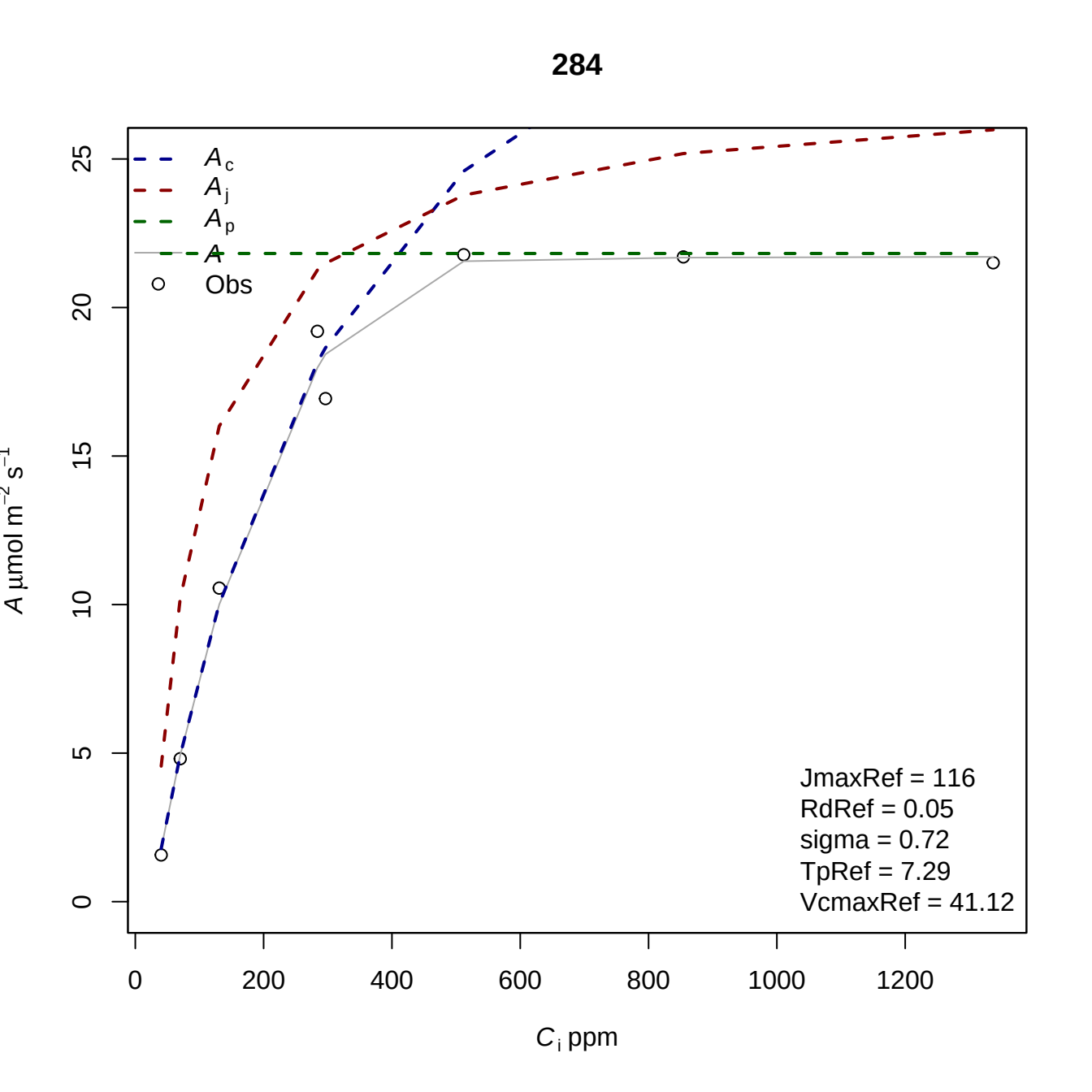

279
A_c
A_j
A_p
A
Obs
JmaxRef = 152.53
RdRef = −0.32
sigma = 0.26
TpRef = 3733529.51
VcmaxRef = 116.68
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
35
200
400
600
800
1000

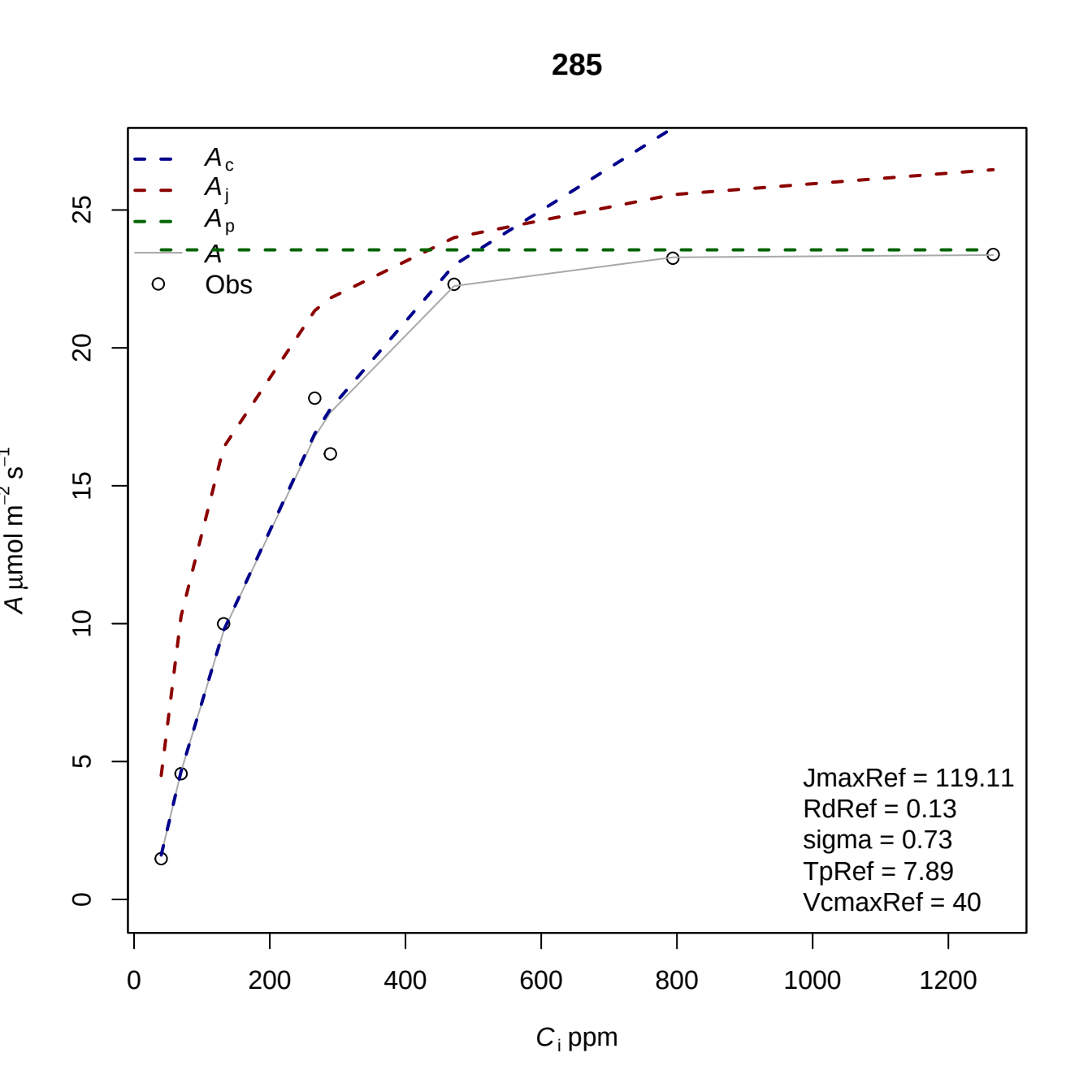

280
A_c
A_j
A_p
A
Obs
JmaxRef = 128.76
RdRef = 0.64
sigma = 0.6
TpRef = 9.37
VcmaxRef = 46.13
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

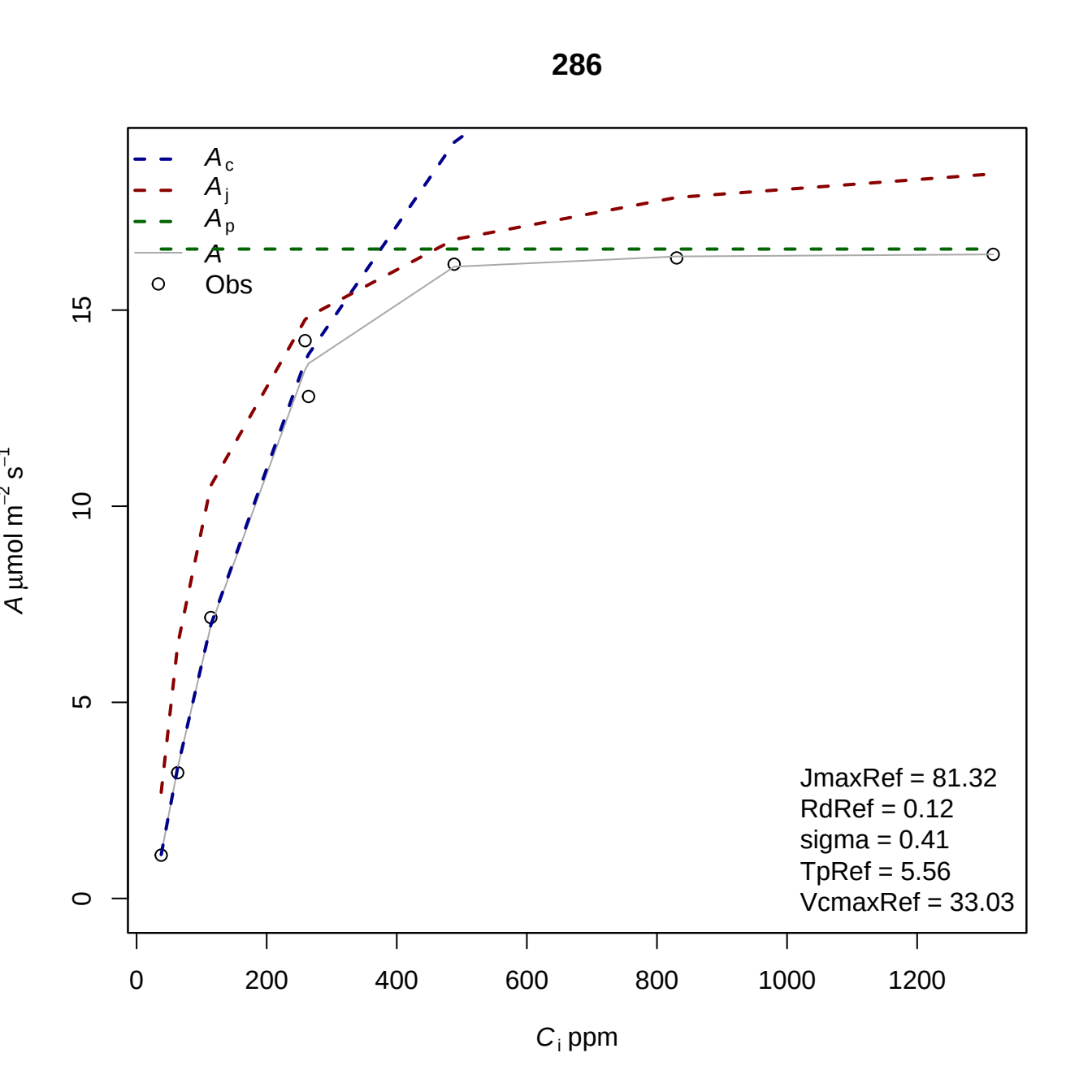

281
A_c
A_j
A_p
A
Obs
JmaxRef = 161.59
RdRef = 0.03
sigma = 0.24
TpRef = 9.62
VcmaxRef = 91.82
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

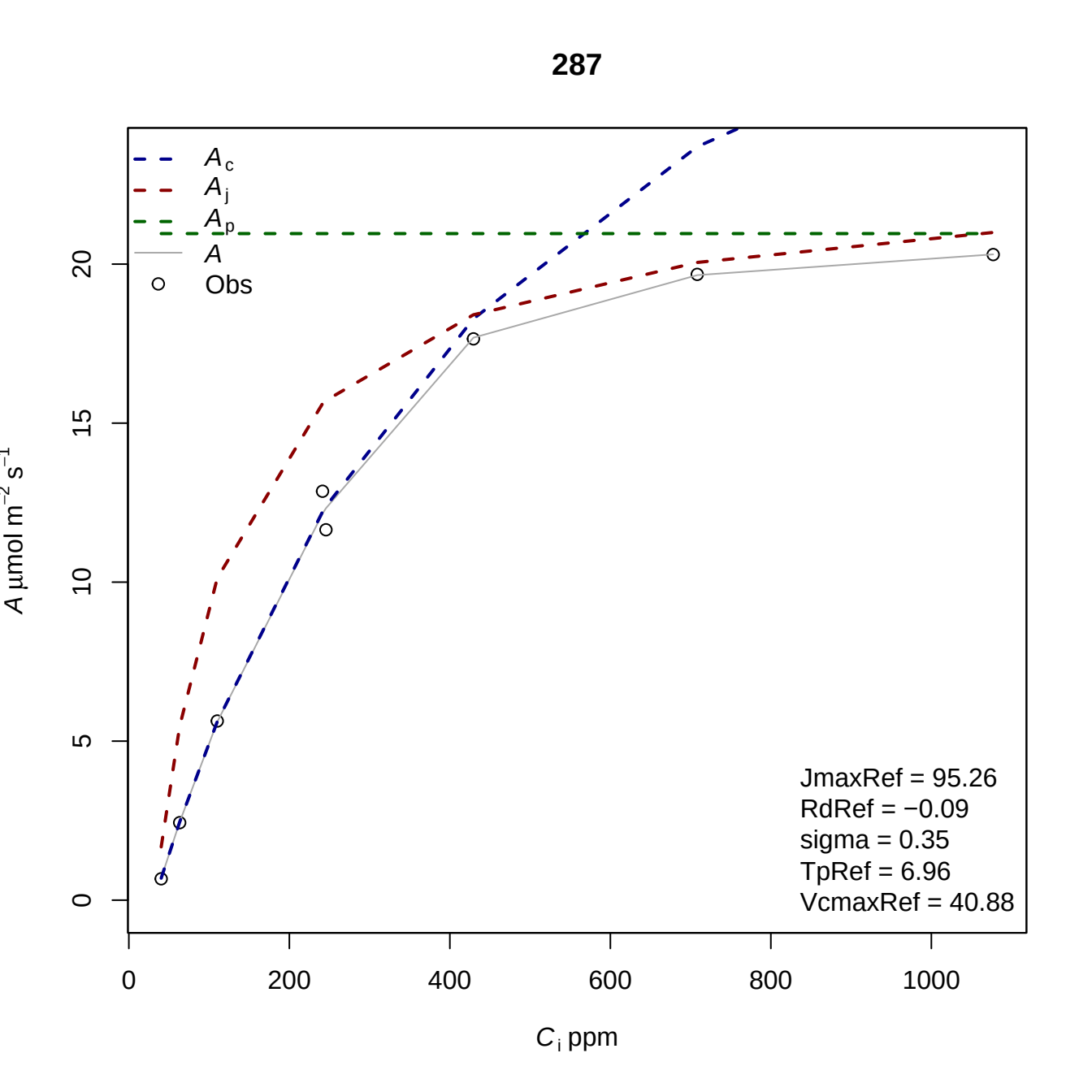

282
A_c
A_j
A_p
A
Obs
JmaxRef = 189.01
RdRef = −0.7
sigma = 0.19
TpRef = 10.92
VcmaxRef = 126.38
0
10
20
30
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

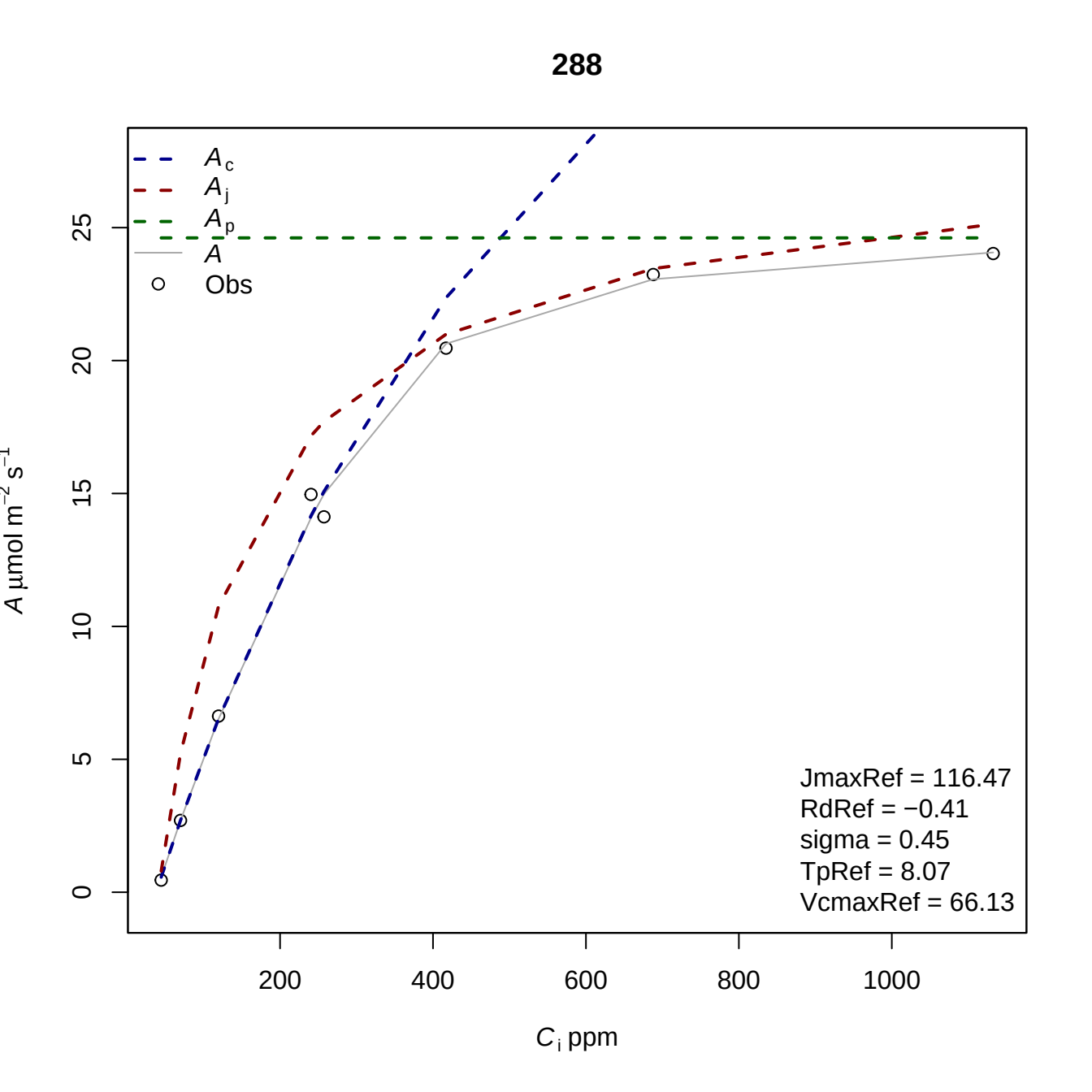

283
A_c
A_j
A_p
A
Obs
JmaxRef = 202.89
RdRef = −0.83
sigma = 0.45
TpRef = 12.51
VcmaxRef = 166.23
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
200
400
600
800
1000
1200

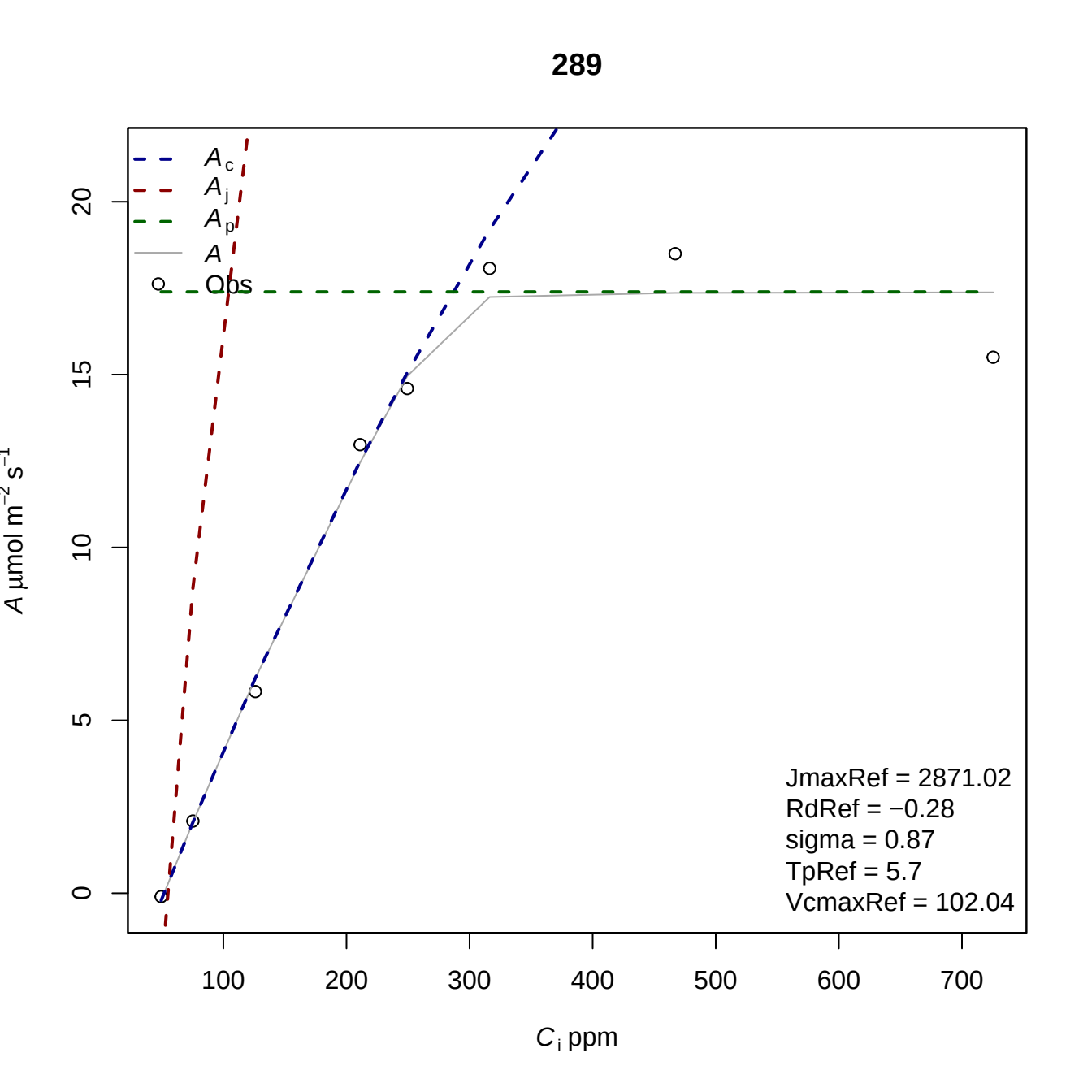

284
A_c
A_j
A_p
A
Obs
JmaxRef = 116
RdRef = 0.05
sigma = 0.72
TpRef = 7.29
VcmaxRef = 41.12
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200

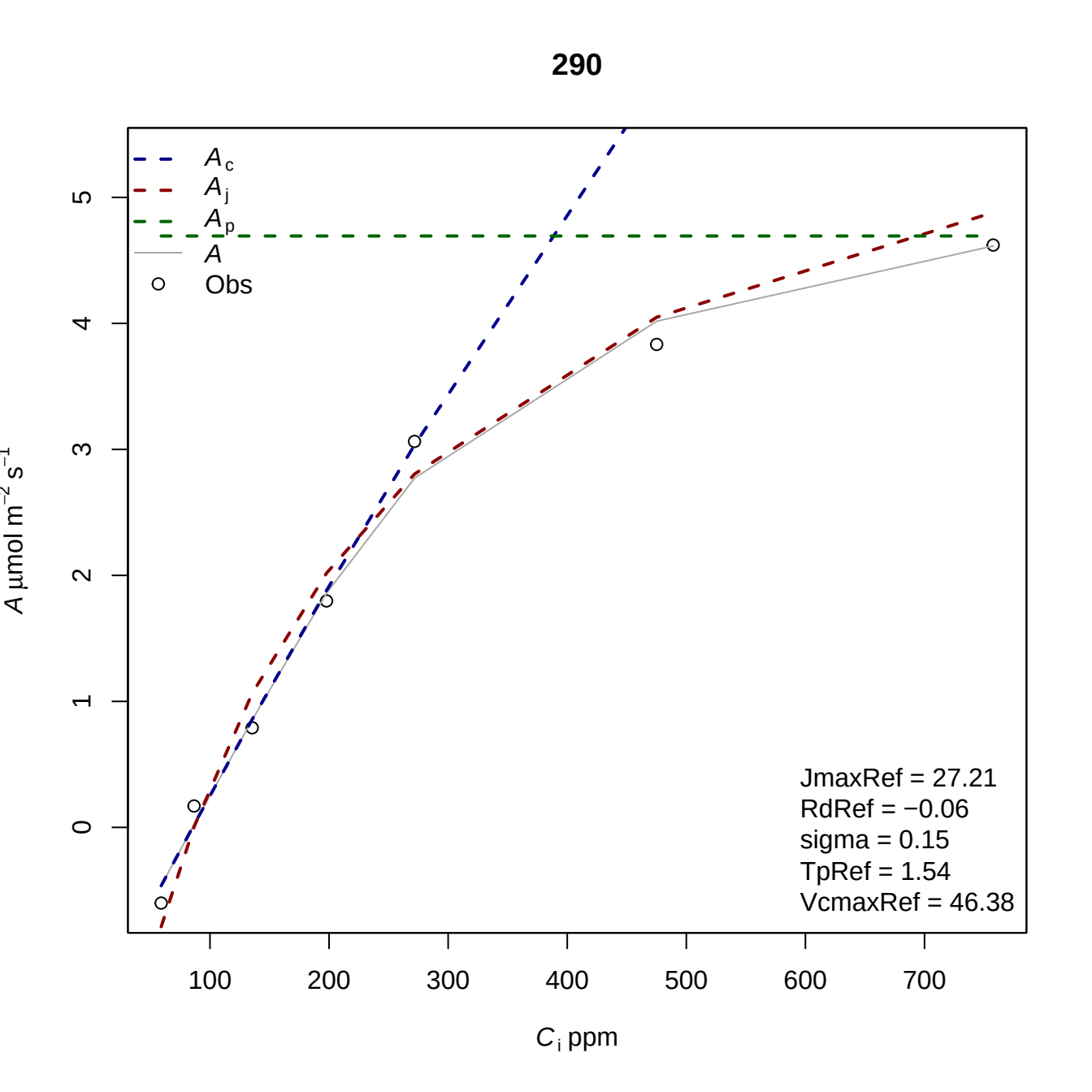

285
A_c
A_j
A_p
A
Obs
JmaxRef = 119.11
RdRef = 0.13
sigma = 0.73
TpRef = 7.89
VcmaxRef = 40
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200

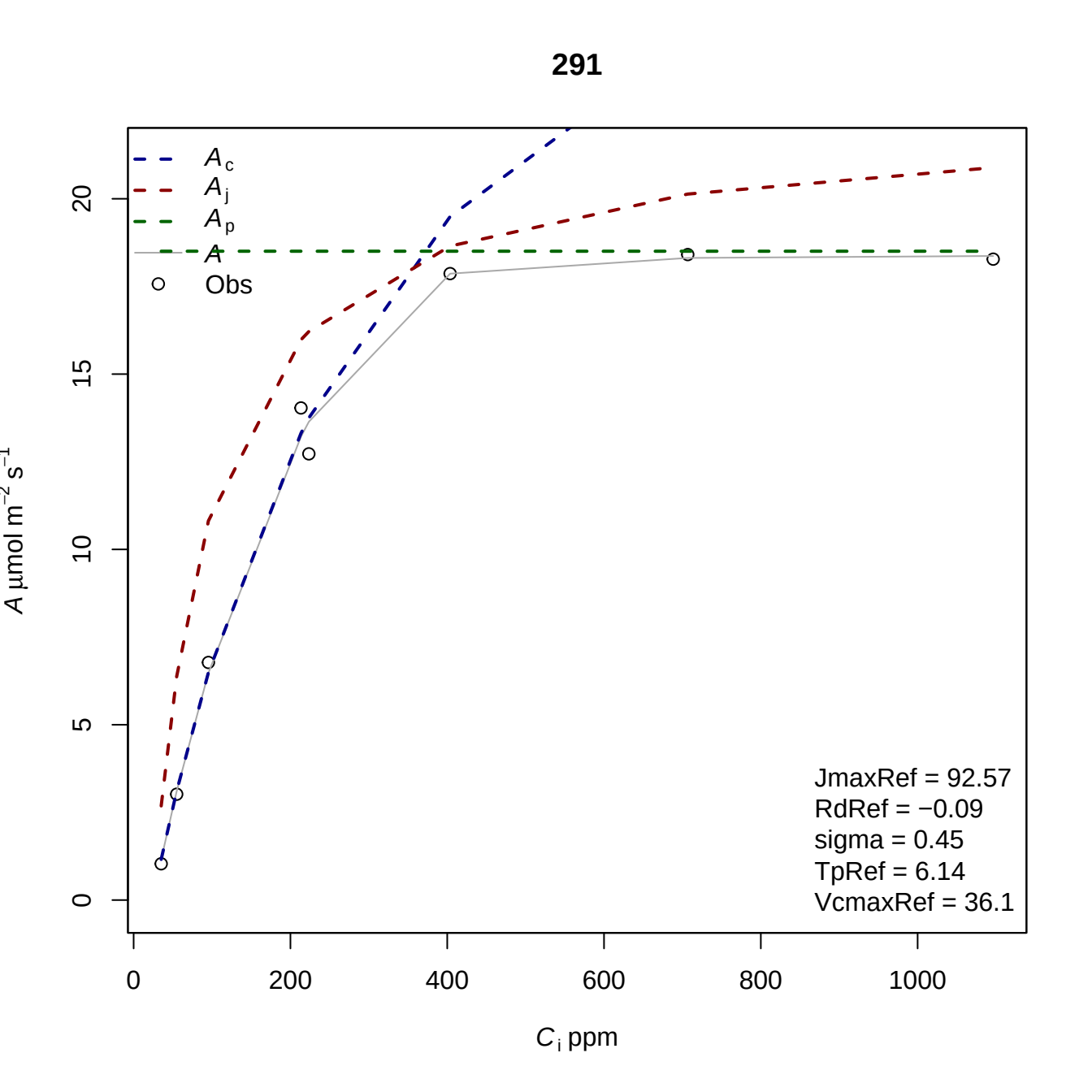

286
A_c
A_j
A_p
A
Obs
JmaxRef = 81.32
RdRef = 0.12
sigma = 0.41
TpRef = 5.56
VcmaxRef = 33.03
0
5
10
15
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
1200
C_i ppm

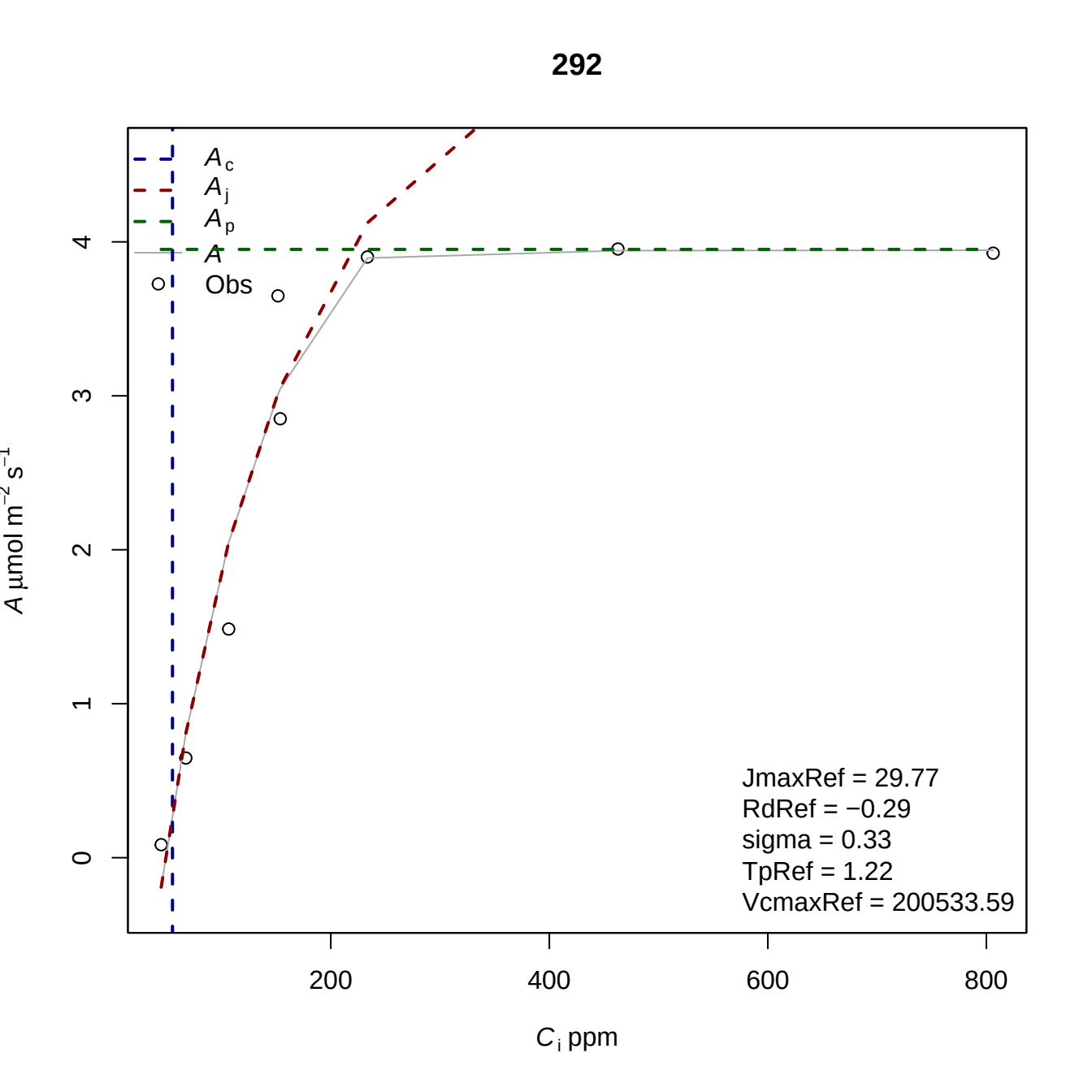

287
A_c
A_j
A_p
A
Obs
JmaxRef = 95.26
RdRef = −0.09
sigma = 0.35
TpRef = 6.96
VcmaxRef = 40.88
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000

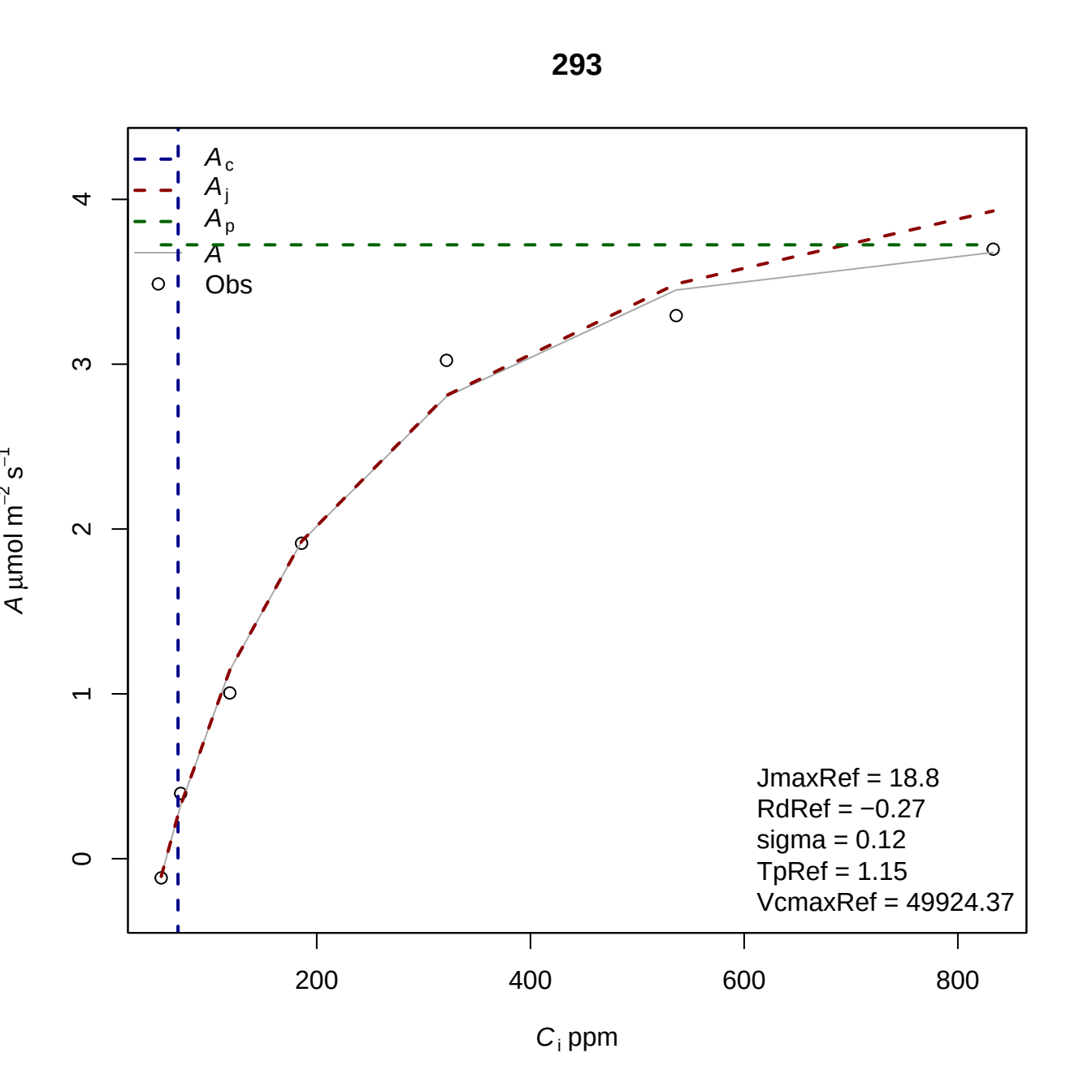

288
A_c
A_j
A_p
A
Obs
JmaxRef = 116.47
RdRef = −0.41
sigma = 0.45
TpRef = 8.07
VcmaxRef = 66.13
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
200
400
600
800
1000

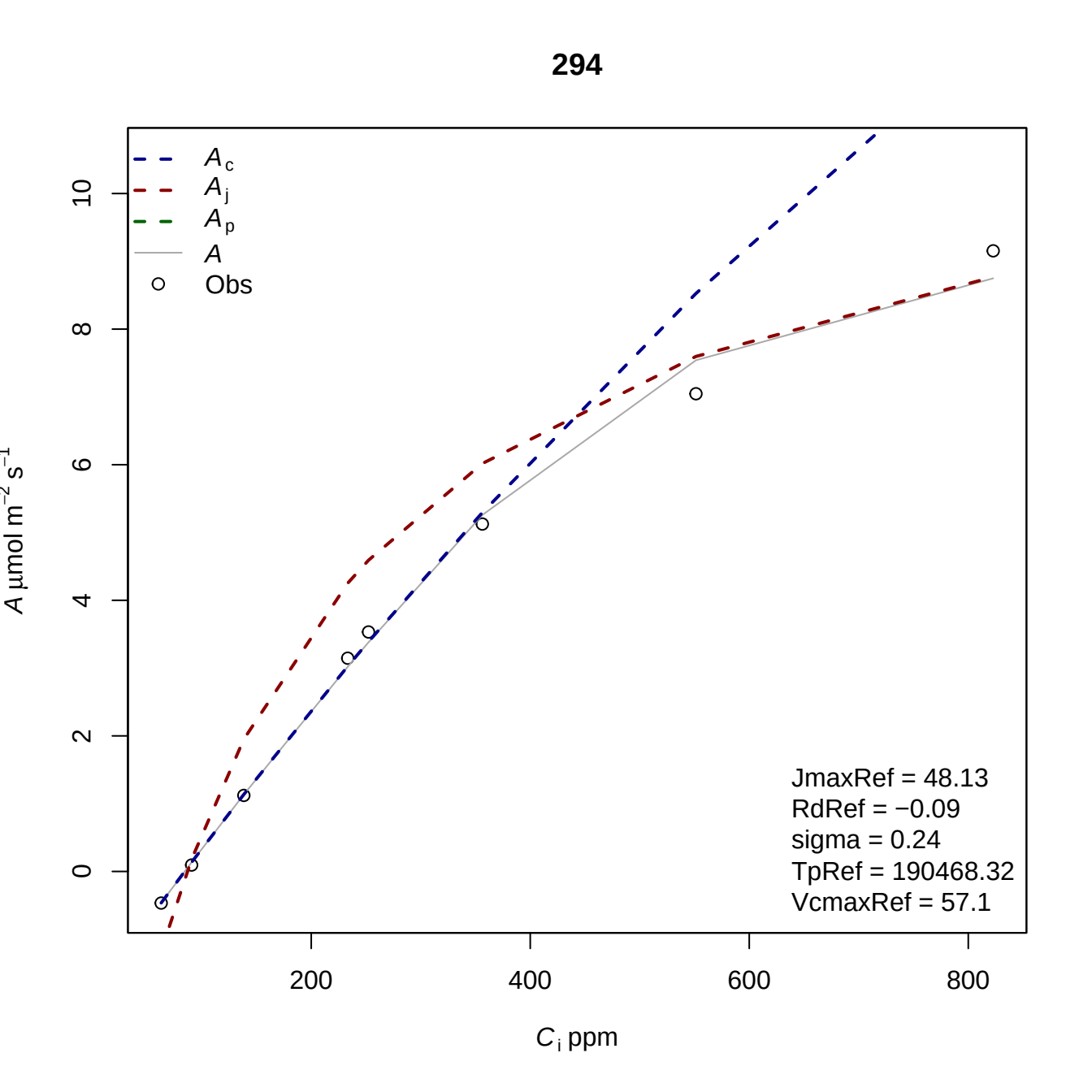

289
A_c
A_j
A_p
A
Obs
JmaxRef = 2871.02
RdRef = −0.28
sigma = 0.87
TpRef = 5.7
VcmaxRef = 102.04
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
100
200
300
400
500
600
700

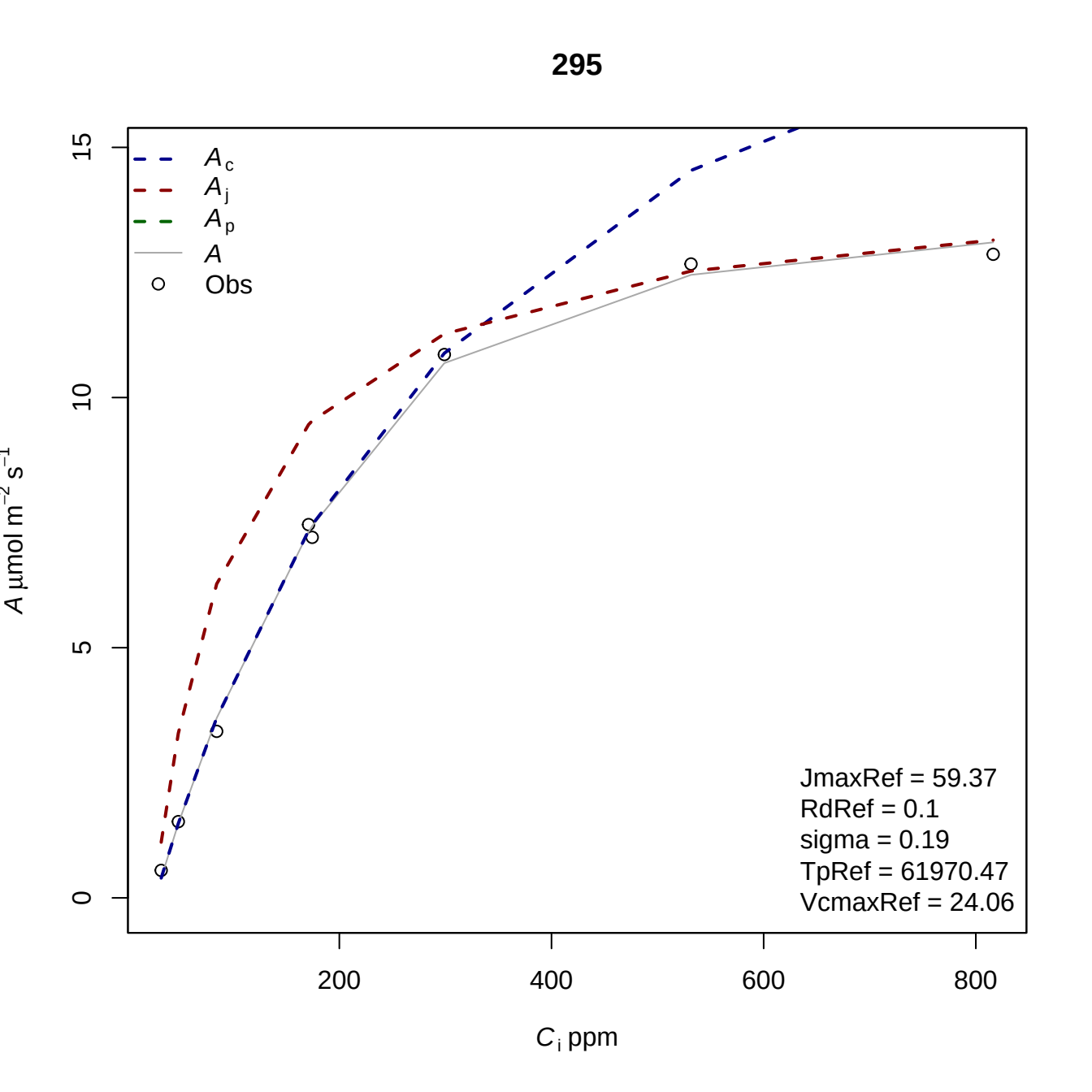

290
A_c
A_j
A_p
A
Obs
JmaxRef = 27.21
RdRef = −0.06
sigma = 0.15
TpRef = 1.54
VcmaxRef = 46.38
A µmol m^{-2} s^{-1}
C_i ppm
0
1
2
3
4
5
100
200
300
400
500
600
700

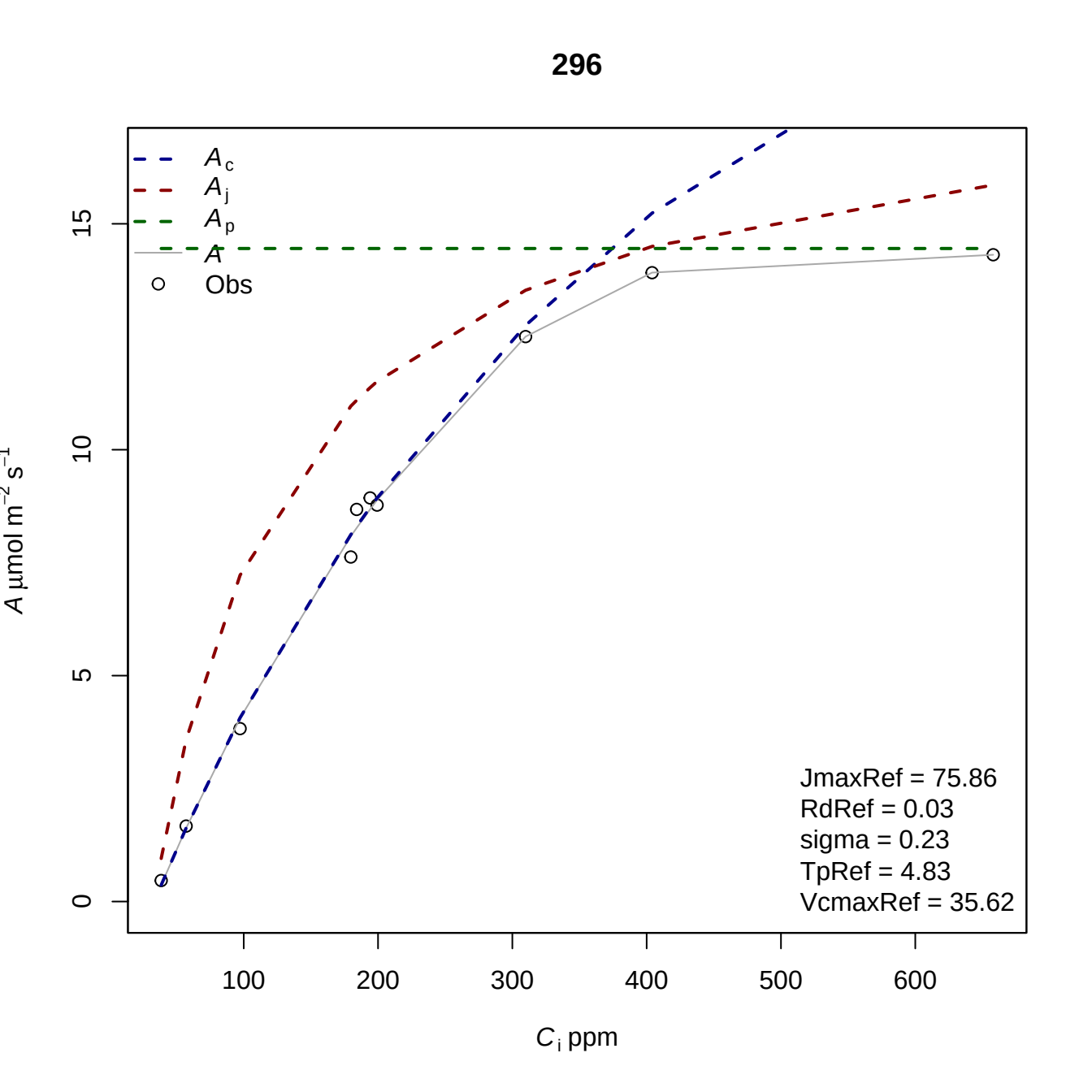

291
A_c
A_j
A_p
A
Obs
JmaxRef = 92.57
RdRef = −0.09
sigma = 0.45
TpRef = 6.14
VcmaxRef = 36.1
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000

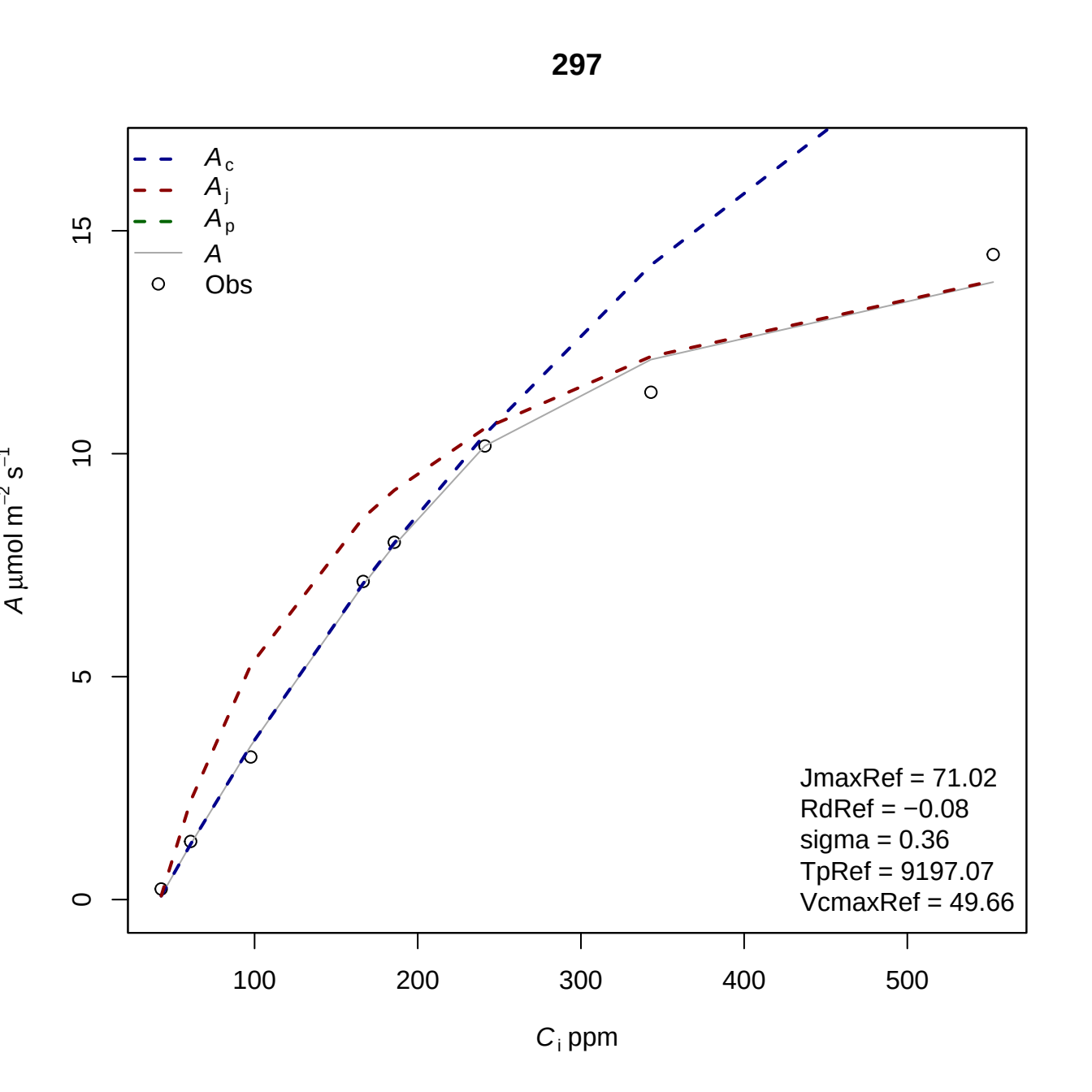

292
A_c
A_j
A_p
A
Obs
JmaxRef = 29.77
RdRef = −0.29
sigma = 0.33
TpRef = 1.22
VcmaxRef = 200533.59
0
1
2
3
4
200
400
600
800
A µmol m^{-2} s^{-1}
C_i ppm

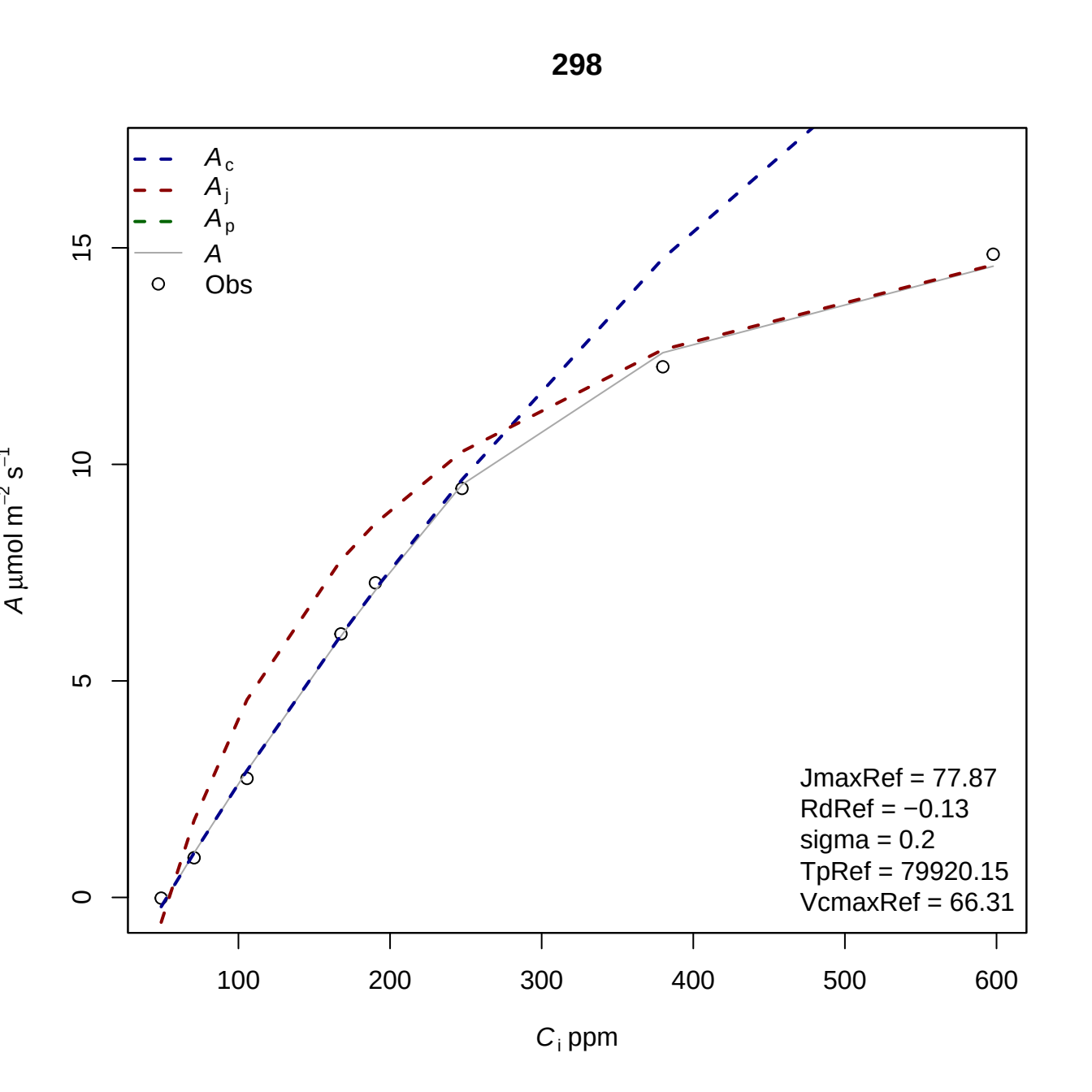

293
A_c
A_j
A_p
A
Obs
JmaxRef = 18.8
RdRef = −0.27
sigma = 0.12
TpRef = 1.15
VcmaxRef = 49924.37
A µmol m^{-2} s^{-1}
C_i ppm
0
1
2
3
4
200
400
600
800

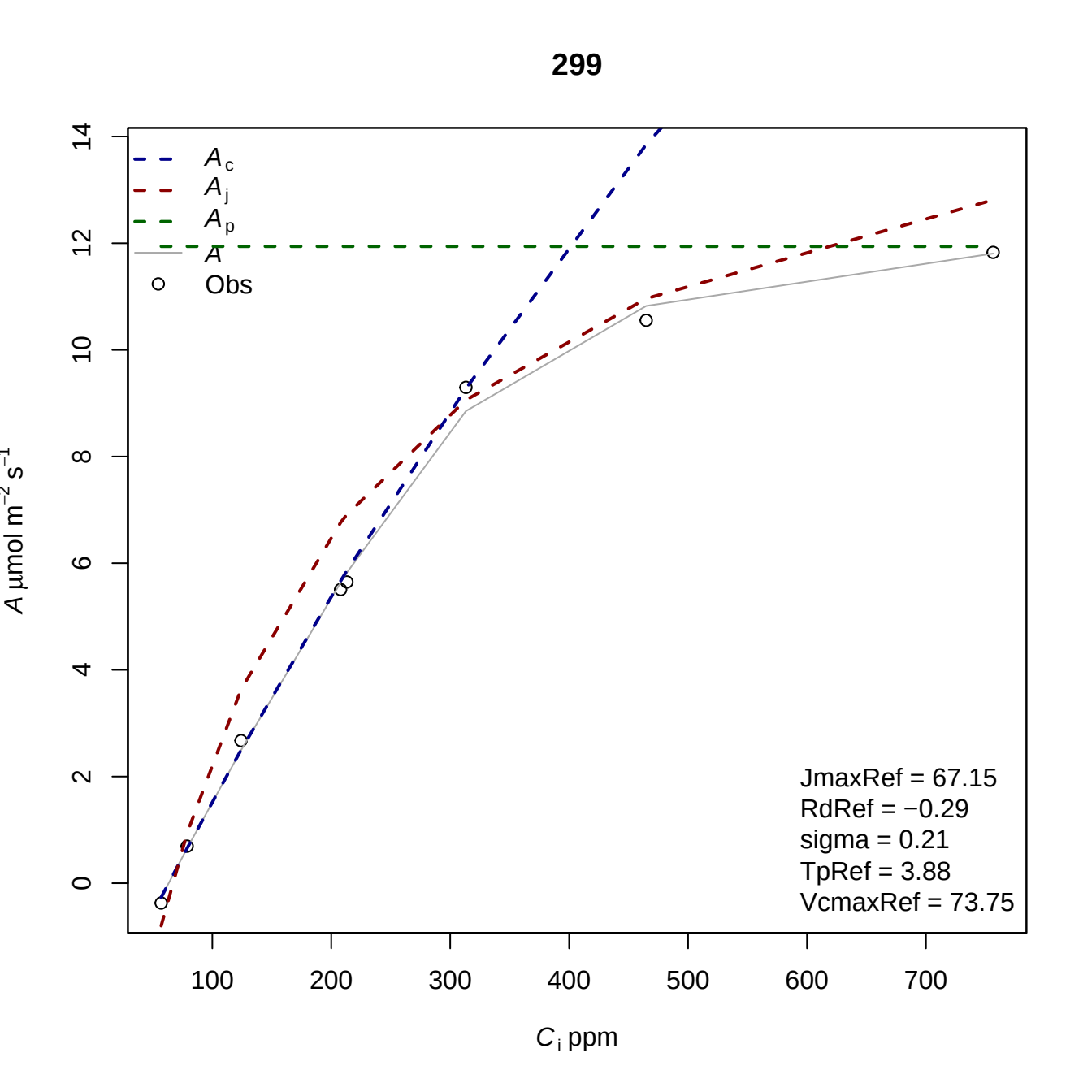

294
A_c
A_j
A_p
A
Obs
JmaxRef = 48.13
RdRef = −0.09
sigma = 0.24
TpRef = 190468.32
VcmaxRef = 57.1
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
200
400
600
800

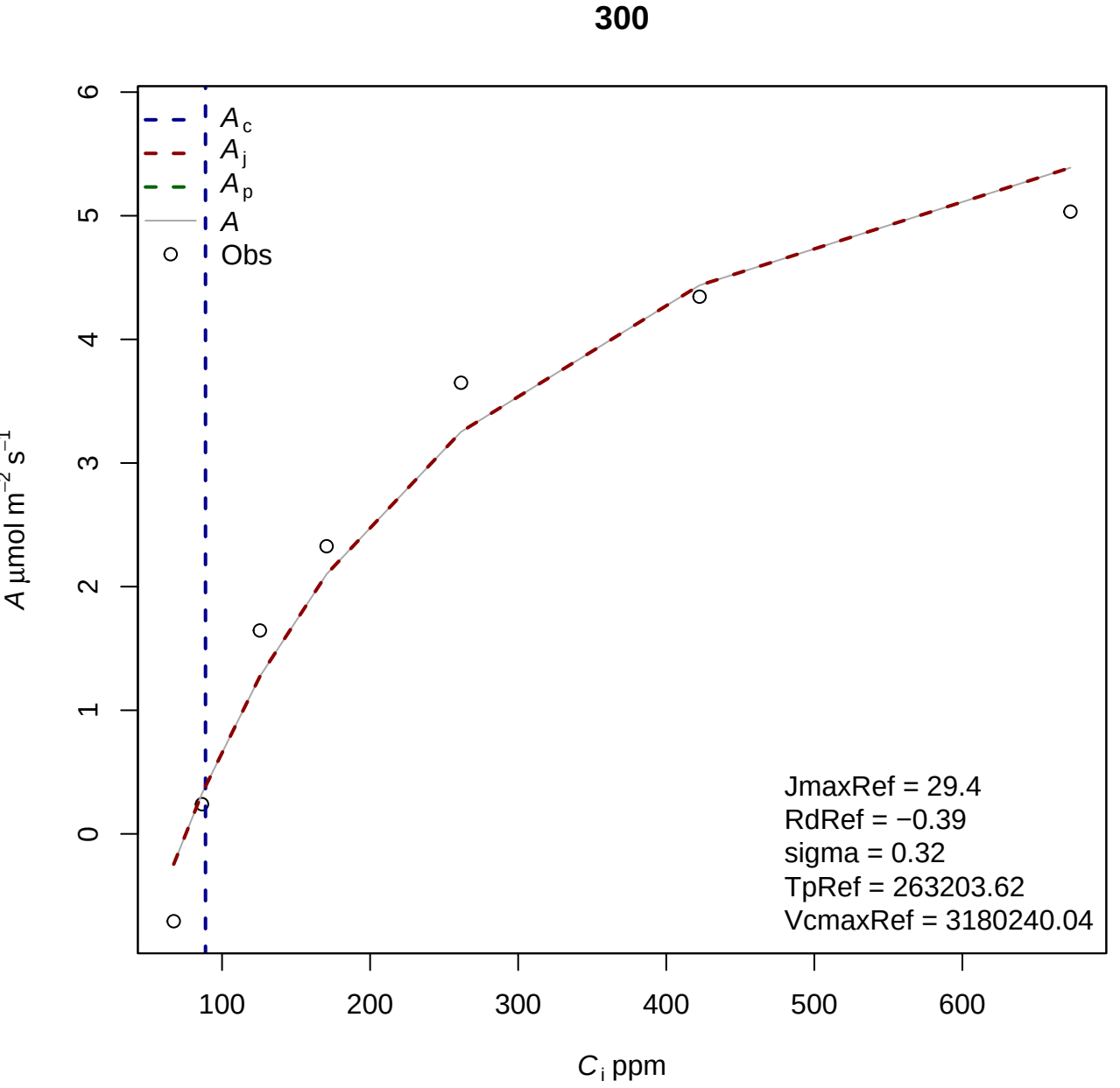

295
A_c
A_j
A_p
A
Obs
JmaxRef = 59.37
RdRef = 0.1
sigma = 0.19
TpRef = 61970.47
VcmaxRef = 24.06
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
200
400
600
800

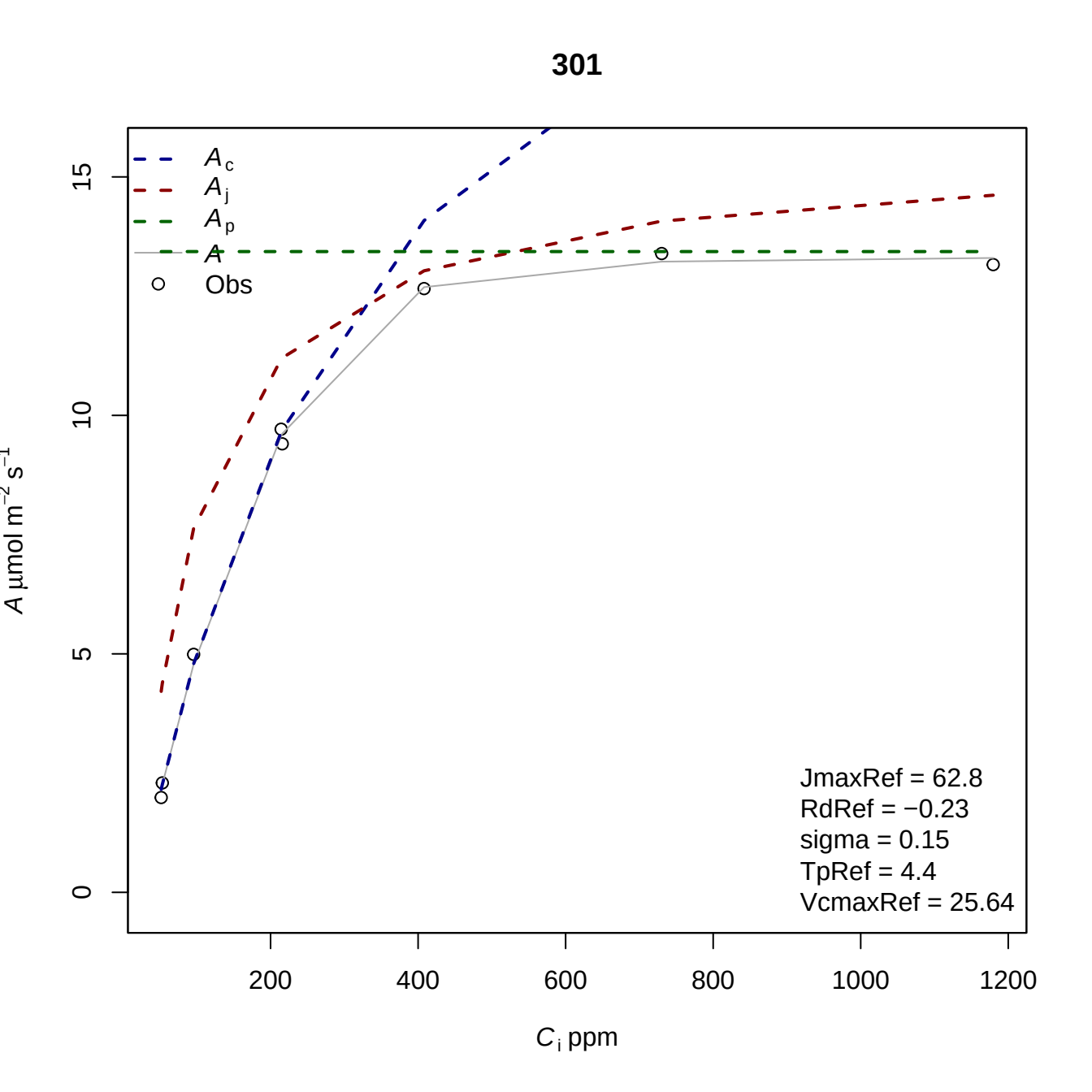

296
A_c
A_j
A_p
A
Obs
JmaxRef = 75.86
RdRef = 0.03
sigma = 0.23
TpRef = 4.83
VcmaxRef = 35.62
0
5
10
15
100
200
300
400
500
600
A µmol m^{-2} s^{-1}
C_i ppm

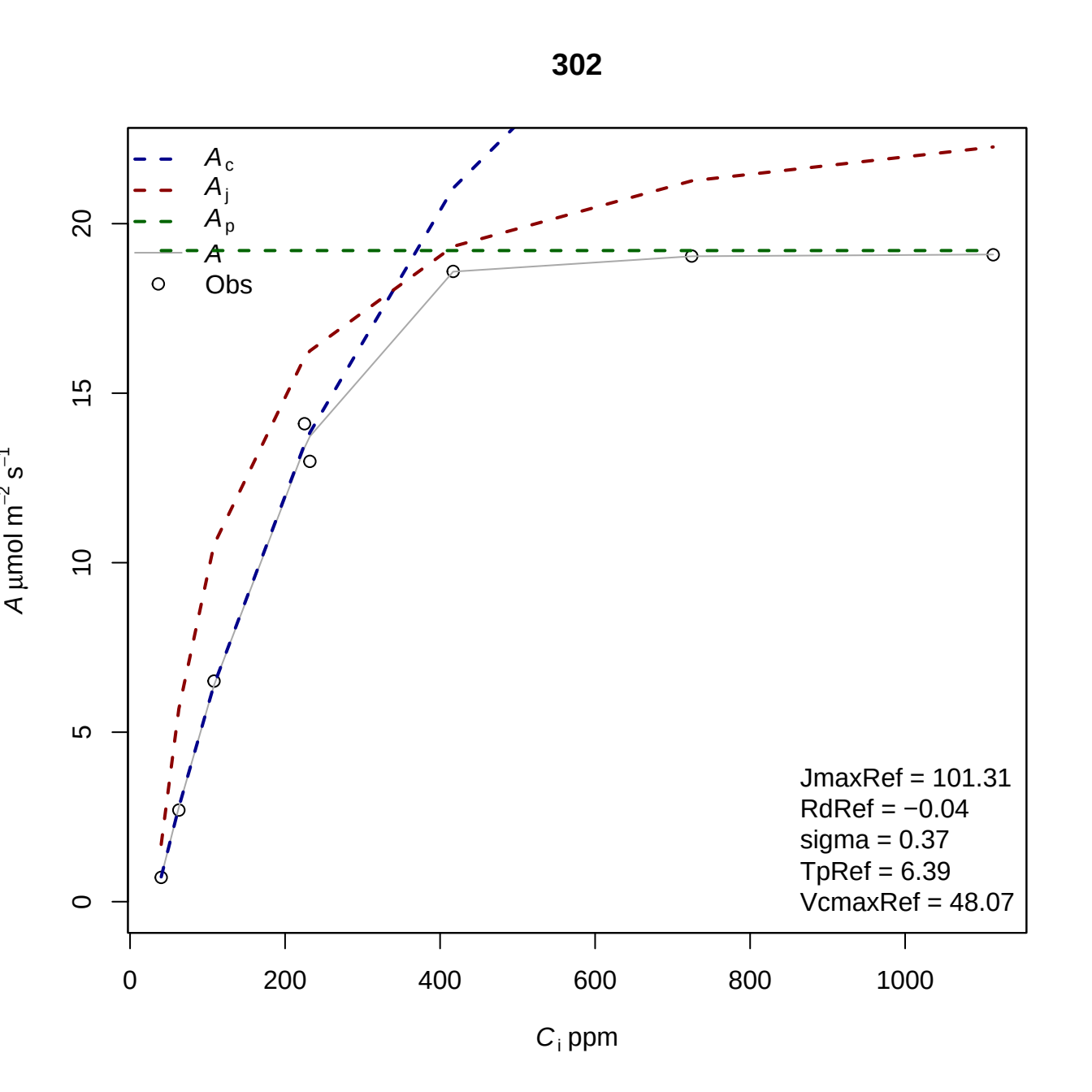

297
A_c
A_j
A_p
A
Obs
JmaxRef = 71.02
RdRef = −0.08
sigma = 0.36
TpRef = 9197.07
VcmaxRef = 49.66
0
5
10
15
100
200
300
400
500
A µmol m^{-2} s^{-1}
C_i ppm

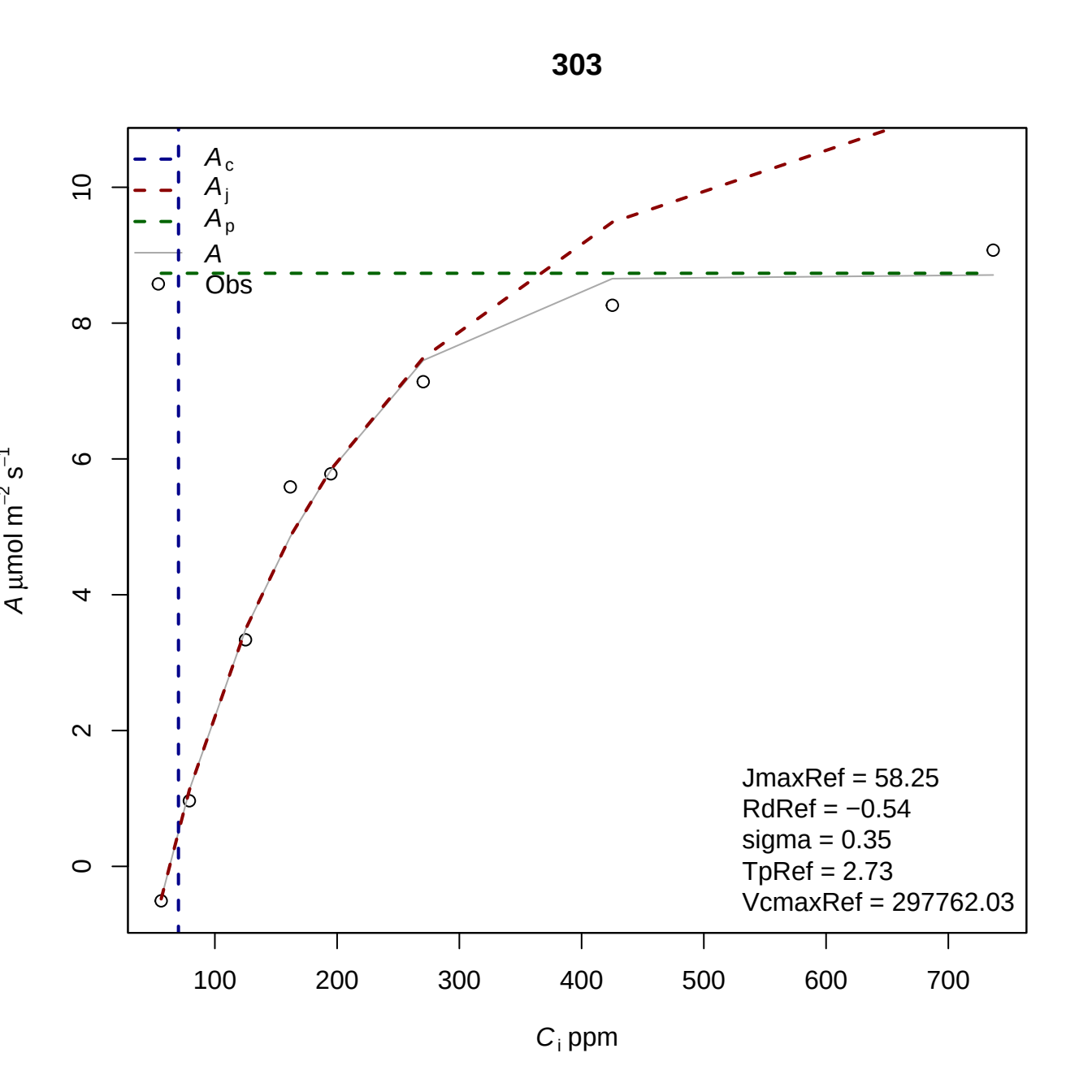

298
A_c
A_j
A_p
A
Obs
JmaxRef = 77.87
RdRef = −0.13
sigma = 0.2
TpRef = 79920.15
VcmaxRef = 66.31
0
5
10
15
100
200
300
400
500
600
A µmol m^{-2} s^{-1}
C_i ppm

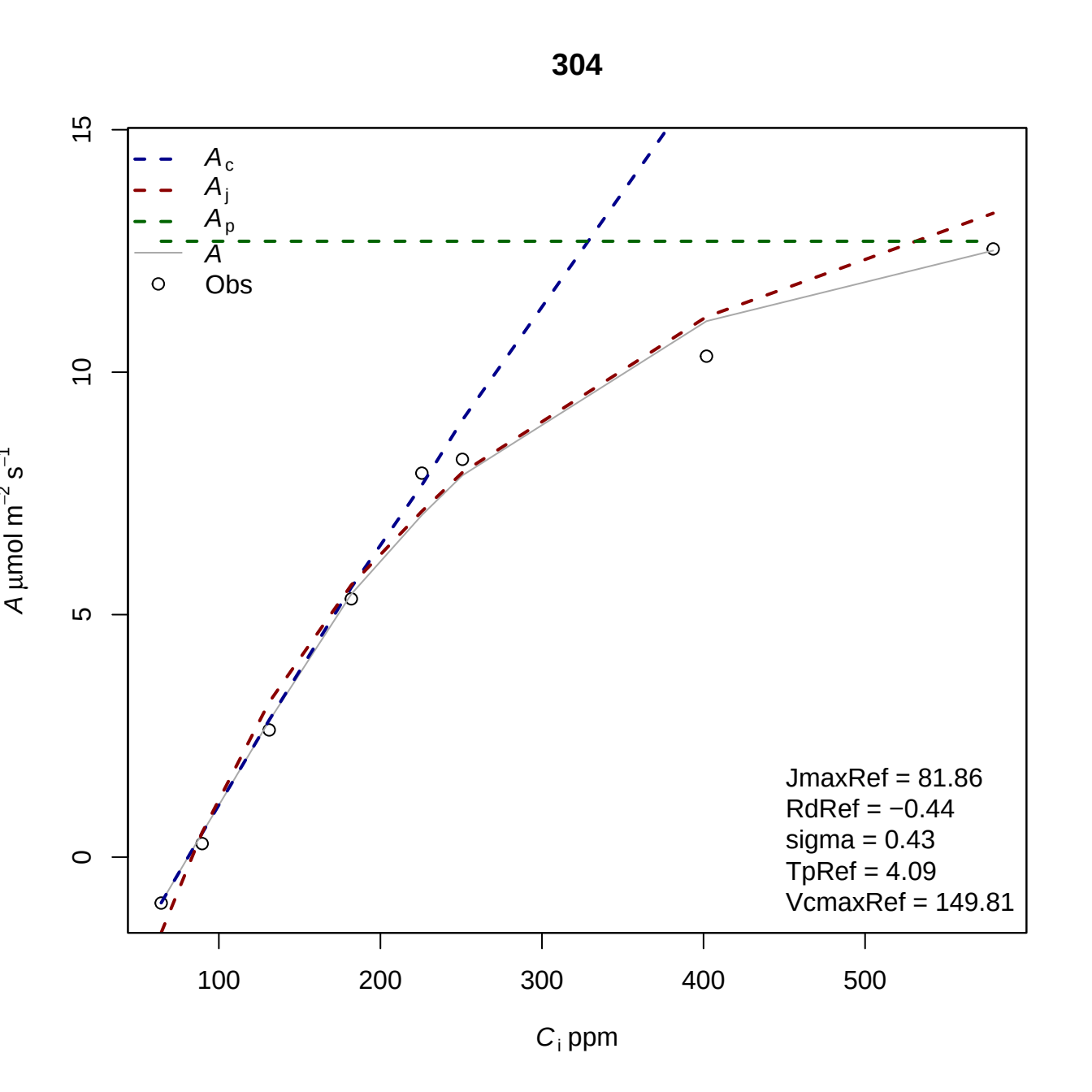

299
A_c
A_j
A_p
A
Obs
JmaxRef = 67.15
RdRef = −0.29
sigma = 0.21
TpRef = 3.88
VcmaxRef = 73.75
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
14
100
200
300
400
500
600
700

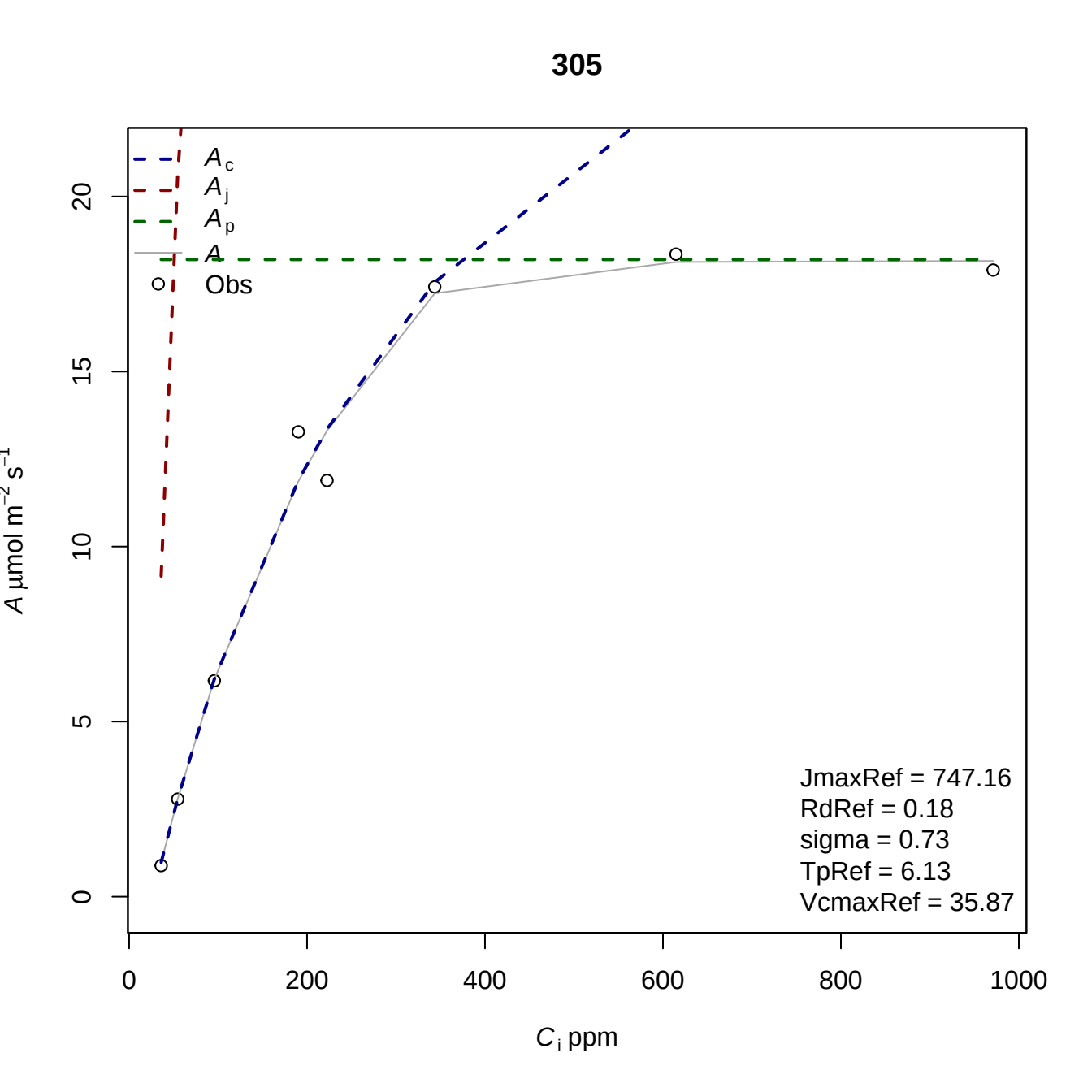

300
A_c
A_j
A_p
A
Obs
JmaxRef = 29.4
RdRef = −0.39
sigma = 0.32
TpRef = 263203.62
VcmaxRef = 3180240.04
A µmol m^{-2} s^{-1}
C_i ppm
0
1
2
3
4
5
6
100
200
300
400
500
600

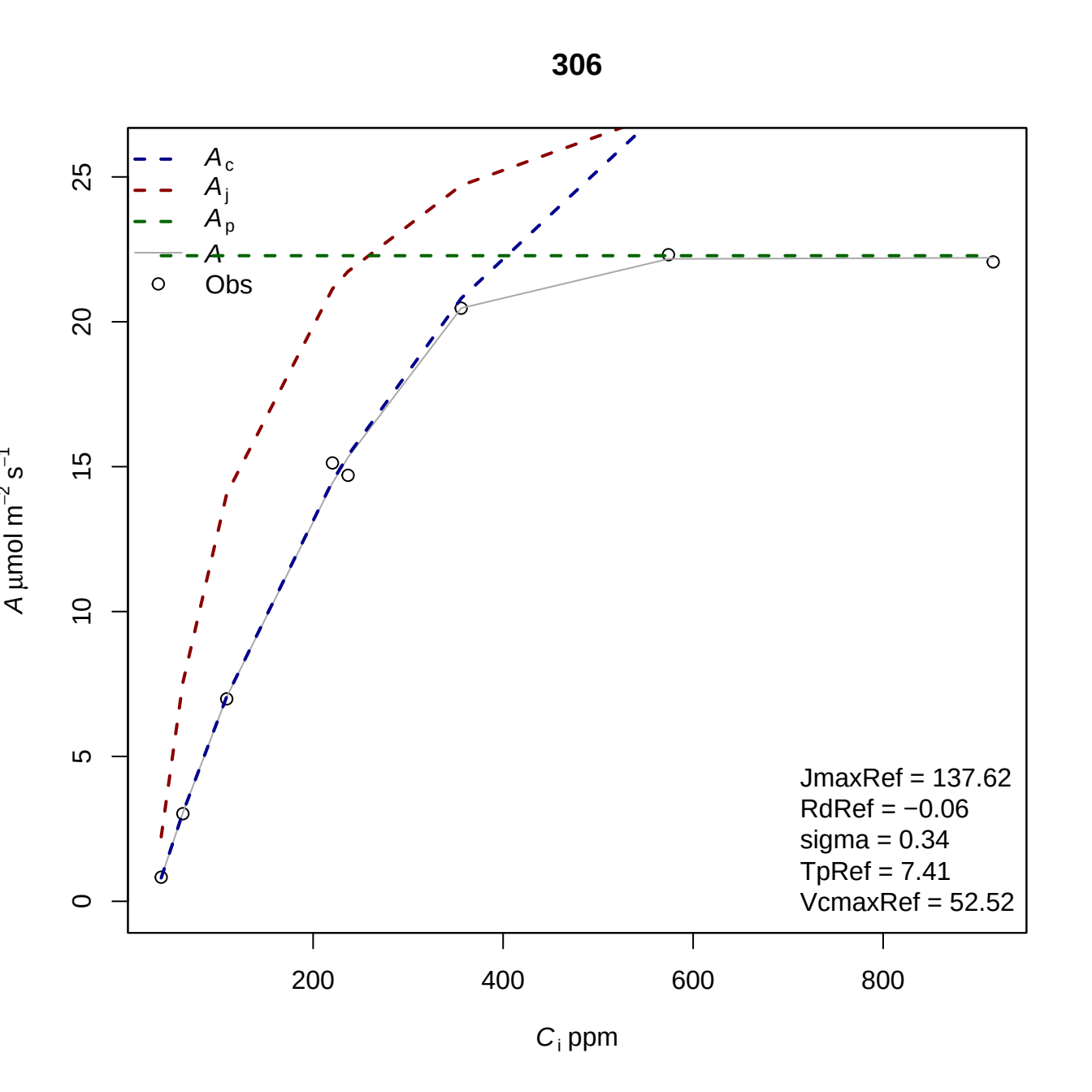

301
A_c
A_j
A_p
A
Obs
JmaxRef = 62.8
RdRef = −0.23
sigma = 0.15
TpRef = 4.4
VcmaxRef = 25.64
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
200
400
600
800
1000
1200

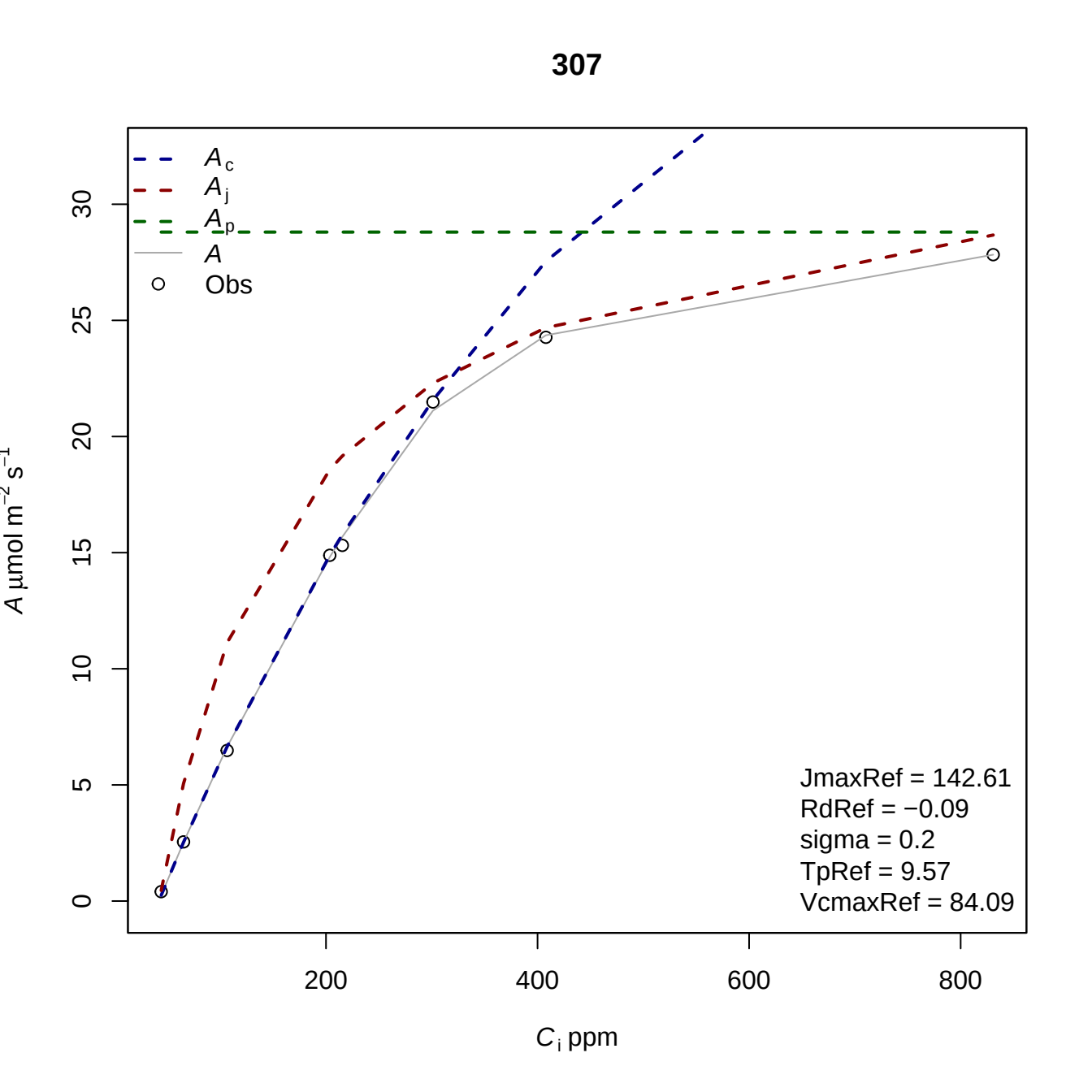

302
A_c
A_j
A_p
A
Obs
JmaxRef = 101.31
RdRef = −0.04
sigma = 0.37
TpRef = 6.39
VcmaxRef = 48.07
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000

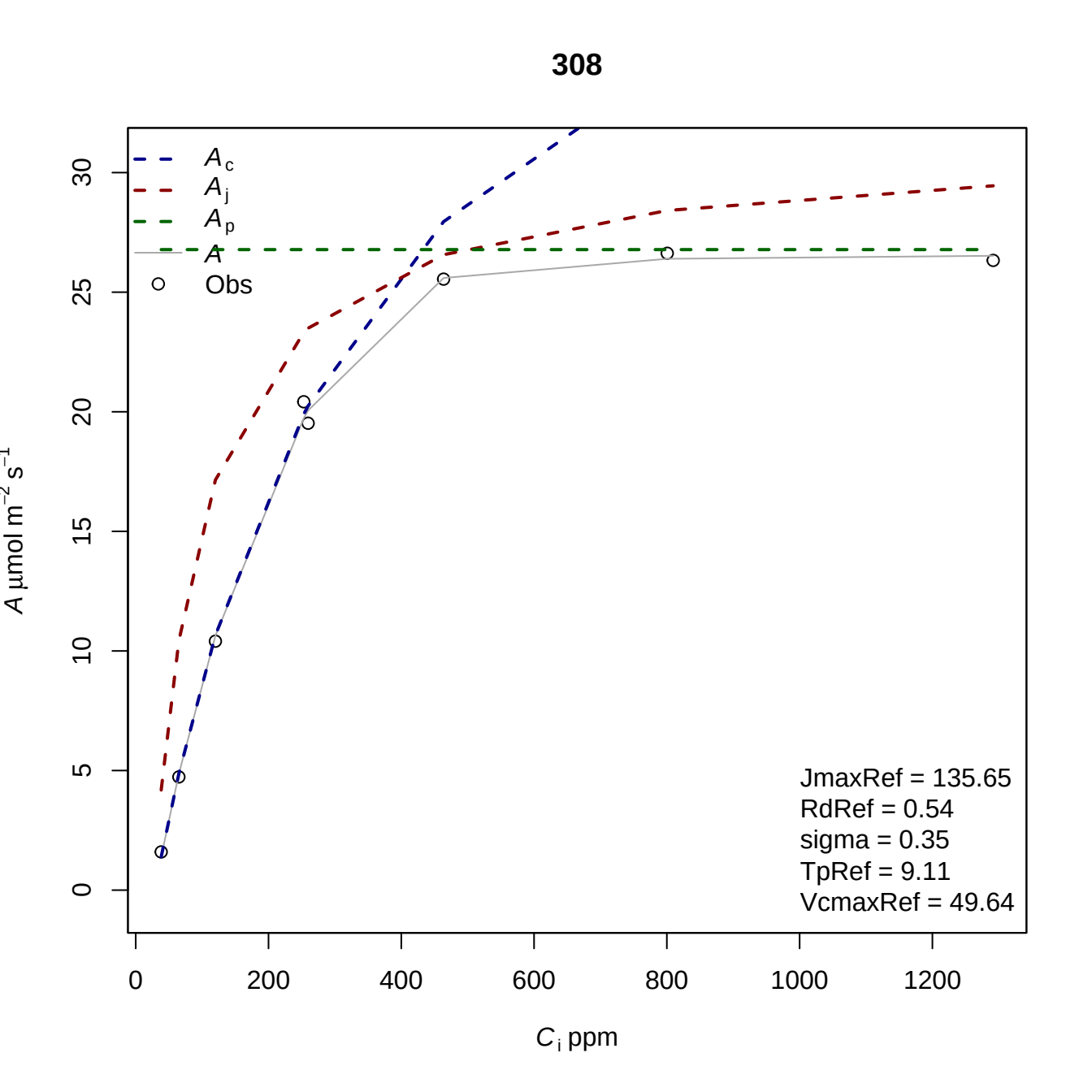

303
A_c
A_j
A_p
A
Obs
JmaxRef = 58.25
RdRef = −0.54
sigma = 0.35
TpRef = 2.73
VcmaxRef = 297762.03
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
100
200
300
400
500
600
700

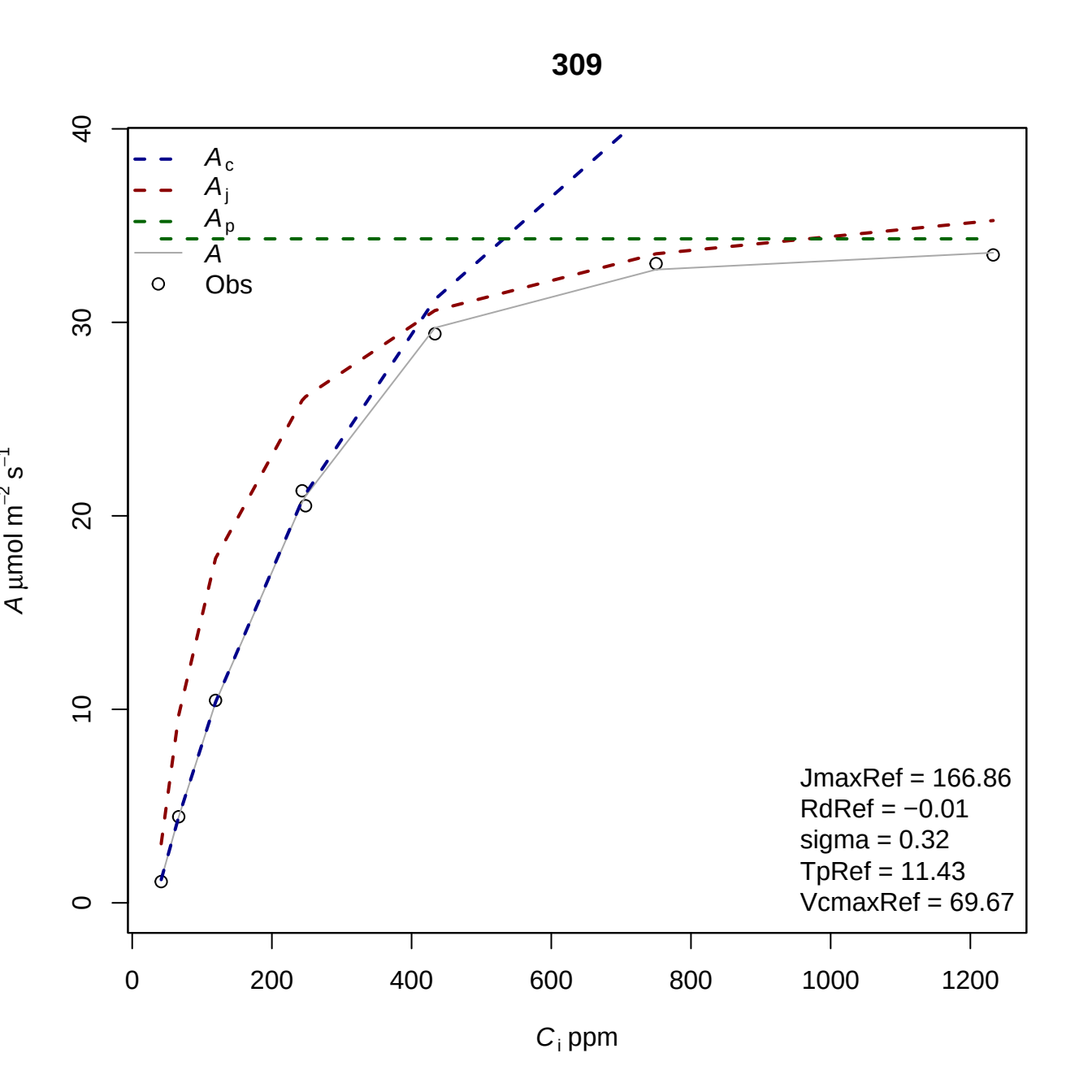

304
A_c
A_j
A_p
A
Obs
JmaxRef = 81.86
RdRef = −0.44
sigma = 0.43
TpRef = 4.09
VcmaxRef = 149.81
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
100
200
300
400
500

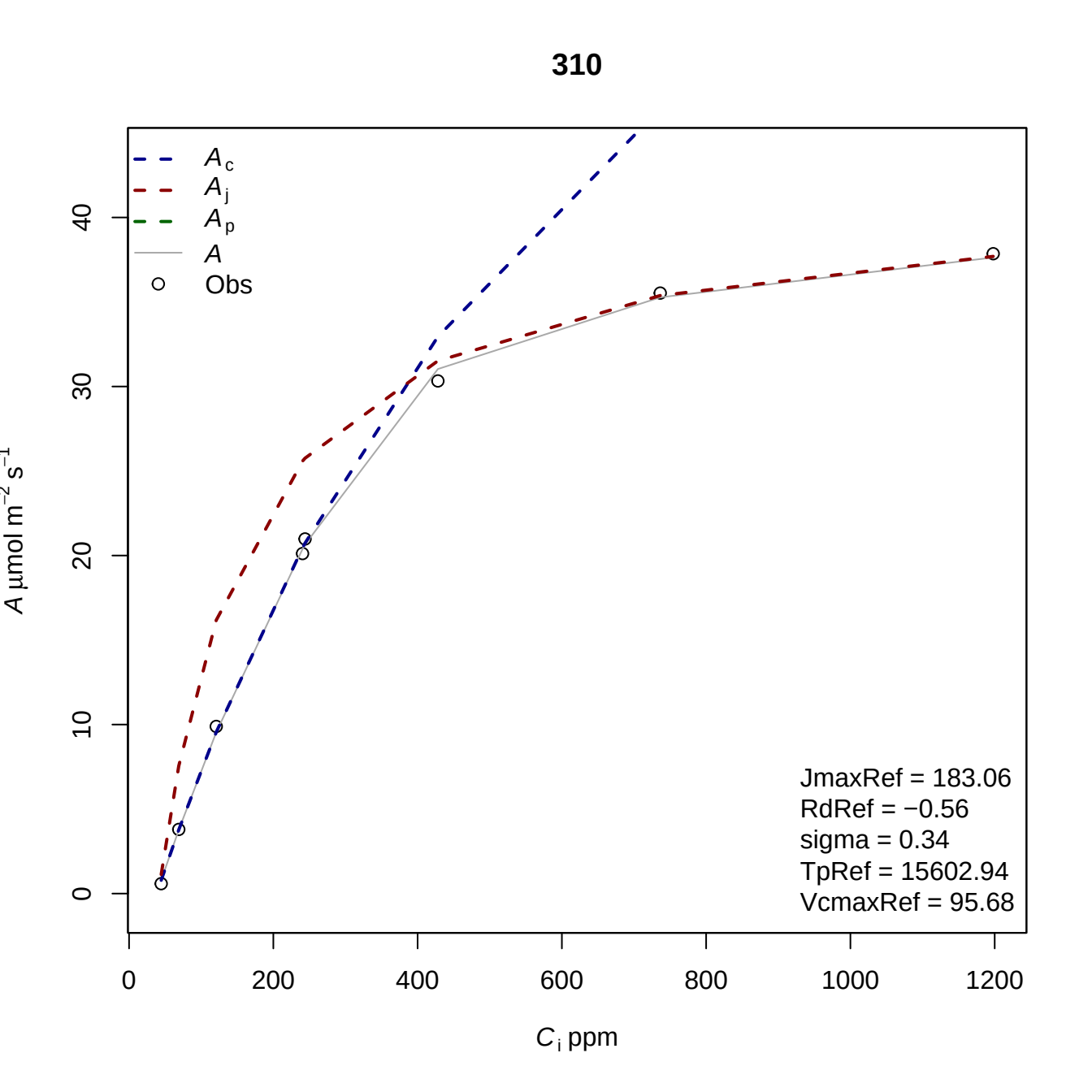

305
A_c
A_j
A_p
A
Obs
JmaxRef = 747.16
RdRef = 0.18
sigma = 0.73
TpRef = 6.13
VcmaxRef = 35.87
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000

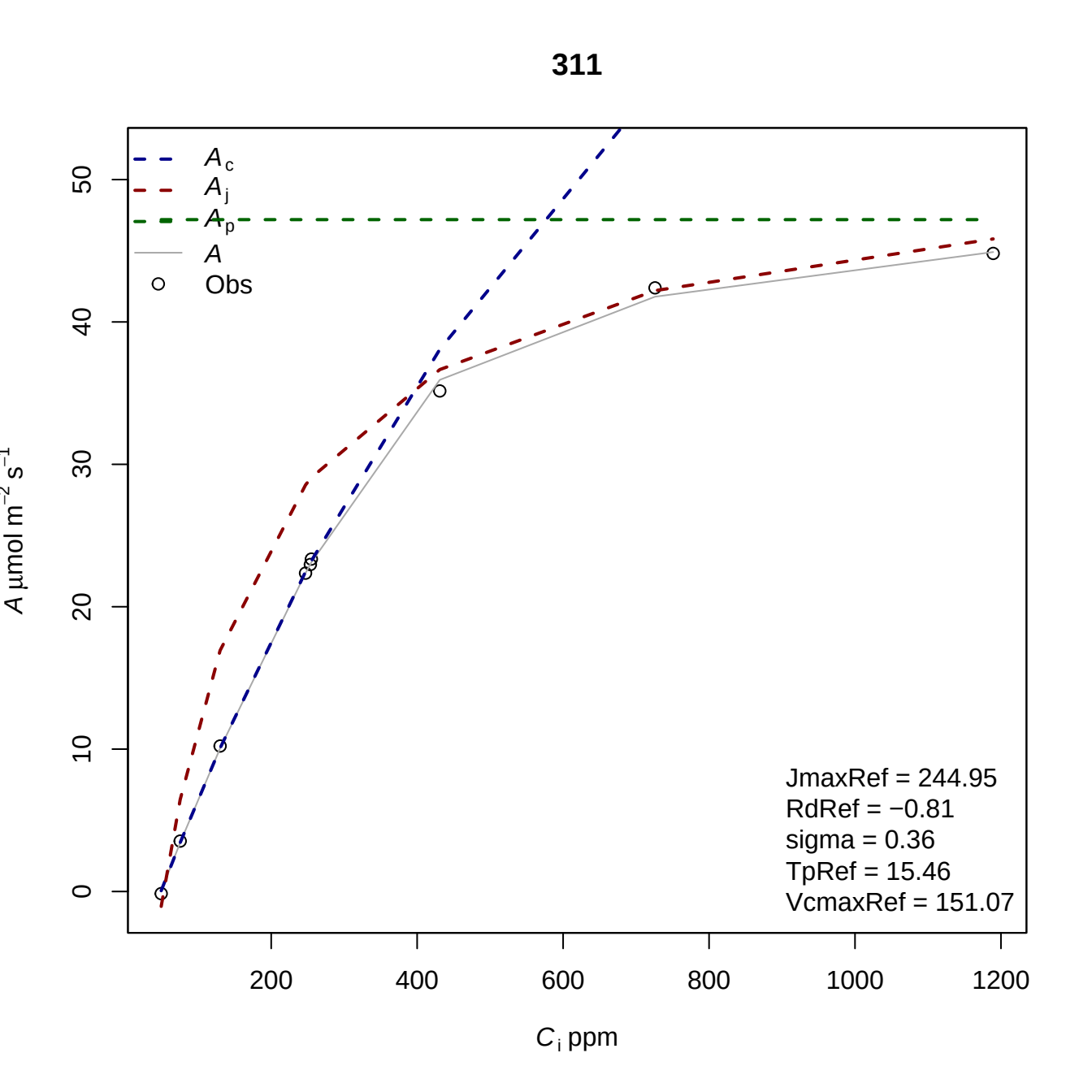

306
A_c
A_j
A_p
A
Obs
JmaxRef = 137.62
RdRef = −0.06
sigma = 0.34
TpRef = 7.41
VcmaxRef = 52.52
0
5
10
15
20
25
200
400
600
800
A µmol m^{-2} s^{-1}
C_i ppm

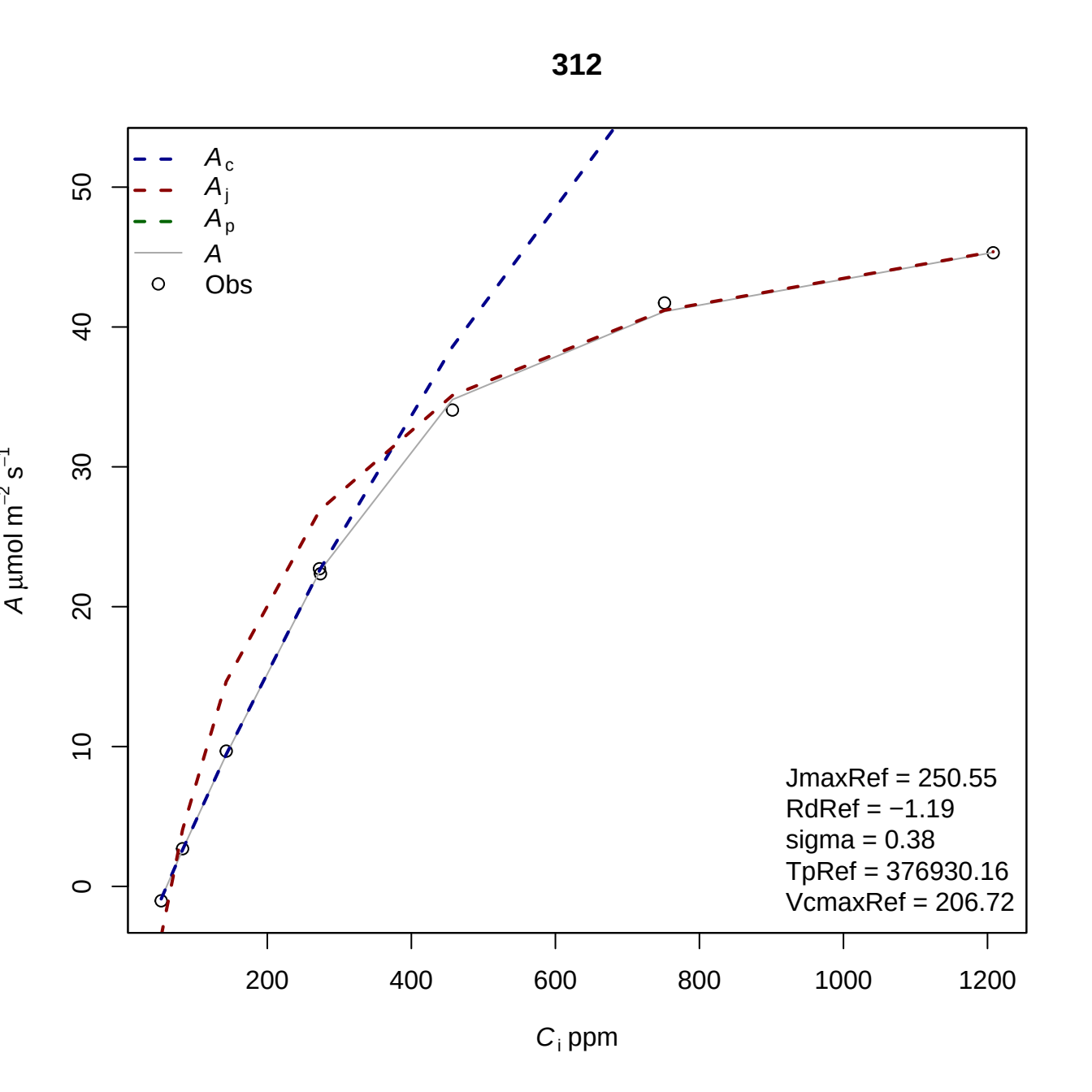

307
A_c
A_j
A_p
A
Obs
JmaxRef = 142.61
RdRef = −0.09
sigma = 0.2
TpRef = 9.57
VcmaxRef = 84.09
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
200
400
600
800

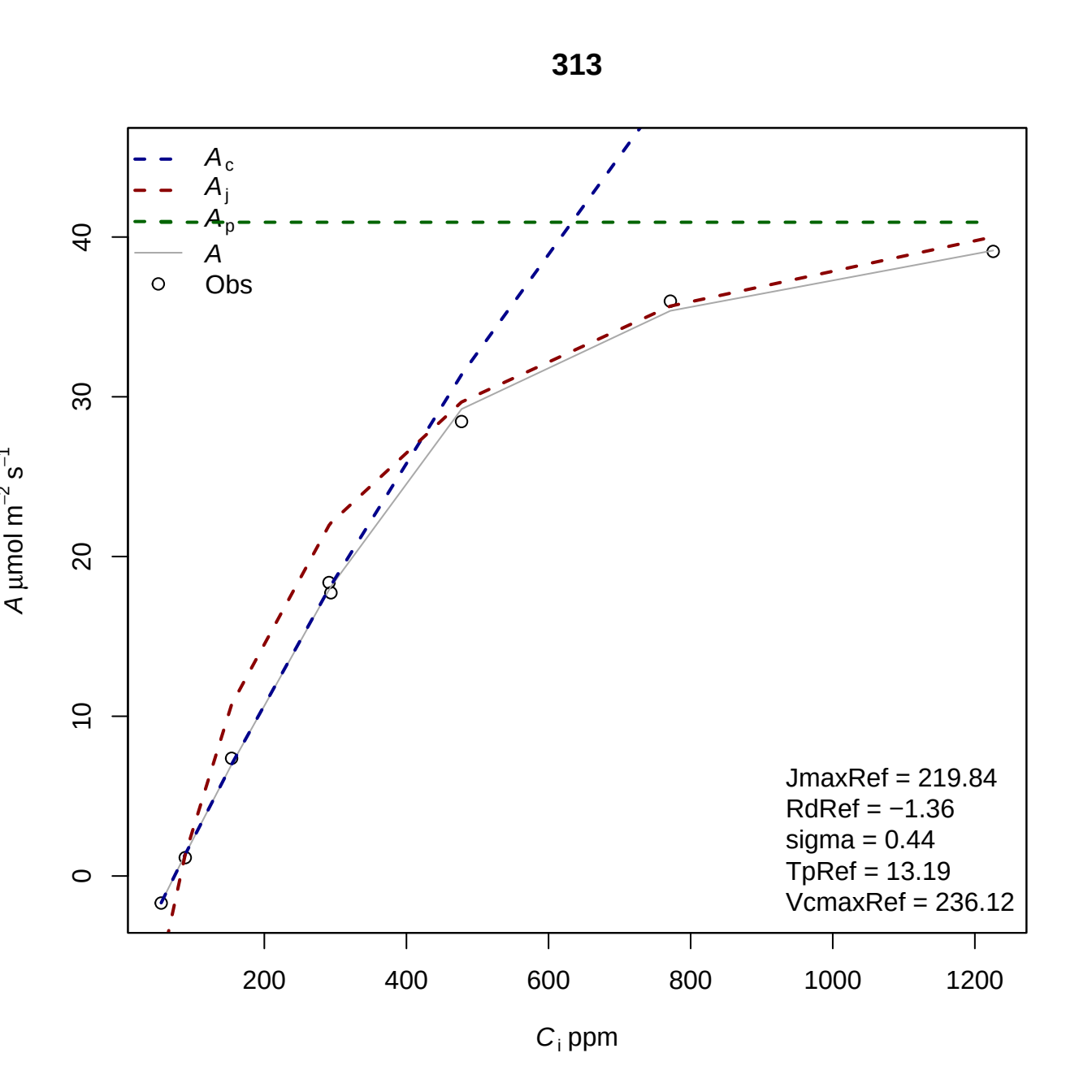

308
A_c
A_j
A_p
A
Obs
JmaxRef = 135.65
RdRef = 0.54
sigma = 0.35
TpRef = 9.11
VcmaxRef = 49.64
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

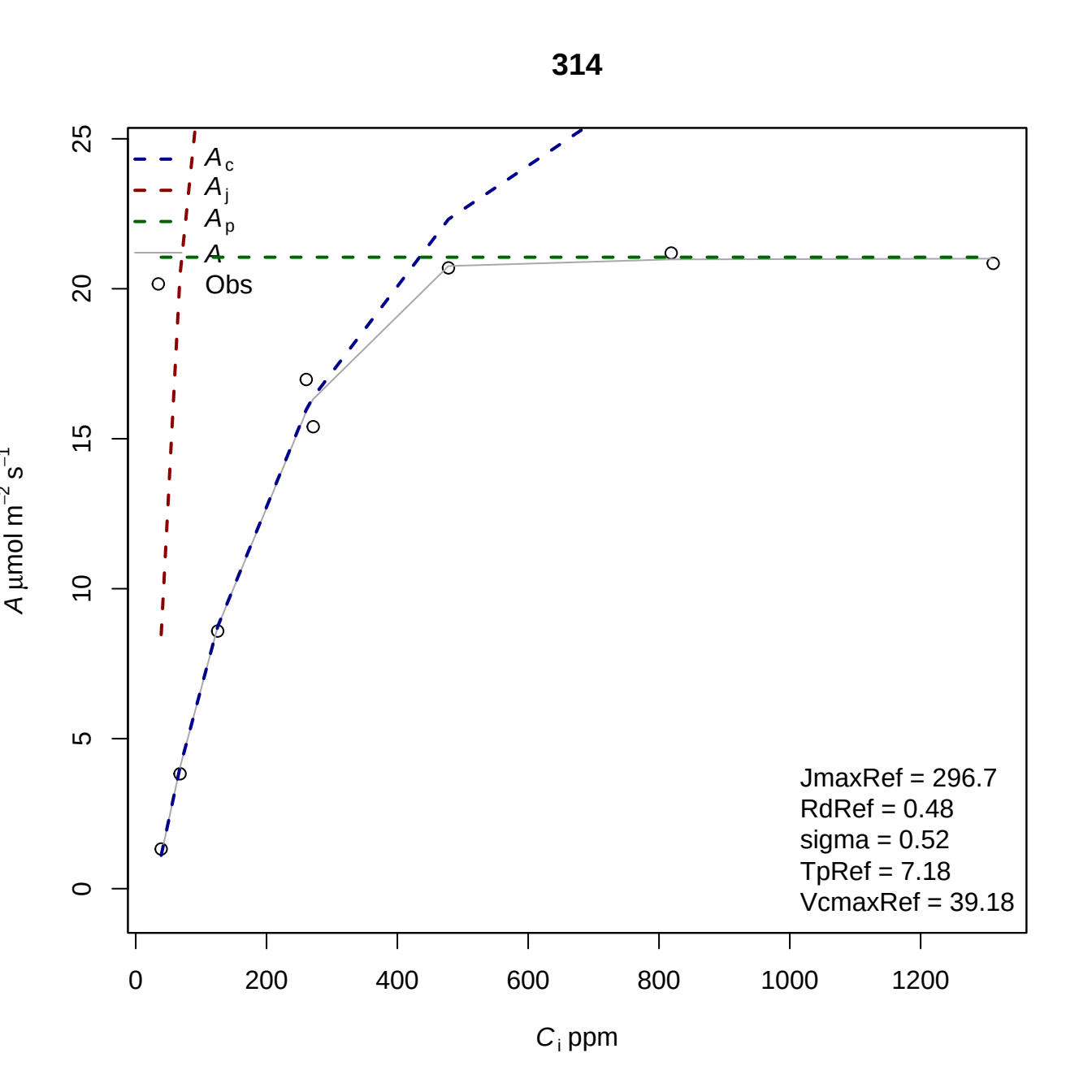

309
A_c
A_j
A_p
A
Obs
JmaxRef = 166.86
RdRef = −0.01
sigma = 0.32
TpRef = 11.43
VcmaxRef = 69.67
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

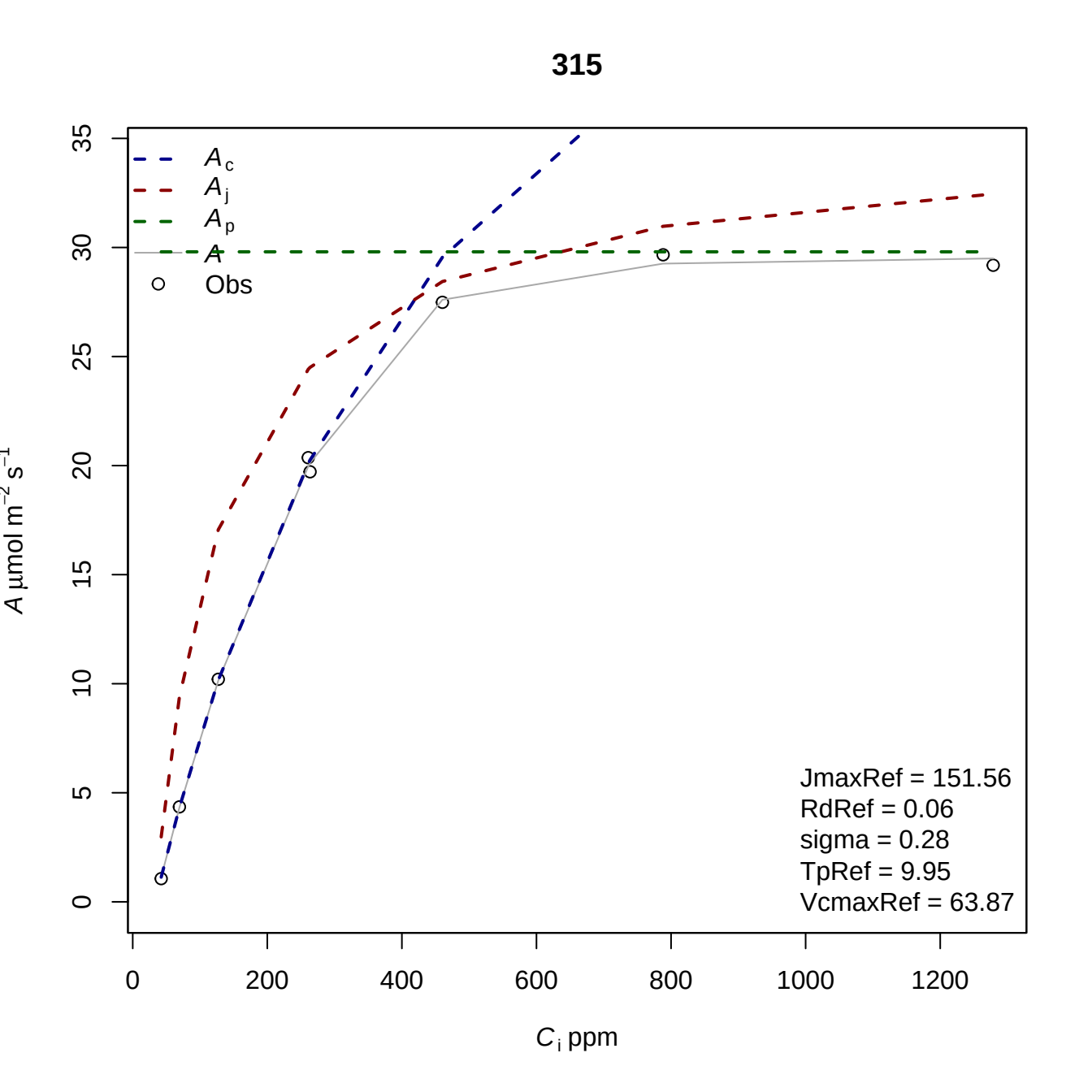

310
A_c
A_j
A_p
A
Obs
JmaxRef = 183.06
RdRef = −0.56
sigma = 0.34
TpRef = 15602.94
VcmaxRef = 95.68
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

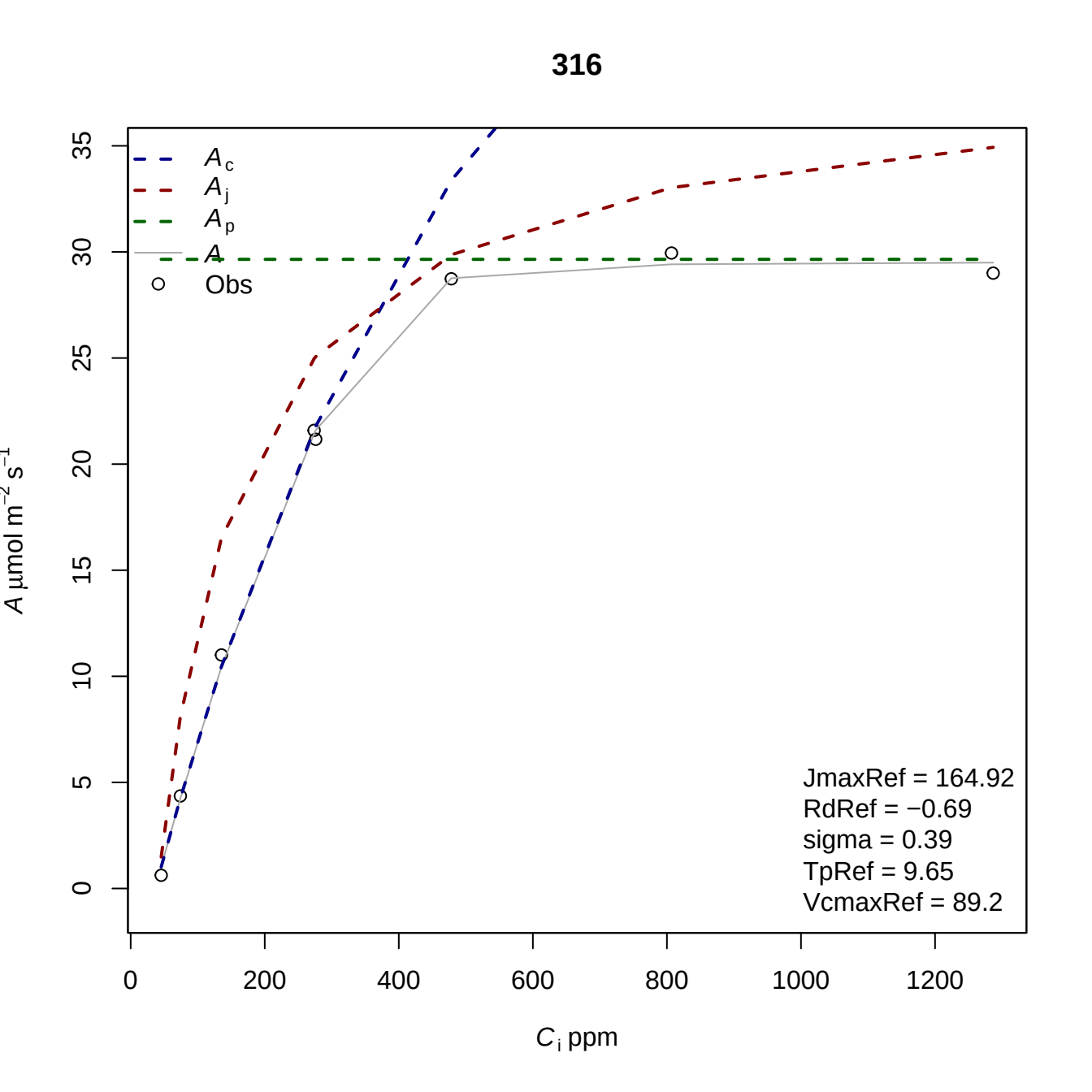

311
A_c
A_j
A_p
A
Obs
JmaxRef = 244.95
RdRef = −0.81
sigma = 0.36
TpRef = 15.46
VcmaxRef = 151.07
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
50
200
400
600
800
1000
1200

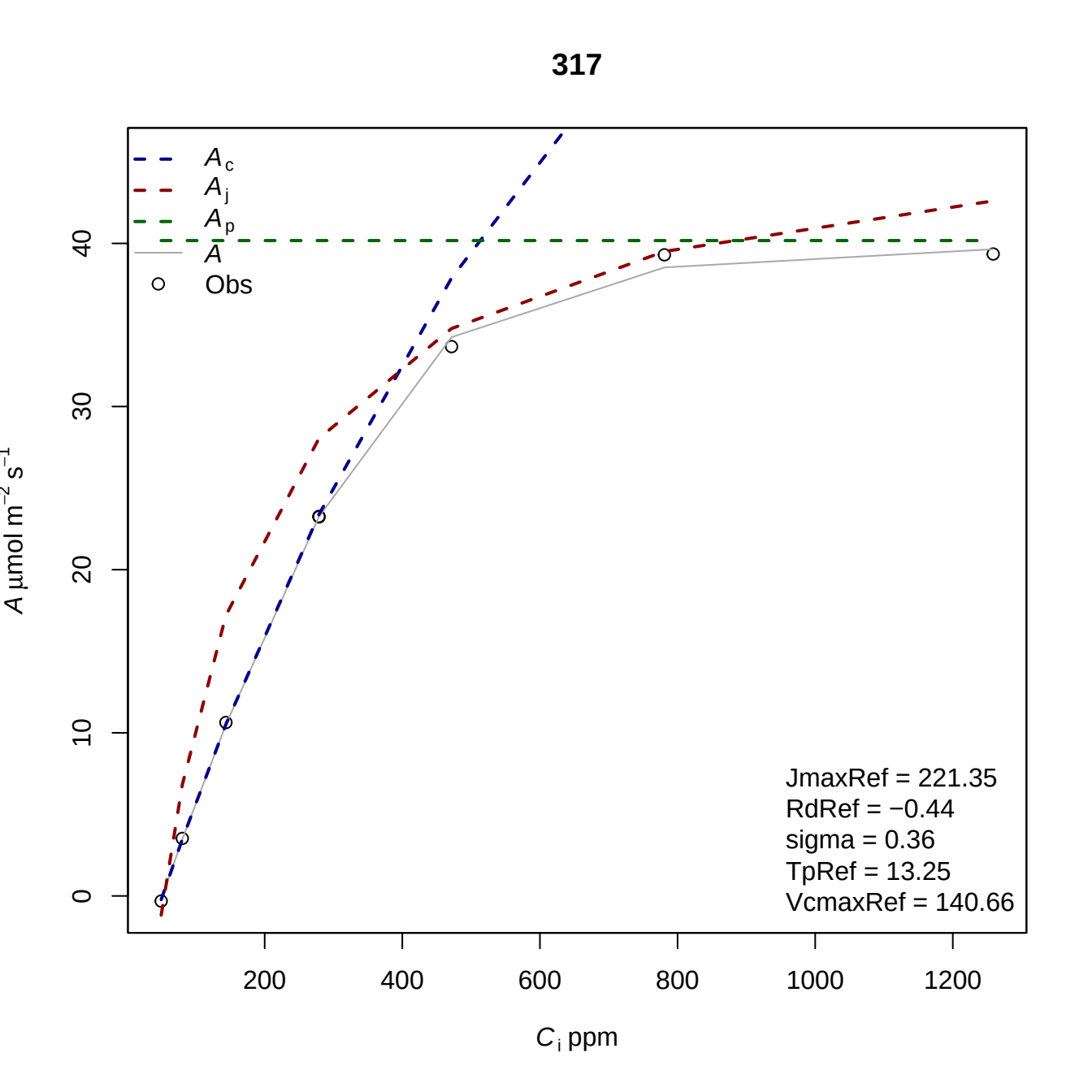

312
A_c
A_j
A_p
A
Obs
JmaxRef = 250.55
RdRef = −1.19
sigma = 0.38
TpRef = 376930.16
VcmaxRef = 206.72
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
50
200
400
600
800
1000
1200

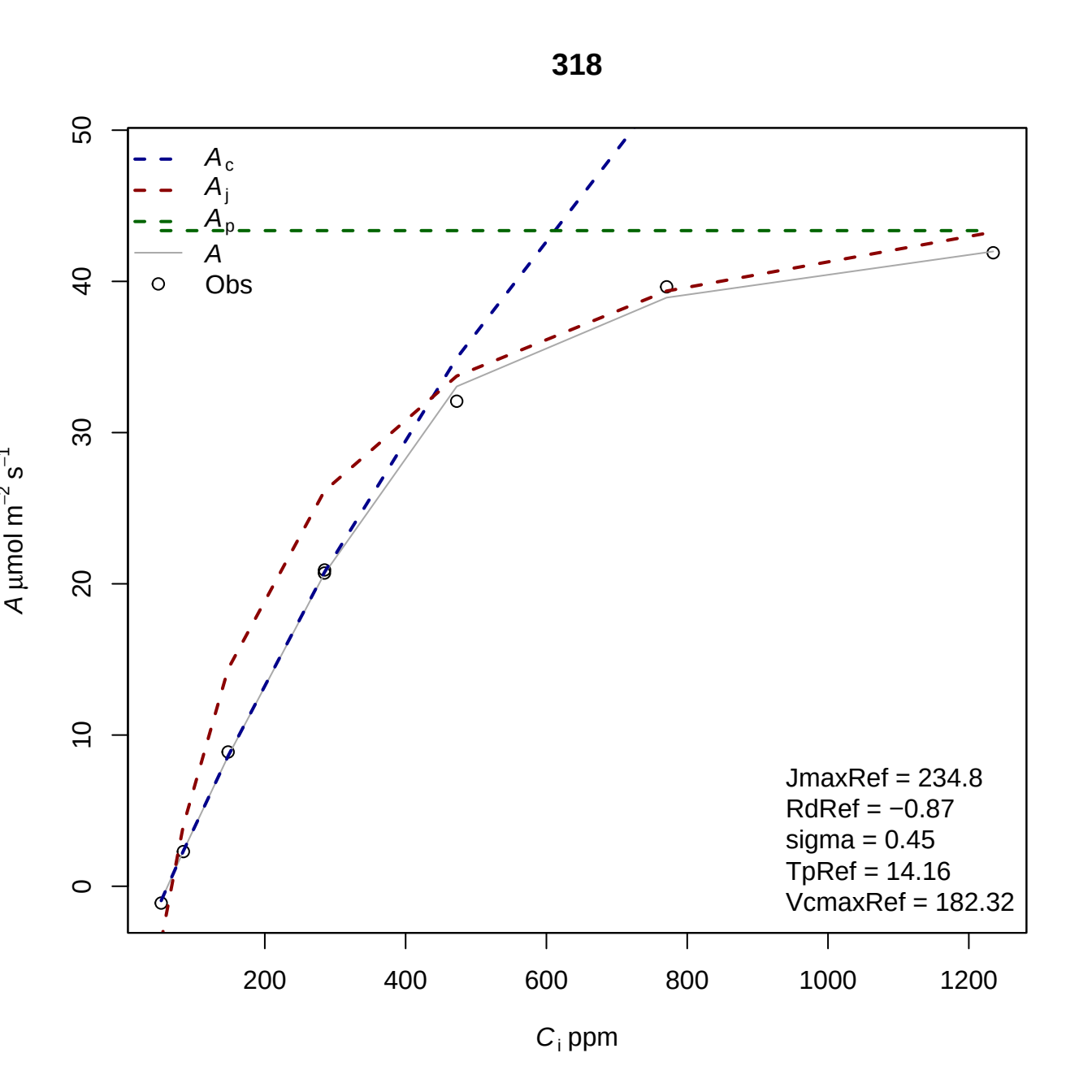

313
A_c
A_j
A_p
A
Obs
JmaxRef = 219.84
RdRef = −1.36
sigma = 0.44
TpRef = 13.19
VcmaxRef = 236.12
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
200
400
600
800
1000
1200

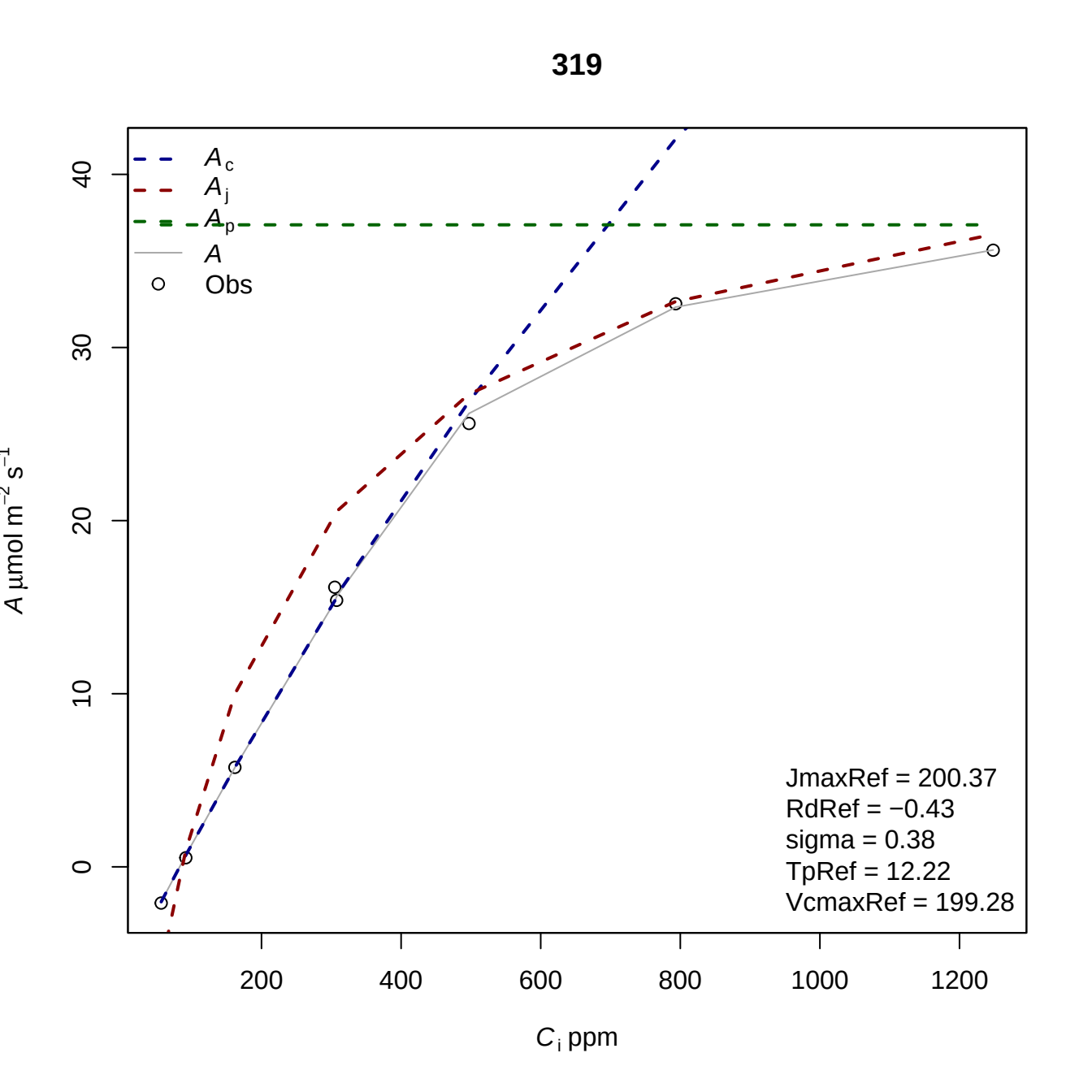

314
A_c
A_j
A_p
A
Obs
JmaxRef = 296.7
RdRef = 0.48
sigma = 0.52
TpRef = 7.18
VcmaxRef = 39.18
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200

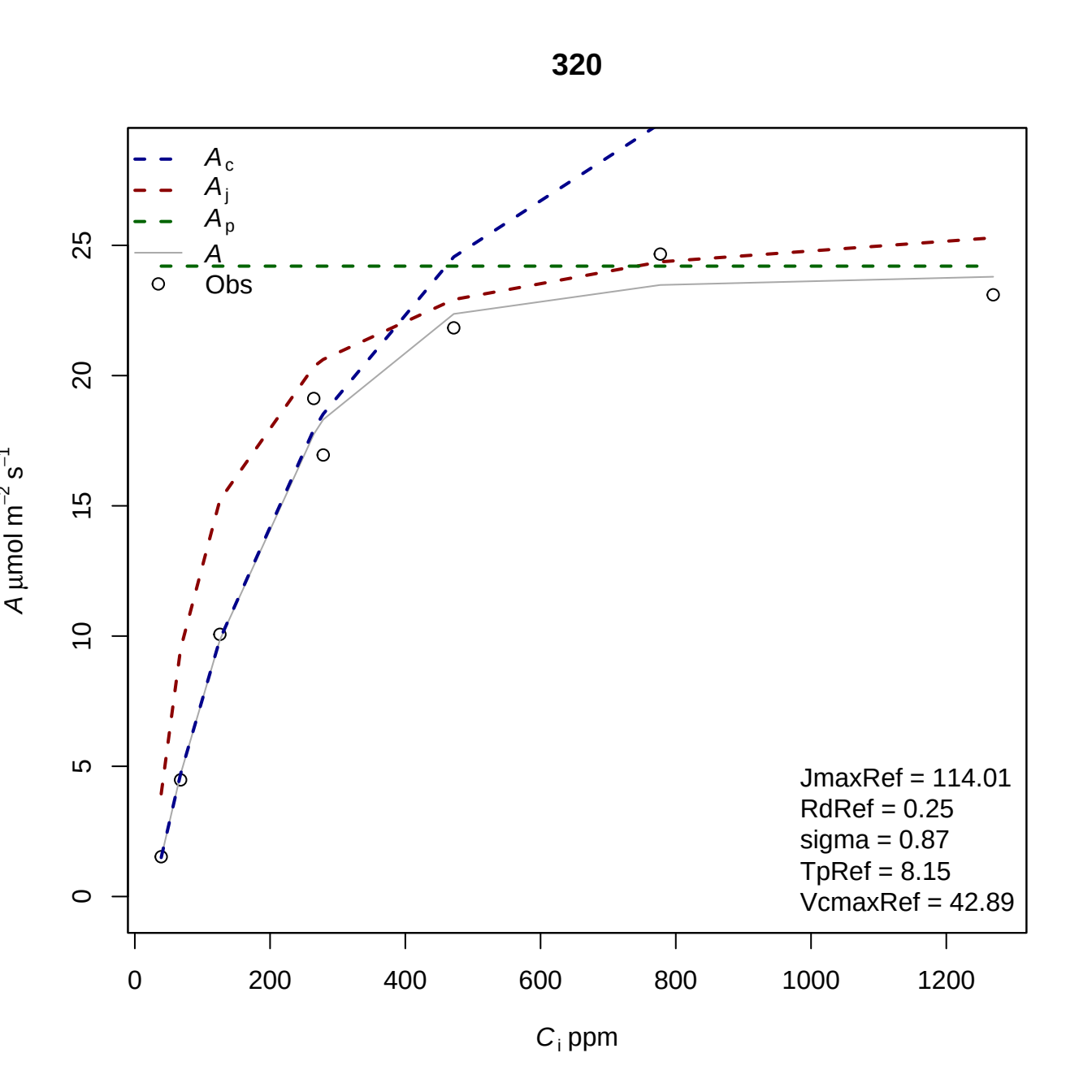

315
A_c
A_j
A_p
A
Obs
JmaxRef = 151.56
RdRef = 0.06
sigma = 0.28
TpRef = 9.95
VcmaxRef = 63.87
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200

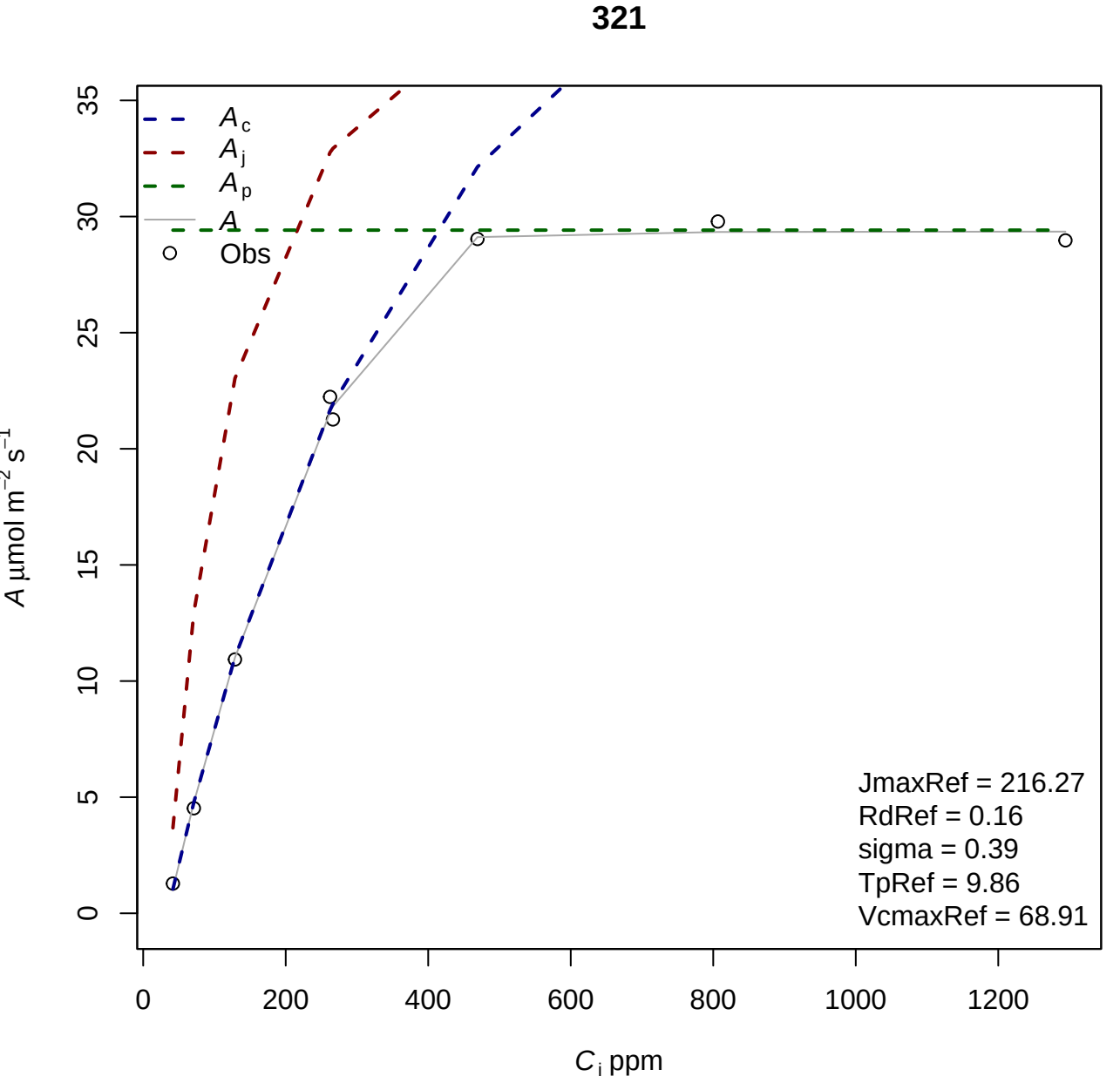

316
A_c
A_j
A_p
A
Obs
JmaxRef = 164.92
RdRef = −0.69
sigma = 0.39
TpRef = 9.65
VcmaxRef = 89.2
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200

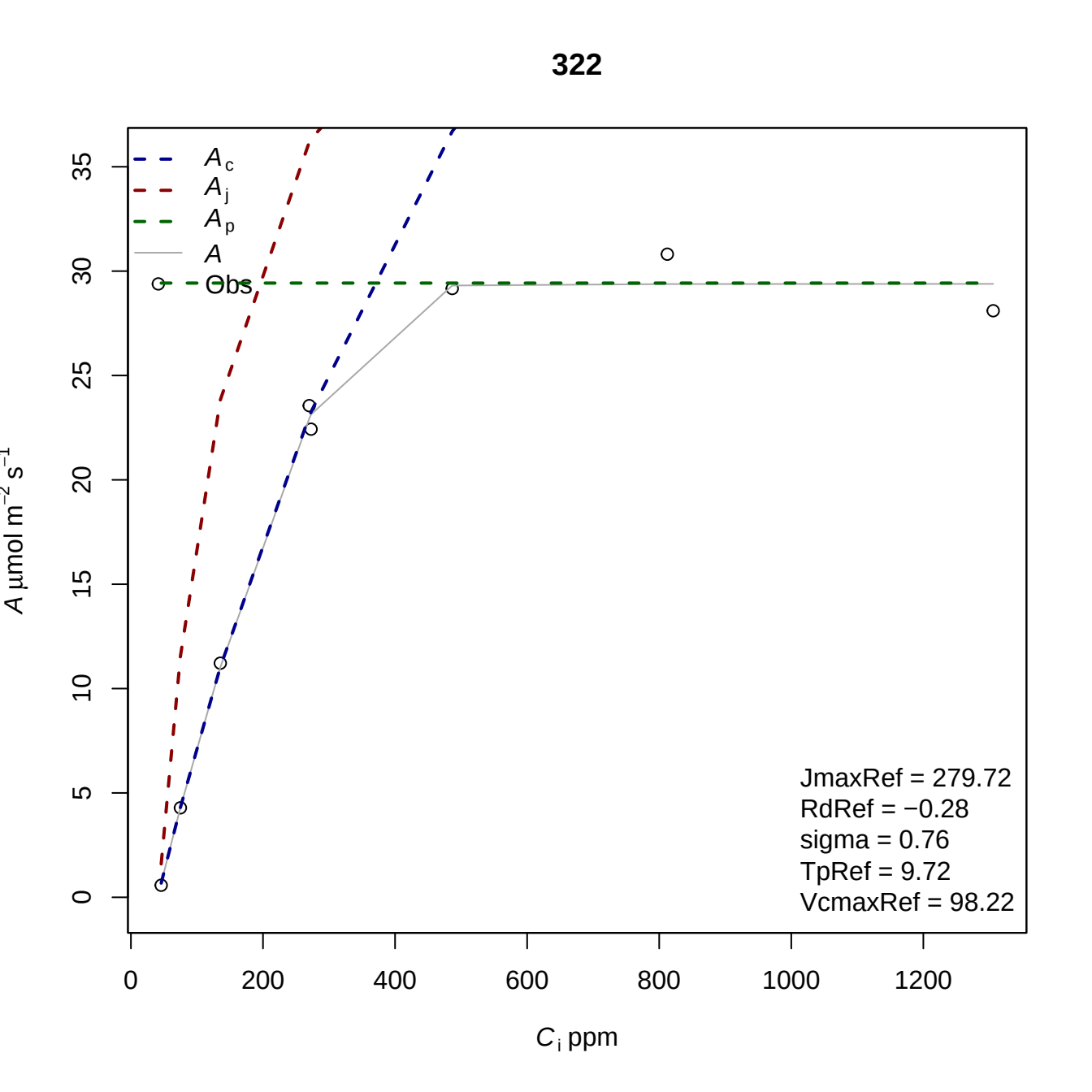

317
A_c
A_j
A_p
A
Obs
JmaxRef = 221.35
RdRef = −0.44
sigma = 0.36
TpRef = 13.25
VcmaxRef = 140.66
0
10
20
30
40
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

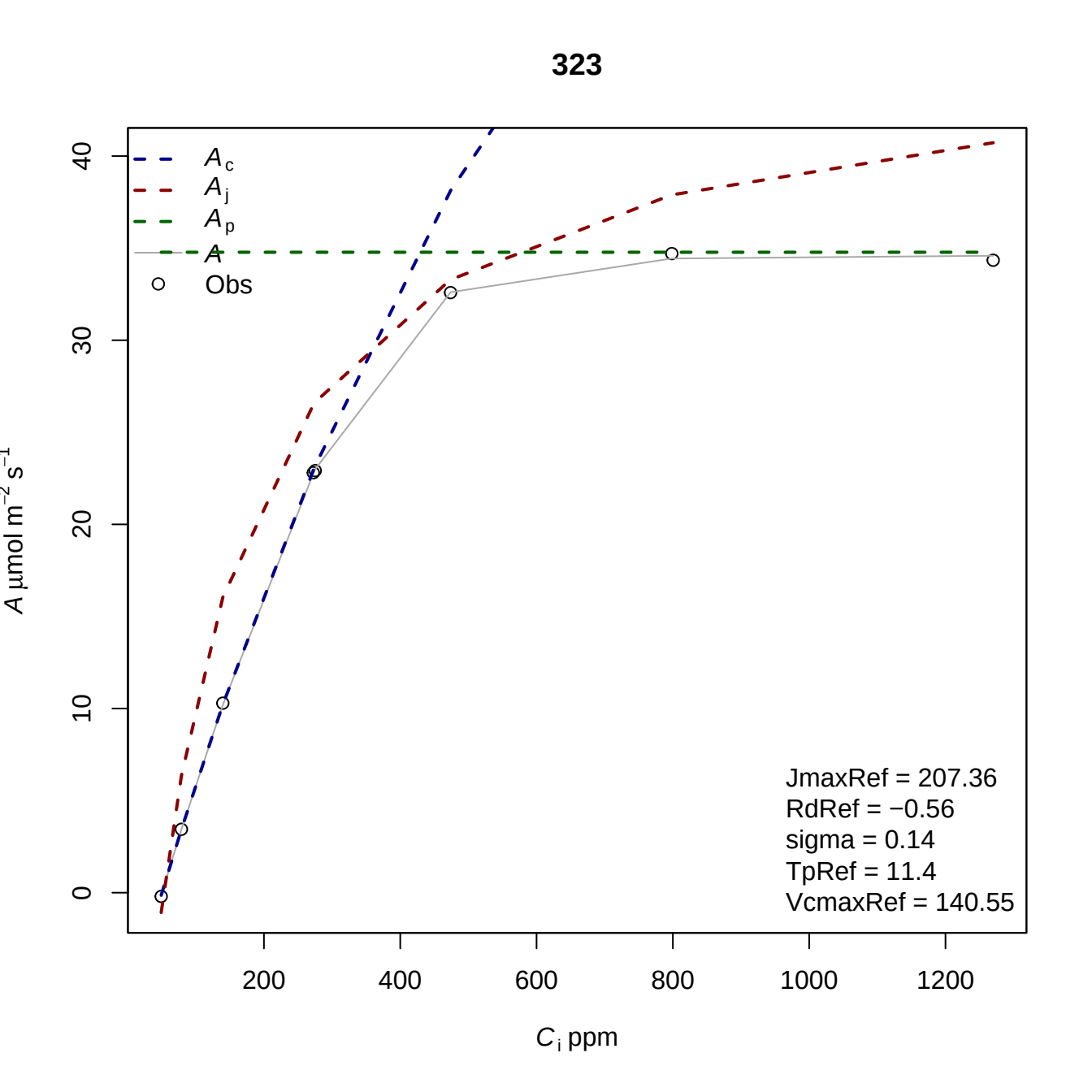

318
A_c
A_j
A_p
A
Obs
JmaxRef = 234.8
RdRef = −0.87
sigma = 0.45
TpRef = 14.16
VcmaxRef = 182.32
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
50
200
400
600
800
1000
1200

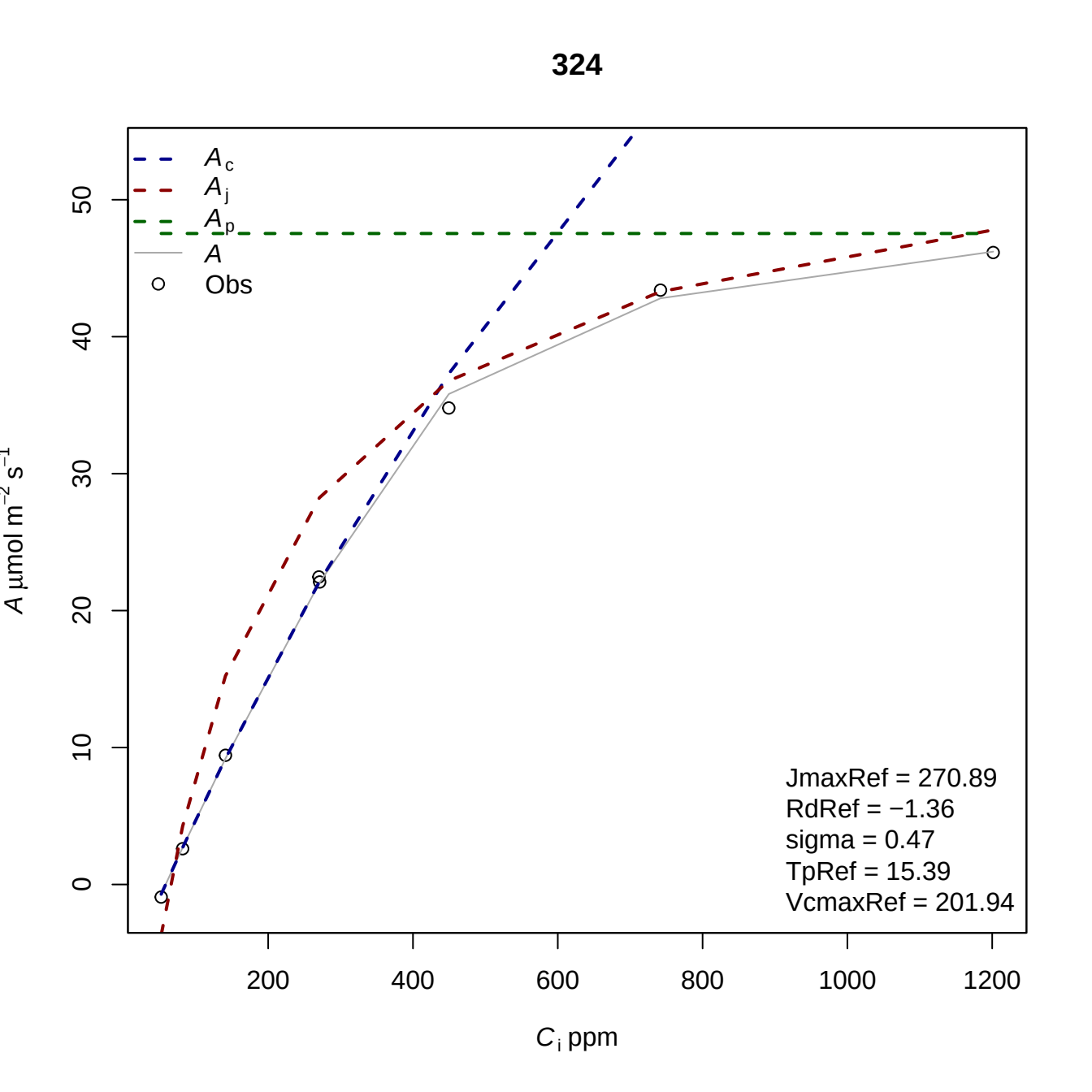

319
A_c
A_j
A_p
A
Obs
JmaxRef = 200.37
RdRef = −0.43
sigma = 0.38
TpRef = 12.22
VcmaxRef = 199.28
0
10
20
30
40
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

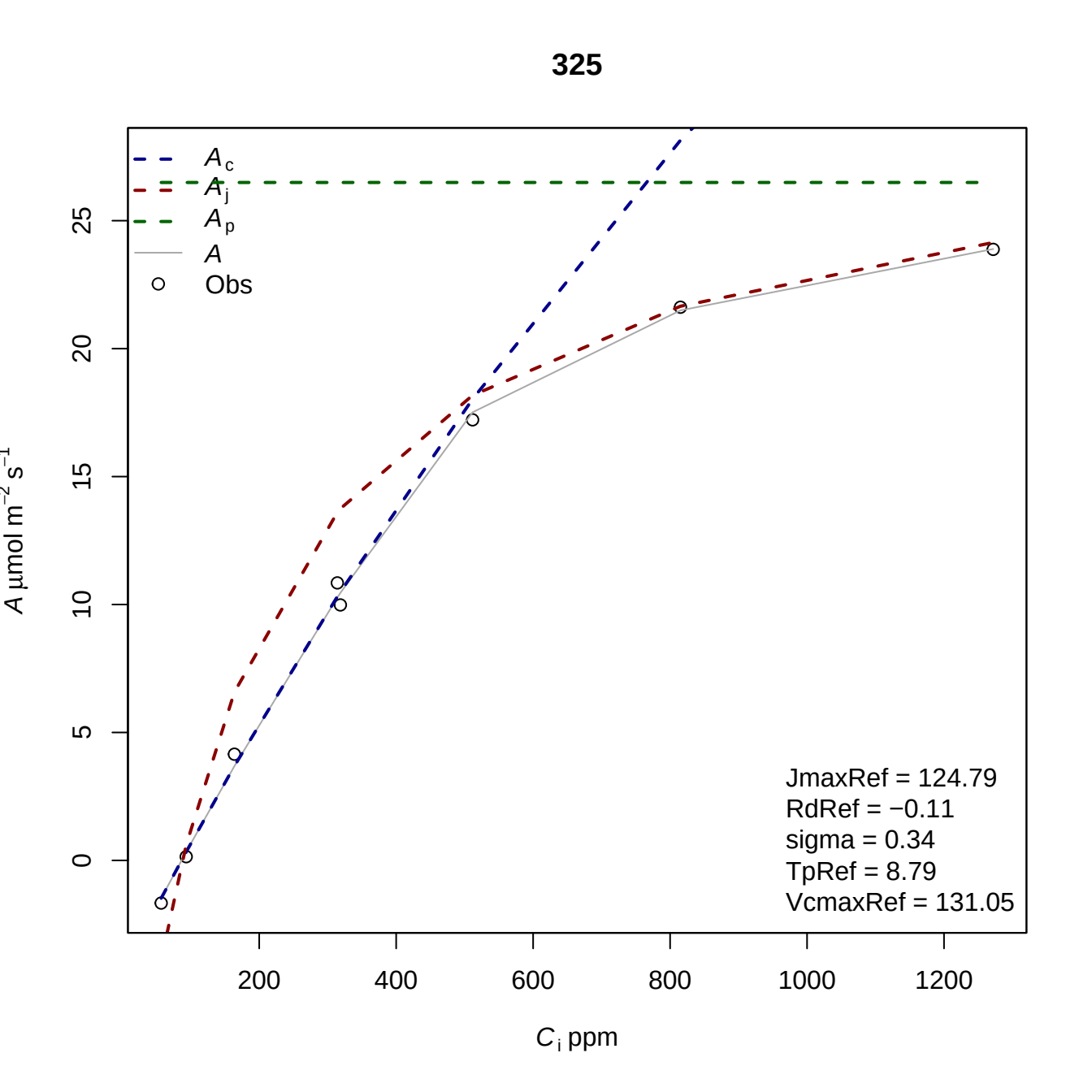

320
A_c
A_j
A_p
A
Obs
JmaxRef = 114.01
RdRef = 0.25
sigma = 0.87
TpRef = 8.15
VcmaxRef = 42.89
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200

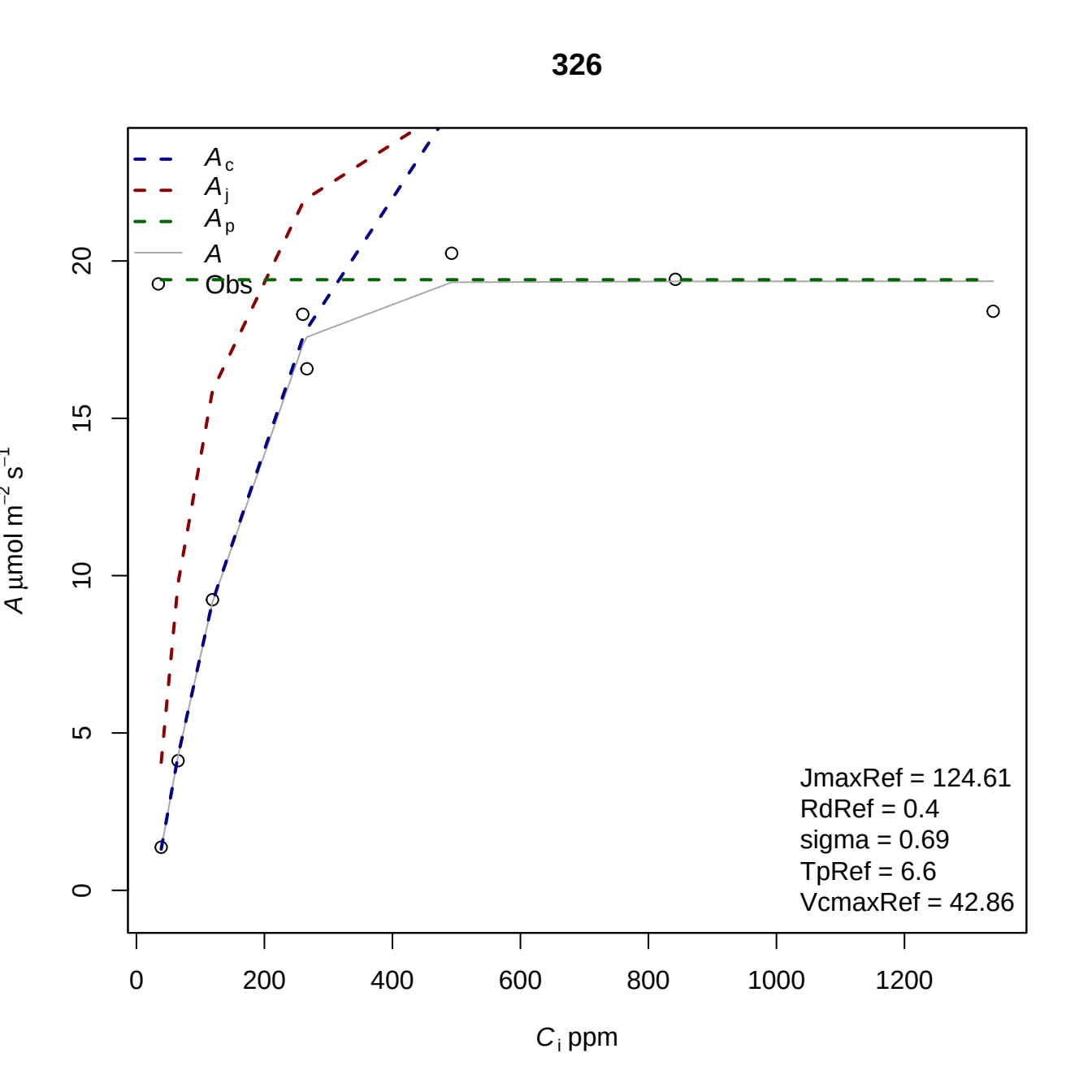

321
A_c
A_j
A_p
A
Obs
JmaxRef = 216.27
RdRef = 0.16
sigma = 0.39
TpRef = 9.86
VcmaxRef = 68.91
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200

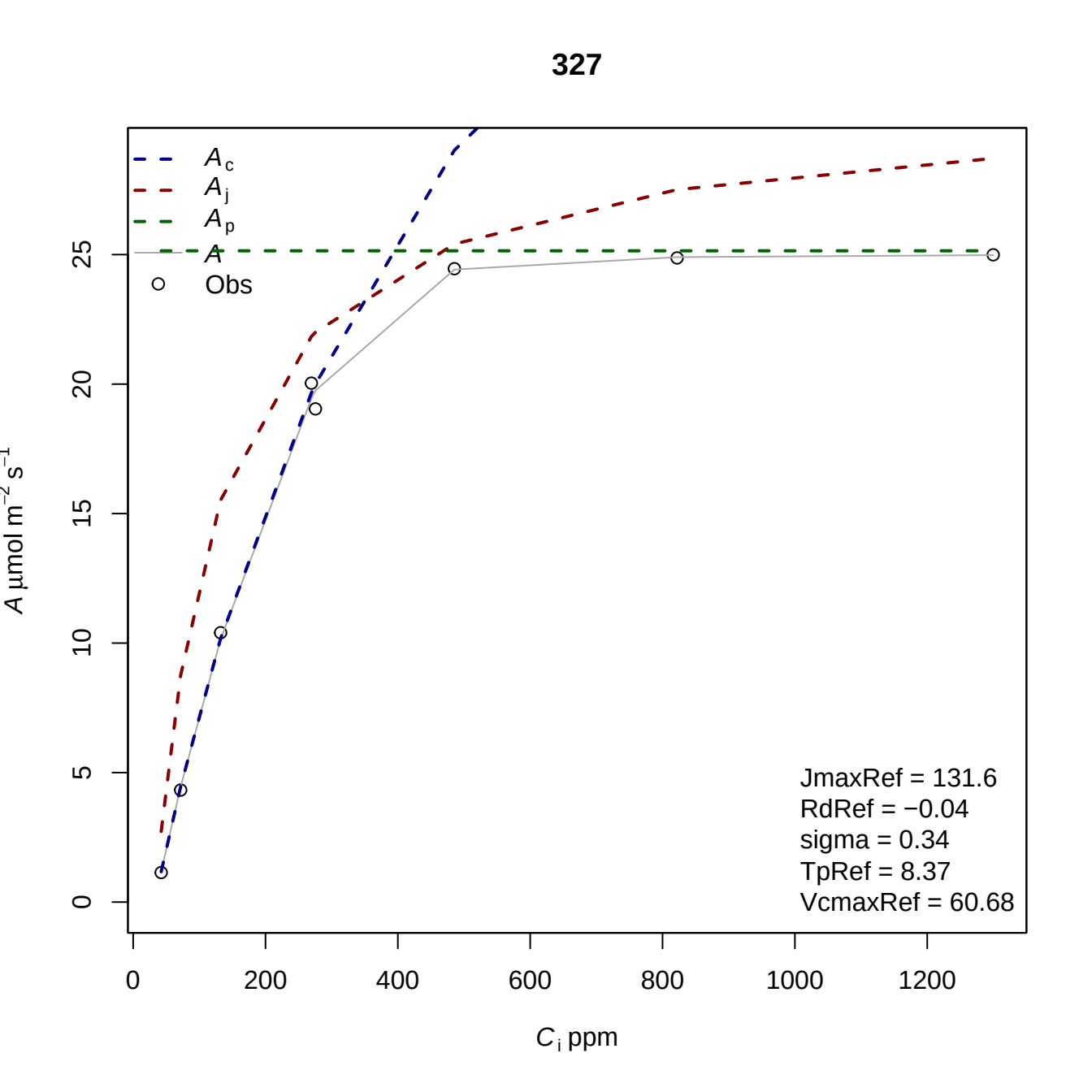

322
A_c
A_j
A_p
A
Obs
JmaxRef = 279.72
RdRef = −0.28
sigma = 0.76
TpRef = 9.72
VcmaxRef = 98.22
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200

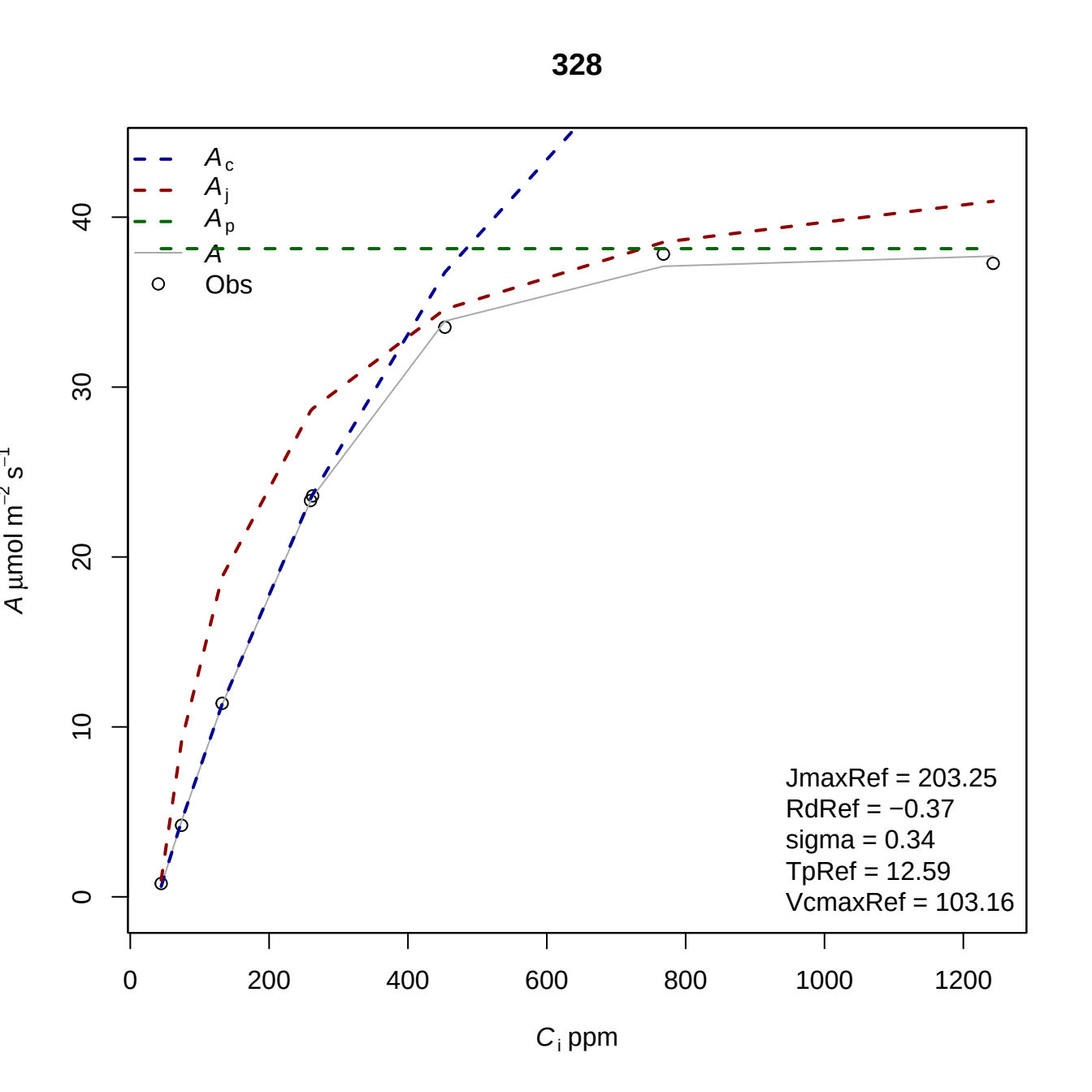

323
A_c
A_j
A_p
A
Obs
JmaxRef = 207.36
RdRef = −0.56
sigma = 0.14
TpRef = 11.4
VcmaxRef = 140.55
0
10
20
30
40
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

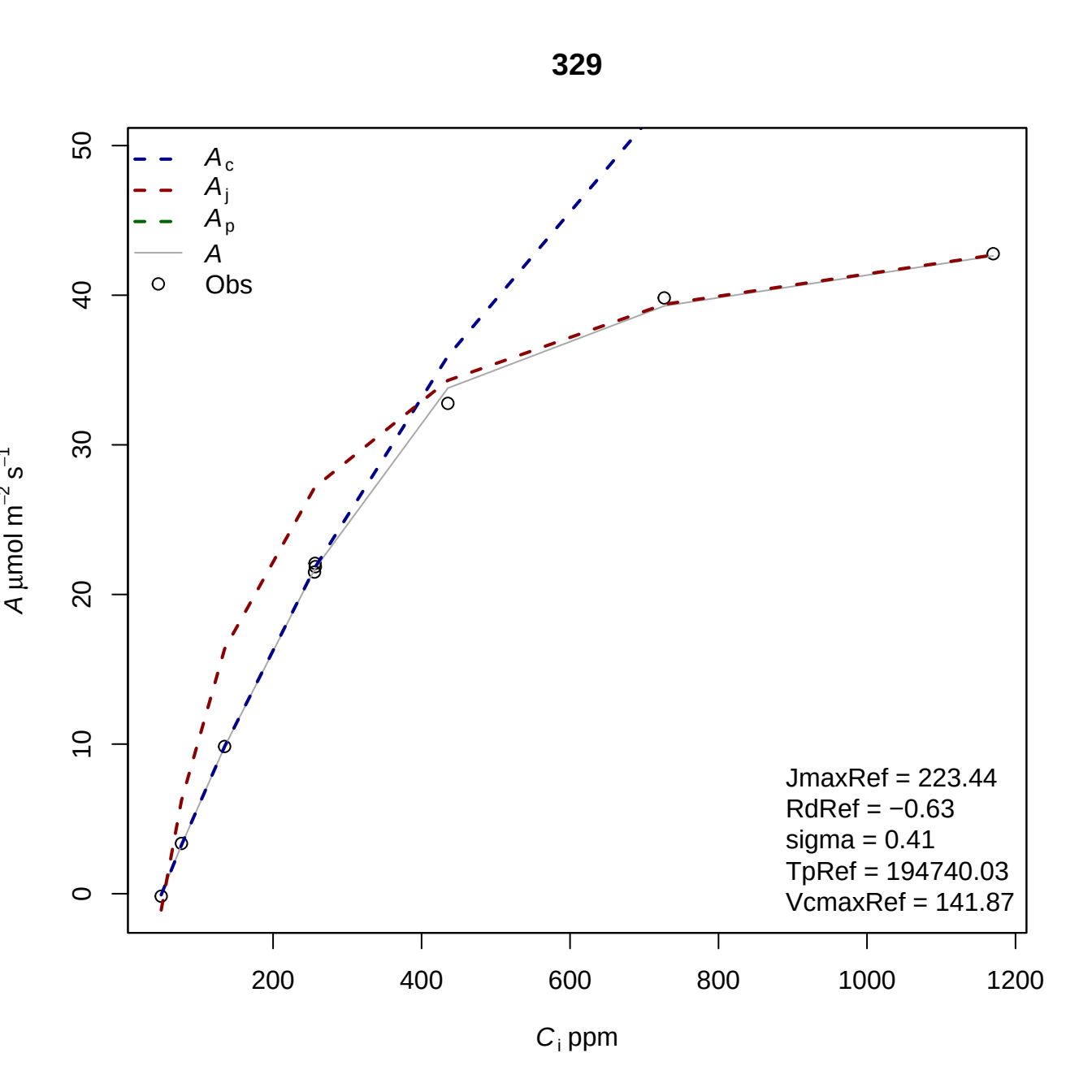

324
A_c
A_j
A_p
A
Obs
JmaxRef = 270.89
RdRef = −1.36
sigma = 0.47
TpRef = 15.39
VcmaxRef = 201.94
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
50
200
400
600
800
1000
1200

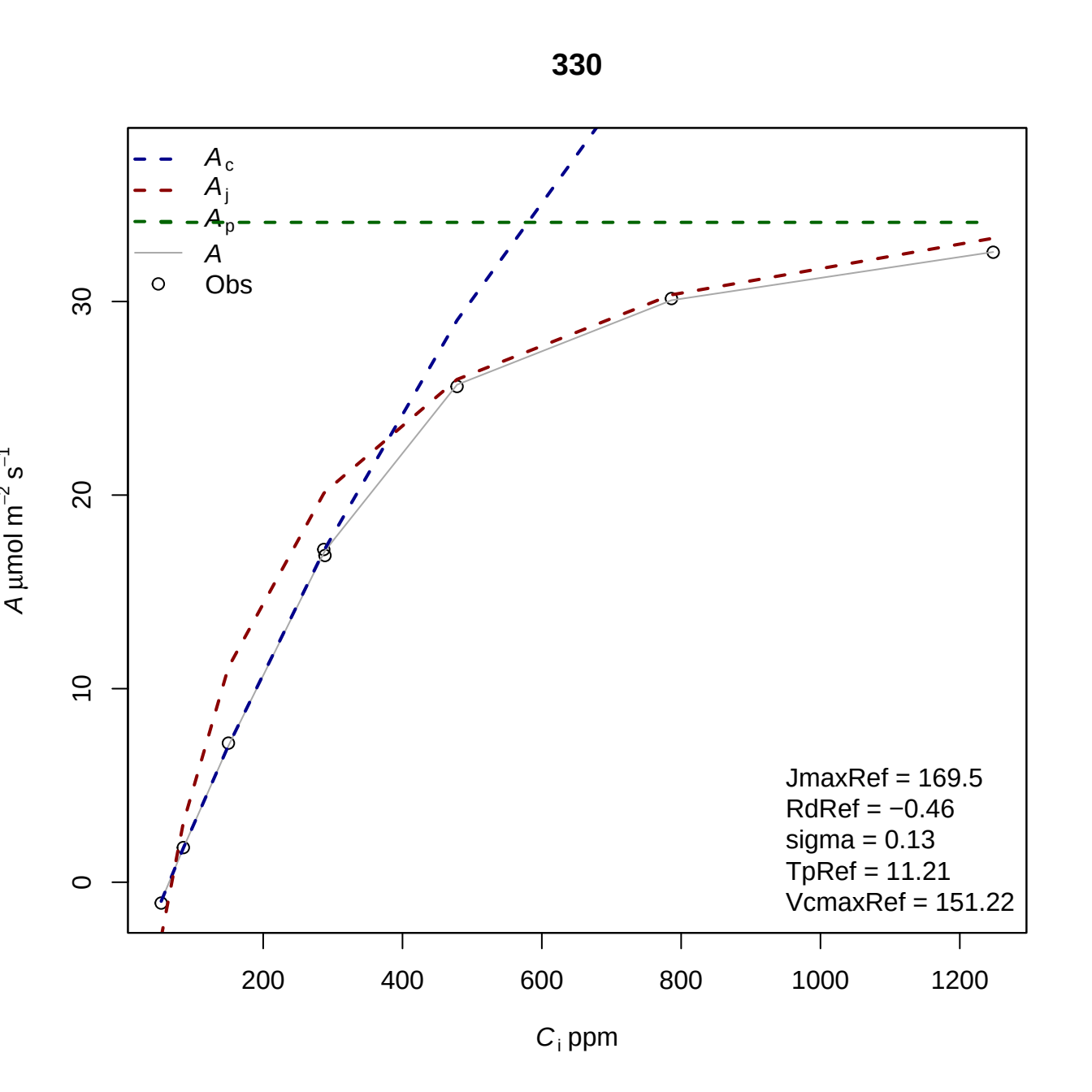

325
A_c
A_j
A_p
A
Obs
JmaxRef = 124.79
RdRef = −0.11
sigma = 0.34
TpRef = 8.79
VcmaxRef = 131.05
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
200
400
600
800
1000
1200

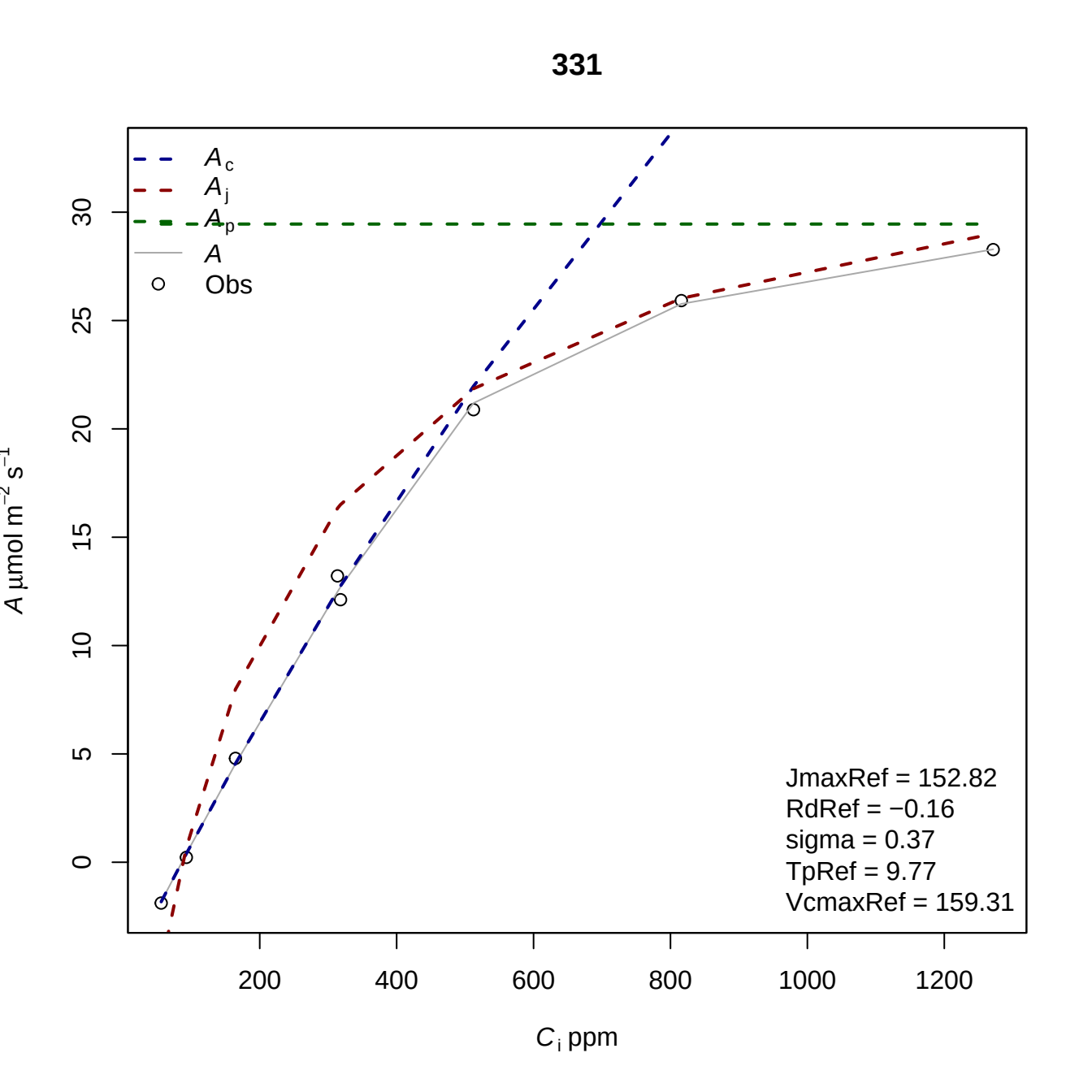

326
A_c
A_j
A_p
A
Obs
JmaxRef = 124.61
RdRef = 0.4
sigma = 0.69
TpRef = 6.6
VcmaxRef = 42.86
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

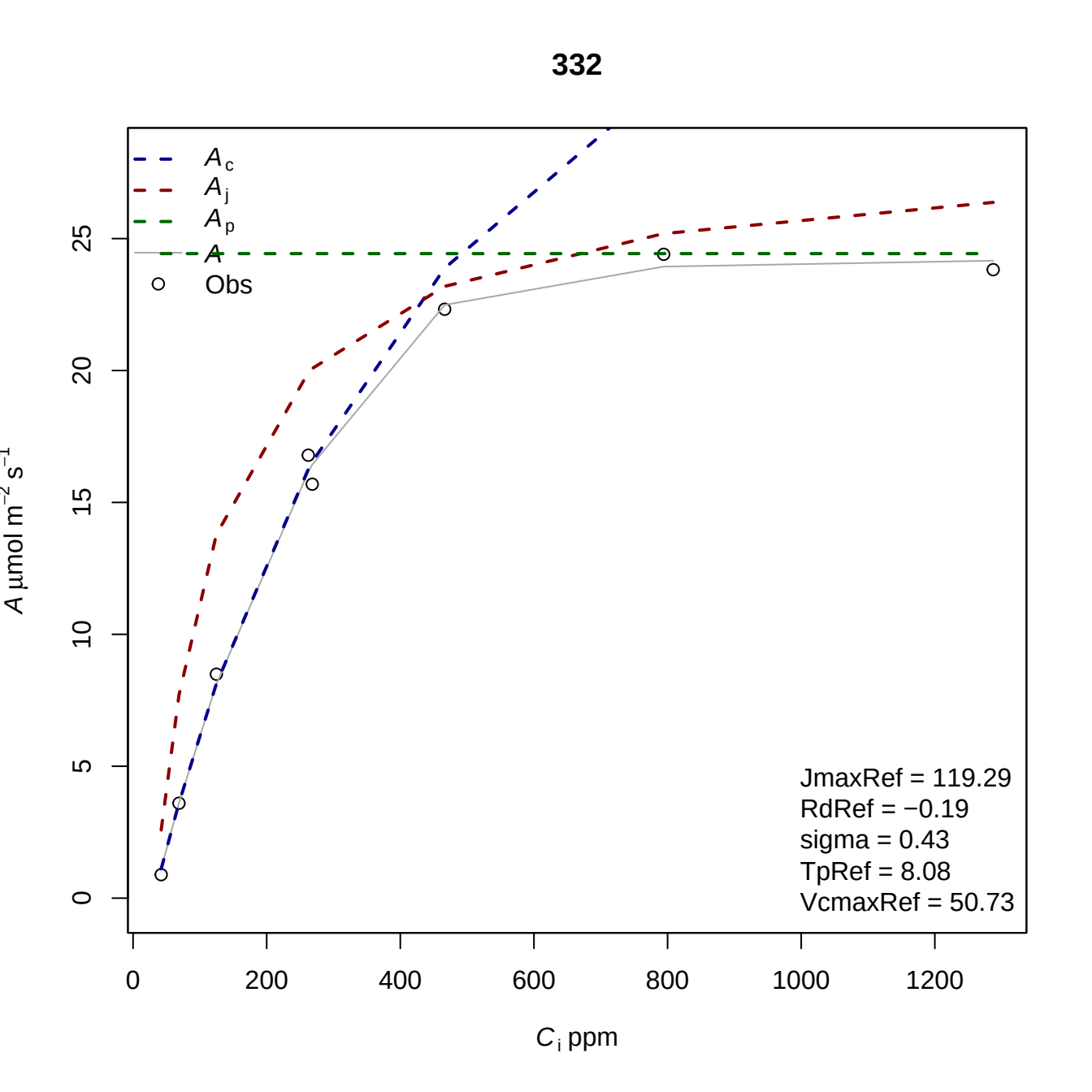

327
A_c
A_j
A_p
A
Obs
JmaxRef = 131.6
RdRef = −0.04
sigma = 0.34
TpRef = 8.37
VcmaxRef = 60.68
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200

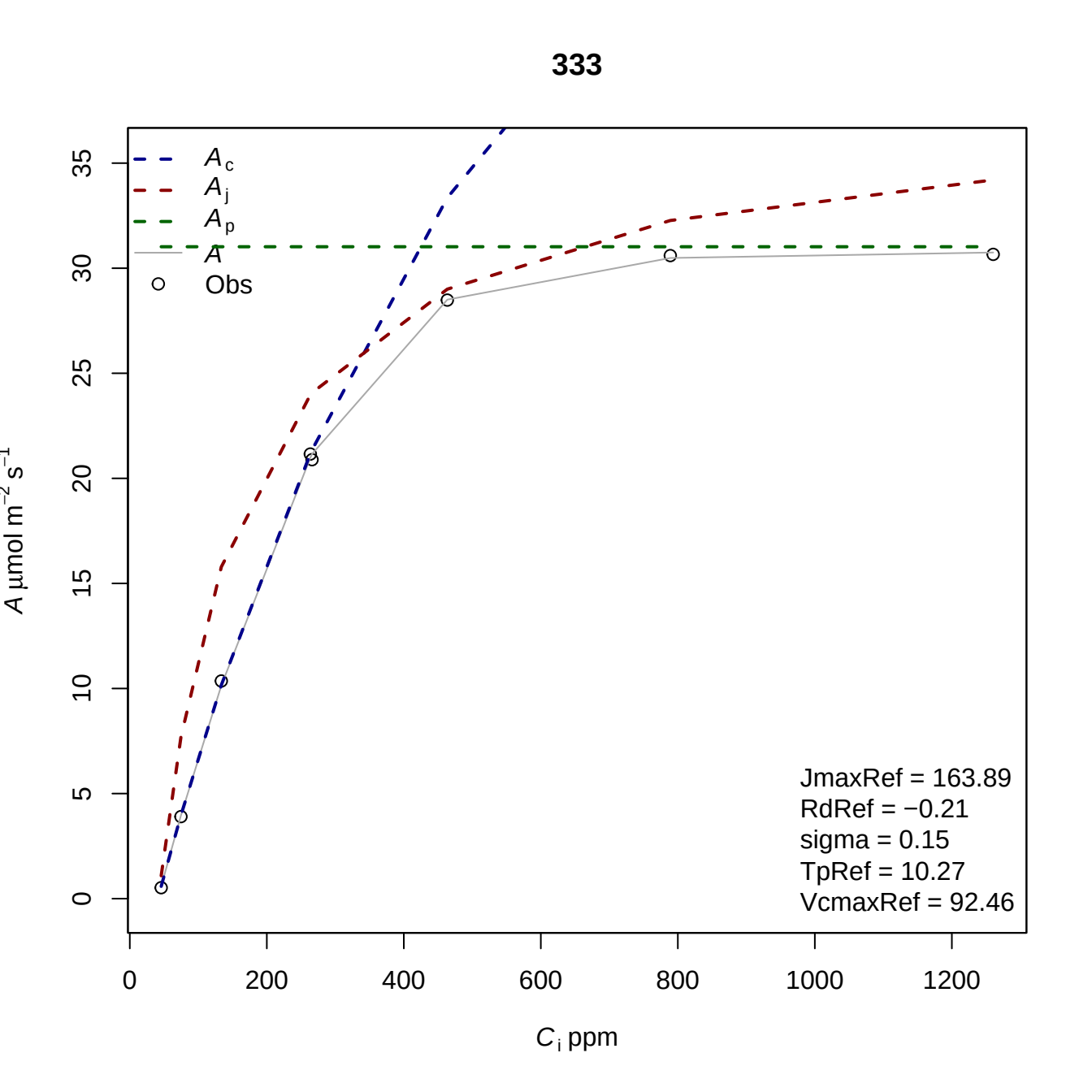

328
A_c
A_j
A_p
A
Obs
JmaxRef = 203.25
RdRef = −0.37
sigma = 0.34
TpRef = 12.59
VcmaxRef = 103.16
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

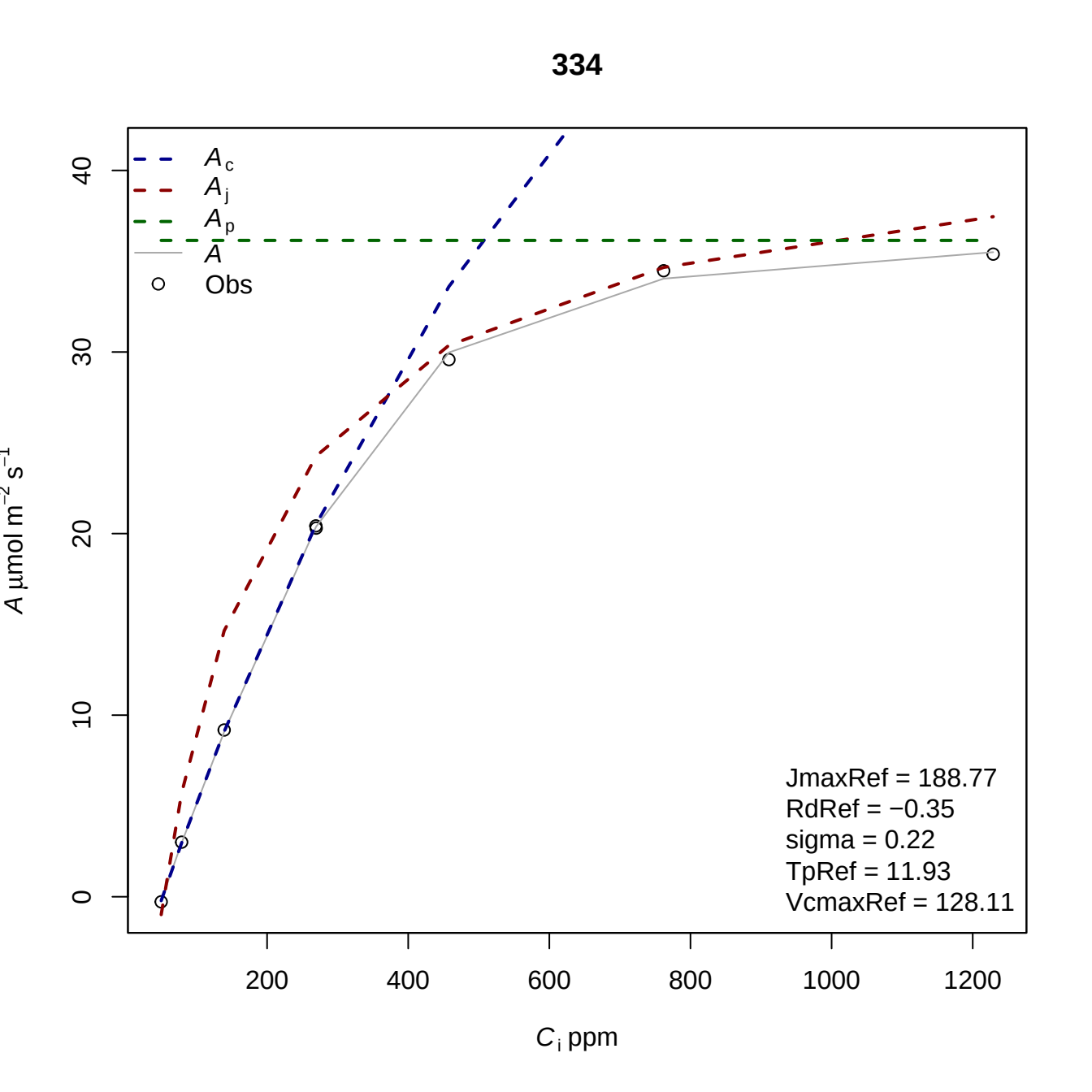

329
A_c
A_j
A_p
A
Obs
JmaxRef = 223.44
RdRef = −0.63
sigma = 0.41
TpRef = 194740.03
VcmaxRef = 141.87
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
50
200
400
600
800
1000
1200

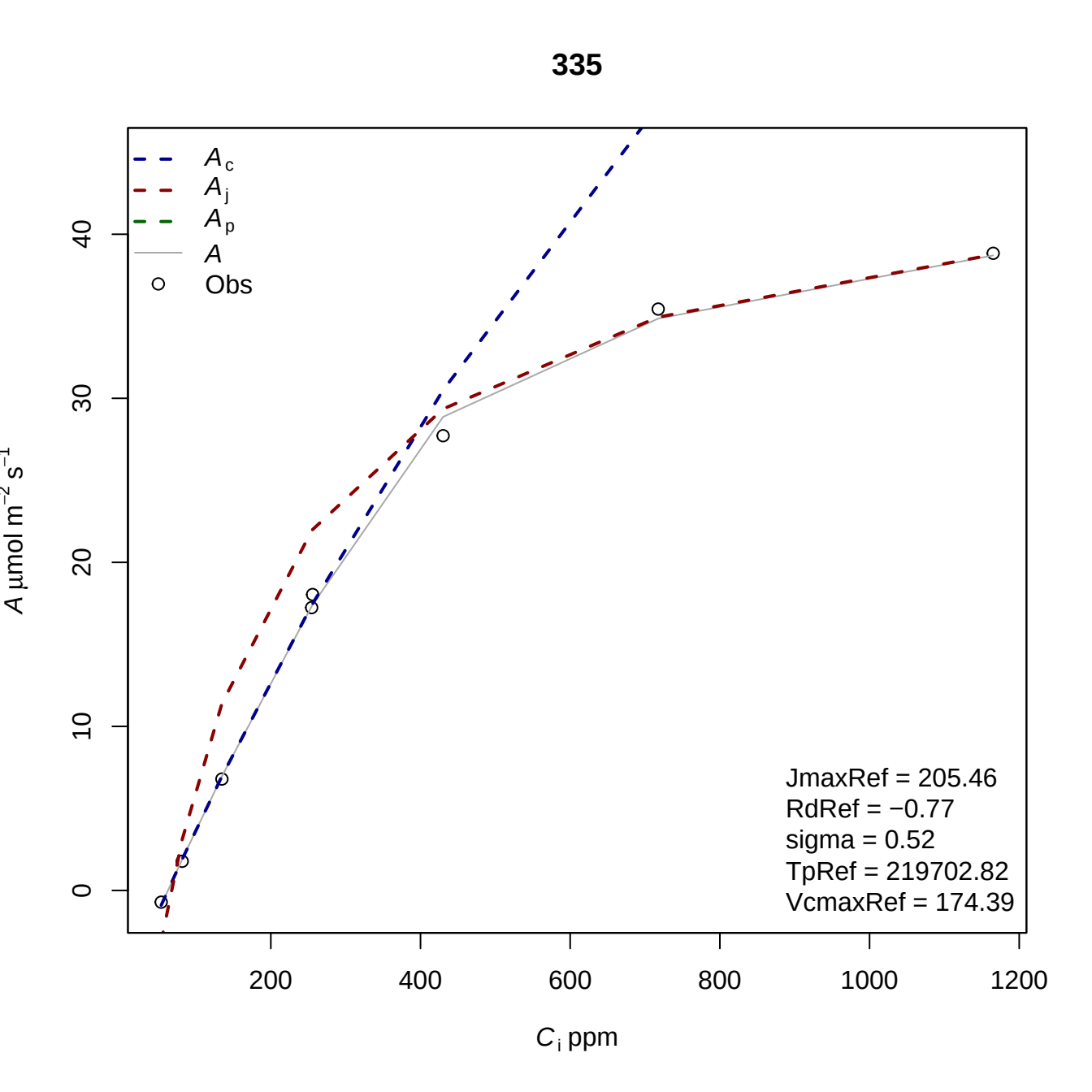

330
A_c
A_j
A_p
A
Obs
JmaxRef = 169.5
RdRef = −0.46
sigma = 0.13
TpRef = 11.21
VcmaxRef = 151.22
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
200
400
600
800
1000
1200

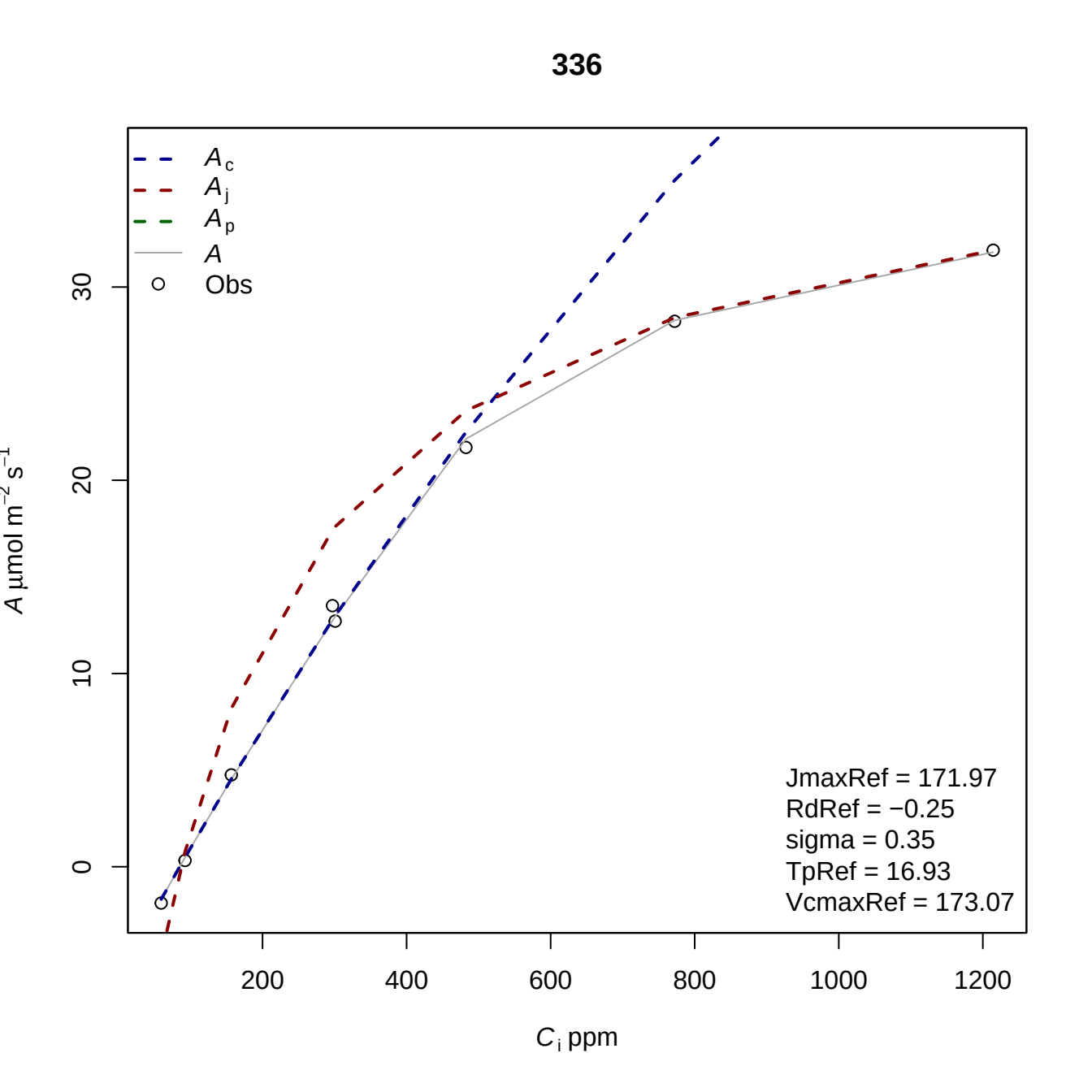

331
A_c
A_j
A_p
A
Obs
JmaxRef = 152.82
RdRef = −0.16
sigma = 0.37
TpRef = 9.77
VcmaxRef = 159.31
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
200
400
600
800
1000
1200

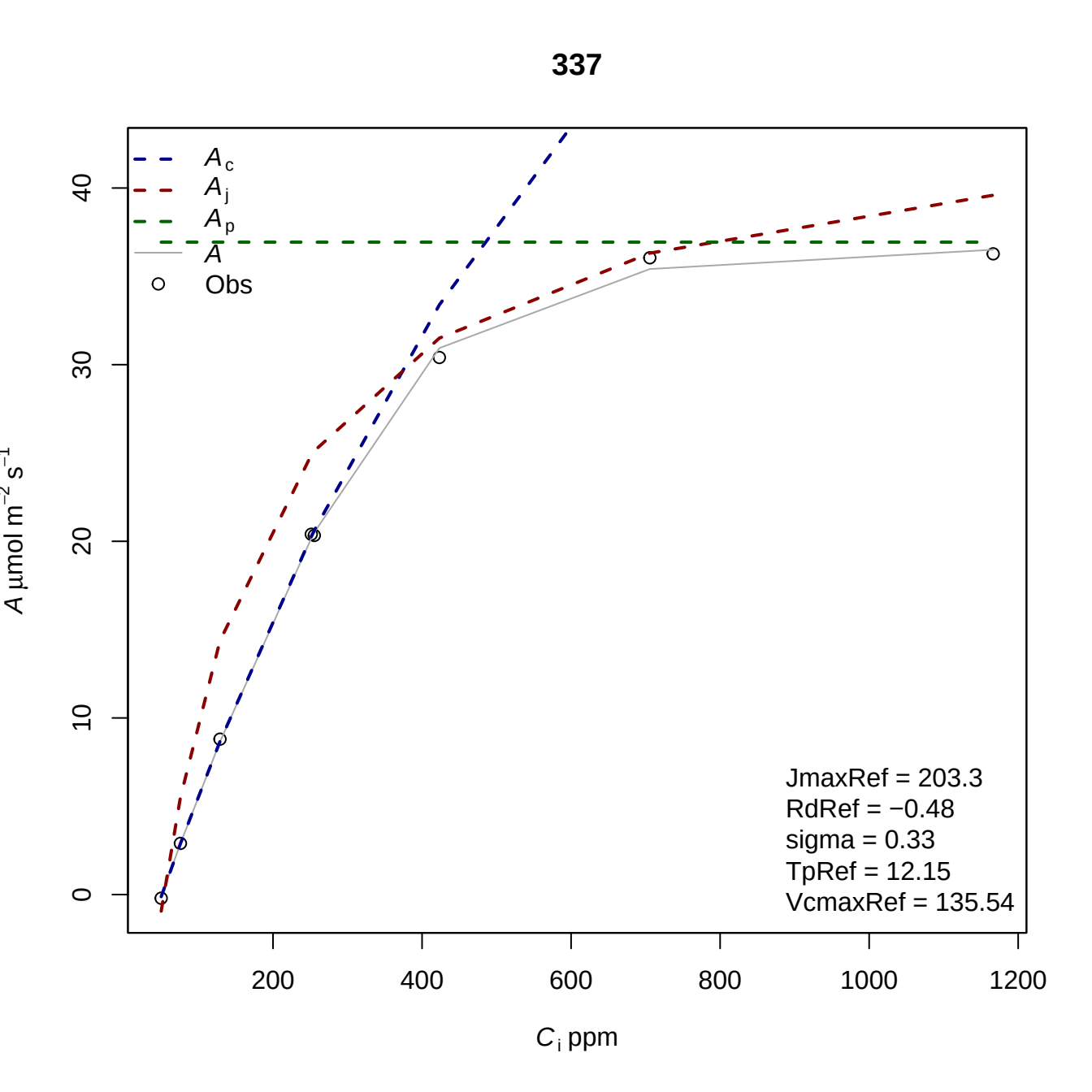

332
A_c
A_j
A_p
A
Obs
JmaxRef = 119.29
RdRef = −0.19
sigma = 0.43
TpRef = 8.08
VcmaxRef = 50.73
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200

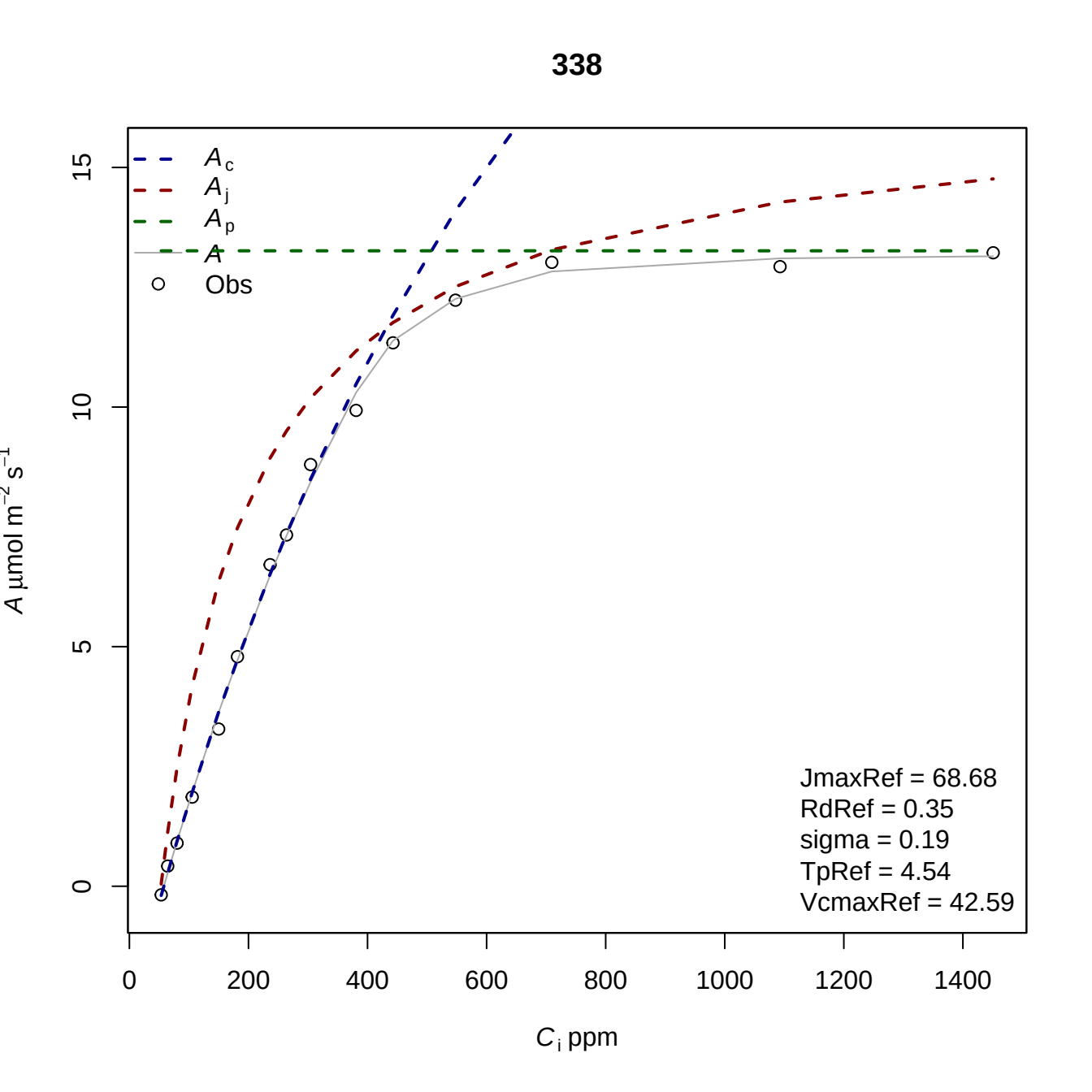

333
A_c
A_j
A_p
A
Obs
JmaxRef = 163.89
RdRef = −0.21
sigma = 0.15
TpRef = 10.27
VcmaxRef = 92.46
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200

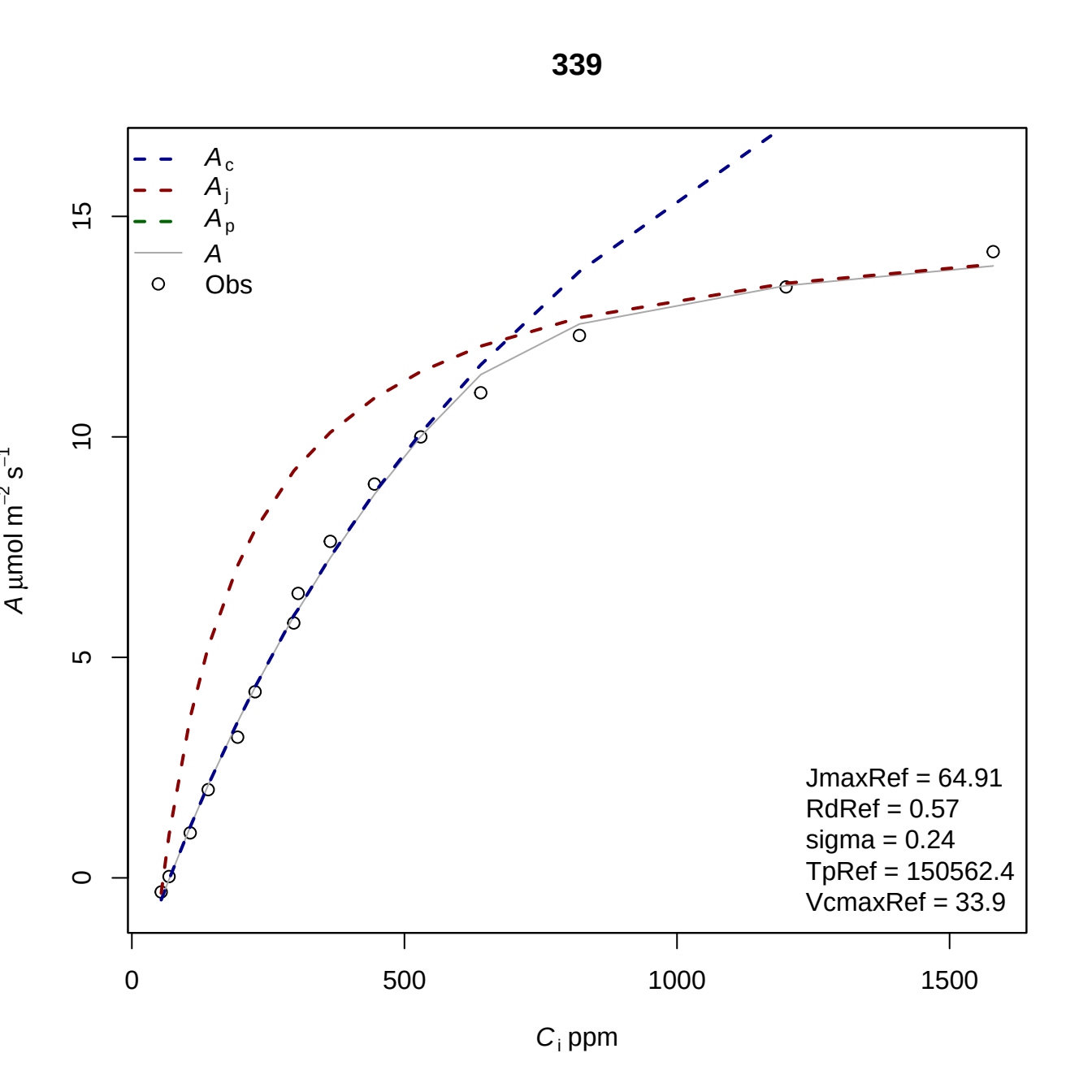

334
A_c
A_j
A_p
A
Obs
JmaxRef = 188.77
RdRef = −0.35
sigma = 0.22
TpRef = 11.93
VcmaxRef = 128.11
0
10
20
30
40
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

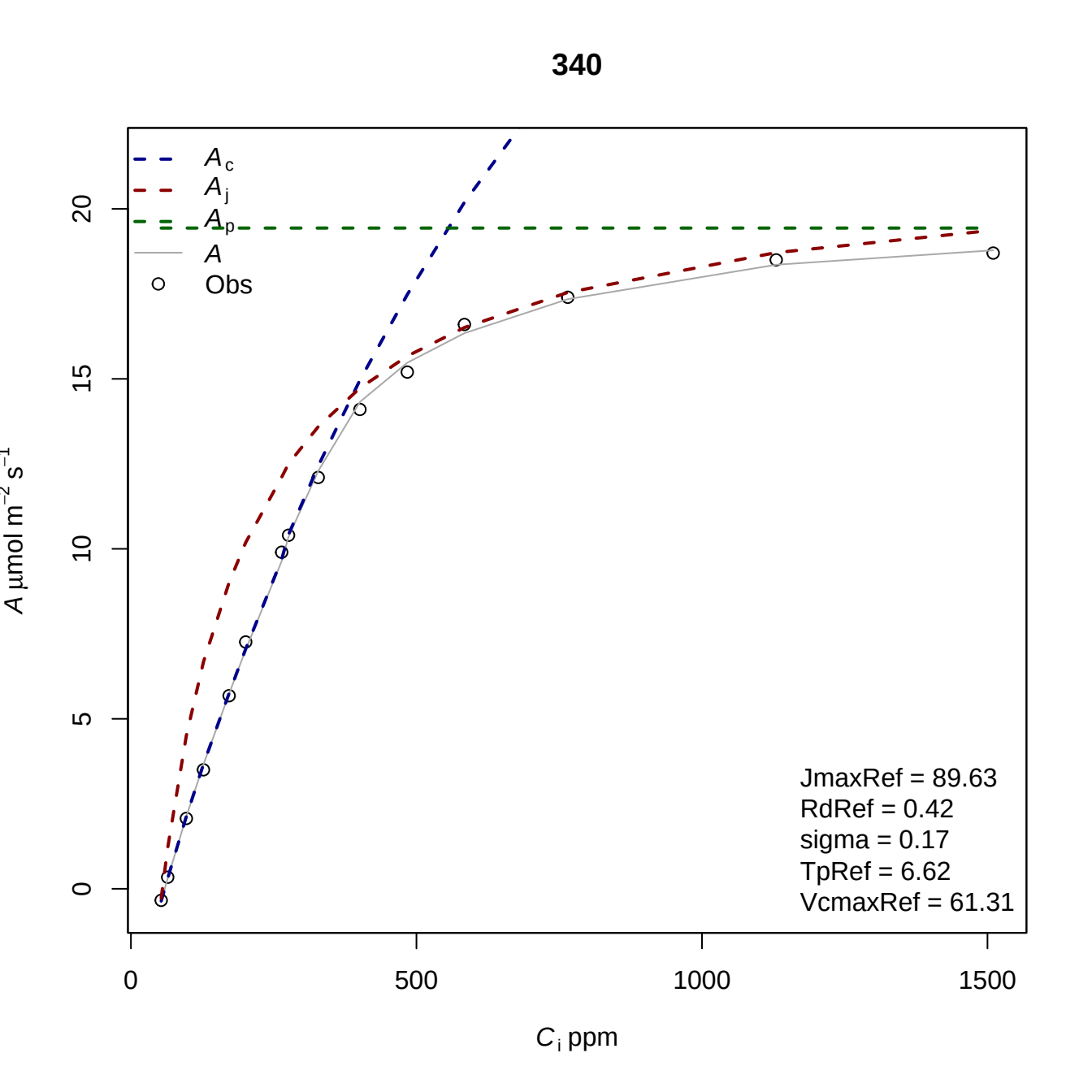

335
A_c
A_j
A_p
A
Obs
JmaxRef = 205.46
RdRef = −0.77
sigma = 0.52
TpRef = 219702.82
VcmaxRef = 174.39
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
200
400
600
800
1000
1200

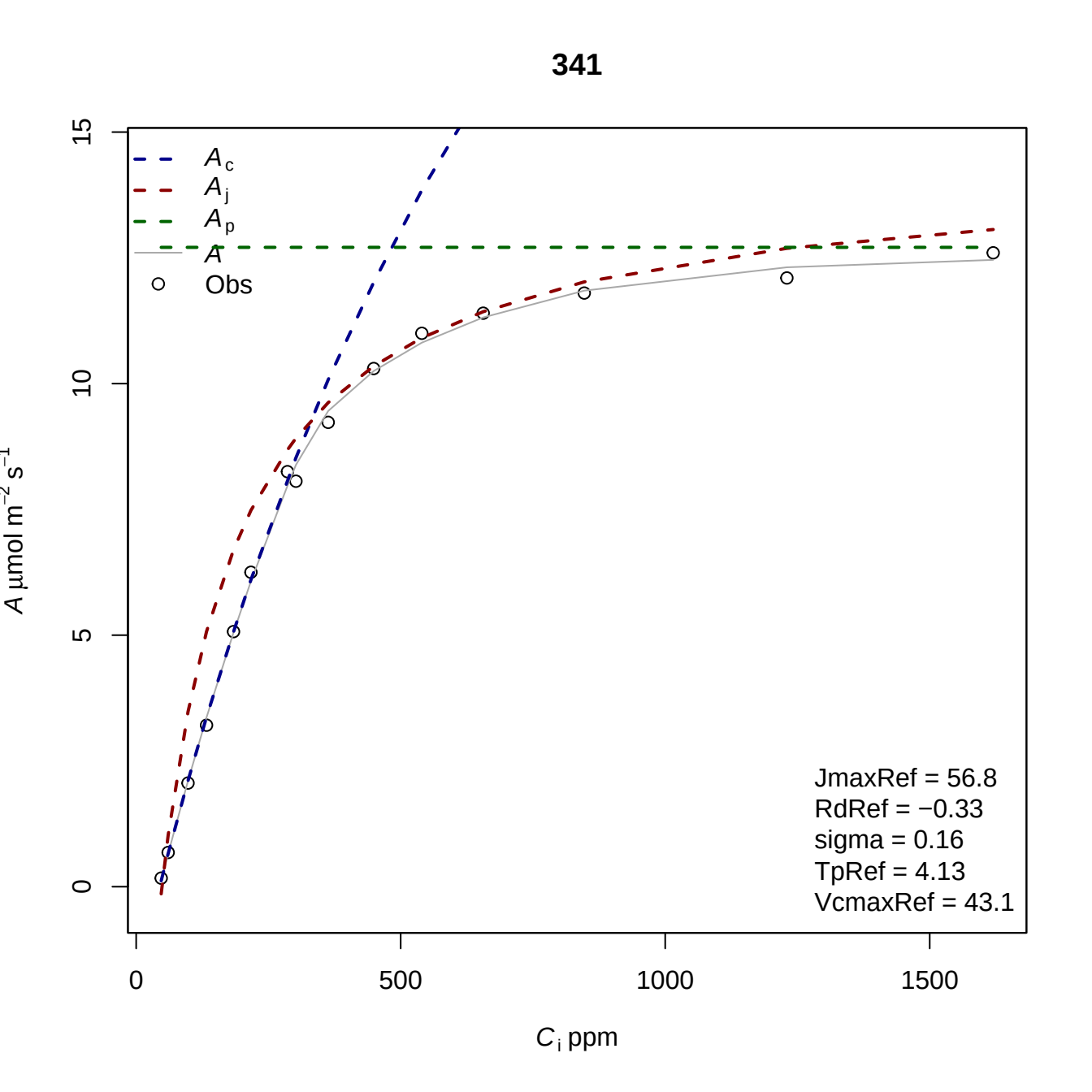

336
A_c
A_j
A_p
A
Obs
JmaxRef = 171.97
RdRef = −0.25
sigma = 0.35
TpRef = 16.93
VcmaxRef = 173.07
0
10
20
30
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

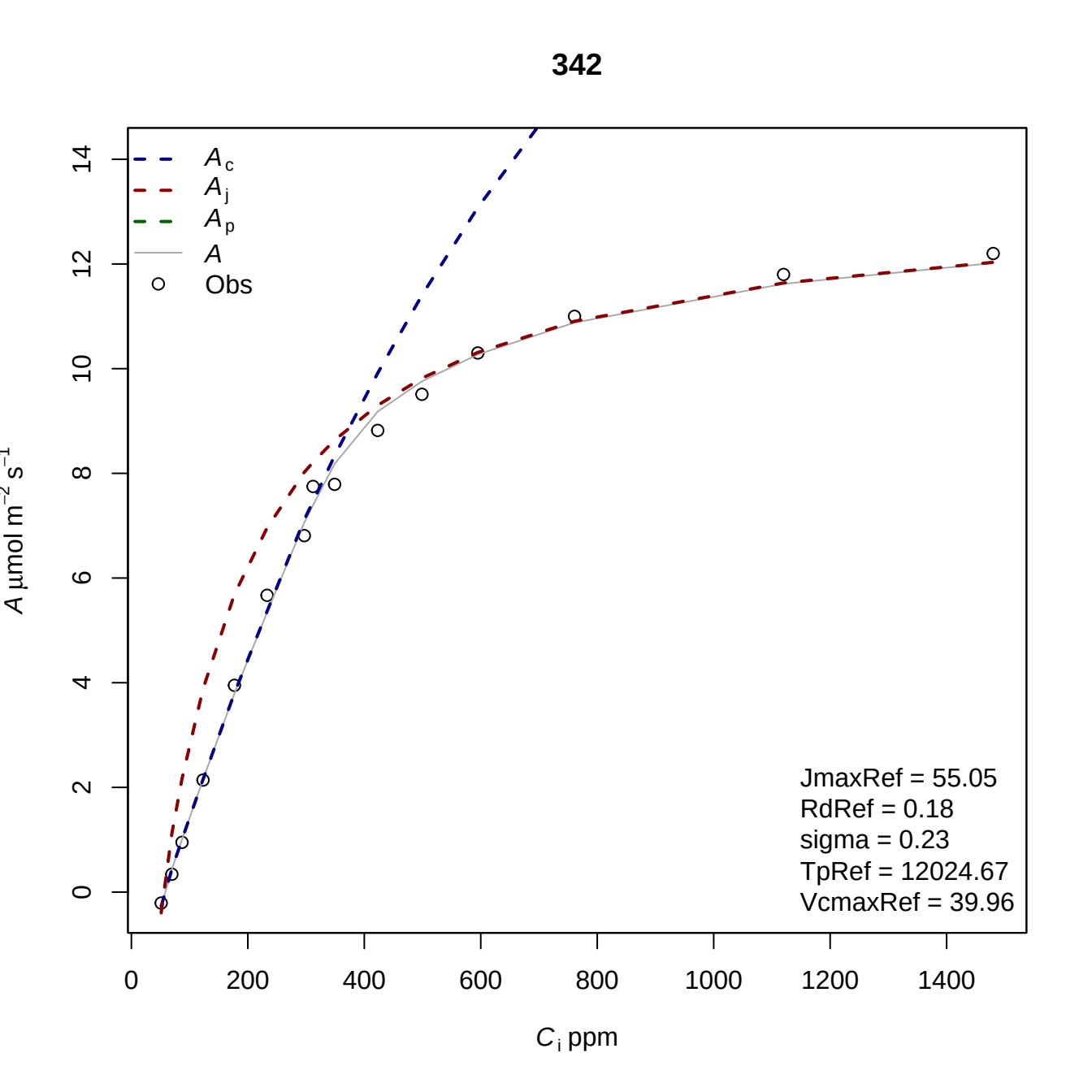

337
A_c
A_j
A_p
A
Obs
JmaxRef = 203.3
RdRef = −0.48
sigma = 0.33
TpRef = 12.15
VcmaxRef = 135.54
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
200
400
600
800
1000
1200

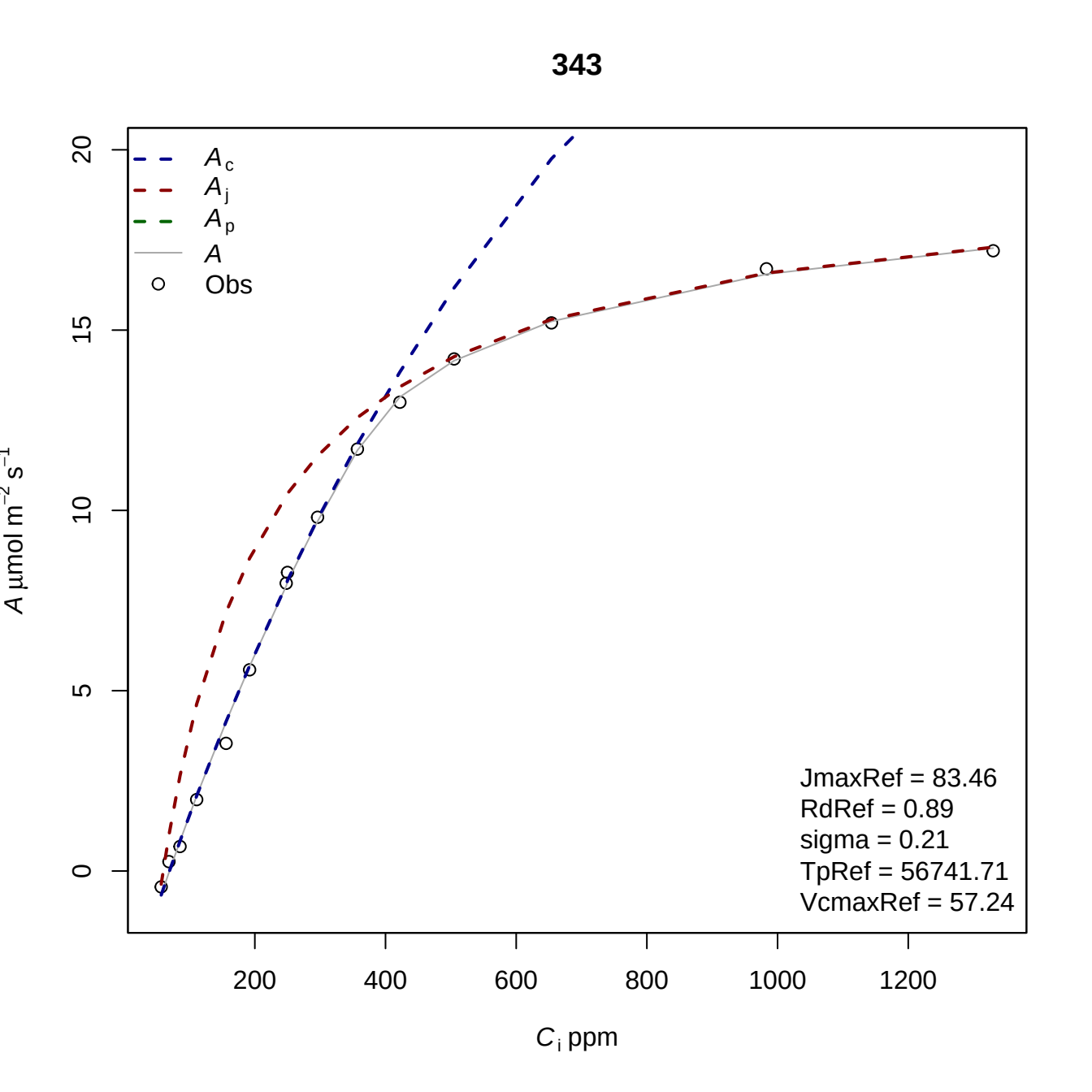

338
A_c
A_j
A_p
A
Obs
JmaxRef = 68.68
RdRef = 0.35
sigma = 0.19
TpRef = 4.54
VcmaxRef = 42.59
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
200
400
600
800
1000
1200
1400

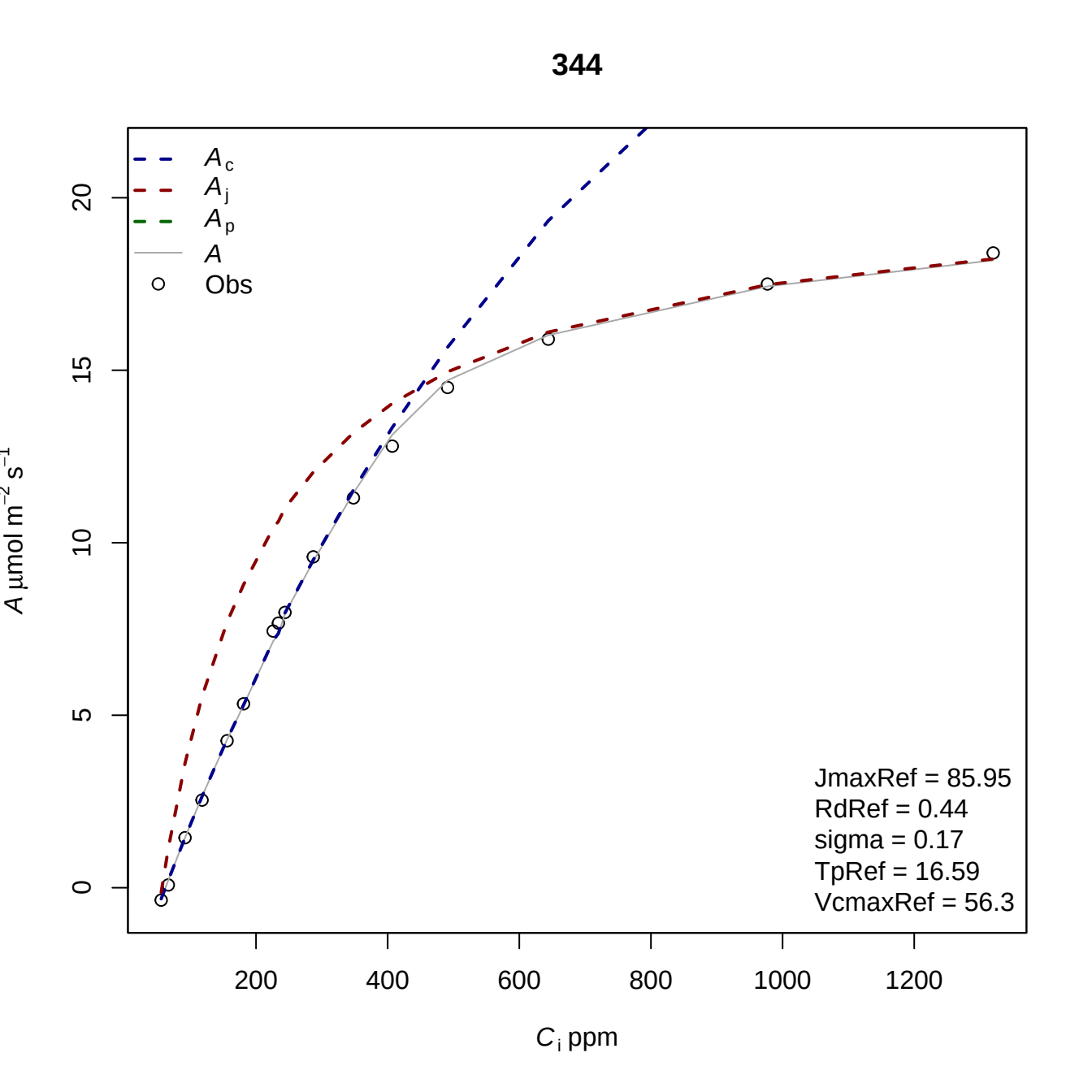

339
A_c
A_j
A_p
A
Obs
JmaxRef = 64.91
RdRef = 0.57
sigma = 0.24
TpRef = 150562.4
VcmaxRef = 33.9
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
C_i ppm

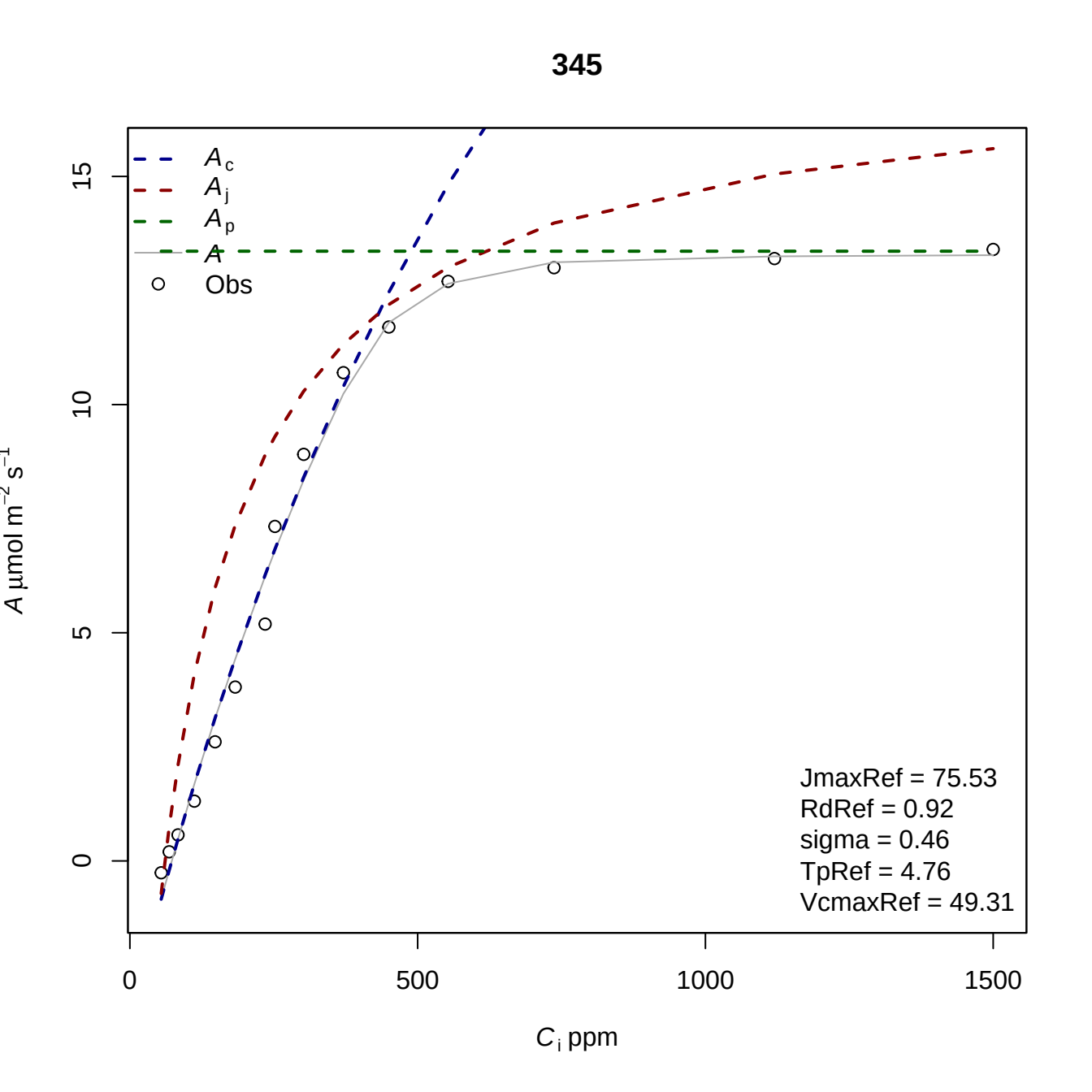

340
A_c
A_j
A_p
A
Obs
JmaxRef = 89.63
RdRef = 0.42
sigma = 0.17
TpRef = 6.62
VcmaxRef = 61.31
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
500
1000
1500

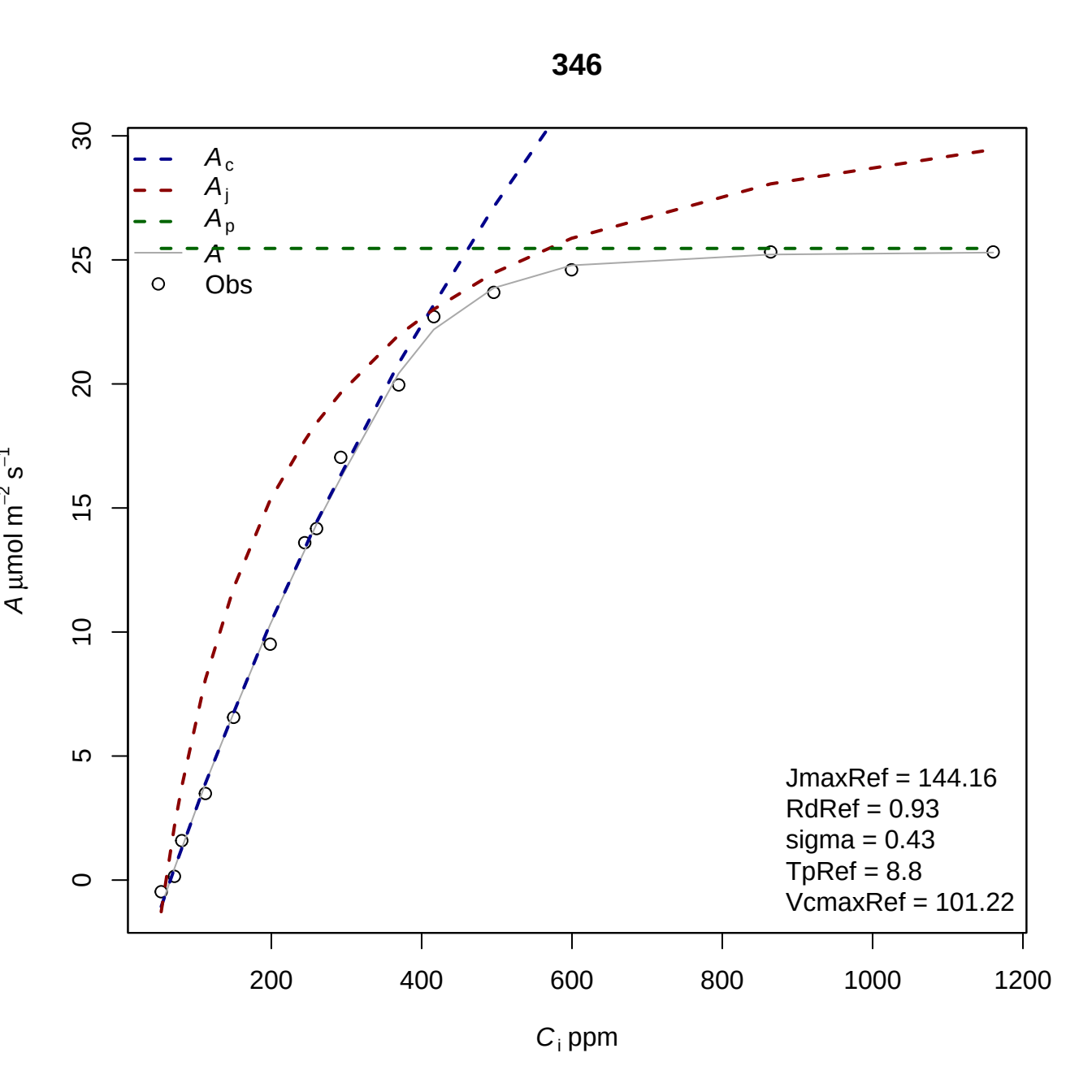

341
A_c
A_j
A_p
A
Obs
JmaxRef = 56.8
RdRef = −0.33
sigma = 0.16
TpRef = 4.13
VcmaxRef = 43.1
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
500
1000
1500

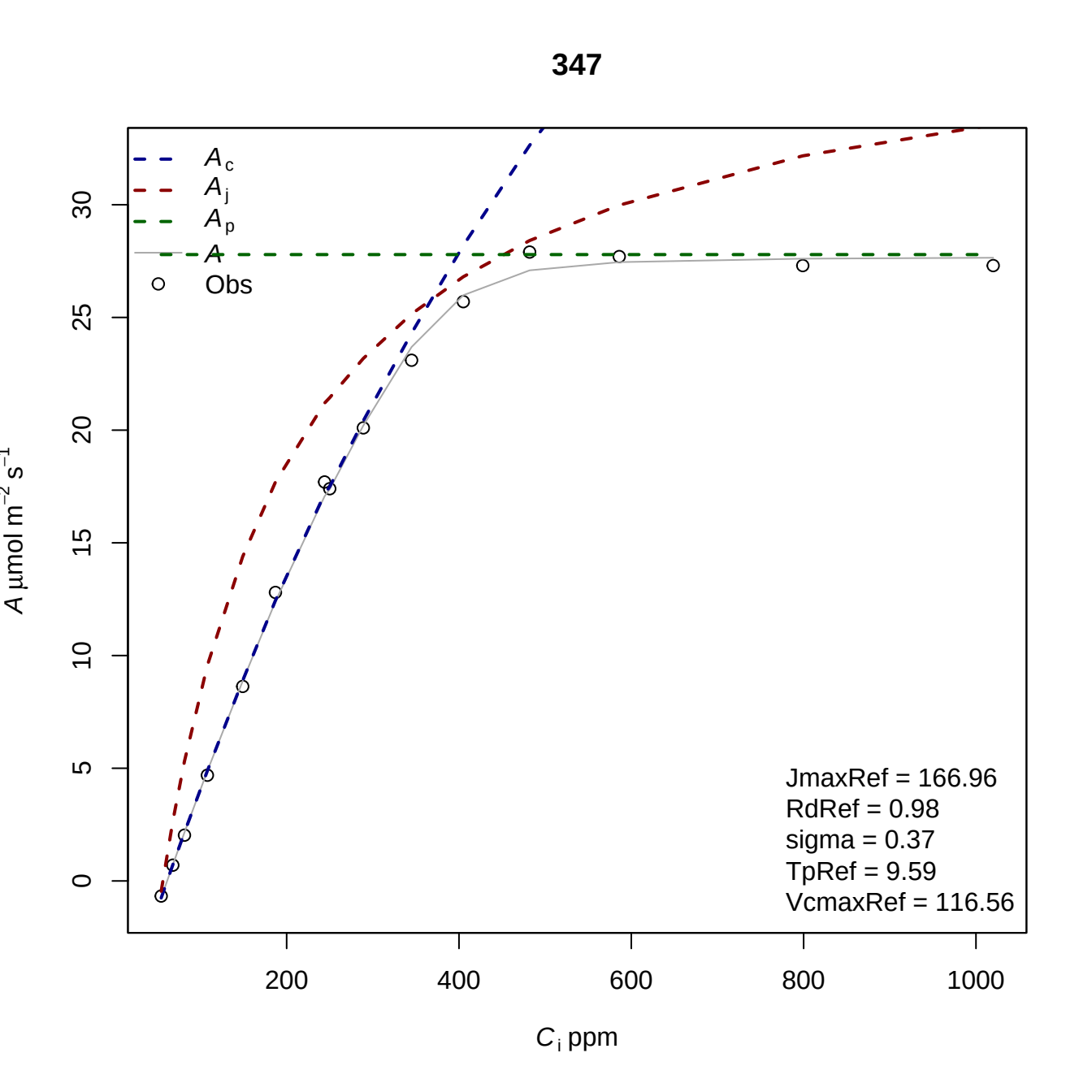

342
A_c
A_j
A_p
A
Obs
JmaxRef = 55.05
RdRef = 0.18
sigma = 0.23
TpRef = 12024.67
VcmaxRef = 39.96
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
14
0
200
400
600
800
1000
1200
1400

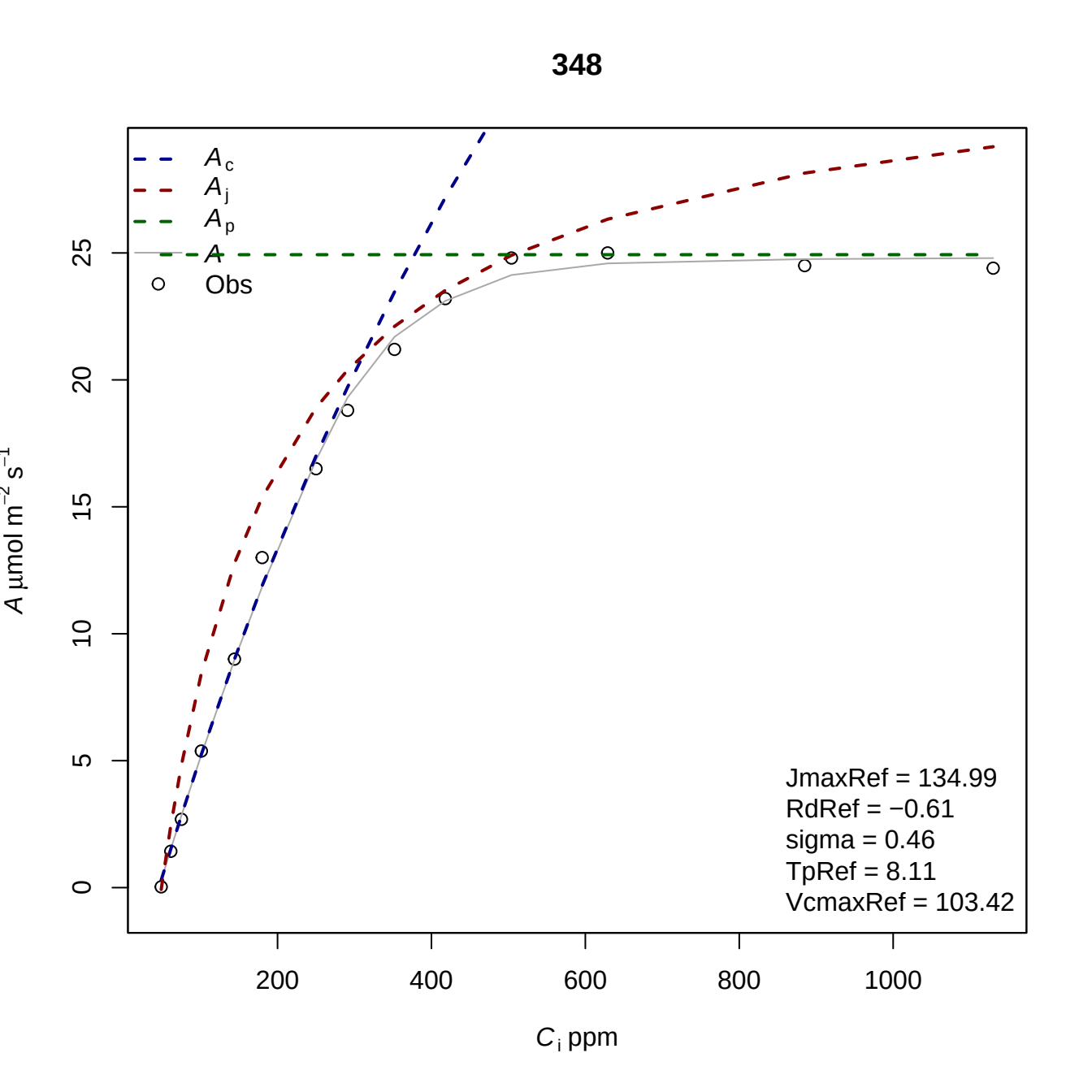

343
A_c
A_j
A_p
A
Obs
JmaxRef = 83.46
RdRef = 0.89
sigma = 0.21
TpRef = 56741.71
VcmaxRef = 57.24
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
200
400
600
800
1000
1200

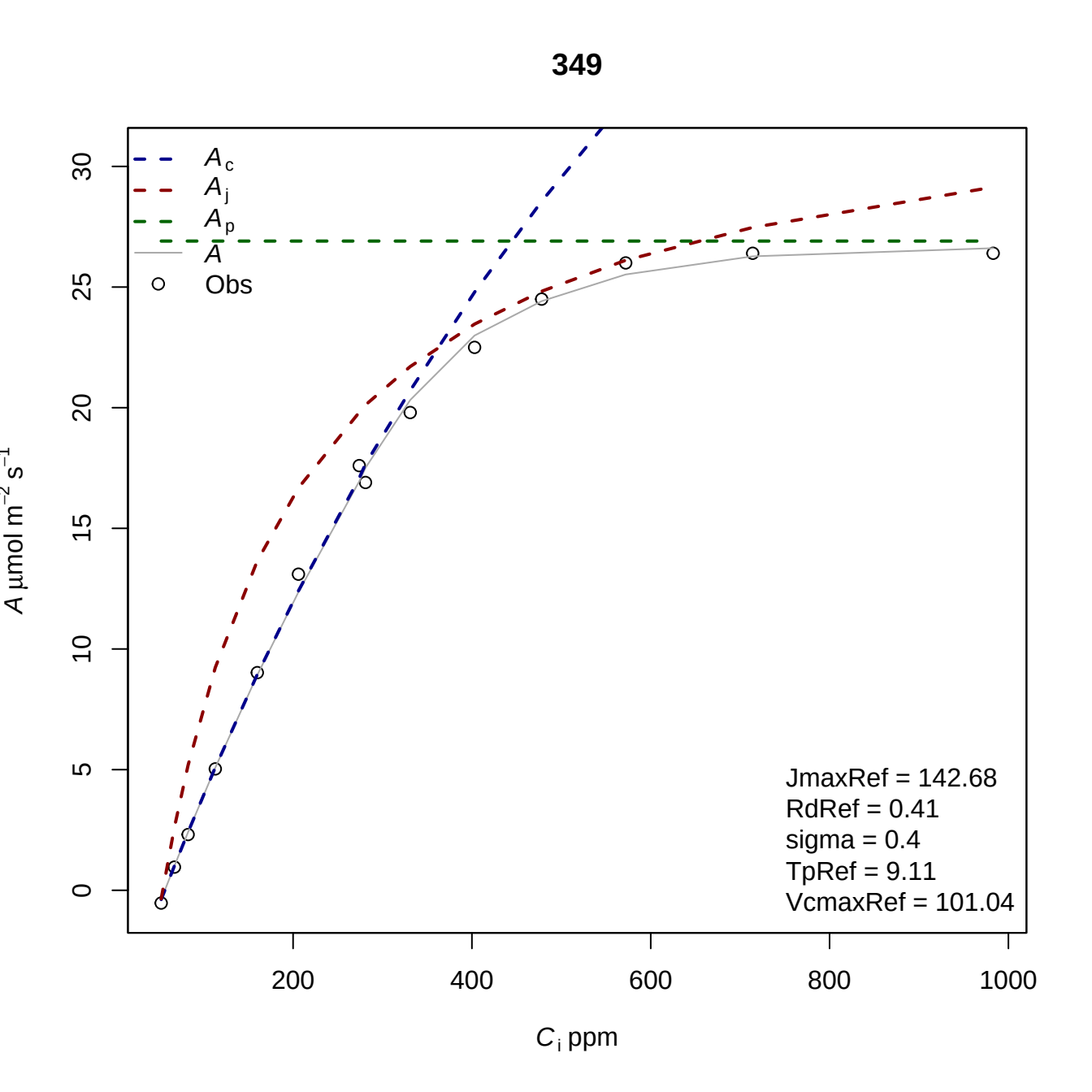

344
A_c
A_j
A_p
A
Obs
JmaxRef = 85.95
RdRef = 0.44
sigma = 0.17
TpRef = 16.59
VcmaxRef = 56.3
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
200
400
600
800
1000
1200

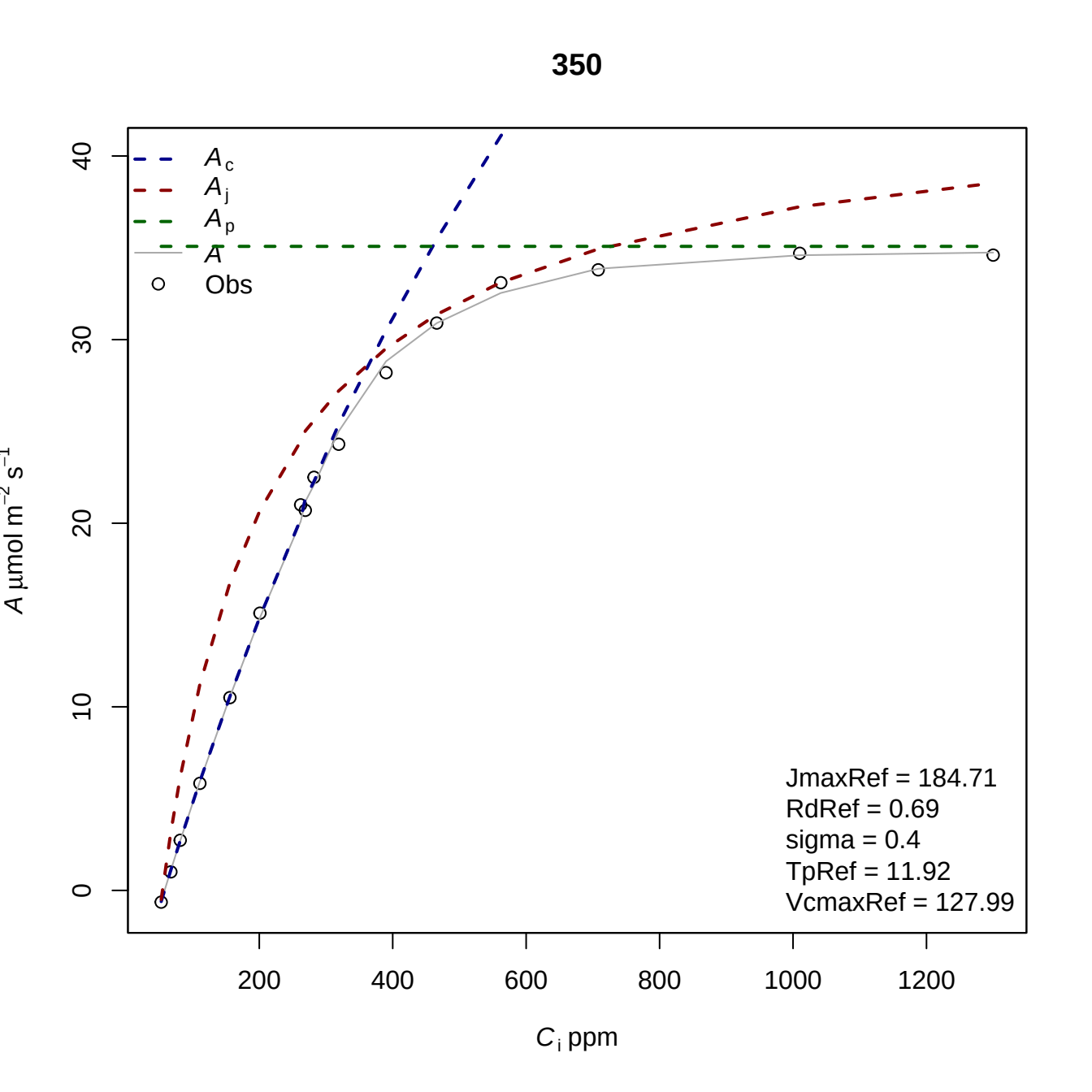

345
A_c
A_j
A_p
A
Obs
JmaxRef = 75.53
RdRef = 0.92
sigma = 0.46
TpRef = 4.76
VcmaxRef = 49.31
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
500
1000
1500

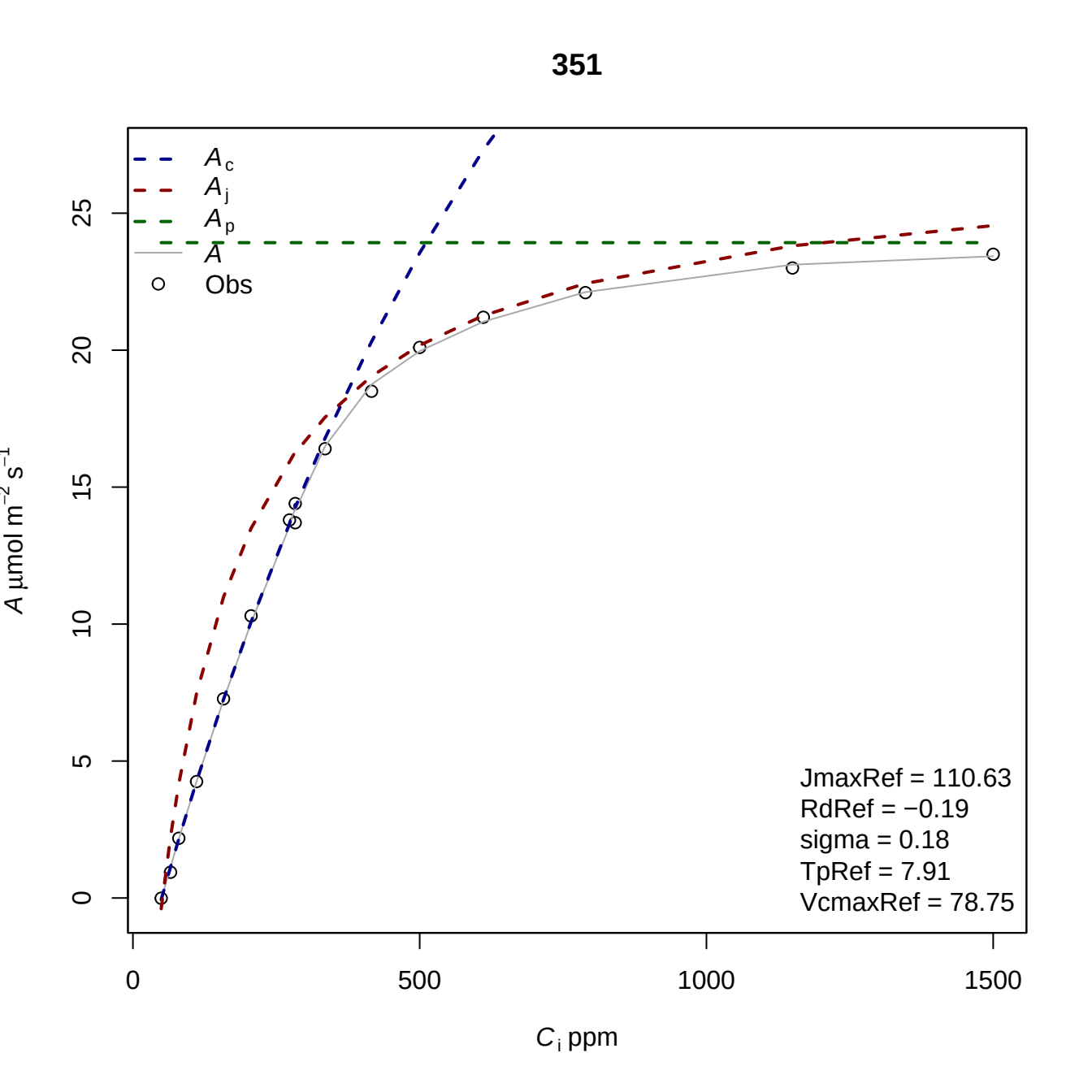

346
A_c
A_j
A_p
A
Obs
JmaxRef = 144.16
RdRef = 0.93
sigma = 0.43
TpRef = 8.8
VcmaxRef = 101.22
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
200
400
600
800
1000
1200

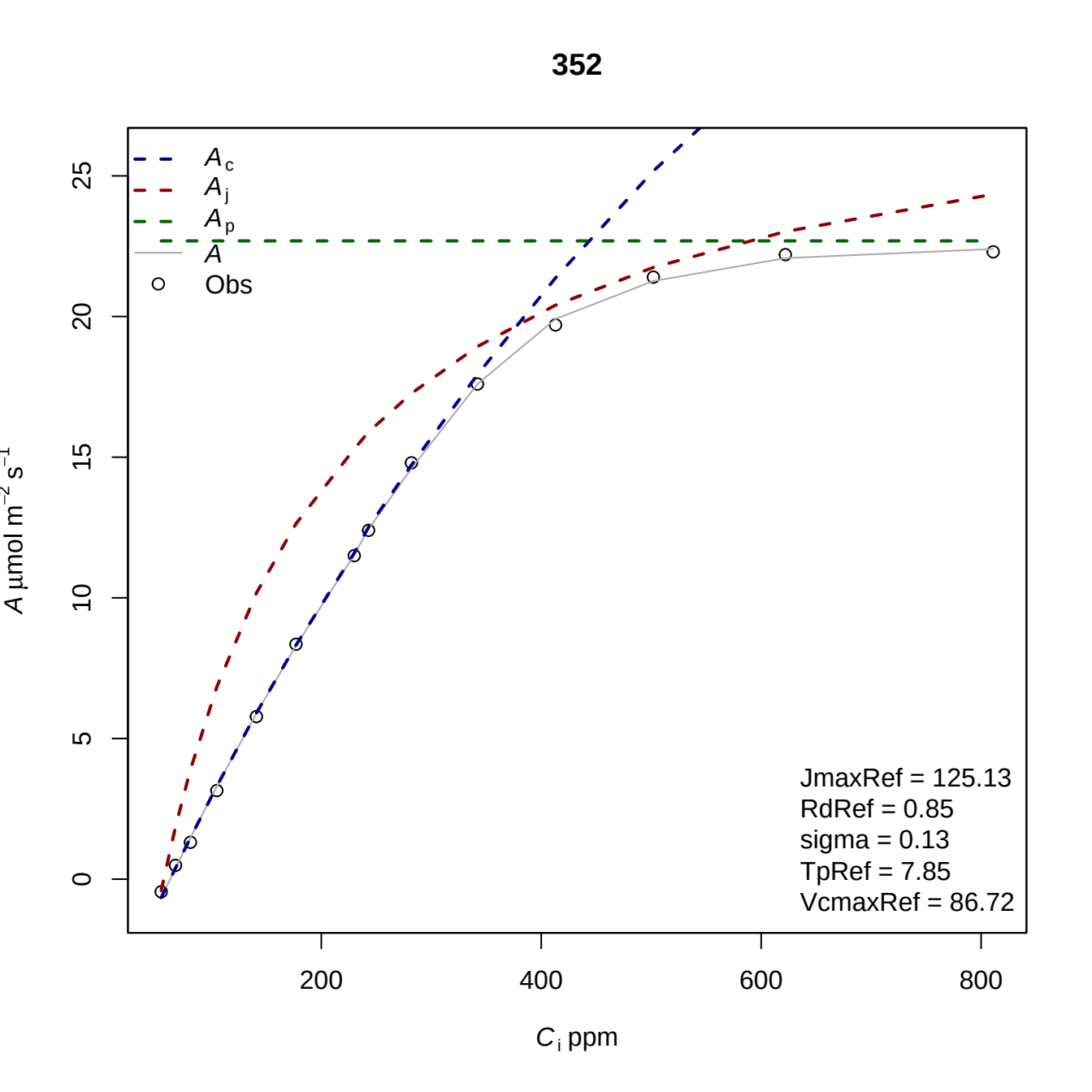

347
A_c
A_j
A_p
A
Obs
JmaxRef = 166.96
RdRef = 0.98
sigma = 0.37
TpRef = 9.59
VcmaxRef = 116.56
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
200
400
600
800
1000

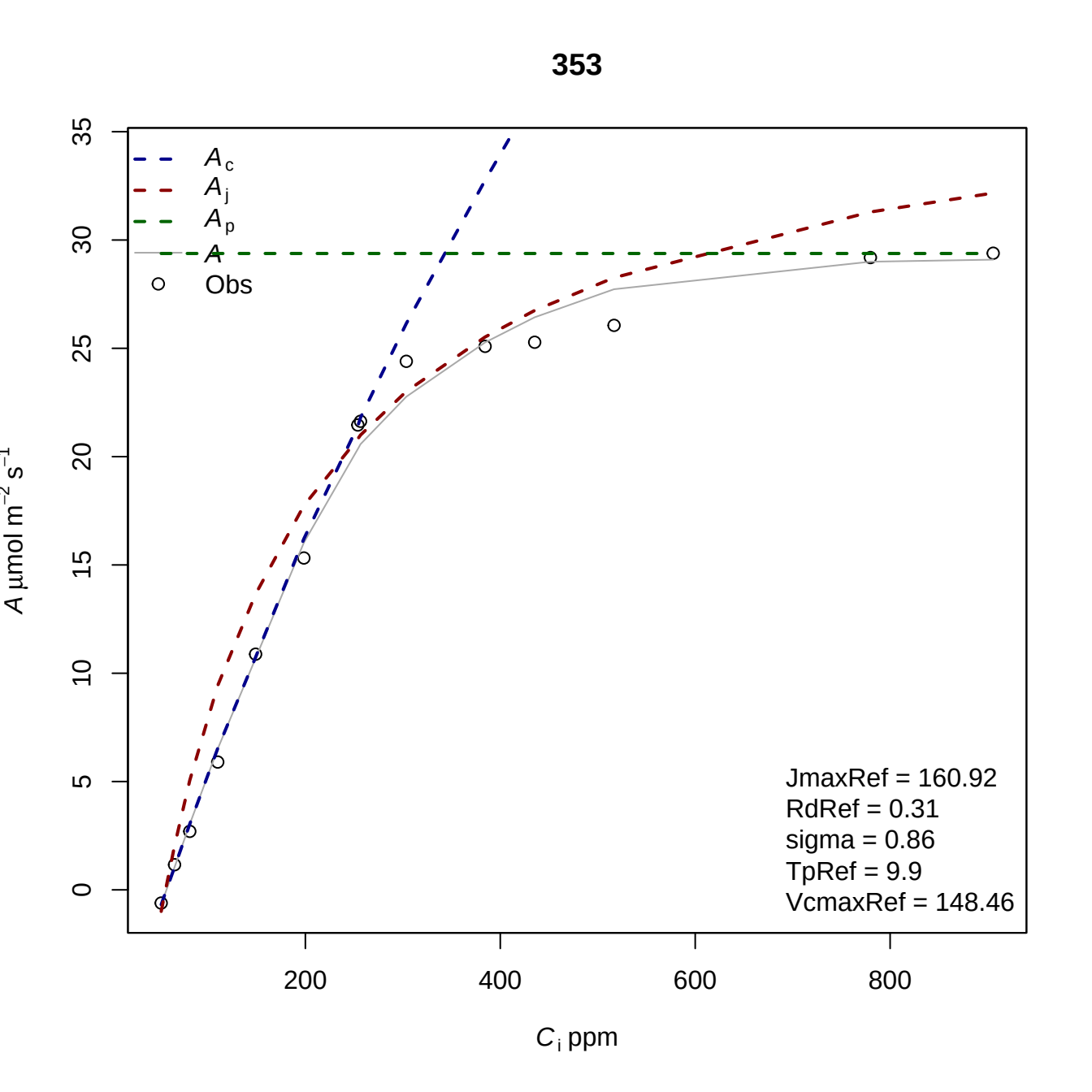

348
A_c
A_j
A_p
A
Obs
JmaxRef = 134.99
RdRef = −0.61
sigma = 0.46
TpRef = 8.11
VcmaxRef = 103.42
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
200
400
600
800
1000

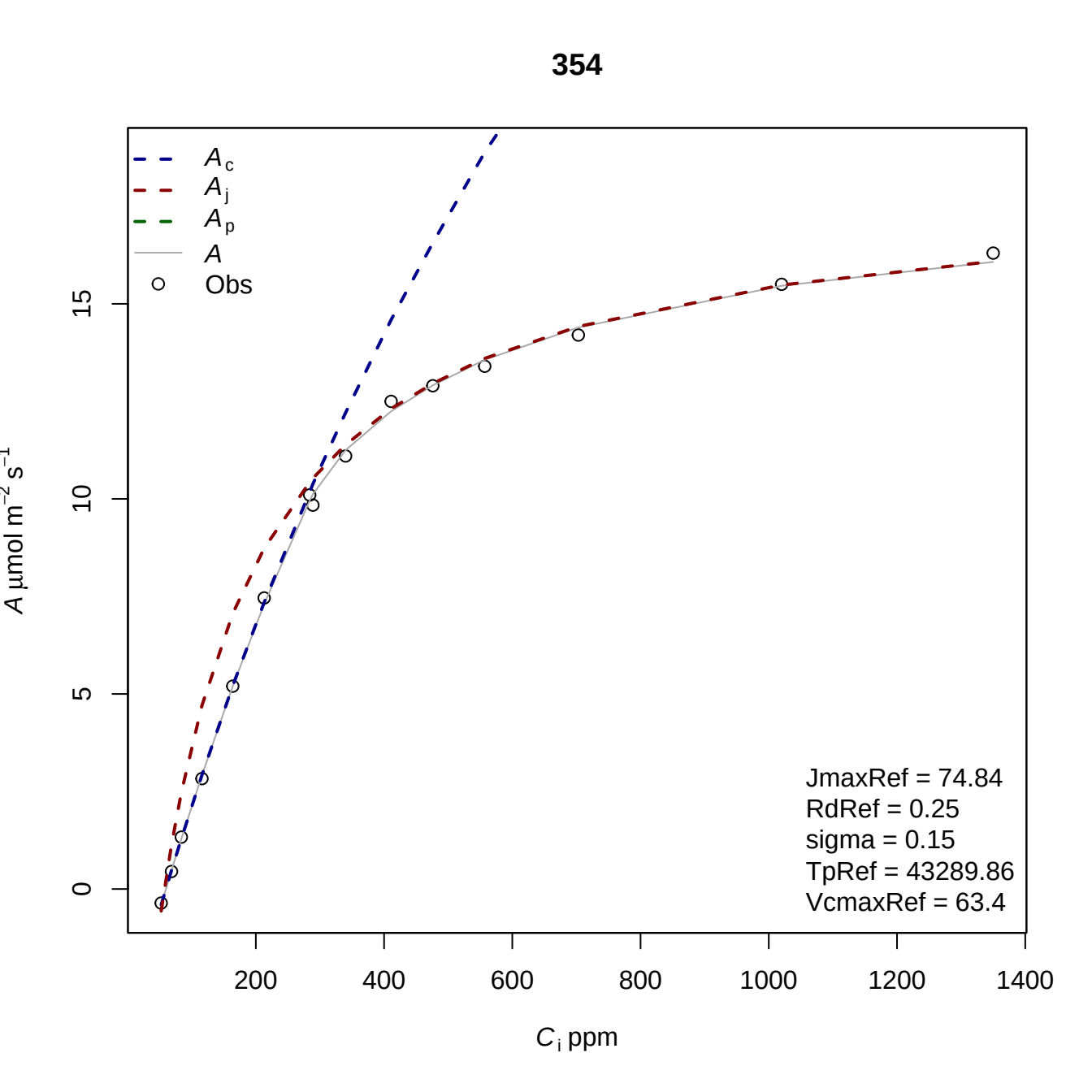

349
A_c
A_j
A_p
A
Obs
JmaxRef = 142.68
RdRef = 0.41
sigma = 0.4
TpRef = 9.11
VcmaxRef = 101.04
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
200
400
600
800
1000

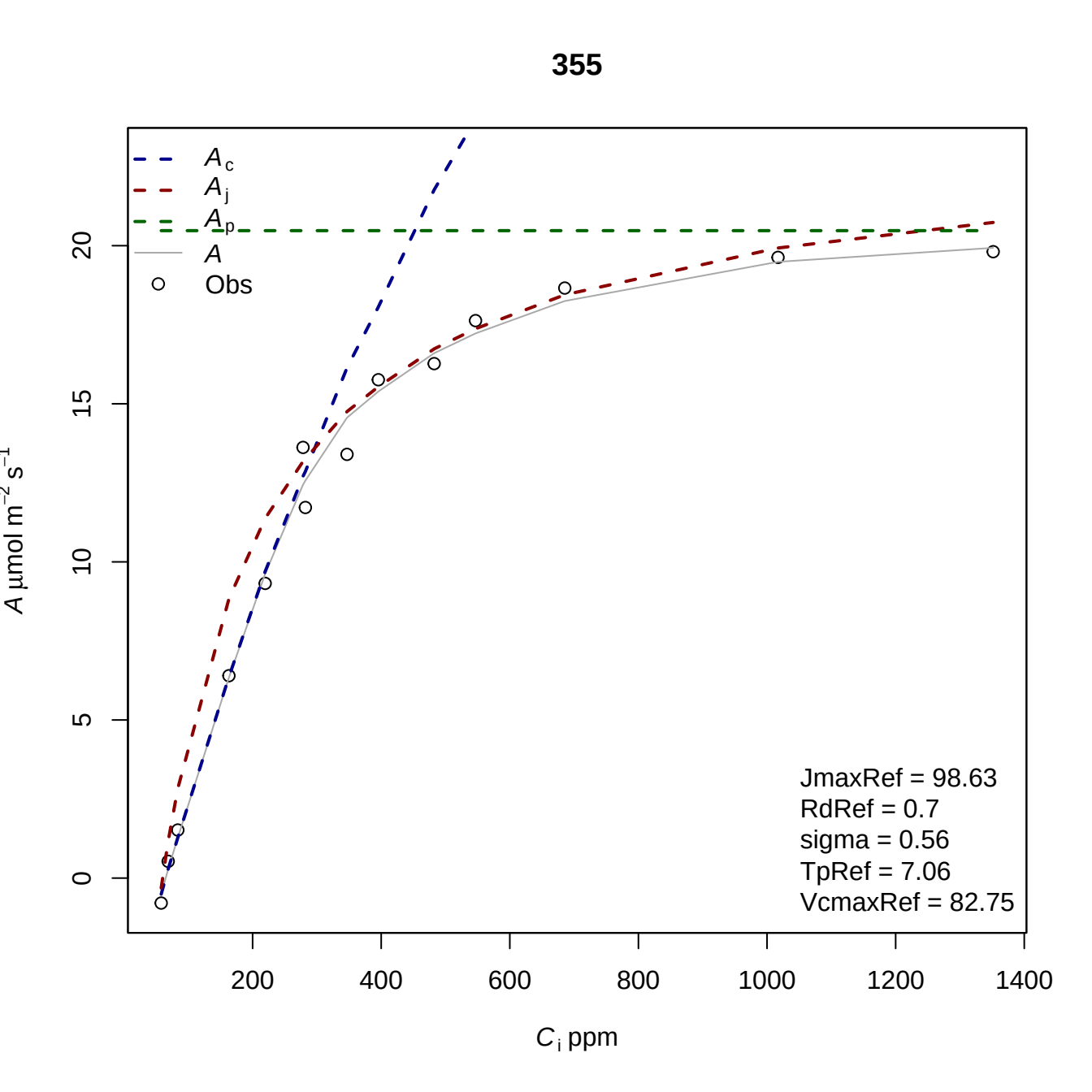

350
A_c
A_j
A_p
A
Obs
JmaxRef = 184.71
RdRef = 0.69
sigma = 0.4
TpRef = 11.92
VcmaxRef = 127.99
0
10
20
30
40
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

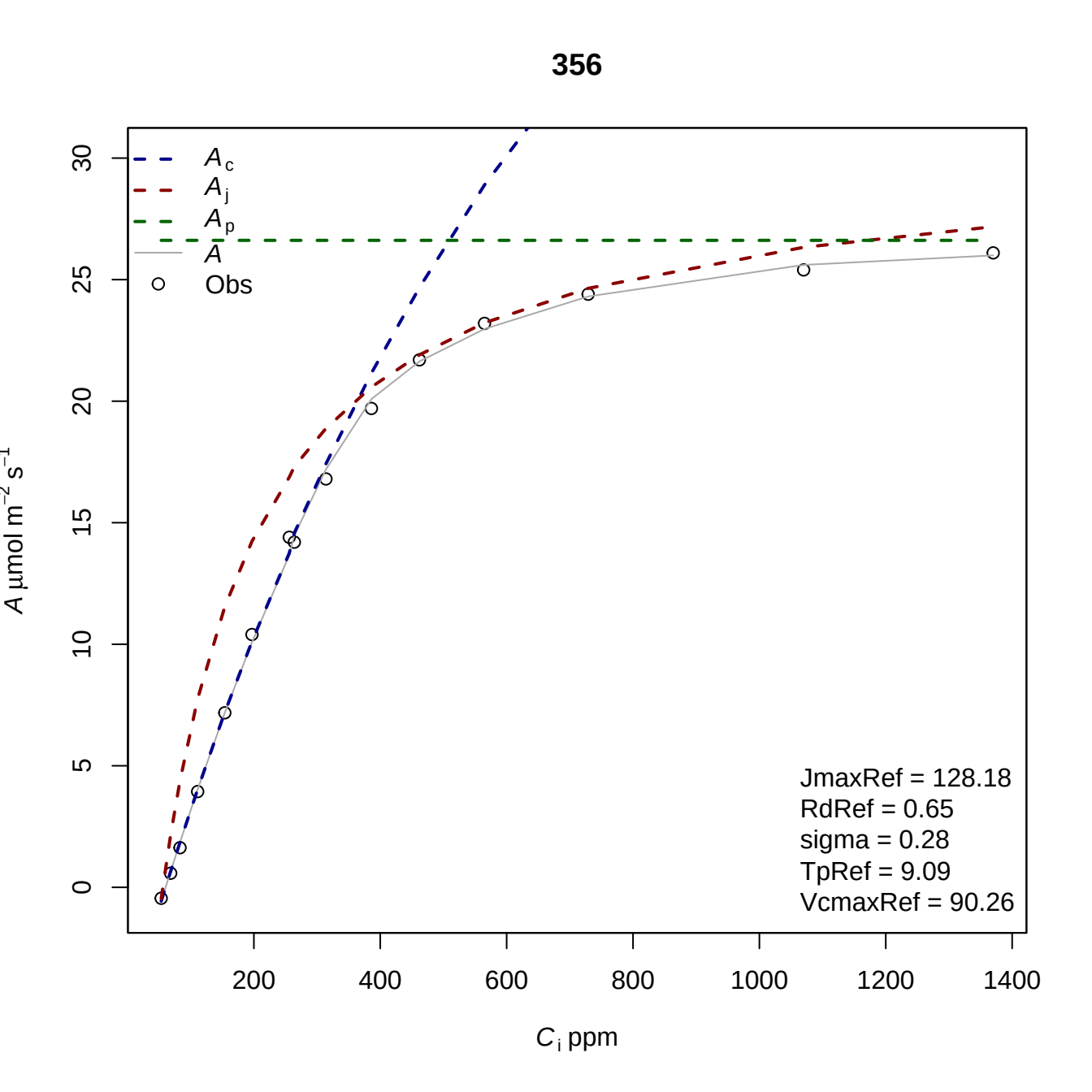

351
A_c
A_j
A_p
A
Obs
JmaxRef = 110.63
RdRef = −0.19
sigma = 0.18
TpRef = 7.91
VcmaxRef = 78.75
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
500
1000
1500

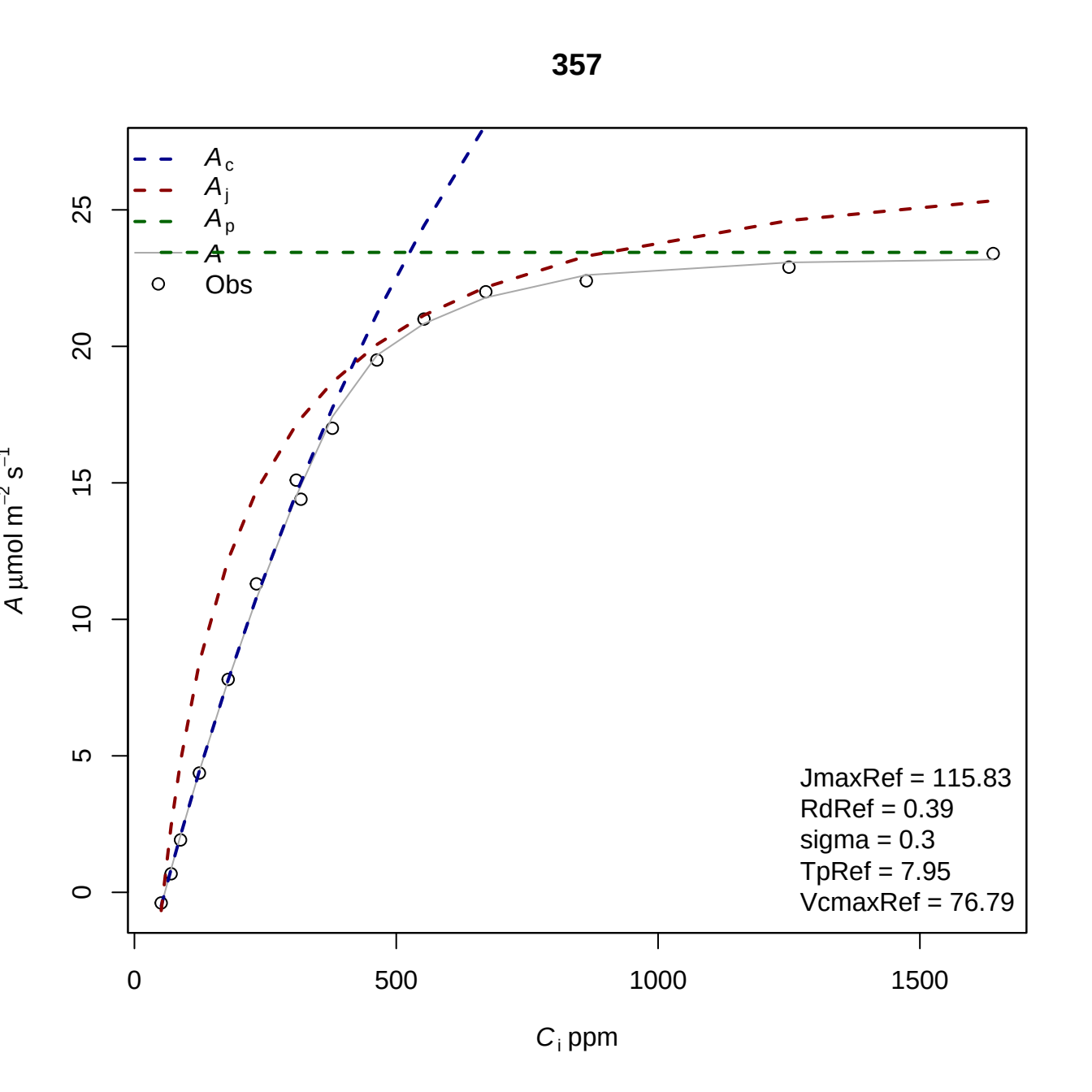

352
A_c
A_j
A_p
A
Obs
JmaxRef = 125.13
RdRef = 0.85
sigma = 0.13
TpRef = 7.85
VcmaxRef = 86.72
0
5
10
15
20
25
200
400
600
800
A µmol m^{-2} s^{-1}
C_i ppm

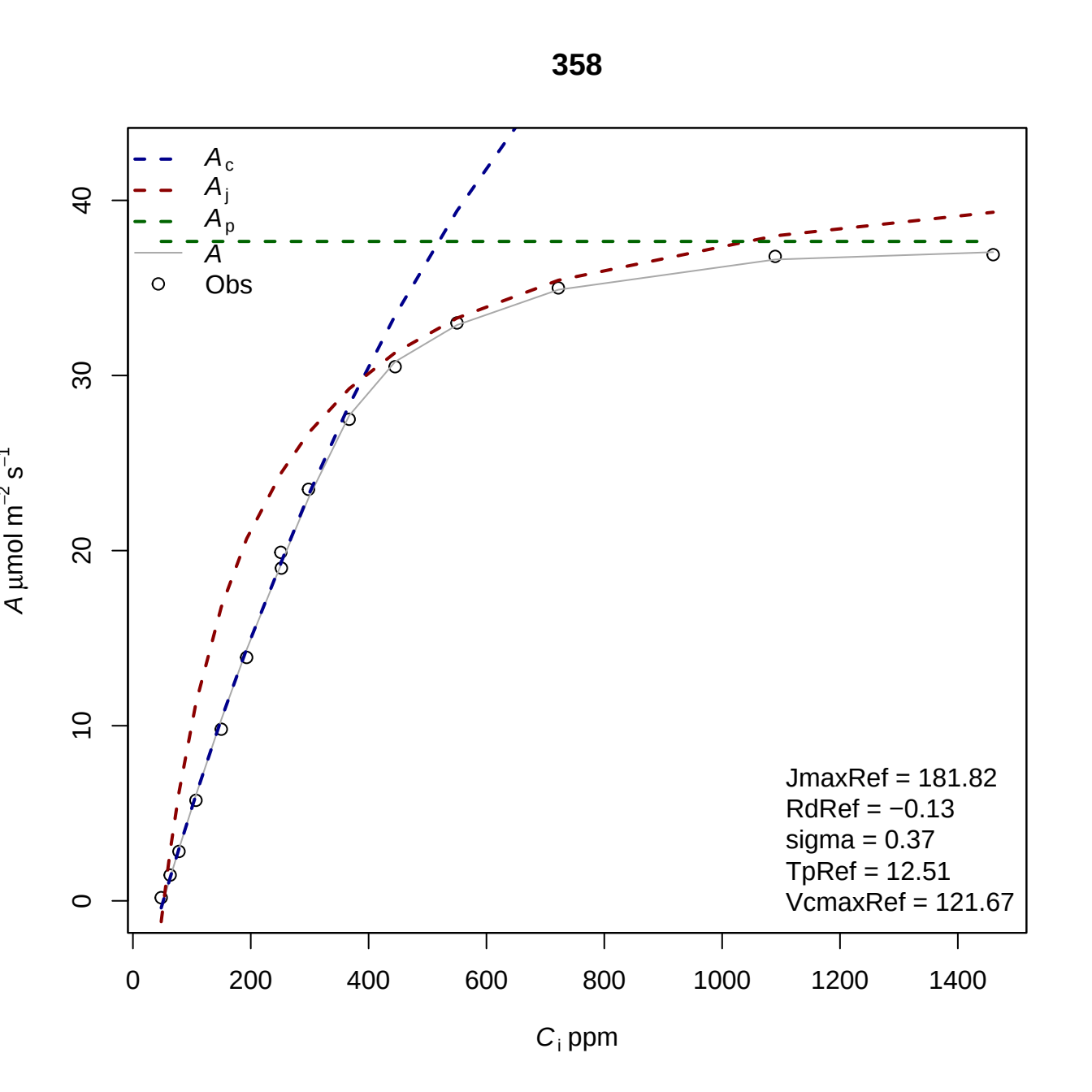

353
A_c
A_j
A_p
A
Obs
JmaxRef = 160.92
RdRef = 0.31
sigma = 0.86
TpRef = 9.9
VcmaxRef = 148.46
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
35
200
400
600
800

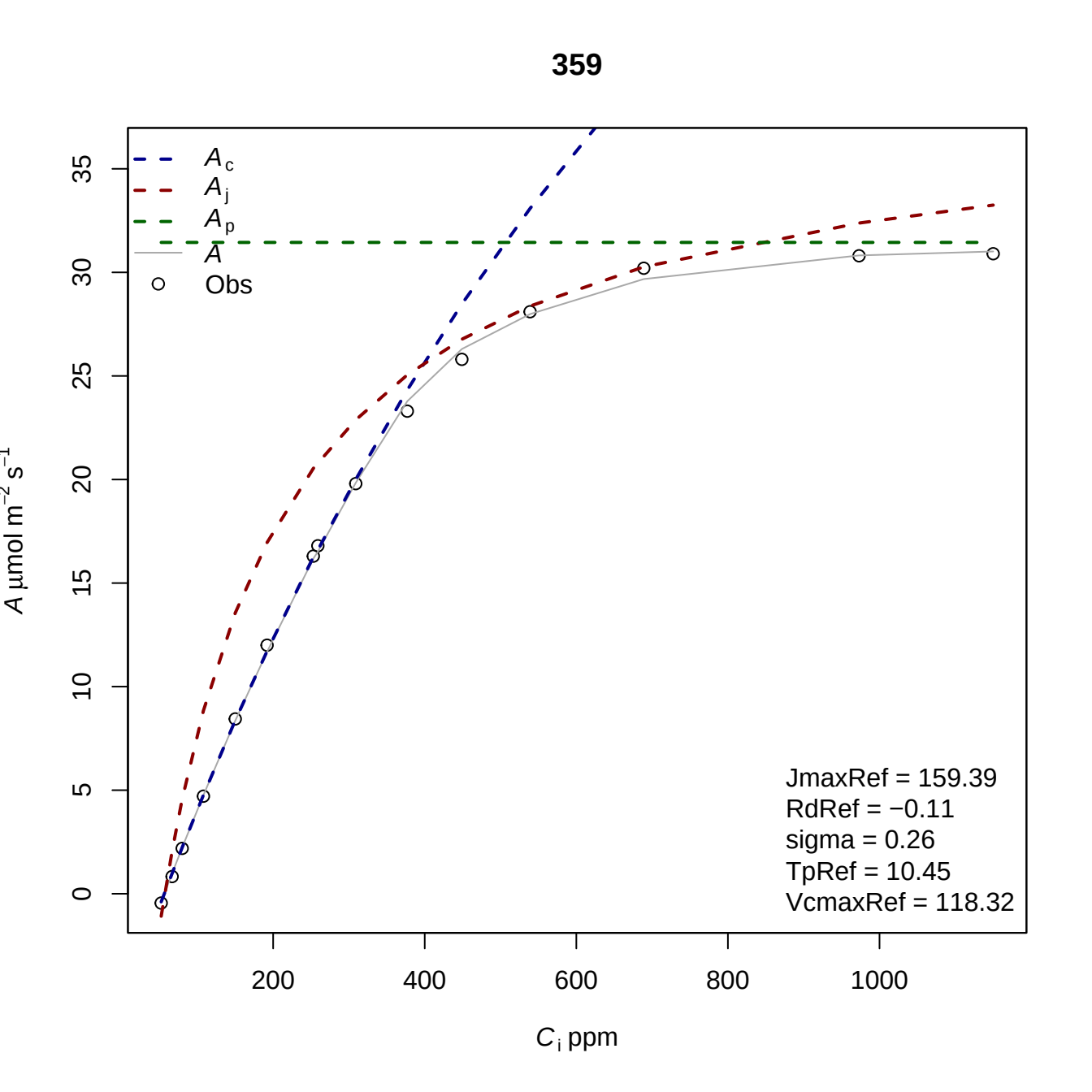

354
A_c
A_j
A_p
A
Obs
JmaxRef = 74.84
RdRef = 0.25
sigma = 0.15
TpRef = 43289.86
VcmaxRef = 63.4
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
200
400
600
800
1000
1200
1400

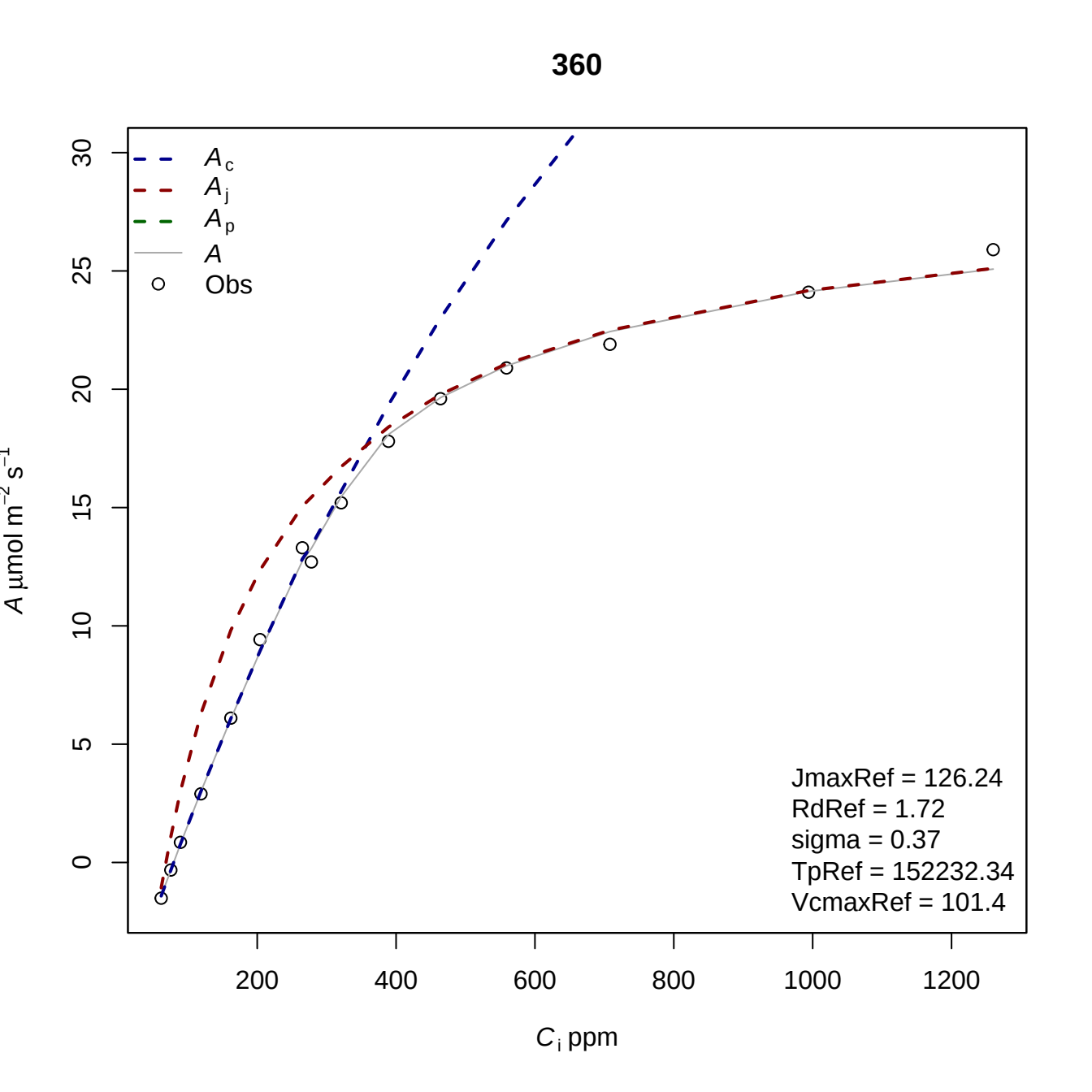

355
A_c
A_j
A_p
A
Obs
JmaxRef = 98.63
RdRef = 0.7
sigma = 0.56
TpRef = 7.06
VcmaxRef = 82.75
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
200
400
600
800
1000
1200
1400

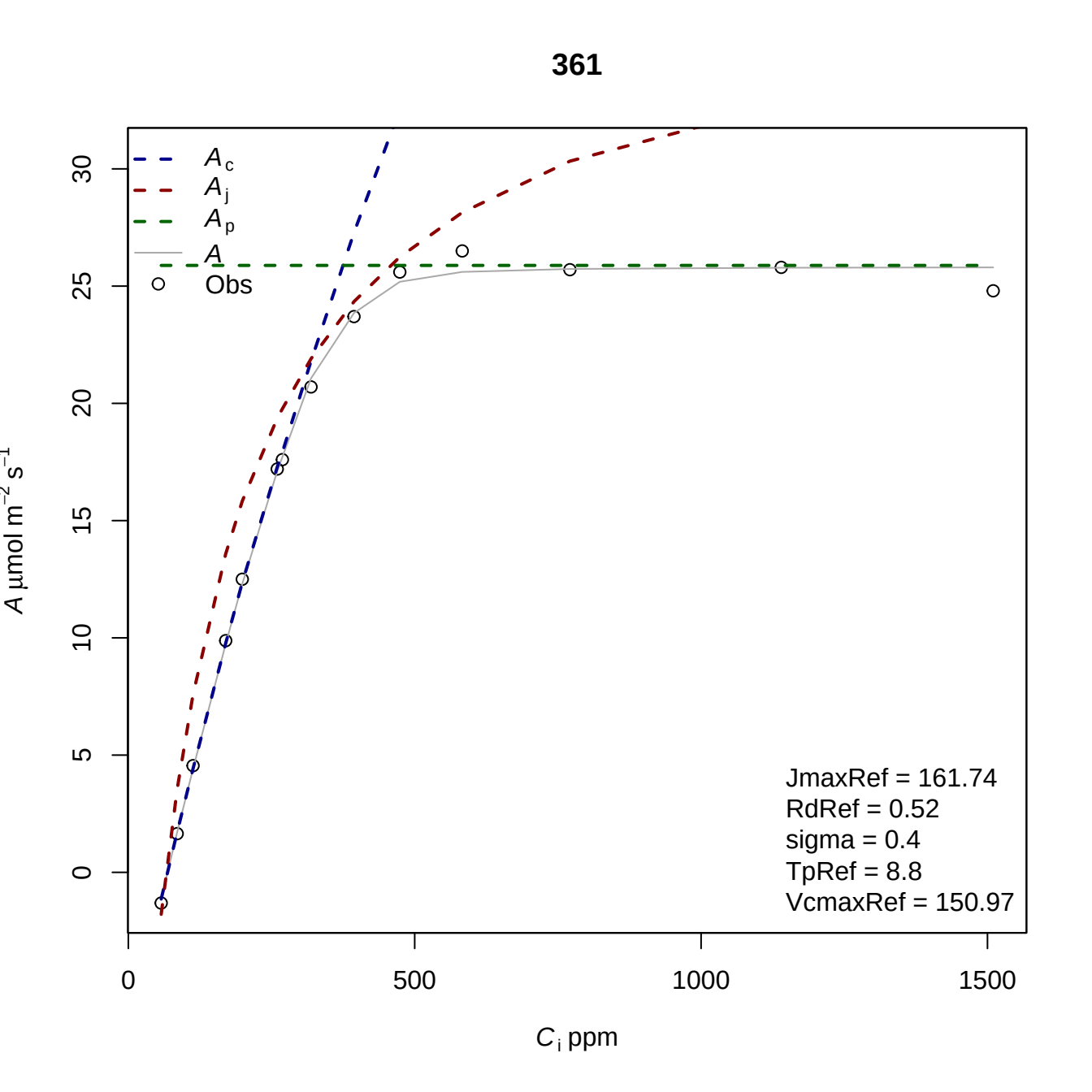

356
A_c
A_j
A_p
A
Obs
JmaxRef = 128.18
RdRef = 0.65
sigma = 0.28
TpRef = 9.09
VcmaxRef = 90.26
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
200
400
600
800
1000
1200
1400

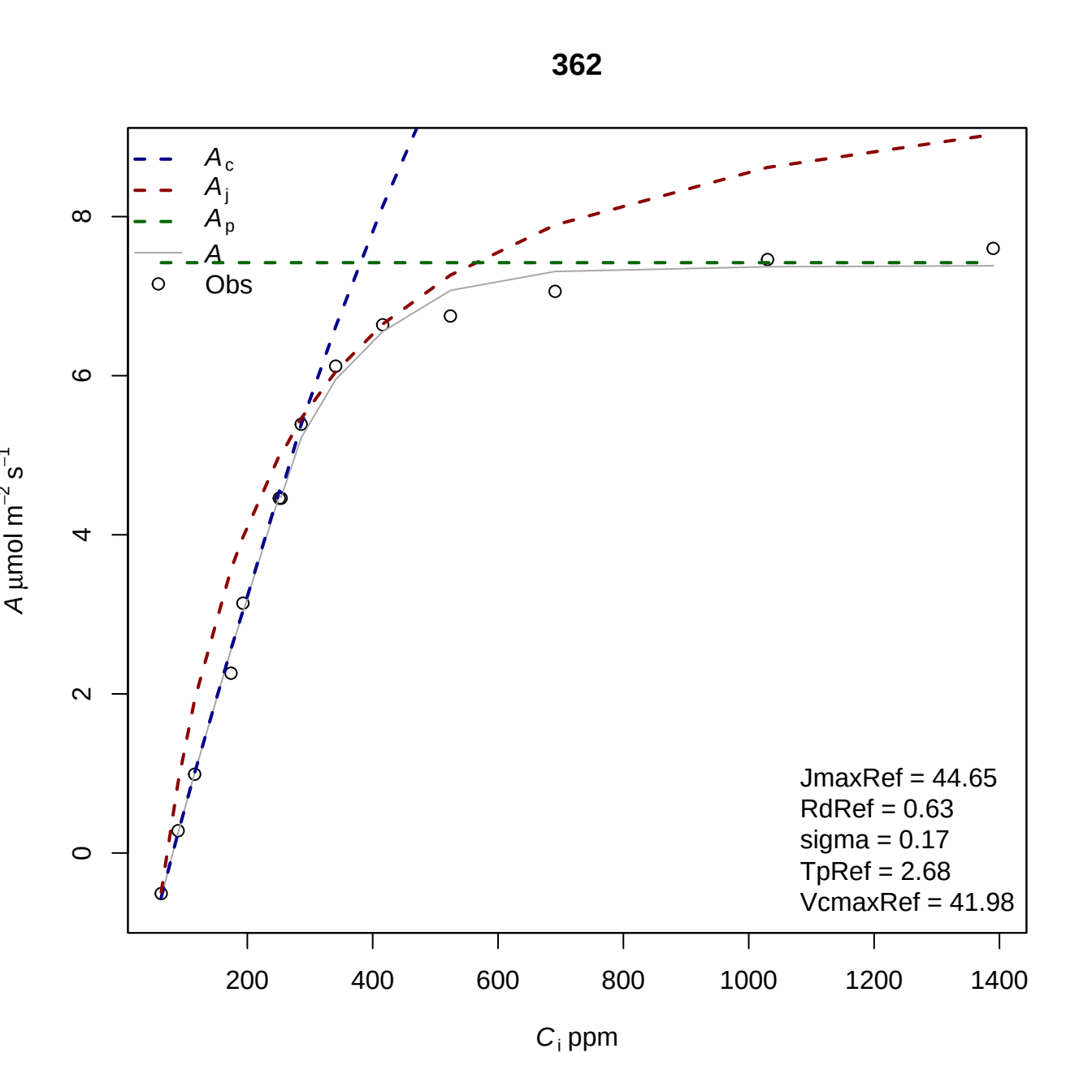

357
A_c
A_j
A_p
A
Obs
JmaxRef = 115.83
RdRef = 0.39
sigma = 0.3
TpRef = 7.95
VcmaxRef = 76.79
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
500
1000
1500

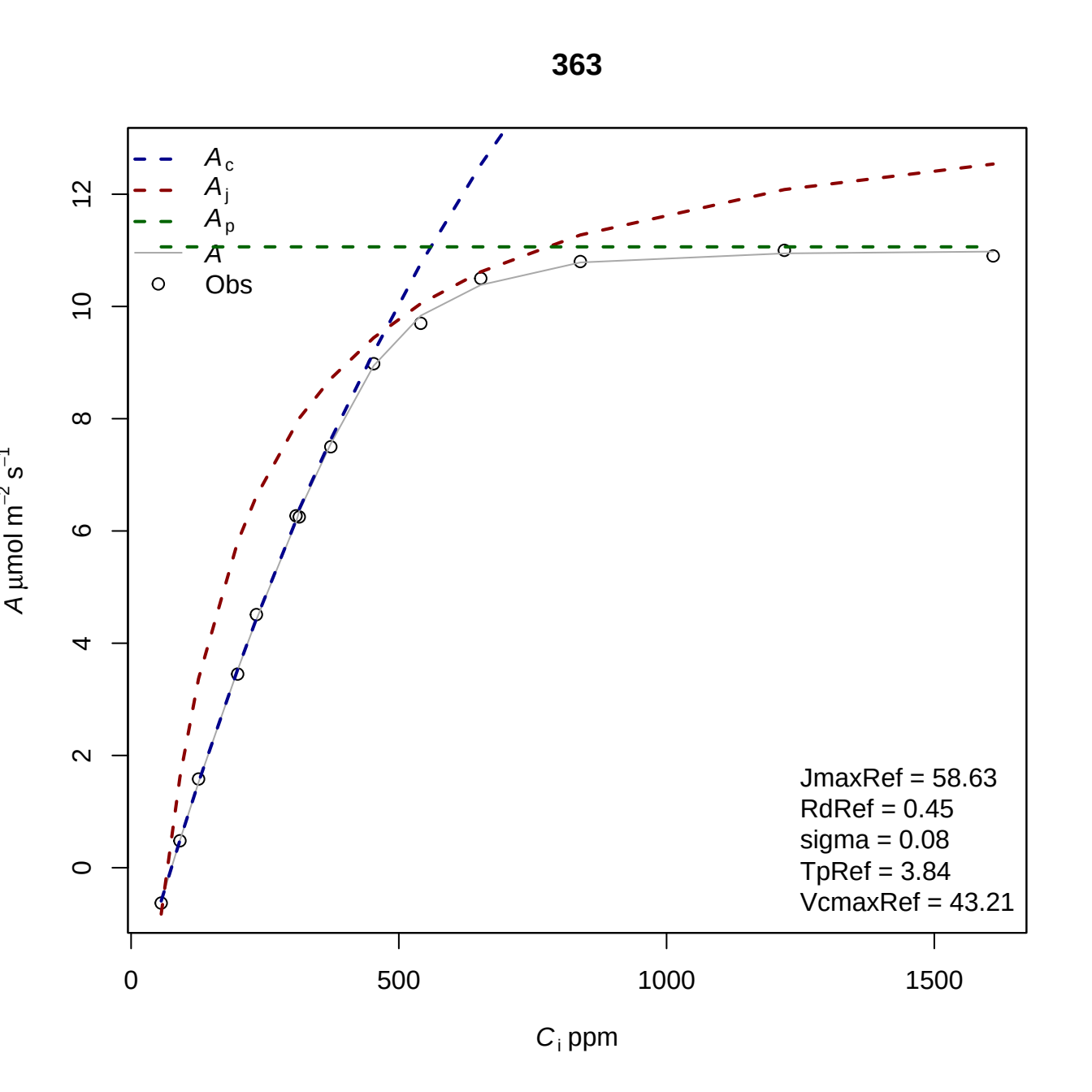

358
A_c
A_j
A_p
A
Obs
JmaxRef = 181.82
RdRef = −0.13
sigma = 0.37
TpRef = 12.51
VcmaxRef = 121.67
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
0
200
400
600
800
1000
1200
1400

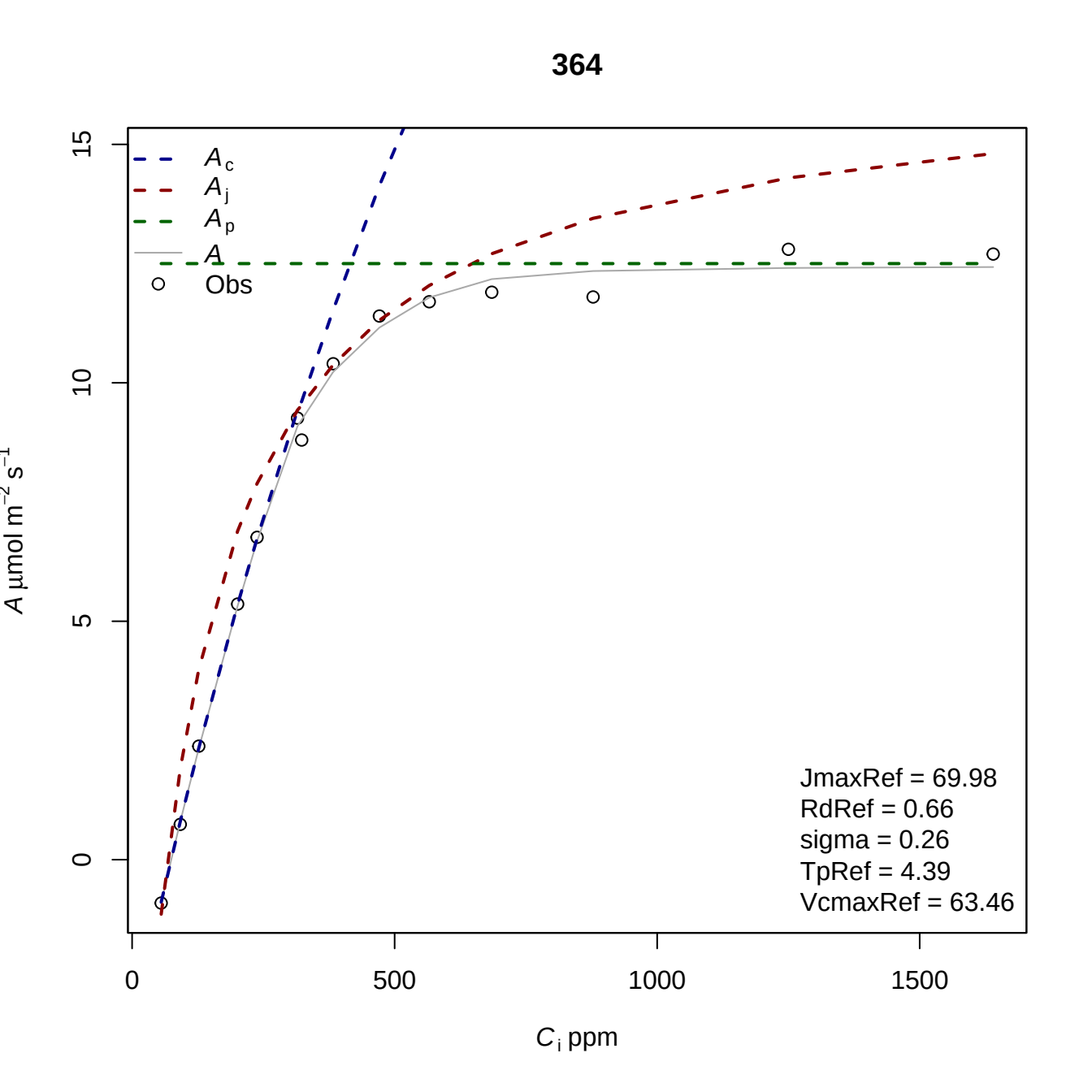

359
A_c
A_j
A_p
A
Obs
JmaxRef = 159.39
RdRef = −0.11
sigma = 0.26
TpRef = 10.45
VcmaxRef = 118.32
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
35
200
400
600
800
1000

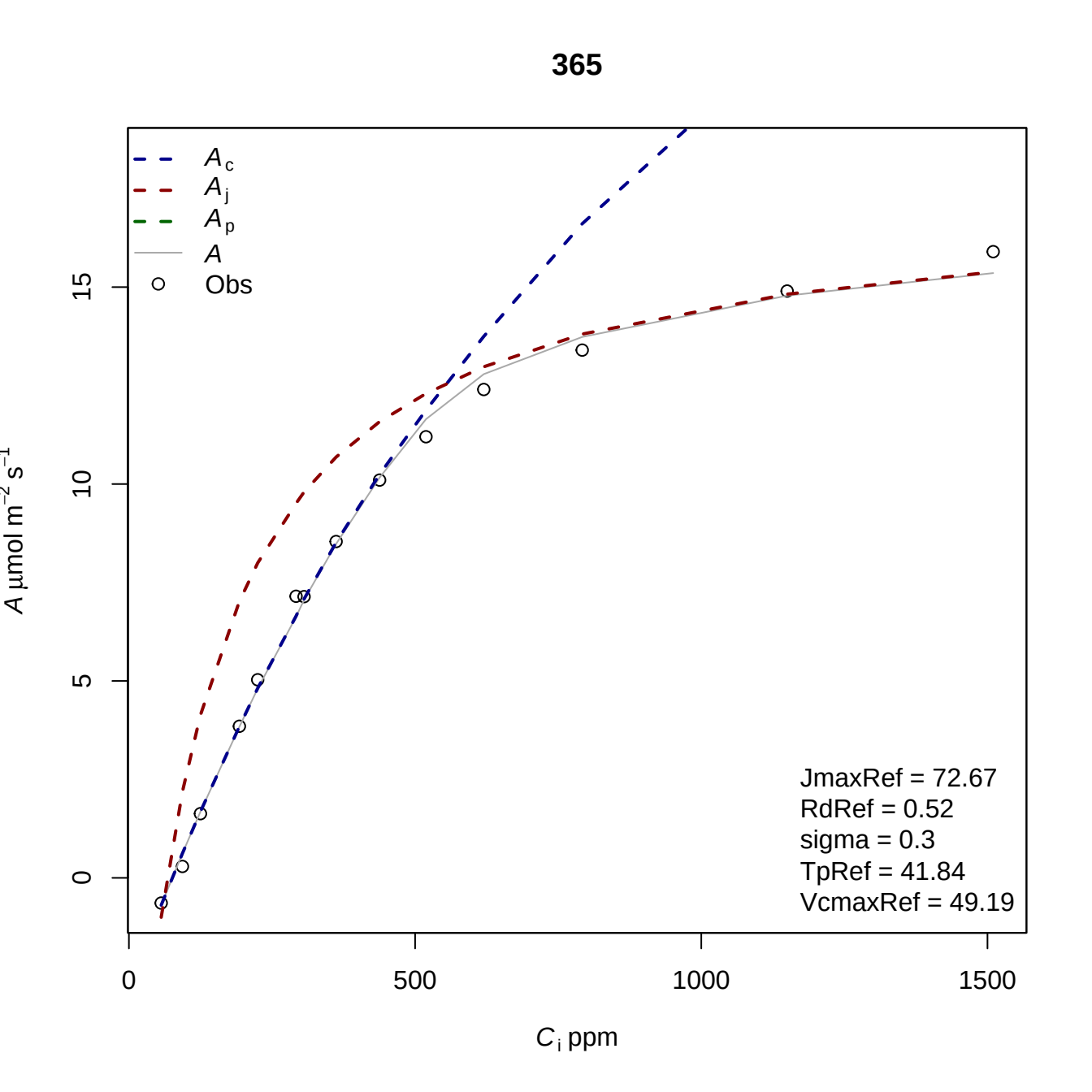

360
A_c
A_j
A_p
A
Obs
JmaxRef = 126.24
RdRef = 1.72
sigma = 0.37
TpRef = 152232.34
VcmaxRef = 101.4
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
200
400
600
800
1000
1200

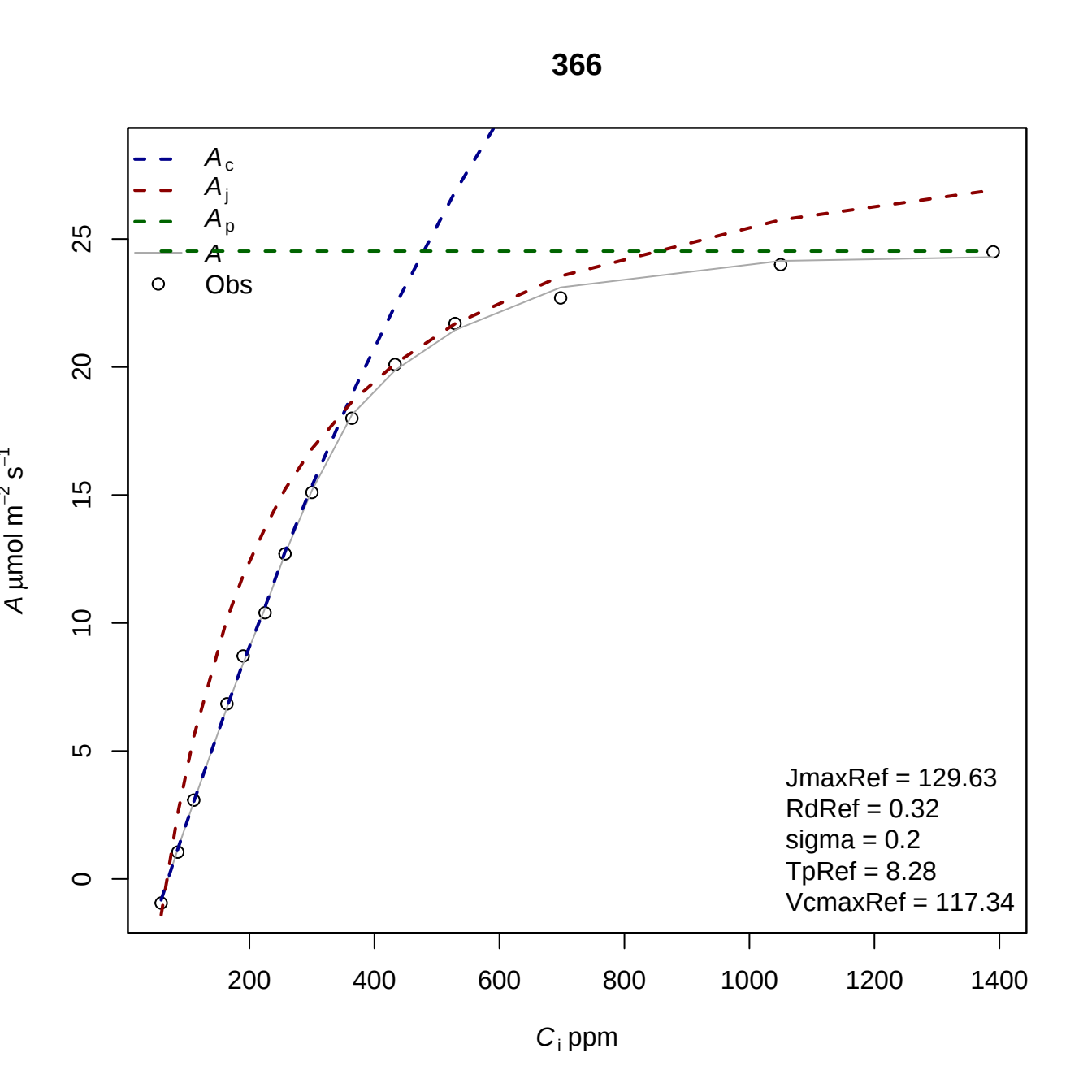

361
A_c
A_j
A_p
A
Obs
JmaxRef = 161.74
RdRef = 0.52
sigma = 0.4
TpRef = 8.8
VcmaxRef = 150.97
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
0
500
1000
1500

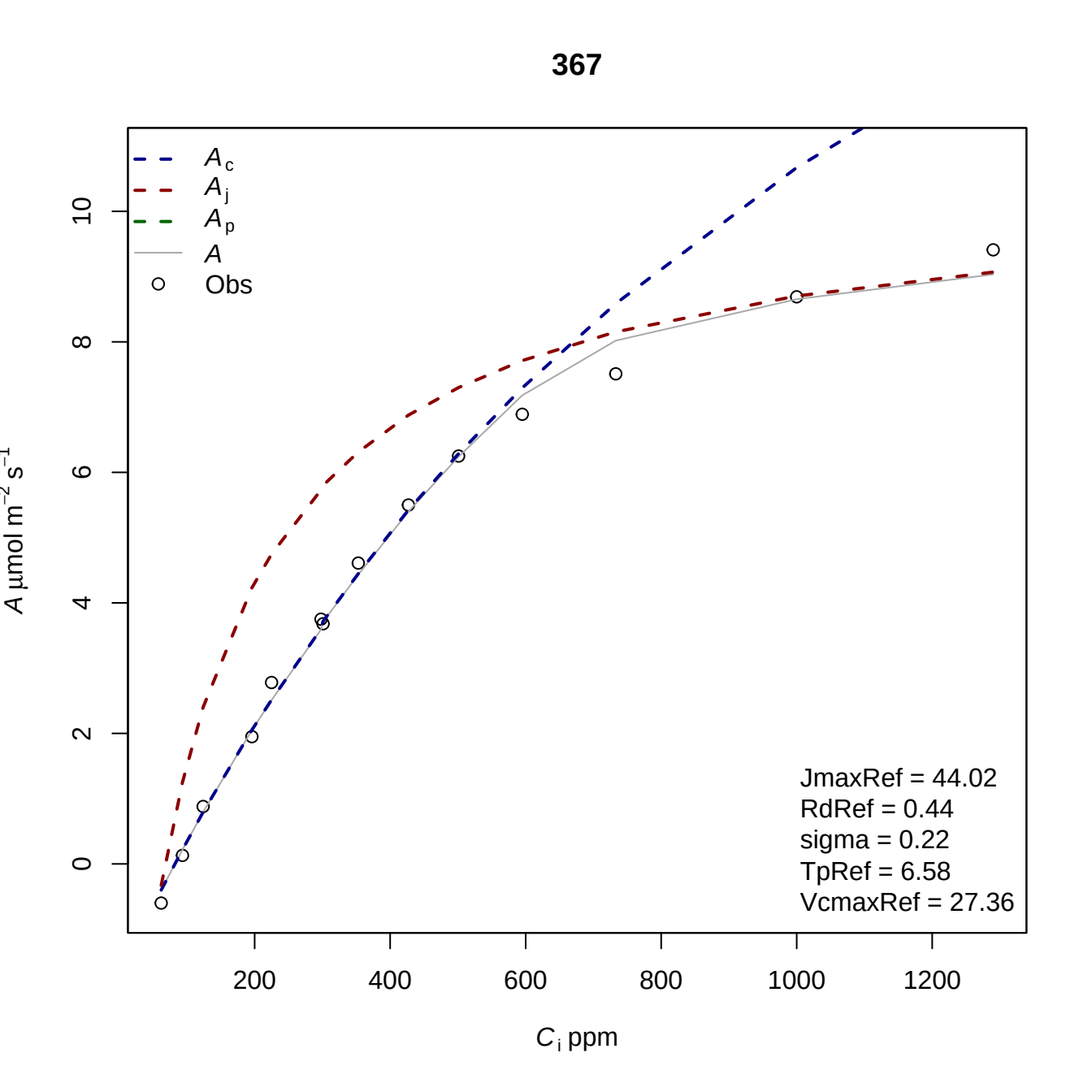

362
A_c
A_j
A_p
A
Obs
JmaxRef = 44.65
RdRef = 0.63
sigma = 0.17
TpRef = 2.68
VcmaxRef = 41.98
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
200
400
600
800
1000
1200
1400

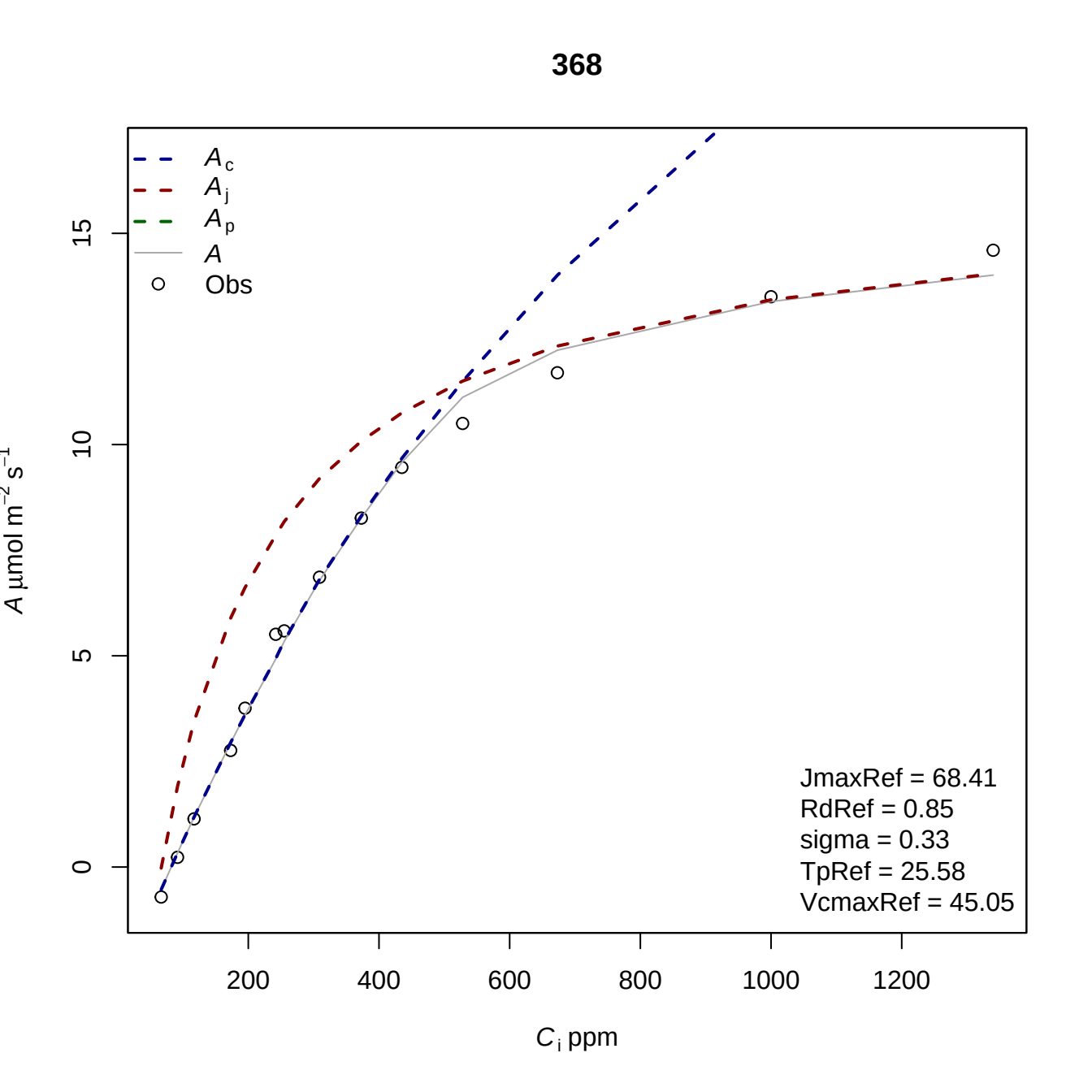

363
A_c
A_j
A_p
A
Obs
JmaxRef = 58.63
RdRef = 0.45
sigma = 0.08
TpRef = 3.84
VcmaxRef = 43.21
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
0
500
1000
1500

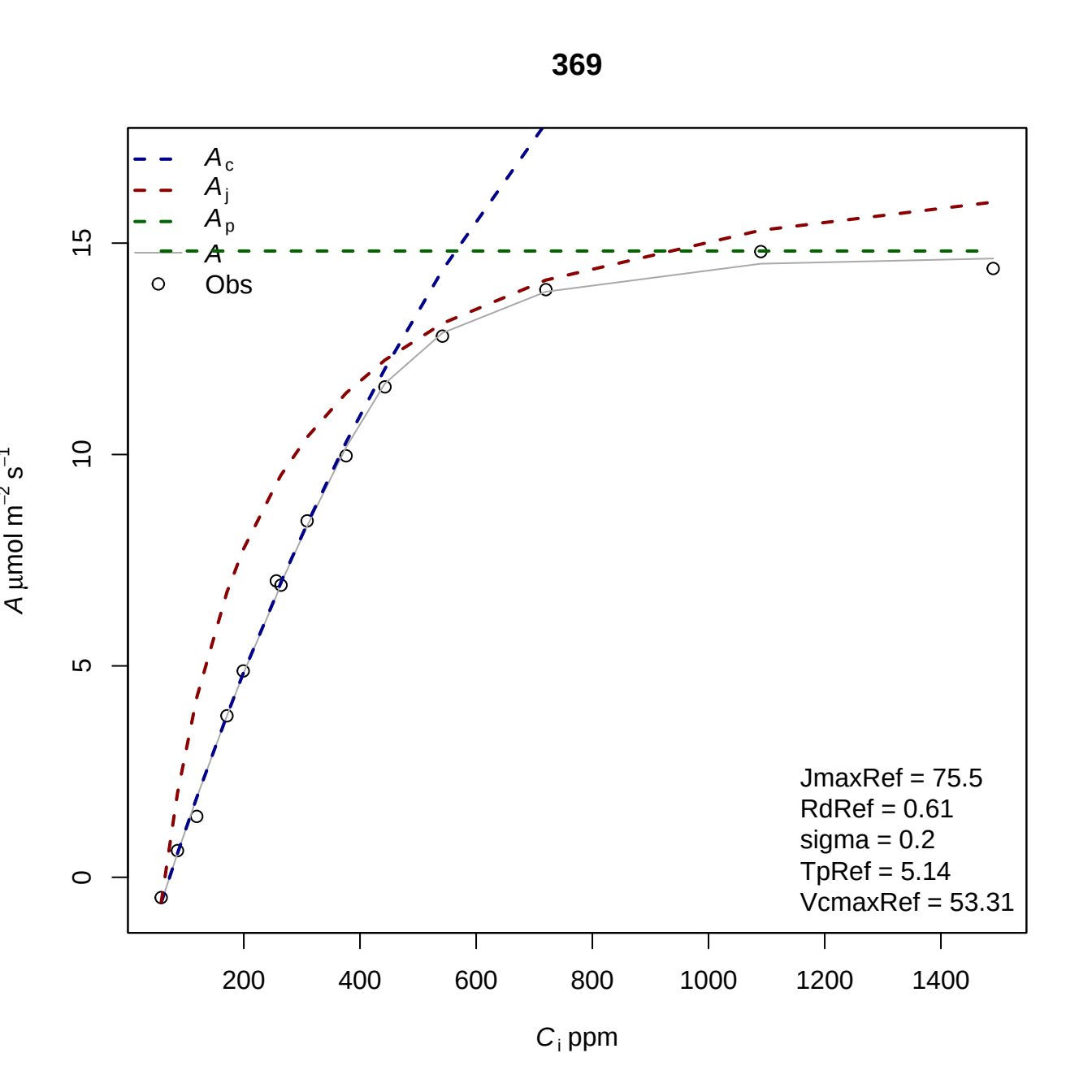

364
A_c
A_j
A_p
A
Obs
JmaxRef = 69.98
RdRef = 0.66
sigma = 0.26
TpRef = 4.39
VcmaxRef = 63.46
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
500
1000
1500

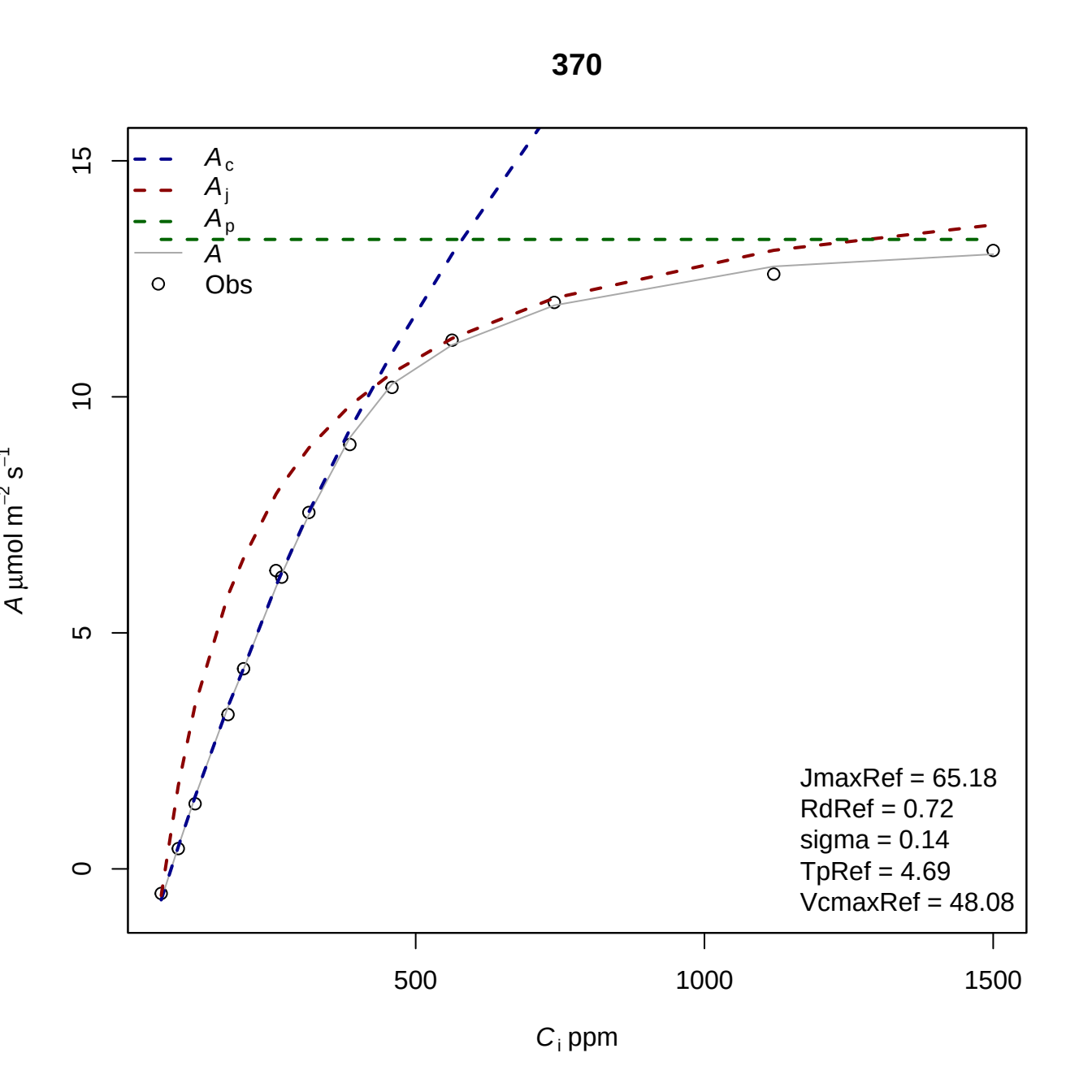

365
A_c
A_j
A_p
A
Obs
JmaxRef = 72.67
RdRef = 0.52
sigma = 0.3
TpRef = 41.84
VcmaxRef = 49.19
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
500
1000
1500

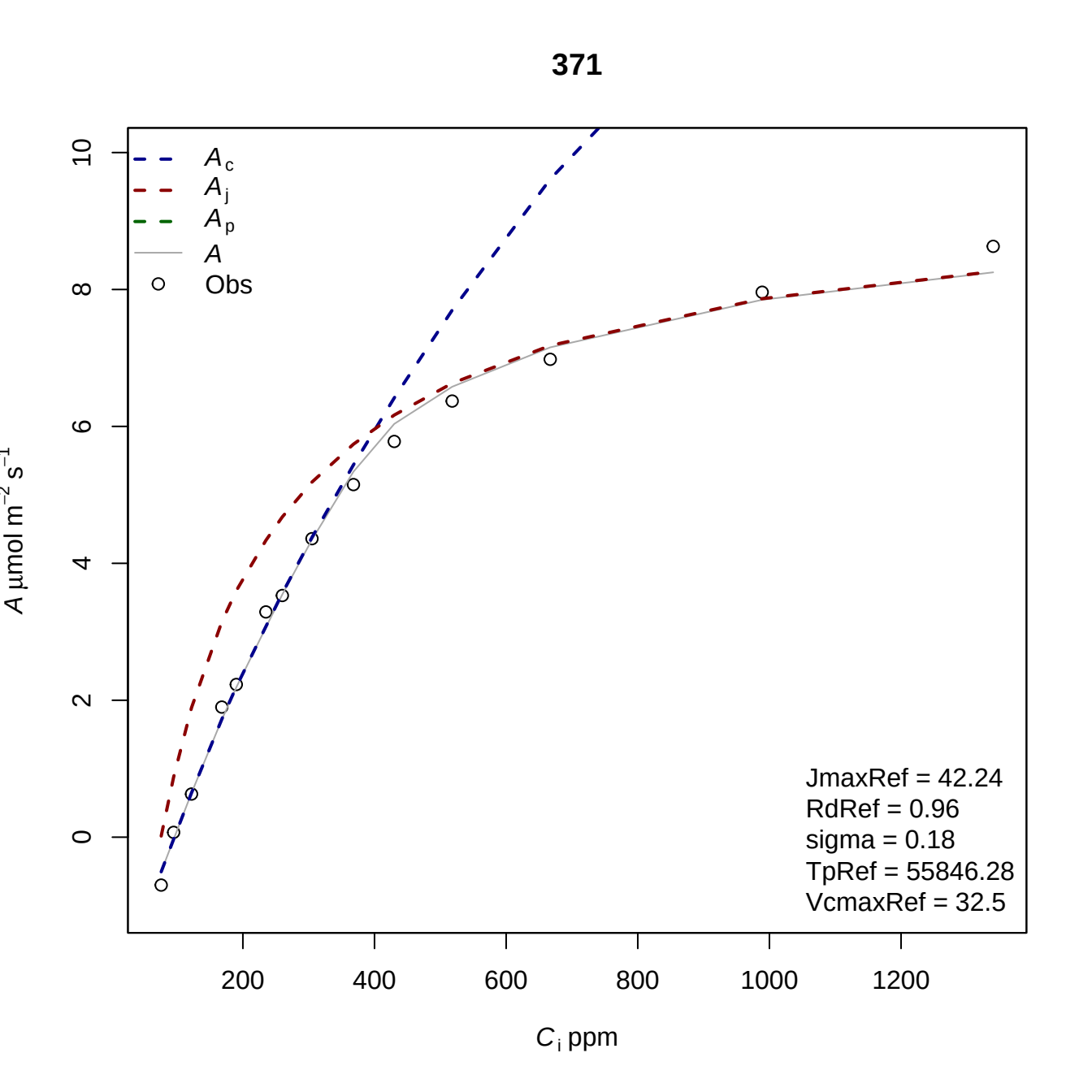

366
A_c
A_j
A_p
A
Obs
JmaxRef = 129.63
RdRef = 0.32
sigma = 0.2
TpRef = 8.28
VcmaxRef = 117.34
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
200
400
600
800
1000
1200
1400

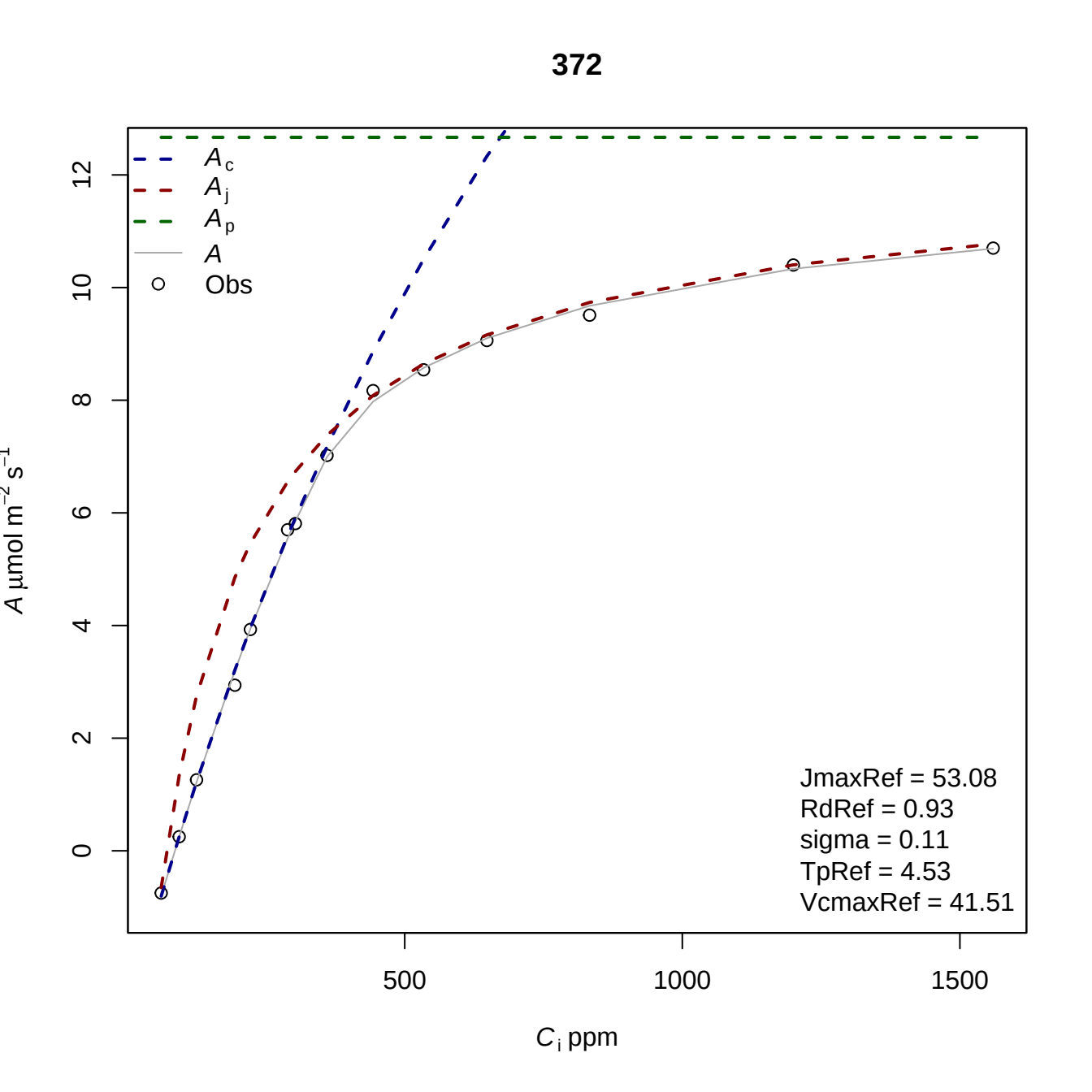

367
A_c
A_j
A_p
A
Obs
JmaxRef = 44.02
RdRef = 0.44
sigma = 0.22
TpRef = 6.58
VcmaxRef = 27.36
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
200
400
600
800
1000
1200

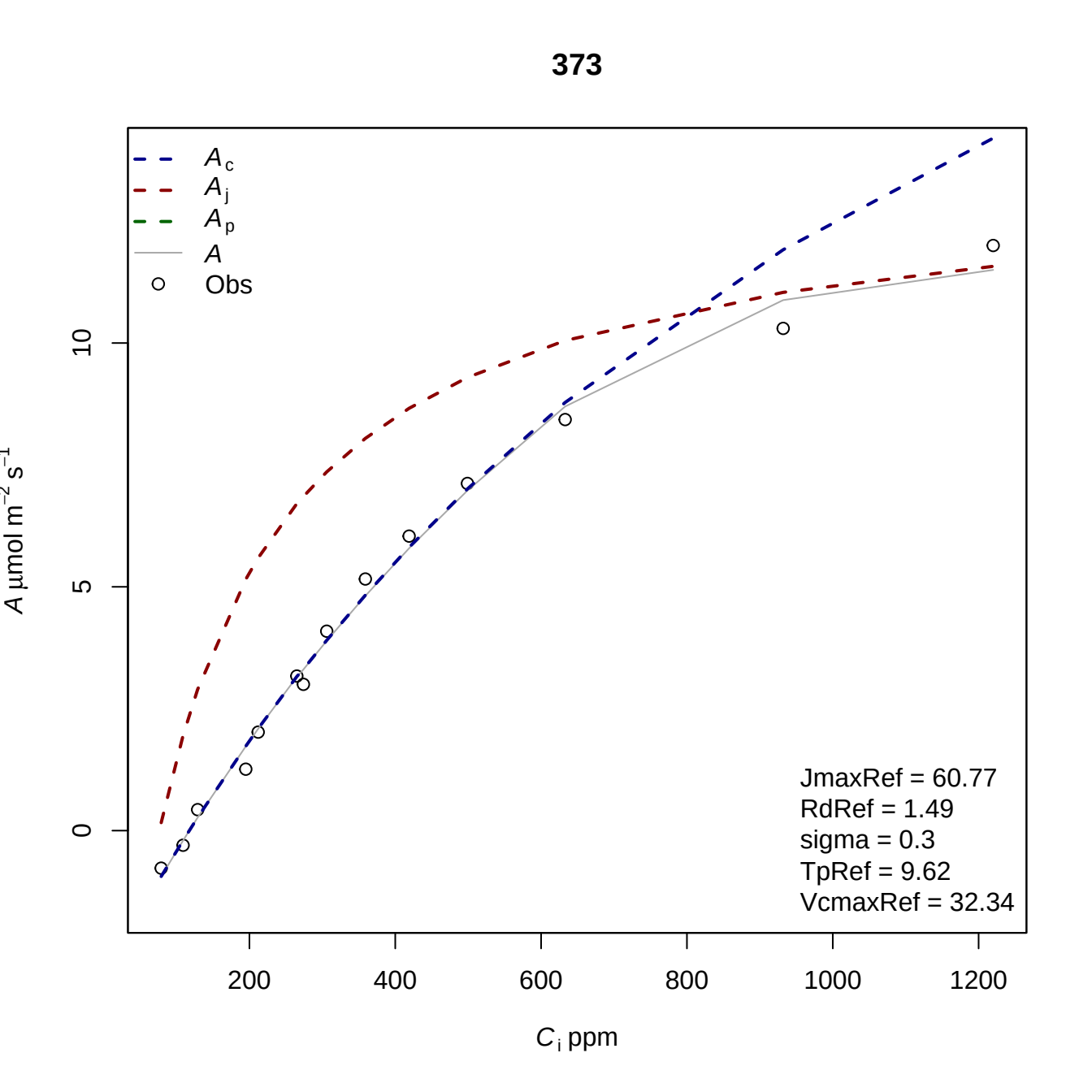

368
A_c
A_j
A_p
A
Obs
JmaxRef = 68.41
RdRef = 0.85
sigma = 0.33
TpRef = 25.58
VcmaxRef = 45.05
0
5
10
15
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

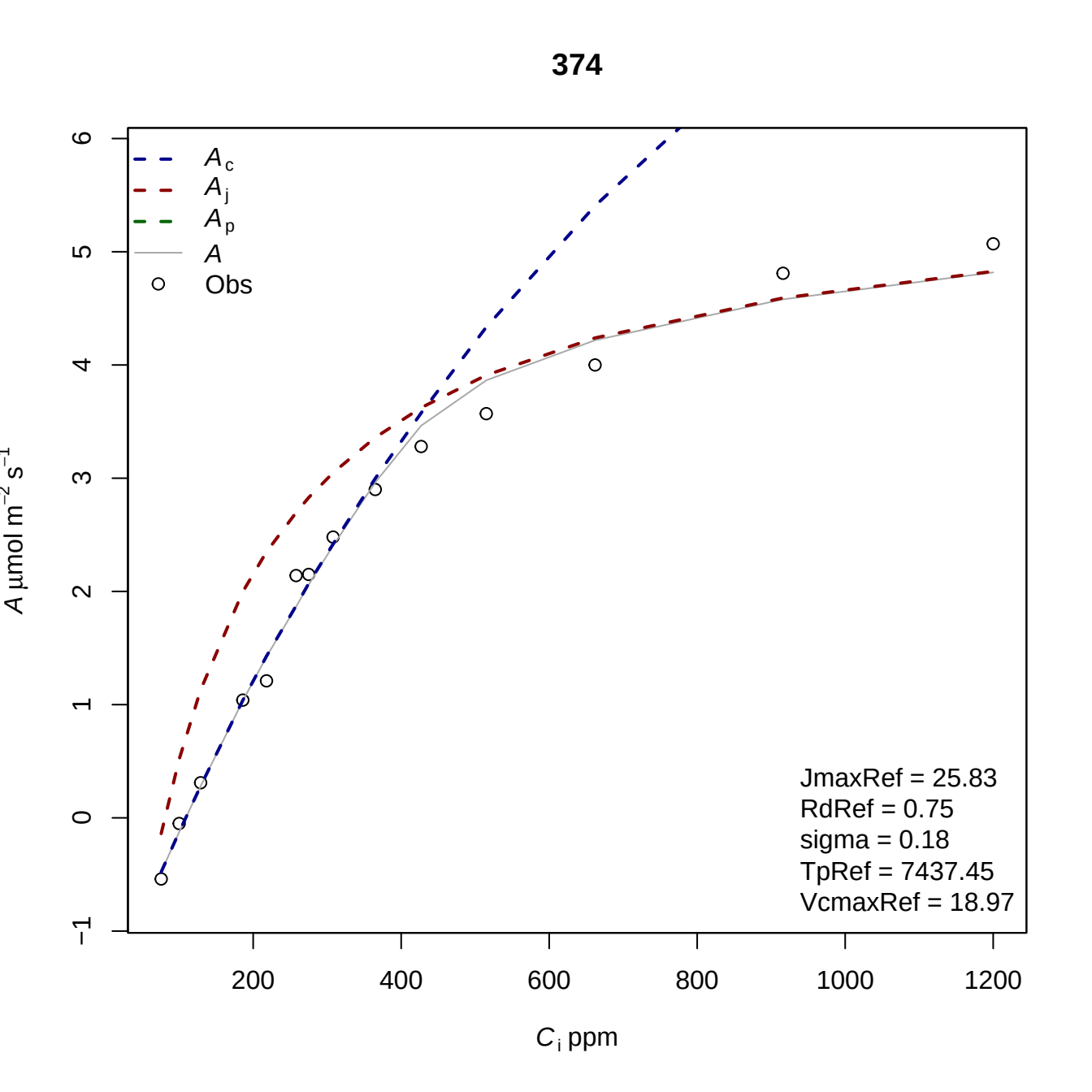

369
A_c
A_j
A_p
A
Obs
JmaxRef = 75.5
RdRef = 0.61
sigma = 0.2
TpRef = 5.14
VcmaxRef = 53.31
0
5
10
15
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
C_i ppm

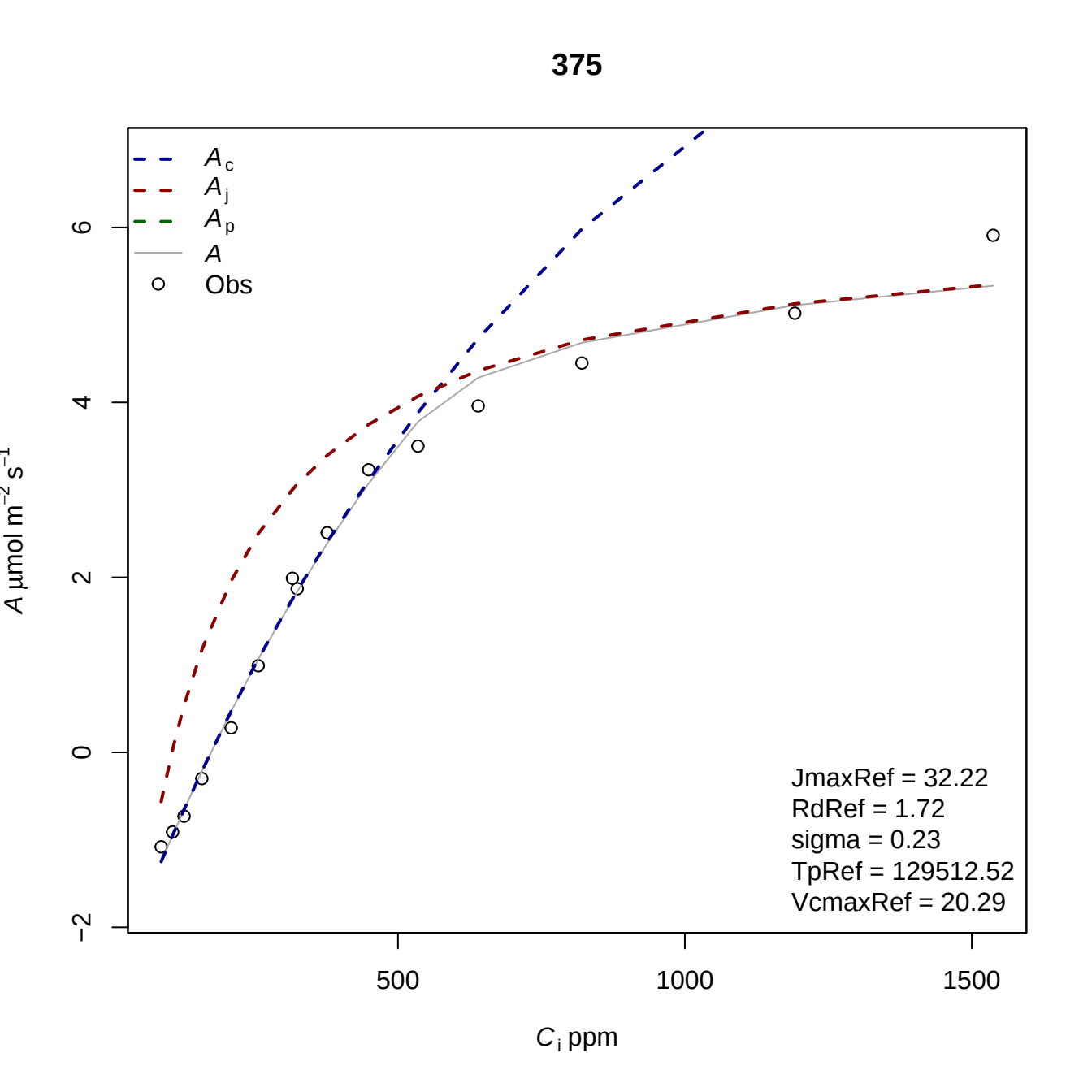

370
A_c
A_j
A_p
A
Obs
JmaxRef = 65.18
RdRef = 0.72
sigma = 0.14
TpRef = 4.69
VcmaxRef = 48.08
0
5
10
15
500
1000
1500
A µmol m^{-2} s^{-1}
C_i ppm

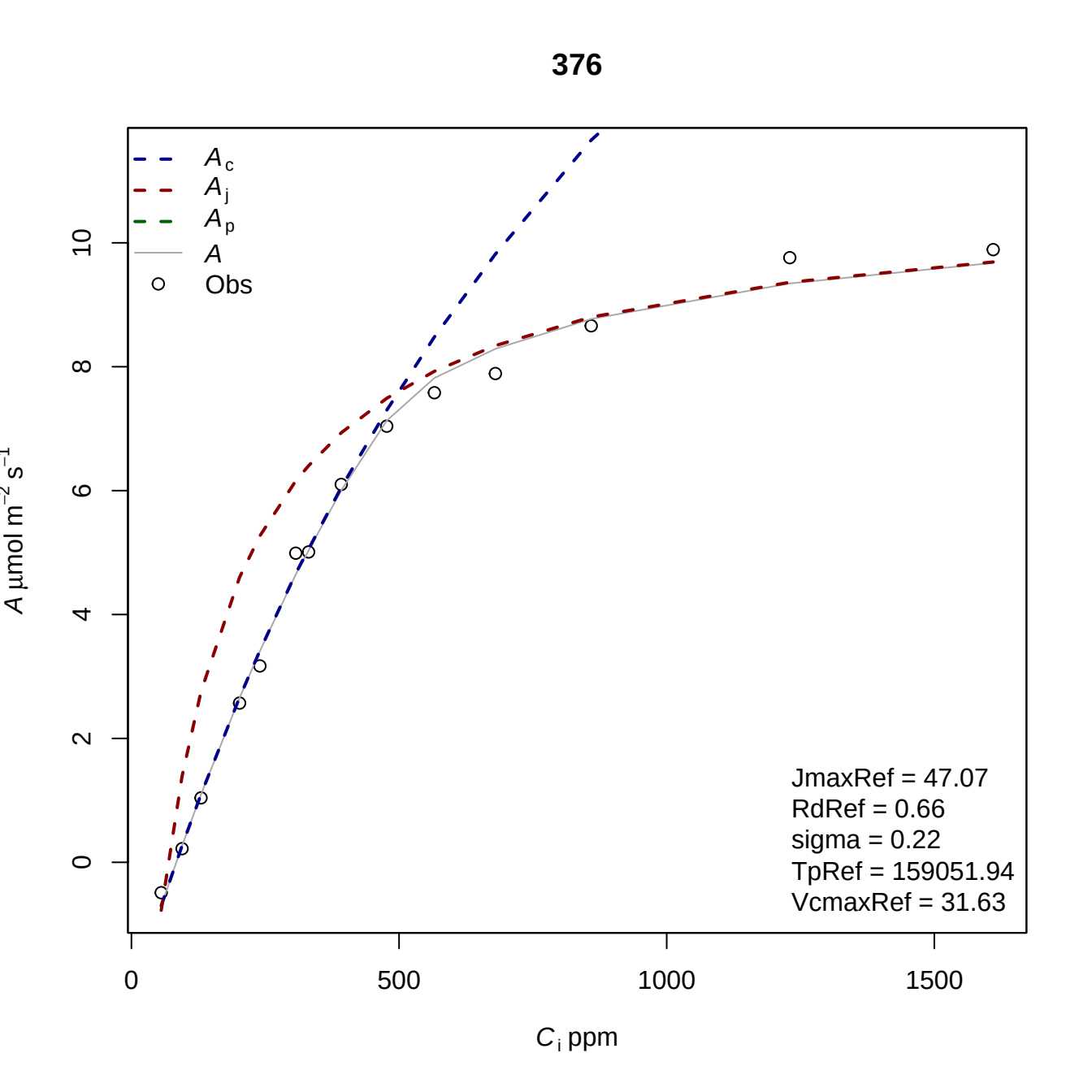

371
A_c
A_j
A_p
A
Obs
JmaxRef = 42.24
RdRef = 0.96
sigma = 0.18
TpRef = 55846.28
VcmaxRef = 32.5
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
200
400
600
800
1000
1200

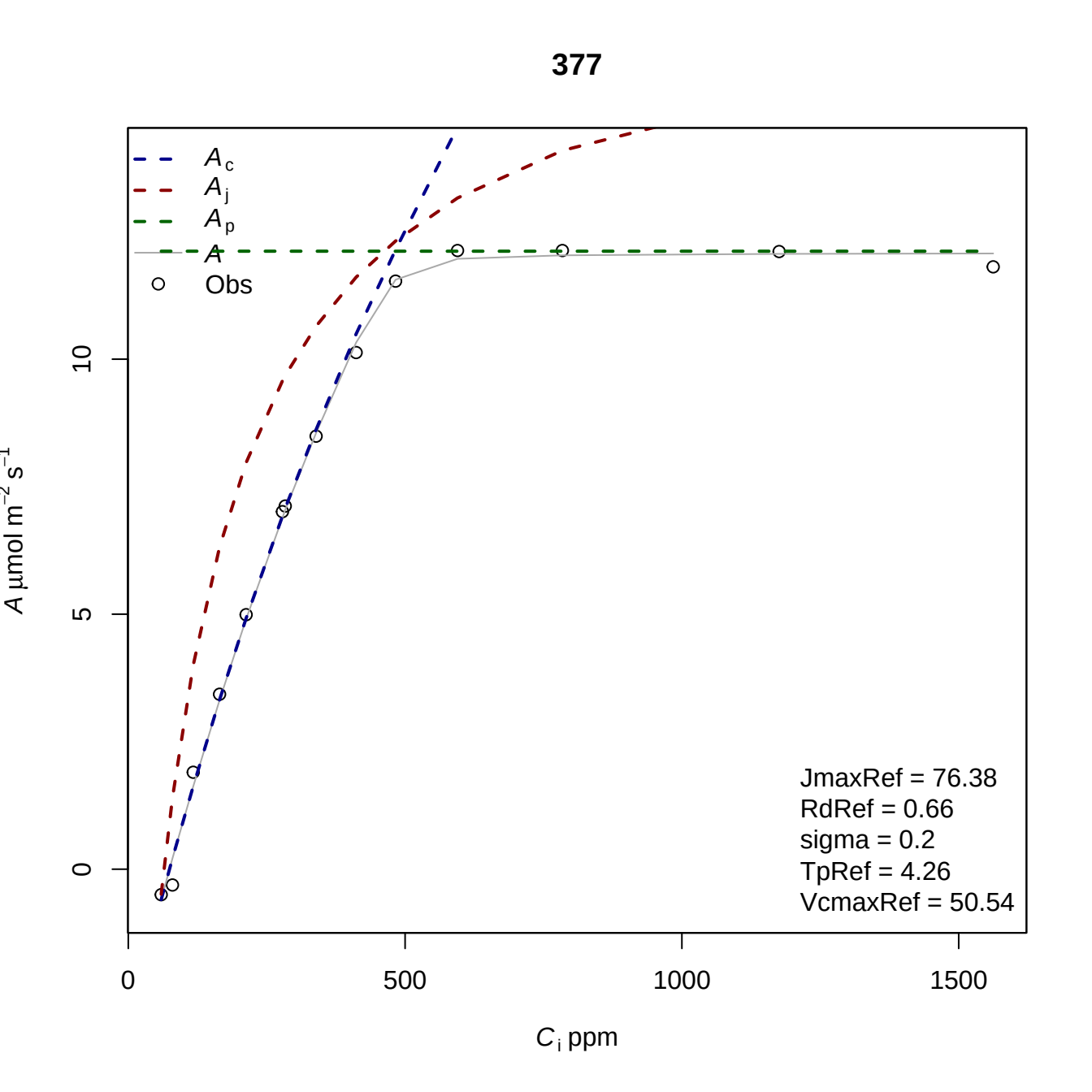

372
A_c
A_j
A_p
A
Obs
JmaxRef = 53.08
RdRef = 0.93
sigma = 0.11
TpRef = 4.53
VcmaxRef = 41.51
0
2
4
6
8
10
12
500
1000
1500
A µmol m^{-2} s^{-1}
C_i ppm

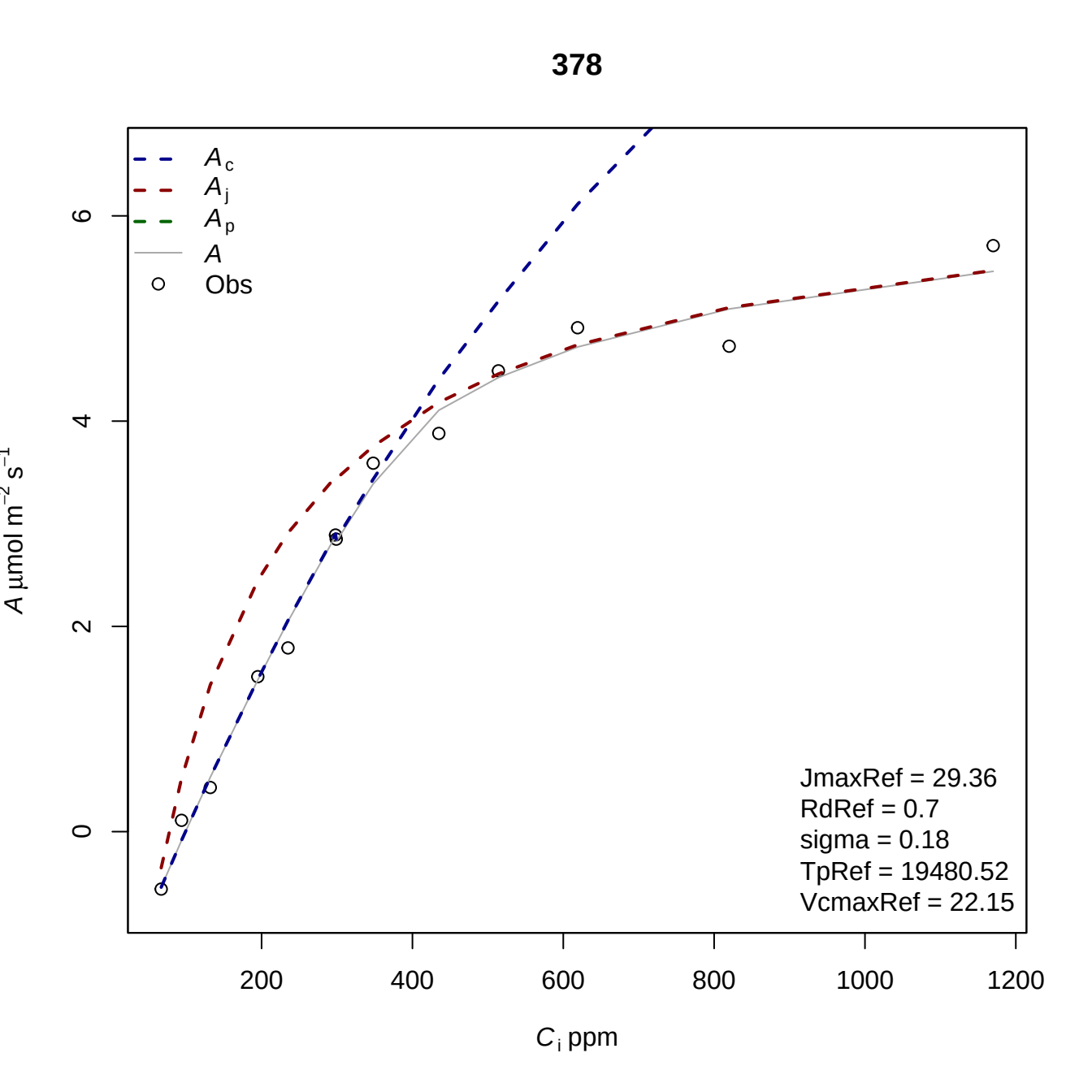

373
A_c
A_j
A_p
A
Obs
JmaxRef = 60.77
RdRef = 1.49
sigma = 0.3
TpRef = 9.62
VcmaxRef = 32.34
0
5
10
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

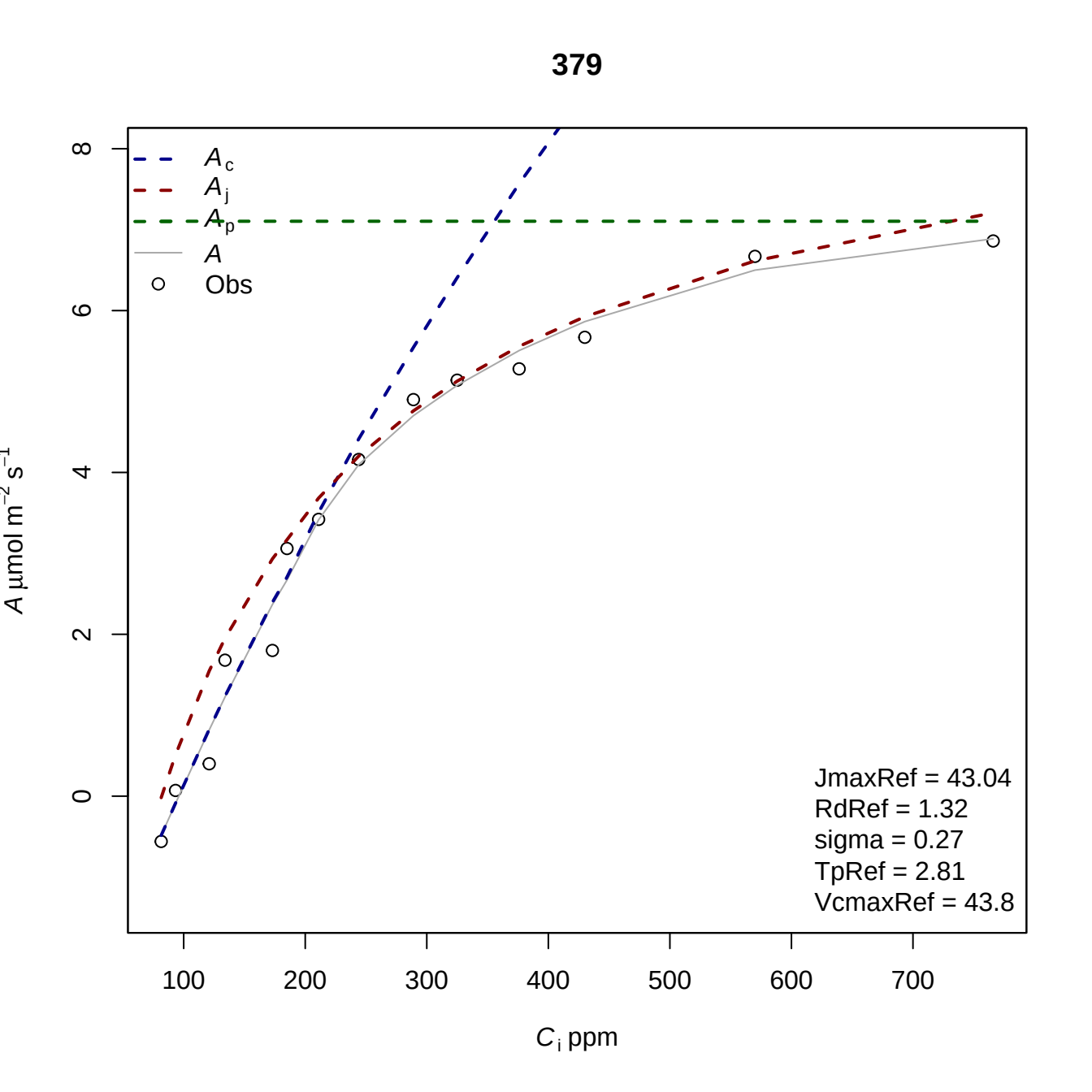

374
A_c
A_j
A_p
A
Obs
JmaxRef = 25.83
RdRef = 0.75
sigma = 0.18
TpRef = 7437.45
VcmaxRef = 18.97
A µmol m^{-2} s^{-1}
C_i ppm
−1
0
1
2
3
4
5
6
200
400
600
800
1000
1200

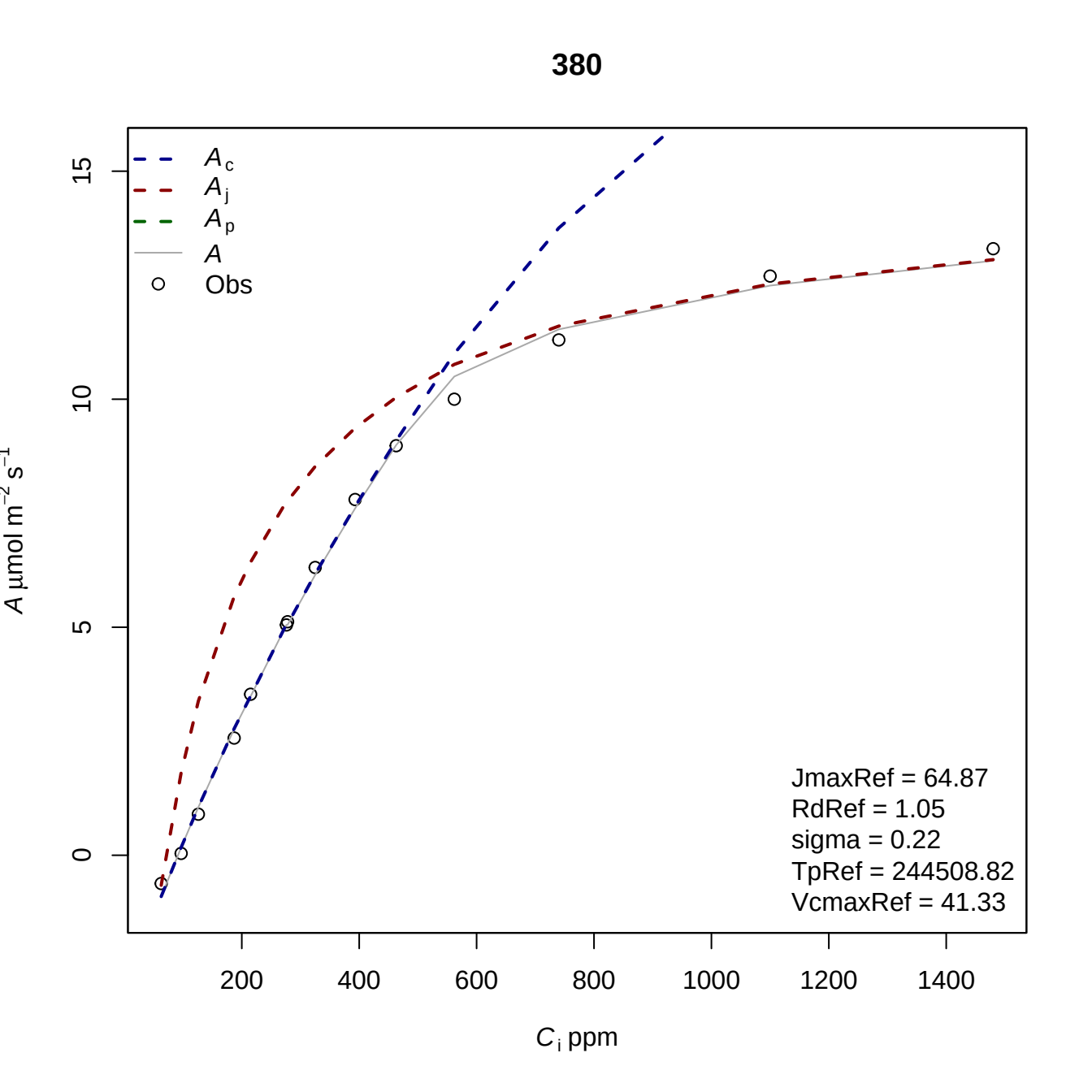

375
A_c
A_j
A_p
A
Obs
JmaxRef = 32.22
RdRef = 1.72
sigma = 0.23
TpRef = 129512.52
VcmaxRef = 20.29
A µmol m^{-2} s^{-1}
C_i ppm
−2
0
2
4
6
500
1000
1500

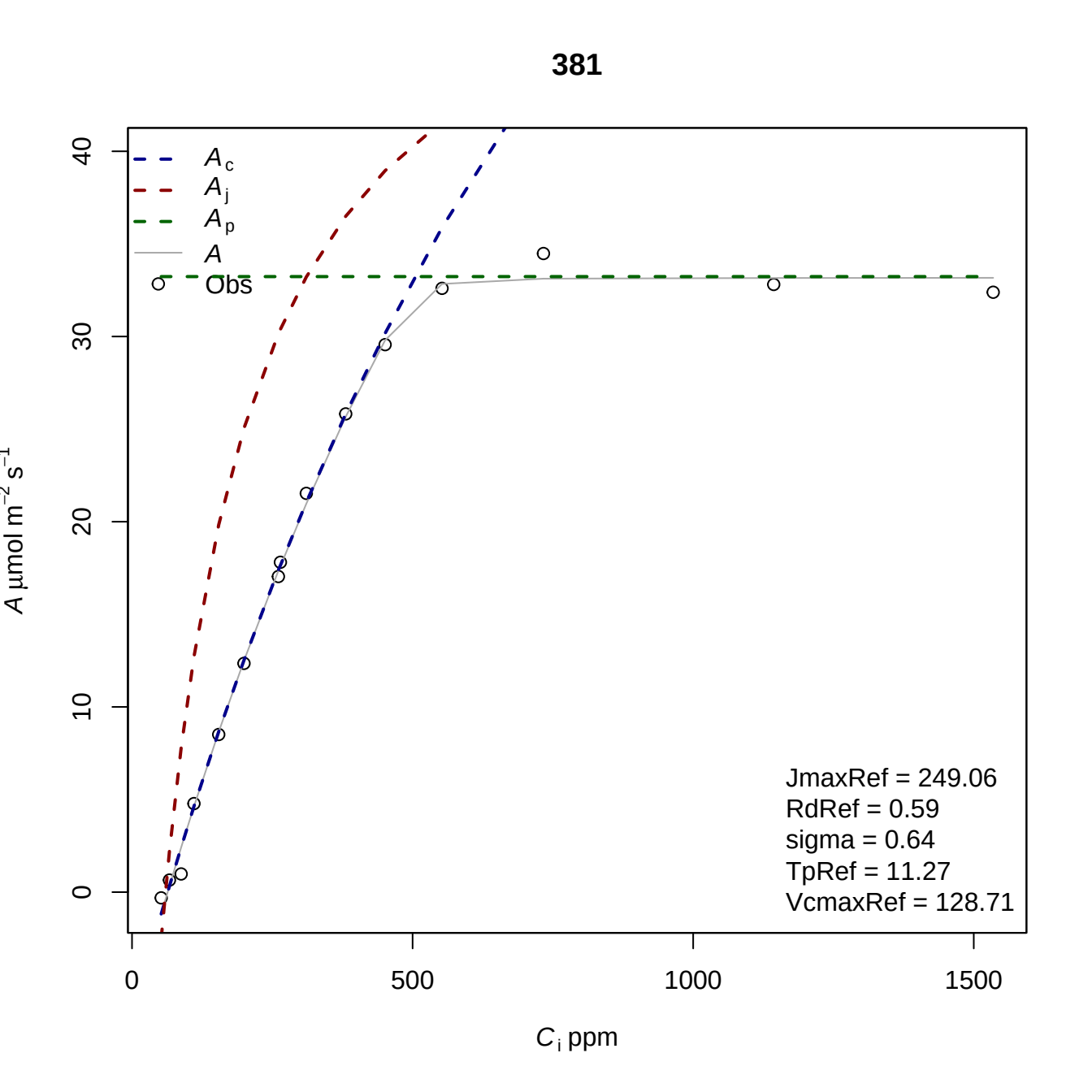

376
A_c
A_j
A_p
A
Obs
JmaxRef = 47.07
RdRef = 0.66
sigma = 0.22
TpRef = 159051.94
VcmaxRef = 31.63
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
0
500
1000
1500

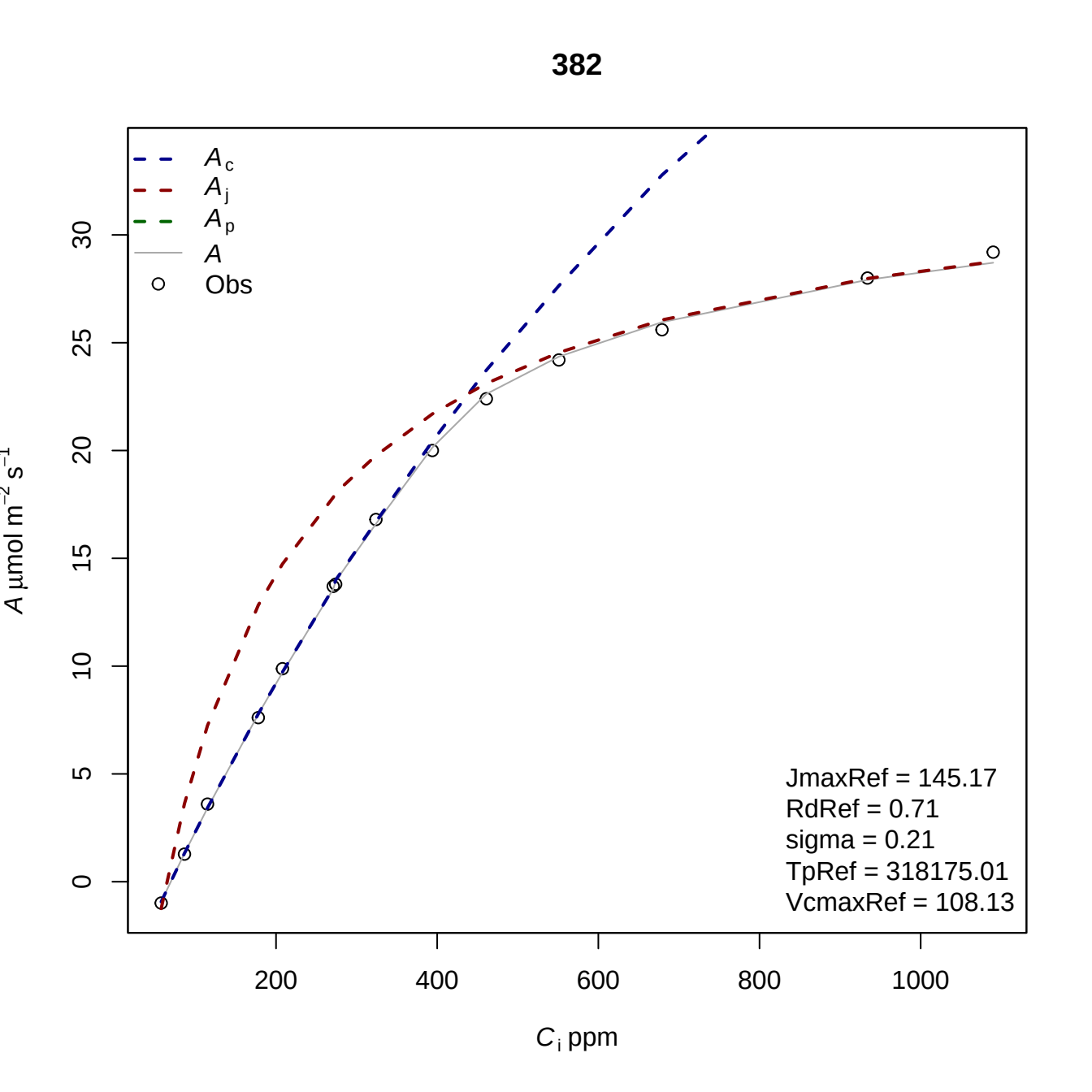

377
A_c
A_j
A_p
A
Obs
JmaxRef = 76.38
RdRef = 0.66
sigma = 0.2
TpRef = 4.26
VcmaxRef = 50.54
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
C_i ppm

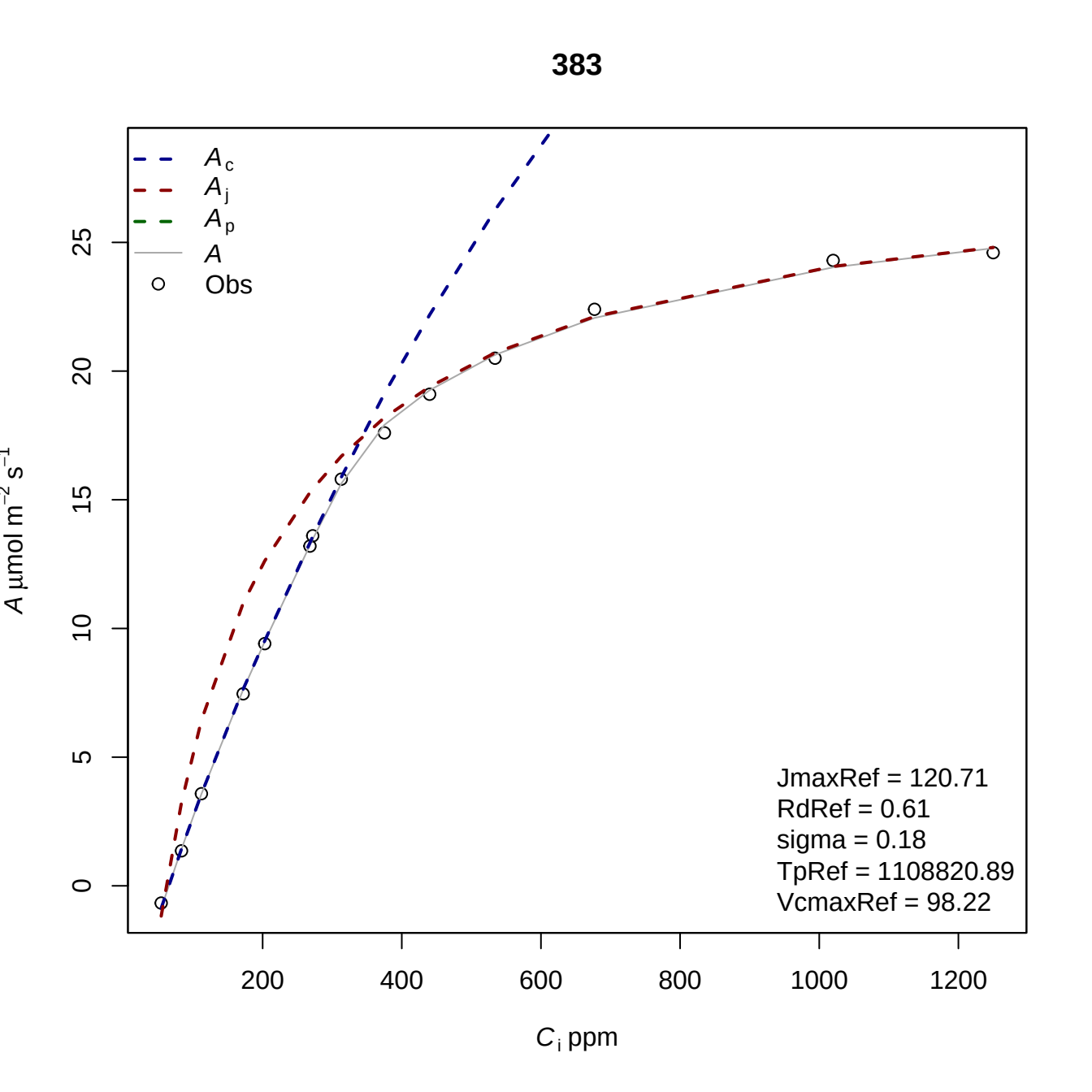

378
A_c
A_j
A_p
A
Obs
JmaxRef = 29.36
RdRef = 0.7
sigma = 0.18
TpRef = 19480.52
VcmaxRef = 22.15
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
200
400
600
800
1000
1200

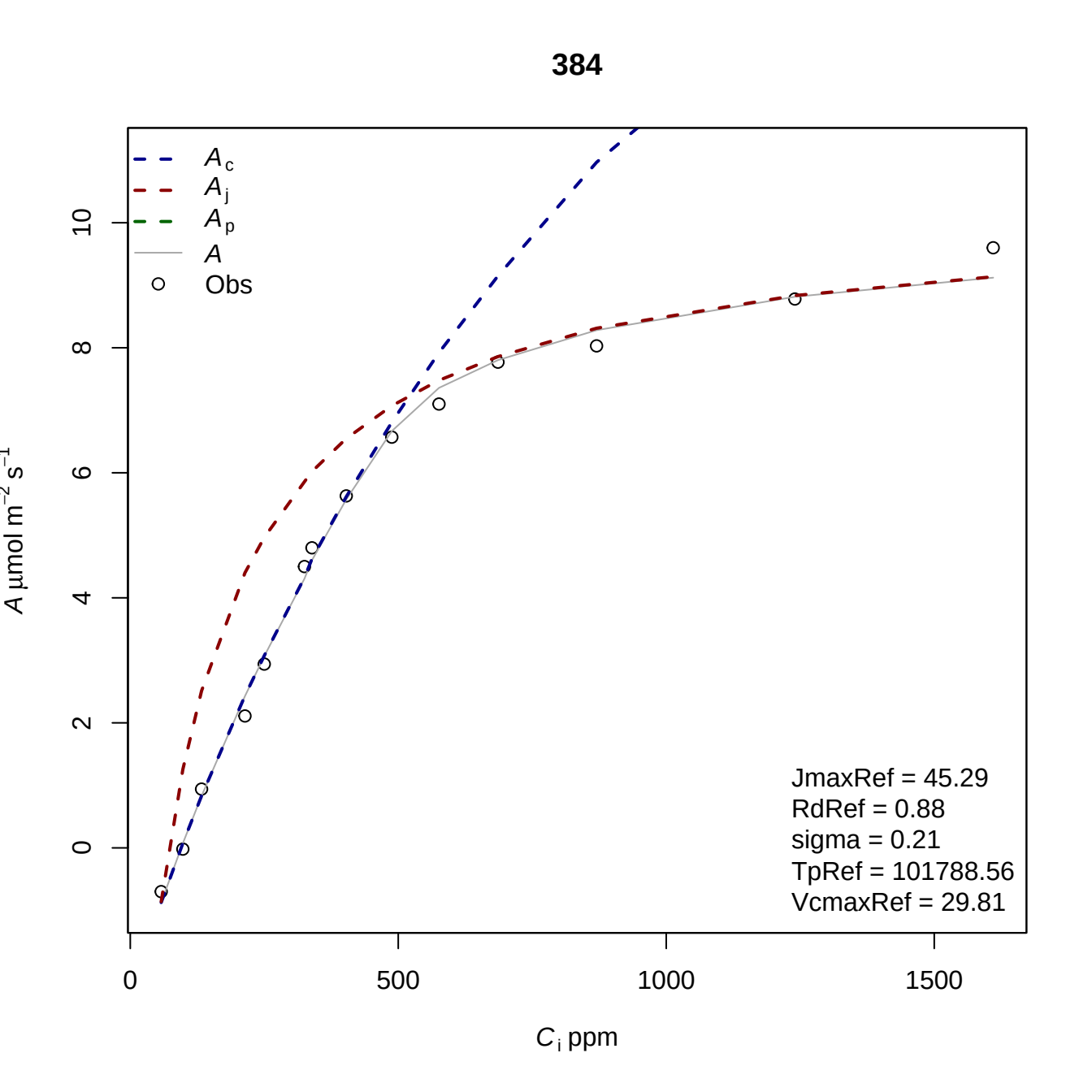

379
A_c
A_j
A_p
A
Obs
JmaxRef = 43.04
RdRef = 1.32
sigma = 0.27
TpRef = 2.81
VcmaxRef = 43.8
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
100
200
300
400
500
600
700

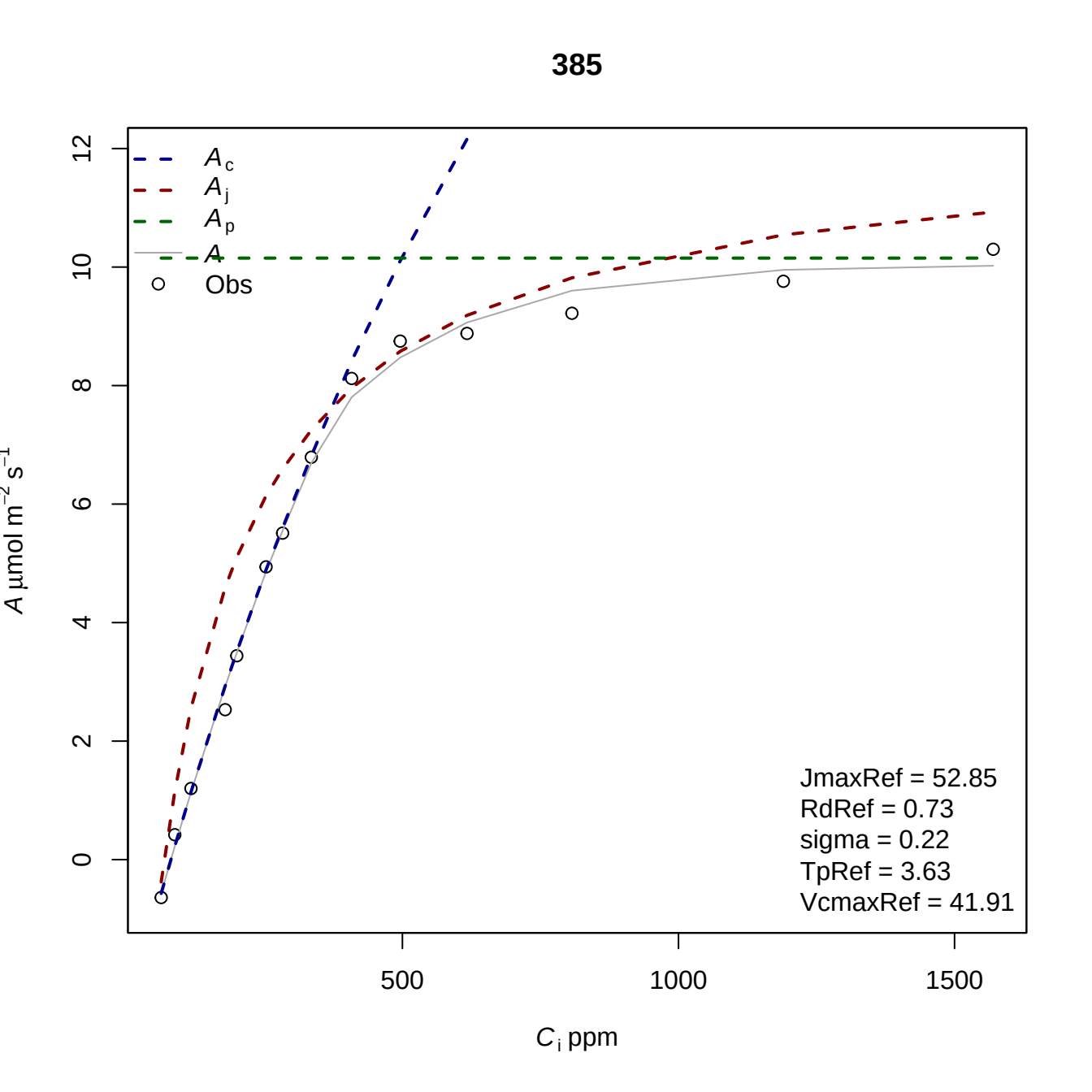

380
A_c
A_j
A_p
A
Obs
JmaxRef = 64.87
RdRef = 1.05
sigma = 0.22
TpRef = 244508.82
VcmaxRef = 41.33
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
200
400
600
800
1000
1200
1400

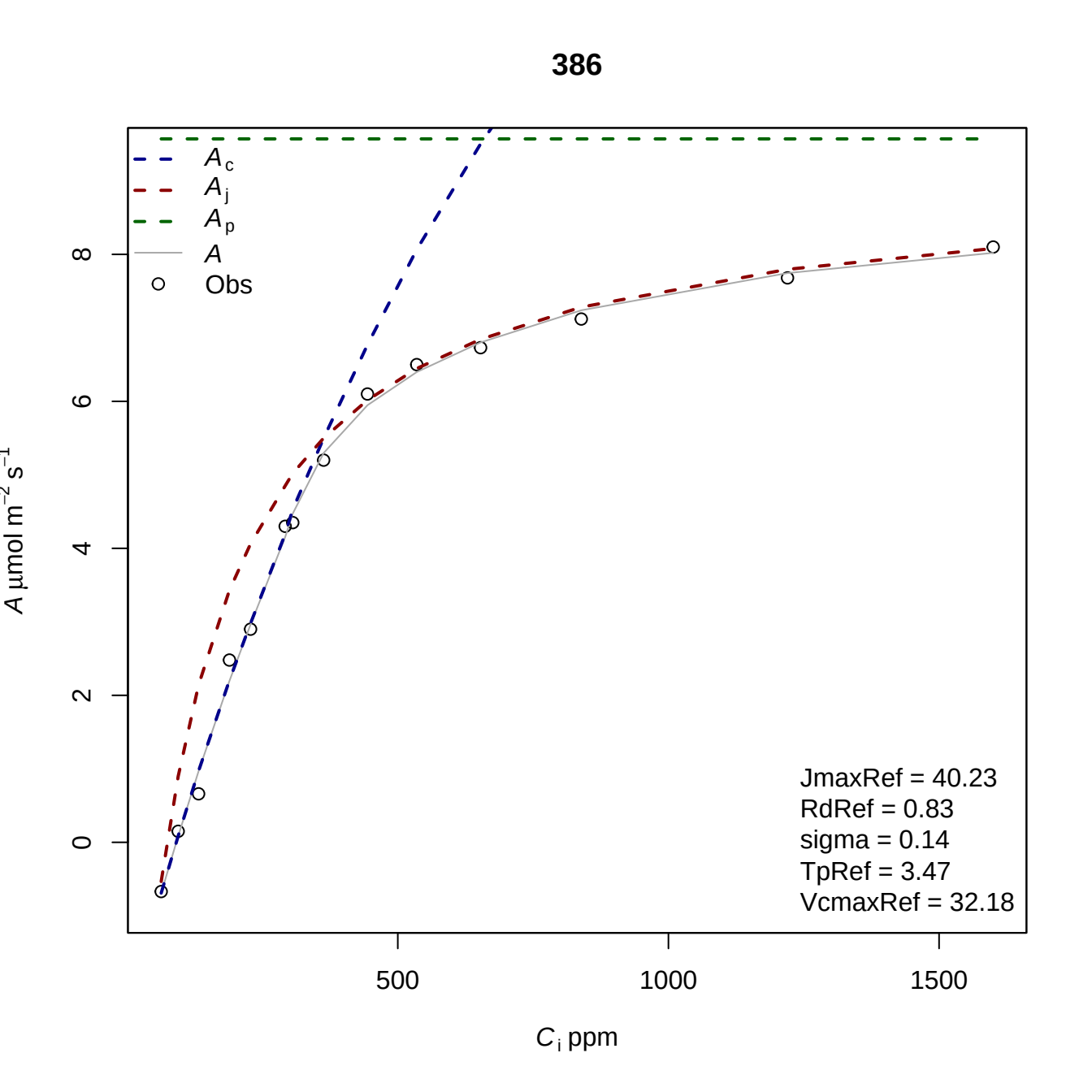

381
A_c
A_j
A_p
A
Obs
JmaxRef = 249.06
RdRef = 0.59
sigma = 0.64
TpRef = 11.27
VcmaxRef = 128.71
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
0
500
1000
1500

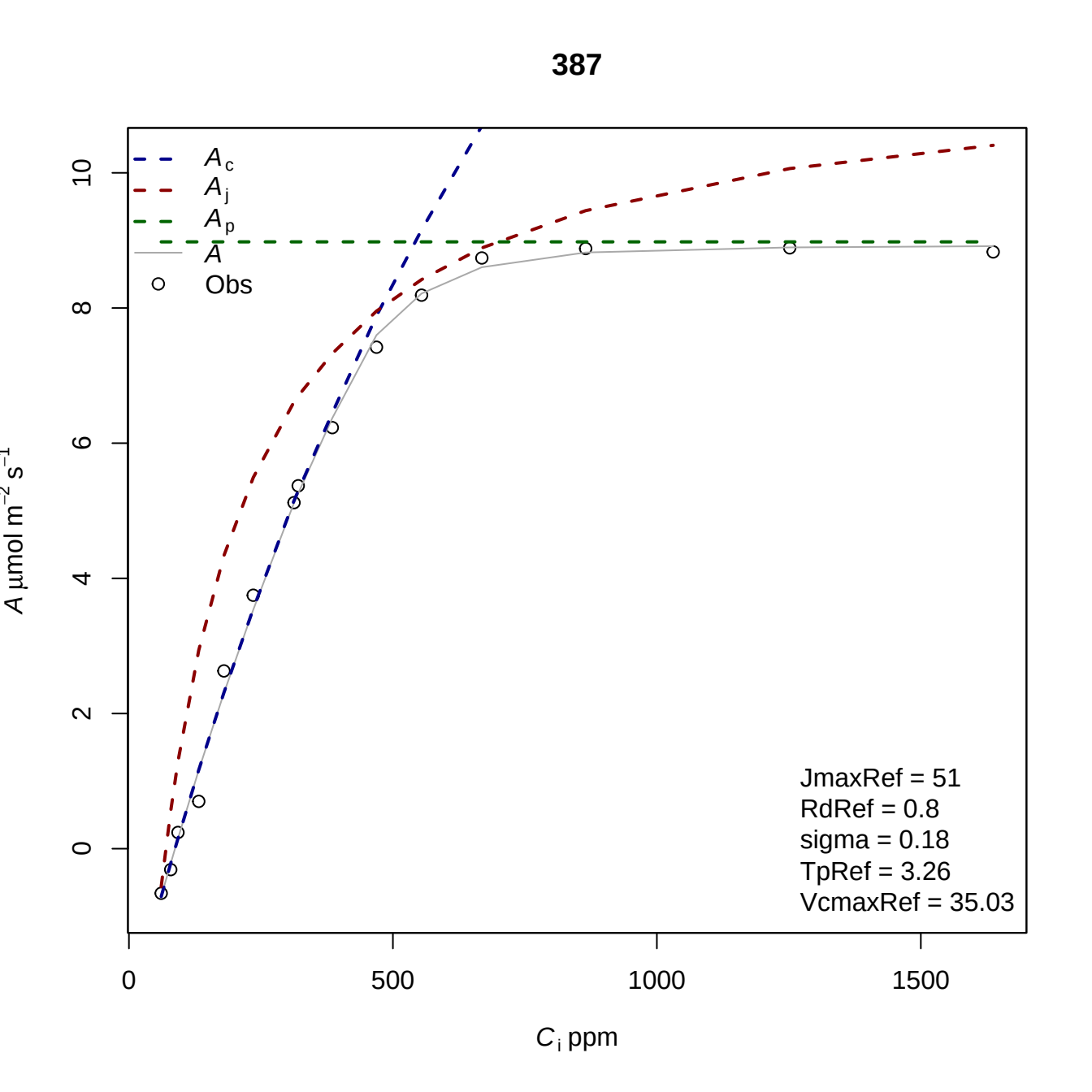

382
A_c
A_j
A_p
A
Obs
JmaxRef = 145.17
RdRef = 0.71
sigma = 0.21
TpRef = 318175.01
VcmaxRef = 108.13
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
200
400
600
800
1000

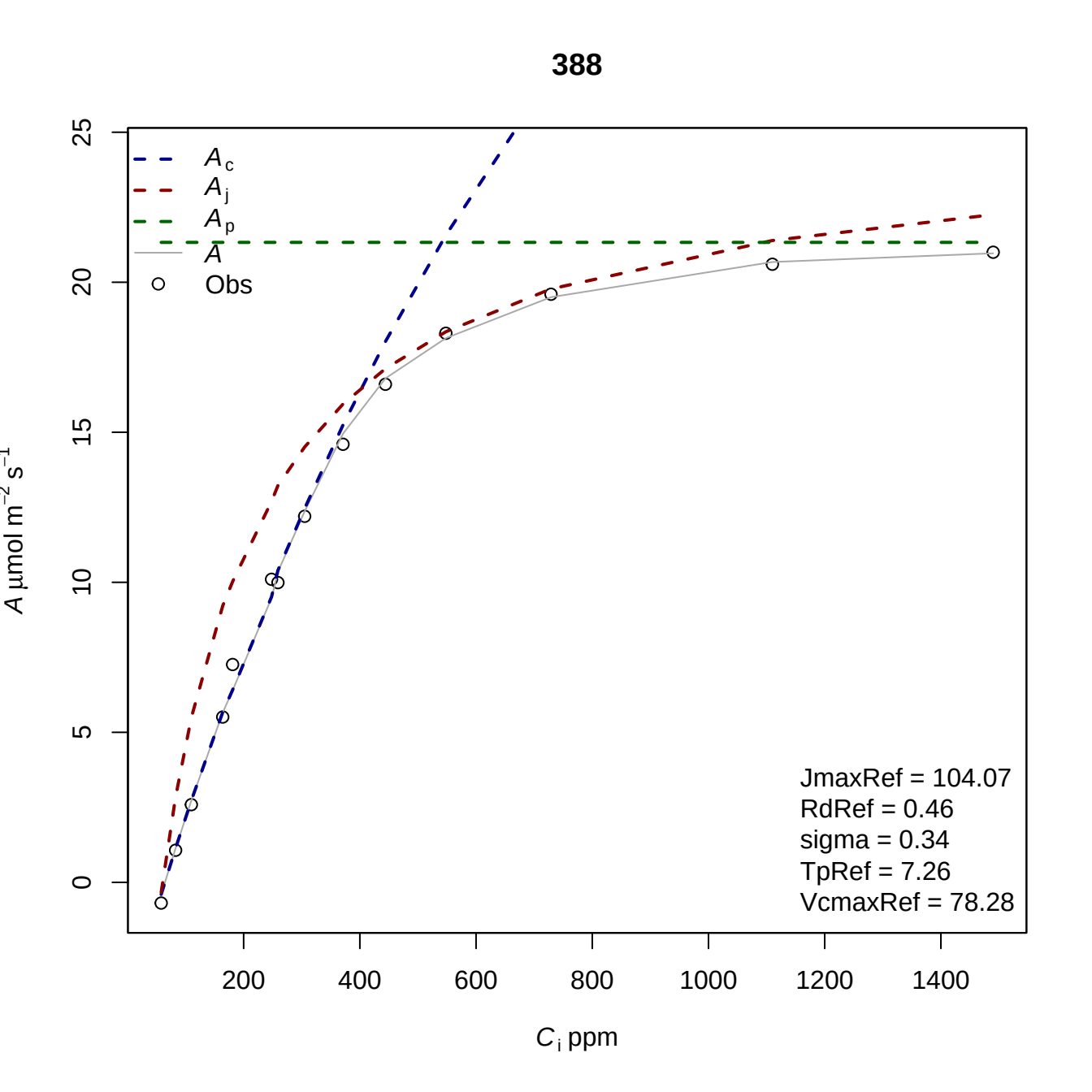

383
A_c
A_j
A_p
A
Obs
JmaxRef = 120.71
RdRef = 0.61
sigma = 0.18
TpRef = 1108820.89
VcmaxRef = 98.22
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
200
400
600
800
1000
1200

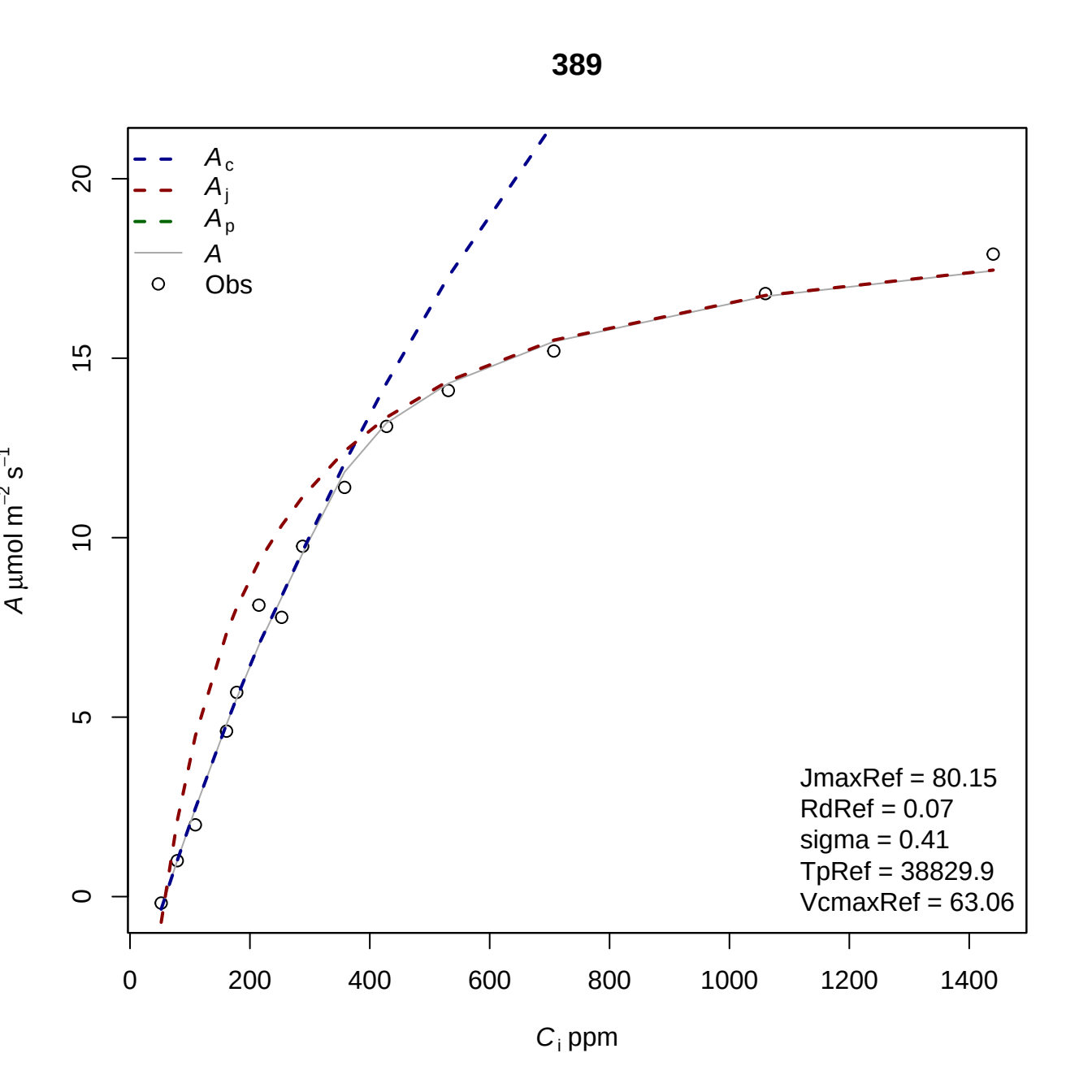

384
A_c
A_j
A_p
A
Obs
JmaxRef = 45.29
RdRef = 0.88
sigma = 0.21
TpRef = 101788.56
VcmaxRef = 29.81
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
0
500
1000
1500

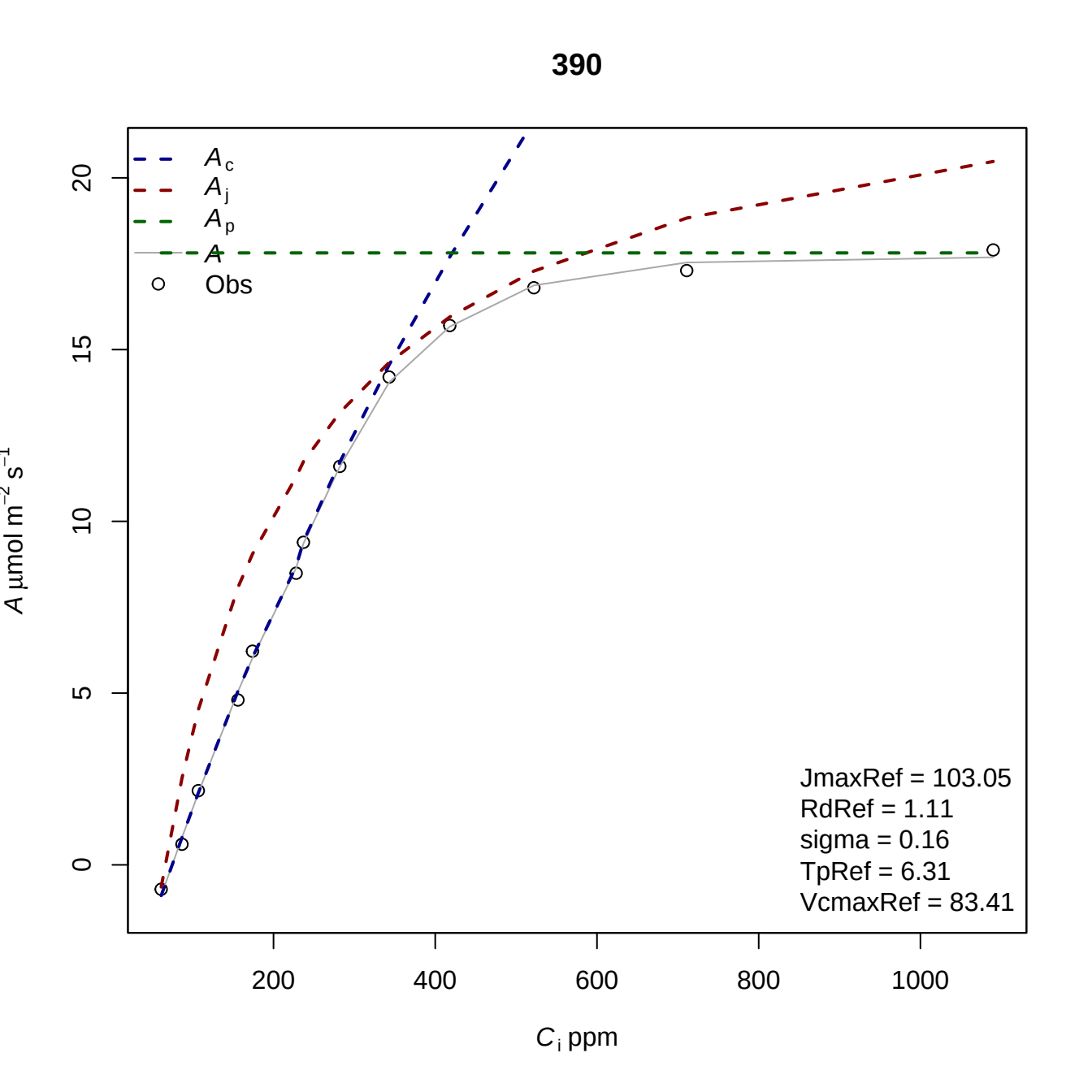

385
A_c
A_j
A_p
A
Obs
JmaxRef = 52.85
RdRef = 0.73
sigma = 0.22
TpRef = 3.63
VcmaxRef = 41.91
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
500
1000
1500

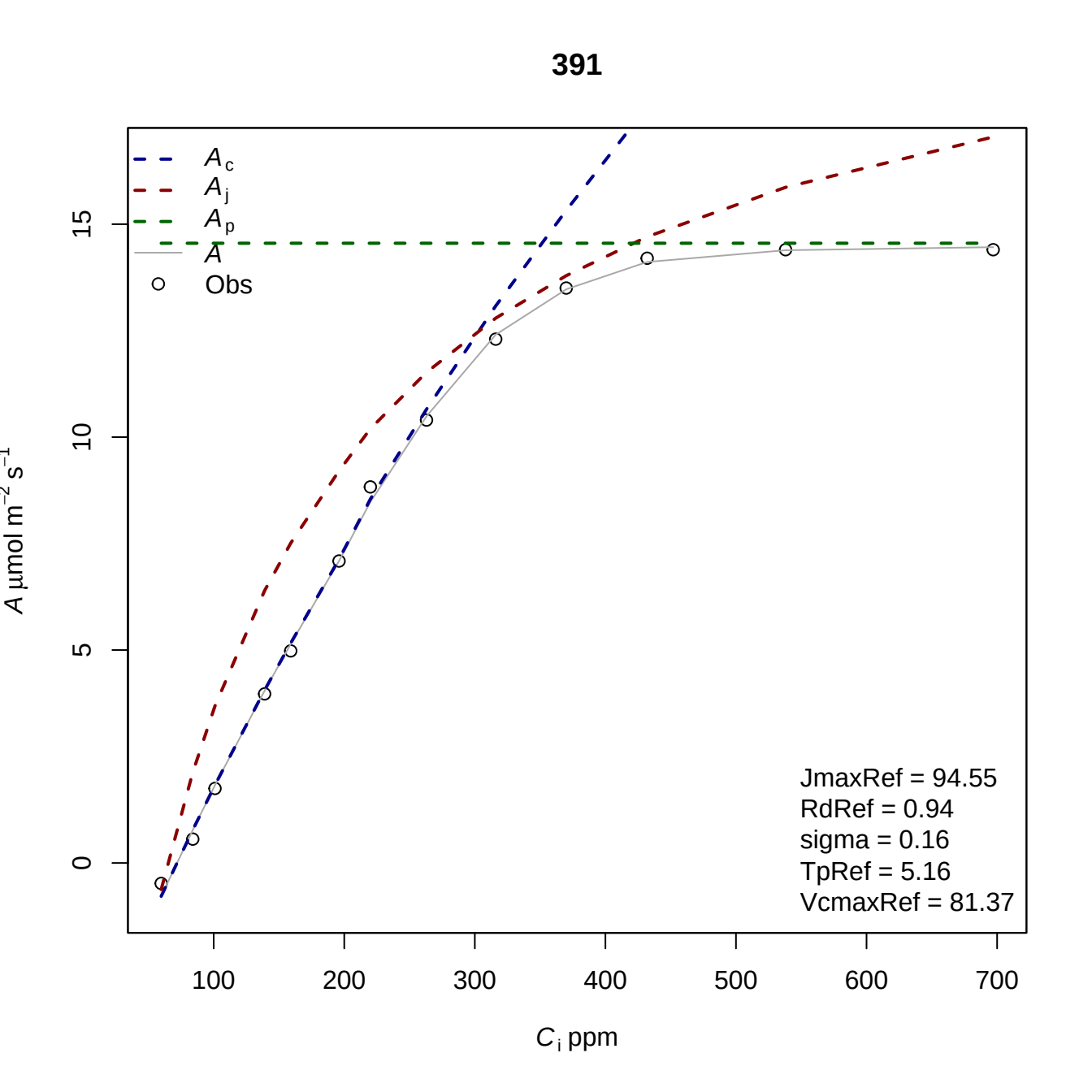

386
A_c
A_j
A_p
A
Obs
JmaxRef = 40.23
RdRef = 0.83
sigma = 0.14
TpRef = 3.47
VcmaxRef = 32.18
0
2
4
6
8
500
1000
1500
A µmol m^{-2} s^{-1}
C_i ppm

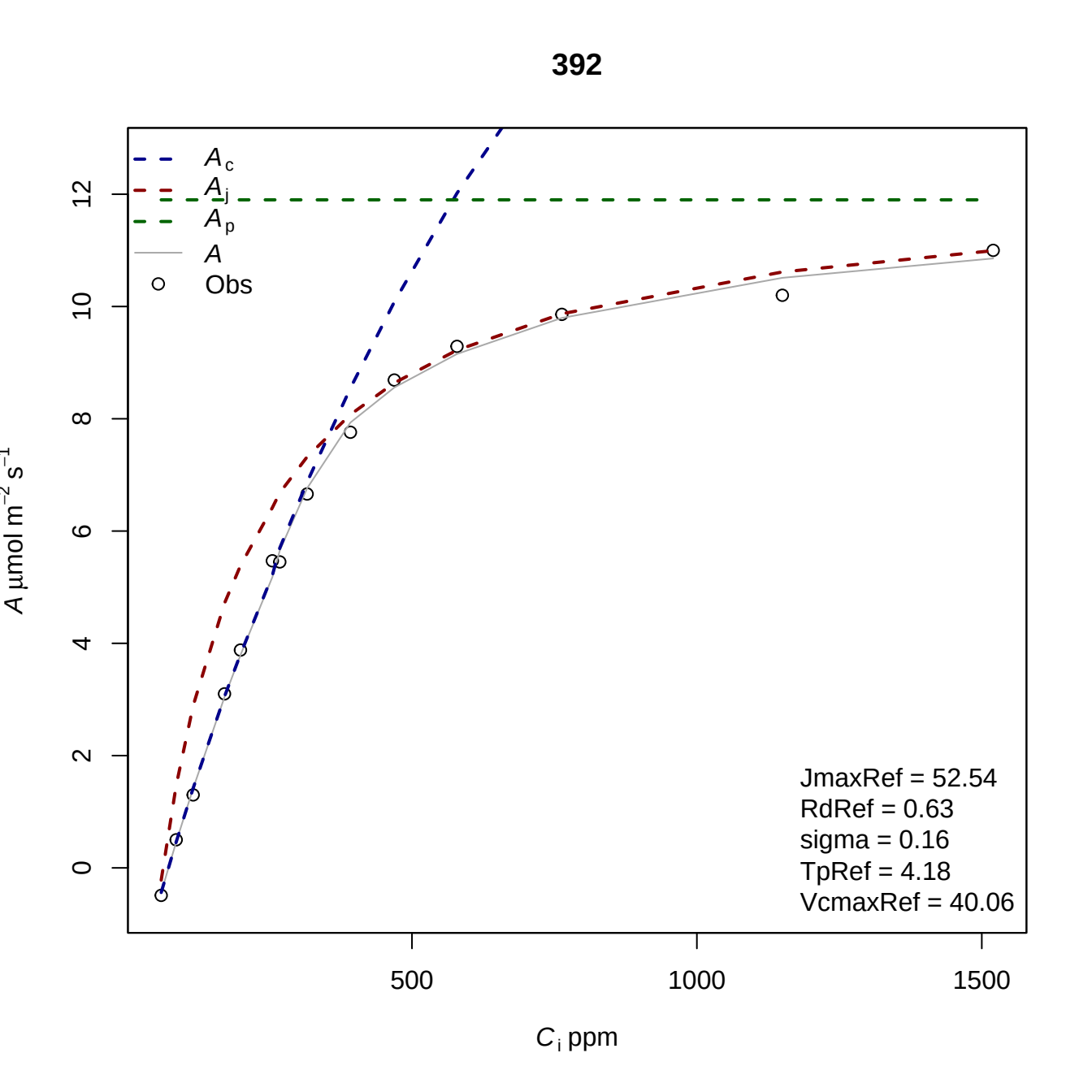

387
A_c
A_j
A_p
A
Obs
JmaxRef = 51
RdRef = 0.8
sigma = 0.18
TpRef = 3.26
VcmaxRef = 35.03
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
0
500
1000
1500

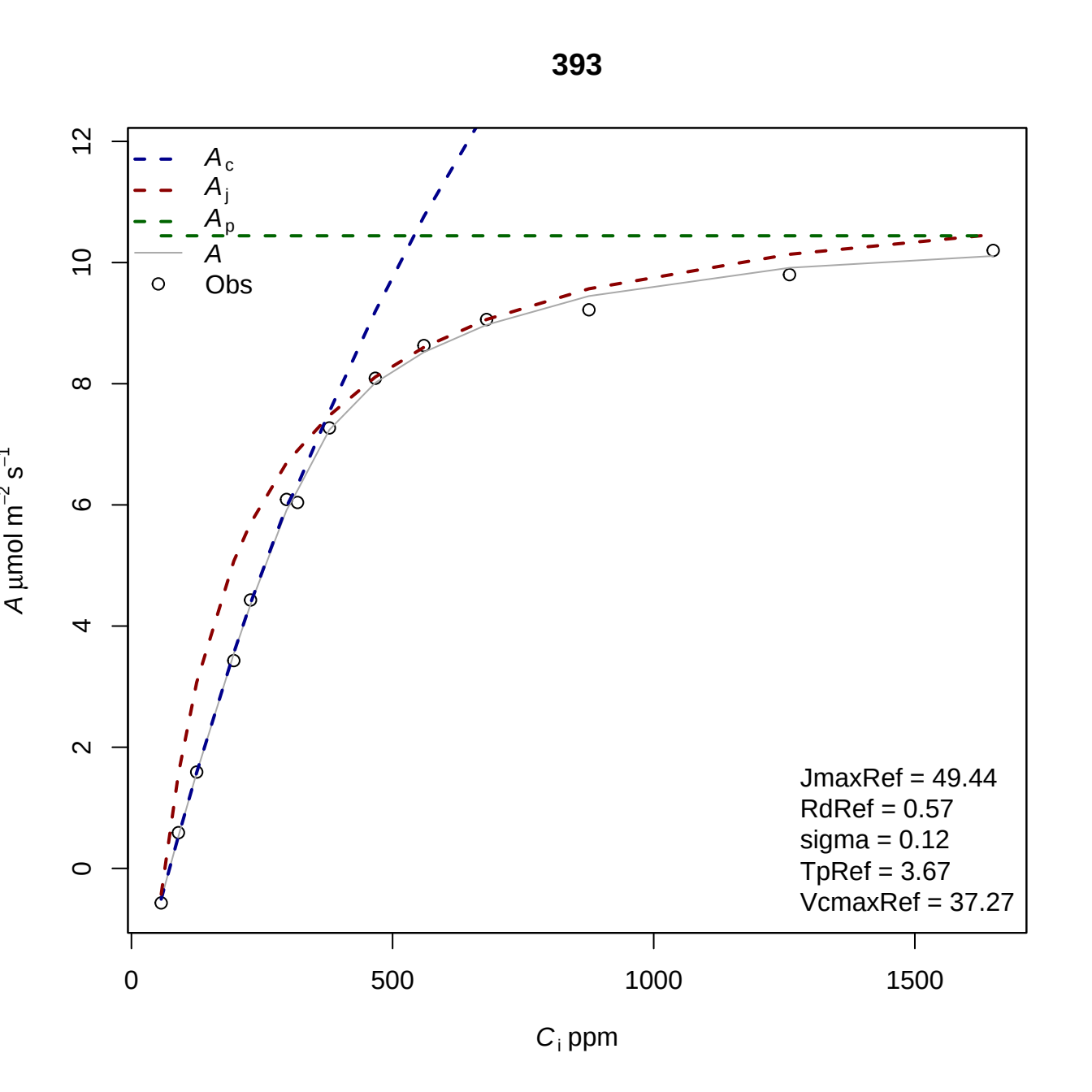

388
A_c
A_j
A_p
A
Obs
JmaxRef = 104.07
RdRef = 0.46
sigma = 0.34
TpRef = 7.26
VcmaxRef = 78.28
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
200
400
600
800
1000
1200
1400

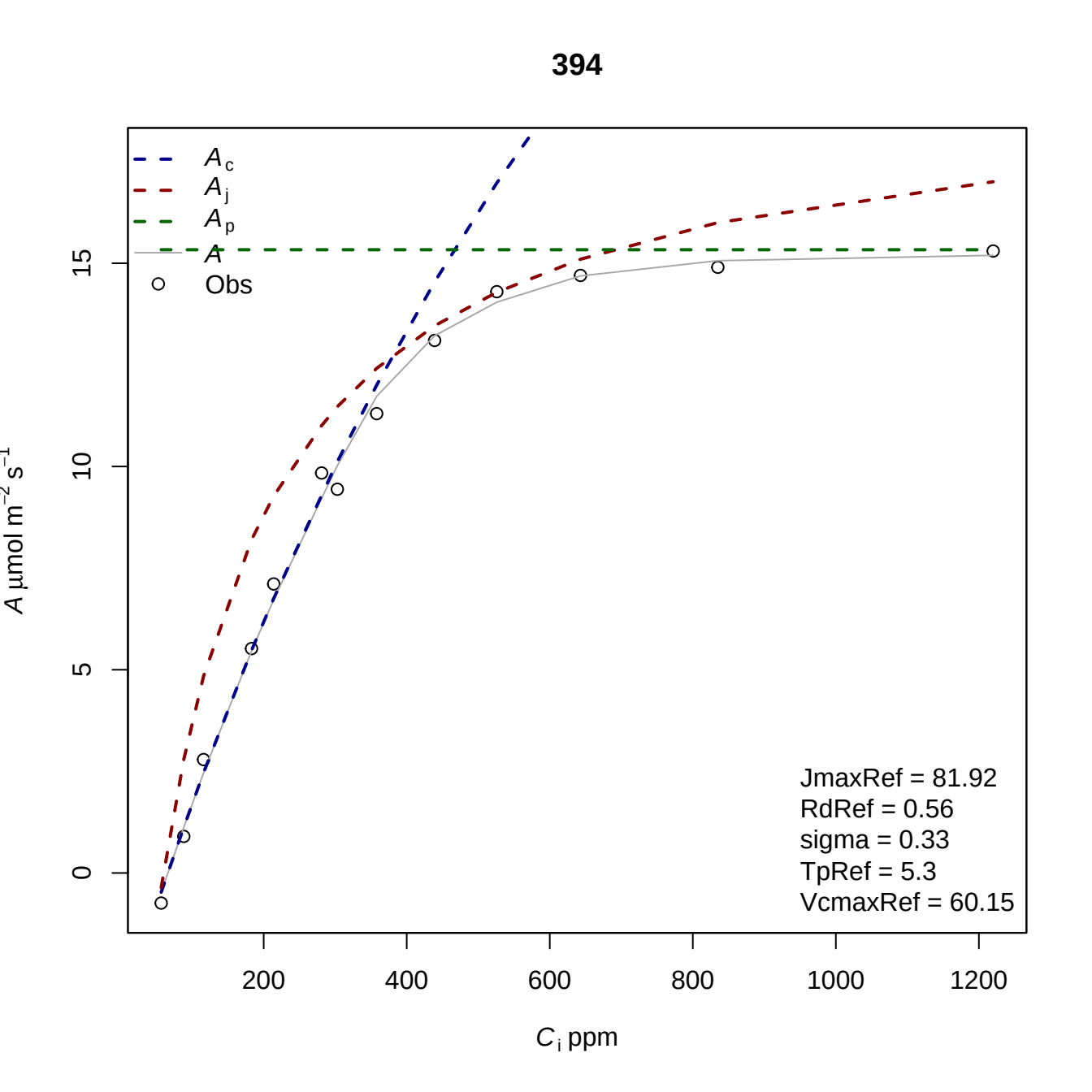

389
A_c
A_j
A_p
A
Obs
JmaxRef = 80.15
RdRef = 0.07
sigma = 0.41
TpRef = 38829.9
VcmaxRef = 63.06
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000
1200
1400

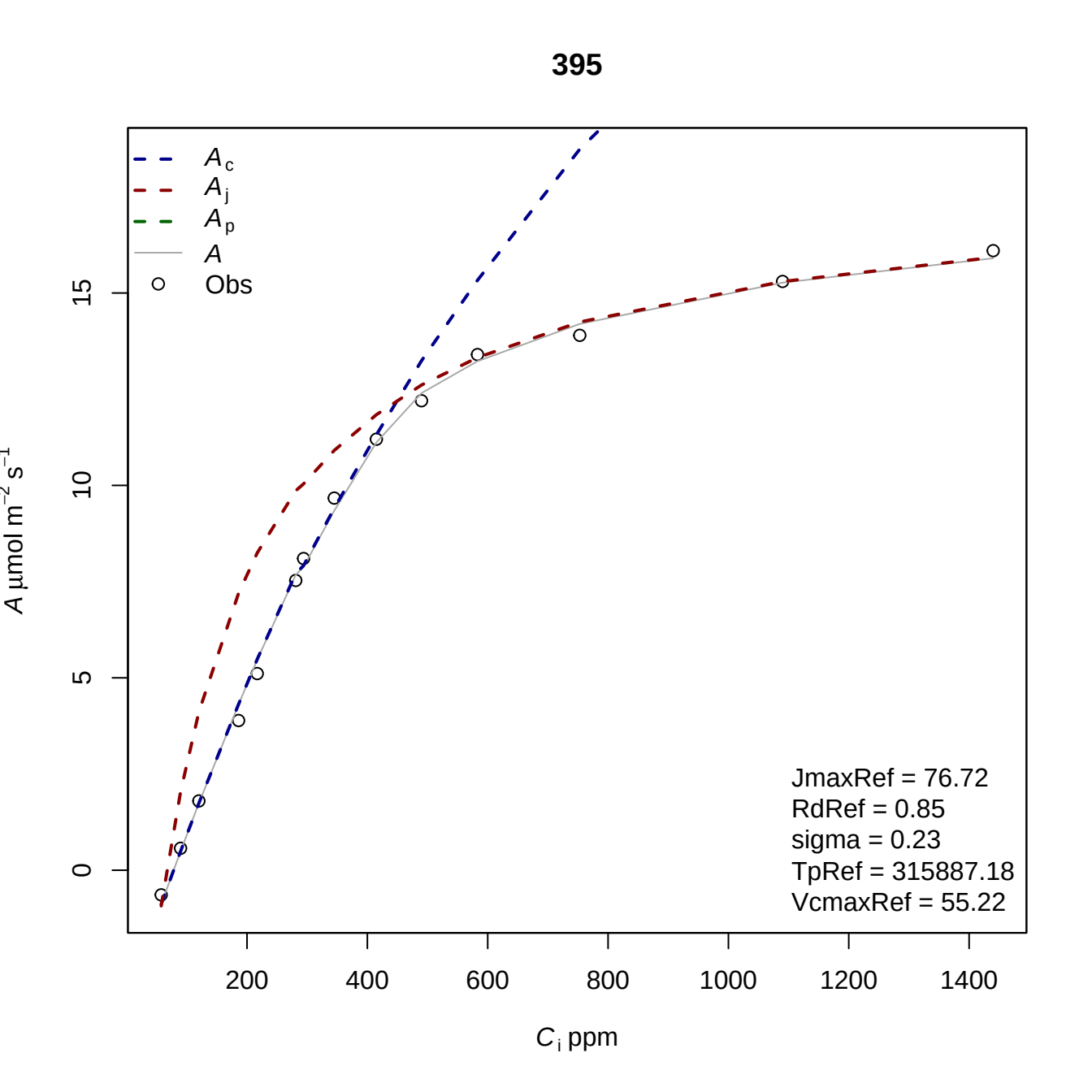

390
A_c
A_j
A_p
A
Obs
JmaxRef = 103.05
RdRef = 1.11
sigma = 0.16
TpRef = 6.31
VcmaxRef = 83.41
0
5
10
15
20
200
400
600
800
1000
A µmol m^{-2} s^{-1}
C_i ppm

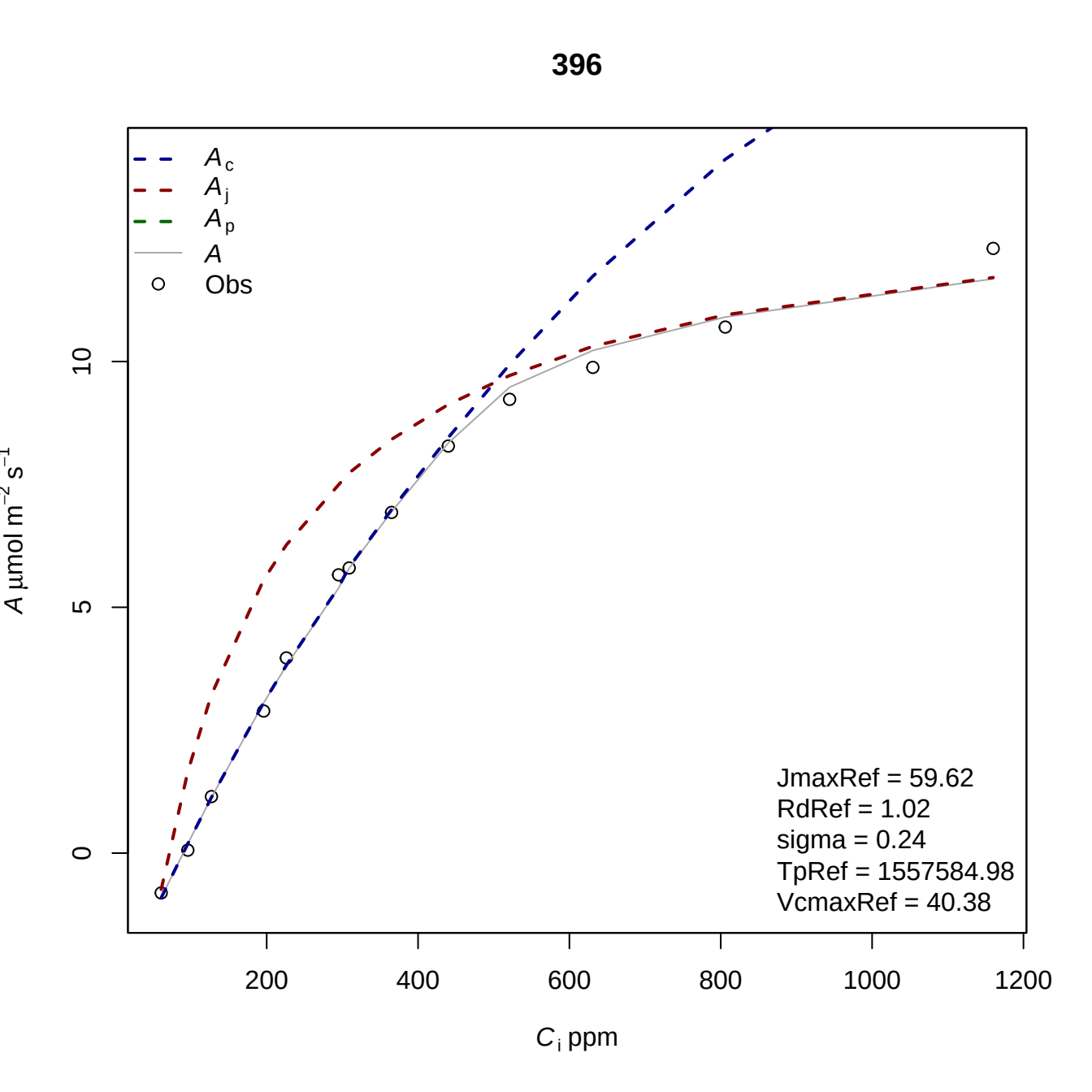

391
A_c
A_j
A_p
A
Obs
JmaxRef = 94.55
RdRef = 0.94
sigma = 0.16
TpRef = 5.16
VcmaxRef = 81.37
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
100
200
300
400
500
600
700

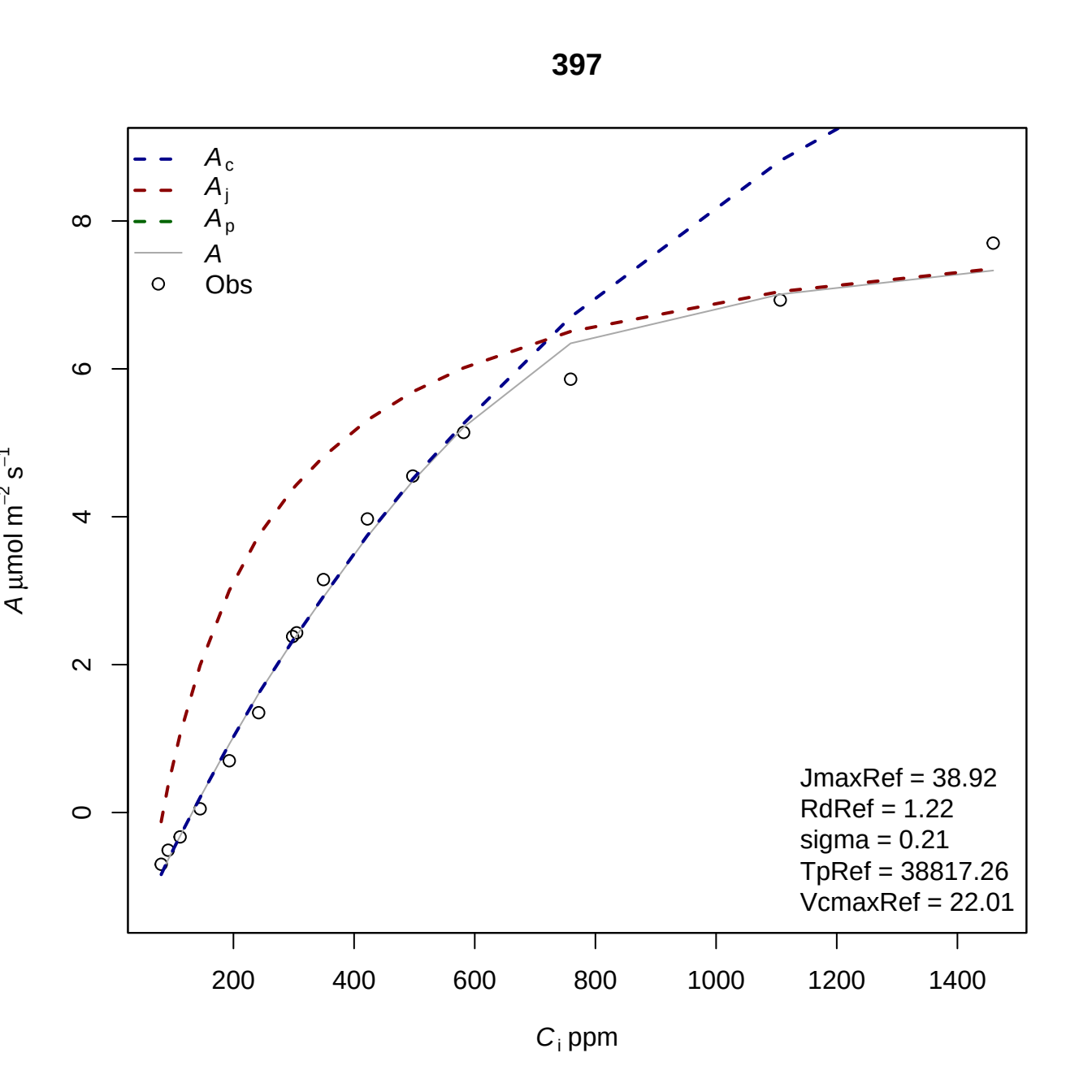

392
A_c
A_j
A_p
A
Obs
JmaxRef = 52.54
RdRef = 0.63
sigma = 0.16
TpRef = 4.18
VcmaxRef = 40.06
0
2
4
6
8
10
12
500
1000
1500
A µmol m^{-2} s^{-1}
C_i ppm

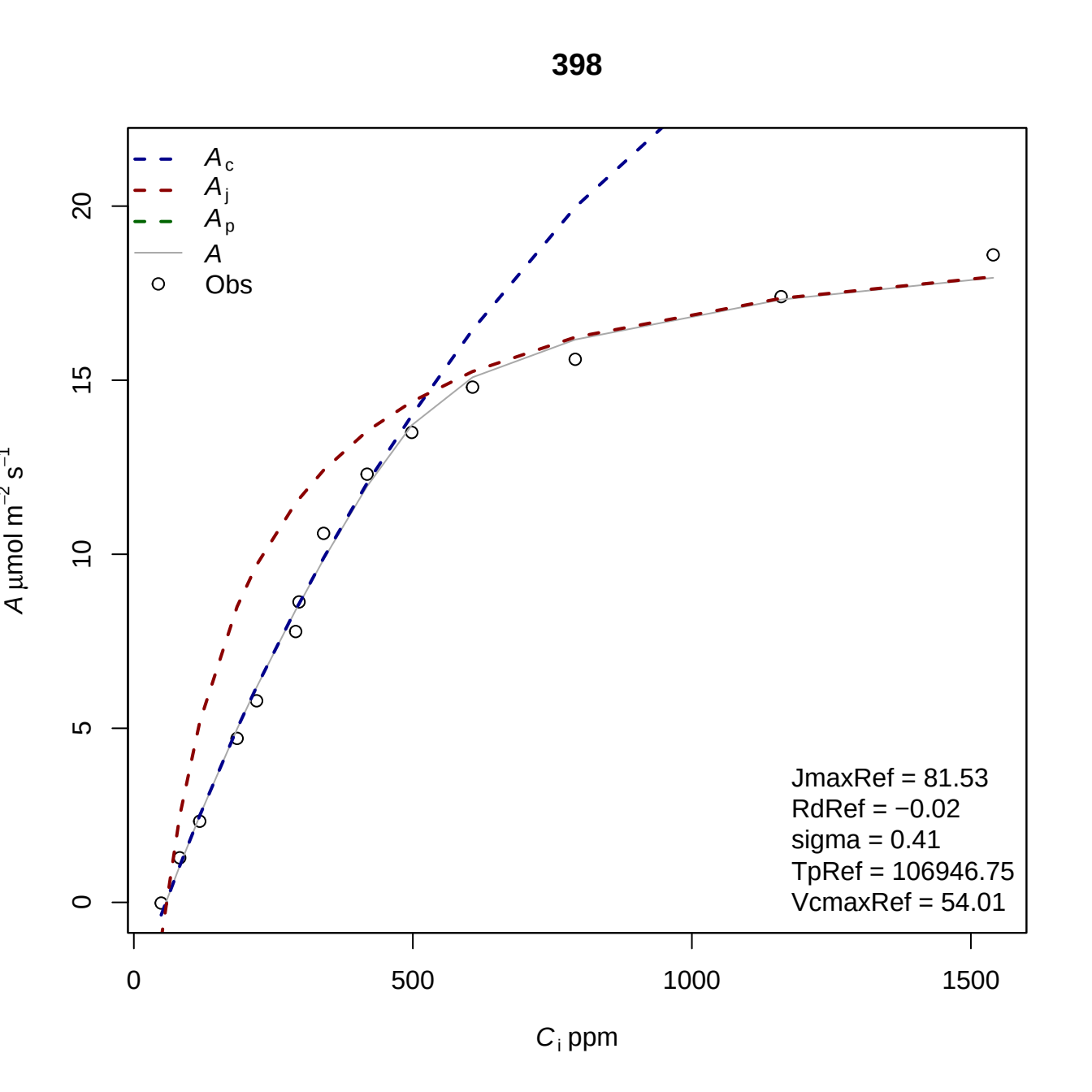

393
A_c
A_j
A_p
A
Obs
JmaxRef = 49.44
RdRef = 0.57
sigma = 0.12
TpRef = 3.67
VcmaxRef = 37.27
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
0
500
1000
1500

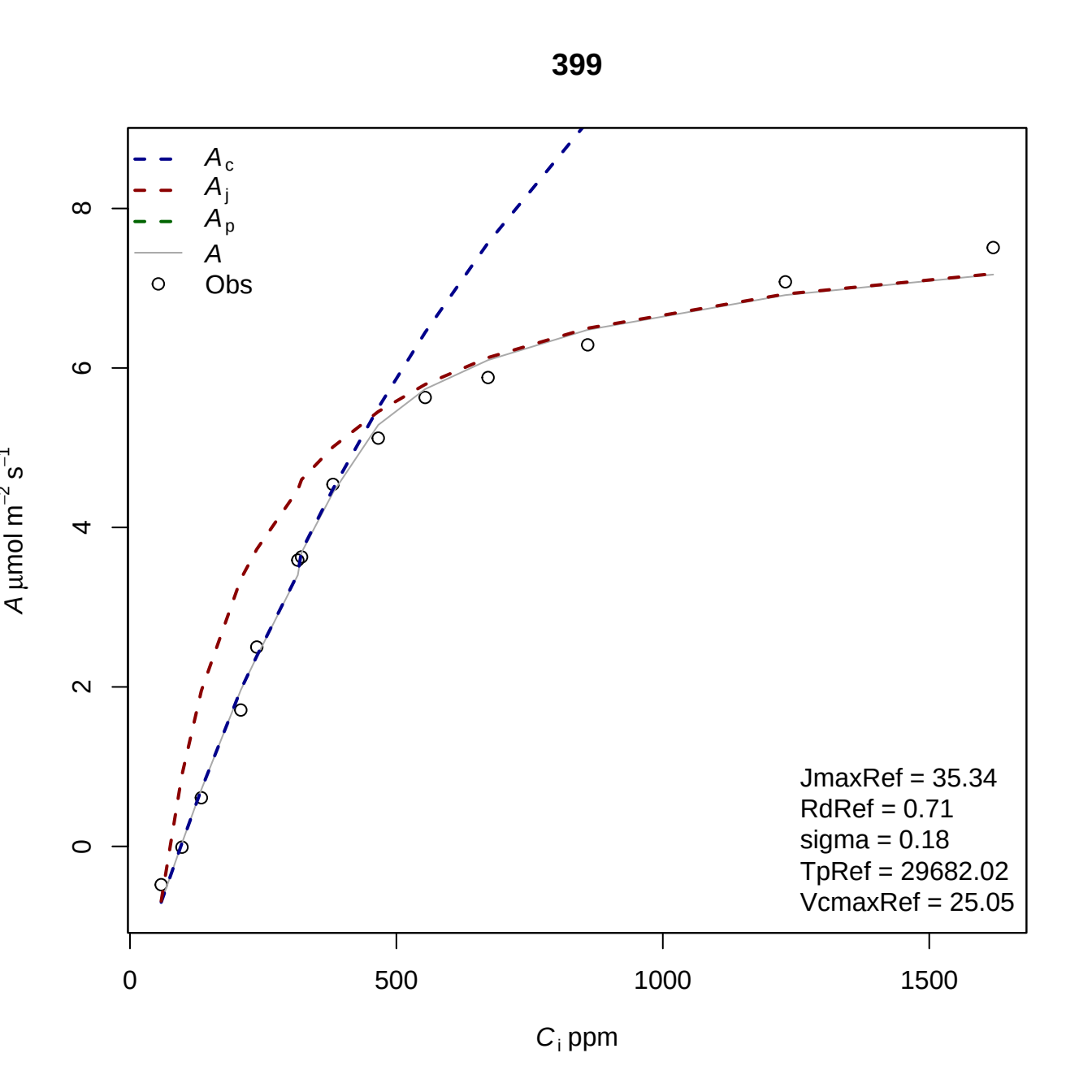

394
A_c
A_j
A_p
A
Obs
JmaxRef = 81.92
RdRef = 0.56
sigma = 0.33
TpRef = 5.3
VcmaxRef = 60.15
0
5
10
15
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

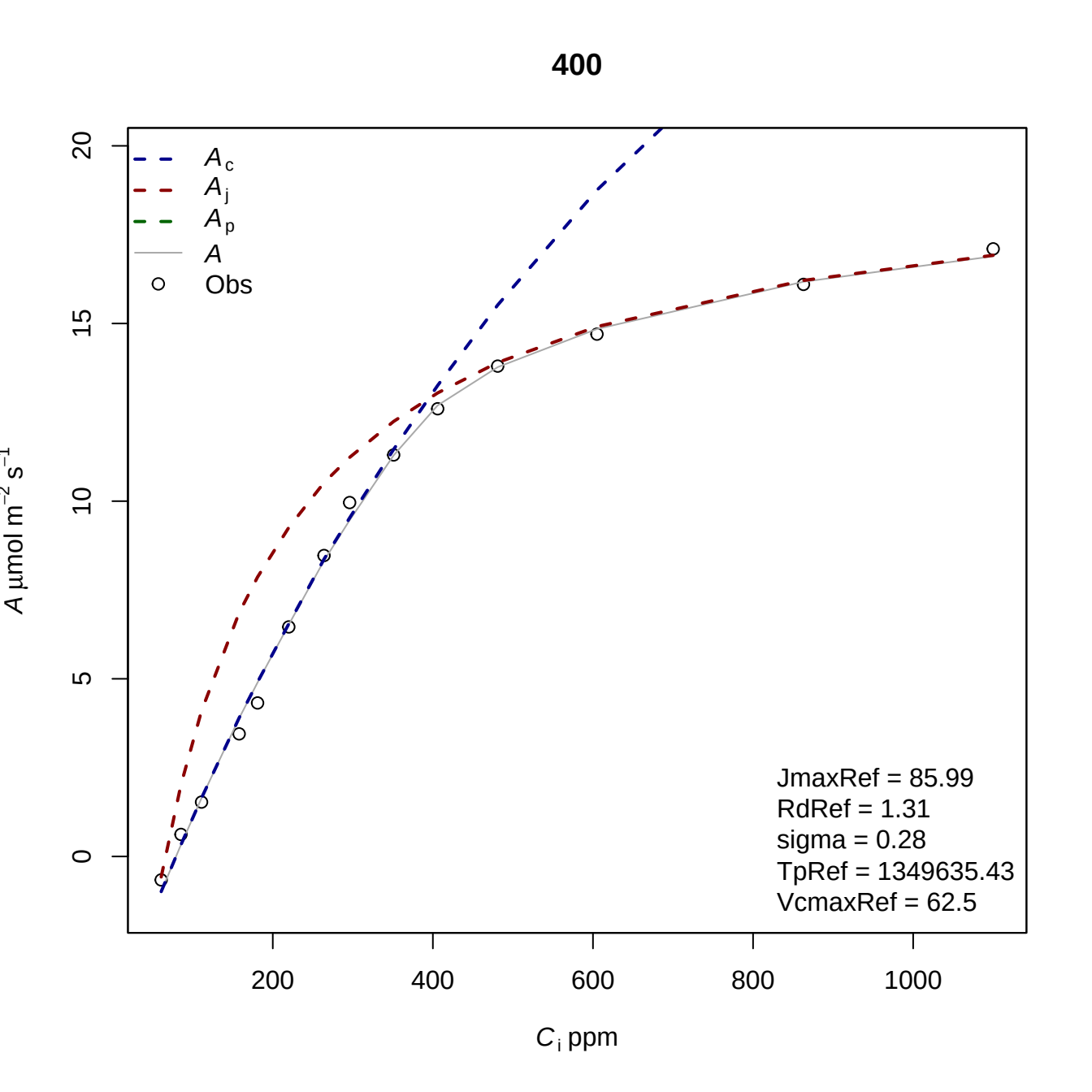

395
A_c
A_j
A_p
A
Obs
JmaxRef = 76.72
RdRef = 0.85
sigma = 0.23
TpRef = 315887.18
VcmaxRef = 55.22
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
200
400
600
800
1000
1200
1400

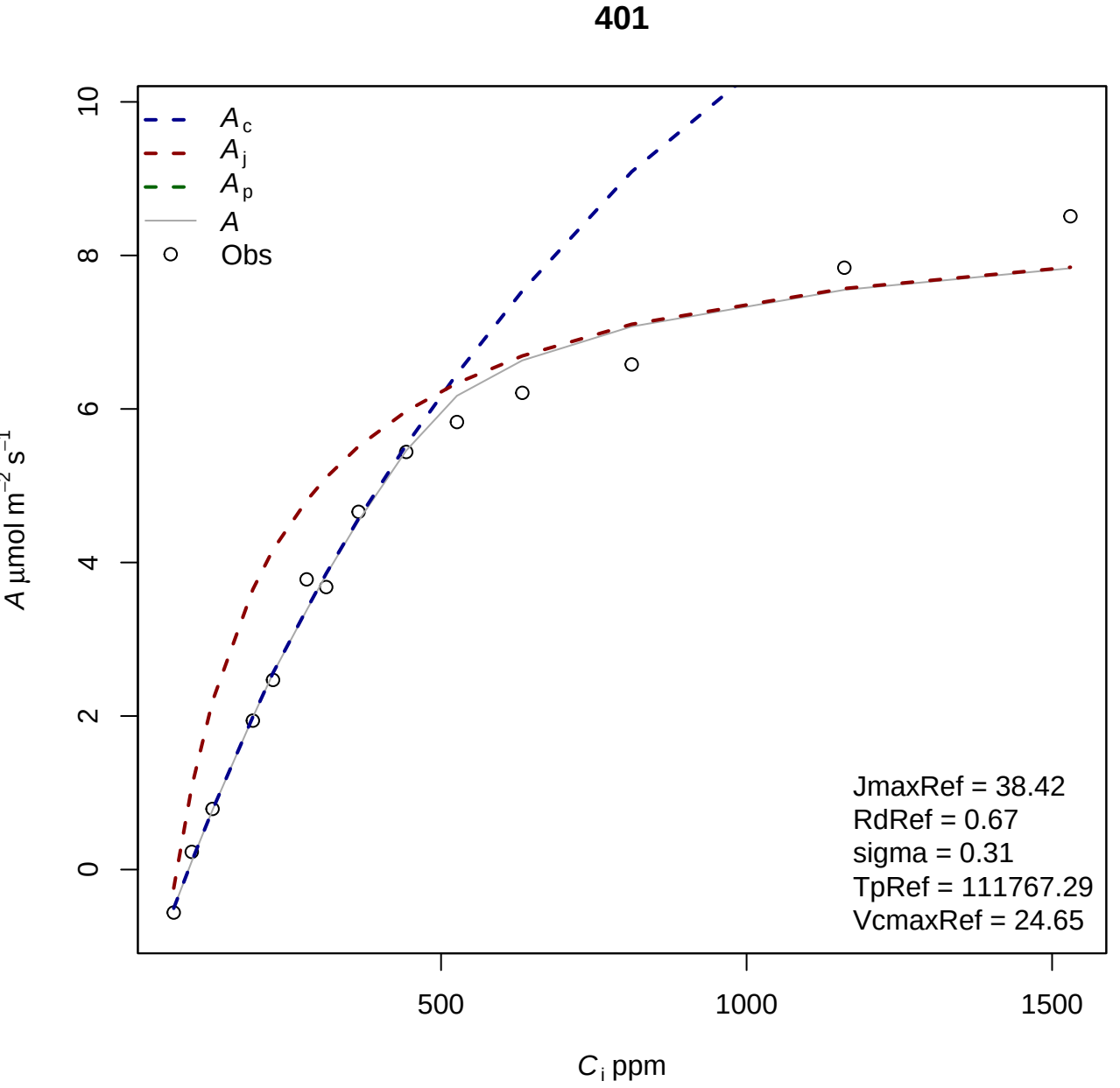

396
A_c
A_j
A_p
A
Obs
JmaxRef = 59.62
RdRef = 1.02
sigma = 0.24
TpRef = 1557584.98
VcmaxRef = 40.38
0
5
10
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

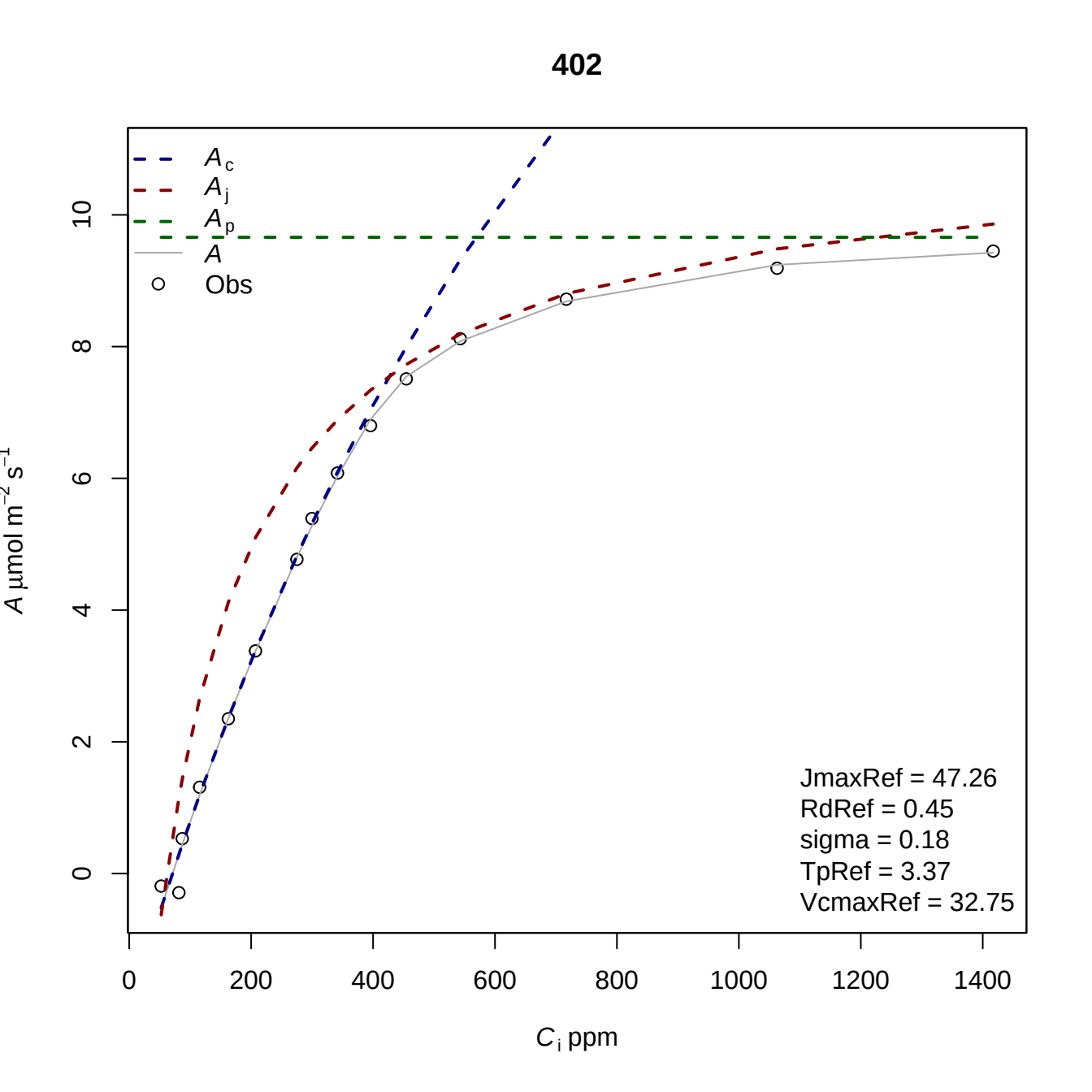

397
A_c
A_j
A_p
A
Obs
JmaxRef = 38.92
RdRef = 1.22
sigma = 0.21
TpRef = 38817.26
VcmaxRef = 22.01
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
200
400
600
800
1000
1200
1400

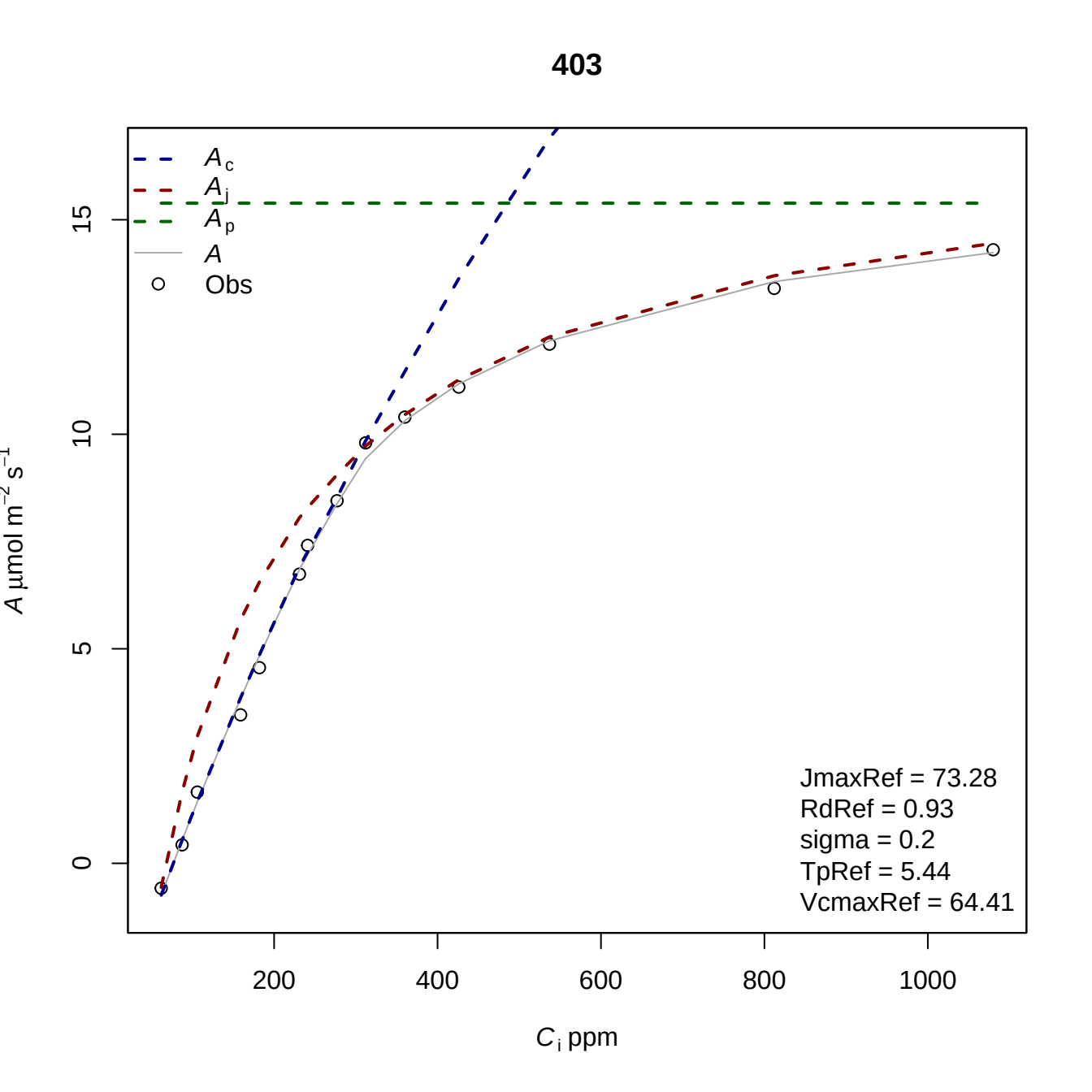

398
A_c
A_j
A_p
A
Obs
JmaxRef = 81.53
RdRef = −0.02
sigma = 0.41
TpRef = 106946.75
VcmaxRef = 54.01
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
500
1000
1500

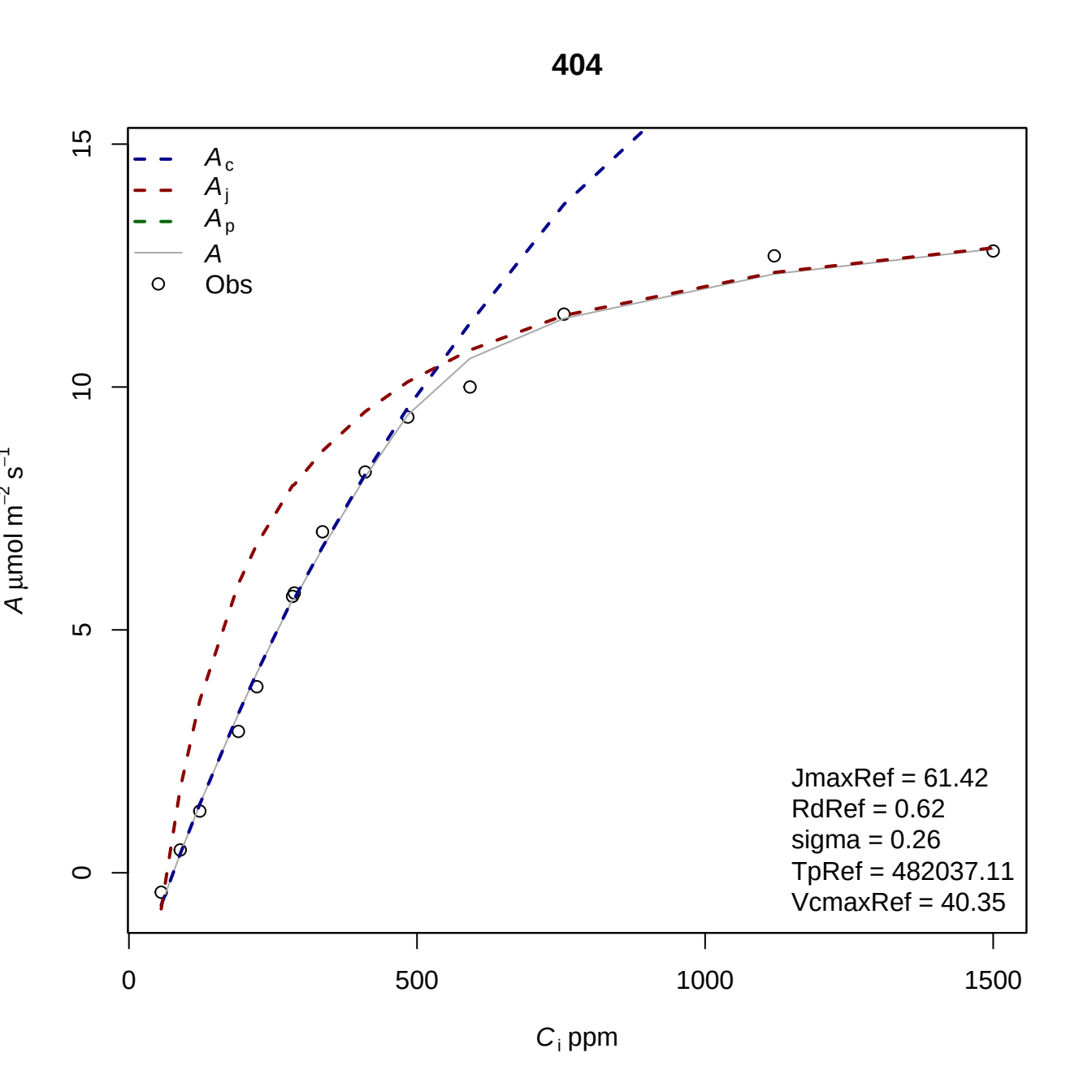

399
A_c
A_j
A_p
A
Obs
JmaxRef = 35.34
RdRef = 0.71
sigma = 0.18
TpRef = 29682.02
VcmaxRef = 25.05
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
0
500
1000
1500

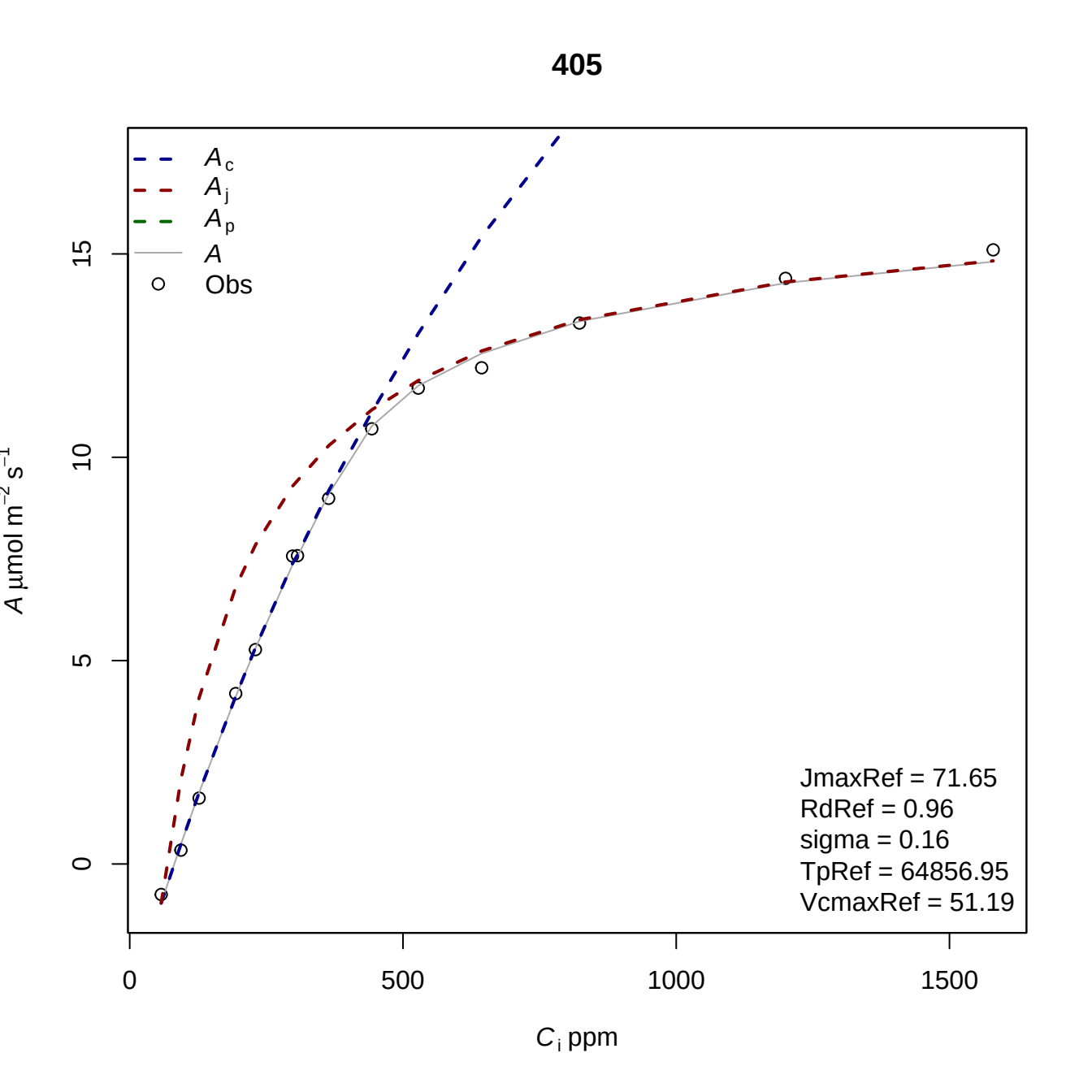

400
A_c
A_j
A_p
A
Obs
JmaxRef = 85.99
RdRef = 1.31
sigma = 0.28
TpRef = 1349635.43
VcmaxRef = 62.5
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
200
400
600
800
1000

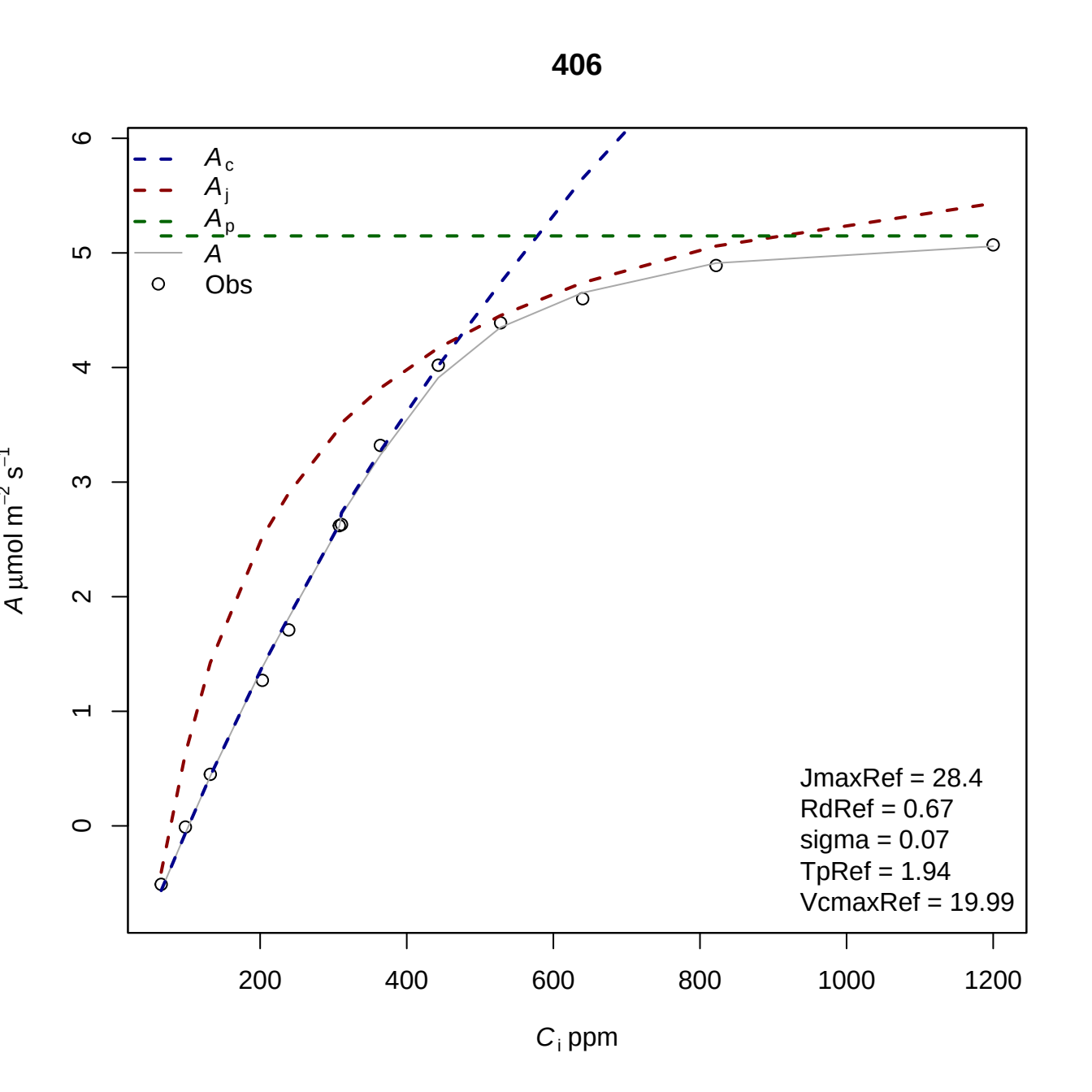

401
A_c
A_j
A_p
A
Obs
JmaxRef = 38.42
RdRef = 0.67
sigma = 0.31
TpRef = 111767.29
VcmaxRef = 24.65
0
2
4
6
8
10
500
1000
1500
A µmol m^{-2} s^{-1}
C_i ppm

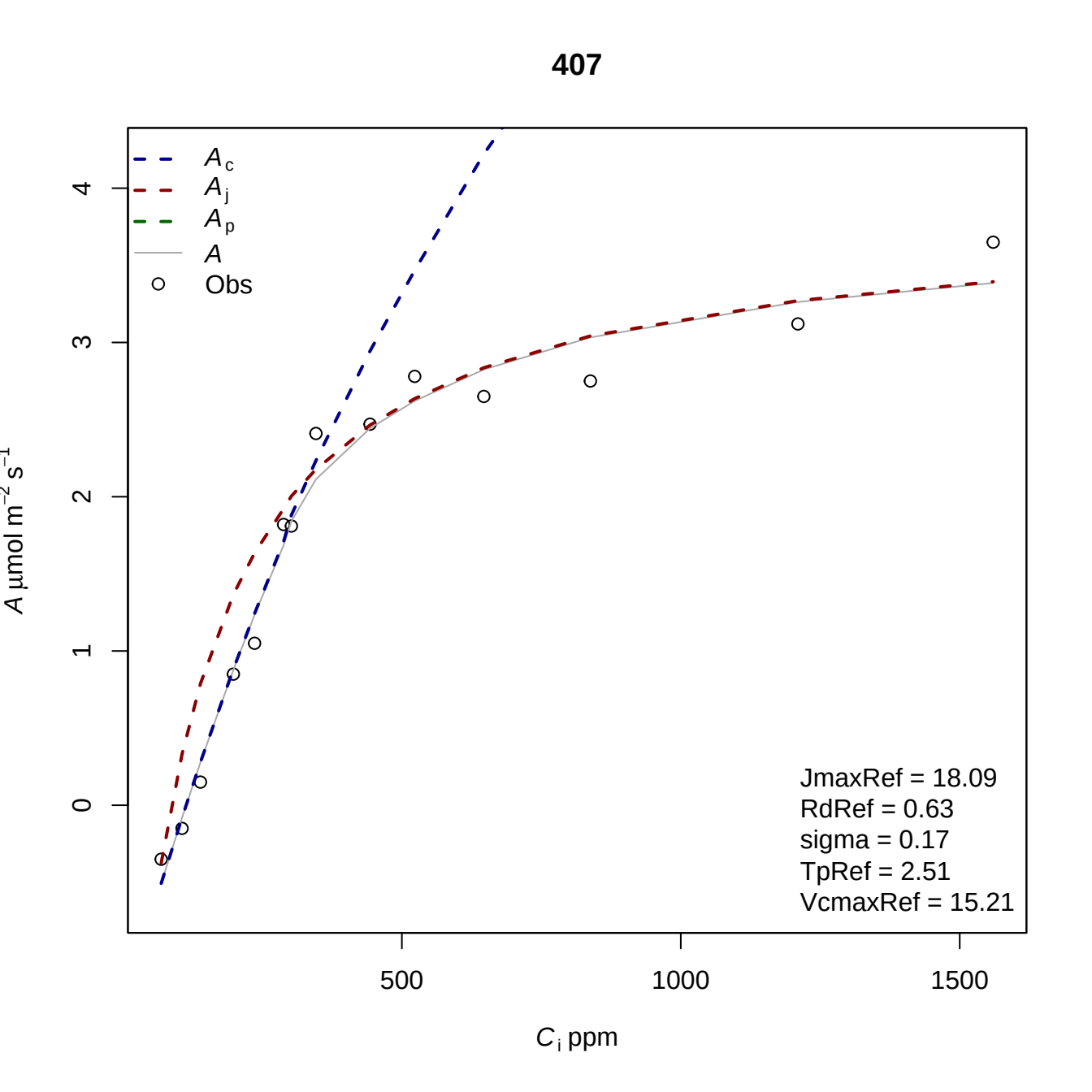

402
A_c
A_j
A_p
A
Obs
JmaxRef = 47.26
RdRef = 0.45
sigma = 0.18
TpRef = 3.37
VcmaxRef = 32.75
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

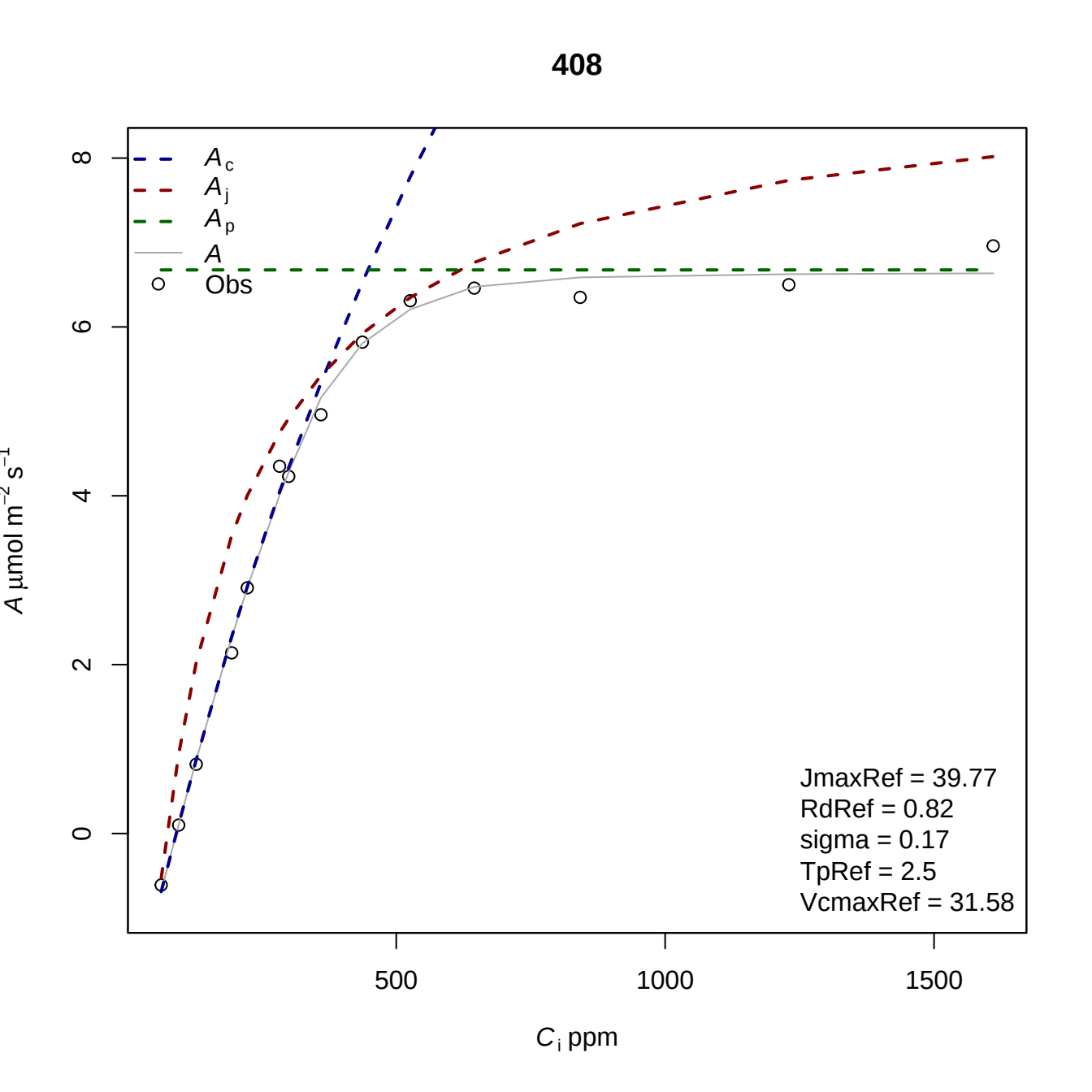

403
A_c
A_j
A_p
A
Obs
JmaxRef = 73.28
RdRef = 0.93
sigma = 0.2
TpRef = 5.44
VcmaxRef = 64.41
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
200
400
600
800
1000

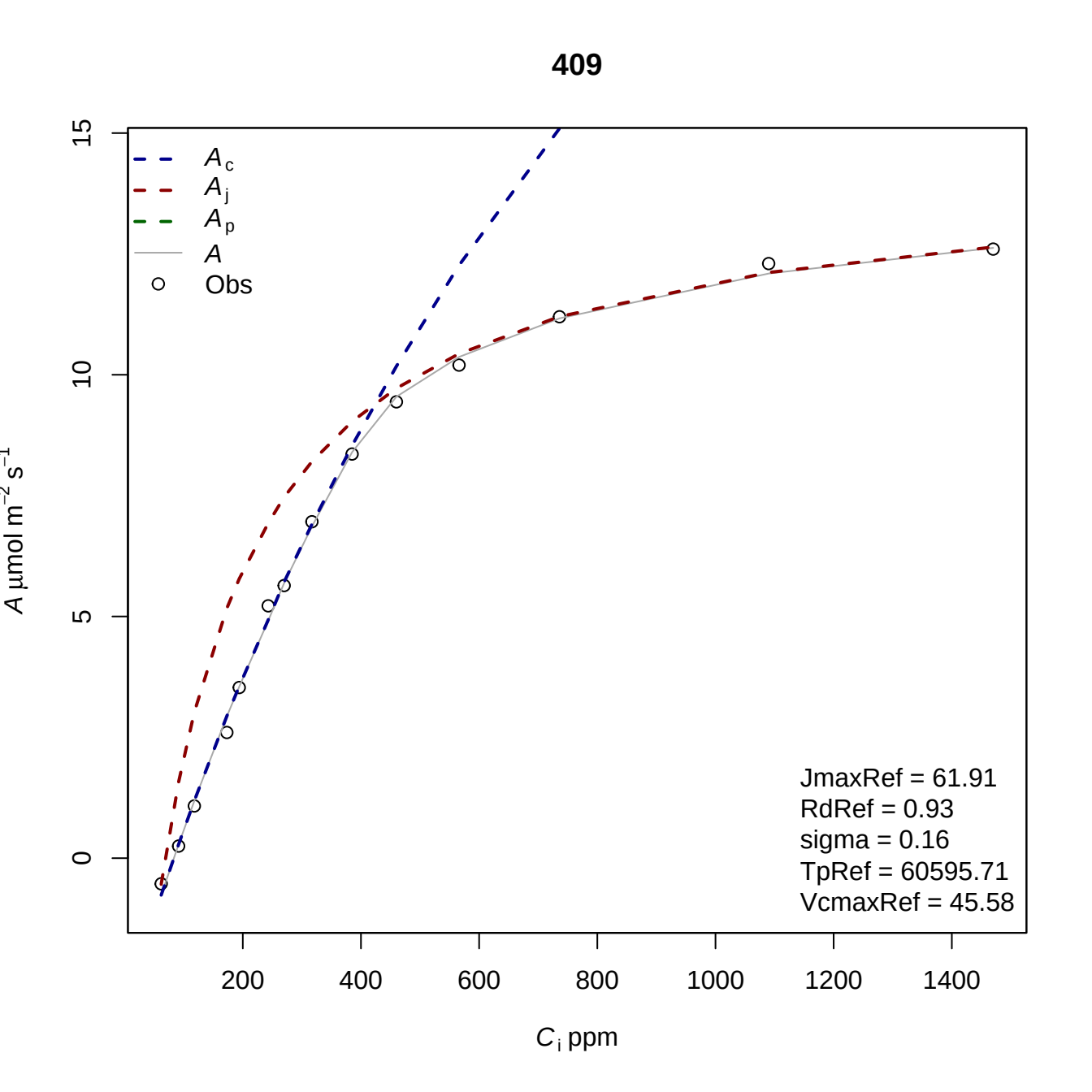

404
A_c
A_j
A_p
A
Obs
JmaxRef = 61.42
RdRef = 0.62
sigma = 0.26
TpRef = 482037.11
VcmaxRef = 40.35
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
500
1000
1500

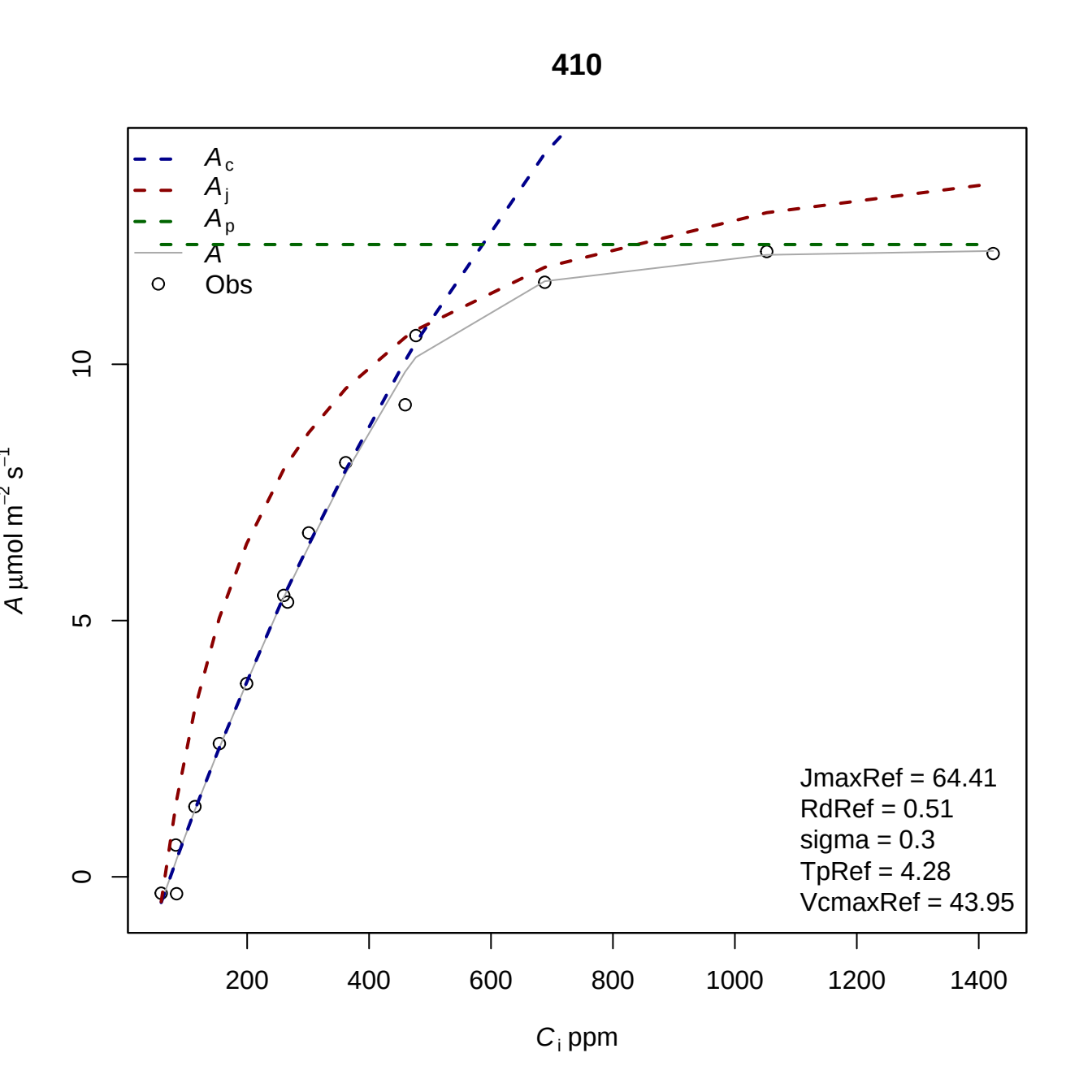

405
A_c
A_j
A_p
A
Obs
JmaxRef = 71.65
RdRef = 0.96
sigma = 0.16
TpRef = 64856.95
VcmaxRef = 51.19
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
500
1000
1500

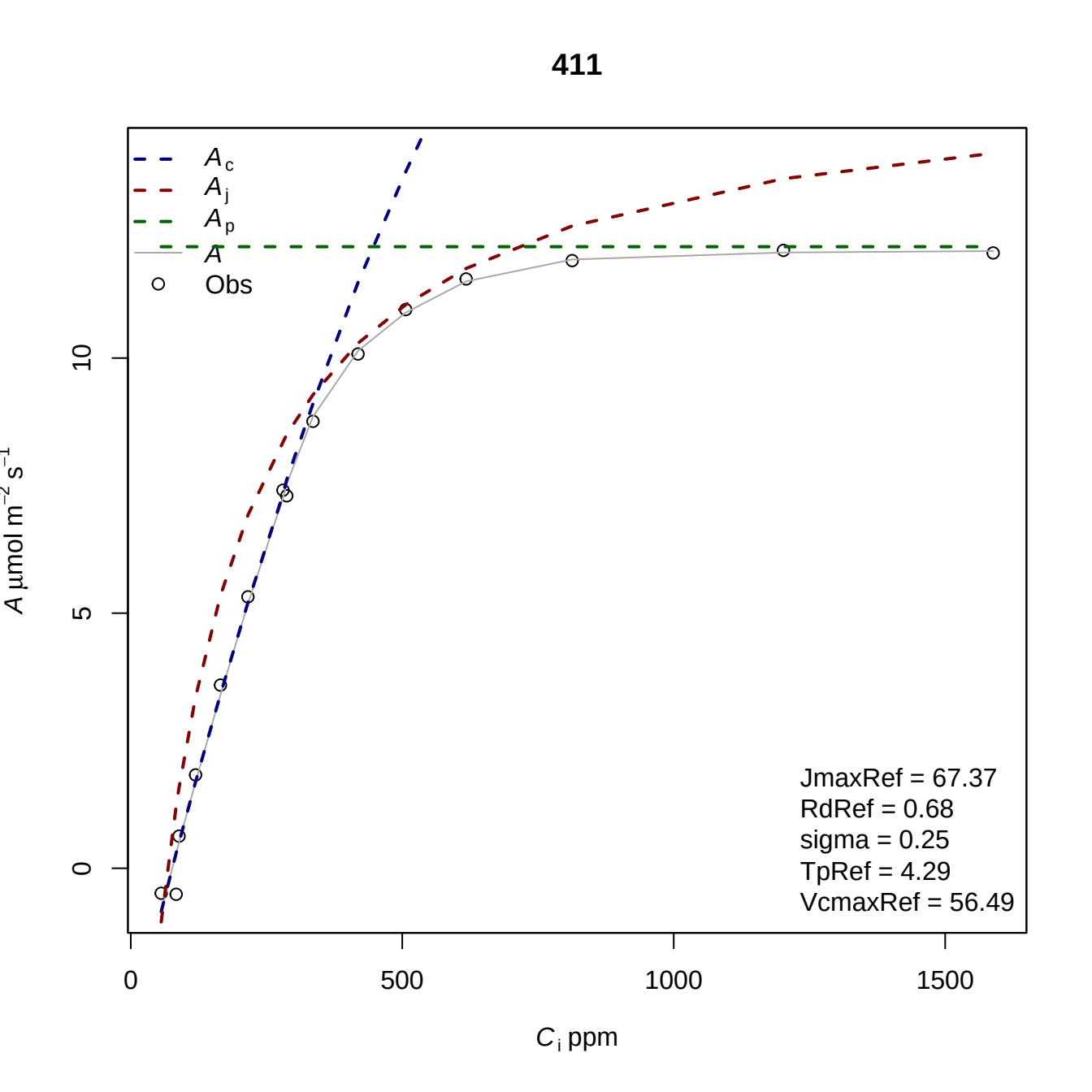

406
A_c
A_j
A_p
A
Obs
JmaxRef = 28.4
RdRef = 0.67
sigma = 0.07
TpRef = 1.94
VcmaxRef = 19.99
A µmol m^{-2} s^{-1}
C_i ppm
0
1
2
3
4
5
6
200
400
600
800
1000
1200

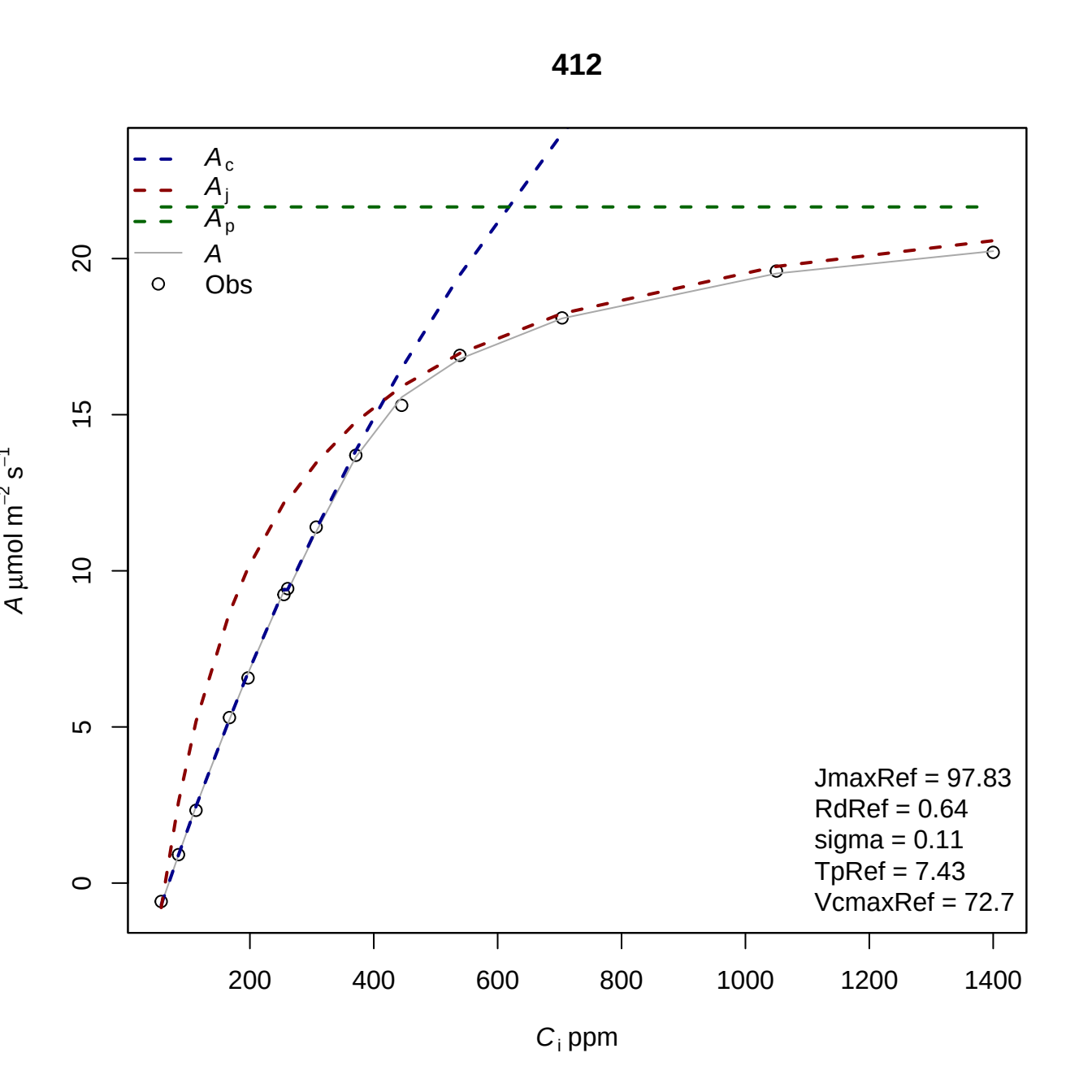

407
A_c
A_j
A_p
A
Obs
JmaxRef = 18.09
RdRef = 0.63
sigma = 0.17
TpRef = 2.51
VcmaxRef = 15.21
0
1
2
3
4
500
1000
1500
A µmol m^{-2} s^{-1}
C_i ppm

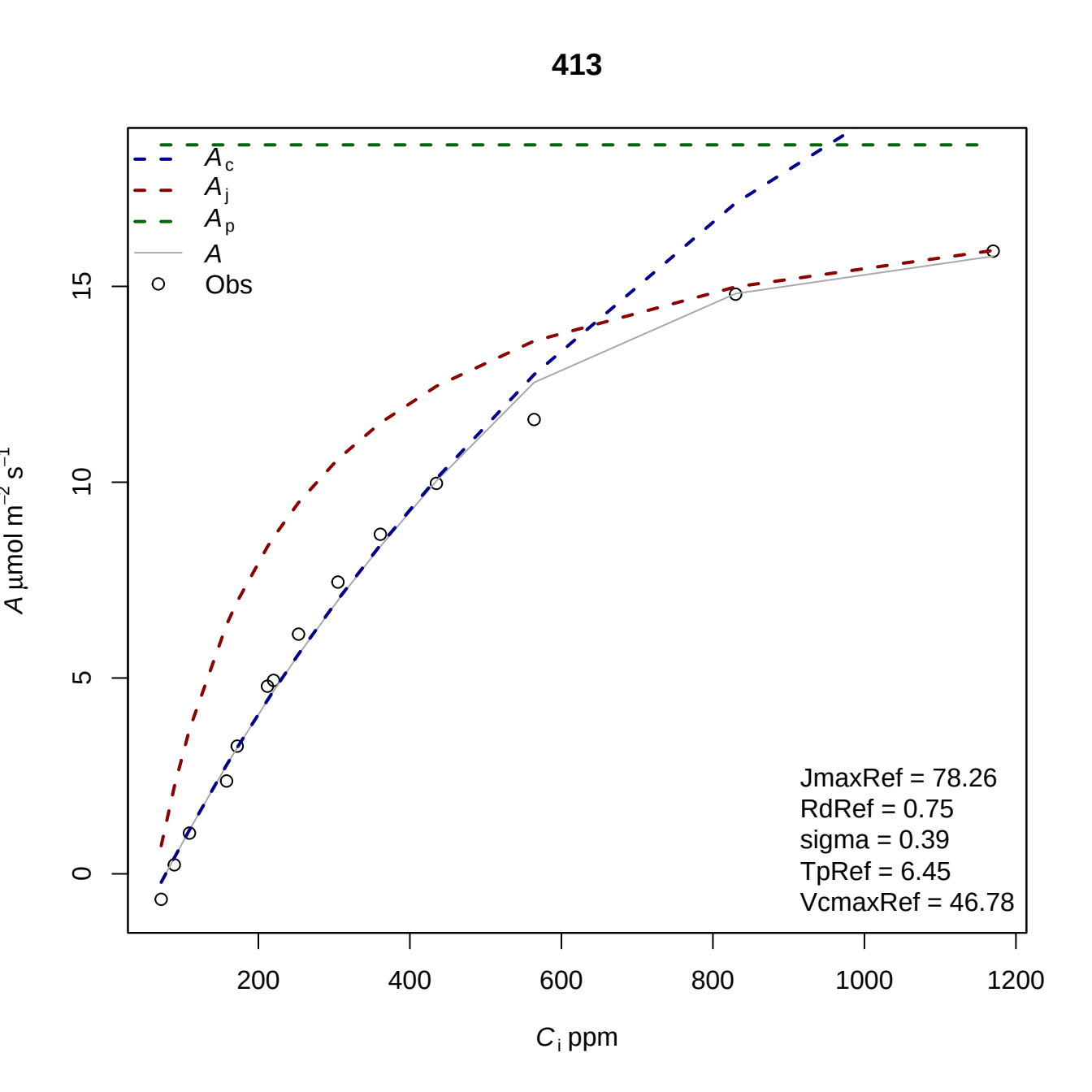

408
A_c
A_j
A_p
A
Obs
JmaxRef = 39.77
RdRef = 0.82
sigma = 0.17
TpRef = 2.5
VcmaxRef = 31.58
0
2
4
6
8
500
1000
1500
A µmol m^{-2} s^{-1}
C_i ppm

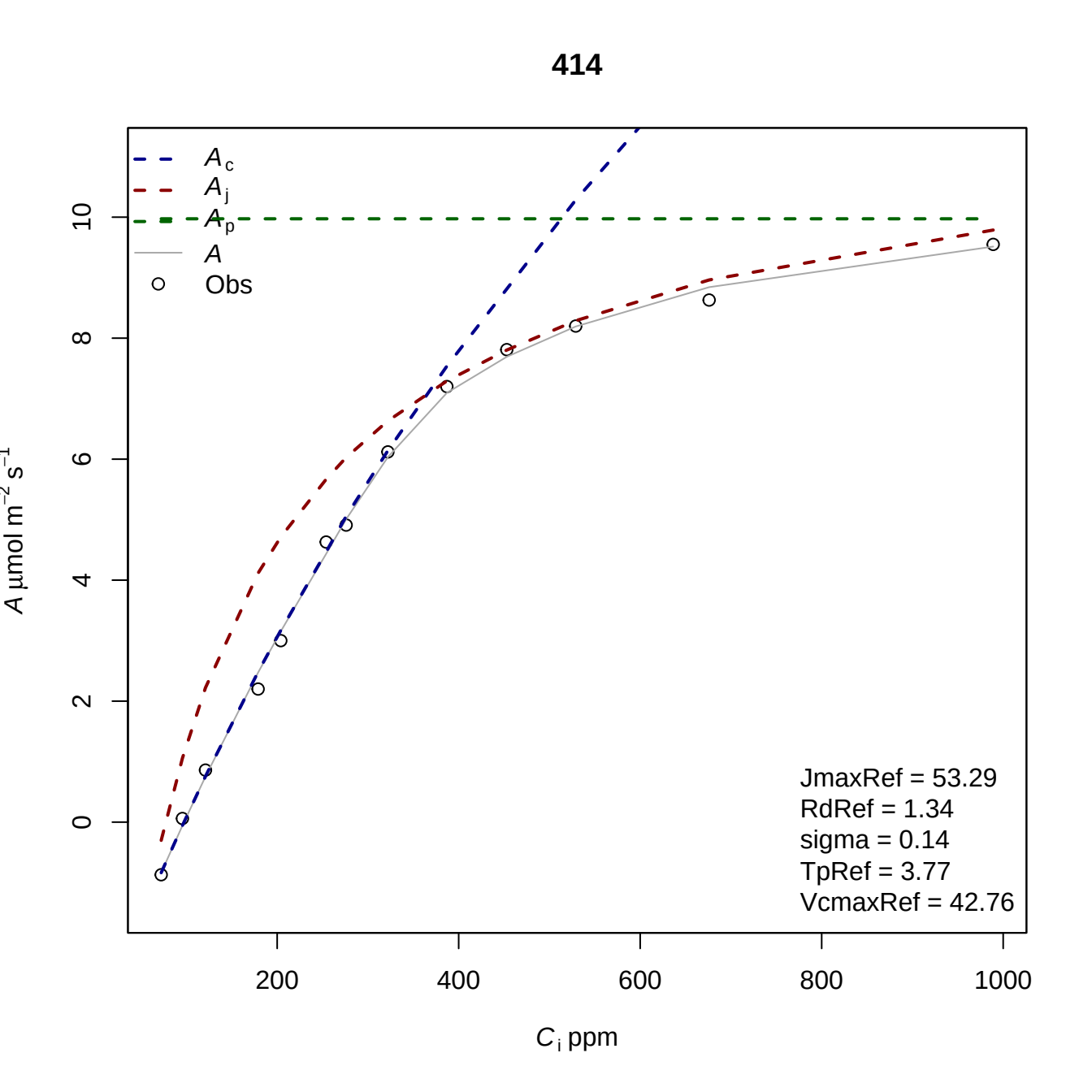

409
A_c
A_j
A_p
A
Obs
JmaxRef = 61.91
RdRef = 0.93
sigma = 0.16
TpRef = 60595.71
VcmaxRef = 45.58
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
200
400
600
800
1000
1200
1400

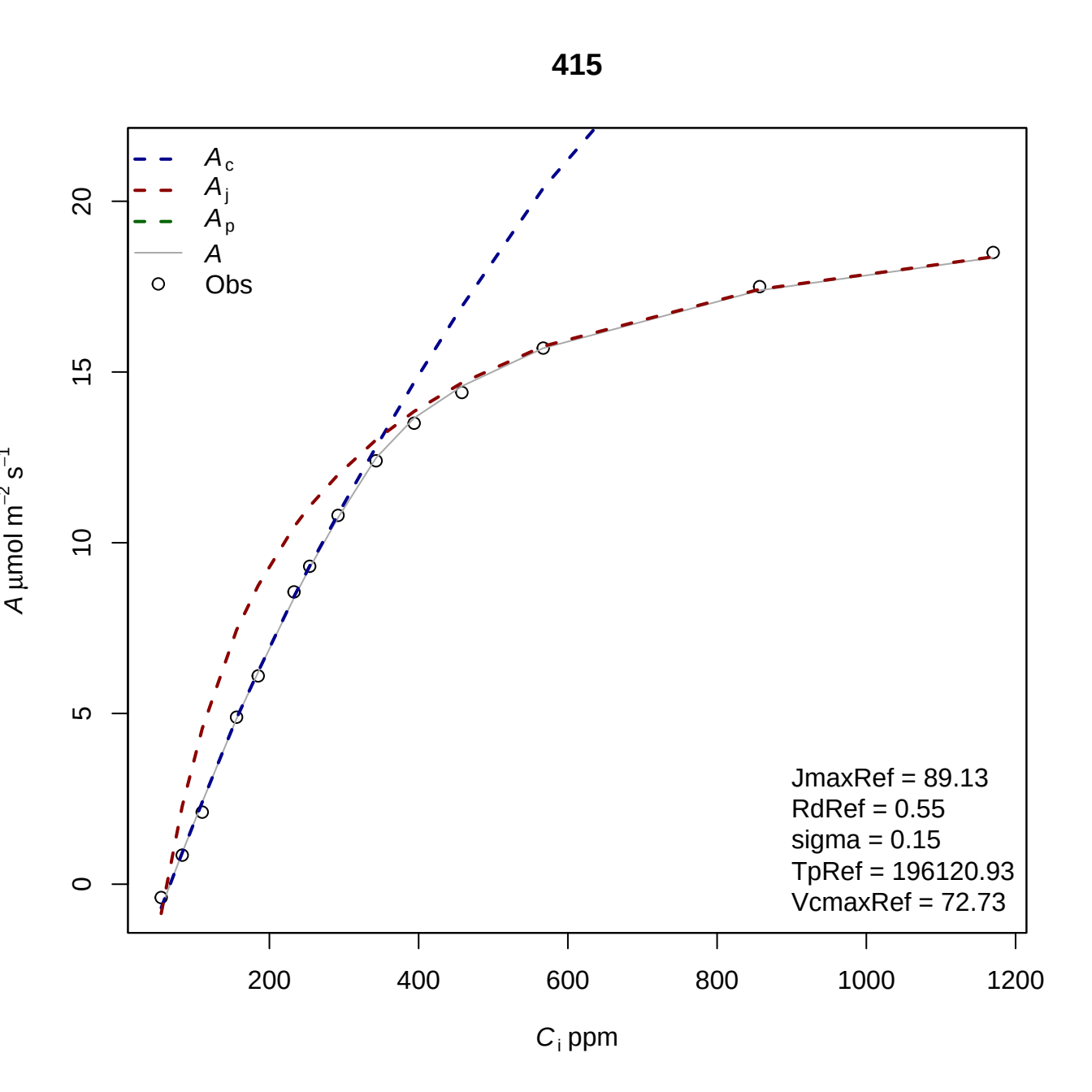

410
A_c
A_j
A_p
A
Obs
JmaxRef = 64.41
RdRef = 0.51
sigma = 0.3
TpRef = 4.28
VcmaxRef = 43.95
0
5
10
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
C_i ppm

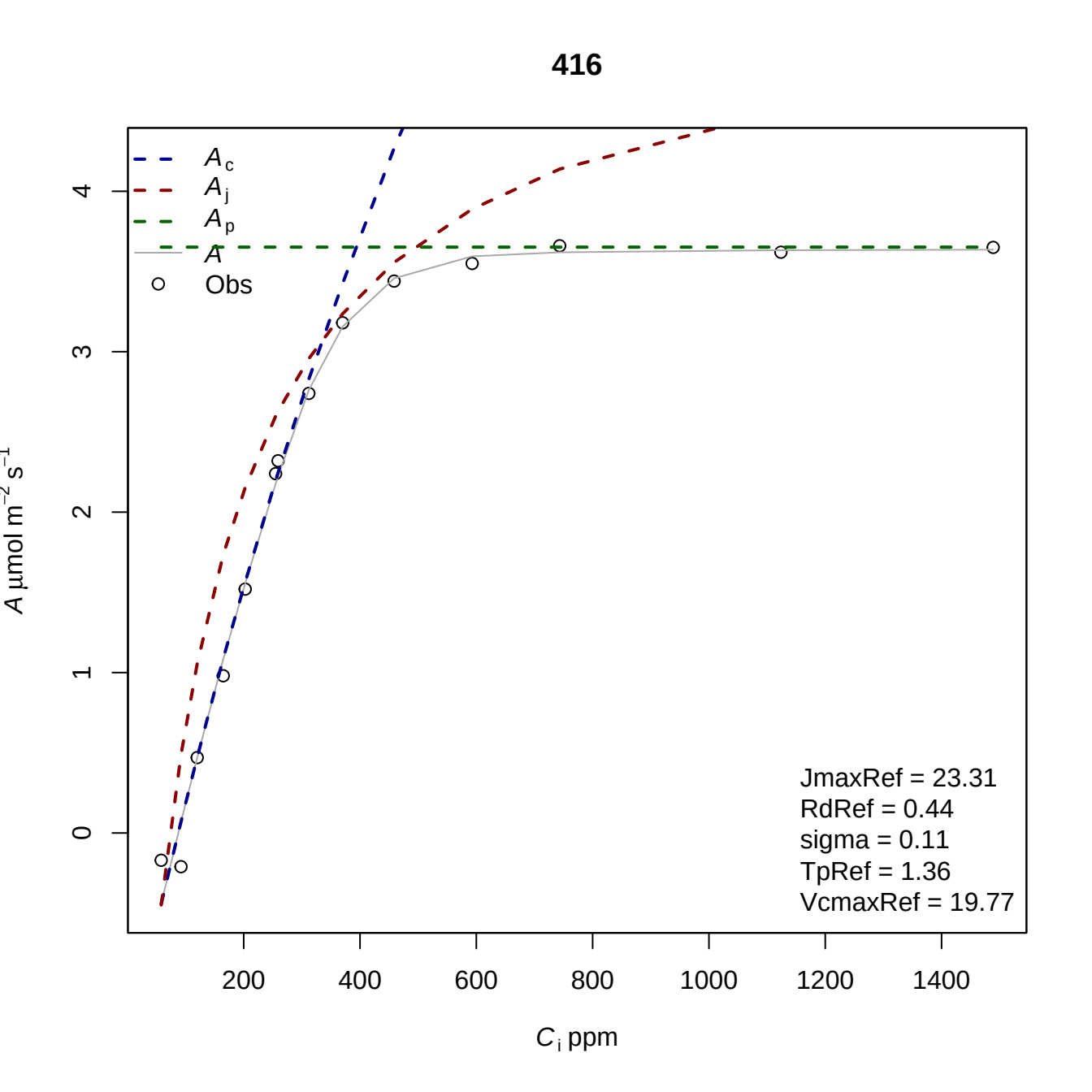

411
A_c
A_j
A_p
A
Obs
JmaxRef = 67.37
RdRef = 0.68
sigma = 0.25
TpRef = 4.29
VcmaxRef = 56.49
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
C_i ppm

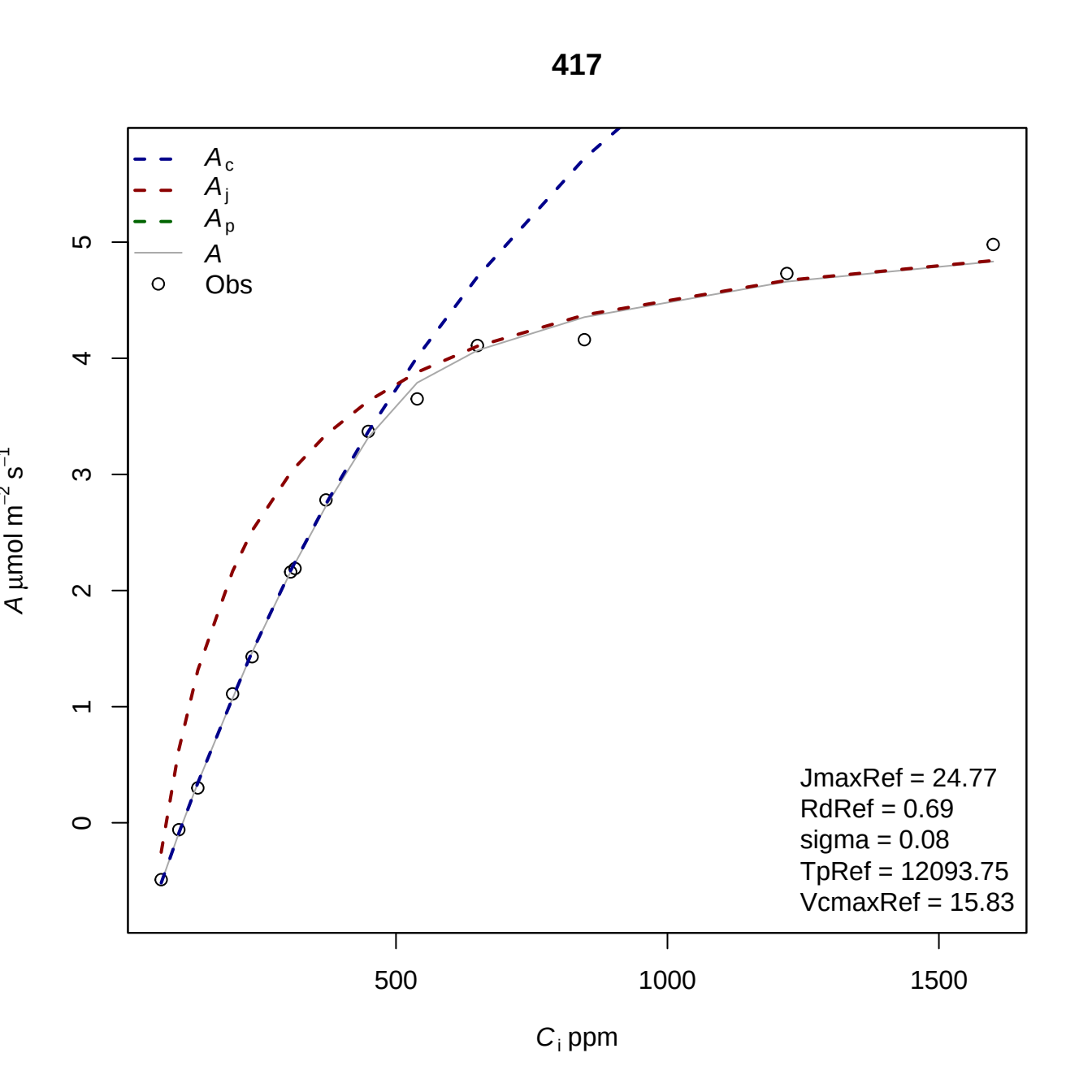

412
A_c
A_j
A_p
A
Obs
JmaxRef = 97.83
RdRef = 0.64
sigma = 0.11
TpRef = 7.43
VcmaxRef = 72.7
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
200
400
600
800
1000
1200
1400

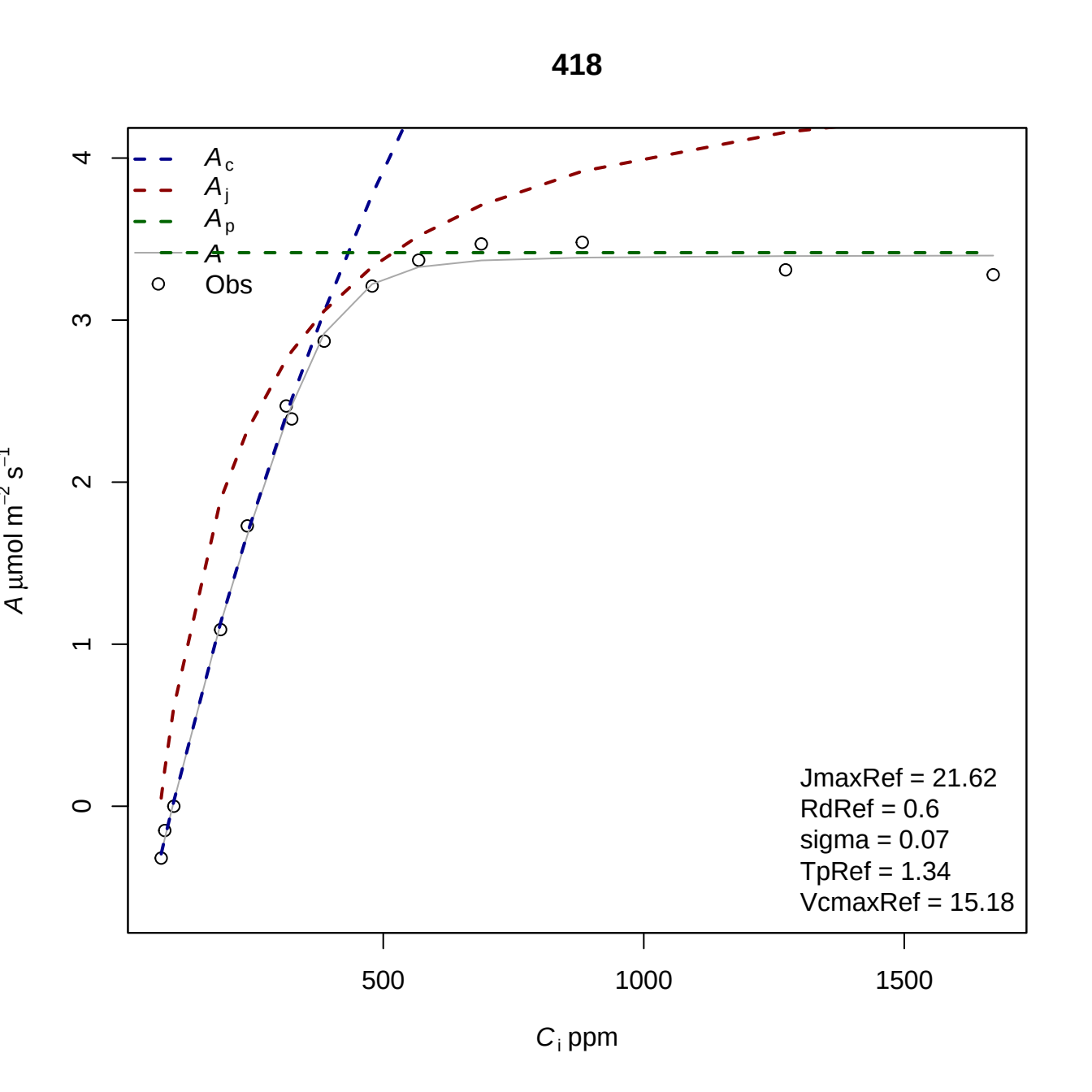

413
A_c
A_j
A_p
A
Obs
JmaxRef = 78.26
RdRef = 0.75
sigma = 0.39
TpRef = 6.45
VcmaxRef = 46.78
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
200
400
600
800
1000
1200

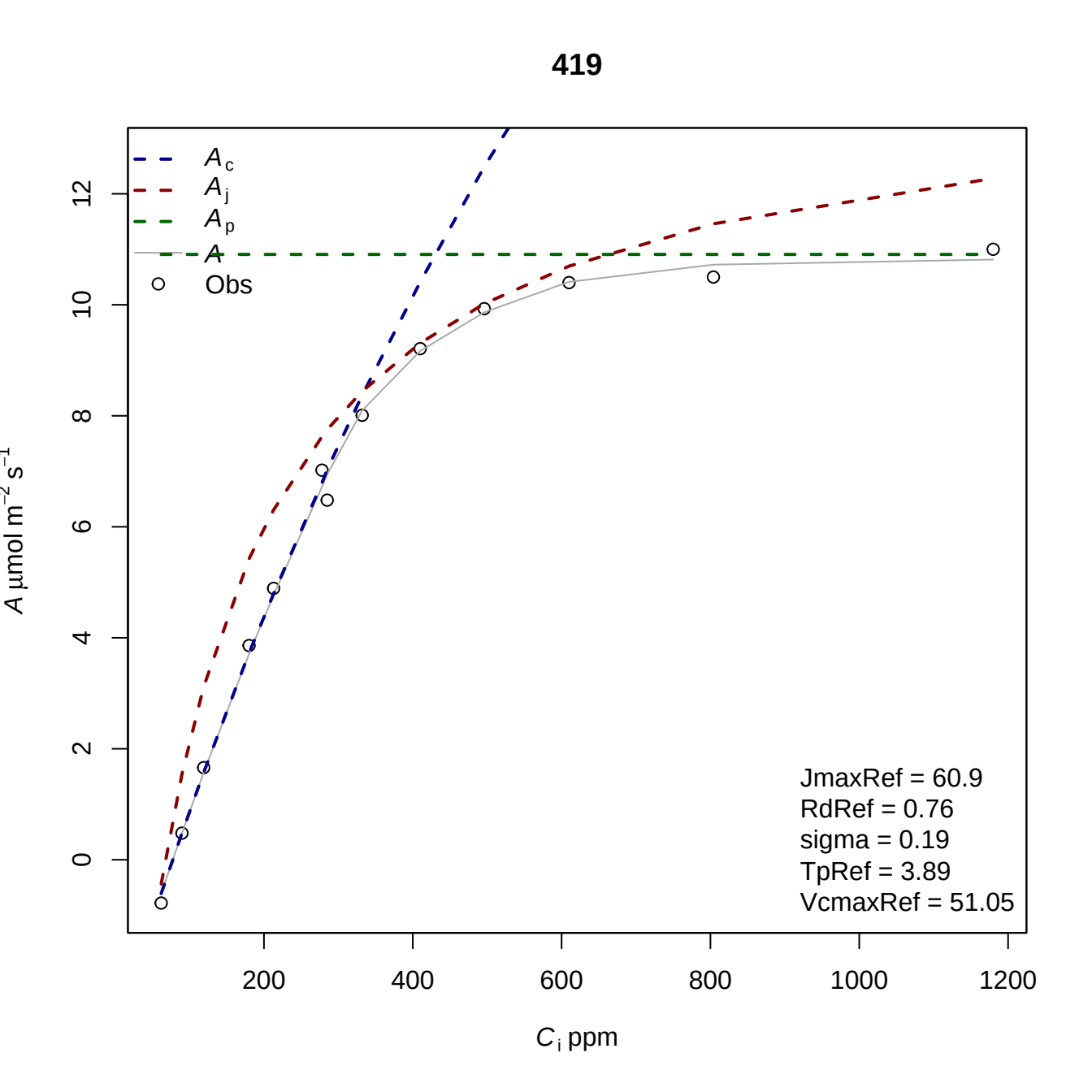

414
A_c
A_j
A_p
A
Obs
JmaxRef = 53.29
RdRef = 1.34
sigma = 0.14
TpRef = 3.77
VcmaxRef = 42.76
0
2
4
6
8
10
200
400
600
800
1000
A µmol m^{-2} s^{-1}
C_i ppm

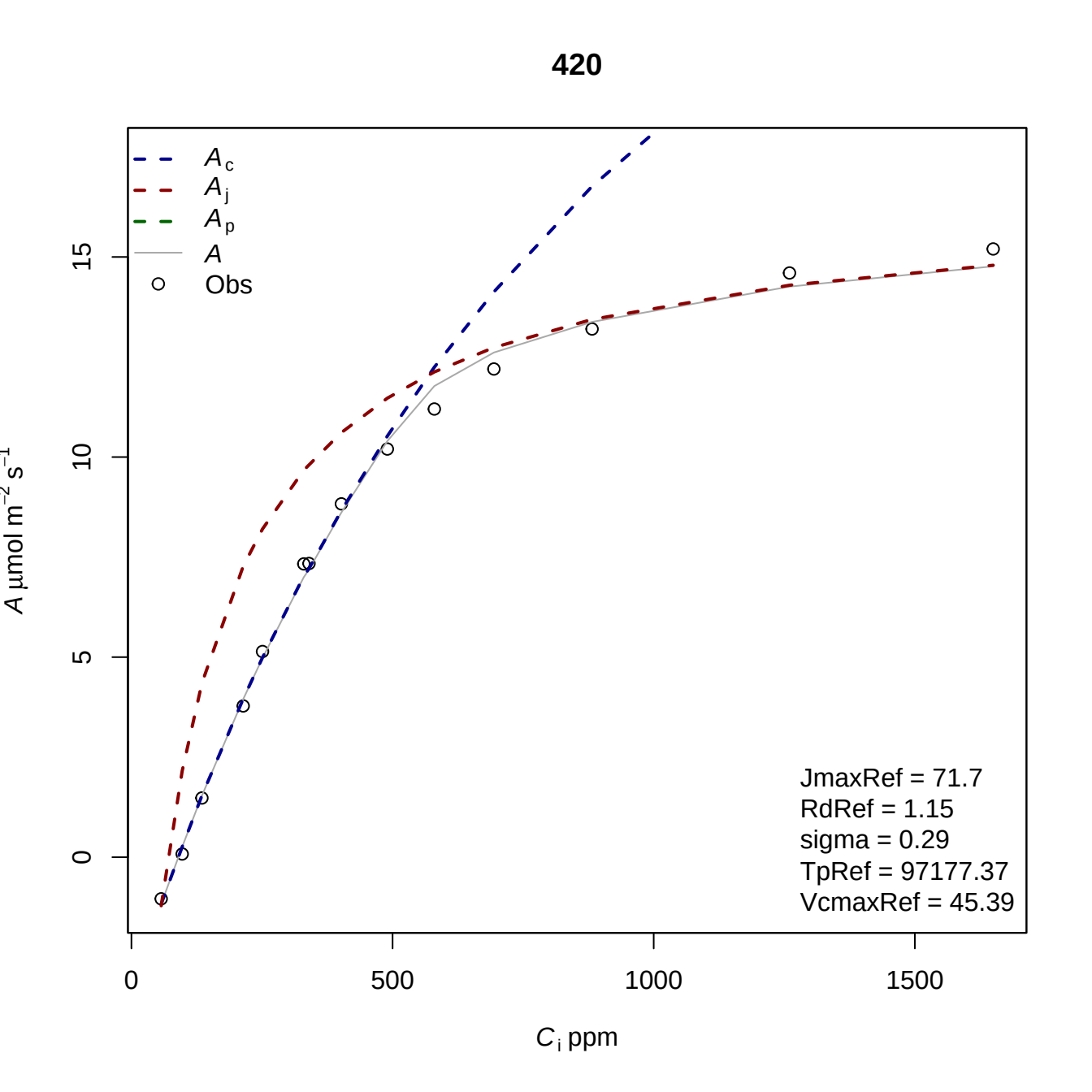

415
A_c
A_j
A_p
A
Obs
JmaxRef = 89.13
RdRef = 0.55
sigma = 0.15
TpRef = 196120.93
VcmaxRef = 72.73
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
200
400
600
800
1000
1200

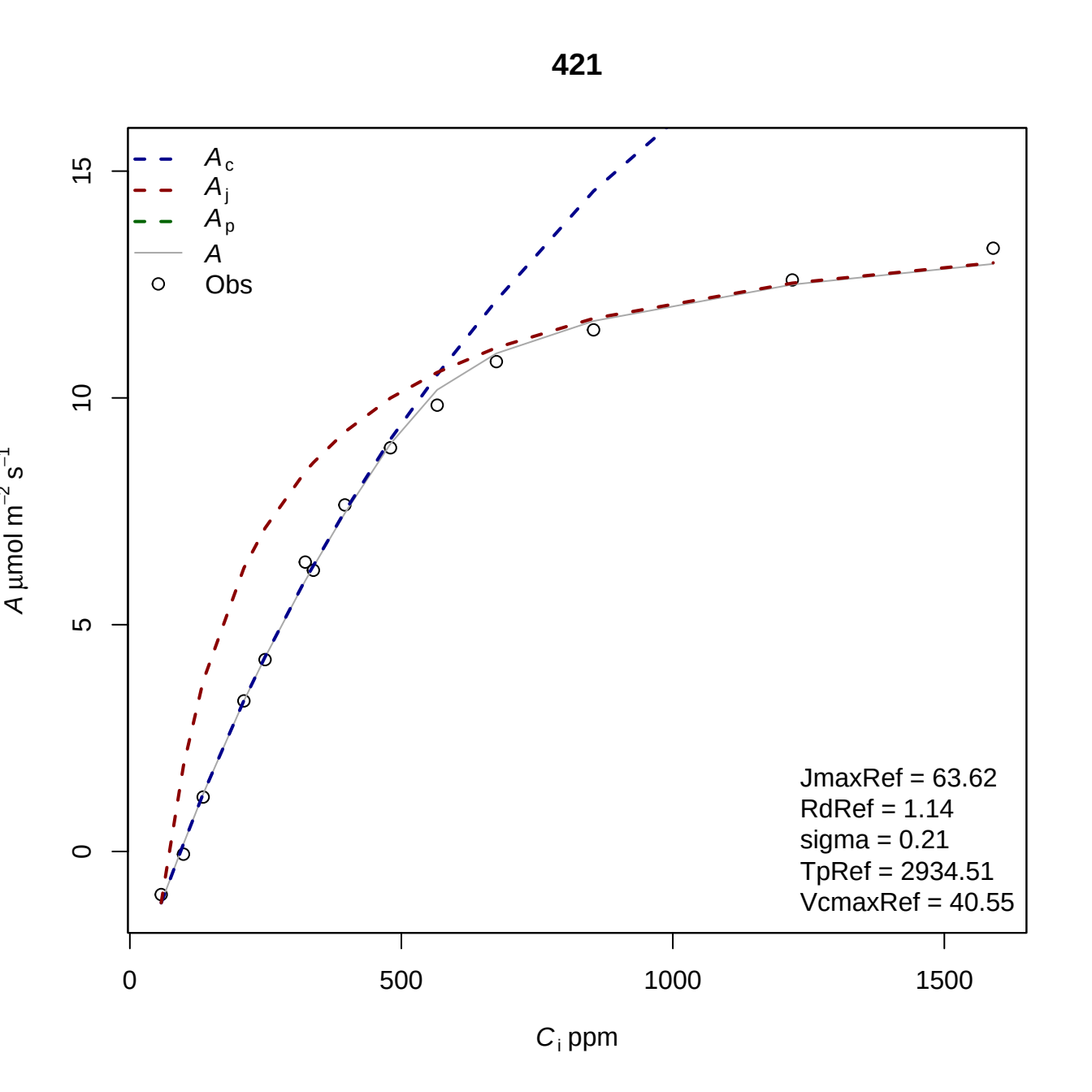

416
A_c
A_j
A_p
A
Obs
JmaxRef = 23.31
RdRef = 0.44
sigma = 0.11
TpRef = 1.36
VcmaxRef = 19.77
A µmol m^{-2} s^{-1}
C_i ppm
0
1
2
3
4
200
400
600
800
1000
1200
1400

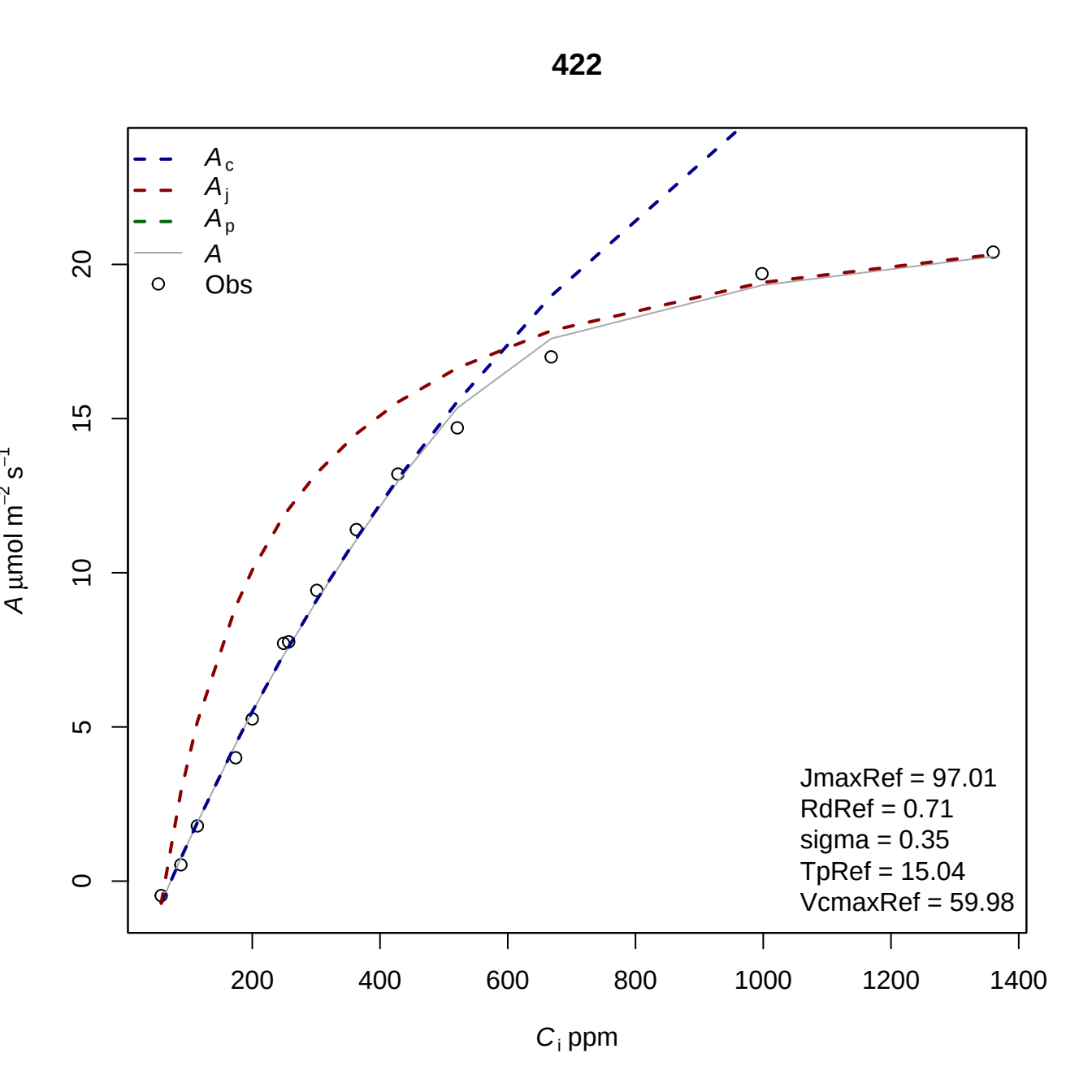

417
A_c
A_j
A_p
A
Obs
JmaxRef = 24.77
RdRef = 0.69
sigma = 0.08
TpRef = 12093.75
VcmaxRef = 15.83
A µmol m^{-2} s^{-1}
C_i ppm
0
1
2
3
4
5
500
1000
1500

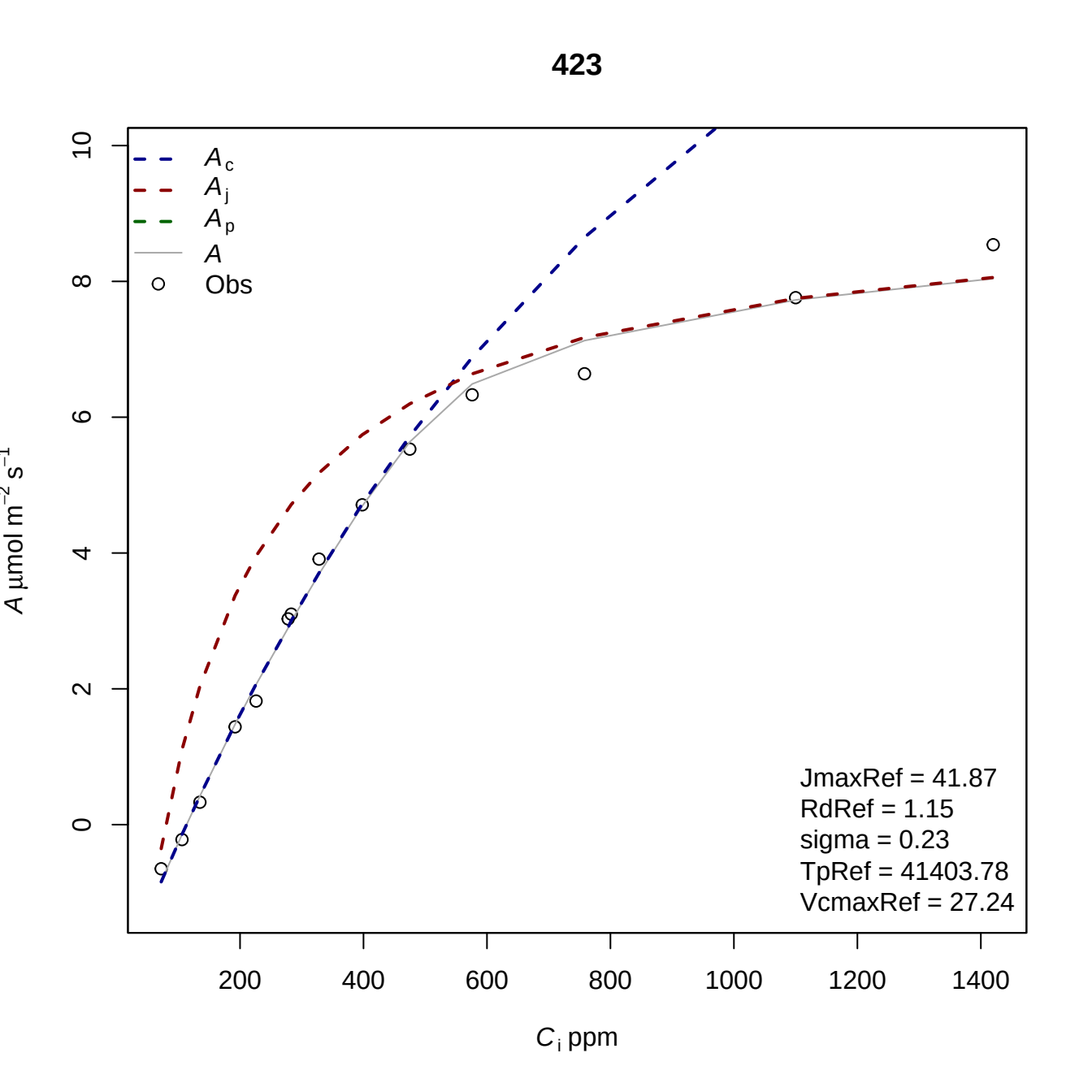

418
A_c
A_j
A_p
A
Obs
JmaxRef = 21.62
RdRef = 0.6
sigma = 0.07
TpRef = 1.34
VcmaxRef = 15.18
0
1
2
3
4
500
1000
1500
A µmol m^{-2} s^{-1}
C_i ppm

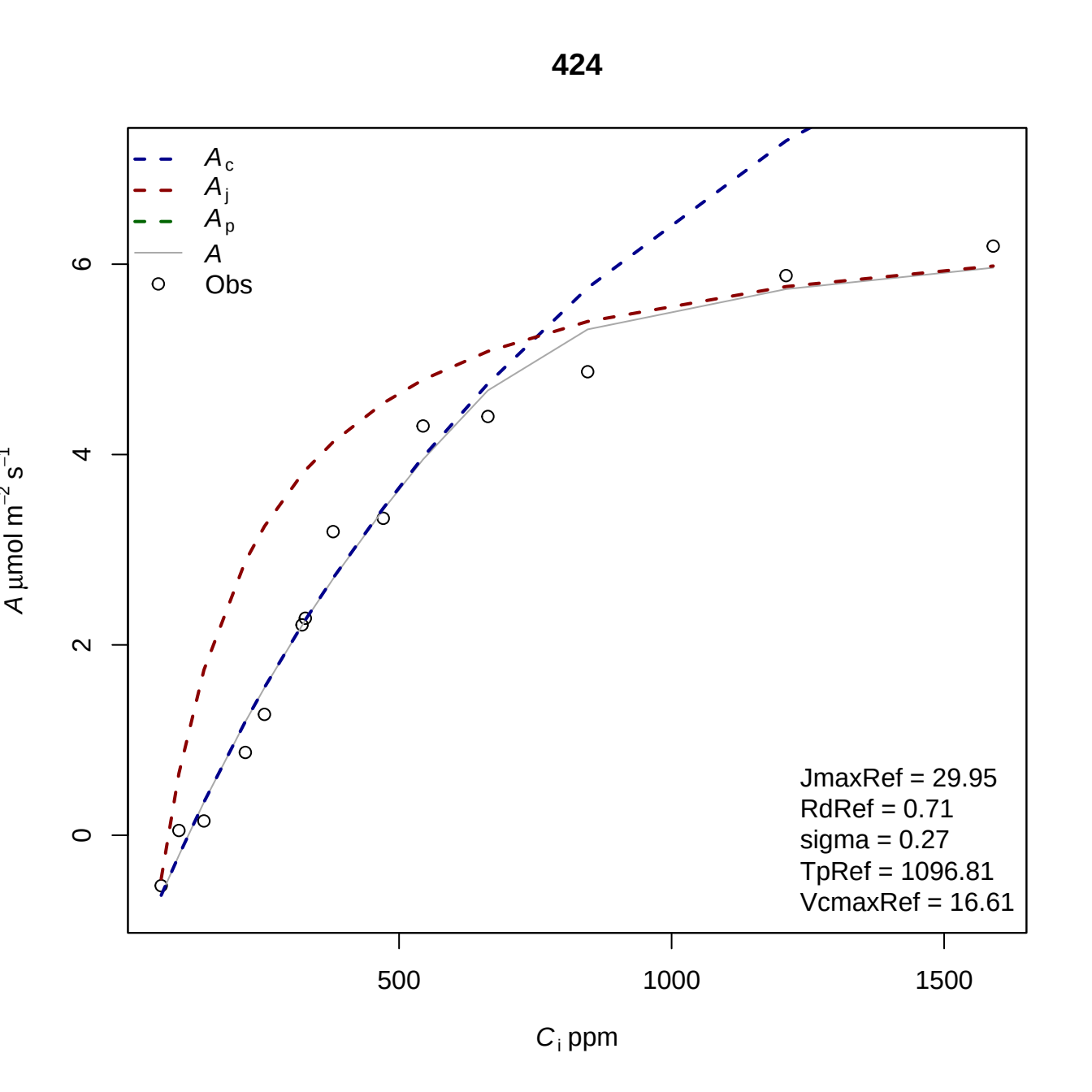

419
A_c
A_j
A_p
A
Obs
JmaxRef = 60.9
RdRef = 0.76
sigma = 0.19
TpRef = 3.89
VcmaxRef = 51.05
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
200
400
600
800
1000
1200

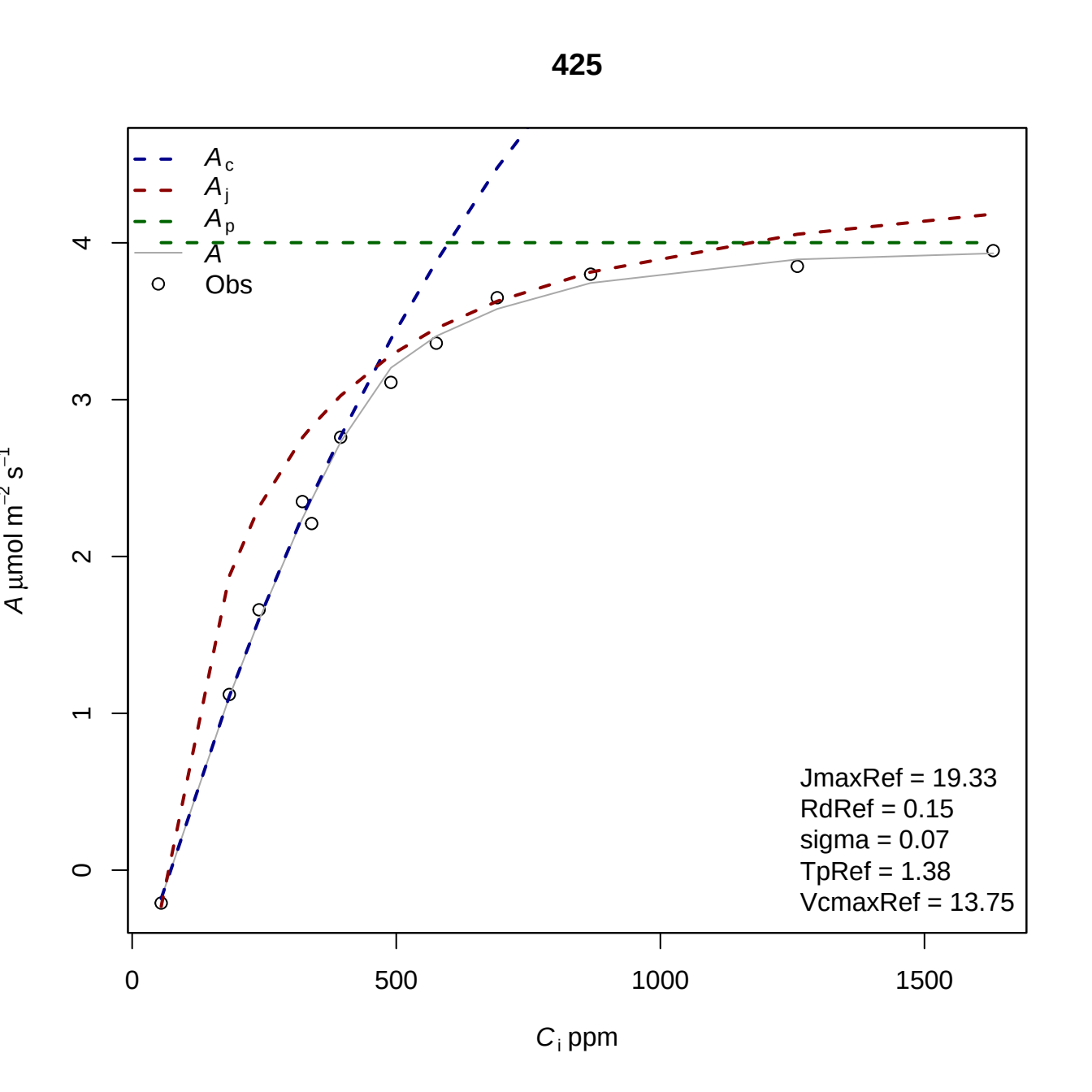

420
A_c
A_j
A_p
A
Obs
JmaxRef = 71.7
RdRef = 1.15
sigma = 0.29
TpRef = 97177.37
VcmaxRef = 45.39
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
500
1000
1500

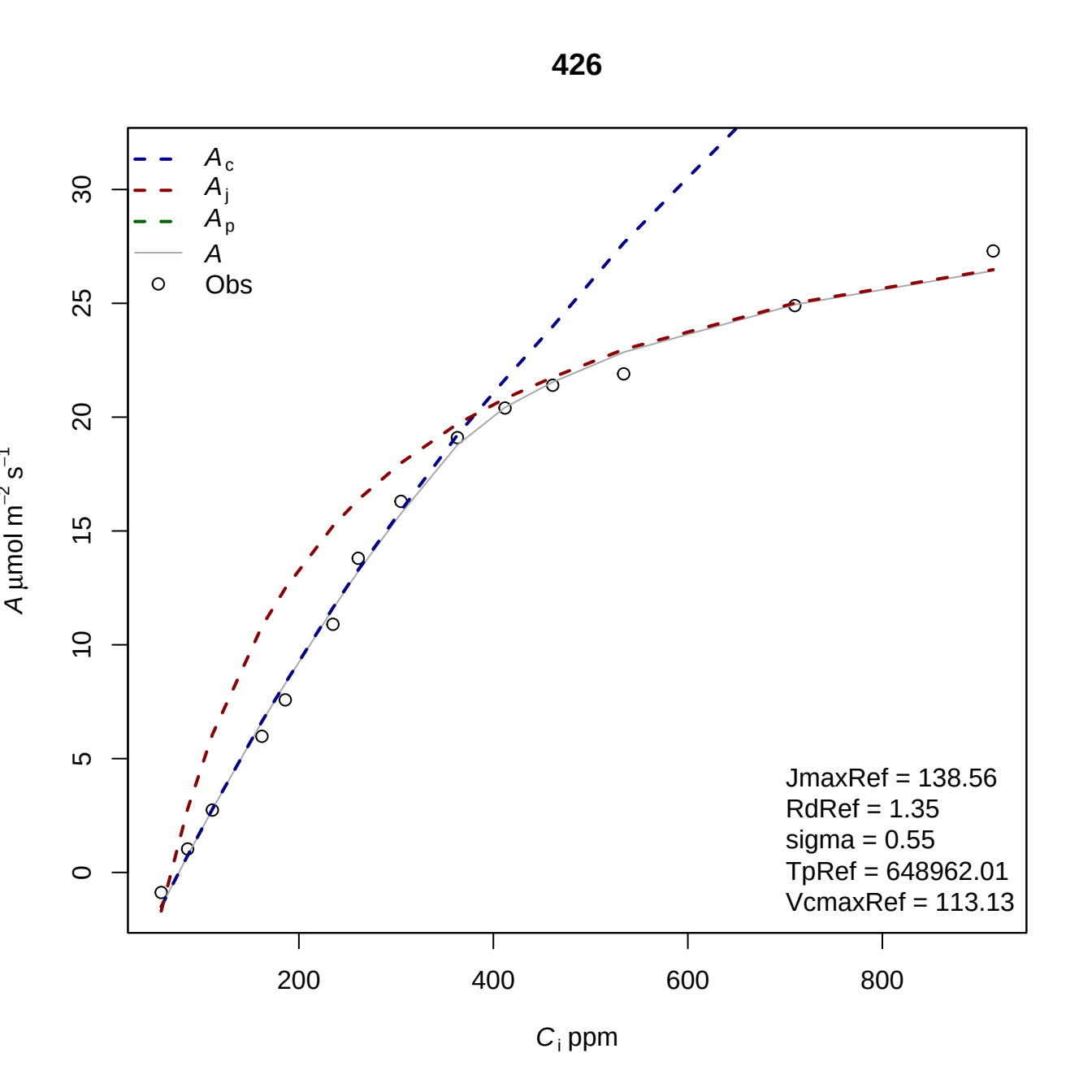

421
A_c
A_j
A_p
A
Obs
JmaxRef = 63.62
RdRef = 1.14
sigma = 0.21
TpRef = 2934.51
VcmaxRef = 40.55
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
500
1000
1500

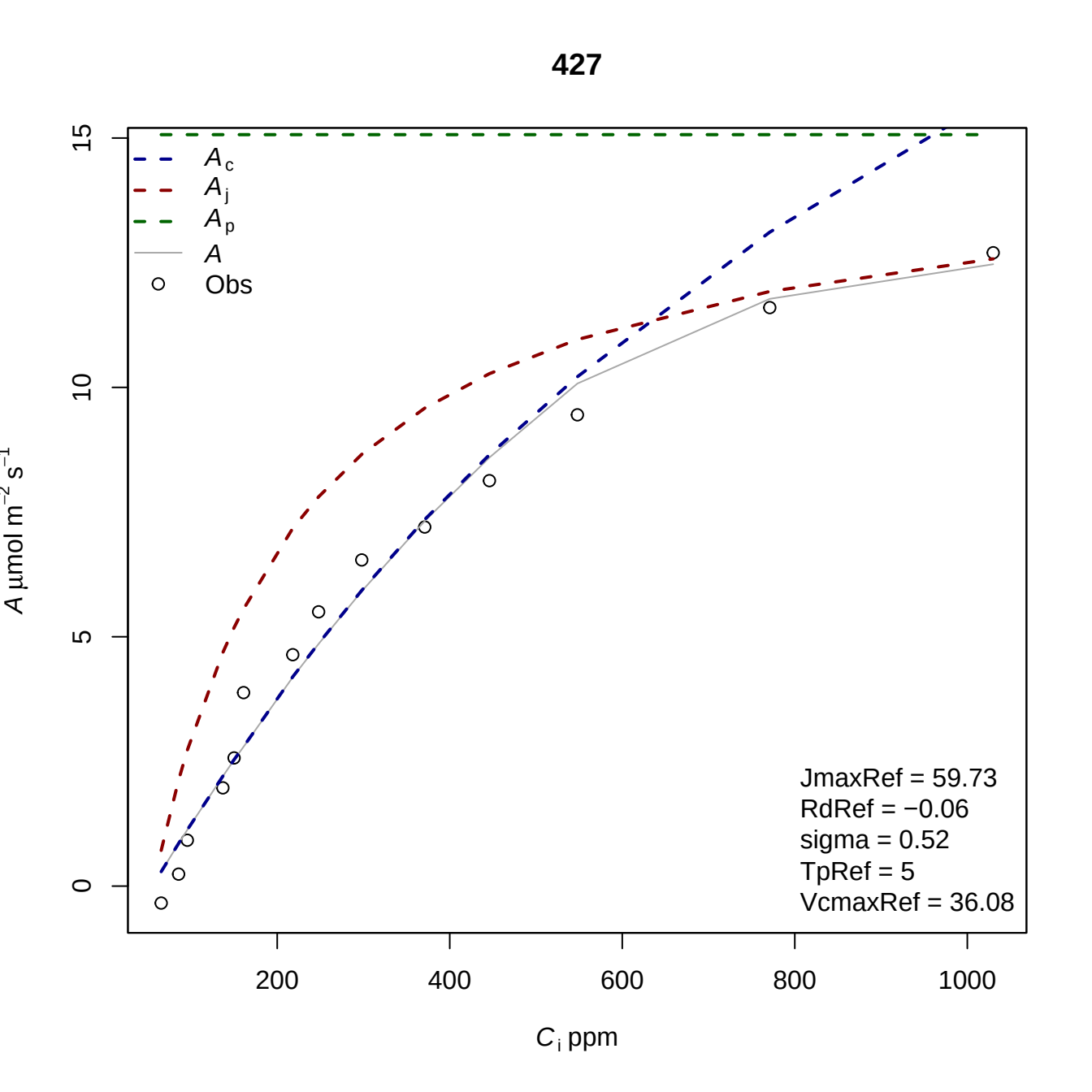

422
A_c
A_j
A_p
A
Obs
JmaxRef = 97.01
RdRef = 0.71
sigma = 0.35
TpRef = 15.04
VcmaxRef = 59.98
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
200
400
600
800
1000
1200
1400

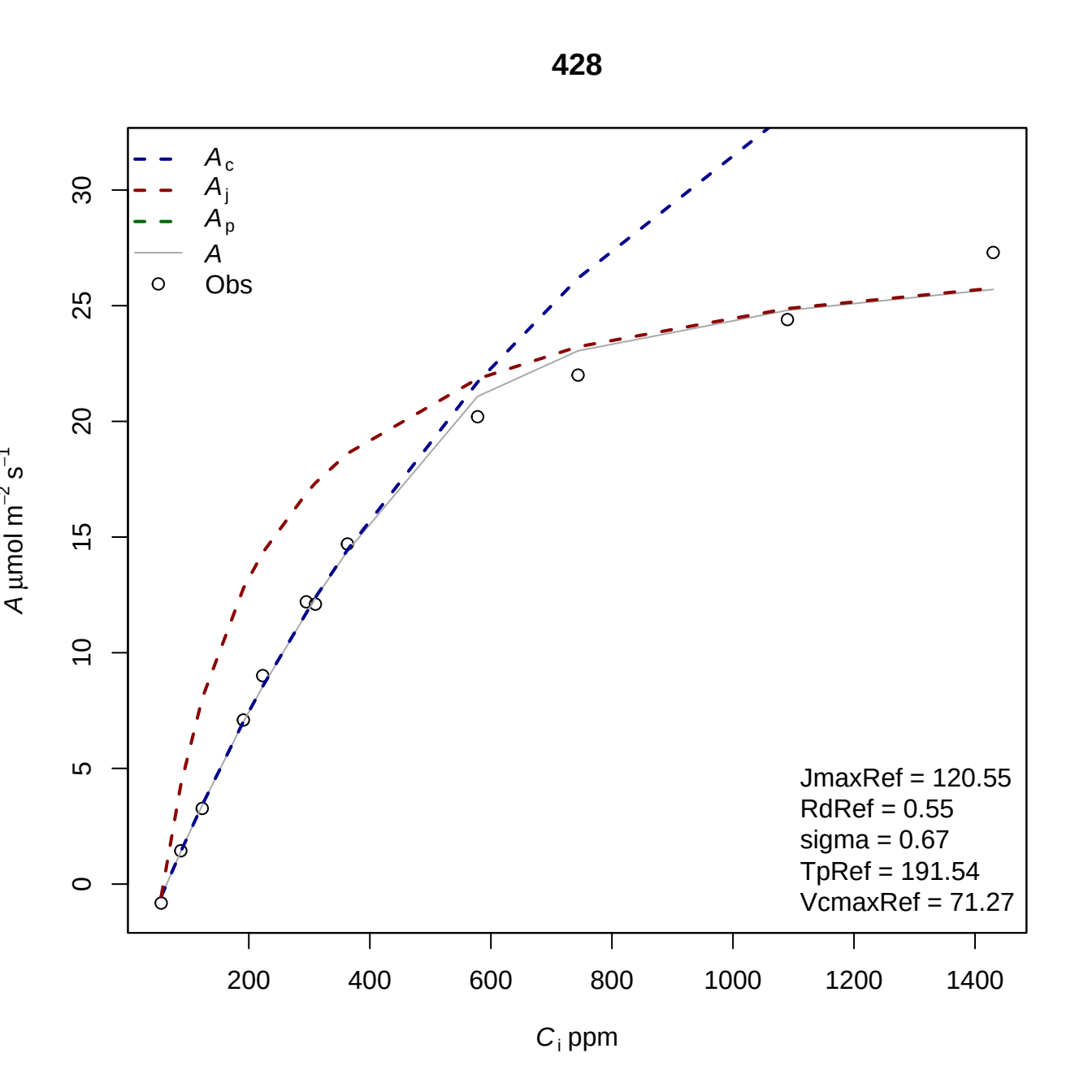

423
A_c
A_j
A_p
A
Obs
JmaxRef = 41.87
RdRef = 1.15
sigma = 0.23
TpRef = 41403.78
VcmaxRef = 27.24
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
200
400
600
800
1000
1200
1400

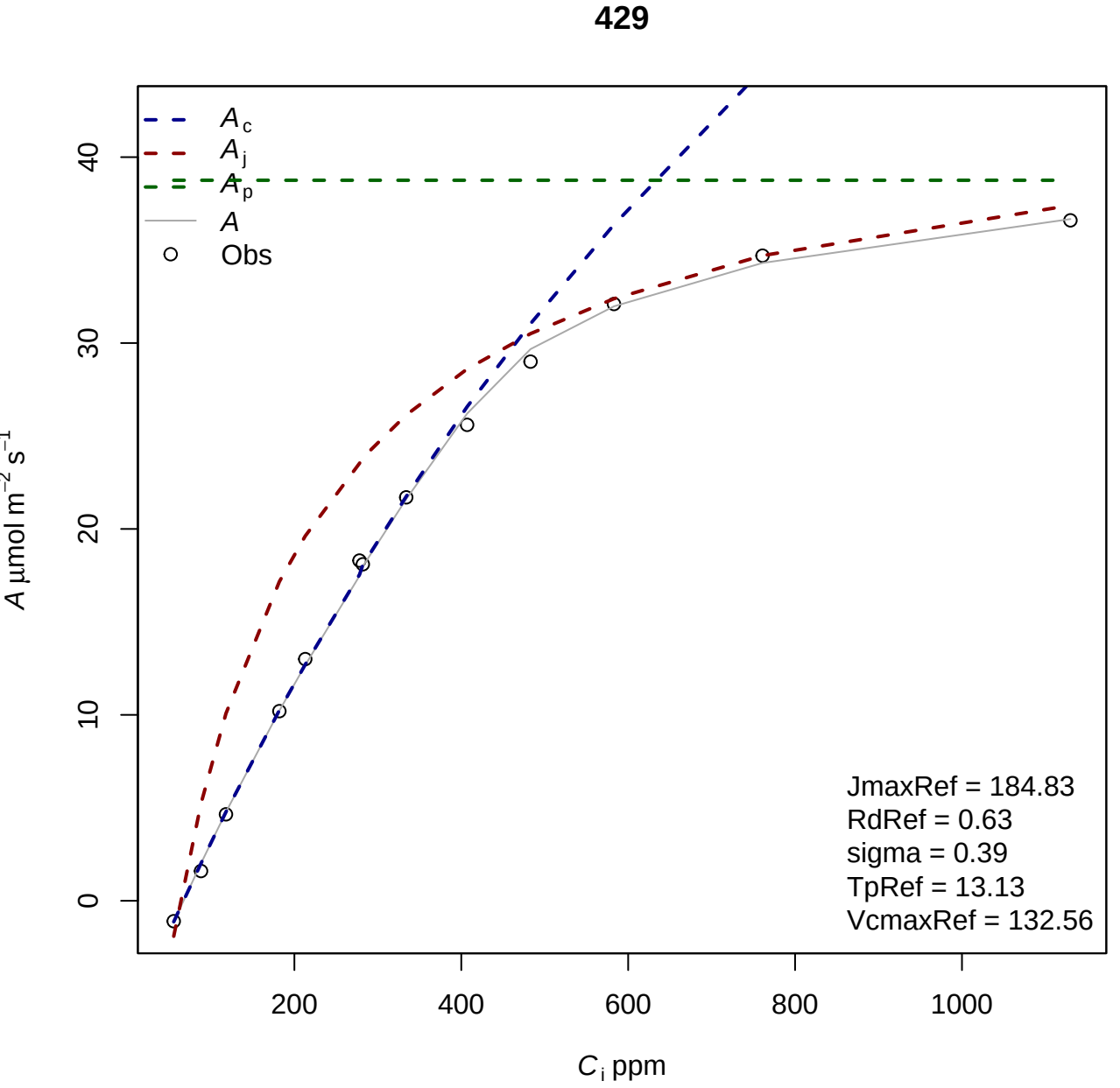

424
A_c
A_j
A_p
A
Obs
JmaxRef = 29.95
RdRef = 0.71
sigma = 0.27
TpRef = 1096.81
VcmaxRef = 16.61
0
2
4
6
500
1000
1500
A µmol m^{-2} s^{-1}
C_i ppm

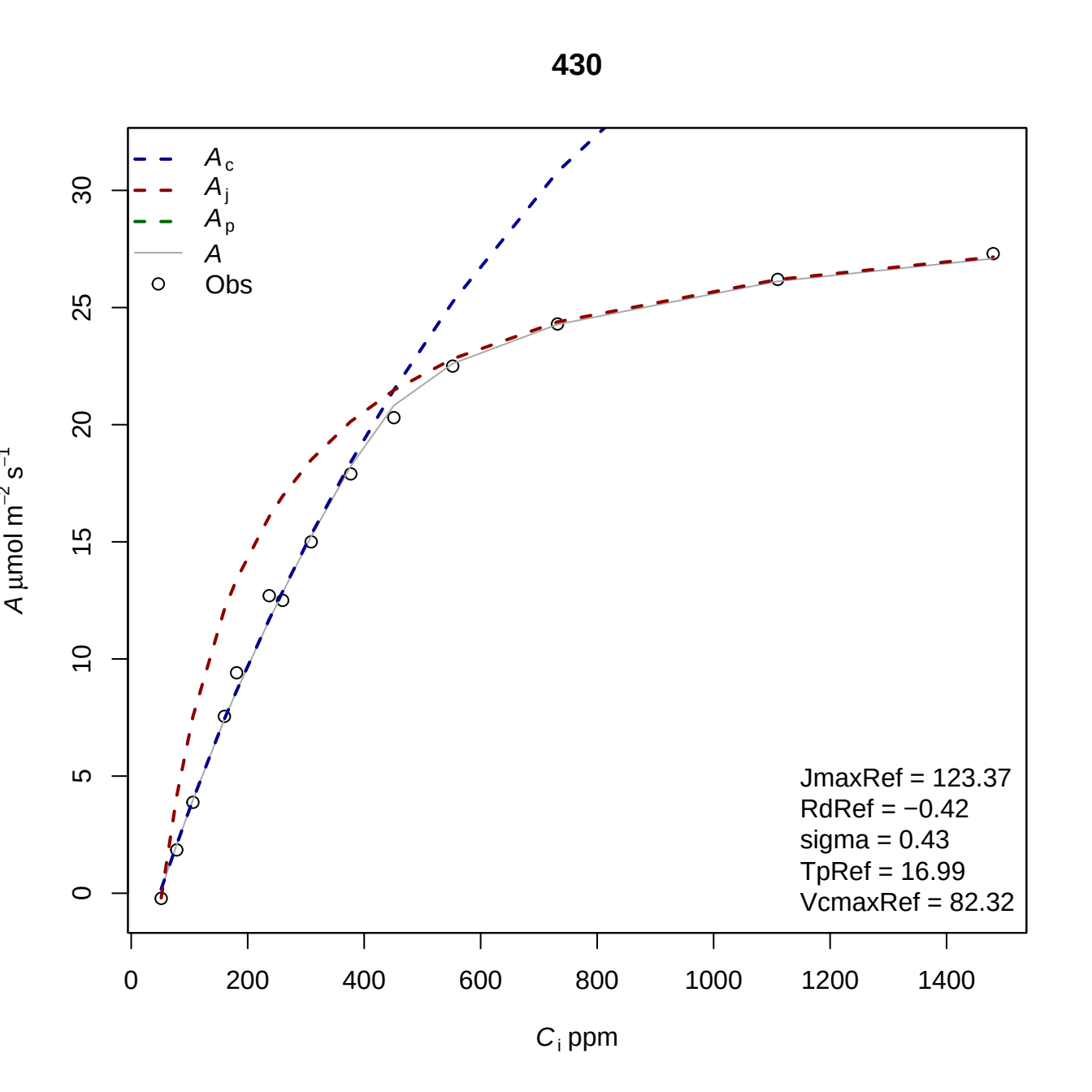

425
A_c
A_j
A_p
A
Obs
JmaxRef = 19.33
RdRef = 0.15
sigma = 0.07
TpRef = 1.38
VcmaxRef = 13.75
A µmol m^{-2} s^{-1}
C_i ppm
0
1
2
3
4
0
500
1000
1500

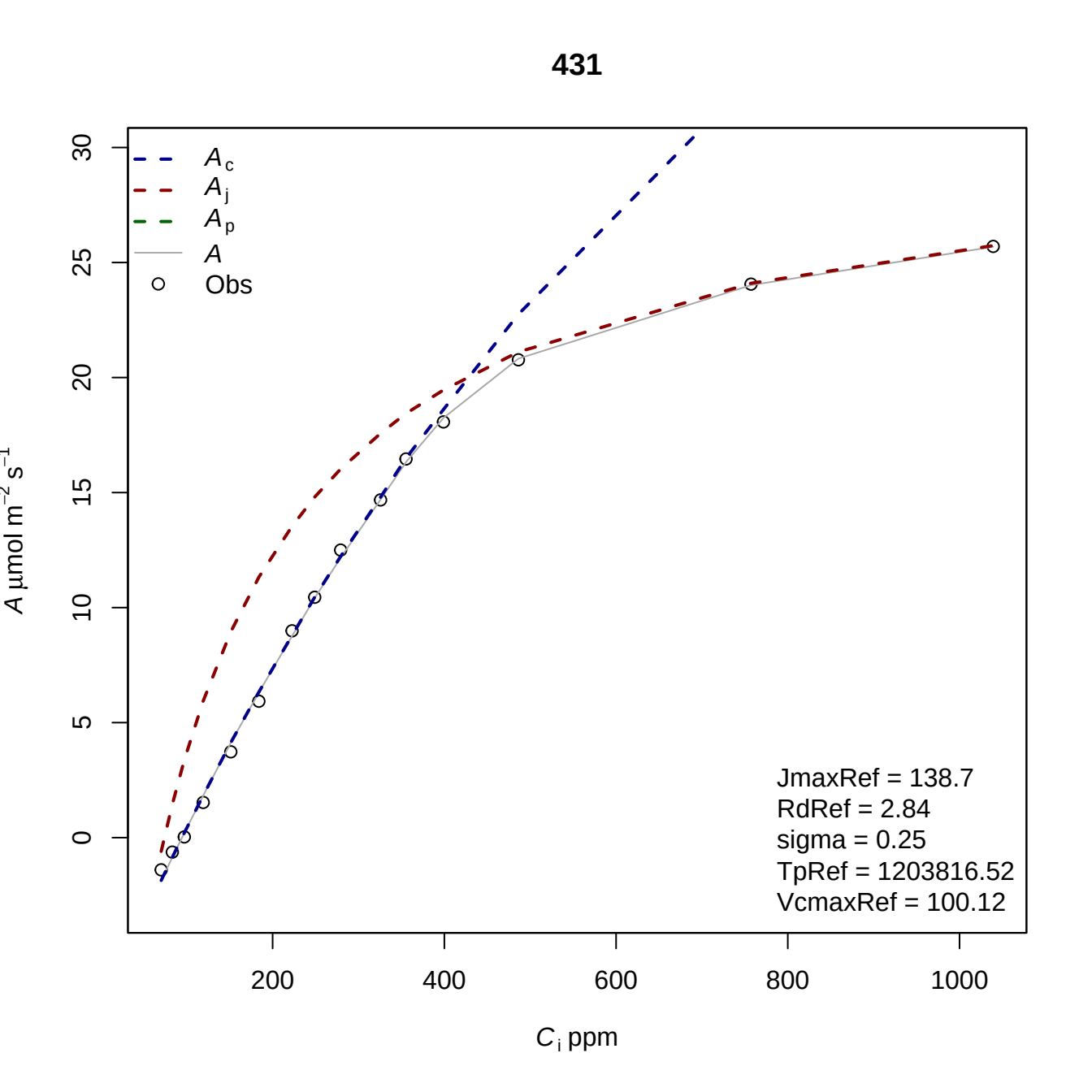

426
A_c
A_j
A_p
A
Obs
JmaxRef = 138.56
RdRef = 1.35
sigma = 0.55
TpRef = 648962.01
VcmaxRef = 113.13
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
200
400
600
800

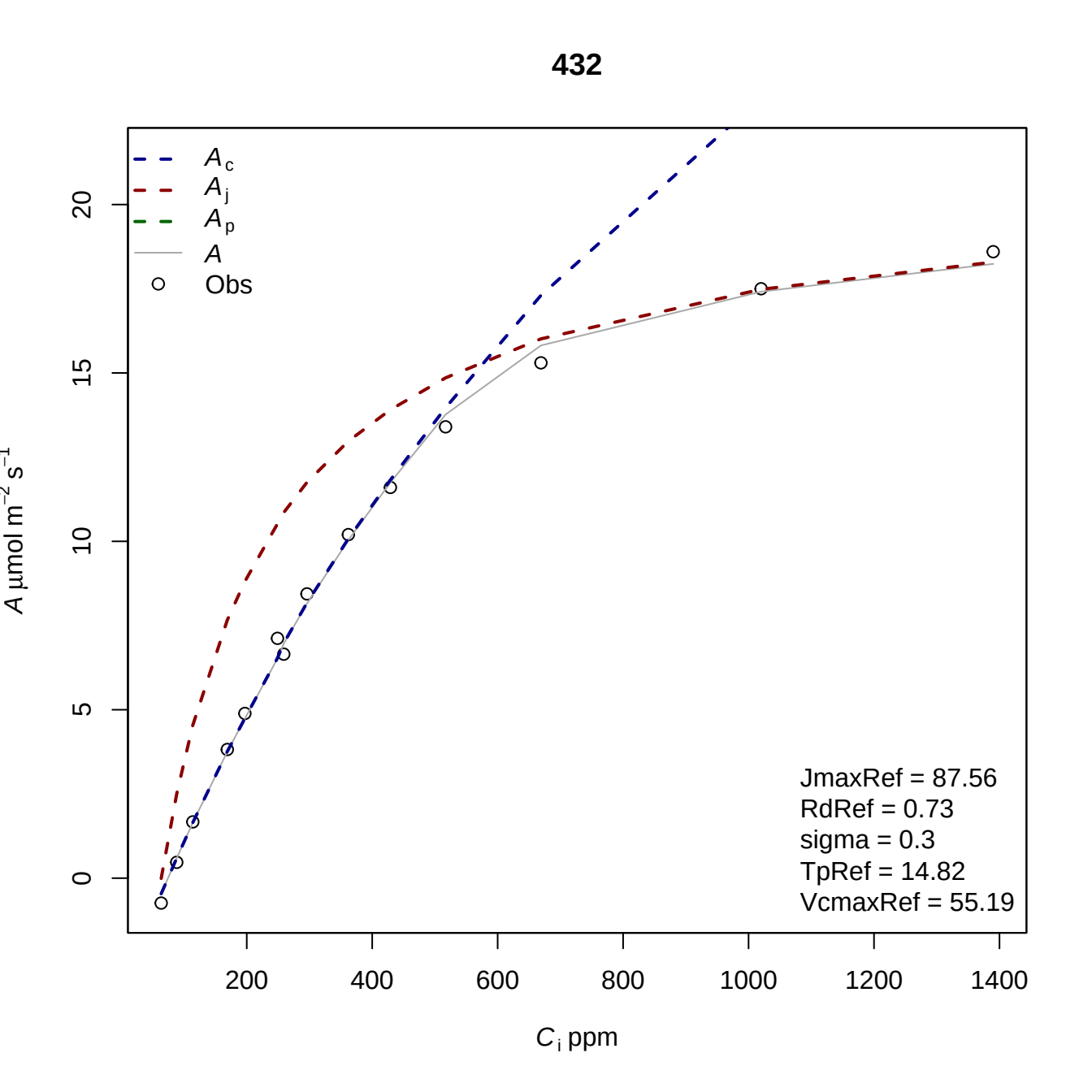

427
A_c
A_j
A_p
A
Obs
JmaxRef = 59.73
RdRef = −0.06
sigma = 0.52
TpRef = 5
VcmaxRef = 36.08
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
200
400
600
800
1000

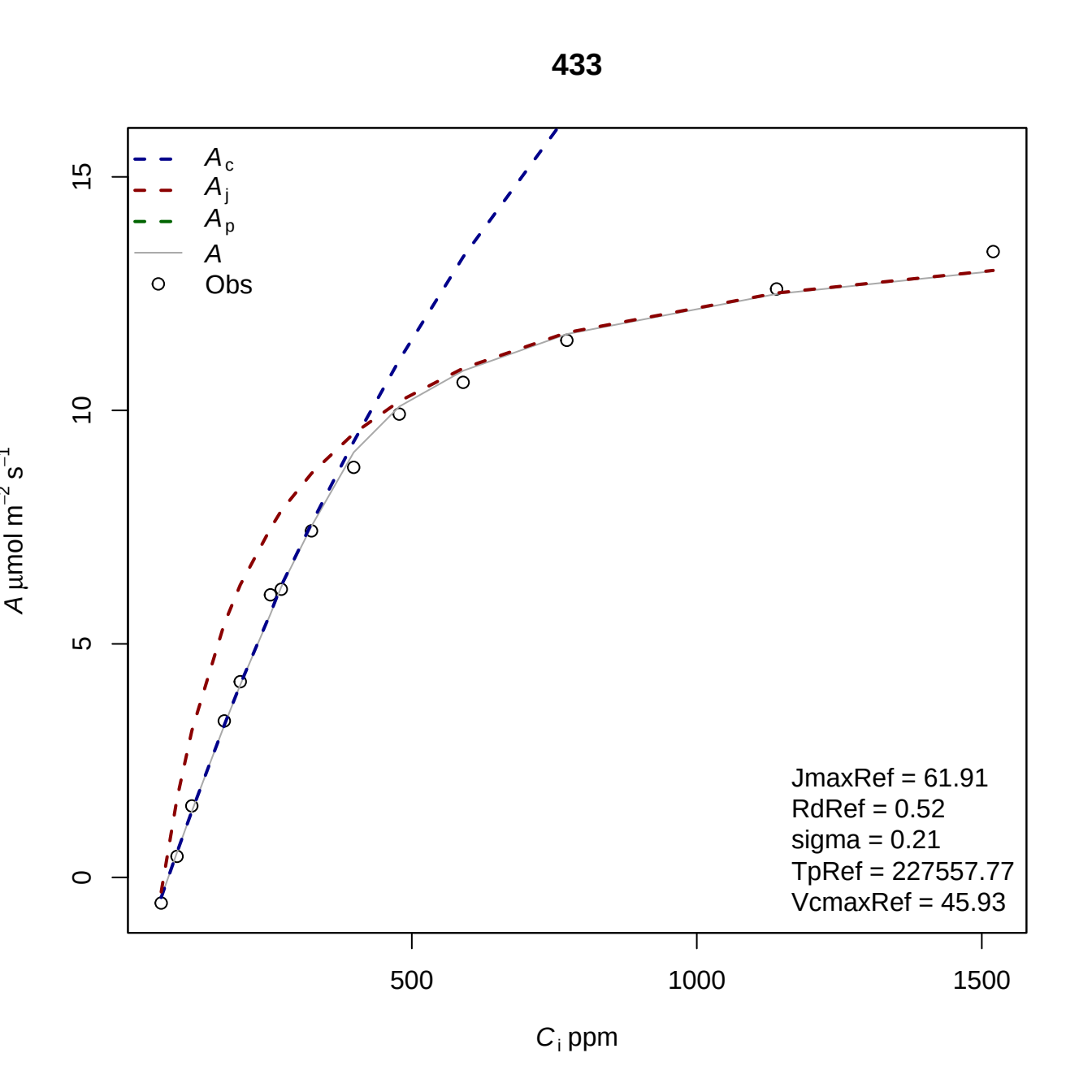

428
A_c
A_j
A_p
A
Obs
JmaxRef = 120.55
RdRef = 0.55
sigma = 0.67
TpRef = 191.54
VcmaxRef = 71.27
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
200
400
600
800
1000
1200
1400

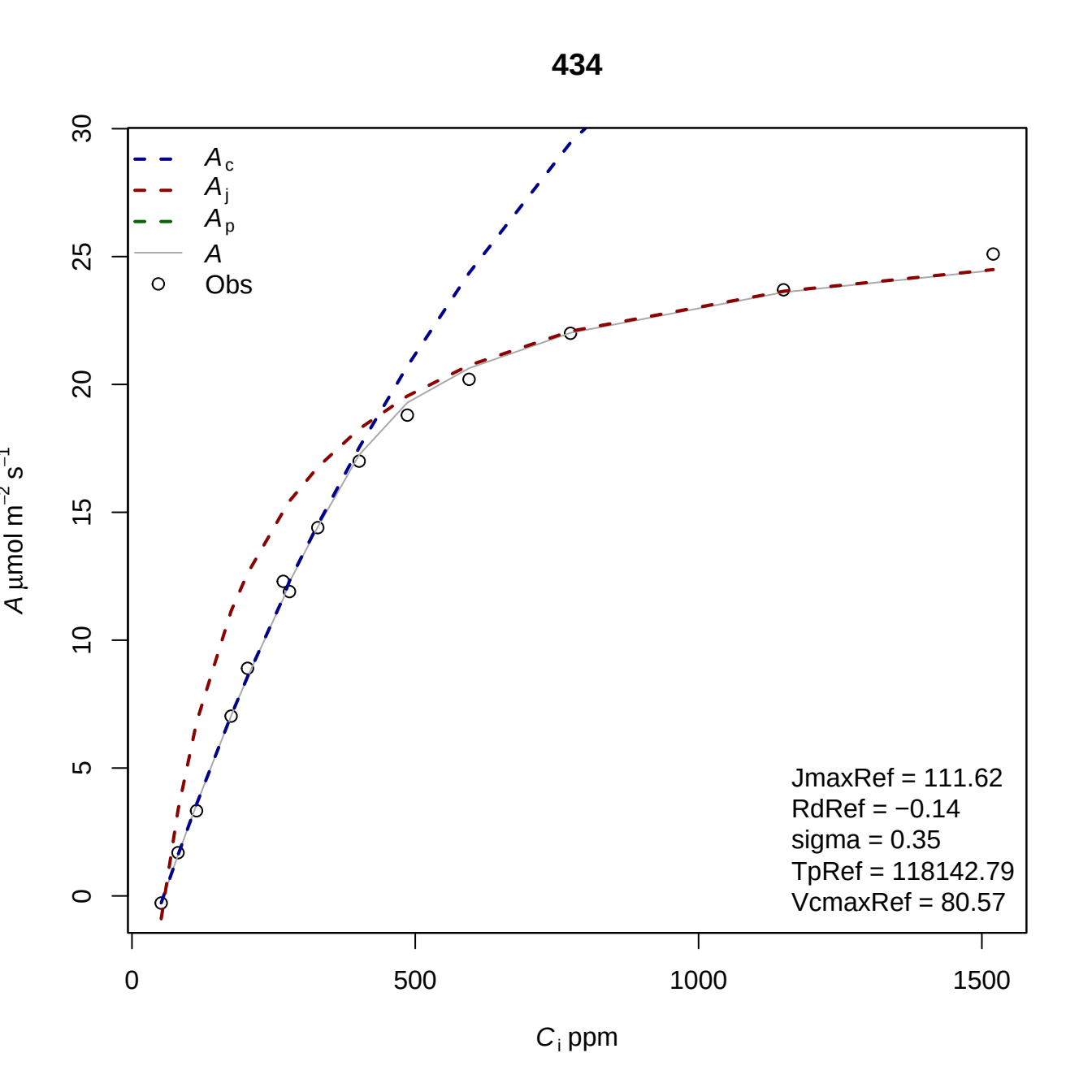

429
A_c
A_j
A_p
A
Obs
JmaxRef = 184.83
RdRef = 0.63
sigma = 0.39
TpRef = 13.13
VcmaxRef = 132.56
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
200
400
600
800
1000

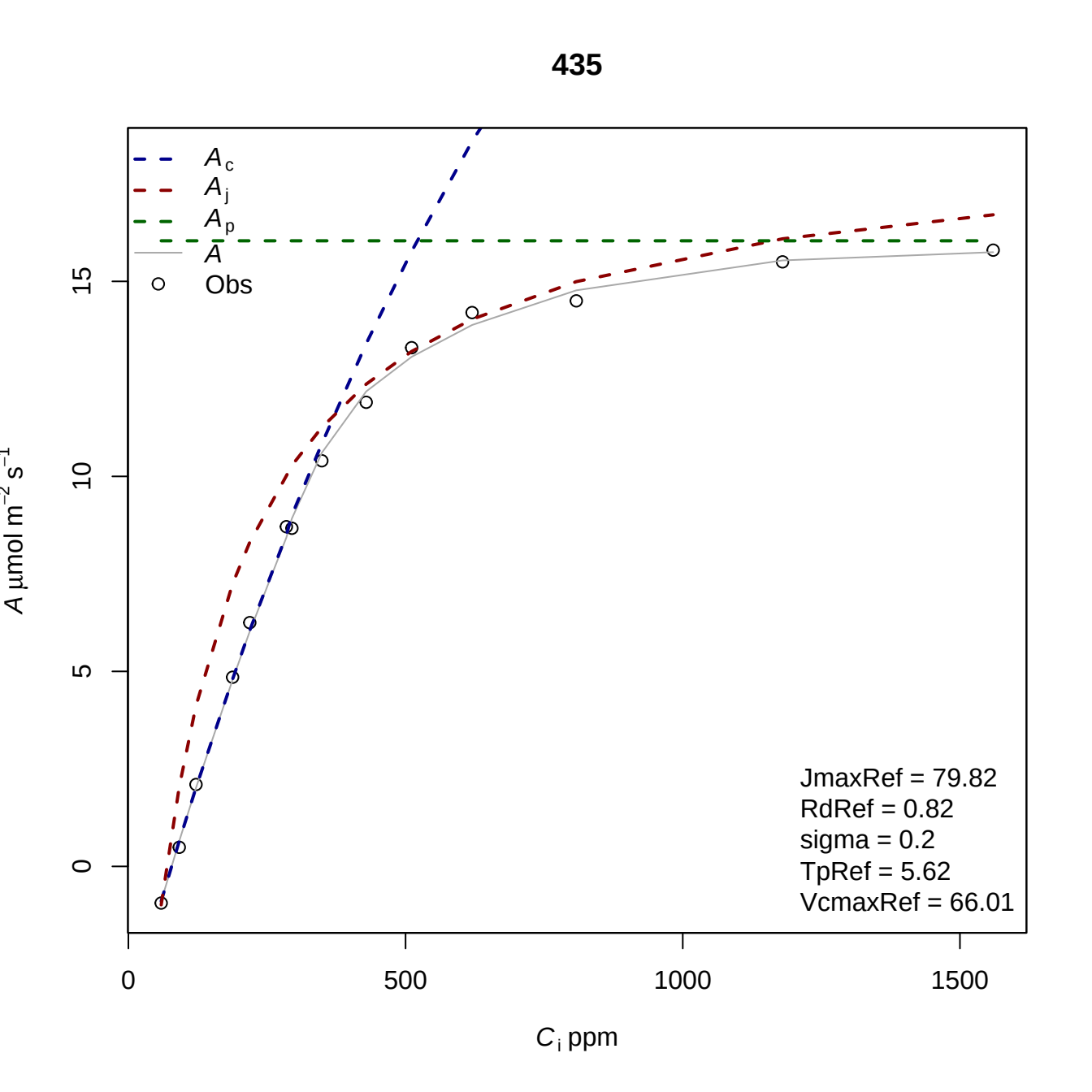

430
A_c
A_j
A_p
A
Obs
JmaxRef = 123.37
RdRef = −0.42
sigma = 0.43
TpRef = 16.99
VcmaxRef = 82.32
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
1400

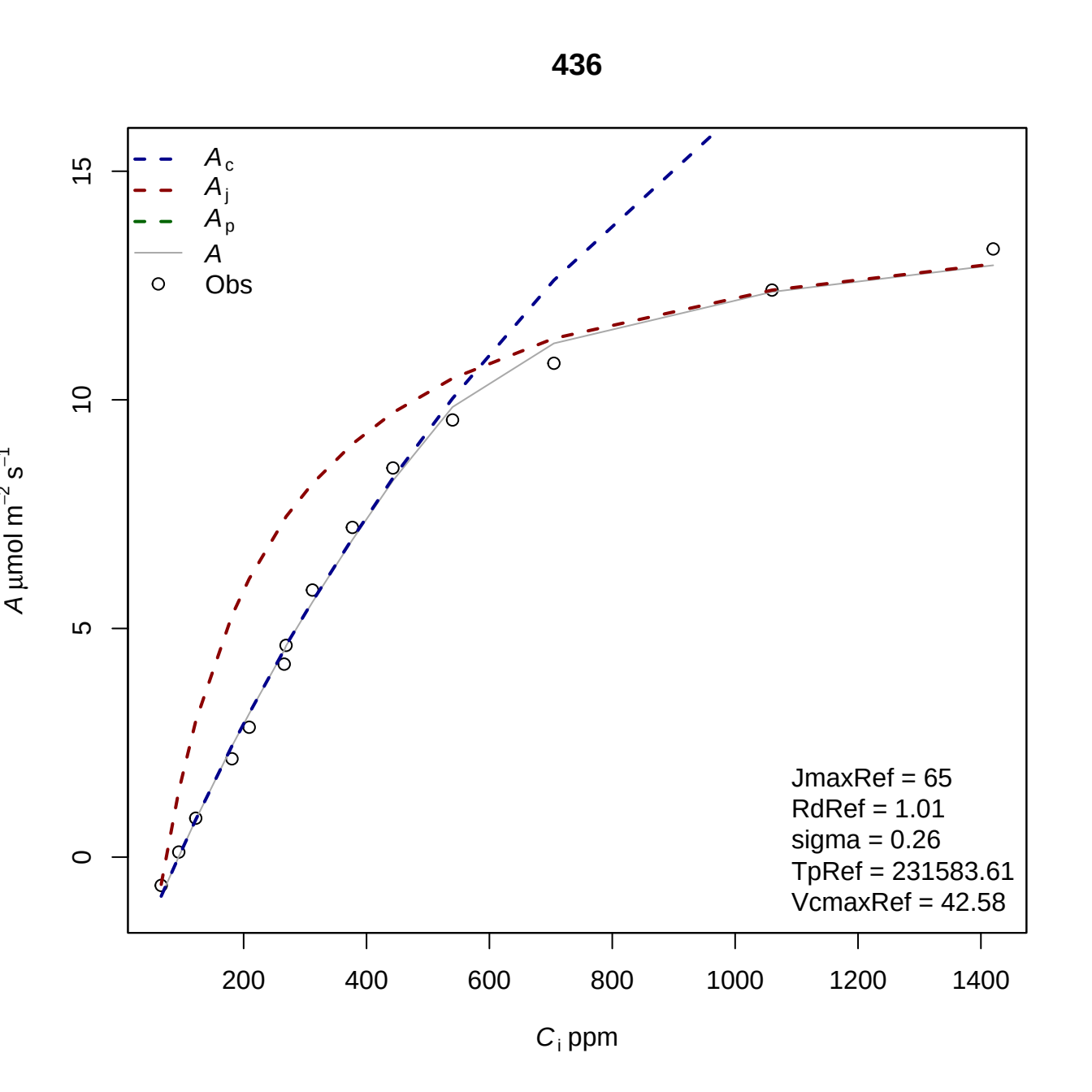

431
A_c
A_j
A_p
A
Obs
JmaxRef = 138.7
RdRef = 2.84
sigma = 0.25
TpRef = 1203816.52
VcmaxRef = 100.12
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
200
400
600
800
1000

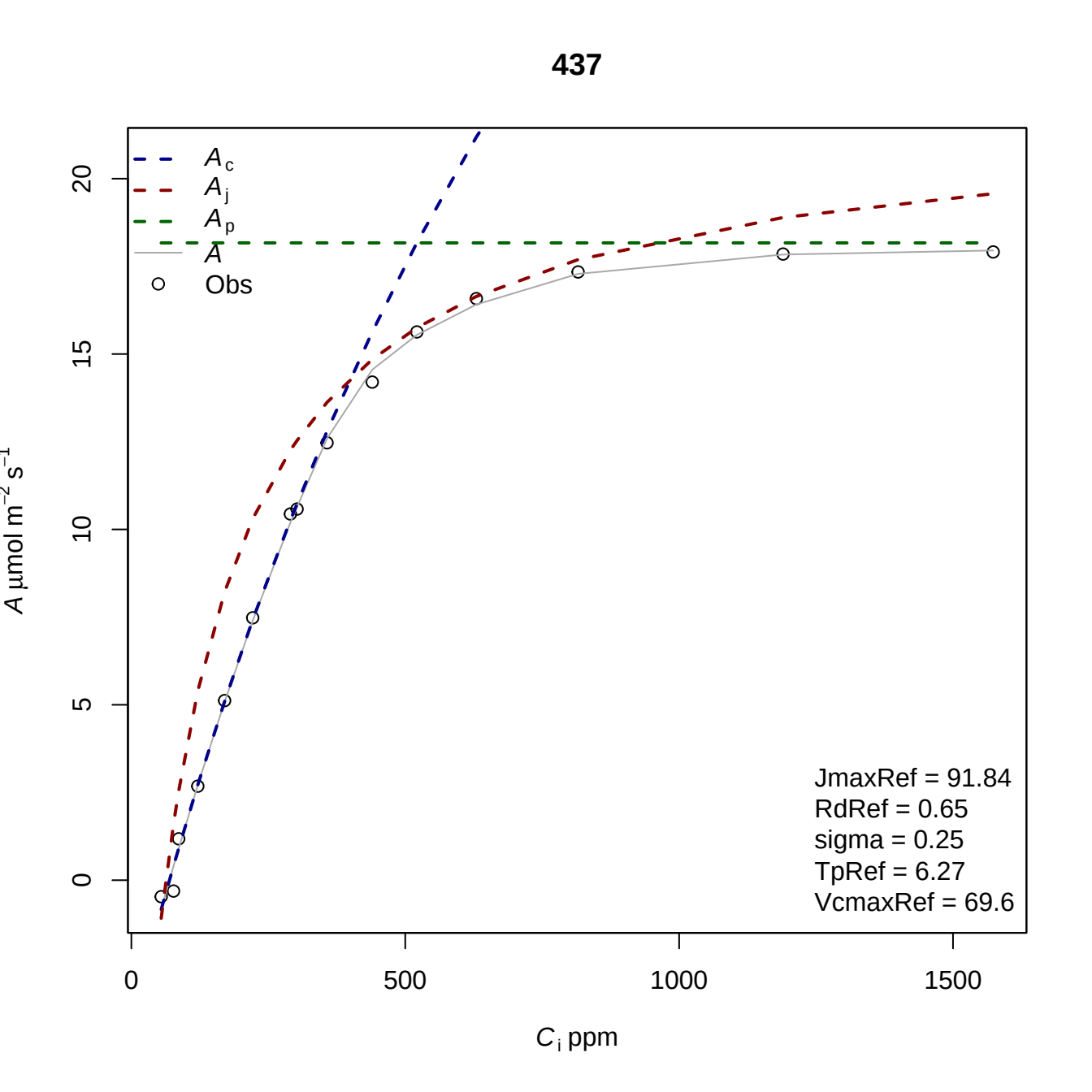

432
A_c
A_j
A_p
A
Obs
JmaxRef = 87.56
RdRef = 0.73
sigma = 0.3
TpRef = 14.82
VcmaxRef = 55.19
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
200
400
600
800
1000
1200
1400

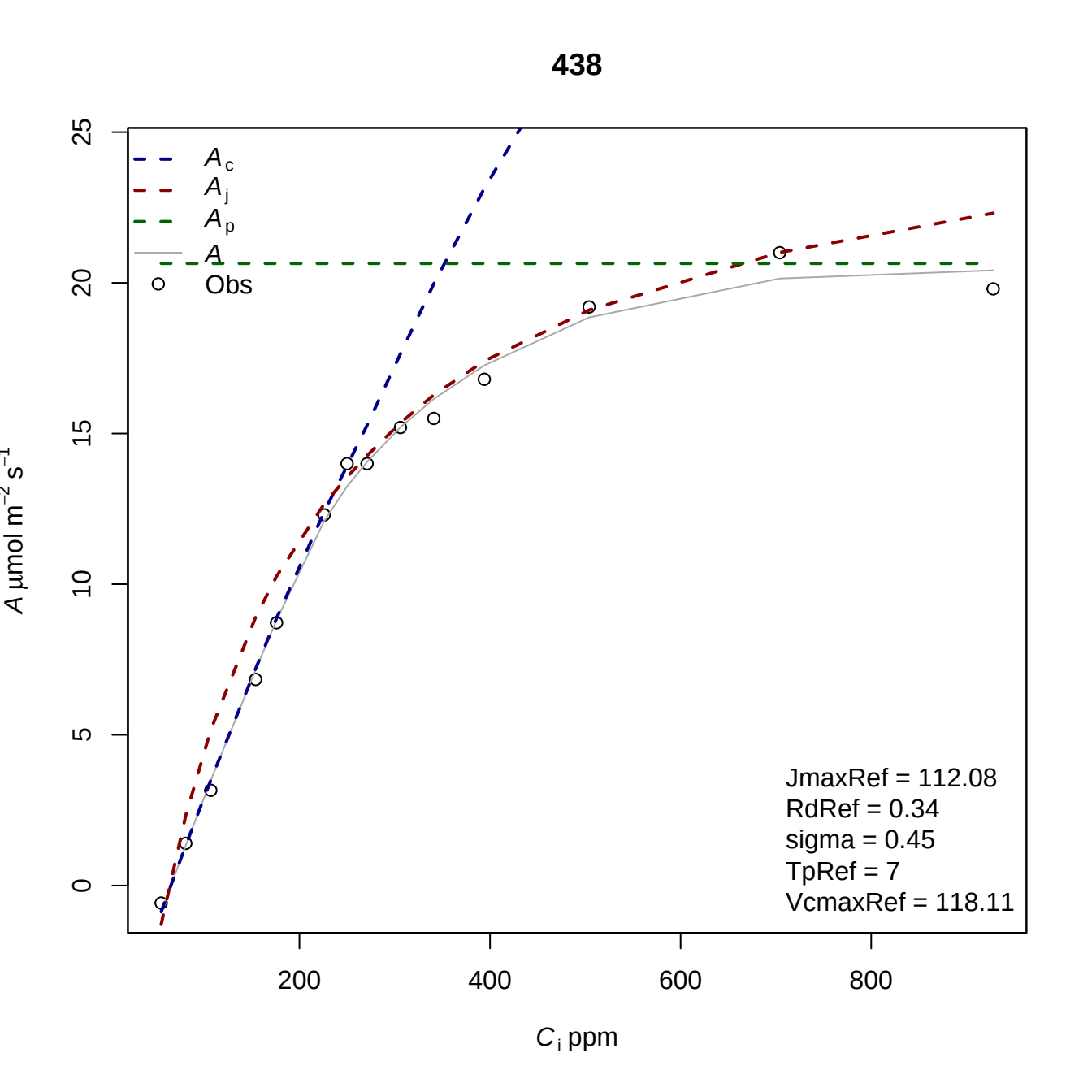

433
A_c
A_j
A_p
A
Obs
JmaxRef = 61.91
RdRef = 0.52
sigma = 0.21
TpRef = 227557.77
VcmaxRef = 45.93
0
5
10
15
500
1000
1500
A µmol m^{-2} s^{-1}
C_i ppm

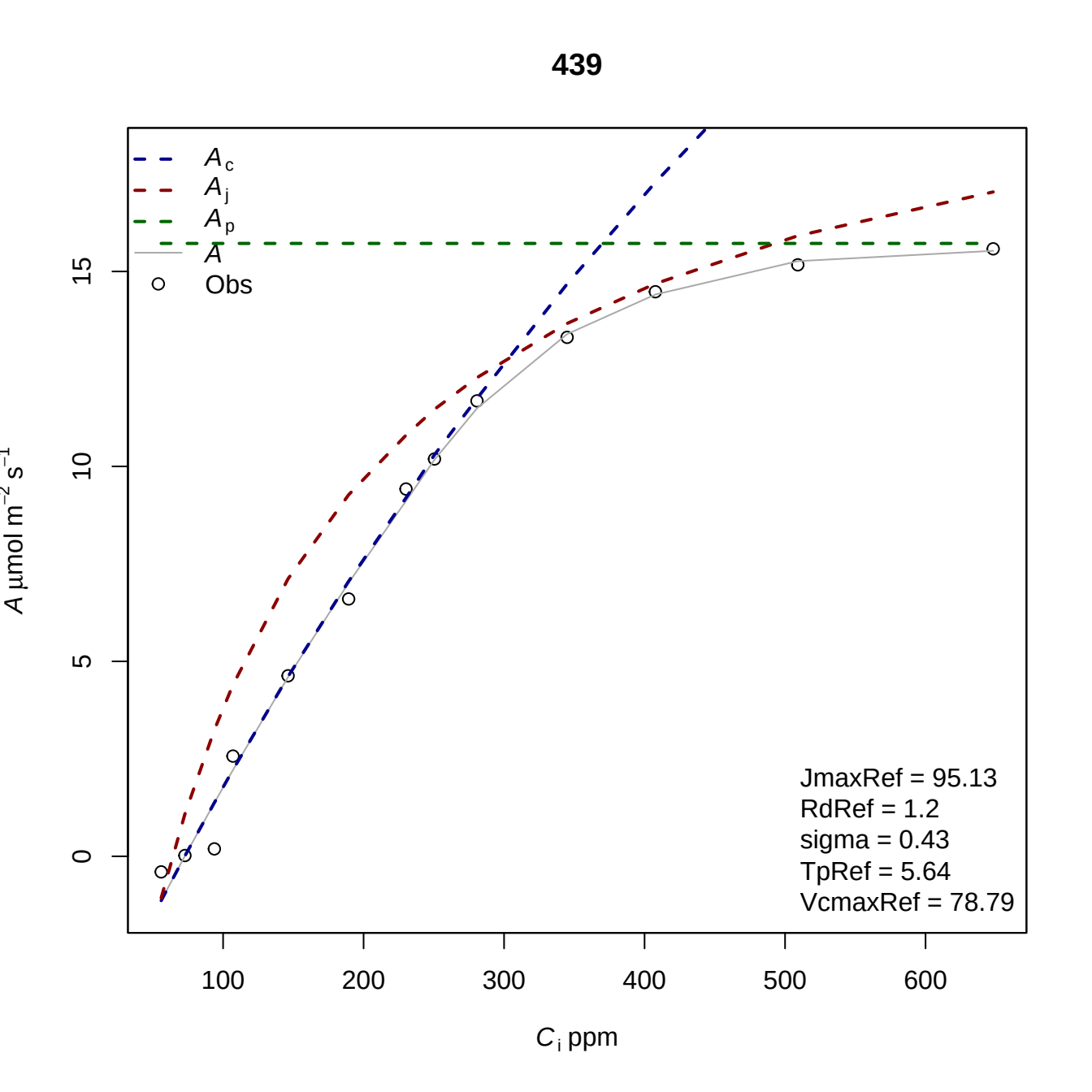

434
A_c
A_j
A_p
A
Obs
JmaxRef = 111.62
RdRef = −0.14
sigma = 0.35
TpRef = 118142.79
VcmaxRef = 80.57
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
0
500
1000
1500

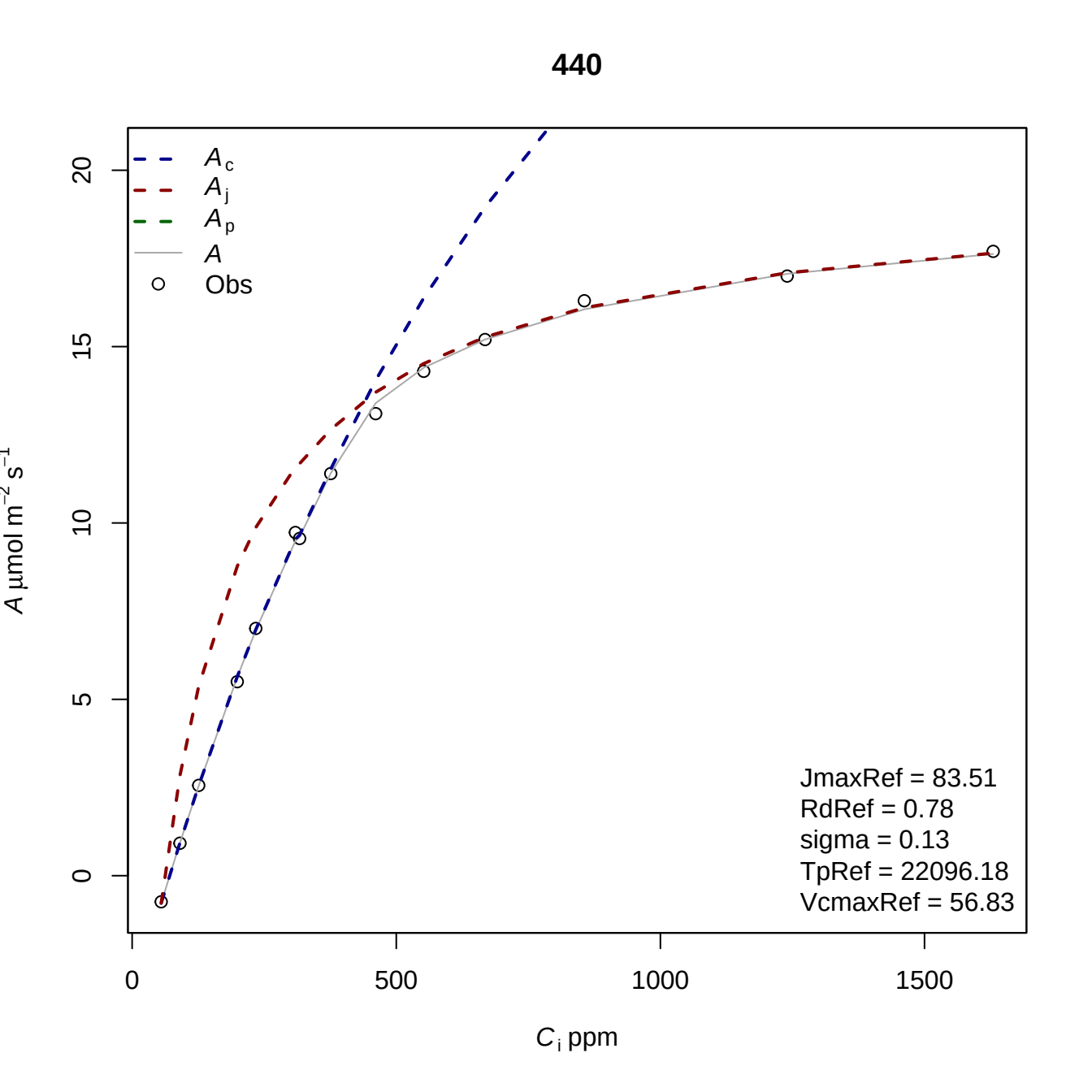

435
A_c
A_j
A_p
A
Obs
JmaxRef = 79.82
RdRef = 0.82
sigma = 0.2
TpRef = 5.62
VcmaxRef = 66.01
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
C_i ppm

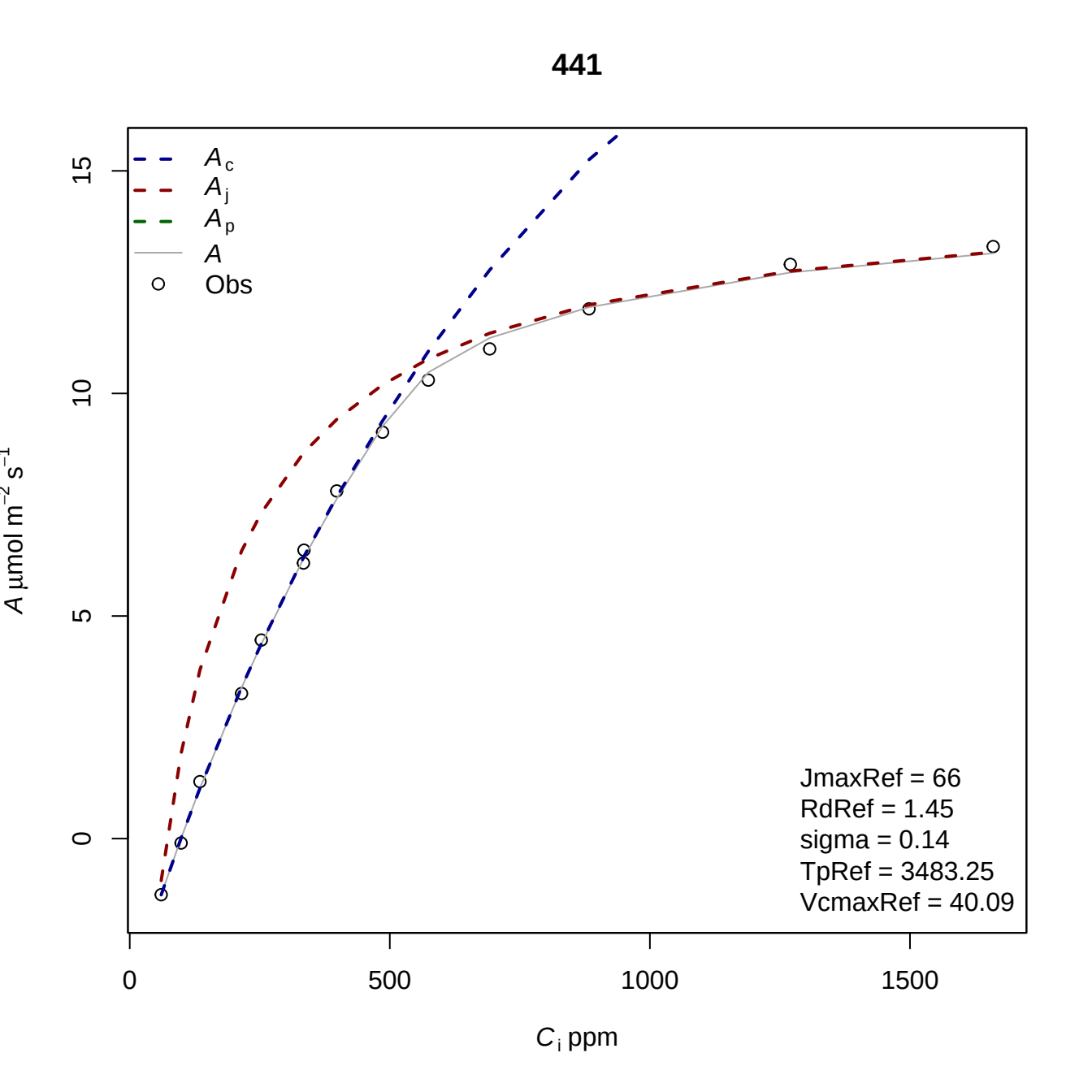

436
A_c
A_j
A_p
A
Obs
JmaxRef = 65
RdRef = 1.01
sigma = 0.26
TpRef = 231583.61
VcmaxRef = 42.58
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
200
400
600
800
1000
1200
1400

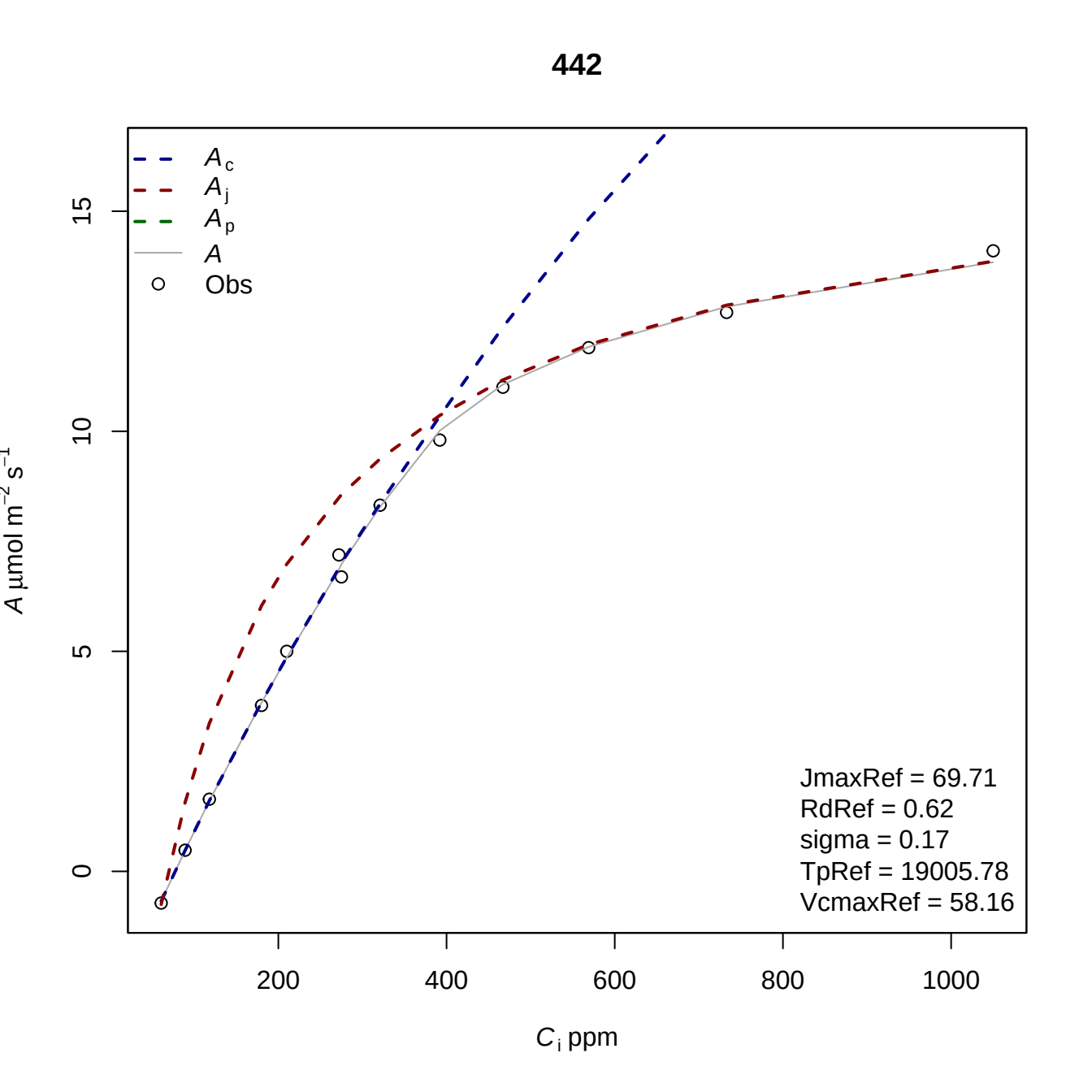

437
A_c
A_j
A_p
A
Obs
JmaxRef = 91.84
RdRef = 0.65
sigma = 0.25
TpRef = 6.27
VcmaxRef = 69.6
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
500
1000
1500

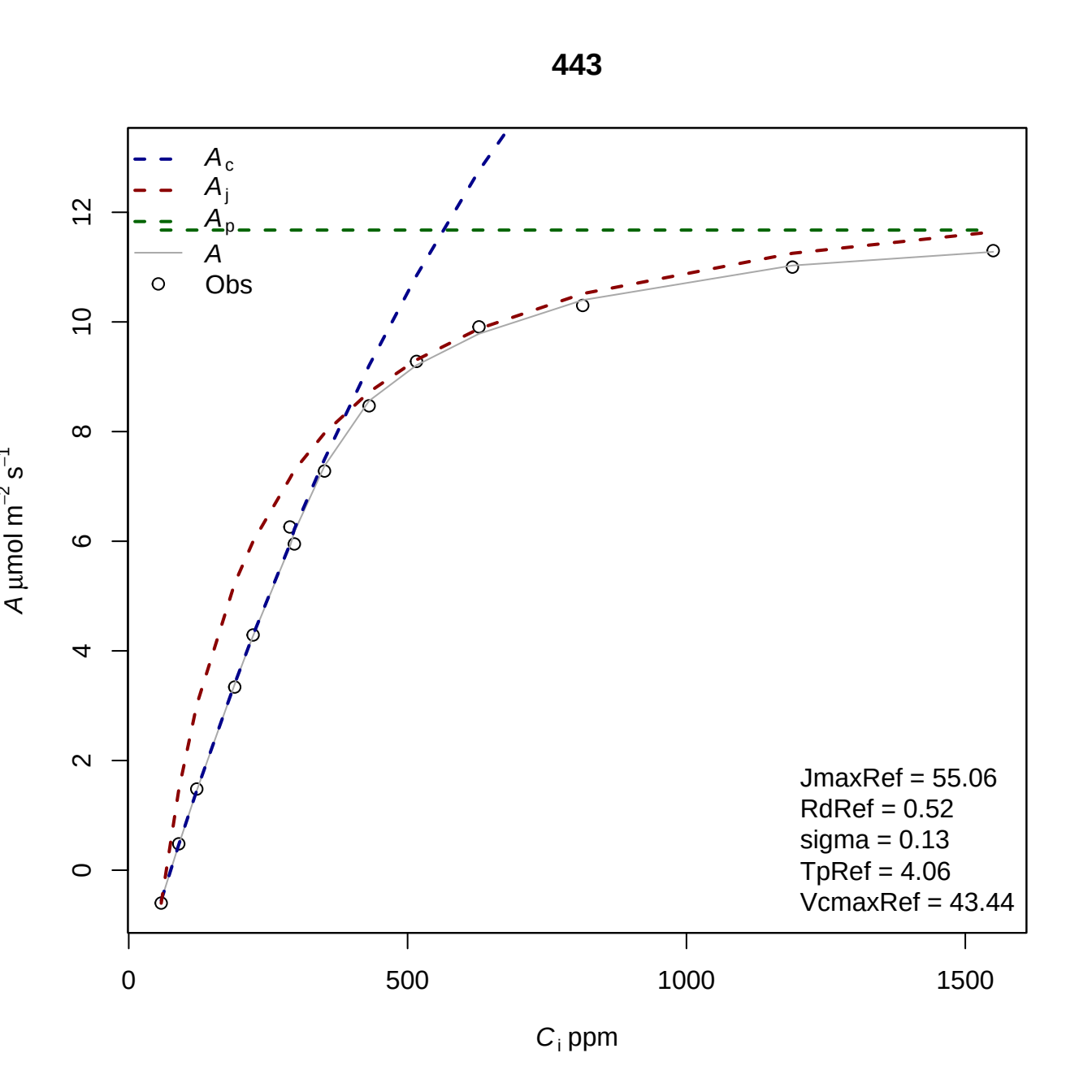

438
A_c
A_j
A_p
A
Obs
JmaxRef = 112.08
RdRef = 0.34
sigma = 0.45
TpRef = 7
VcmaxRef = 118.11
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
200
400
600
800

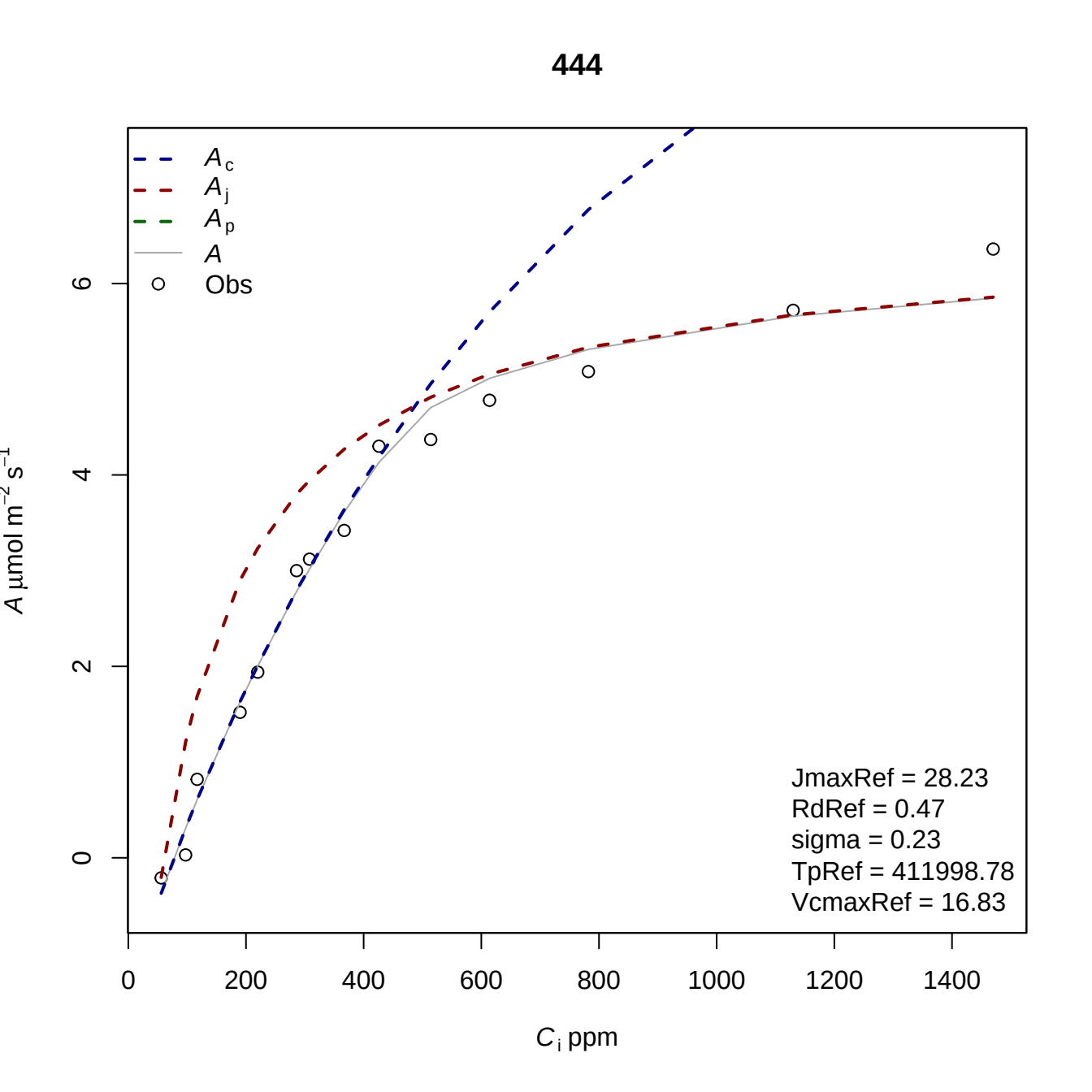

439
A_c
A_j
A_p
A
Obs
JmaxRef = 95.13
RdRef = 1.2
sigma = 0.43
TpRef = 5.64
VcmaxRef = 78.79
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
100
200
300
400
500
600

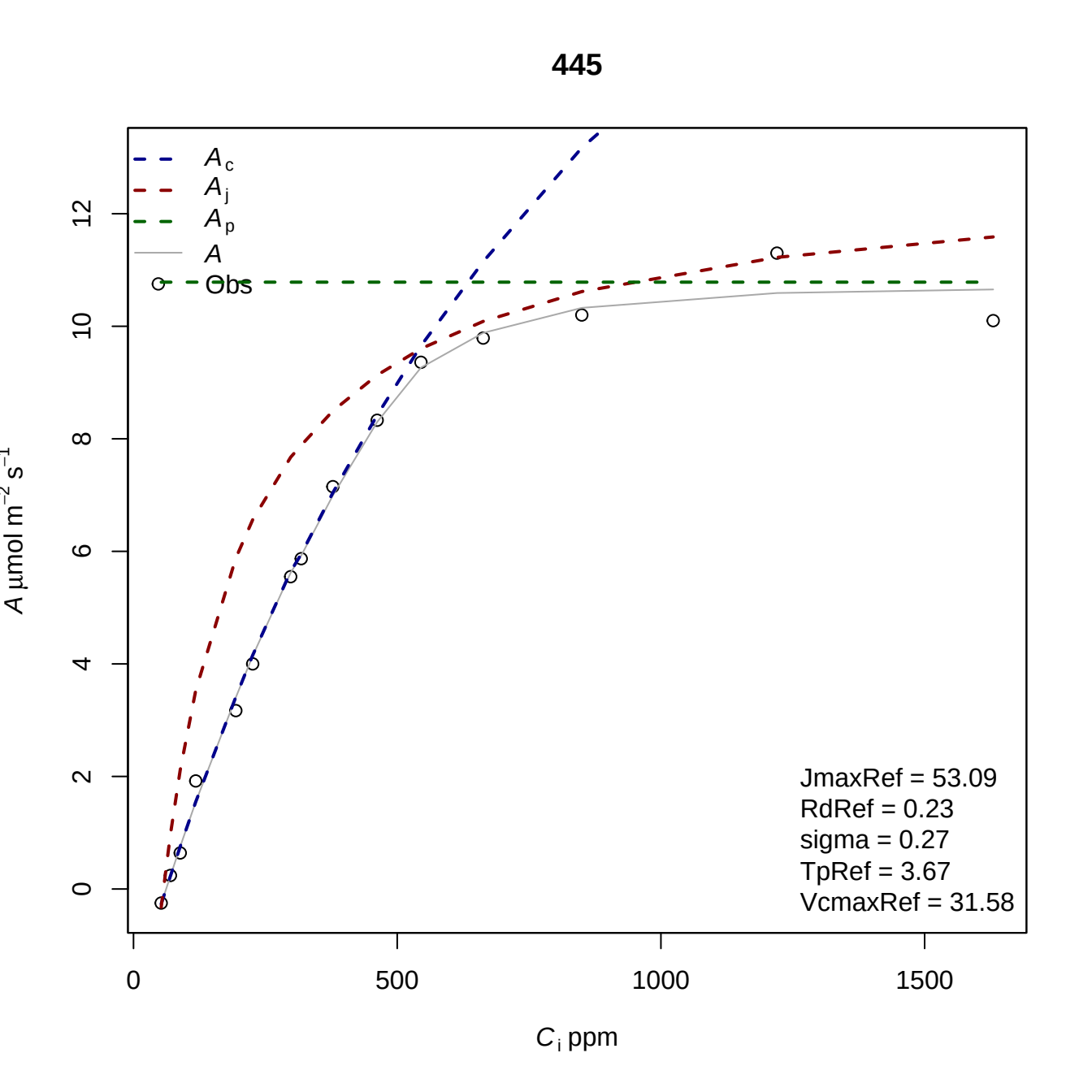

440
A_c
A_j
A_p
A
Obs
JmaxRef = 83.51
RdRef = 0.78
sigma = 0.13
TpRef = 22096.18
VcmaxRef = 56.83
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
500
1000
1500

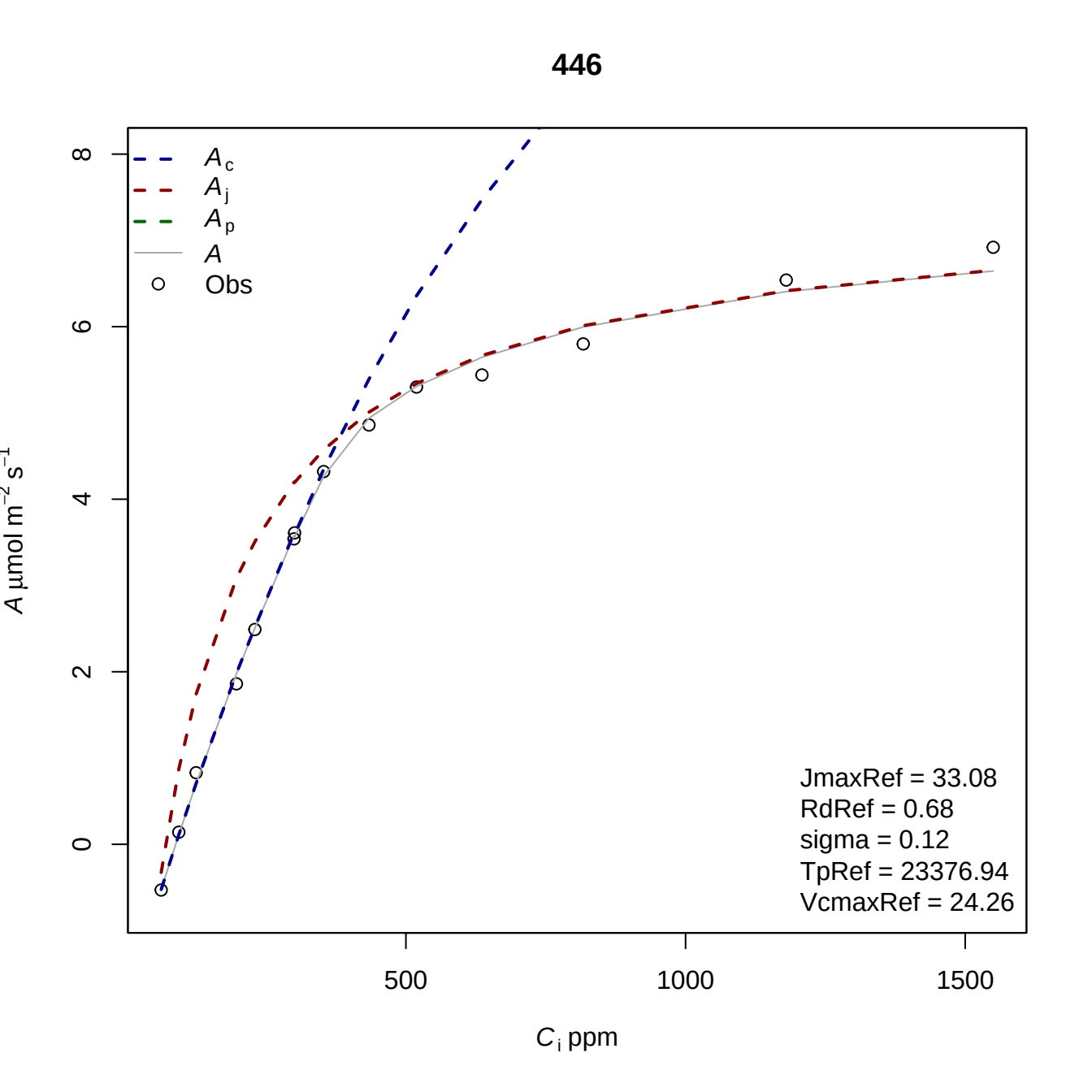

441
A_c
A_j
A_p
A
Obs
JmaxRef = 66
RdRef = 1.45
sigma = 0.14
TpRef = 3483.25
VcmaxRef = 40.09
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
500
1000
1500

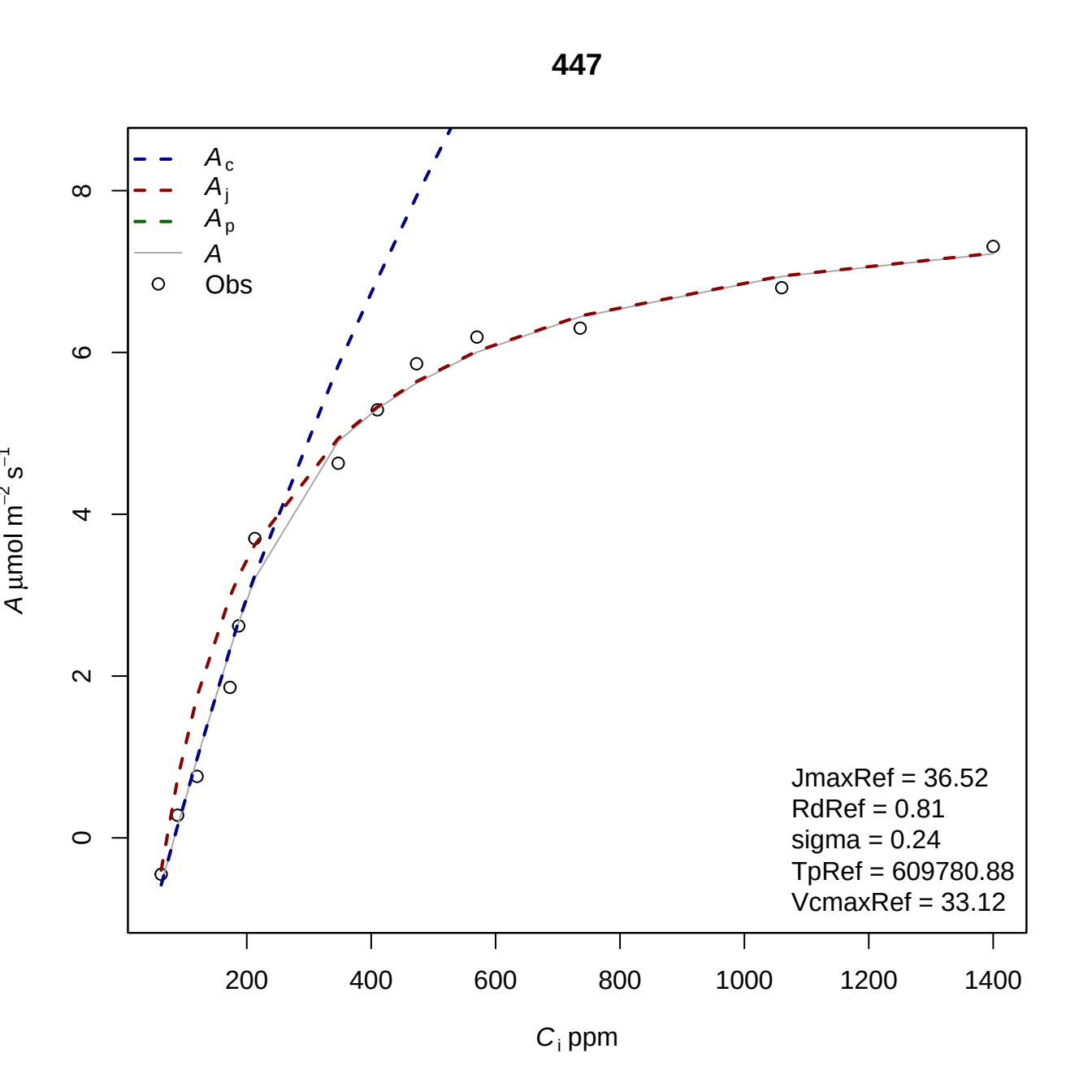

442
A_c
A_j
A_p
A
Obs
JmaxRef = 69.71
RdRef = 0.62
sigma = 0.17
TpRef = 19005.78
VcmaxRef = 58.16
0
5
10
15
200
400
600
800
1000
A µmol m^{-2} s^{-1}
C_i ppm

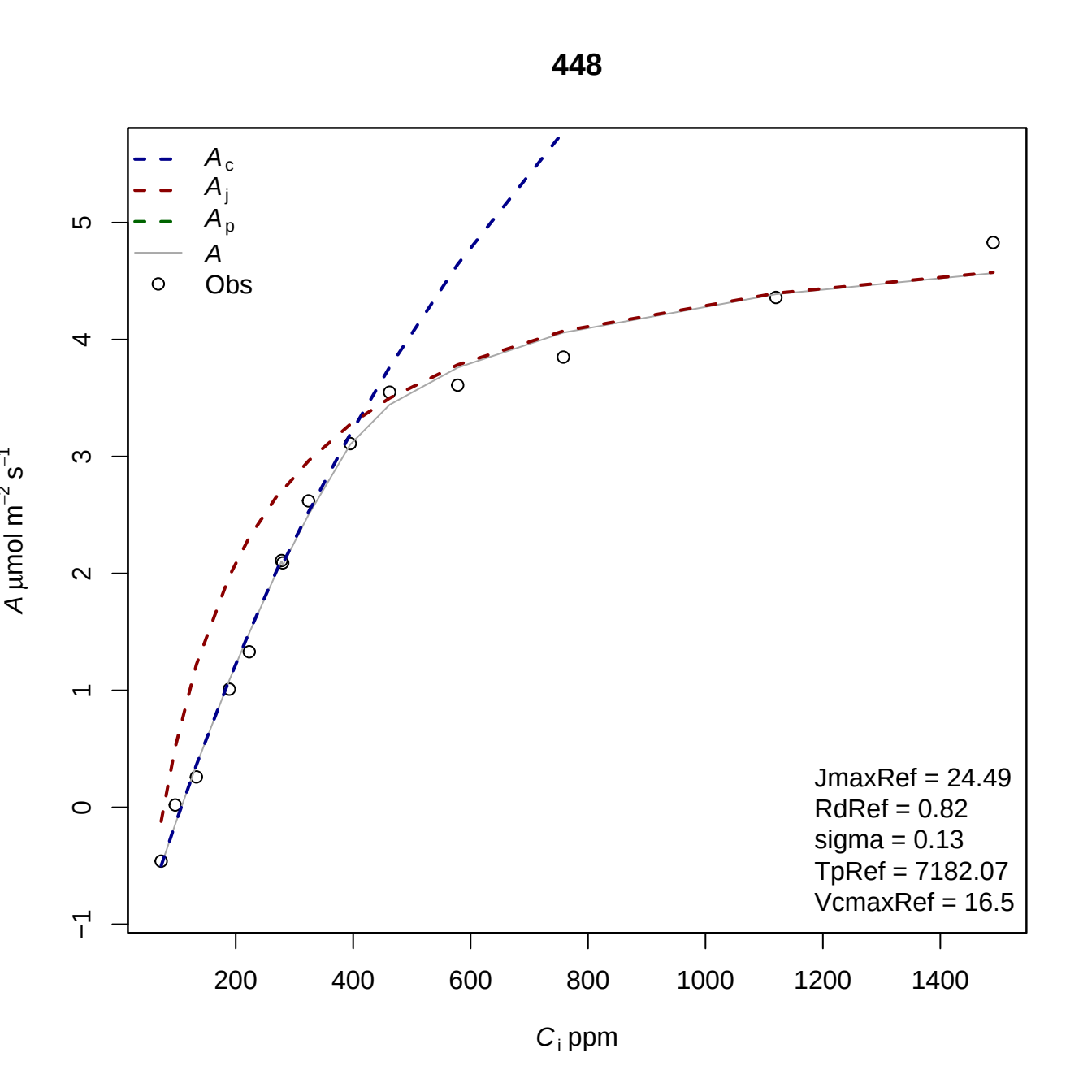

443
A_c
A_j
A_p
A
Obs
JmaxRef = 55.06
RdRef = 0.52
sigma = 0.13
TpRef = 4.06
VcmaxRef = 43.44
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
0
500
1000
1500

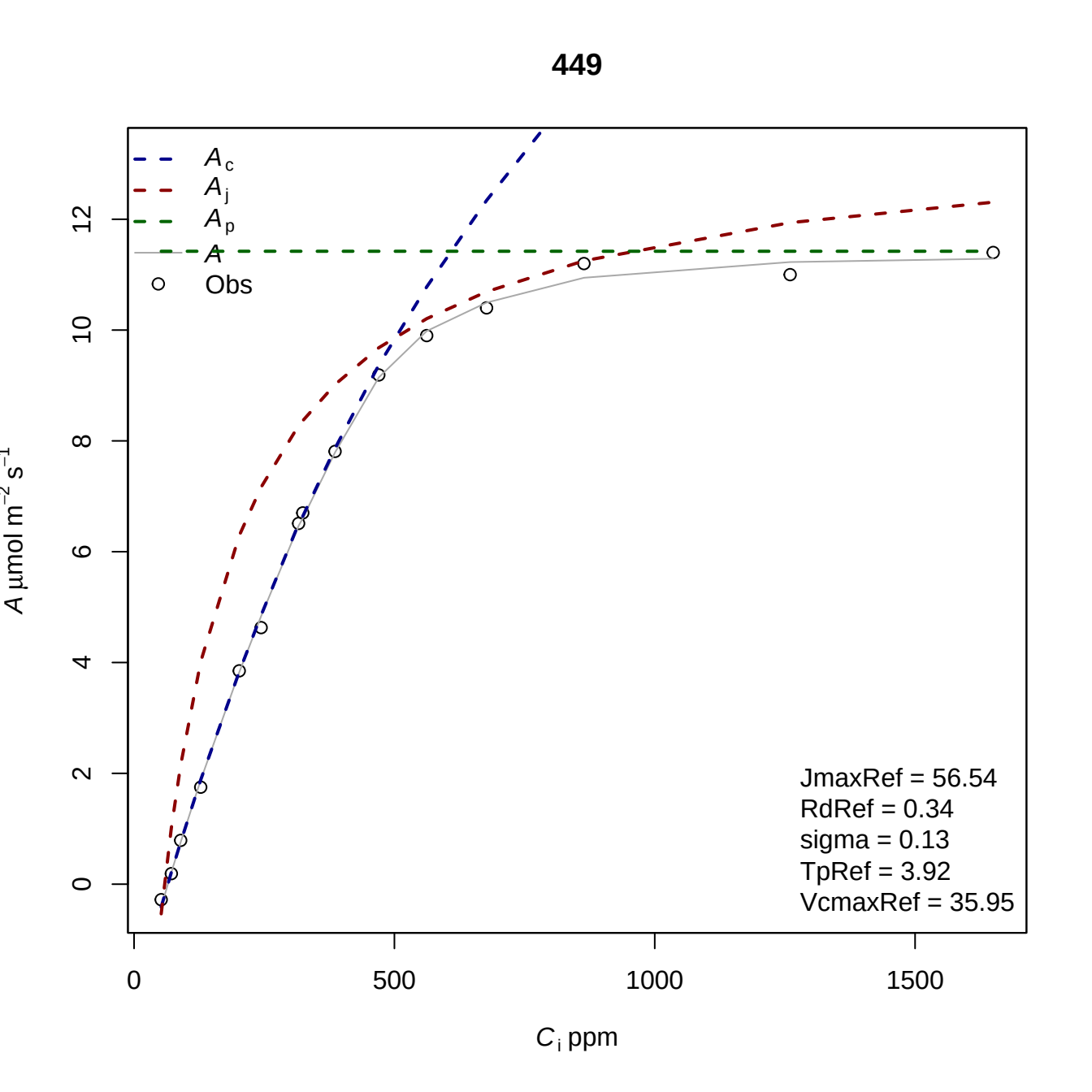

444
A_c
A_j
A_p
A
Obs
JmaxRef = 28.23
RdRef = 0.47
sigma = 0.23
TpRef = 411998.78
VcmaxRef = 16.83
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
0
200
400
600
800
1000
1200
1400

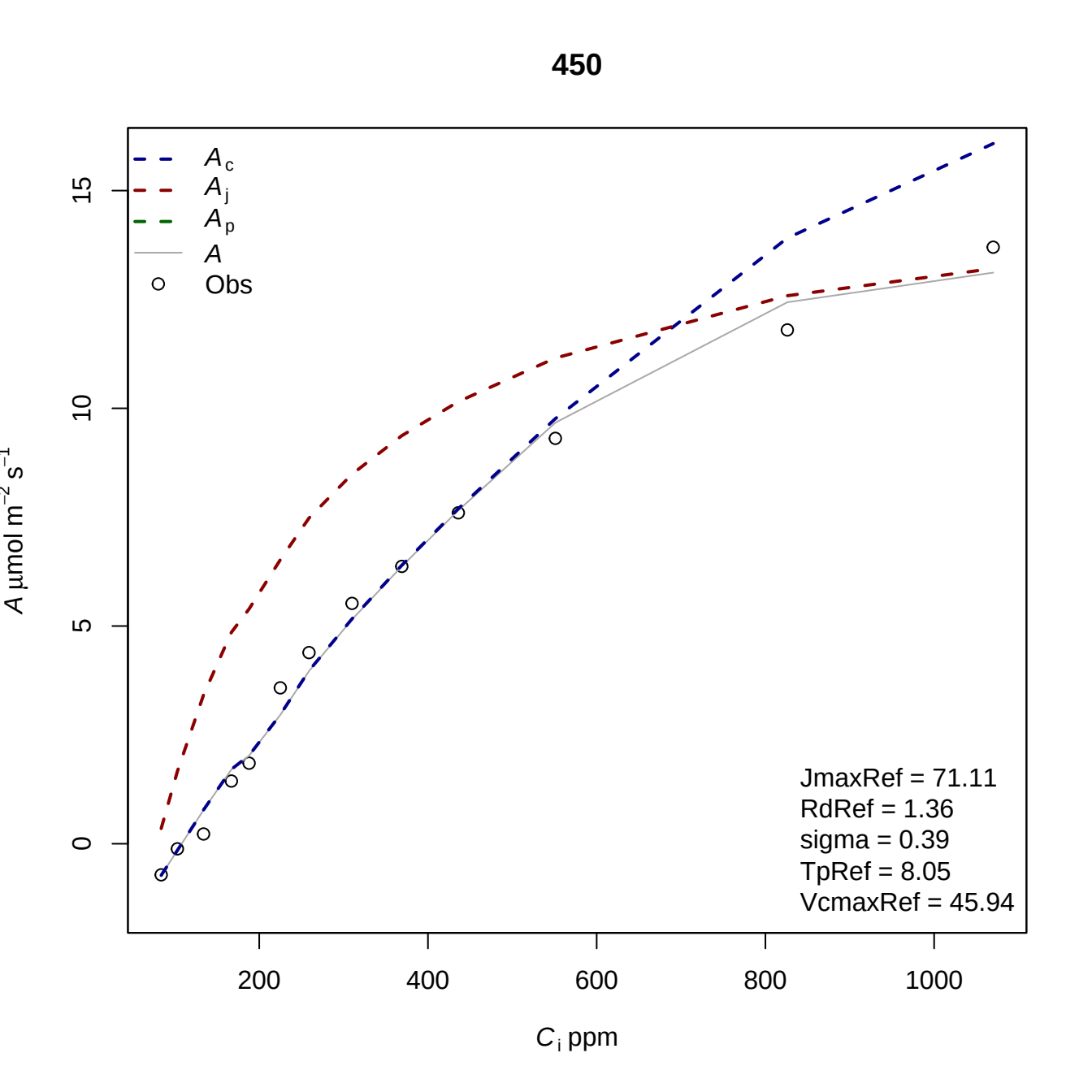

445
A_c
A_j
A_p
A
Obs
JmaxRef = 53.09
RdRef = 0.23
sigma = 0.27
TpRef = 3.67
VcmaxRef = 31.58
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
0
500
1000
1500

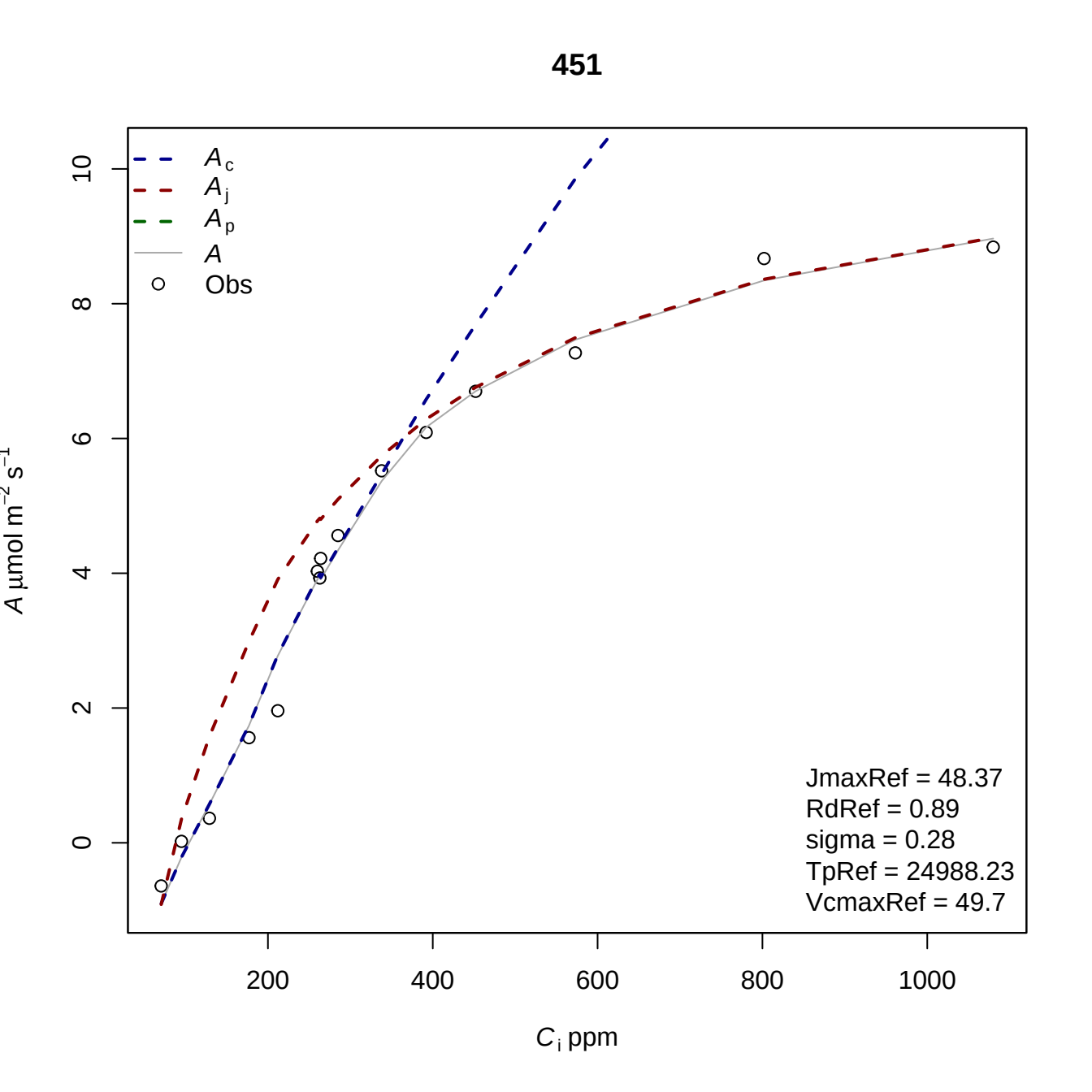

446
A_c
A_j
A_p
A
Obs
JmaxRef = 33.08
RdRef = 0.68
sigma = 0.12
TpRef = 23376.94
VcmaxRef = 24.26
0
2
4
6
8
500
1000
1500
A µmol m^{-2} s^{-1}
C_i ppm

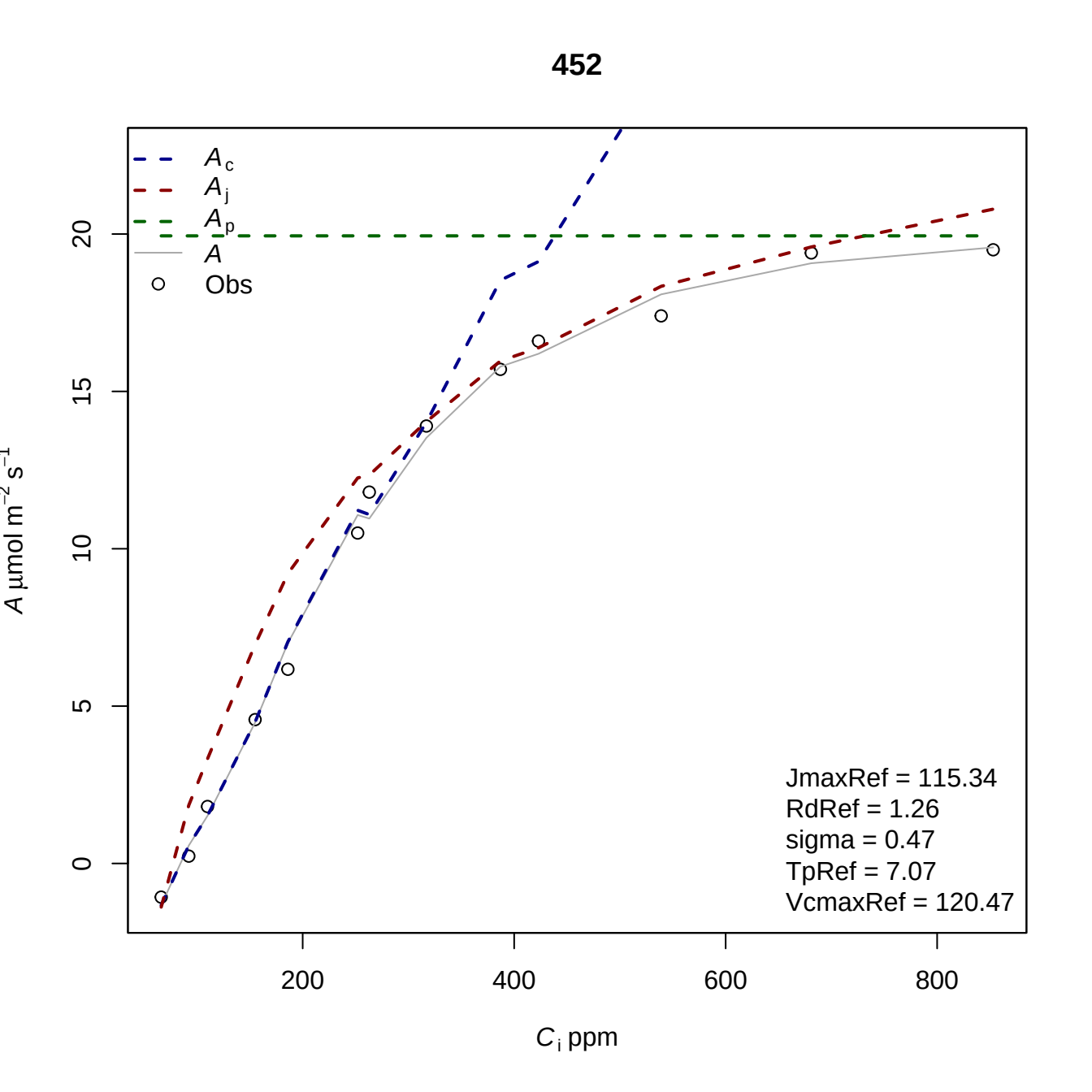

447
A_c
A_j
A_p
A
Obs
A µmol m^{-2} s^{-1}
0
2
4
6
8
200
400
600
800
1000
1200
1400
C_i ppm
JmaxRef = 36.52
RdRef = 0.81
sigma = 0.24
TpRef = 609780.88
VcmaxRef = 33.12

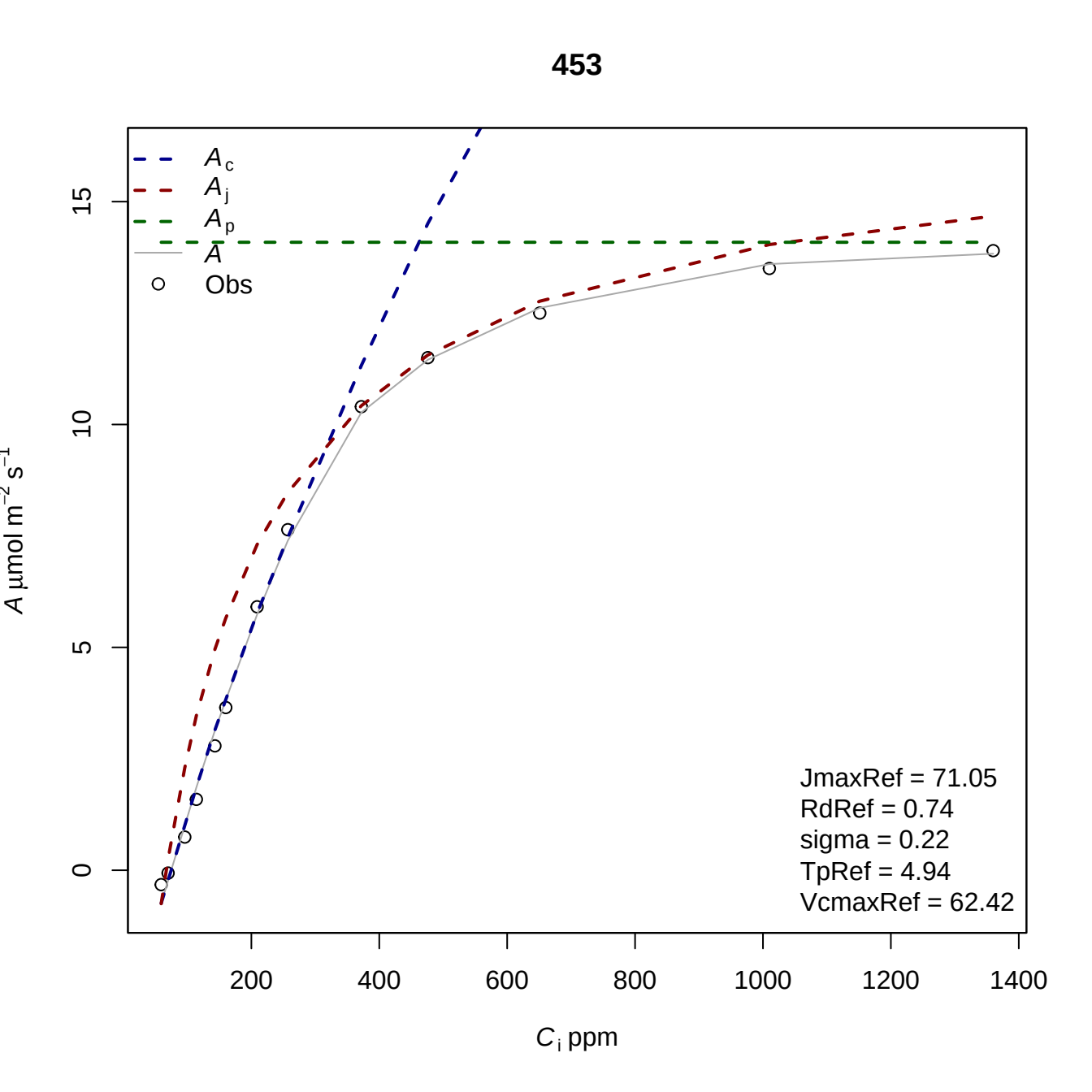

448
A_c
A_j
A_p
A
Obs
JmaxRef = 24.49
RdRef = 0.82
sigma = 0.13
TpRef = 7182.07
VcmaxRef = 16.5
A µmol m^{-2} s^{-1}
C_i ppm
−1
0
1
2
3
4
5
200
400
600
800
1000
1200
1400

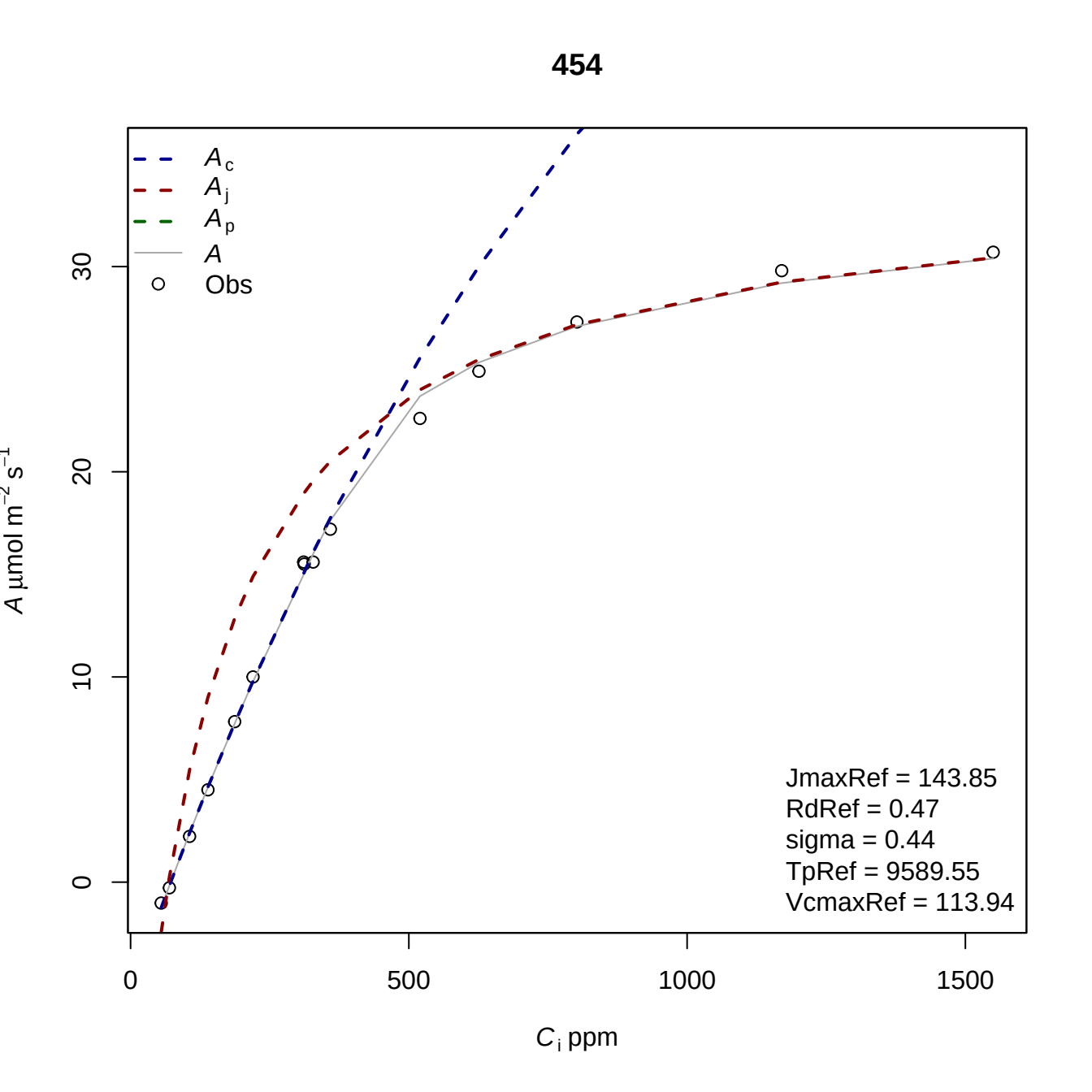

449
A_c
A_j
A_p
A
Obs
JmaxRef = 56.54
RdRef = 0.34
sigma = 0.13
TpRef = 3.92
VcmaxRef = 35.95
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
0
500
1000
1500

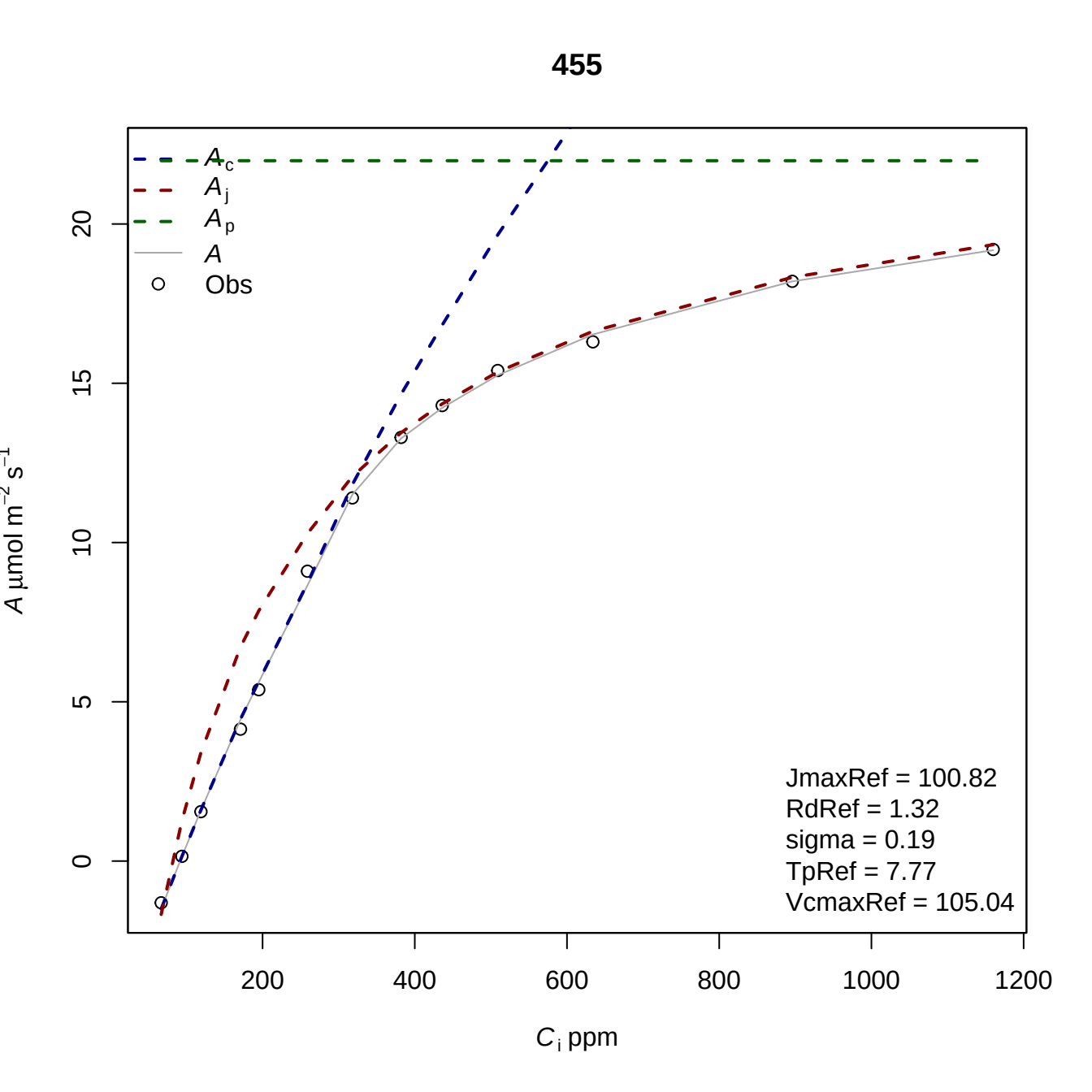

450
A_c
A_j
A_p
A
Obs
JmaxRef = 71.11
RdRef = 1.36
sigma = 0.39
TpRef = 8.05
VcmaxRef = 45.94
0
5
10
15
200
400
600
800
1000
A µmol m^{-2} s^{-1}
C_i ppm

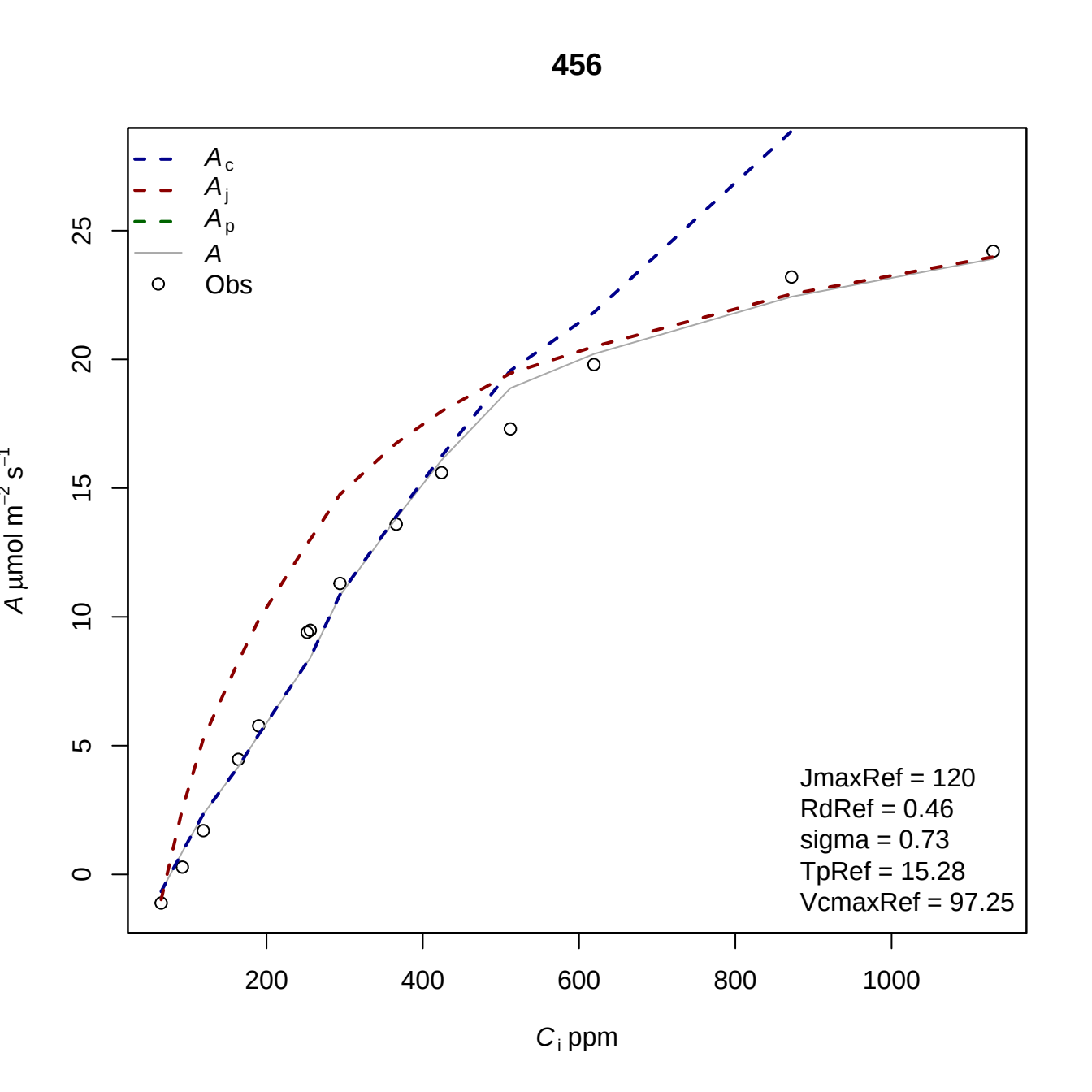

451
A_c
A_j
A_p
A
Obs
JmaxRef = 48.37
RdRef = 0.89
sigma = 0.28
TpRef = 24988.23
VcmaxRef = 49.7
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
200
400
600
800
1000

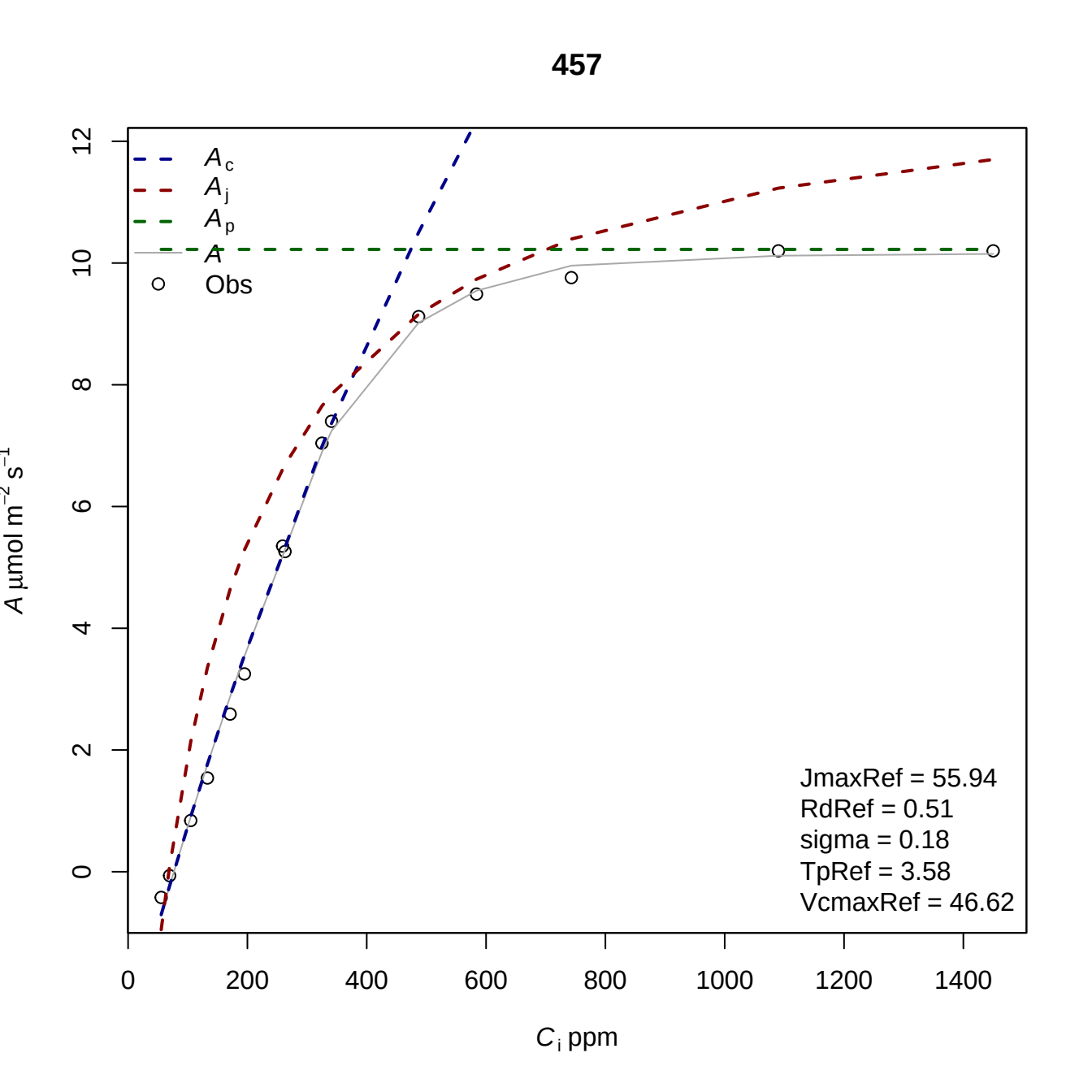

452
A_c
A_j
A_p
A
Obs
JmaxRef = 115.34
RdRef = 1.26
sigma = 0.47
TpRef = 7.07
VcmaxRef = 120.47
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
200
400
600
800

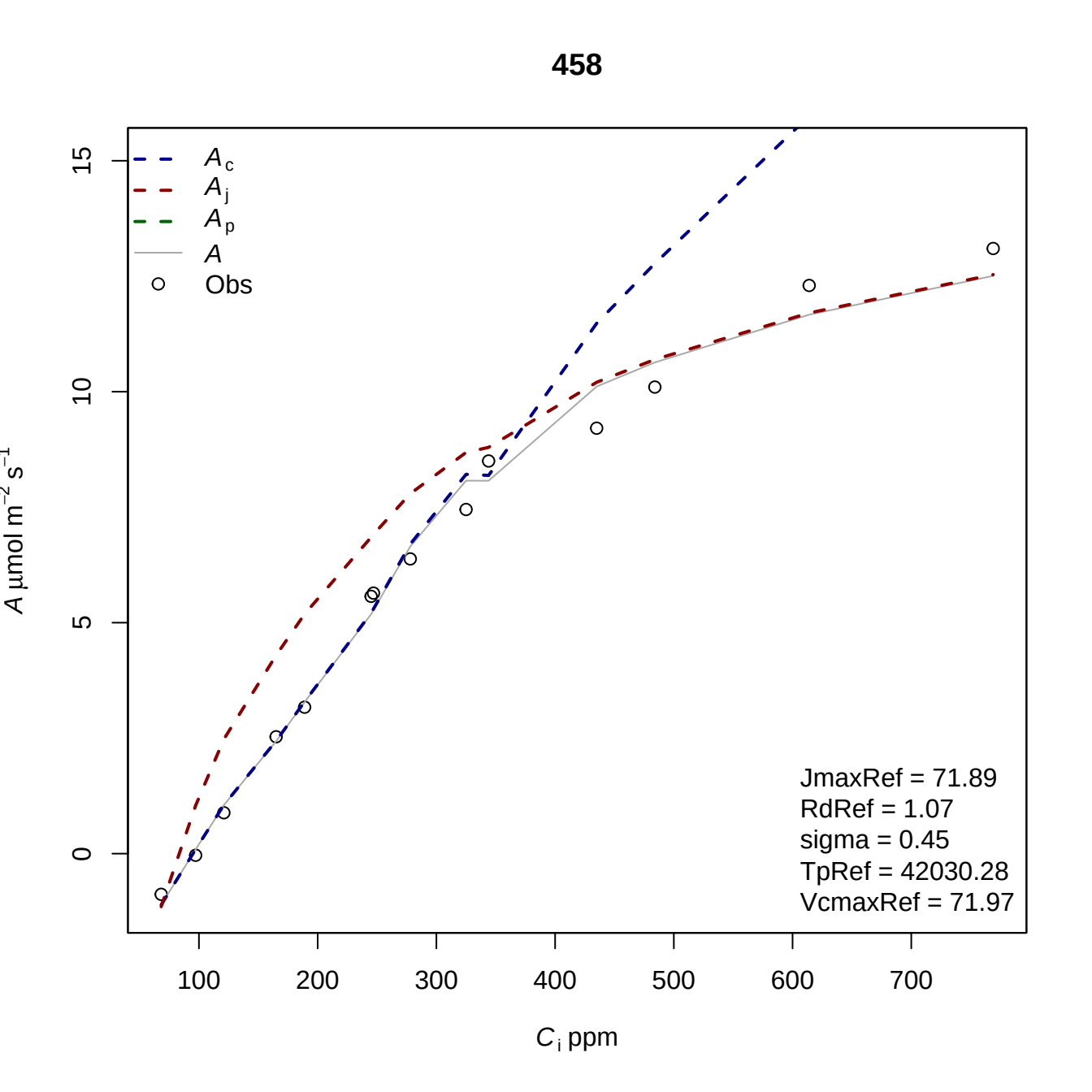

453
A_c
A_j
A_p
A
Obs
JmaxRef = 71.05
RdRef = 0.74
sigma = 0.22
TpRef = 4.94
VcmaxRef = 62.42
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
200
400
600
800
1000
1200
1400

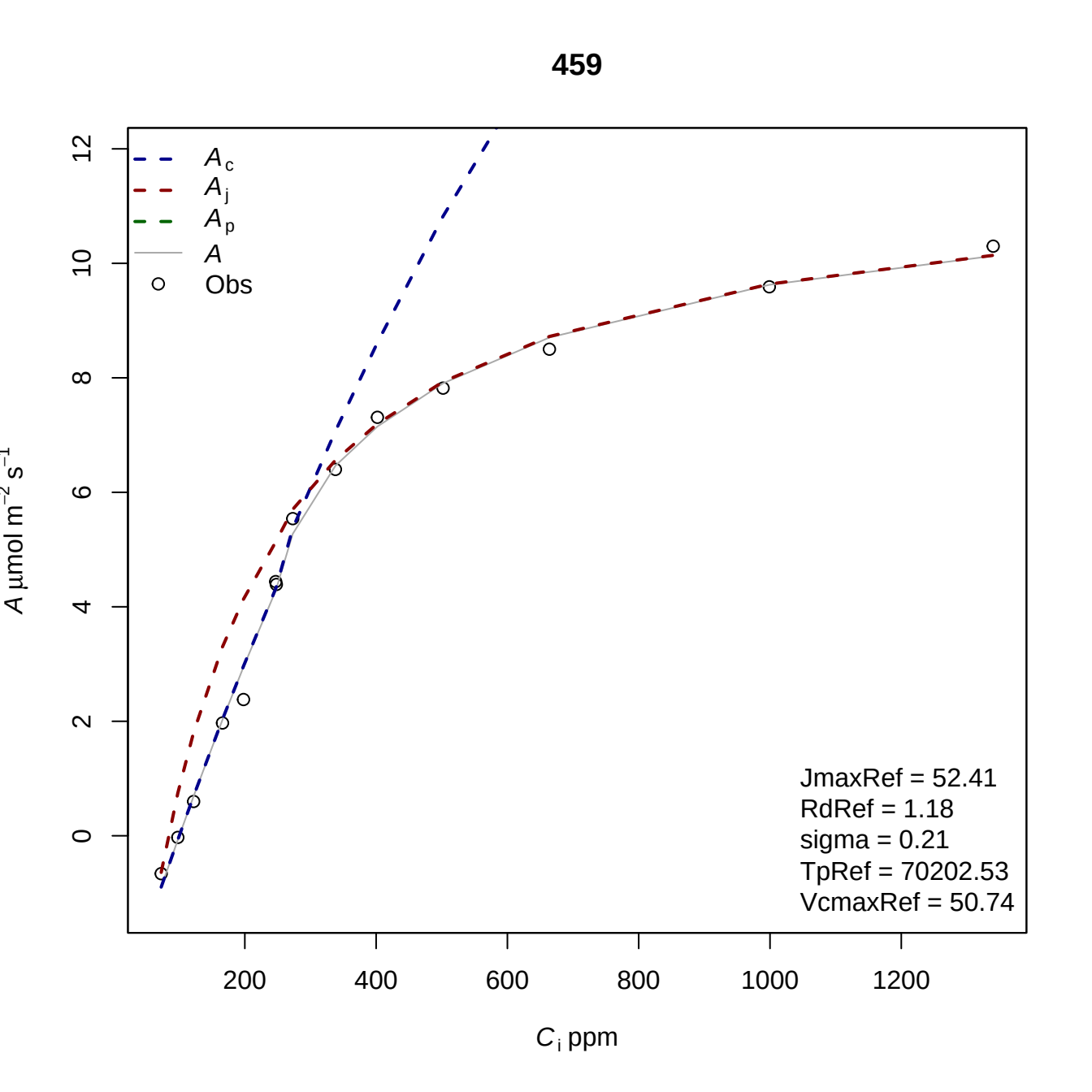

454
A_c
A_j
A_p
A
Obs
JmaxRef = 143.85
RdRef = 0.47
sigma = 0.44
TpRef = 9589.55
VcmaxRef = 113.94
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
C_i ppm

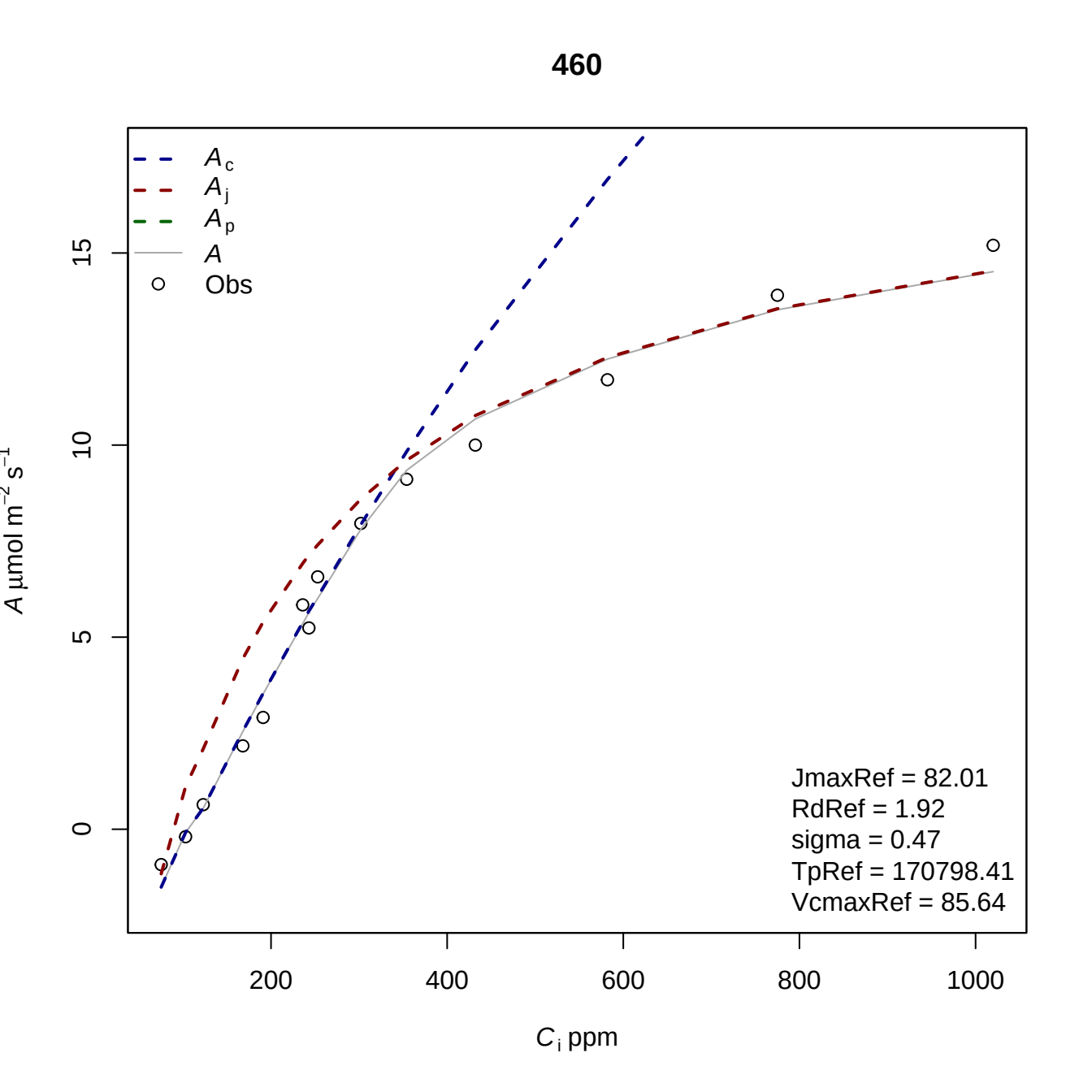

455
A_c
A_j
A_p
A
Obs
JmaxRef = 100.82
RdRef = 1.32
sigma = 0.19
TpRef = 7.77
VcmaxRef = 105.04
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
200
400
600
800
1000
1200

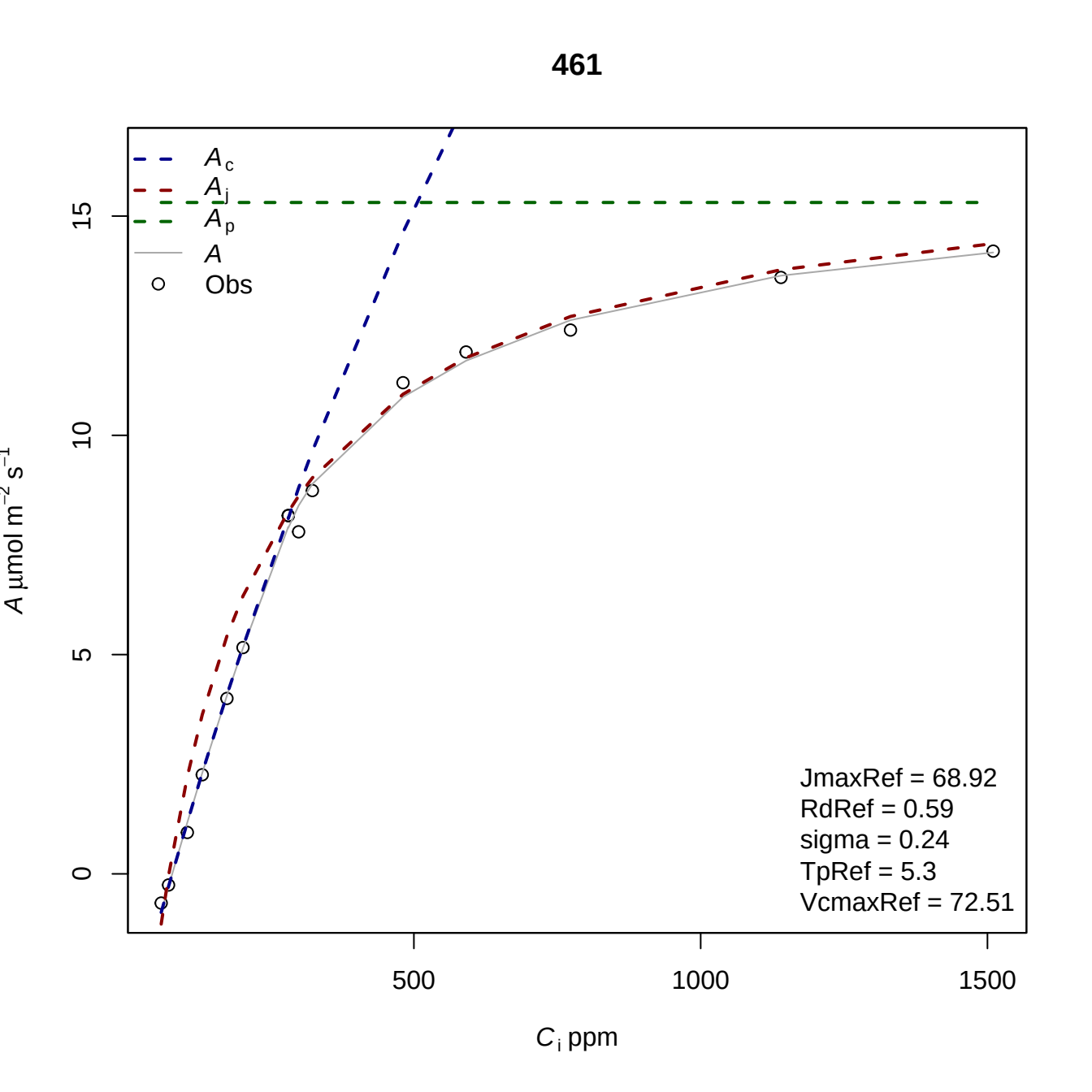

456
A_c
A_j
A_p
A
Obs
JmaxRef = 120
RdRef = 0.46
sigma = 0.73
TpRef = 15.28
VcmaxRef = 97.25
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
200
400
600
800
1000

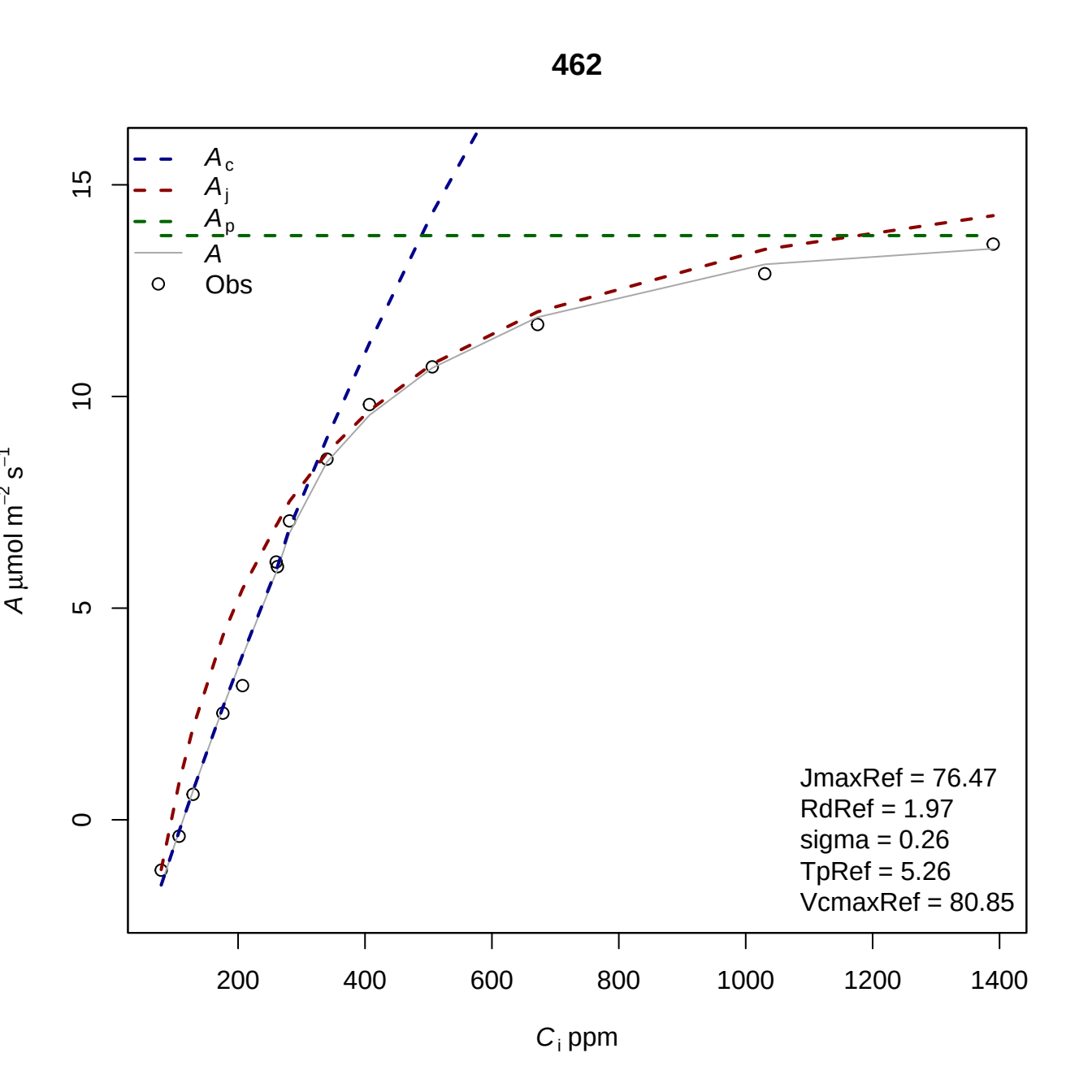

457
A_c
A_j
A_p
A
Obs
JmaxRef = 55.94
RdRef = 0.51
sigma = 0.18
TpRef = 3.58
VcmaxRef = 46.62
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

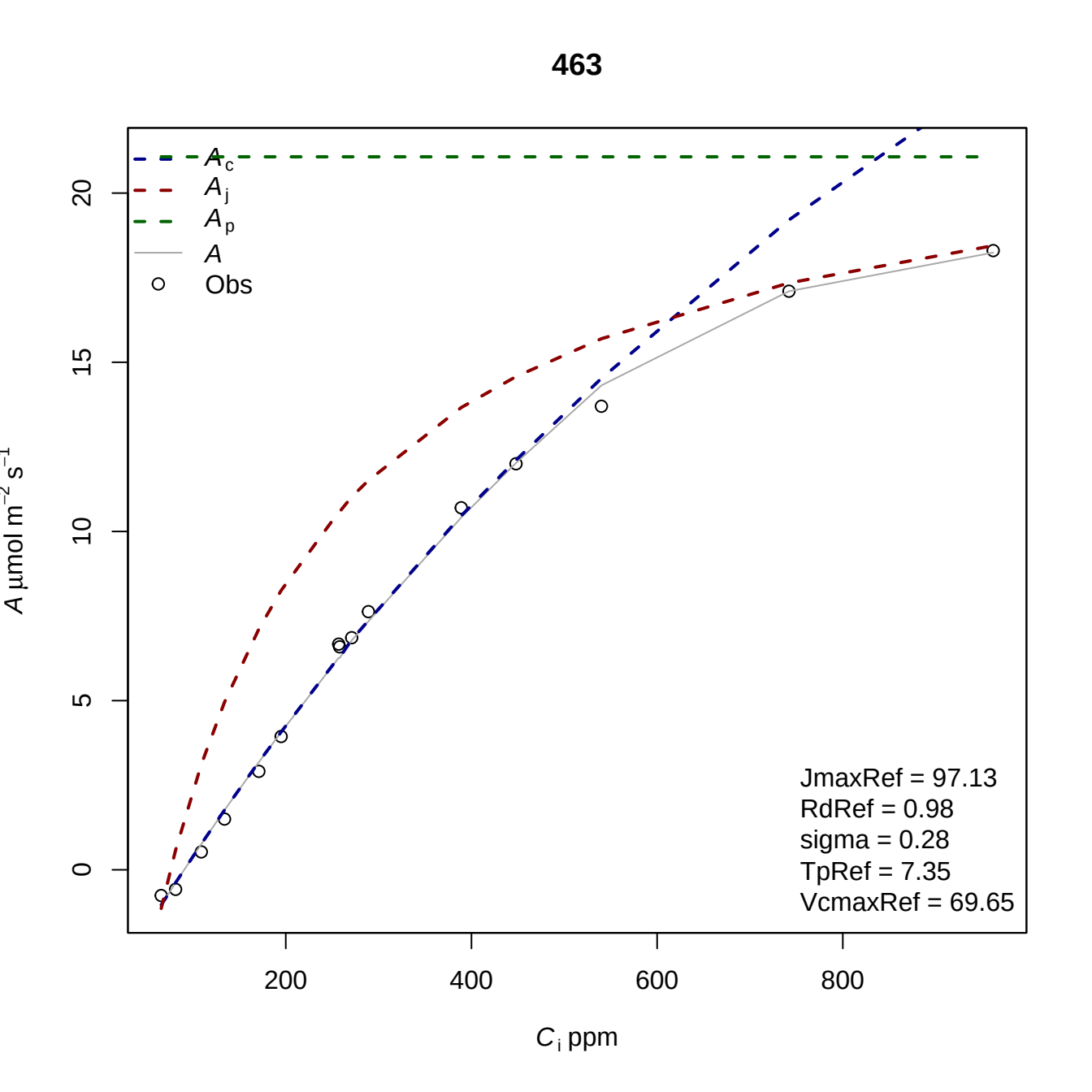

458
A_c
A_j
A_p
A
Obs
JmaxRef = 71.89
RdRef = 1.07
sigma = 0.45
TpRef = 42030.28
VcmaxRef = 71.97
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
100
200
300
400
500
600
700

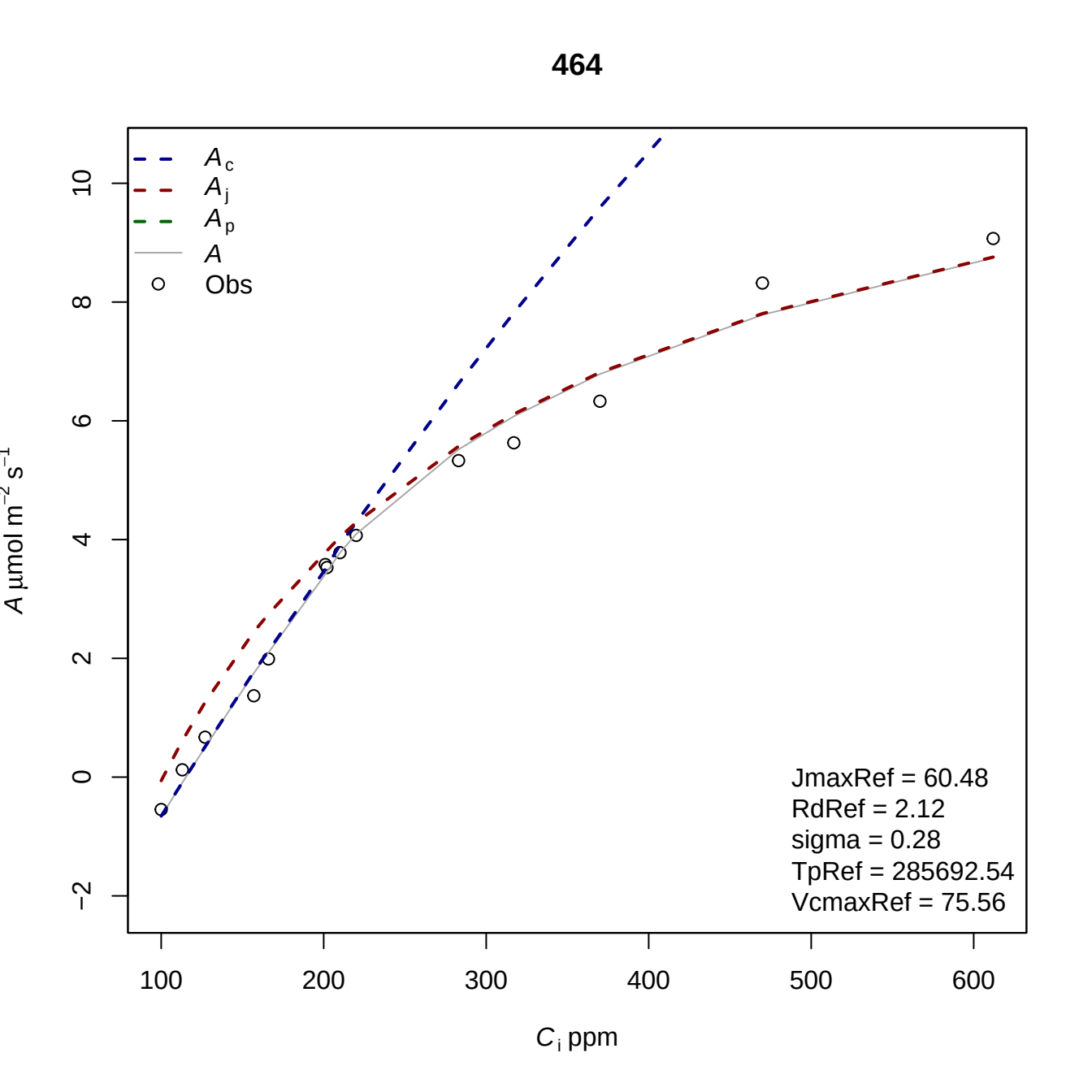

459
A_c
A_j
A_p
A
Obs
JmaxRef = 52.41
RdRef = 1.18
sigma = 0.21
TpRef = 70202.53
VcmaxRef = 50.74
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
200
400
600
800
1000
1200

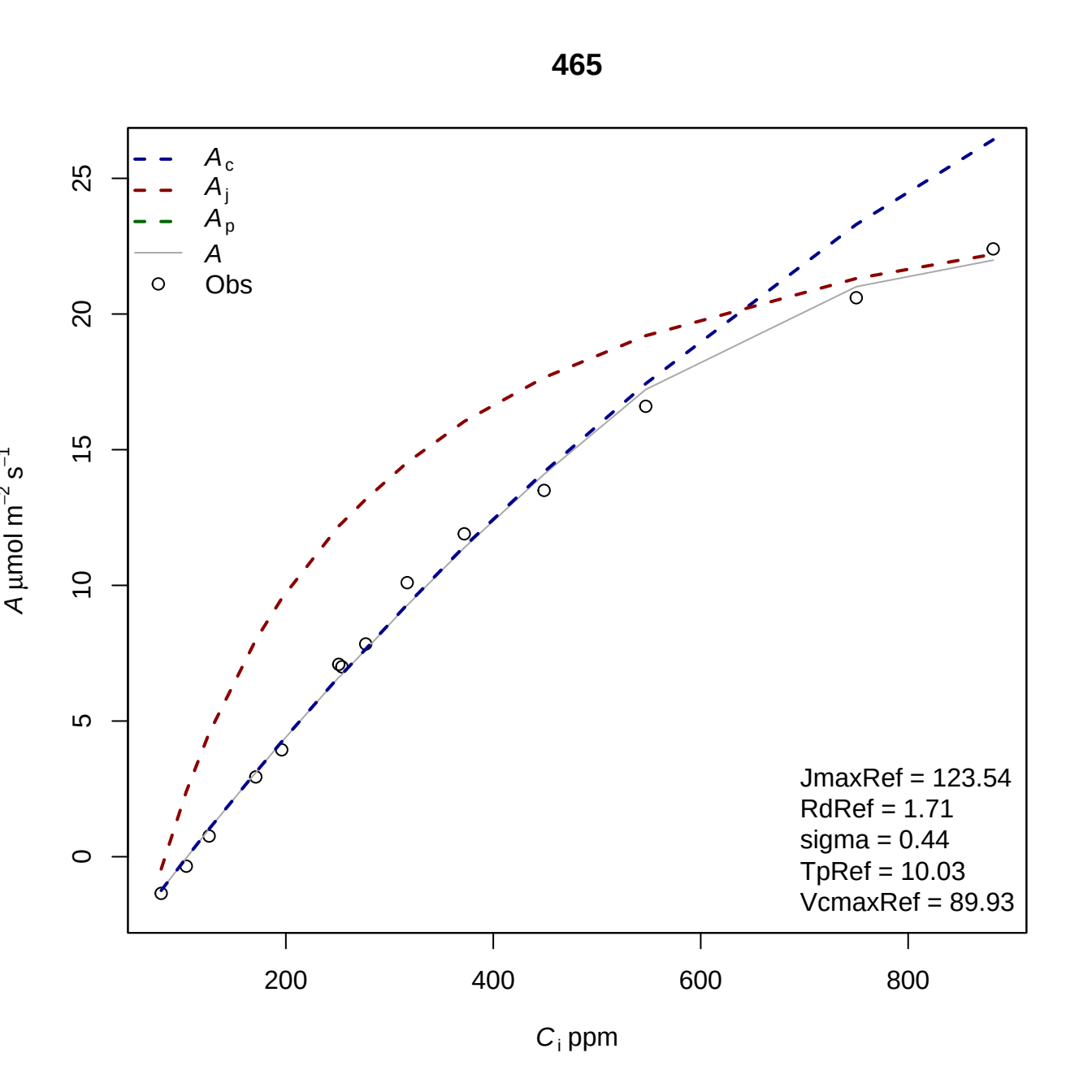

460
A_c
A_j
A_p
A
Obs
JmaxRef = 82.01
RdRef = 1.92
sigma = 0.47
TpRef = 170798.41
VcmaxRef = 85.64
0
5
10
15
200
400
600
800
1000
A µmol m^{-2} s^{-1}
C_i ppm

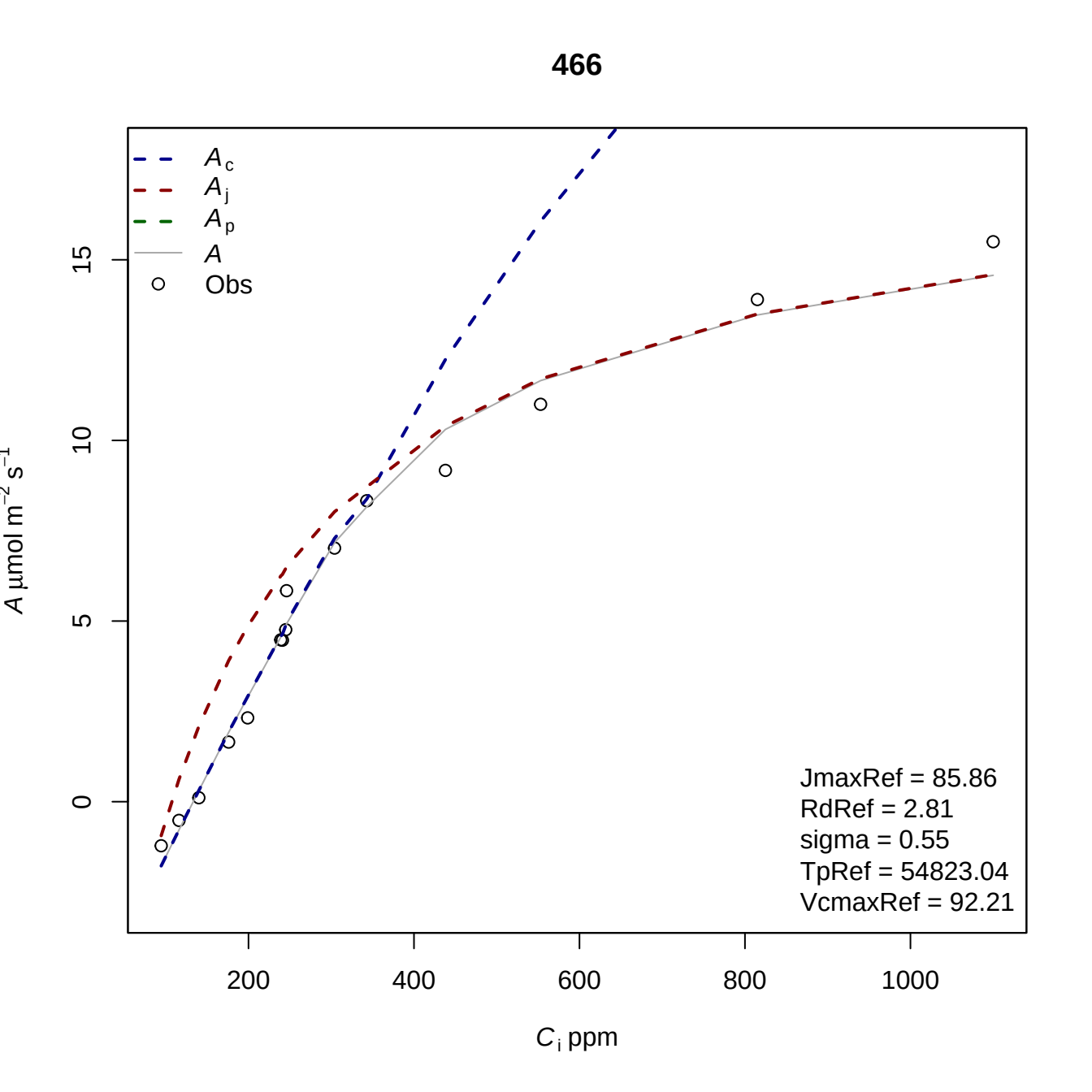

461
A_c
A_j
A_p
A
Obs
JmaxRef = 68.92
RdRef = 0.59
sigma = 0.24
TpRef = 5.3
VcmaxRef = 72.51
0
5
10
15
500
1000
1500
A µmol m^{-2} s^{-1}
C_i ppm

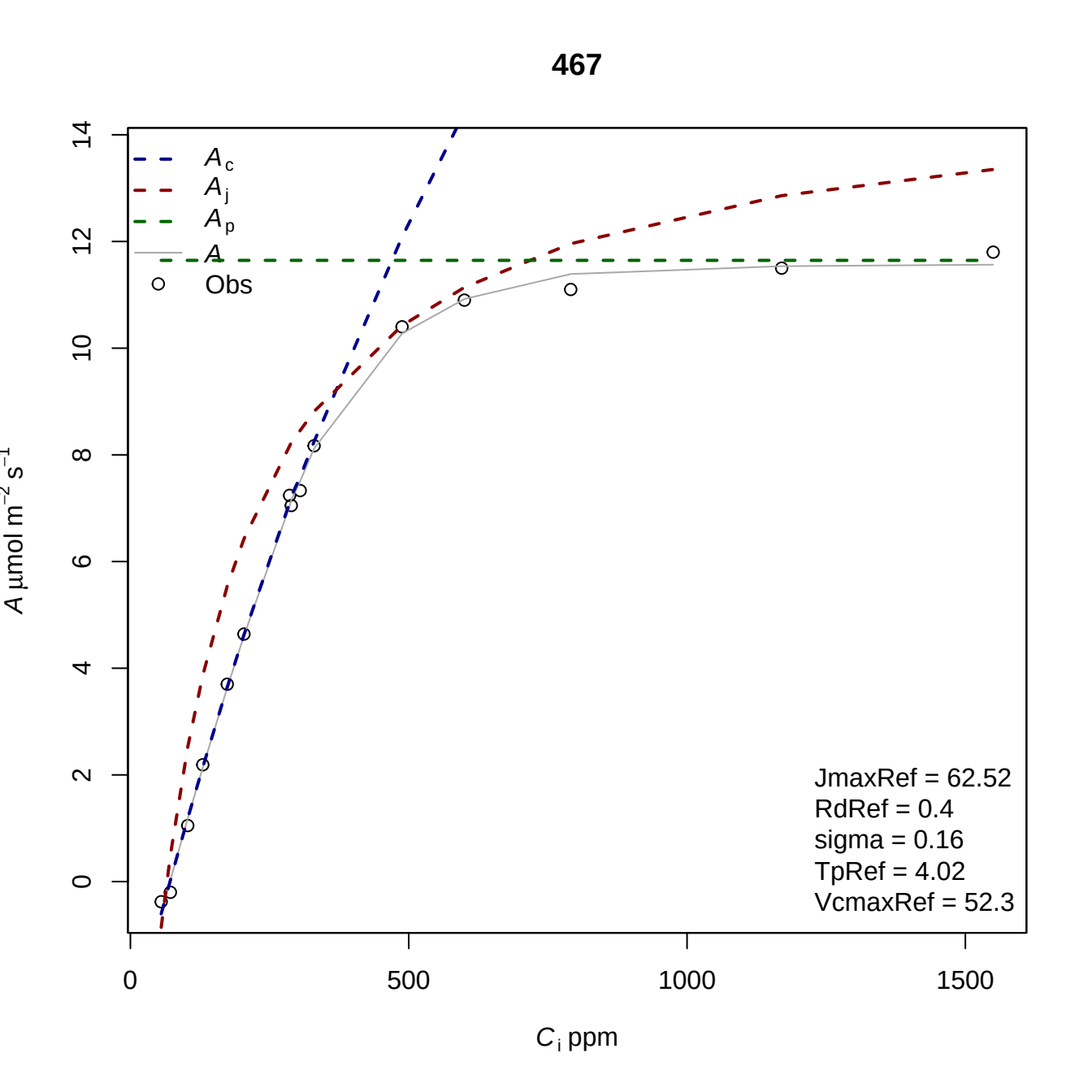

462
A_c
A_j
A_p
A
Obs
JmaxRef = 76.47
RdRef = 1.97
sigma = 0.26
TpRef = 5.26
VcmaxRef = 80.85
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
200
400
600
800
1000
1200
1400

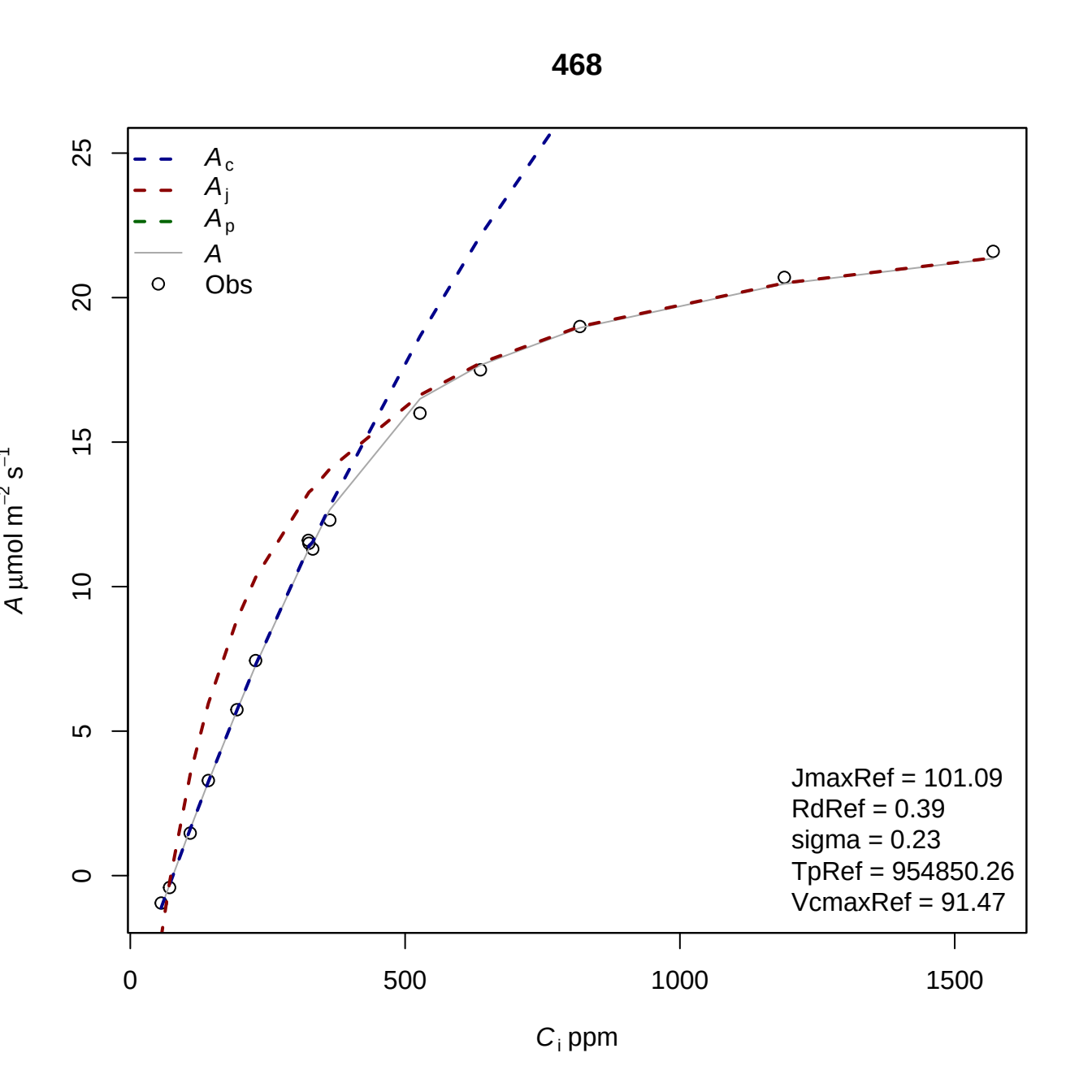

463
A_c
A_j
A_p
A
Obs
JmaxRef = 97.13
RdRef = 0.98
sigma = 0.28
TpRef = 7.35
VcmaxRef = 69.65
0
5
10
15
20
200
400
600
800
A µmol m^{-2} s^{-1}
C_i ppm

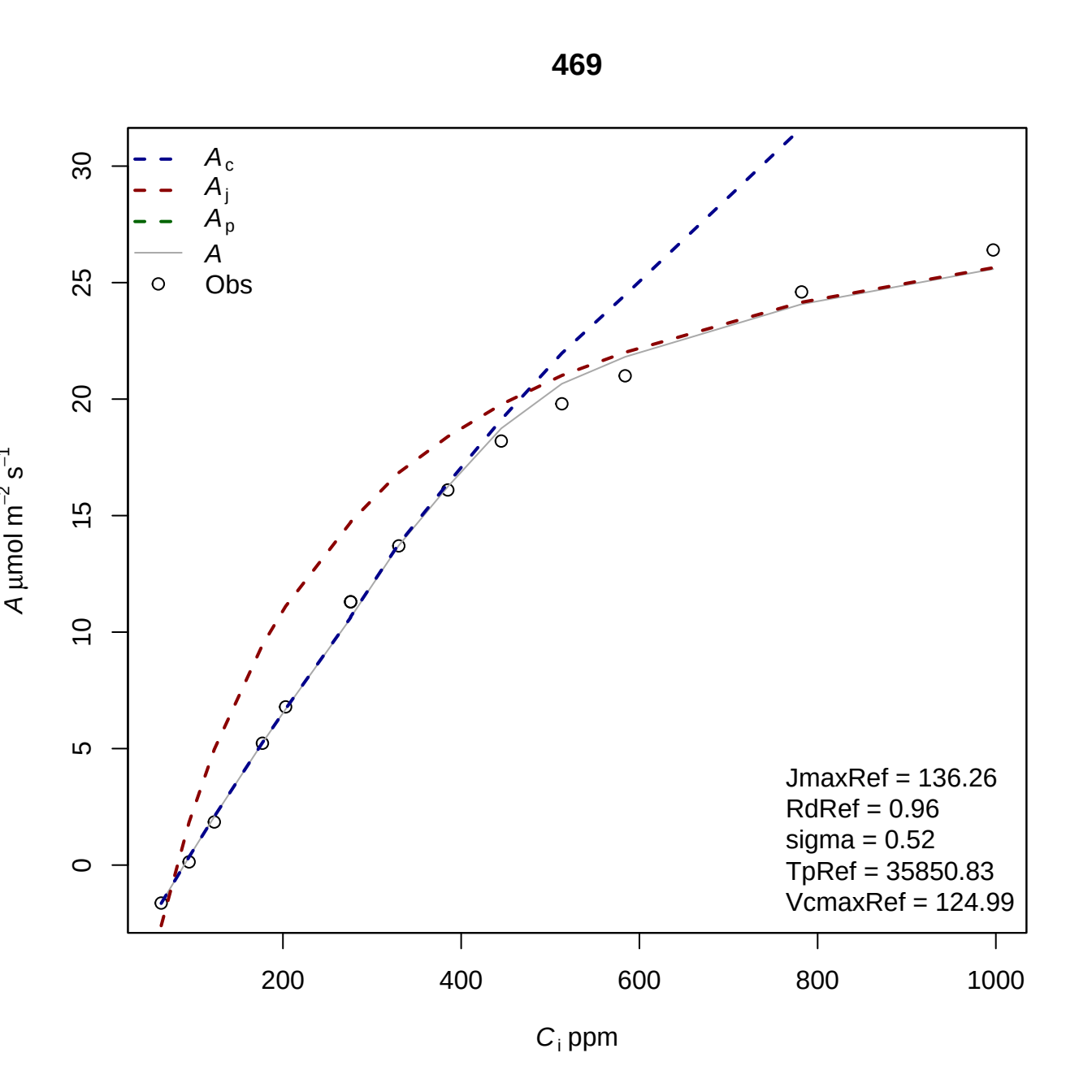

464
A_c
A_j
A_p
A
Obs
JmaxRef = 60.48
RdRef = 2.12
sigma = 0.28
TpRef = 285692.54
VcmaxRef = 75.56
A µmol m^{-2} s^{-1}
C_i ppm
−2
0
2
4
6
8
10
100
200
300
400
500
600

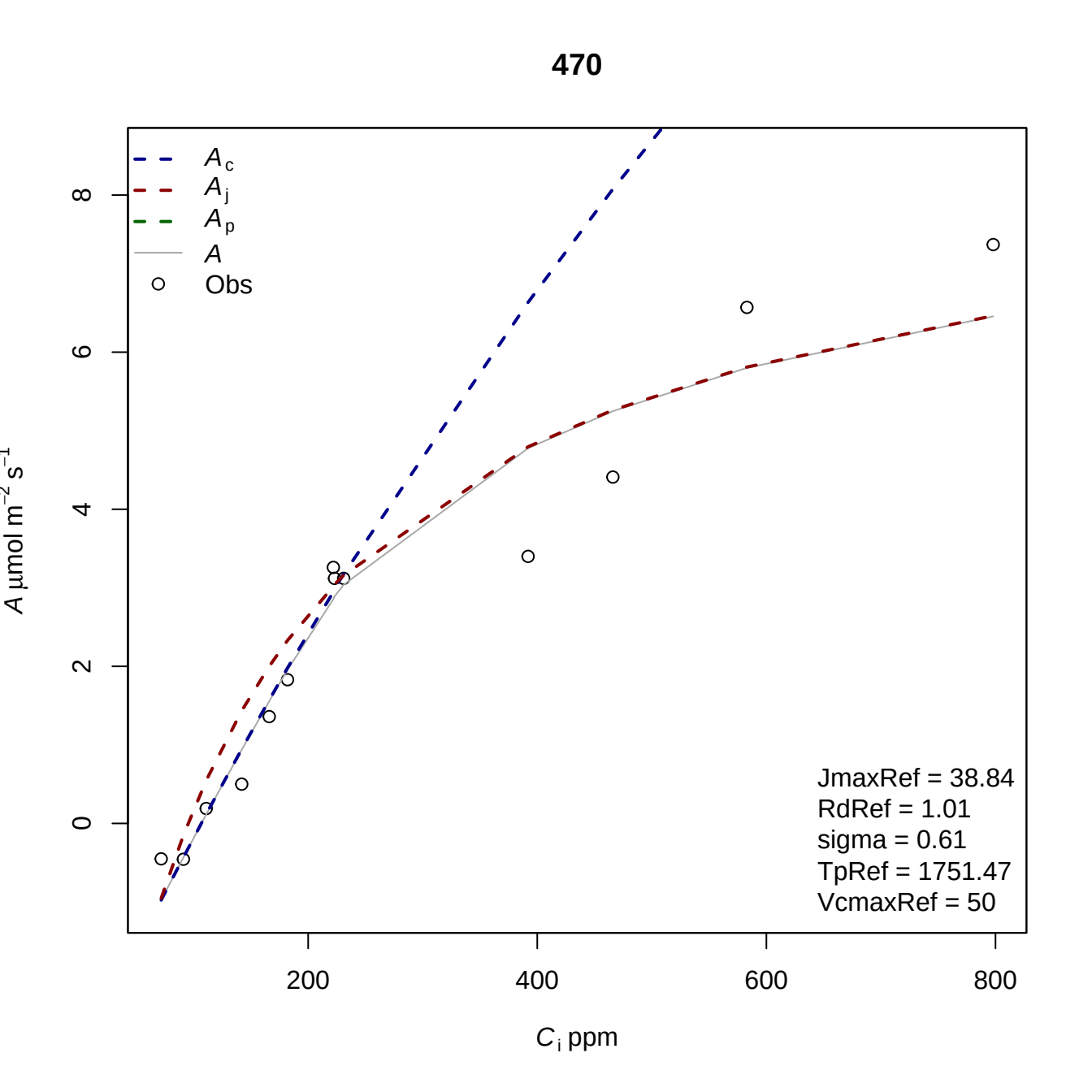

465
A_c
A_j
A_p
A
Obs
JmaxRef = 123.54
RdRef = 1.71
sigma = 0.44
TpRef = 10.03
VcmaxRef = 89.93
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
200
400
600
800

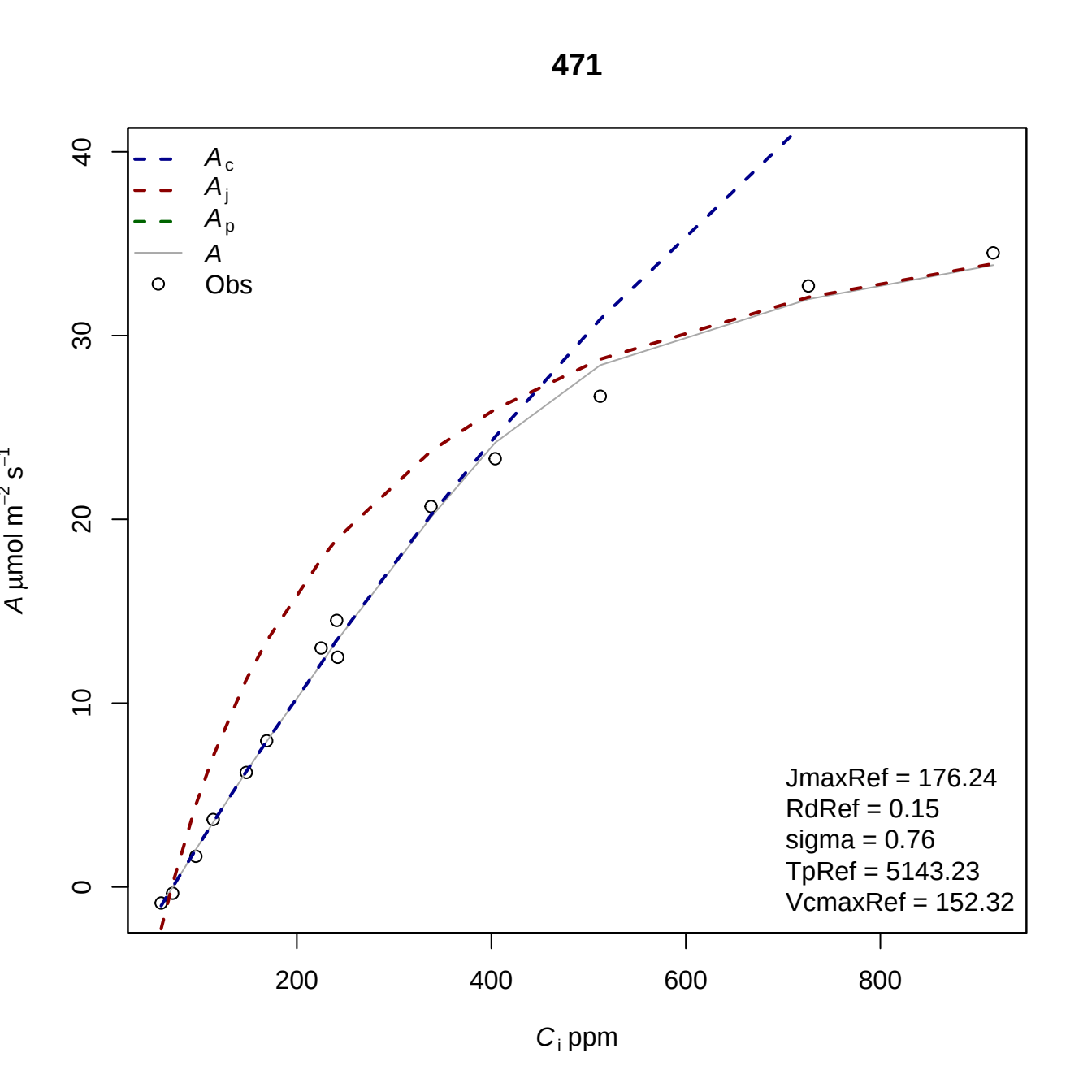

466
A_c
A_j
A_p
A
Obs
JmaxRef = 85.86
RdRef = 2.81
sigma = 0.55
TpRef = 54823.04
VcmaxRef = 92.21
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
200
400
600
800
1000

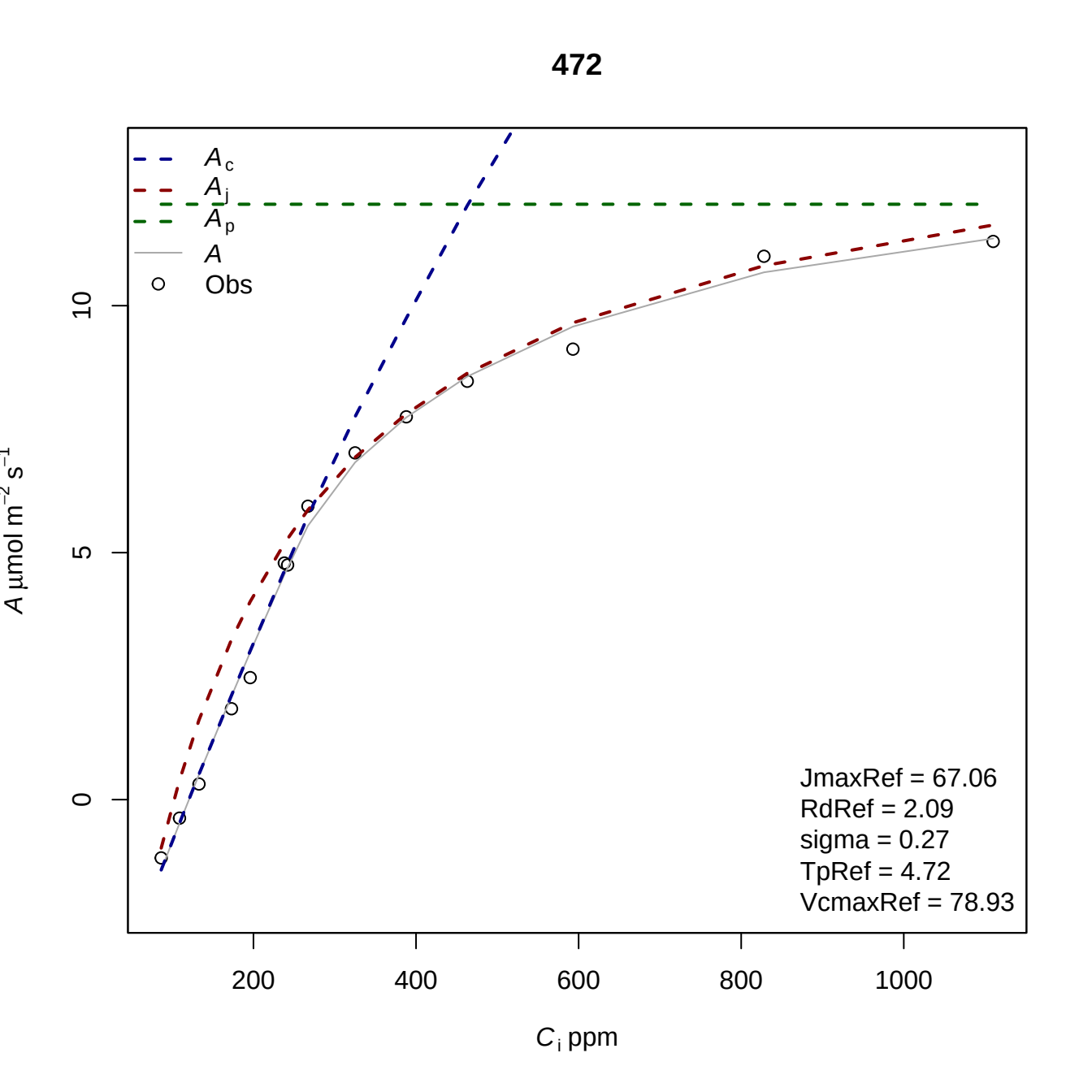

467
A_c
A_j
A_p
A
Obs
JmaxRef = 62.52
RdRef = 0.4
sigma = 0.16
TpRef = 4.02
VcmaxRef = 52.3
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
14
0
500
1000
1500

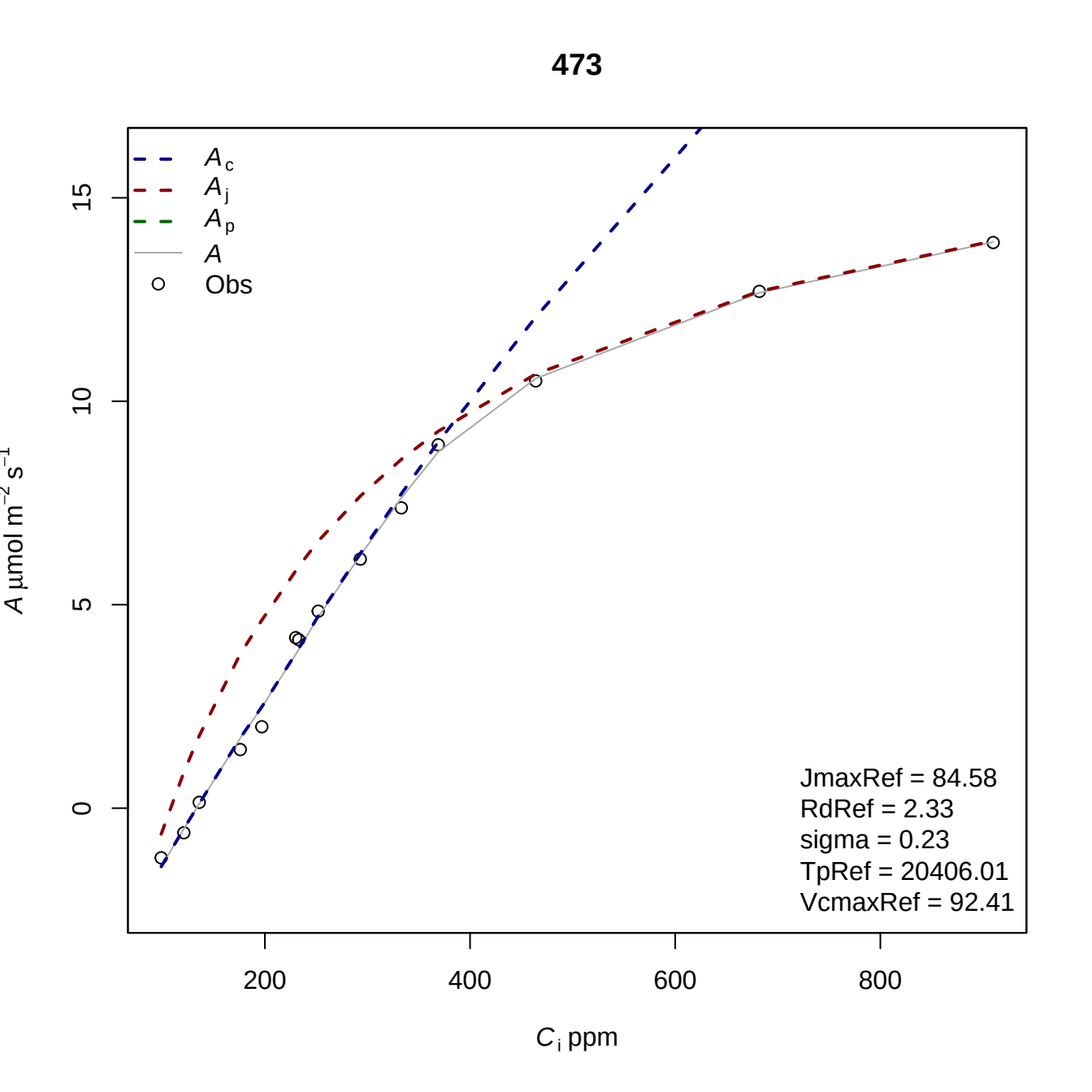

468
A_c
A_j
A_p
A
Obs
JmaxRef = 101.09
RdRef = 0.39
sigma = 0.23
TpRef = 954850.26
VcmaxRef = 91.47
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
500
1000
1500

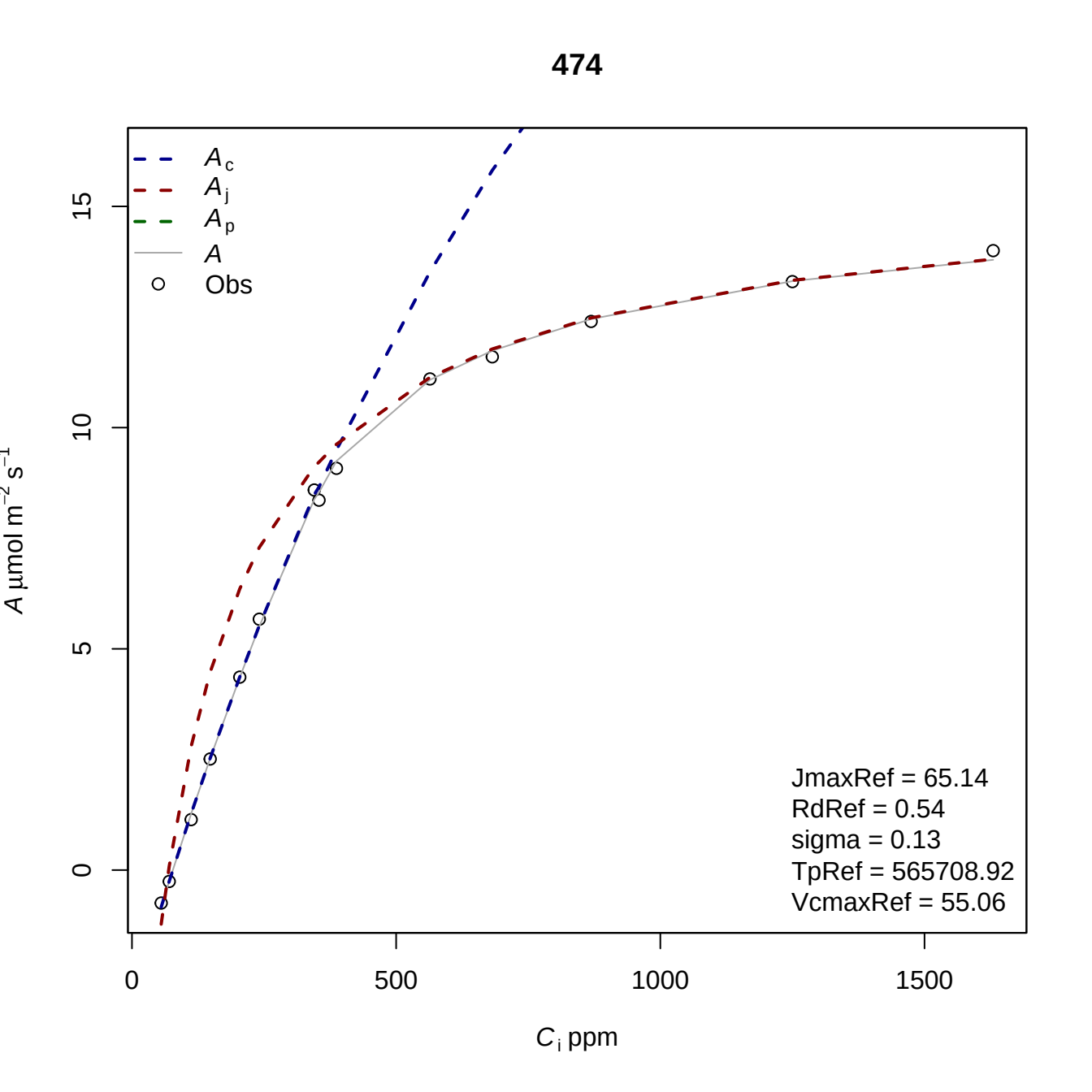

469
A_c
A_j
A_p
A
Obs
JmaxRef = 136.26
RdRef = 0.96
sigma = 0.52
TpRef = 35850.83
VcmaxRef = 124.99
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
200
400
600
800
1000

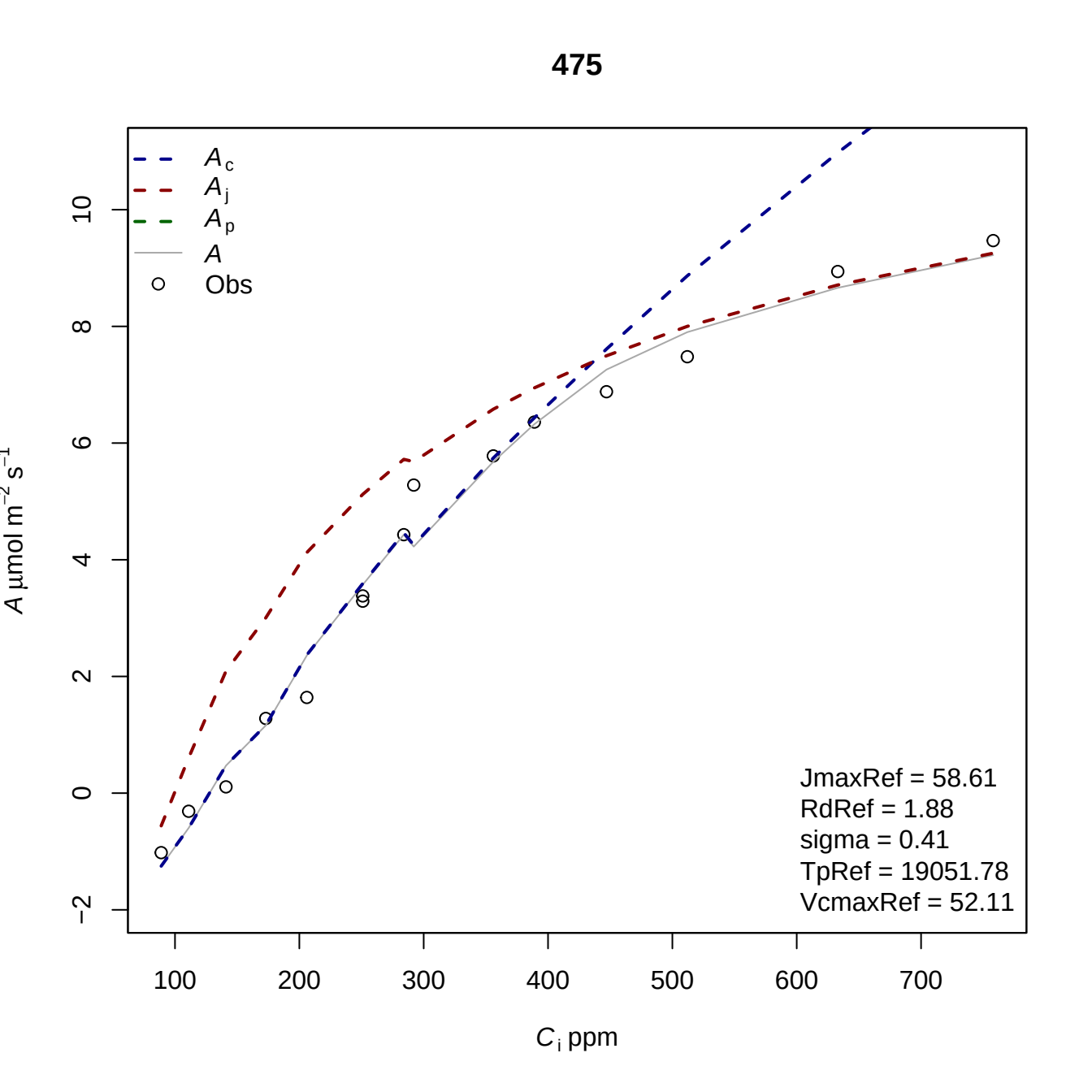

470
A_c
A_j
A_p
A
Obs
JmaxRef = 38.84
RdRef = 1.01
sigma = 0.61
TpRef = 1751.47
VcmaxRef = 50
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
200
400
600
800

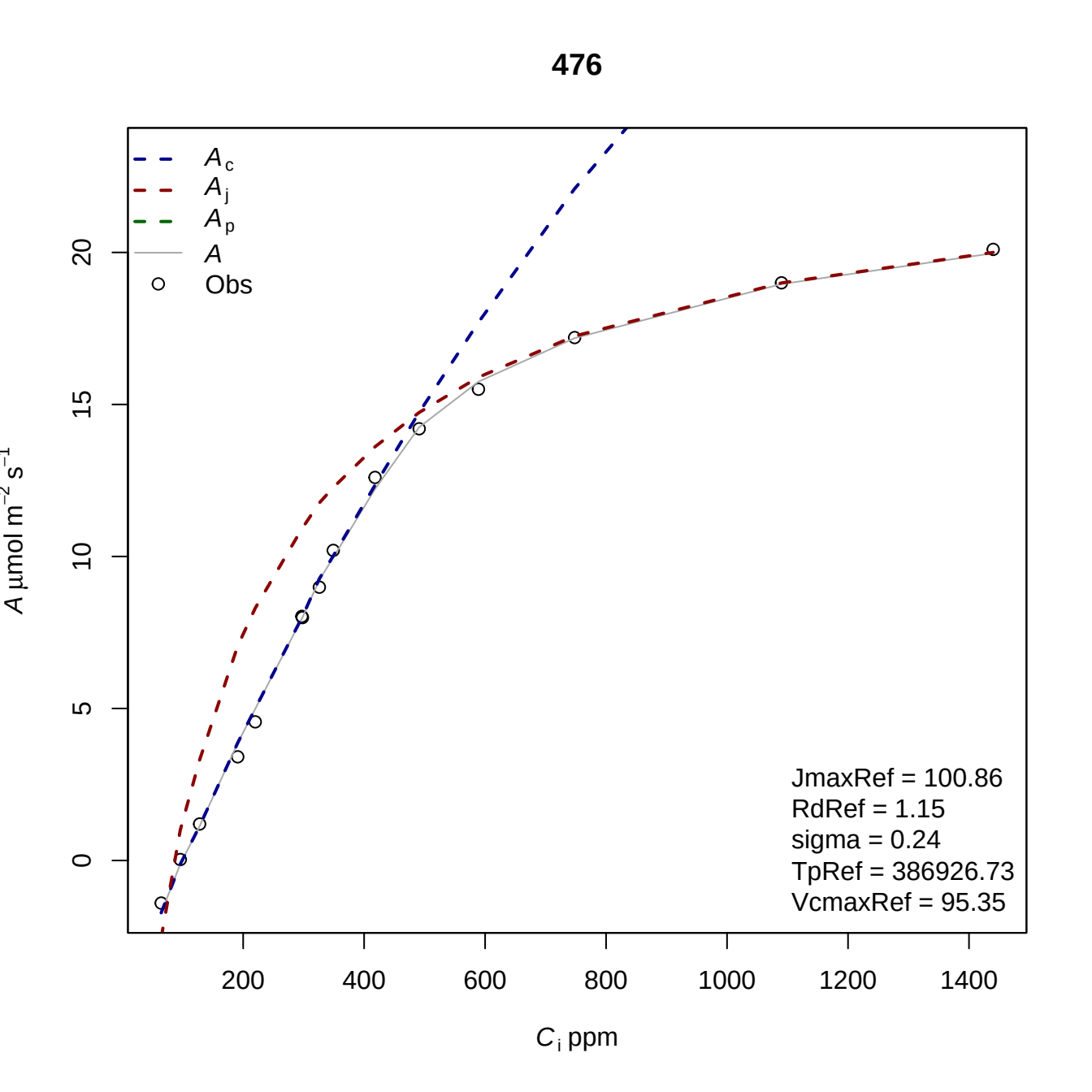

471
A_c
A_j
A_p
A
Obs
JmaxRef = 176.24
RdRef = 0.15
sigma = 0.76
TpRef = 5143.23
VcmaxRef = 152.32
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
200
400
600
800

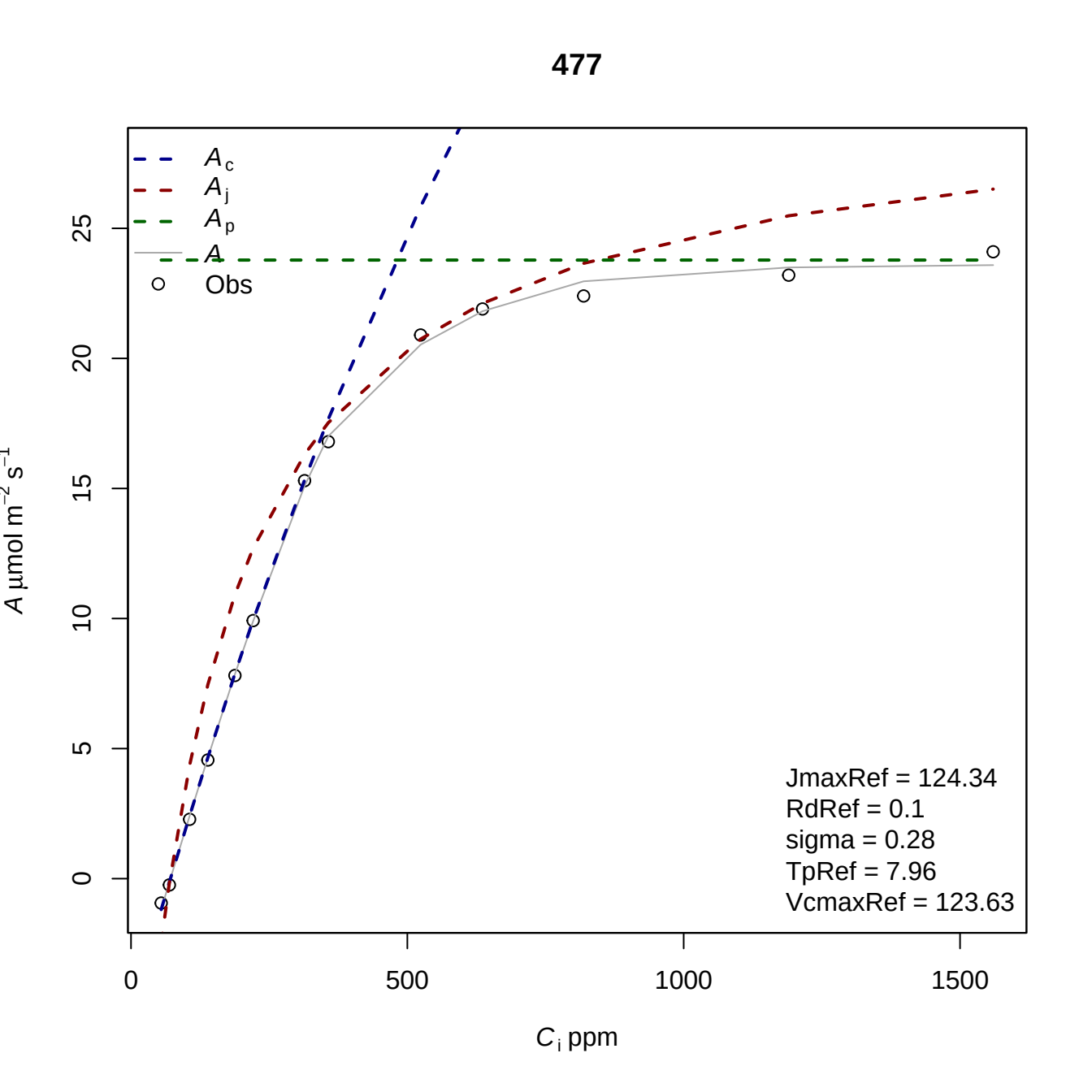

472
A_c
A_j
A_p
A
Obs
JmaxRef = 67.06
RdRef = 2.09
sigma = 0.27
TpRef = 4.72
VcmaxRef = 78.93
0
5
10
200
400
600
800
1000
A µmol m^{-2} s^{-1}
C_i ppm

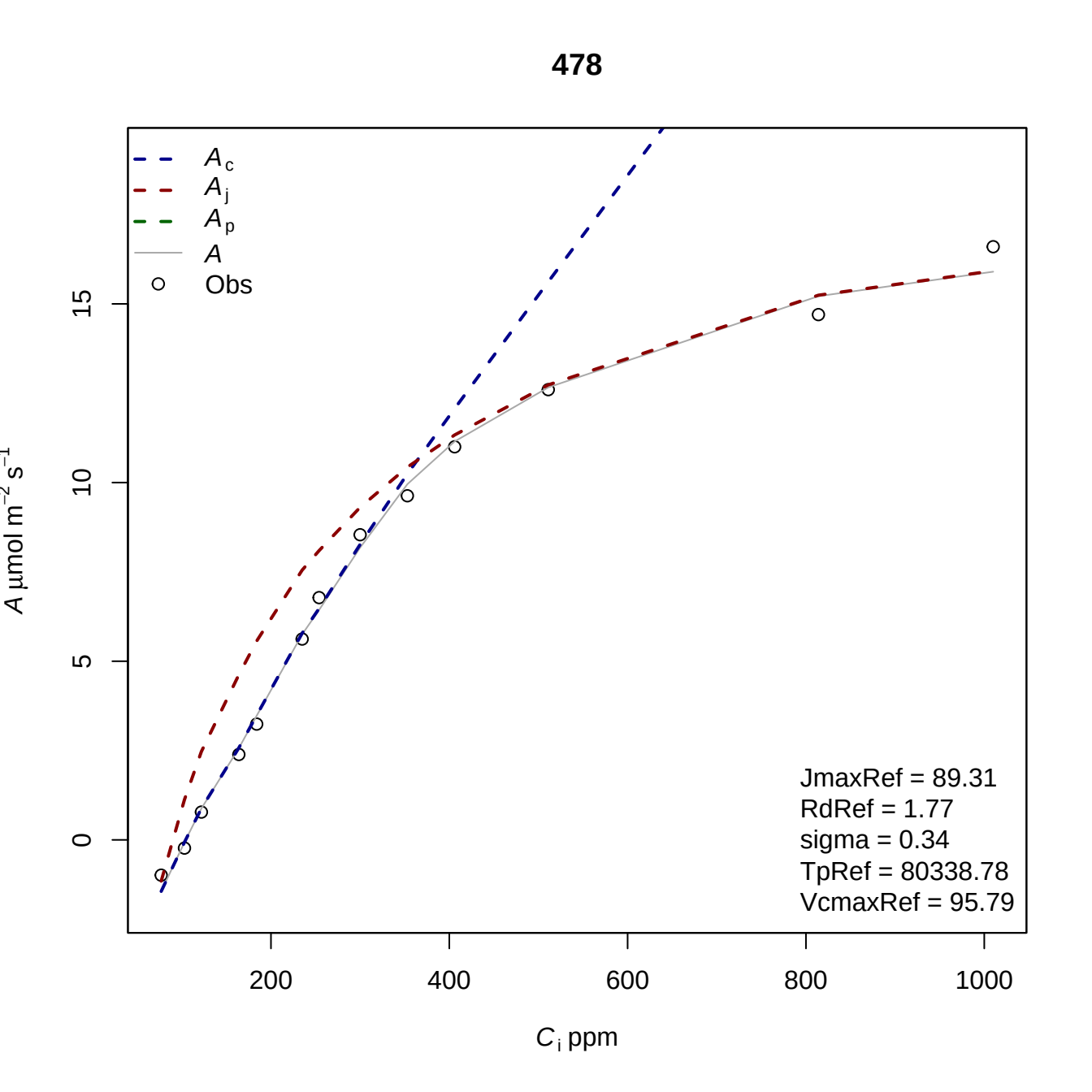

473
A_c
A_j
A_p
A
Obs
JmaxRef = 84.58
RdRef = 2.33
sigma = 0.23
TpRef = 20406.01
VcmaxRef = 92.41
0
5
10
15
200
400
600
800
A µmol m^{-2} s^{-1}
C_i ppm

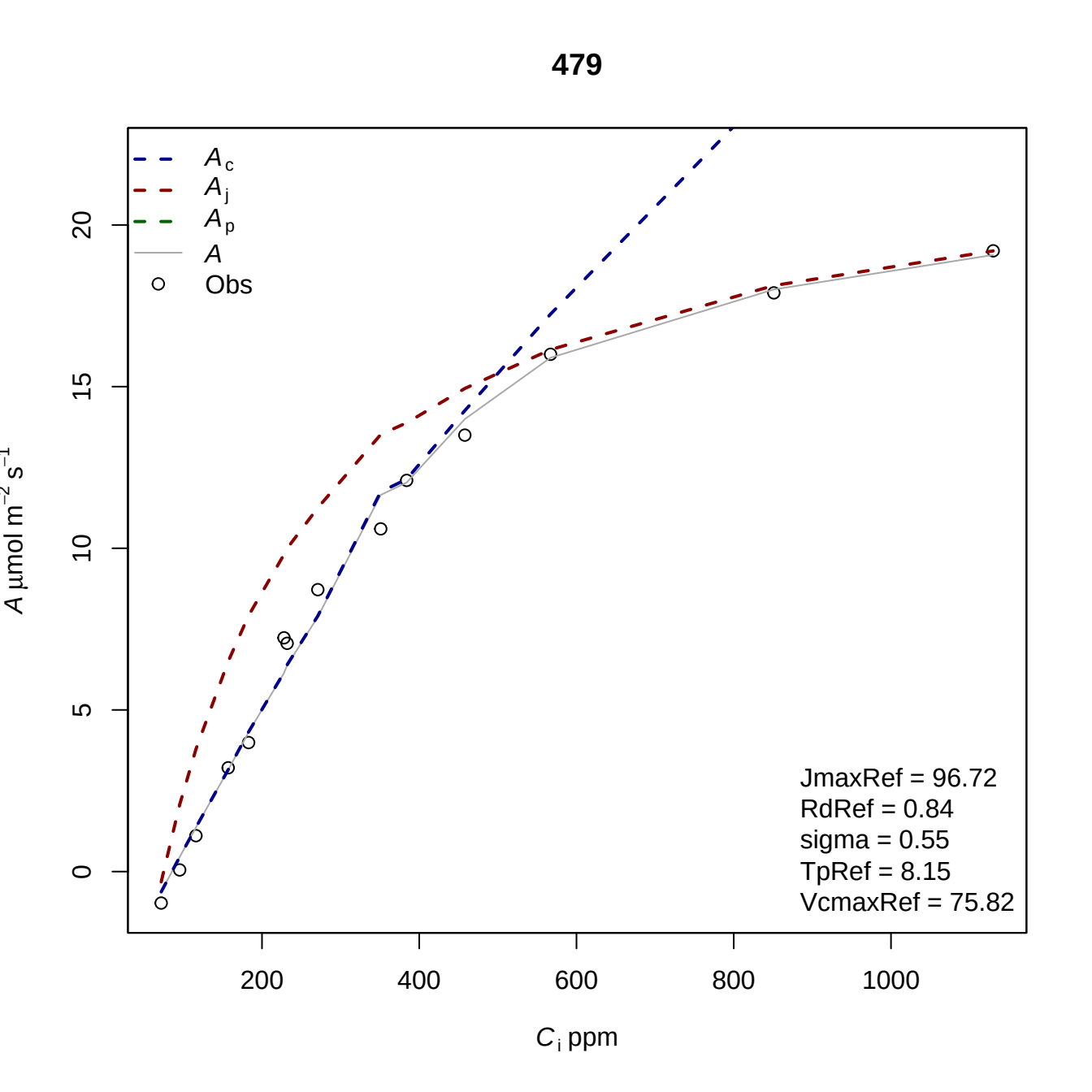

474
A_c
A_j
A_p
A
Obs
JmaxRef = 65.14
RdRef = 0.54
sigma = 0.13
TpRef = 565708.92
VcmaxRef = 55.06
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
500
1000
1500

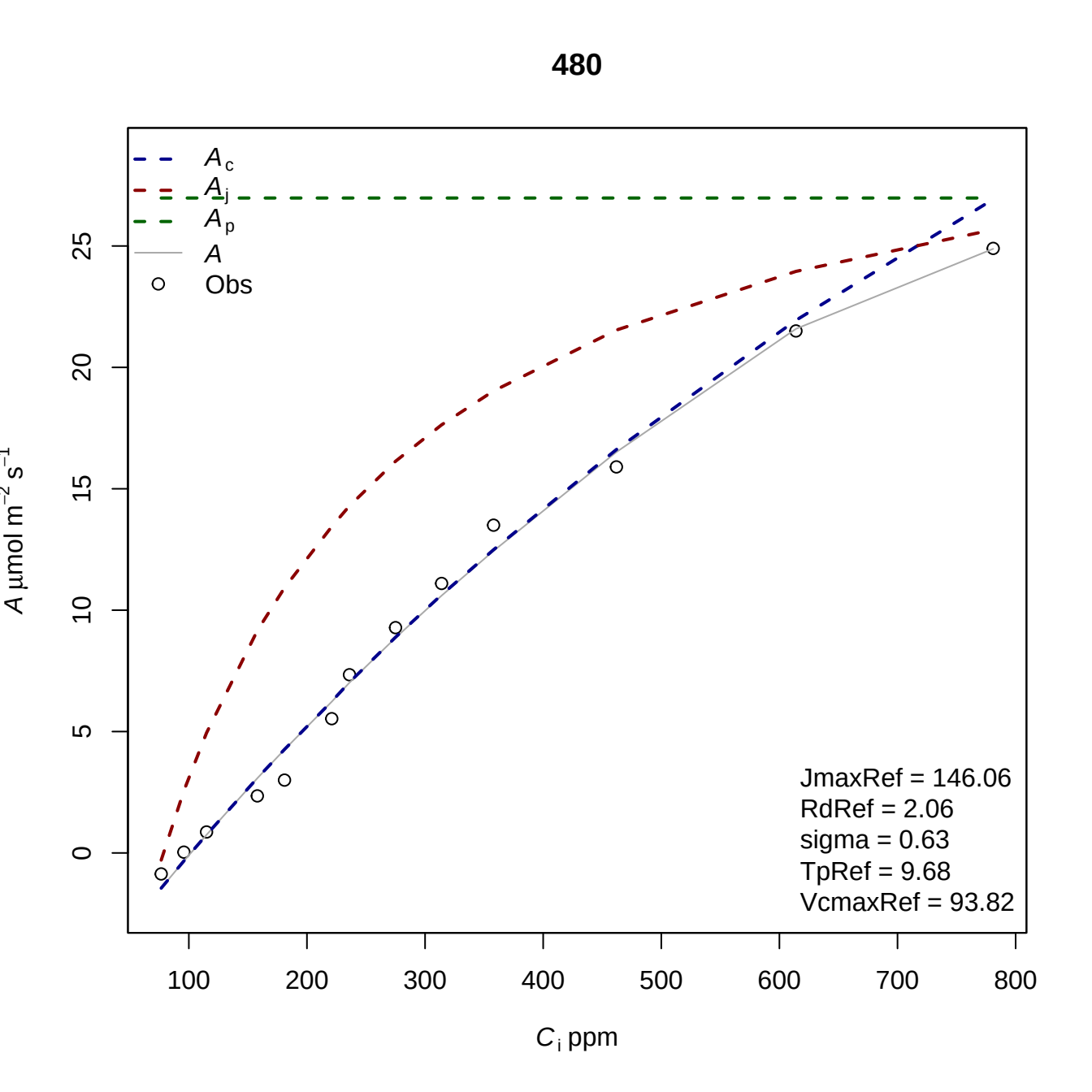

475
A_c
A_j
A_p
A
Obs
JmaxRef = 58.61
RdRef = 1.88
sigma = 0.41
TpRef = 19051.78
VcmaxRef = 52.11
A µmol m^{-2} s^{-1}
C_i ppm
−2
0
2
4
6
8
10
100
200
300
400
500
600
700

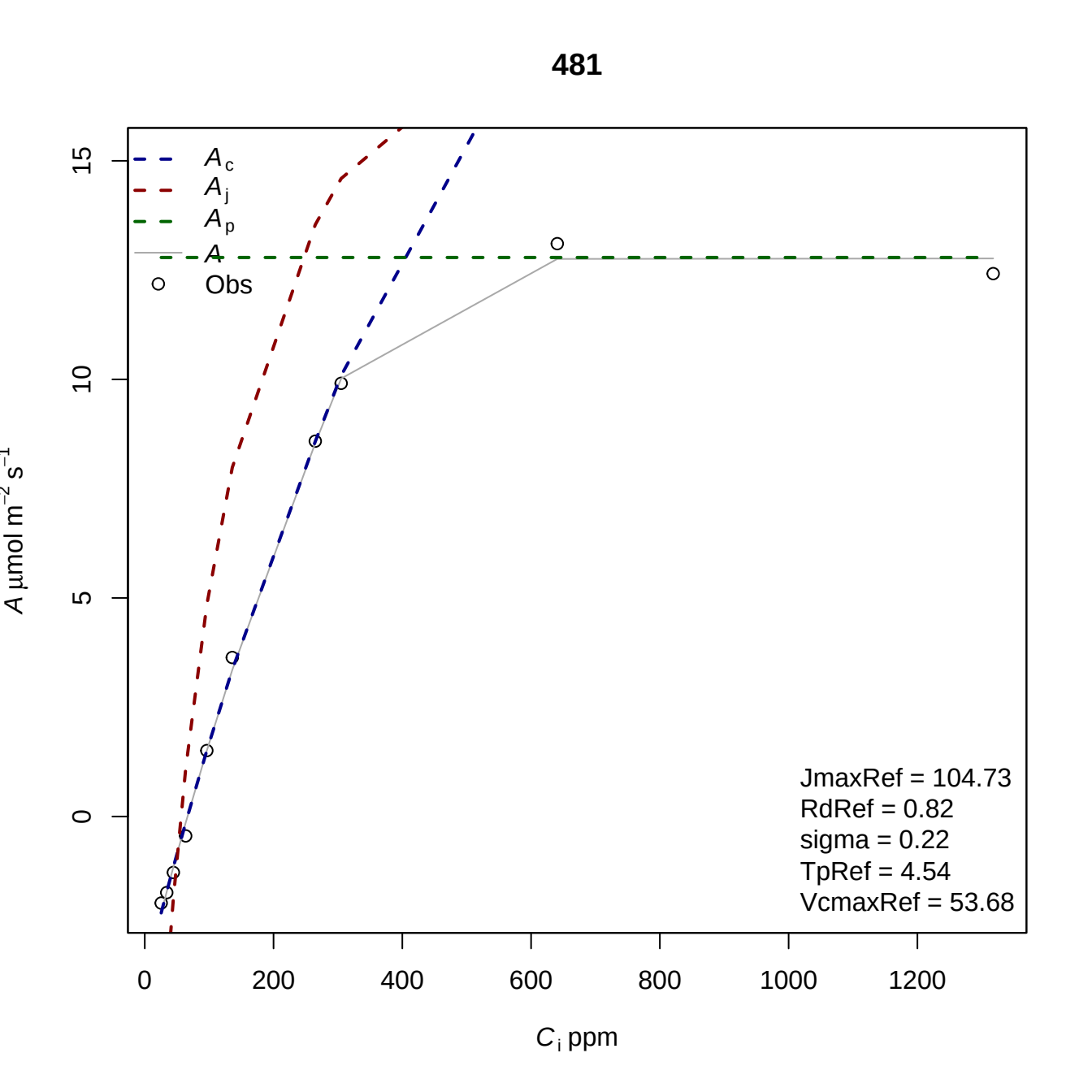

476
A_c
A_j
A_p
A
Obs
JmaxRef = 100.86
RdRef = 1.15
sigma = 0.24
TpRef = 386926.73
VcmaxRef = 95.35
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
200
400
600
800
1000
1200
1400

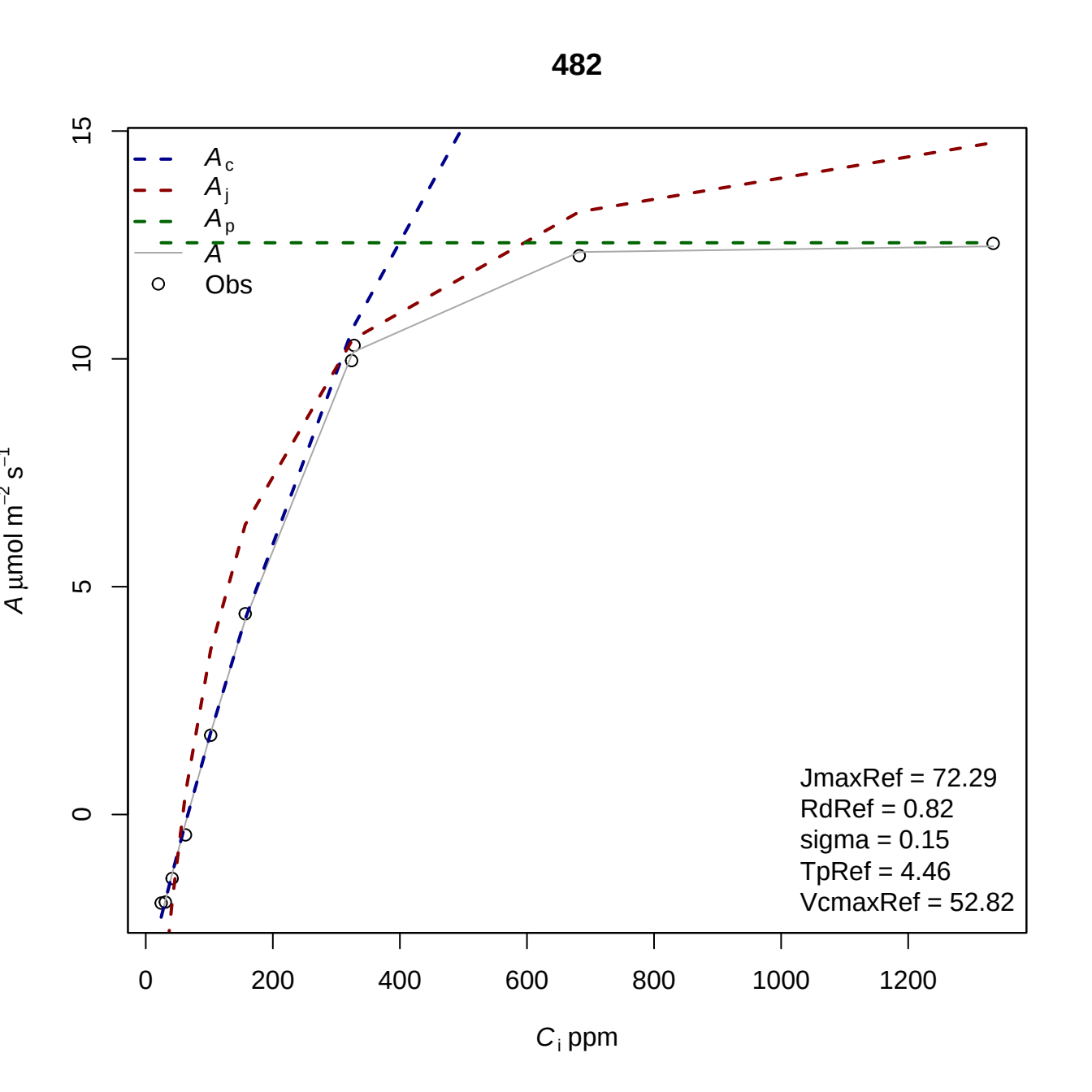

477
A_c
A_j
A_p
A
Obs
JmaxRef = 124.34
RdRef = 0.1
sigma = 0.28
TpRef = 7.96
VcmaxRef = 123.63
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
500
1000
1500

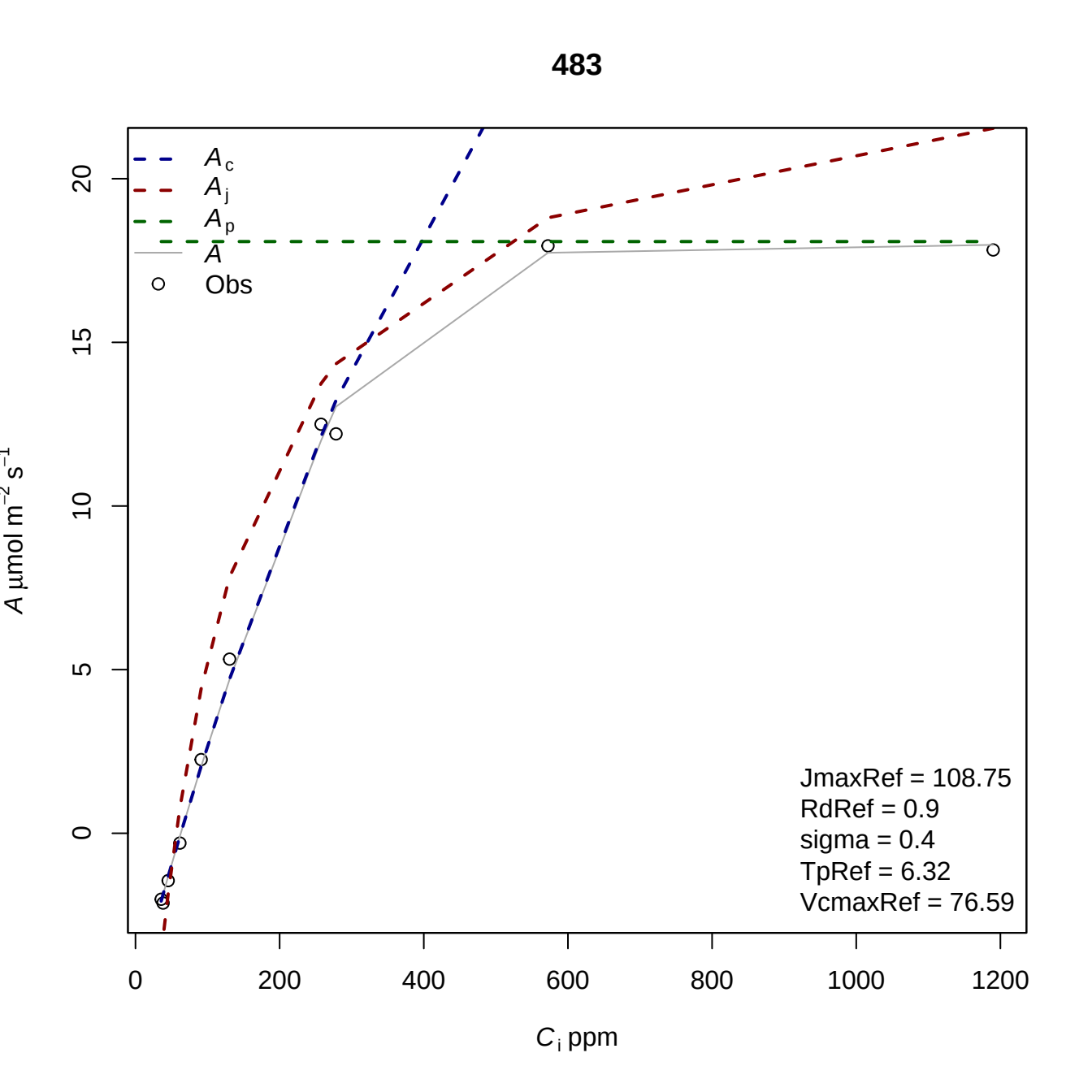

478
A_c
A_j
A_p
A
Obs
JmaxRef = 89.31
RdRef = 1.77
sigma = 0.34
TpRef = 80338.78
VcmaxRef = 95.79
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
200
400
600
800
1000

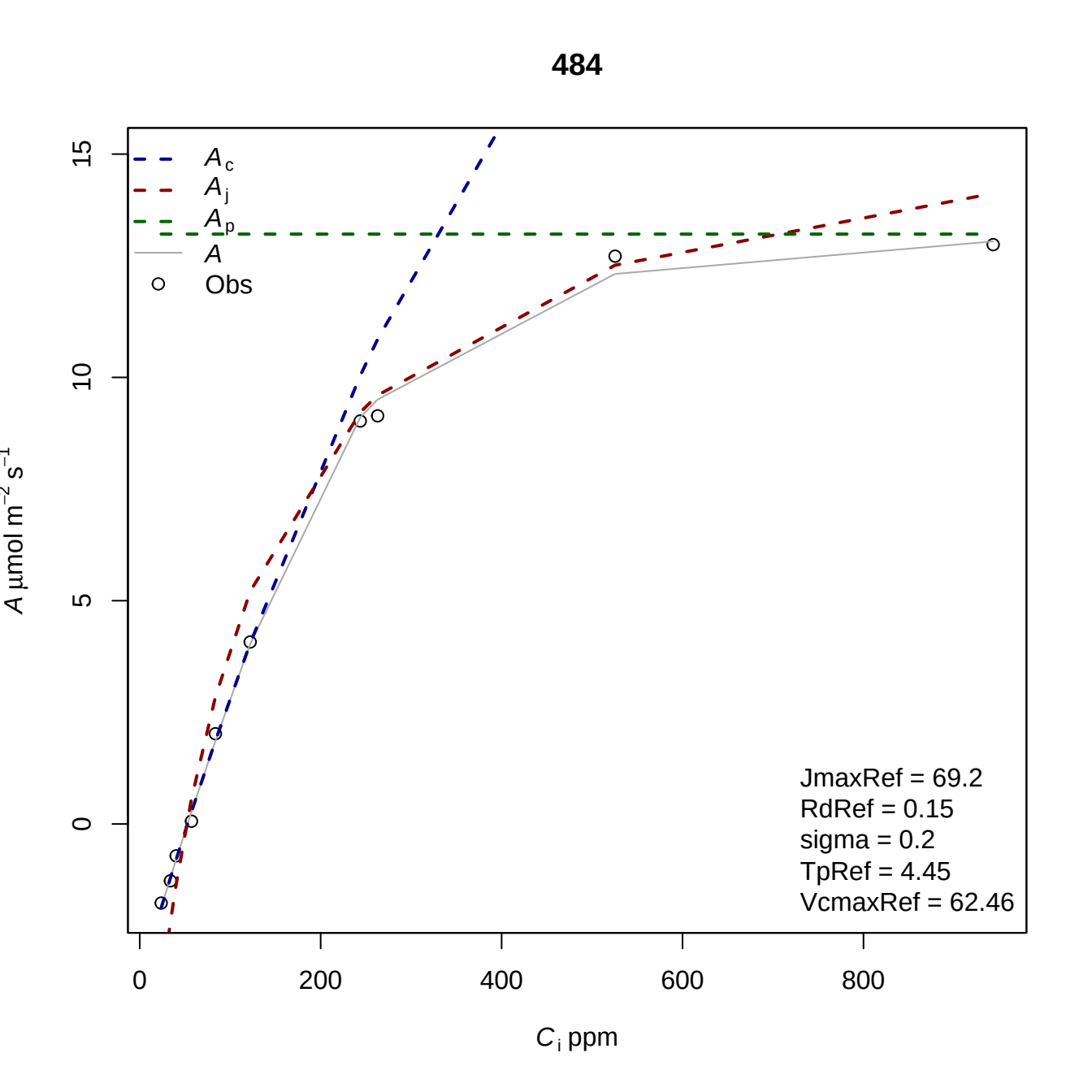

479
A_c
A_j
A_p
A
Obs
JmaxRef = 96.72
RdRef = 0.84
sigma = 0.55
TpRef = 8.15
VcmaxRef = 75.82
0
5
10
15
20
200
400
600
800
1000
A µmol m^{-2} s^{-1}
C_i ppm

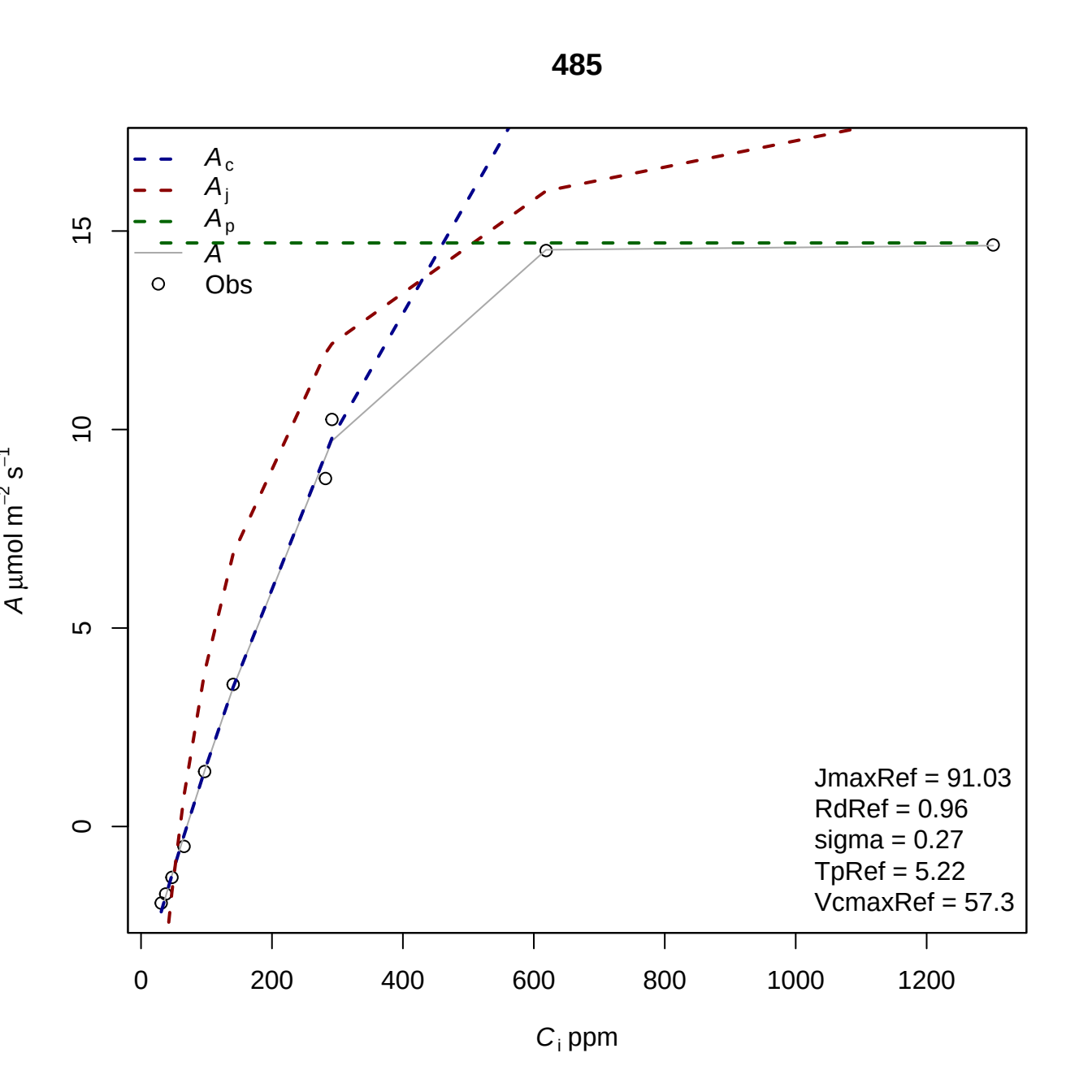

480
A_c
A_j
A_p
A
Obs
JmaxRef = 146.06
RdRef = 2.06
sigma = 0.63
TpRef = 9.68
VcmaxRef = 93.82
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
100
200
300
400
500
600
700
800

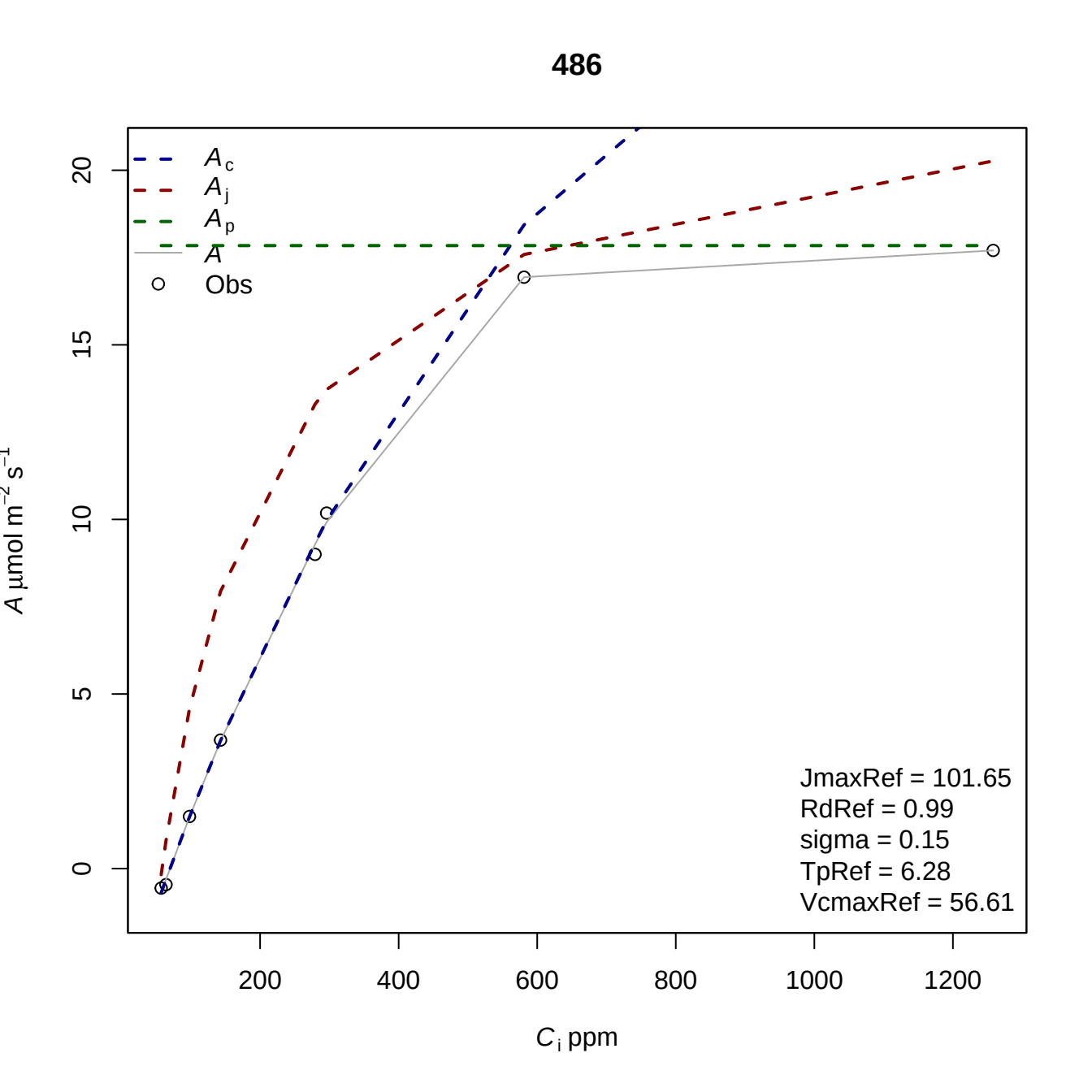

481
A_c
A_j
A_p
A
Obs
JmaxRef = 104.73
RdRef = 0.82
sigma = 0.22
TpRef = 4.54
VcmaxRef = 53.68
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
200
400
600
800
1000
1200

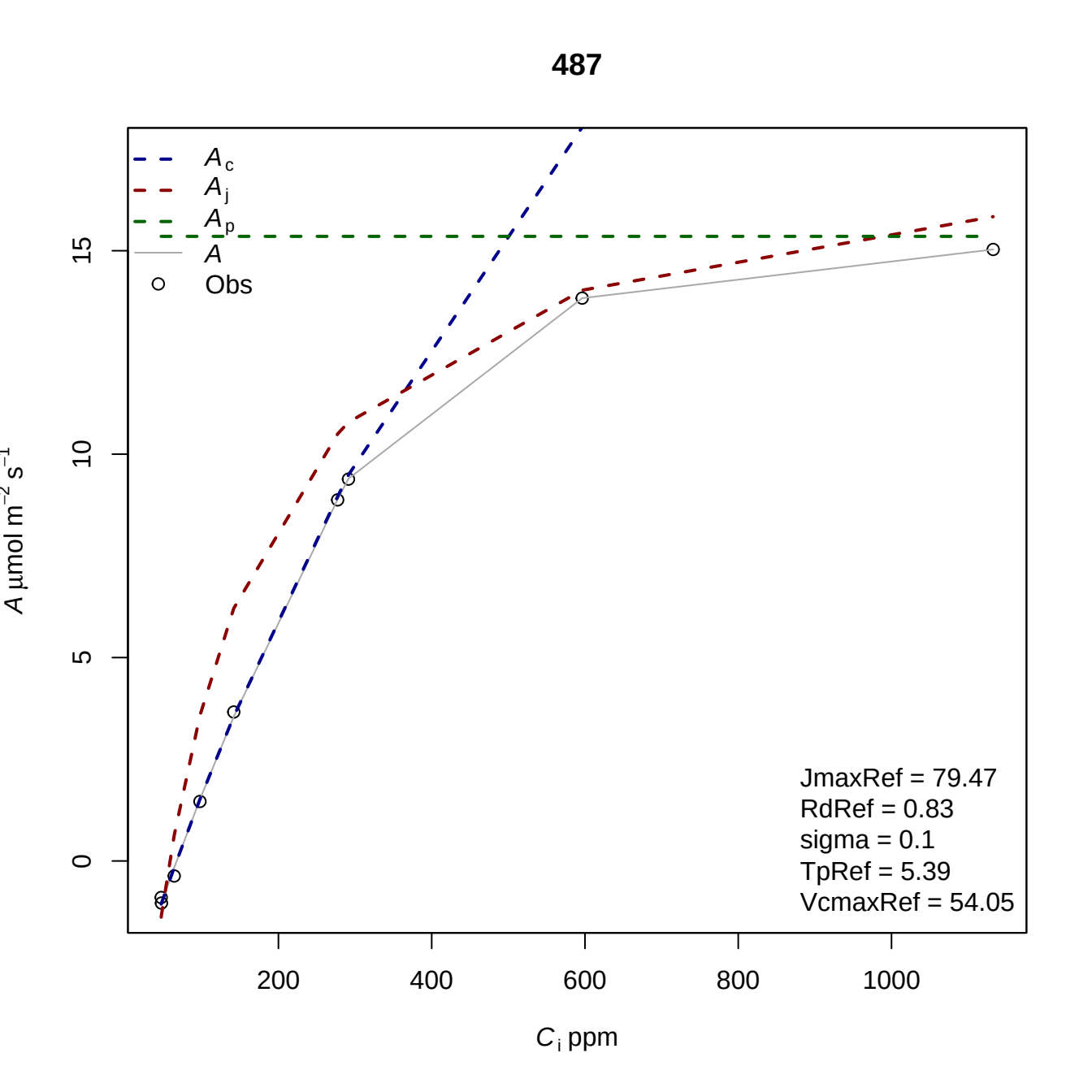

482
A_c
A_j
A_p
A
Obs
JmaxRef = 72.29
RdRef = 0.82
sigma = 0.15
TpRef = 4.46
VcmaxRef = 52.82
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
200
400
600
800
1000
1200

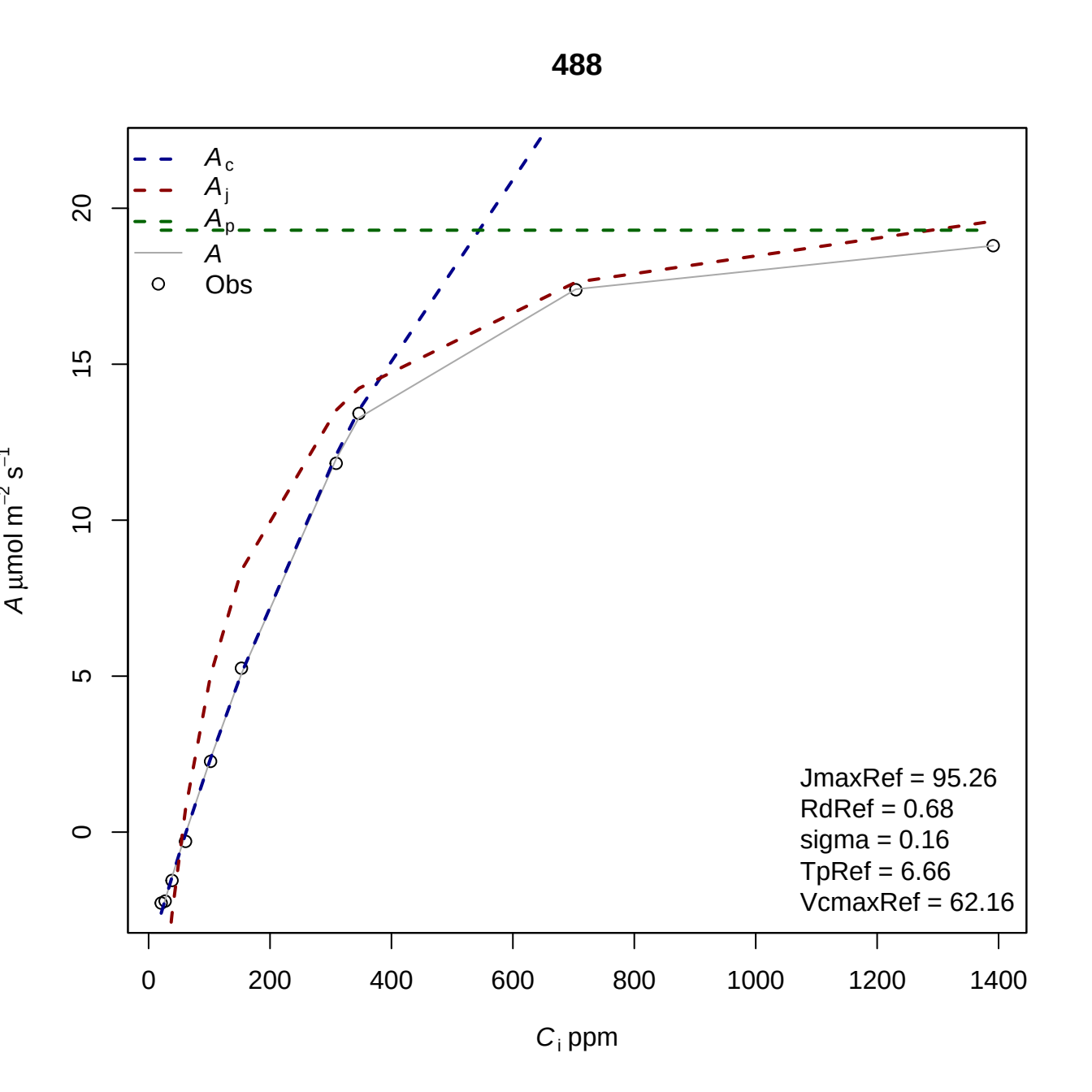

483
A_c
A_j
A_p
A
Obs
JmaxRef = 108.75
RdRef = 0.9
sigma = 0.4
TpRef = 6.32
VcmaxRef = 76.59
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

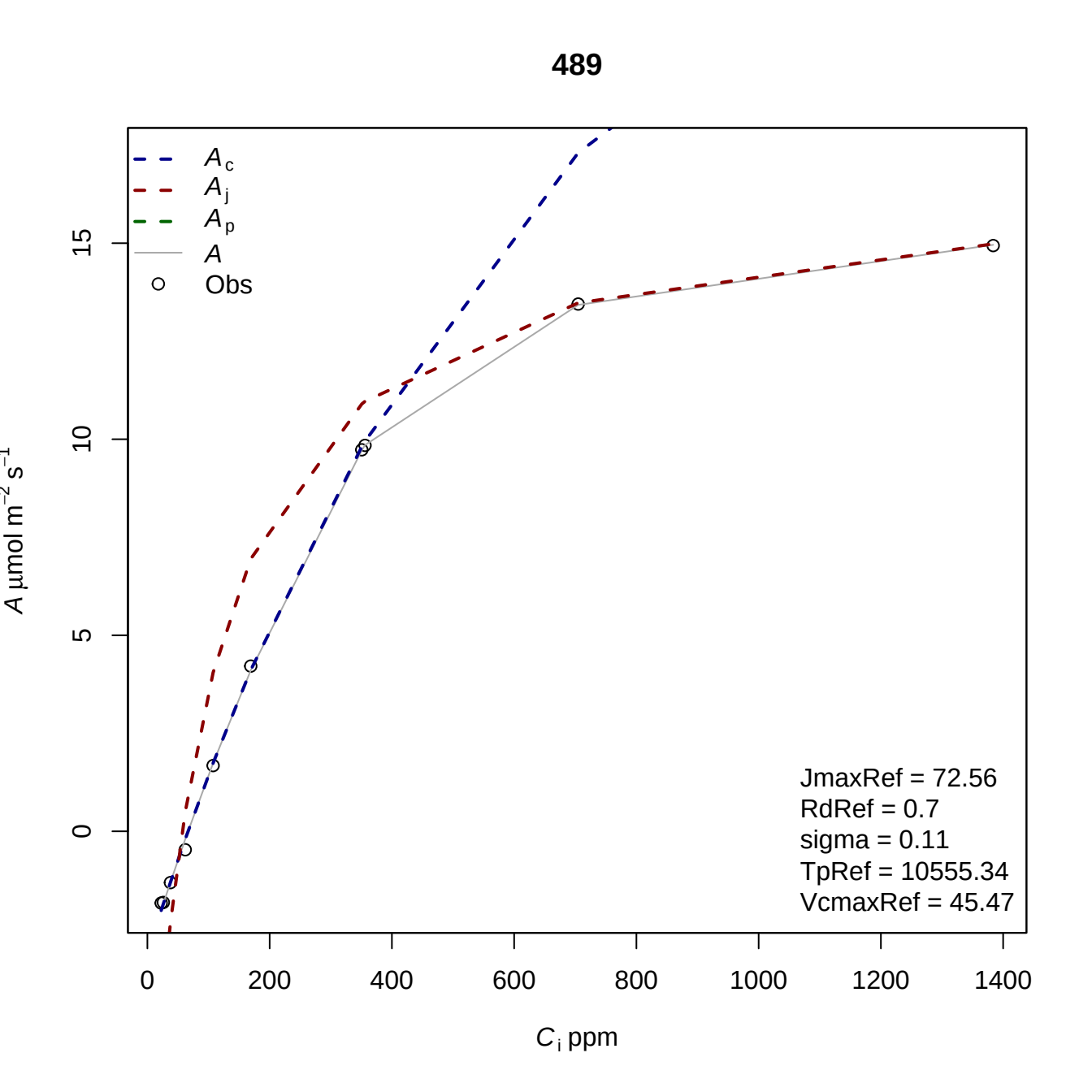

484
A_c
A_j
A_p
A
Obs
JmaxRef = 69.2
RdRef = 0.15
sigma = 0.2
TpRef = 4.45
VcmaxRef = 62.46
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
200
400
600
800

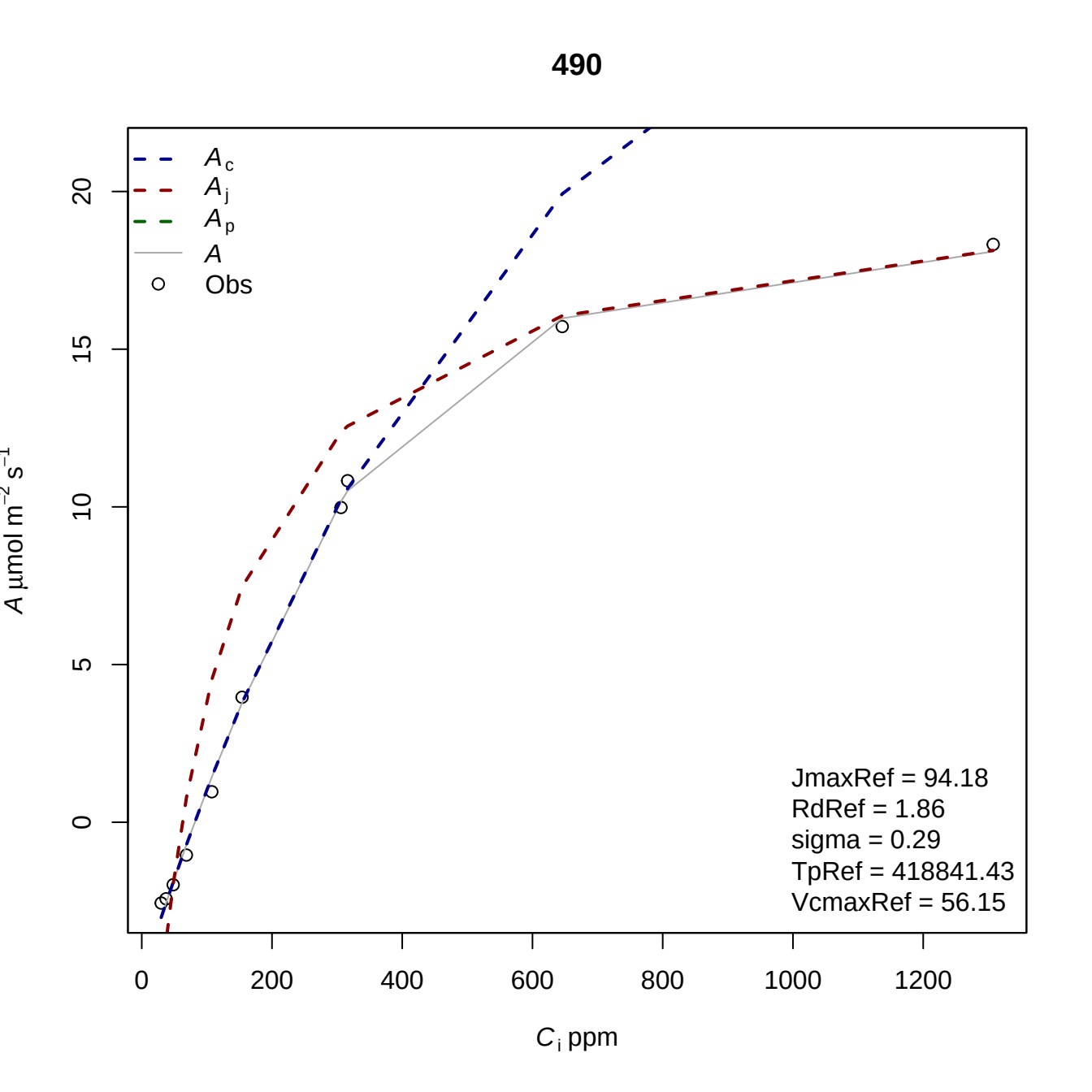

485
A_c
A_j
A_p
A
Obs
JmaxRef = 91.03
RdRef = 0.96
sigma = 0.27
TpRef = 5.22
VcmaxRef = 57.3
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
200
400
600
800
1000
1200

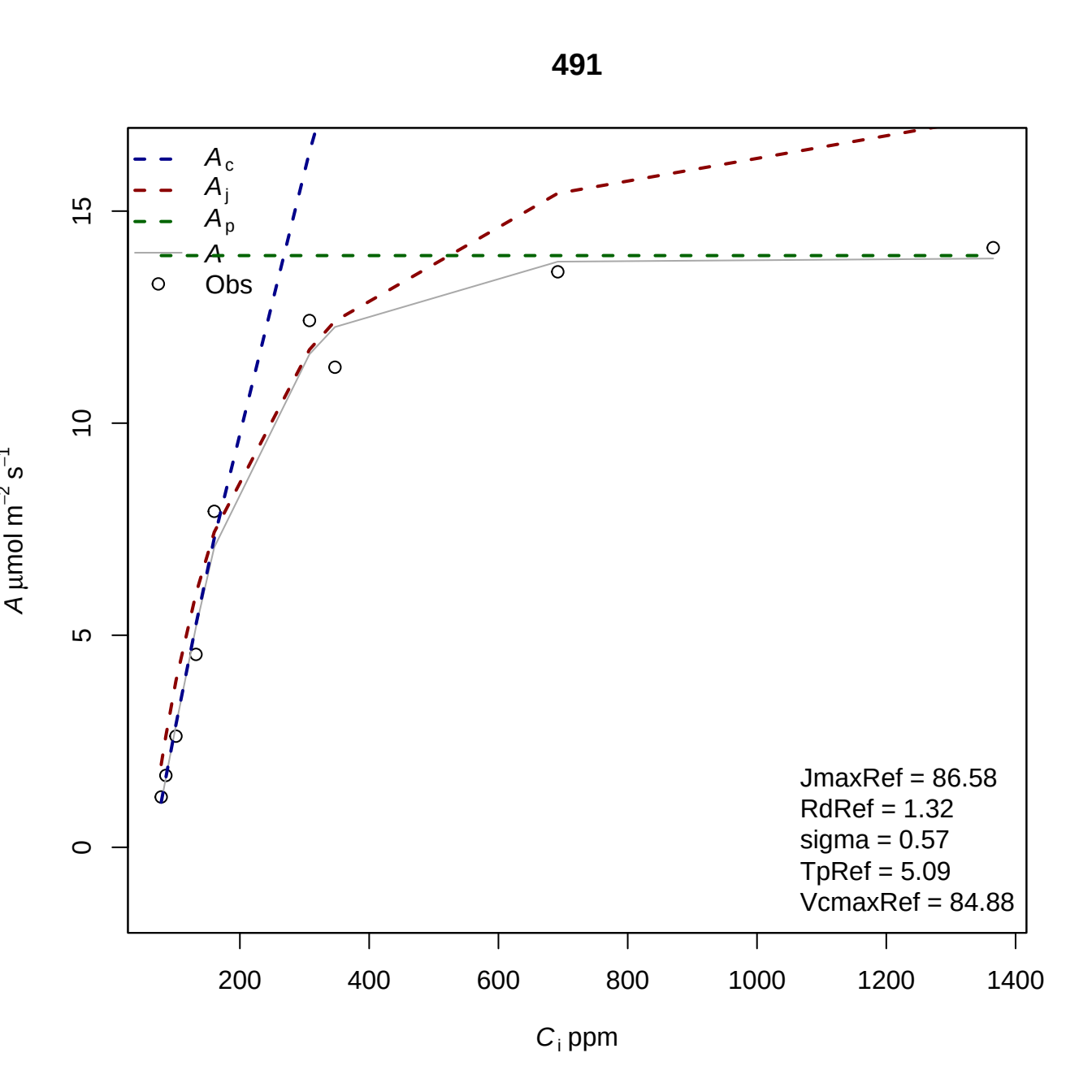

486
A_c
A_j
A_p
A
Obs
JmaxRef = 101.65
RdRef = 0.99
sigma = 0.15
TpRef = 6.28
VcmaxRef = 56.61
0
5
10
15
20
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

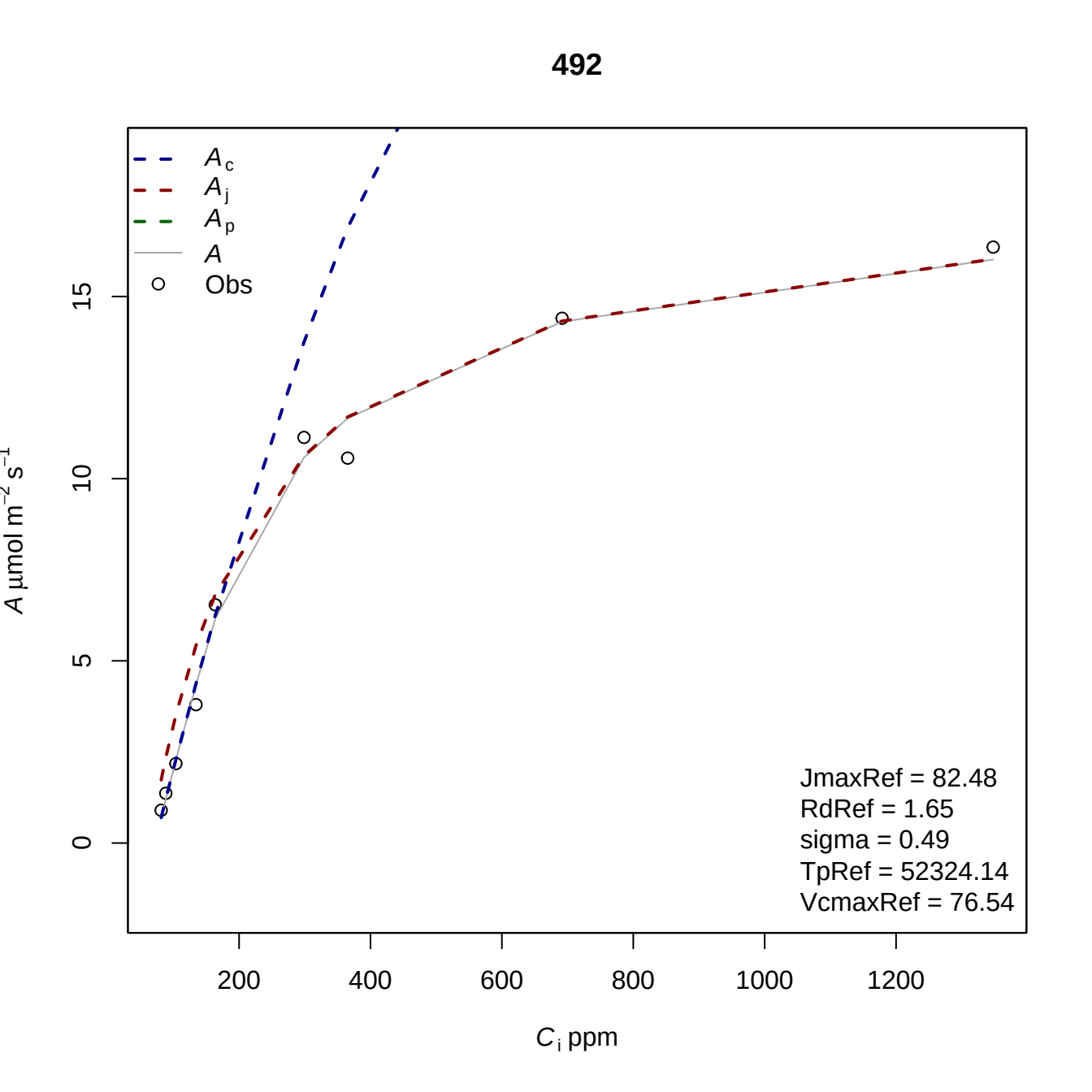

487
A_c
A_j
A_p
A
Obs
JmaxRef = 79.47
RdRef = 0.83
sigma = 0.1
TpRef = 5.39
VcmaxRef = 54.05
0
5
10
15
200
400
600
800
1000
A µmol m^{-2} s^{-1}
C_i ppm

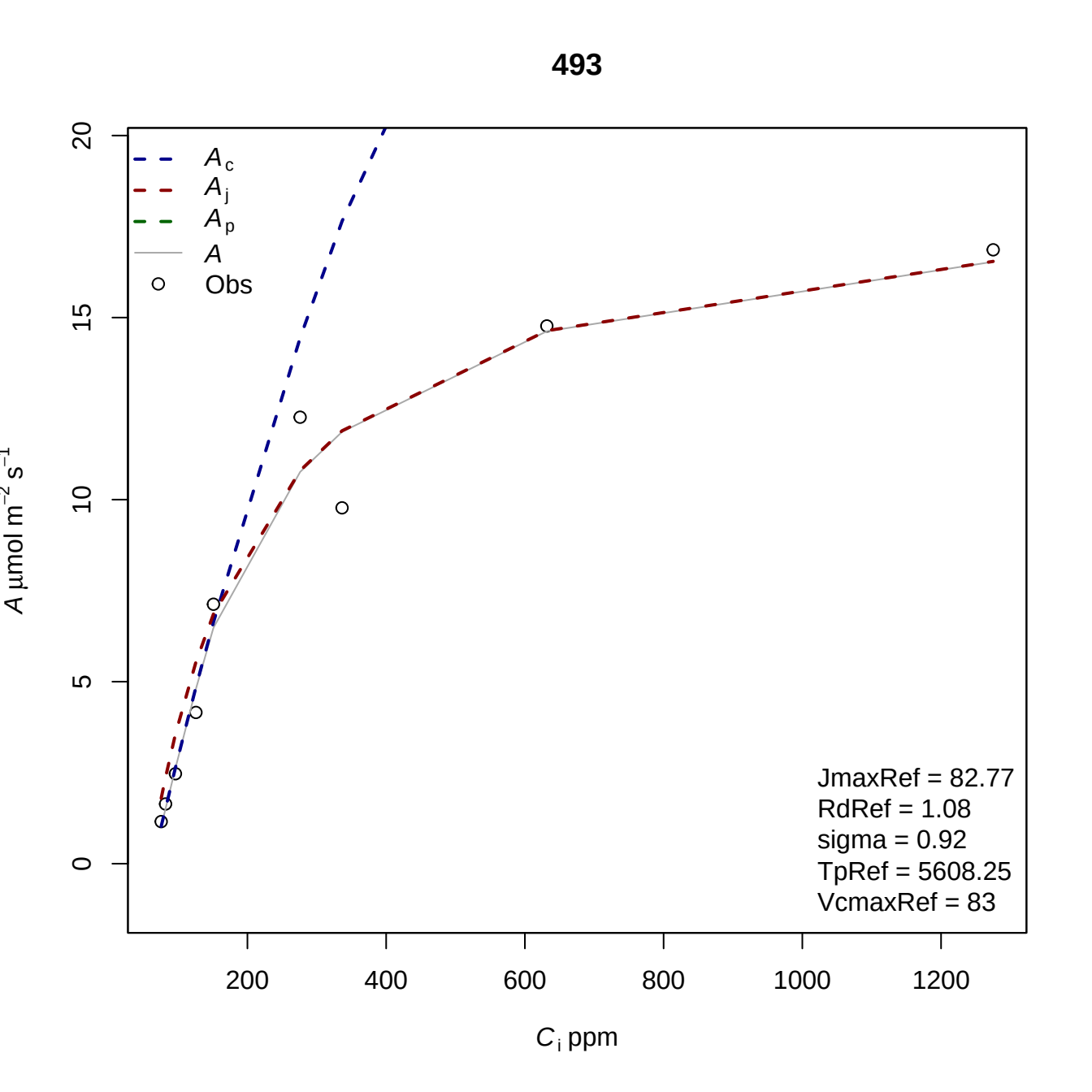

488
A_c
A_j
A_p
A
Obs
JmaxRef = 95.26
RdRef = 0.68
sigma = 0.16
TpRef = 6.66
VcmaxRef = 62.16
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000
1200
1400

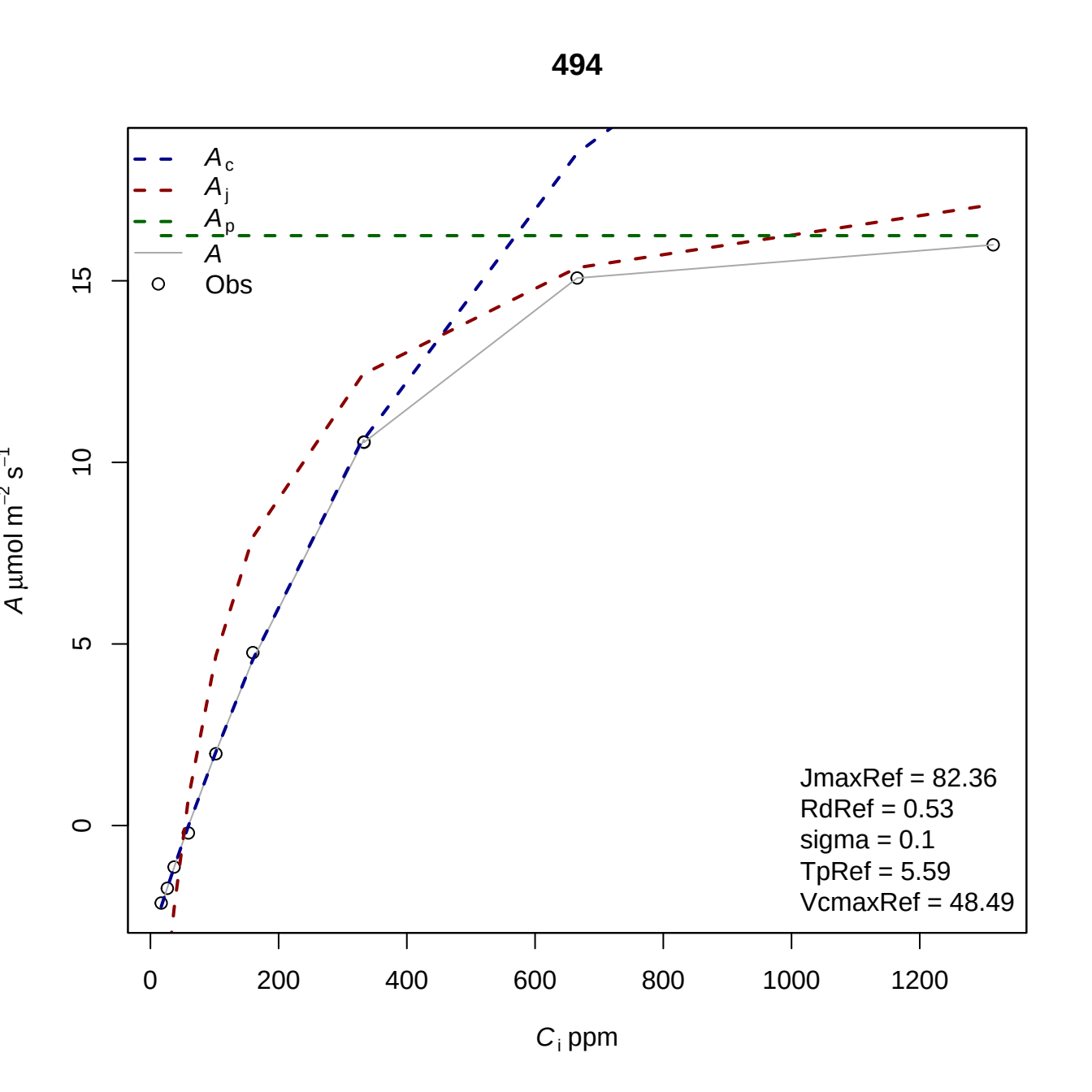

489
A_c
A_j
A_p
A
Obs
JmaxRef = 72.56
RdRef = 0.7
sigma = 0.11
TpRef = 10555.34
VcmaxRef = 45.47
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
200
400
600
800
1000
1200
1400

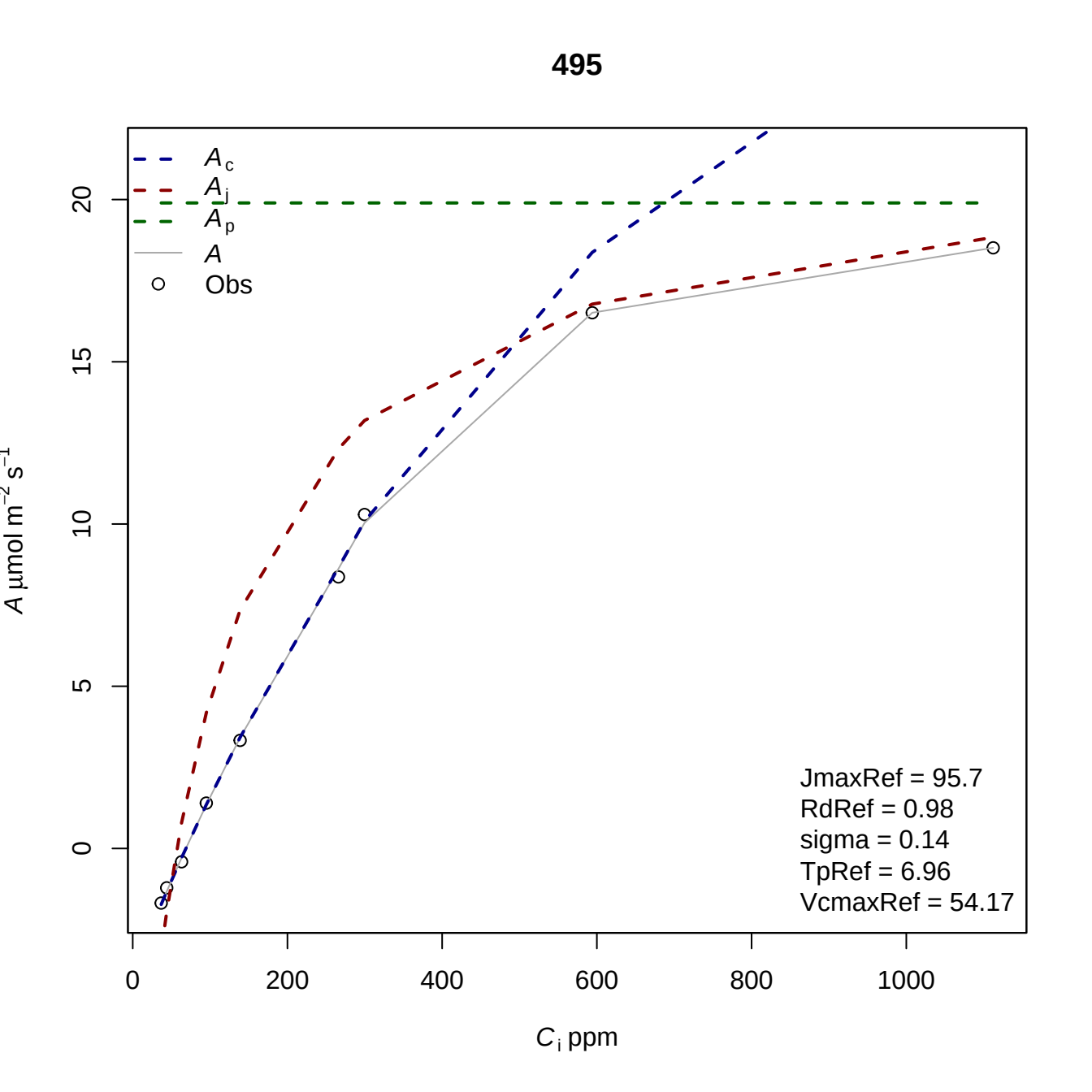

490
A_c
A_j
A_p
A
Obs
JmaxRef = 94.18
RdRef = 1.86
sigma = 0.29
TpRef = 418841.43
VcmaxRef = 56.15
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

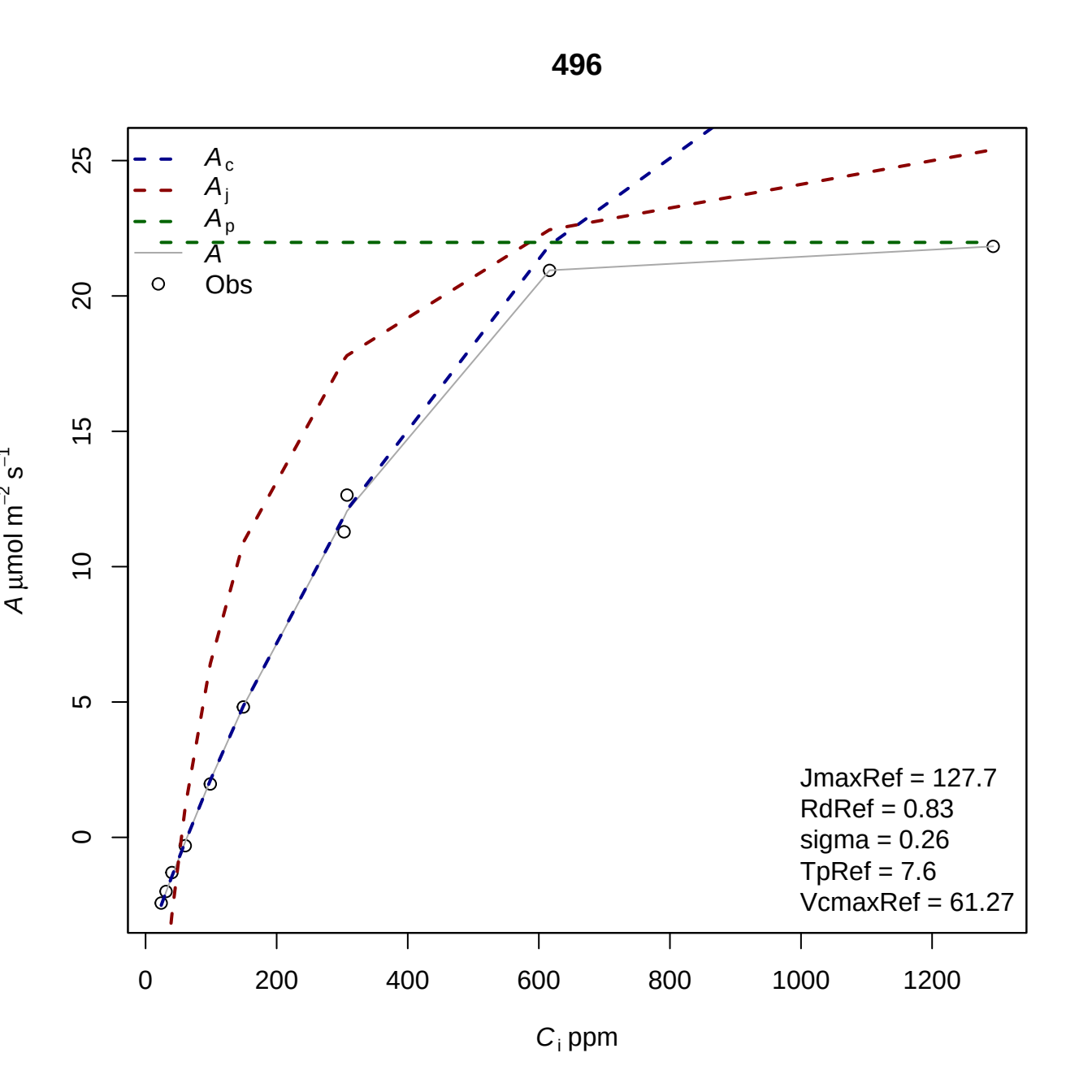

491
A_c
A_j
A_p
A
Obs
JmaxRef = 86.58
RdRef = 1.32
sigma = 0.57
TpRef = 5.09
VcmaxRef = 84.88
0
5
10
15
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
C_i ppm

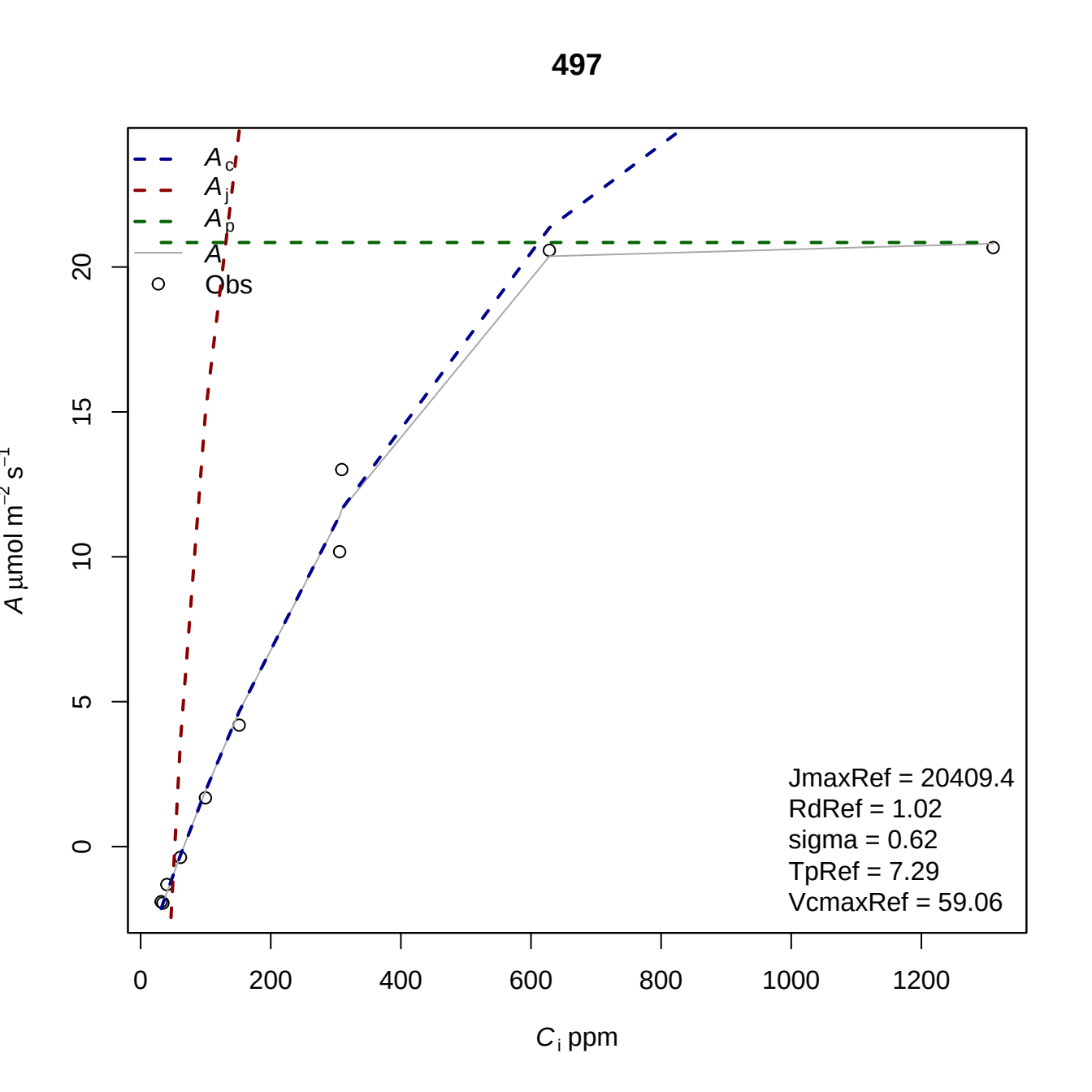

492
A_c
A_j
A_p
A
Obs
JmaxRef = 82.48
RdRef = 1.65
sigma = 0.49
TpRef = 52324.14
VcmaxRef = 76.54
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
200
400
600
800
1000
1200

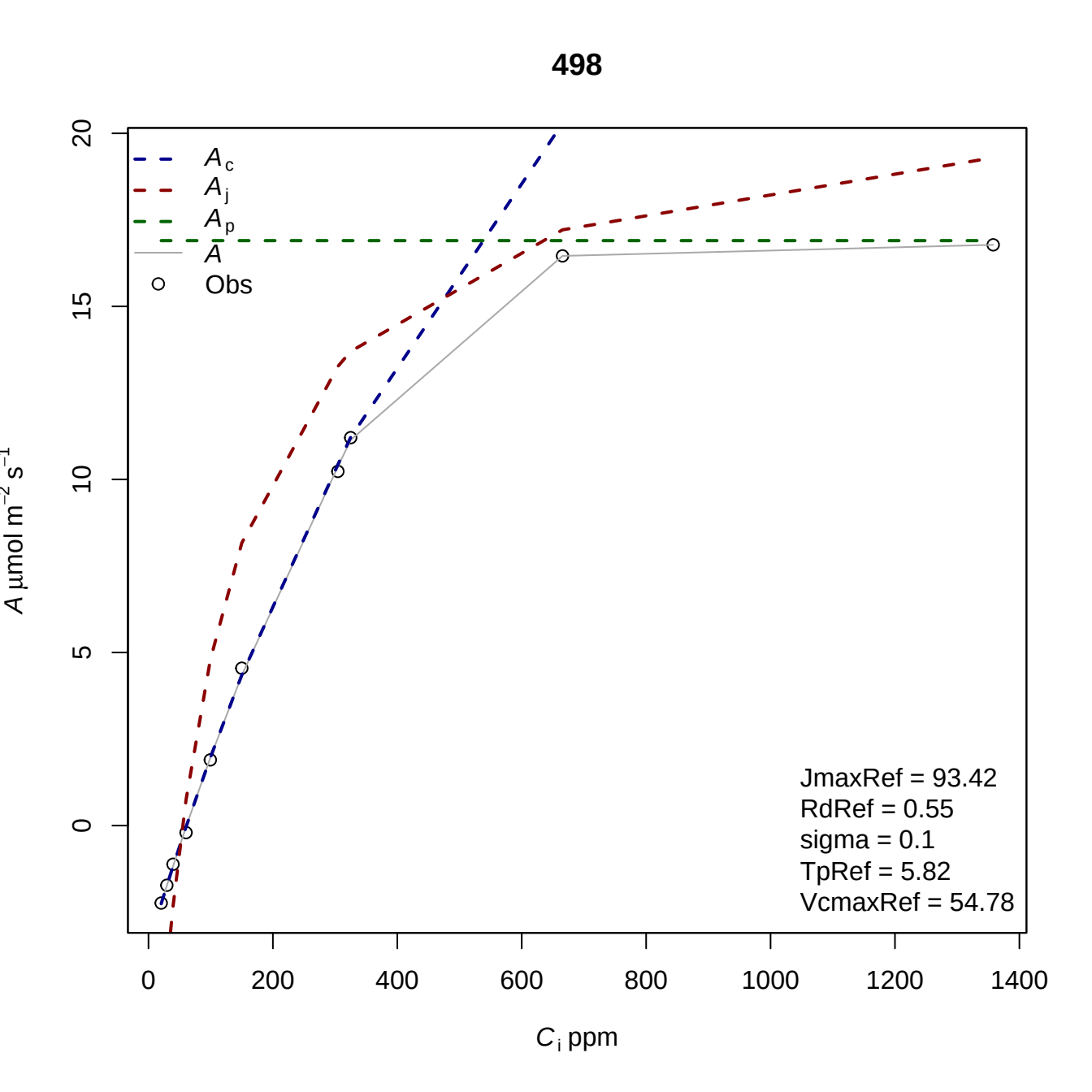

493
A_c
A_j
A_p
A
Obs
JmaxRef = 82.77
RdRef = 1.08
sigma = 0.92
TpRef = 5608.25
VcmaxRef = 83
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
200
400
600
800
1000
1200

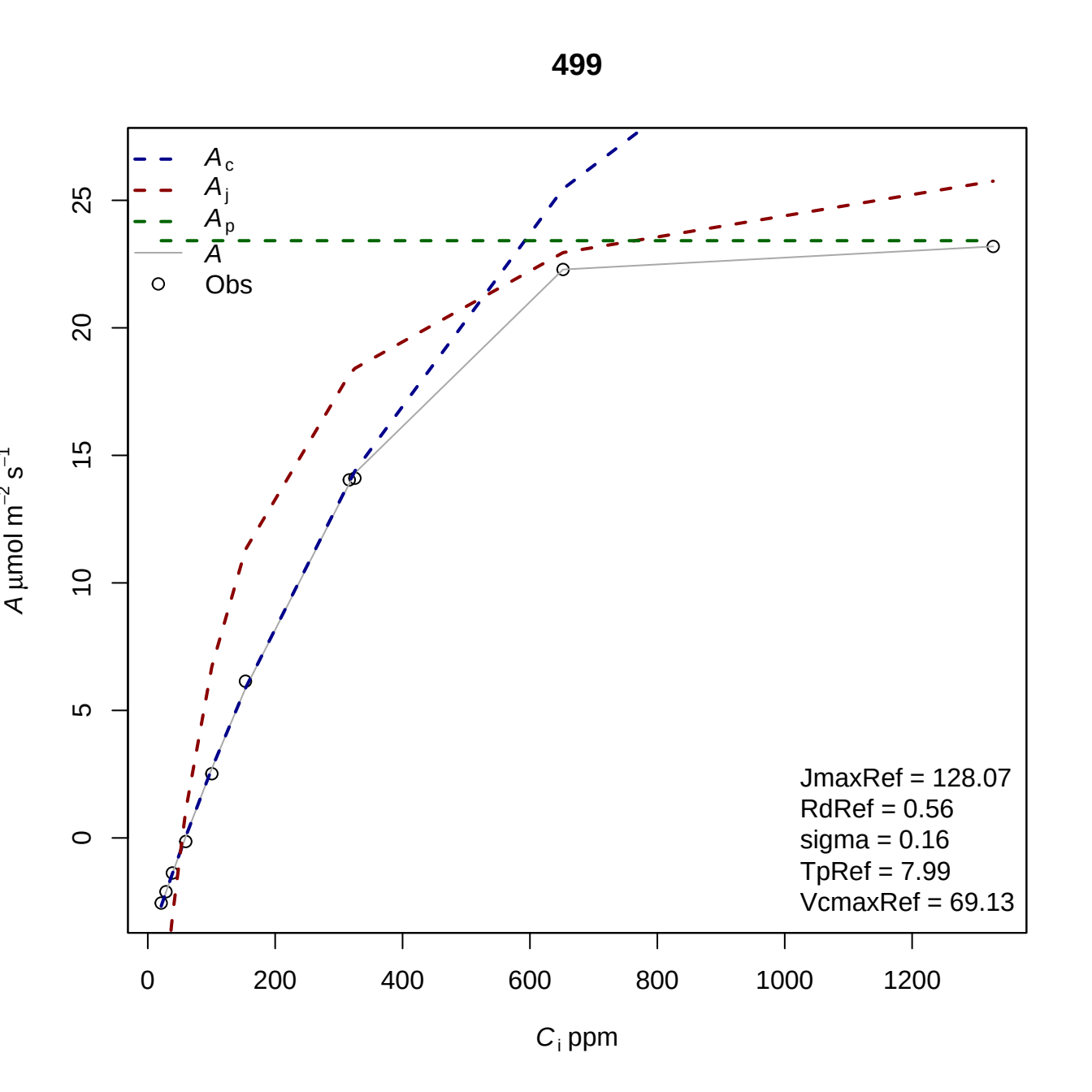

494
A_c
A_j
A_p
A
Obs
JmaxRef = 82.36
RdRef = 0.53
sigma = 0.1
TpRef = 5.59
VcmaxRef = 48.49
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
200
400
600
800
1000
1200

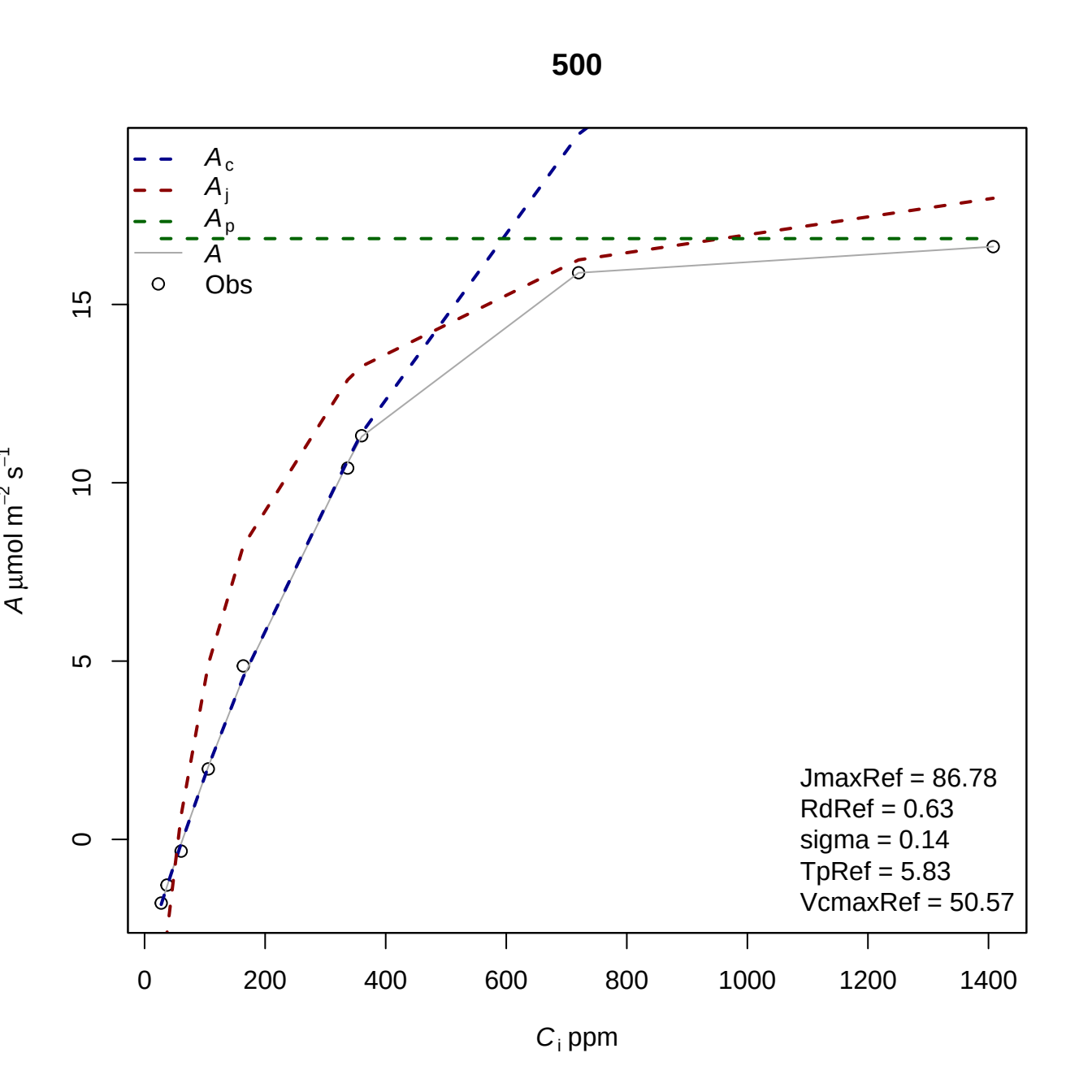

495
A_c
A_j
A_p
A
Obs
JmaxRef = 95.7
RdRef = 0.98
sigma = 0.14
TpRef = 6.96
VcmaxRef = 54.17
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000

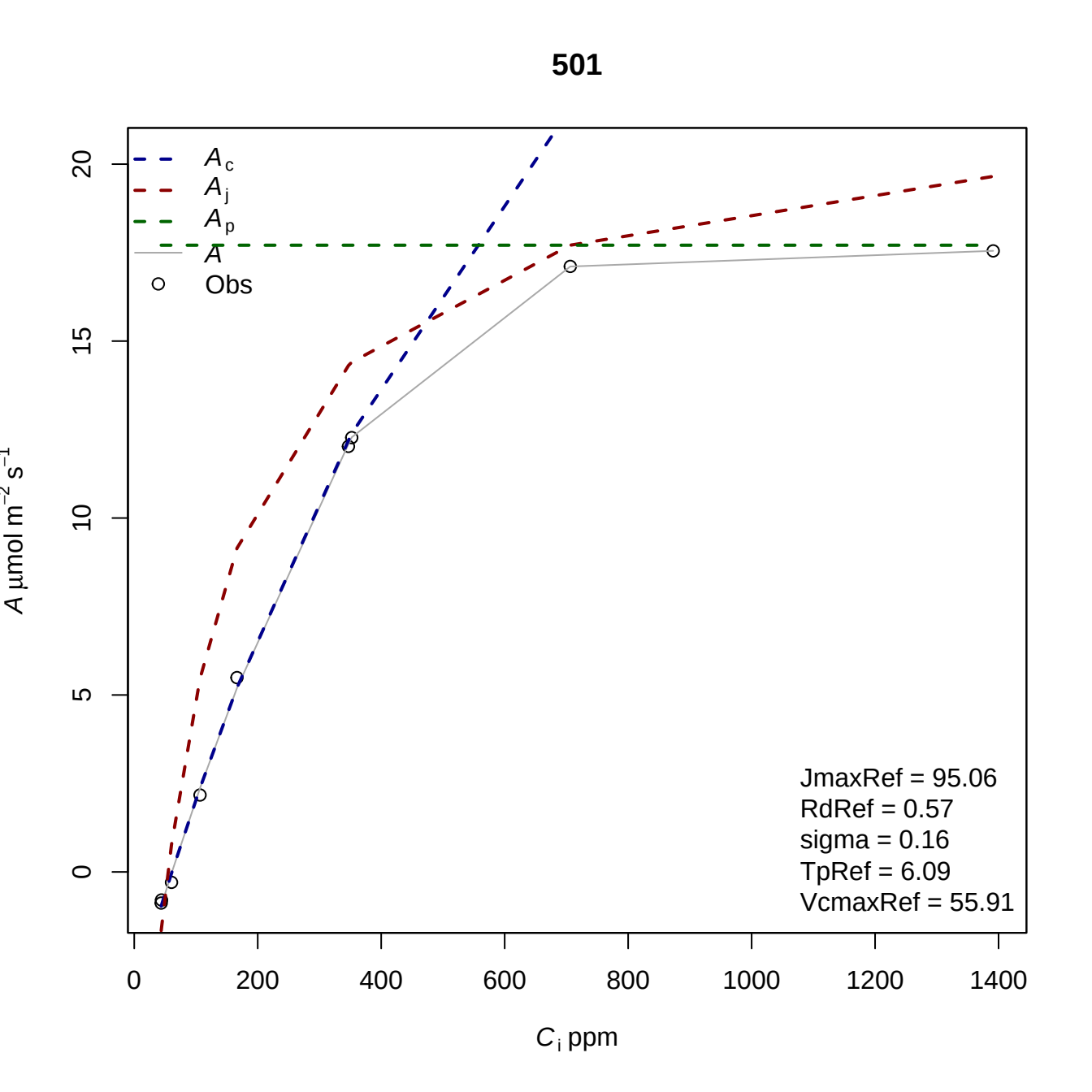

496
A_c
A_j
A_p
A
Obs
JmaxRef = 127.7
RdRef = 0.83
sigma = 0.26
TpRef = 7.6
VcmaxRef = 61.27
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200

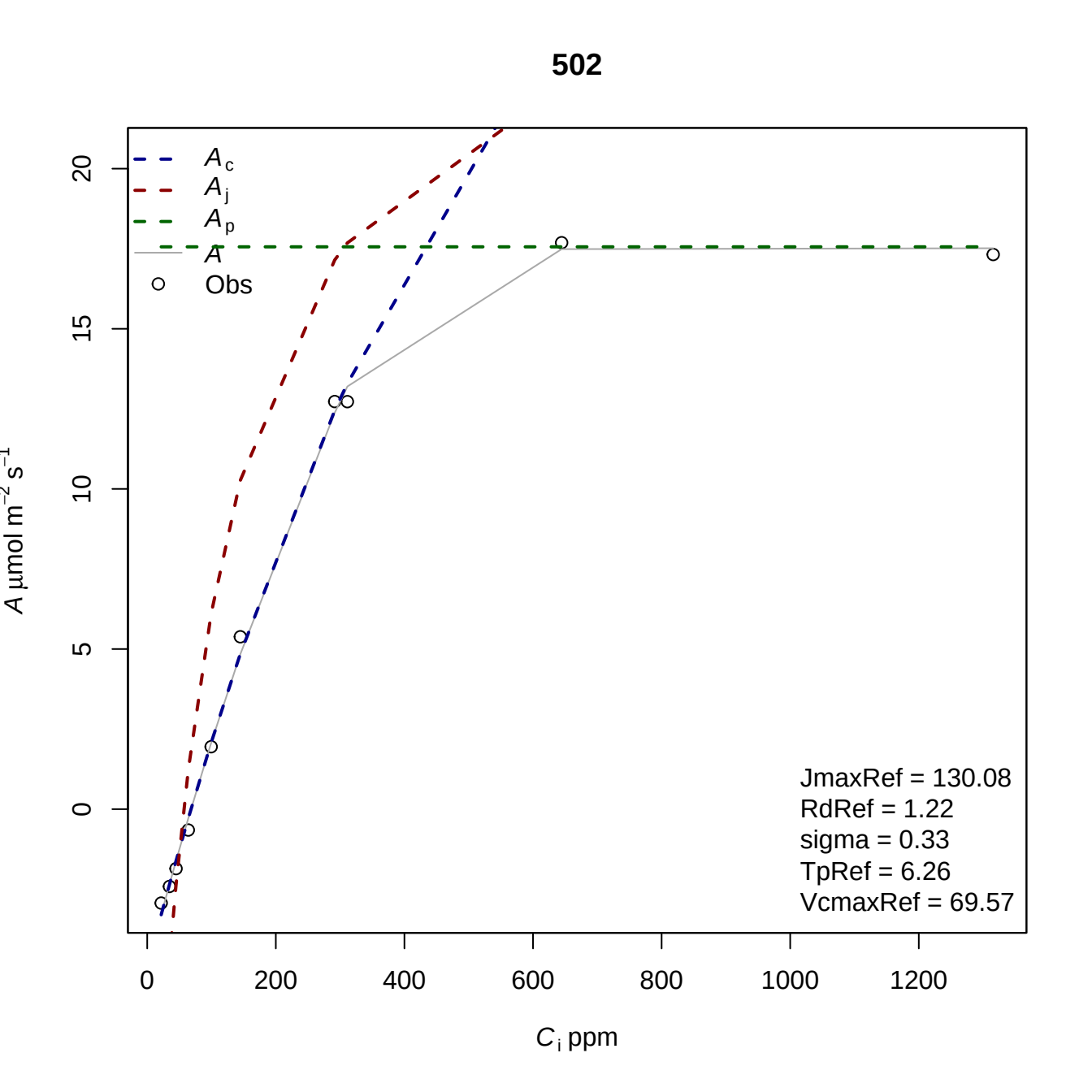

497
A_c
A_j
A_p
A
Obs
JmaxRef = 20409.4
RdRef = 1.02
sigma = 0.62
TpRef = 7.29
VcmaxRef = 59.06
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

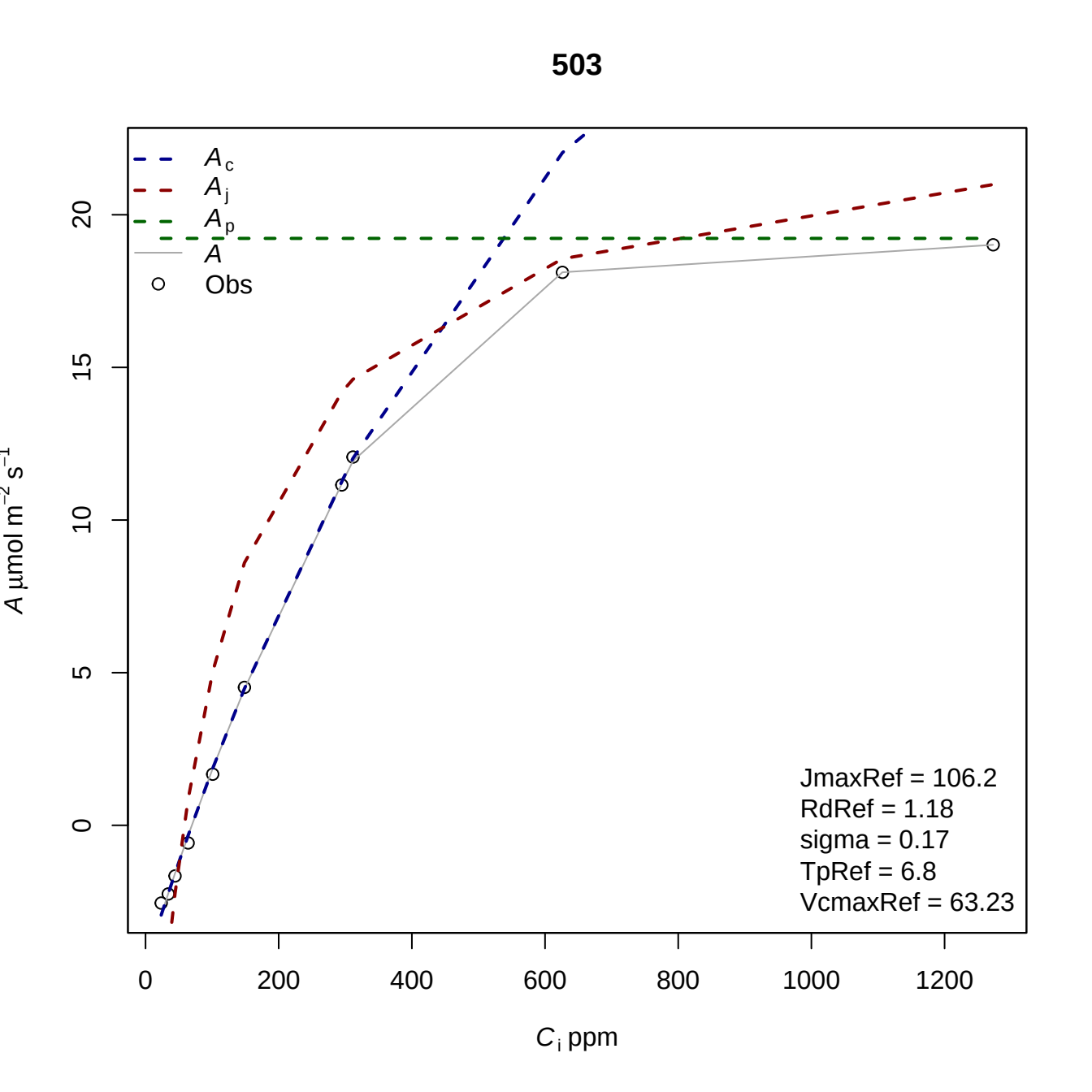

498
A_c
A_j
A_p
A
Obs
JmaxRef = 93.42
RdRef = 0.55
sigma = 0.1
TpRef = 5.82
VcmaxRef = 54.78
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000
1200
1400

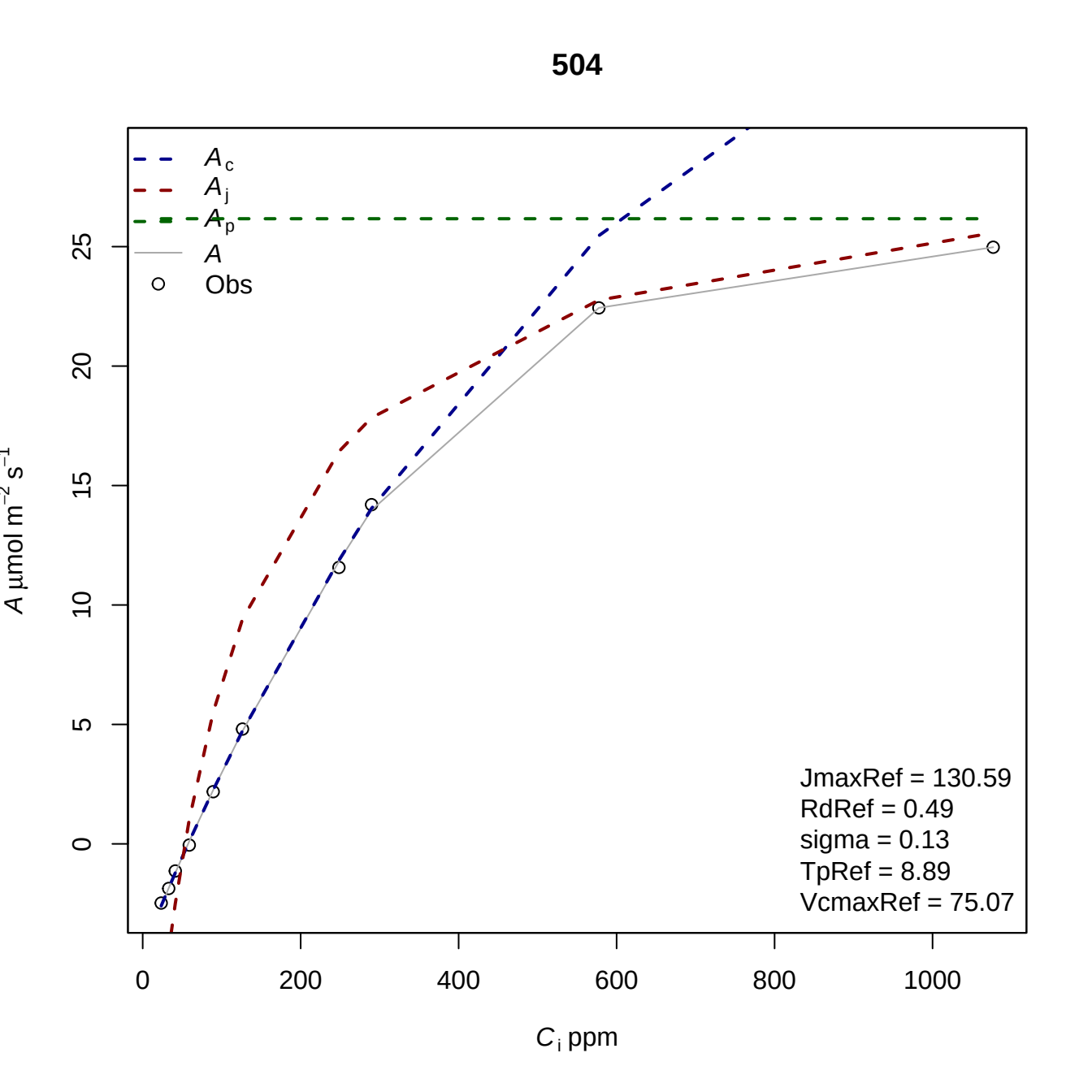

499
A_c
A_j
A_p
A
Obs
JmaxRef = 128.07
RdRef = 0.56
sigma = 0.16
TpRef = 7.99
VcmaxRef = 69.13
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200

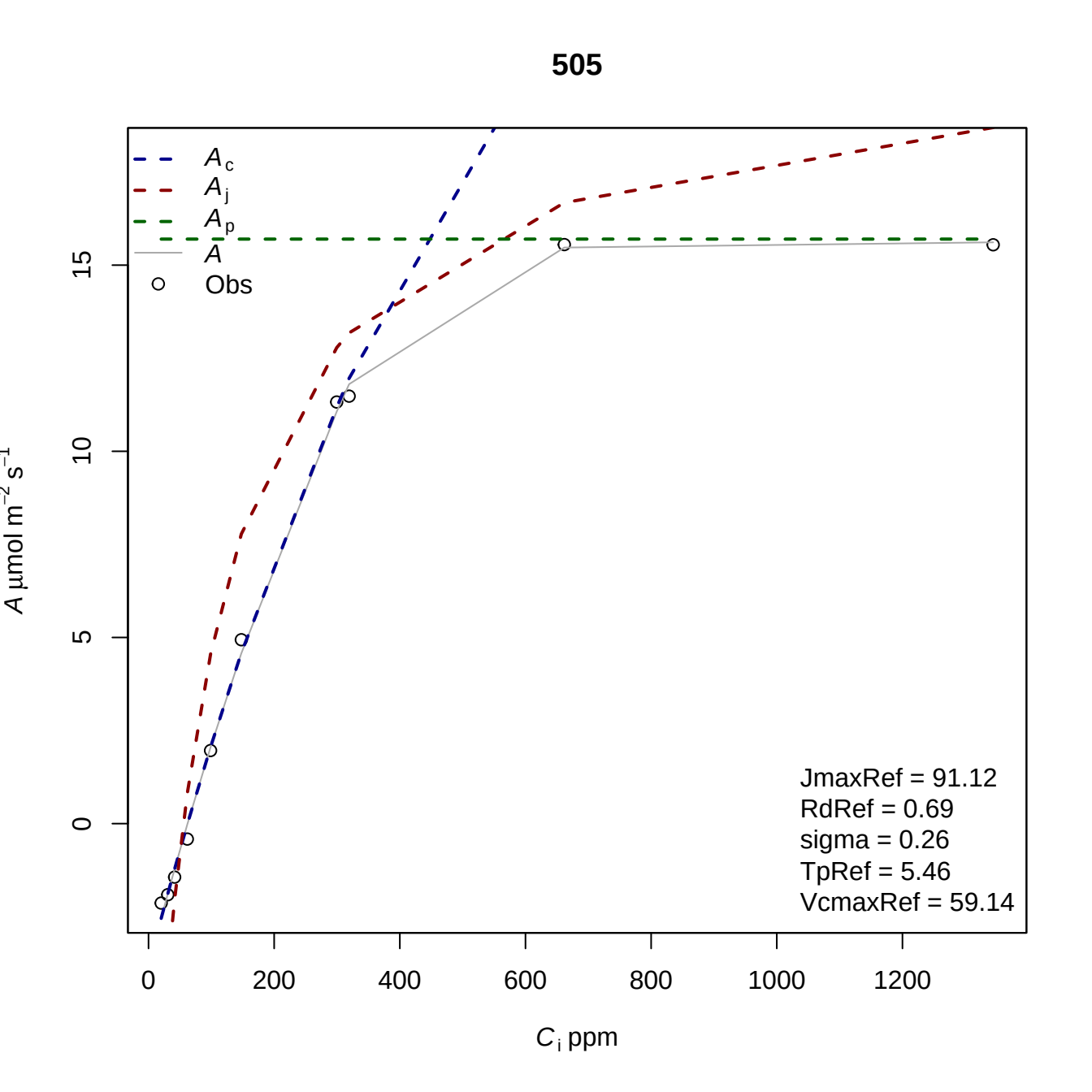

500
A_c
A_j
A_p
A
Obs
JmaxRef = 86.78
RdRef = 0.63
sigma = 0.14
TpRef = 5.83
VcmaxRef = 50.57
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
200
400
600
800
1000
1200
1400

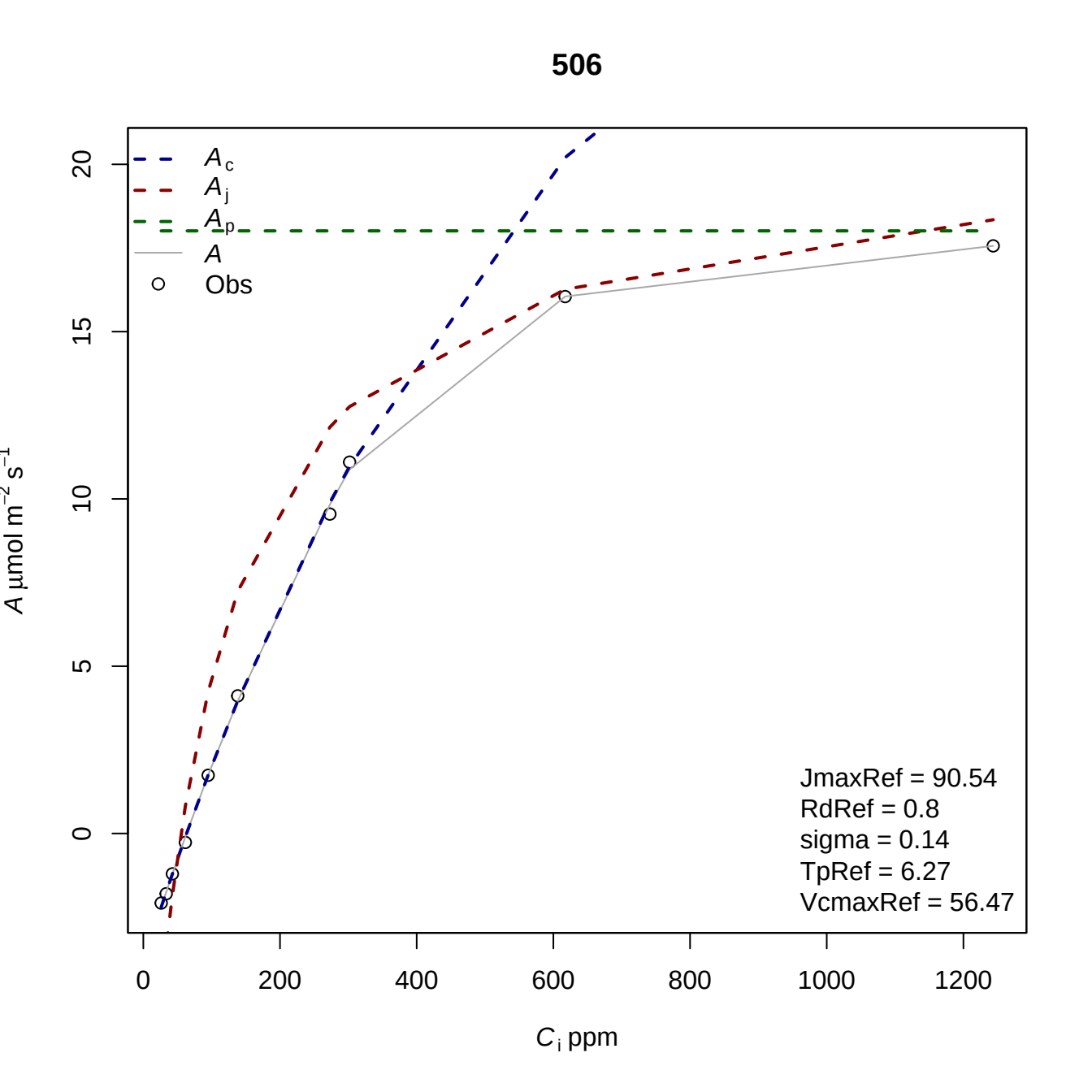

501
A_c
A_j
A_p
A
Obs
JmaxRef = 95.06
RdRef = 0.57
sigma = 0.16
TpRef = 6.09
VcmaxRef = 55.91
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000
1200
1400

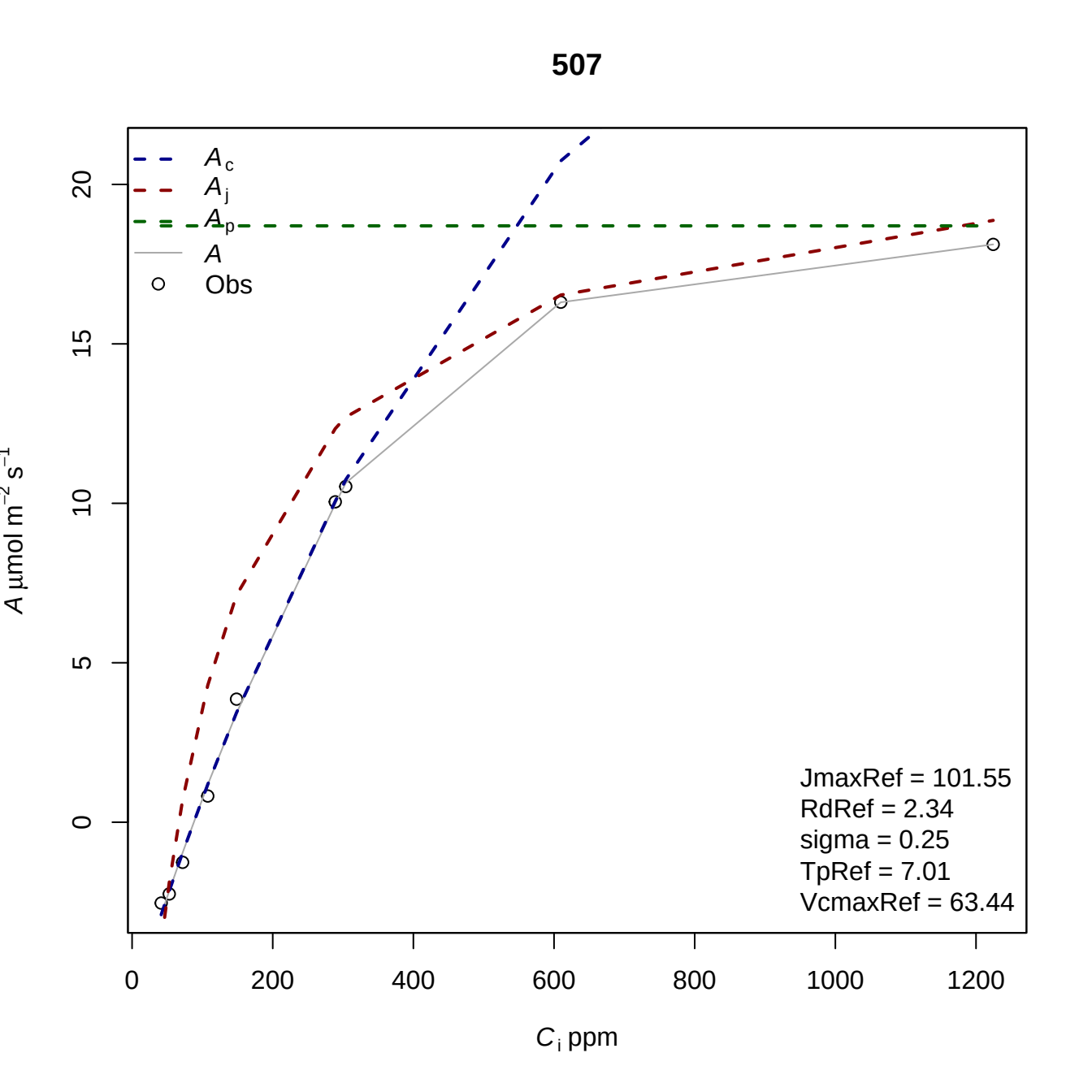

502
A_c
A_j
A_p
A
Obs
JmaxRef = 130.08
RdRef = 1.22
sigma = 0.33
TpRef = 6.26
VcmaxRef = 69.57
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

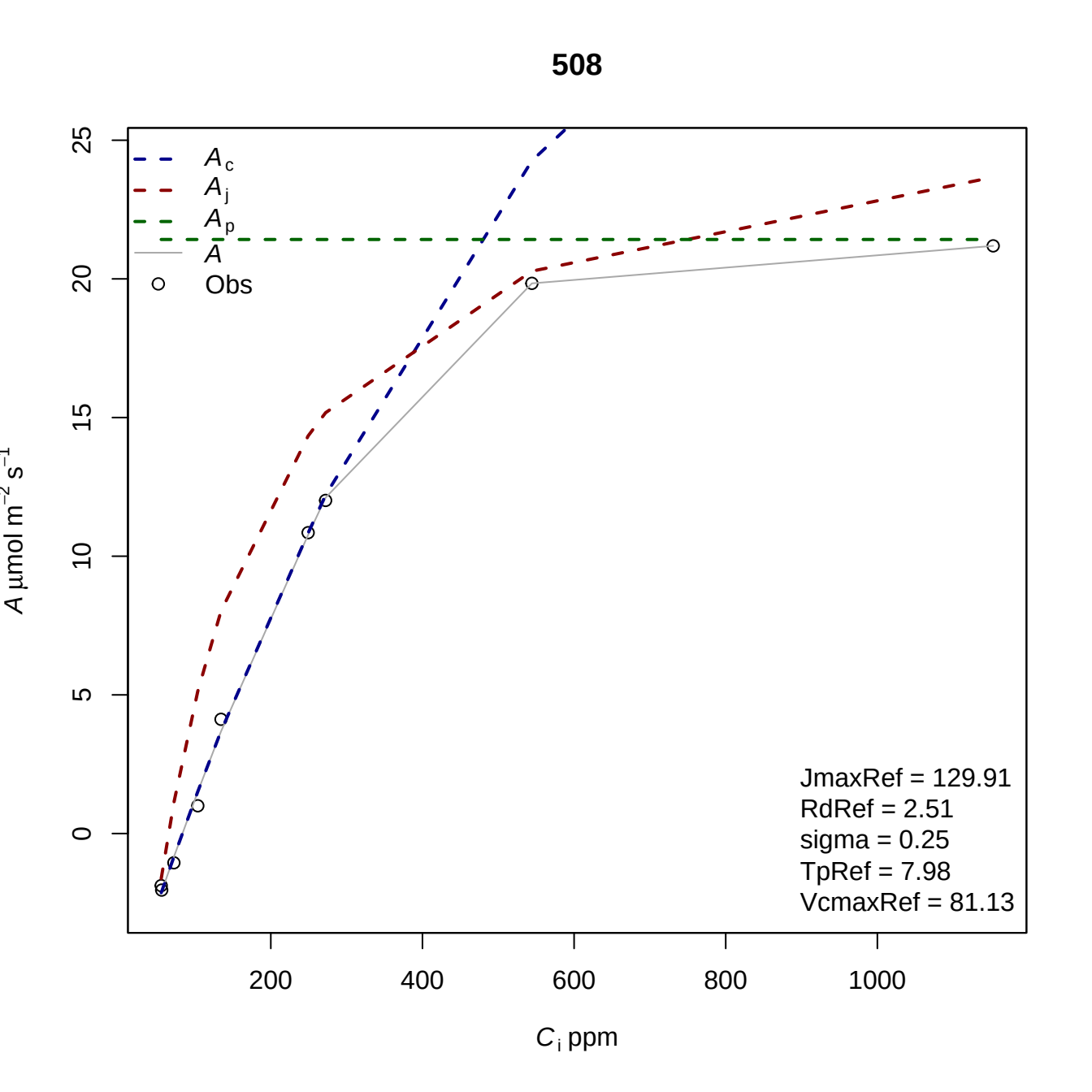

503
A_c
A_j
A_p
A
Obs
JmaxRef = 106.2
RdRef = 1.18
sigma = 0.17
TpRef = 6.8
VcmaxRef = 63.23
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

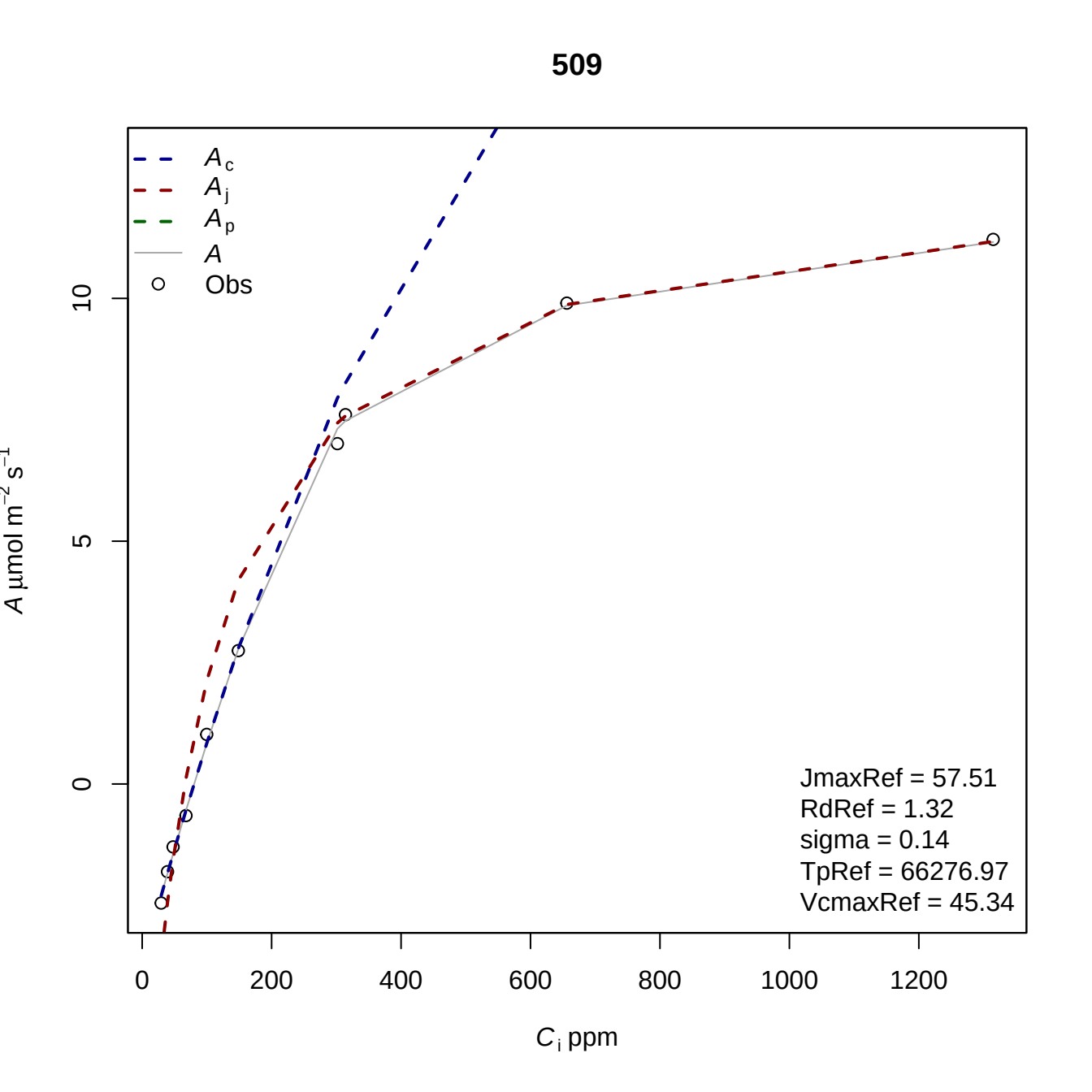

504
A_c
A_j
A_p
A
Obs
JmaxRef = 130.59
RdRef = 0.49
sigma = 0.13
TpRef = 8.89
VcmaxRef = 75.07
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
200
400
600
800
1000

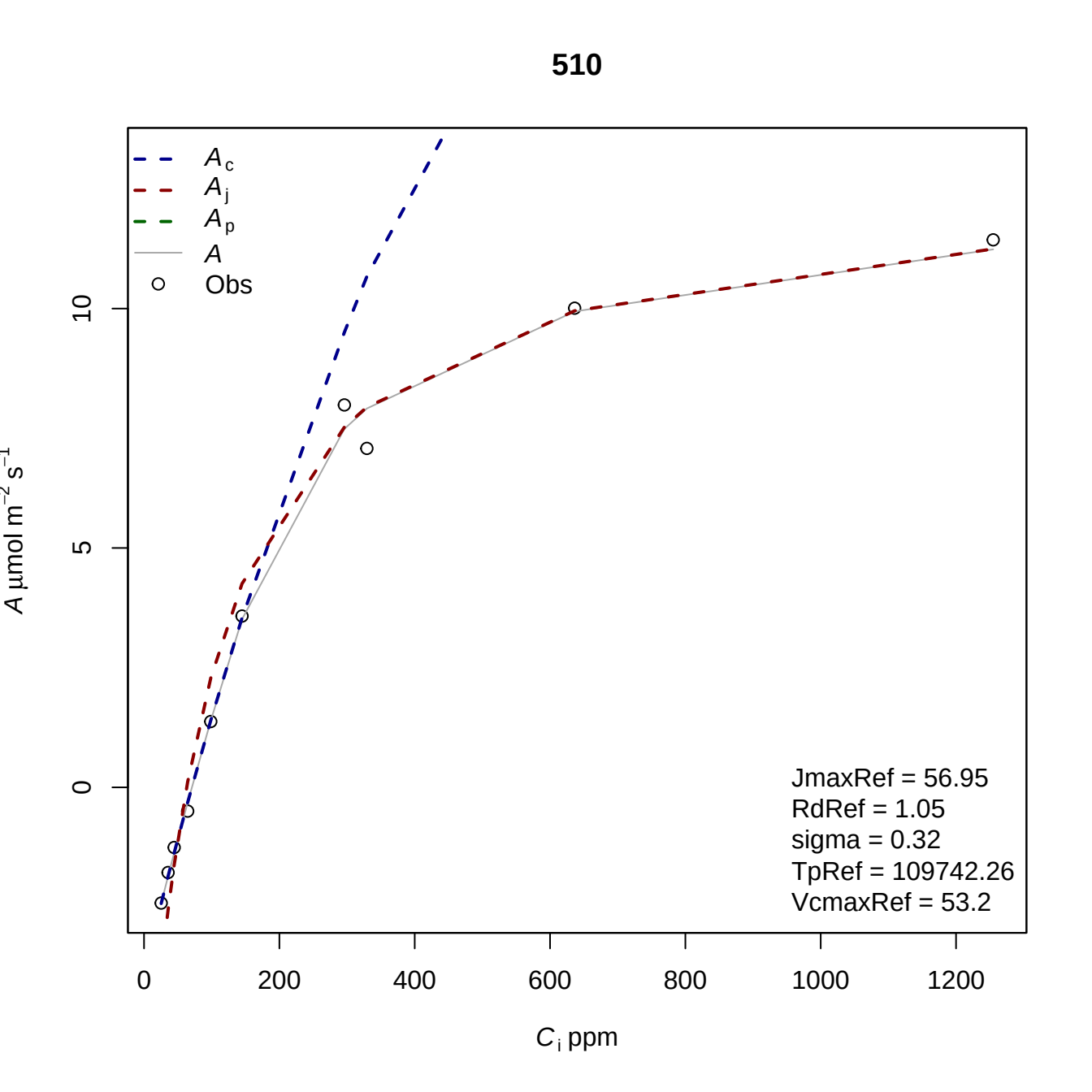

505
A_c
A_j
A_p
A
Obs
JmaxRef = 91.12
RdRef = 0.69
sigma = 0.26
TpRef = 5.46
VcmaxRef = 59.14
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
200
400
600
800
1000
1200

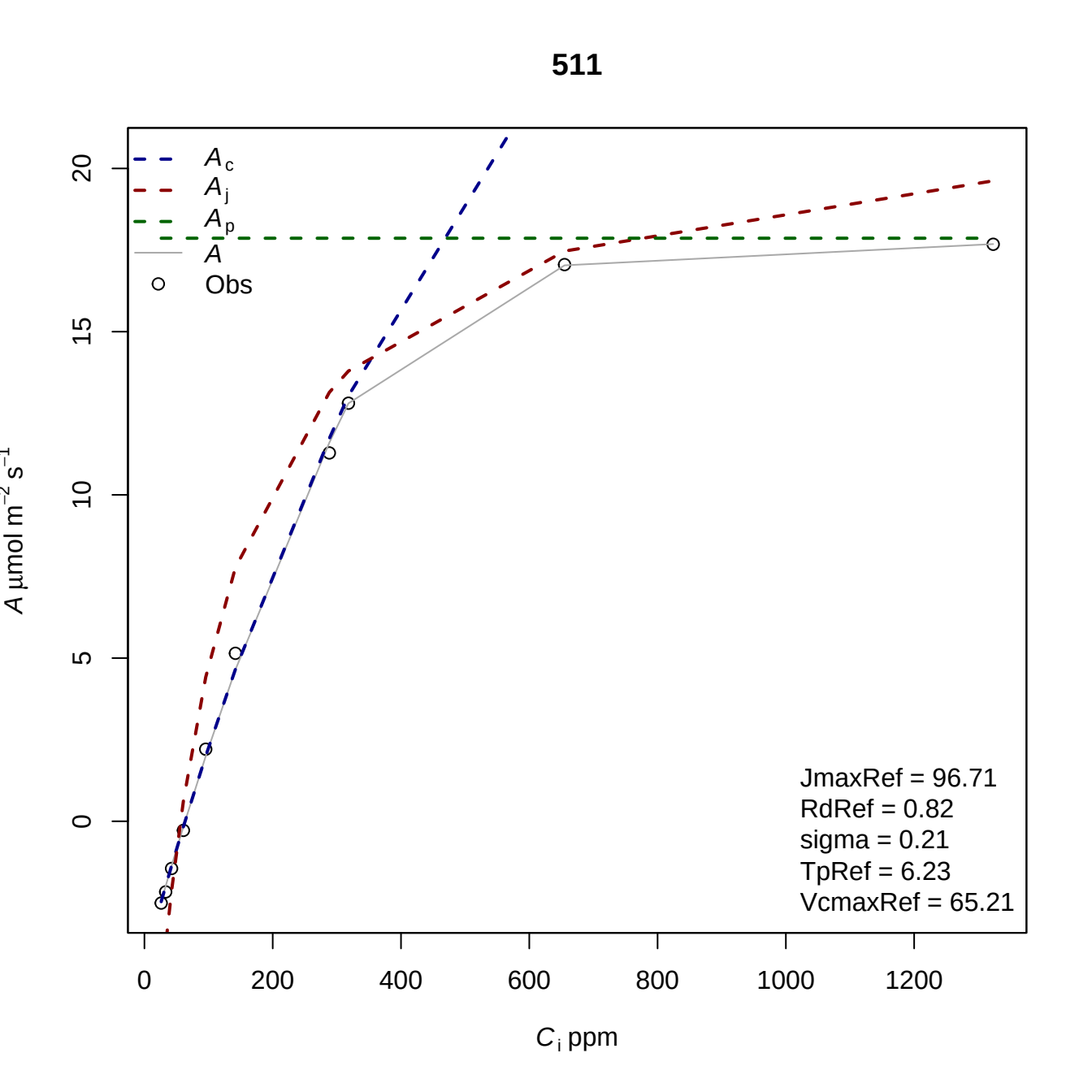

506
A_c
A_j
A_p
A
Obs
JmaxRef = 90.54
RdRef = 0.8
sigma = 0.14
TpRef = 6.27
VcmaxRef = 56.47
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

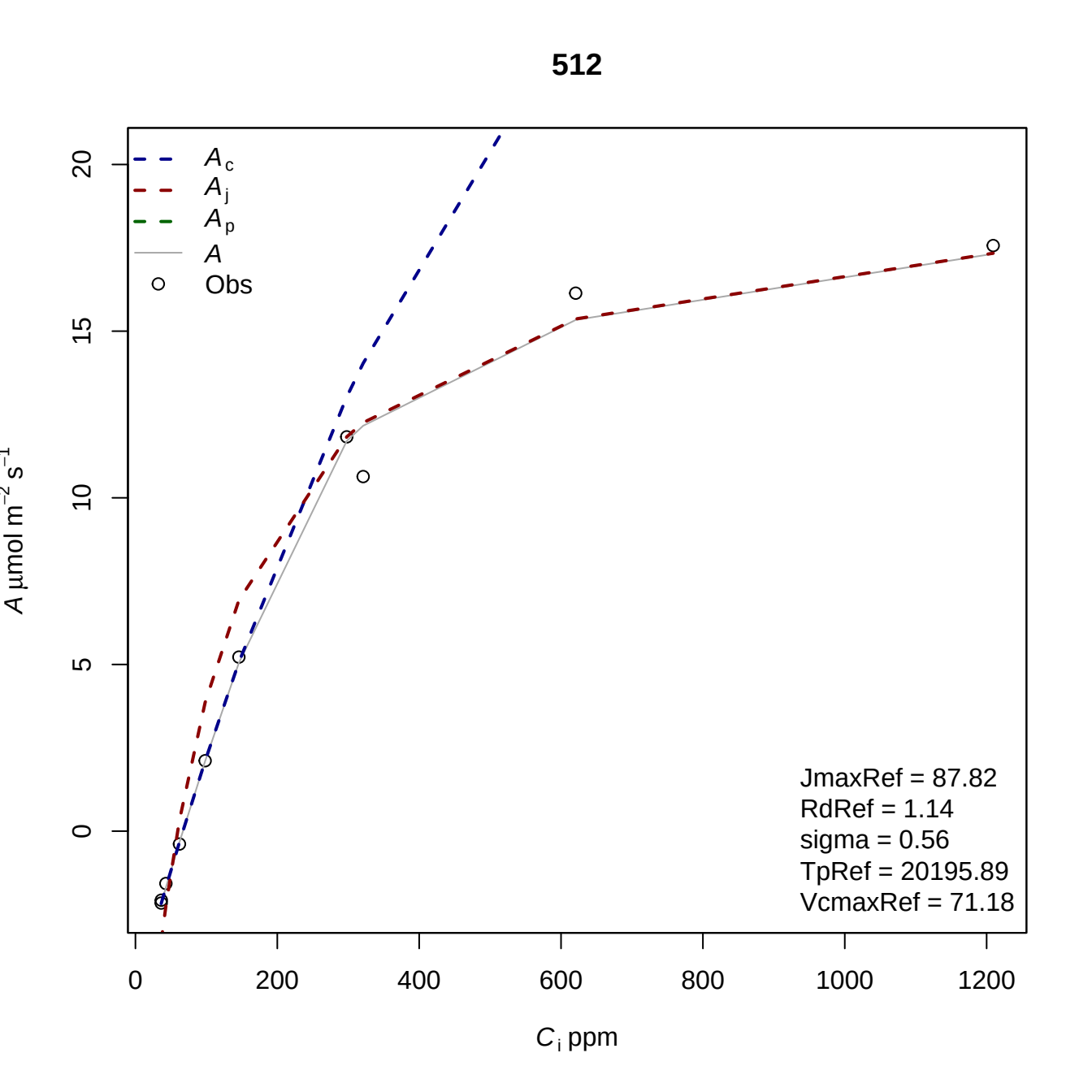

507
A_c
A_j
A_p
A
Obs
JmaxRef = 101.55
RdRef = 2.34
sigma = 0.25
TpRef = 7.01
VcmaxRef = 63.44
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

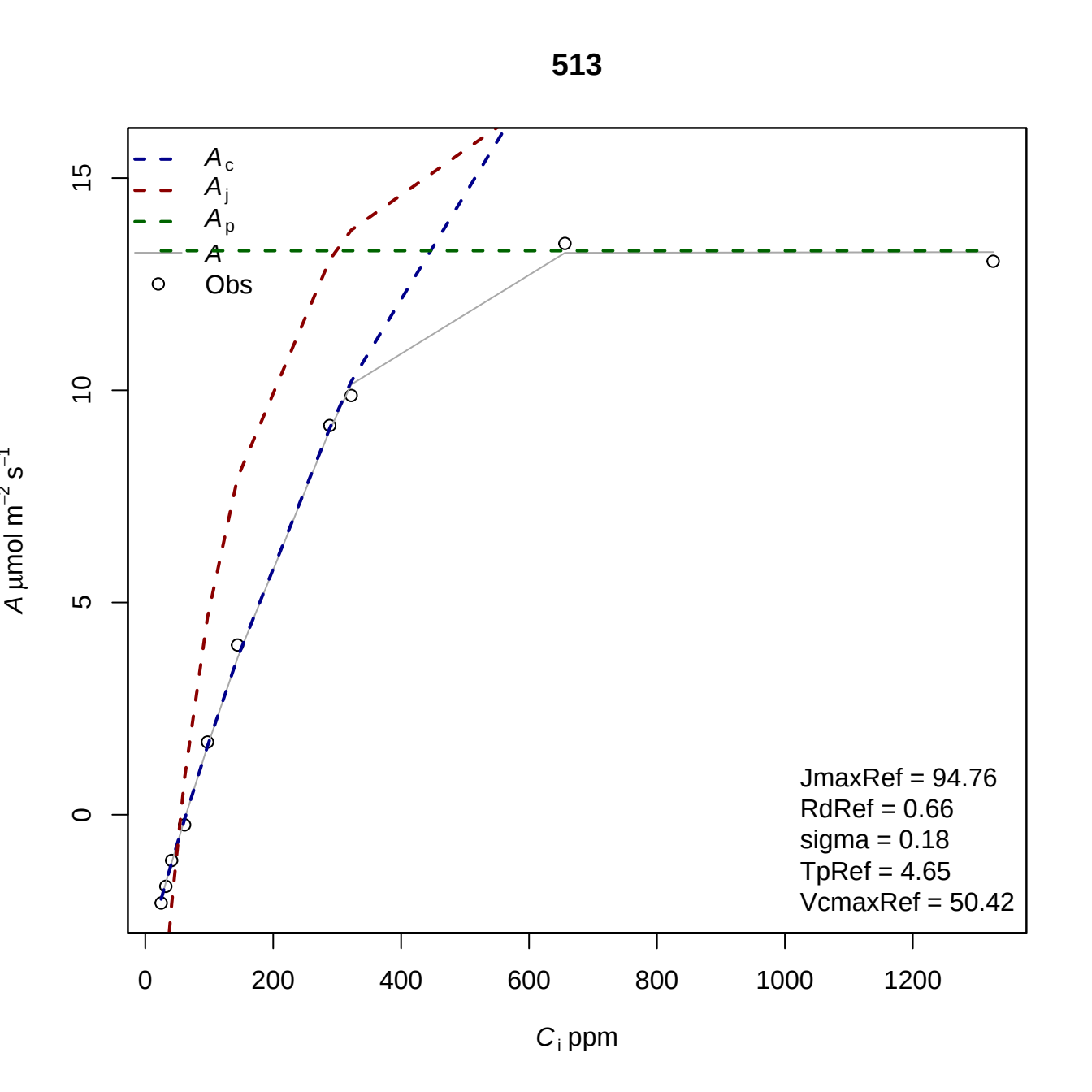

508
A_c
A_j
A_p
A
Obs
JmaxRef = 129.91
RdRef = 2.51
sigma = 0.25
TpRef = 7.98
VcmaxRef = 81.13
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
200
400
600
800
1000

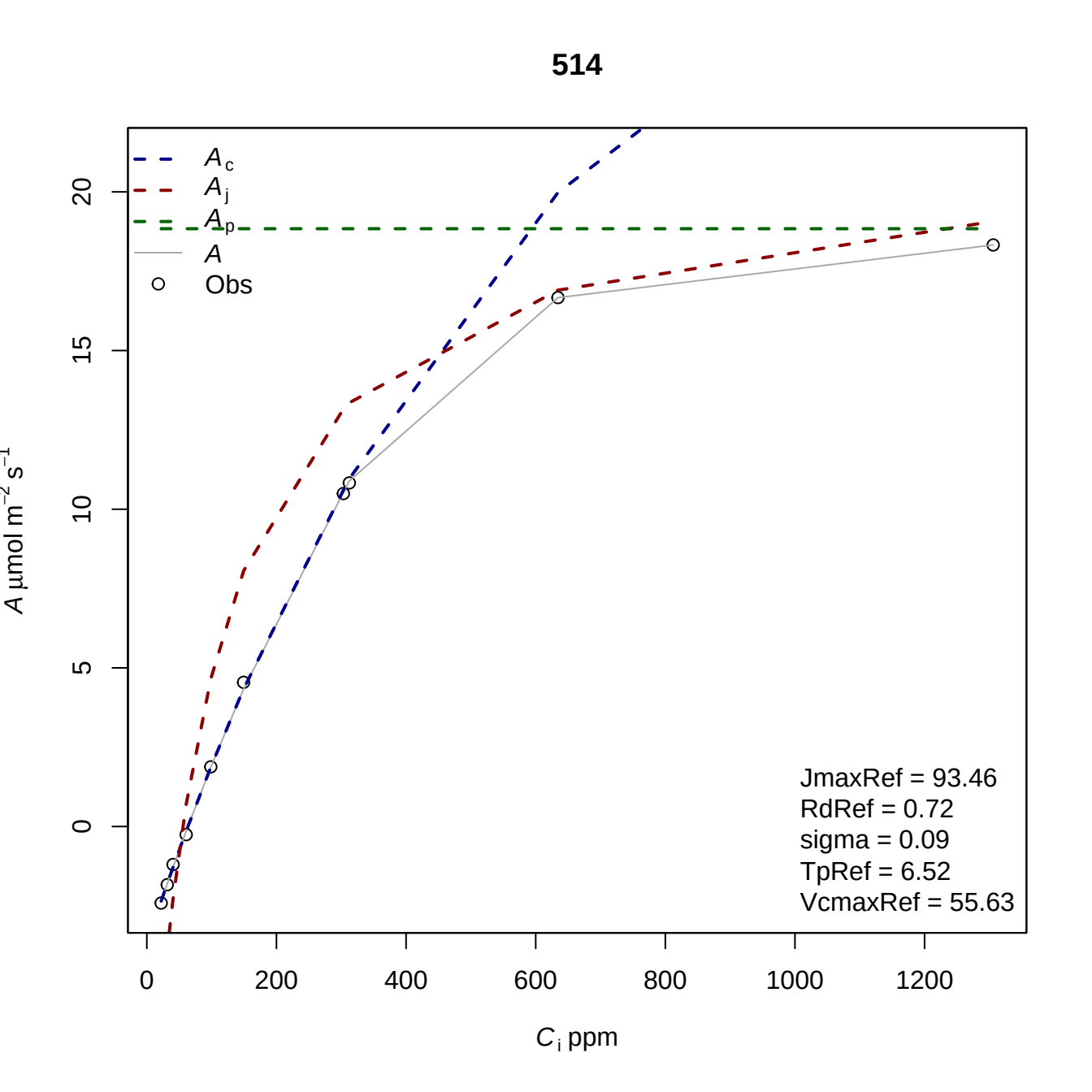

509
A_c
A_j
A_p
A
Obs
JmaxRef = 57.51
RdRef = 1.32
sigma = 0.14
TpRef = 66276.97
VcmaxRef = 45.34
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
0
200
400
600
800
1000
1200

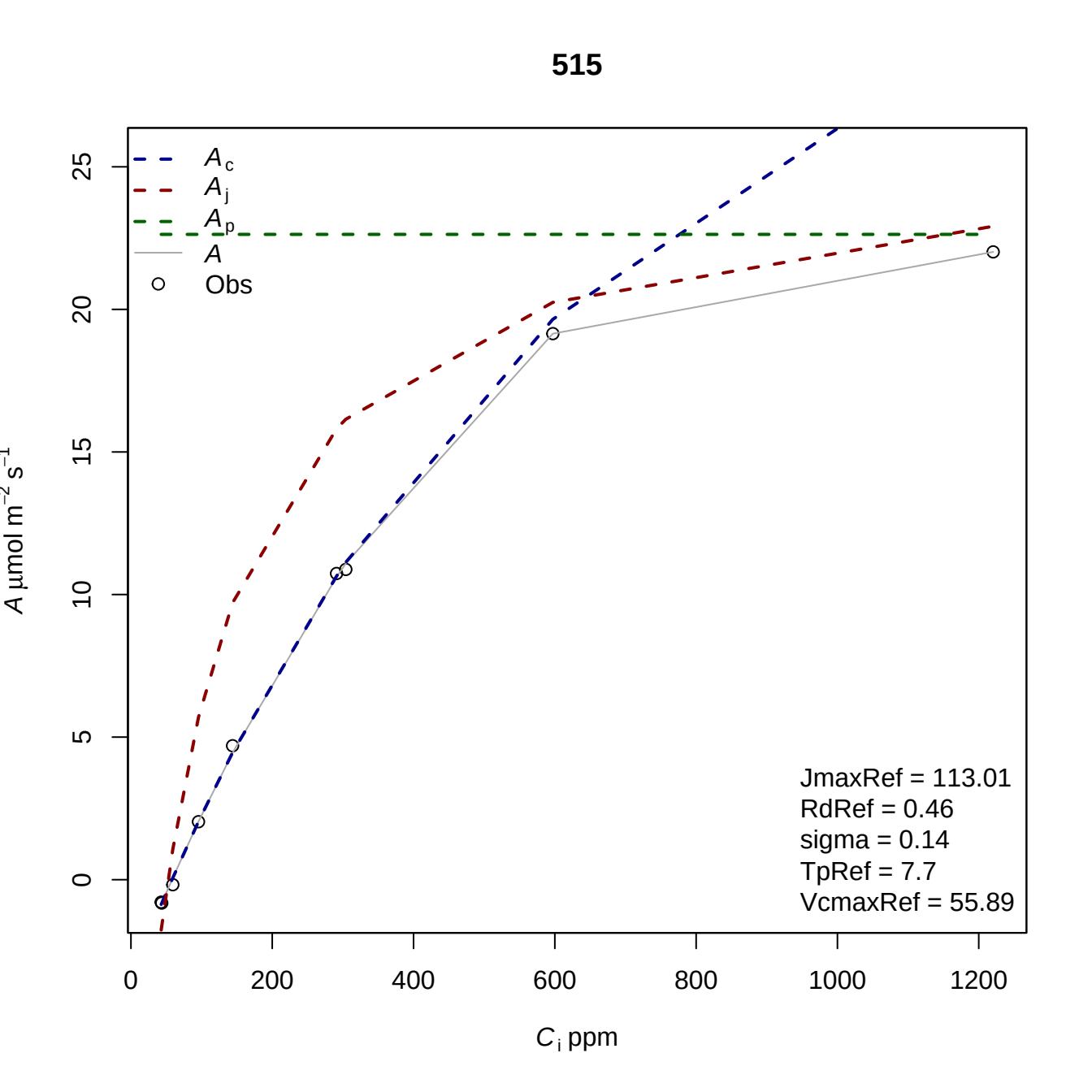

510
A_c
A_j
A_p
A
Obs
JmaxRef = 56.95
RdRef = 1.05
sigma = 0.32
TpRef = 109742.26
VcmaxRef = 53.2
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
0
200
400
600
800
1000
1200

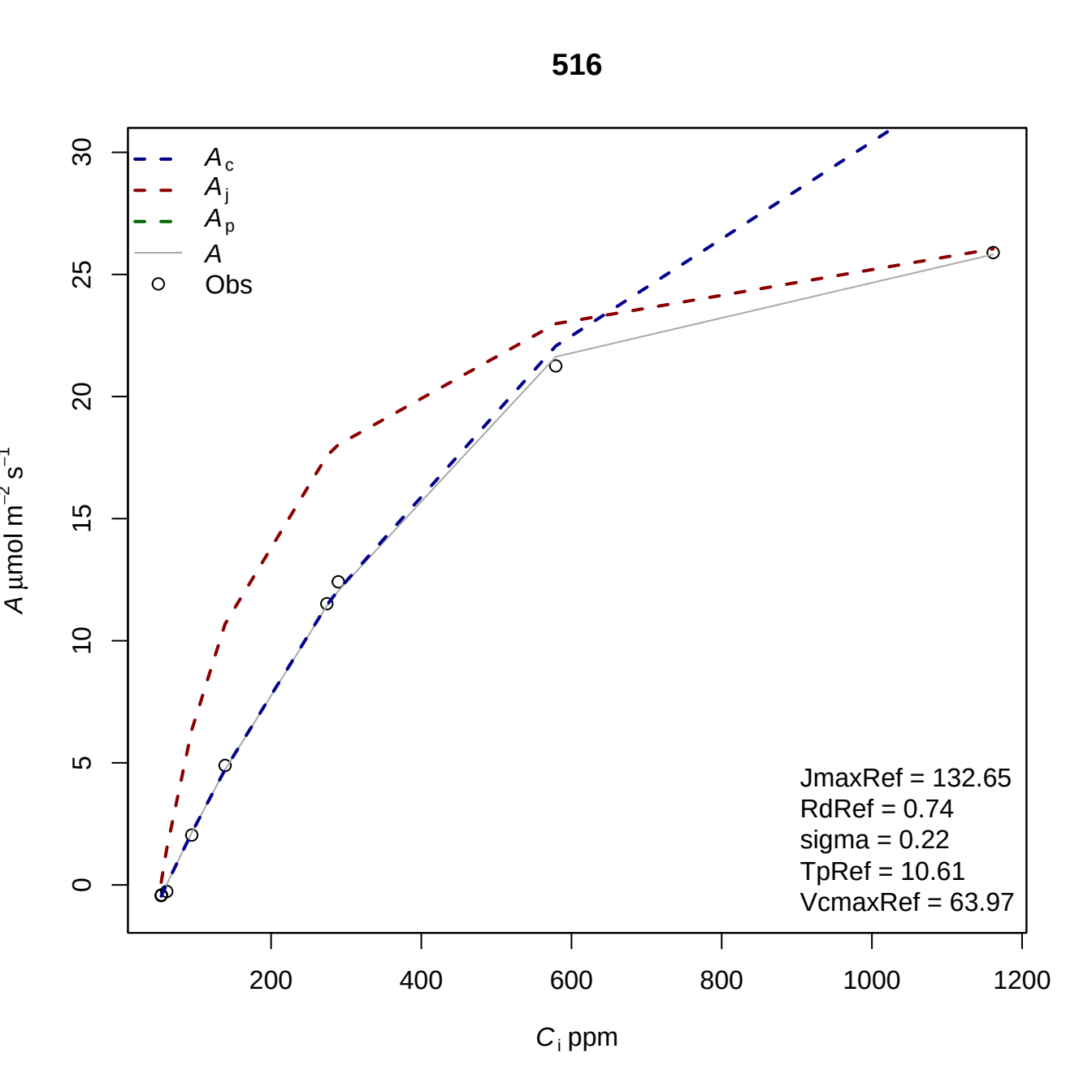

511
A_c
A_j
A_p
A
Obs
JmaxRef = 96.71
RdRef = 0.82
sigma = 0.21
TpRef = 6.23
VcmaxRef = 65.21
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

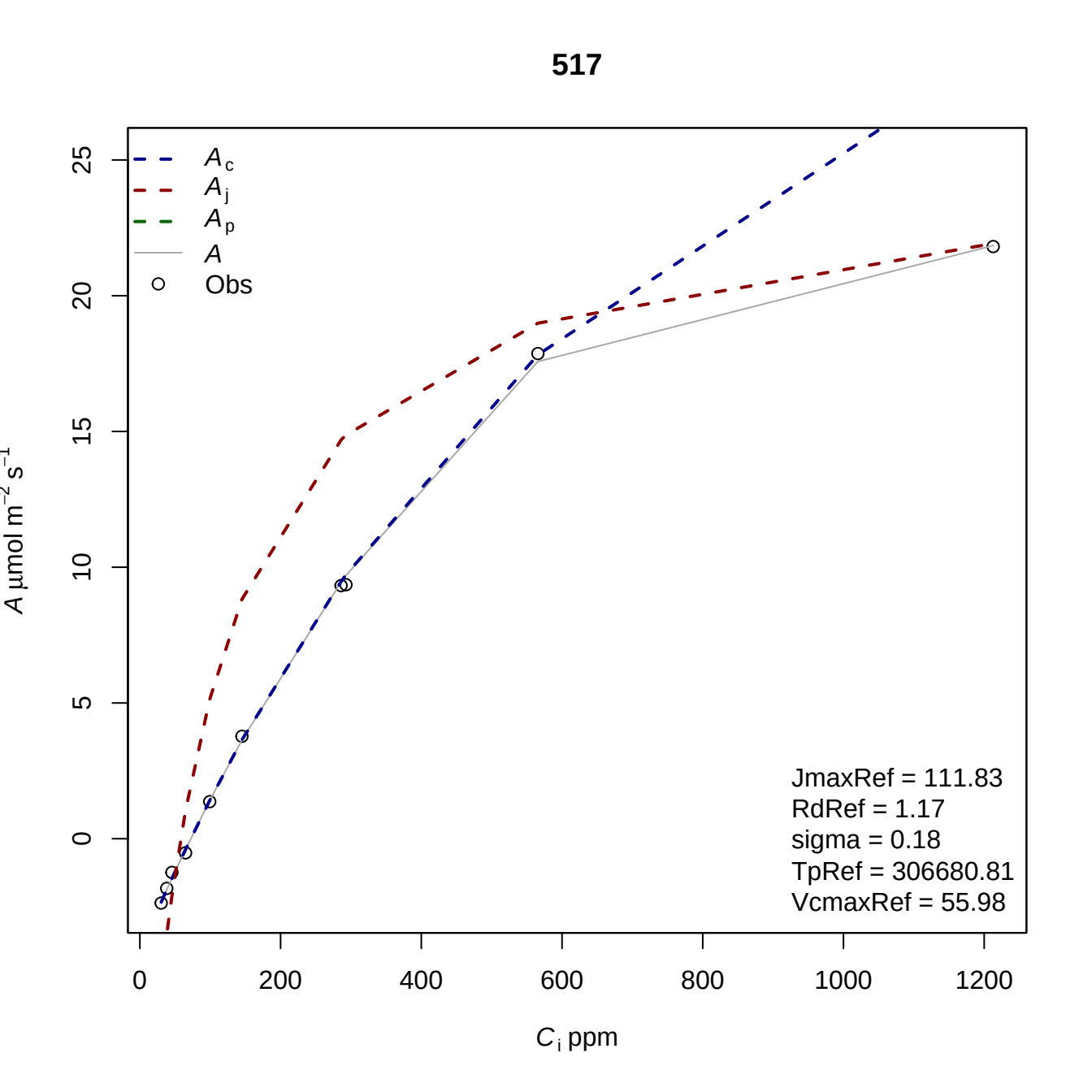

512
A_c
A_j
A_p
A
Obs
JmaxRef = 87.82
RdRef = 1.14
sigma = 0.56
TpRef = 20195.89
VcmaxRef = 71.18
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

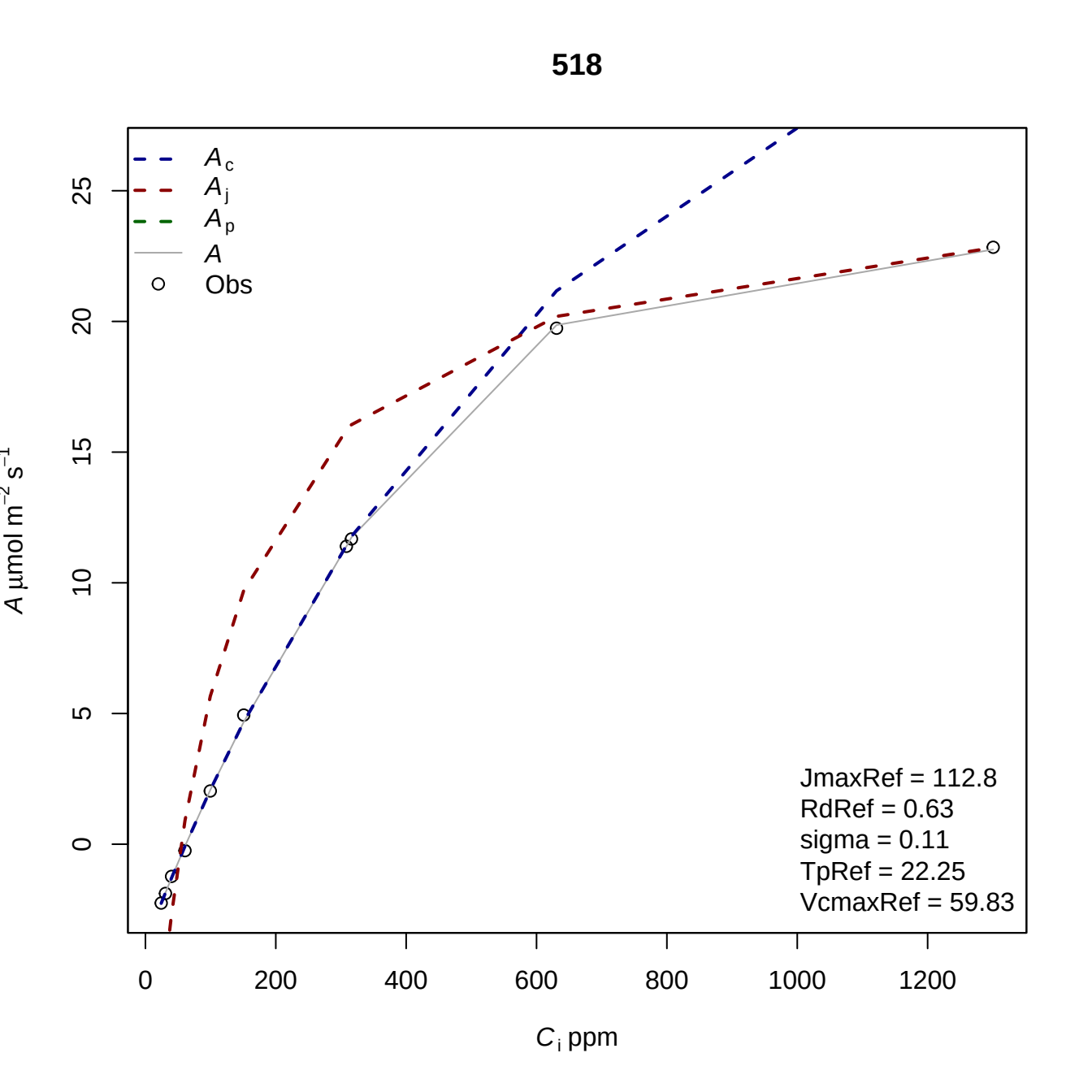

513
A_c
A_j
A_p
A
Obs
JmaxRef = 94.76
RdRef = 0.66
sigma = 0.18
TpRef = 4.65
VcmaxRef = 50.42
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
200
400
600
800
1000
1200

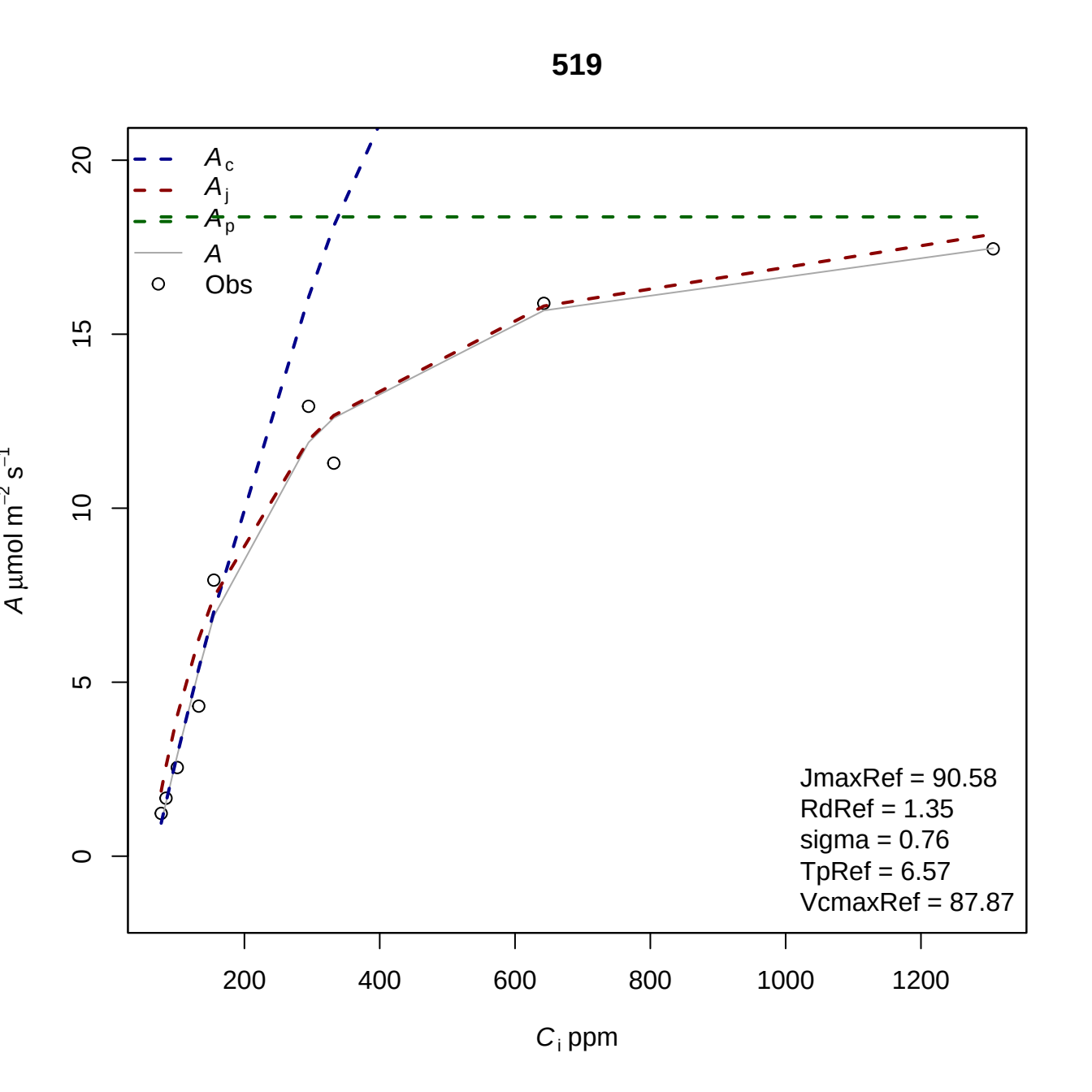

514
A_c
A_j
A_p
A
Obs
JmaxRef = 93.46
RdRef = 0.72
sigma = 0.09
TpRef = 6.52
VcmaxRef = 55.63
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

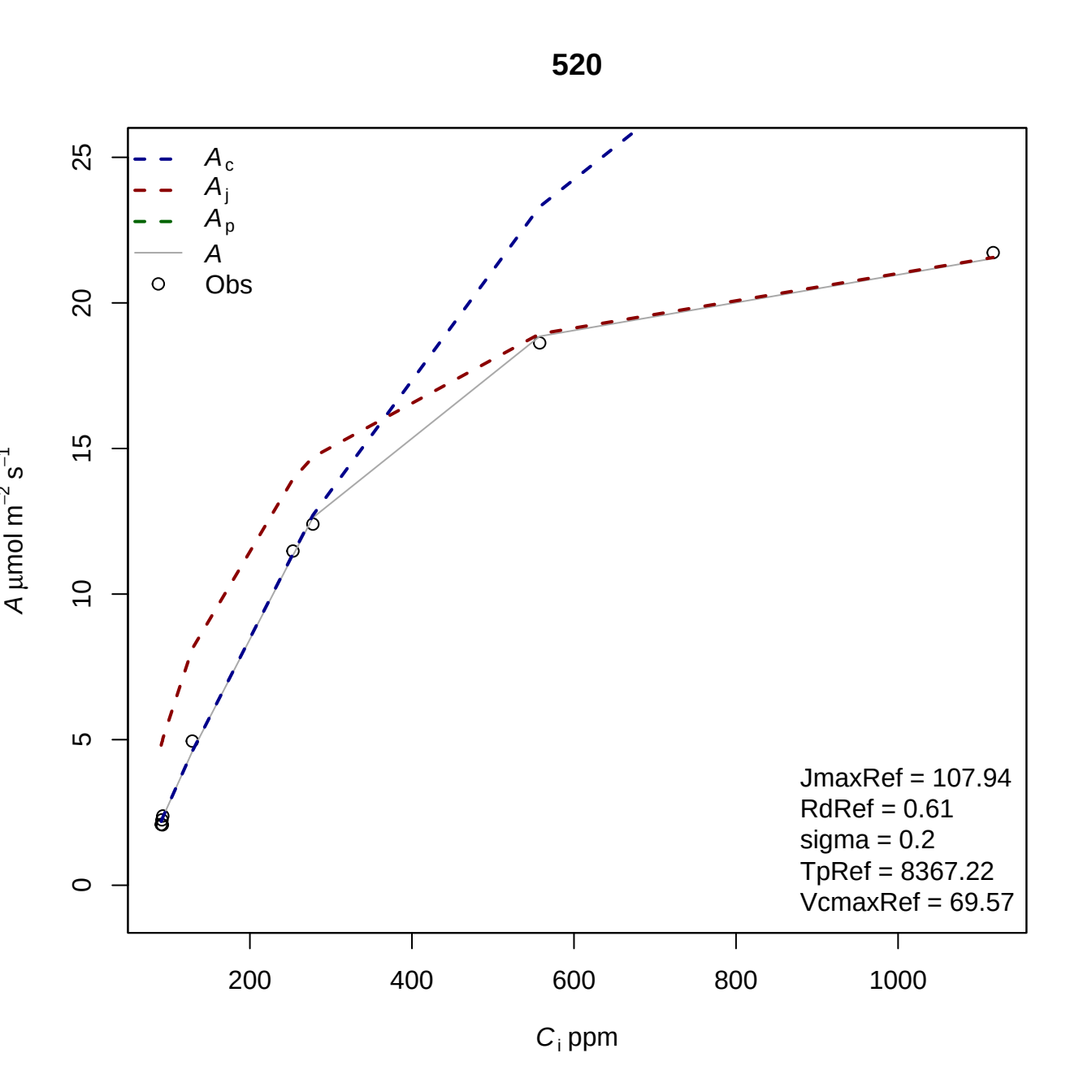

515
A_c
A_j
A_p
A
Obs
JmaxRef = 113.01
RdRef = 0.46
sigma = 0.14
TpRef = 7.7
VcmaxRef = 55.89
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200

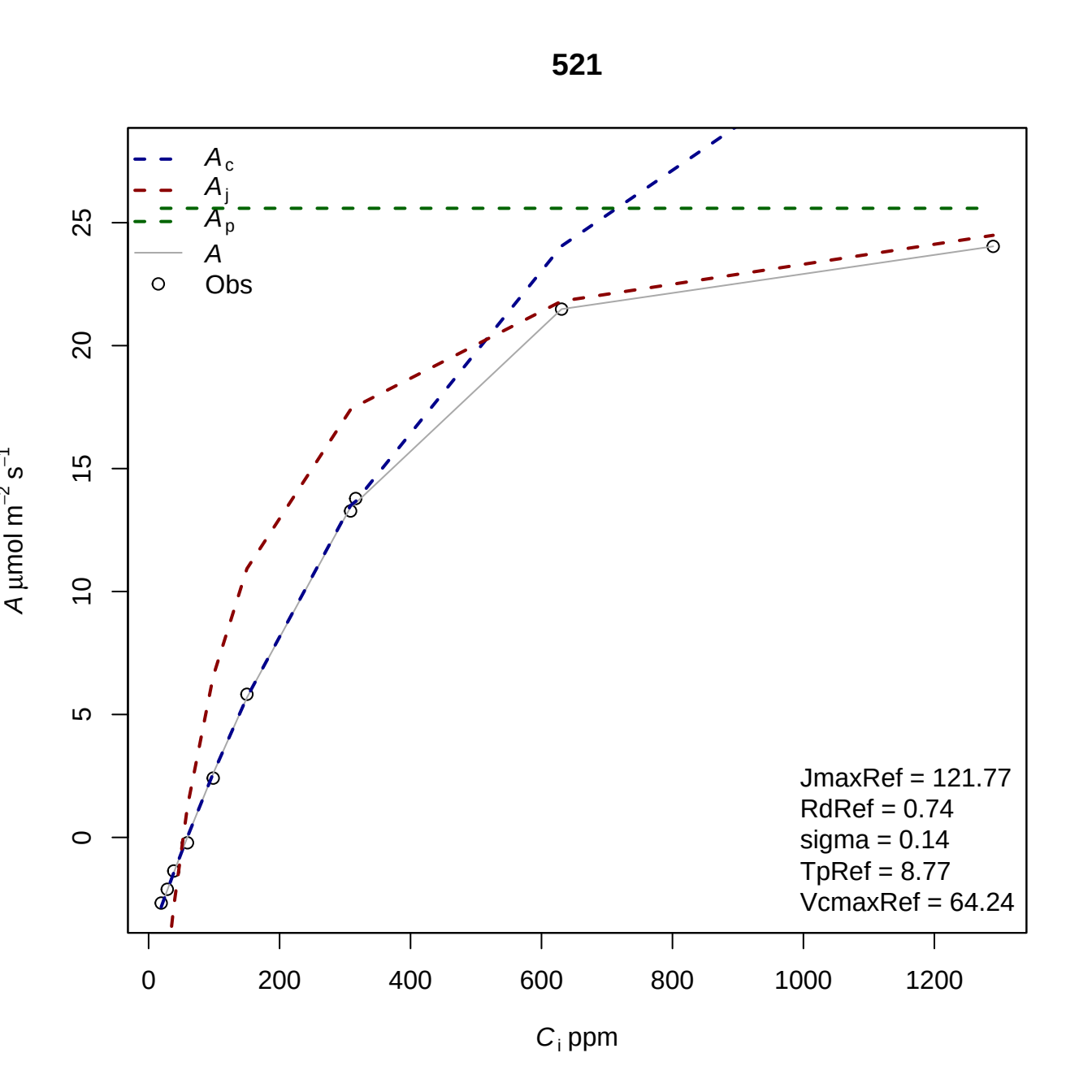

516
A_c
A_j
A_p
A
Obs
JmaxRef = 132.65
RdRef = 0.74
sigma = 0.22
TpRef = 10.61
VcmaxRef = 63.97
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
200
400
600
800
1000
1200

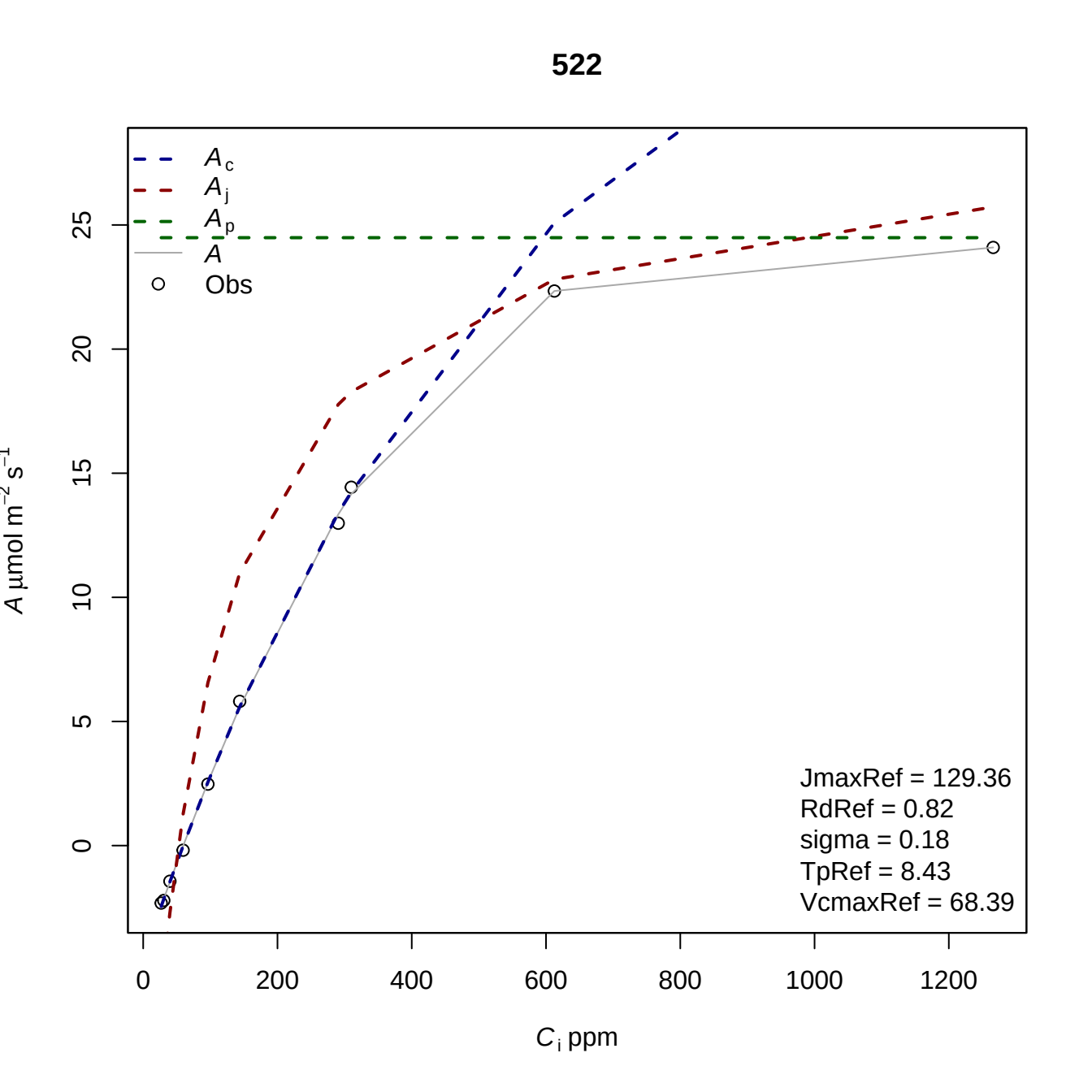

517
A_c
A_j
A_p
A
Obs
JmaxRef = 111.83
RdRef = 1.17
sigma = 0.18
TpRef = 306680.81
VcmaxRef = 55.98
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200

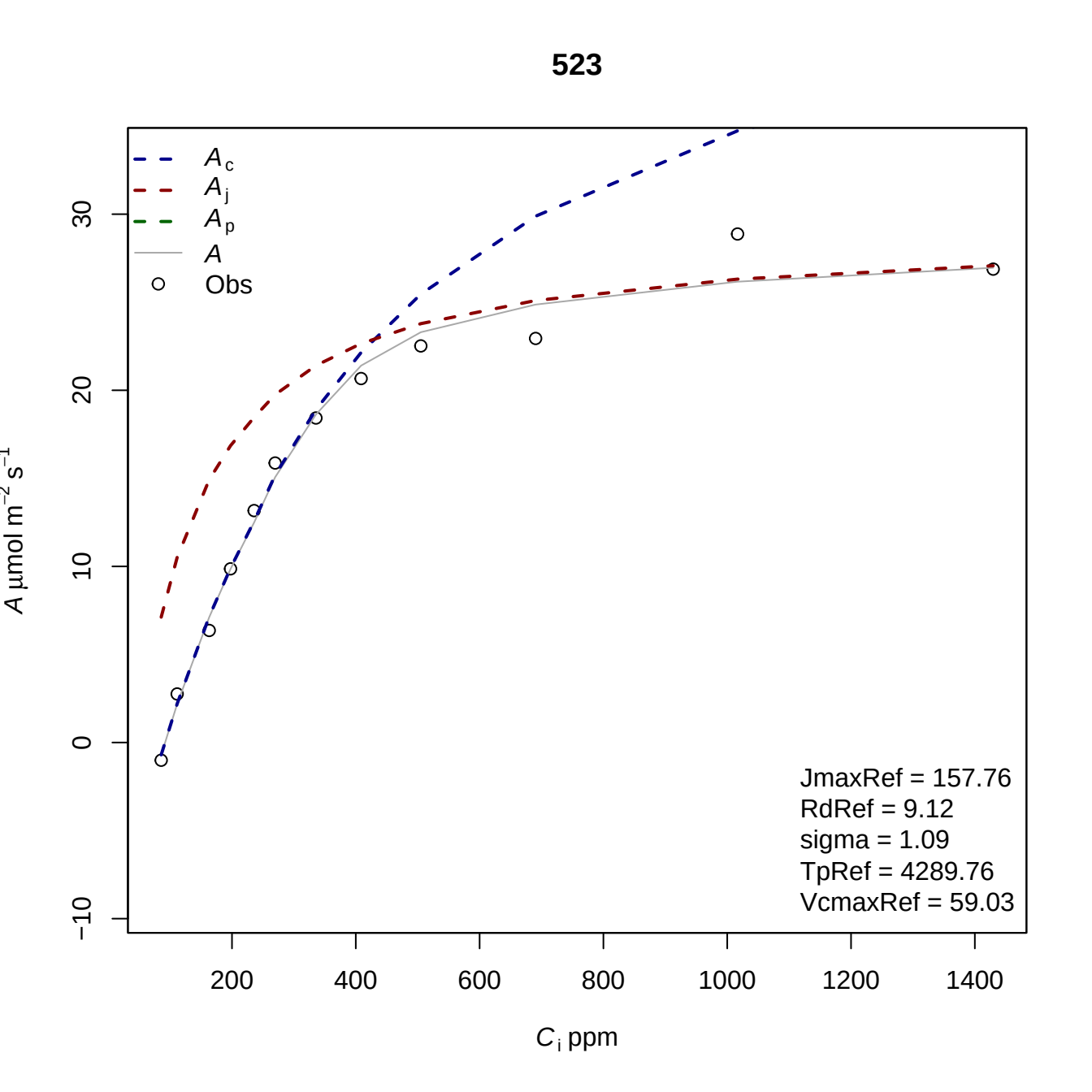

518
A_c
A_j
A_p
A
Obs
JmaxRef = 112.8
RdRef = 0.63
sigma = 0.11
TpRef = 22.25
VcmaxRef = 59.83
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200

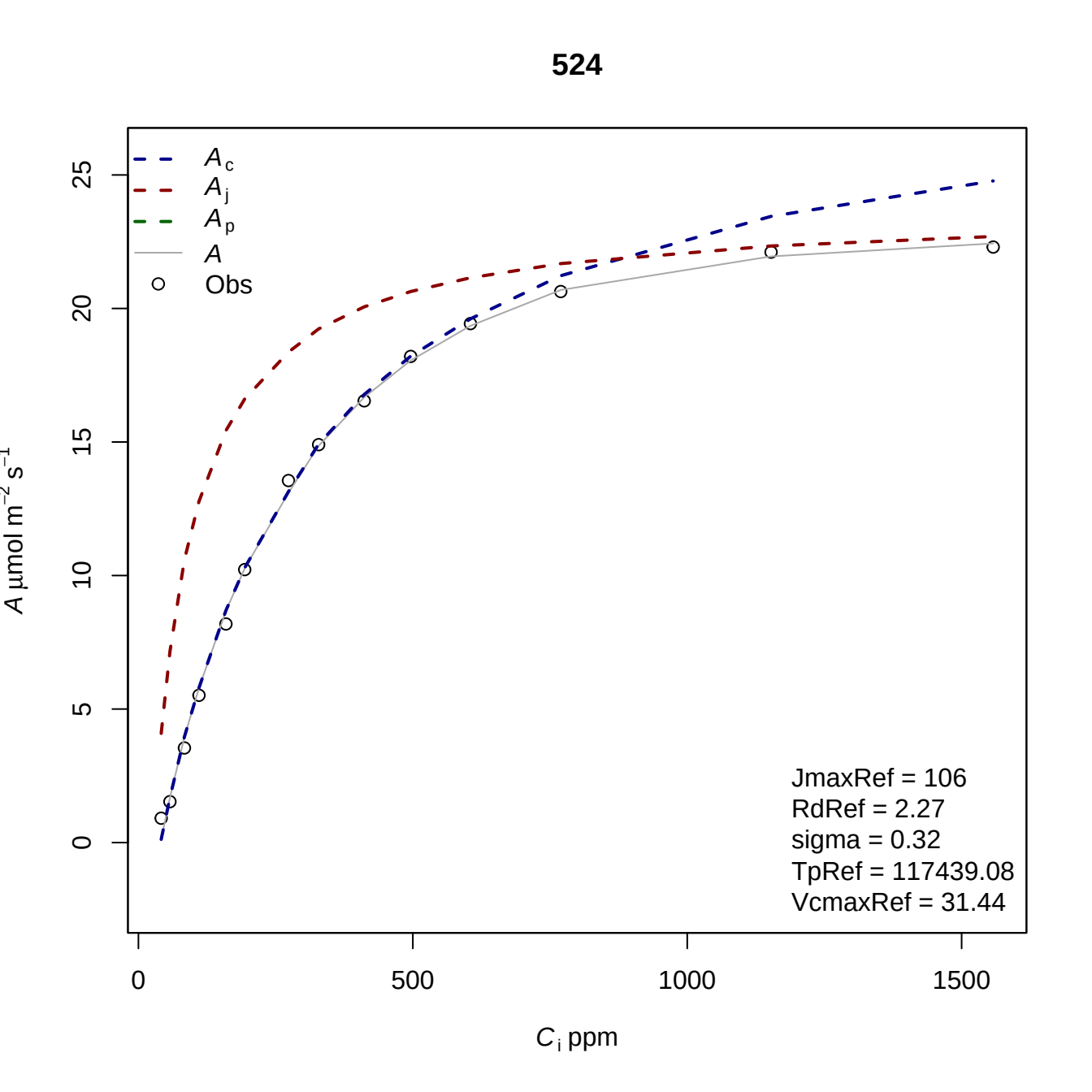

519
A_c
A_j
A_p
A
Obs
JmaxRef = 90.58
RdRef = 1.35
sigma = 0.76
TpRef = 6.57
VcmaxRef = 87.87
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
200
400
600
800
1000
1200

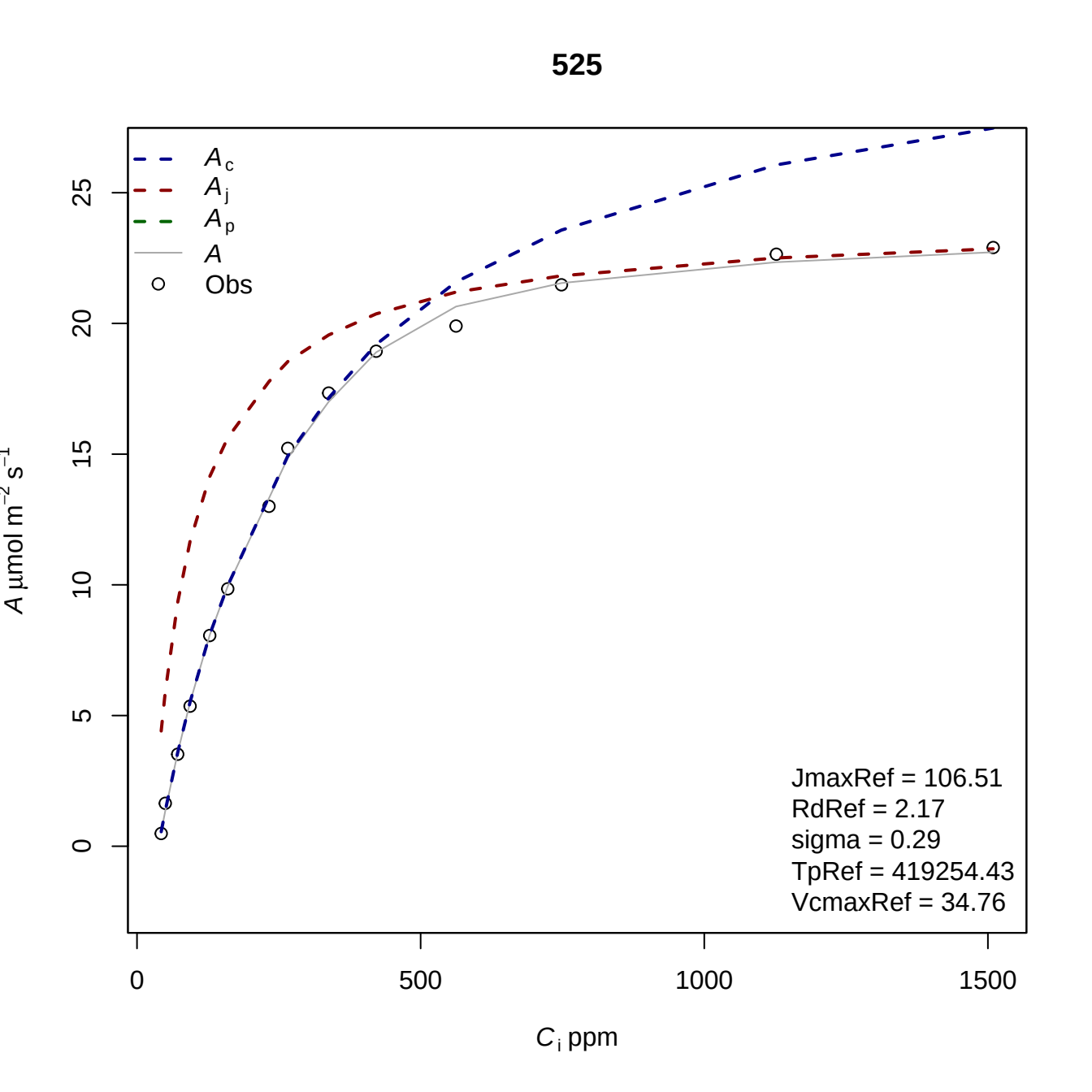

520
A_c
A_j
A_p
A
Obs
JmaxRef = 107.94
RdRef = 0.61
sigma = 0.2
TpRef = 8367.22
VcmaxRef = 69.57
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
200
400
600
800
1000

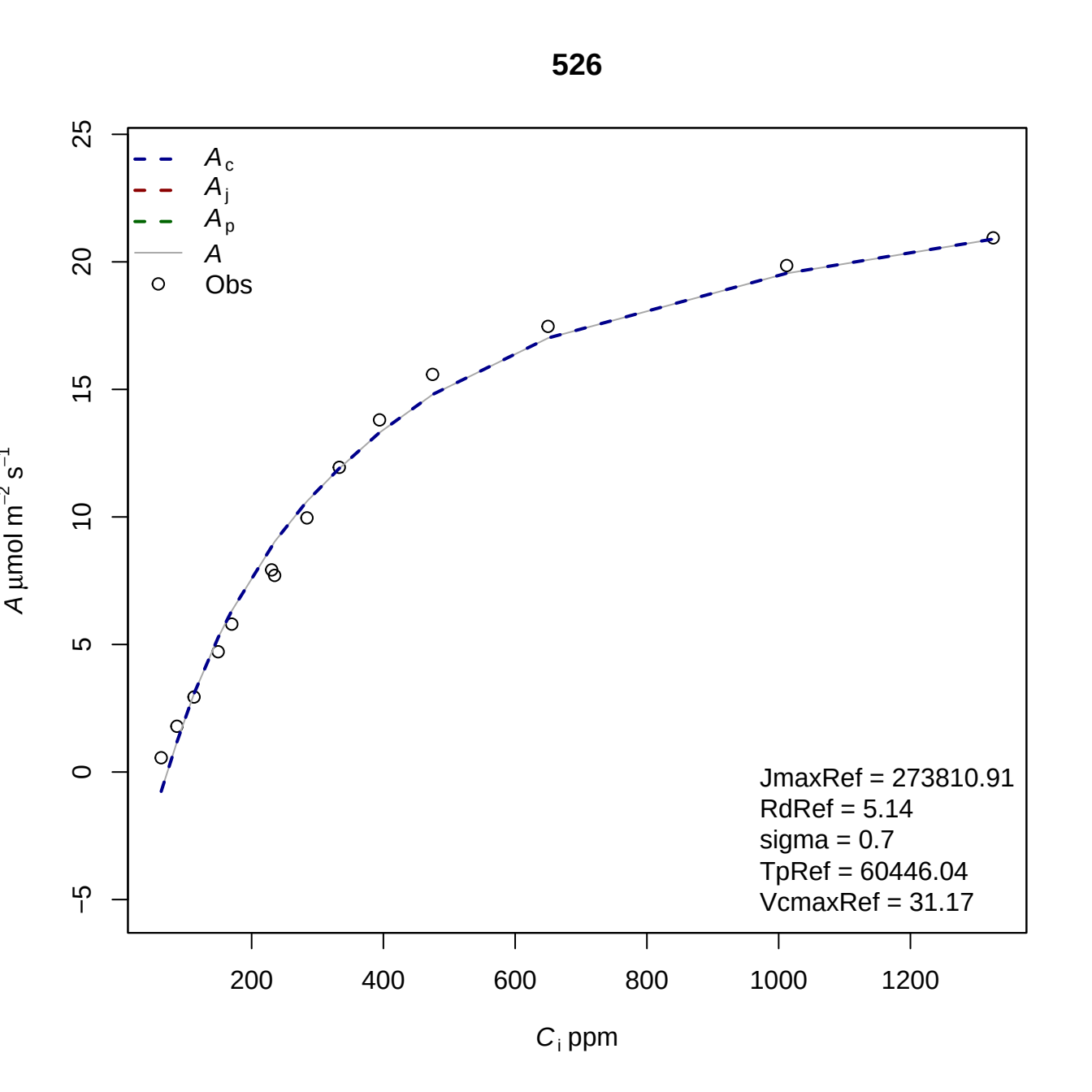

521
A_c
A_j
A_p
A
Obs
JmaxRef = 121.77
RdRef = 0.74
sigma = 0.14
TpRef = 8.77
VcmaxRef = 64.24
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200

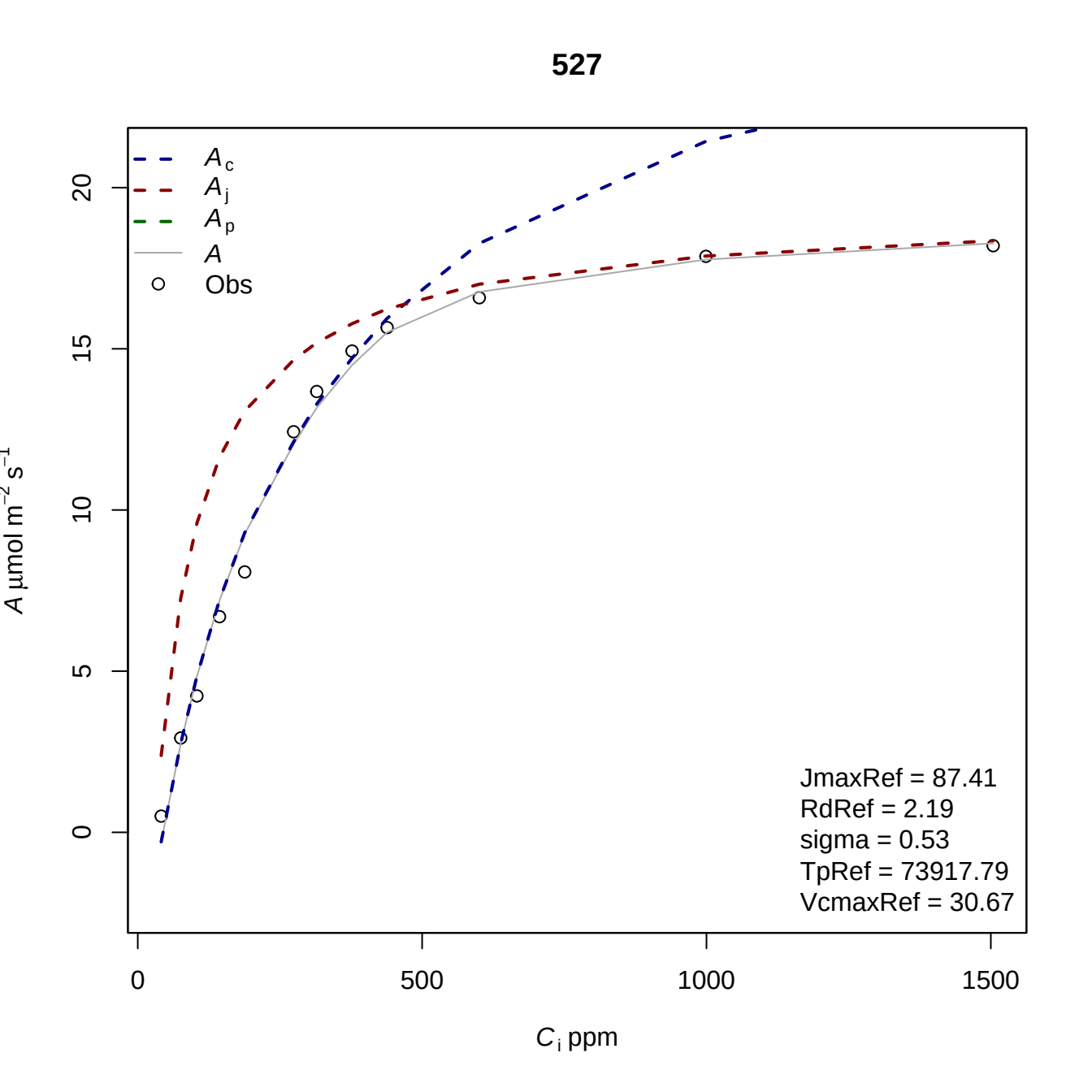

522
A_c
A_j
A_p
A
Obs
JmaxRef = 129.36
RdRef = 0.82
sigma = 0.18
TpRef = 8.43
VcmaxRef = 68.39
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200

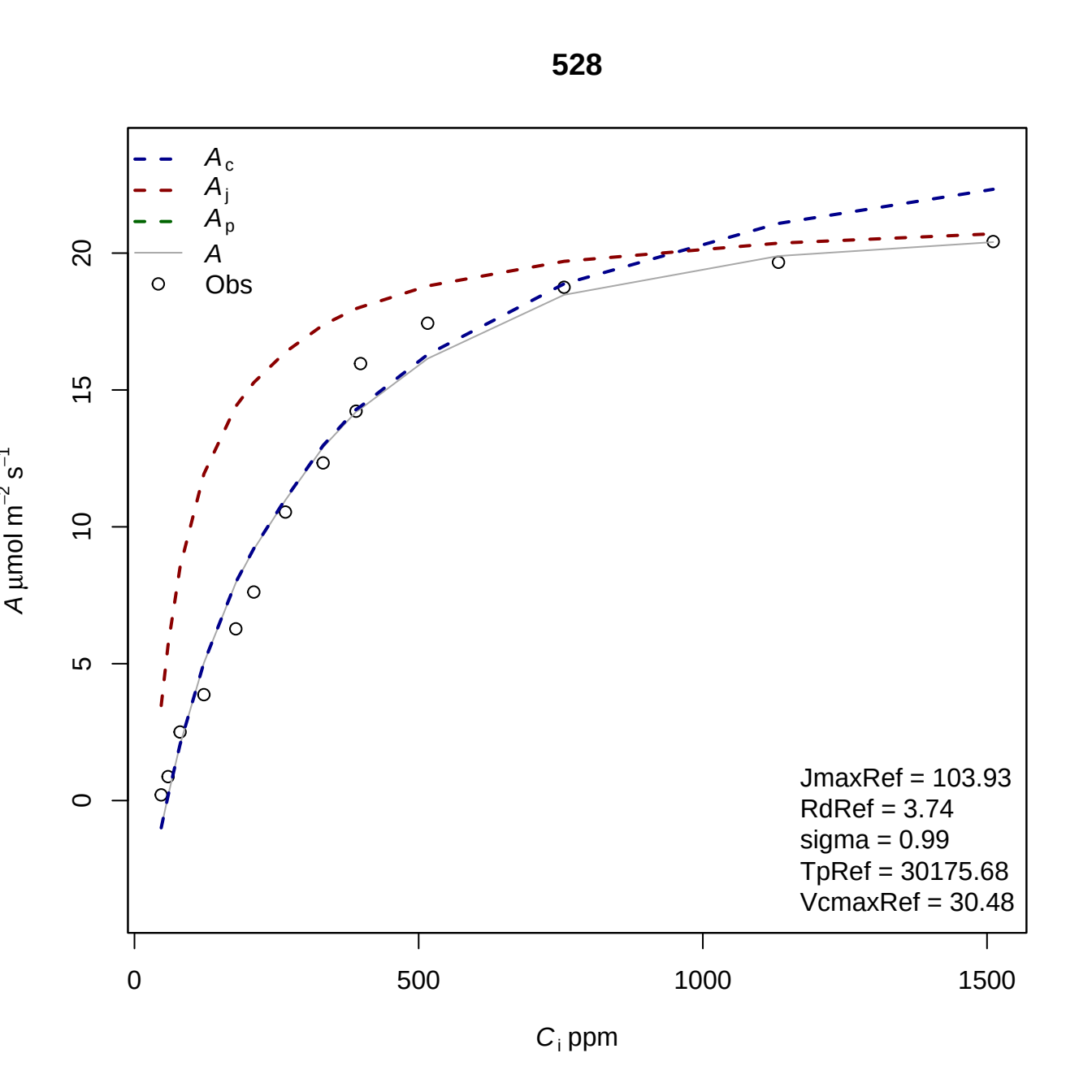

523
A_c
A_j
A_p
A
Obs
JmaxRef = 157.76
RdRef = 9.12
sigma = 1.09
TpRef = 4289.76
VcmaxRef = 59.03
A µmol m^{-2} s^{-1}
C_i ppm
−10
0
10
20
30
200
400
600
800
1000
1200
1400

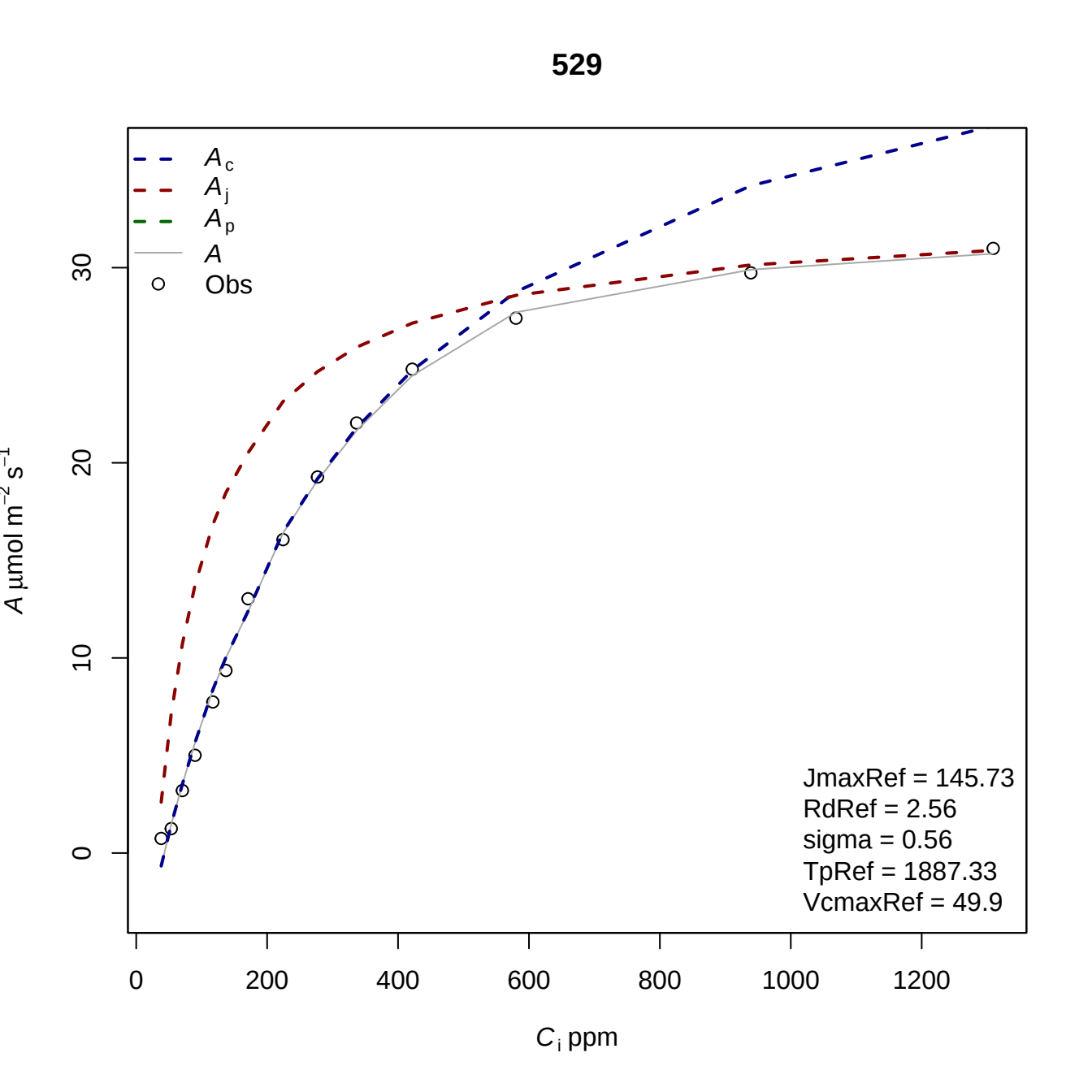

524
A_c
A_j
A_p
A
Obs
JmaxRef = 106
RdRef = 2.27
sigma = 0.32
TpRef = 117439.08
VcmaxRef = 31.44
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
500
1000
1500

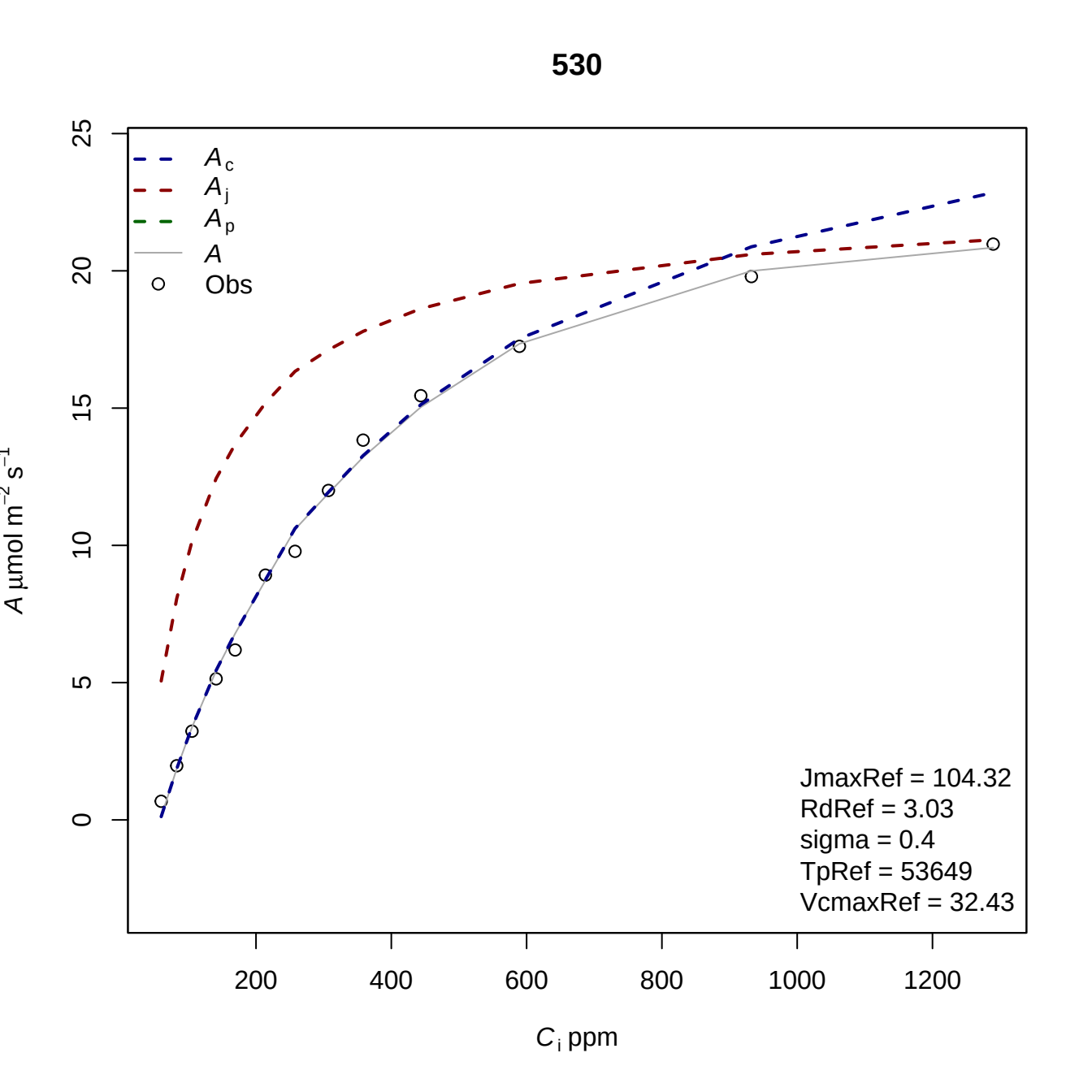

525
A_c
A_j
A_p
A
Obs
JmaxRef = 106.51
RdRef = 2.17
sigma = 0.29
TpRef = 419254.43
VcmaxRef = 34.76
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
500
1000
1500

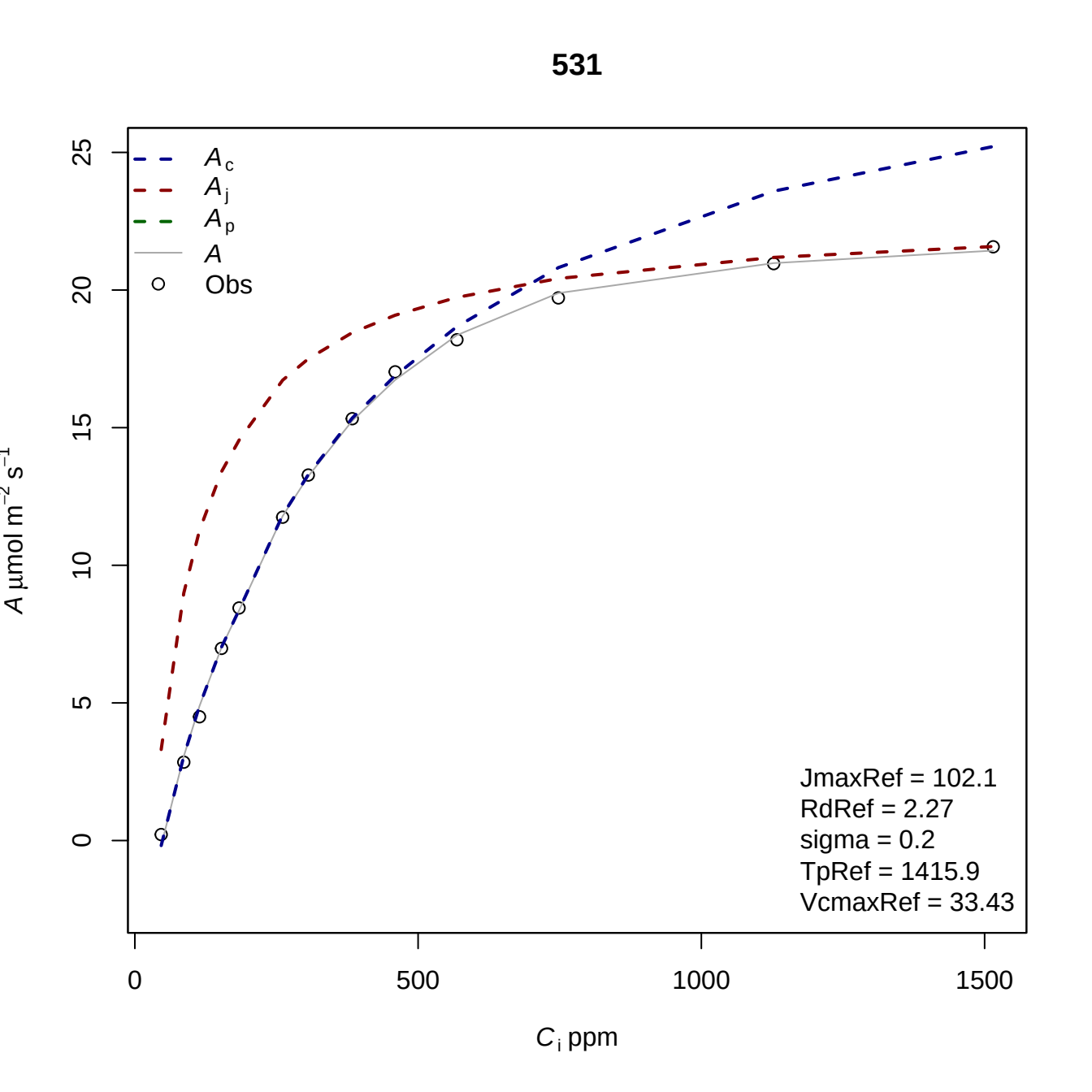

526
A_c
A_j
A_p
A
Obs
JmaxRef = 273810.91
RdRef = 5.14
sigma = 0.7
TpRef = 60446.04
VcmaxRef = 31.17
A µmol m^{-2} s^{-1}
C_i ppm
25
20
15
10
5
0
−5
200
400
600
800
1000
1200

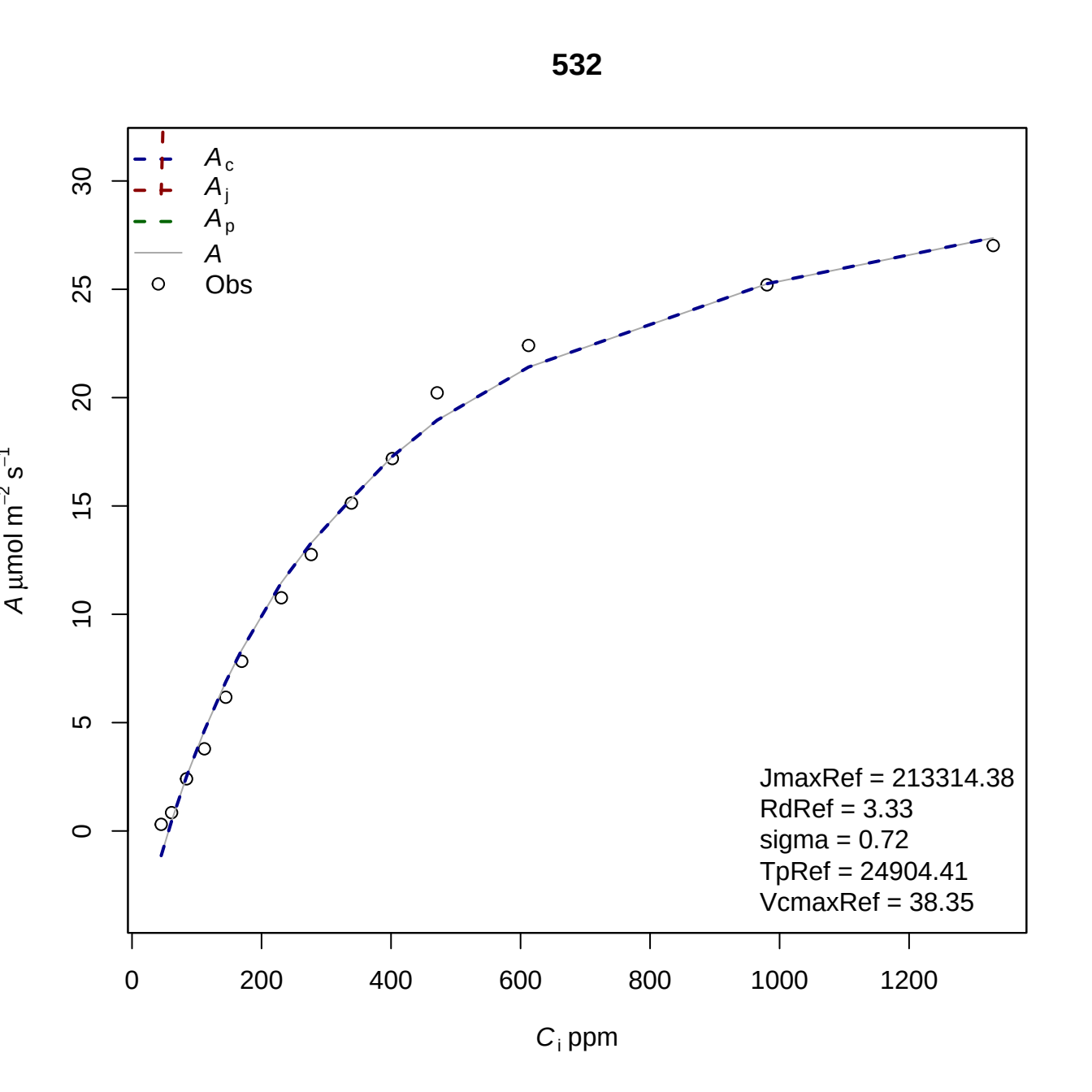

527
A_c
A_j
A_p
A
Obs
JmaxRef = 87.41
RdRef = 2.19
sigma = 0.53
TpRef = 73917.79
VcmaxRef = 30.67
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
500
1000
1500

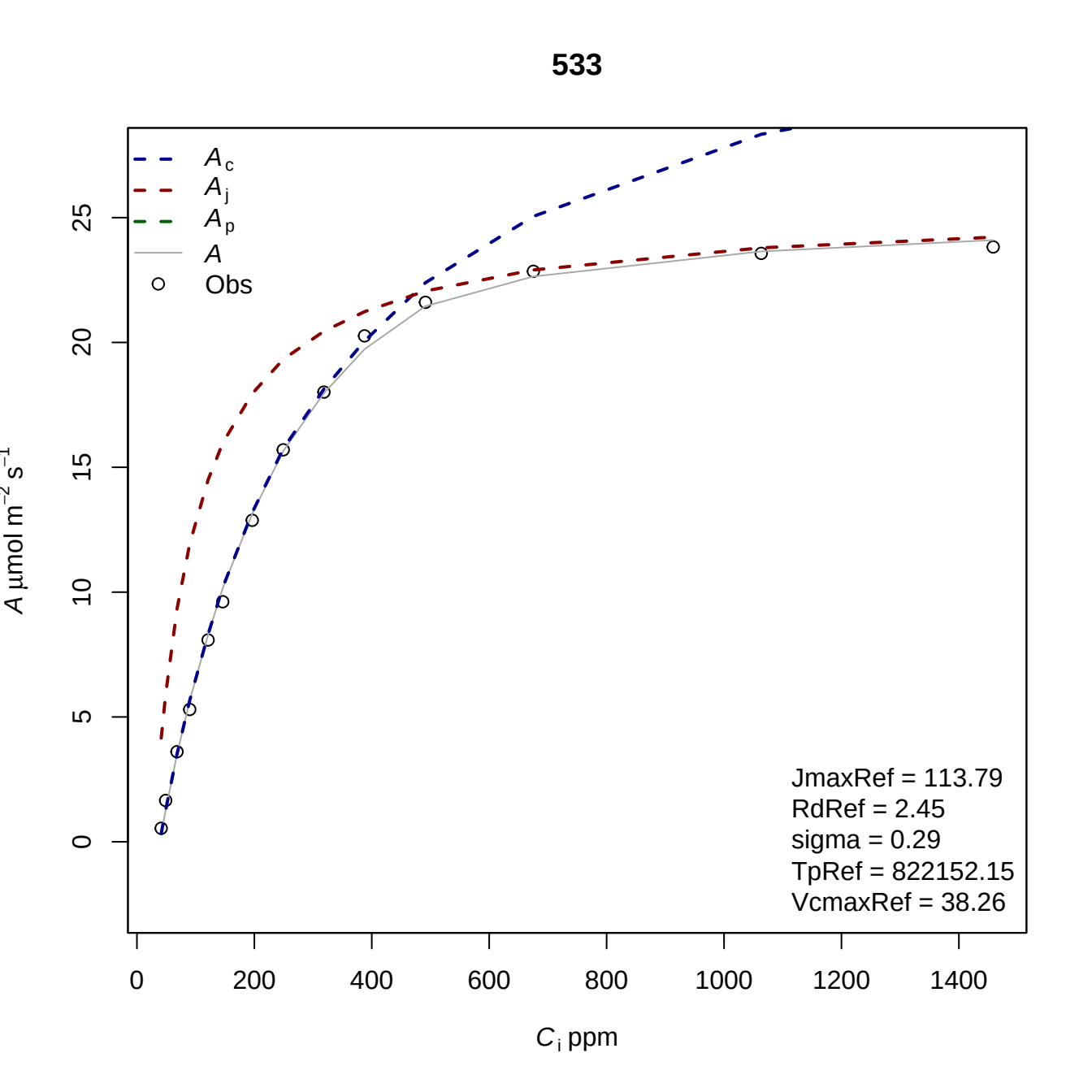

528
A_c
A_j
A_p
A
Obs
JmaxRef = 103.93
RdRef = 3.74
sigma = 0.99
TpRef = 30175.68
VcmaxRef = 30.48
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
500
1000
1500

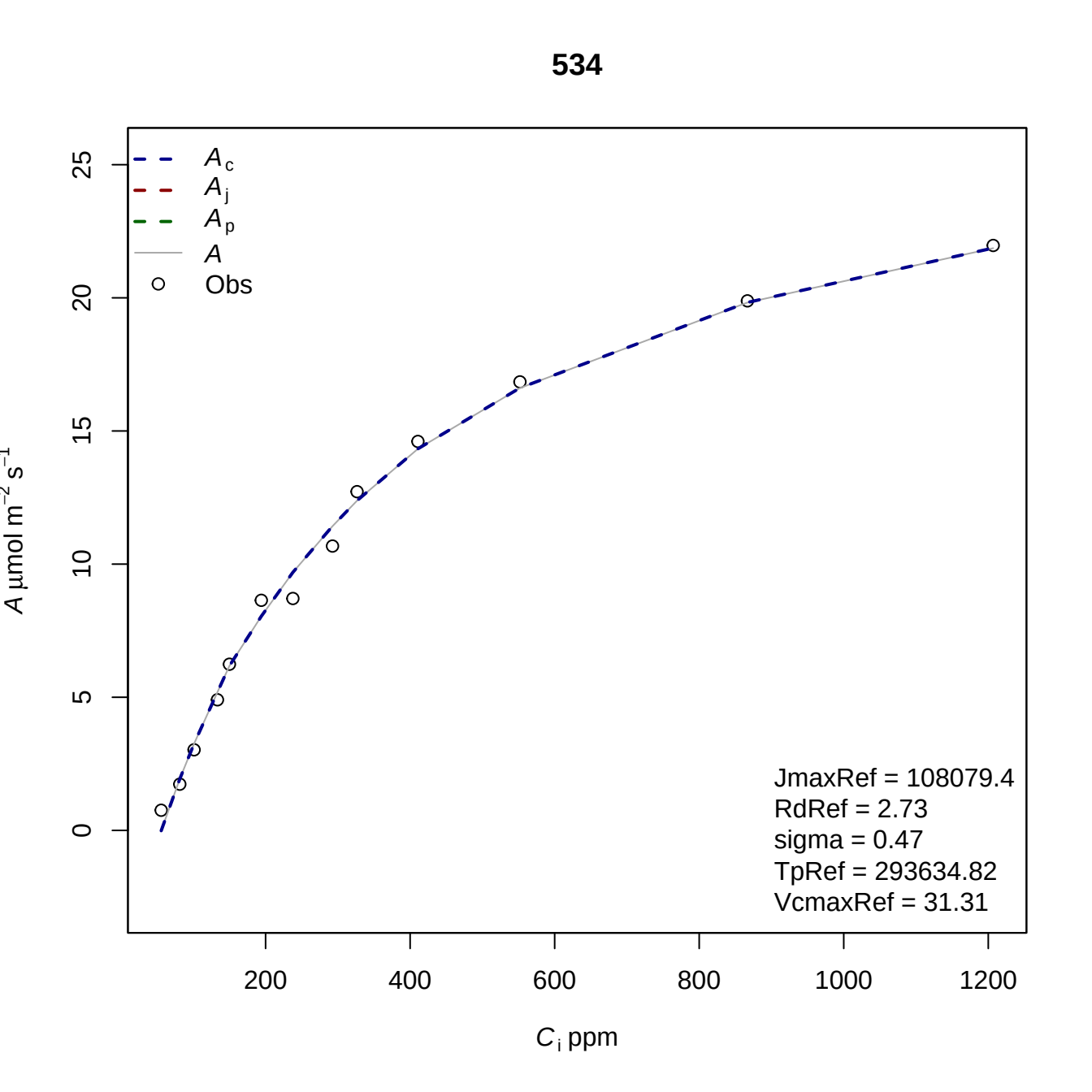

529
A_c
A_j
A_p
A
Obs
JmaxRef = 145.73
RdRef = 2.56
sigma = 0.56
TpRef = 1887.33
VcmaxRef = 49.9
0
10
20
30
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
1200
C_i ppm

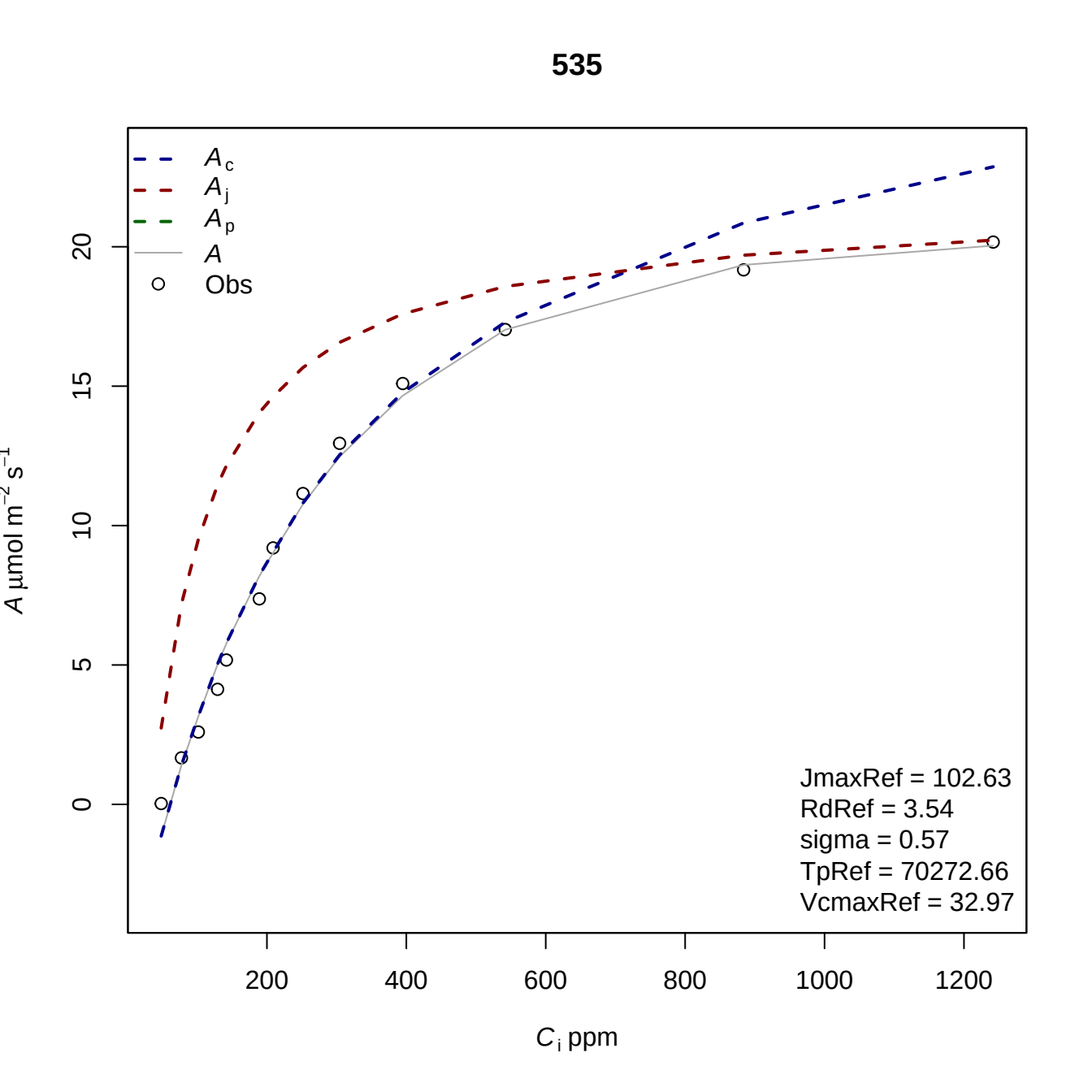

530
A_c
A_j
A_p
A
Obs
JmaxRef = 104.32
RdRef = 3.03
sigma = 0.4
TpRef = 53649
VcmaxRef = 32.43
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
200
400
600
800
1000
1200

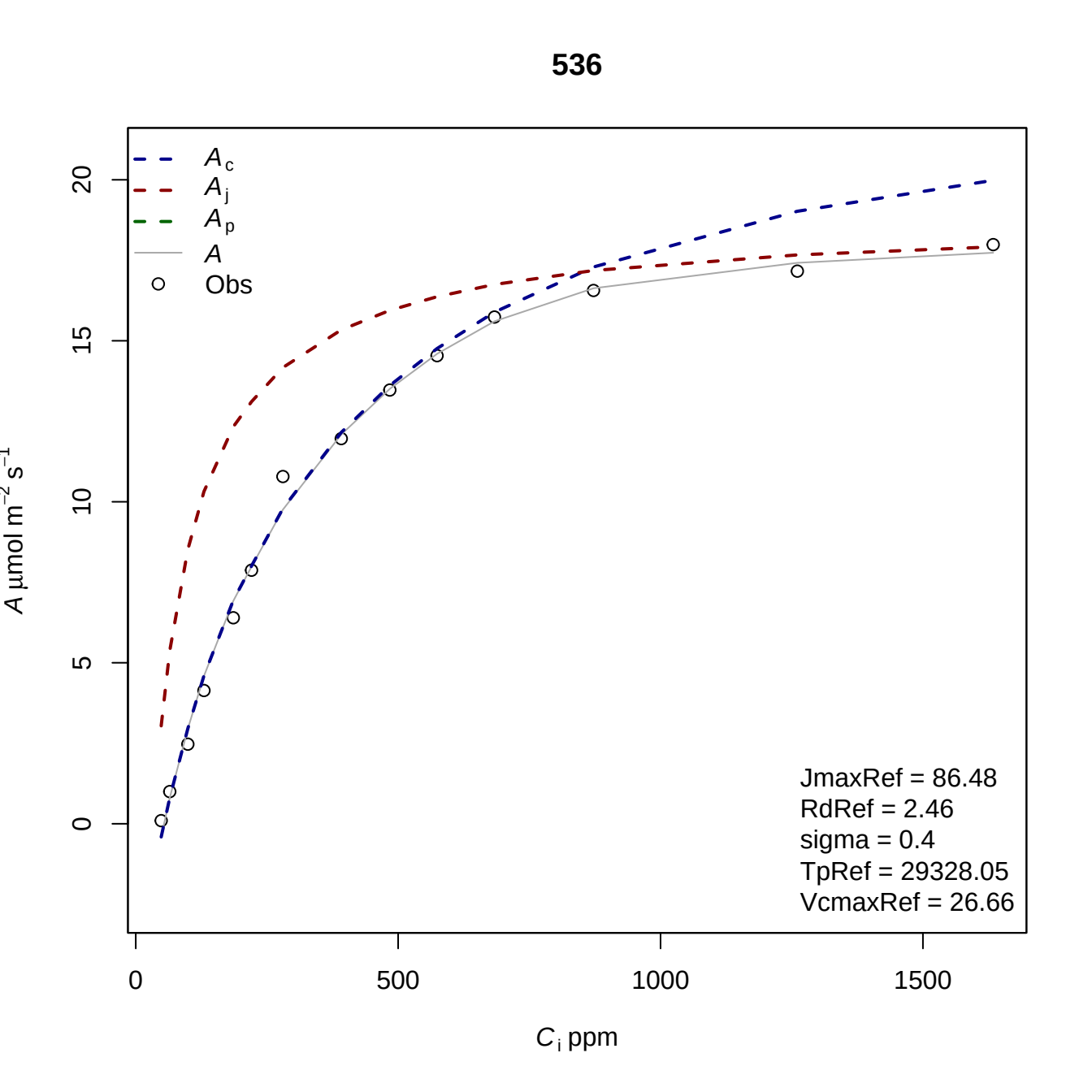

531
A_c
A_j
A_p
A
Obs
JmaxRef = 102.1
RdRef = 2.27
sigma = 0.2
TpRef = 1415.9
VcmaxRef = 33.43
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
500
1000
1500

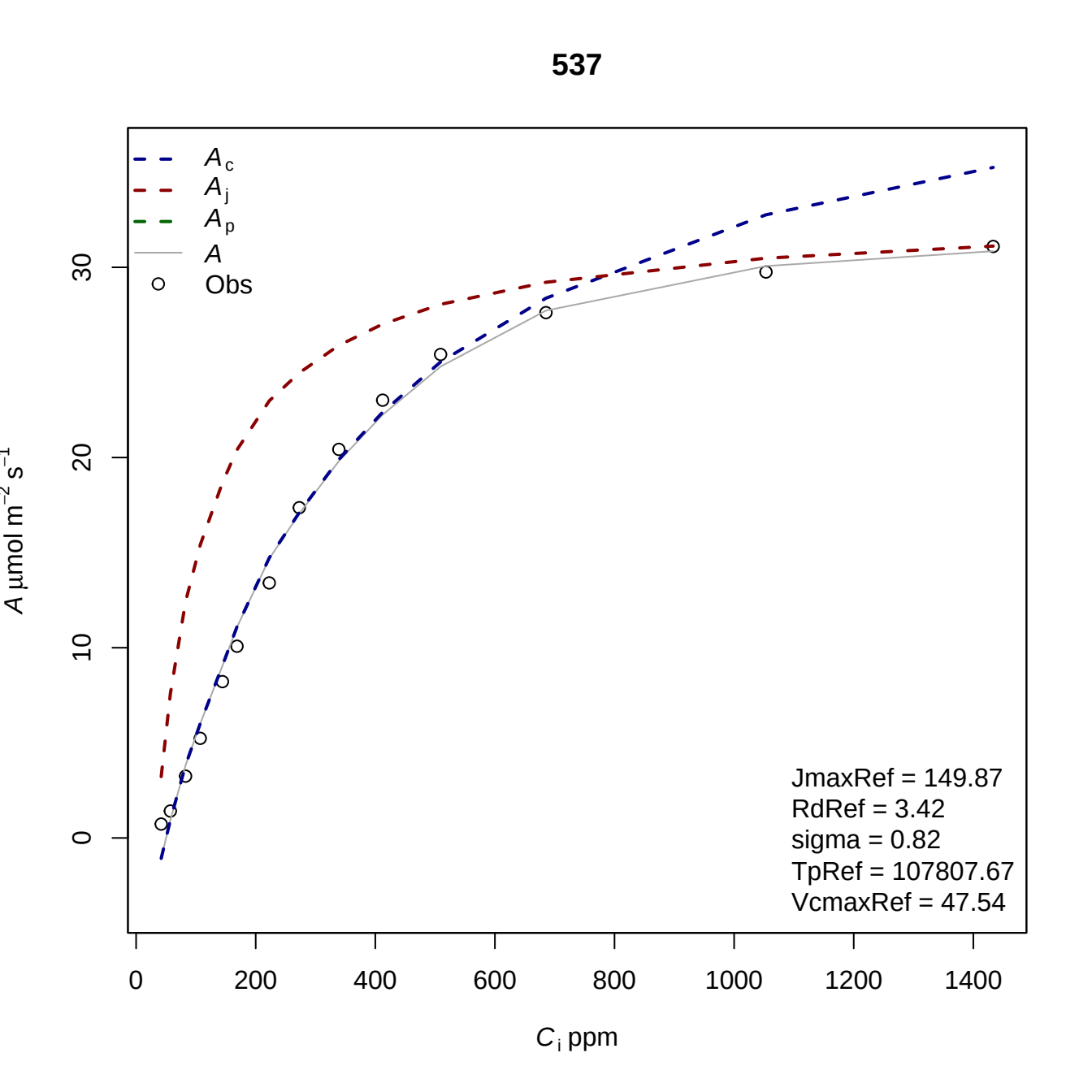

532
A_c
A_j
A_p
A
Obs
JmaxRef = 213314.38
RdRef = 3.33
sigma = 0.72
TpRef = 24904.41
VcmaxRef = 38.35
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200

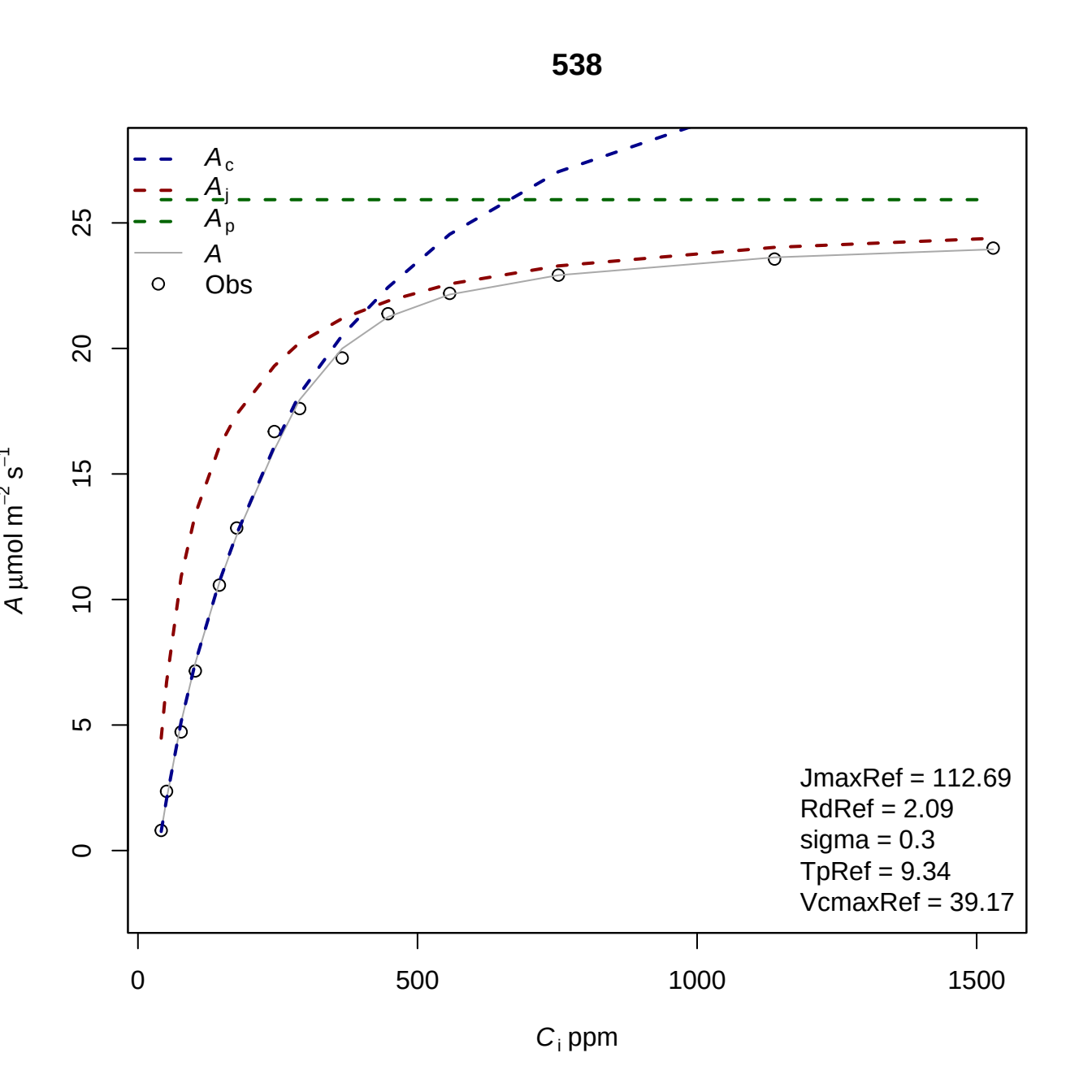

533
A_c
A_j
A_p
A
Obs
JmaxRef = 113.79
RdRef = 2.45
sigma = 0.29
TpRef = 822152.15
VcmaxRef = 38.26
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200
1400

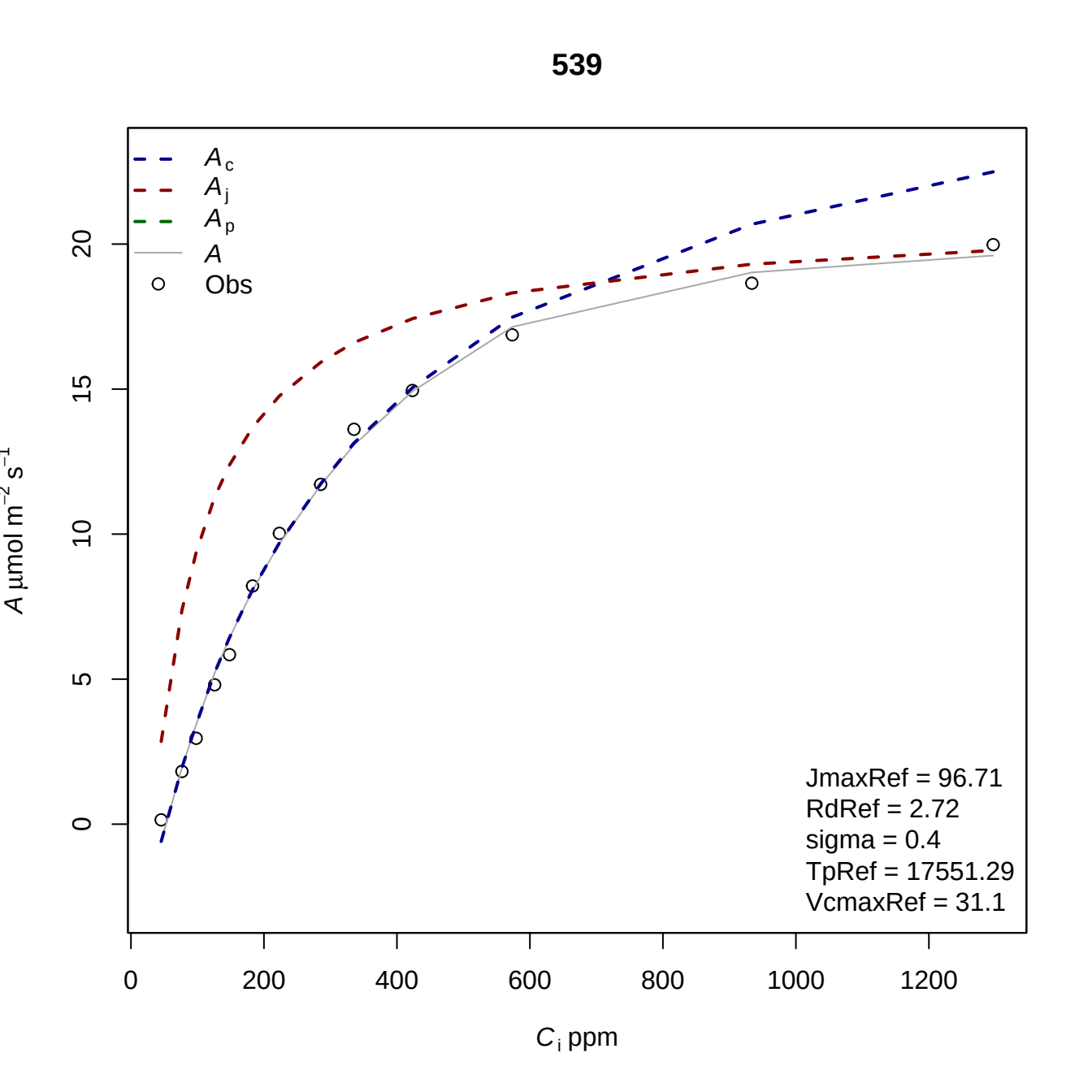

534
A_c
A_j
A_p
A
Obs
JmaxRef = 108079.4
RdRef = 2.73
sigma = 0.47
TpRef = 293634.82
VcmaxRef = 31.31
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
200
400
600
800
1000
1200

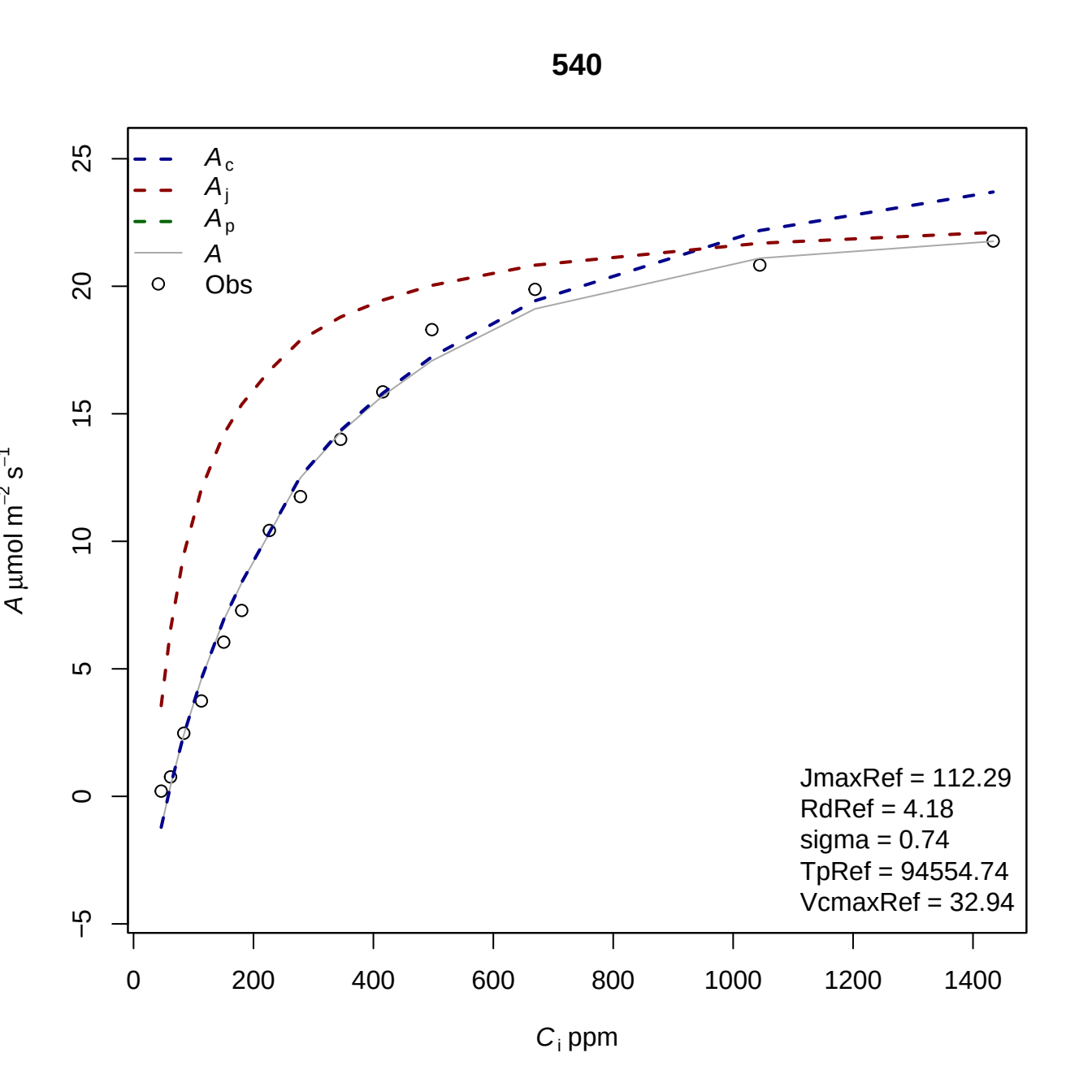

535
A_c
A_j
A_p
A
Obs
JmaxRef = 102.63
RdRef = 3.54
sigma = 0.57
TpRef = 70272.66
VcmaxRef = 32.97
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
200
400
600
800
1000
1200

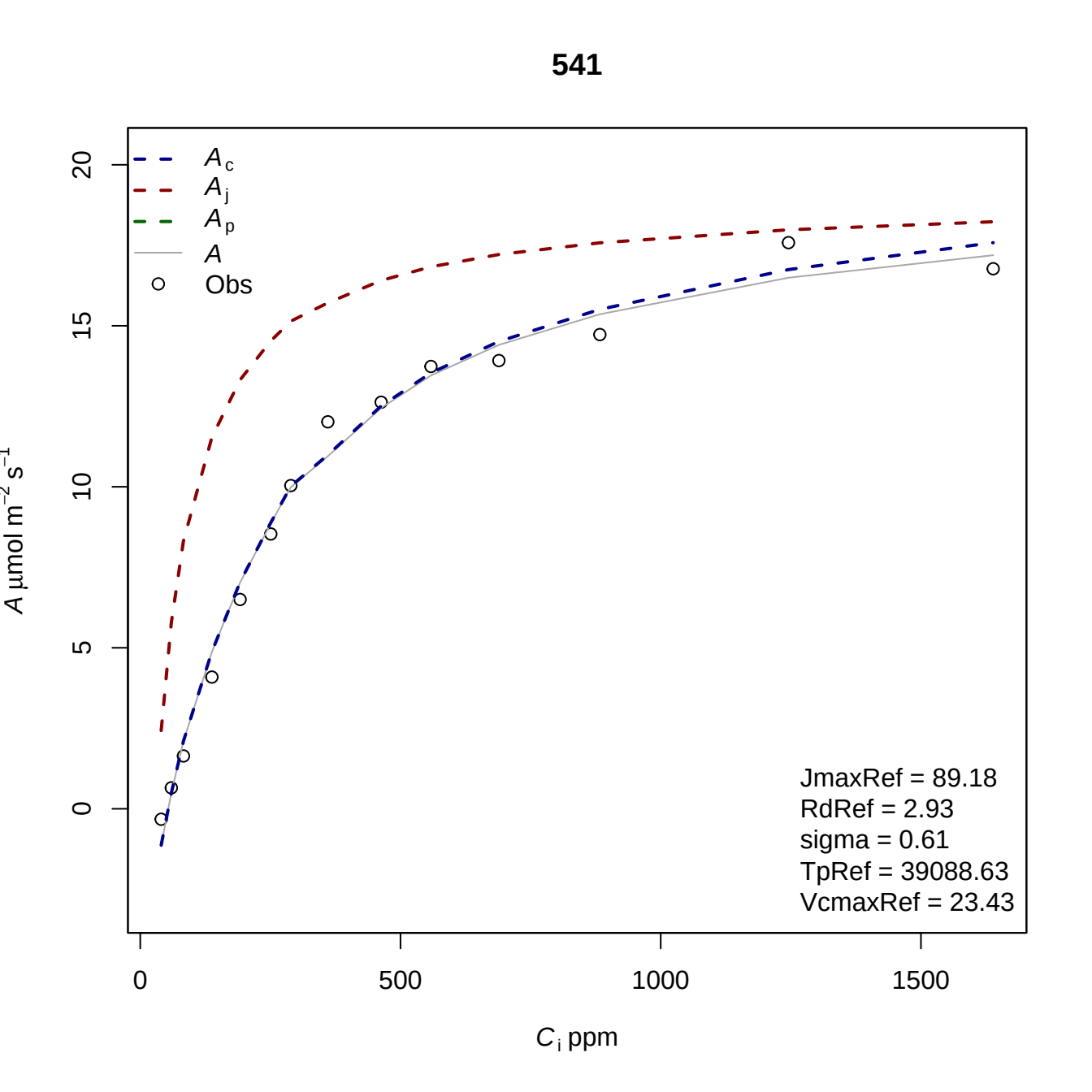

536
A_c
A_j
A_p
A
Obs
JmaxRef = 86.48
RdRef = 2.46
sigma = 0.4
TpRef = 29328.05
VcmaxRef = 26.66
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
500
1000
1500

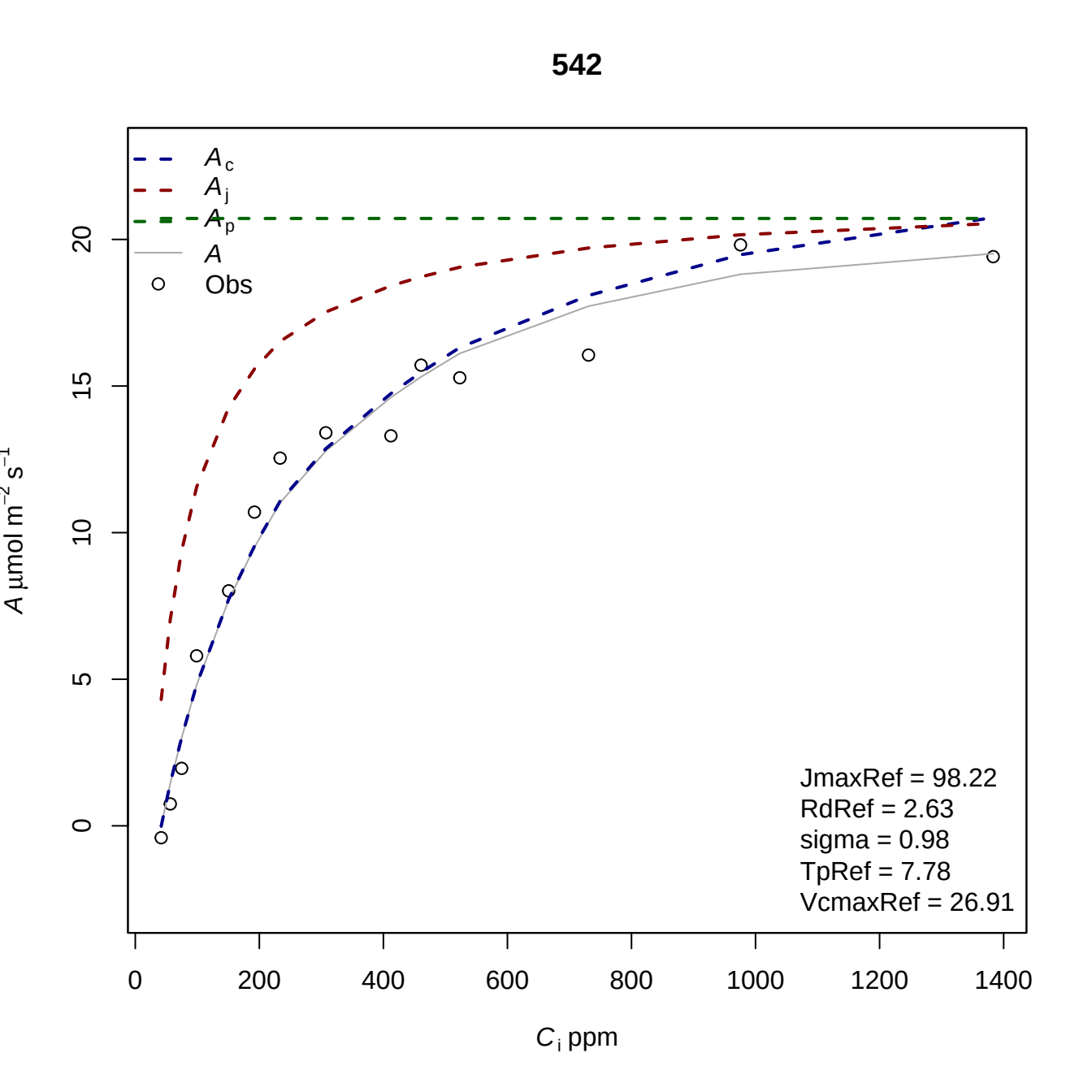

537
A_c
A_j
A_p
A
Obs
JmaxRef = 149.87
RdRef = 3.42
sigma = 0.82
TpRef = 107807.67
VcmaxRef = 47.54
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
0
200
400
600
800
1000
1200
1400

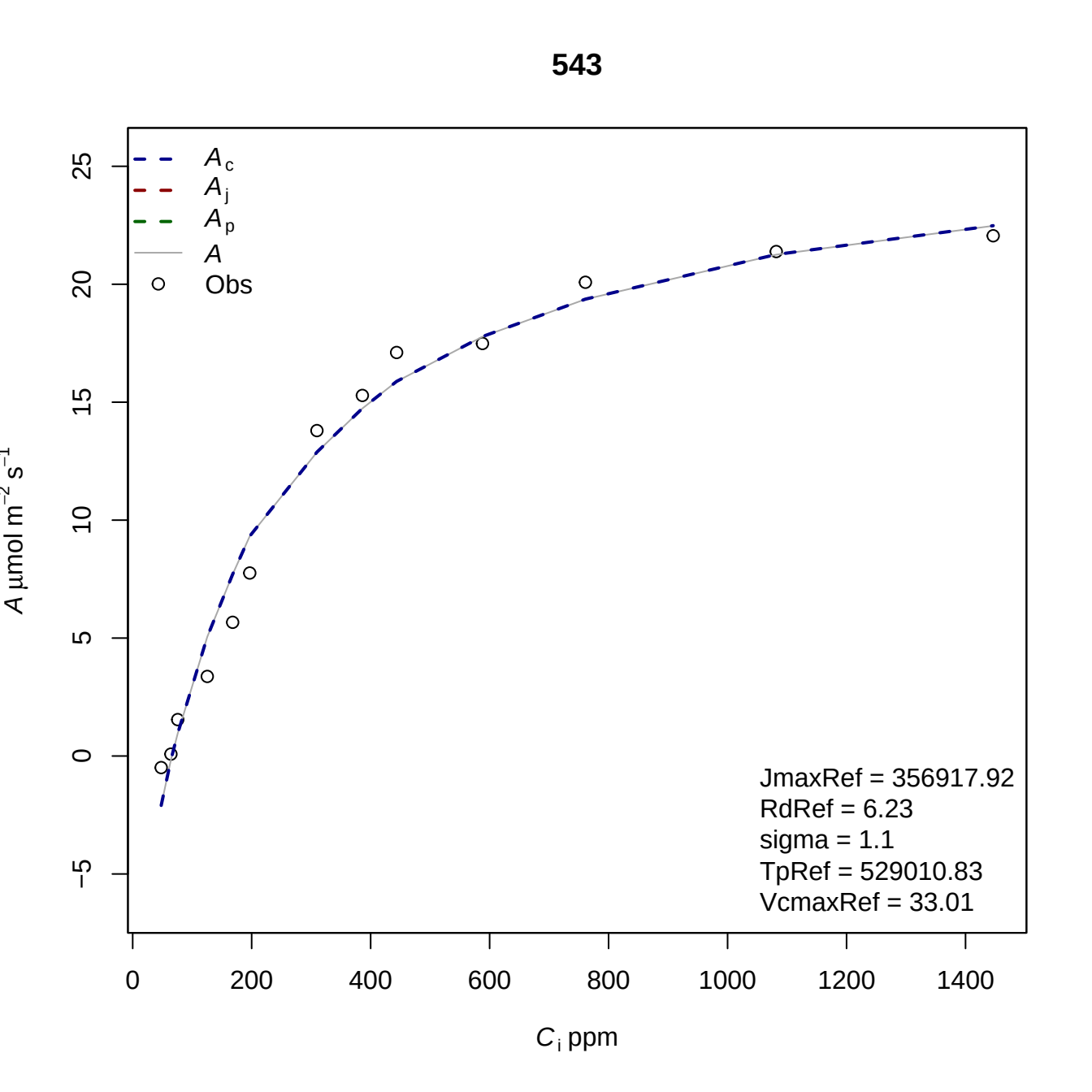

538
A_c
A_j
A_p
A
Obs
JmaxRef = 112.69
RdRef = 2.09
sigma = 0.3
TpRef = 9.34
VcmaxRef = 39.17
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
500
1000
1500

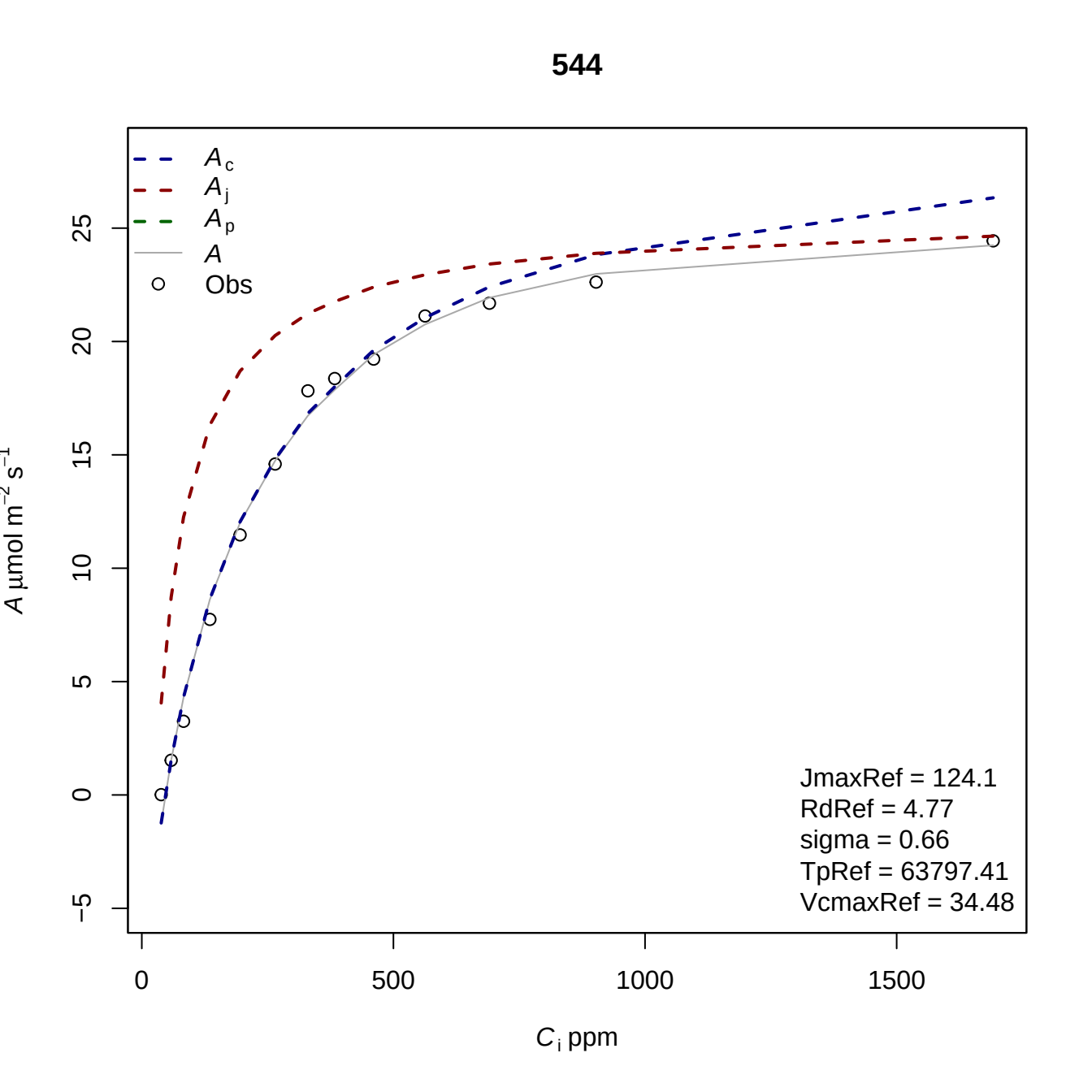

539
A_c
A_j
A_p
A
Obs
JmaxRef = 96.71
RdRef = 2.72
sigma = 0.4
TpRef = 17551.29
VcmaxRef = 31.1
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

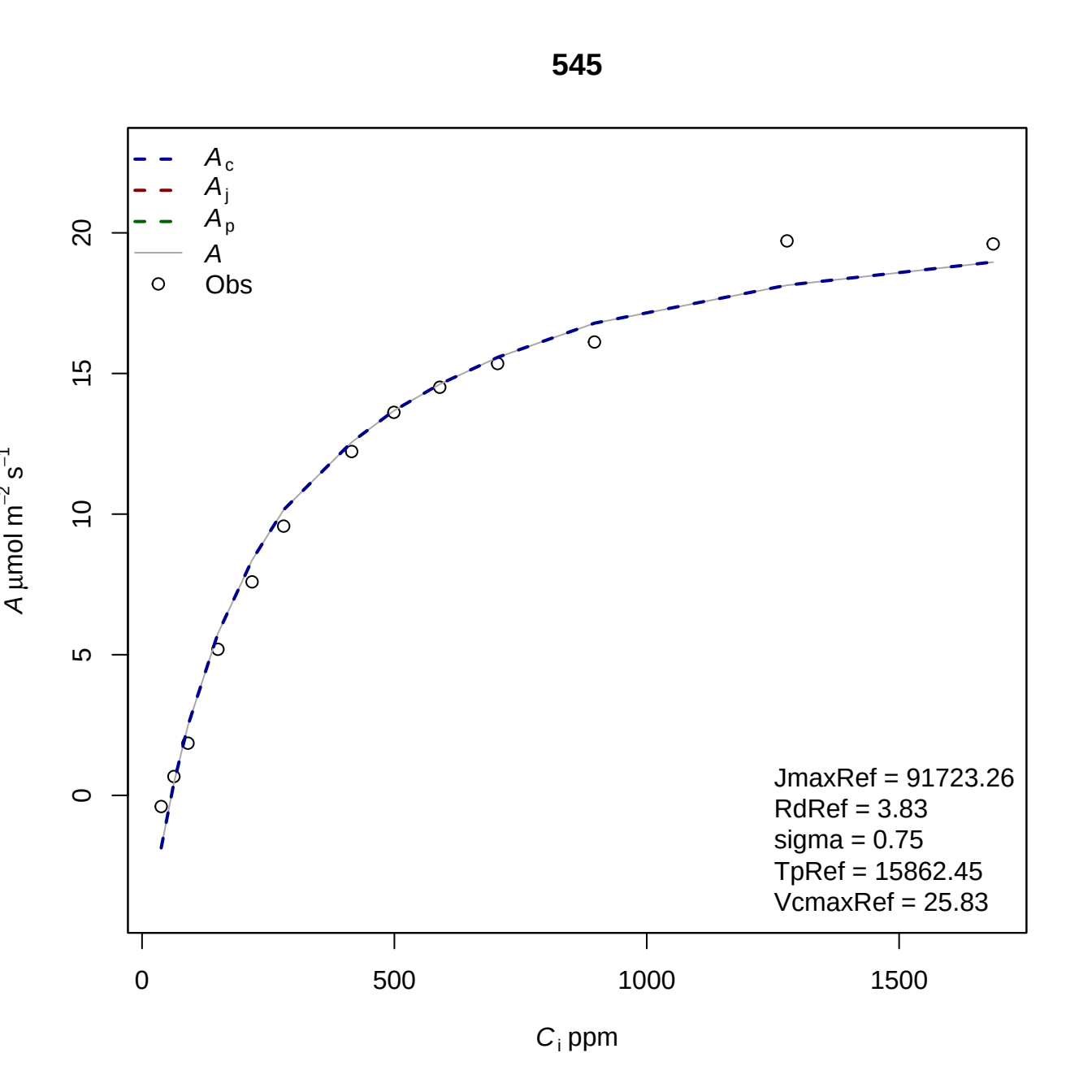

540
A_c
A_j
A_p
A
Obs
JmaxRef = 112.29
RdRef = 4.18
sigma = 0.74
TpRef = 94554.74
VcmaxRef = 32.94
A µmol m^{-2} s^{-1}
C_i ppm
25
20
15
10
5
0
−5
0
200
400
600
800
1000
1200
1400

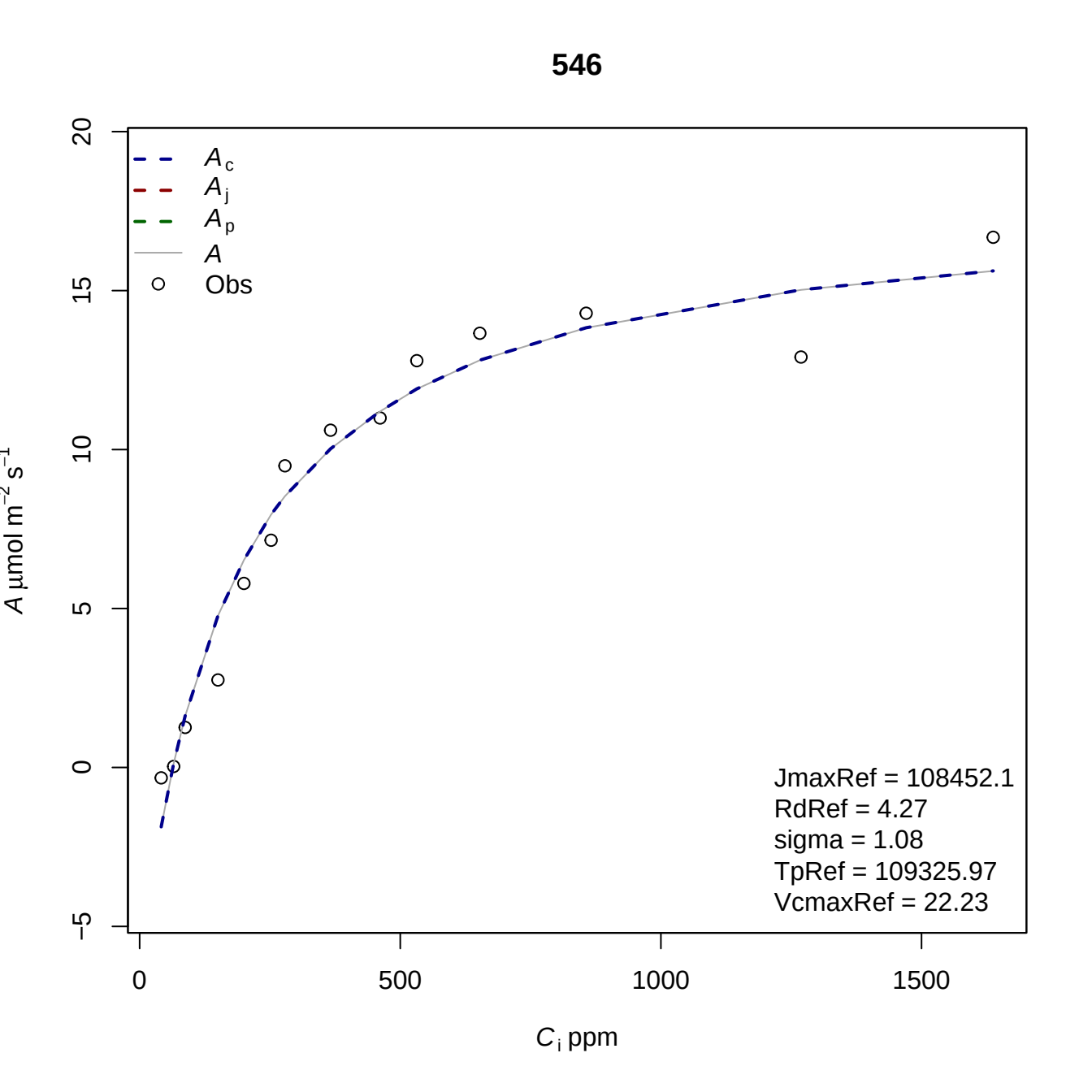

541
A_c
A_j
A_p
A
Obs
JmaxRef = 89.18
RdRef = 2.93
sigma = 0.61
TpRef = 39088.63
VcmaxRef = 23.43
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
500
1000
1500

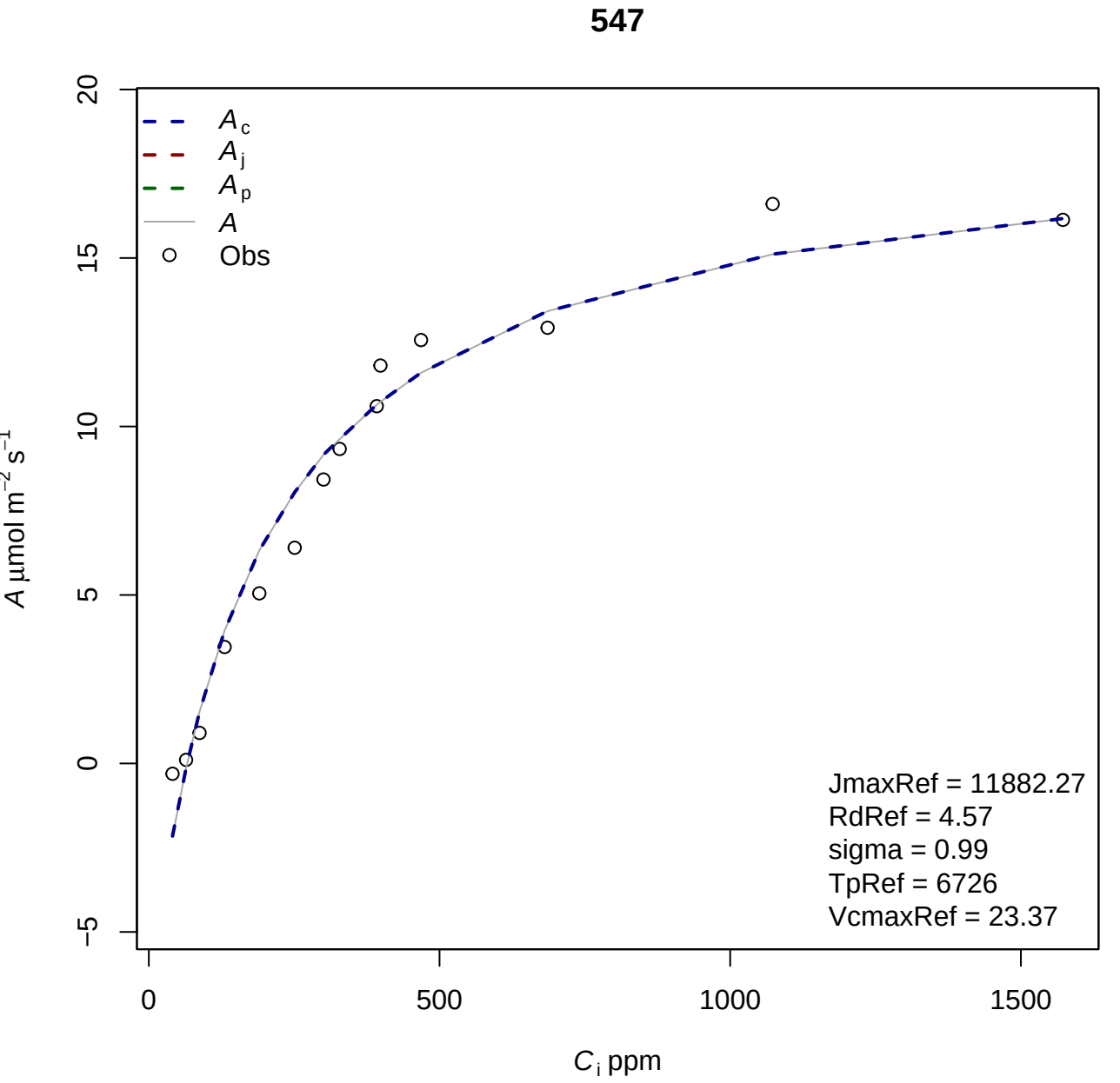

542
A_c
A_j
A_p
A
Obs
JmaxRef = 98.22
RdRef = 2.63
sigma = 0.98
TpRef = 7.78
VcmaxRef = 26.91
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000
1200
1400

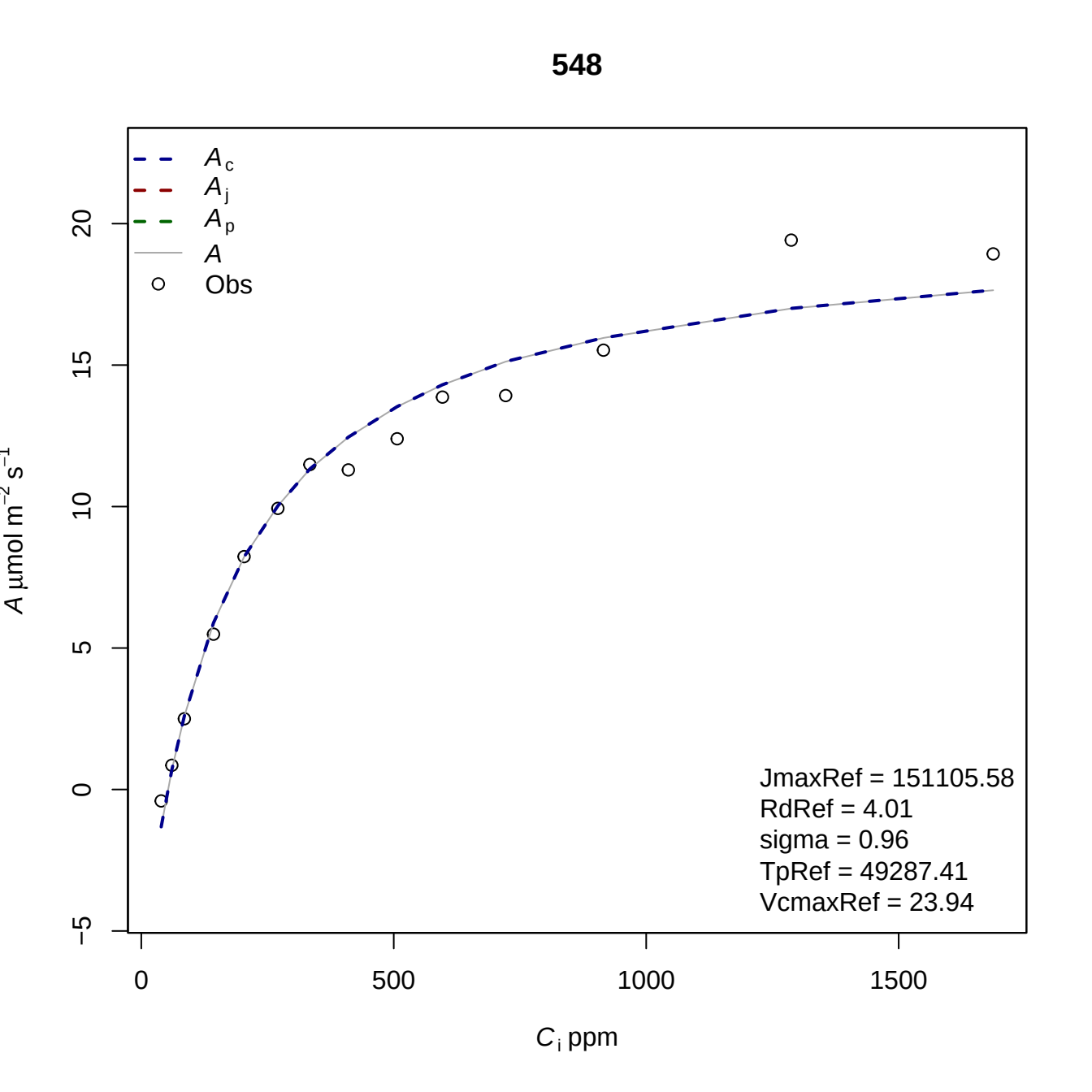

543
A_c
A_j
A_p
A
Obs
JmaxRef = 356917.92
RdRef = 6.23
sigma = 1.1
TpRef = 529010.83
VcmaxRef = 33.01
A µmol m^{-2} s^{-1}
C_i ppm
25
20
15
10
5
0
−5
0
200
400
600
800
1000
1200
1400

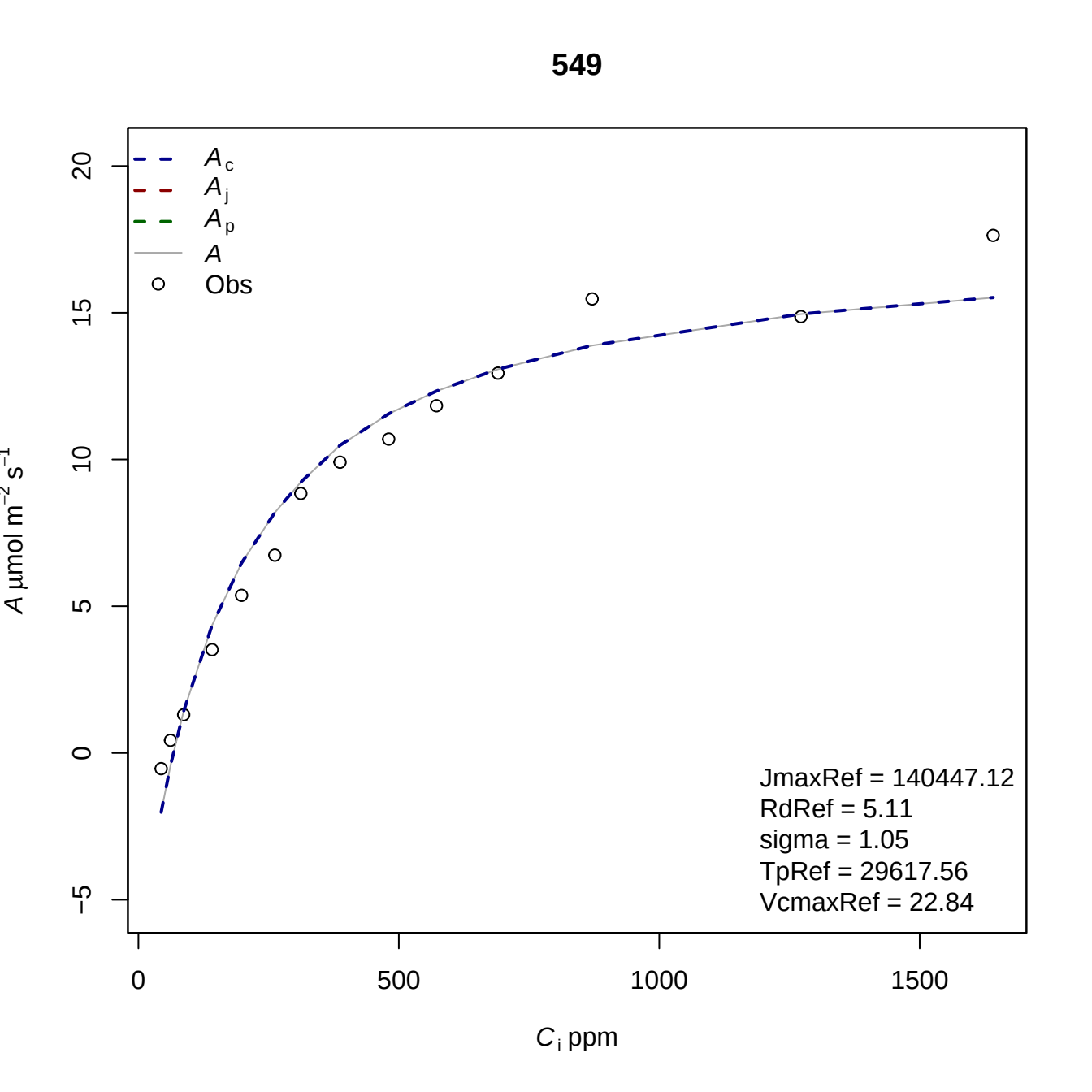

544
A_c
A_j
A_p
A
Obs
JmaxRef = 124.1
RdRef = 4.77
sigma = 0.66
TpRef = 63797.41
VcmaxRef = 34.48
A µmol m^{-2} s^{-1}
C_i ppm
−5
0
5
10
15
20
25
0
500
1000
1500

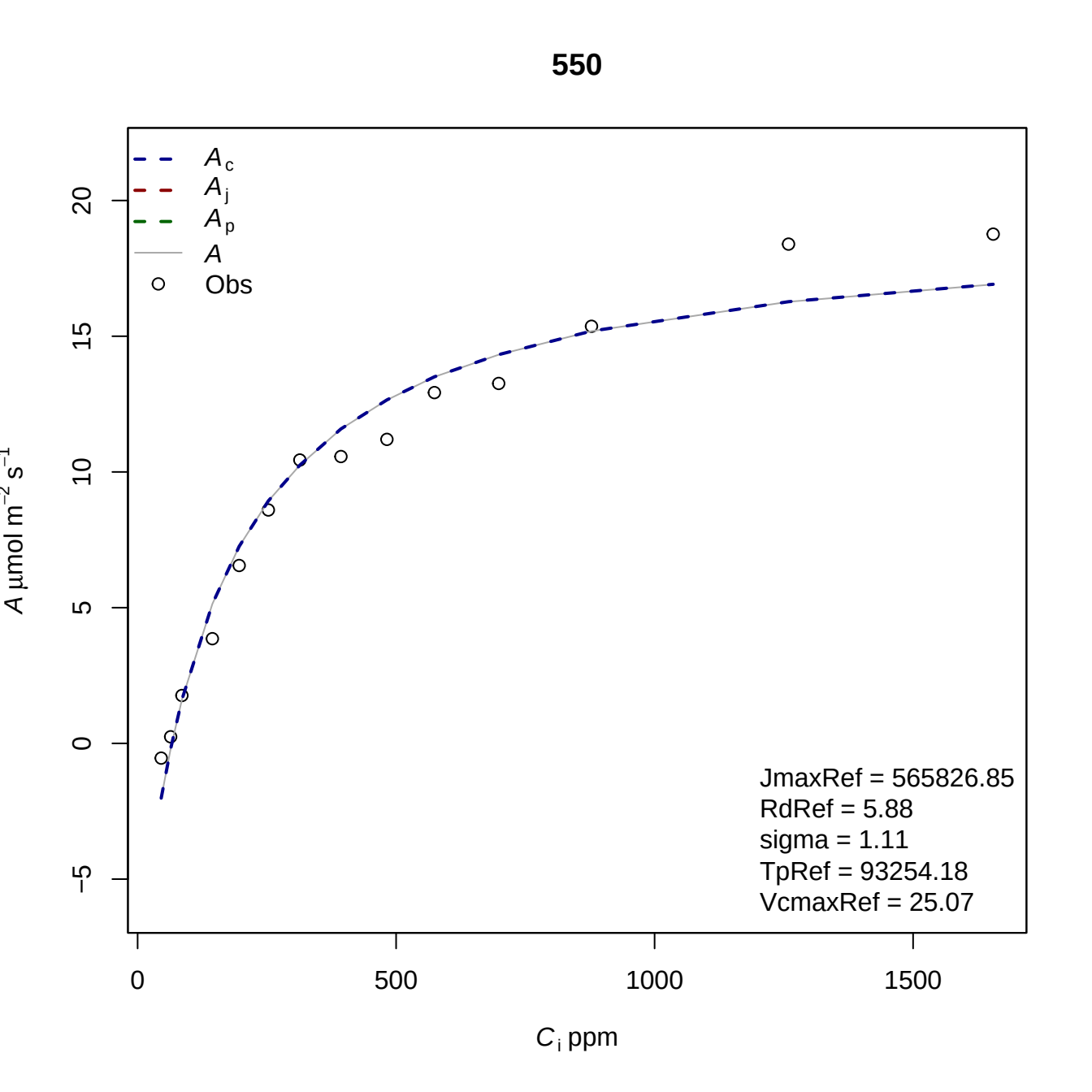

545
A_c
A_j
A_p
A
Obs
JmaxRef = 91723.26
RdRef = 3.83
sigma = 0.75
TpRef = 15862.45
VcmaxRef = 25.83
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
500
1000
1500

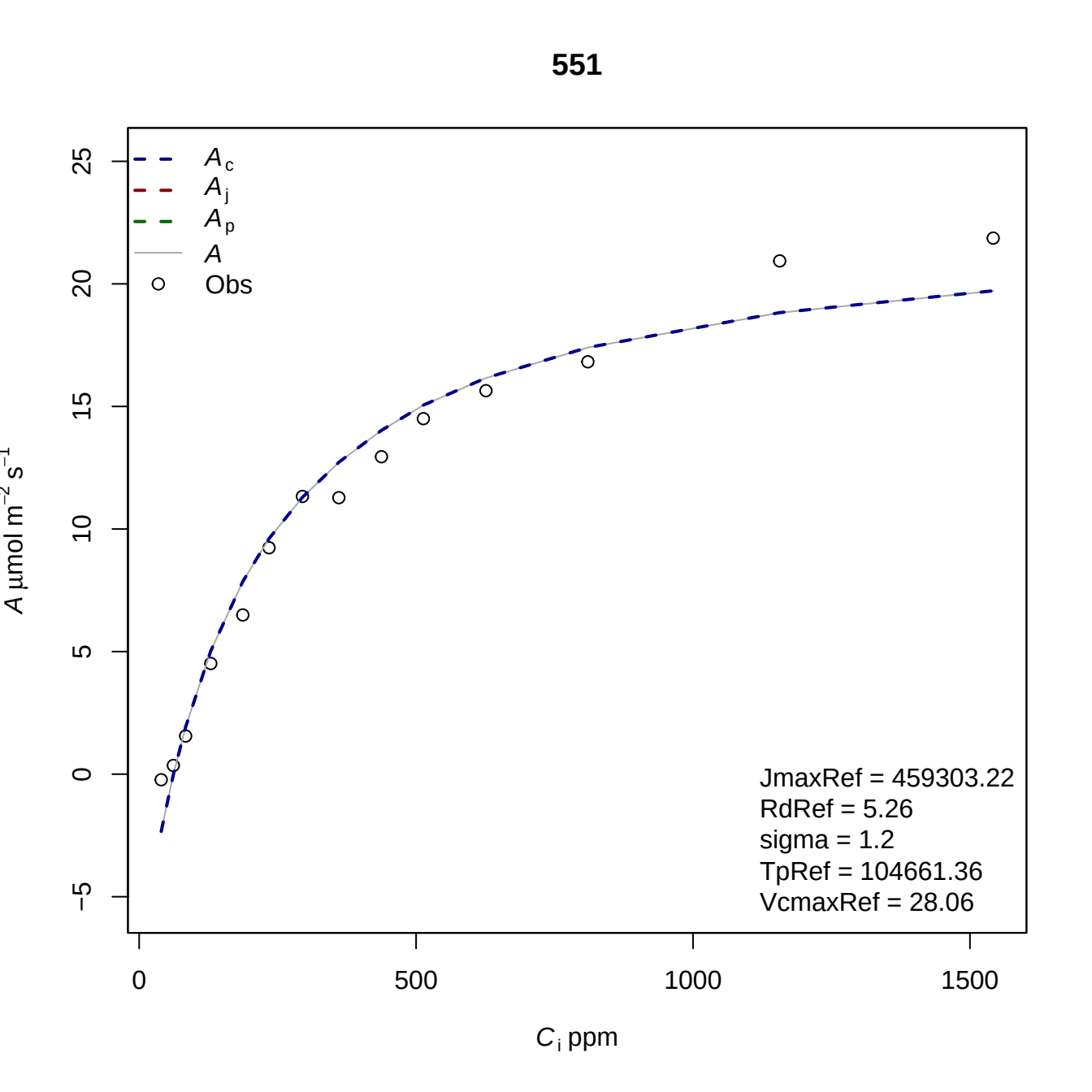

546
A_c
A_j
A_p
A
Obs
JmaxRef = 108452.1
RdRef = 4.27
sigma = 1.08
TpRef = 109325.97
VcmaxRef = 22.23
A µmol m^{-2} s^{-1}
C_i ppm
20
15
10
5
0
−5
0
500
1000
1500

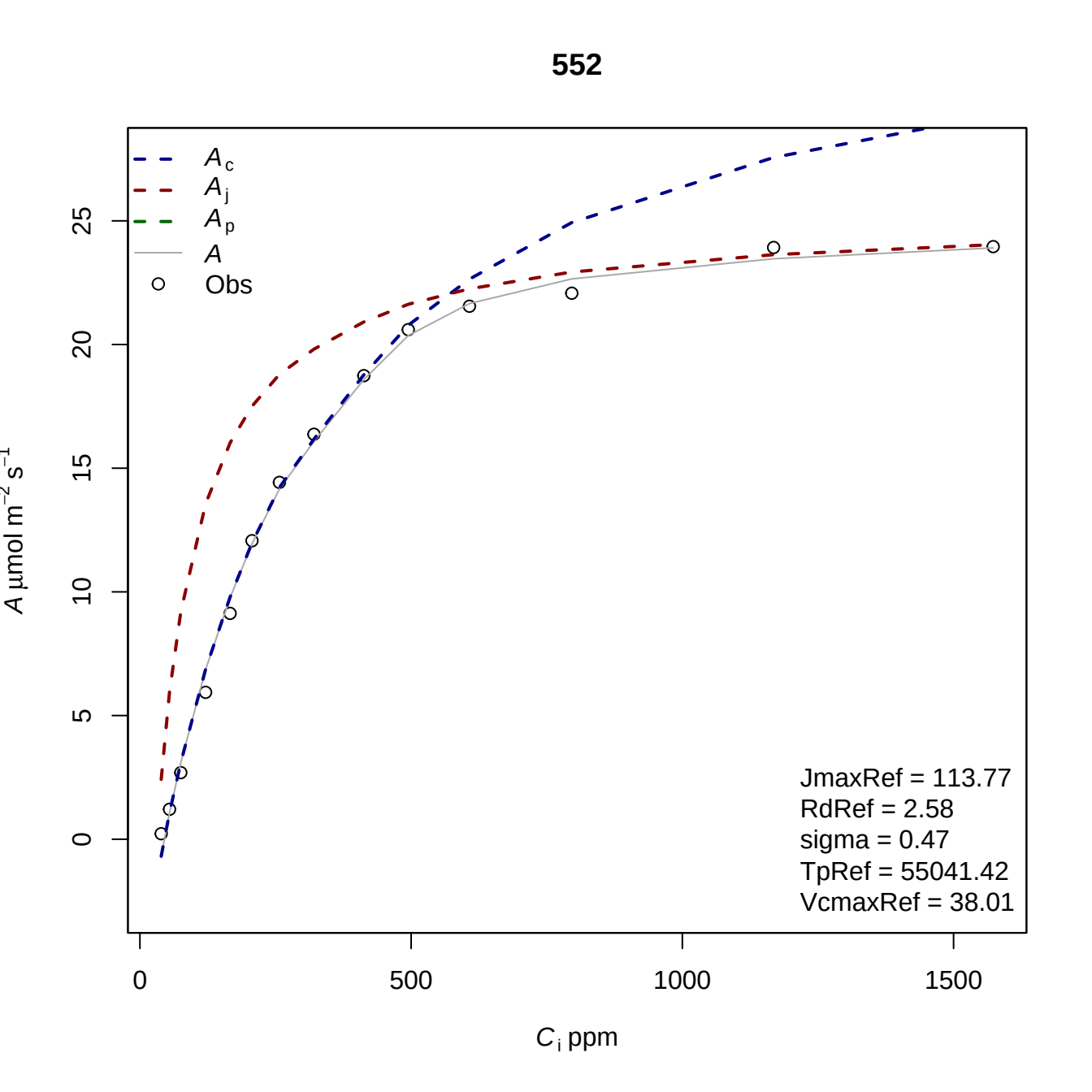

547
A_c
A_j
A_p
A
Obs
JmaxRef = 11882.27
RdRef = 4.57
sigma = 0.99
TpRef = 6726
VcmaxRef = 23.37
A µmol m^{-2} s^{-1}
C_i ppm
20
15
10
5
0
−5
0
500
1000
1500

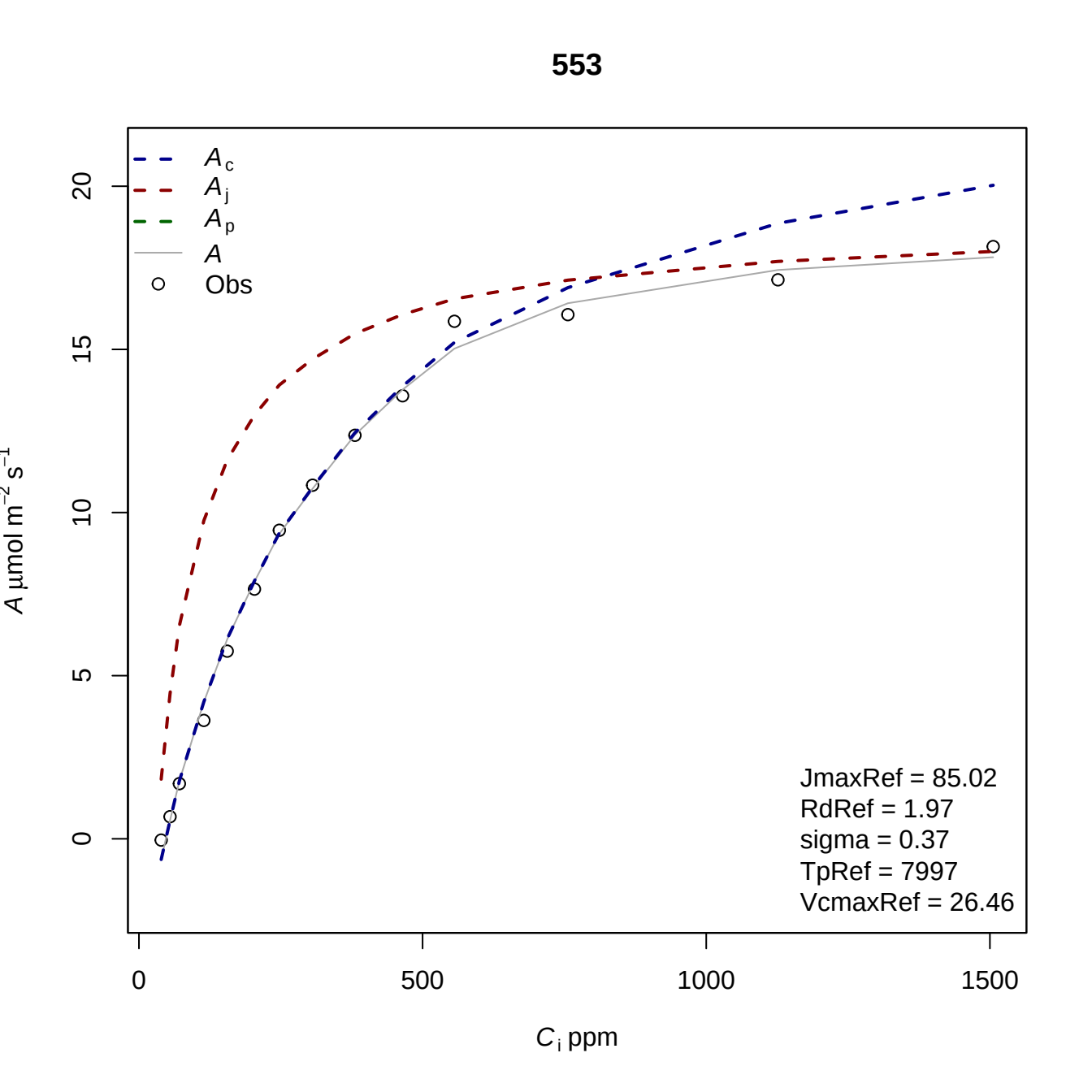

548
A_c
A_j
A_p
A
Obs
JmaxRef = 151105.58
RdRef = 4.01
sigma = 0.96
TpRef = 49287.41
VcmaxRef = 23.94
A µmol m^{-2} s^{-1}
C_i ppm
−5
0
5
10
15
20
0
500
1000
1500

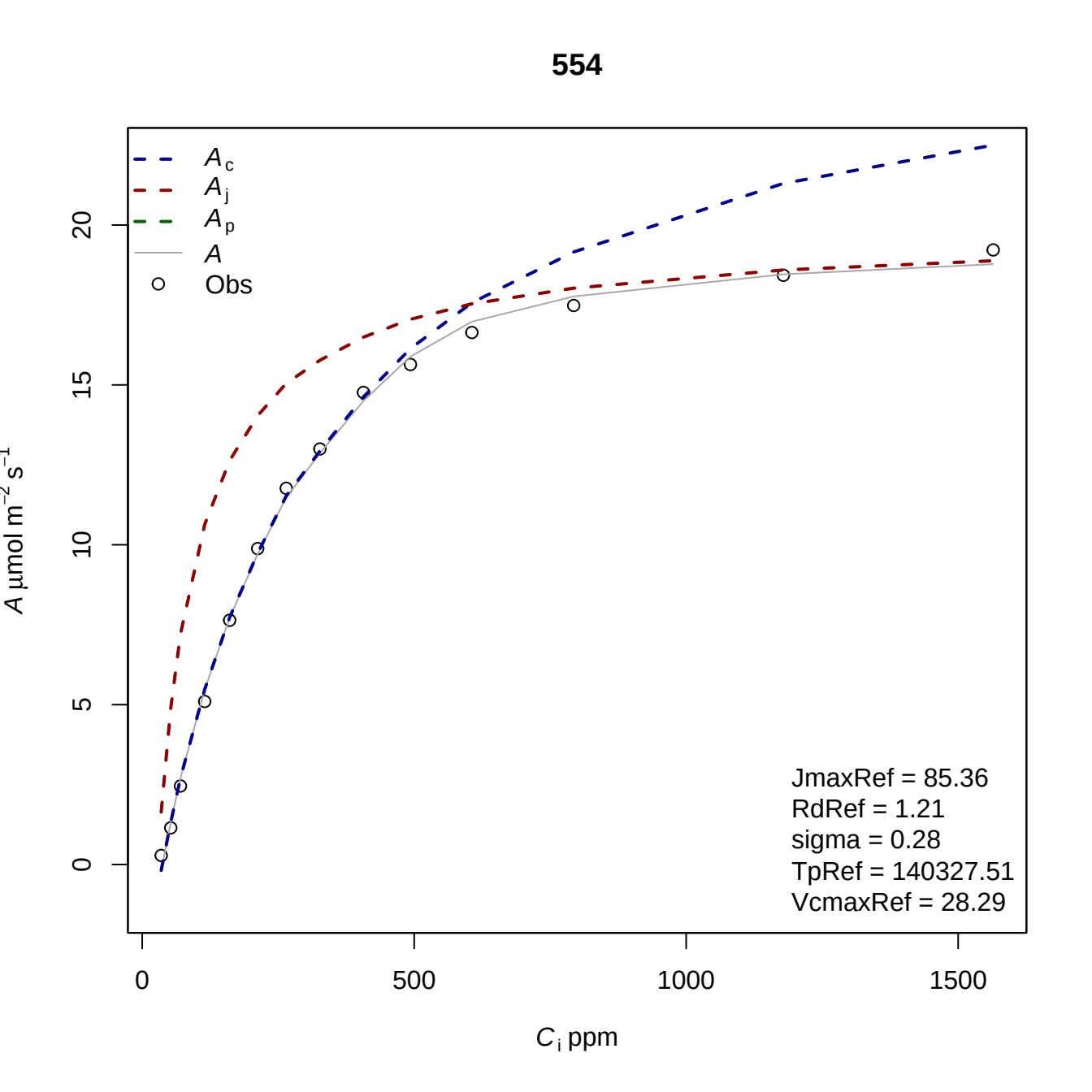

549
A_c
A_j
A_p
A
Obs
JmaxRef = 140447.12
RdRef = 5.11
sigma = 1.05
TpRef = 29617.56
VcmaxRef = 22.84
A µmol m^{-2} s^{-1}
C_i ppm
20
15
10
5
0
−5
0
500
1000
1500

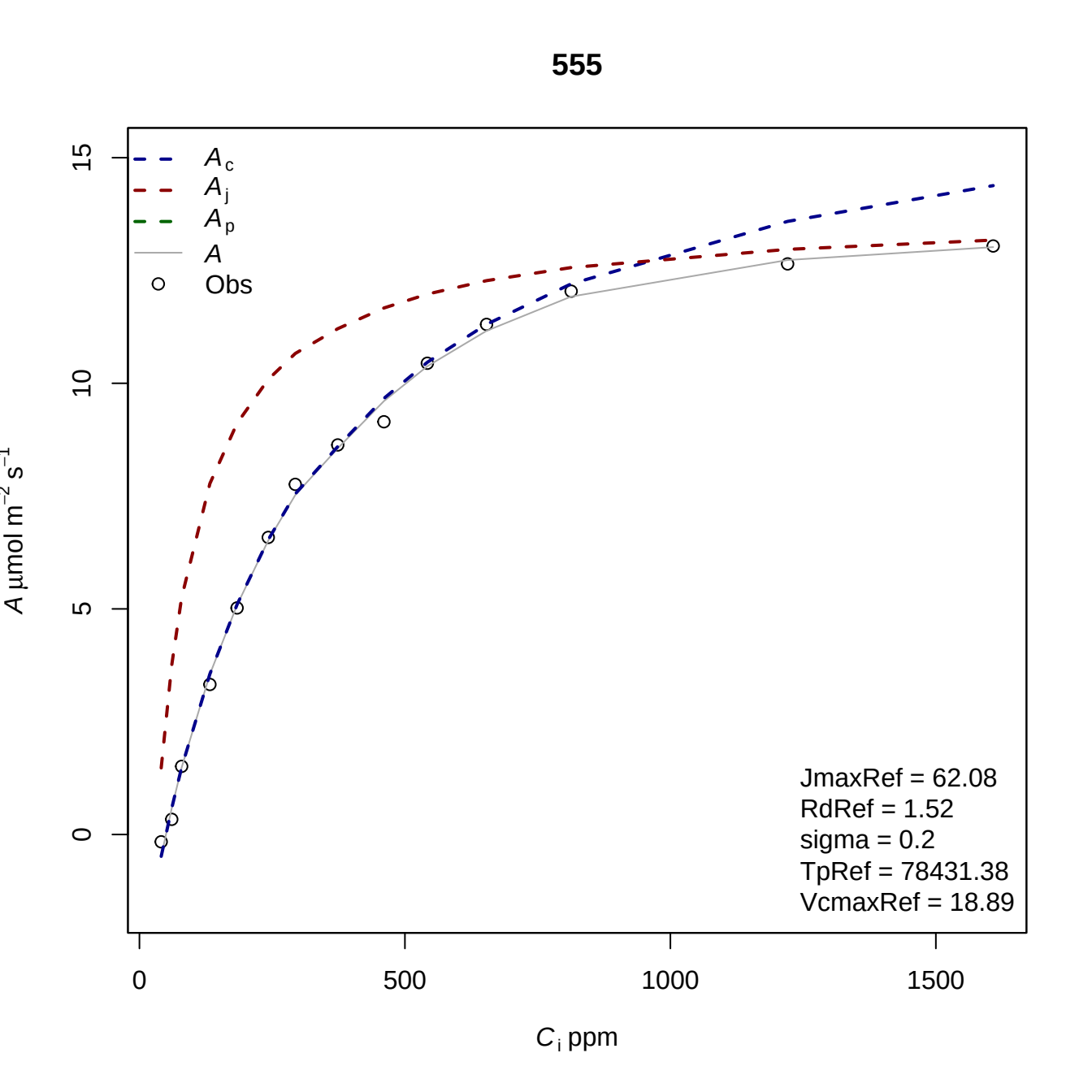

550
A_c
A_j
A_p
A
Obs
JmaxRef = 565826.85
RdRef = 5.88
sigma = 1.11
TpRef = 93254.18
VcmaxRef = 25.07
A µmol m^{-2} s^{-1}
C_i ppm
−5
0
5
10
15
20
0
500
1000
1500

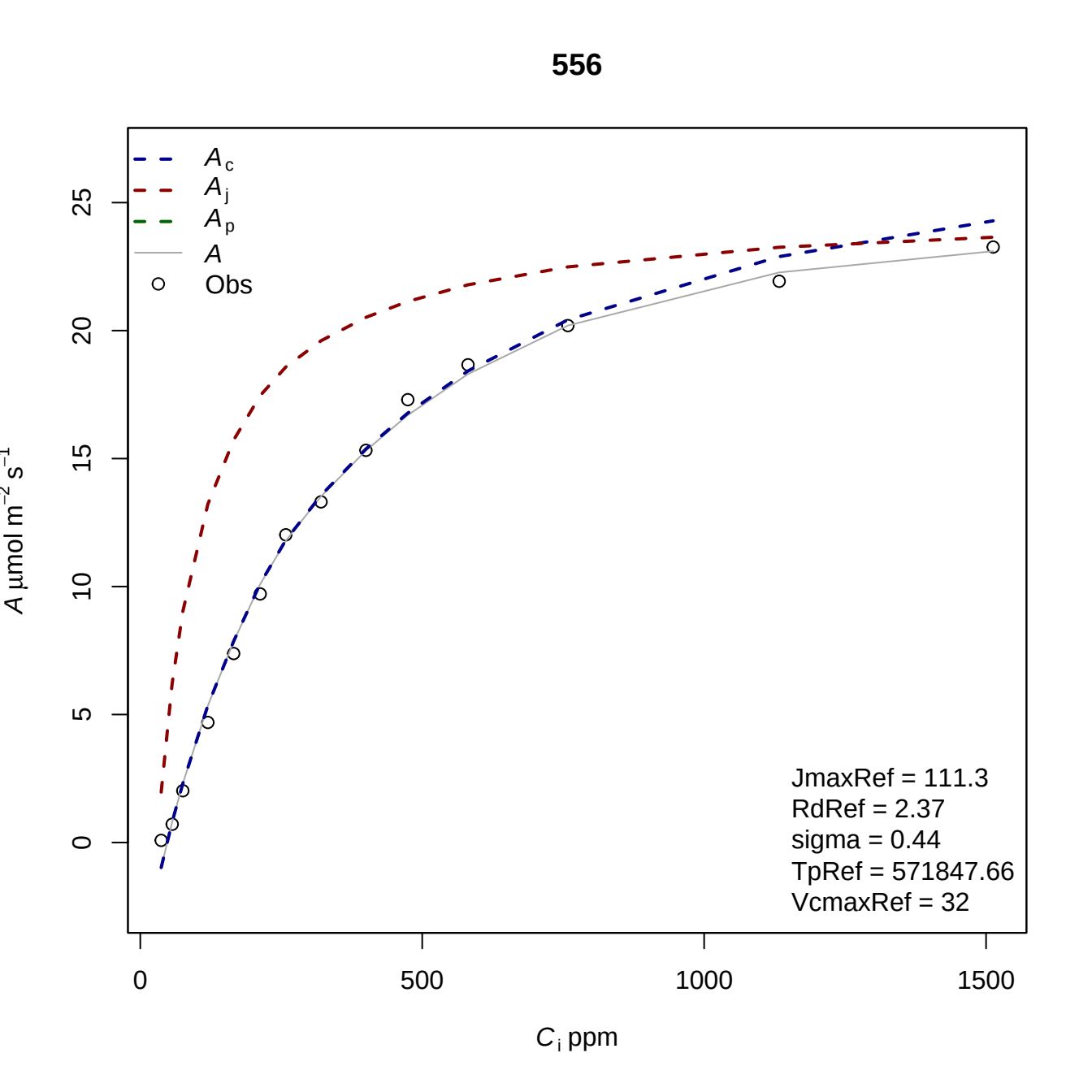

551
A_c
A_j
A_p
A
Obs
JmaxRef = 459303.22
RdRef = 5.26
sigma = 1.2
TpRef = 104661.36
VcmaxRef = 28.06
A µmol m^{-2} s^{-1}
C_i ppm
25
20
15
10
5
0
−5
0
500
1000
1500

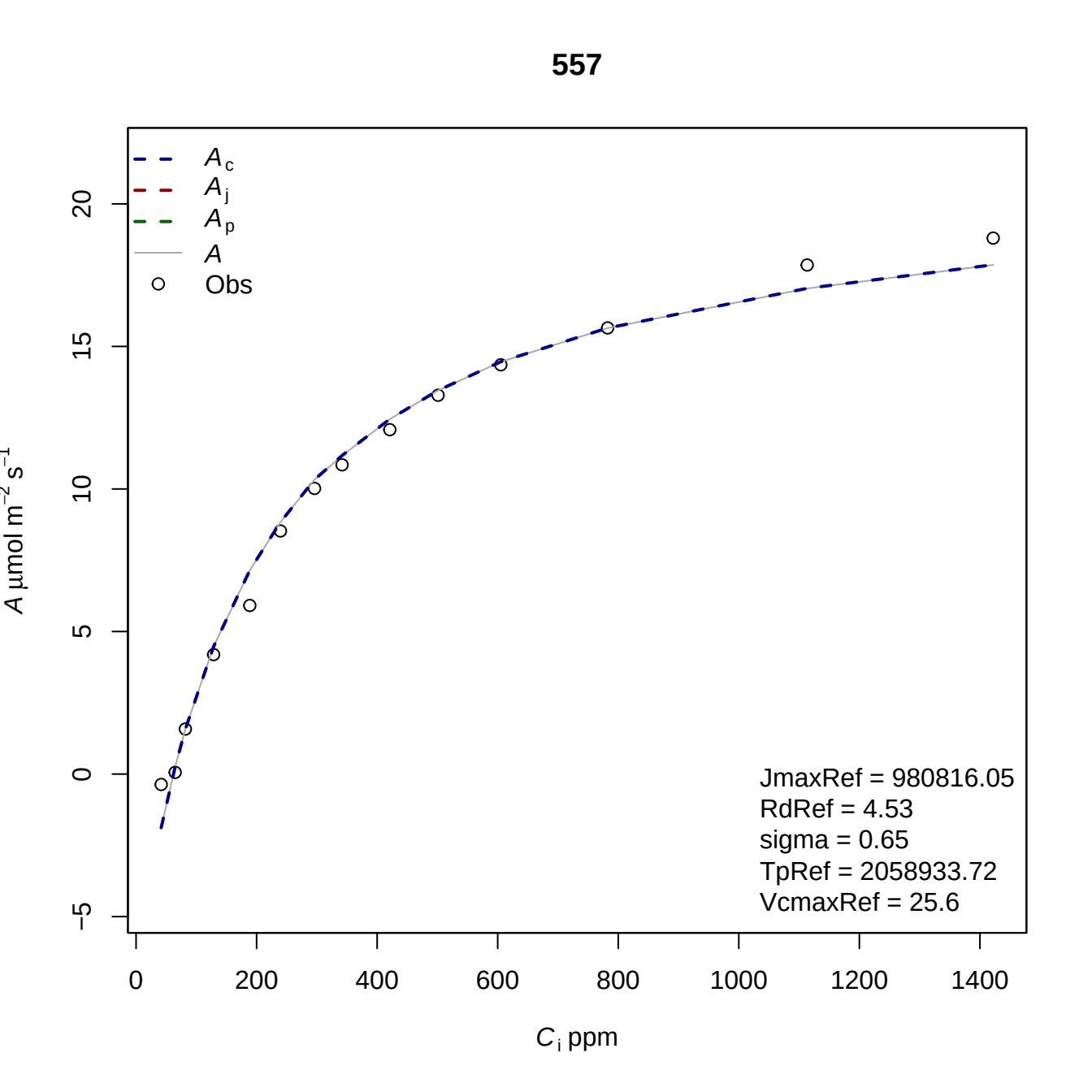

552
A_c
A_j
A_p
A
Obs
JmaxRef = 113.77
RdRef = 2.58
sigma = 0.47
TpRef = 55041.42
VcmaxRef = 38.01
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
500
1000
1500

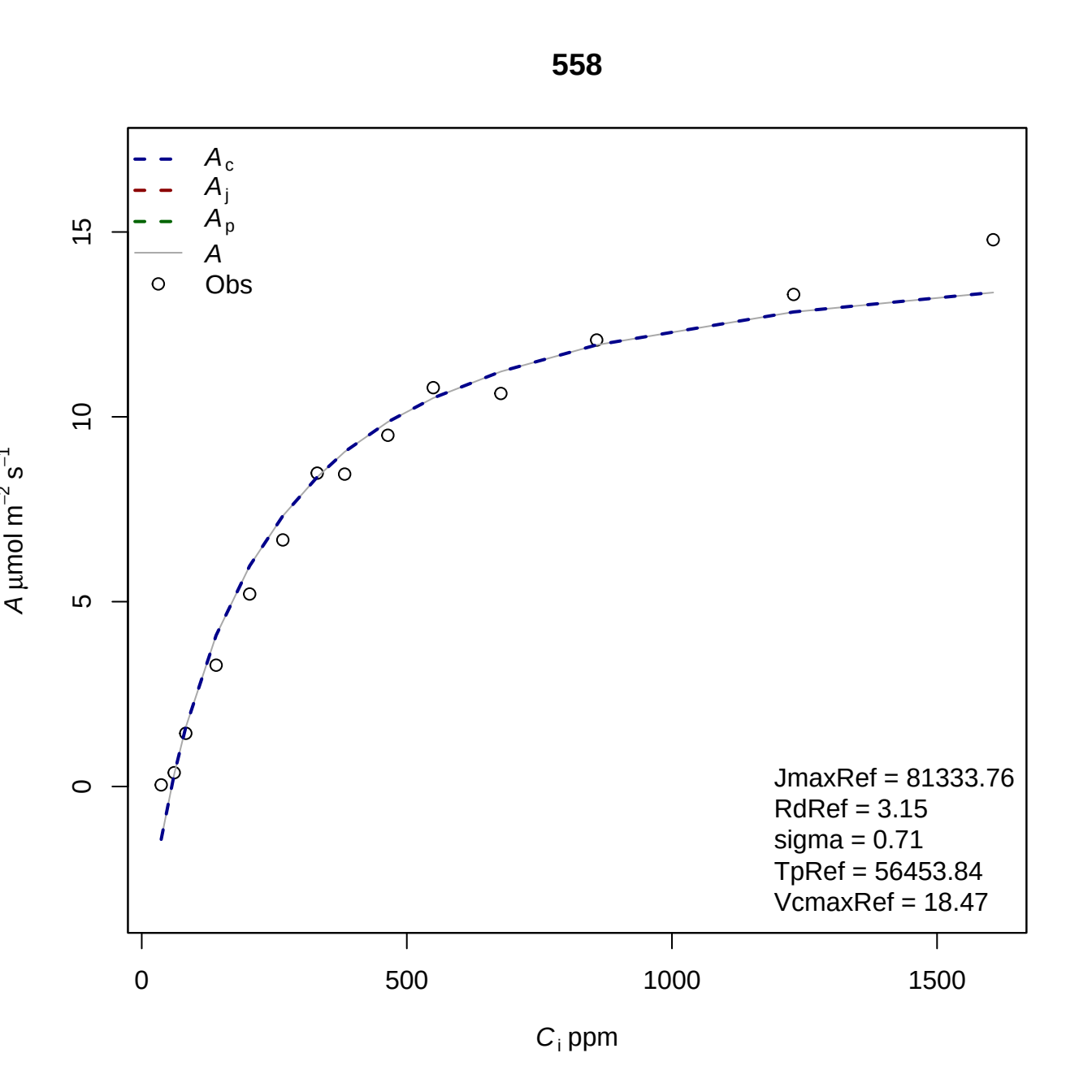

553
A_c
A_j
A_p
A
Obs
JmaxRef = 85.02
RdRef = 1.97
sigma = 0.37
TpRef = 7997
VcmaxRef = 26.46
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
500
1000
1500

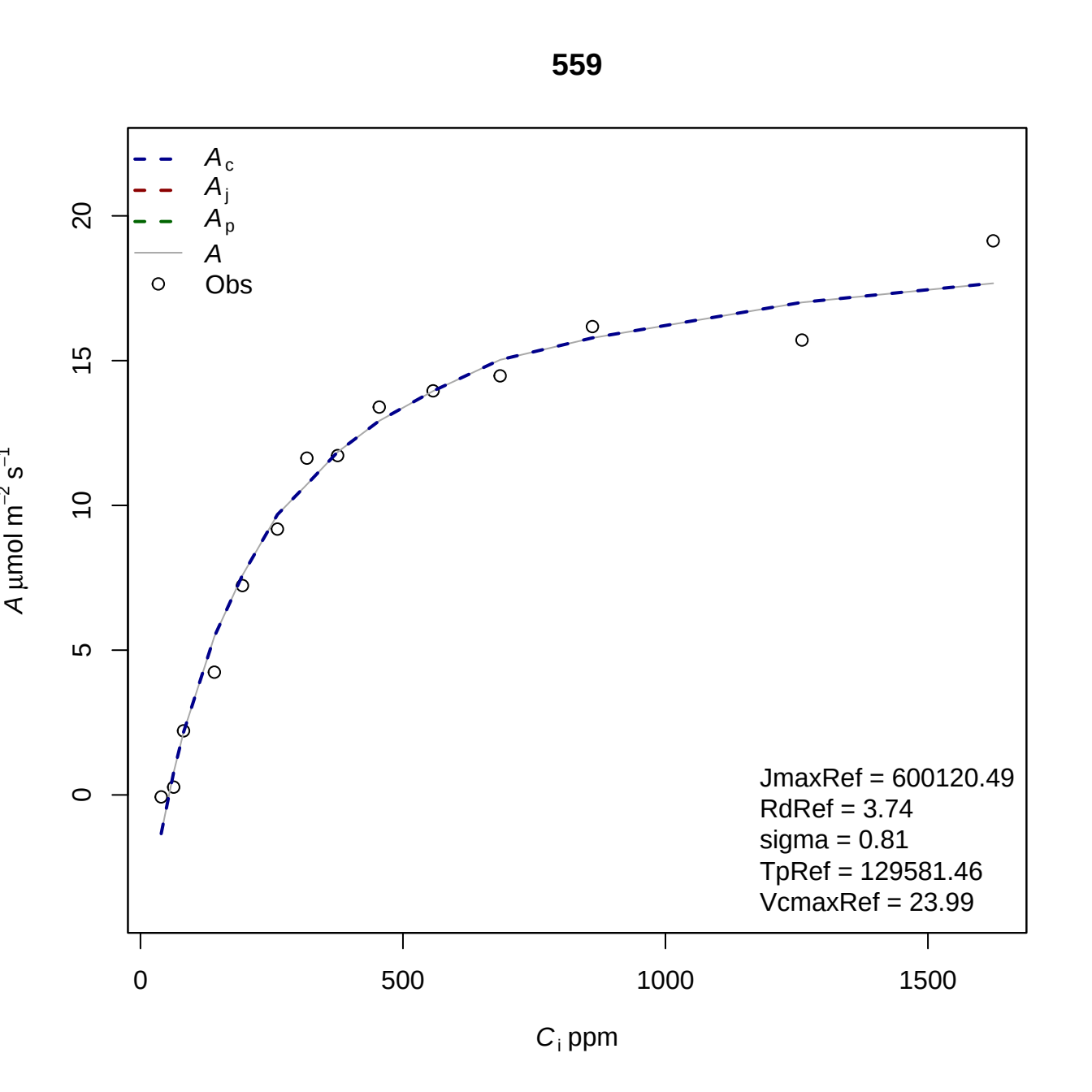

554
A_c
A_j
A_p
A
Obs
JmaxRef = 85.36
RdRef = 1.21
sigma = 0.28
TpRef = 140327.51
VcmaxRef = 28.29
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
500
1000
1500

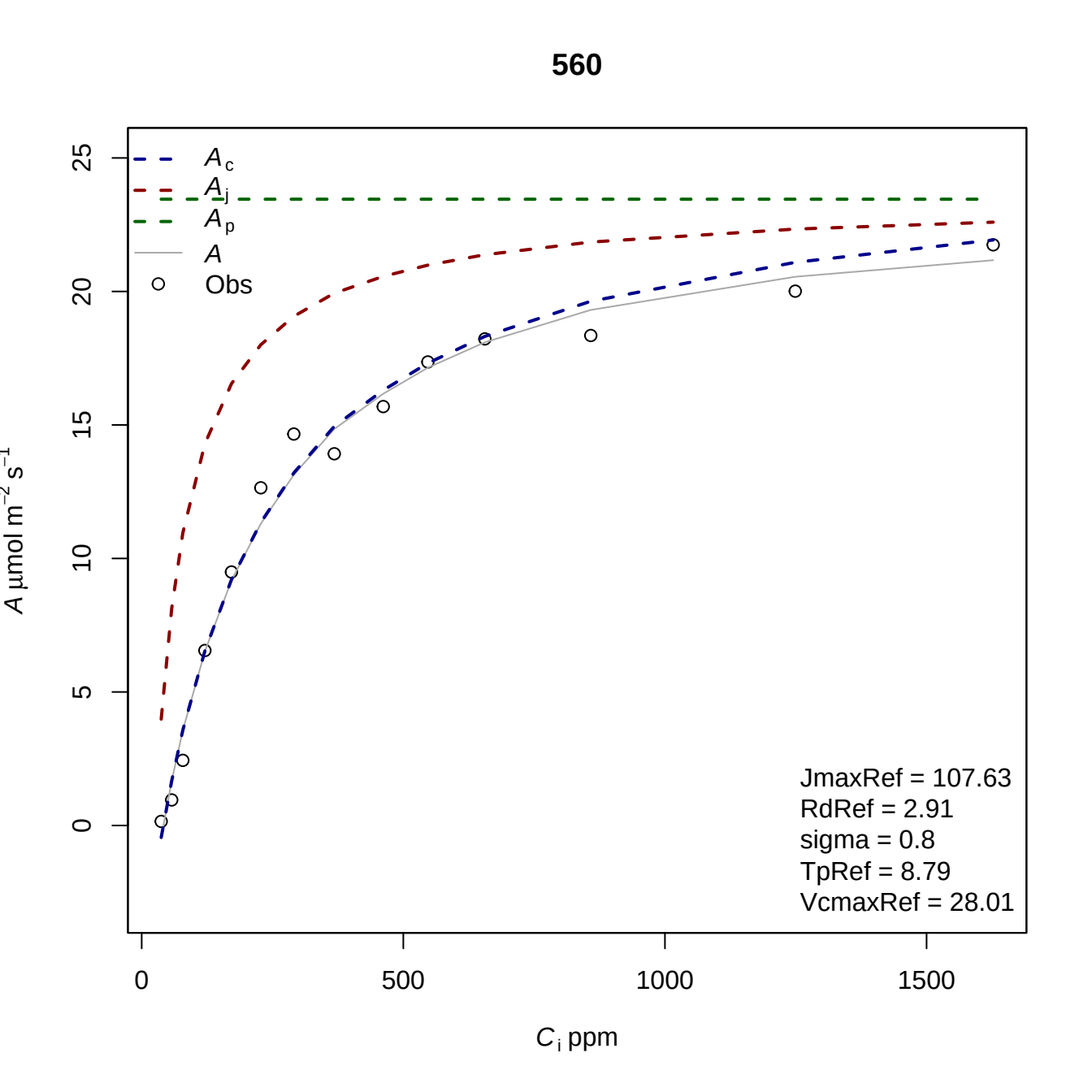

555
A_c
A_j
A_p
A
Obs
JmaxRef = 62.08
RdRef = 1.52
sigma = 0.2
TpRef = 78431.38
VcmaxRef = 18.89
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
500
1000
1500

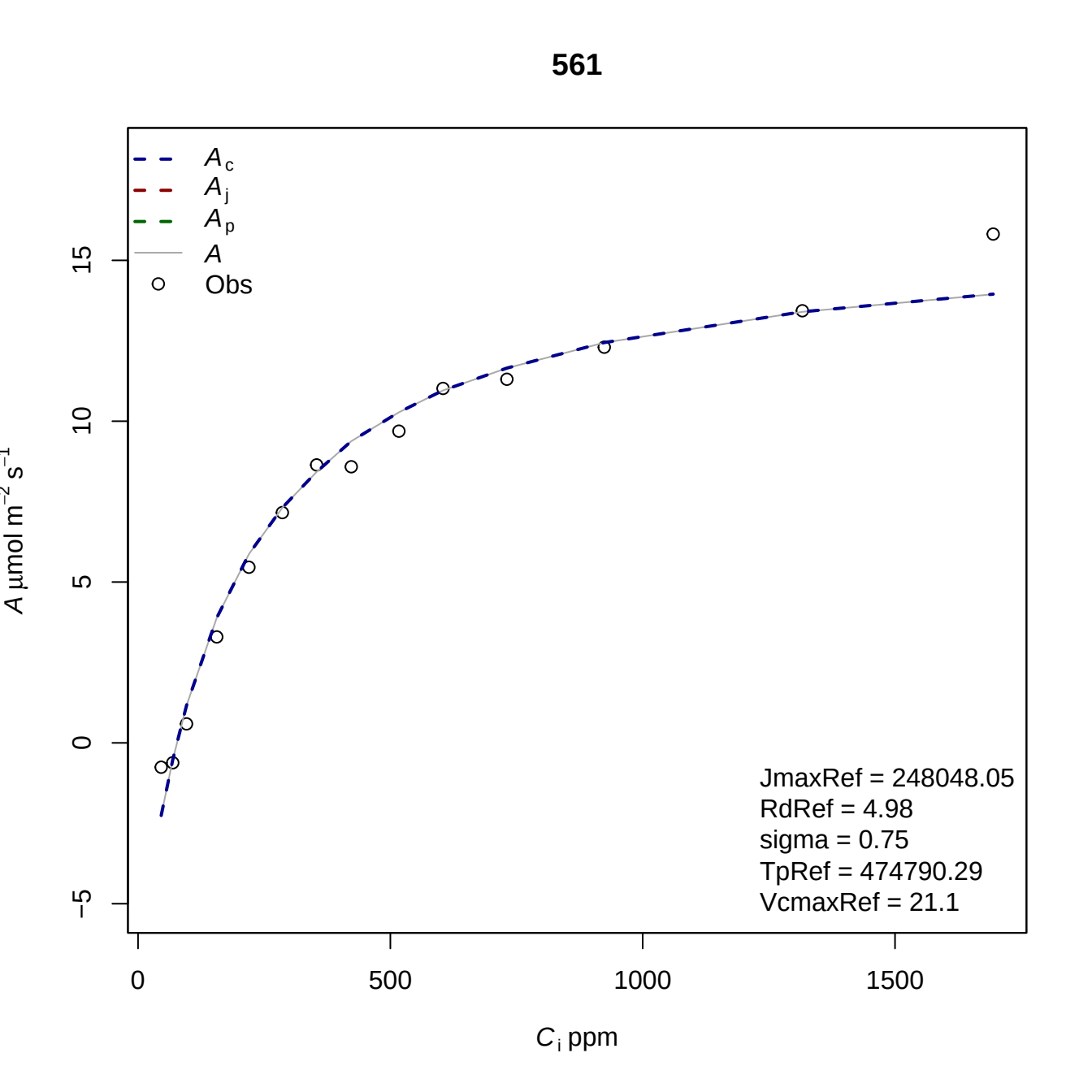

556
A_c
A_j
A_p
A
Obs
JmaxRef = 111.3
RdRef = 2.37
sigma = 0.44
TpRef = 571847.66
VcmaxRef = 32
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
500
1000
1500

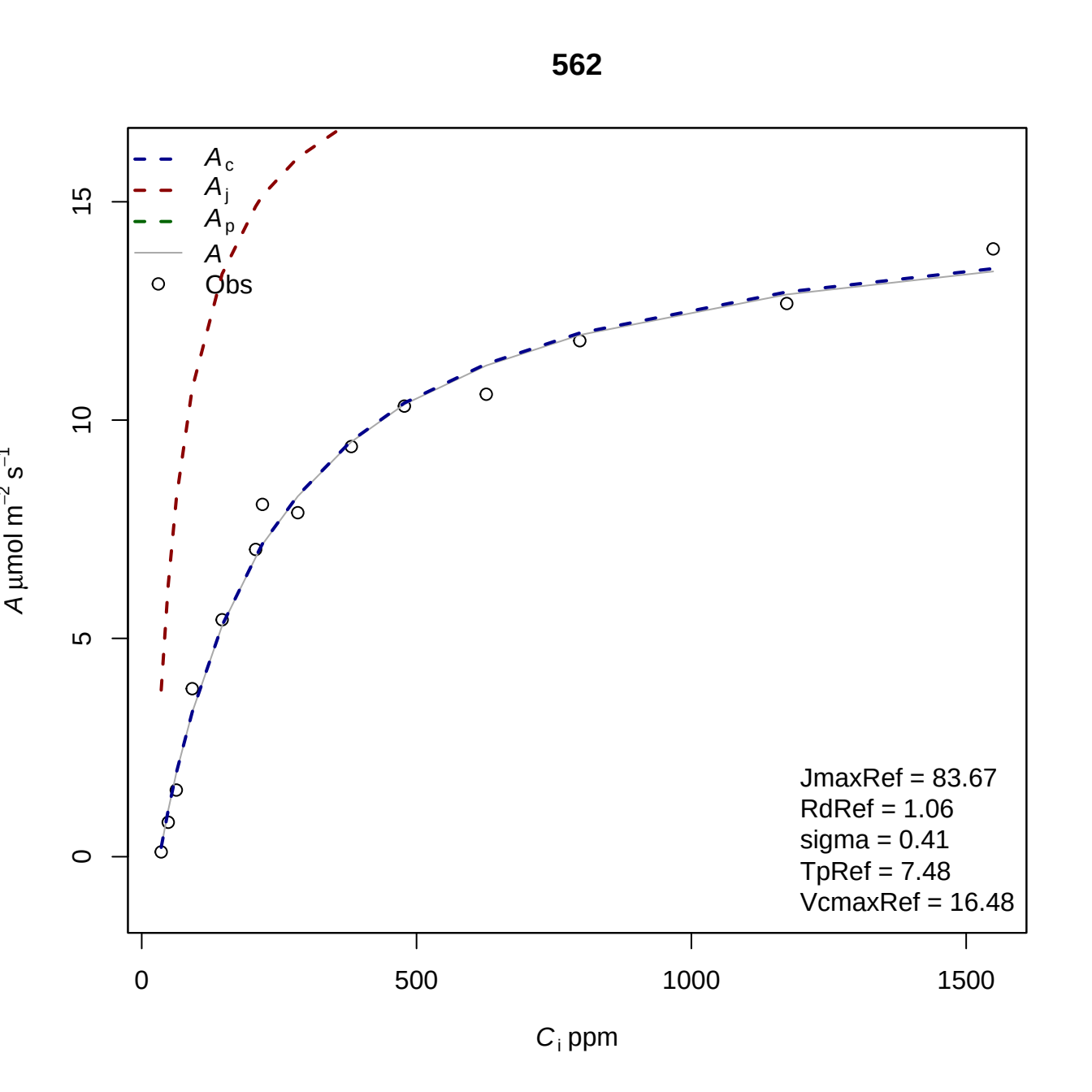

557
A_c
A_j
A_p
A
Obs
JmaxRef = 980816.05
RdRef = 4.53
sigma = 0.65
TpRef = 2058933.72
VcmaxRef = 25.6
A µmol m^{-2} s^{-1}
C_i ppm
−5
0
5
10
15
20
0
200
400
600
800
1000
1200
1400

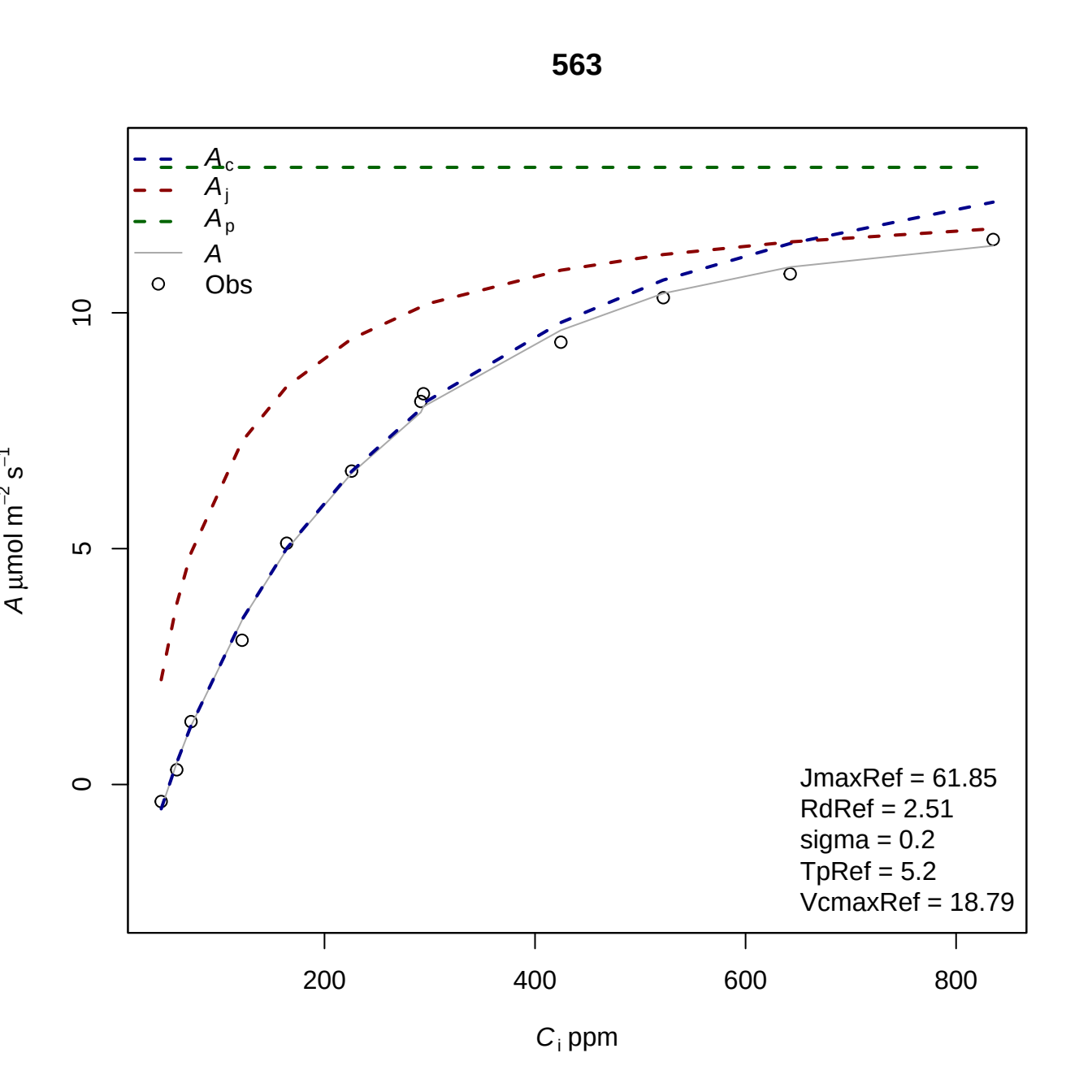

558
A_c
A_j
A_p
A
Obs
JmaxRef = 81333.76
RdRef = 3.15
sigma = 0.71
TpRef = 56453.84
VcmaxRef = 18.47
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
500
1000
1500

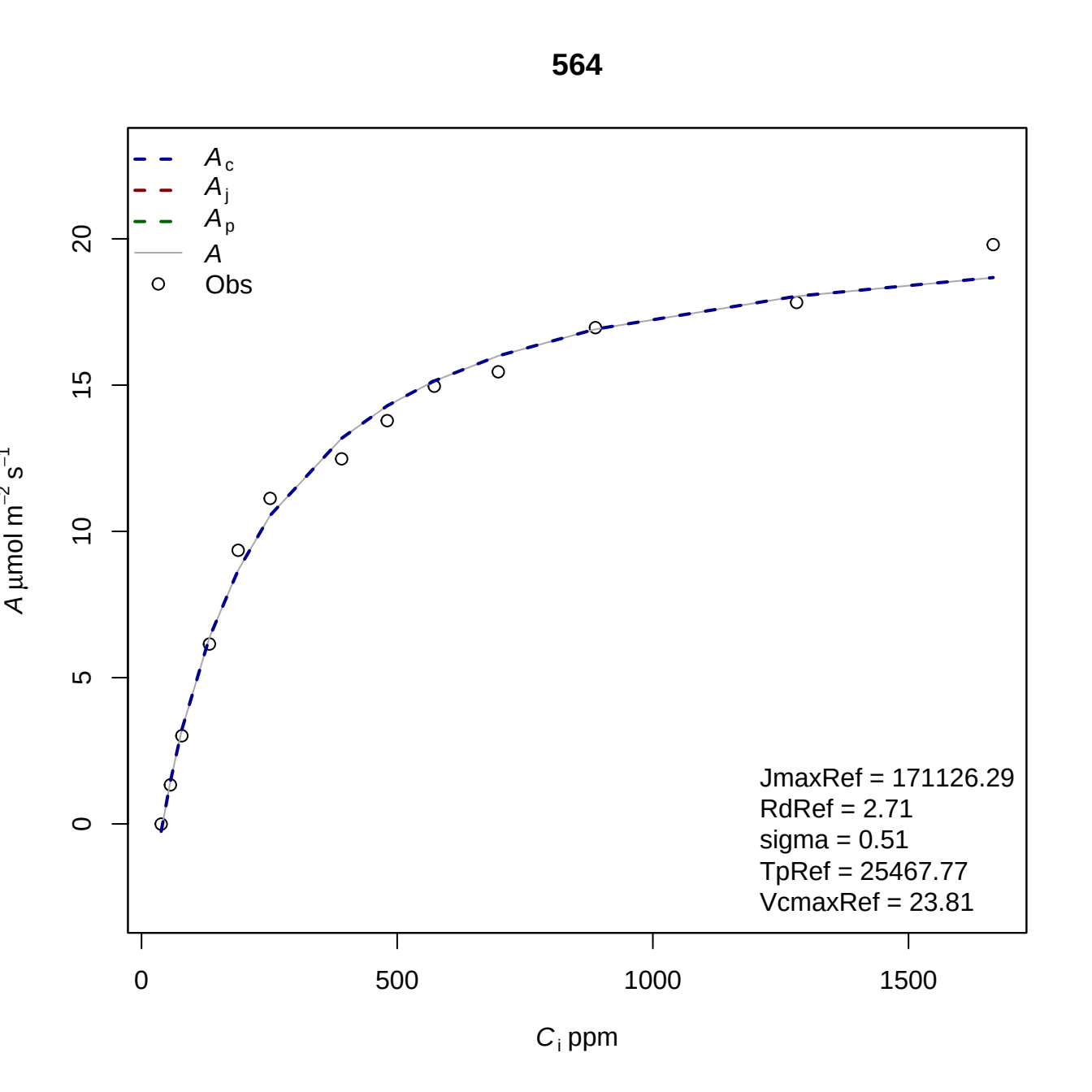

559
A_c
A_j
A_p
A
Obs
JmaxRef = 600120.49
RdRef = 3.74
sigma = 0.81
TpRef = 129581.46
VcmaxRef = 23.99
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
500
1000
1500

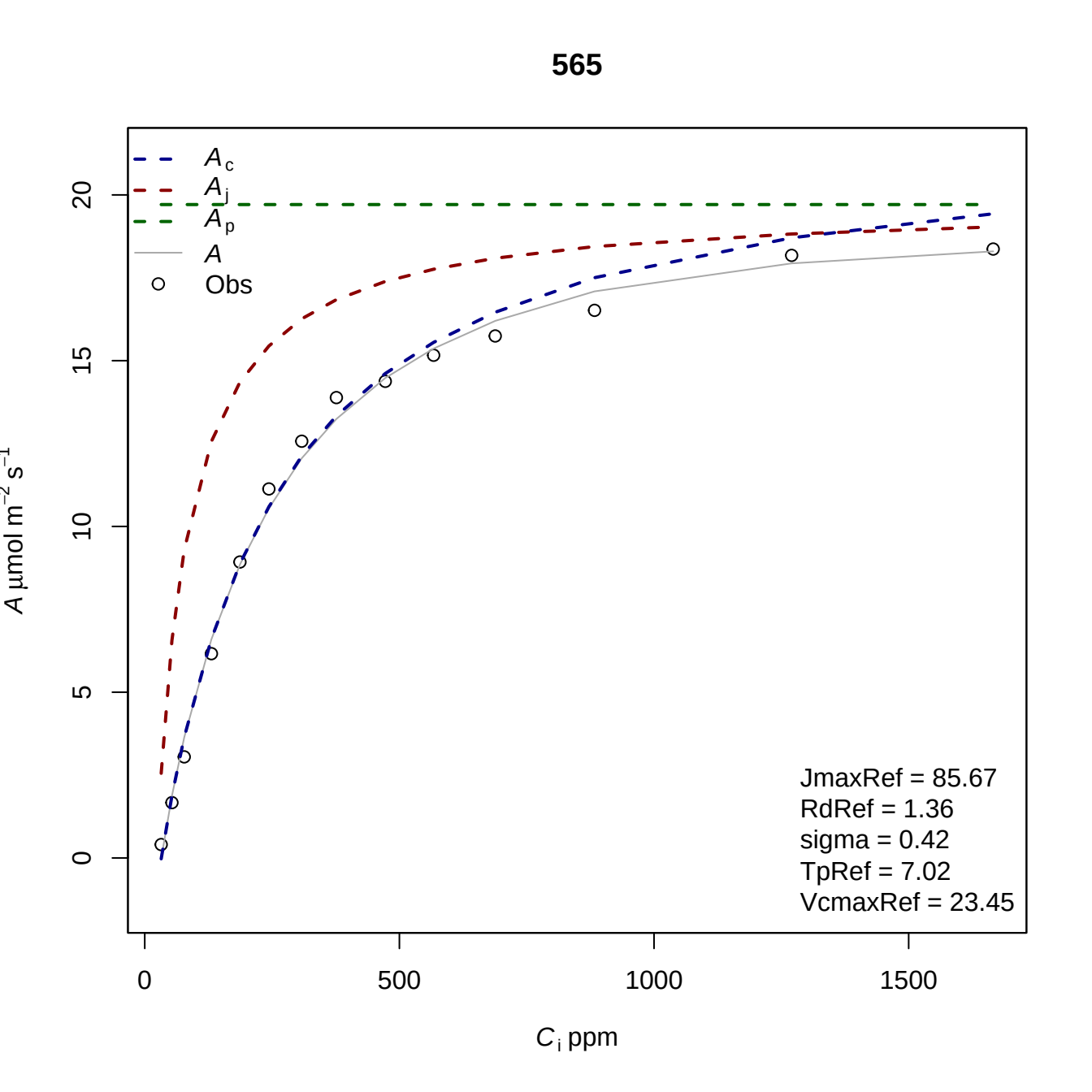

560
A_c
A_j
A_p
A
Obs
JmaxRef = 107.63
RdRef = 2.91
sigma = 0.8
TpRef = 8.79
VcmaxRef = 28.01
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
500
1000
1500

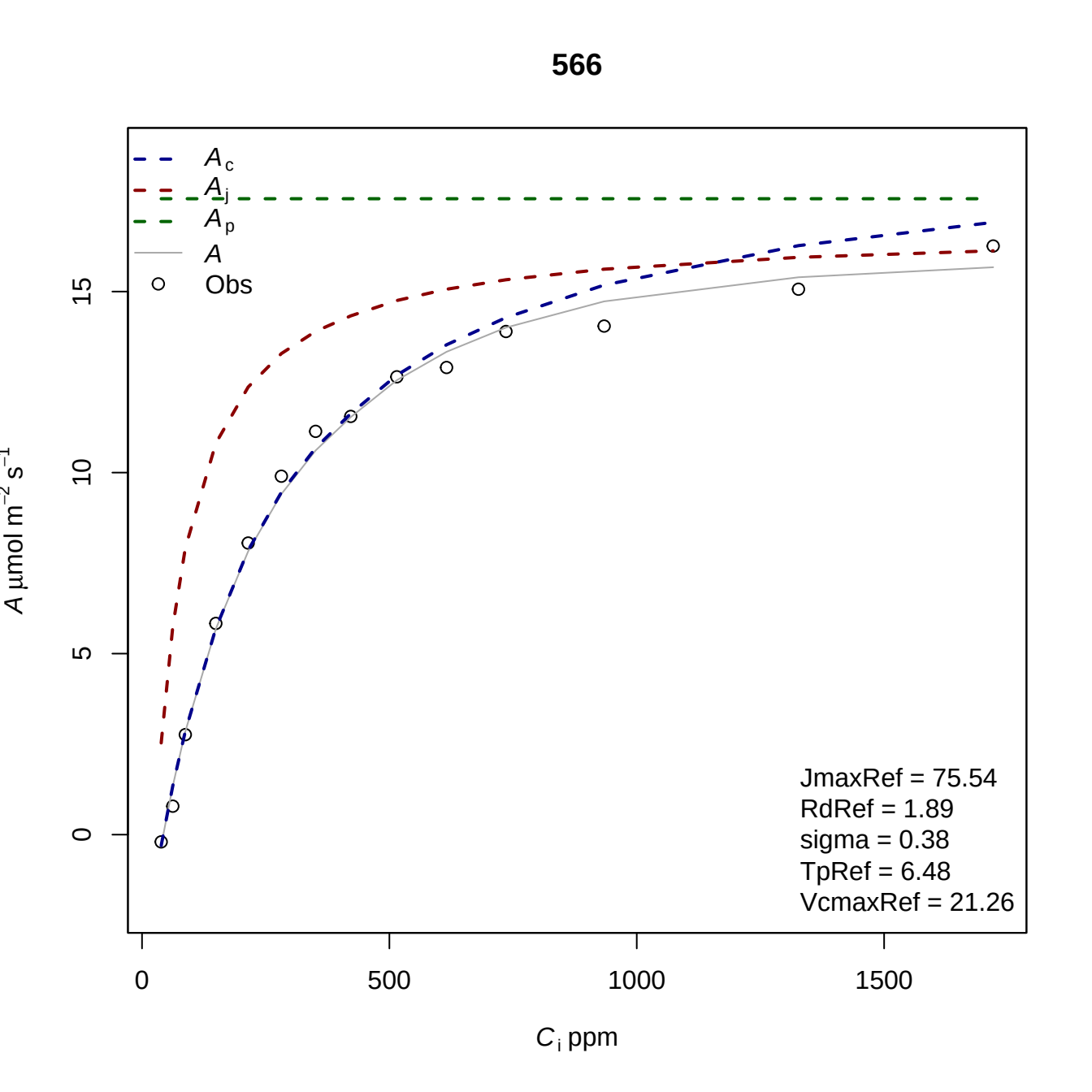

561
A_c
A_j
A_p
A
Obs
JmaxRef = 248048.05
RdRef = 4.98
sigma = 0.75
TpRef = 474790.29
VcmaxRef = 21.1
A µmol m^{-2} s^{-1}
C_i ppm
−5
0
5
10
15
0
500
1000
1500

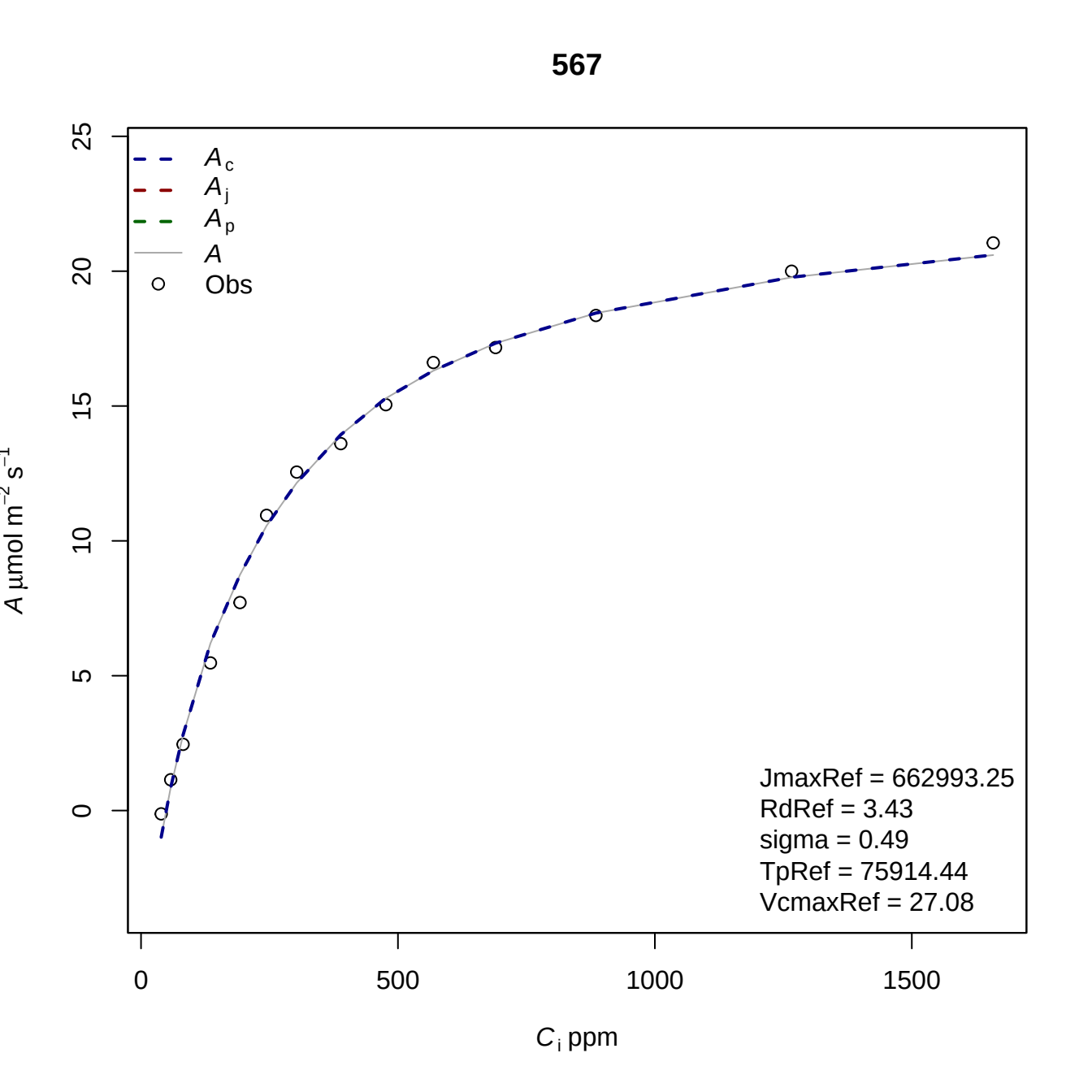

562
A_c
A_j
A_p
A
Obs
JmaxRef = 83.67
RdRef = 1.06
sigma = 0.41
TpRef = 7.48
VcmaxRef = 16.48
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
500
1000
1500

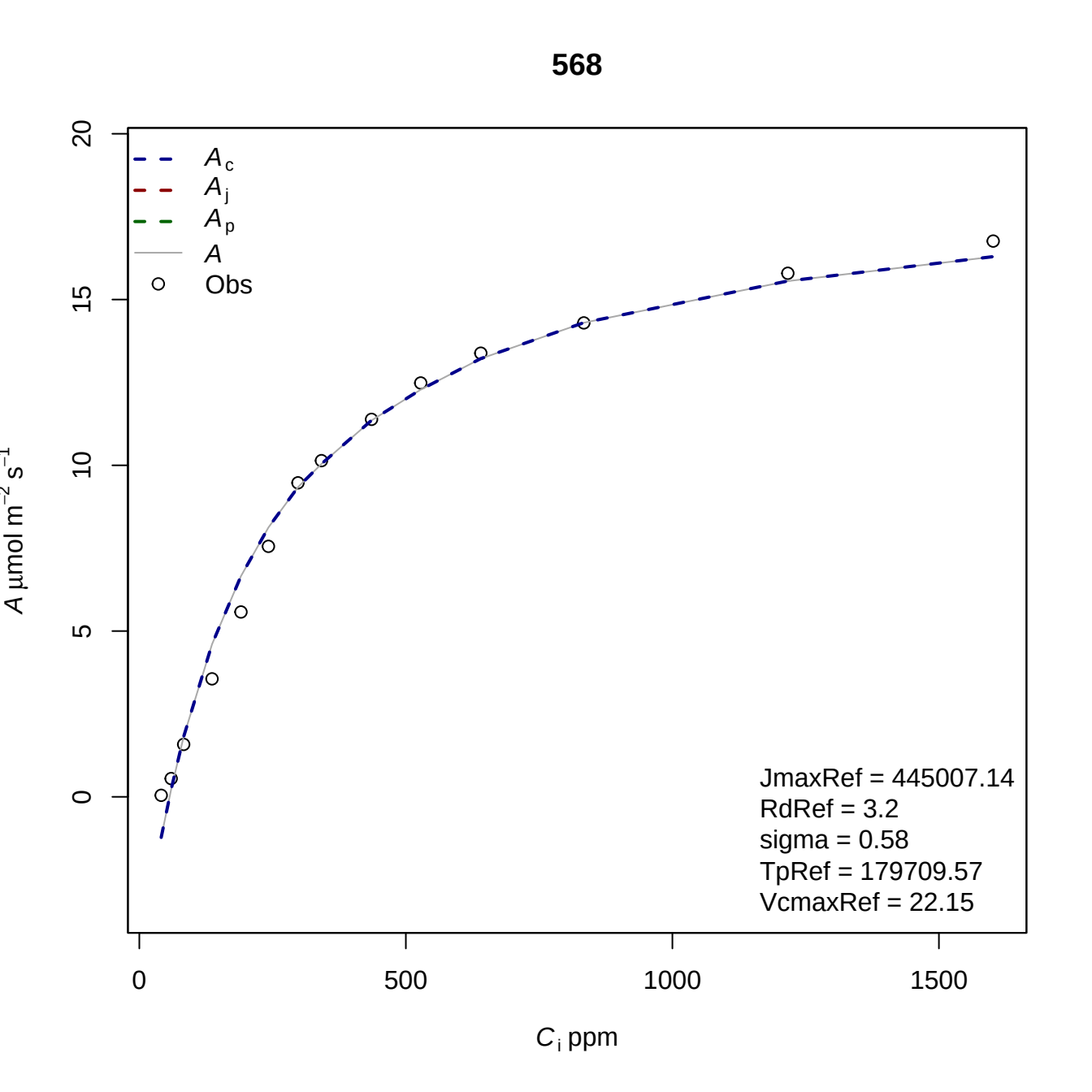

563
A_c
A_j
A_p
A
Obs
JmaxRef = 61.85
RdRef = 2.51
sigma = 0.2
TpRef = 5.2
VcmaxRef = 18.79
0
5
10
200
400
600
800
A µmol m^{-2} s^{-1}
C_i ppm

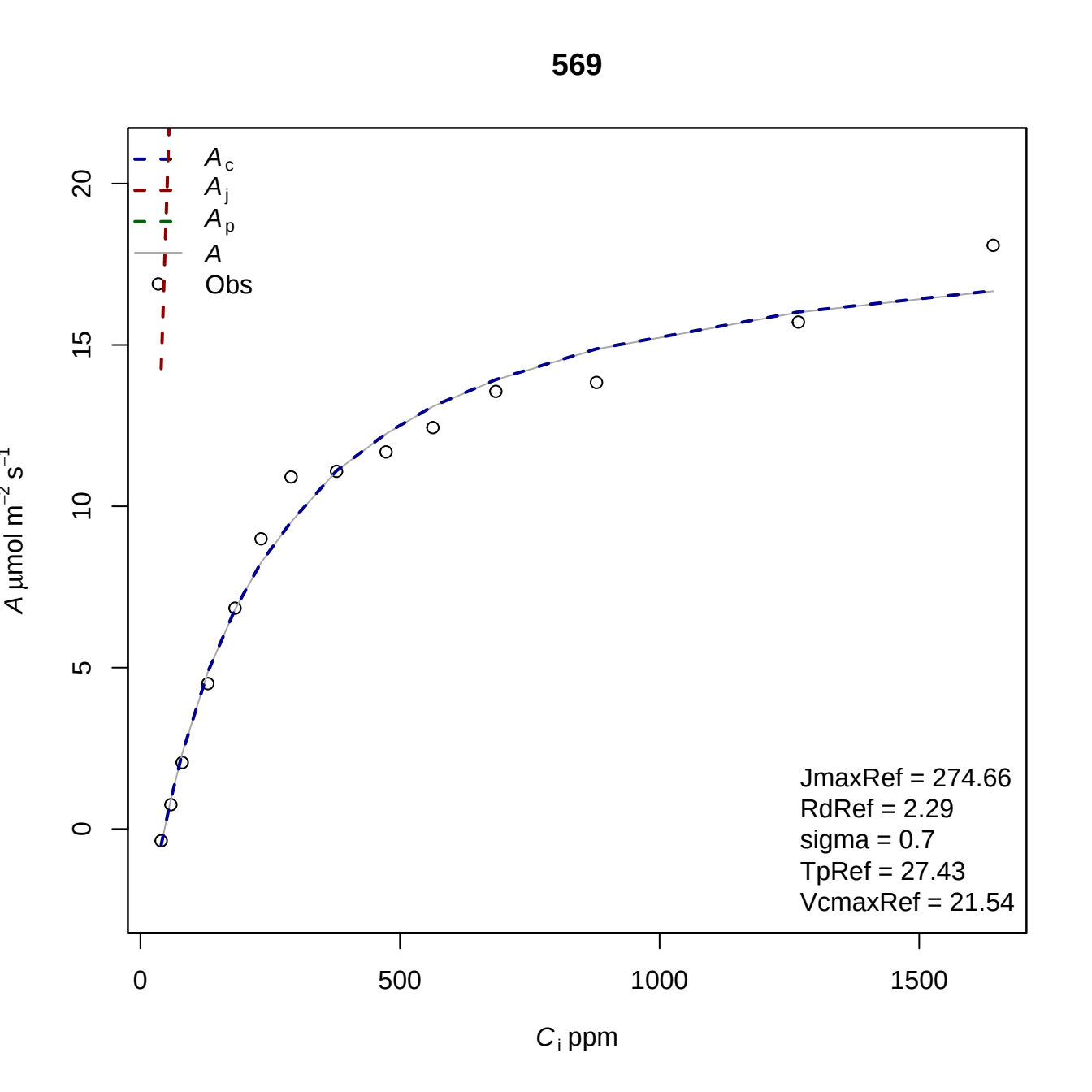

564
A_c
A_j
A_p
A
Obs
JmaxRef = 171126.29
RdRef = 2.71
sigma = 0.51
TpRef = 25467.77
VcmaxRef = 23.81
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
500
1000
1500

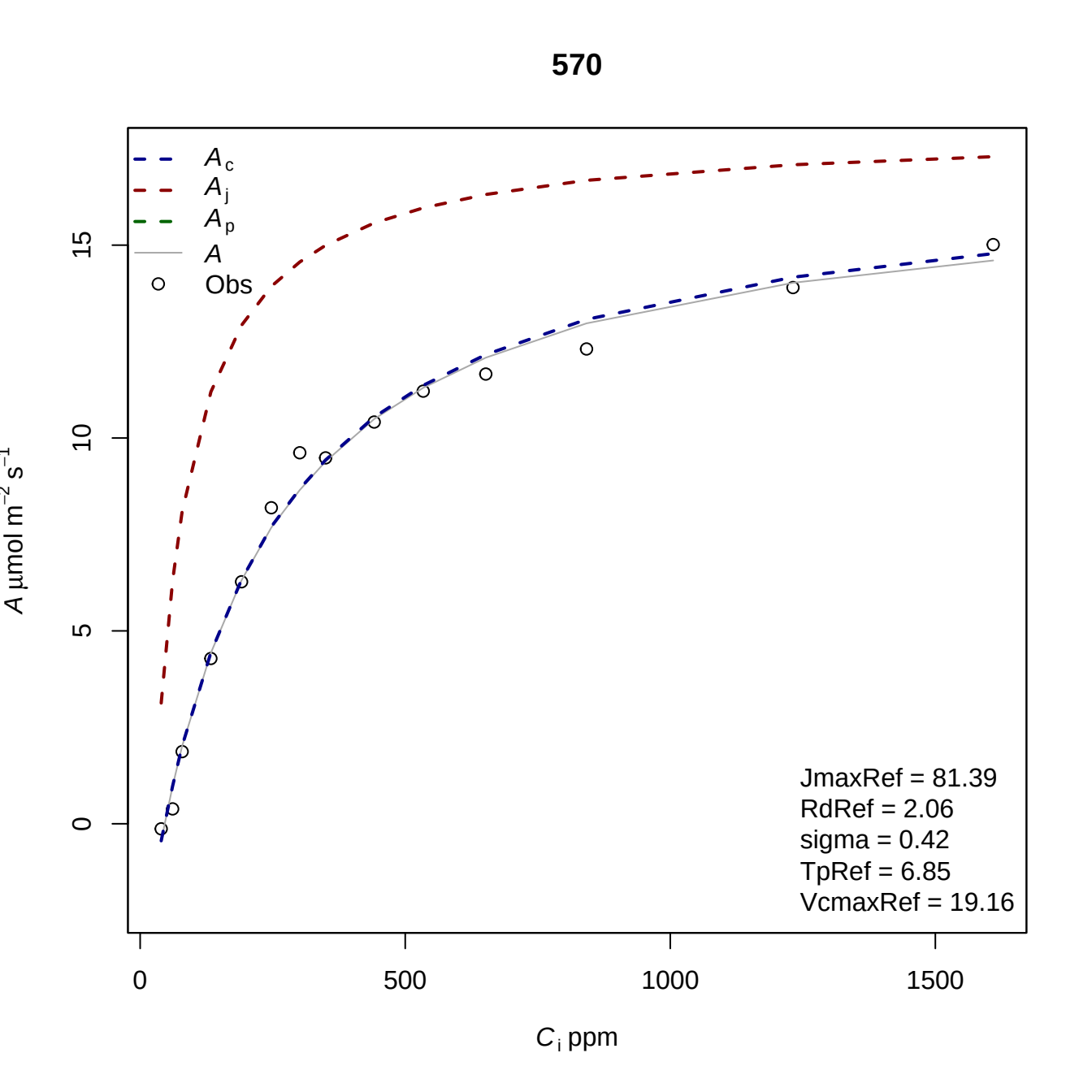

565
A_c
A_j
A_p
A
Obs
JmaxRef = 85.67
RdRef = 1.36
sigma = 0.42
TpRef = 7.02
VcmaxRef = 23.45
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
C_i ppm

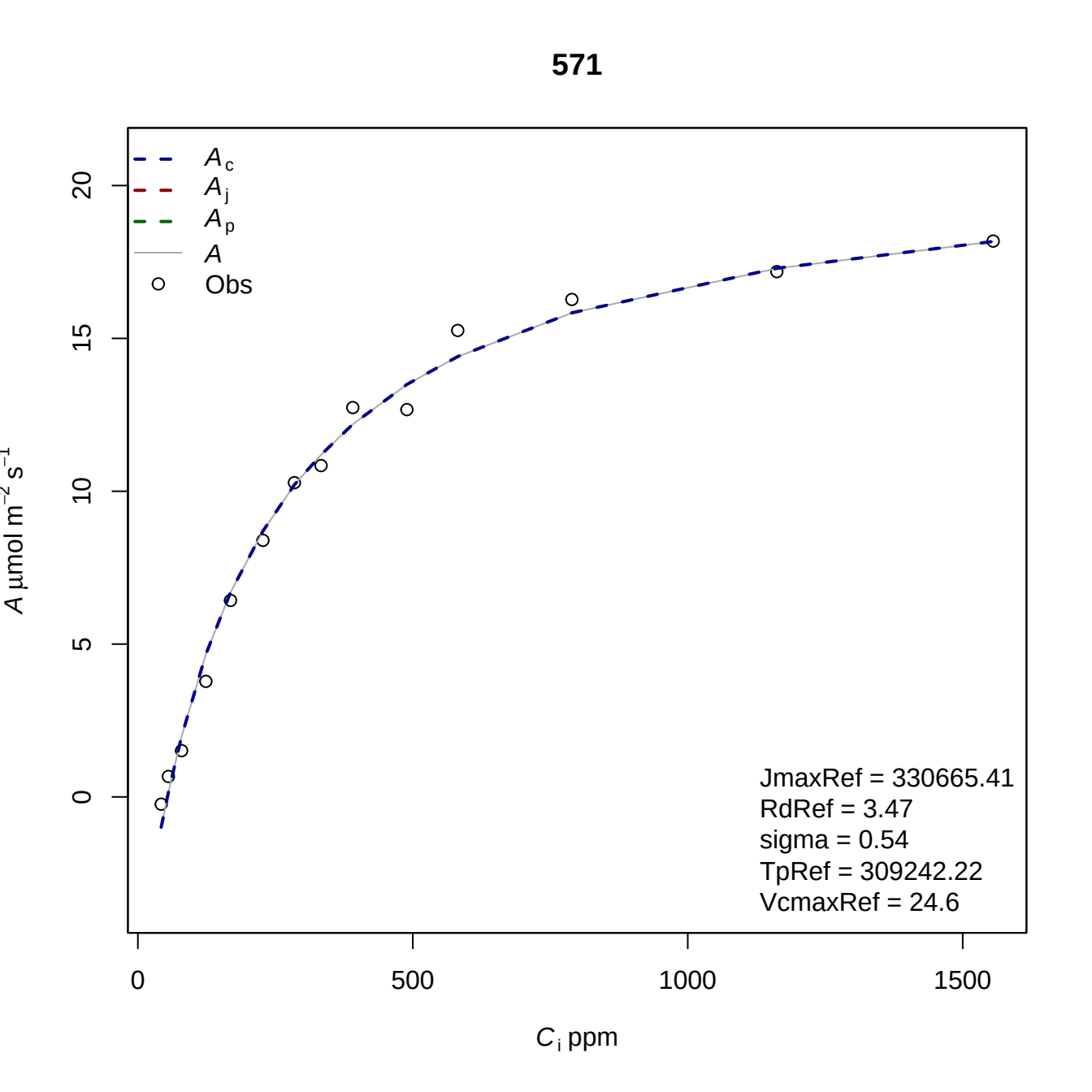

566
A_c
A_j
A_p
A
Obs
JmaxRef = 75.54
RdRef = 1.89
sigma = 0.38
TpRef = 6.48
VcmaxRef = 21.26
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
500
1000
1500

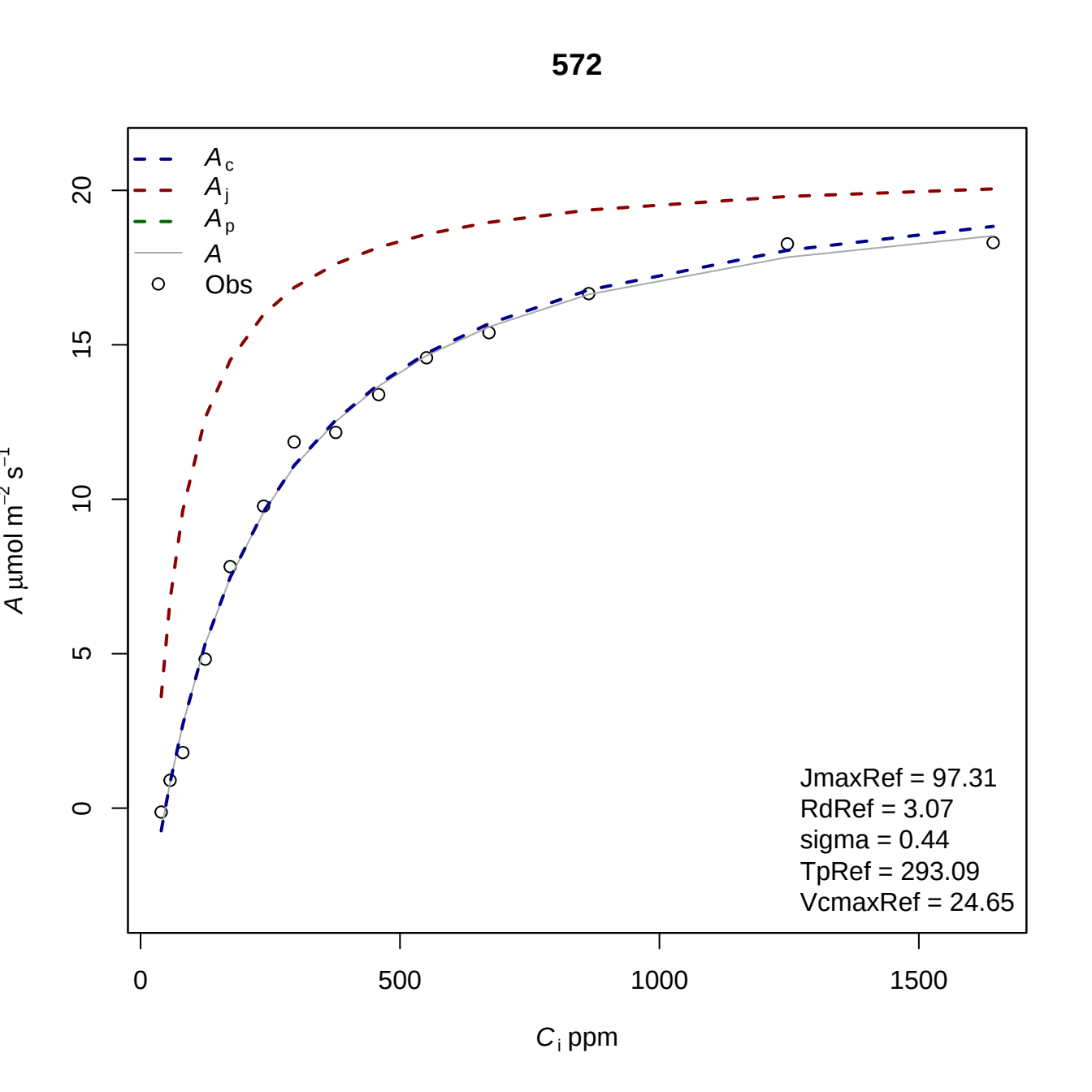

567
A_c
A_j
A_p
A
Obs
JmaxRef = 662993.25
RdRef = 3.43
sigma = 0.49
TpRef = 75914.44
VcmaxRef = 27.08
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
500
1000
1500

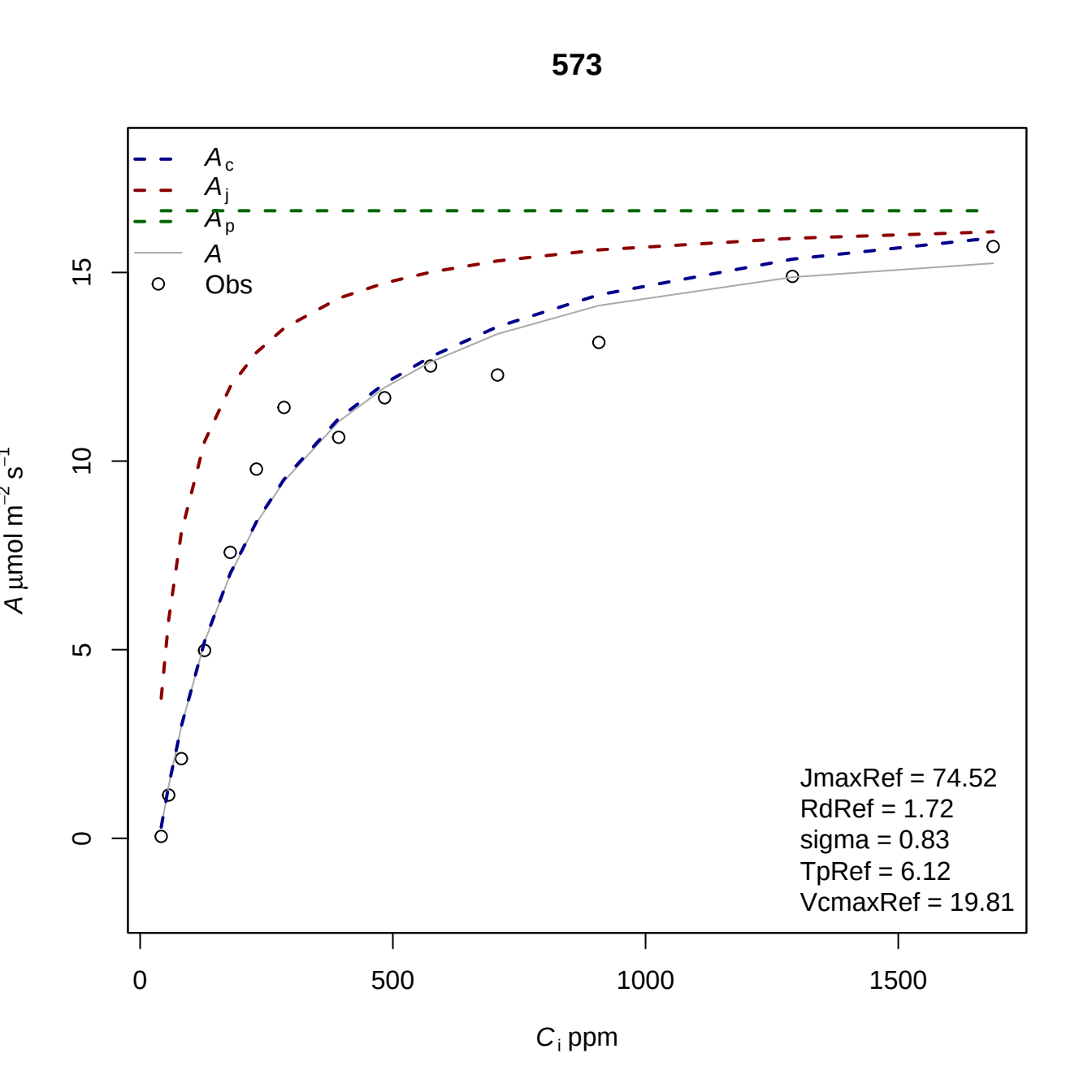

568
A_c
A_j
A_p
A
Obs
JmaxRef = 445007.14
RdRef = 3.2
sigma = 0.58
TpRef = 179709.57
VcmaxRef = 22.15
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
500
1000
1500

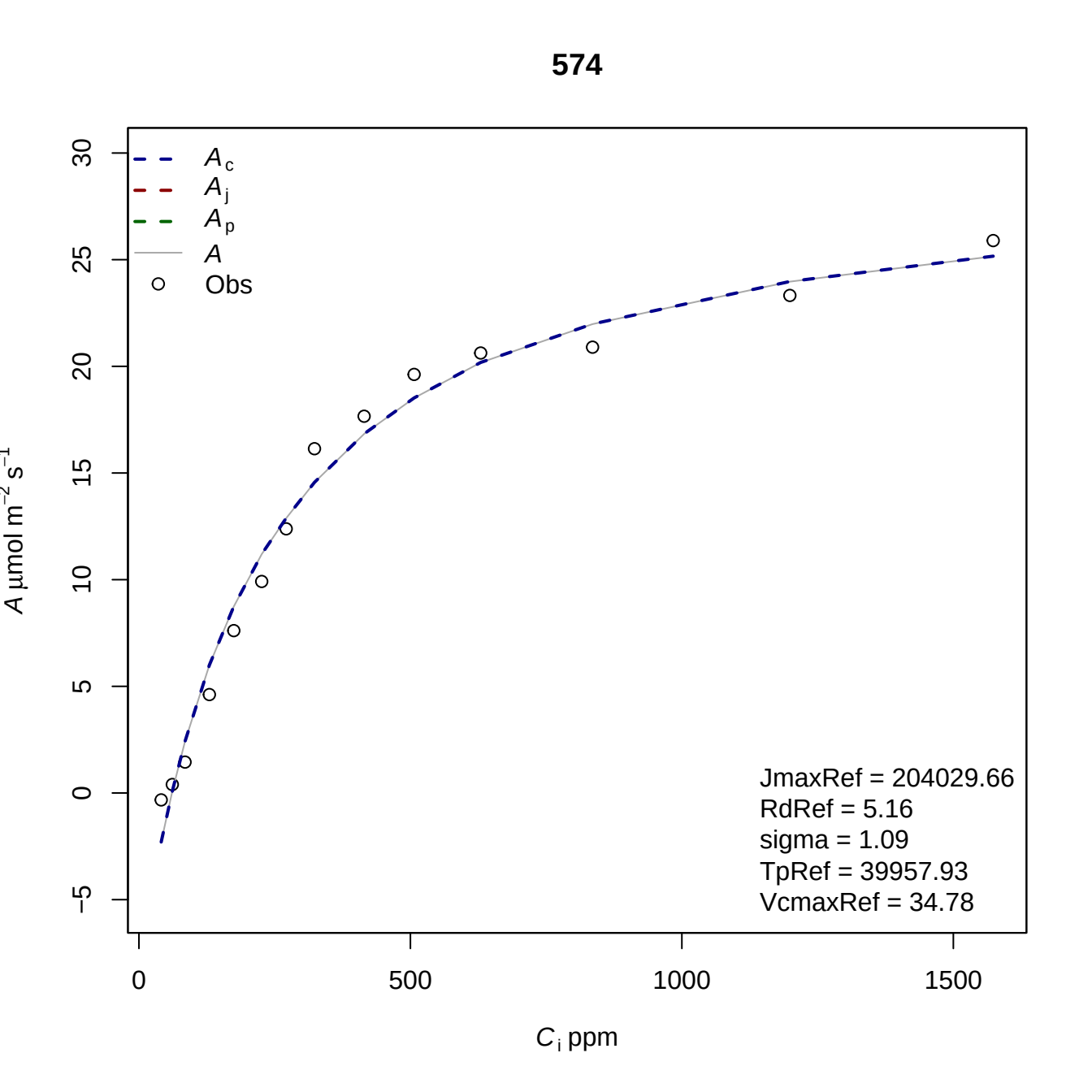

569
A_c
A_j
A_p
A
Obs
JmaxRef = 274.66
RdRef = 2.29
sigma = 0.7
TpRef = 27.43
VcmaxRef = 21.54
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
500
1000
1500

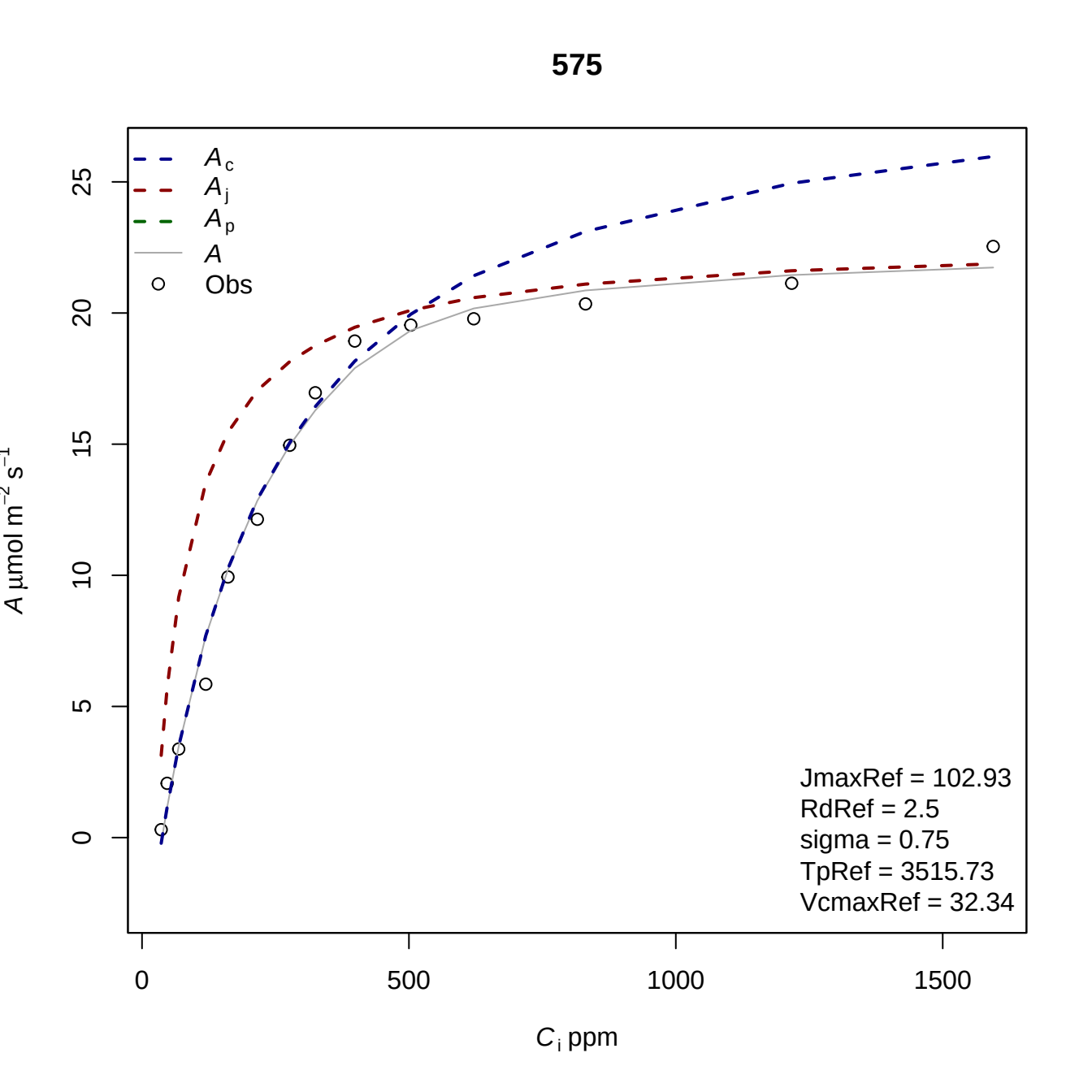

570
A_c
A_j
A_p
A
Obs
JmaxRef = 81.39
RdRef = 2.06
sigma = 0.42
TpRef = 6.85
VcmaxRef = 19.16
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
500
1000
1500

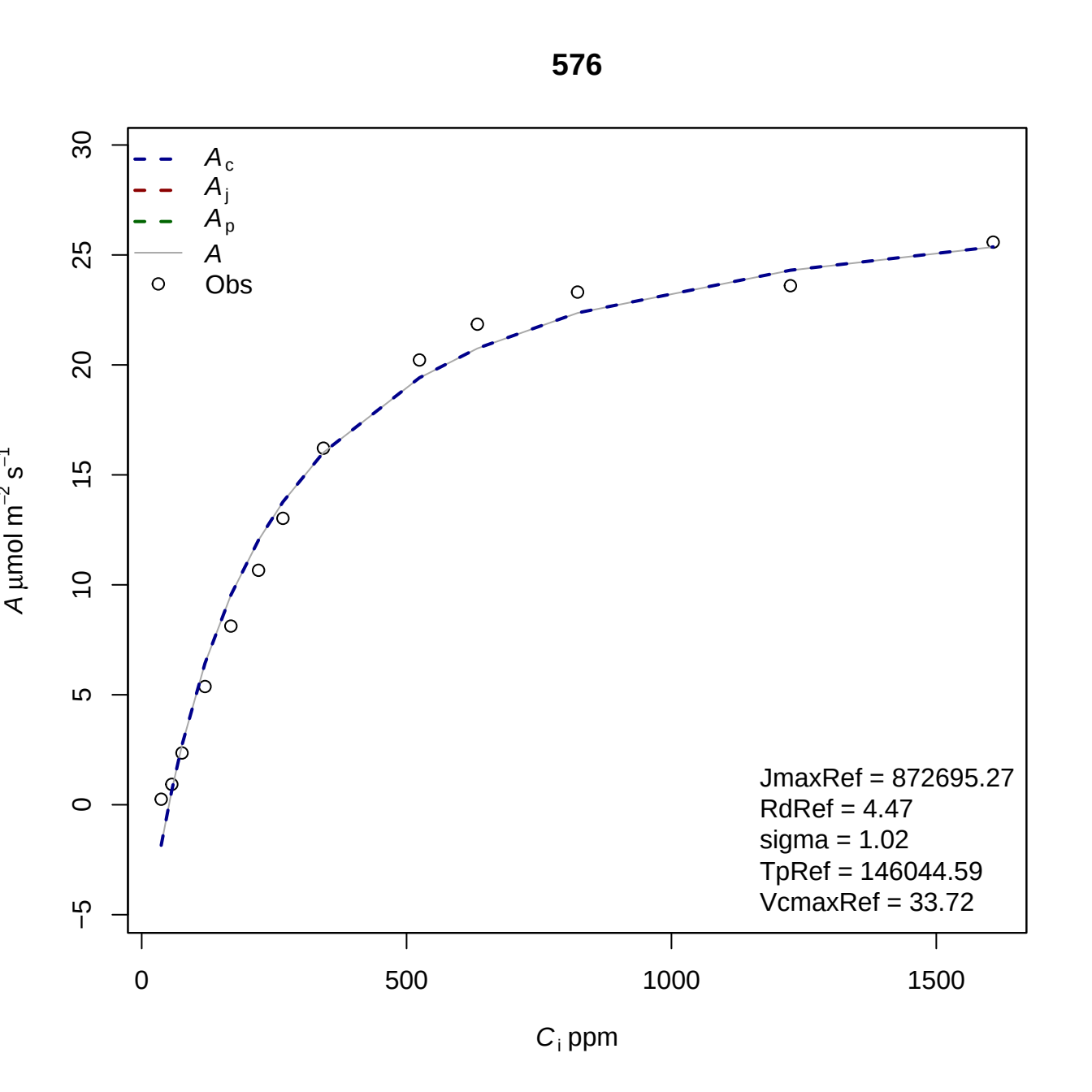

571
A_c
A_j
A_p
A
Obs
JmaxRef = 330665.41
RdRef = 3.47
sigma = 0.54
TpRef = 309242.22
VcmaxRef = 24.6
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
500
1000
1500

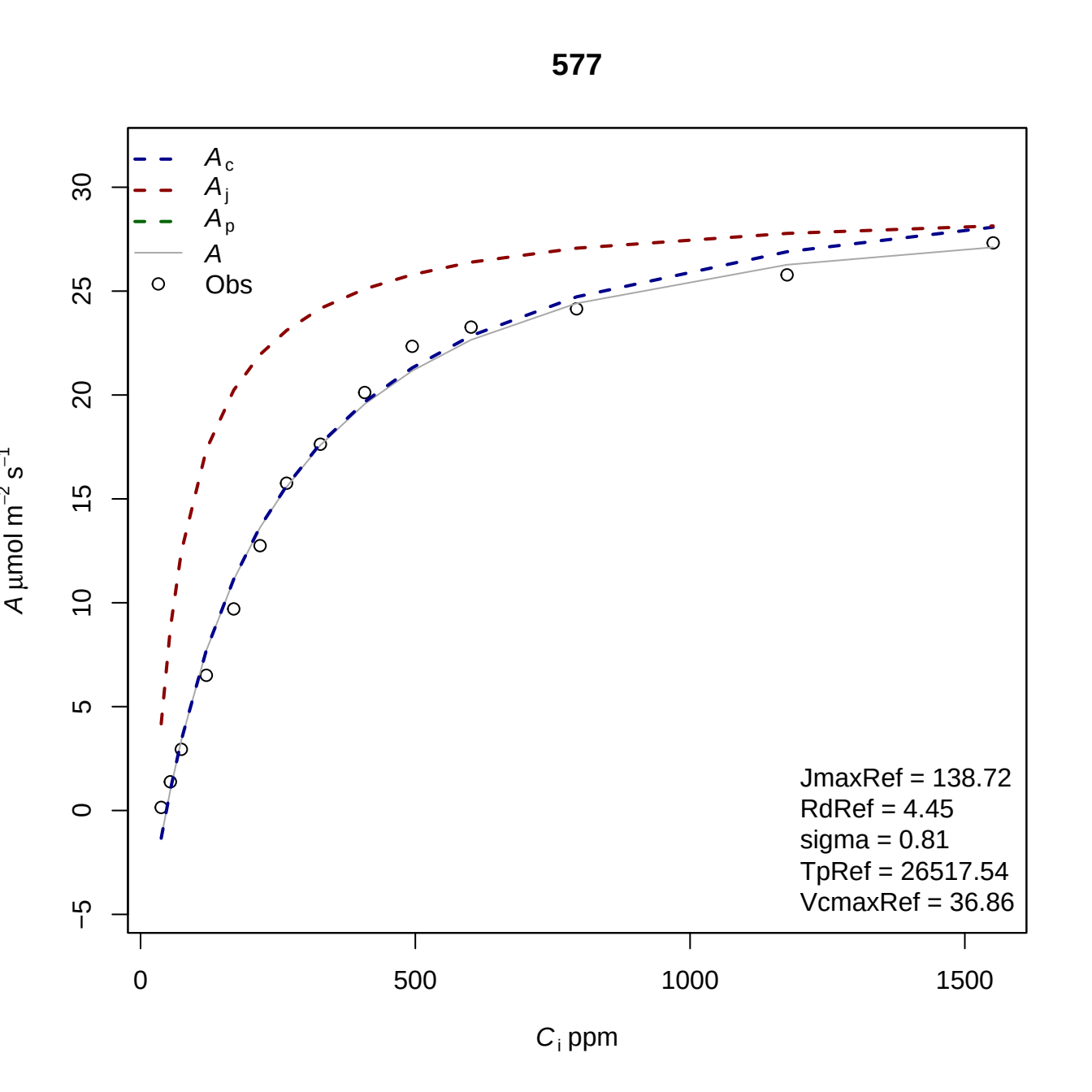

572
A_c
A_j
A_p
A
Obs
JmaxRef = 97.31
RdRef = 3.07
sigma = 0.44
TpRef = 293.09
VcmaxRef = 24.65
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
500
1000
1500

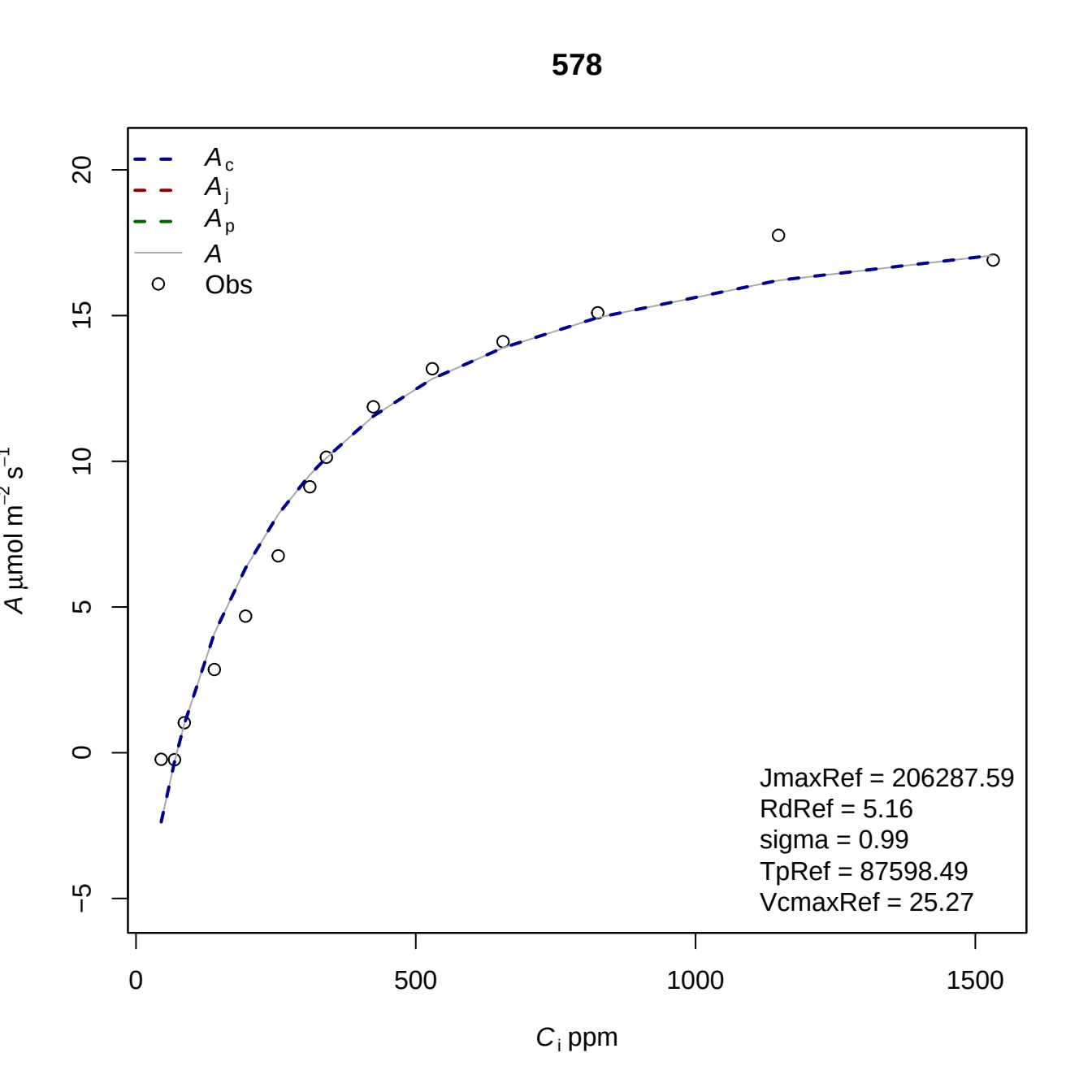

573
A_c
A_j
A_p
A
Obs
JmaxRef = 74.52
RdRef = 1.72
sigma = 0.83
TpRef = 6.12
VcmaxRef = 19.81
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
C_i ppm

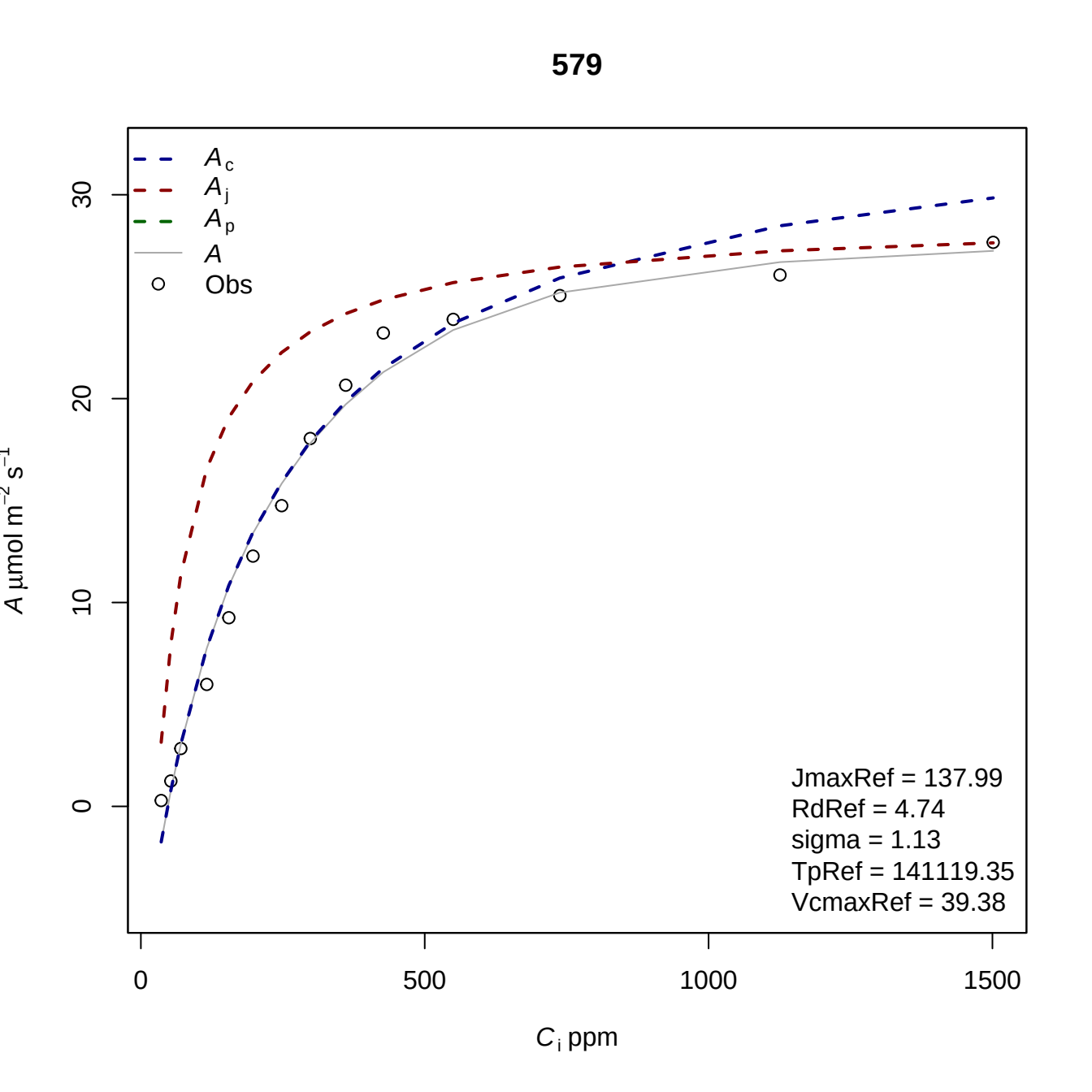

574
A_c
A_j
A_p
A
Obs
JmaxRef = 204029.66
RdRef = 5.16
sigma = 1.09
TpRef = 39957.93
VcmaxRef = 34.78
A µmol m^{-2} s^{-1}
C_i ppm
30
25
20
15
10
5
0
−5
0
500
1000
1500

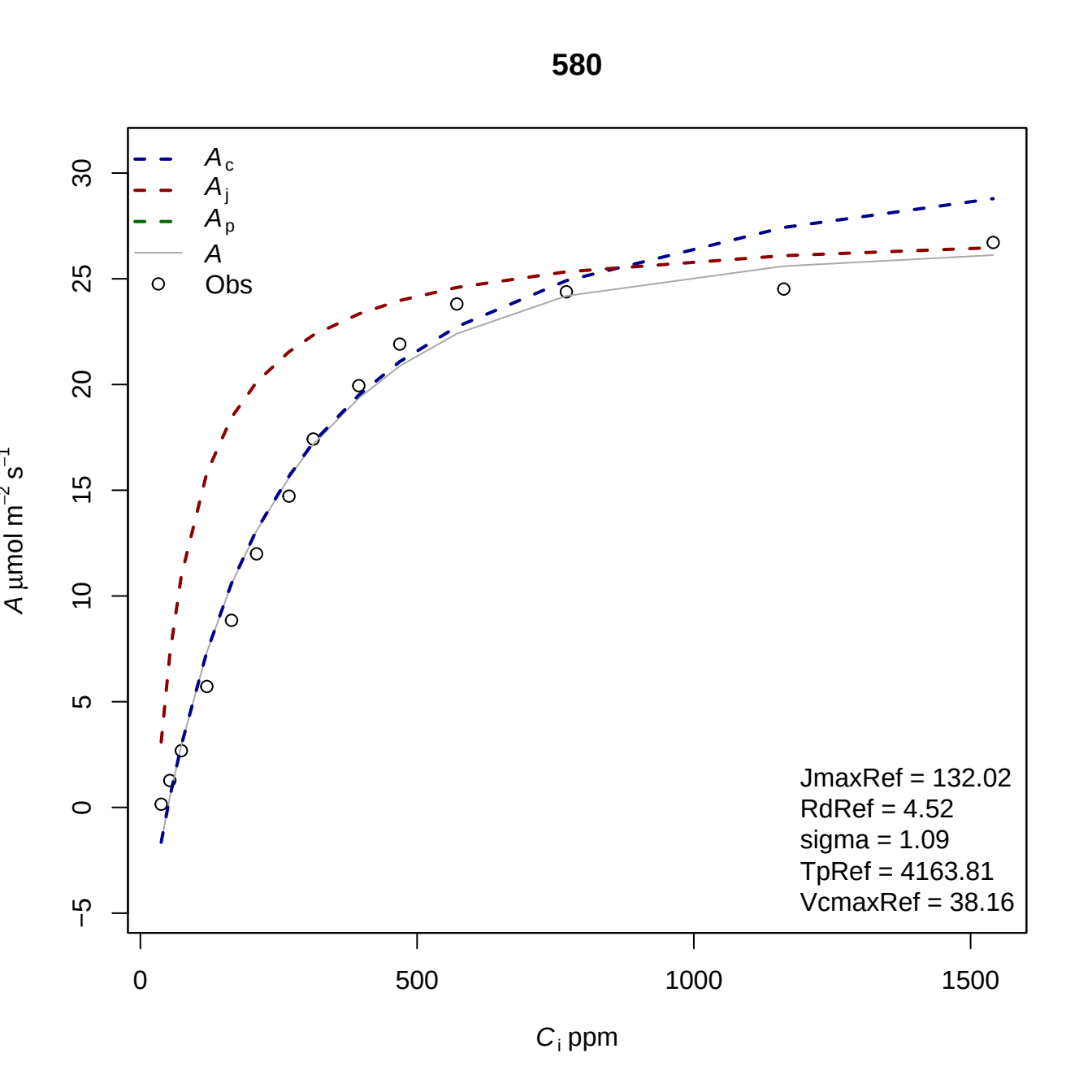

575
A_c
A_j
A_p
A
Obs
JmaxRef = 102.93
RdRef = 2.5
sigma = 0.75
TpRef = 3515.73
VcmaxRef = 32.34
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
500
1000
1500

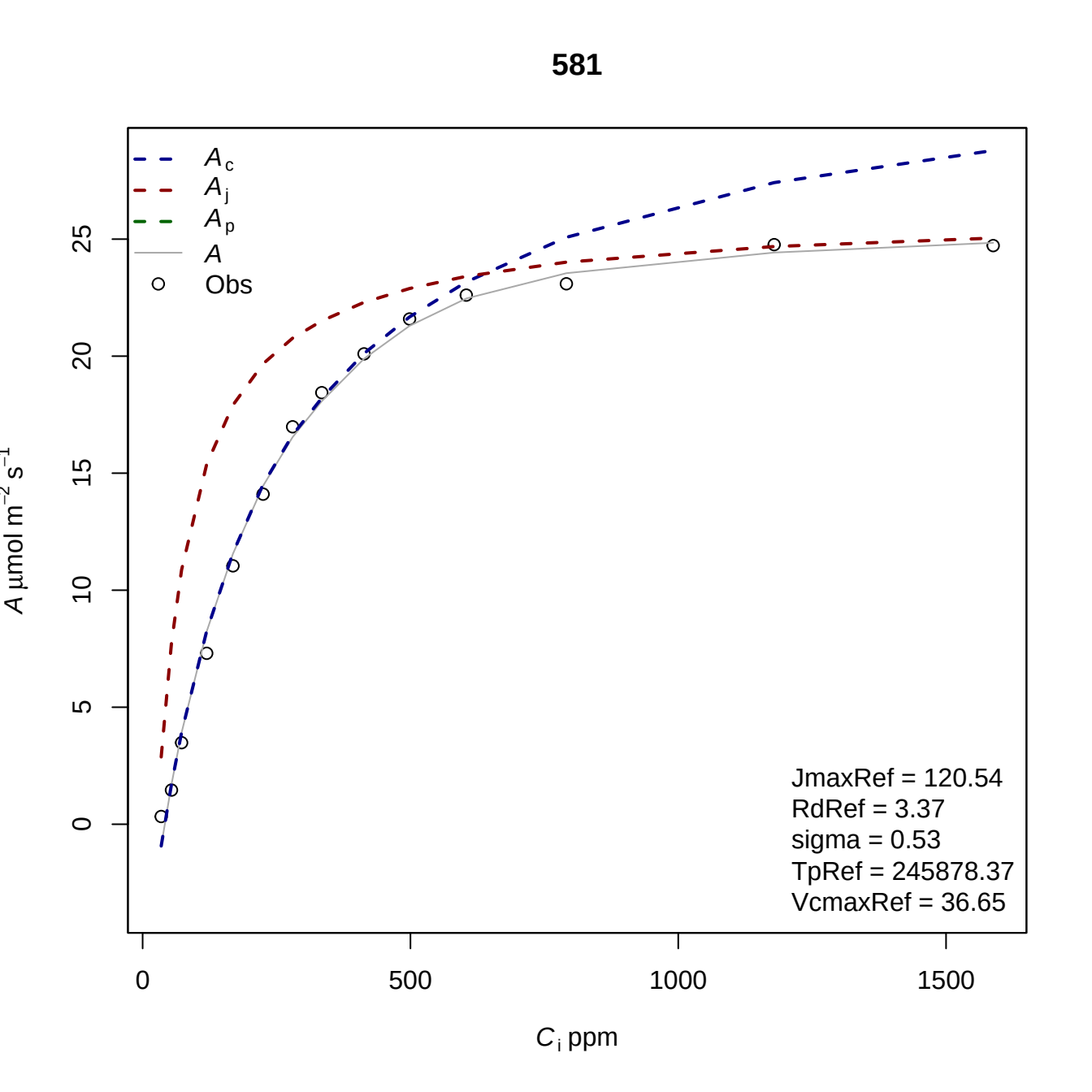

576
A_c
A_j
A_p
A
Obs
JmaxRef = 872695.27
RdRef = 4.47
sigma = 1.02
TpRef = 146044.59
VcmaxRef = 33.72
A µmol m^{-2} s^{-1}
C_i ppm
−5
0
5
10
15
20
25
30
0
500
1000
1500

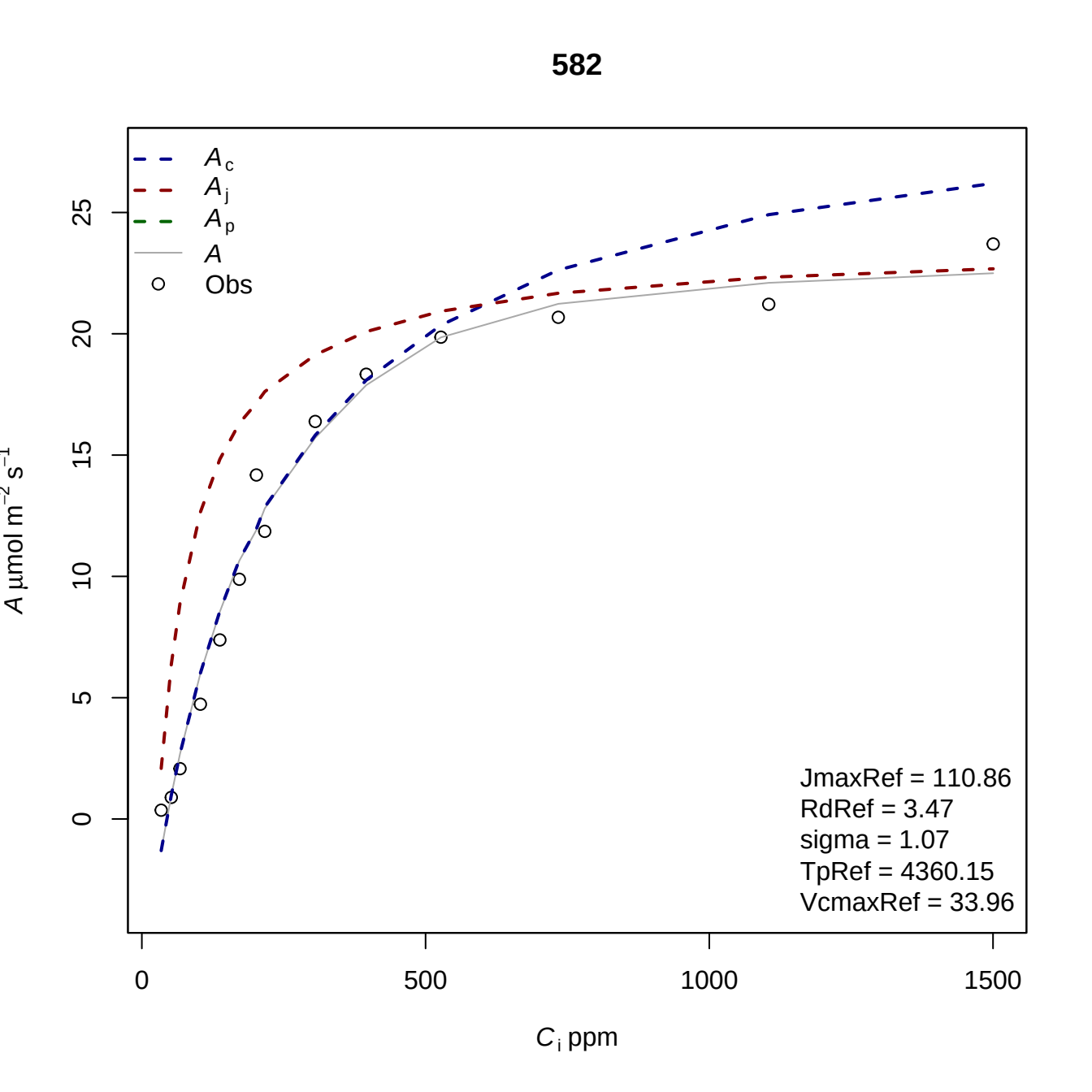

577
A_c
A_j
A_p
A
Obs
JmaxRef = 138.72
RdRef = 4.45
sigma = 0.81
TpRef = 26517.54
VcmaxRef = 36.86
A µmol m^{-2} s^{-1}
C_i ppm
−5
0
5
10
15
20
25
30
0
500
1000
1500

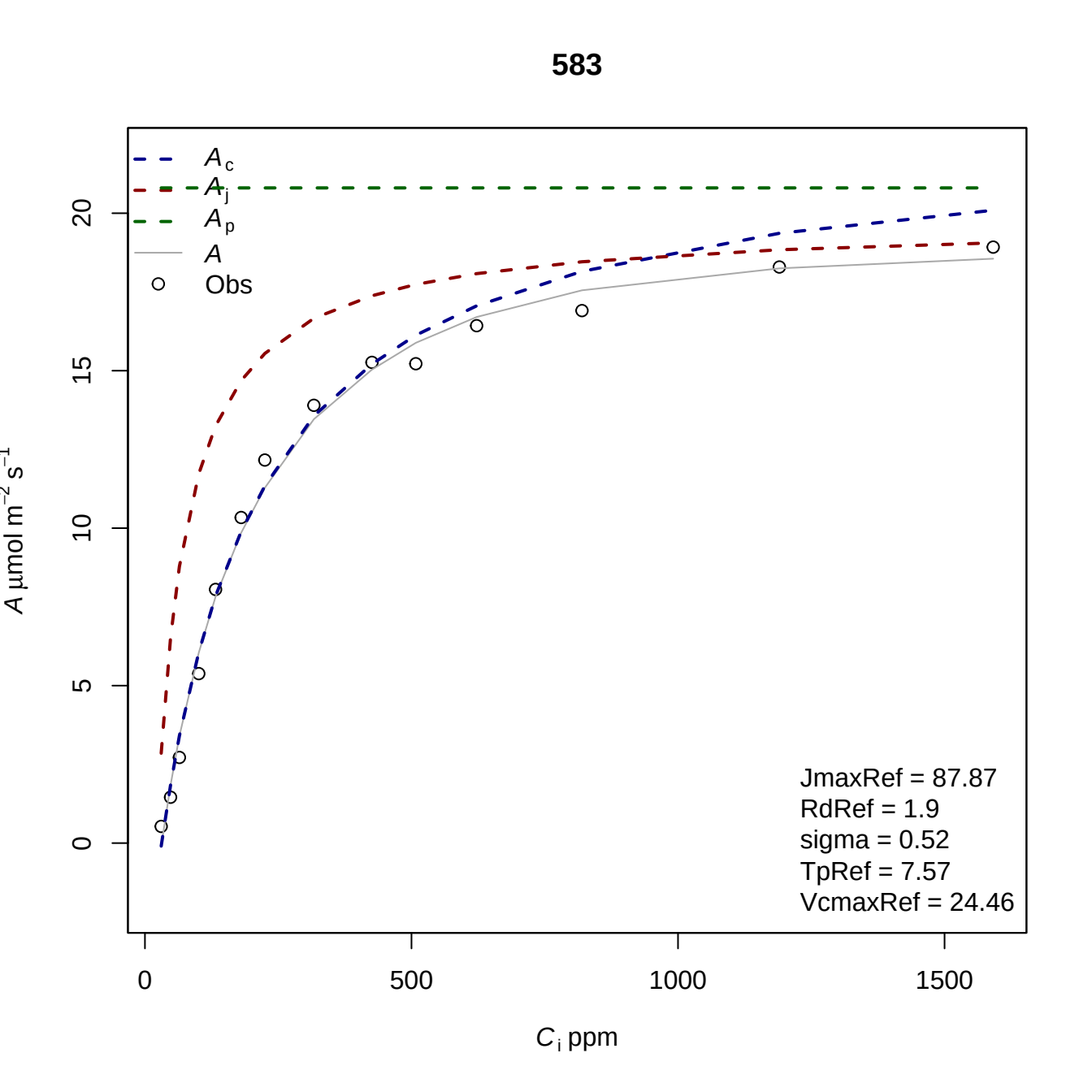

578
A_c
A_j
A_p
A
Obs
JmaxRef = 206287.59
RdRef = 5.16
sigma = 0.99
TpRef = 87598.49
VcmaxRef = 25.27
A µmol m^{-2} s^{-1}
C_i ppm
20
15
10
5
0
−5
0
500
1000
1500

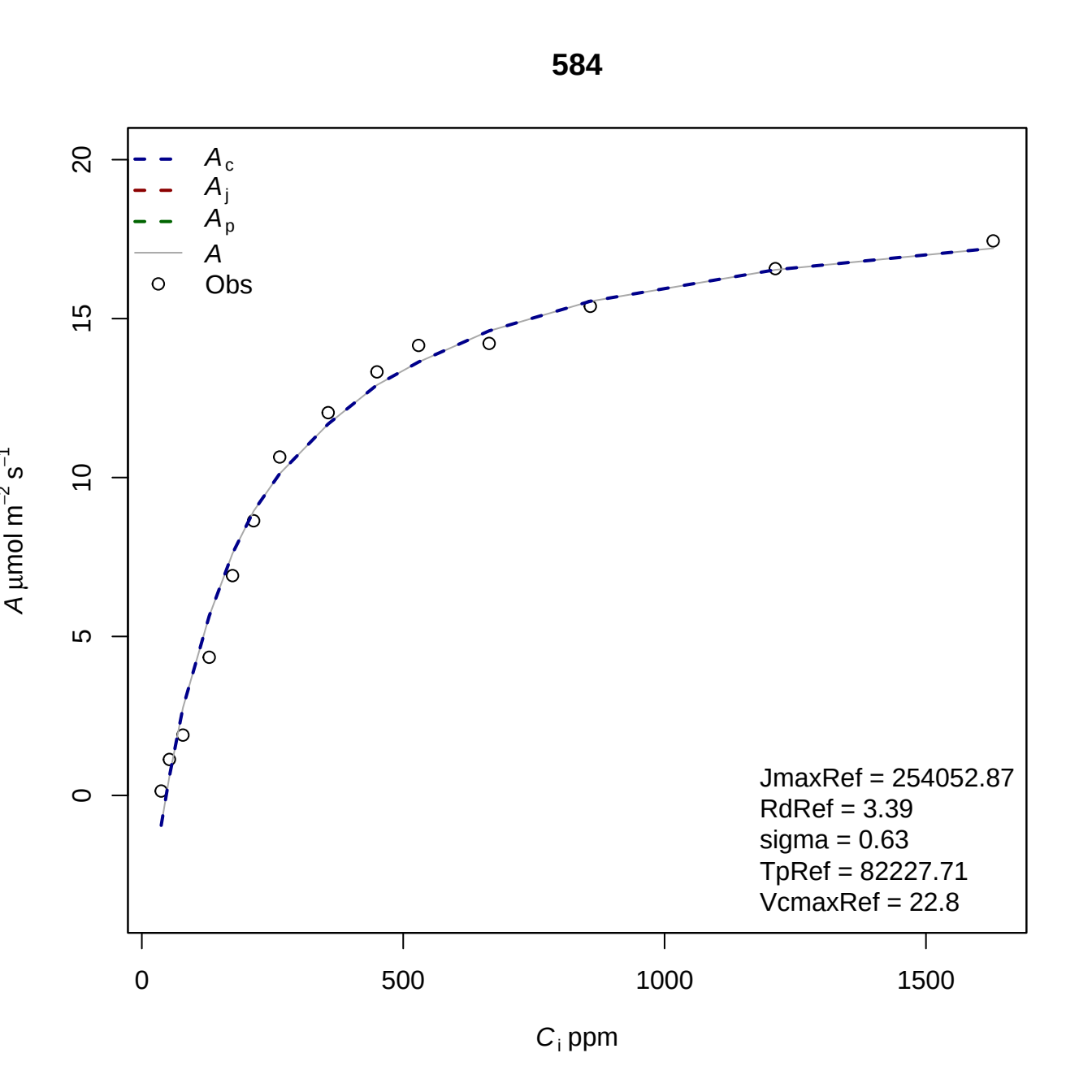

579
A_c
A_j
A_p
A
Obs
JmaxRef = 137.99
RdRef = 4.74
sigma = 1.13
TpRef = 141119.35
VcmaxRef = 39.38
A µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
C_i ppm

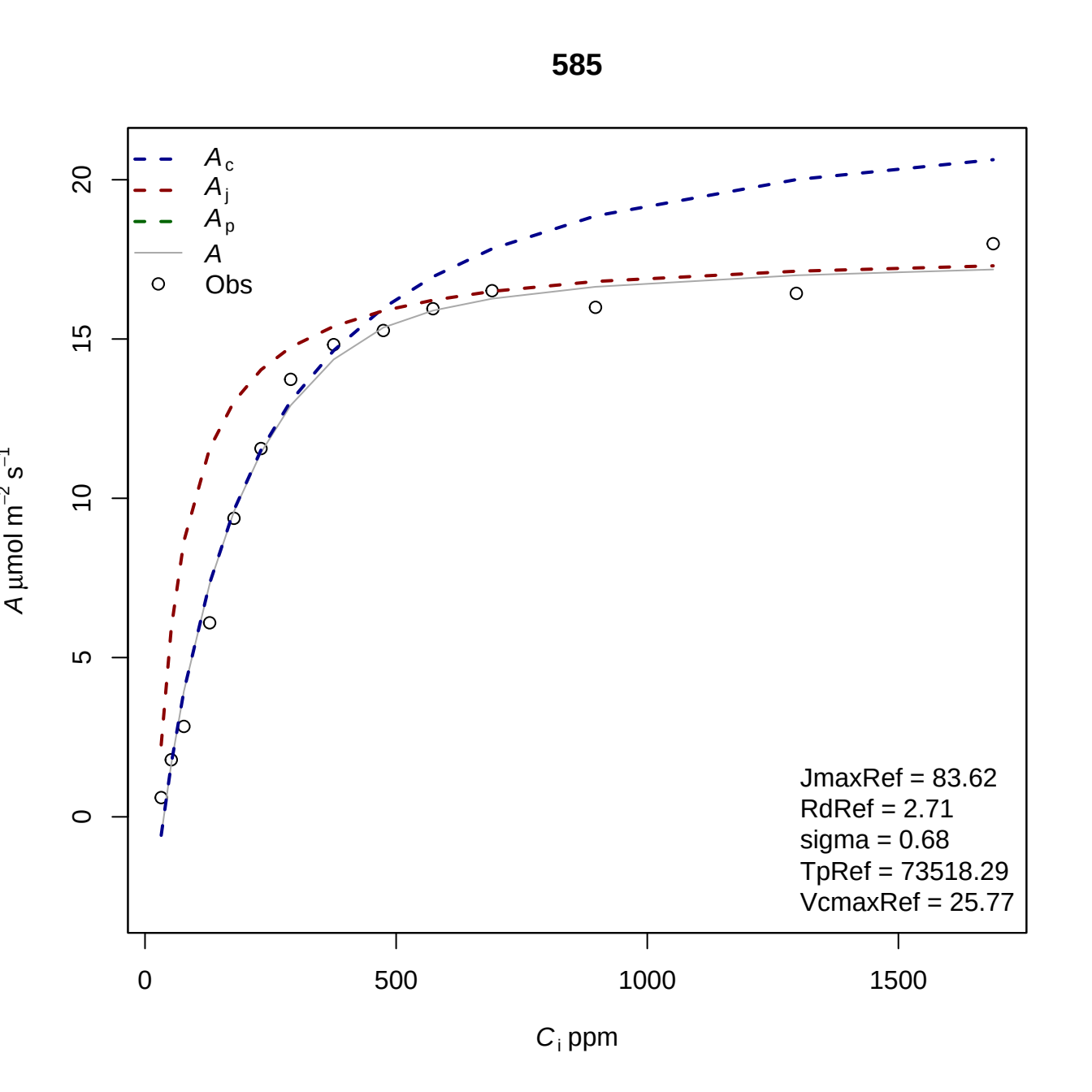

580
A_c
A_j
A_p
A
Obs
JmaxRef = 132.02
RdRef = 4.52
sigma = 1.09
TpRef = 4163.81
VcmaxRef = 38.16
A µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
30
0
500
1000
1500
C_i ppm

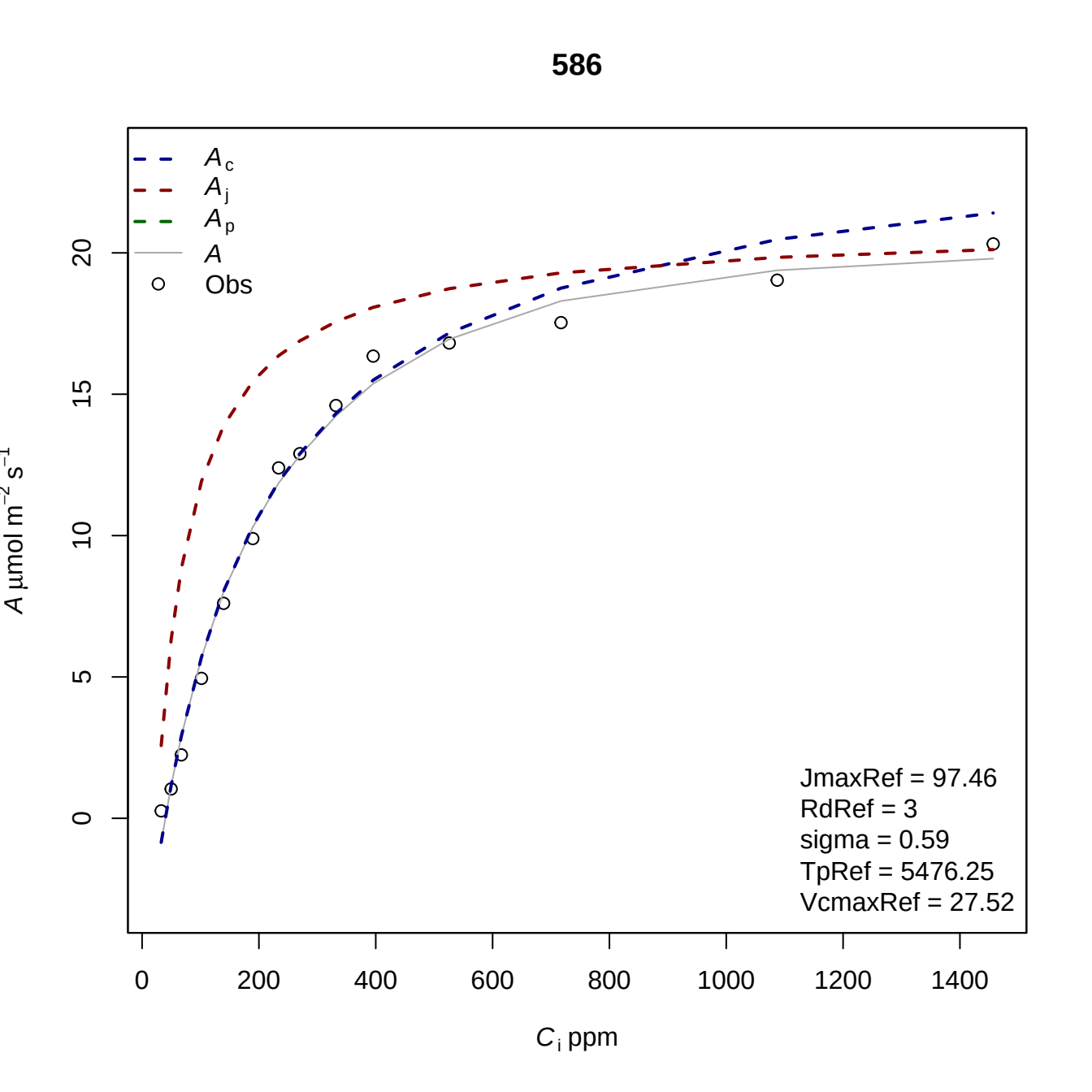

581
A_c
A_j
A_p
A
Obs
JmaxRef = 120.54
RdRef = 3.37
sigma = 0.53
TpRef = 245878.37
VcmaxRef = 36.65
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
500
1000
1500

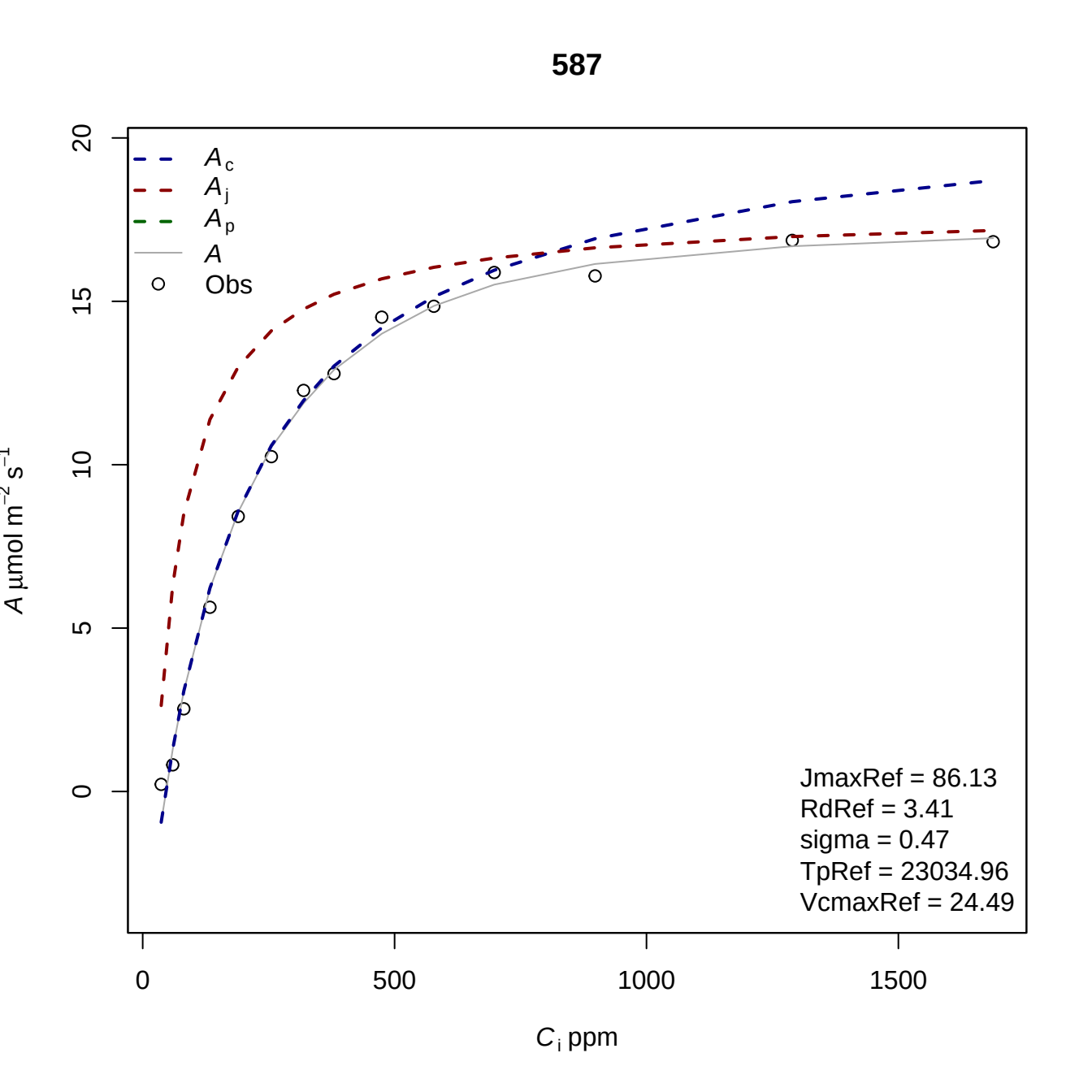

582
A_c
A_j
A_p
A
Obs
JmaxRef = 110.86
RdRef = 3.47
sigma = 1.07
TpRef = 4360.15
VcmaxRef = 33.96
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
500
1000
1500

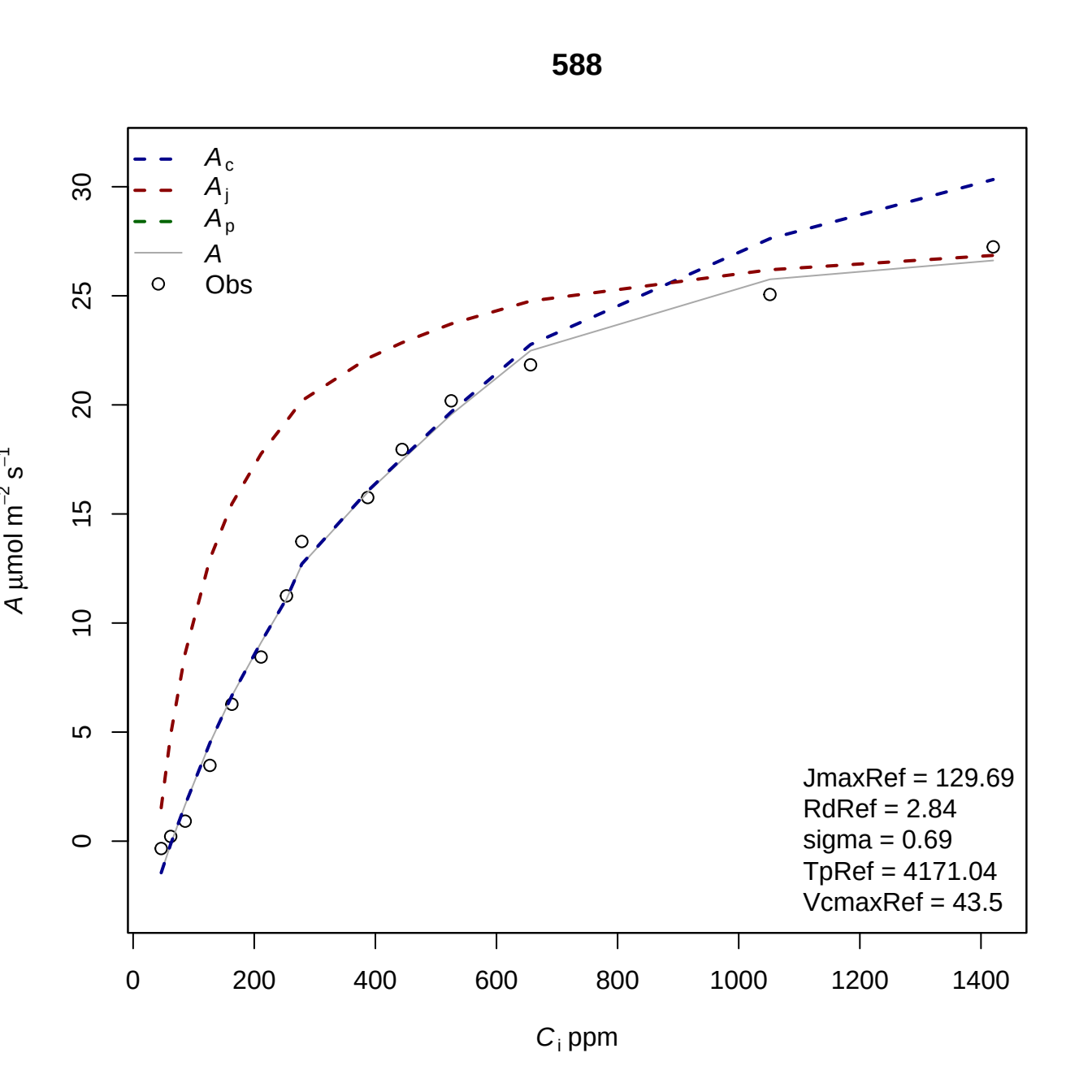

583
A_c
A_j
A_p
A
Obs
JmaxRef = 87.87
RdRef = 1.9
sigma = 0.52
TpRef = 7.57
VcmaxRef = 24.46
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
C_i ppm

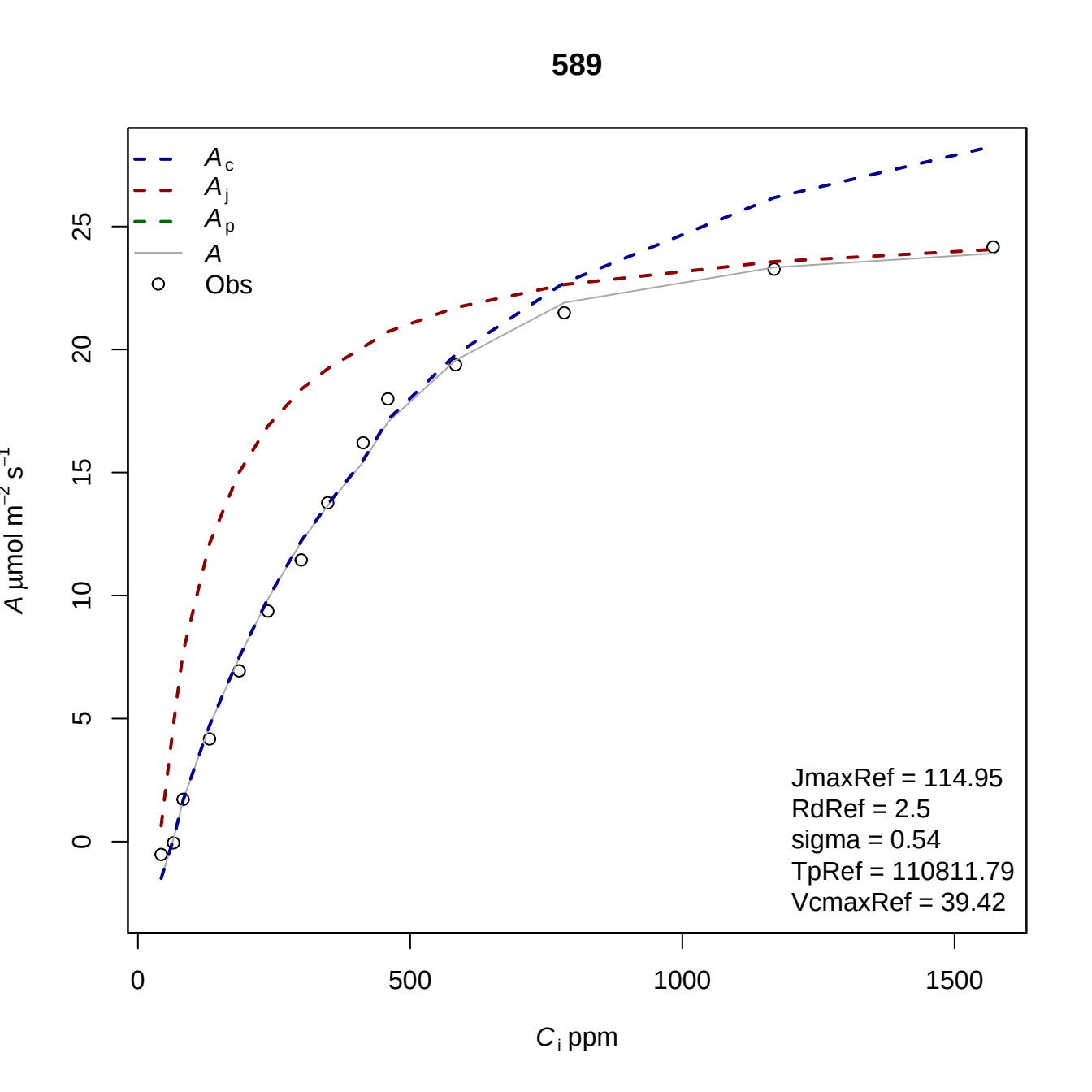

584
A_c
A_j
A_p
A
Obs
JmaxRef = 254052.87
RdRef = 3.39
sigma = 0.63
TpRef = 82227.71
VcmaxRef = 22.8
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
500
1000
1500

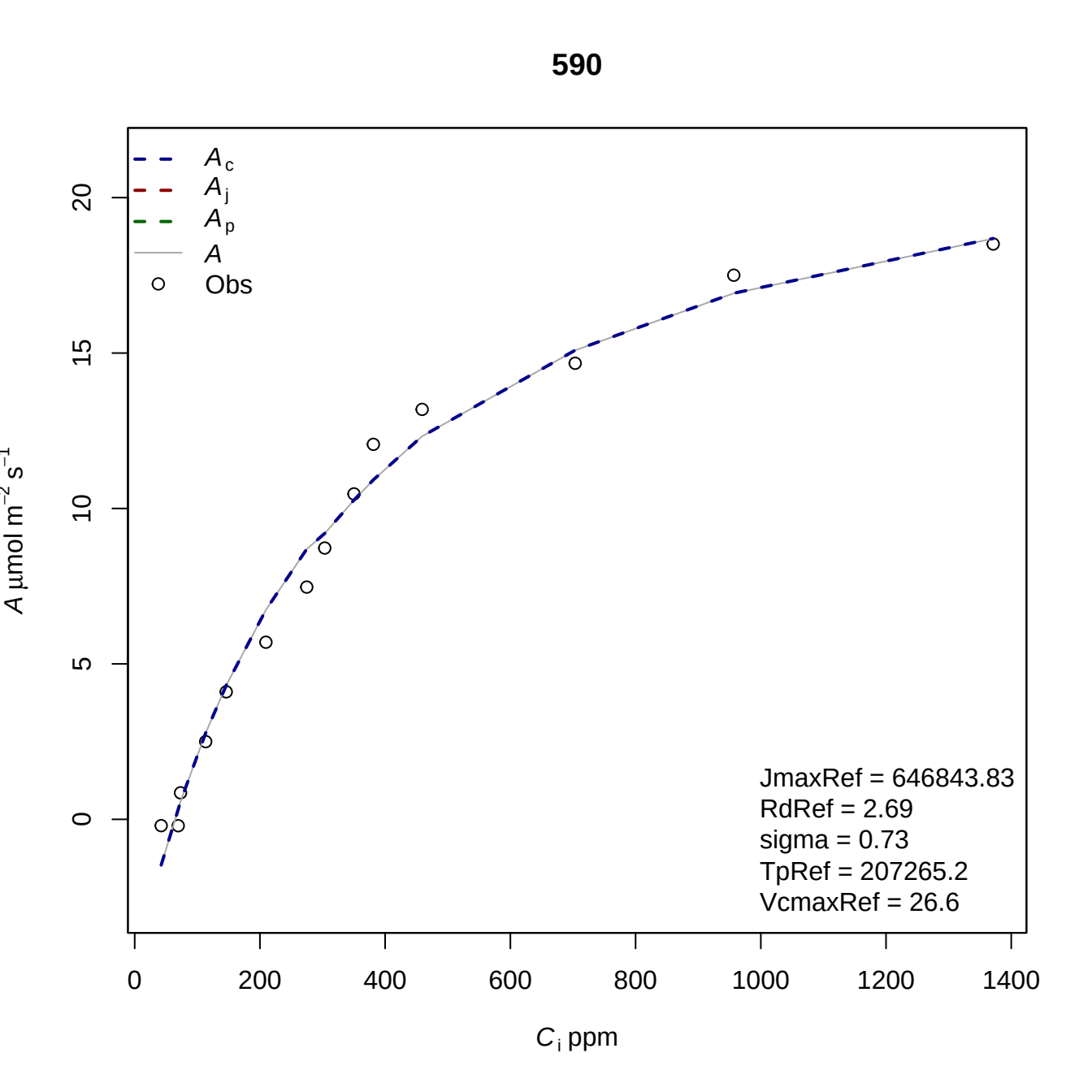

585
A_c
A_j
A_p
A
Obs
JmaxRef = 83.62
RdRef = 2.71
sigma = 0.68
TpRef = 73518.29
VcmaxRef = 25.77
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
500
1000
1500

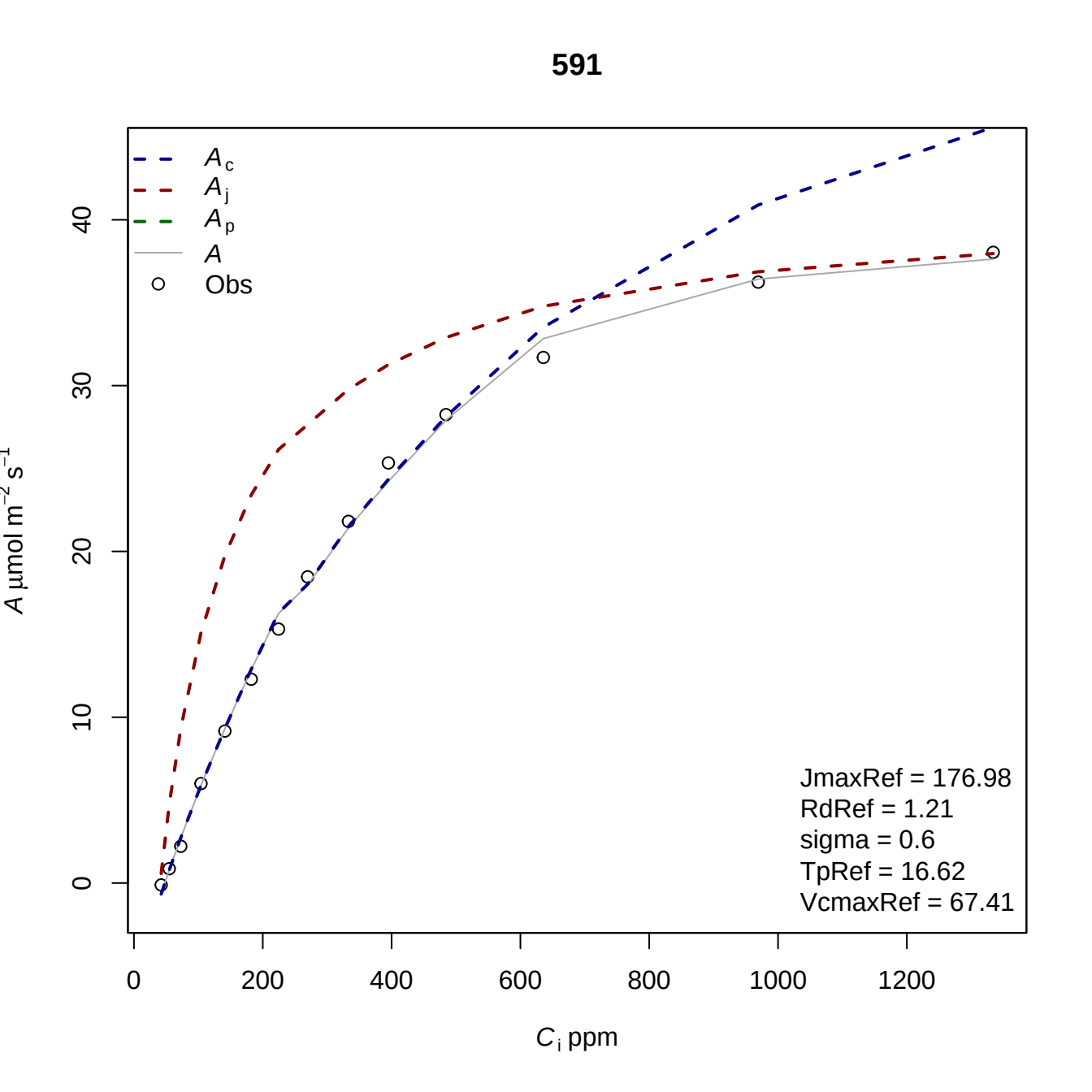

586
A_c
A_j
A_p
A
Obs
JmaxRef = 97.46
RdRef = 3
sigma = 0.59
TpRef = 5476.25
VcmaxRef = 27.52
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000
1200
1400

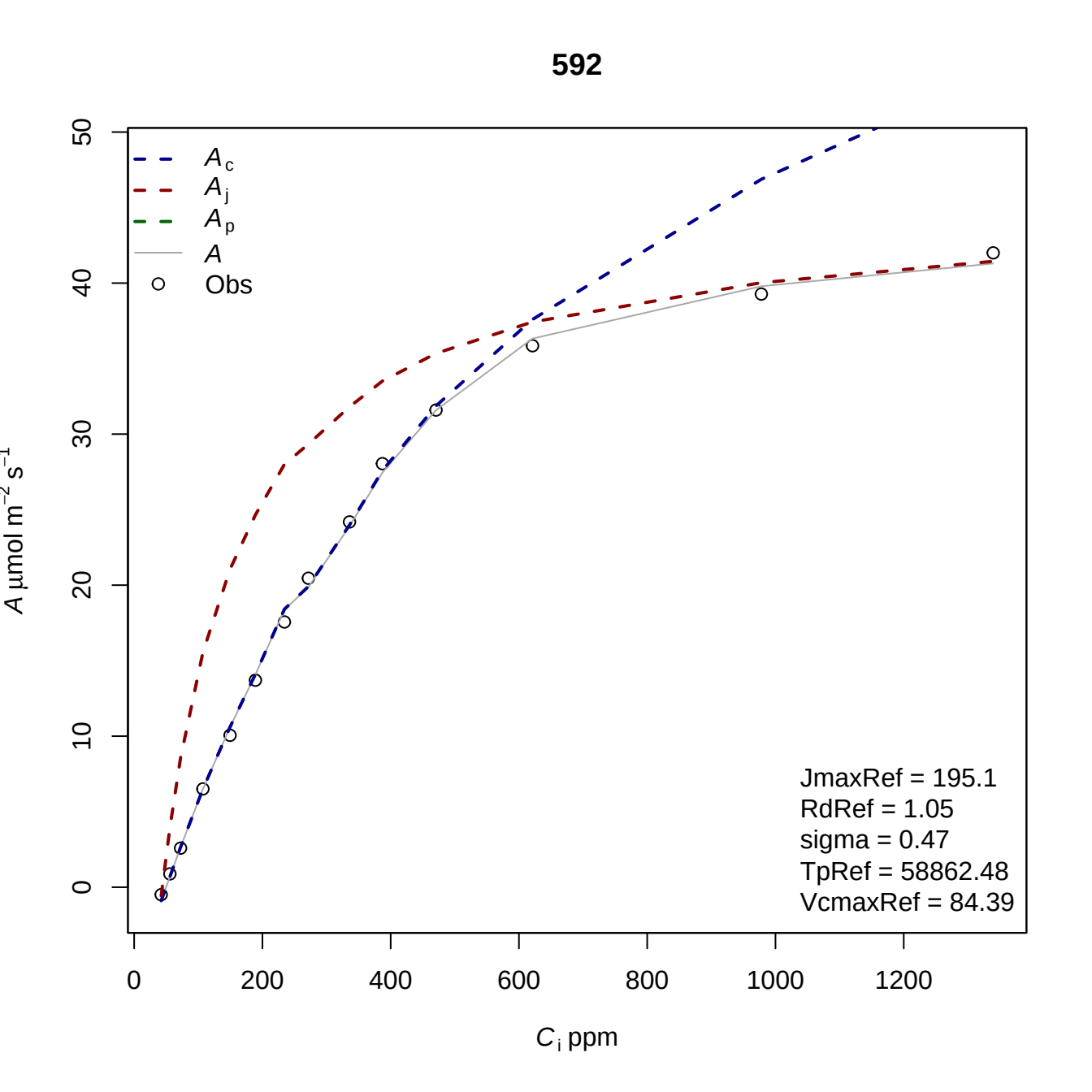

587
A_c
A_j
A_p
A
Obs
JmaxRef = 86.13
RdRef = 3.41
sigma = 0.47
TpRef = 23034.96
VcmaxRef = 24.49
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
500
1000
1500

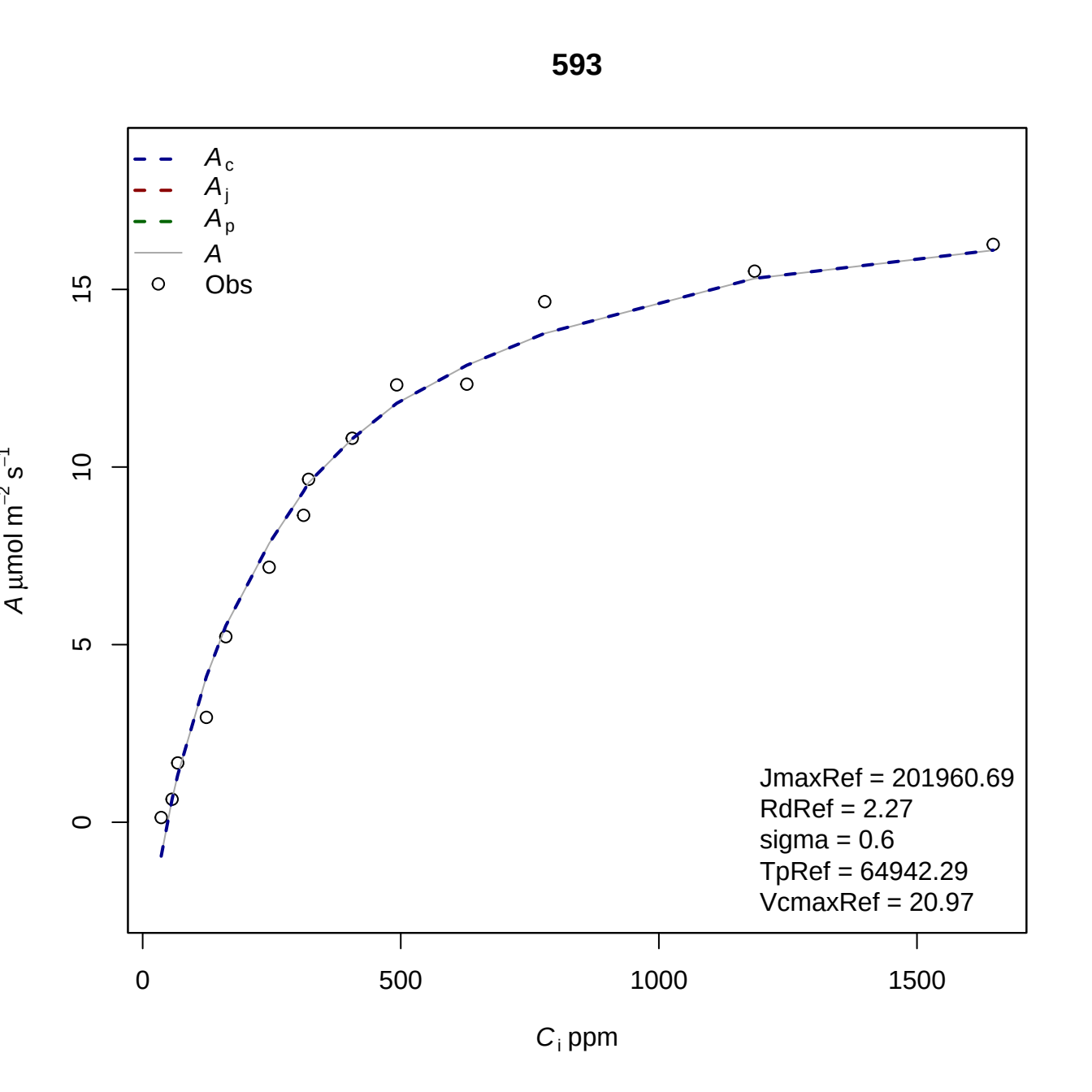

588
A_c
A_j
A_p
A
Obs
JmaxRef = 129.69
RdRef = 2.84
sigma = 0.69
TpRef = 4171.04
VcmaxRef = 43.5
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
1400

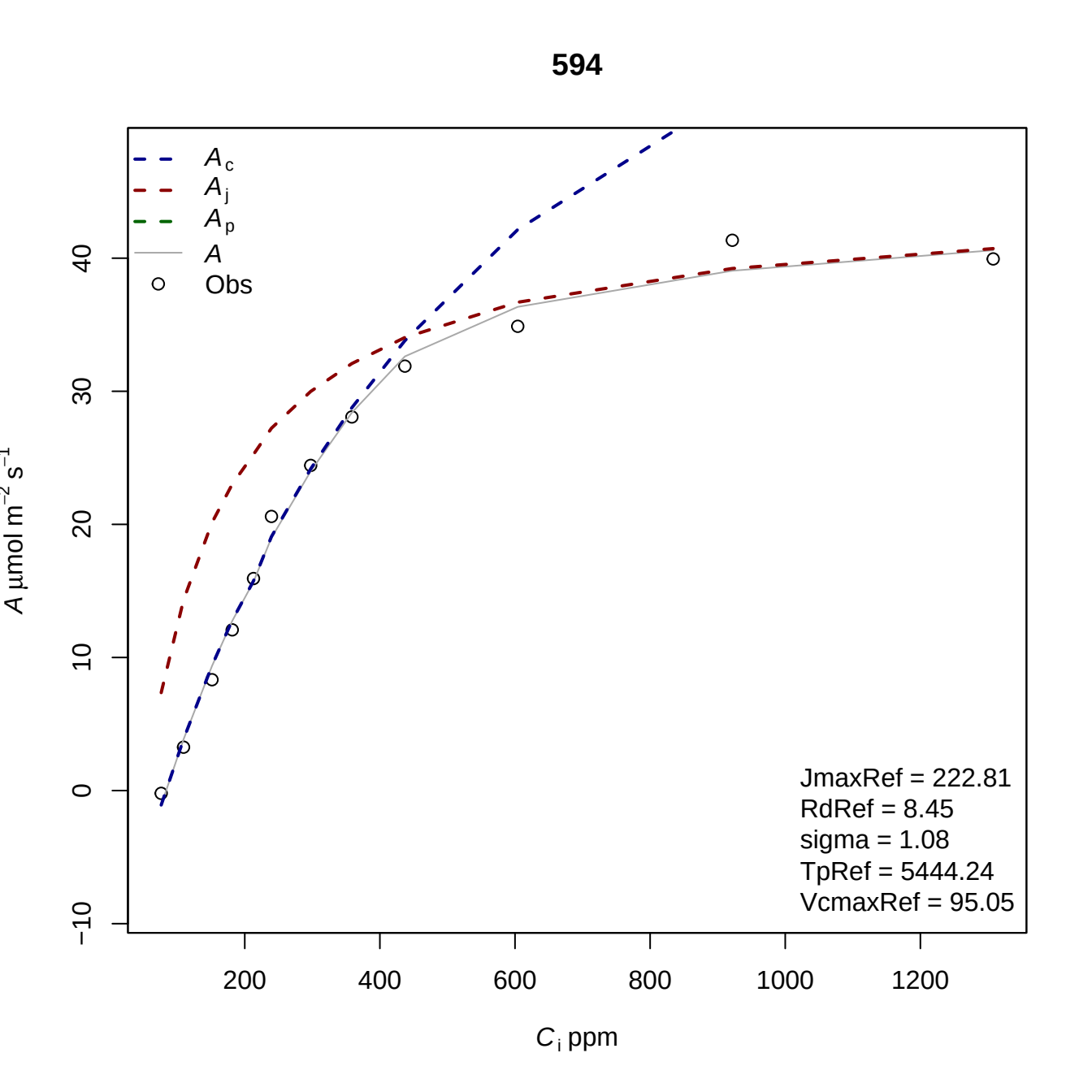

589
A_c
A_j
A_p
A
Obs
JmaxRef = 114.95
RdRef = 2.5
sigma = 0.54
TpRef = 110811.79
VcmaxRef = 39.42
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
500
1000
1500

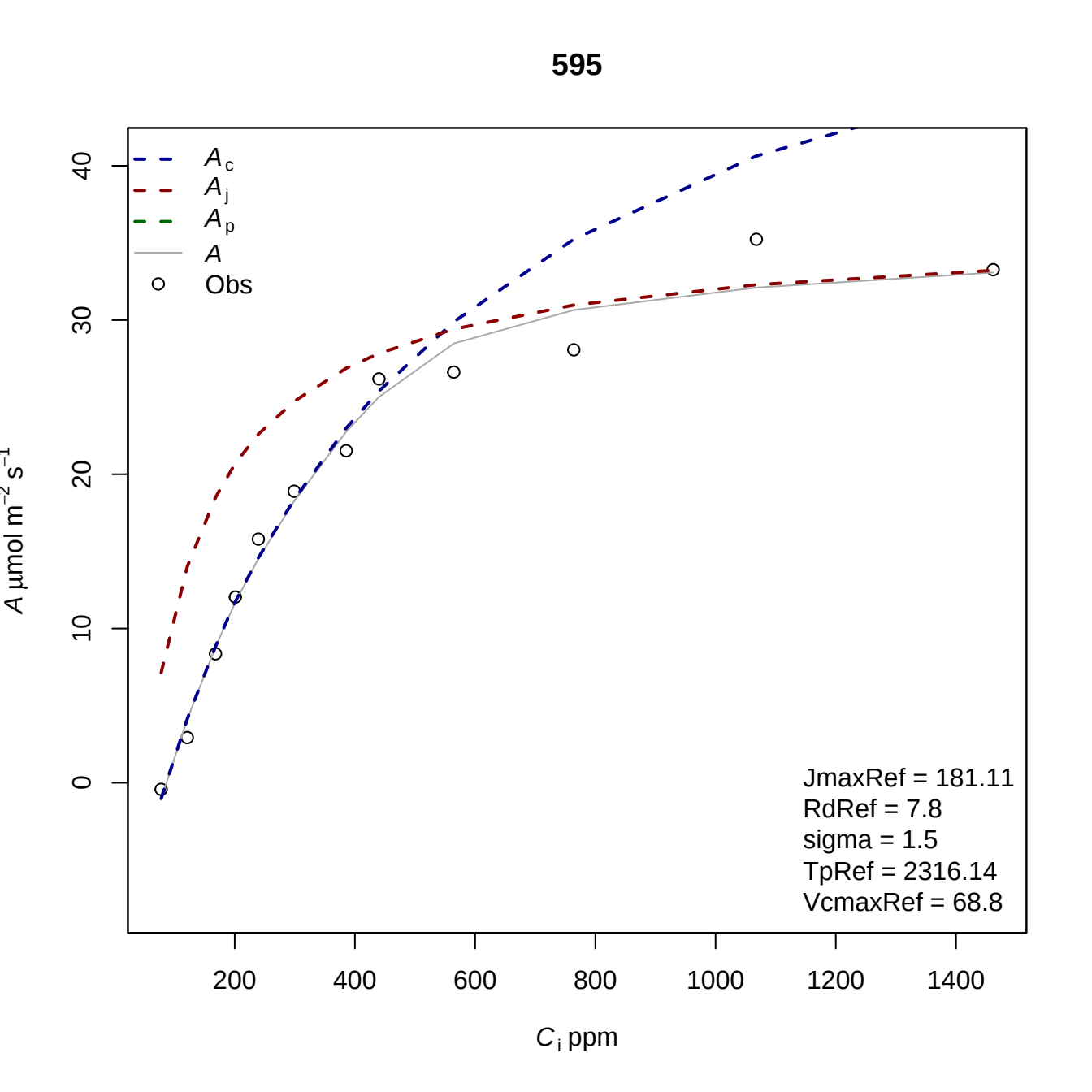

590
A_c
A_j
A_p
A
Obs
JmaxRef = 646843.83
RdRef = 2.69
sigma = 0.73
TpRef = 207265.2
VcmaxRef = 26.6
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000
1200
1400

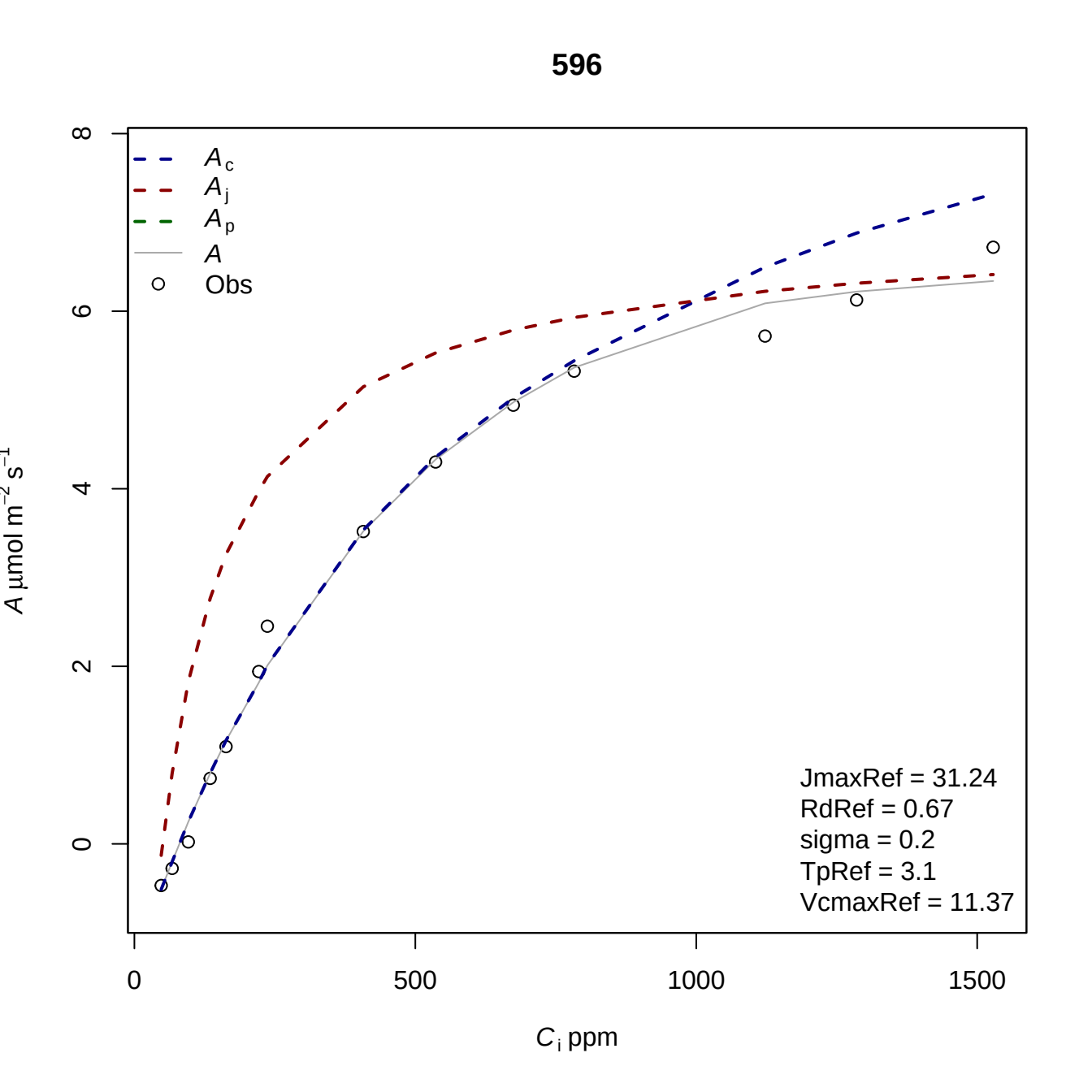

591
A_c
A_j
A_p
A
Obs
JmaxRef = 176.98
RdRef = 1.21
sigma = 0.6
TpRef = 16.62
VcmaxRef = 67.41
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
0
200
400
600
800
1000
1200

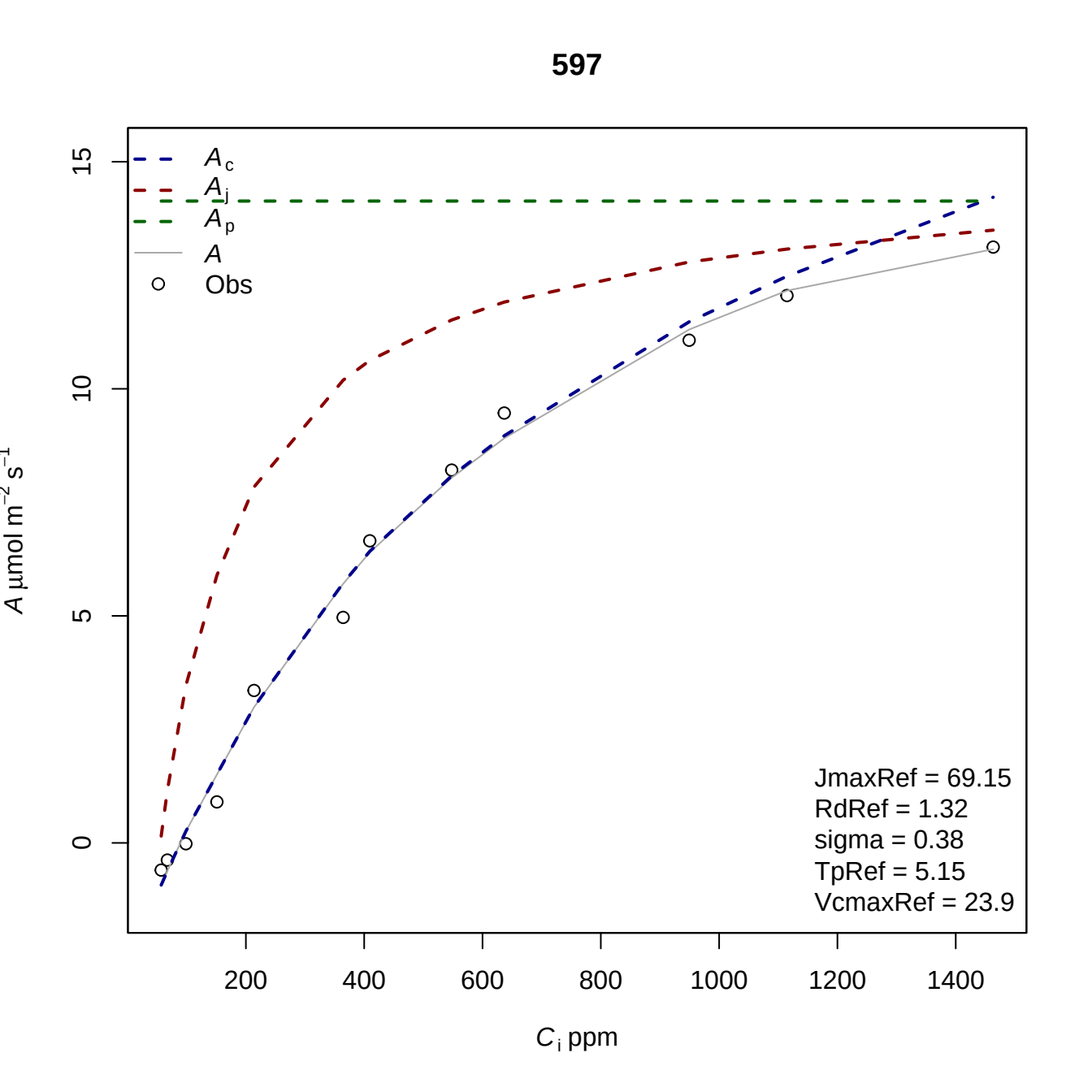

592
A_c
A_j
A_p
A
Obs
JmaxRef = 195.1
RdRef = 1.05
sigma = 0.47
TpRef = 58862.48
VcmaxRef = 84.39
A µmol m^{-2} s^{-1}
C_i ppm
0
10
20
30
40
50
0
200
400
600
800
1000
1200

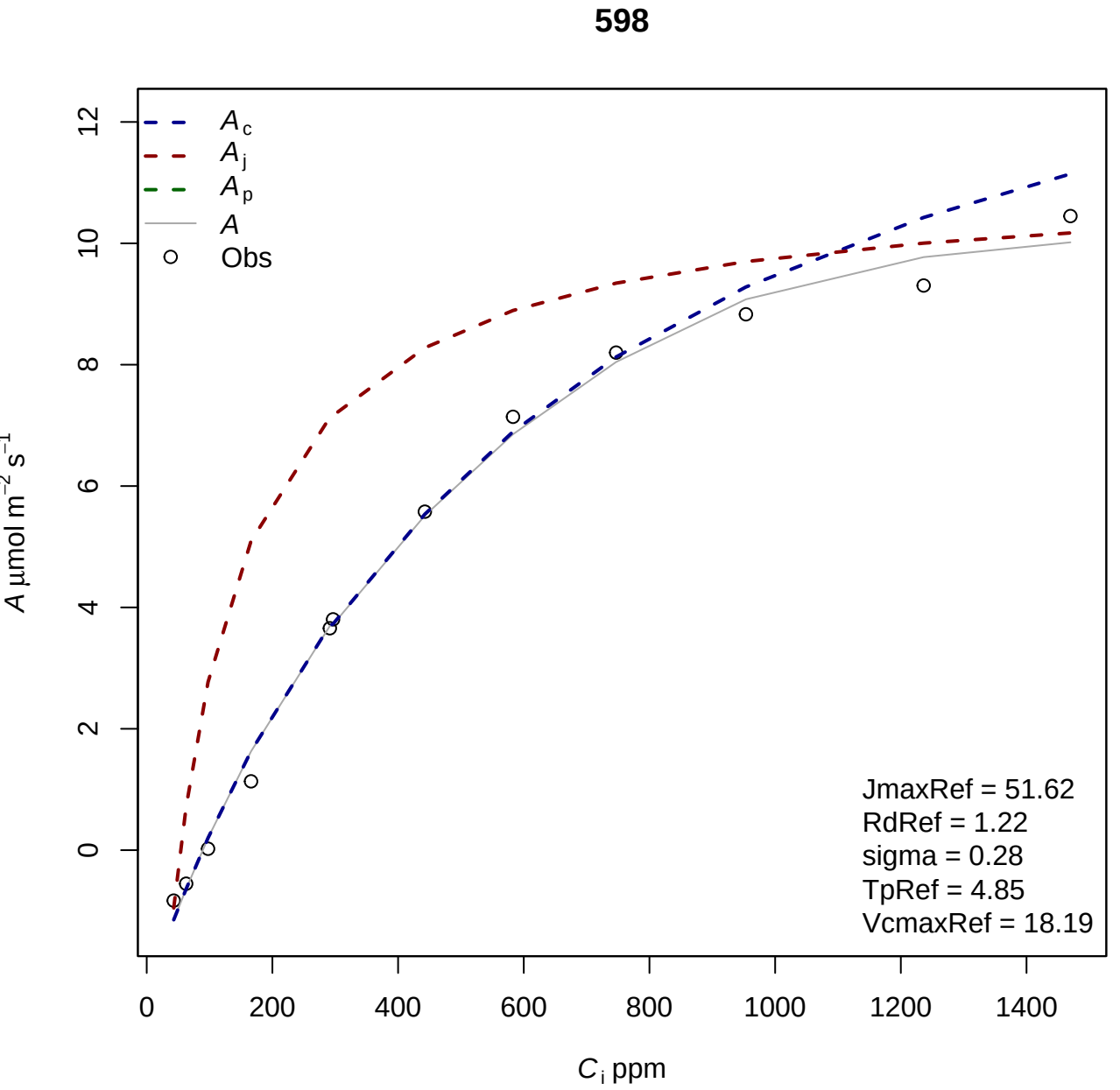

593
A_c
A_j
A_p
A
Obs
JmaxRef = 201960.69
RdRef = 2.27
sigma = 0.6
TpRef = 64942.29
VcmaxRef = 20.97
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
500
1000
1500

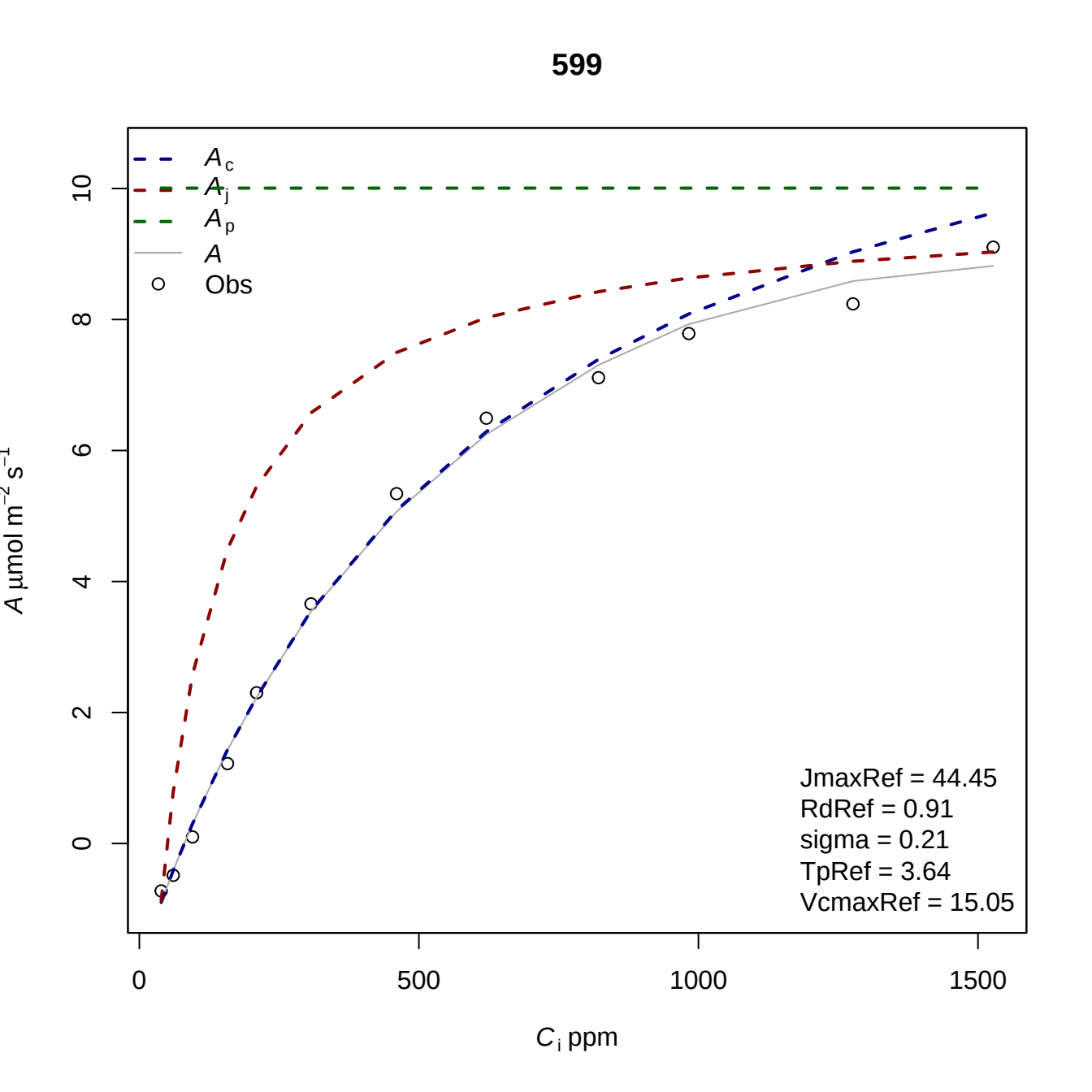

594
A_c
A_j
A_p
A
Obs
JmaxRef = 222.81
RdRef = 8.45
sigma = 1.08
TpRef = 5444.24
VcmaxRef = 95.05
A µmol m^{-2} s^{-1}
−10
0
10
20
30
40
200
400
600
800
1000
1200
C_i ppm

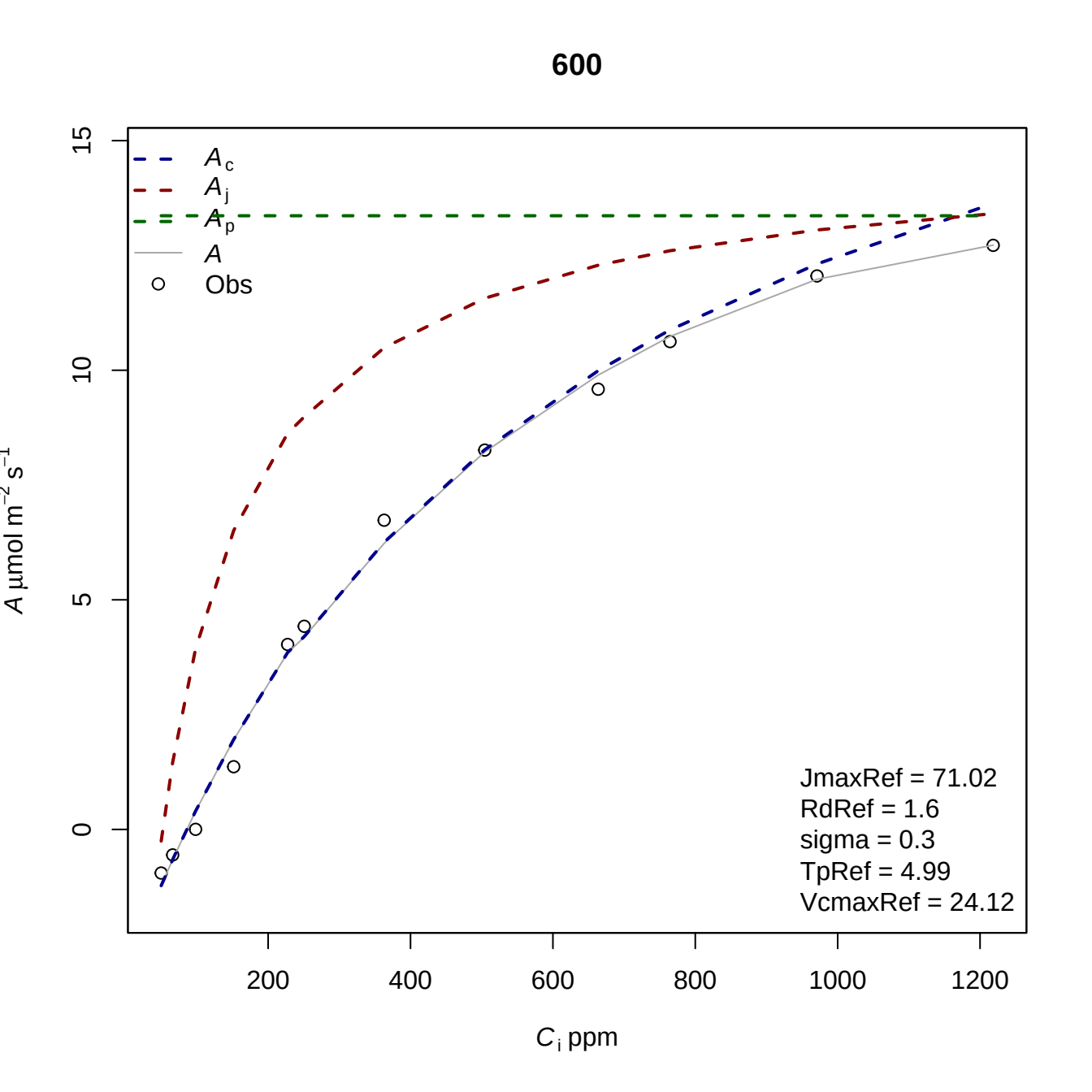

595
A_c
A_j
A_p
A
Obs
JmaxRef = 181.11
RdRef = 7.8
sigma = 1.5
TpRef = 2316.14
VcmaxRef = 68.8
0
10
20
30
40
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
C_i ppm

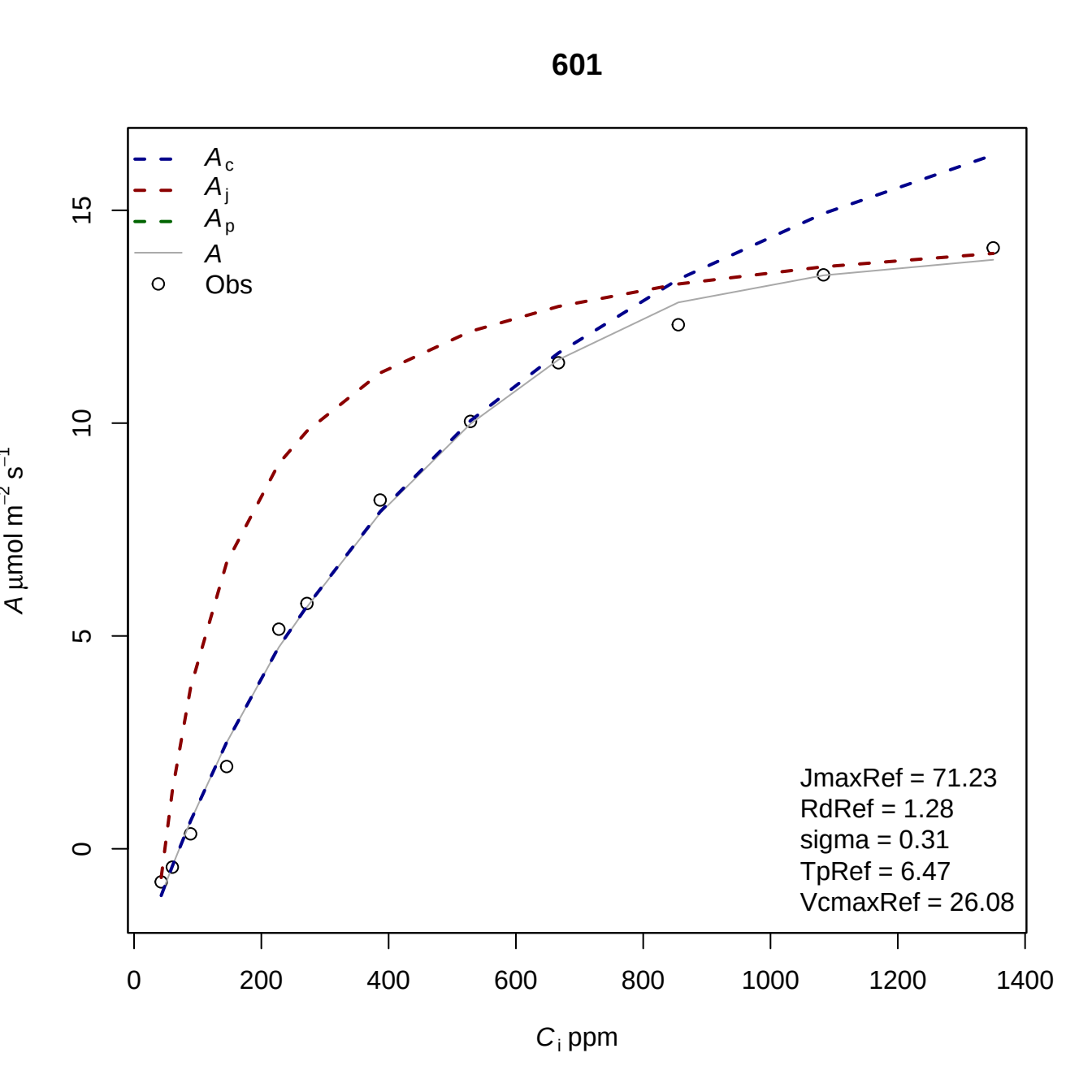

596
A_c
A_j
A_p
A
Obs
JmaxRef = 31.24
RdRef = 0.67
sigma = 0.2
TpRef = 3.1
VcmaxRef = 11.37
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
0
500
1000
1500

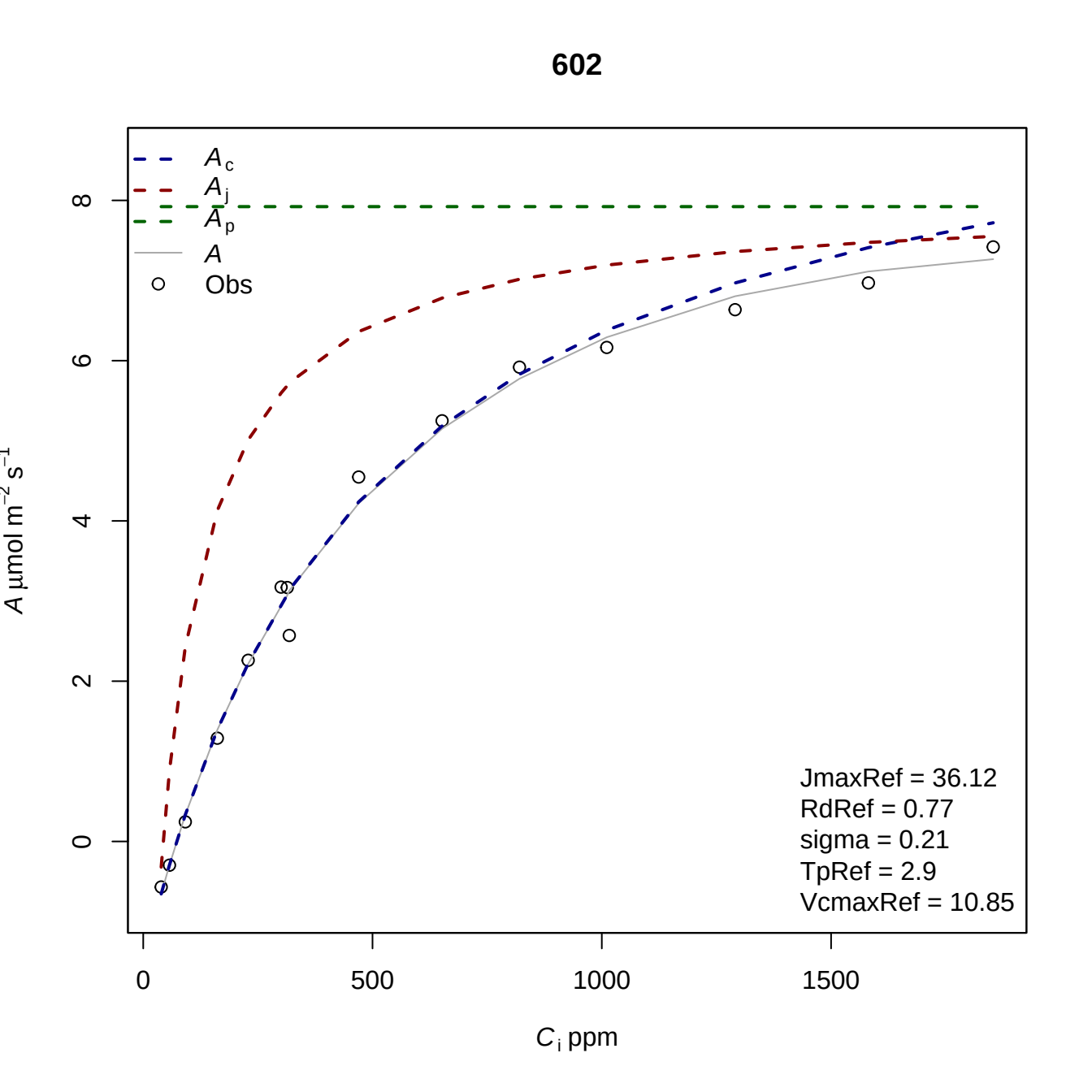

597
A_c
A_j
A_p
A
Obs
JmaxRef = 69.15
RdRef = 1.32
sigma = 0.38
TpRef = 5.15
VcmaxRef = 23.9
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
200
400
600
800
1000
1200
1400

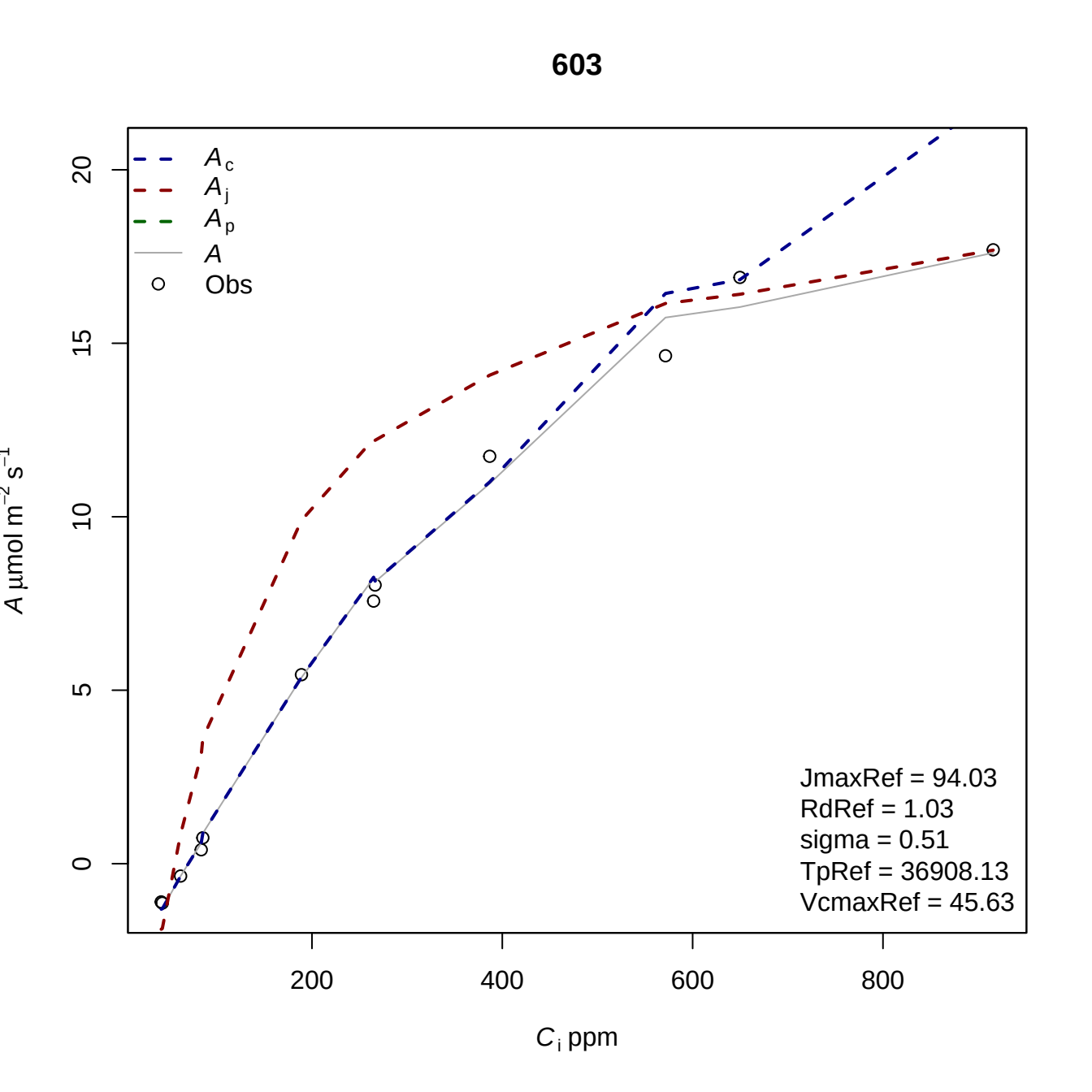

598
A_c
A_j
A_p
A
Obs
JmaxRef = 51.62
RdRef = 1.22
sigma = 0.28
TpRef = 4.85
VcmaxRef = 18.19
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

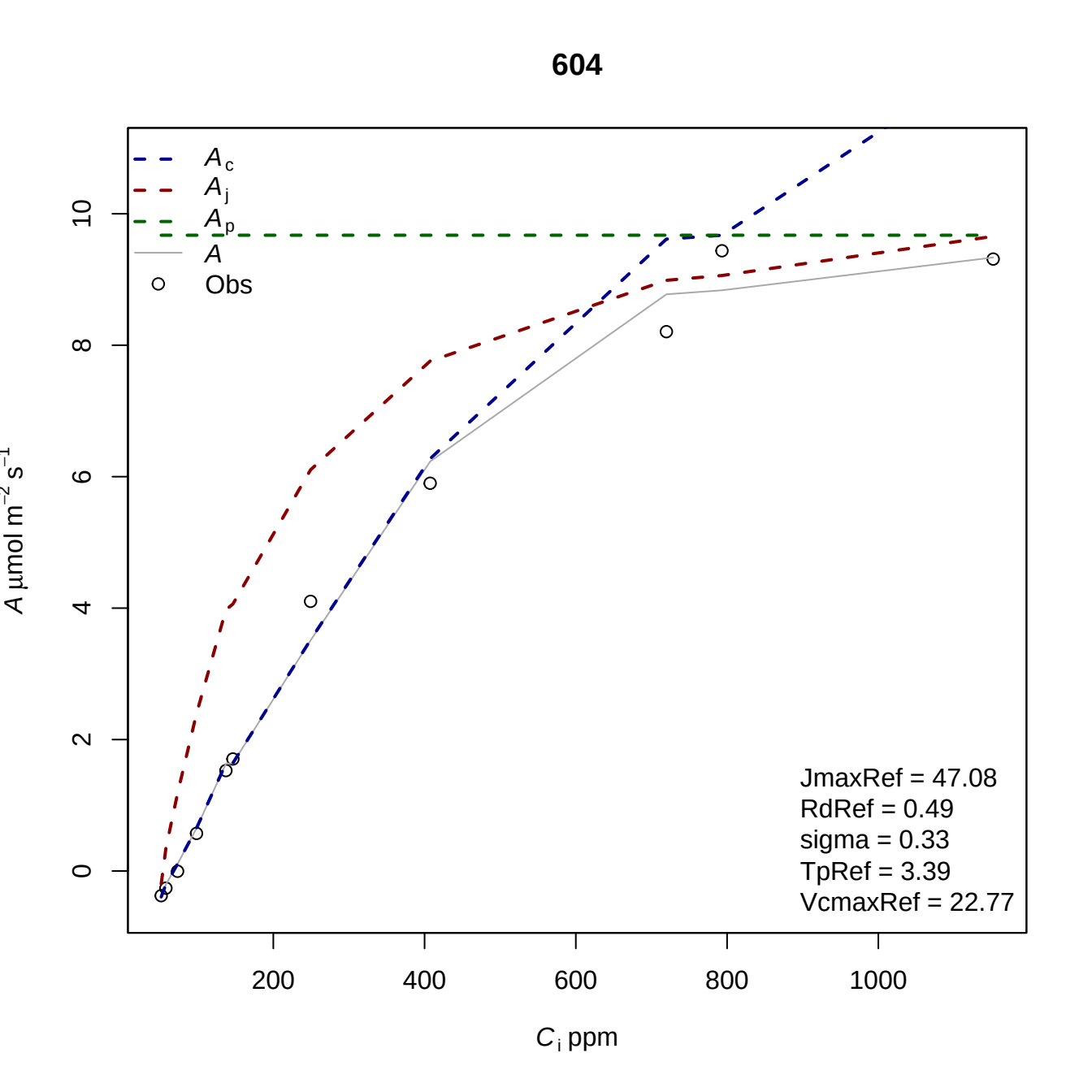

599
A_c
A_j
A_p
A
Obs
JmaxRef = 44.45
RdRef = 0.91
sigma = 0.21
TpRef = 3.64
VcmaxRef = 15.05
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
0
500
1000
1500

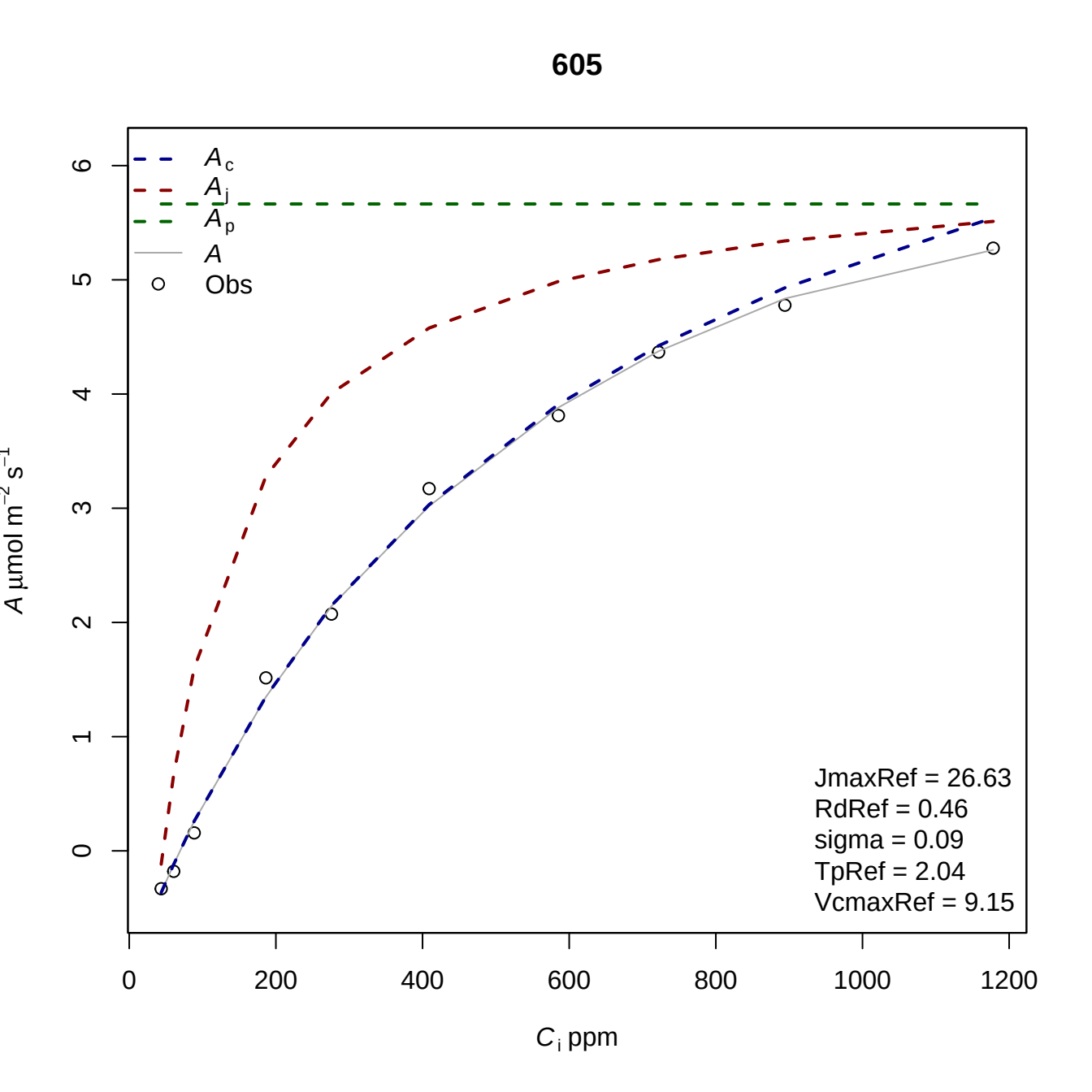

600
A_c
A_j
A_p
A
Obs
JmaxRef = 71.02
RdRef = 1.6
sigma = 0.3
TpRef = 4.99
VcmaxRef = 24.12
0
5
10
15
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
C_i ppm

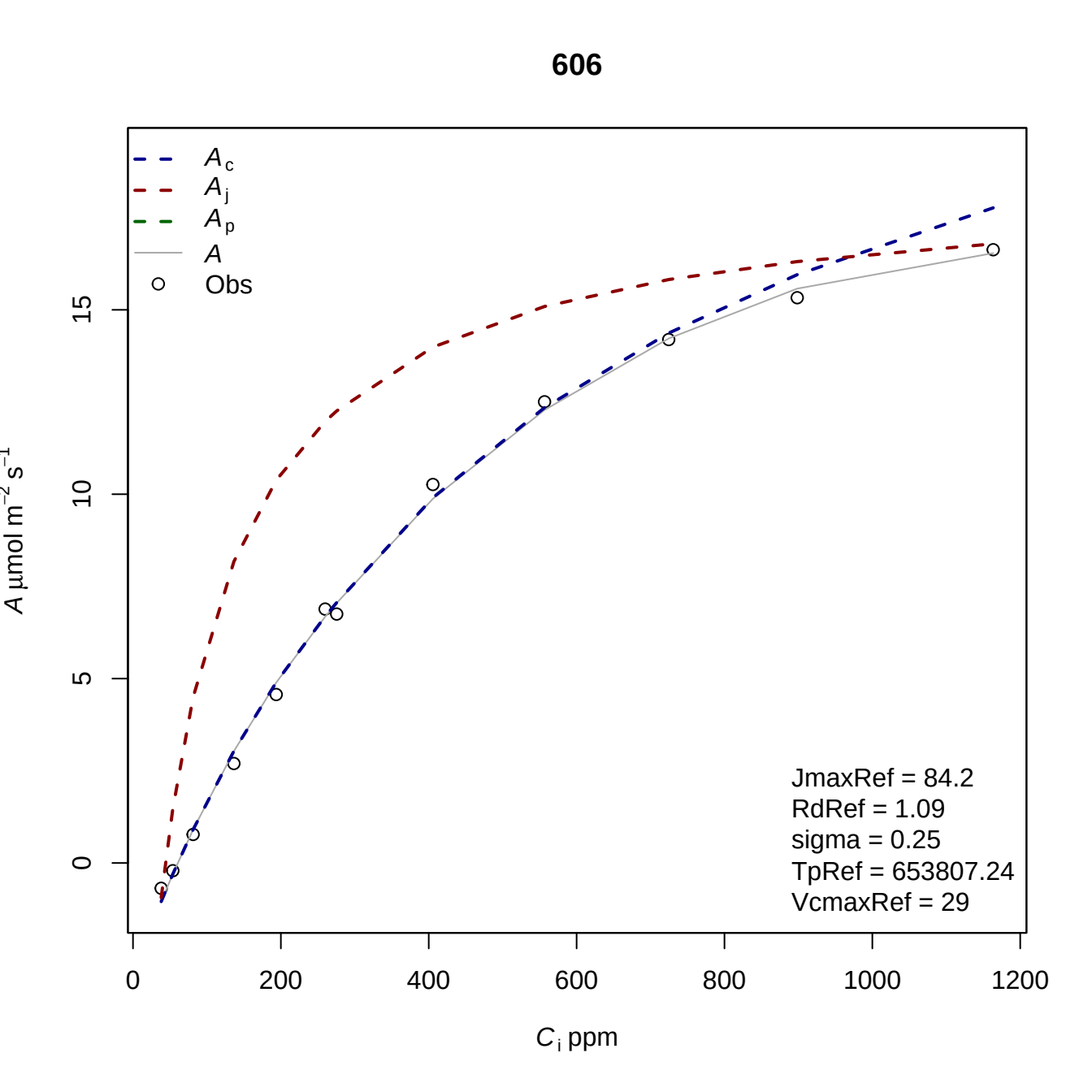

601
A_c
A_j
A_p
A
Obs
JmaxRef = 71.23
RdRef = 1.28
sigma = 0.31
TpRef = 6.47
VcmaxRef = 26.08
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
200
400
600
800
1000
1200
1400

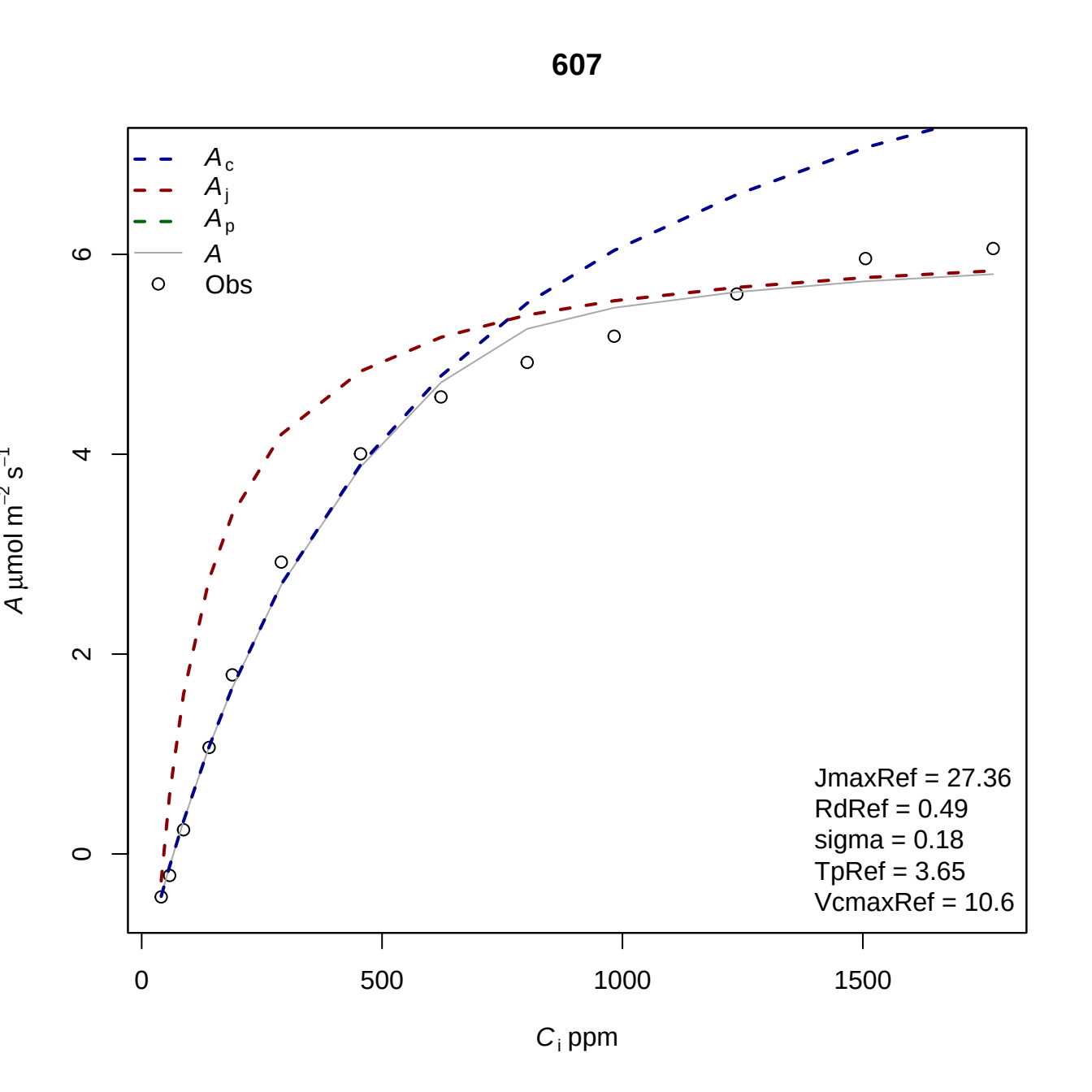

602
A_c
A_j
A_p
A
Obs
JmaxRef = 36.12
RdRef = 0.77
sigma = 0.21
TpRef = 2.9
VcmaxRef = 10.85
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
0
500
1000
1500

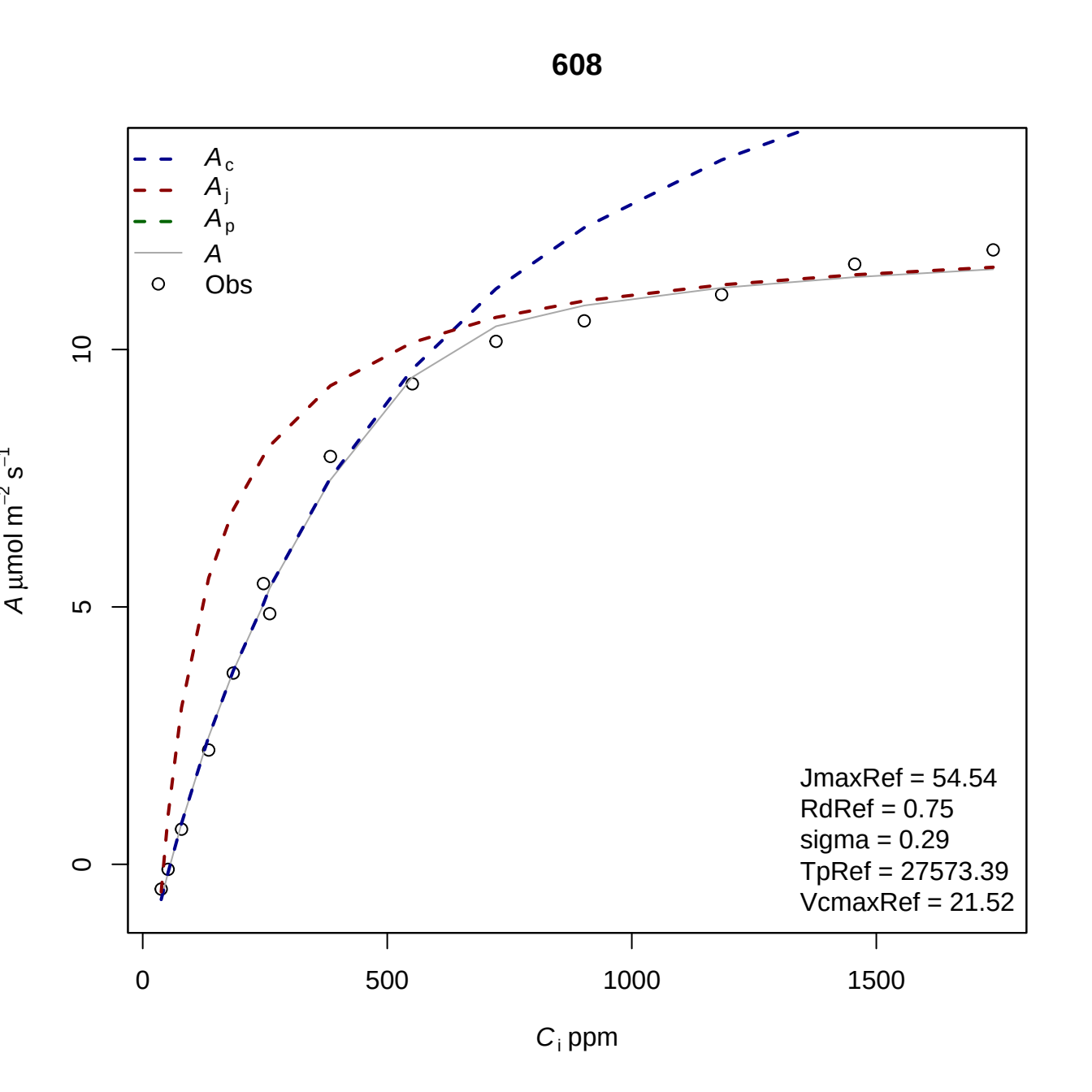

603
A_c
A_j
A_p
A
Obs
JmaxRef = 94.03
RdRef = 1.03
sigma = 0.51
TpRef = 36908.13
VcmaxRef = 45.63
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
200
400
600
800

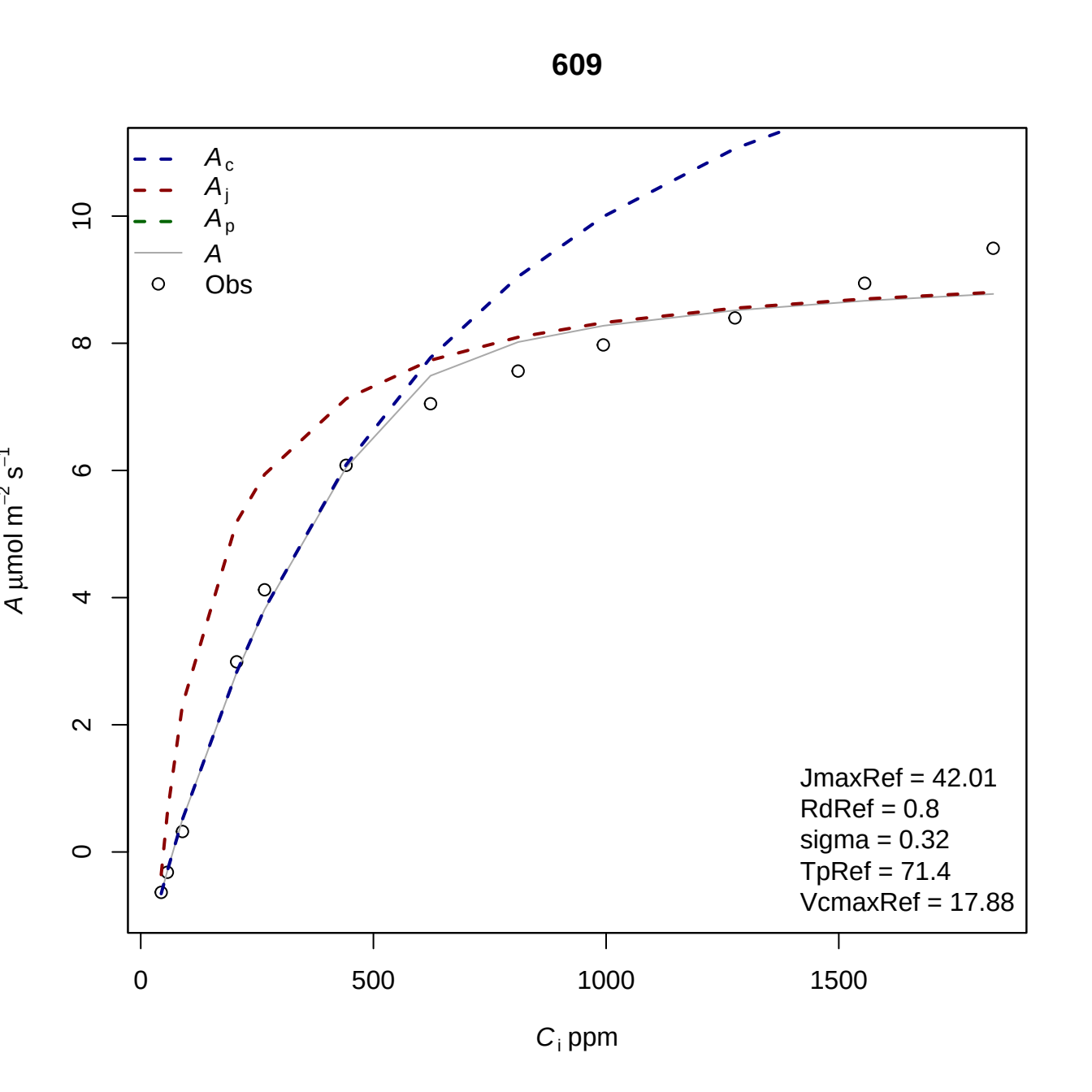

604
A_c
A_j
A_p
A
Obs
JmaxRef = 47.08
RdRef = 0.49
sigma = 0.33
TpRef = 3.39
VcmaxRef = 22.77
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
200
400
600
800
1000

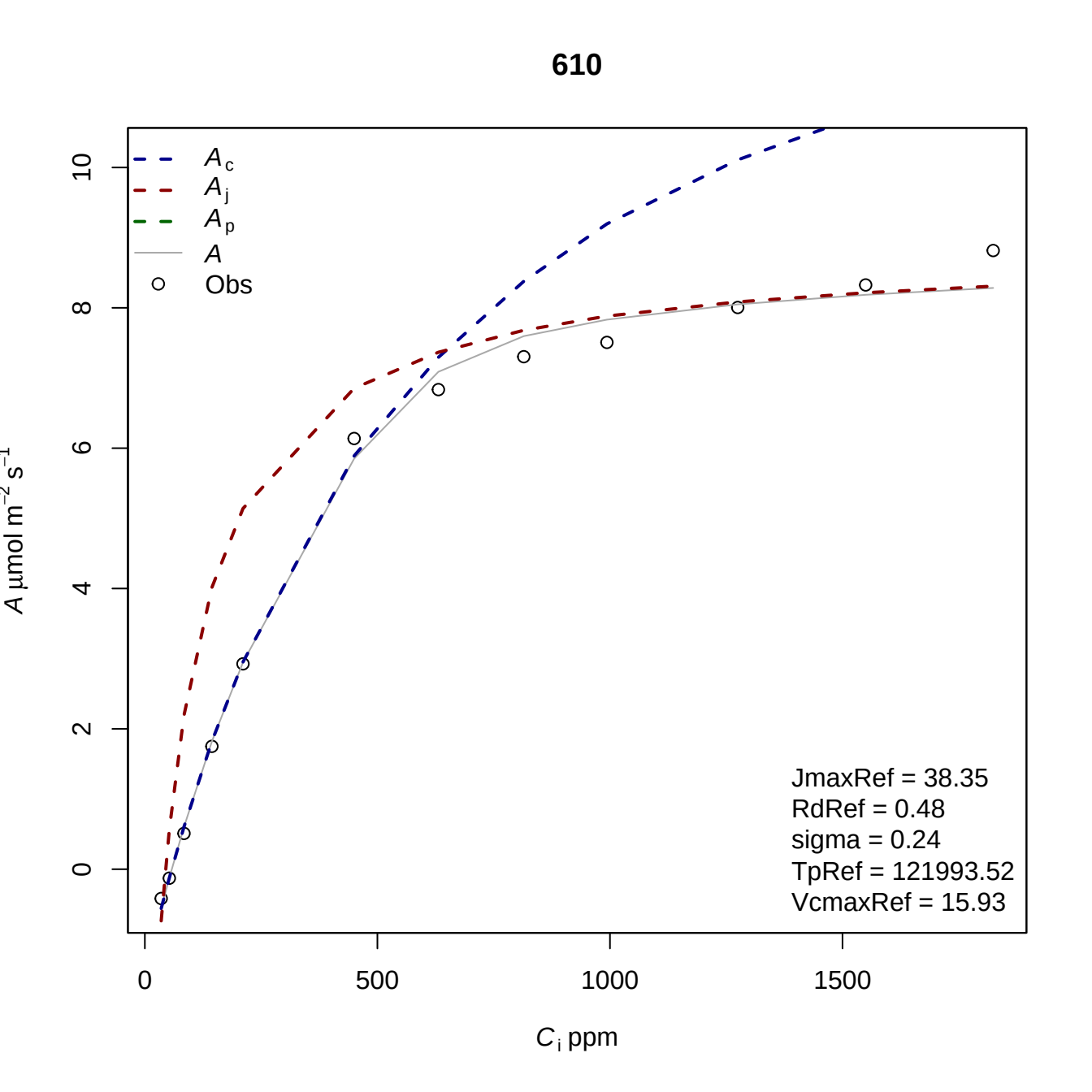

605
A_c
A_j
A_p
A
Obs
JmaxRef = 26.63
RdRef = 0.46
sigma = 0.09
TpRef = 2.04
VcmaxRef = 9.15
A µmol m^{-2} s^{-1}
C_i ppm
0
1
2
3
4
5
6
0
200
400
600
800
1000
1200

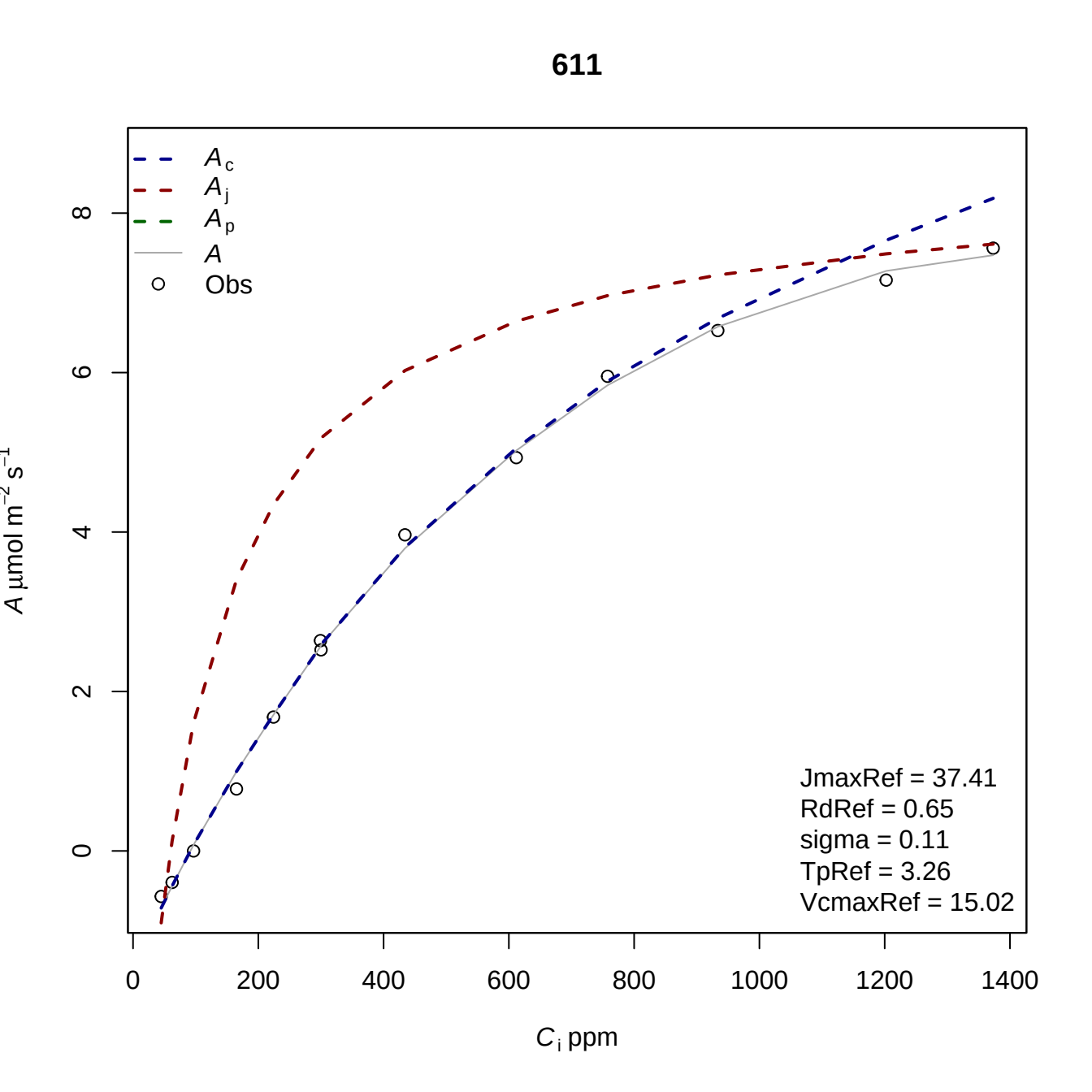

606
A_c
A_j
A_p
A
Obs
JmaxRef = 84.2
RdRef = 1.09
sigma = 0.25
TpRef = 653807.24
VcmaxRef = 29
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
200
400
600
800
1000
1200

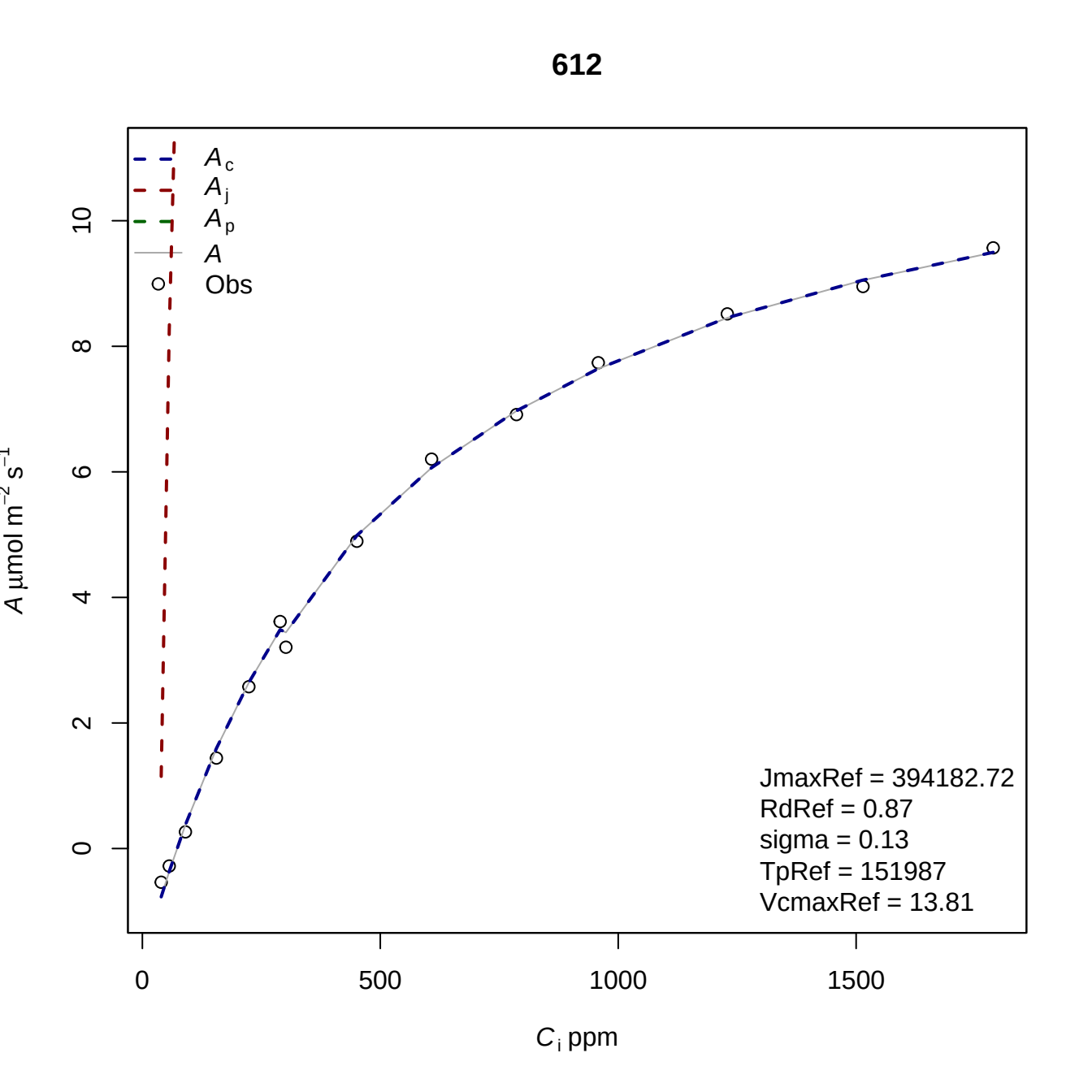

607
A_c
A_j
A_p
A
Obs
JmaxRef = 27.36
RdRef = 0.49
sigma = 0.18
TpRef = 3.65
VcmaxRef = 10.6
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
C_i ppm

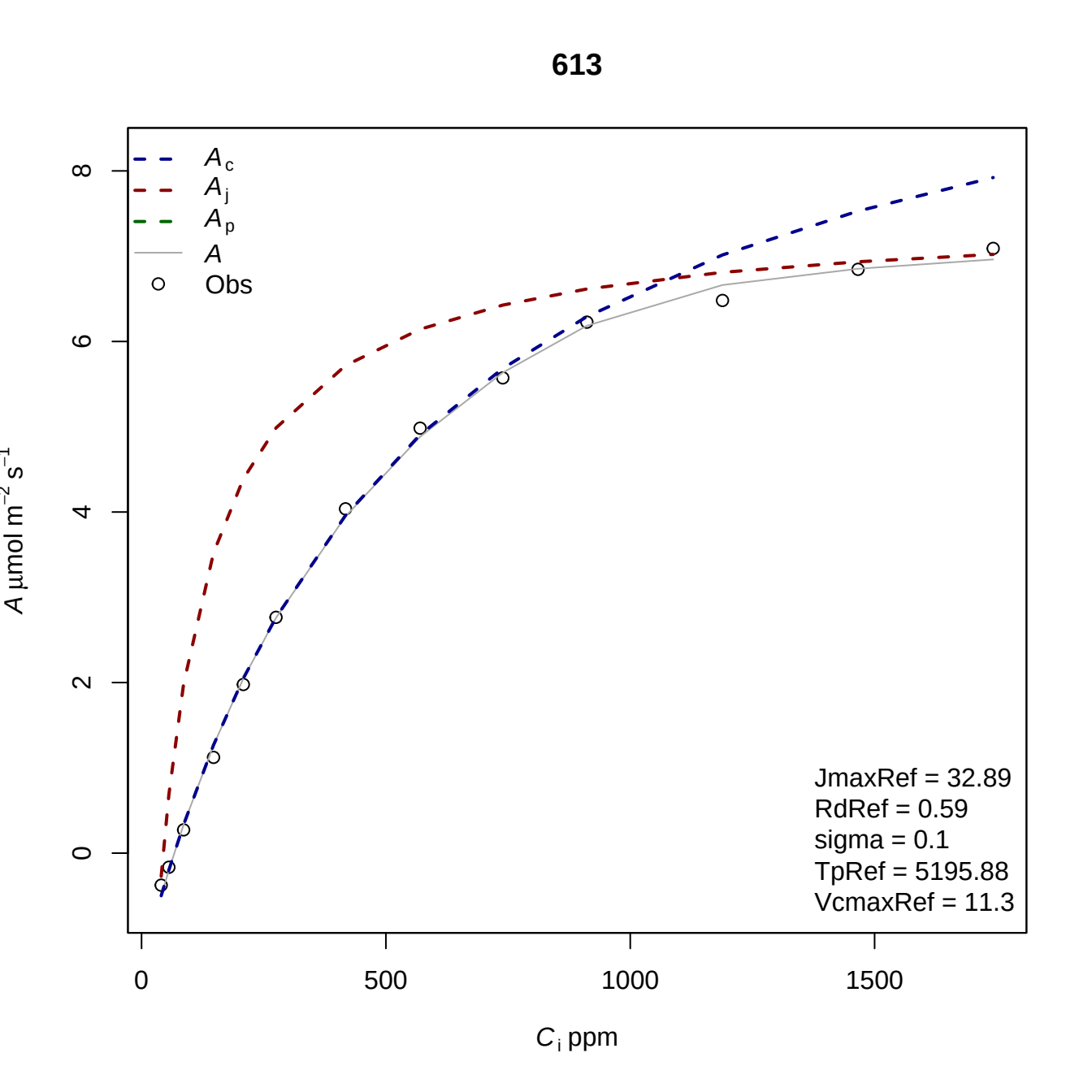

608
A_c
A_j
A_p
A
Obs
JmaxRef = 54.54
RdRef = 0.75
sigma = 0.29
TpRef = 27573.39
VcmaxRef = 21.52
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
C_i ppm

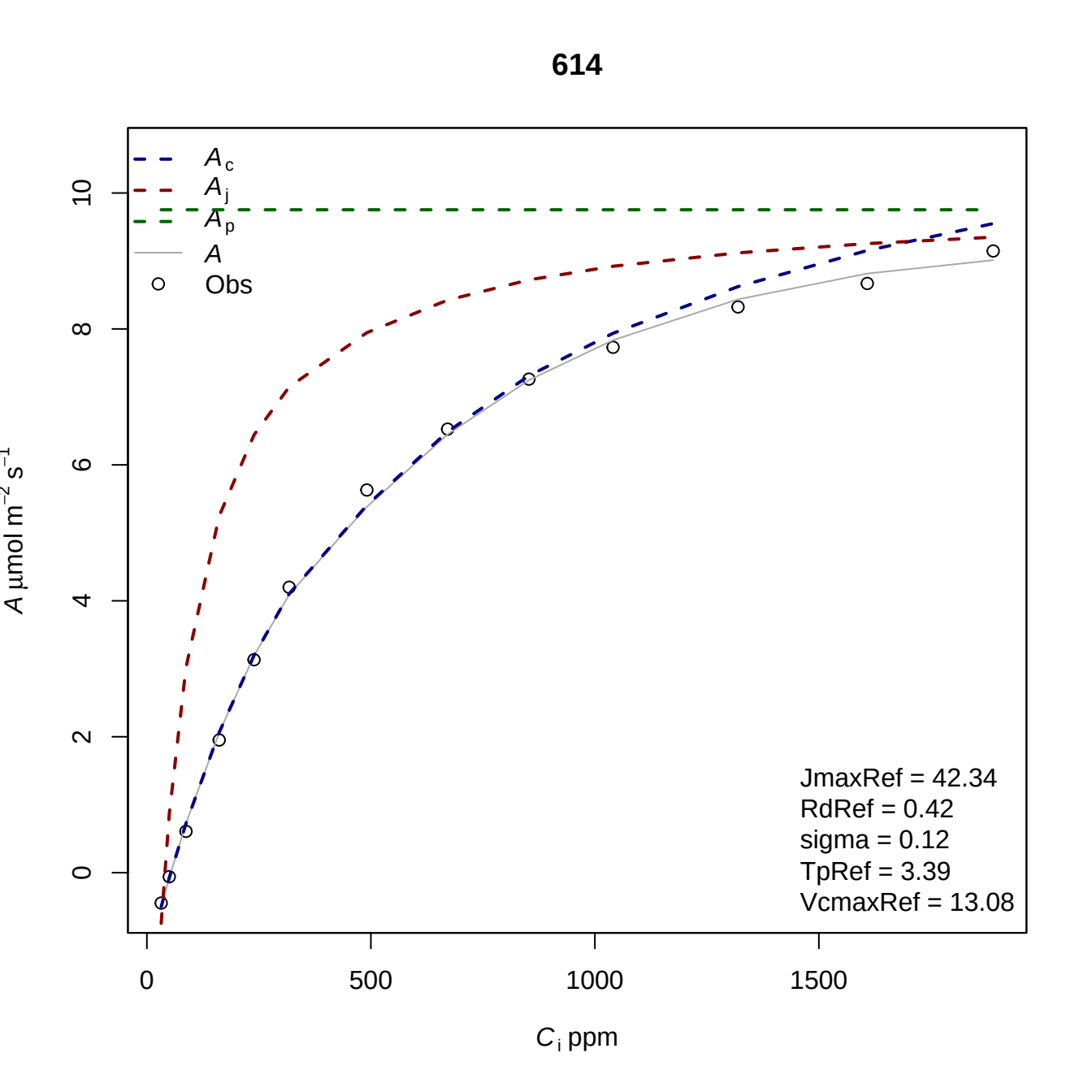

609
A_c
A_j
A_p
A
Obs
JmaxRef = 42.01
RdRef = 0.8
sigma = 0.32
TpRef = 71.4
VcmaxRef = 17.88
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
0
500
1000
1500

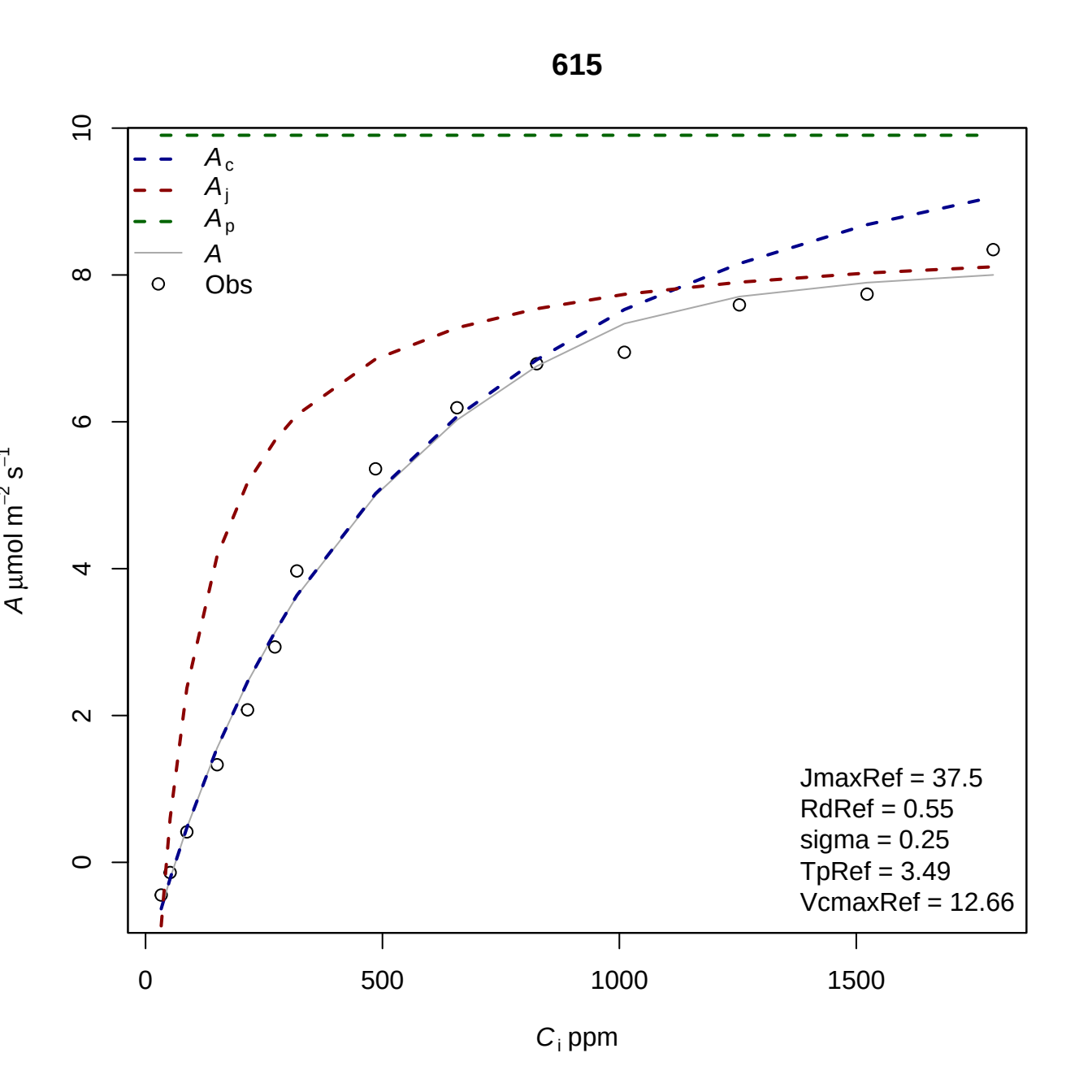

610
A_c
A_j
A_p
A
Obs
JmaxRef = 38.35
RdRef = 0.48
sigma = 0.24
TpRef = 121993.52
VcmaxRef = 15.93
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
0
500
1000
1500

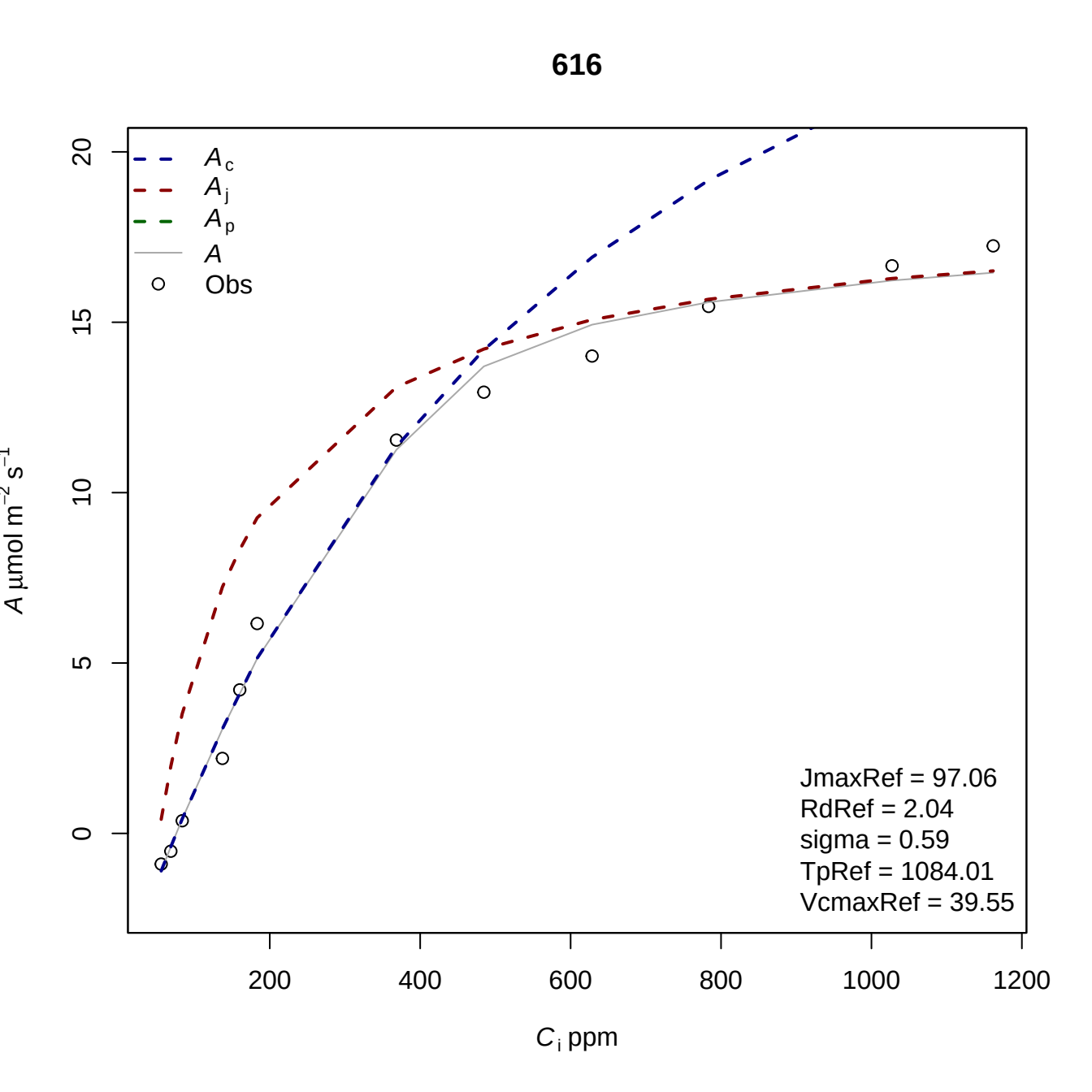

611
A_c
A_j
A_p
A
Obs
JmaxRef = 37.41
RdRef = 0.65
sigma = 0.11
TpRef = 3.26
VcmaxRef = 15.02
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
0
200
400
600
800
1000
1200
1400

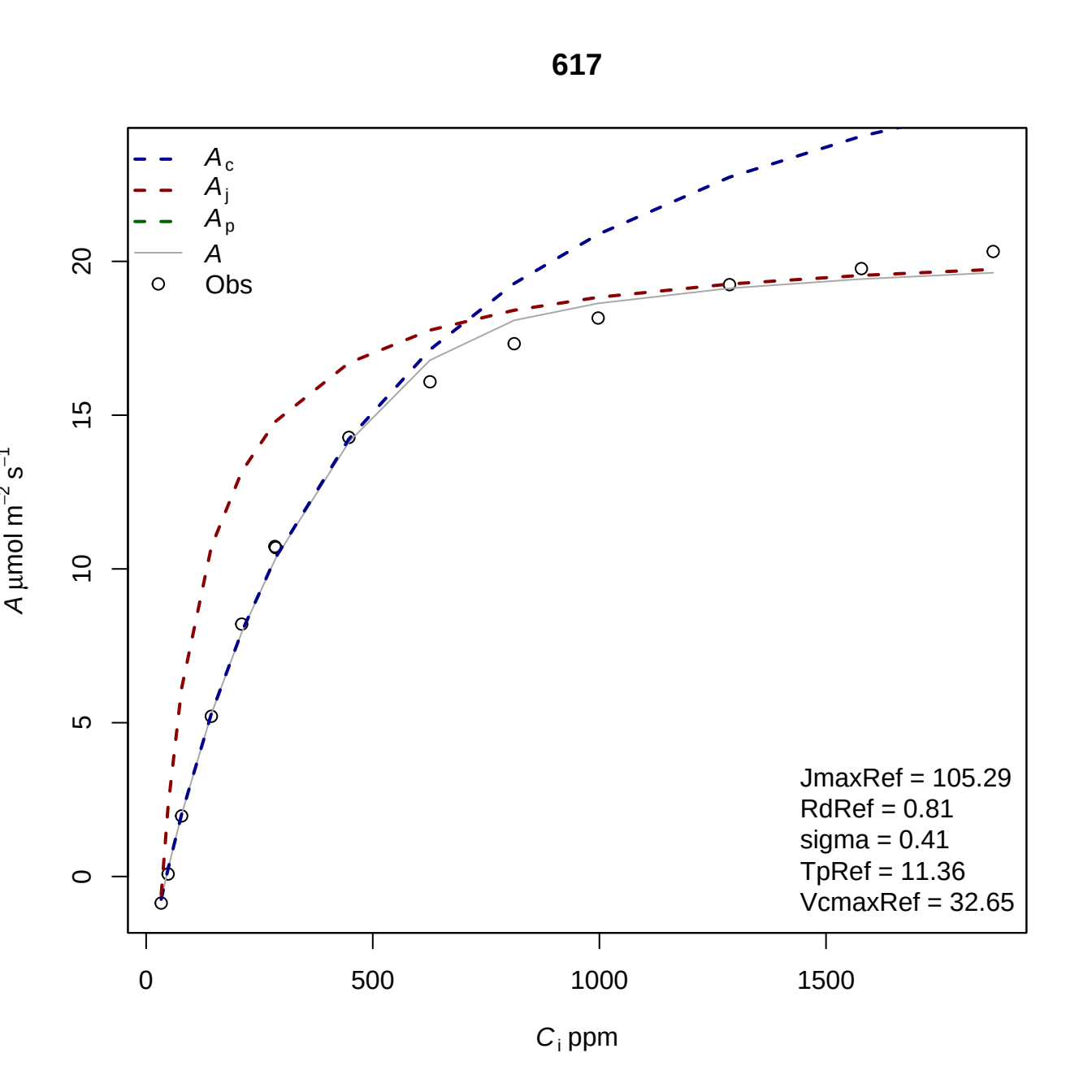

612
A_c
A_j
A_p
A
Obs
JmaxRef = 394182.72
RdRef = 0.87
sigma = 0.13
TpRef = 151987
VcmaxRef = 13.81
A µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
C_i ppm

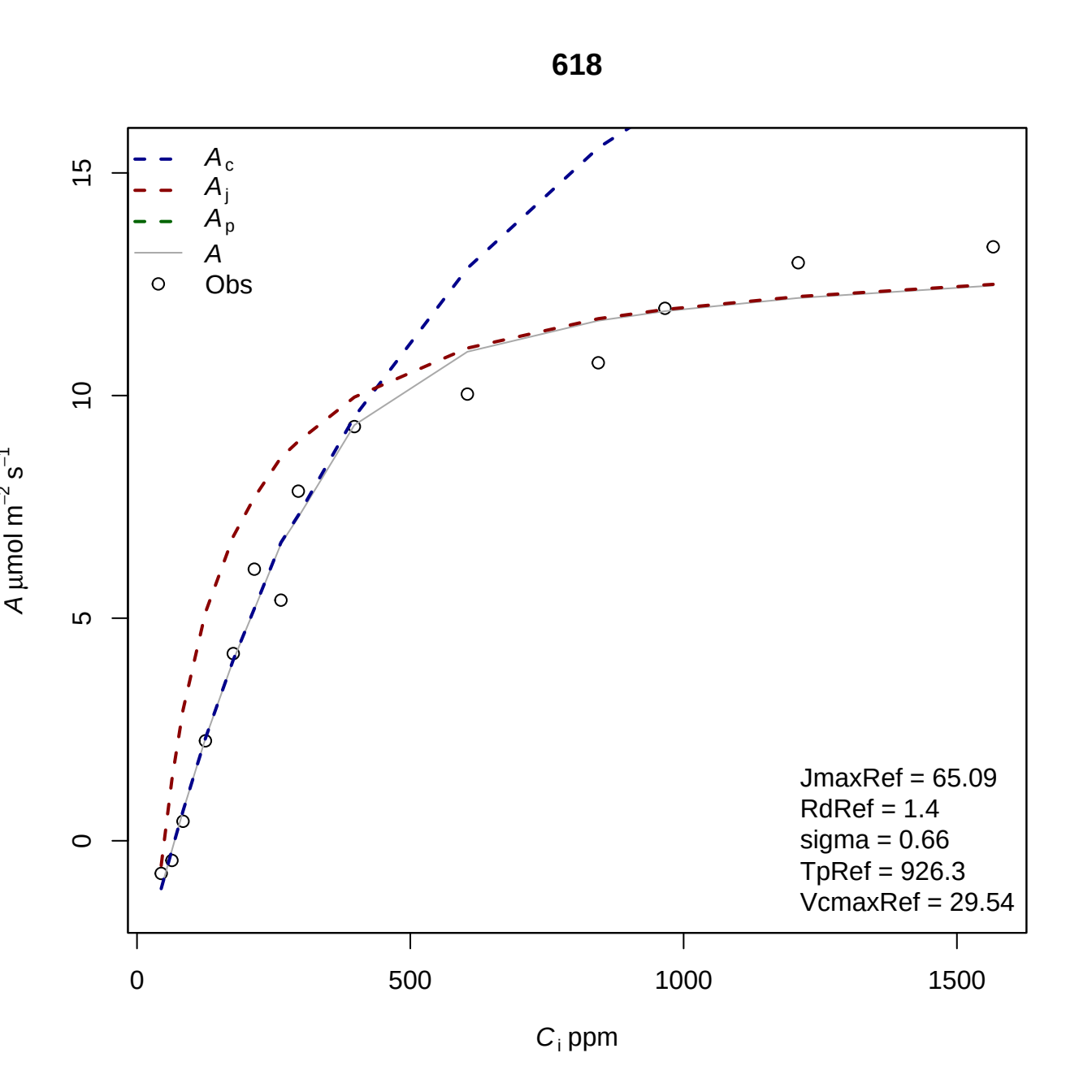

613
A_c
A_j
A_p
A
Obs
JmaxRef = 32.89
RdRef = 0.59
sigma = 0.1
TpRef = 5195.88
VcmaxRef = 11.3
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
0
500
1000
1500

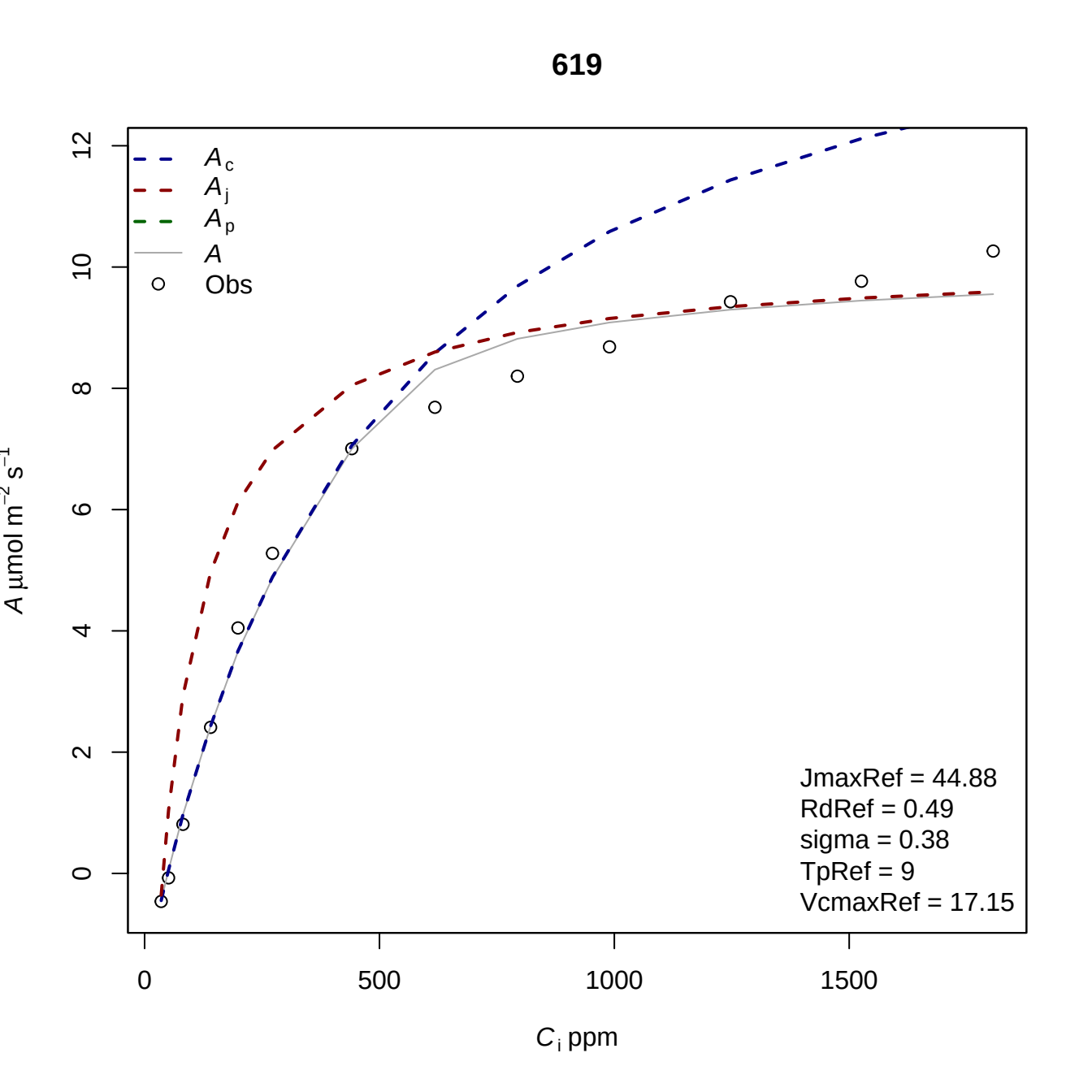

614
A_c
A_j
A_p
A
Obs
JmaxRef = 42.34
RdRef = 0.42
sigma = 0.12
TpRef = 3.39
VcmaxRef = 13.08
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
0
500
1000
1500

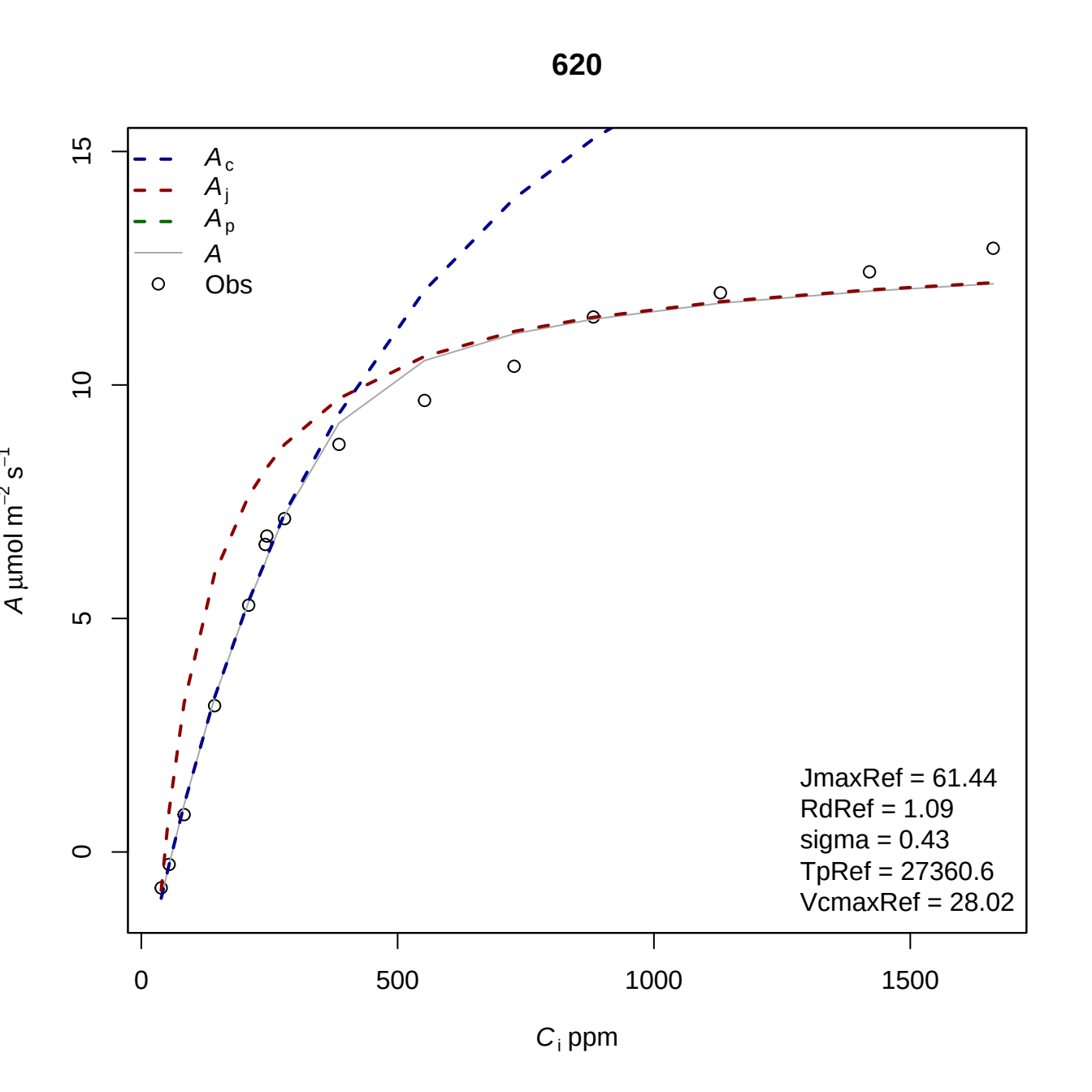

615
A_c
A_j
A_p
A
Obs
JmaxRef = 37.5
RdRef = 0.55
sigma = 0.25
TpRef = 3.49
VcmaxRef = 12.66
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
0
500
1000
1500

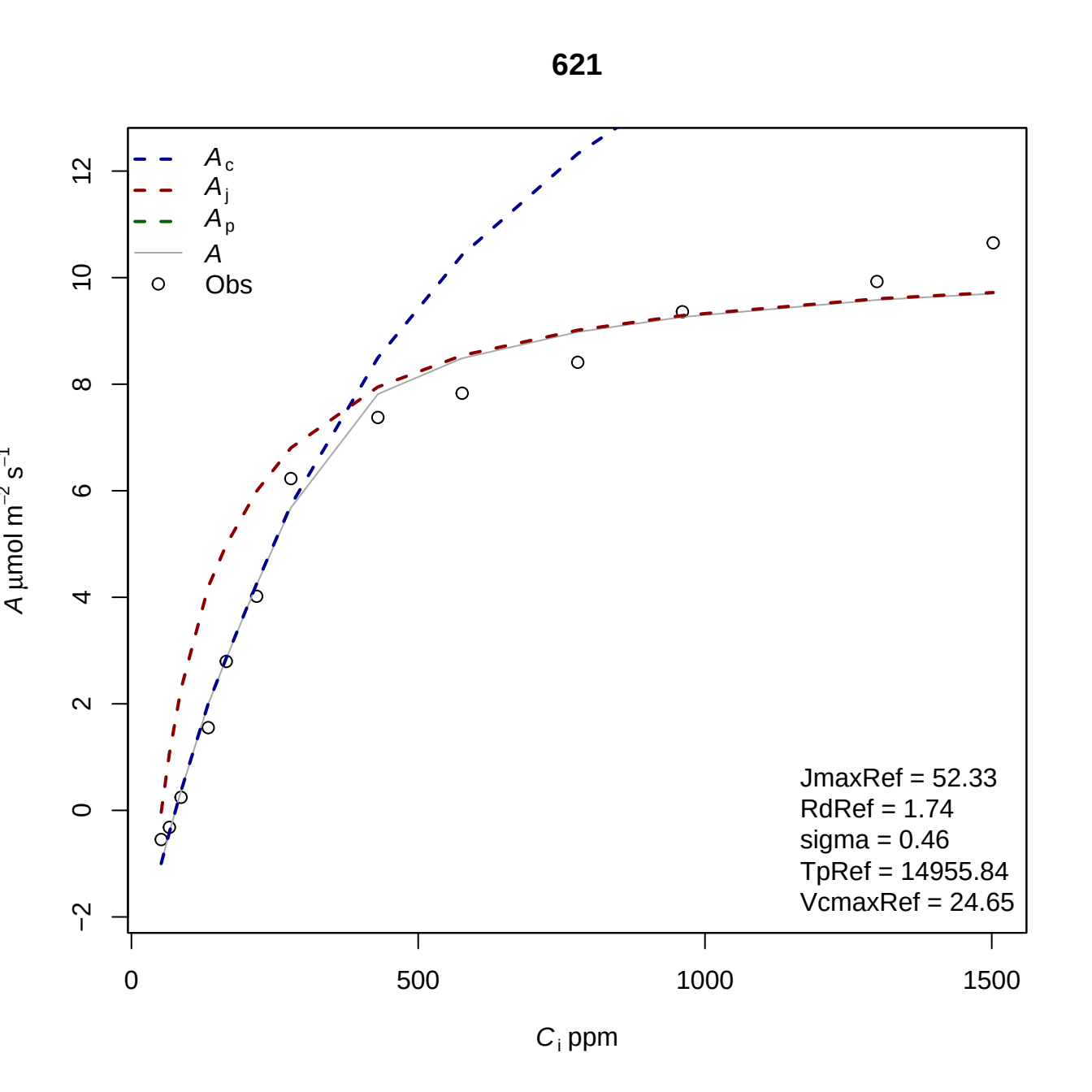

616
A_c
A_j
A_p
A
Obs
JmaxRef = 97.06
RdRef = 2.04
sigma = 0.59
TpRef = 1084.01
VcmaxRef = 39.55
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
200
400
600
800
1000
1200

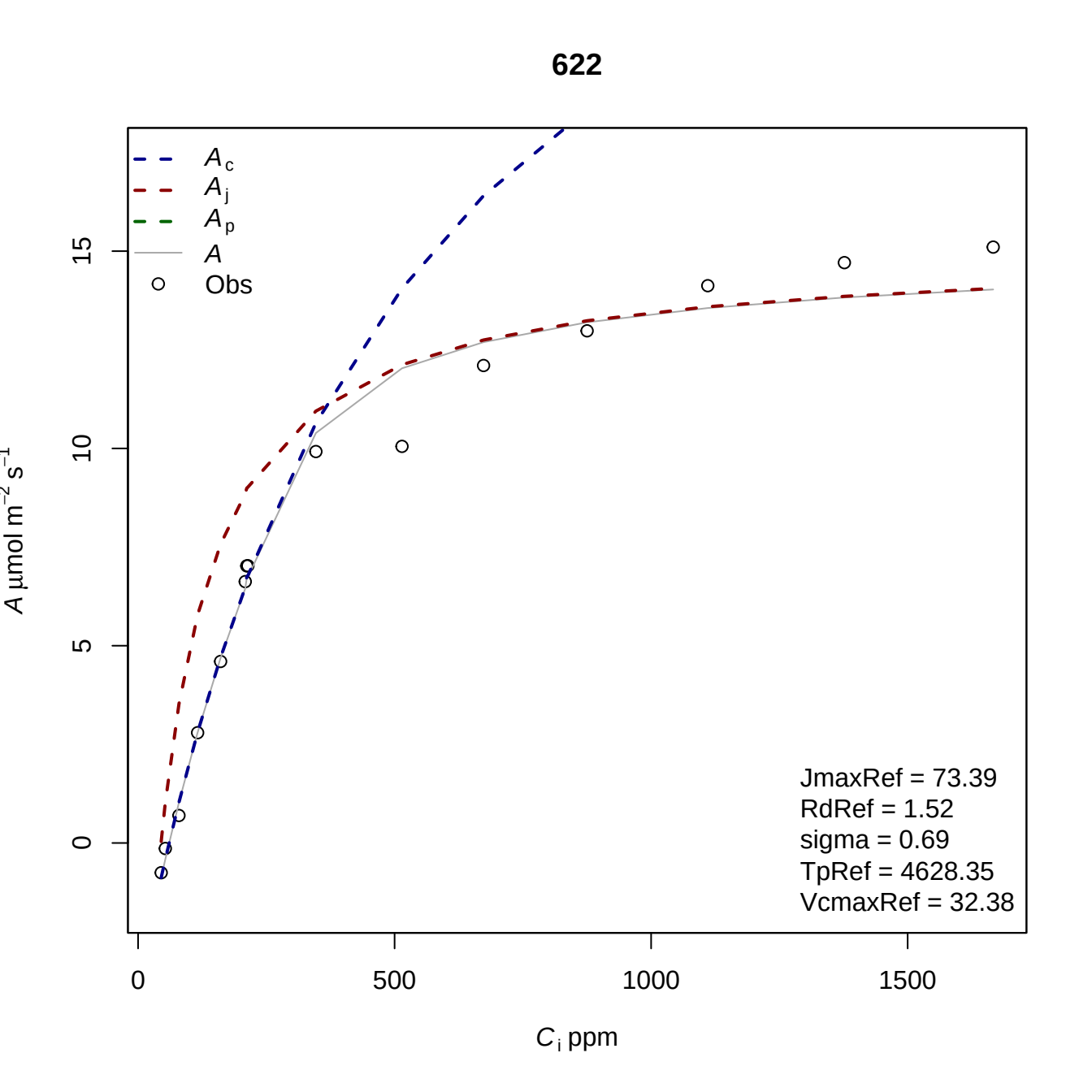

617
A_c
A_j
A_p
A
Obs
JmaxRef = 105.29
RdRef = 0.81
sigma = 0.41
TpRef = 11.36
VcmaxRef = 32.65
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
C_i ppm

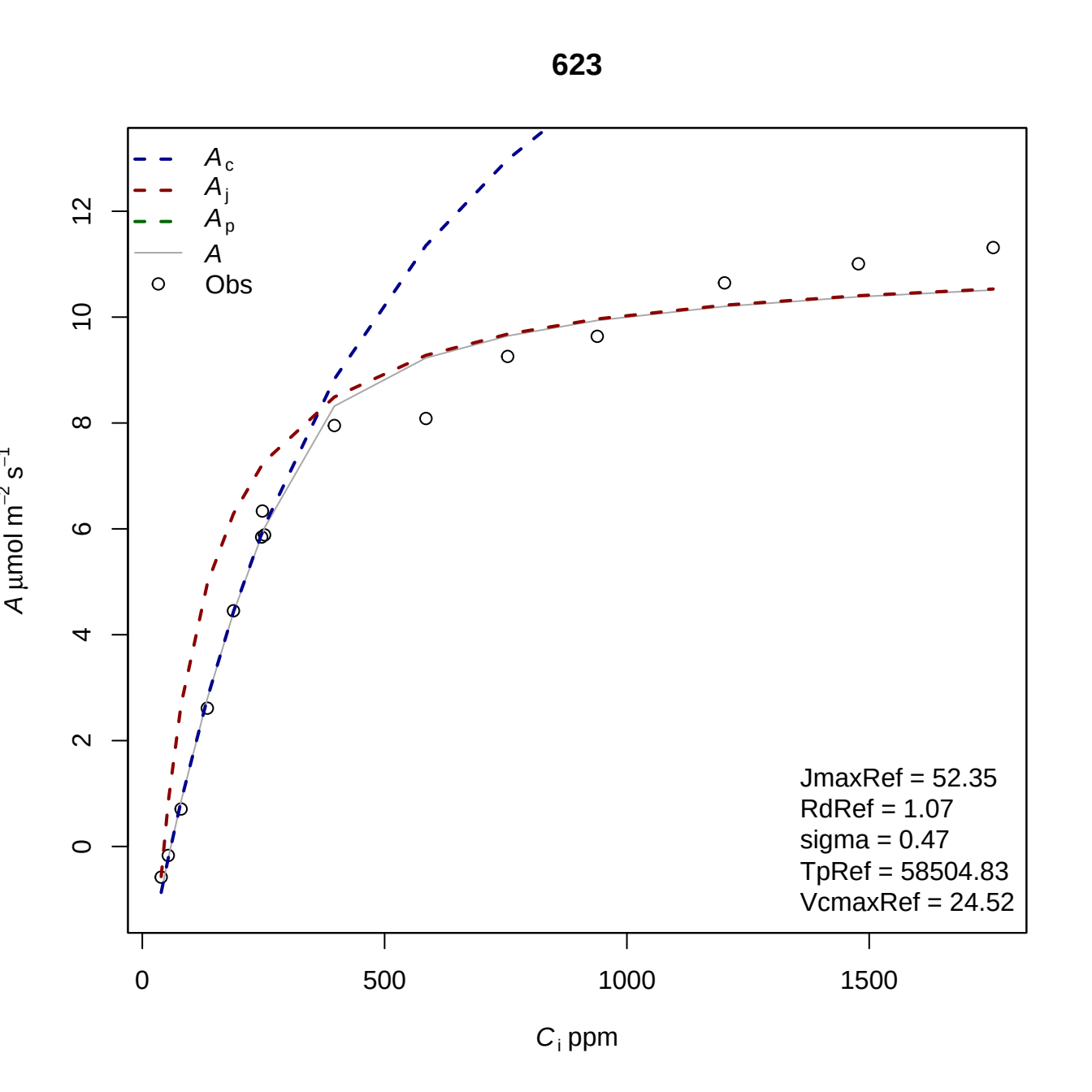

618
A_c
A_j
A_p
A
Obs
JmaxRef = 65.09
RdRef = 1.4
sigma = 0.66
TpRef = 926.3
VcmaxRef = 29.54
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
500
1000
1500

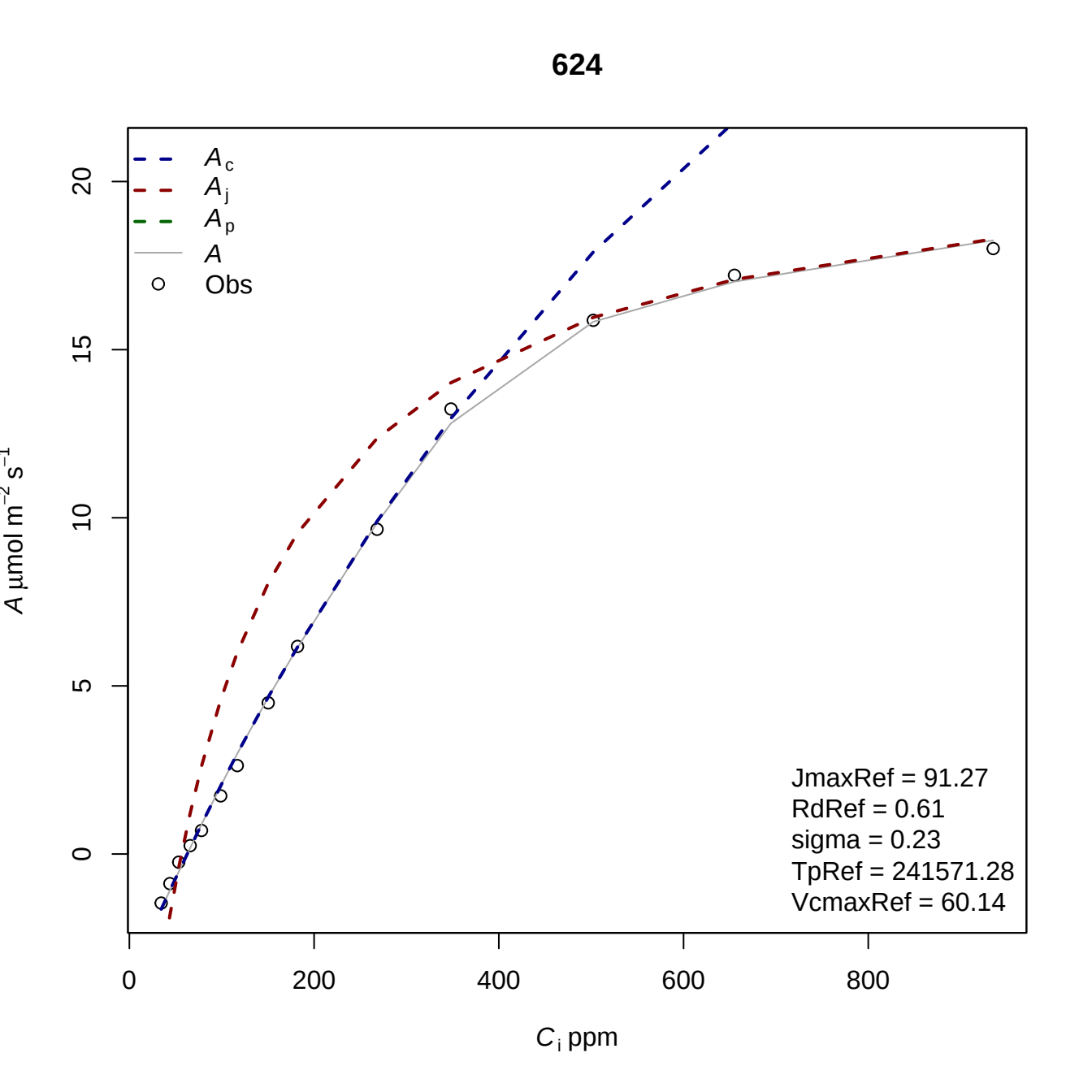

619
A_c
A_j
A_p
A
Obs
JmaxRef = 44.88
RdRef = 0.49
sigma = 0.38
TpRef = 9
VcmaxRef = 17.15
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
0
500
1000
1500

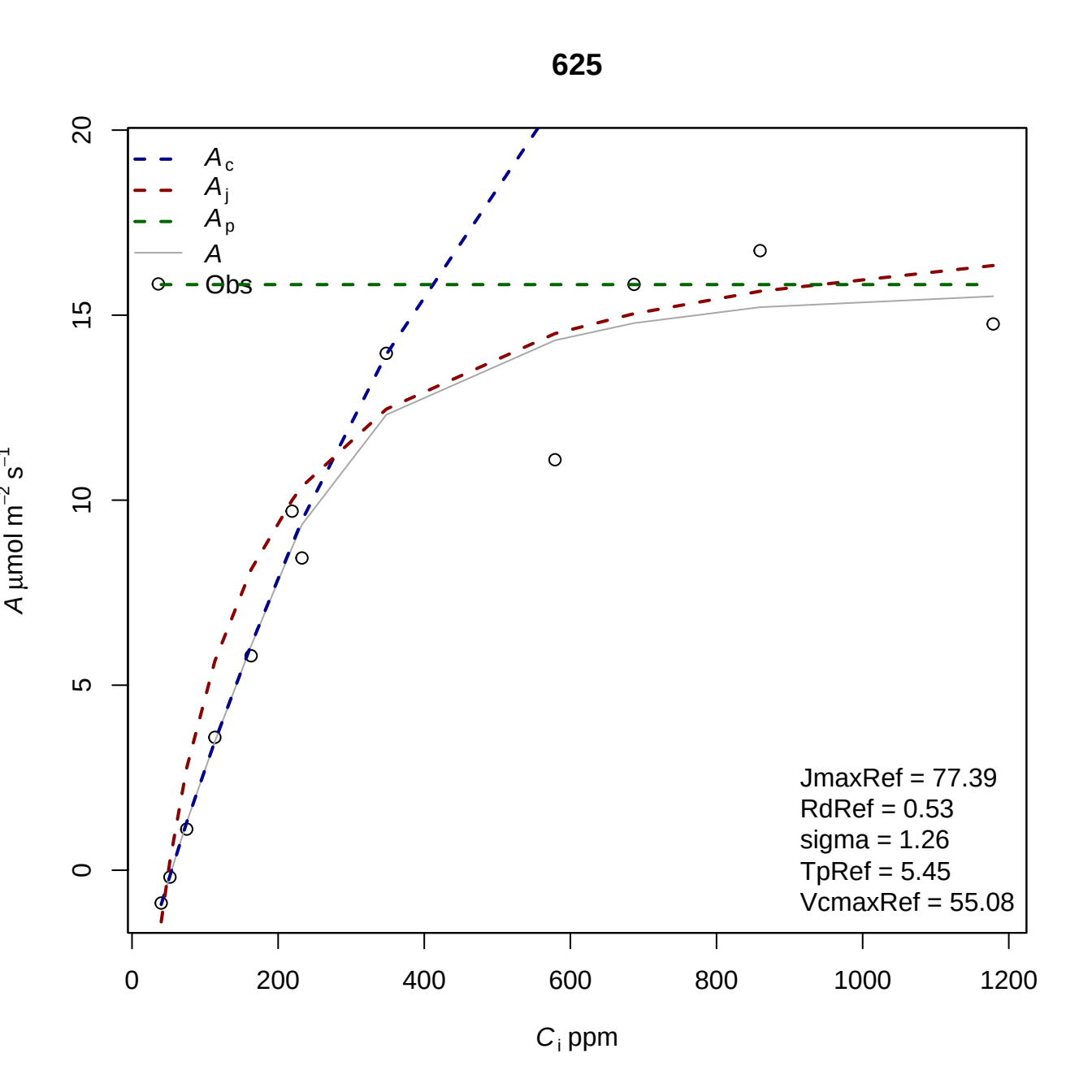

620
A_c
A_j
A_p
A
Obs
JmaxRef = 61.44
RdRef = 1.09
sigma = 0.43
TpRef = 27360.6
VcmaxRef = 28.02
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
500
1000
1500

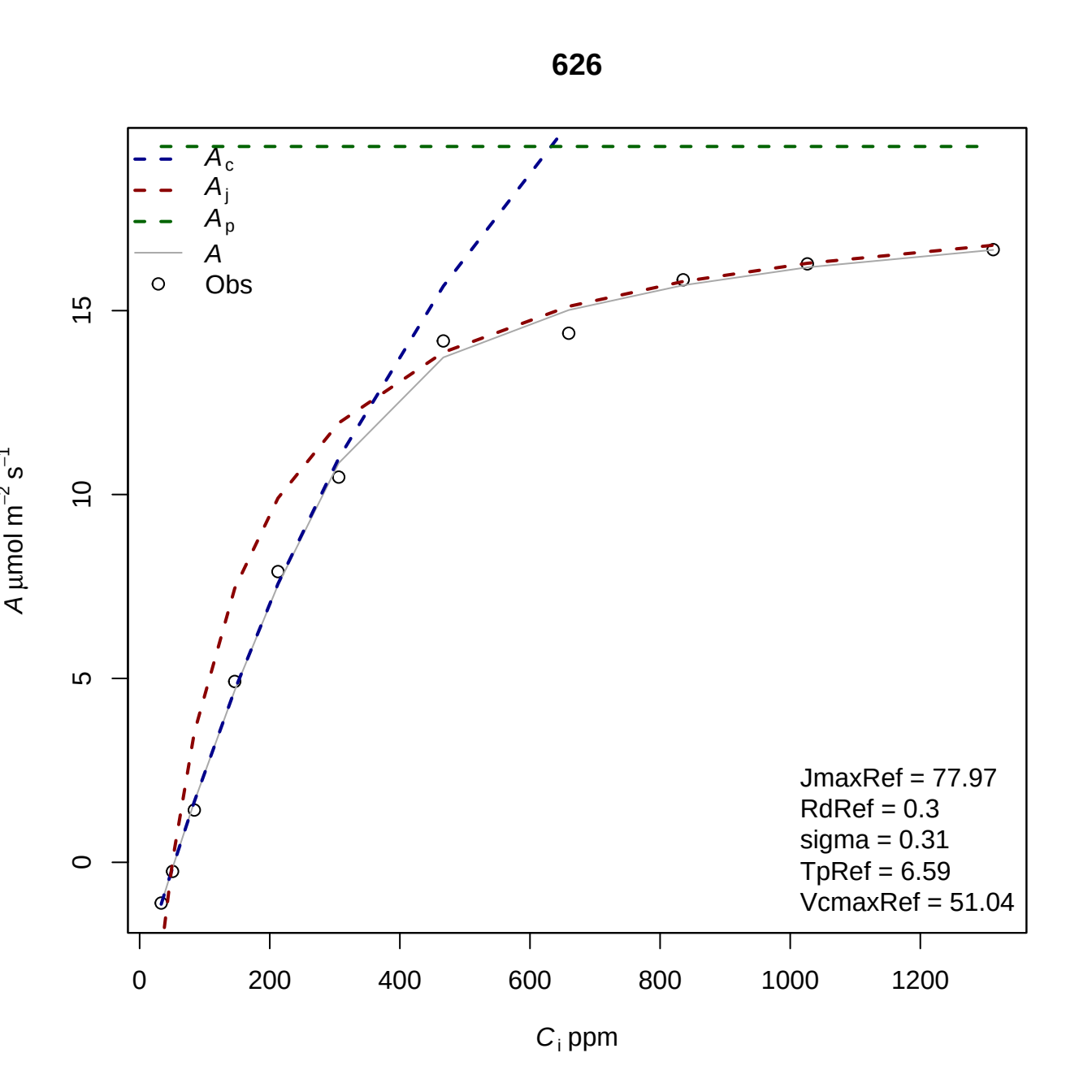

621
A_c
A_j
A_p
A
Obs
JmaxRef = 52.33
RdRef = 1.74
sigma = 0.46
TpRef = 14955.84
VcmaxRef = 24.65
A µmol m^{-2} s^{-1}
C_i ppm
−2
0
2
4
6
8
10
12
0
500
1000
1500

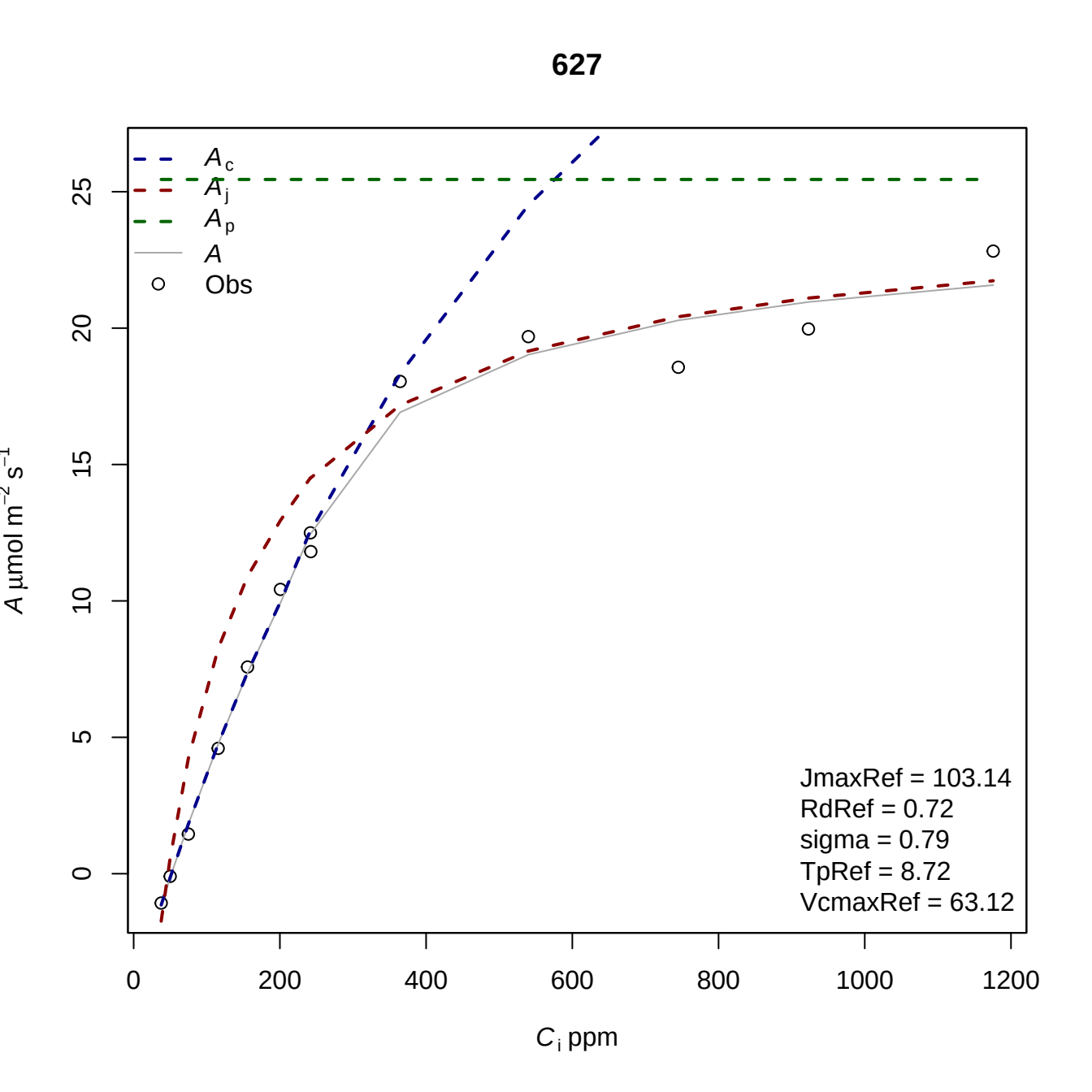

622
A_c
A_j
A_p
A
Obs
JmaxRef = 73.39
RdRef = 1.52
sigma = 0.69
TpRef = 4628.35
VcmaxRef = 32.38
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
500
1000
1500

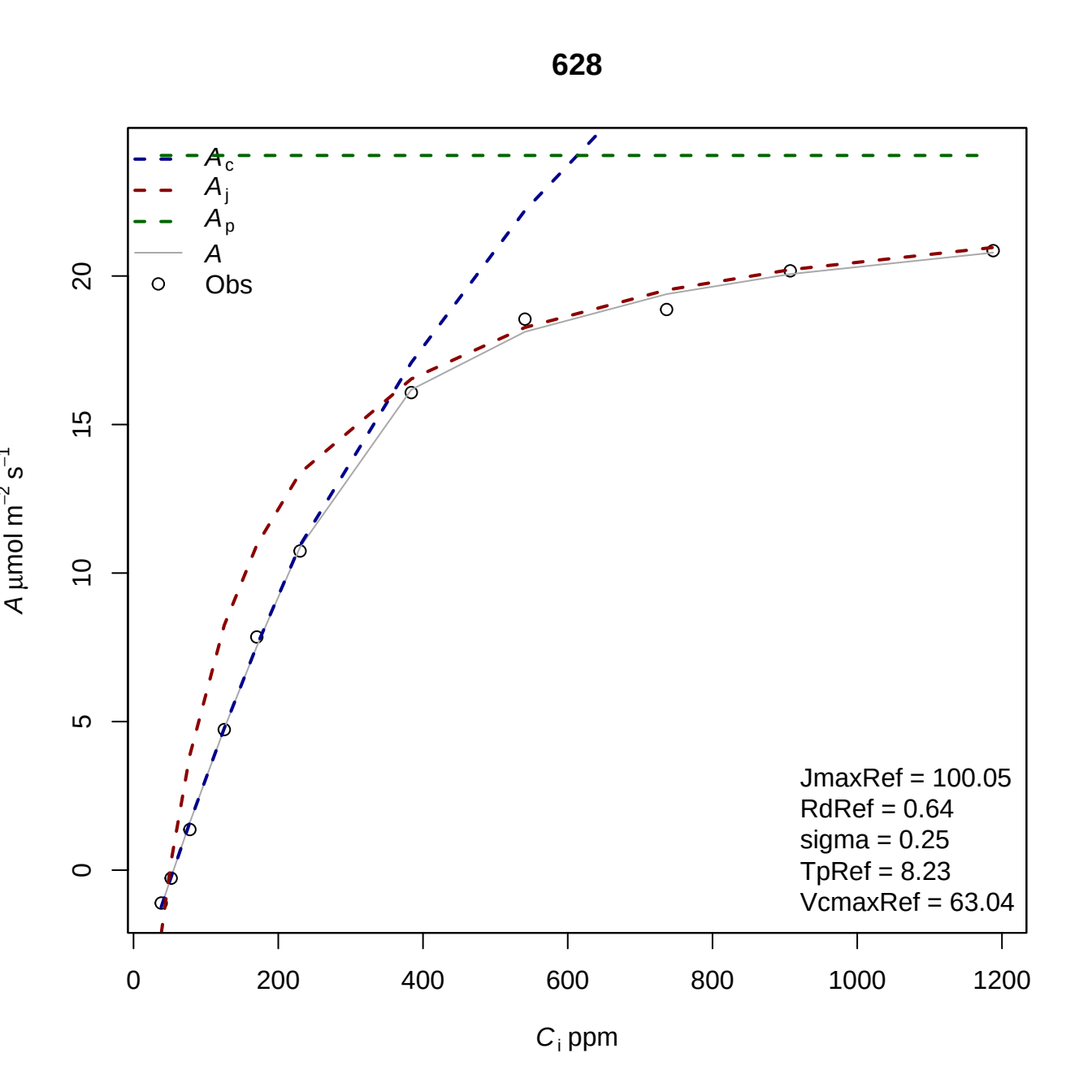

623
A_c
A_j
A_p
A
Obs
JmaxRef = 52.35
RdRef = 1.07
sigma = 0.47
TpRef = 58504.83
VcmaxRef = 24.52
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
0
500
1000
1500

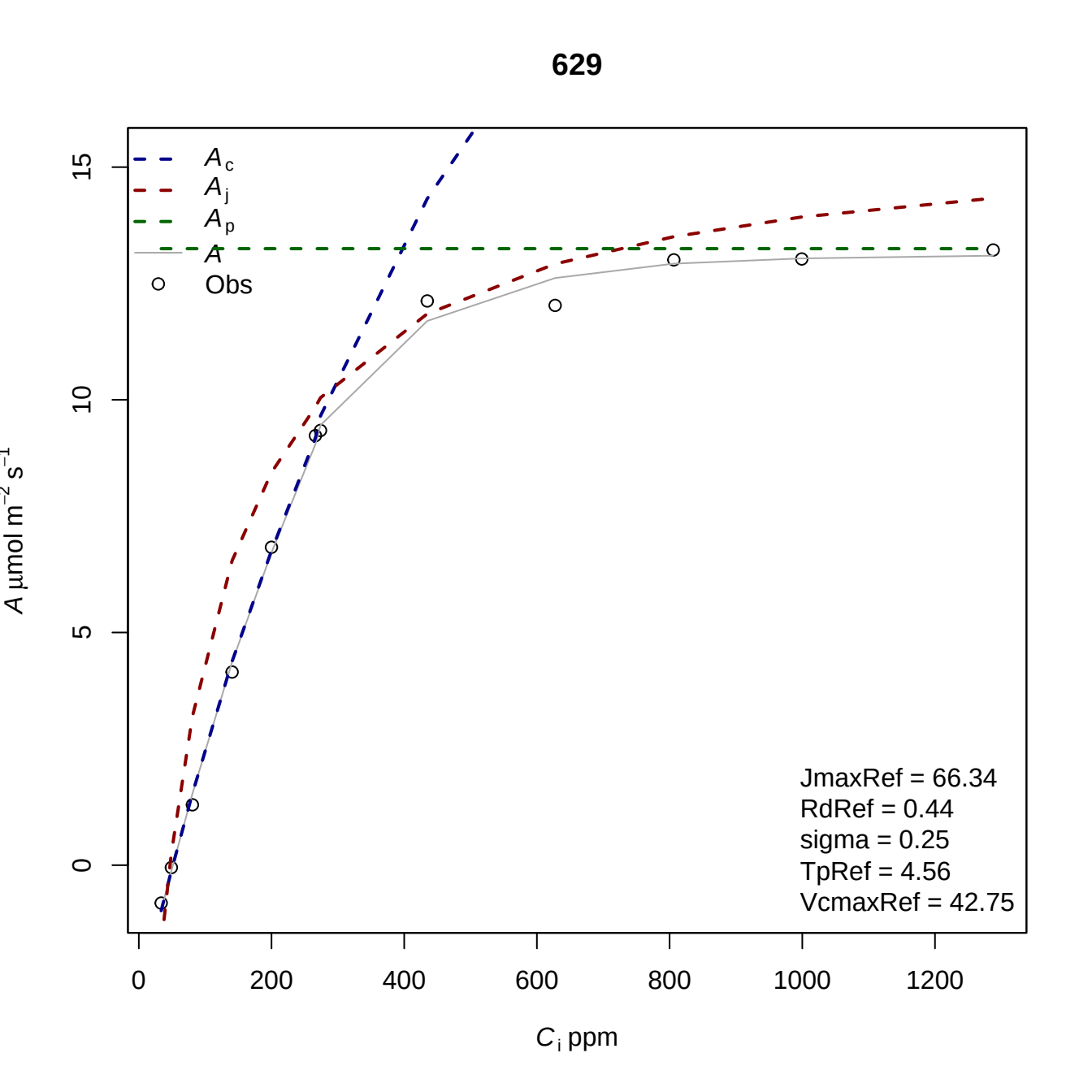

624
A_c
A_j
A_p
A
Obs
JmaxRef = 91.27
RdRef = 0.61
sigma = 0.23
TpRef = 241571.28
VcmaxRef = 60.14
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800

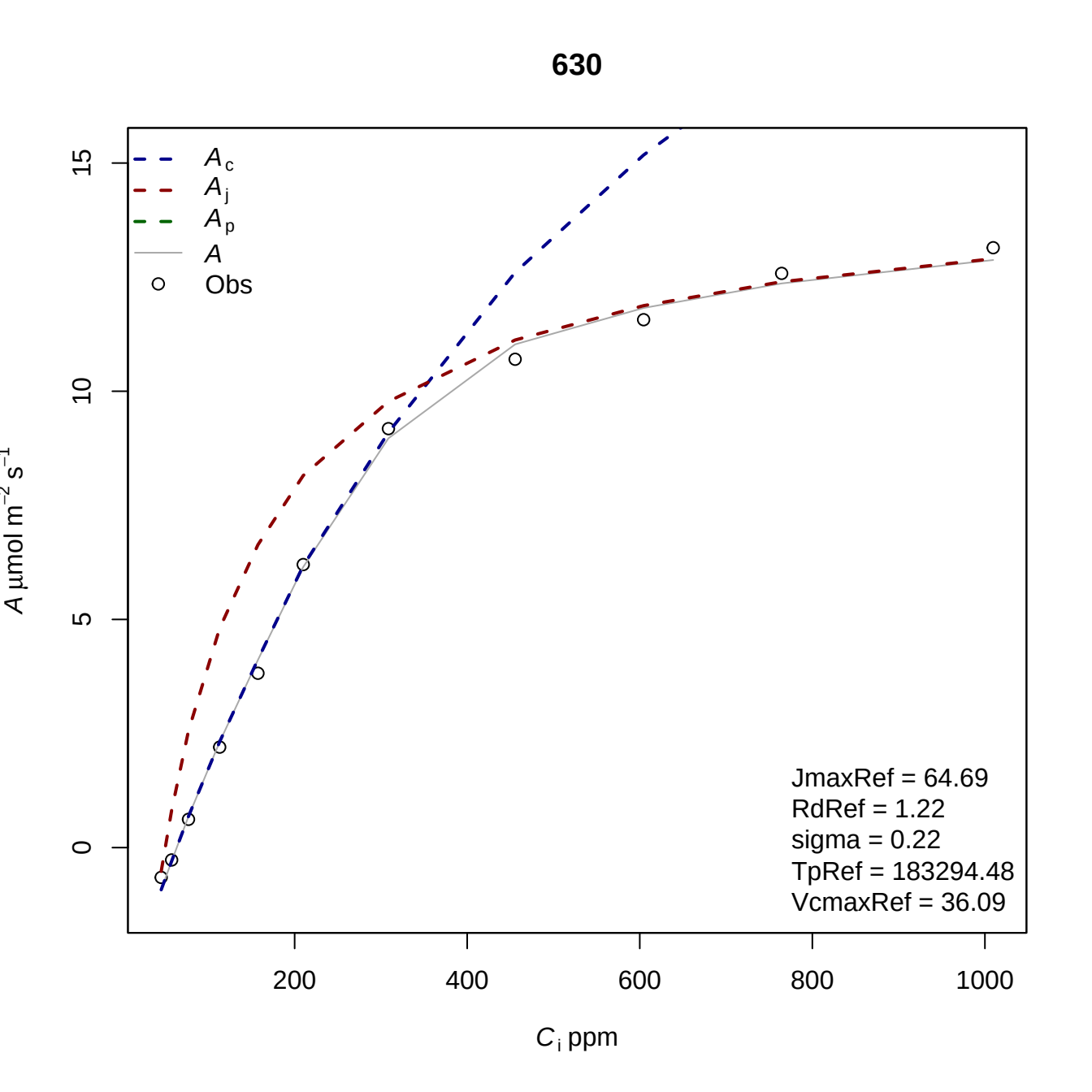

625
A_c
A_j
A_p
A
Obs
JmaxRef = 77.39
RdRef = 0.53
sigma = 1.26
TpRef = 5.45
VcmaxRef = 55.08
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

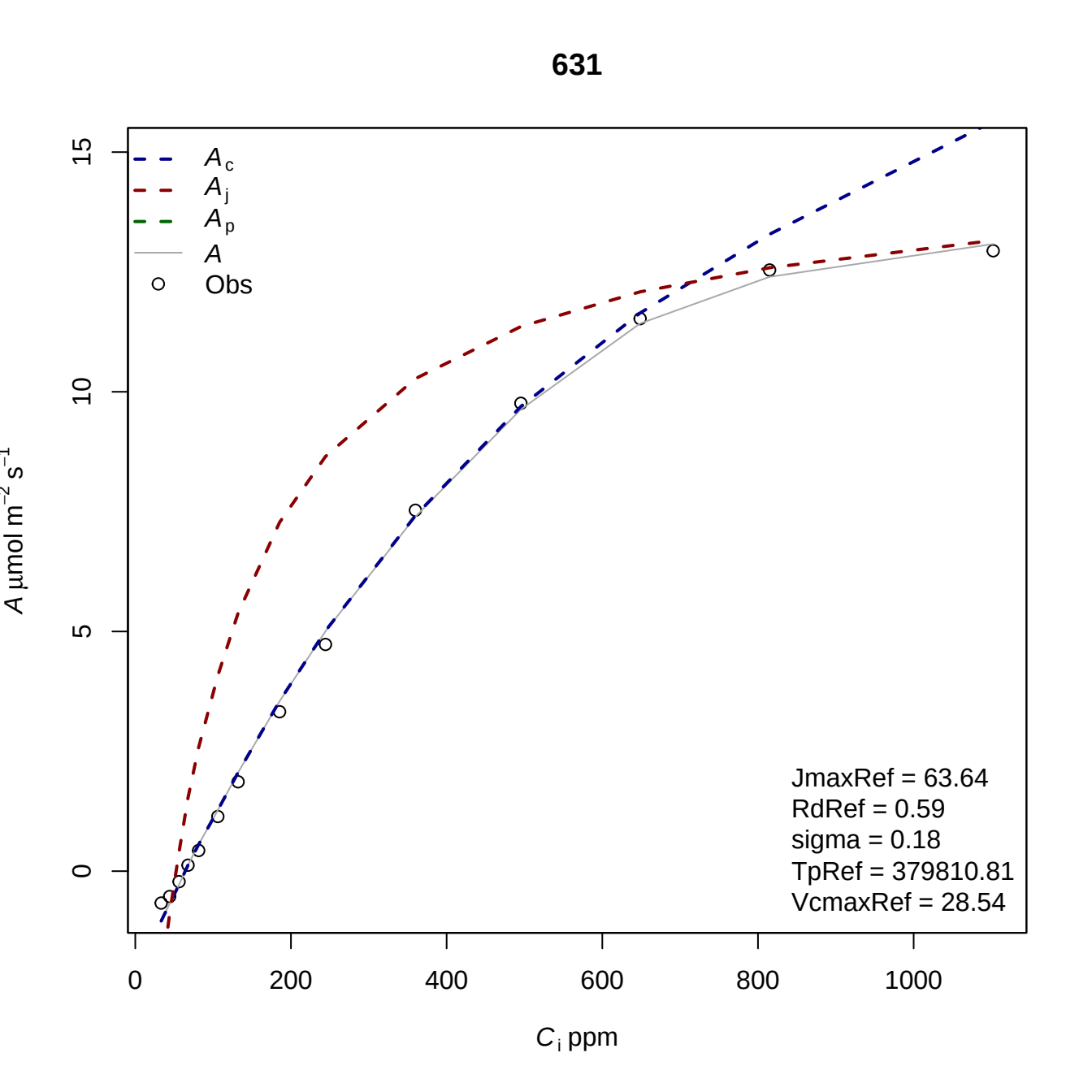

626
A_c
A_j
A_p
A
Obs
JmaxRef = 77.97
RdRef = 0.3
sigma = 0.31
TpRef = 6.59
VcmaxRef = 51.04
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
200
400
600
800
1000
1200

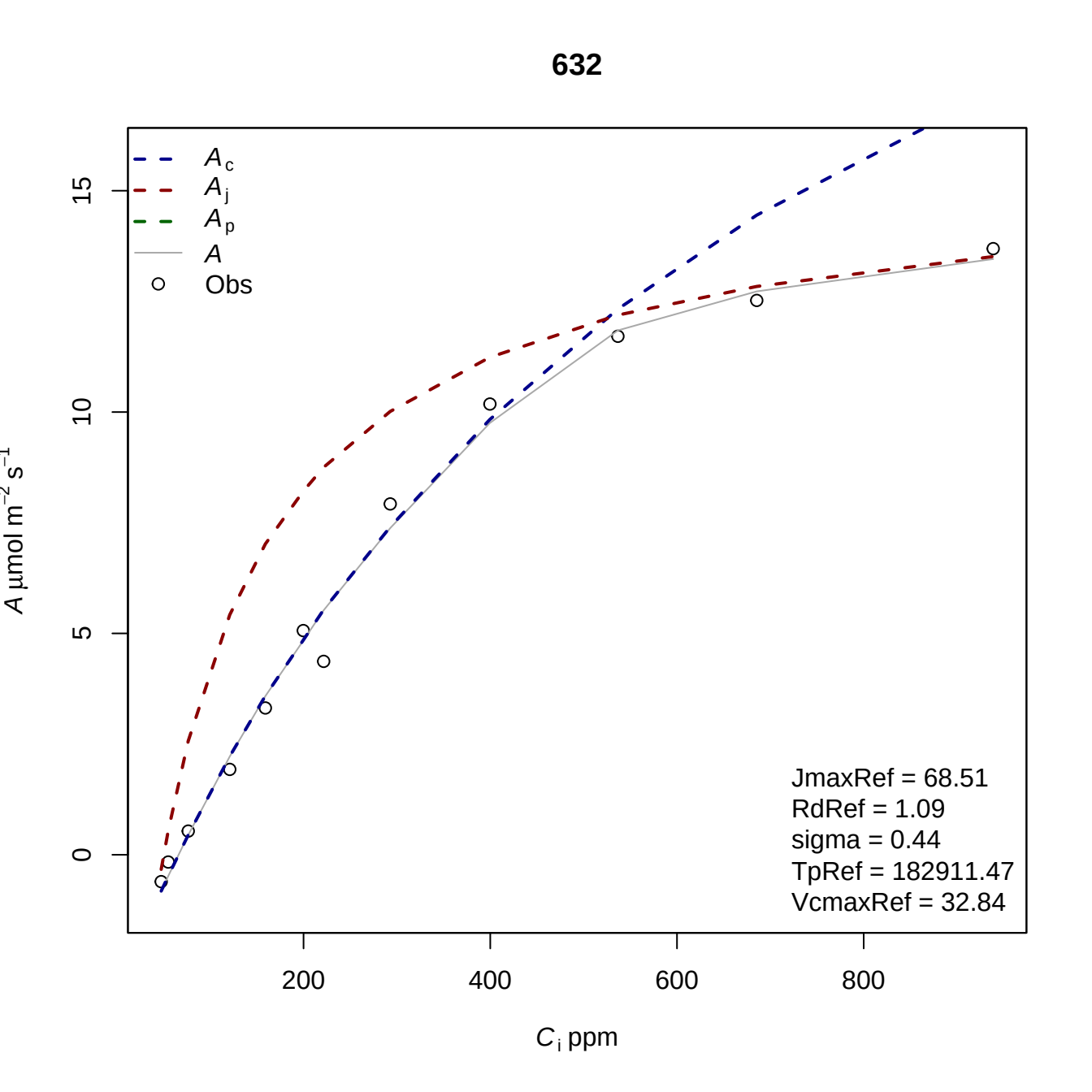

627
A_c
A_j
A_p
A
Obs
JmaxRef = 103.14
RdRef = 0.72
sigma = 0.79
TpRef = 8.72
VcmaxRef = 63.12
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
25
0
200
400
600
800
1000
1200

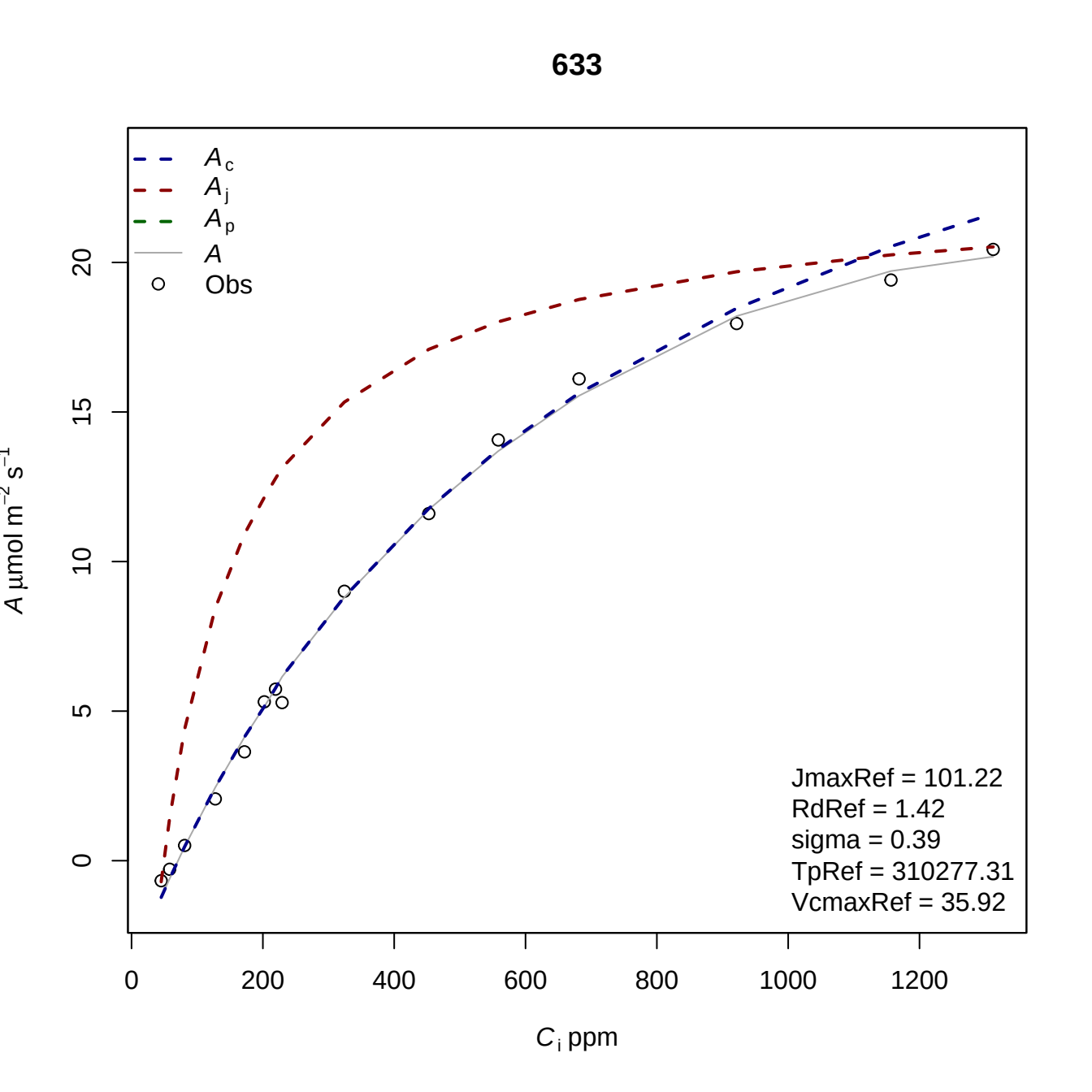

628
A_c
A_j
A_p
A
Obs
JmaxRef = 100.05
RdRef = 0.64
sigma = 0.25
TpRef = 8.23
VcmaxRef = 63.04
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

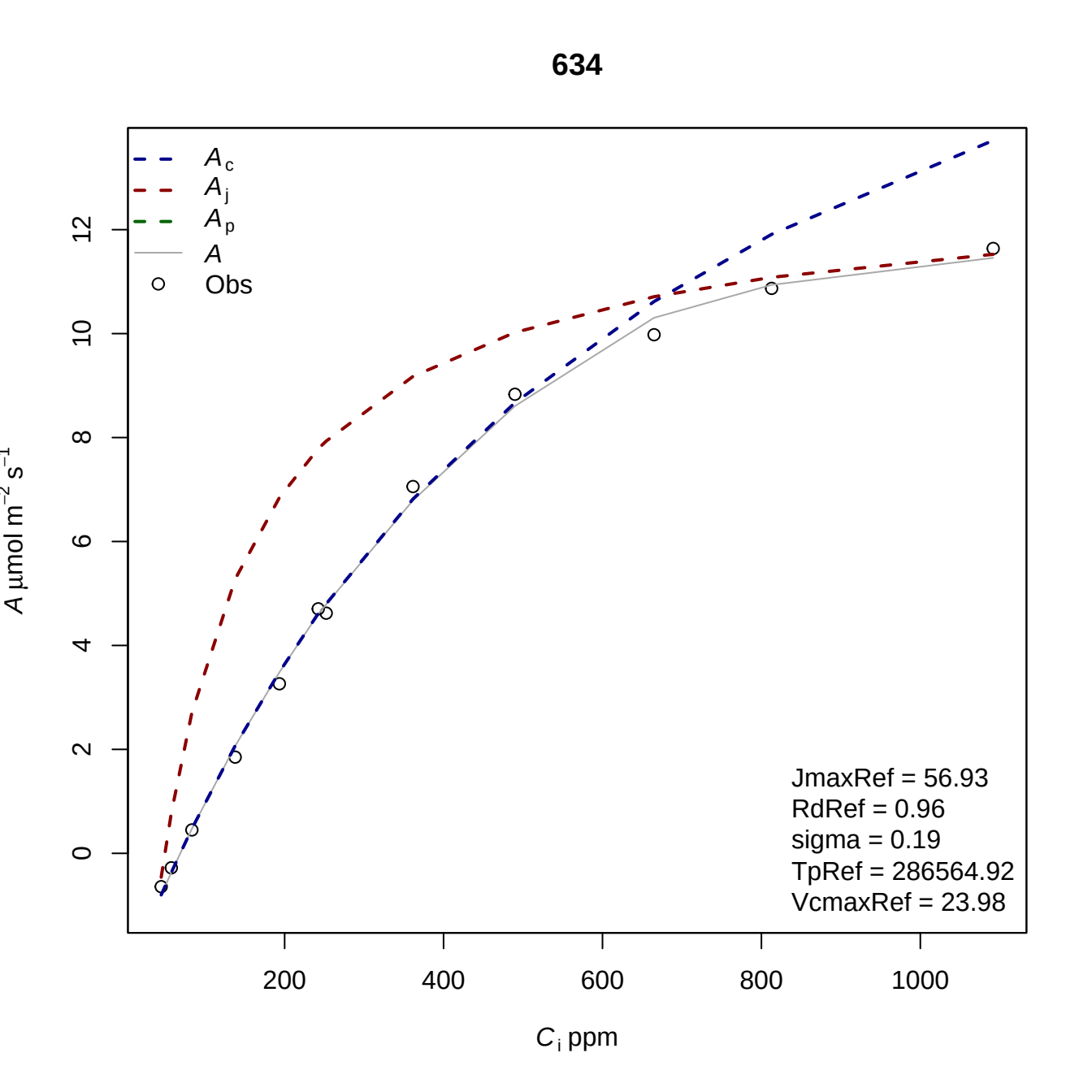

629
A_c
A_j
A_p
A
Obs
JmaxRef = 66.34
RdRef = 0.44
sigma = 0.25
TpRef = 4.56
VcmaxRef = 42.75
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
200
400
600
800
1000
1200

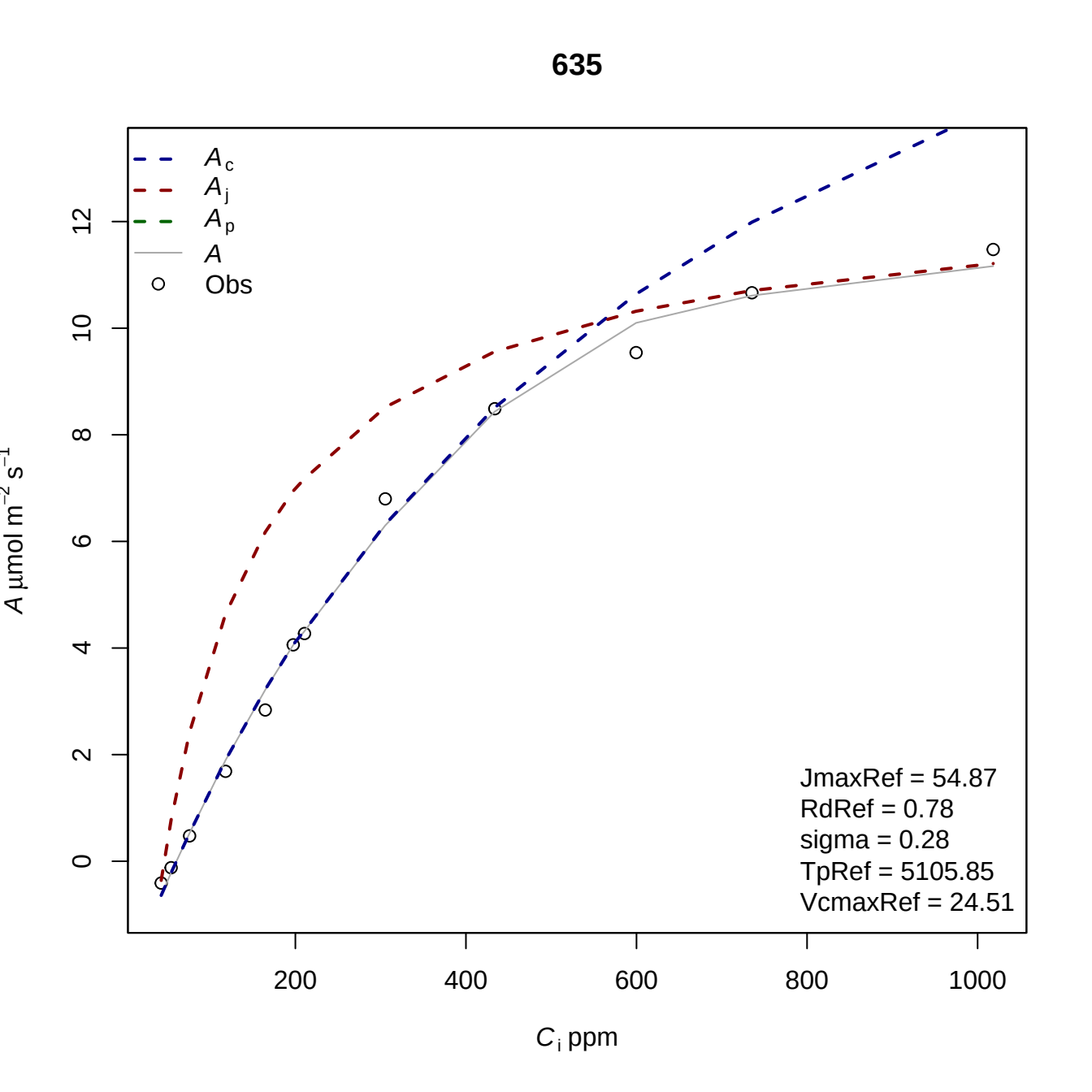

630
A_c
A_j
A_p
A
Obs
JmaxRef = 64.69
RdRef = 1.22
sigma = 0.22
TpRef = 183294.48
VcmaxRef = 36.09
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
200
400
600
800
1000

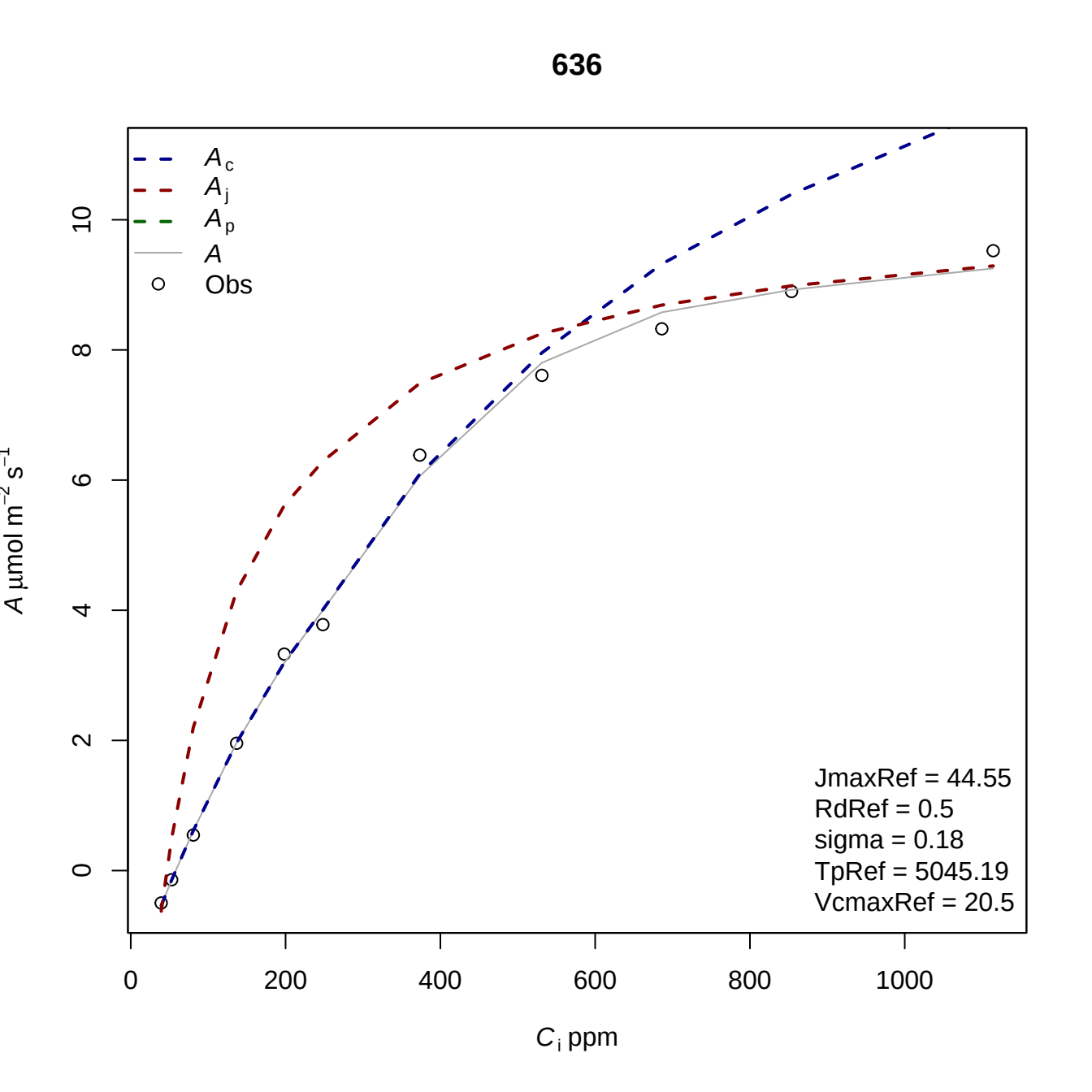

631
A_c
A_j
A_p
A
Obs
JmaxRef = 63.64
RdRef = 0.59
sigma = 0.18
TpRef = 379810.81
VcmaxRef = 28.54
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
0
200
400
600
800
1000

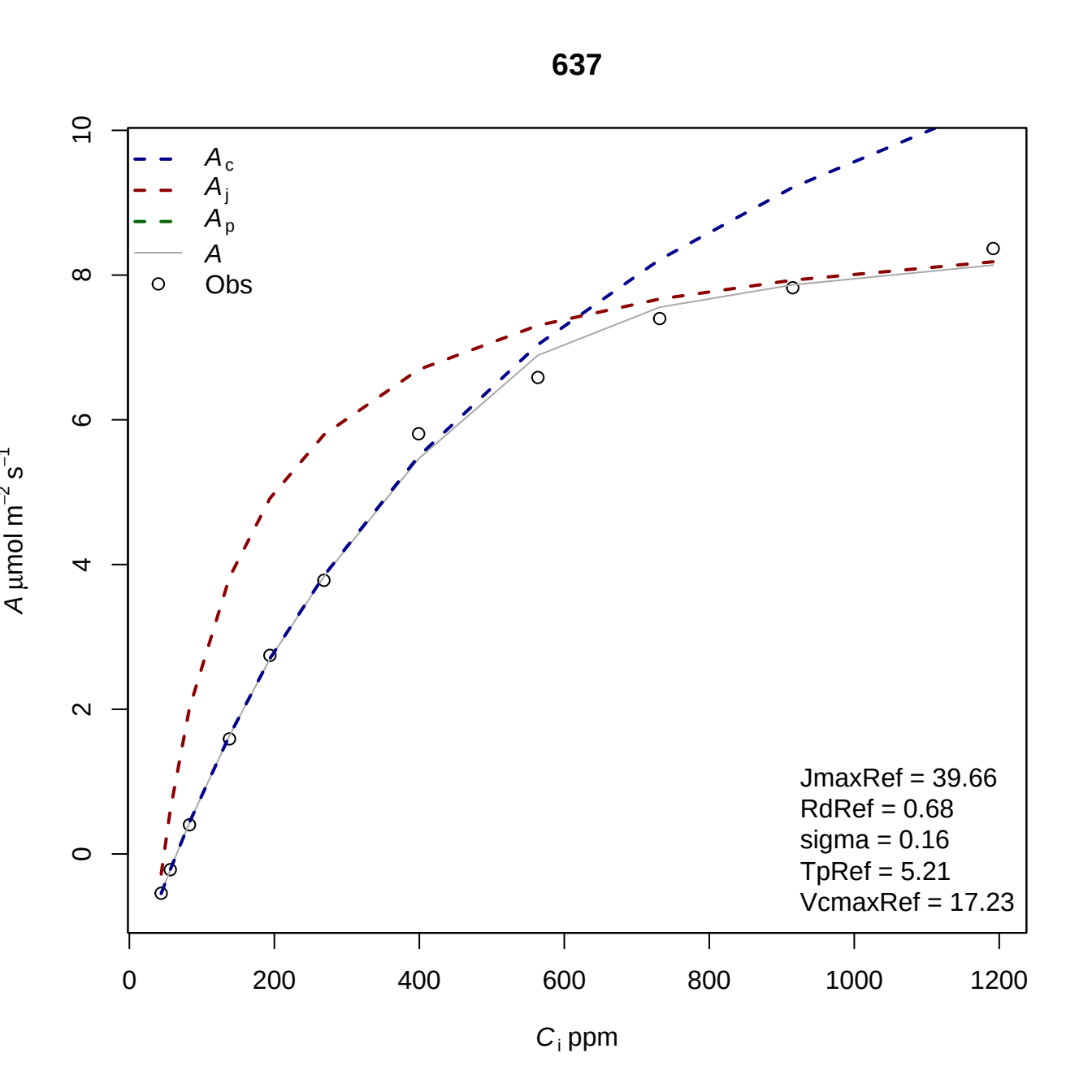

632
A_c
A_j
A_p
A
Obs
JmaxRef = 68.51
RdRef = 1.09
sigma = 0.44
TpRef = 182911.47
VcmaxRef = 32.84
0
5
10
15
200
400
600
800
A µmol m^{-2} s^{-1}
C_i ppm

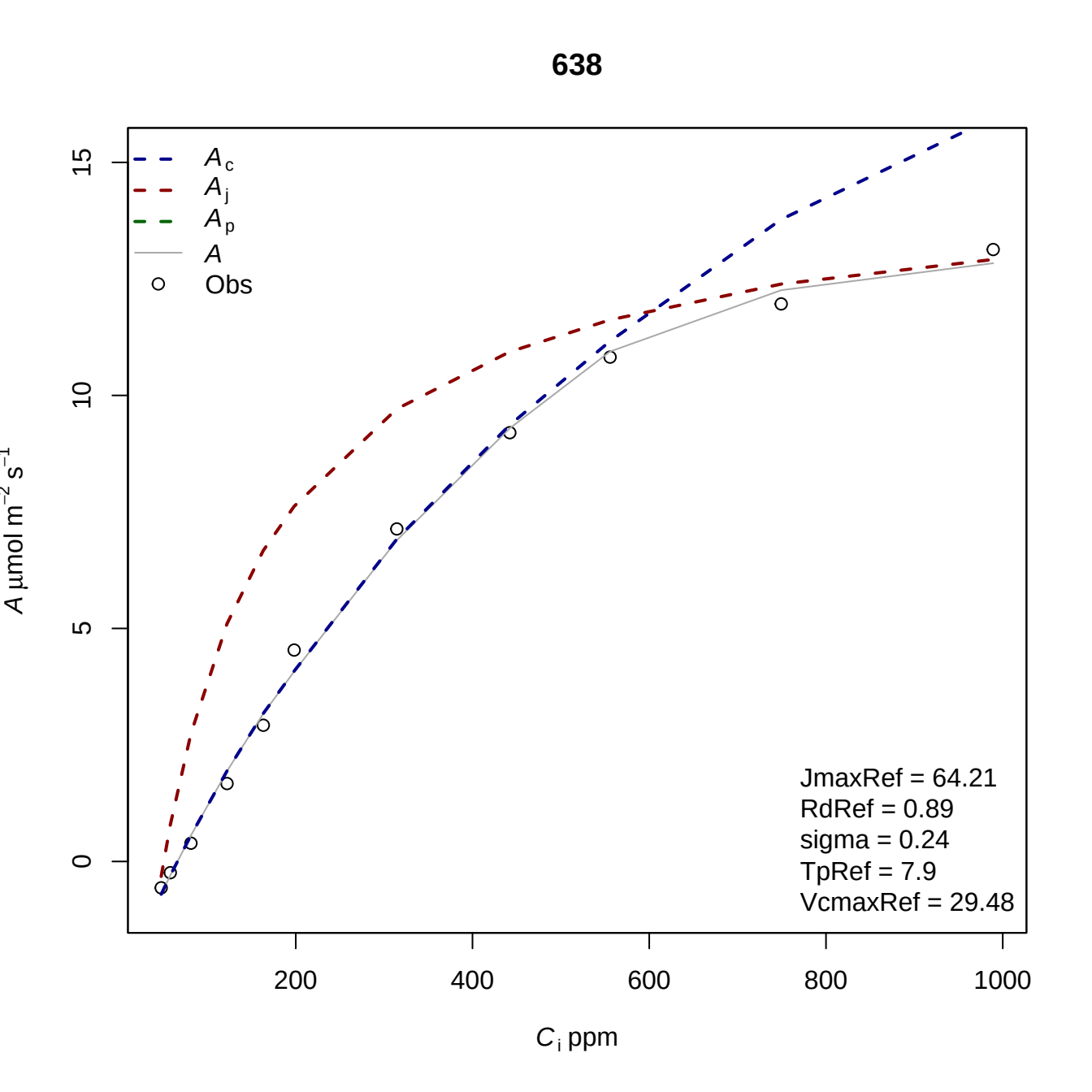

633
A_c
A_j
A_p
A
Obs
JmaxRef = 101.22
RdRef = 1.42
sigma = 0.39
TpRef = 310277.31
VcmaxRef = 35.92
A µmol m^{-2} s^{-1}
C_i ppm
0
5
10
15
20
0
200
400
600
800
1000
1200

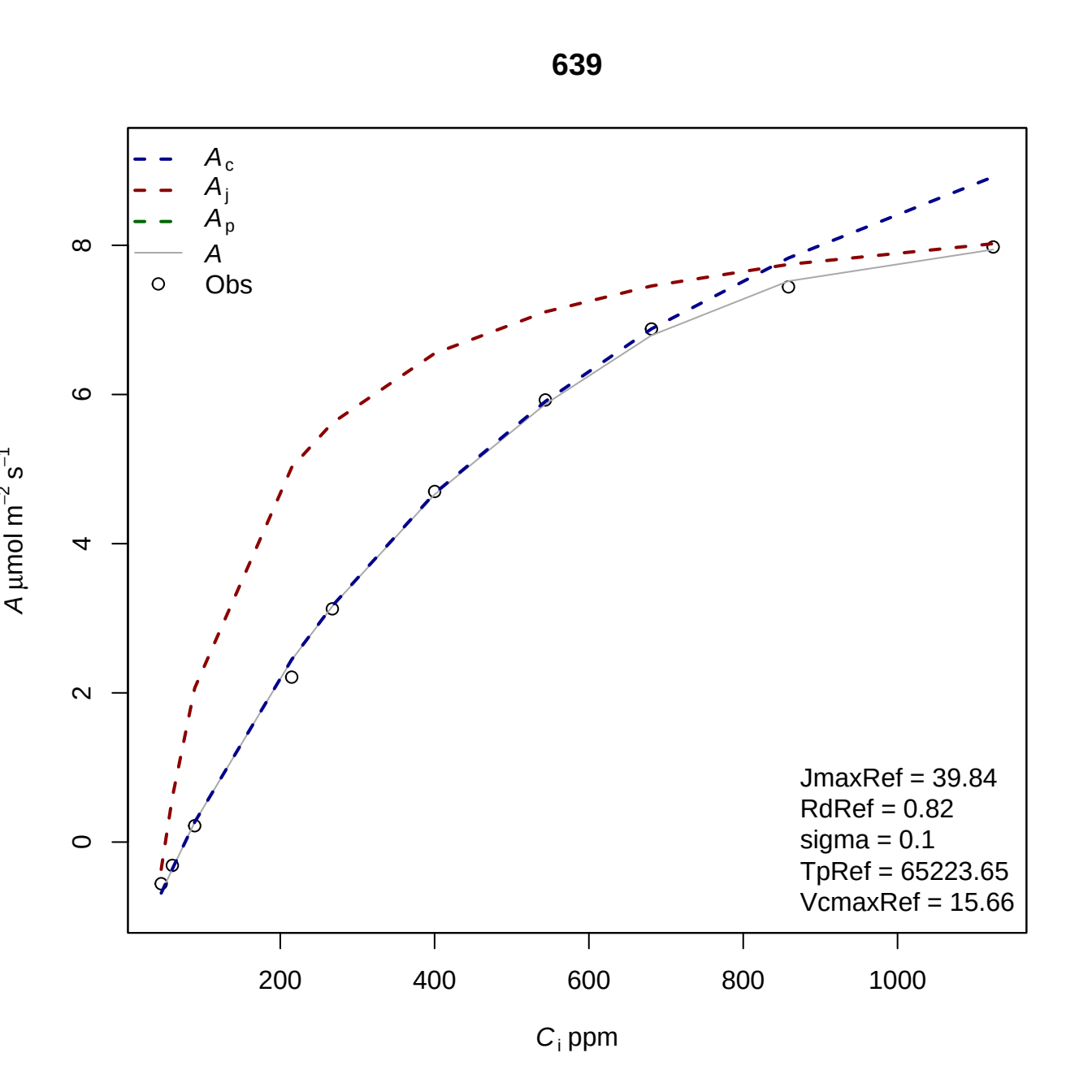

634
A_c
A_j
A_p
A
Obs
JmaxRef = 56.93
RdRef = 0.96
sigma = 0.19
TpRef = 286564.92
VcmaxRef = 23.98
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
200
400
600
800
1000

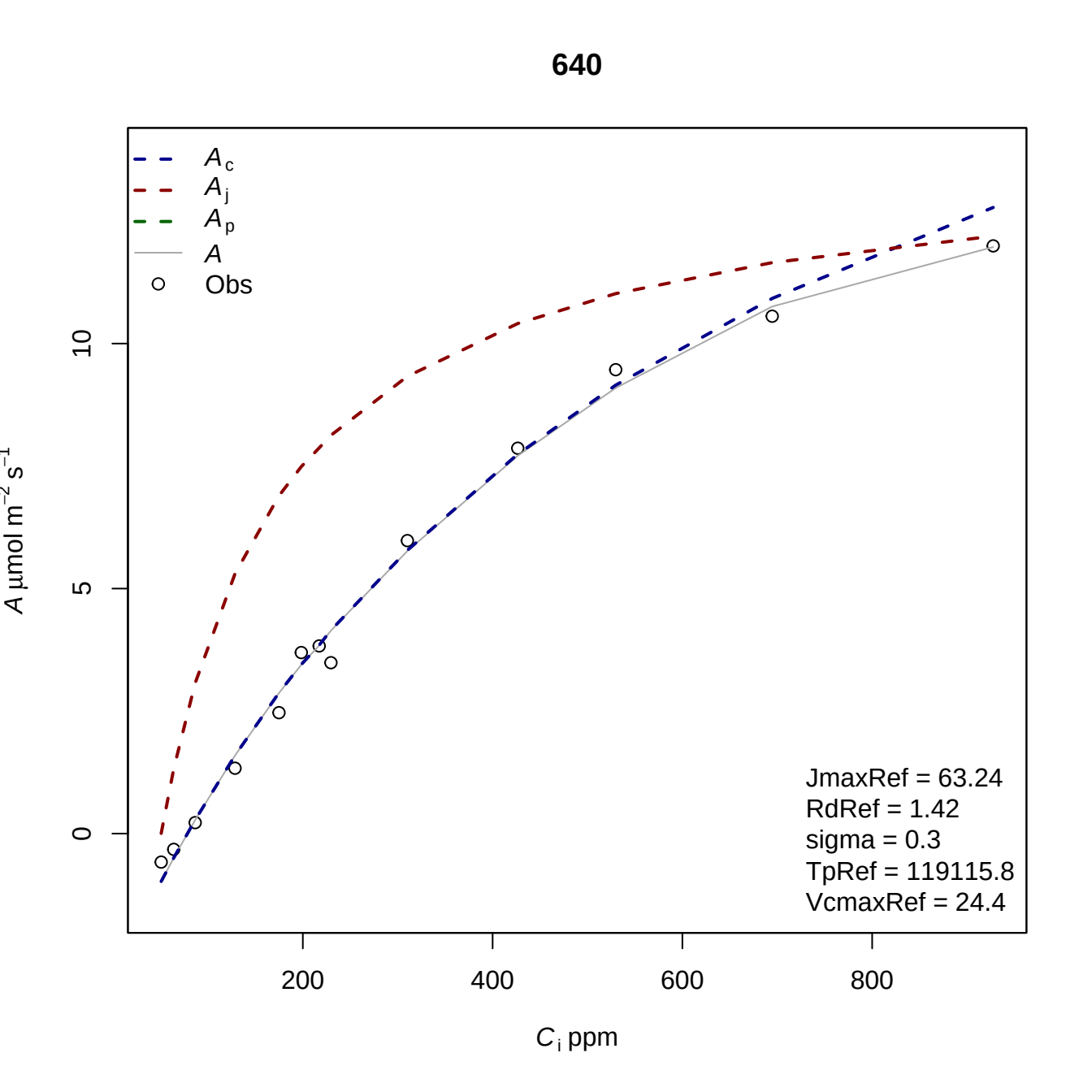

635
A_c
A_j
A_p
A
Obs
JmaxRef = 54.87
RdRef = 0.78
sigma = 0.28
TpRef = 5105.85
VcmaxRef = 24.51
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
200
400
600
800
1000

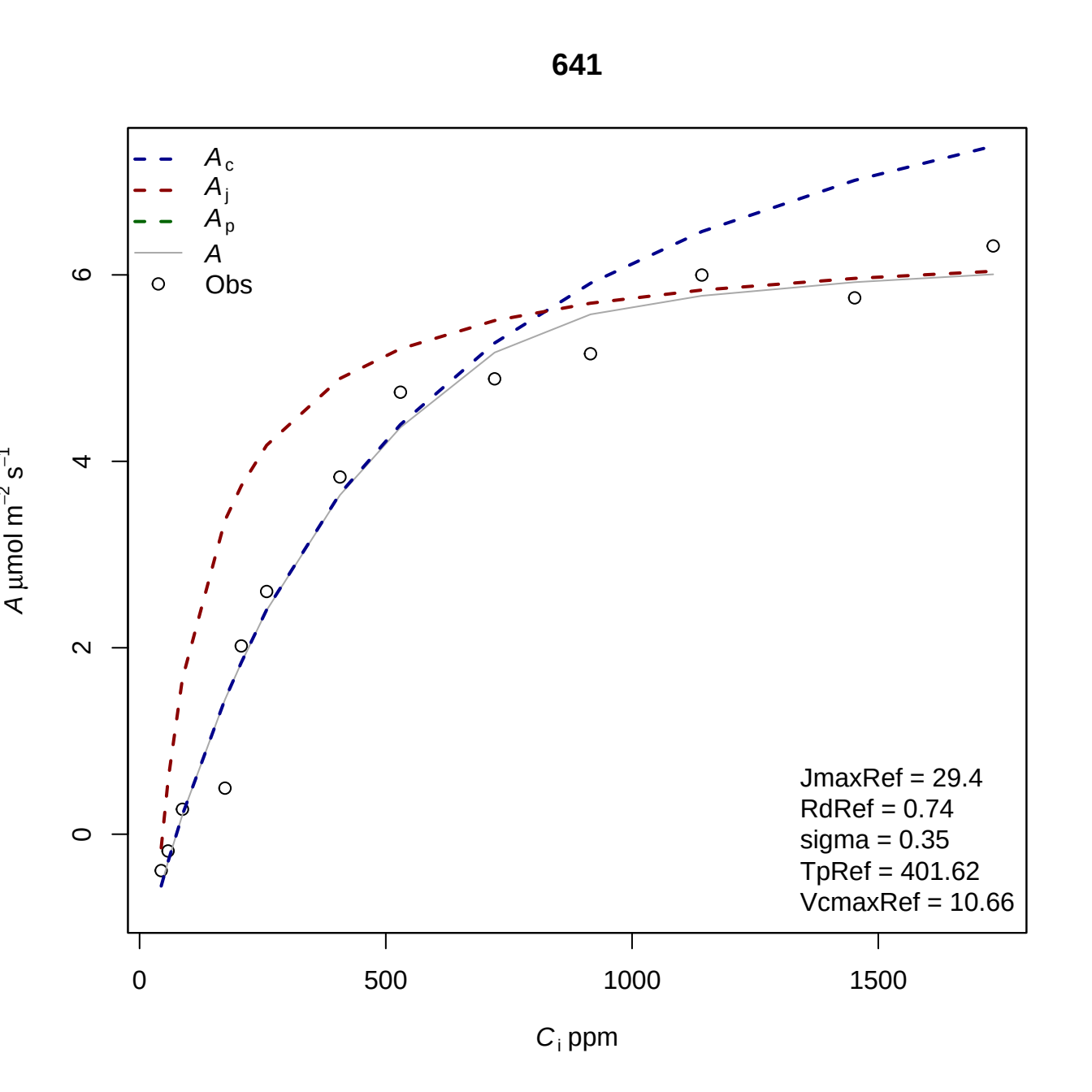

636
A_c
A_j
A_p
A
Obs
JmaxRef = 44.55
RdRef = 0.5
sigma = 0.18
TpRef = 5045.19
VcmaxRef = 20.5
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
0
200
400
600
800
1000

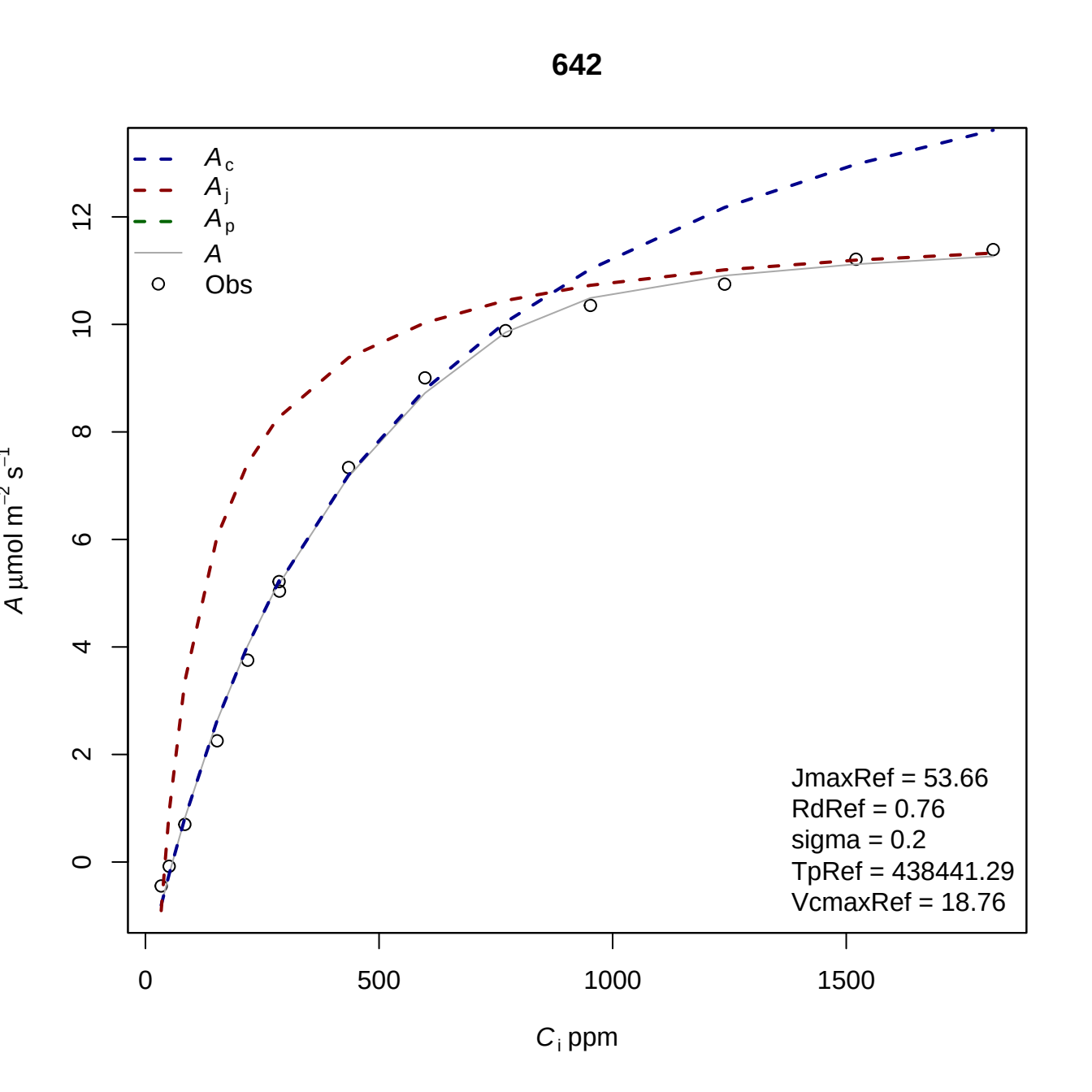

637
A_c
A_j
A_p
A
Obs
JmaxRef = 39.66
RdRef = 0.68
sigma = 0.16
TpRef = 5.21
VcmaxRef = 17.23
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
0
200
400
600
800
1000
1200

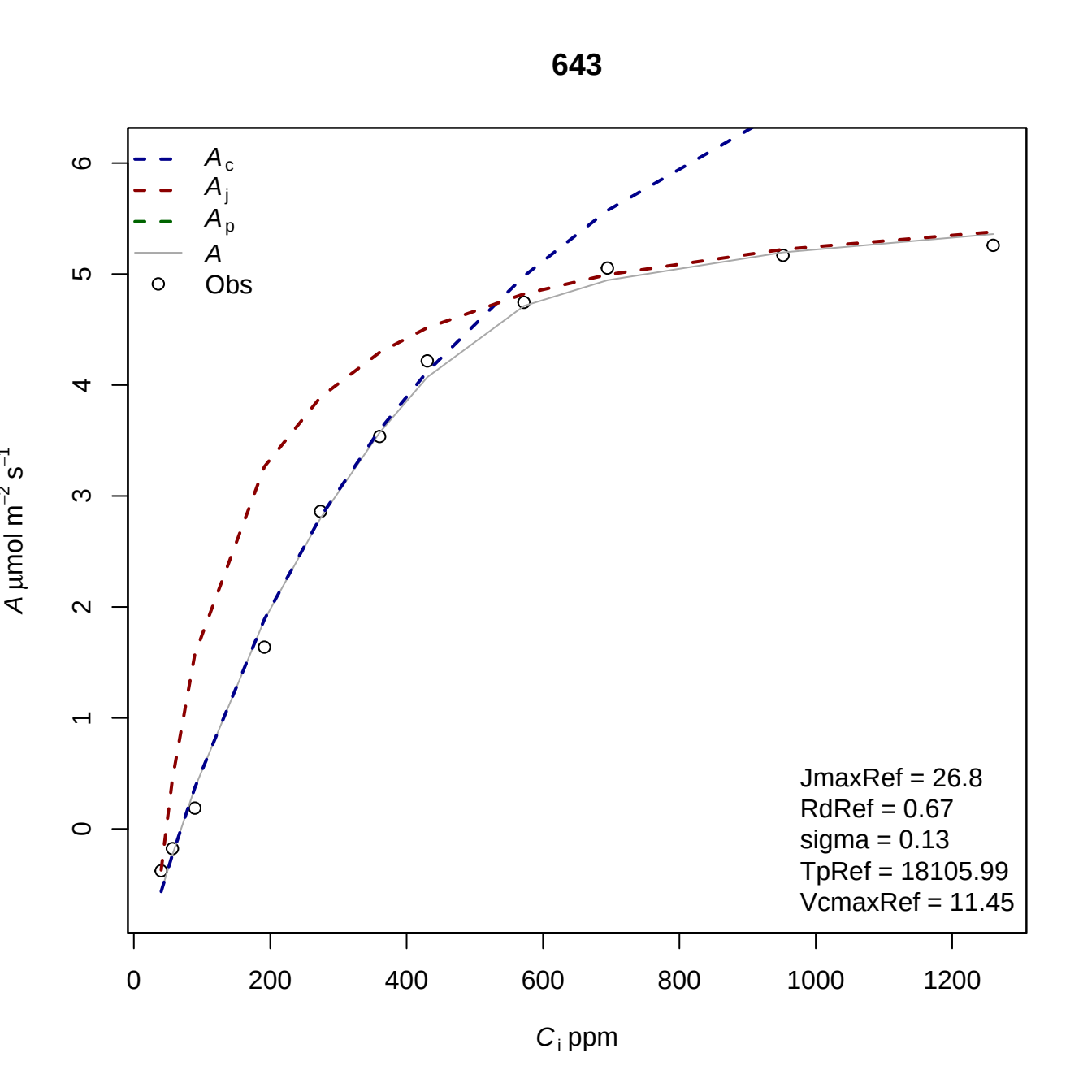

638
A_c
A_j
A_p
A
Obs
JmaxRef = 64.21
RdRef = 0.89
sigma = 0.24
TpRef = 7.9
VcmaxRef = 29.48
0
5
10
15
200
400
600
800
1000
A µmol m^{-2} s^{-1}
C_i ppm


639
A_c
A_j
A_p
A
Obs
JmaxRef = 39.84
RdRef = 0.82
sigma = 0.1
TpRef = 65223.65
VcmaxRef = 15.66
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
200
400
600
800
1000


640
A_c
A_j
A_p
A
Obs
JmaxRef = 63.24
RdRef = 1.42
sigma = 0.3
TpRef = 119115.8
VcmaxRef = 24.4
0
5
10
200
400
600
800
A µmol m^{-2} s^{-1}
C_i ppm


641
A_c
A_j
A_p
A
Obs
JmaxRef = 29.4
RdRef = 0.74
sigma = 0.35
TpRef = 401.62
VcmaxRef = 10.66
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
C_i ppm


642
A_c
A_j
A_p
A
Obs
JmaxRef = 53.66
RdRef = 0.76
sigma = 0.2
TpRef = 438441.29
VcmaxRef = 18.76
A µmol m^{-2} s^{-1}
C_i ppm
0
2
4
6
8
10
12
0
500
1000
1500


643
A_c
A_j
A_p
A
Obs
JmaxRef = 26.8
RdRef = 0.67
sigma = 0.13
TpRef = 18105.99
VcmaxRef = 11.45
A µmol m^{-2} s^{-1}
C_i ppm
0
1
2
3
4
5
6
0
200
400
600
800
1000
1200